LISTED

Architecting Large-Scale Classifieds and Marketplace Platforms

Željko Obrenović

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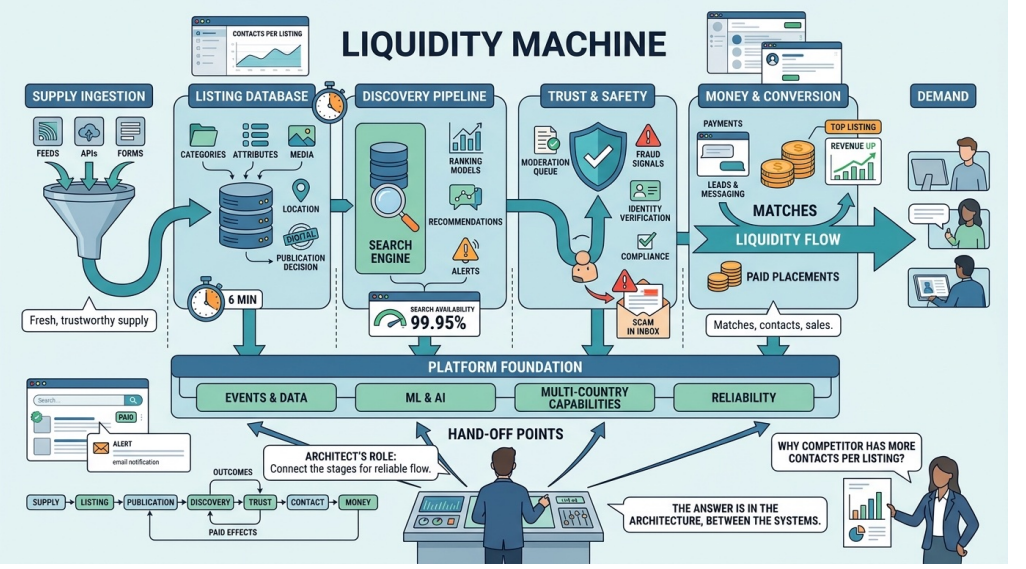
LISTED: Architecting Large-Scale Classifieds and Marketplace Platforms

Željko Obrenović

How architects and engineering leaders design, scale, govern and evolve the systems behind classifieds and listing marketplaces — listings, ingestion, search, trust, monetization and the multi-country platforms beneath them. This is a living journal and a work in progress: I revise it as I learn more through my work.

Preface

Introduction: Architecture for Liquidity



IN THIS BOOK, YOU WILL: Learn how to design, scale, govern and evolve the systems behind classifieds and listing marketplaces from the position of a Chief Architect of a multi-country, multi-vertical marketplace group. This page explains the book's argument, its nine parts, how to find the chapter for the decision already in front of you, and how to tell its documented facts from industry practice, labeled inference, recommendation and the fictional Veldmark.

WHY THE BUSINESS CARES: A marketplace earns its revenue from matches: contacts, leads and sales. Most of the reasons a marketplace loses matches (slow publication, sold items that stay visible, duplicates, scams in the inbox, paid placements that push out relevant results) sit in the systems, and they rarely appear in any one team's report.

WHY ARCHITECTS SHOULD CARE: In a marketplace group the most expensive architecture problems sit between systems and between brands: in

the hand-off from feed to index, from moderation to alerts, from package to ranking, from an acquired stack to the group platform. The Chief Architect is one of the few people whose job is to look at all of those boundaries.

KEY POINTS:

- **A listing marketplace is a liquidity machine.** Listings, search, trust and monetization are not separate products; they are stages of one pipeline whose throughput is liquidity. Judge the architecture by how reliably it turns fresh, trustworthy supply into matches with demand, in every country, category and vertical, while each marketplace evolves at its own pace.
- **Nine parts follow the machine from business model to practice.** The book moves from the business behind the listings, through modeling, supply, discovery, trust, money and the platform underneath, to the Chief Architect's long-term agenda and six publicly documented cases. The routing table on this page sends you straight to the chapter for the decision on your desk.
- **Every statement says what kind of statement it is.** Company-specific claims are documented facts with a source and a publication date, because marketplace groups reorganize, rebrand and change owners faster than books are revised. Industry practice, labeled inference, the author's recommendations and the fictional Veldmark are each marked so you can weigh them differently.

The question usually comes at the end of a quarterly business review, after the revenue slides. The general manager of the country's largest classifieds site puts up one more chart: **contacts per live listing**, theirs against a younger competitor's estimate. The competitor has a third of the listings and, by every indication, more buyers contacting sellers per listing. *Why?*

Every team in the room has a good answer for its own part. Search availability was 99.95% for the quarter. The moderation queue met its service level. Feed imports from professional sellers were green every night. Monetization had its best quarter, with paid "top listing" sales up a fifth. Nobody owns the question, because the question is not about any one of those systems.

If you are a Chief Architect, you know what to do next: follow one listing from the seller's first tap to the buyer's message. Usually the reasons turn up quickly. A new listing takes six minutes to become searchable because moderation and indexing run in batches, and buyers who check the app at 8 p.m. see the competitor's listings first. A bike sold on Monday still appears in Tuesday morning's alert e-mails. The same flat appears three times because three agencies listed it. The first page of results is half paid placements that the ranking cannot tell apart from relevance. **The general manager's**

question is an architecture question. Its answer runs through events that were never published, a search index that was never designed to forget, a ranking boost added years ago that nobody measured, and a messaging feature that was never treated as a trust surface.

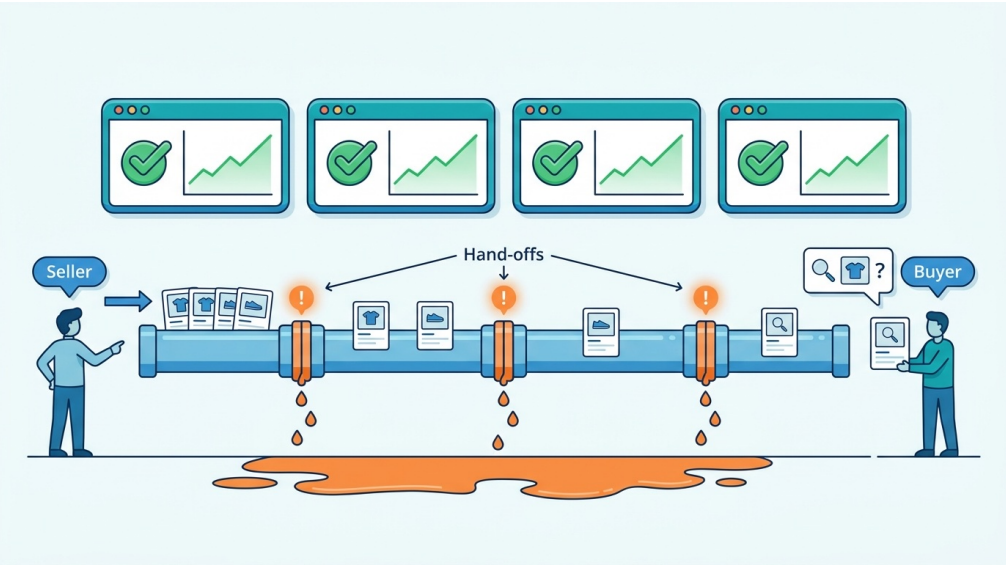


Figure 1: *Every team's own report can be green while matches leak at the hand-offs between systems, which is why the general manager's question is an architecture question.*

This book is about answering that kind of question, and about designing marketplaces where it is asked less often.

Who This Book Is For

The book is written for people who are accountable for how a listing marketplace's systems fit together:

- **Chief architects and enterprise architects of marketplace groups**, who must make one coherent system out of many marketplaces, several of which arrived by acquisition.
- **Principal and staff engineers** who own a stage of the pipeline (ingestion, search, trust and safety, messaging, monetization) and need to understand the stages on either side of it.
- **Search, data and machine-learning leads** whose ranking, recommendation, fraud and valuation models make decisions that sellers, buyers and regulators will ask about.

- **Platform-engineering leads** building capabilities that several brands and countries must adopt.
- **Product and engineering directors** of classifieds, real-estate, automotive, jobs and recommerce marketplaces.
- **Technology leaders integrating acquired marketplaces**, who must decide what to consolidate, in what order, and what to leave alone.

The book takes the perspective of a **Chief Architect of a multi-country, multi-vertical marketplace group**: several brands in several countries, each with its own history, traffic, sellers and stack, and an owner who wants growth and lower run costs at the same time. A single-country, single-vertical marketplace will find almost all of the reasoning applies; the trade-offs are drawn at group scale, where the cost of building a capability four times becomes visible.

What it assumes. Architecture literacy: you know what an API, a database, an event stream, a search engine and Kubernetes are, and you have made and defended architecture decisions. The book does not explain them.

What it does not assume. Marketplace economics, search and ML vocabulary, regulatory detail and vertical specifics. Liquidity, learning to rank, percolation, statements of reasons under the EU Digital Services Act (DSA) and vehicle identification numbers (VINs) are each defined where they first matter, in the chapter that owns them.

The Argument: A Liquidity Machine

A listing marketplace sells one thing: **matches**. A private seller wants a buyer for a city bike before Saturday. A dealer wants qualified leads for the cars on the lot. An agency wants mandates and viewings. A buyer wants the right item, car or home before someone else gets it. The marketplace does not usually own the goods, set the price or, in classic classifieds, touch the money. What it provides is the probability that the two sides find each other.

That probability is **liquidity**: the likelihood that a listing finds a buyer, or a search finds a listing worth contacting, within the time window that matters in its category. Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics defines it precisely. Two properties matter for architecture from the first page. Liquidity is **local**: it is measured per cell (a region crossed with a category), so a group can be liquid in Amsterdam bikes and illiquid in rural Portuguese furniture at the same time. And liquidity is **what compounds**: buyers come back where they find things, sellers list where buyers are, and every network effect a marketplace has is liquidity accumulated over years.

The book's central claim follows: **a listing marketplace is a liquidity machine, and its architecture should be judged by how reliably it turns fresh, trustworthy supply**

into matches with demand, in every country, category and vertical, while letting each marketplace evolve at its own pace. Listings, search, trust and monetization are not separate products; they are stages of one pipeline whose throughput is liquidity.

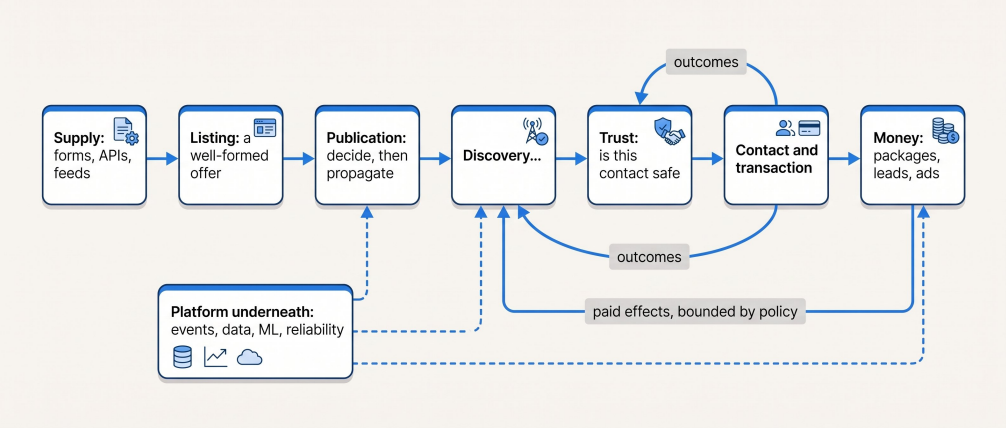


Figure 2: *The liquidity machine. Supply becomes listings that are published, found, trusted, contacted and paid for; outcomes flow back into discovery and trust, and paid effects flow back into discovery under policy. Each part of the book designs one stage, and the platform carries them all.*

Read that way, every architectural choice is judged by the liquidity it protects or spends. A freshness objective is no longer an engineering preference; it is justified by the liquidity metric it protects, such as time to first contact. A removal that reaches search but not the alert digest is not a minor bug; it spends buyers' trust on an item that no longer exists. A paid placement is not only a product line; it is a sale of liquidity that must earn more than it costs the match rate. And a shared platform in a group is not good because it is shared; it is good if every brand's machine runs faster on it than without it.

The chapters work on one stage at a time and do not stop to restate the argument. The last chapter of Part VIII, Built to Be Replaced: Evolving a Marketplace Platform Over a Decade, draws it together: it maps each part to the stage of the machine it designs, the guarantee that stage must give, the liquidity metric it protects and the failure pattern that appears when it does not, and turns that map into marketplace fitness functions a group can measure every quarter.

Why This Book Exists

Good material exists on each side of this subject. Search-engine, streaming and cloud documentation explains the engines. Marketplace economics, from two-sided market theory to practitioner writing on network effects, explains why liquidity matters.

Marketplace companies have published years of engineering posts, talks and annual reports, some remarkably candid.

What I have found missing, working with marketplace architecture, is the material **between** them. Engine documentation explains how to run an index; it does not say that the index must never become the system of record for listings, or what to do when a dealer's truncated file unpublishes tens of thousands of cars at three in the morning. Economics explains liquidity; it does not say which latency target, event contract or duplicate policy protects it. Company engineering posts describe one team's solution at one moment, usually without the constraints, the alternatives or what happened two years later.

The gap is most expensive at **group scale**. The classifieds industry has spent twenty years assembling portfolios: country marketplaces launched from a shared template, vertical portals bought one at a time, groups merged, demerged, taken private and partly sold again. From Newspaper Columns to Portfolios: How the Classifieds Industry Evolved tells that history from the public record. Each wave left systems behind: in many groups, several copies of identity, search, moderation and data sit beneath brands that buyers see as unrelated. **The failures appear at the boundaries**, between stages of the pipeline and between brands, and the boundaries belong to no project, no vendor and often no team. The Chief Architect is one of very few people with a mandate to look at all of them, between a board asking about synergies and the engine room where the costs sit. This book is written from that position.

How This Book Is Written

Company-aware but company-neutral. The book draws on the public engineering and investor record of Adevinta, OLX Group, eBay Classifieds Group, AVIV Group, Kleinanzeigen, Marktplaats, leboncoin, mobile.de, Auto Trader, Rightmove, Zillow, Idealista, CoStar and others, because architects work with real precedents and generic advice does not survive contact with them. But it organizes the argument around architectural questions, not company profiles. Engines and vendors (Elasticsearch, OpenSearch, Solr, Vespa, Kafka, Adyen and others) appear where the choice between them is the decision; no chapter recommends a product in general.

Five kinds of statement. Architecture writing becomes unreliable when facts, habits, guesses, opinions and examples blur into one confident voice. Every chapter keeps them apart, and the wording and the sources show which is which:

Kind of statement	What it is	How to recognize it
Documented fact	A company's public statement, a filing, a regulation, a standard, a date	Linked inline to the source the first time it matters, with the **year of publication** in the sentence ("in a 2025 engineering post…", "the 2023 annual report states…"), and listed in the chapter's Sources and Evidence and in the Bibliography: Sources and Evidence.
Industry practice	What marketplaces commonly do	Phrased as "commonly", "a typical pattern is"; no invented percentages. It describes the field, not a rule.
Reference-architecture inference	How a system like this is plausibly built, reasoned from engineering principles	When it concerns a real company, it is labeled explicitly, as a callout marked **INFERENCE, NOT DOCUMENTED** or as "a plausible reference architecture, not a description of that company's systems". It is never presented as a company's internals.
Recommendation	My judgment as a Chief Architect	First person ("I would", "the rule I use"), with the reasoning and the conditions under which it does not hold. No source validates it; argue with it.

Kind of statement	What it is	How to recognize it
Illustration	An example set at Veldmark	Labeled fictional the first time Veldmark appears in each chapter; its numbers come from one canonical record and agree across the book.

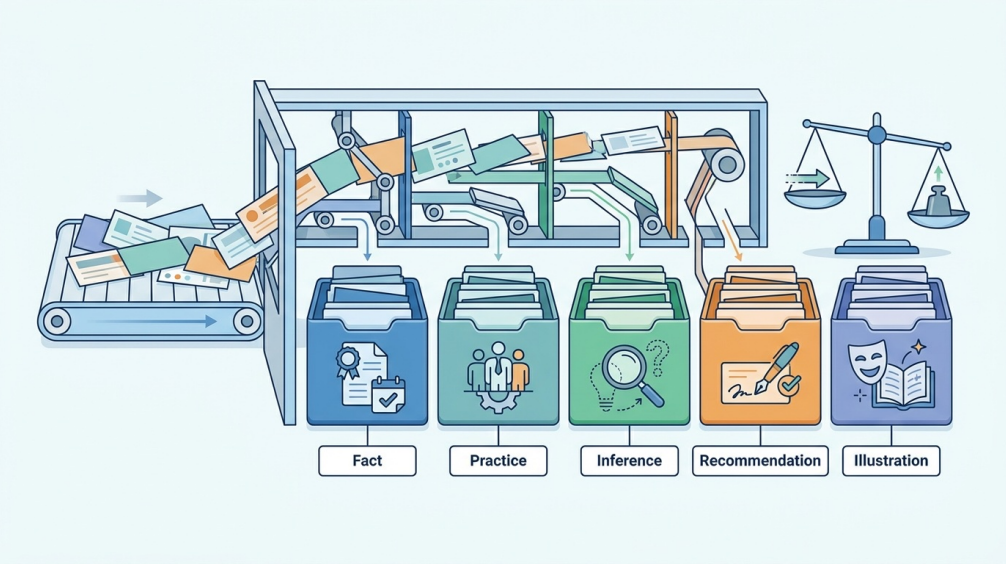


Figure 3: *The book keeps five kinds of statement apart, from dated documented fact to fictional illustration, so each can be weighed for what it is.*

Three of these distinctions are easy to lose. A Veldmark figure, such as six minutes from submission to searchable, is an illustration, not a benchmark. An inference about how a public marketplace probably indexes its listings is not evidence of how it does. And a recommendation that sounds like a fact, such as "the search index is not the system of record", is still a recommendation, which is why each one comes with its reasoning and its exceptions.

Why company claims carry dates. Marketplace groups change faster than their engineering blogs. Adevinta's 2023 annual report (published 2024) described turning the leboncoin and Kleinanzeigen platforms into "a unified global platform" for its generalist marketplaces; the same report records the rebrand of eBay Kleinanzeigen as Kleinanzeigen. After a 2024 take-private, the group's 2025 strategic report (published 2026) describes a move to "a decentralised, standalone operating model". Both statements were true when they were written. A chapter that cited the first without its date would describe a strategy the company had since reversed. So every company-specific

claim says when it was published, and where the evidence is more than about three years old, the chapter says that the current state may differ. Where a brand has been renamed or a business sold, the chapter uses the current name and mentions the former one once.

As of October 2026. Facts in this book were checked against public sources in October 2026 unless a chapter says otherwise. Treat any date-sensitive statement as a pointer to its source, and check the source before relying on it in a board paper or a contract.

What the Book Draws On

The book uses three kinds of material, and no other.

- **Two domain models.** The book started from two Productscape domain models in my sibling repository: a model of a horizontal general-listings marketplace and a model of a residential real-estate marketplace, each covering customers and their jobs, business capabilities, product streams, data assets and the stressors that test the architecture. Both are generic models written in the style of public operators, not descriptions of any company; the book never presents their contents as facts about a real marketplace or quotes their illustrative metrics as benchmarks.
- **The public record of the named companies.** Engineering blogs and talks, annual and strategic reports, filings, press releases and help pages, each cited with its publication year. Where the record stops, the chapter says so.
- **Standards, regulation and research.** EU regulations from the Official Journal, industry data standards for property and vehicle feeds, and published research and practitioner books on search, experimentation, machine learning and architecture.

No internal material. Nothing in this book comes from confidential documents, private conversations or inside knowledge of any company, and nothing describes a real company's internal architecture beyond what that company has published. Where a chapter needs a realistic situation the public record cannot supply, it uses Veldmark.

How the Book Is Organized

The book opens with three preface pages, continues with eight parts that form a learning path along the machine, tests the patterns against public cases in Part IX, and ends with reference material.

Section	Question it answers	Chapters
Preface	What is this book for, how do listing systems fail, and which group do the examples follow?	This page, Unlisted: Fifteen Ways Listing Systems Go Wrong, Meet Veldmark: Four Brands, One Platform to Build
Part I — MARKETPLACE: The Business Behind the Listings MARKETPLACE	What business is behind the listings, and why does it decide the architecture?	1–6: taxonomy, liquidity, industry history, capabilities, bounded contexts, verticals
Part II — LISTING: Modeling the Thing Being Sold LISTING	What is the thing being sold, and how should it be modeled?	7–12: listing aggregate, categories, attributes, media, location, sellers and accounts
Part III — SUPPLY: Getting Inventory In SUPPLY	How does inventory get in, and how fast does it become searchable?	13–17: ingestion, normalization and enrichment, entity resolution, publication, professional tooling
Part IV — DISCOVERY: Matching Intent to Inventory DISCOVERY	How is buyers' intent matched to inventory that changes by the minute?	18–23: search, query understanding, ranking, recommendations, map search, alerts
Part V — TRUST: Keeping the Marketplace Safe TRUST	How does a marketplace keep contact between strangers safe?	24–27: trust and safety, fraud, identity, regulation
Part VI — MONEY: Conversion and Monetization MONEY	How do contacts become deals, and deals revenue, without selling out liquidity?	28–32: messaging and leads, payments, monetization, entitlements, advertising
Part VII — PLATFORM: The Distributed System Underneath PLATFORM	What distributed system do all of these stand on?	33–40: events, data management, web and SEO, data, experimentation, ML, multi-country platforms, reliability

Section	Question it answers	Chapters
Part VIII — LEADERSHIP: The Chief Architect's Agenda LEADERSHIP	How does a Chief Architect keep the machine coherent over a decade?	41–45: consolidation, organization, modernization, AI, long-term evolution
Part IX — PRACTICE: How Others Built It PRACTICE	What do publicly documented groups teach, and where does the evidence stop?	46–51: Adevinta, OLX, eBay Classifieds, AVIV, property portals, automotive
Reference	Where are the templates, definitions and sources?	Field Kit: Templates for the Marketplace Architect, Glossary: The Language of Listing Systems, Bibliography: Sources and Evidence

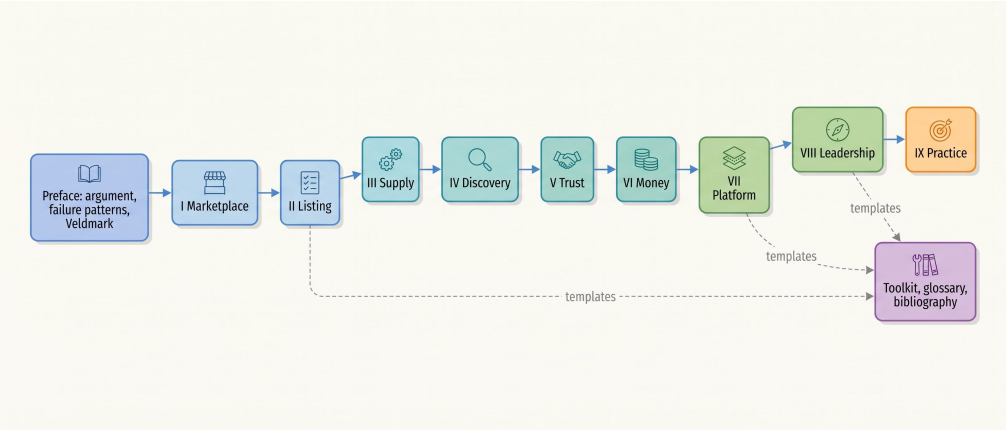


Figure 4: *Parts II to VI follow a listing through the machine, Part I sets the business terms they rely on, Part VII builds the platform beneath them and Part VIII keeps it coherent over time; Part IX tests the patterns against public evidence.*

Part I, MARKETPLACE, starts with the business, because in a listing marketplace the business model decides the architecture: a site whose job ends at the contact, a portal that sells leads and a marketplace that holds buyers' money need different identity, trust, payment and data systems. The part defines liquidity and the marketplace models, tells the industry's history from the public record, and describes the marketplace in capabilities and bounded contexts that outlive reorganizations. Do not skip it; later chapters justify their targets by pointing back to it.

Part II, LISTING, models the thing being sold. Its argument is that a listing is **an**

offer about a thing, not the thing itself: a small, strict aggregate with one owner and a lifecycle, surrounded by categories, attributes, media, location and accounts, each governed as data with its own owner and rate of change.

Part III, **SUPPLY**, puts that model under load: supply arrives through forms, partner APIs and bulk feeds from systems the marketplace does not control. Ingestion, normalization, duplicate resolution, publication and professional tooling form one chain with contracts, provenance and a freshness target, in which removal must be faster than publication.

Part IV, **DISCOVERY**, turns to the buyer. Listing search looks like web search but behaves like a high-write operational system: documents live for days, most queries carry no text, and quality is judged by how fast a new listing appears and a sold one disappears. Search, ranking, recommendations, map search and alerts are one pipeline with several surfaces.

Part V, **TRUST**, treats trust and safety as a platform capability, not a team with a queue: shared decision services, signals and evidence; local policy, thresholds and legal grounds. It covers fraud, identity and the regulations, chiefly the DSA, that become system requirements.

Part VI, **MONEY**, follows the conversion from first message to invoice. Its argument is that **conversion and monetization are one pipeline with a conflict built in**: many revenue lines earn more when buyers see more pages or when paid listings displace relevant ones, so every sale of attention must be measured, and capped, against what it costs the match rate.

Part VII, **PLATFORM**, builds the distributed system underneath, from listing events and systems of record to experimentation in a two-sided market, machine learning, multi-country tenancy and reliability at the weekly peak. Its claim is that a shared platform earns trust through contracts, not mandates.

Part VIII, **LEADERSHIP**, is the Chief Architect's agenda: consolidation after acquisitions in an order that protects liquidity, an organization in which every journey, context and shared capability has one owning team, modernizing a live marketplace without losing traffic, governing AI, and evolving the platform over a decade against stressors rather than toward a target state. Its claim is that the hard problems of a marketplace group are problems of change, not of design.

Part IX, **PRACTICE**, steps outside the fiction. Six cases (Adevinta, OLX Group, eBay Classifieds Group, AVIV Group, the major property portals and the automotive marketplaces) separate what each group has documented, with dates, from my interpretation, say where the public evidence stops, and test the book's patterns against what actually happened.

Each part opens with a short introduction. Each chapter follows the same shape: opening callouts and three key points; a Veldmark scene that poses the question; concepts, architecture, documented practice and a worked example; trade-offs, vertical differences

and failure modes; the Chief Architect's view; questions for a design review; and Sources and Evidence.

Start With the Decision in Front of You

The parts are a learning sequence for a first reading. If a decision is already waiting, find the row that matches it, read the chapter in the **Start with** column, and follow the **Then** column in order if you need more. Return to Part I whenever a marketplace term is unfamiliar.

Your immediate need	Start with	Then
The board asks whether the marketplace works, and the dashboards count listings and visits	Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics	Data Platform for Marketplaces: Events, Lakehouse and Data Products, Testing a Two-Sided Market: Experimentation Platforms, Built to Be Replaced: Evolving a Marketplace Platform Over a Decade
Sold or removed listings still appear in search, alerts or recommendations	From Submitted to Searchable: The Publication Pipeline	Every Change Is an Event: Event-Driven Listing Architecture, Tell Me When It Appears: Saved Searches and Alerts, System of Record and Read Models: Listing Data Management
Adding or splitting a category takes weeks and a cross-team release	Category Trees That Can Change: Designing Marketplace Taxonomies	Attributes Without Chaos: Schemas, Dictionaries and Validation, Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models, The Listing Aggregate: Identity, Lifecycle and Invariants

Your immediate need	Start with	Then
A partner's or dealer's feed breaks, and imports are a surprise every night	Many Doors In: Forms, APIs and Bulk Feeds	From Raw Supply to Usable Listings: Normalization and Enrichment, Serving the Pros: Dealer, Agent and Merchant Tooling
The same property or car appears many times in results	One Property, Three Listings: Duplicates and Entity Resolution	Ranking for Two Sides: Relevance, Freshness and Paid Visibility, Photos Sell: Media Pipelines for Listings, Zillow, Rightmove, Idealista and REA: What Property Portals Teach
We are choosing a search engine, planning a reindex or failing at the weekly peak	Search Architecture for Listings: Indices, Freshness and Scale	Sunday Evening Peaks: Reliability, Performance and Cost, What Did They Mean? Query Understanding and Autocomplete, Changing the Engine in Flight: Modernizing a Live Marketplace
Paid placements crowd the first page, and nobody can explain position 3	Ranking for Two Sides: Relevance, Freshness and Paid Visibility	Packages, Bumps and Quotas: Products, Pricing and Entitlements, How Listing Marketplaces Make Money, Testing a Two-Sided Market: Experimentation Platforms
Scams close in our messaging, or fake listings reach buyers within minutes	Scams, Fakes and Account Takeovers: Fraud Prevention for Listings	From Interest to Contact: Messaging and Lead Management, Trust and Safety as a Platform, Who Is on the Other Side? Identity and Verification

Your immediate need	Start with	Then
A regulator, an auditor or the DSA transparency database needs evidence we cannot reproduce	Rules of the Marketplace: DSA, DAC7 and Vertical Regulation	Trust and Safety as a Platform, System of Record and Read Models: Listing Data Management
A brand wants to add payments, shipping or buyer protection	Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy	Beyond the Contact: Payments, Shipping and Buyer Protection, Payments for Finance Architects: Eleven Reasons the Numbers Disagree, Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform
An acquisition just closed, and it arrives with its own stack	After the Deal: Consolidating Platforms After Acquisitions	Map What the Marketplace Does Before What Runs It, Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models, AVIV Group: Building a European Real-Estate Platform from Acquisitions
We want one platform for several countries, brands or verticals	Same Platform, Different Physics: How Listing Verticals Differ	One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture, Who Owns What: Organizing Marketplace Engineering, eBay Classifieds Group: Shared Platforms, Local Brands and a Sale

Your immediate need	Start with	Then
A legacy monolith blocks every team, and we must retire it without losing traffic	Changing the Engine in Flight: Modernizing a Live Marketplace	Every Change Is an Event: Event-Driven Listing Architecture, Found by Google: Web, App and SEO Architecture for Listings, Search Architecture for Listings: Indices, Freshness and Scale
We are piloting a photo-to-listing assistant or conversational search	AI Across the Listing Lifecycle	Models in Production: ML Platforms and Use Cases, What Did They Mean? Query Understanding and Autocomplete, Trust and Safety as a Platform
The owner asks what it would cost to sell or carve out one brand	Built to Be Replaced: Evolving a Marketplace Platform Over a Decade	One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture, After the Deal: Consolidating Platforms After Acquisitions, OLX Group: Regional Platforms, Verticalization and Retreat

Two rows start somewhere unexpected, on purpose. The payments row starts with the marketplace taxonomy, not with payments: moving from contact to transaction changes who holds the money and the risk, and that decides identity depth, trust and regulation before it decides a payment provider. And the single-platform row starts with what differs between verticals, because the most common failure I see is a shared platform designed for one vertical's physics and then imposed on the others.

The Failure Patterns

Listing marketplaces rarely fail through one bad decision. They lose liquidity through recurring patterns that each looked reasonable when it was chosen: a category tree hard-coded at launch, a dealer feed accepted as it was, one more ranking boost. The

next page, Unlisted: Fifteen Ways Listing Systems Go Wrong, names fifteen of them, from **The Category Straitjacket** and **Feed Roulette** through **Ghost Inventory**, **Search as System of Record** and **Multiplier Creep** to **The Copy-Paste Country**, **Deal-Shaped Architecture** and **Set-and-Forget AI**. Each comes with its warning signs, its mechanism, its cost and the chapters that address it, so a design review can say "this is Feed Roulette" and move straight to the remedy. Every chapter in Parts I to VIII opens by naming the patterns it works on.

Veldmark and the Toolkit

Veldmark Group is fictional. It is a privately held marketplace group headquartered in Rotterdam, with €1.31 billion of revenue in 2026 and four brands in eight European countries: Tradeyard, a horizontal classifieds site with paid visibility and an in-platform payment service; Motorvane, an automotive marketplace that sells dealer packages and leads; Hearthly, a real-estate portal assembled from acquisitions in France and Spain; and Keybridge, an acquired CRM for estate agents that also publishes their listings to Hearthly's competitors. It runs three clouds, three search engines, four identity systems and a PHP monolith from 2006, and in January 2027 its board approves Common Ground, a €140 million platform investment governed by one principle: **share platforms, not products**. Meet Veldmark: Four Brands, One Platform to Build introduces its brands, systems, people and the worked examples the chapters reuse, such as the €240 Rotterdam city bike that is sold on Monday and still advertised in 214 alert e-mails on Tuesday. Veldmark is a composite of patterns common in European marketplace groups, not a disguise for any of them. Use it as a rehearsal space: when a chapter shows a decision that Ingrid Halvorsen, its Group Chief Architect, makes, ask what the equivalent decision would be in your marketplace and what would change it.

The **Field Kit: Templates for the Marketplace Architect** collects the book's instruments as twelve templates, from the capability map and the feed contract to the listing event specification, the entitlement matrix and the architecture decision record. Do not adopt them all at once; start with the one that changes the decision on your desk this week. The Glossary: The Language of Listing Systems defines the book's terms, grouped by part, and names the chapter that owns each one.

A Living Journal

This book is a living journal, and a work in progress, that I revise as I learn more through my work. Every chapter has a "View spec" link in its byline that opens the specification it was written against: its intent, its audience, its success criteria, what it deliberately leaves to other chapters, and the open questions it has not settled. Each spec ends with a changelog recording what changed and why. Read the book as current thinking with its limits stated, not as a finished text, and use the specs to see where those limits are.

I developed the book through AI-mediated authoring: I set the concept, the structure, the chapter contracts and the arguments; AI tools supported research against public sources, drafting from those contracts, consistency checks across chapters and editing; I reviewed and decided. The specs and the scripts used to build this site are in the book's GitHub repository. The content, its judgments and its errors are my responsibility.

Where to Go Next

If you read in order, turn next to Unlisted: Fifteen Ways Listing Systems Go Wrong. It names the fifteen failure patterns the rest of the book keeps returning to, and will help you recognize which are already in your marketplace. Then read Meet Veldmark: Four Brands, One Platform to Build, so that every chapter's opening scene lands in a group you know. If a decision is waiting, use the routing table and come back to the preface later. Either way, keep the general manager's question in mind: why does a smaller competitor turn its listings into more contacts? Every chapter is part of the answer, and part of the design of a marketplace that turns fresh, trustworthy supply into matches with demand.

Sources and Evidence

Company engineering and investor sources

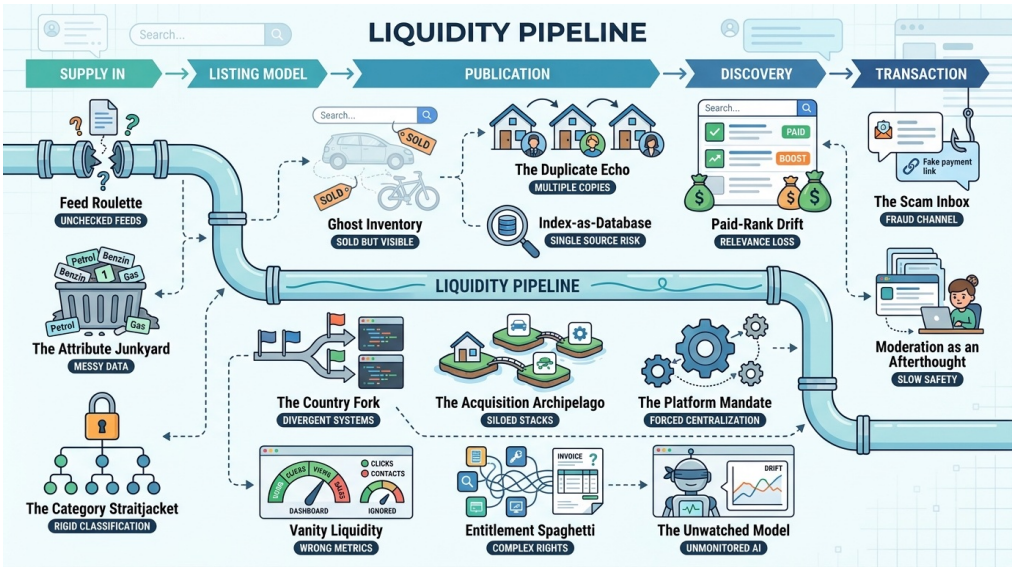
- Annual Report 2023, Adevinta, 2024: the plan to turn the leboncoin and Kleinanzeigen platforms into "a unified global platform", and the rebrand of eBay Kleinanzeigen as Kleinanzeigen.
- Strategic Report 2025, Aurelia UK Feederco Ltd (Adevinta), Adevinta, 2026: the move to "a decentralised, standalone operating model".

Research and practitioner sources

- Productscape general-listings and real-estate marketplace domain models (author's sibling repository, 2026): the generic domain models the book started from; summarized for writers in the book's research digest. Not descriptions of any company.

Company, regulatory and research sources for each topic are listed in the chapter that owns the topic and collected in the Bibliography: Sources and Evidence. Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

Unlisted: Fifteen Ways Listing Systems Go Wrong



IN THIS SECTION, YOU WILL: Learn fifteen named failure patterns that recur in classifieds and listing marketplaces, recognize their warning signs, understand the incentives that produce them and what they cost in liquidity, trust, speed and money, and find the chapters whose designs prevent or repair each one.

KEY POINTS:

- **Each pattern is locally rational.** A product team that hard-codes the first category tree, an integration engineer who accepts a dealer's feed as it is, a monetization team that adds one more ranking boost: each made a sensible choice under its own target. The failure is the accumulation, and only someone accountable for the whole marketplace sees it.
- **The cost shows up as lost liquidity, rarely in a project budget.** These patterns are paid for in contacts that never happen, alerts that point to sold items, scams that close in the inbox, and features built four times, long after the project that created them was declared a success.

- **The patterns travel in chains, and most chains start with a missing contract or a vanity metric**. Feed Roulette feeds Ghost Inventory and The Attribute Junkyard; Vanity Liquidity licenses Multiplier Creep and Set-and-Forget AI. Fixing the downstream symptom rarely holds; breaking the chain at its origin usually does.

The Introduction: Architecture for Liquidity treats a listing marketplace as a machine that turns fresh, trustworthy supply into matches with demand, and judges its architecture by **liquidity**: the probability that a listing finds a buyer, and a buyer finds the right listing, quickly (see Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics). Sellers and partners put listings in; the platform models, enriches, checks and publishes them; buyers search, get alerted, contact and, in some models, pay. This page names the ways that machine loses liquidity, so the rest of the book can point back to them.

Such a machine rarely fails dramatically. What fails is **coherence**: the marketplace slowly loses the ability to say what a listing is, whether it is still available, who it was shown to and why, and what the seller paid for. Sold bikes stay in alert e-mails. One flat appears three times. A dealer's stock vanishes overnight. A ranking boost added in 2021 is still there, and nobody can say what it costs.

This guide names the **fifteen failure patterns** I see most often in listing systems. The names are shorthand for observable behavior, so a design review or a steering committee can say "this is Feed Roulette" and move straight to the mechanism and the remedy. They are the book's shared vocabulary: every chapter in Parts I to VIII opens with a callout naming the patterns it addresses, linked back here, and the matrix at the end of this page lists every chapter that addresses each one.

Each pattern is described the same way: a characteristic quote, the warning signs, the mechanism that produces it, what it costs (with a documented public example where one exists), an illustration from **Veldmark, the fictional marketplace group this book follows** (see Meet Veldmark: Four Brands, One Platform to Build), the direction of the remedy, and the chapters that address it. The quotes are composites attributed to roles, not to people. The Veldmark illustrations are invented; they are not accounts of any real company.

None of these patterns implies incompetence. **Most are the residue of reasonable decisions** taken under deadline, by teams accountable for one stage of the pipeline. That is why they need a name and an owner.

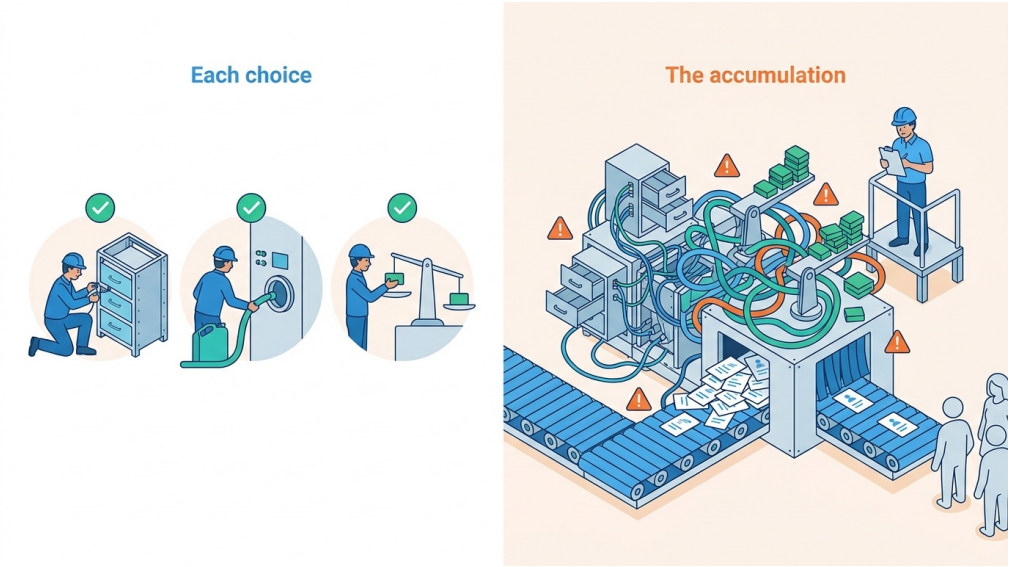


Figure 1: *Each pattern starts as a locally rational choice under one team's target; the failure is the accumulation, and only someone accountable for the whole marketplace sees it.*

Fifteen Patterns to Recognize

The patterns fall into four loose families. The **data** family (the first six) damages the platform's ability to know what a listing is and whether it still exists. The **marketplace** family (the next three) damages the order and safety of what buyers see. The **portfolio** family (three) is about how a group of marketplaces shares, or fails to share, its systems. The **measurement and control** family (the last three) is about what the business counts, sells and automates.

Pattern	The warning sign
The Category Straitjacket	Adding, splitting or renaming a category needs a code release, a data migration and a cross-team meeting.
The Attribute Junkyard	Attributes are free-form strings with no dictionary, so filters, facets and feeds disagree about what a listing is.
Ghost Inventory	Sold, expired or removed listings stay searchable, alertable and contactable long after they are gone.

Pattern	The warning sign
Seeing Double	The same item or property appears many times, from reposts, multiple feeds or multiple agencies, and crowds out everything else.
Feed Roulette	Partner and professional feeds are ingested without contracts, validation or feedback, so every import is a surprise.
Search as System of Record	The search index has quietly become the system of record, and nobody can rebuild it from anything else.
Multiplier Creep	Paid boosts and promotions accumulate in the ranking until relevance degrades and buyers stop trusting the order.
Moderation as an Afterthought	Trust and safety is bolted on after launch, as a queue that grows faster than the team that empties it.
Fast Mail for Fraudsters	Messaging and contact features ship without safety signals and become the main channel for fraud.
The Copy-Paste Country	Each market runs its own copy of the codebase, and every improvement has to be built several times.
Deal-Shaped Architecture	Acquired marketplaces keep their entire stacks indefinitely, and the group pays for every capability several times.
The Forced Fit	A central platform is imposed on markets that it does not fit, and local teams route around it.
Vanity Liquidity	The marketplace measures listings and visits instead of matches, contacts and sales, and optimizes the wrong thing.
Billing by Guesswork	What a seller has paid for is checked ad hoc in many services, so nobody can say what was sold versus delivered.
Set-and-Forget AI	A ranking, pricing, valuation or fraud model makes consequential decisions without evaluation, monitoring or an owner.

The patterns are not spread evenly along the pipeline. Most of the data family strikes on the way in and in the copies that publication creates; the marketplace family strikes where buyers meet listings; the portfolio and measurement families sit underneath every stage.

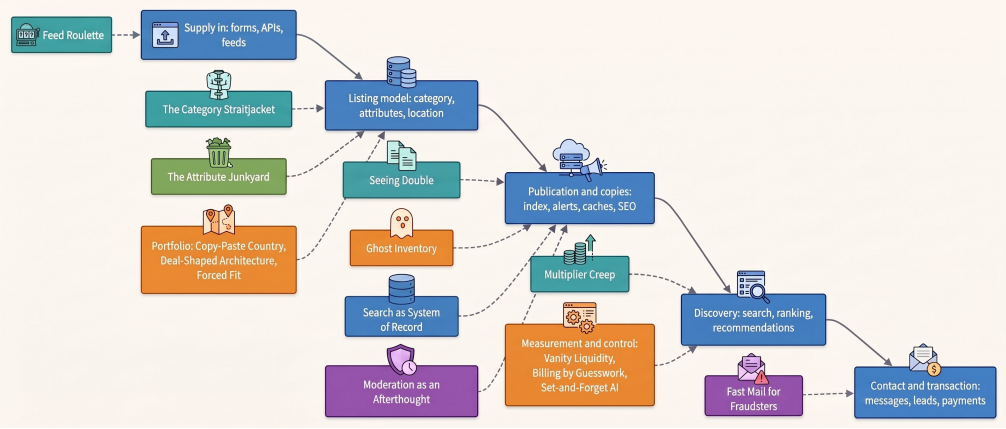


Figure 2: *Where each pattern strikes along the liquidity pipeline; the portfolio and measurement families are not stages but conditions that make every stage worse.*

Costs below are described along up to four dimensions: **liquidity** (matches lost or delayed), **trust** (of buyers, sellers and regulators), **speed** (how fast the marketplace can change) and **cost** (money spent or refunded).

The Category Straitjacket

"We can have the new category live in about six weeks, once search, SEO and the app team have each done their part." — a category product manager

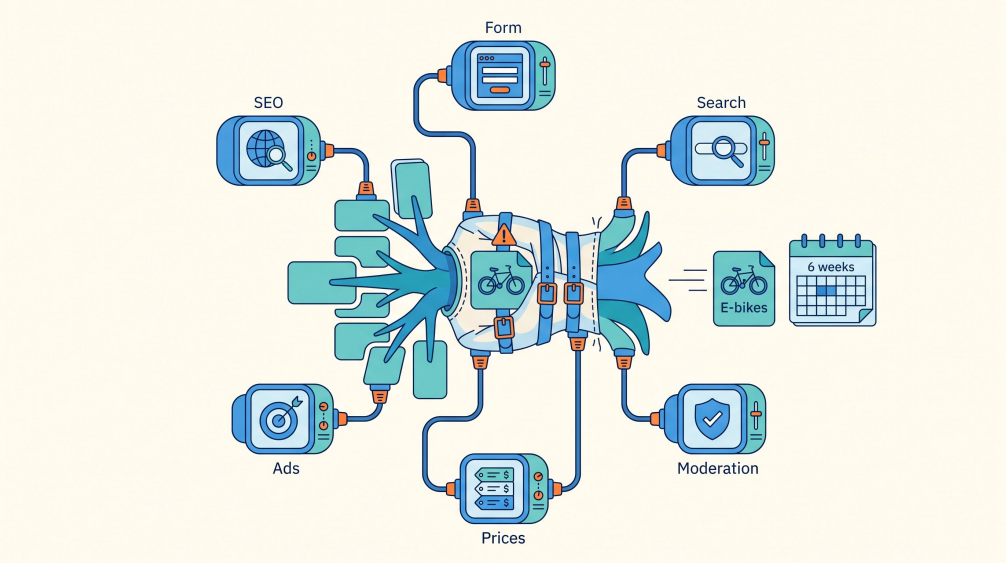


Figure 3: *The Category Straitjacket: category identifiers are wired into every consumer, so a new leaf waits about six weeks while each team does its part.*

What it looks like. A **category taxonomy** is the tree of categories sellers list into and buyers browse (see Category Trees That Can Change: Designing Marketplace Taxonomies). In a straitjacketed marketplace, category identifiers are embedded in code: in the insertion form, the search mapping, the moderation rules, the price lists, the ad-targeting setup and the SEO URL tables. Nobody versions the tree, so a change has no "from" and "to", only a release. Observable signs: lead time for a new leaf category is measured in weeks; catch-all "Other" leaves grow year after year; country trees drift apart because changing them together is too hard; taxonomy changes are bundled into one risky annual release.

Why it happens. The first tree is designed as product UI, at a time when it changes rarely, so hard-coding it is cheap. Then every consumer copies the identifiers: search for facets, ads for targeting, analytics for reporting, SEO for landing pages. A category is in fact a **schema key**: it selects which attributes apply, which rules run and which prices are charged, so every change ripples through all of them. Because nobody owns the taxonomy as versioned reference data, nobody designs change as a migration with mappings, re-classification and redirects.

What it costs. **Liquidity**: buyers cannot find inventory that sits in the wrong leaf, and new demand (e-bikes, heat pumps, cargo bikes) waits months for a home. **Trust**: prohibited items hide in generic leaves where category rules do not reach. **Speed**: product changes queue behind taxonomy changes. **Cost**: every re-categorization becomes a program. Even a well-run marketplace pays heavily for a large change: in a 2024 engineering post, Mercari described renewing its category structure and re-assigning more than 3 billion items, and estimated that labeling every item with a large

language model would cost about US$1 million and take about 1.9 years, which led it to label a sample with the LLM and propagate labels with a cheaper nearest-neighbor model.

An illustration. At Veldmark, Tradeyard, the general-classifieds brand, runs four country trees in 2027: 1,420 leaf categories in the Netherlands, 1,380 in Belgium, 980 in Poland and 760 in Portugal. A new category takes about **six weeks**. Bikes sit in 19 different country leaves holding about 640,000 live listings. When the group taxonomy service goes live and the bikes split ships on **Monday 1 November 2027**, it requires re-classifying about 410,000 listings, rewriting about 1.3 million saved searches, and creating about 2,300 redirects and 2,900 landing pages. A month later a "Fatbikes" leaf is added in **two days**, which is what the target was meant to look like.

The remedy direction. Treat the taxonomy as versioned reference data owned by one team; make consumers reference categories by stable identifier and subscribe to releases; publish every change with from-to mappings; and run splits, merges and re-classifications through a tested migration runbook that covers listings, saved searches, redirects and models.

Where this book helps: Category Trees That Can Change: Designing Marketplace Taxonomies designs the versioned tree and the migration runbook; Attributes Without Chaos: Schemas, Dictionaries and Validation ties attributes to categories without hard-coding; What Did They Mean? Query Understanding and Autocomplete keeps query classifiers in step with tree releases; Found by Google: Web, App and SEO Architecture for Listings owns the redirect registry that makes category changes SEO-safe; Same Platform, Different Physics: How Listing Verticals Differ explains why verticals should not be forced into one horizontal tree.

The Attribute Junkyard

"Just put it in the extra-fields column for now; we'll clean the data up after launch." — a feed-integration engineer

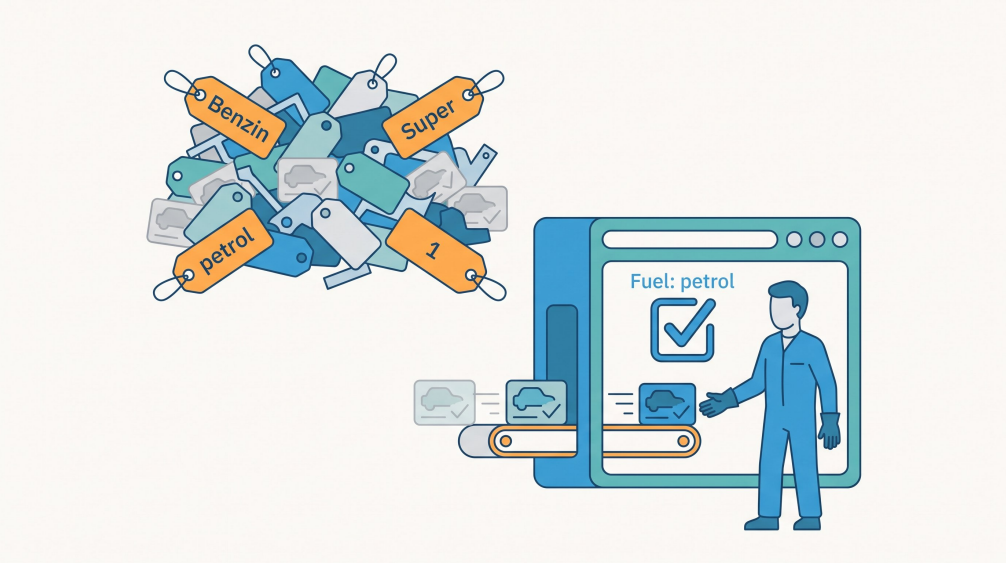


Figure 4: *The Attribute Junkyard: the same fact under four spellings, so a buyer filtering for petrol never sees most of the cars.*

What it looks like. **Attributes** are the structured facts on a listing: fuel type, mileage, number of rooms, frame size, energy class (see Attributes Without Chaos: Schemas, Dictionaries and Validation). In the Junkyard they are free-form key-value pairs with no **attribute dictionary**, the governed list of attributes with their types, units, allowed values and requiredness per category. The same fact arrives under several keys and in several formats: fuel as "Benzin", "Super", "petrol" and "1". Observable signs: the number of distinct keys grows faster than the number of categories; nobody can say which attributes a category requires; filters return different counts than facets; every partner export needs its own mapping table; the search mapping is so large that adding a field is risky.

Why it happens. Flexible storage (a JSON document or an entity-attribute-value table) makes adding a key free, and that is the point of choosing it. Each country, team, feed adapter and campaign adds keys for its own need. The web form and the feed importer validate against different rules, or none. No team owns the dictionary, because attributes look like a detail of each feature. Generative listing assistants now add a new source: an assistant that writes free text into attribute fields fills the Junkyard faster than any seller could.

What it costs. **Liquidity**: a buyer filtering for petrol cars does not see the listings stored as "Benzin", and the listing that is invisible to filters gets fewer contacts. **Trust**: rules that depend on attributes, such as mandatory energy-class display in property ads, cannot be enforced reliably. **Speed**: every new filter, facet or feed starts with a cleaning project. **Cost**: index mappings explode and enrichment work is repeated. No named marketplace has published a measurement of its own junkyard, but the public contracts of large

marketplaces show what the remedy looks like: eBay's taxonomy API publishes, per leaf category, each aspect's data type, whether it is required, whether values are controlled or free, its cardinality and, for new mandatory aspects, an `expectedRequiredByDate` (eBay Taxonomy API specification, n.d.).

An illustration. In 2026 Tradeyard carries **3,900** distinct attribute keys across its four countries, and **41%** of them are used on fewer than 0.1% of listings. When the group search platform defines its first schema in 2027, only about 610 keys are worth promoting to typed index fields; the rest go into a flattened field. Before an extraction pilot in 2027, frame size was present on only 23% of Dutch bike listings, so the frame-size filter quietly hid three quarters of the inventory.

The remedy direction. Build one attribute dictionary per vertical, with types, units, enumerations and requiredness by category; validate every channel (web, app, feed, AI assistant) against the same schema; record the provenance of every value; and promote, merge and retire keys deliberately rather than letting them accumulate.

Where this book helps: Attributes Without Chaos: Schemas, Dictionaries and Validation designs the dictionary and its governance; From Raw Supply to Usable Listings: Normalization and Enrichment maps raw values to canonical ones with provenance; Category Trees That Can Change: Designing Marketplace Taxonomies scopes schemas to categories; Search Architecture for Listings: Indices, Freshness and Scale decides which attributes become typed index fields; AI Across the Listing Lifecycle constrains generative assistants to the schema.

Ghost Inventory

"It's marked sold in the database. Whatever the alert e-mail said this morning isn't our service." — a listing-service engineer

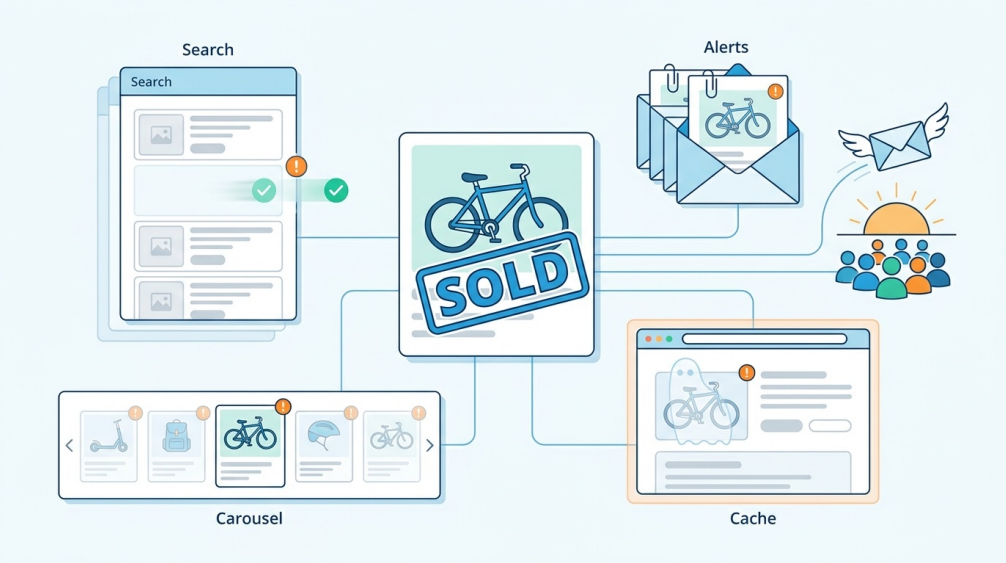


Figure 5: *Ghost Inventory: Femke's bike leaves search a minute after it sells, but an alert digest assembled earlier from a snapshot still sends it to 214 users the next morning.*

What it looks like. Sold, expired, withdrawn or removed listings remain visible somewhere: in search results and facet counts, in recommendation carousels computed overnight, in alert digests assembled from snapshots, in cached map tiles, on SEO pages that still answer `200 OK`, in partner feeds, and behind contact buttons that still work. Observable signs: the most common first message from buyers is "Is this still available?"; professionals complain about paying for leads on stock they sold days ago; nobody can state removal latency per consumer, only for the search index.

Why it happens. Removal is the change teams design least. Publication gets a pipeline, a latency target and a dashboard; unpublication gets a flag. Each consumer copies listing state on its own schedule, and some read a database replica or a nightly dump rather than lifecycle events. Feeds send full snapshots irregularly, or deltas without deletes. Private sellers who sell offline do not come back to mark the item sold. Every copy is someone else's system.

What it costs. **Liquidity**: contacts and alerts are spent on items that cannot be bought, live supply is diluted, and buyers learn that alerts cannot be trusted. **Trust**: a listing that stays up after the sale is indistinguishable, to the buyer, from bait advertising or a scam. **Cost**: professionals are charged for leads that cannot convert, and crawl budget is spent on pages for gone inventory. No named marketplace has published a ghost-listing prevalence figure. Downstream channels, however, already punish staleness: Google's job-posting guidance warns: "Failure to take timely action on expired jobs may result in a manual action" (Google Search Central, page updated 2026), and Google Merchant Center expires product data 30 days after its last refresh (Google Merchant Center Help, n.d.).

An illustration. Femke, a private seller on Tradeyard Netherlands, sells her €240 city bike on **Monday 8 March 2027**; it is marked sold at 18:10 and removed from search at 18:11. The evening alert digest, assembled at 17:30 from a database replica snapshot and never re-checked, goes out on Tuesday at 07:00 to 214 users with the bike in it. A March 2027 audit finds about **41,000** Tradeyard Dutch listings a week still present in at least one downstream copy 24 hours after they ended. At Motorvane, the automotive brand, the Essen Volkswagen Golf sells on Thursday 29 April; the dealer feed's removal arrives **26 hours** later, and three e-mail enquiries arrive in between, two of them through a similar-cars carousel that was rebuilt only nightly.

The remedy direction. Make removal a first-class lifecycle event with its own latency objective for every registered consumer, not just the search index; re-check availability at send time in alerts and digests; reconcile feeds and expire listings that are not renewed; and serve correct status codes for gone pages.

Where this book helps: From Submitted to Searchable: The Publication Pipeline sets removal objectives and the consumer registry; Every Change Is an Event: Event-Driven Listing Architecture carries removals as ordered events; System of Record and Read Models: Listing Data Management puts staleness budgets on read models and caches; Tell Me When It Appears: Saved Searches and Alerts re-checks availability before sending; Many Doors In: Forms, APIs and Bulk Feeds reconciles feed state; Found by Google: Web, App and SEO Architecture for Listings handles sold and expired pages; Recommendations for Inventory That Disappears filters status at serving time instead of trusting a nightly job.

Seeing Double

"Three agencies hold the mandate, so three listings is correct. Isn't it?" — a real-estate portal product owner

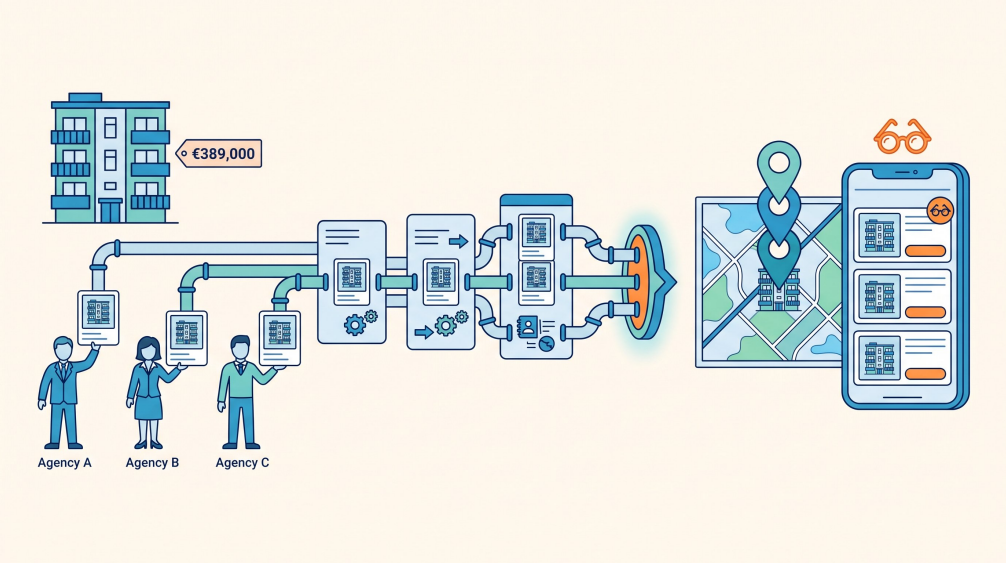


Figure 6: *Seeing Double: one 68 m² Lyon flat at €389,000 arrives three times, from three agencies, and stacks three pins on one building.*

What it looks like. The same item or property appears many times: reposted by a seller to look new, pushed through two channels by one dealer, listed by several agencies for the same flat, or copied by fraudsters with stolen photos. Observable signs: result pages repeat the same photos; listing counts grow faster than the number of distinct properties or vehicles; agencies complain that a competitor's copy outranks theirs; one alert sends the same flat three times; three pins stack on one building on the map.

Why it happens. Duplicates are rewarded. Reposting resets a listing's freshness; each agency with a mandate pays for its own exposure; a dealer integrated through two software vendors gets twice the visibility. Feeds often lack stable external keys, and imprecise locations defeat the first test any matcher applies. Most of all, nobody owns the decision of what to show once, because deduplication looks like a cosmetic search feature rather than **entity resolution**, the work of deciding which records describe the same real-world thing (see One Property, Three Listings: Duplicates and Entity Resolution). And abusive duplicates, which call for enforcement, are mixed up with legitimate ones, which call for presentation.

What it costs. Liquidity: demand for one property is split across copies, supply looks deeper than it is, and buyers wade through repeats. **Trust**: fake listings built from stolen photos hide among the legitimate copies. **Cost**: storage, indexing, alerts and moderation are multiplied. The presentation problem is common enough in property that portals productize it: in 2021, Idealista launched a duplicate-control tool that groups listings of the same property, including listings on other portals, and notifies users of price changes, removals and new duplicates (Online Marketplaces, 2021, a trade-press report).

An illustration. At Hearthly France, Veldmark's real-estate portal, **1.9 million** listings

describe about **1.12 million** distinct properties in 2026, **1.7 listings per property** on average. In May 2027 a 68 m² flat in Lyon's 7th arrondissement at €389,000 arrives three times, from three agencies through two software vendors and Keybridge, the group's agent CRM. Two months later, in the Barcelona fake-rental ring, 11 photo-match alerts are closed by reviewers as "legitimate multi-agency duplicates"; they were stolen photos.

The remedy direction. Separate abusive duplicates (route to trust and safety and enforce) from legitimate ones (cluster under a canonical entity and show once, with "also listed by"); keep cluster identity and lineage in a registry outside the search index; and choose the representative listing by quality, never by paid status.

Where this book helps: One Property, Three Listings: Duplicates and Entity Resolution designs matching, clustering and merge policies; Photos Sell: Media Pipelines for Listings provides perceptual hashes; Where Is It? Location, Geocoding and Privacy supplies the precise location that matching relies on; Many Doors In: Forms, APIs and Bulk Feeds recognizes the same partner record arriving twice; Searching by Place: Map, Polygon and Commute Search collapses clusters on the map; Scams, Fakes and Account Takeovers: Fraud Prevention for Listings treats stolen photos as fraud signals.

Feed Roulette

"The dealer's file looked fine last night. We'll find out what happened when support opens." — an ingestion on-call engineer

Figure 7: *Feed Roulette: at 02:40 a truncated nightly file for 286 dealers unpublishes 38,400 Motorvane listings, because nothing guards a full snapshot.*

What it looks like. Professional and partner feeds (exports from a dealer management system, or DMS; real-estate software XML; bulk CSV files) are ingested without a **feed contract**, the agreed schema, semantics, validation rules and error reporting for a channel (see Many Doors In: Forms, APIs and Bulk Feeds). Records are not validated against the same rules as the web form, writes are not idempotent, partners get no per-run error report, and nothing guards against a truncated file. Observable signs: partners learn about missing stock from their own customers; a bad file unpublishes thousands of listings overnight; listing identifiers change on re-import; support calls spike on Monday mornings; the importer is full of per-vendor special cases.

Why it happens. Feeds are integrated one partner at a time, by whichever team won the commercial deal, as code adapters. The partner is a paying customer, so rejecting their data feels commercially risky. Full-snapshot semantics ("absence means delete") are easy to implement and dangerous without a guard. Error reporting to partners has no product owner. And the software vendors in the middle serve many portals at once, so they do the minimum each one requires.

What it costs. **Liquidity**: professional stock, the bulk of supply in automotive and real estate, goes missing or stale overnight. **Trust**: dealers and agents stop believing the platform's statistics and its invoices. **Speed**: onboarding a new vendor takes months of special cases. **Cost**: support, credits for lapsed paid placements, and lost renewals. In feed-driven verticals inventory is not exclusive: in 2026 Germany's competition authority noted that "many platforms offer dealers data export to other platforms" and treated dealer multi-homing as a competitive constraint (Bundeskartellamt, 2026), so a dealer who loses confidence in one platform's feed can shift attention to another. No named marketplace has published a feed incident, but public partner APIs show the remedy: mobile.de's Seller API makes creation idempotent through an insertion request ID, answering a retry with `303` to the existing ad, and returns errors as stable keys with arguments (mobile.de Seller API documentation, n.d.); RFC 9457 standardizes machine-readable problem details for such errors (IETF, 2023).

An illustration. On **Monday 8 March 2027 at 02:40 CET**, one transfer service provider sends Motorvane cut-short nightly full-stock files for 286 dealers. By 02:55 the importer has unpublished **38,400** listings, about 1,900 of them with paid top placements that lapse. Complete files arrive at 09:40, but about a fifth of the cars come back with new listing identifiers, because the importer had purged the external-key mapping. Dealer support takes more than 600 calls. At Hearthly, of about 160 software vendors, roughly 30 cause most incidents, including some that never send deletes.

The remedy direction. Give every channel a contract: an ingestion envelope, schema versions, a catalog of stable error codes, per-run error reports, stable external keys, a mass-change guard on every snapshot, and periodic reconciliation of delta channels.

Where this book helps: Many Doors In: Forms, APIs and Bulk Feeds designs feed contracts, idempotency and mass-change guards; Attributes Without Chaos: Schemas, Dictionaries and Validation provides the shared validation rules; Private Sellers, Dealers

and Agencies: Modeling Who Lists models integration partners and scoped grants; Serving the Pros: Dealer, Agent and Merchant Tooling shows partners their errors; mobile.de, Auto Trader and AutoScout24: Lessons from Automotive Marketplaces compares how automotive marketplaces expose dealer APIs.

Search as System of Record

"Don't touch the Solr collection. Some of those fields only exist there." — a search engineer

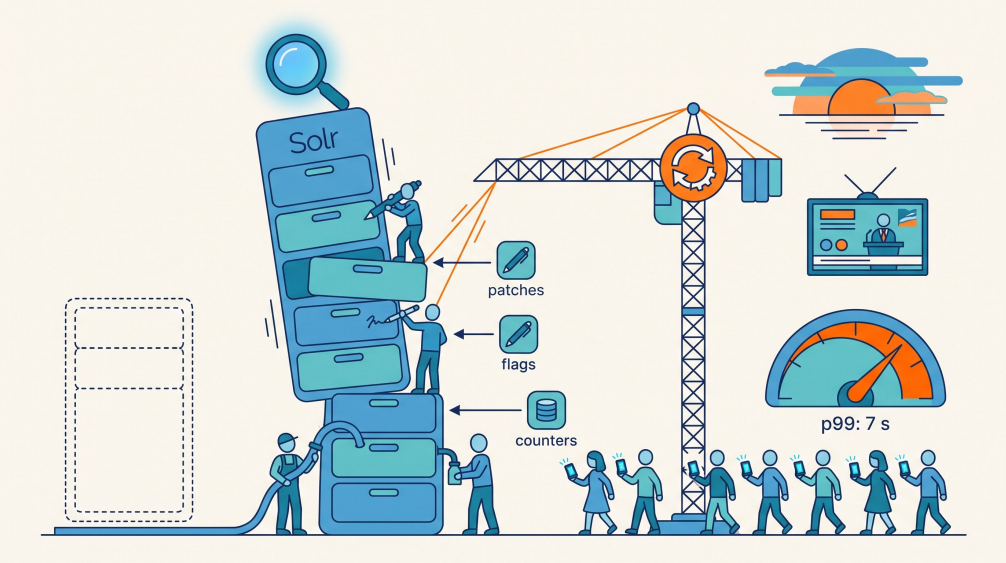


Figure 8: *Search as System of Record: teams write view counters and fixes straight into Solr, so a Sunday-evening reindex pushes p99 latency to about 7 seconds.*

What it looks like. The search index holds fields, corrections or enrichment results that exist nowhere else. "Published" is defined as "present in the index". Other services query the index because it is the only complete, denormalized view of a listing. Rebuilding it takes days, needs a frozen weekend, or is not possible at all. Observable signs: reindexing is scheduled for a "quiet" hour and treated as an event; mapping changes are postponed for quarters; nobody has run a rebuild drill; the answer to "why is this listing missing?" starts with a query against search.

Why it happens. The index is the fastest, most complete view in the company, so teams write to it directly: view counters, enrichment outputs, moderation flags, manual fixes. Each shortcut skips the system of record. The event stream that could rebuild the index either does not exist or does not carry everything.

What it costs. **Liquidity**: reindexing collides with traffic peaks, and search degrades exactly when buyers are most active. **Trust**: the platform cannot prove whether a

missing listing was never approved, failed to index or was removed. **Speed**: schema evolution stalls and engine migrations become impossible. **Cost**: capacity is sized for risky rebuilds instead of normal load. The documented alternative is old and proven: in a 2019 engineering post, eBay Kleinanzeigen (now Kleinanzeigen) described rebuilding its Elasticsearch cluster, then holding more than 34 million live ads, from a compacted Kafka topic in under 30 minutes, for both recovery and schema changes (Berlin Tech Blog, 2019; the current state is not public).

An illustration. On **Sunday 17 January 2027 at 20:35 CET**, Tradeyard Netherlands' search p99 latency rises from about 400 ms to about **7 seconds** for 48 minutes, and about 9% of searches fail. A scheduled reindex is writing into the live Solr collection, and every commit clears the caches the Sunday browse traffic depends on, while a TV campaign pushes traffic above the usual peak. The review finds that Yardstick, Tradeyard's PHP monolith, writes about 9 million document updates a day to Solr, about 6.4 million of them view counters, and that the publication pipeline defines "published" as "present in Solr".

The remedy direction. Treat the index as a read model built from the listing system of record and its event stream, with monotonic versions; keep data that exists nowhere else out of it; prove rebuilds with a regular drill; and put a stable search contract in front of the engine.

Where this book helps: Search Architecture for Listings: Indices, Freshness and Scale designs the rebuildable index and the search contract; System of Record and Read Models: Listing Data Management separates system of record from read models; Every Change Is an Event: Event-Driven Listing Architecture provides the replayable stream; From Submitted to Searchable: The Publication Pipeline defines "published" independently of the index; Sunday Evening Peaks: Reliability, Performance and Cost moves rebuilds off the peak; Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models names the owner of listing truth.

Multiplier Creep

"It's only a small multiplier, and top-listing customers expect to see results for their money." — a monetization product manager

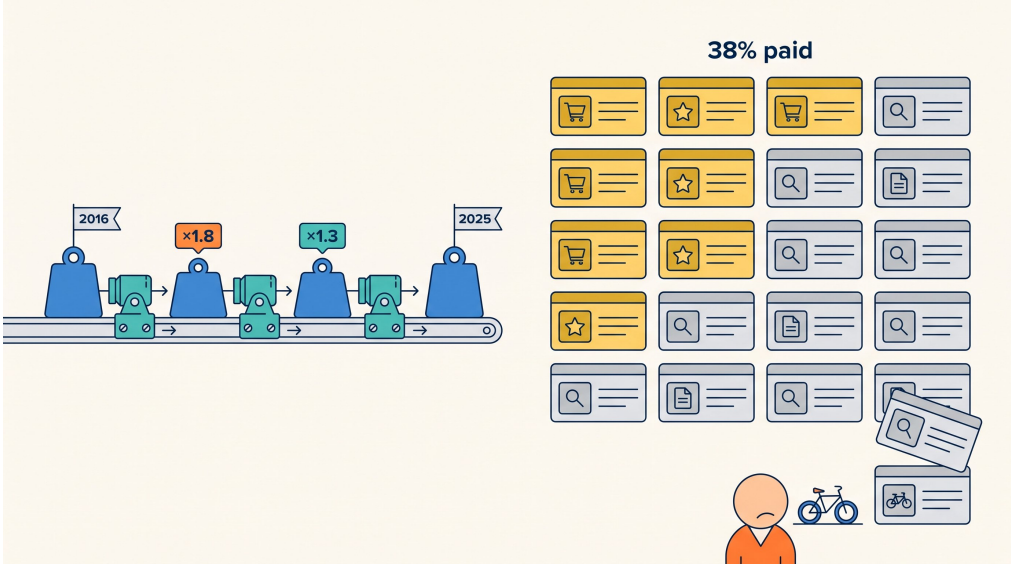


Figure 9: *Multiplier Creep: five paid layers added since 2016 leave paid listings holding 38% of Tradeyard Netherlands' first-page positions.*

What it looks like. Each visibility product (bump, top listing, highlight, a "plus" package for professionals) is implemented as another multiplier, re-dated timestamp or reserved slot in ranking. Nobody caps the total, no relevance floor applies, and the organic cost of each boost is never measured. Over the years the default sort turns into a price list. Observable signs: the share of first-page positions held by paid listings rises year after year; contacts per search fall; new private listings increasingly get no contact at all; the ranking configuration contains boosts nobody can explain; paid labels are missing on some surfaces.

Why it happens. Visibility products are sold by a monetization team with a revenue target; ranking belongs to another team with a relevance target; neither owns the trade-off. Each boost is added incrementally and never reviewed as a whole. Experiments that randomize buyers measure the revenue gain while the cost lands on sellers outside the test (see Testing a Two-Sided Market: Experimentation Platforms).

What it costs. **Liquidity**: relevance degrades, buyers stop trusting the order, and organic sellers lose exposure. **Trust**: legal as well as commercial. Since 28 May 2022, point 11a of Annex I to the Unfair Commercial Practices Directive, added by the Omnibus Directive (EU) 2019/2161, treats as always unfair "providing search results in response to a consumer's online search query without clearly disclosing any paid advertisement or payment specifically for achieving higher ranking"; the Platform-to-Business Regulation (EU) 2019/1150, applicable since 12 July 2020, requires the main ranking parameters, including paid influence, to be disclosed to business users (a repeal was proposed in November 2025; status as of October 2026 is a proposal). **Speed**: every ranking change becomes a revenue negotiation. **Cost**: visibility revenue that depends on

relevance falls with it. Two public designs show the alternative. In 2020, eBay described ranking promoted listings together with organic results, so that the top of the page can show no ads at all when few relevant promoted items exist. In 2025, OLX described a separate "VAS LTR" ranking objective that balances relevance for users against visibility for paid listings, rather than multiplying scores.

An illustration. In April 2027, a review of Tradeyard's ranking finds five layers added since 2016: the top-listing slots (2016), bumps that re-date the sort timestamp (2019), a ×1.8 top-listing multiplier in relevance sort (2021), a bump recency boost (2023) and a ×1.3 "Visibility Plus" multiplier for professionals (2025). In the Netherlands, paid listings hold **38%** of first-page positions in 2026, up from 12% in 2021; search sessions with a contact fell from 10.6% to 9.4%, and new private listings with no contact in 48 hours rose from 24% to 33%. Ranking policy v1, live on 1 December 2027 after a cluster-randomized test, puts paid slots under a relevance floor and caps paid listings at four in the top 20: the paid first-page share falls to 17%, sessions with a contact rise to 9.9%, and visibility revenue per search falls 2.5%.

The remedy direction. Carry paid effects as explicit, bounded fields from the entitlement service into ranking; place paid listings under a relevance floor with caps per page and per seller; evaluate ranking changes with two-sided guardrail metrics; and label paid placement end to end.

Where this book helps: Ranking for Two Sides: Relevance, Freshness and Paid Visibility designs paid-organic blending and its guardrails; How Listing Marketplaces Make Money explains the liquidity-revenue tension; Packages, Bumps and Quotas: Products, Pricing and Entitlements makes paid effects explicit and capped; Selling Attention: Display, Native and Retail-Media Advertising keeps sponsored units inside the same rules; Testing a Two-Sided Market: Experimentation Platforms measures the cost to sellers outside the test; Rules of the Marketplace: DSA, DAC7 and Vertical Regulation covers the disclosure obligations.

Moderation as an Afterthought

"Launch the category first. Trust and safety can add a queue once we see what comes in." — a growth lead

Figure 10: *Moderation as an Afterthought: alerts fan new listings out before any risk check, and a review queue added later sits behind publication.*

What it looks like. **Trust and safety** (T&S) — policies, detection, enforcement and appeals — is added after launch as a queue in front of or behind publication (see Trust and Safety as a Platform). Checks run only when a listing is created, not when it is edited; image checks run as a nightly batch after the listing is live; alerts fan out new listings before any risk check; the statements of reasons that the EU Digital Services Act (DSA) requires are produced by exporters after the fact. Observable signs: the review backlog grows faster than the team; time to action on reports is measured in hours; nobody knows the share of live listings that violate policy; each brand's transparency numbers are computed differently.

Why it happens. T&S is seen as a cost center, while launches belong to growth teams with growth targets. Checks are bolted on serially, so every new one slows publication, which makes teams reluctant to add them. Policies are not versioned, so decisions cannot be explained later. Regulation is treated as reporting rather than as a design requirement.

What it costs. **Liquidity**: fast fan-out delivers fraud to the most motivated buyers first, and serial queues slow every honest listing. **Trust**: the gaps are visible from outside. In December 2015 a Paris court found that leboncoin had engaged in misleading commercial practices because it claimed that all ads passed a validation process when not all were reviewed before going live; the court also confirmed its hosting-provider status, under which it was not obliged to pre-check every ad (Clubic, 2015). In a December 2024 undercover test, the UK consumer group Which? found that Gumtree blocked its first scam-style ad, but that edited versions then passed review "in minutes" (Which?, 2024): checks at creation only are not enough. **Cost**: headcount grows with volume instead of with risk.

An illustration. In 2027, Tradeyard automates 94% of new-listing decisions and sends 6%, about 138,000 a week, to human review; about 210 moderators work across the group's brands. A blind weekly sample of 4,000 live Dutch listings, started in April 2027, puts the prevalence of policy-violating listings at about **0.6%**, three times the assumed 0.2%. Median time to action on fraud reports is 3 hours 10 minutes. A new anti-phishing rule written on Sunday 7 March reaches Belgium only on Wednesday with the next Yardstick deploy, and never reaches Poland or Portugal. In July, the Barcelona fake-rental ring's 180 listings produce about 41,000 alert notifications before takedown.

The remedy direction. Build T&S as a platform inside the flow: risk-tiered checks that run in parallel with publication, re-scoring on every material edit, a risk gate in front of alert fan-out, enforcement propagated as events, decisions recorded as statements of reasons, and prevalence measured by sampling.

Where this book helps: Trust and Safety as a Platform designs the shared moderation pipeline and its metrics; From Submitted to Searchable: The Publication Pipeline tiers checks by risk instead of queuing them serially; Tell Me When It Appears: Saved Searches and Alerts places the risk gate before fan-out; Rules of the Marketplace: DSA, DAC7 and Vertical Regulation turns DSA obligations into records; Who Is on the Other Side? Identity and Verification builds verification into onboarding; Photos Sell: Media Pipelines for Listings screens images before publication; AI Across the Listing Lifecycle keeps T&S inside AI-assisted flows.

Fast Mail for Fraudsters

"Messaging is a delivery problem. Fraud is for trust and safety, once users report it." — a messaging-team lead

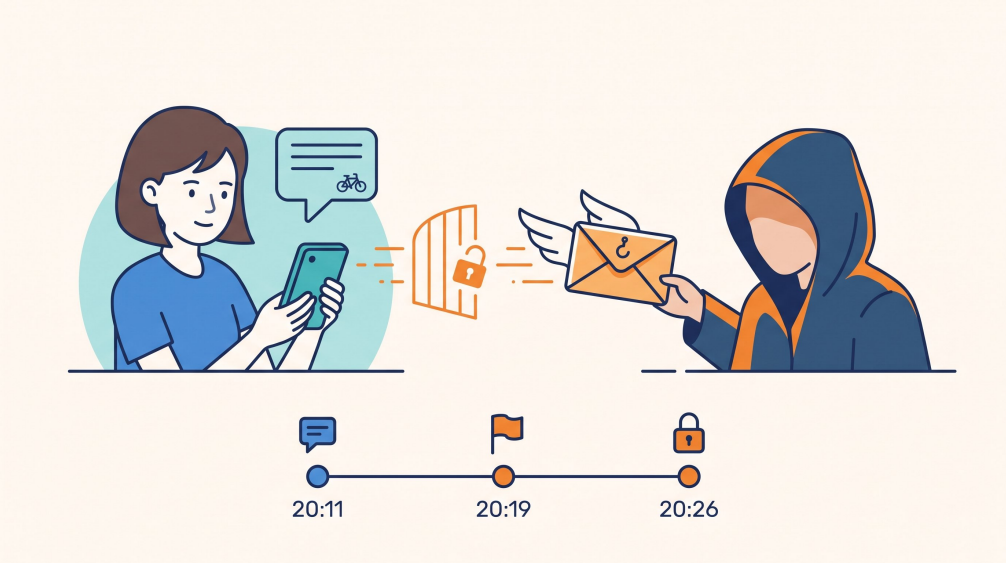


Figure 11: *Fast Mail for Fraudsters: a fake payment link from a domain registered that day reaches Femke at 20:11; only her report at 20:19 stops it.*

What it looks like. Chat and contact features are built for delivery speed alone. Messages arrive instantly with clickable links; nobody checks whether a domain was registered this morning; anyone with a fresh e-mail address can message any seller; payment is negotiated in the conversation and made off the platform; user reports are the only control. Observable signs: rising reports of fake "payment received" or courier links; scam messages that impersonate the platform's own brand; banks, not the marketplace, holding the best data on losses.

Why it happens. In classifieds and lead-generation models the platform's job ends at the contact, so money moves off the platform and fraud closes in the conversation (see Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy). Messaging teams are measured on delivery and latency. Scanning private messages raises legitimate privacy concerns, so teams avoid it rather than design it lawfully. Identity checks happen at sign-up, if at all, not at the first contact, where the risk is.

What it costs. **Liquidity**: sellers stop answering unknown buyers, and buyers fear paying. **Trust**: victims blame the platform whatever the terms say. **Cost**: support load, and growing pressure to fund reimbursement or verification. The UK has the best public data: UK Finance's 2026 Annual Fraud Report counts 175,809 purchase-scam cases and £118.1 million of losses in 2025, 71% of all authorised push payment scam cases, and asks government to "legislate to mandate seller verification and secure payment protections on online marketplaces" (a lobbying position, not law, as of October 2026). Platforms that fight back put controls before delivery: in 2026, OLX reported that fraudulent images make up about 30% of fraud cases in Poland and that it blocks suspicious chat images before they reach users.

An illustration. On Sunday 7 March 2027, Femke's bike attracts 14 conversations in 24 hours. At 20:11 a "buyer" sends a fake Tradeyard Safe Deal payment link; it is delivered because the domain was registered that day. Femke reports it at 20:19; the account is suspended at 20:26; a new rule then withholds 37 similar messages to other sellers. Tradeyard receives about **41,000** user reports of payment-link phishing in the Netherlands and Belgium in 2026.

The remedy direction. Put safety in the delivery path: tokenize links and payment details, check domain age, warn in context, hold or delay risky first contacts from new accounts, add step-up verification where the risk score warrants it, and steer payment into a protected on-platform flow where the model allows.

Where this book helps: From Interest to Contact: Messaging and Lead Management designs safety scanning into the messaging architecture; Scams, Fakes and Account Takeovers: Fraud Prevention for Listings provides risk scoring and step-up friction; Who Is on the Other Side? Identity and Verification decides how much to know about who sends the first message; Beyond the Contact: Payments, Shipping and Buyer Protection moves money into a protected flow; Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy explains why the contact-ending model concentrates fraud in the inbox.

The Copy-Paste Country

"Poland has its own release train. If you want the fix there, file a ticket with their team."
— a group product director

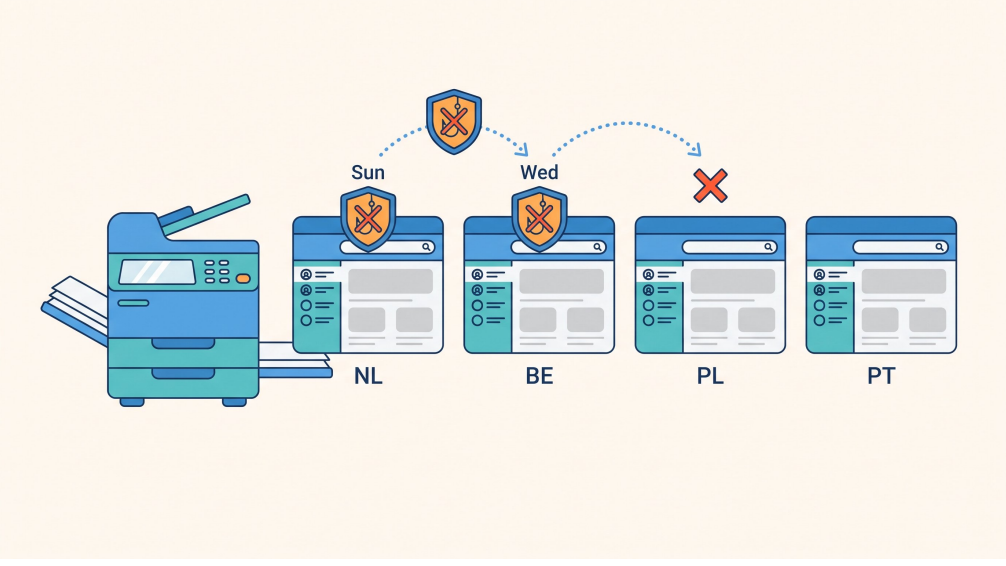


Figure 12: *The Copy-Paste Country: a March 2027 anti-phishing rule reaches the Netherlands on Sunday, Belgium on Wednesday, and Poland and Portugal never.*

What it looks like. Each market runs its own copy of the codebase, or its own copy of the reference data that the code depends on: category tree, address model, rule set, ranking configuration. Improvements are built several times or not at all; a fix in one country does not reach the others; versions drift until they can no longer share anything. Observable signs: the same feature appears in several backlogs; regulatory changes are implemented as country code paths; country category trees have different leaves for the same goods; a group-wide fraud rule needs one deploy per country.

Why it happens. Copying a working site is the fastest way to enter a market, and for years it worked. Local autonomy is a genuine strength: local teams know their buyers, sellers and laws, and they are accountable for a local profit-and-loss statement. Sharing code requires a configuration design that nobody was funded to build.

What it costs. **Speed**: every improvement is built several times. **Cost**: duplicated teams, tools and licenses. **Trust**: fraud rules and safety fixes reach some markets late or never. **Liquidity**: smaller markets cannot afford the features that larger ones build. The public record shows the pattern and its price. Adevinta's 2019 annual report described "local autonomy and decentralised capacity in product and technology" as "one of our historic success formulas", even as it centralized data and analytics (Adevinta Annual Report 2019). Its engineering blog later recorded that in 2021 its 22 marketplaces used ten different A/B testing tools: "Every team had its own tool for A/B testing, resulting in inefficiencies, redundancy and escalating costs" (Adevinta Tech Blog, 2024). In 2025, OLX described how learning-to-rank projects spread across teams and repositories caused "duplicated efforts" and were centralized into one repository with brand-level configuration (OLX Tech Blog, 2025).

An illustration. Tradeyard's four countries share a brand but not a platform. They run four category trees of different sizes (1,420, 1,380, 980 and 760 leaves), two search engines (Solr in the Netherlands, Belgium and Portugal; Elasticsearch in Poland), and country-specific address handling. The March 2027 anti-phishing rule shows the cost in one week: written for the Netherlands on Sunday, in Belgium on Wednesday, never in Poland or Portugal.

The remedy direction. Converge on one codebase per capability with country differences expressed as configuration over shared components; keep local policy local and local code rare; and harvest good local inventions into the shared platform rather than suppressing them.

Where this book helps: One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture designs tenancy and configuration over code; Category Trees That Can Change: Designing Marketplace Taxonomies harmonizes country trees; Where Is It? Location, Geocoding and Privacy replaces country address models with one place model; Rules of the Marketplace: DSA, DAC7 and Vertical Regulation puts country rules in configuration; From Newspaper Columns to Portfolios: How the

Classifieds Industry Evolved explains where the copy-paste model came from; Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform and OLX Group: Regional Platforms, Verticalization and Retreat show two groups working their way out of it.

Deal-Shaped Architecture

"They're profitable and the deal closed last month. Let's not touch their stack for now."
— a corporate-development lead

Figure 13: *Deal-Shaped Architecture: Veldmark enters 2027 with four identity systems, four trust-and-safety stacks and three data warehouses, so a phisher banned on one brand stays active on another.*

What it looks like. Acquired marketplaces keep their whole technology stack: identity, trust and safety, search, product catalog, billing, data warehouse. "For now" has no end date. The group pays for every capability several times and cannot answer group-level questions: how many businesses it serves, whether a fraudster banned on one brand is active on another, what a "contact" is. Observable signs: several logins, several ban lists, several warehouses and several contact definitions; synergy targets that appear in board decks but not in any funded roadmap.

Why it happens. Deal teams protect revenue during the integration period, and the acquired leadership is rewarded for its own growth, not the group's. The synergy case assumes integration but does not fund a sequence. Sometimes regulators set the limits: competition remedies can require that overlapping businesses be sold rather than integrated.

What it costs. **Cost**: duplicated run cost, licenses and on-call rotations. **Trust**: fraudsters move between brands that cannot share signals. **Speed**: every group capability has to integrate with each stack separately. **Liquidity**: cross-brand features (one login, syndication between brands, shared alerts) are impossible. The size of the prize is public: when Adevinta completed its acquisition of eBay Classifieds Group in June 2021, it targeted run-rate cost synergies of **€130–165 million** after three years (Adevinta, 2021), and its later consolidation of ten experimentation tools into one shows the kind of work that synergy required (Adevinta Tech Blog, 2024).

An illustration. Veldmark enters 2027 with **four of everything**: four identity systems, four trust-and-safety stacks, three search engines, three clouds, four product catalogs and three data warehouses. About 2,900 Polish dealers have active accounts on both Tradeyard Poland and Motorvane Poland, and about 40% of them pay both. The phone number behind the account suspended for phishing Femke had also registered a Motorvane private-seller account, which no system connected. The **Common Ground** program, approved in January 2027 with €140 million for 2027–2030, targets a run-cost reduction of €38 million a year.

The remedy direction. Decide, per capability, whether to keep it separate, share it or replatform it, with a date and an owner; consolidate the generic capabilities (identity, trust and safety, payments, data) first and the product surfaces last; and design shared services for separability, because portfolios change owners.

Where this book helps: After the Deal: Consolidating Platforms After Acquisitions sets the consolidation sequence and tests synergy cases; Map What the Marketplace Does Before What Runs It makes the duplication visible; Who Is on the Other Side? Identity and Verification designs one group identity; Packages, Bumps and Quotas: Products, Pricing and Entitlements replaces several catalogs with one model; Data Platform for Marketplaces: Events, Lakehouse and Data Products governs shared definitions; Built to Be Replaced: Evolving a Marketplace Platform Over a Decade keeps every acquisition separable over a decade; Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform and eBay Classifieds Group: Shared Platforms, Local Brands and a Sale show the public record of one large integration.

The Forced Fit

"The group platform is the standard from the first quarter. Your roadmap items go into our backlog." — a central platform director

Figure 14: *The Forced Fit: a list-shaped search platform would have missed Hearthly, where 61% of French search sessions open the map, and pushed its teams to keep their own geo stack.*

What it looks like. A central platform is imposed on markets or verticals it was not designed for. Adoption is by decree and measured by migrations completed. Local teams comply on paper and route around it in practice: a shadow stack behind an adapter, an "exception" that becomes permanent, features waiting in the central backlog. Observable signs: low voluntary adoption; a growing exceptions list; duplicate systems kept "temporarily" alongside the platform; local roadmaps blocked on a central team.

Why it happens. Consolidation is under pressure from a synergy target, and a mandate looks like the fastest way to hit it. The platform was designed around the physics of its first or largest tenant (list-shaped horizontal classifieds, for example), and the differences of other verticals look like edge cases. Sharing infrastructure, where technology is uniform, is confused with sharing the product itself, where markets differ. The platform team is rewarded for migrations, not for fit.

What it costs. **Speed**: local roadmaps stall behind the central backlog. **Liquidity**: vertical-specific features (map search, vehicle identification, rental applications) degrade or never ship. **Cost**: shadow stacks and parallel running double the bill the mandate was meant to cut. The public record shows mandates that reversed. eBay's Product-Based Shopping Experience required sellers in many categories to associate listings with catalog products from 2018; eBay's developer documentation now states that the mandate was rolled back, without giving a date (eBay PBSE playbook, n.d.). Adevinta's 2023 annual report announced turning the leboncoin and Kleinanzeigen platforms "into a unified global platform" (Adevinta Annual Report 2023); its 2025 strategic report describes a move "to a decentralised, standalone operating model" with "the ramp-down

of central functions" and "the localisation of intellectual property" (Adevinta Strategic Report 2025). Whether the unified platform was completed is not publicly documented, and the reversal followed a change of owner as much as any technical lesson. The contrast is Adevinta's own earlier platform-as-a-service, which a 2020 talk described as reaching about 1,500 engineers through voluntary adoption (Navarro, 2020).

An illustration. In September 2027, Veldmark decides to move Tradeyard and Hearthly onto one shared search platform. Hearthly's numbers show what a list-shaped platform would miss: 61% of its French search sessions open the map, about 2,700 of its 4,200 peak search requests per second are map requests, and drawn search areas have a median of 146 vertices. Delivered as a list-search API, the shared platform would have pushed Hearthly's teams to keep their own geo stack. Instead the geo query compiler becomes part of the shared platform, configured per brand. The same reasoning shapes the November 2027 ML platform decision: models must be registered in one group registry, but Hearthly keeps Vertex AI and Keybridge keeps Azure Machine Learning until the golden path is better than what they have.

The remedy direction. Run platforms as products whose adoption is earned; mandate only narrow contracts (event schemas, identity, a model registry), not whole stacks; share platforms, not products; and express vertical differences as configuration of shared components rather than as exceptions to them.

Where this book helps: One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture distinguishes shared cores from imposed products; Who Owns What: Organizing Marketplace Engineering designs platform teams and governance; After the Deal: Consolidating Platforms After Acquisitions sequences adoption; Same Platform, Different Physics: How Listing Verticals Differ decides what verticals can share; Searching by Place: Map, Polygon and Commute Search and Models in Production: ML Platforms and Use Cases show two platforms built to be adopted; Built to Be Replaced: Evolving a Marketplace Platform Over a Decade explains why a ten-year plan must not fix the split between central and local; Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform and AVIV Group: Building a European Real-Estate Platform from Acquisitions test the pendulum against the public record.

Vanity Liquidity

"Live listings are up eight percent and sessions are at a record. The marketplace has never been healthier." — a marketplace general manager

Figure 15: *Vanity Liquidity: live listings rise while Tradeyard's contact index falls from 100 to 37 once only qualified contacts count.*

What it looks like. The marketplace reports activity: live listings, visits, page views, messages sent, alerts sent, indexed URLs. It does not report outcomes: qualified contacts per listing, time to first contact, share of listings that sell, search sessions that end in a contact (see Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics). Each team optimizes the activity metric it controls. Observable signs: "contact" is defined differently by each brand; dashboards show growth while contacts per listing fall; ranking and recommendation models are trained on clicks; alerts are measured by volume sent; professionals' impressions rise while their renewals fall.

Why it happens. Activity metrics are easy to count, always grow with traffic, and sit inside one team's system. Outcome metrics need joins across systems and agreed definitions, and in classifieds the transaction itself is often invisible to the platform.

What it costs. **Liquidity**: the marketplace optimizes the wrong thing, so changes that create noise are approved and changes that create matches are not. **Trust**: professionals pay for exposure they cannot convert, and renewals fail. **Cost**: investment follows the metrics that are easiest to grow. Two public examples show the alternative and the trap. In 2022, OLX described training its ranking on seller contacts (messages or calls) because transactions are not always confirmed, and checking that the number of unique contacted sellers rose. In 2026, OLX described how naive comparisons suggested that premium boost packages underperformed, while matched estimates validated by an A/B test showed uplifts of 17.6% to 97.4%: a metric without a causal design can point in the wrong direction.

An illustration. In March 2027, Veldmark's three brands count contacts three ways: Tradeyard counts every buyer message plus phone-number reveals; Motorvane counts

leads per channel, including calls of any length; Hearthly deduplicates forms and phone reveals per user, listing and day. When "qualified contact v1.0" is replayed over one March week, Tradeyard's contact index falls from 100 to **37**, Motorvane's to 86, and Hearthly's rises to 106. Femke's bike had 52 "contacts" under the old definition and 13 qualified ones, with the phisher excluded. Earlier that year, a cheaper bump price was approved because bumps rose 48% in a seller-randomized test; nobody watched the control sellers, whose contacts per bump fell, and after rollout contacts per bump fell 31%.

The remedy direction. Define liquidity metrics once, as governed metrics with owners; require every latency, freshness or revenue objective to name the liquidity metric it protects; and judge experiments, models and products on qualified contacts and sales, with guardrails for both sides of the market.

Where this book helps: Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics defines the liquidity metrics; Data Platform for Marketplaces: Events, Lakehouse and Data Products governs the contact definition; Testing a Two-Sided Market: Experimentation Platforms judges tests on outcomes with guardrails; Ranking for Two Sides: Relevance, Freshness and Paid Visibility trains on two-sided contact labels; How Listing Marketplaces Make Money builds revenue plans on contacts, not page views; From Interest to Contact: Messaging and Lead Management counts qualified, deduplicated leads; Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy explains why each model ends at a different outcome; Built to Be Replaced: Evolving a Marketplace Platform Over a Decade replaces program milestones with marketplace fitness functions.

Billing by Guesswork

"Is this dealer allowed 150 cars or 163? It depends which service you ask." — a dealer-support agent

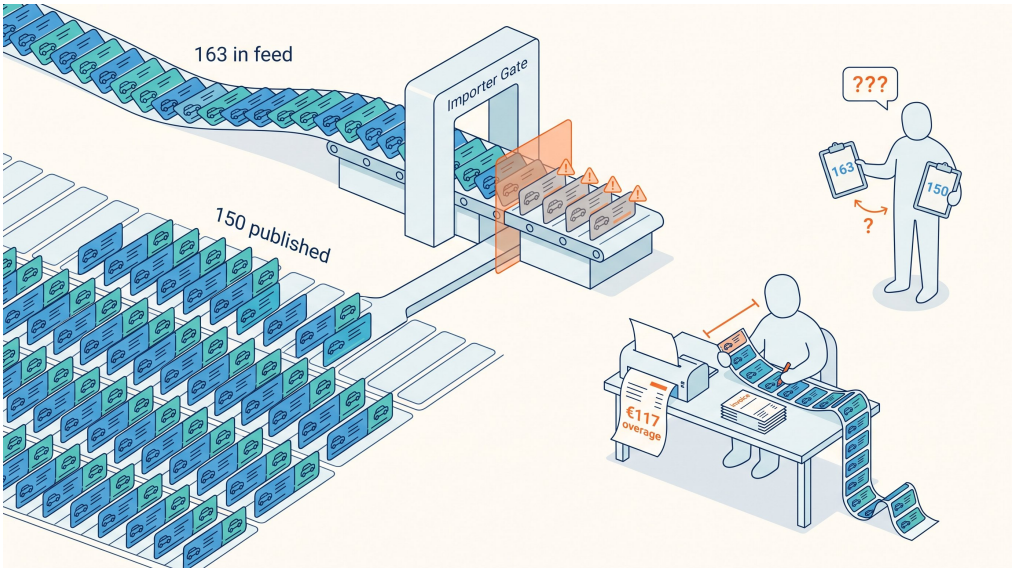


Figure 16: *Billing by Guesswork: Autohaus Ruhrblick's feed has 163 cars, 150 are published, and the billing export still charges €117 of overage for the 13 that never went live.*

What it looks like. An **entitlement** is what a seller has the right to receive because of what they bought: listing slots, bumps, top positions, statistics, features (see Packages, Bumps and Quotas: Products, Pricing and Entitlements). In Billing by Guesswork those rights are checked ad hoc: package conditions in the insertion service, in the search query, in the listing-card template, in the billing export and in the CRM, each with its own logic. Paid flags are stored as listing fields. Observable signs: credits issued after disputes; nobody can produce "sold versus delivered" for one seller; a new product needs changes in five services; promotions are implemented as code branches.

Why it happens. The first visibility product is implemented as a flag on the listing, which is cheap and correct for one product. Each later product adds a condition wherever it is needed. Catalog, entitlements and billing belong to different teams, and acquired brands bring their own.

What it costs. **Trust**: overbilling, or delivering less than was sold, damages the relationship with the professional sellers who provide most of the revenue. **Speed**: launching a product or a price change takes months. **Cost**: credits, audits and support. **Liquidity**: paid effects leak into ranking without bounds, which is one route to Multiplier Creep. No marketplace has published an entitlement failure, but published results show how precise entitlements must be. Rightmove reported that "more than 50% of independent agents purchased incremental product above their package commitment" in 2025 (Rightmove full-year results, 2026), which means committed bundles and metered extras on one invoice. Autotrader's 2026 results explain revenue per retailer partly through a "stock" lever, the number of paid vehicles advertised (Autotrader Group, 2026): when billing

follows live inventory, inventory accuracy is billing accuracy.

An illustration. Motorvane's "Pro 150" package gives a dealer 150 listing slots and 20 boosts a month for €1,450, with €9 per car per month for overage. Autohaus Ruhrblick's August 2027 feed contains 163 cars. The feed importer publishes 150 and skips the rest; the billing export counts feed rows and charges €117 of overage for 13 cars that were never published. A 14-month audit in October 2027 finds **1,140** dealers overbilled by about **€212,000**, all credited.

The remedy direction. Model products, entitlements and their effects in one service with an append-only ledger; make every enforcement point (insertion, ranking, display, billing) read the same effects; carry paid effects as events; and bill from what was delivered.

Where this book helps: Packages, Bumps and Quotas: Products, Pricing and Entitlements designs the catalog, entitlement ledger and enforcement points; How Listing Marketplaces Make Money shows what each revenue model requires; Private Sellers, Dealers and Agencies: Modeling Who Lists says which organization and branch holds an entitlement; Serving the Pros: Dealer, Agent and Merchant Tooling shows professionals what they hold and used; Selling Attention: Display, Native and Retail-Media Advertising keeps sponsored placements in the same ledger; The Listing Aggregate: Identity, Lifecycle and Invariants keeps paid state out of the listing itself.

Set-and-Forget AI

"The model has been running since 2021, and nobody has complained." — a data-science manager

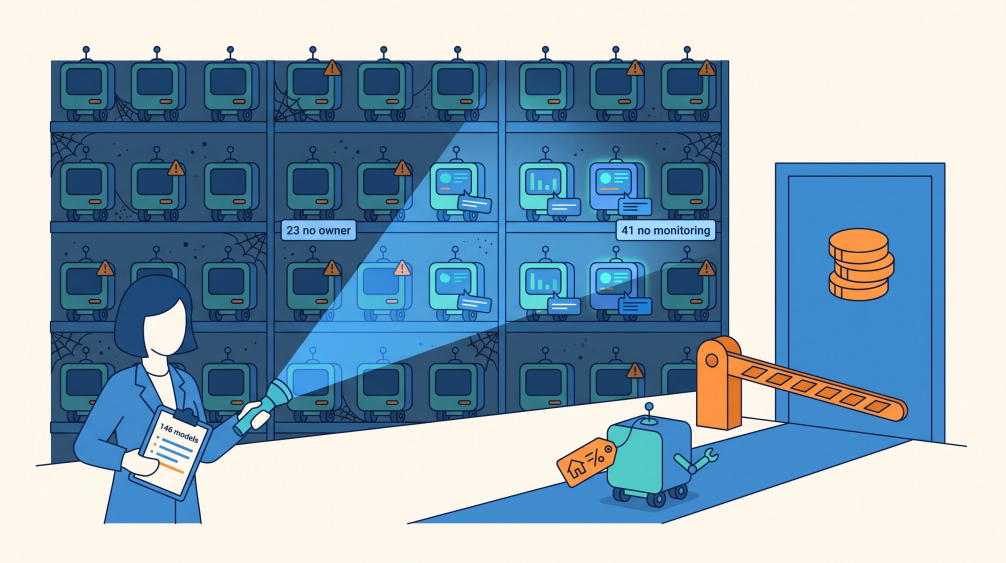


Figure 17: *Set-and-Forget AI: Veldmark's first model inventory finds 146 production models, 23 without a named owner and 41 without monitoring.*

What it looks like. Models make consequential decisions: they order results, approve or hold listings, merge duplicates, fill attributes, estimate prices and values, score fraud. Some run without an evaluation set, without drift monitoring, without prediction logs, without a named owner or a way to roll back. Observable signs: nobody can list all production models; a model's use expands (from informing a badge to deciding a price) without new validation; labels are years old; an AI feature's prompt and model version change weekly with no regression test.

Why it happens. Models are shipped as projects, and projects end. Platforms for registering, monitoring and retiring models arrive later than the models themselves. Uses expand quietly: a valuation built to inform a listing page is proposed for an offer or a loan. Adversaries adapt, so a fraud model's labels go stale while its accuracy on old data still looks good.

What it costs. **Cost**: when a model commits money, its errors become losses. **Trust**: a wrong merge, a hidden listing or a biased ranking cannot be explained to the seller or the regulator. **Liquidity**: silently degrading ranking and matching models erode match quality. The canonical public case is Zillow Offers. On 2 November 2021, Zillow's board decided to wind down its home-buying business, citing "home pricing unpredictability, capacity constraints and other operational challenges"; the third quarter included a $304.4 million inventory write-down, and the company announced a workforce reduction of about 25% (Zillow Form 8-K, 2021). Its shareholder letter said: "We have been unable to accurately forecast future home prices at different times in both directions" (Zillow shareholder letter, 2021). The lesson is not that the valuation model was bad at informing buyers; it is that forecasting prices to commit capital is a different

use, and needs a different validation regime than the one that serves a listing page.

An illustration. In September 2027, Priya Raman, Veldmark's Head of Data and ML, presents the group's first model inventory: **146** production models on four training stacks, of which **23** have no named owner, **41** have no monitoring and **12** cannot be reproduced. The same autumn, a "Hearthly Direct" proposal to make guaranteed offers to homeowners at 94% of Valuo's estimate is declined: Valuo, Hearthly's valuation model, was validated to inform, and the proposal would have used it to commit money.

The remedy direction. Keep one model registry with an owner, an evaluation set, a monitoring standard and a rollback path for every production model; classify uses by decision tier (inform, order, gate, commit) and let governance follow the use, not the model; and re-validate whenever the use moves up a tier.

Where this book helps: Models in Production: ML Platforms and Use Cases designs the registry, monitoring and decision tiers; Ranking for Two Sides: Relevance, Freshness and Paid Visibility owns learning-to-rank evaluation and guardrails; Scams, Fakes and Account Takeovers: Fraud Prevention for Listings handles stale labels and adversarial drift; From Raw Supply to Usable Listings: Normalization and Enrichment gates extraction models; One Property, Three Listings: Duplicates and Entity Resolution evaluates matchers; AI Across the Listing Lifecycle governs generative features; Zillow, Rightmove, Idealista and REA: What Property Portals Teach puts Zillow's case in context; mobile.de, Auto Trader and AutoScout24: Lessons from Automotive Marketplaces shows price ratings that move search position and dealer pricing.

How the Patterns Reinforce Each Other

The patterns rarely appear alone. Most marketplaces show several, and they feed each other in recognizable chains. The chain matters because **fixing a downstream symptom while the upstream pattern continues produces the same symptom again**, usually within a year.

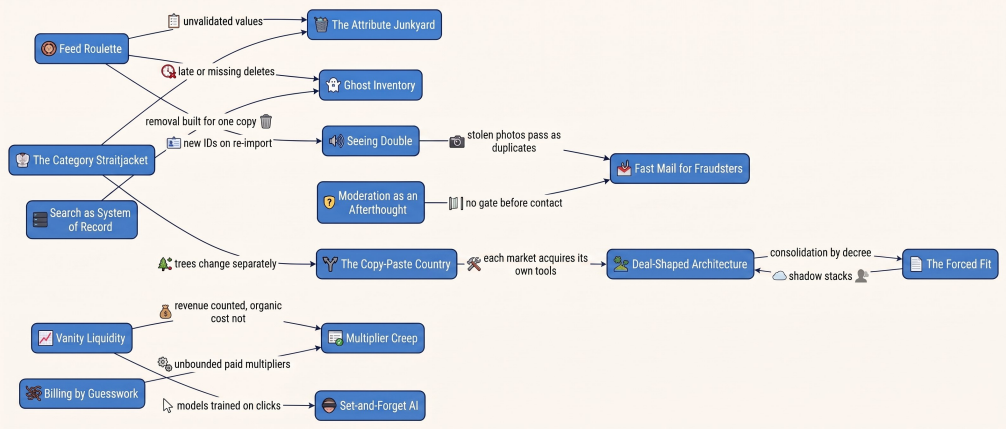


Figure 18: *The patterns form chains; most downstream symptoms, such as ghost listings or paid results crowding the first page, trace back to a missing contract, a missing owner or the wrong metric.*

Four chains appear most often.

The supply chain: Feed Roulette → The Attribute Junkyard, Ghost Inventory and Seeing Double. A feed without a contract delivers values nobody validated, so the Junkyard grows; it delivers removals late or not at all, so ghosts accumulate; and when it re-creates listings under new identifiers, the same car or flat appears twice. Teams then fund three downstream fixes for one upstream gap. At Veldmark, the truncated-snapshot incident of 8 March 2027 produced all three in one morning. **Break the chain at the contract**, not at the index.

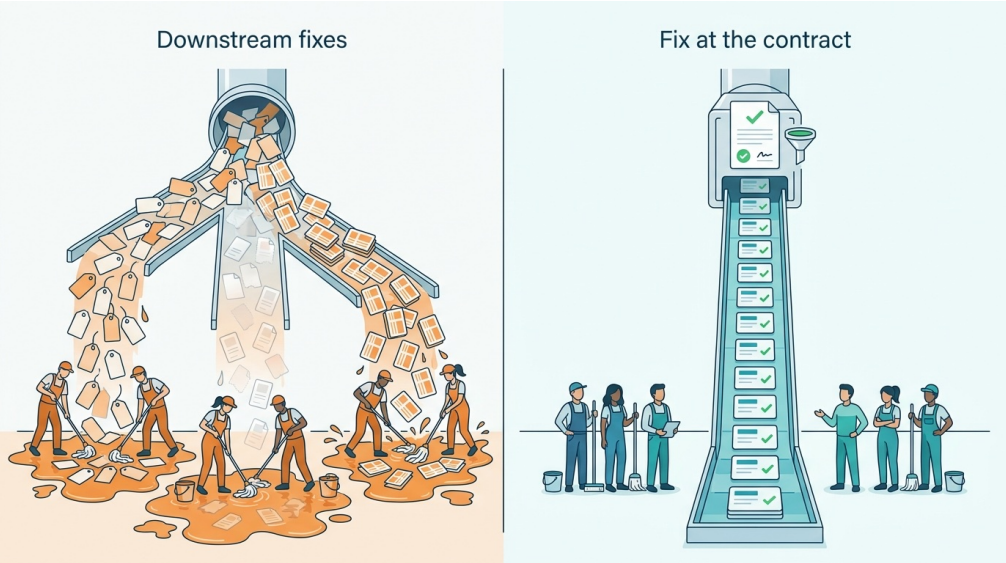


Figure 19: *Funding three downstream fixes while the upstream gap stays open brings the symptoms back; breaking the chain at its origin, a contract, an owner or a metric definition, holds.*

The monetization chain: **Vanity Liquidity and Billing by Guesswork → Multiplier Creep**. When the business measures revenue and clicks but not contacts per search, every boost looks like a gain. When paid effects are conditions scattered across services rather than bounded fields, nothing stops another multiplier being added. The drift then degrades the relevance that paid products depend on. The remedy is not a ranking rewrite alone but a measurement change (governed contact metrics with two-sided guardrails) and an entitlement change (explicit, capped paid effects).

The trust chain: **Moderation as an Afterthought and Seeing Double → Fast Mail for Fraudsters**. A platform that checks listings only at creation, and treats copied photos as harmless duplicates, lets a fake listing reach the inbox, where the scam closes. Link scanning in the inbox helps; re-scoring on edit, routing photo matches to fraud review and gating alerts help earlier and more cheaply.

The portfolio pendulum: **The Copy-Paste Country and Deal-Shaped Architecture → The Forced Fit → back again**. A group that grew by copying and acquiring ends up paying for everything several times. Under synergy pressure it mandates a central platform that does not fit everyone; local teams route around it, and the shadow stacks recreate the archipelago. The public record of large classifieds groups shows this pendulum swinging in both directions within a few years. The way off the pendulum is to **share platforms, not products**: centralize the uniform, expensive layers, configure the differences, and earn adoption.

Two patterns sit at the roots of most chains. **Vanity Liquidity** is a measurement failure: without a governed definition of a match, every team optimizes something easier. **Feed Roulette** and **The Category Straitjacket** are contract failures: without a governed contract for what enters the platform and how it is classified, every downstream system compensates on its own. That is why this book defines liquidity metrics in Part I and contracts in Parts II and III before it discusses search, ranking or AI.

First Indicators for Each Pattern

Pattern	First indicator to ask for	Typical owner of the number
The Category Straitjacket	Lead time to add or split a leaf category, end to end	Taxonomy owner, product
The Attribute Junkyard	Distinct attribute keys per category, and the share used on fewer than 0.1% of listings	Attribute dictionary owner
Ghost Inventory	Removal latency per registered consumer; share of contacts and clicks on unavailable listings	Listing platform team
Seeing Double	Listings per distinct real-world entity, per vertical	Listing data domain owner
Feed Roulette	Share of partner runs with a machine-readable error report; mass-change holds per month	Ingestion team
Search as System of Record	Time of the last successful full rebuild drill, and its duration	Search platform
Multiplier Creep	Paid share of first-page positions, trended; contacts per search	Ranking owner with monetization
Moderation as an Afterthought	Sampled prevalence of violating listings; median time to action on reports	Head of trust and safety
Fast Mail for Fraudsters	Fraud reports per thousand conversations; share of risky messages held before delivery	Messaging and fraud teams
The Copy-Paste Country	Features implemented more than once across markets in the last year	Group architecture

Pattern	First indicator to ask for	Typical owner of the number
Deal-Shaped Architecture	Capabilities served by more than one stack, with run cost	Group architecture, finance
The Forced Fit	Voluntary adoption rate and open exceptions to the platform	Platform product manager
Vanity Liquidity	Whether "qualified contact" has one governed definition, and which dashboards use it	Data governance, CPO
Billing by Guesswork	Number of services that evaluate package rules; credits issued for billing disputes	Monetization platform
Set-and-Forget AI	Production models without an owner, an evaluation set or monitoring	Head of data and ML

These are not targets in themselves. Duplicate rates can fall because clustering hides legitimate listings; moderation backlogs can shrink because thresholds were loosened. **Read each indicator with the mechanism in mind**.

Using the Patterns

The names are meant to be used in rooms where decisions are made. In a design review, ask which pattern a proposal risks and what would stop it. In a steering committee, ask which pattern a delay or a falling metric is a symptom of. In a portfolio review, ask which patterns each brand shows and which root they trace back to.

Three habits keep the names useful rather than accusatory: **name the mechanism, not the person**; **pair every pattern with a measure** from the table above; and **look for the upstream pattern** before funding a downstream remedy, because the cheapest fix is usually a contract, an owner or a metric definition, not a new system.

Some duplication, divergence and payment for visibility is legitimate. Three agencies listing one flat is the market working, not Seeing Double, as long as the platform shows it once. A country-specific rental rule required by law is not The Copy-Paste Country, if it is configuration over shared code. Paid placement is not Multiplier Creep when it is

labeled, bounded and measured. The test is whether the variation was **decided, owned and measured**, or simply accumulated.

The Patterns Across the Book

The matrix below maps each chapter to the patterns it addresses. ● means the pattern is one of the chapter's main concerns: it is named in the chapter's opening callout. ○ means the chapter treats the pattern in passing. The column codes are: **CS** The Category Straitjacket, **AJ** The Attribute Junkyard, **GI** Ghost Inventory, **SD** Seeing Double, **FR** Feed Roulette, **SR** Search as System of Record, **MC** Multiplier Creep, **MA** Moderation as an Afterthought, **FM** Fast Mail for Fraudsters, **CP** The Copy-Paste Country, **DA** Deal-Shaped Architecture, **FF** The Forced Fit, **VL** Vanity Liquidity, **BG** Billing by Guesswork, **SF** Set-and-Forget AI. The "Where this book helps" lines above name the chapters that do the main work for each pattern; the matrix lists every chapter that names it.

Chapter	CS	AJ	GI	SD	FR	SR	MC	MA	FM	CP	DA	FF	VL	BG	SF
Part I — Marketplace															
1 Marketplace taxonomy									●			●	●		
2 Liquidity and network effects			●	●									●		

Chapter	CR	AJ	GI	SD	FR	SR	MC	MA	FM	CP	DA	FF	VL	BG	SF
3 Industry evolution										●	●	●			
4 Business capabilities											●	●	●		
5 Bounded contexts			●			●								●	
6 Vertical requirements	●	●										●			
Part II — Listing															
7 Listing aggregate			●			●								●	
8 Categories and taxonomies	●	●								●					

Chapter	CS	AJ	GI	SD	FR	SR	MC	MA	FM	CP	DA	FF	VL	BG	SF
9 Attribute schemas	●	●			●										
10 Media pipeline			●	●				●							
11 Location and geo		●		●						●					
12 Sellers and accounts					●						●			●	
Part III — Supply															
13 Listing ingestion		○	●	●	●										
14 Normalization and enrichment		●			○	●									●

Chapter	CS	AJ	GI	SD	FR	SR	MC	MA	FM	CP	DA	FF	VL	BG	SF
15 Entity resolution			●	●											●
16 Publication pipeline			●			●		●							
17 Professional seller tooling					●								●	●	
Part IV — Discovery															
18 Search architecture		●	●			●									
19 Query understanding	●	●													●
20 Ranking							●						●	○	●

Chapter	CR	AJ	GI	SD	FR	SR	MC	MA	FM	CP	DA	FF	VL	BG	SF
21 Recommendations			●										●		●
22 Map search			●	●								●			
23 Saved searches and alerts			●					●					●		
Part V — Trust															
24 Trust and safety architecture			●					●							●
25 Fraud prevention				●					●						●
26 Identity and verification								●	●		●				

Chapter	GS	AJ	GI	SD	FR	SR	MC	MA	FM	CP	DA	FF	VL	BG	SF
27 Marketplace regulation							●	●		●					
Part VI — Money															
28 Messaging and leads			●						●				●		
29 Transactions and payments			●						●			●		○	
30 Monetization models							●						●	●	
31 Products and entitlements							●				●			●	

Chapter	GR	AJ	GI	SD	FR	SR	MC	MA	FM	CP	DA	FF	VL	BG	SF
32 Advertising platforms							●						●	●	
Part VII — Platform															
33 Event-driven listings			●			●					●				
34 Listing data management			●			●					●				
35 Web and SEO architecture	●		●										●		
36 Data platform											●		●		
37 Experimentation							●						●		

Chapter	CS	AJ	GI	SD	FR	SR	MC	MA	FM	CP	DA	FF	VL	BG	SF
38 ML platform												●	●		●
39 Multi-country platforms										●	●	●			
40 Reliability and scale						●					●		●		
Part VIII — Leadership															
41 Platform consolidation			●							○	●	●			
42 Organizing marketplace engineering										●	●	●			

Chapter	GS	AJ	GI	SD	FR	SR	MC	MA	FM	CP	DA	FF	VL	BG	SF
43 Modernization and migration	○		●			●					○				
44 AI in listing systems		●						●							●
45 Long-term evolution			○	○							●	●	●		○
Part IX — Practice															
46 Adevinta										●	●	●			
47 OLX Group							●			●	●	○	○		
48 eBay Classifieds Group					●	●					●	○			
49 AVIV Group				○							●	●	●		

Chapter	GS	AJ	GI	SD	FR	SR	MC	MA	FM	CP	DA	FF	VL	BG	SF
50 Property portals				●	●										●
51 Automotive marketplaces					●								○	●	●

Read down a column to find every chapter that helps with one pattern; read across a row to see what a chapter is defending against. Three columns are dense: **Ghost Inventory**, **Vanity Liquidity** and **Set-and-Forget AI** appear in almost every part, because removal, measurement and automated decisions cut across the whole pipeline. That density is itself a finding: these three need an owner at group level, not only in one team.

The rest of the book works through the remedies. Before that, meet the group whose marketplaces show every one of these patterns in 2027, and the program that sets out to fix them: Meet Veldmark: Four Brands, One Platform to Build.

Sources and Evidence

Company engineering and investor sources

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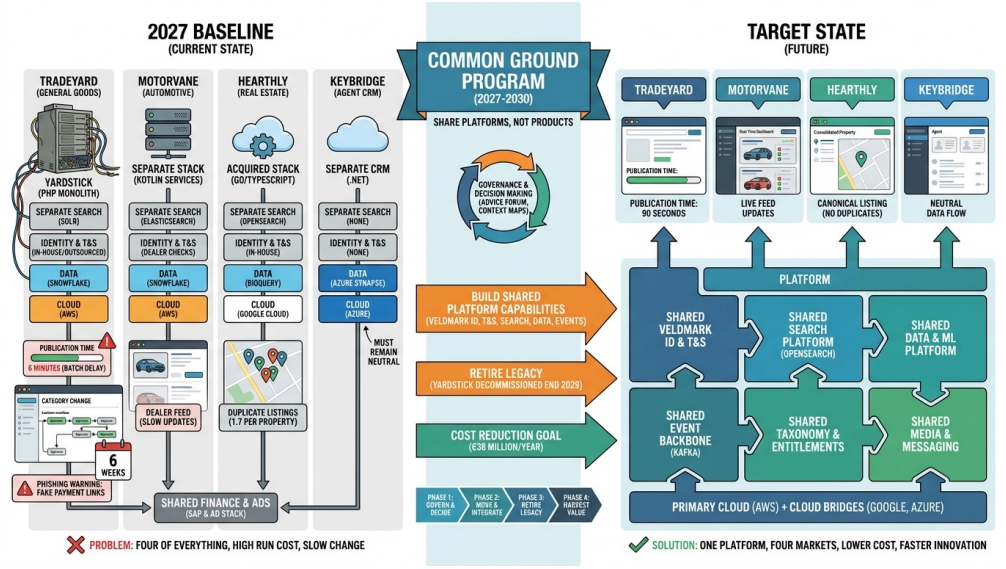
The pattern names and their descriptions are the author's synthesis of common industry practice. Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

Meet Veldmark: Four Brands, One Platform to Build



IN THIS CHAPTER, YOU WILL: Meet Veldmark Group, the fictional marketplace group this book follows: its four brands and their business models, its volumes, its 2027 systems landscape, the people who run it, the eight problems it starts 2027 with, the Common Ground platform program approved to solve them, and the recurring worked examples whose numbers every chapter shares.

WHY THE BUSINESS CARES: Veldmark earns €1.31 billion a year from four different ways of turning listings into money. Its owner wants growth and a lower run cost at the same time, and the group pays for most of its capabilities four times.

WHY ARCHITECTS SHOULD CARE: Veldmark's difficulty is variety, not scale. Three clouds, three search engines, four identity systems and a PHP monolith from 2006 must serve private sellers, dealers and agencies in eight countries without slowing any marketplace down, and that is the Chief Architect's problem in miniature.

KEY POINTS:

- **Veldmark is a fictional composite, not a disguised company.** It generalizes patterns that commonly appear in European marketplace groups, so the book can show real architectural tensions with consistent numbers without describing any real group's internals. Every name, figure, system choice and event on this page is invented.
- **One group, four marketplace physics.** Tradeyard sells visibility to private sellers and takes a fee on payments, Motorvane sells dealer packages and leads, Hearthly sells agency subscriptions and valuation data, and Keybridge sells software to agents who list on Hearthly's competitors. The same word, "listing", carries different supply, money and risk in each.
- **Common Ground shares platforms, not products.** The €140 million program (2027–2030) builds identity, trust and safety, search, events, taxonomy, entitlements and data beneath the brands, retires the Yardstick monolith by the end of 2029 and must cut run costs by €38 million a year by 2030, while every brand keeps its own name, user experience and go-to-market.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Deal-Shaped Architecture** — Hearthly, its Spanish predecessor Casavista and Keybridge each arrived with a complete stack, and in 2027 the group still runs four identity systems, four trust-and-safety stacks and three data warehouses.
- **The Copy-Paste Country** — Tradeyard's Belgian and Portuguese sites run on their own copies of Yardstick, and all four countries keep separate category trees and rule sets, so a fraud rule written on a Sunday night reaches Belgium on Wednesday and Poland and Portugal never.
- **Ghost Inventory** — the Rotterdam city bike, sold on Monday evening, still appears in 214 alert e-mails on Tuesday morning, and it is the book's spine example for a reason.

This book tests its argument against **one fictional group, detailed enough to be realistic and consistent enough to be followed across fifty chapters.** The usual alternatives fail in opposite ways: a toy marketplace, one listing table and a search box, is too simple to show why the hard decisions are hard, and a real company's internals cannot be verified and its choices do not transfer. This page is the group's reference. The failure patterns of Unlisted: Fifteen Ways Listing Systems Go Wrong appear here as Veldmark's 2027 problems, and every chapter from Part I on opens with a scene whose people, systems and numbers come from this page.

Veldmark Group is fictional. It is a composite that borrows *patterns* common in the classifieds industry: a horizontal brand that grew country by country on one codebase, a vertical brand built for professional supply, acquisitions that arrived with their own clouds, a transactional product added on top of a contact marketplace, and a private-equity owner who wants both growth and efficiency. It is **not a portrait of any real company**. Nothing on this page or in any chapter is a claim about how a real group is built; the vendor products it runs are real, but its use of them is invented. The real companies the book discusses, such as Adevinta, OLX, Kleinanzeigen, Rightmove or Auto Trader, appear under their own names, with dated sources, and never as Veldmark.

The Group at a Glance

Veldmark is a privately held, multi-country, multi-vertical marketplace group headquartered in **Rotterdam**, with engineering hubs in **Lisbon**, **Warsaw**, **Lyon** and **Düsseldorf**. Since **2025** a private-equity investor holds the majority; a founding-family trust holds a minority stake. The owner wants growth and a lower run cost at once, and will one day sell the group or parts of it, which makes clean boundaries between brands a strategic asset.

Measure (fiscal 2026 = calendar 2026)	**Value**
Revenue	**€1.31 billion**
Adjusted EBITDA	about **€470 million** (about 36% of revenue)
Employees	about **5,200**, of whom about **1,450** in product and technology, including about **1,150 engineers**
Countries and marketplaces	**8** countries: Netherlands, Belgium, Poland, Portugal, Germany, Austria, France and Spain; **ten** country marketplaces (Tradeyard 4, Motorvane 4, Hearthly 2) plus Keybridge in three countries. The Netherlands, Belgium, Poland, France and Spain each host more than one brand
Live listings	about **17.25 million**
Monthly active users	about **57 million** across the three marketplaces (31M + 14M + 12M; users of several brands are counted once per brand)

Measure (fiscal 2026 = calendar 2026)	Value
Shared group services	**Veldmark Ads**, group finance on SAP S/4HANA, and little else

The margin is typical of mature classifieds: the marketplaces never hold the goods, and most revenue is subscriptions and visibility fees with low marginal cost. The last row matters most to this book. With almost nothing shared beneath its brands, the group pays for search, moderation, identity, media and data several times over.

How Veldmark Came to Be

Veldmark's history, like everything else about it, is fictional, but its shape follows the industry's (the documented history is in From Newspaper Columns to Portfolios: How the Classifieds Industry Evolved).

Year	Event
2004	Tradeyard launches in the Netherlands as a free general-classifieds site.
2006	Yardstick, Tradeyard's PHP application on MySQL, replaces the first version; it is still the core of Tradeyard in 2027.
2008	Motorvane launches in Germany, its main market, as an automotive marketplace for dealers and private sellers; the Netherlands, Poland and Austria follow later.
2009–2014	Tradeyard expands by copying itself: Belgium (2009) and Portugal (2014) launch on their own copies of Yardstick, each with its own database and deployment; Poland (2012) comes through the purchase of a local classifieds site, moved onto Yardstick's code but keeping its own Elasticsearch search. Each country has its own category tree.
2023	Veldmark acquires the French real-estate portal that becomes **Hearthly**.
2024	Veldmark acquires **Keybridge**, an agent CRM in France, Belgium and Spain. **Tradeyard Safe Deal** launches in the Netherlands.
2025	Veldmark acquires **Casavista** in Spain. The private-equity investor takes a majority stake. Safe Deal launches in Belgium.

Year	Event
2026	Casavista is rebranded **Hearthly.es**. Safe Deal launches in Poland.
January 2027	The board approves the **Common Ground** program.

Each phase left a layer in the systems: four category trees in country copies of one monolith, a separate automotive stack in another AWS region, Google Cloud and Azure from the acquisitions, and a payments product on top of a contact marketplace that was never designed to know when something was sold.

Four Brands, Four Business Models

Veldmark runs four businesses and one shared advertising unit. They differ in who supplies the listings, where the marketplace's job ends and how the money is made, the classification Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy develops.

Brand	What it is	Markets	Business model	2026 revenue
Tradeyard	Horizontal general classifieds: goods, plus small cars, property and jobs sections in Poland and Portugal	NL (since 2004), BE (2009), PL (2012), PT (2014)	Free private listings with paid visibility (value-added services, VAS: bumps, top listing, highlight), professional-seller subscriptions, insertion fees in some categories (Polish vehicles and property), transactional fees from **Safe Deal**	**€410M**: VAS €150M, pro subscriptions €135M, Safe Deal €75M, insertion fees €50M
Motorvane	Automotive marketplace: cars, vans, motorbikes	DE (main, since 2008), NL, PL, AT	Dealer subscriptions (packages with listing slots), depth products, lead and financing referral fees, private-seller listing fees	**€360M**: dealer subscriptions €238M, depth/VAS €52M, leads and financing referrals €40M, private-seller fees €30M

Brand	What it is	Markets	Business model	2026 revenue
Hearthly	Real-estate portal: sale, rent, new builds	FR (acquired 2023), ES (former Casavista, acquired 2025, rebranded 2026)	Agency subscriptions, new-build developer campaigns, valuation data and seller leads through **Valuo**, rental services (tenant application files)	**€395M**: agency subscriptions €270M, new-build campaigns €55M, Valuo data and leads €40M, rental services €30M
Keybridge	CRM and listing-management software for real-estate agents; publishes agency listings to many portals, including Hearthly's competitors ("multi-diffusion")	FR, BE, ES (acquired 2024)	SaaS subscriptions from **9,800** agency customers	**€45M**
Veldmark Ads	Group advertising unit selling across all brands	all	Display, native and sponsored listings; retail media for car makers, banks and insurers	**€100M**

The five lines add up to €1.31 billion: Tradeyard about 31% of revenue, Hearthly 30%,

Motorvane 27%, Veldmark Ads 8% and Keybridge 3%.

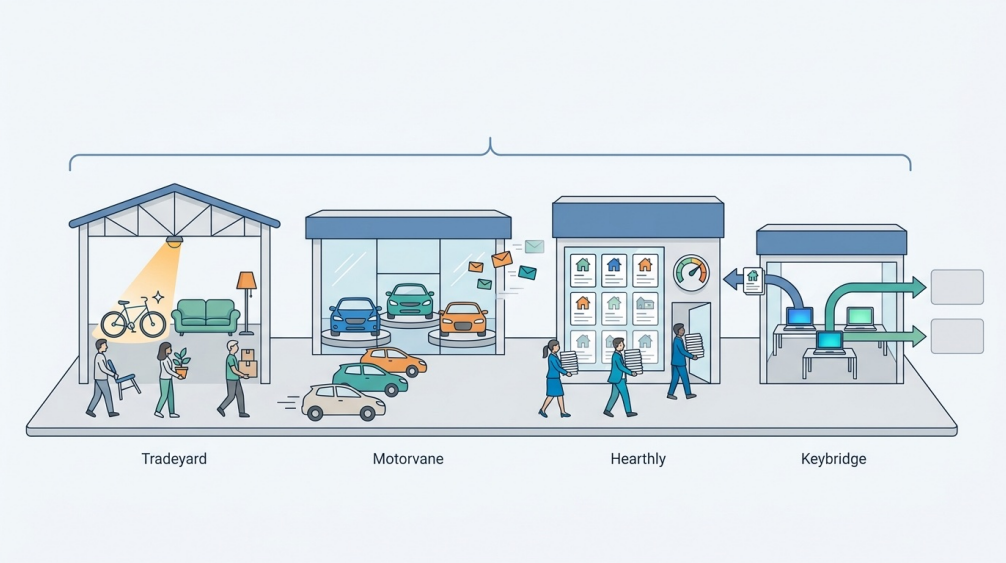


Figure 1: *One group, four marketplace physics: Tradeyard sells visibility to private sellers, Motorvane packages and leads to dealers, Hearthly subscriptions and valuation data to agencies, and Keybridge software to agents who also list on Hearthly's competitors.*

Four features of the brand portfolio recur throughout the book.

- **Tradeyard is mostly a contact marketplace with a transactional product inside it.** Most listings end at a message; Tradeyard never learns whether the sofa was sold. **Tradeyard Safe Deal** adds in-platform payment with shipping or "pay at pickup" for private sales in the Netherlands, Belgium and Poland, but not Portugal. Payments run on **Adyen for Platforms**: Adyen is the licensed party, and Veldmark holds no payment licence (Beyond the Contact: Payments, Shipping and Buyer Protection). The buyer pays a protection fee of **5% of the item price plus €0.70**. In 2026 Safe Deal handled **11.6 million** transactions worth about **€1.02 billion** of gross merchandise value (GMV), and earned **€75 million**: about €59 million of buyer fees and about €16 million of shipping-label margin.
- **Motorvane and Hearthly are professional-supply businesses.** Motorvane's **21,400 dealers** supply 91% of its listings, 88% of them through feeds from dealer management systems (DMS); Hearthly's **36,500 agencies** supply 84% of its listings through feeds from about **160 real-estate software vendors**. Their customers are businesses with contracts, invoices and lead expectations, which makes much of their architecture business-to-business software.
- **Valuo turns listings into a data product.** Valuo, Hearthly's automated valuation model (AVM) in France, is trained on France's public transaction records (the

DVF open data, *demandes de valeurs foncières*, published by the French tax administration) together with Hearthly's own listing data; it earns money through data products and seller leads routed to agencies, and a Spanish version is planned for 2028.

- **Keybridge must stay neutral**. Keybridge publishes its customers' listings to Hearthly and to Hearthly's competitors. Agencies fear that Veldmark will use their CRM data to favor Hearthly, which makes neutrality a design constraint on every data flow between Keybridge and the group.

Veldmark Ads is the exception to "four of everything": one ad stack, on Google Ad Manager with Prebid header bidding, sells inventory on all three marketplaces (Selling Attention: Display, Native and Retail-Media Advertising).

Volumes Every Design Is Sized Against

2026 figures, rounded.

Metric	Value
Live listings	**17.25M**: Tradeyard 13.2M (NL 6.1M, BE 2.4M, PL 3.6M, PT 1.1M); Motorvane 1.45M; Hearthly 2.6M (FR 1.9M, ES 0.7M)
New listings per week	Tradeyard **2.3M** (NL 1.05M, BE 0.42M, PL 0.62M, PT 0.21M); Motorvane 210,000; Hearthly 240,000
Monthly active users	Tradeyard 31M (NL 14.2M, BE 5.1M, PL 8.9M, PT 2.8M); Motorvane 14M; Hearthly 12M
Supply mix	Tradeyard: 78% of new listings from private sellers. Motorvane: 91% from 21,400 dealers, 88% of dealer listings via DMS feeds. Hearthly: 84% via feeds from about 160 software vendors, 36,500 agencies
Search	Group average **9,000 queries/s**; peak **31,000 queries/s** on Sunday evenings, 20:00–22:00 CET; Tradeyard carries about 70% of queries
Saved searches and alerts	**46 million** active saved searches (Tradeyard 31M, Motorvane 8M, Hearthly 7M); about **38 million** alert notifications a day by push and e-mail
Media	about **6.4 billion** stored images, **1.8 billion** of them at Hearthly; about 9.5 million new images a day

Metric	Value
Organic search	Tradeyard: about **40 million** indexed URLs; **46%** of web sessions start there
Moderation	Tradeyard: 94% of new-listing decisions automated, 6% (about 138,000 a week) to human review; about **210 moderators** across the brands, in-house and outsourced
Duplicates	Hearthly France: 1.9M listings describe about **1.12 million** distinct properties, **1.7** listings per property
Taxonomy	Tradeyard: four country trees with 1,420 (NL), 1,380 (BE), 980 (PL) and 760 (PT) leaf categories; **3,900** distinct attribute keys, 41% of them used on fewer than 0.1% of listings

Two caveats. The **Sunday-evening peak is a Veldmark illustration**, not an industry statistic; the documented public example of a weekly peak, Rightmove's, is in Sunday Evening Peaks: Reliability, Performance and Cost. And ratios matter more than totals: Tradeyard Netherlands and Portugal have almost the same monthly users per live listing (2.3 and 2.5), yet a new Dutch listing gets its first contact in a median of 2.6 hours and a Portuguese one in 19. Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics explains why, and why every latency target in this book names the liquidity metric it protects.

The Systems Landscape

The 2027 baseline estate is the product of that history, with almost nothing shared between the brands' stacks.

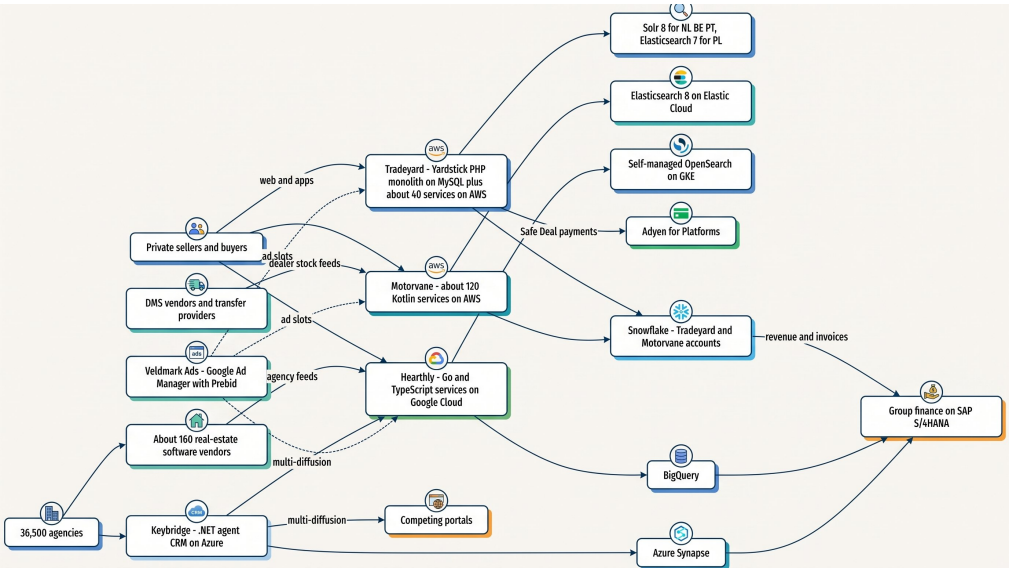


Figure 2: *Veldmark's 2027 baseline: three marketplaces on three stacks and two clouds, an agent CRM on a third cloud, and only advertising and finance shared beneath them.*

Area	Tradeyard	Motorvane	Hearthly	Keybridge
Core application	**Yardstick**: PHP monolith (since 2006) on MySQL in a co-located Amsterdam data center; about 40 microser-vices on AWS (eu-west-1, EKS) built since 2019 around it	About 120 Kotlin/JVM microser-vices on AWS (eu-central-1, EKS)	Go and TypeScript services on **Google Cloud** (GKE, europe-west1); Hearthly.es still partly on Casavista's code, including its login and review tool	.NET on **Microsoft Azure**

Area	Tradeyard	Motorvane	Hearthly	Keybridge
Search	**Solr 8**, self-hosted, for NL, BE and PT; **Elasticsearch 7** for PL	**Elasticsearch 8** on Elastic Cloud	**OpenSearch**, self-managed on GKE	—
Events	RabbitMQ inside Yardstick; Kafka (Amazon MSK) for the new services	Kafka (Amazon MSK)	Google Pub/Sub	Azure Service Bus
Identity	Own accounts	Own accounts; separate dealer-portal login	Own accounts; Hearthly.es on Casavista's login	Own accounts
Trust and safety	In-house rules engine and ML models (NL/BE); outsourced moderation queues (PL/PT)	Dealer business checks (KYB) at contract signature only; private-seller checks	In-house rules; manual review of new agencies	—
Data	Snowflake	Snowflake (separate account and model)	BigQuery	Azure Synapse
Media	Own image service plus CloudFront	S3 plus CloudFront	Google Cloud Storage plus Cloud CDN	—

Three facts in this landscape shape most of the book.

- **Yardstick does almost everything for Tradeyard.** Insertion, moderation hooks, category trees, fee rules, Solr indexing, alert crons and most web pages live in one

PHP application and one MySQL schema; its `ads` table has 214 columns read by 38 modules. It deploys **twice a week** and blocks about **140 Tradeyard engineers**. The newer AWS services read its data through RabbitMQ, replica queries and nightly exports, and nobody had a full list of those readers until 2027 (Every Change Is an Event: Event-Driven Listing Architecture).

- **Motorvane is modern but self-contained.** Its risk is the opposite of Tradeyard's: it has little reason to share, and a group platform that serves it worse than its own stack would be a regression.
- **Hearthly and Keybridge kept their clouds.** Any group platform must work across clouds or accept that some brands reach it through a bridge.

The People

Each person owns something the architecture depends on, and most of the book's tensions are between their legitimate goals. All are fictional.

Name	Role	What they own and want
Ingrid Halvorsen (she/her)	Group Chief Architect	The book's point of view; joined on **Monday 1 February 2027**. Reports to Mateus; chairs the architecture advice forum; owns the capability map, the context map and the cross-brand decision records. Has influence, not command, over the brands' teams.
Henrik de Boer (he/him)	Group CEO	Answers to the private-equity owner; wants growth without eating the Common Ground budget.

Name	Role	What they own and want
Mateus Carvalho (he/him)	Group CTO	Sponsor of Common Ground's engineering work; Ingrid's line manager; owns the cloud decision.
Noor Hassan (she/her)	Group Chief Product Officer	Owns the cross-brand product roadmap, from the bikes split to the AI proposals.
Olivier Petit (he/him)	VP Monetization	Owns pricing and paid products: VAS, dealer packages, agency tiers, Valuo leads.
Daan Visser (he/him)	Tradeyard engineering director	Owns Yardstick; knows where every workaround lives and why.
Katarzyna Nowak (she/her)	Tradeyard Poland general manager	Runs Tradeyard's most varied country: goods, cars, property and jobs.
Felix Brandt (he/him)	Motorvane CTO	Owns Motorvane's stack, dealer feeds and Dealer Hub, the dealer portal; skeptical of shared platforms built for Tradeyard's sofas.
Claire Fontaine (she/her)	Hearthly CEO	Owns Hearthly France and Spain; wants agencies' trust and Valuo's growth.
Julien Marchand (he/him)	Hearthly CTO	Owns Hearthly's Google Cloud estate, its feeds and its duplicate problem.
Emma Laurent (she/her)	Keybridge founder and general manager	Owns Keybridge's neutrality promise to 9,800 agencies; resists any data flow that could break it.

Name	Role	What they own and want
Aisha Rahman (she/her)	Group Head of Trust & Safety	Owns policy, moderation and fraud response; builds the shared trust-and-safety platform.
Tomasz Wiśniewski (he/him)	Head of the group search platform	Formerly Tradeyard's search lead; owns the move to a shared search platform.
Priya Raman (she/her)	Head of Data & ML	Owns the data platforms, metric definitions and model governance.
Lucía Ortega (she/her)	Group Data Protection Officer and regulatory lead	Owns GDPR, Digital Services Act (DSA) reporting and the privacy side of every design.
Sven Lindqvist (he/him)	Principal engineer, event platform	Owns the group event backbone and the Listing event contract.
Yara Mendes (they/them)	Head of Platform Engineering	Owns the shared platform teams: publication, taxonomy and attributes, entitlements, notifications.

The recurring customers, private seller **Femke** (she/her) in Rotterdam, dealer **Autohaus Ruhrblick** in Essen and agency network **Agence Rhône Habitat** in Lyon, are fictional too.

Eight Problems Veldmark Starts 2027 With

When Ingrid arrives, the marketplaces work and the group is profitable, but its effort and duplication grow faster than the business. Eight problems define her starting position; most are instances of patterns from Unlisted: Fifteen Ways Listing Systems Go Wrong.

1. **Four of everything**. Four identity systems, four trust-and-safety stacks, three search engines, three clouds, four product catalogs, three data warehouses. *(Deal-Shaped Architecture.)*

2. **Six minutes to searchable**. On Tradeyard, a new listing takes **6 minutes** at the 95th percentile (p95) from submission to searchable, because moderation and indexing work in batches. The target is **90 seconds** p95 by the end of 2028.
3. **Six weeks for a category**. Tradeyard's four country trees cannot share features, and a new category takes about **6 weeks**. The target is **2 days**. *(The Category Straitjacket, The Attribute Junkyard.)*
4. **Duplicates at Hearthly**. With **1.7 listings per property** in France, buyers see the same flat repeatedly and agencies argue about whose copy ranks first. *(Seeing Double.)*
5. **Payment-link phishing on Tradeyard**. Scammers posing as buyers send fake Safe Deal payment pages through messaging in the Netherlands and Belgium. *(Fast Mail for Fraudsters.)*
6. **Yardstick blocks Tradeyard**. Every Common Ground capability Tradeyard needs must first be extracted from it.
7. **Keybridge neutrality**. Agencies fear that Veldmark uses their Keybridge data to favor Hearthly. If they believe it, they leave, and the group loses a business and its standing with agencies at once.
8. **Overlap in Poland**. Tradeyard Poland's cars section competes with Motorvane Poland for the same dealers: about 2,900 Polish dealers hold accounts on both, and about 40% pay both. *(Deal-Shaped Architecture, inside the group.)*

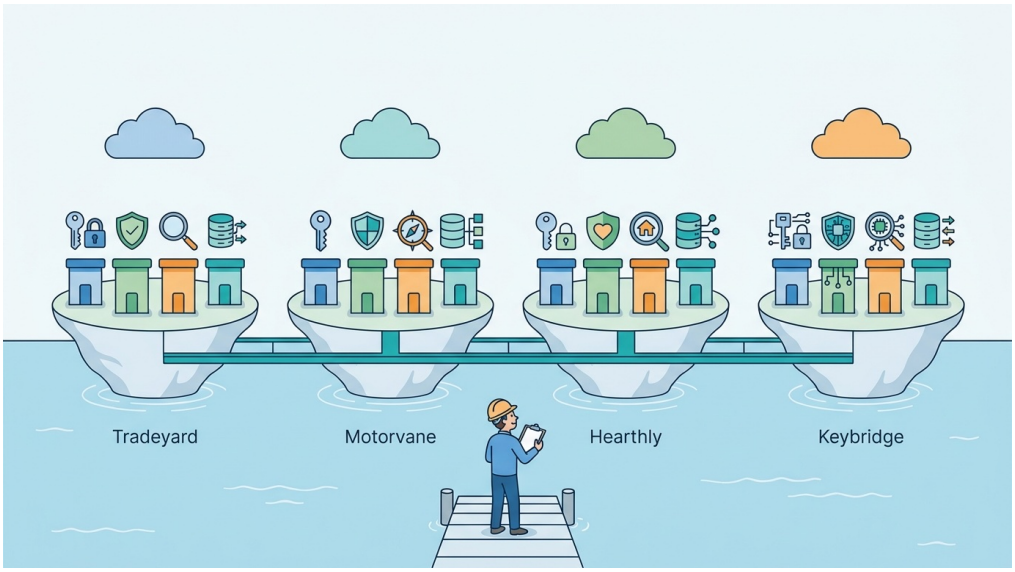


Figure 3: *Four of everything: in 2027 the brands run four identity systems, four trust-and-safety stacks, three search engines and three clouds, with little more than advertising and finance shared beneath them.*

None of these is a technology problem alone. The phishing exists because messaging was built as a feature, not as a trust surface; the neutrality problem because a commercial promise has no architecture yet; and the Polish overlap is a portfolio decision that systems can make easier or harder, but cannot make.

The Common Ground Program

In **January 2027** Veldmark's board approved **Common Ground**: **€140 million** over **2027–2030** to build shared platform capabilities beneath the brands. It has three goals.

- **Shared platform capabilities**: a group identity (**Veldmark ID**), trust and safety, media, a search platform, messaging, a product catalog and entitlements, an event backbone, a data platform, and a taxonomy and attribute service.
- **Retire Yardstick by the end of 2029**, through a strangler migration: the insertion flow moves out in the second quarter of 2028, search in the fourth quarter of 2028, and the monolith is decommissioned at the end of 2029.
- **Cut run costs by €38 million a year by 2030**.

Its governing principle is **share platforms, not products**. The marketplaces keep their brands, user experience and go-to-market; the group shares the capabilities beneath them. The hard design work is deciding which *part* of each capability is shared. A shared trust-and-safety pipeline can serve local policies; a shared search engine can serve brand-specific relevance; a shared entitlement ledger can serve brand-owned packages and prices.

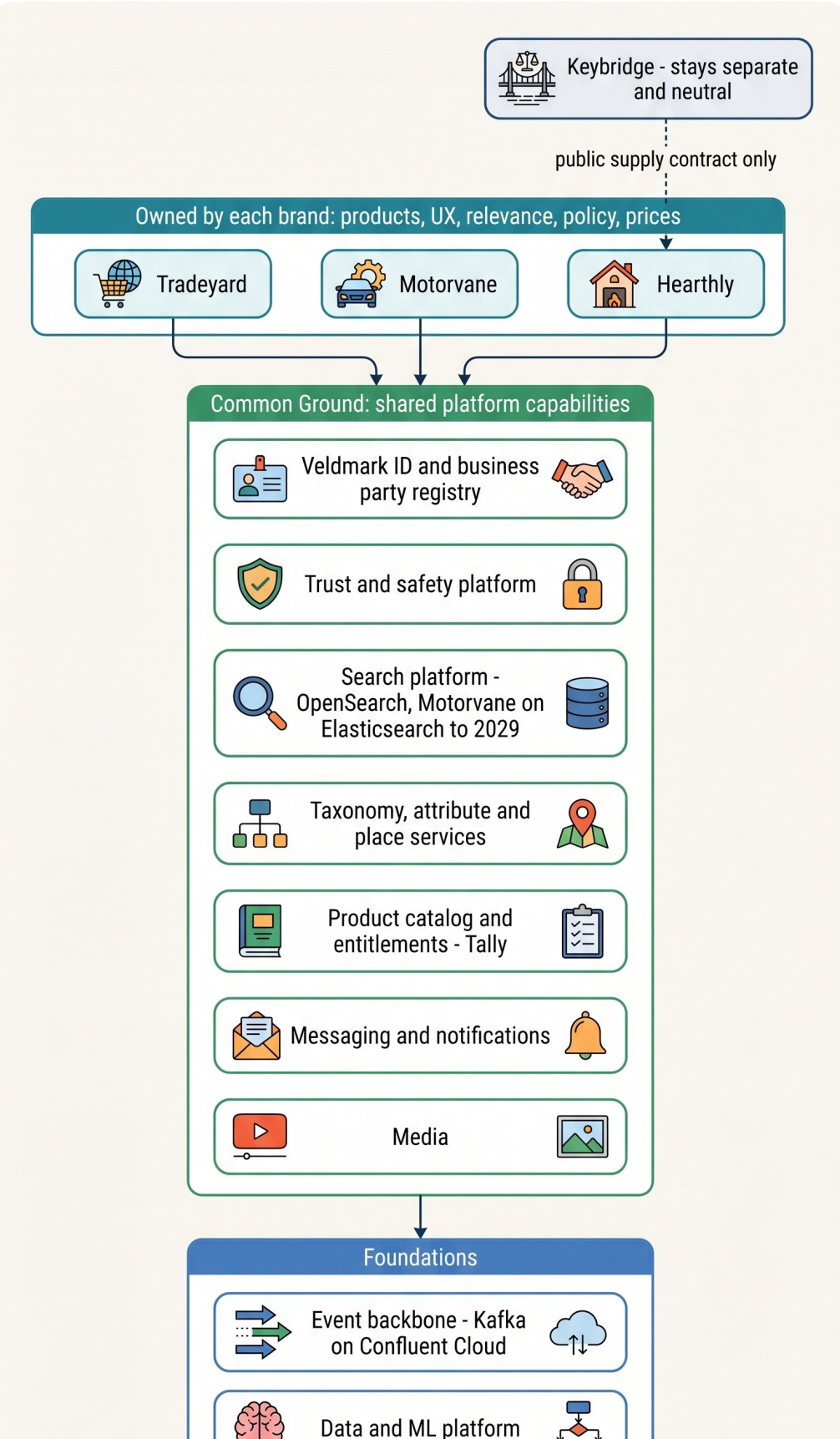

Keybridge - stays separate and neutral
public supply contract only
Owned by each brand: products, UX, relevance, policy, prices
Tradeyard
Motorvane
Hearthly
Common Ground: shared platform capabilities
Veldmark ID and business party registry
Trust and safety platform
Search platform - OpenSearch, Motorvane on Elasticsearch to 2029
Taxonomy, attribute and place services
Product catalog and entitlements - Tally
Messaging and notifications
Media
Foundations
Event backbone - Kafka on Confluent Cloud
Data and ML platform

Figure 4: *"Share platforms, not products": the brands keep what buyers and sellers see, Common Ground builds what lies beneath, and Keybridge stays outside by design.*

Common Ground is also a governance change. In **April 2027** the group adopted architecture principles and set up the **architecture advice forum**, a weekly, open meeting that gives advice rather than approvals, and made **decision records mandatory for cross-brand decisions**. Much of the book's Veldmark story happens in that forum.

The decisions the chapters rely on, in order:

Date	Decision or milestone	Chapter that works it
Jan 2027	Common Ground approved	this page
31 Mar 2027	Group capability map: 10 level-1 areas, 54 level-2 and 212 level-3 capabilities; five overlaps	Map What the Marketplace Does Before What Runs It
Mar 2027	Keybridge stays a separate, neutral SaaS: no Keybridge agency data used by Hearthly for ranking or sales targeting; annual independent review	Private Sellers, Dealers and Agencies: Modeling Who Lists, Serving the Pros: Dealer, Agent and Merchant Tooling
Apr 2027	Architecture principles and the advice forum; decision records mandatory	Who Owns What: Organizing Marketplace Engineering
May 2027	Common Ground context map: 19 bounded contexts, one owning team each	Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models
Jun 2027	Veldmark ID approved (one login, brand-specific profiles; rollout 2028); Ranking policy v1	Who Is on the Other Side? Identity and Verification, Ranking for Two Sides: Relevance, Freshness and Paid Visibility
1 Jul 2027	Group event backbone on **Kafka on Confluent Cloud** (AWS); **Listing event contract v1** live with a schema registry	Every Change Is an Event: Event-Driven Listing Architecture

Date	Decision or milestone	Chapter that works it
14 Sep 2027	Shared search platform on **Amazon OpenSearch Service** for Tradeyard (all countries) and Hearthly by the end of 2028; Motorvane stays on Elasticsearch until a 2029 review	Search Architecture for Listings: Indices, Freshness and Scale
Oct 2027	Group taxonomy and attribute service live, Tradeyard first; entitlement service **Tally** designed; group retention schedule	Category Trees That Can Change: Designing Marketplace Taxonomies, Packages, Bumps and Quotas: Products, Pricing and Entitlements, System of Record and Read Models: Listing Data Management
1 Nov 2027	The **bikes split** (city, e-bikes, cargo, kids) goes live	Category Trees That Can Change: Designing Marketplace Taxonomies, What Did They Mean? Query Understanding and Autocomplete
Nov 2027	Shared trust-and-safety platform live for Tradeyard NL and BE	Trust and Safety as a Platform
2028	Veldmark ID rollout; Hearthly Spain on the shared trust-and-safety platform (Q1); one group product catalog and entitlement service (Q2); Yardstick's insertion flow moved out (Q2) and search moved (Q4); 90-second publication target	Changing the Engine in Flight: Modernizing a Live Marketplace, From Submitted to Searchable: The Publication Pipeline
End 2029	Yardstick decommissioned	Changing the Engine in Flight: Modernizing a Live Marketplace
2030	Run-cost reduction of €38 million a year	After the Deal: Consolidating Platforms After Acquisitions

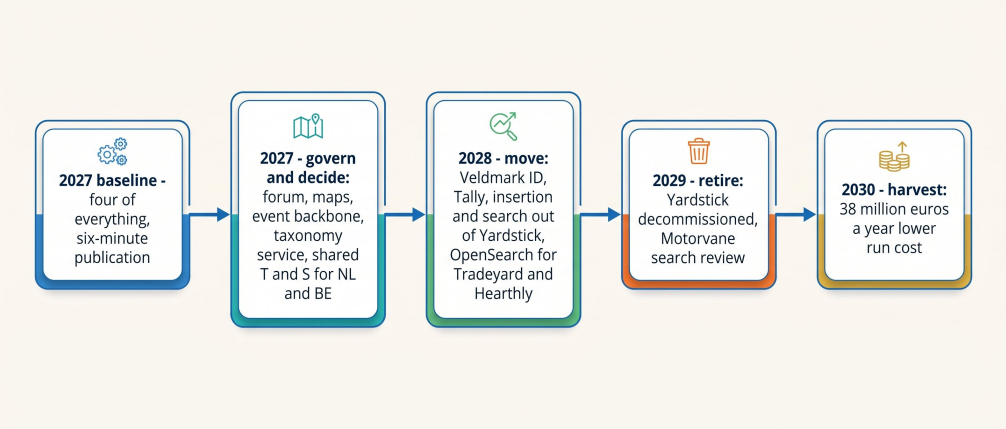


Figure 5: *Common Ground's four plateaus; each must leave every marketplace working on its own, not only as a step toward the next.*

AWS is the group's primary cloud from 2027. **Hearthly stays on Google Cloud through 2028** and **Keybridge stays on Azure**; the book treats a multi-cloud group as normal, not as a failure.

The Recurring Worked Examples

Five stories recur throughout the book. Their facts never change, so a reader can follow one listing across many chapters.

Example	Canonical facts	Where it does most work
The Rotterdam city bike (Tradeyard NL)	Femke lists a used city bike for **€240** on **Sunday 7 March 2027 at 19:42 CET**. Searchable at **19:48**, the six-minute baseline. Matches **312** saved searches; first contact **19:55**; **14** conversations within 24 hours. At **20:11** a "buyer" sends a fake Safe Deal payment link from a domain registered that day; Femke reports it at **20:19**; the account is suspended at **20:26**; a new rule withholds **37** similar messages. On **Monday 8 March** a Safe Deal "pay at pickup" order reserves the bike at **14:02**; it sells for **€225** with a buyer fee of **€11.95** (5% × €225 + €0.70). Marked sold at **18:10**, removed from search at **18:11** — but the alert digest assembled at 17:30 still features it in **214** e-mails at 07:00 on Tuesday.	The Listing Aggregate: Identity, Lifecycle and Invariants, From Submitted to Searchable: The Publication Pipeline, Tell Me When It Appears: Saved Searches and Alerts, From Interest to Contact: Messaging and Lead Management, Every Change Is an Event: Event-Driven Listing Architecture

Example	Canonical facts	Where it does most work
The Essen Volkswagen Golf (Motorvane DE)	A 2021 Golf 1.5 TSI, **48,000 km**, **€21,900**, from dealer **Autohaus Ruhrblick** through its DMS feed on **Wednesday 14 April 2027**. VIN decoding fills **31** attributes; the market-value model estimates **€22,800** ("good price"). **23 raw leads in 10 days** (14 e-mails, 7 calls, 2 test-drive bookings), **19** after removing repeat contacts from the same buyers. Price cut to **€21,400** on **Thursday 22 April** triggers price-drop alerts to **1,180** watchers. Sold **Thursday 29 April**; the feed's removal arrives **26 hours** later.	Many Doors In: Forms, APIs and Bulk Feeds, Recommendations for Inventory That Disappears, Packages, Bumps and Quotas: Products, Pricing and Entitlements

Example	Canonical facts	Where it does most work
The Lyon apartment (Hearthly FR)	A three-room (T3) **68 m²** flat in Lyon 7e for sale at **€389,000**, energy class (DPE, the French energy performance diagnostic) **D**. Listed in **May 2027** by **three agencies**, through two real-estate software vendors and Keybridge; one is **Agence Rhône Habitat**, a network of **14 branches**. Entity resolution clusters the three listings into one canonical property. Valuo estimate: **€372,000–€401,000**.	One Property, Three Listings: Duplicates and Entity Resolution, Private Sellers, Dealers and Agencies: Modeling Who Lists, Searching by Place: Map, Polygon and Commute Search, Models in Production: ML Platforms and Use Cases
The Barcelona fake-rental ring (Hearthly ES)	In **July 2027**, **26** accounts post **180** fake rental listings in Eixample using photos stolen from real listings, priced about **35%** below market, asking for deposits by bank transfer outside the platform. Detected through perceptual-hash matches against existing listings plus price anomalies; **63** people report losses of about **€71,000** before takedown.	Scams, Fakes and Account Takeovers: Fraud Prevention for Listings, Photos Sell: Media Pipelines for Listings, Who Is on the Other Side? Identity and Verification, Tell Me When It Appears: Saved Searches and Alerts

Example	Canonical facts	Where it does most work
The Sunday-evening brown-out (Tradeyard NL)	**Sunday 17 January 2027, 20:35 CET**: search p99 latency rises from about **400 ms** to about **7 s** for **48 minutes**, and about **9%** of searches fail. Causes: a TV campaign, a scheduled reindex job and Solr cache eviction at the weekly peak.	Search Architecture for Listings: Indices, Freshness and Scale, Sunday Evening Peaks: Reliability, Performance and Cost

The **Rotterdam bike is the book's spine**. In about 24 hours one €240 listing passes through a dozen systems and exposes a different weakness in almost every one: batch delays behind the six minutes, a messaging check that ran after delivery, a rule that reached Belgium three days late, an analytics extract that recorded the sale as "expired", and a digest that never asked whether the bike was still for sale. Six months later, Femke's erasure request finds **seven** copies of the listing. None of this is exotic; it is what a successful classifieds platform looks like after twenty years of reasonable local decisions.

The **Golf** plays the same role for professional supply: a complete but slow feed, a lead count the dealer disputes, a carousel that recommends a sold car, and a "Pro 150" package (150 listing slots for €1,450 a month) that billed Ruhrblick for cars the platform never showed. The **Lyon apartment** carries property identity, agency structure and valuation; the **Barcelona ring** shows signals that existed from the first hour failing because nothing combined them; the **brown-out** shows three harmless things becoming an incident at peak.

Smaller stories are introduced by the chapters that own them: the Motorvane morning when a vendor's truncated files unpublished **38,400** cars (Many Doors In: Forms, APIs and Bulk Feeds), the Tradeyard boost archaeology behind paid visibility (Ranking for Two Sides: Relevance, Freshness and Paid Visibility), Priya Raman's inventory of **146** production models (Models in Production: ML Platforms and Use Cases), and an Utrecht electric cargo bike listed through the photo-to-listing assistant pilot in December 2027 (AI Across the Listing Lifecycle).

How the Chapters Use Veldmark

Each chapter opens with a 2027 scene at Veldmark that poses its question, then answers it with concepts, dated public evidence from real marketplaces and a Veldmark worked

example. The table maps the eight problems to the chapters that pose them and the chapters that do most of the work.

Veldmark problem	Where it is posed	Where it is worked
Four of everything	Map What the Marketplace Does Before What Runs It, Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models	Who Is on the Other Side? Identity and Verification, Trust and Safety as a Platform, Search Architecture for Listings: Indices, Freshness and Scale, Data Platform for Marketplaces: Events, Lakehouse and Data Products, One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture, After the Deal: Consolidating Platforms After Acquisitions
Six minutes to searchable	From Submitted to Searchable: The Publication Pipeline	Every Change Is an Event: Event-Driven Listing Architecture, Search Architecture for Listings: Indices, Freshness and Scale, Tell Me When It Appears: Saved Searches and Alerts, Sunday Evening Peaks: Reliability, Performance and Cost

Veldmark problem	Where it is posed	Where it is worked
Six weeks for a category	Category Trees That Can Change: Designing Marketplace Taxonomies	Attributes Without Chaos: Schemas, Dictionaries and Validation, From Raw Supply to Usable Listings: Normalization and Enrichment, What Did They Mean? Query Understanding and Autocomplete, Found by Google: Web, App and SEO Architecture for Listings
Duplicates at Hearthly	One Property, Three Listings: Duplicates and Entity Resolution	Photos Sell: Media Pipelines for Listings, Searching by Place: Map, Polygon and Commute Search, Many Doors In: Forms, APIs and Bulk Feeds
Payment-link phishing	From Interest to Contact: Messaging and Lead Management	Trust and Safety as a Platform, Scams, Fakes and Account Takeovers: Fraud Prevention for Listings, Who Is on the Other Side? Identity and Verification
Yardstick	The Listing Aggregate: Identity, Lifecycle and Invariants, Every Change Is an Event: Event-Driven Listing Architecture	System of Record and Read Models: Listing Data Management, Who Owns What: Organizing Marketplace Engineering, Changing the Engine in Flight: Modernizing a Live Marketplace

Veldmark problem	Where it is posed	Where it is worked
Keybridge neutrality	Private Sellers, Dealers and Agencies: Modeling Who Lists	Serving the Pros: Dealer, Agent and Merchant Tooling, One Property, Three Listings: Duplicates and Entity Resolution, Models in Production: ML Platforms and Use Cases, Rules of the Marketplace: DSA, DAC7 and Vertical Regulation
Overlap in Poland	Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy	Same Platform, Different Physics: How Listing Verticals Differ, Private Sellers, Dealers and Agencies: Modeling Who Lists, How Listing Marketplaces Make Money

Part VIII follows Common Ground as a program, through to Built to Be Replaced: Evolving a Marketplace Platform Over a Decade; Part IX leaves Veldmark and tests the book against the public record of real groups, starting with Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform.

Three conventions hold throughout. **Veldmark is always labeled fictional** the first time it appears in a chapter. **Its facts come from one canonical record**, so the six minutes, the 1.7 listings per property and the €240 bike mean the same thing everywhere. And a chapter that needs a new detail adds it without contradicting what is established here.

With the group, its problems and its program introduced, Part I begins where Ingrid began in February 2027: by asking what kind of marketplace each part of Veldmark actually is, and why the answer decides the architecture. Continue with Part I — MARKETPLACE: The Business Behind the Listings.

Sources and Evidence

Company engineering and investor sources

- None. Veldmark Group, its brands (Tradeyard, Motorvane, Hearthly, Keybridge, Casavista, Veldmark Ads, Valuo, Safe Deal, Tally), its people, customers, figures,

systems and events are fictional and invented for teaching; they are not drawn from any single real company.

Standards, regulation and documentation

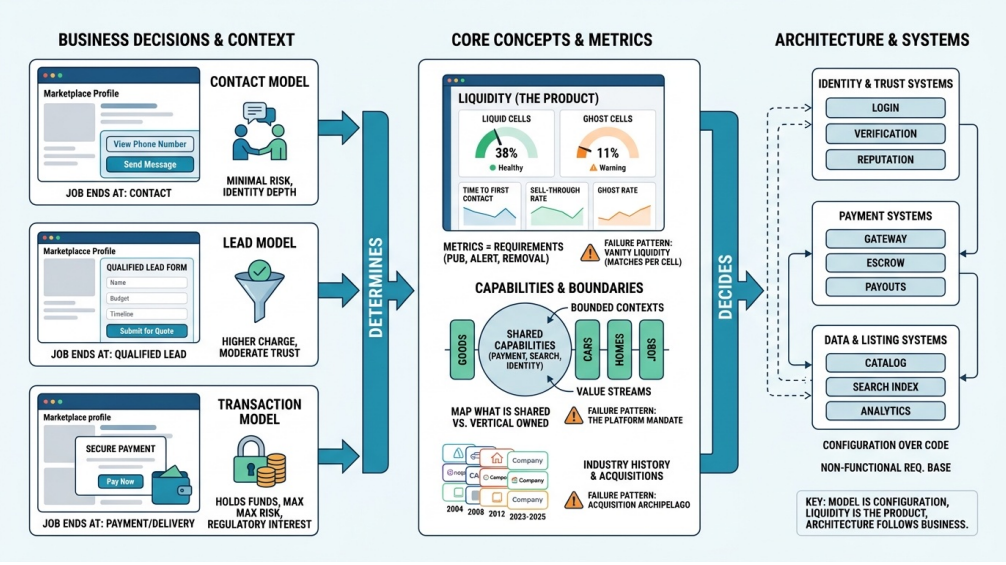
- Demandes de valeurs foncières (DVF), Direction générale des finances publiques, data.gouv.fr, n.d. (updated twice a year) — the real French open dataset of property transactions on which the fictional Valuo is described as trained.

Research and practitioner sources

- The scenario's shape draws on the generalized marketplace domain models described in this book's provenance note and on the industry history covered, with dated sources, in From Newspaper Columns to Portfolios: How the Classifieds Industry Evolved and the Part IX case studies.

The vendor products named (Solr, Elasticsearch, OpenSearch, Kafka, Confluent Cloud, RabbitMQ, Google Pub/Sub, Azure Service Bus, Snowflake, BigQuery, Azure Synapse, Adyen for Platforms, Google Ad Manager, Prebid, SAP S/4HANA) are real; their capabilities are described, with sources, in the chapters that use them. Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

Part I — MARKETPLACE: The Business Behind the Listings



IN THIS PART, YOU WILL: Describe a listing marketplace as a business before describing it as software: its model, its liquidity, its history, its capabilities, its bounded contexts and the physics of each vertical it serves.

WHY THE BUSINESS CARES: The model decides what the marketplace can charge for, which risks it carries and which regulators take an interest. Liquidity decides whether professionals renew and buyers come back. Both are business facts that the systems either serve or work against.

WHY ARCHITECTS SHOULD CARE: Every boundary, shared platform and non-functional requirement later in this book is justified by something in this part. Without it, platform debates become contests between system owners.

KEY POINTS:

- **The model is an architecture decision, made per country and category**. Where the marketplace's job ends (contact, lead, payment or delivery) decides identity depth, payments, trust and data. One brand usually holds several models, so the model belongs in configuration, not in code.
- **Liquidity is the product, and its metrics are the requirements**. Network effects work per place and category. Time to first contact, sell-through and ghost rates become publication, alert, removal and capacity targets.
- **Map what the marketplace does before deciding what to share**. Capabilities, value streams, bounded contexts and vertical profiles turn "should this be one platform?" into a decision that can be reviewed: share the machinery, let each vertical own its content.

FAILURE PATTERNS THIS PART ADDRESSES:

- **Vanity Liquidity** — Defines success as matches per cell, not listings and visits.
- **The Forced Fit** — Shows why a platform built around one model or vertical fails the others.
- **Deal-Shaped Architecture** — Gives a group the capability vocabulary to see what it pays for several times.

This book starts with the business, not the systems, because **in a listing marketplace the business model decides the architecture** more directly than in most industries. A classifieds site whose job ends at the contact, a portal that sells qualified leads and a marketplace that holds buyers' money need different identity, trust, payment and data systems, even when their search pages look alike.

Sharing decisions need three pieces of business knowledge, and the part supplies them in order. First, know what kind of marketplace you run and what it sells: **liquidity**, the likelihood that a listing finds a buyer in a given place and category fast enough that both come back. Second, know how the industry got here, because today's portfolios carry the systems of every earlier wave. Third, describe the marketplace in terms that outlive systems and reorganizations, so that sharing decisions rest on evidence.

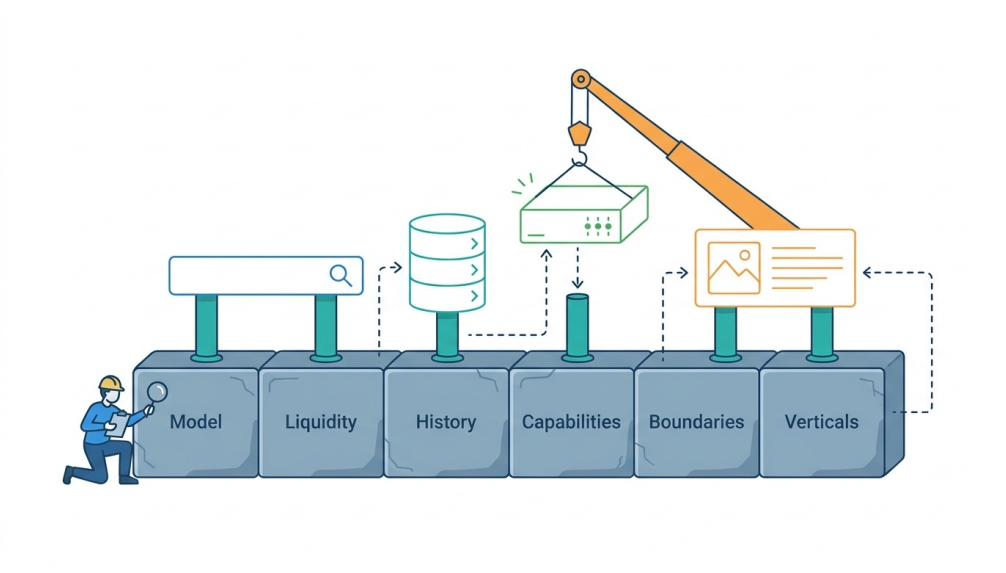


Figure 1: *Part I lays the foundation first: the marketplace model, liquidity, history, capabilities, boundaries and vertical physics that every later system decision rests on.*

Chapters in this part

Chapter	The question it answers	Read it when…
Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy	What kind of marketplace is this, and why does the answer decide the architecture?	A brand plans to add payments, leads or guarantees.
Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics	What does a marketplace sell, and how do we know whether it works?	Dashboards count listings and visits, or nobody can say why a latency target exists.
From Newspaper Columns to Portfolios: How the Classifieds Industry Evolved	How did today's marketplace groups form, and what did each of the five waves, from print columns to private-equity break-ups, leave in their systems?	You inherit a portfolio of local stacks, or the owner may sell a brand.

Chapter	The question it answers	Read it when…
Map What the Marketplace Does Before What Runs It	How do we describe what the marketplace does independently of its systems?	You must find what the group pays for several times.
Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models	Where do the domain boundaries go, and who owns the listing's truth?	Every change to the listing table needs a cross-team meeting.
Same Platform, Different Physics: How Listing Verticals Differ	What genuinely differs between verticals, and what can be shared?	A program proposes one platform for goods, cars, homes and jobs.

How this part connects

This part supplies the vocabulary the rest of the book uses: marketplace profile, liquidity cell, capability, bounded context and vertical requirement profile. It builds on the Preface's failure patterns (Unlisted: Fifteen Ways Listing Systems Go Wrong) and running example (Meet Veldmark: Four Brands, One Platform to Build). Part II — LISTING: Modeling the Thing Being Sold designs the listing domain model inside the boundaries drawn here. Later parts justify their freshness, latency and sharing decisions by pointing back to this part's liquidity metrics and share test.

The Veldmark thread

Veldmark, the fictional marketplace group this book follows, approved its €140 million Common Ground program in January 2027. Ingrid Halvorsen joined as Group Chief Architect in February and spent her first months testing it against the business. In this part her team:

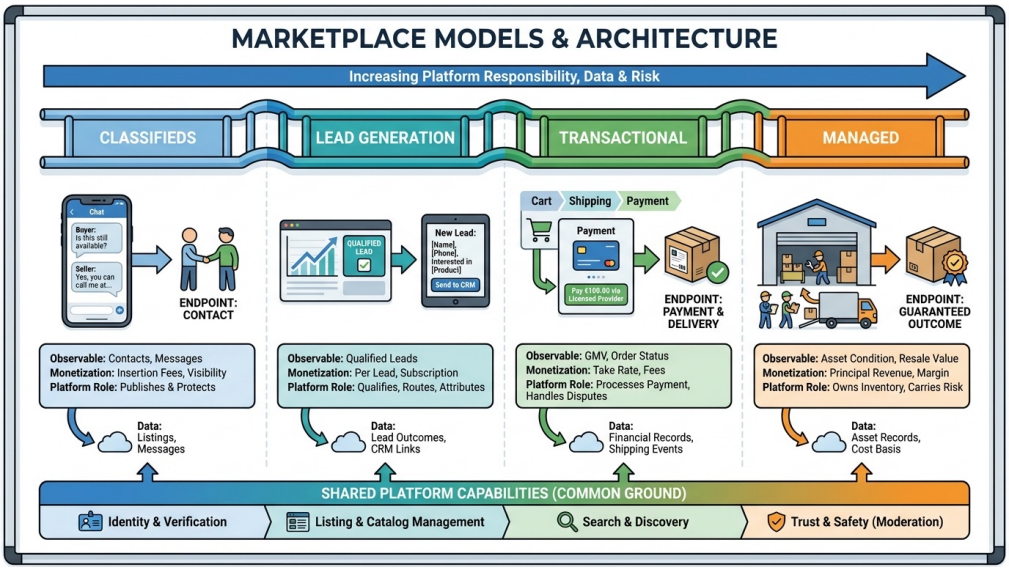
- Classifies the group into **20 marketplace profiles** (brand × country × category group) and resolves the Poland overlap between Tradeyard's cars section and Motorvane.
- Measures liquidity per cell (region × leaf category): **38%** of Tradeyard Netherlands' cells are liquid, against **11%** in Portugal. From then on, every new latency or freshness objective must name the liquidity metric it protects.
- Reads the group's history (Tradeyard launched in 2004, Motorvane in 2008, a Polish purchase in 2012, three acquisitions in 2023–2025) as sediment left by each wave,

and records in March 2027 that any brand may be sold within five years, so every shared platform needs a documented exit path.

- Presents a capability map on 31 March 2027 with **five overlaps**, one commercial overlap and one deliberate duplicate.
- Gets a shared-versus-specific table ratified in April and a context map of **19 bounded contexts** approved by the architecture advice forum in May 2027.

Begin with Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy.

1. Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy



IN THIS CHAPTER, YOU WILL: Classify a marketplace along six independent axes, starting with the one that matters most: where the platform's job ends. You will see what each answer adds to the architecture (identity depth, payments, trust, search, monetization and data) and how documented marketplaces moved between models, sometimes successfully and sometimes at great cost.

WHY THE BUSINESS CARES: The marketplace model decides what the business can charge for, which risks it carries on its balance sheet and which regulators take an interest. A general manager who moves from "we connect buyers and sellers" to "we guarantee the purchase" changes the company's margins, licences and liabilities, not only its checkout.

WHY ARCHITECTS SHOULD CARE: Most expensive marketplace platform mistakes come from building one model's assumptions into systems

that must serve another. A platform built for classifieds cannot simply be switched to transactions, and a transactional platform imposed on a lead-generation portal fails just as badly. The taxonomy is the first input to every boundary drawn later in this book.

KEY POINTS:

- **A marketplace is defined by where its job ends.** Classifieds end at the contact, lead-generation portals at a qualified lead, transactional marketplaces at a completed payment and delivery, and managed marketplaces at a guaranteed outcome. Each step adds bounded contexts, data the platform must hold and liabilities it must carry.
- **Classify along independent axes, and per category rather than per brand.** The endpoint, horizontal or vertical scope, seller type, item identity, money flow and risk ownership vary separately. One brand usually contains several profiles, so the profile belongs in configuration per country and category, not in a brand-wide assumption buried in code.
- **Moving toward the transaction is a category-gated bet, not a destination.** leboncoin, Kleinanzeigen and Marktplaats added payments and shipping for goods but not for cars or homes. OLX deprioritized its pay-and-ship flow in Europe, and OLX Autos, Zillow Offers and CarGurus' CarOffer retreated from inventory risk. Build so the endpoint can move per category in both directions.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Forced Fit** — A shared platform designed around one marketplace model (usually transactional goods) is imposed on brands whose job ends somewhere else, and their teams route around it.
- **Fast Mail for Fraudsters** — When the platform's job ends at the contact, messaging becomes the place where the money is negotiated, so it is also where fraud concentrates unless the model's endpoint is designed into trust and safety.
- **Vanity Liquidity** — A marketplace that does not know where its job ends cannot say what success is, so it counts listings and visits instead of the contacts, leads or completed sales its model actually delivers.

Ask five people in a marketplace group what kind of business it runs, and you usually get five answers that are each partly right. The general manager says "classifieds". The product director says "we are becoming transactional". The sales director says "we sell dealer subscriptions". The finance team says "most of our money is advertising and visibility". The trust and safety lead says "we are a messaging platform with listings

attached". **Each answer describes a different end point of the marketplace's job**, and each one implies a different architecture.

This chapter gives that conversation a shared vocabulary, and opens Part I because every later boundary in the book depends on it. It defines four marketplace models (classifieds, lead-generation portals, transactional marketplaces and managed marketplaces), the axes along which they vary, and what each one asks of the systems underneath. It owns the terms *classifieds*, *lead-generation marketplace*, *transactional marketplace* and *horizontal* versus *vertical*, and it introduces the *marketplace profile* that later chapters build on: Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics measures how well each profile's job is done, Map What the Marketplace Does Before What Runs It and Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models map what each one needs, and One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture makes the profile a configuration scope. The running example is Veldmark's four brands from Meet Veldmark: Four Brands, One Platform to Build. How the industry arrived at today's portfolios belongs to From Newspaper Columns to Portfolios: How the Classifieds Industry Evolved, and the detailed differences between verticals to Same Platform, Different Physics: How Listing Verticals Differ.

The model is an architecture decision, not a pricing decision. Moving the endpoint by one step, from contact to lead or from lead to payment, adds new bounded contexts, new data with new retention rules, new identity requirements, new regulators and new failure modes. The decision therefore needs to be explicit, recorded per country and category, and revisable, because the documented record shows marketplaces moving in both directions.

Setting the Scene at Veldmark: One Group, Four Models

Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), approved its Common Ground program in January 2027: €140 million over four years to build shared platform capabilities beneath four brands. Ingrid Halvorsen joined as Group Chief Architect in February. In her second week, Noor Hassan, the Group Chief Product Officer, brought her a slide titled "One listing service, one checkout, four brands". Tradeyard Safe Deal, the in-platform payment and shipping product, had processed **11.6 million** transactions and about **€1.02 billion** of gross merchandise value (GMV) in 2026, earning about €75 million in buyer fees and shipping-label margin, a take rate of about 7%. Why not extend it to Motorvane's cars and Hearthly's apartments and build a single listing and order platform for the whole group?

Ingrid did not argue with the slide. She asked Mateus Carvalho, the Group CTO, for one

afternoon with the brand leaders and put a grid on the wall: brands and countries down the side, five questions across the top.

1. Where does our job end: at the contact, at a lead delivered to a professional, at a payment, or at a delivery?
2. Who supplies the listings: private people, professionals, or both?
3. Is each item unique, or could it be matched to a catalog product?
4. Who holds the buyer's money, and when?
5. If the deal goes wrong, who loses money first?

By the end of the afternoon the grid had twenty rows, not four, and the answers did not line up with the brands. Tradeyard's goods categories in the Netherlands, Belgium and (since 2026) Poland ended at the contact for most listings but at a completed payment for those sold through Safe Deal. Tradeyard Portugal had no Safe Deal at all. Tradeyard Poland's car and property sections charged insertion fees and ended at the contact. Motorvane ended at a lead delivered to one of **21,400** dealers, and sold financing referrals on the side. Hearthly ended at a lead delivered to an agency, but its Valuo valuation product also generated seller leads before a property was even listed. Keybridge, the agent CRM acquired in 2024, was not a marketplace at all: it was software that published agencies' listings to many portals, including Hearthly's competitors.

The most uncomfortable row was Poland. **Tradeyard Poland's cars section and Motorvane Poland served the same dealers under two different models**: per-listing insertion fees on one side, dealer subscriptions with listing slots on the other. Both brands' sales teams were calling on the same showrooms.

Ingrid's conclusion, recorded in the first decision record of her tenure, was not that Noor's proposal was wrong. A shared order platform might be exactly right for Tradeyard's shippable goods. It was that **"one checkout for four brands" assumed one model, and the group ran at least four**. The rest of this chapter is the vocabulary she used to explain why.

Where the Marketplace Ends: Four Models

A **listing marketplace** is an intermediary that publishes offers from many suppliers and helps buyers find them. Listing marketplaces differ most in how far along the buyer's journey the platform's responsibility reaches, so this book classifies them first by **endpoint**: the last step of the transaction for which the platform takes responsibility.

Classifieds

Classifieds are marketplaces whose job ends when a buyer contacts a seller: a message, a revealed phone number, an e-mail form. Price negotiation, payment and handover

happen between the parties, usually off the platform. The platform publishes, ranks and protects; it does not take part in the sale. Facebook's announcement of Marketplace in October 2016 states the model in a single sentence: "Facebook does not facilitate the payment or delivery of items in Marketplace" (Meta Newsroom, 2016).

Because the platform never sees the money, classifieds monetize **visibility and access**: insertion fees in some categories, paid visibility products such as bumps and top placements (often called value-added services, VAS, a term How Listing Marketplaces Make Money owns), professional-seller subscriptions and display advertising. Adevinta's 2023 accounts describe the core of the business as "sales of classified ad listings and related features" and report classifieds revenue of €1,399 million out of €1,826 million group revenue, with advertising at €312 million and transactional revenue at €103 million (Adevinta Annual Report 2023, published 2024). That is about 77% classifieds, 17% advertising and 6% transactions, years into a declared transactional push.

Lead-generation marketplaces

A **lead-generation marketplace** (or lead-generation portal) ends when a qualified enquiry is delivered to a professional: a dealer, an estate agent, an employer. Both models deliver contacts, so the difference looks commercial rather than technical. But the professional pays because the lead has value, so the platform must **qualify**, **count**, **route and attribute leads**, and prove their value at renewal time.

The documented models differ in how professionals pay and who receives the lead:

- **Subscriptions priced on value delivered**. Rightmove's revenue is per-advertiser subscriptions plus products: average revenue per advertiser (ARPA) was £1,621 a month in 2025 (Rightmove full-year results, 2026). Autotrader reports average revenue per retailer forecourt (ARPR) of £2,995 a month for the year to March 2026 (Autotrader FY26 results, 2026).
- **Share of voice**. Zillow's market-based Premier Agent pricing distributes connections "in proportion to their share of voice, or a Premier Agent partner's share of total advertising purchased in a particular zip code" (Zillow 10-K for 2025, filed 2026).
- **Pay at close**. Zillow Preferred agents "pay a performance advertising fee when a real estate transaction is closed with one of the leads, generally within two years" (same filing). AVIV France "monetises seller leads through a commission-share model" (OnlineMarketplaces, 2023).
- **The lead belongs to the lister**. CoStar's Homes.com advertises a "your-listing your lead basis", a deliberate contrast with portals that resell buyer leads to agents who did not list the property (CoStar 10-K for 2025, filed 2026).

Pay-at-close pricing pulls the portal into the transaction's *outcome* without touching its *money*. The platform must learn, months later, whether a lead became a sale, which is a matching and reconciliation problem rather than a payments problem.

Transactional marketplaces

A **transactional marketplace** ends when payment and delivery are complete on the platform. The buyer pays through the platform (or, more precisely, through its payment provider), the item ships with a platform-integrated carrier or is handed over against a platform-confirmed payment, and disputes are resolved by the platform. Revenue becomes a **take rate** on GMV. eBay defines take rate as "net revenues divided by GMV" and reports 13.94% for 2025 on GMV of $79.6 billion (eBay 10-K for 2025, filed 2026). Andreessen Horowitz's 2020 marketplace metrics distinguish GMV, "the total sales dollar volume of merchandise transacting through the marketplace", from revenue, the part the marketplace "takes" (a16z, 2020). Classifieds usually have no observable GMV at all, which is why they report revenue per advertiser or per listing instead. Bill Gurley's 2012 list of marketplace factors puts the commercial attraction of the transactional rung bluntly: "if you bring the cash, you are king", because it is "much easier to extract reasonable economics when you are in the flow of payment" (Gurley, 2012).

Vinted is the clearest example of a marketplace built transactional from the start. It describes "its proven marketplace model in an ecosystem of vertically-integrated shipping and payment infrastructure" and reported GMV of €10.8 billion and revenue of €1.1 billion for 2025 (Vinted financial results 2025, published 2026). Sellers list free; the main fee is paid by the buyer at checkout, now called "the Vinted fee" (formerly the "Buyer Protection fee"), "usually 5% + $0.70" in the US (Vinted US price list, updated 2026).

Managed marketplaces

A **managed marketplace** goes one step further: the platform takes inventory, quality or outcome risk itself. It buys and resells, guarantees the result, or curates supply so tightly that it acts as a retailer. Autotrader's Autorama leasing business is a documented example inside a classifieds group: about 1,350 vehicles were "temporarily taken on balance sheet" in the year to March 2026 (Autotrader FY26 results). iBuying, in which the portal buys homes directly from sellers, is the best-known example from real estate; Pattern 3 below covers its failure.

The endpoint ladder

The four models form a ladder. Each rung adds what the platform must do, know and carry.

Model	Platform's job ends at	What the platform observes	Primary monetization	Who holds money and risk	Documented examples
Classifieds	Contact	Views, contacts, conversations; *not* whether a sale happened	Insertion fees, VAS, pro subscriptions, display ads	Parties transact off-platform; platform holds no funds	Facebook Marketplace (2016), leboncoin, Kleinanzeigen and Marktplaats for most categories
Lead generation	Qualified lead delivered to a professional	Leads, lead channel, sometimes the outcome via CRM or public records	Subscriptions priced on leads, share of voice, pay-per-lead, pay-at-close	Off-platform transaction; platform may carry estimation risk on success fees	Rightmove, Autotrader, Zillow Premier Agent, CarGurus, Homes.com
Transactional	Completed payment and delivery	Order, payment, shipping and dispute events; true GMV	Take rate: buyer fees, commissions, shipping margin	Licensed payment provider holds funds; platform carries buyer-protection losses	Vinted, eBay; transactional layers at leboncoin, Kleinanzeigen, Marktplaats

Model	Platform's job ends at	What the platform observes	Primary monetization	Who holds money and risk	Documented examples
Managed	Guaranteed outcome or owned inventory	Everything above, plus condition, cost and resale data	Spread, high take rate, principal revenue	Platform carries inventory and price risk on its balance sheet	Autorama (Autotrader), Zillow Offers (closed 2021), OLX Autos (exited 2023–2025)

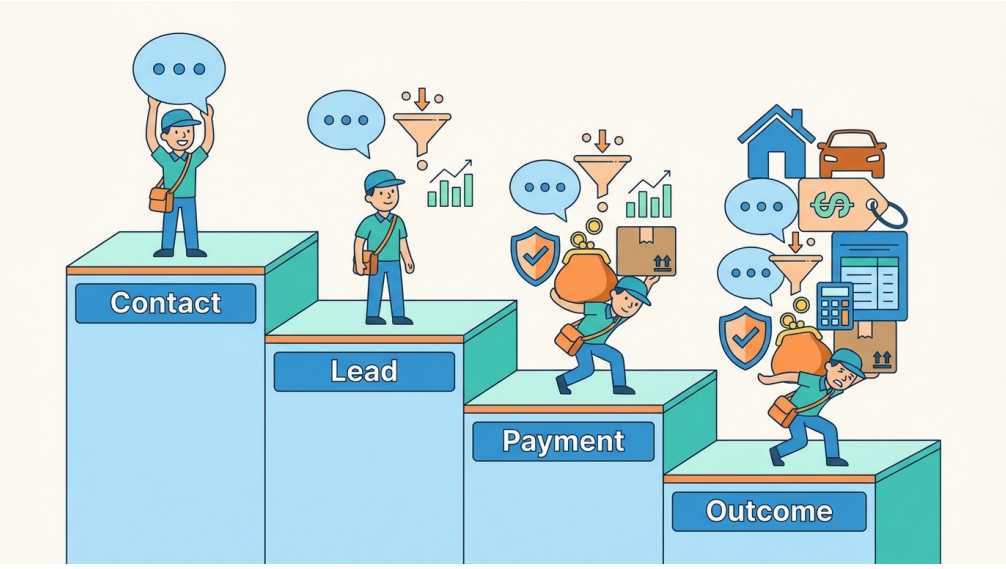


Figure 1: *Each rung of the endpoint ladder adds what the platform must do, know and carry: contacts, then leads, then payments and disputes, then inventory and price risk.*

One adjacent business model is not on the ladder but appears in almost every marketplace group: **software for professionals**. Agent CRMs, dealer management add-ons and listing-distribution tools serve the supply side directly. Scout24 reports "Transaction Enablement" revenue of €106.6 million for 2025 from CRM, data and valuation products (Scout24 preliminary 2025 results, published 2026), and Zillow owns the Follow Up Boss CRM. These are SaaS products whose customer is the professional and whose architecture follows SaaS rules: tenancy, data ownership and, where they publish to

rival portals, neutrality. Serving the Pros: Dealer, Agent and Merchant Tooling covers them.

Six Axes, One Profile

Six axes vary independently, and together they form what this book calls a **marketplace profile**: the combination of answers that describes one marketplace in one country and category. The endpoint is the most consequential axis, but not the only one.

Axis	Values	What it decides first
Endpoint	Contact · lead · payment · delivery · guaranteed outcome	Bounded contexts, observable outcomes, regulatory obligations
Scope	Horizontal (many categories) · vertical (one domain)	Taxonomy breadth, attribute depth, search tuning, brand strategy
Seller type	C2C (consumer to consumer) · B2C (business to consumer) · B2B · mixed	Supply channels, account model, trader obligations, monetization unit
Item identity	Unique items · items with a strong external identity (VIN, address) · catalog products	Enrichment, deduplication, entity resolution, price guidance
Money flow	Off-platform · via the platform's payment provider · via the platform's own licensed entity	Payments architecture, ledger, licensing, tax reporting
Risk ownership	Parties · platform for buyer protection · platform for inventory	Provisions, dispute systems, fraud appetite, balance sheet

Horizontal marketplaces cover many unrelated categories in one place: Kleinanzeigen, leboncoin, Marktplaats, OLX. They win on breadth and habit, and their supply is mostly private people. **Vertical** marketplaces specialize in one domain, usually a high-value one with professional supply: property (Rightmove, Idealista, SeLoger), vehicles (mobile.de, Autotrader, AutoScout24), jobs (StepStone, InfoJobs). The economics differ sharply. Rightmove and Autotrader both report operating margins of about 70% on professional subscriptions (sources above), while Adevinta's 2023 report shows the margin of its French business falling two points "mainly reflecting the evolution of the business

mix with an increasing share of transaction services and a decreasing share of highly profitable advertising revenues".

Horizontal groups also contain verticals. Andrew Parker's much-cited 2010 post on "the spawn of craigslist" showed how startups carved niches out of Craigslist's categories (Parker, 2010). The same unbundling happens inside groups: Adevinta reorganized in 2023 around "Re-Commerce, Mobility, and Real Estate & Emerging Verticals", and Prosus now reports OLX by vertical, with motors, real estate and jobs at 71% of OLX revenue in the year to March 2026 (OLX Group, 2026). A horizontal brand's car section and a sister vertical brand can end up competing for the same dealers, which is Veldmark's Polish problem.

Seller type matters because consumers and professionals respond to different prices and carry different legal status. Consumers buy per item and per event: an insertion fee, a bump, a buyer fee. Professionals buy monthly contracts with quotas and upgrades. Legally, the line between a consumer and a **trader** (a person acting for business purposes) decides which consumer-protection rules apply, what the platform must collect about the seller and what it must disclose to buyers. Private Sellers, Dealers and Agencies: Modeling Who Lists owns the trader-status model.

Item identity decides how much of the listing can be derived rather than typed. Second-hand goods are usually unique items with weak structure. Vehicles have a strong external identity (the vehicle identification number, VIN) that resolves to manufacturer data. Properties have a durable address and parcel identity that outlives any single listing. Catalog goods, such as a specific phone model, can be matched to a product record, which favors transactional models with standardized fulfillment.

Money flow and **risk ownership** are separate questions, and the rest of this chapter keeps them separate. A platform can route money through a licensed provider without holding it, hold it through its own licensed entity, or never touch it. Independently, it can carry buyer-protection losses without owning any inventory, or carry inventory risk without processing consumer payments. eBay books losses "resulting from our buyer protection programs, chargebacks" as transaction losses (10-K, above): **buyer protection is a liability to provision, not only a fee to collect**.

A brand is not a profile. Tradeyard is one brand with ten profiles across four countries. A platform that encodes the profile as "this is Tradeyard, so it is classifieds" will be wrong for Safe Deal categories in the Netherlands and right for Portugal, and will need code changes every time the business moves a category.

What Each Model Asks of the Architecture

Moving down the endpoint ladder changes nearly every system, not just one. The table summarizes the consequences; the three sections after it explain those that most often

surprise architects.

Concern	Classifieds	Lead generation	Transactional	Managed
Identity depth	E-mail or phone; risk-tiered checks	Verified business for professionals (KYB, know your business); consumer contact verified	KYC (know your customer) on payees through the payment provider; trader traceability	Full customer due diligence; credit and contract checks
Core new contexts	Listing, search, messaging, moderation	Lead, lead routing, attribution, pro CRM integration	Order, payment, payout, shipping, dispute, ledger	Inventory, valuation, reconditioning, resale
Trust focus	Fake listings, off-platform payment scams in messages	Fake leads, lead fraud, misrepresentation by professionals	Account takeover, payment and refund fraud, phishing of the platform's own payment brand	Valuation error, adverse selection, condition fraud
Search and ranking signal	Contact as success label	Lead quality and conversion	Purchase and completion	Margin and turn time
Monetization system	VAS catalog, insertion pricing, entitlements	Contracts, quotas, lead metering, attribution and true-ups	Fee pricing on the read path, checkout, take-rate accounting	Pricing authority, principal revenue

Concern	Classifieds	Lead generation	Transactional	Managed
Data the platform must keep	Listings, messages, reports	Leads, lead outcomes, call records	Orders, payment and shipping events, financial records under statutory retention	Asset records, cost basis, condition evidence
Typical regulatory switch	DSA notice and action, ranking disclosure	Professional seller traceability, lead-data consent	Payment licensing (usually via partner), DAC7 seller reporting, buyer-protection terms	Consumer sale law as seller, financial reporting of inventory

Figure 2 shows the same idea as a pipeline: every rung adds contexts and none replaces the ones before it.

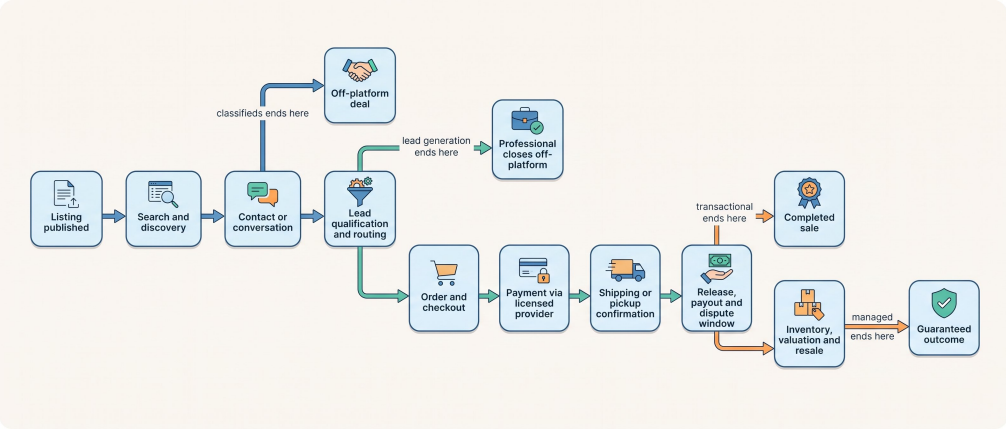


Figure 2: *The endpoint ladder as a pipeline. Each model inherits every context to its left and adds new ones; a marketplace that moves its endpoint for one category must run the additional contexts for that category alone.*

What the platform can observe

The least obvious consequence is **observability of outcomes**: a classifieds platform does not know whether a sale happened. It sees a contact and, if it is lucky, a seller marking the item sold. OLX's 2022 description of its learning-to-rank model is explicit about this: training labels were purchases *or* "messages/calls to sellers", "because transactions aren't always confirmed" (OLX Tech, 2022). A lead-generation portal sees the lead and, with CRM integration or public records, sometimes the outcome. A transactional marketplace sees the completed order.

Observability decides **what the ranking optimizes, what the experiments measure and what can be sold**. A portal that cannot observe sales cannot price on sales without building an attribution pipeline first; Zillow's pay-at-close model treats the fee as "variable consideration" and estimates "the conversion rate of a lead to a real estate transaction" (10-K, above). Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics turns this into a metric design.

Identity and regulation deepen with the endpoint

Each rung down switches on more identity and regulatory obligations. In the EU, the Digital Services Act (DSA) requires platforms that allow consumers to conclude distance contracts with traders to collect trader information, and to make best efforts to assess whether it is reliable and complete, before those traders can offer products (Article 30 of Regulation (EU) 2022/2065). The Omnibus Directive defines an "online marketplace" as a service that "allows consumers to conclude distance contracts with other traders or consumers" and makes the main ranking parameters material information (Directive (EU) 2019/2161, 2019).

Adding payments switches on seller reporting that a pure classifieds site may not have. The EU's DAC7 directive makes platform operators collect and report seller data to tax authorities when they know, or can reasonably know, what sellers were paid (Belastingdienst, n.d.). Commentary on DAC7 indicates that platforms engaged solely in listing or advertising fall outside the reporting definition. This is a reading of the rules to confirm with counsel, not settled law for every classifieds case. Rules of the Marketplace: DSA, DAC7 and Vertical Regulation owns these obligations; Who Is on the Other Side? Identity and Verification owns the identity levels.

Money flows through a licensed partner, until it does not

Classifieds groups that add payments almost always keep the money with a licensed payment service provider (PSP). Kleinanzeigen's "Sicher bezahlen" holds funds with its provider for up to 14 days until the buyer confirms receipt (Kleinanzeigen, n.d.). Marktplaats withdrew its earlier escrow-style "Gelijk Oversteken" in April 2022 as "not safe enough"; its successor, "Kopersbescherming" (buyer protection), runs through the

PSP Online Payment Platform, which holds the money until a week after receipt, with a 5% buyer fee at the time (AVROTROS Radar, 2022). The marketplace owns the offer and acceptance flow, the fee pricing, the shipping labels, the release timer and the disputes; the provider owns the regulated money.

The step beyond that is a licence of one's own. Vinted Pay holds an e-money institution (EMI) licence from the Bank of Lithuania and began onboarding members onto its own wallet in January 2026, "expected to reduce payment-related costs and dependencies" (Vinted, 2026). With a licence comes safeguarding of client funds, a ledger that must reconcile to the cent and supervisory reporting. RECONCILED's Payments for Finance Architects: Eleven Reasons the Numbers Disagree explains why those obligations reshape the finance architecture.

In Practice: Moving Along the Ladder

The documented history of European and American marketplaces shows movement along the ladder in both directions. Four patterns recur.

Pattern 1: classifieds add a transactional layer, category by category

The large European horizontals added payments and shipping for shippable goods while keeping classifieds for everything else.

- **leboncoin** "launched a transactional business model in some of the generalist categories" in 2019, according to Adevinta's 2019 annual report, and its transactional revenue grew from €53 million in 2022 to €73 million in 2023 (Adevinta 2023 report, above).
- **Marktplaats** launched transactional services in 2017, shipping in 2019 and buyer protection, adding a second shipping company in 2022 (same report).
- **Kleinanzeigen** stated "strong ambitions to become a full transactional leader in German Re-Commerce" after its 2023 rebrand from eBay Kleinanzeigen (same report).

None of these brands moved cars or homes to on-platform payment. mobile.de, inside the same group, earned essentially no transactional revenue in 2023. **The transactional layer was category-gated**: it applied where items are shippable, prices are moderate and the buyer's main fear is non-delivery.

The transactional layer also cost margin. Transactional revenue was 5.6% of Adevinta's 2023 revenue, it diluted the French margin, and it came with rising "direct costs from transactional services (such as payment and shipping service provider costs)". It may still be the right strategy, because buyer protection changes trust and frequency, but it is not a free revenue line.

Pattern 2: transactional pushes are scaled back

Some groups pushed further and pulled back.

- **OLX** described its Pay&Ship flow in 2023: "OLX holds the money of the transaction and transfers it to the seller when the package is delivered" (OLX Tech, 2023). Prosus's results for the year to March 2025 then record "the de-prioritisation of pay-and-ship in Europe" in the second half of the year, while OLX concentrated on motors, real estate and jobs.
- **Adevinta**'s 2023 plan to turn the leboncoin and Kleinanzeigen platforms into "a unified global platform" for generalist marketplaces, including "the full transactional journey", was followed in 2025 by a move "from a centralised structure to a decentralised, standalone operating model" (Adevinta strategic report 2025, published 2026). No public source confirms whether the unified transactional platform was delivered. Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform tells that story in full.

Pattern 3: managed models retreat when prices turn

The most expensive moves were down to the managed rung.

- **Zillow Offers**, Zillow's iBuying business, was wound down on 2 November 2021, with a "$304.4 million write-down of inventory" in the quarter and the admission: "We have been unable to accurately forecast future home prices at different times in both directions" (Zillow 8-K and shareholder letter, 2021).
- **OLX Autos**, an inventory-holding "buy, recondition, sell" business, reached $1.6 billion of revenue in the year to March 2022, just over half of OLX's classifieds revenue, before losses of $206 million in the following half-year led Prosus to announce its exit in March 2023 (News24/Fin24, 2023). All operations were disposed of, held for sale or closed by March 2025.
- **CarGurus** decided on 6 August 2025 to wind down CarOffer, its digital wholesale business, which had "proven less effective in today's more volatile and unpredictable pricing environment", and to focus on "technology and analytics that will enable smarter sourcing and pricing decisions rather than facilitating the transactions themselves" (CarGurus 10-K for 2025, filed 2026).

The common thread is that **the managed rung changes what a model error costs**, not that transactions are bad. A valuation that is 7% off is useful information on a listing page; the same error is a balance-sheet loss when the platform buys at the estimate and sellers whose homes are overvalued accept while the rest walk away. Classifieds groups have strong audience economics and weak experience of holding inventory, and the documented retreats show that the systems, the processes and the capital structure were built for the rungs above.

Pattern 4: lead generation moves closer to the deal

Vertical portals have found a middle path: taking part in more of the deal without holding the money or the asset. Autotrader's Deal Builder lets buyers start a deal and reserve a car online; Autotrader reports that such enquiries "convert at double the rate". The same results attribute part of a 1.7% drop in average retailer forecourts in the second half of the year to "concerns with the speed and nature of our Deal Builder product roll-out" (FY26 results, above). mobile.de's 2026 platform update added an AI call assistant and a lead API "so that leads are transferred directly and securely into existing CRM systems" (mobile.de, 2026). **The lead-generation endpoint is being extended toward reservations and deal steps**, and the Autotrader case shows that professionals can push back when the platform's new step changes their own workflow.

Veldmark: four brands, twenty profiles

Ingrid's grid became the **marketplace profile register**, a short, versioned table that the architecture advice forum reviews whenever a brand proposes to change its model. Each row is one profile: brand × country × category group. The summary:

Brand	Profiles	Endpoint	Seller type	Money flow	Risk carried by Veld-mark	Monetization (2026)
Tradeyard goods (NL, BE, PL)	3	Contact; payment and pickup or delivery for Safe Deal orders	78% private	Adyen for Platforms for Safe Deal; otherwise off-platform	Buyer protection on Safe Deal orders	VAS, pro subscriptions, Safe Deal buyer fee of 5% + €0.70
Tradeyard goods (PT)	1	Contact	Mostly private	Off-platform	None beyond trust and safety	VAS, pro subscriptions

Brand	Profiles	Endpoint	Seller type	Money flow	Risk carried by Veldmark	Monetization (2026)
Tradeyard	6 vehicles, property, jobs (PL, PT)	Contact	Mixed	Off-platform	None	Insertion fees (PL vehicles and property), VAS
Motorvane	4 (DE, NL, PL, AT)	Lead delivered to a dealer	91% of listings from 21,400 dealers	Off-platform; financing referred to partners	Lead-billing disputes	Dealer subscriptions with slots, depth products, lead and financing referral fees, private-seller fees
Hearthly	6 sale, rent, new builds (FR, ES)	Lead delivered to an agency or developer; tenant application file for rentals	84% of listings via feeds from about 160 software vendors	Off-platform; settles at the notary or lease signing	None on transactions	Agency subscriptions, new-build campaigns, Valuo data and seller leads, rental services

Brand	Profiles	Endpoint	Seller type	Money flow	Risk carried by Veldmark	Monetization (2026)
Keybridge (FR, BE, ES)	—	Not a marketplace: SaaS for 9,800 agencies	Professionals only	SaaS billing	Data-custody and neutrality obligations	SaaS subscriptions

Read across the rows and Noor's proposal shrinks: a shared order and payment platform serves three Tradeyard profiles today and might serve others later, for example Tradeyard Portugal's goods categories, where Safe Deal is not yet live. It does not serve Motorvane, whose dealers want qualified leads in their own systems, or Hearthly, whose transactions settle at a notary. What *every* profile needs is shared beneath the endpoint: identity, listing management, media, search, messaging, trust and safety and a product catalog with entitlements. That is the list the Common Ground program already contained, and the register made the reason explicit.

The book's two running examples, Femke's city bike and the Essen Golf, sit on different rungs. When Femke sold her **€240** Rotterdam city bike on Monday 8 March 2027, the sale went through Safe Deal "pay at pickup" for **€225**, with a buyer fee of **€11.95** (5% × €225 = €11.25, plus €0.70): a transactional endpoint inside a classifieds brand. When Autohaus Ruhrblick in Essen listed a 2021 Volkswagen Golf for **€21,900** on Motorvane on Wednesday 14 April 2027, the platform's job ended at **23 leads in 10 days** (14 e-mails, 7 calls and 2 test-drive bookings; 19 once repeat contacts from the same buyers are merged); the sale on 29 April happened in the showroom, and Motorvane learned of it only when the dealer's feed removed the car 26 hours later. Both were successes of different models, measured by different events and served by different systems above a common listing core.

Figure 3: *Two successes on two rungs: Femke's €225 bike sale completed through Safe Deal, while the Essen Golf succeeded as 23 leads in 10 days, both served from the same listing core.*

The Polish overlap was resolved as a commercial decision first and an architectural one second. In the register, Tradeyard Poland's vehicles profile stayed classifieds with insertion fees for private sellers. Dealers with more than a small number of listings were offered Motorvane packages, with an option to show their stock on Tradeyard through Motorvane's feed. The architecture consequence was a **syndication contract** between two brands of the same group rather than two competing dealer sales motions, and a requirement that the shared listing core should know which brand owns the commercial relationship for each listing.

Trade-offs: Where Should the Marketplace End?

Every marketplace leadership team keeps returning to one question: should we move the endpoint? The options and their costs:

Option	What you gain	What you take on	When it is right	When it is wrong
Stop at the contact (classifieds)	Lowest friction for supply; small set of contexts; high margins in verticals	No outcome data; fraud concentrates in messaging; monetization limited to visibility and access	Local pickup goods, high-value unique items, thin markets where friction kills supply	Shippable goods where buyers fear non-delivery more than they value low fees
Own the lead (lead generation)	Pricing on value delivered; renewal arguments backed by data	Lead qualification, routing, deduplication, attribution; pro CRM integrations; disputes over lead counts	Professional supply with high ticket sizes: cars, property, jobs, services	Private C2C supply, where there is no one to pay for a lead
Own the transaction (transactional)	GMV visibility, buyer protection, take rate, fraud converted into disputes	Orders, payments via partner, shipping, disputes, ledger, provisioning, DAC7 and trader checks	Shippable goods with moderate prices and repeat buyers	Homes and most cars, where money moves through notaries, banks or dealers
Own the outcome (managed)	High revenue per transaction; control of quality and price	Inventory and price risk, operations, capital, valuation accuracy under adverse selection	Narrow, well-understood segments with predictable resale prices and sufficient capital	As a growth bolt-on to a high-margin classifieds business when prices are volatile

Option	What you gain	What you take on	When it is right	When it is wrong
Hybrid, category-gated	Each category ends where its buyers need it to	A platform that must run several endpoints and switch them by configuration	Almost every horizontal group of meaningful size	When the gate lives in code, so every change needs a release

Two judgements follow from this table and from the documented cases.

First, **I would treat every move down the ladder as a bet with an exit plan**. The bet is reasonable: buyer protection measurably reduces some fraud, and shipping widens the market beyond the buyer's city. But the decision should define the categories, the metrics that will show whether it works and the conditions under which the business would retreat, and the architecture should make the retreat cheap. OLX's de-prioritization of pay-and-ship is what a cheap retreat looks like from the outside. Adevinta's parallel running of central and local services during its 2025 decentralization, which more than doubled its cloud-provider emissions temporarily according to its 2025 strategic report, is what an expensive reversal looks like.

Second, **the endpoint should be data, not code**. A category gate expressed as configuration ("Safe Deal: enabled for NL goods, BE goods, PL goods; not PT; not vehicles") lets the business move categories without a program. A gate expressed as `if brand == 'tradeyard' and country != 'PT'` in a checkout service, a ranking feature and a notification template turns each commercial decision into a cross-team release. The exception is a marketplace that will only ever run one model, such as a pure jobs board; it gains little from generalizing the endpoint. Few groups stay that simple for long.

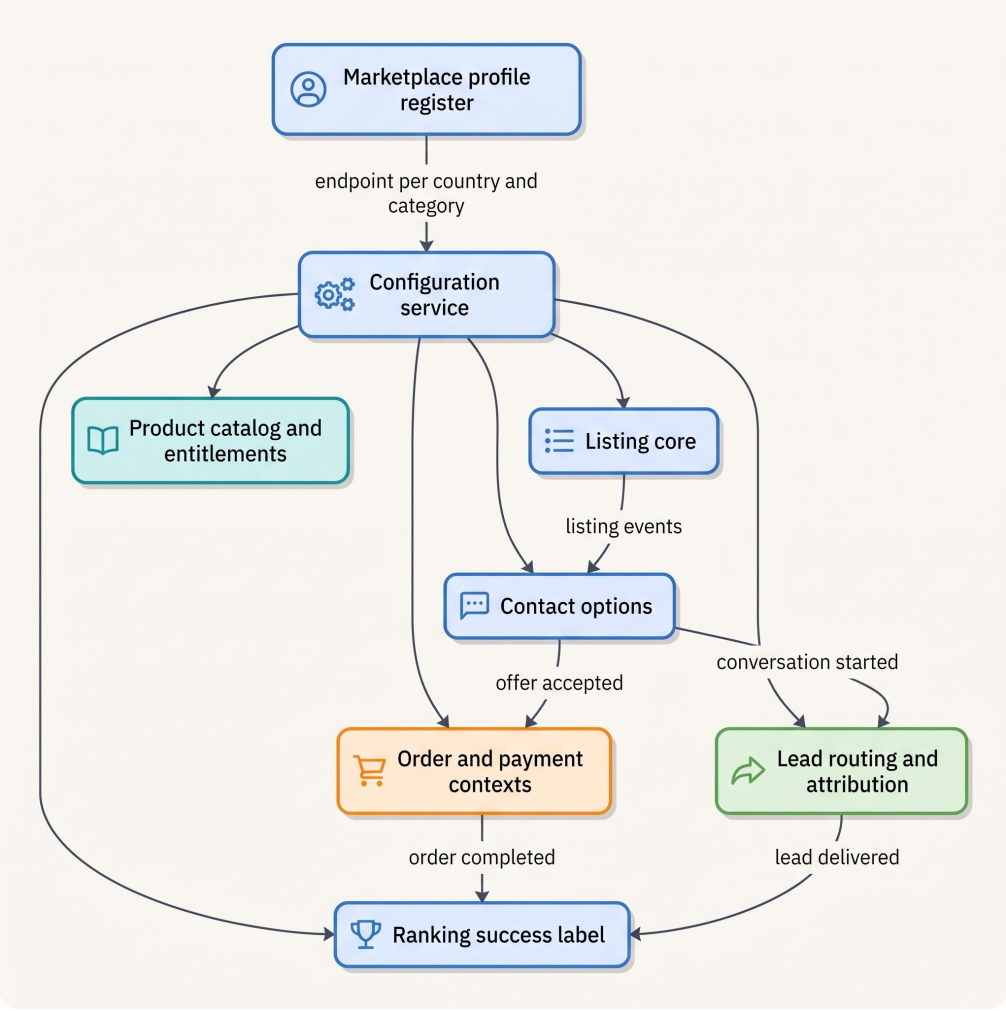


Figure 4: *The endpoint as configuration. The profile register decides, per country and category, which contexts a listing flows through after contact and which event counts as success; the listing core and the contact surface stay the same for every profile.*

Vertical Differences

The endpoint and the other axes take typical values in each vertical. Same Platform, Different Physics: How Listing Verticals Differ covers the full comparison; this table shows only the taxonomy view.

Vertical	Typical end-point	Supply	Item identity	Money flow	Monetization unit	Taxonomy trap
General classifieds	Contact; payment for shippable goods	Mostly private, long tail of small businesses	Weak; free text and photos	Off-platform, or via the PSP for protected sales	Per listing, per bump, per buyer fee	Treating a hybrid as one model and forcing checkout on pickup categories
Real estate	Lead to an agency or developer; rental applications	Agencies via feeds from software vendors; some owners	Strong: address and parcel; many listings per property	Off-platform, at notary or lease signing	Per advertiser subscription (Rightmove), per listing paid by the seller (Hemnet), commission share (AVIV France)	Mistaking a seller-lead valuation business for a feature

Vertical	Typical end-point	Supply	Item identity	Money flow	Monetization unit	Taxonomy trap
Automotive	Lead to a dealer; increasingly reservation or deal steps	Dealers via dealer management system (DMS) feeds; private sellers	Strong: VIN or plate	Off-platform, with financing referrals; managed models where the platform buys cars	Per dealer forecourt with stock-based slots (Autotrader ARPR)	Drifting into inventory risk (CarOffer, OLX Autos)
Jobs	Application delivered to an employer	Employers, agencies, career-site and applicant-tracking feeds	Probabilistic: titles mapped to occupation taxonomies	No consumer payment	Packages per posting, or pay per click or started application (Indeed)	Becoming an ad-tech auction without the systems for it
Recommerce	Completed payment and delivery	Private sellers at very high volume; separate professional channel	Weak, with brand, size and condition attributes	Through a PSP or the platform's own licensed entity	Buyer fee, shipping, promoted items	Underestimating dispute, ledger and logistics load

Indeed's pricing page shows what the jobs endpoint becomes when it is monetized by performance: billing on "job seeker interactions like clicks or started applications", with some jobs available only as sponsored listings (Indeed, n.d.). Vinted's separation of

commercial sellers into a Vinted Pro channel, because "businesses have legal obligations that private sellers don't" (Vinted, 2026), shows the seller-type axis enforced in product and software.

Failure Modes

The taxonomy fails in practice in seven recognizable ways.

One model assumed for the whole group. The most common mistake is the one in Noor's first slide: a shared platform designed for the model of the largest or loudest brand. A listing service with a mandatory order object, a checkout embedded in the contact flow or an identity system that requires payment-grade KYC before posting will fit transactional goods and hurt everyone else. This is how **The Forced Fit** begins. Brands whose job ends at a lead or a contact find that the platform makes them slower, and they route around it.

The endpoint hidden in code. When the model is never written down, it is encoded implicitly in dozens of places: a ranking feature that assumes purchases exist, a notification that promises buyer protection in a category without it, a billing rule that counts contacts as leads. Changing a category's endpoint then means first finding every place that assumes it. The symptom is that nobody can answer "which categories in which countries have protected payment?" without reading code.

A transactional pivot launched as a feature. Payments look like a checkout screen to product teams and like a ledger, a dispute desk, a provisioning policy and a set of tax reports to finance and compliance. A pivot that ships the screen before the rest produces payouts that cannot be reconciled, disputes handled by e-mail and reporting obligations discovered after the first year. Beyond the Contact: Payments, Shipping and Buyer Protection describes the minimum set.

Contact-stage fraud ignored because "we do not handle money". A classifieds platform does not handle money, but its messaging is where money is negotiated. Fraud therefore concentrates at the endpoint, including fake payment links that impersonate the platform's own protected-payment brand. Veldmark's first worked example contains one: at 20:11 on Sunday 7 March 2027, a "buyer" sent Femke a fake Safe Deal payment link from a domain registered that day. Designing the endpoint without designing its abuse is **Fast Mail for Fraudsters**. From Interest to Contact: Messaging and Lead Management and Scams, Fakes and Account Takeovers: Fraud Prevention for Listings own the countermeasures.

Revenue from different rungs treated as comparable. Transaction GMV, take-rate revenue and principal revenue from selling inventory are not comparable with subscription revenue. OLX Autos briefly made up just over half of OLX's classifieds revenue while losing money. An architecture that reports all of them in one "revenue" field makes portfolio decisions harder.

The success event borrowed from the wrong model. A lead-generation portal that optimizes for page views, or a transactional marketplace that optimizes for contacts, optimizes the wrong thing. That is **Vanity Liquidity**, and its root cause is often a taxonomy that was never agreed.

Neutrality conflicts in professional software. A group that owns software used by professionals to publish to rival marketplaces holds data it must not use for its own portal. Keybridge publishes Veldmark's agencies' listings to Hearthly's competitors. Veldmark's March 2027 decision that no Keybridge agency data is used by Hearthly for ranking or sales targeting, with access controls and an annual independent review, is a taxonomy decision as much as a governance one: Keybridge is a SaaS business with SaaS obligations, not a supply channel for Hearthly.

The Chief Architect's View

At group scale, the taxonomy is less a classification exercise than a set of standing decisions. Four matter most.

Make the profile register an architecture artifact. Every marketplace in the group should have a written profile per country and category, reviewed when it changes: endpoint, seller mix, item identity, money flow, risk owner and the event that counts as success. At Veldmark it is a table of twenty rows that fits on one page and is versioned like any other decision record. Its job is to stop shared platforms from being designed around an assumed model, and to make the cost of a model change visible before the commercial decision is final.

Separate the decision rights. Moving an endpoint is a commercial decision owned by the brand's general manager and the Group CPO, with finance and legal involved because it changes licences, provisions and reporting. The architecture function does not own the decision, but it owns the **cost of change** and should be consulted early enough to influence it. A rule I use: no move down the ladder is approved without a named owner for disputes, reconciliation and the regulatory switches it turns on.

Share beneath the endpoint first. The capabilities every profile needs (identity, listing management, media, search, messaging, trust and safety, the product catalog and entitlements, the event backbone) are the natural candidates for shared platforms. Contexts that exist only for one rung (orders and payouts for transactional goods, lead attribution for verticals) should be shared only among the profiles that use them. Adevinta's 2019 rule, which centralized areas "where investment requirements are large and the technology is uniform", is a good default; its 2023–2025 swing from converging whole product platforms to decentralizing again shows why the rule should be applied per capability, not per brand (Adevinta Annual Report 2019, published 2020). One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture and After the Deal: Consolidating Platforms After Acquisitions develop this.

Design for movement in both directions. The documented record contains as many retreats as advances. Endpoint configuration, category-scoped contexts and separable data (orders and payments kept apart from the listing core, leads kept apart from conversations) make a retreat a configuration change and a data-retention plan, not a migration program.

What to watch at group level: the share of contacts, leads and orders per profile; whether the success event in ranking and experimentation matches the profile; the regulatory switches each profile has turned on; and the cost per profile of the contexts it uses. When a profile's endpoint changes, all four should change with it.

Questions to Ask

Take these into a design review or a strategy discussion about a marketplace model:

1. For each country and category, where does our job end today, and where do we intend it to end in three years?
2. Which event counts as success in our ranking, our experiments and our sales materials, and is it the same event in all three?
3. Who holds the buyer's money between payment and handover, and which licensed entity is responsible for it?
4. If a deal goes wrong, who loses money first, and where is that provision recorded?
5. Is the category gate for protected payment, lead products or reservations configuration or code? How long would it take to move one category?
6. Which regulatory obligations switch on if we move a category one step down the ladder (trader traceability, seller tax reporting, payment licensing), and who owns each?
7. Which of our shared platforms assume a specific endpoint, and which brands does that assumption hurt?
8. If we had to retreat from a transactional or managed model in one category next year, what data would we have to keep, for how long, and in which system?
9. Do we own software used by professionals to publish elsewhere, and what prevents our marketplaces from using its data?

From Model to Liquidity

The taxonomy tells a marketplace where its job ends. It does not tell it whether the job is being done well. A classifieds platform, a lead-generation portal and a transactional marketplace all sell the same thing underneath: the probability that a buyer and a seller find each other, in a given place and category, quickly enough that both come back. Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics defines that probability, explains why it is local and winner-take-most, and turns it into the metrics that later chapters convert into freshness, latency and alerting requirements.

Sources and Evidence

Company engineering and investor sources

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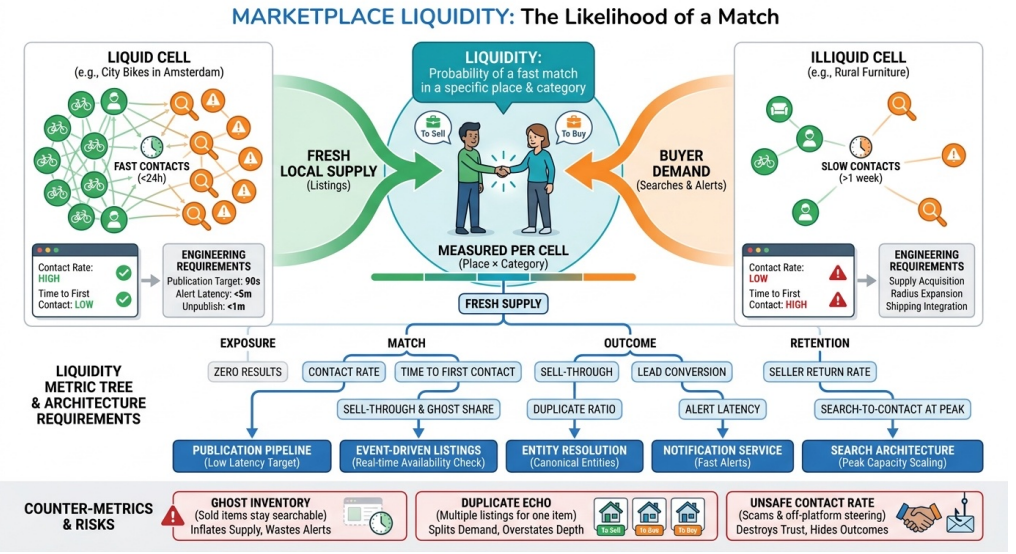
Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

2. Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics



IN THIS CHAPTER, YOU WILL: Define liquidity precisely, see why network effects in listing marketplaces work per place and category and produce winner-take-most markets, and build a liquidity metric tree that runs from fresh supply to completed matches. You will then translate those metrics into architecture requirements: publication time, alert latency, unpublish propagation and peak-hour search capacity.

WHY THE BUSINESS CARES: Professionals pay for leads, and consumers come back only when they find what they want quickly. Both depend on liquidity, not on listing counts or traffic. A marketplace that measures liquidity per city and category knows where it is winning, where it is merely busy and where a competitor can take the market.

WHY ARCHITECTS SHOULD CARE: Liquidity metrics are where business goals turn into engineering requirements. "Sellers get a first contact

within hours" implies a publication-time target, an alert-latency budget, an unpublish guarantee and a peak-hour capacity plan. Architects who do not take part in defining the metrics end up building to numbers that nobody can trace back to a business reason.

KEY POINTS:

- **Liquidity is local, so measure it per cell.** A buyer in Rotterdam needs Rotterdam supply. Network effects, competition and churn all work per place × category × time window. National averages can look healthy while most cells are empty, which is why leadership is won market by market and rarely transfers between countries.
- **Measure matches, not inventory.** Listings and visits are inputs. The liquidity metric tree runs from fresh, deduplicated, available supply through exposure and contact to outcomes such as sell-through and lead conversion. Each metric needs a governed definition, a cell key and a counter-metric that catches gaming, ghosts and duplicates.
- **Liquidity metrics become architecture requirements.** Time to first contact sets the publication and alert budgets. Ghost inventory sets the unpublish guarantee. Search-to-contact at the weekly peak sets capacity. Multi-homing and disintermediation decide what the platform must own and what it can only measure.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Vanity Liquidity** — Listing counts, sessions and page views rise while contacts per listing, time to first contact and sell-through get worse, and the dashboards reward the wrong teams.
- **Ghost Inventory** — Sold and expired listings stay searchable and alertable, inflating supply metrics and wasting the contacts and alerts that liquidity depends on.
- **Seeing Double** — The same item or property counted several times makes supply look deeper than it is and splits the demand that should have gone to one listing.

A listing marketplace does not sell listings. Sellers can list anywhere, often for free. Buyers can browse anywhere, always for free. What a marketplace offers both sides is **the likelihood that a match will happen**: that a seller who lists a city bike on a Sunday evening will hear from a serious buyer within the hour, and that a buyer searching for a two-bedroom flat near a particular metro station will find one that is still available. That likelihood is **liquidity**, and it is the product every other system in this book exists to produce.

Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy showed that marketplaces differ in where their job ends: at a contact, a lead, a payment or a delivery. This chapter is about how well that job is done. It owns four terms: *liquidity*, *network effect*, *multi-homing* and *liquidity metrics*. It also sets up an argument that runs through the book: **liquidity metrics**, **defined carefully**, **are the source of most non-functional requirements in a listing platform**. Publication time, alert latency, unpublish propagation, search capacity at the weekly peak and duplicate rates are not engineering preferences. They are the terms of the marketplace's promise to its users, and the chapter ends by tracing each one to the later chapter that owns its design.

Setting the Scene at Veldmark: Same Brand, Different Markets

In April 2027, three months into the Common Ground program, Henrik de Boer, Veldmark's Group CEO, asked a question at the quarterly business review that nobody could answer with the existing dashboards. Tradeyard Portugal had 1.1 million live listings, and its monthly active users had grown faster than the Netherlands' for three years. Yet Portuguese professional sellers renewed less often, revenue per listing was lower and the local team kept asking for marketing budget. **Was Tradeyard Portugal a smaller version of Tradeyard Netherlands, or a different kind of market?**

Priya Raman, Head of Data & ML, was given four weeks to answer. Veldmark, the fictional group this book follows, had three data warehouses and no common definition of "contact" (a problem Data Platform for Marketplaces: Events, Lakehouse and Data Products takes on). Priya's team built one liquidity view for Tradeyard's four countries from raw events, with one definition per metric. The national comparison is below.

Tradeyard, 2026	Netherlands	Portugal
Live listings	6.1M	1.1M
New listings per week	1.05M	0.21M
Monthly active users	14.2M	2.8M
Monthly active users per live listing	2.3	2.5
New listings with at least one contact within 48 hours	**71%**	**46%**
Median time to first contact (contacted listings)	**2.6 hours**	**19 hours**
Listings marked sold within 30 days	48%	29%
Search sessions that lead to a contact	9.4%	5.1%
Searches returning zero results	3.8%	11.6%

The national ratio of buyers to listings was almost the same: 2.3 users per live listing

in the Netherlands, 2.5 in Portugal. On that measure Portugal looked as liquid as the Netherlands, if not more so. Every outcome metric said the opposite. A listing in Portugal was much less likely to be contacted at all, waited seven times longer when it was, and sold far less often.

The explanation was geography. When Priya's team split the data into **cells** (a district or province × a leaf category), the difference became clear. In the Netherlands, 38% of cells met the liquidity threshold the team chose, and those cells held 86% of active listings. In Portugal, only 11% of cells did, and they held 52% of listings. The Lisbon and Porto metropolitan areas held 64% of Portuguese listings and behaved much like Dutch cities. The rest of the country was a long tail of cells with a few dozen listings and a handful of searches a week. **The national average blended two different markets and described neither**.

The meeting changed its question. It was no longer "how do we grow Tradeyard Portugal?" but "which Portuguese cells can become liquid, and what would it take?" The rest of this chapter explains the concepts behind that question and how Veldmark turned the answer into engineering requirements.

Two-Sided Markets and Network Effects

A **two-sided market** is one in which a platform serves two distinct groups whose value depends on each other: buyers and sellers, renters and landlords, job seekers and employers. Rochet and Tirole's 2003 paper framed the platform's problem as having to "get both sides on board", and studied how the *allocation* of prices between the sides, not just their level, affects volume (Rochet and Tirole, 2003). In classifieds, that insight shows up as a familiar price structure. The side whose participation creates the most value for the other side (buyers, and in horizontal marketplaces most private sellers) pays little or nothing. Professionals, who capture the most value from buyers, pay.

A **network effect** exists when a participant's value from the platform depends on how many other participants use it. Listing marketplaces have two kinds.

- **Cross-side network effects** link the two sides. More relevant supply attracts more buyers, and more buyers attract more supply. This is the main engine of a listing marketplace, and it is almost always positive.
- **Same-side network effects** link participants on the same side. Among sellers they are usually *negative*, because more sellers of similar items compete for the same buyers' attention. Among buyers they can be negative too, where supply is scarce: more renters chasing the same apartments make each renter worse off. Scout24's results show that consumers will pay for an advantage in such a market. Its private-segment subscriptions for renters, buyers and owners averaged 506,937 customers in 2025, up 14% (Scout24 preliminary 2025 results, published 2026).

Same-side effects can also be positive indirectly, for example when more buyers' searches improve ranking and recommendations for everyone.

Network effects in marketplaces are **asymptotic**: the benefit of adding participants flattens once a market is deep enough. NFX's widely read essay on network effects defines them as "network effects with diminishing returns" and observes that "in nearly every marketplace, one side, or one type of node, is harder to acquire than the other" (NFX, The Network Effects Bible, 2019, updated 2024). Both hold in listing marketplaces. A buyer looking for a used city bike in Rotterdam gains a lot when the supply grows from five listings to fifty, and little more when it grows from five hundred to five thousand. The constraint is almost always supply in the long tail of cells, not in the dense centre.

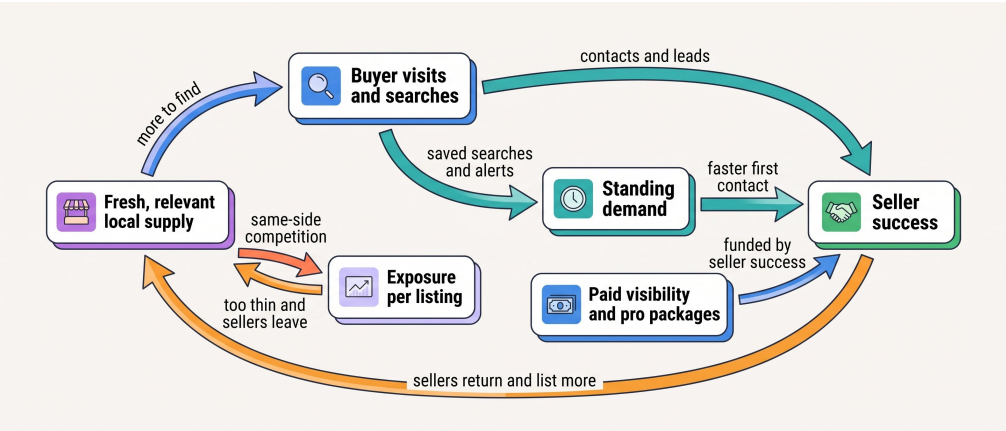


Figure 1: *The liquidity loop. Cross-side effects (supply attracts demand, demand produces seller success, success brings supply back) are the engine; same-side competition among sellers limits each listing's exposure, and paid visibility is only worth buying while the loop is turning.*

Liquidity, defined

This book uses a precise definition. **Liquidity** is the probability that, within a time window that matters to users, a listing receives a qualified contact and a buyer finds a suitable, available listing, measured for a given place and category. Three parts of that definition do most of the work.

- **Within a time window**. A first contact after three days is worth much less than one after three hours for a city bike, and about the same for a house. The window is set per vertical and category.
- **Qualified and available**. A contact from a scammer or about an item that was sold yesterday is not liquidity; it is noise that costs both sides time. Ghost listings and fake buyers inflate raw activity while lowering liquidity.

- **Place and category**. Liquidity is not a site property. It is a property of a **cell**: a geographic area that buyers will actually travel or search across, crossed with a category whose items substitute for each other.

The productscape model behind this book describes the same idea in plain terms: value is "enough fresh, relevant supply in a given place and category", and "success in one metro/category does not transfer automatically". Its buyer journey adds the classifieds-specific detail: "Recency and distance matter more than absolute relevance in classifieds: a perfect match 300 km away is useless to a pickup buyer."

Why the cell is the unit

Cells are not a reporting convenience; they describe how the market works. Three properties follow.

1. **Leadership is won cell by cell**. A competitor does not need to beat a national marketplace everywhere. It needs to become the better place to buy and sell in a few dense cells (a city's apartments to rent, a region's used vans), after which its own loop starts turning. Andrew Parker's 2010 picture of startups carving categories out of Craigslist is cell-level competition seen from the outside (Parker, 2010).
2. **Network effects rarely cross borders**. A buyer in Lisbon gains nothing from Dutch supply. This is why the large classifieds groups are portfolios of local market leaders rather than single global sites, a structure From Newspaper Columns to Portfolios: How the Classifieds Industry Evolved traces through Schibsted, eBay Classifieds, OLX and Adevinta.
3. **Cell definitions differ by vertical**. For general goods with pickup, a cell is a metro area or a region within reasonable driving distance. For shippable goods, the place dimension widens to the shipping zone, which is one reason shipping changes liquidity. For cars, buyers travel further, so cells are larger. For rentals, a cell can be a few neighborhoods. Zillow's share-of-voice pricing allocates leads per zip code (Zillow 10-K for 2025, filed 2026), which is a cell definition expressed as a pricing rule.

Competing for Liquidity

The chicken-and-egg problem and seeding

A new marketplace, or an existing one entering a new cell, faces the **chicken-and-egg problem**: buyers do not come without supply, and sellers do not list without buyers. In classifieds, supply is usually the harder side early on, and the documented solutions all lower the cost of listing or bring supply in from somewhere else:

- **Free private listings**. Horizontal classifieds almost everywhere charge private sellers nothing in most categories. It is the price structure Rochet and Tirole describe, applied to the side that has to come first.
- **Professional feeds**. In verticals, supply already exists in professionals' systems. A portal that integrates the main dealer-management and real-estate software vendors gets most of a market's stock without a single manual listing. That is how Motorvane receives 88% of its dealer listings and Hearthly 84% of its listings.
- **One dense cell first**. Starting in one city or one category and making it liquid before expanding reflects asymptotic network effects: the first hundred listings in a cell are worth far more than the next thousand.
- **Distribution from an existing network**. Facebook launched Marketplace in October 2016 on top of community buy-and-sell groups that, by its own account, about 450 million people visited each month (Meta Newsroom, 2016). Distribution of this kind is not available to a standalone marketplace, and it has its own regulatory exposure: the European Commission's 2024 decision against Meta concerned tying Facebook Marketplace to Facebook and imposing unfair trading conditions on competing classified-ads providers (European Commission, IP/24/5801, 2024).

Seeding is a phase, not a strategy. Once a cell is liquid, the problem changes from attracting supply to keeping its quality high enough that buyers keep trusting the results.

Winner-take-most, documented

Once a marketplace leads a country and vertical, the loop tends to widen its lead. Vertical portals document their dominance, each in its own measure:

- **Rightmove** says "over 80% of all time spent on property portals is on Rightmove", 89% in December 2025 on Comscore data but 75% on SimilarWeb, Data.ai and SensorTower data, with more than 85% of traffic direct and organic (Rightmove full-year results, 2026). The two panels differ by 14 points, so any comparison must name the panel and the competitor set.
- **Autotrader** reports 11 times more time spent than its nearest competitor in the year to March 2026, with 67% of its audience unique to it and over 80% of visits direct (Autotrader FY26 results, 2026).
- **REA Group** reports 6.3 million people using realestate.com.au exclusively out of 12.7 million average monthly visitors in its year to June 2026 (REA Group FY26 results, 2026).
- **Hemnet** states that about 90% of property sold in Sweden is listed on it (Hemnet Group, read 2026).

The mechanism behind these numbers is a combination of local cross-side effects and an asymmetry in **multi-homing**. The leader captures disproportionate *attention*, and attention is what turns listings into leads.

Bill Gurley's 2012 list of marketplace factors explains why these markets tip: "High buyer and supplier fragmentation is a huge positive", higher frequency is "obviously better", and "network effects are rare but golden" (Gurley, 2012). Fragmented agents and dealers cannot easily coordinate against a leading portal, which is part of what makes the leader's position durable, and concentrated chains and dealer groups are where pushback starts.

Dominance has limits, and they are also documented. Autotrader's average retailer forecourts fell 0.5% in the year to March 2026 and were 1.7% lower in the second half, which management attributed partly to "concerns with the speed and nature of our Deal Builder product roll-out". Its "stock" lever, the number of paid vehicles retailers advertise, subtracted £48 from average revenue per retailer, as retailers cut paid stock without leaving (same source). On 29 September 2026, Hemnet announced "free entry-level formats" to gather all publicly available listings, after a quarter in which net sales fell 23.1% (Hemnet investor page, read 2026); the cause has not been documented, and reading it as a response to supply moving to another channel is an inference, not a reported fact. **A leader's share of attention protects it from buyers leaving; it does not protect it from suppliers reducing what they pay or where they list**.

Multi-homing

Multi-homing means participating in several competing platforms at once. In listing marketplaces it is asymmetric.

- **Professional sellers multi-home cheaply**. A dealer's management system or an agency's CRM can publish the same stock to every portal through feeds; Veldmark's own Keybridge does exactly this for 9,800 agencies, including to Hearthly's competitors. Germany's competition authority, clearing mobile.de's acquisition of Carwow's German business in October 2026, found that mobile.de has a "starke Marktposition" (strong market position) in used-car brokerage, and that dealers' multi-homing and their ability to export data to other platforms constrain that power (Bundeskartellamt, 2026).
- **Buyers multi-home at first, then concentrate**. Browsing several sites costs a buyer only time, but habit, saved searches and alerts concentrate attention on the site with the most complete and fresh supply.
- **Private sellers multi-home occasionally**. They cross-post when an item does not sell, which is one source of duplicates in horizontal marketplaces.

NFX describes multi-tenanting as occurring when there are "low costs or no costs to simultaneously participating in competing networks", and notes that it can "harm the defensibility of a network and mitigate network effects, but only up to a point" (NFX, above). The architecture consequence is direct. When supply multi-homes, **inventory is not exclusive, so differentiation must come from what the platform does with**

the inventory: freshness, data quality, deduplication, ranking, alerts, trust and the workflow tools professionals use every day. Serving the Pros: Dealer, Agent and Merchant Tooling and One Property, Three Listings: Duplicates and Entity Resolution are therefore liquidity chapters as much as engineering chapters. a16z's 2020 list of marketplace metrics includes the "prevalence of multi-tenanting" as a defensibility measure (a16z, 2020).

Disintermediation

Disintermediation (or leakage) happens when parties who met on a platform complete their deal, or their next deal, outside it. For classifieds this is structural: the business model *is* to let parties meet, which is why classifieds monetize visibility and access rather than the transaction. Disintermediation becomes a liquidity problem in three situations.

1. **When it carries fraud**. Scammers steer buyers off-platform, to messaging apps or fake payment pages, precisely to escape the platform's protections. Leakage of this kind destroys trust, which is the precondition for liquidity.
2. **When it hides the outcome**. Once a deal moves off-platform, the marketplace does not know whether it succeeded, so its outcome metrics weaken.
3. **When monetization depends on the transaction or on repeat contact**. A transactional fee leaks if parties move to cash after the first contact. A lead fee leaks if professionals can identify and contact buyers outside the paid channel.

Hagiu and Rothman's 2016 *Harvard Business Review* article "Network Effects Aren't Enough" is commonly summarized as listing disintermediation among the traps that network effects do not protect a marketplace against (HBR, 2016). The practical responses in classifieds are product and trust measures rather than prohibitions: on-platform messaging that screens for scam patterns before contact details are exposed, masked phone numbers, buyer protection for shippable goods and lead tooling that makes the platform's channel the most convenient one for professionals.

The Liquidity Metric Tree

Liquidity metrics are the measurements that tell a marketplace whether matches are happening, per cell, fast enough. They are best organized as a tree that follows the path from supply to outcome, with counter-metrics that detect inflation.

Level	Metric	Definition (per cell and period)	Typical pitfall
Supply	Fresh available supply	Active listings that are available, deduplicated and updated within the vertical's freshness window	Counting ghosts and duplicates as supply
Supply	Listing completeness	Share of listings with the attributes that the cell's main filters use	Optional attributes left empty drop listings out of filtered results
Exposure	Early exposure	Share of new listings that receive at least N search impressions within 24 hours	Paid boosts concentrating exposure on a few sellers
Exposure	Zero-result rate	Share of searches returning no results	Fallbacks that hide zero results without improving them
Match	Contact rate	Share of new listings with at least one qualified contact within the window (48 hours for goods, longer for property and cars)	Counting scam and spam contacts
Match	Time to first contact	Median time from *searchable* to first qualified contact, for contacted listings	Measuring from submission and mixing publication delay with demand

Level	Metric	Definition (per cell and period)	Typical pitfall
Match	Search-to-contact rate	Share of search sessions that produce a contact	Sessions inflated by bots and scrapers
Outcome	Sell-through	Share of listings marked sold (or removed by the seller as sold) within N days	Relying on sellers to mark items sold; most do not
Outcome	Lead conversion	Share of leads that become a sale, viewing, test drive or application, where the outcome is observable	Outcome data arriving months later, or only for some professionals
Outcome	Days to sell	Median time from listing to sale for sold listings	Survivorship: unsold listings are invisible in the median
Retention	Seller return rate	Share of sellers who list again within N days of a successful sale	Professional and private sellers mixed in one figure
Retention	Standing demand	Active saved searches and alert-driven sessions per cell	Saved searches that never match anything
Counter-metrics	Ghost share	Share of contacts and alerts that reach listings already sold or removed	Measured only in search, not in alerts, digests and recommen-dations
Counter-metrics	Duplicate ratio	Listings per distinct real-world item or property	Not measured at all in horizontal goods

Level	Metric	Definition (per cell and period)	Typical pitfall
Counter-metrics	Unsafe contact rate	Share of conversations flagged as scam or reported	Measured only on reports, which arrive after the harm

Three design rules make the tree usable.

Measure from "searchable", not from "submitted". Time to first contact measured from submission mixes two problems: how fast the platform publishes and how much demand exists. Splitting them separates an engineering problem from a market problem. The Rotterdam city bike below shows why it matters.

Use contacts as the success event where outcomes are invisible. OLX described in 2022 how its learning-to-rank model was trained on purchases or "messages/calls to sellers", "because transactions aren't always confirmed", and reported that after the change unique contacted sellers *rose*, a check that ranking did not concentrate demand on a few sellers (OLX Tech, 2022). The a16z metrics frame the same idea as "match rate": "How successfully can the two sides of the marketplace find each other?", including searches that return nothing (a16z, above).

Give every metric a counter-metric. Each metric in the tree can be improved dishonestly. Contact rate rises if scam messages count. Fresh supply rises if expired listings are kept live. Exposure rises if duplicates multiply. The counter-metrics at the bottom of the table exist to make those improvements visible.

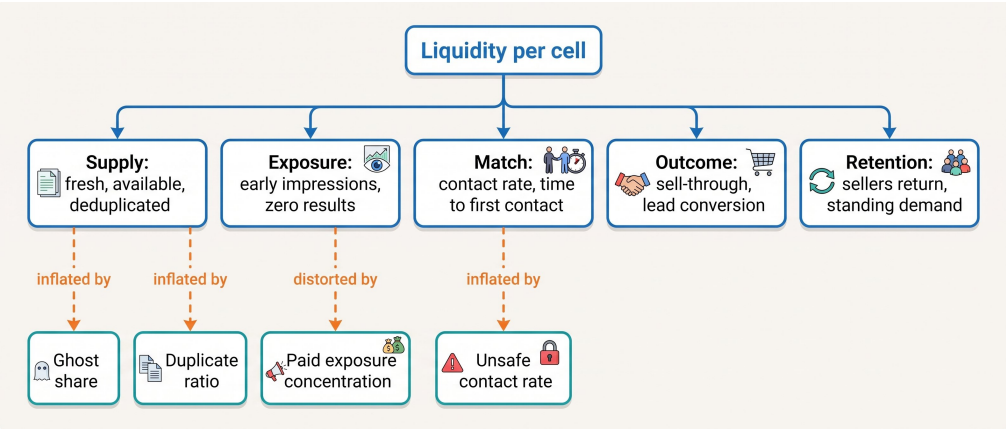


Figure 2: *The liquidity metric tree with its counter-metrics. The solid branches measure whether matches happen; the dotted branches detect the ways each branch can be inflated without any real improvement.*

Contacts, leads and outcomes by model

The success event depends on the marketplace model. For classifieds, the qualified contact is the most reliable event the platform can observe. For lead-generation portals it is the lead delivered to the professional and, increasingly, what happened next: Rightmove's reported lead products, Zillow's pay-at-close fees and AVIV France's commission share all depend on outcomes recorded outside the portal (see Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy). For transactional categories it is the completed order without a dispute. A group that runs several models needs one metric tree with **model-specific success events at the outcome level** and common definitions everywhere else.

Measuring paid products against these metrics requires care. OLX reported in 2026 that its dashboards showed ads in premium boost packages performing *worse* than ads in cheaper ones, because professional sellers self-select into packages; propensity-score matching, validated by a randomized experiment, found uplifts of 17.6% to 97.4% between package tiers (OLX Tech, 2026). **Liquidity dashboards answer "where are matches happening?"; they do not answer "what caused them?"** That second question belongs to Testing a Two-Sided Market: Experimentation Platforms.

In Practice: Veldmark's Liquidity Measures

The cell analysis

Priya's team defined a **liquid cell** for Tradeyard goods as one with at least 25 active listings, at least 50 searches a week and a median time to first contact under 24 hours. The cell key was the administrative region a seller chose (12 provinces in the Netherlands, 20 districts and autonomous regions in Portugal) crossed with the leaf category. The definition is a judgement, and the team said so. Its value lay in being applied the same way in every country.

Tradeyard, 2026	Netherlands	Portugal
Leaf categories	1,420	760
Cells (region × leaf category)	17,040	15,200
Share of cells that are liquid	38%	11%
Share of active listings in liquid cells	86%	52%
Share of active listings in the two largest metro areas	—	64% (Lisbon and Porto)

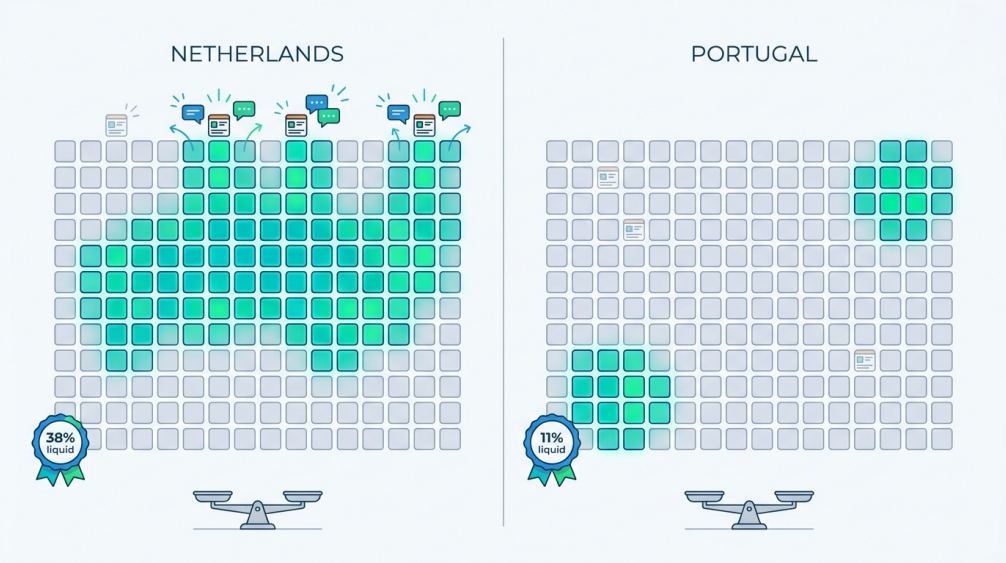


Figure 3: *Liquidity lives in cells: 38% of Tradeyard Netherlands' cells were liquid against 11% in Portugal, a gap the near-identical national ratio of users per listing hid.*

The analysis changed three decisions. Marketing spend in Portugal moved from national brand campaigns to supply acquisition in the twenty largest illiquid cells around Lisbon and Porto. The product team proposed Safe Deal shipping for Portuguese goods. Shipping widens a goods cell from a district to the whole country, so 760 national cells would replace 15,200 regional ones for shippable items. And the search team added a radius-expansion step for sparse cells, with a visible "nearby" label rather than silent filler. Each decision addressed a different branch of the tree: supply, the cell definition itself and exposure.

The Rotterdam city bike, decomposed

Veldmark's most-used worked example shows what time to first contact contains. Femke listed a used city bike for **€240** on Tradeyard NL on **Sunday 7 March 2027 at 19:42 CET**. It became searchable at **19:48**, matched **312** saved searches, and received its first contact at **19:55**. Thirteen minutes from submission to first contact sounds excellent, and in liquidity terms it was: the listing sat in one of the densest cells Tradeyard has.

Split at the "searchable" event, **six of the thirteen minutes were the platform's own publication delay**: Tradeyard's 2027 baseline of a six-minute 95th-percentile time from submission to searchable. Demand took seven. In a cell where "the first buyer to message usually wins", as the productscape model's buyer journey puts it, the platform spent almost half of the seller's wait. At the Common Ground target of **90 seconds** at the 95th percentile, the same bike would have been visible at about 19:43:30, and the 312 saved-search alerts could have gone out before most competing listings.

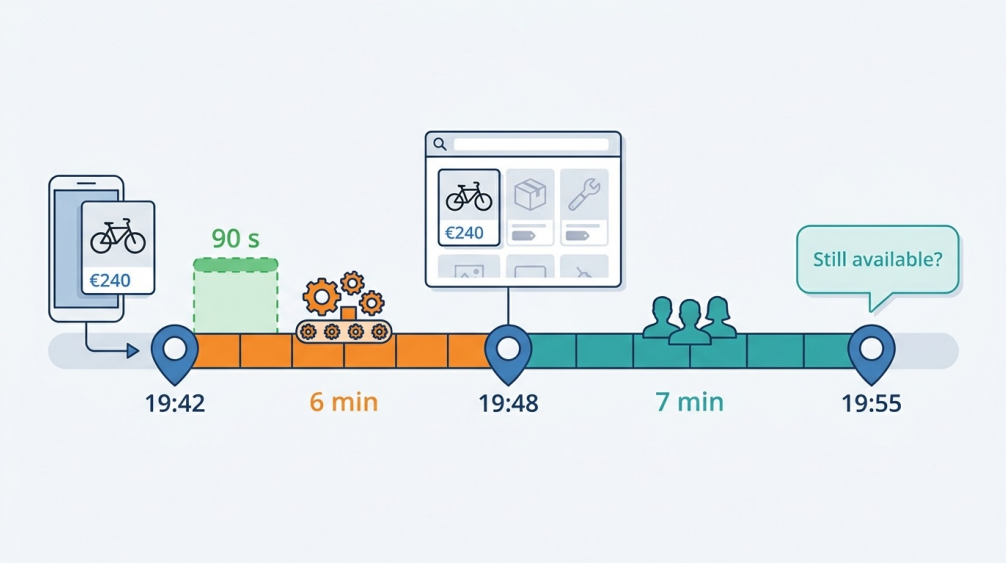


Figure 4: *Measured from submission, the bike's 13 minutes to first contact looked excellent; split at searchable, six of them were the platform's own publication delay that the 90-second target removes.*

The same listing also shows ghost inventory. Femke's bike sold on **Monday 8 March** for **€225** through Safe Deal "pay at pickup" and was marked sold at **18:10**. It left search at **18:11**, but the evening alert digest still featured it the next morning. Every buyer who opened that alert and found the bike gone counted as a session in the dashboards and as a disappointment in reality. Measured per listing, this was one ghost. Measured per cell, it was a steady stream of alerts and recommendations pointing at sold items, and it was only visible once the ghost-share counter-metric included digests and not just search.

Ghosts and duplicates in the other brands

The two other recurring examples show the counter-metrics in verticals.

- **Motorvane**. Autohaus Ruhrblick's 2021 Volkswagen Golf sold in Essen on **Thursday 29 April 2027**, but the dealer's feed removal arrived **26 hours** later. For 26 hours the car was searchable, alertable and contactable. For a dealer-heavy marketplace, ghost share is mostly a feed-reconciliation problem. Many Doors In: Forms, APIs and Bulk Feeds owns the fix, and From Submitted to Searchable: The Publication Pipeline owns unpublish propagation.
- **Hearthly**. In France, **1.9 million** listings described about **1.12 million** distinct properties, **1.7** listings per property on average. Counting listings as supply overstated depth by 70% and split each property's enquiries across several listings and agencies. When Hearthly's liquidity view switched its unit from listings to canonical properties, its contact rate per supply unit rose and its supply fell, both

closer to the truth. One Property, Three Listings: Duplicates and Entity Resolution covers the clustering, illustrated by the Lyon apartment that three agencies listed in May 2027.

From Liquidity Metrics to Architecture Requirements

The liquidity tree is where most of a listing platform's non-functional requirements come from. Each requirement below can be traced to a branch of the tree, and each chapter named in the last column owns the design.

Liquidity mechanism	Metric	Architecture requirement (Veldmark, 2027–2028)	Owner chapter
First buyer to message wins in dense goods cells	Time to first contact from searchable	Submission to searchable: p95 from 6 minutes to **90 seconds** by end of 2028	From Submitted to Searchable: The Publication Pipeline
Standing demand converts fastest	Alert-driven contacts	Saved-search alert sent within **5 minutes** of a listing becoming searchable	Tell Me When It Appears: Saved Searches and Alerts
Contacts wasted on sold items	Ghost share	Sold or removed listings leave search within a minute and are rechecked for availability at the moment an alert, digest or recommendation is sent	From Submitted to Searchable: The Publication Pipeline, Tell Me When It Appears: Saved Searches and Alerts

Liquidity mechanism	Metric	Architecture requirement (Veldmark, 2027–2028)	Owner chapter
Feed-driven supply goes stale	Fresh available supply	Delta feeds plus periodic full-snapshot reconciliation per dealer and agency; stale listings flagged by age of last confirmation	Many Doors In: Forms, APIs and Bulk Feeds
Duplicates split demand	Duplicate ratio	Canonical property and vehicle entities; search collapses duplicates and counts supply per entity	One Property, Three Listings: Duplicates and Entity Resolution
Searches at the weekly peak decide the week's contacts	Search-to-contact at peak	Search capacity and latency objectives set for the peak hour, not the daily average	Search Architecture for Listings: Indices, Freshness and Scale, Sunday Evening Peaks: Reliability, Performance and Cost
Fraud drives buyers and sellers away	Unsafe contact rate	Message screening before delivery; account and link checks within seconds	From Interest to Contact: Messaging and Lead Management, Scams, Fakes and Account Takeovers: Fraud Prevention for Listings
Thin cells return nothing	Zero-result rate	Query relaxation and radius expansion with labeled results	What Did They Mean? Query Understanding and Autocomplete

Liquidity mechanism	Metric	Architecture requirement (Veldmark, 2027–2028)	Owner chapter
Metrics compared across brands	All	One governed definition of contact, lead and sold, with cell keys, in the group data platform	Data Platform for Marketplaces: Events, Lakehouse and Data Products

The peak-hour row deserves a worked example, because it is where liquidity and reliability meet. Veldmark's search traffic peaks on Sunday evenings at about **31,000 queries per second** group-wide, about 70% of it on Tradeyard. The documented public example of a predictable, marketing-driven peak is Rightmove's: its busiest minute was "8:48 pm on Wednesday, during the final ad break on prime-time TV", according to a 2021 Google Cloud announcement (Google Cloud, 2021); Veldmark's Sunday peak is an illustration, not an industry statistic. On **Sunday 17 January 2027 at 20:35 CET**, Tradeyard NL's search 99th-percentile latency rose from about 400 milliseconds to about 7 seconds for **48 minutes**, and about **9%** of searches failed, after a TV campaign, a scheduled reindex job and a Solr cache eviction coincided with the weekly peak.

Priya's team later estimated the liquidity cost. Tradeyard NL carries about 45% of Tradeyard's search traffic, or roughly 9,000 queries per second at the peak. The incident review put the damage at about **3.4 million** failed search requests in the 48 minutes (the campaign added about 30% to the traffic, and apps retried failed requests), many more slow ones, and about **9,800 fewer first contacts** than on a comparable Sunday evening, concentrated in the densest cells where speed matters most. The estimate is rough, but it turned the incident from "search was slow" into "sellers in our best cells lost a Sunday evening's contacts", which is what persuaded the brand to fund the capacity work described in Sunday Evening Peaks: Reliability, Performance and Cost.

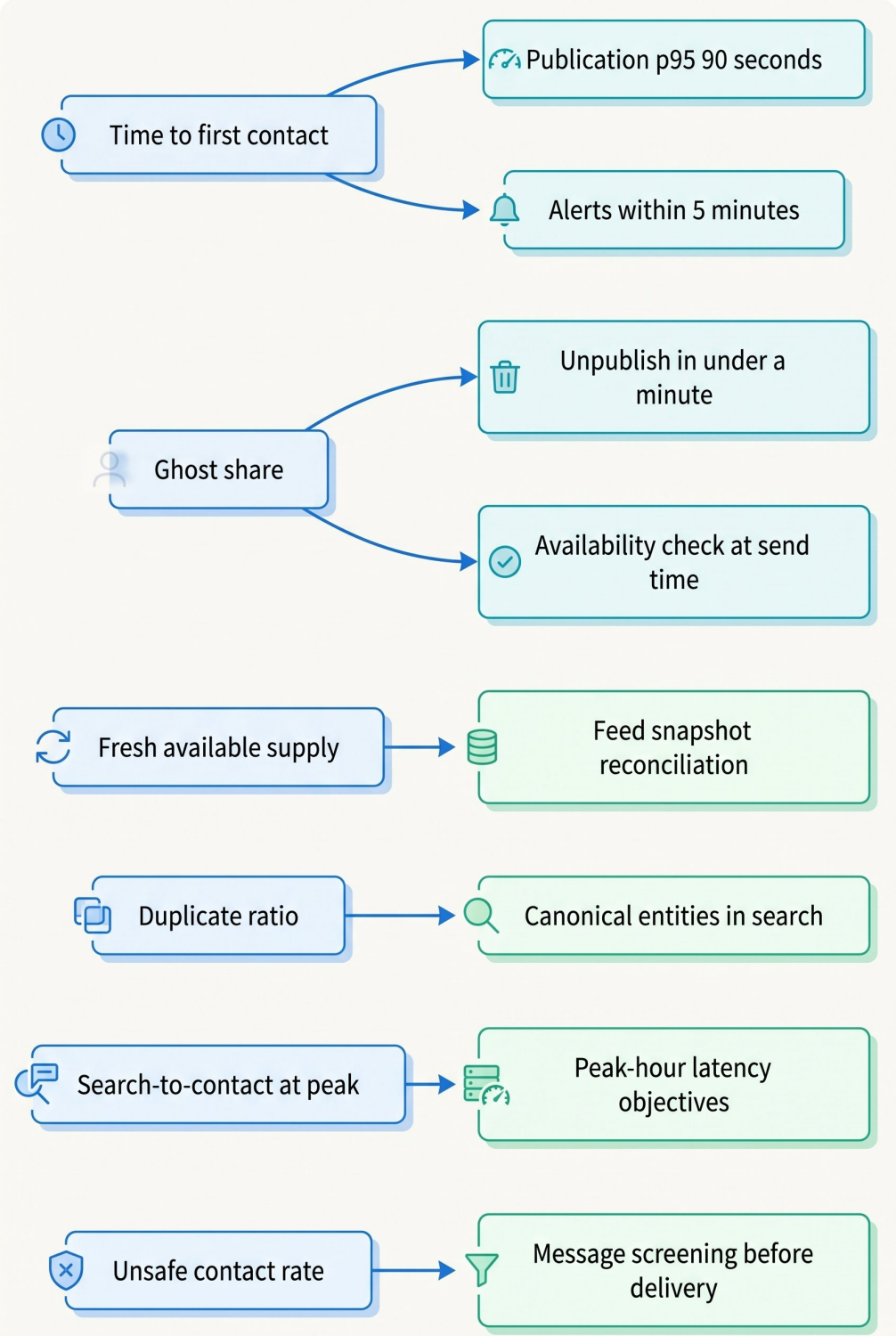

Time to first contact
Publication p95 90 seconds
Alerts within 5 minutes
Ghost share
Unpublish in under a minute
Availability check at send time
Fresh available supply
Feed snapshot reconciliation
Duplicate ratio
Canonical entities in search
Search-to-contact at peak
Peak-hour latency objectives
Unsafe contact rate
Message screening before delivery

Figure 5: *Liquidity metrics on the left, the requirements they generate on the right. Every target in the second column should be traceable to the metric in the first; a requirement that cannot be traced is either missing its business reason or not worth its cost.*

Trade-offs in Measuring and Managing Liquidity

Choice	Option A	Option B	When A is right	When B is right
Unit of measurement	Site or country averages	Cells (place × category)	Board-level trend reporting, once cells are understood	Every operational and investment decision
Success event	Contact	Observed outcome (sale, lead conversion, completed order)	Classifieds and categories where outcomes are invisible	Transactional categories and lead products priced on outcomes
Sold signal	Seller-reported	Inferred (feed removal, messages, payment, price drop then removal)	Professional feeds with reliable removals	Private goods, where most sellers never mark items sold
Cell definition	Administrative regions	Travel-time or shipping zones	Simple, stable reporting and SEO landing pages	Categories where buyers travel far or items ship
Contact channel	Allow contact details in listings	On-platform messaging and masked numbers by default	Low-fraud professional verticals where calls convert best	Private goods and rentals, where off-platform steering carries scams

Choice	Option A	Option B	When A is right	When B is right
Seeding thin cells	Marketing to buyers	Supply acquisition and feed integrations	Cells with supply but low awareness	Cells with demand but no supply, the usual case
Paid visibility	Allowed wherever sold	Limited by exposure caps per cell	Dense cells, where organic competition is strong	Thin cells, where paid boosts crowd out the only organic results

The most consequential row is the sold signal. A marketplace that relies on sellers to mark items sold will undercount outcomes and overcount supply, which inflates the very ghost metrics it is trying to reduce. **Inferring the outcome from several weak signals is less precise for any one listing but more honest for a cell**. It needs the event model described in Every Change Is an Event: Event-Driven Listing Architecture, and it should be labeled as an inference in every dashboard that shows it.

Vertical Differences

Vertical	Liquidity cell	Time window that matters	Success event the platform sees	Multi-homing	Disintermediation risk
General goods (pickup)	Metro or region × leaf category	Hours	Contact; sometimes a protected payment	Private sellers cross-post when items do not sell	High: scams steer buyers off-platform
General goods (shipped)	Country or shipping zone × category	Hours to days	Completed order without dispute	Moderate	Lower once payment is on-platform

Vertical	Liquidity cell	Time window that matters	Success event the platform sees	Multi-homing	Disintermediation risk
Real estate (sale)	Neighborhoods or towns × property type	Weeks to months	Lead to the agency; outcome via agency or public records	High: agencies list everywhere via software	Moderate: buyers contact agencies directly
Real estate (rent)	Neighborhood × size band	Hours in tight markets	Contact, viewing request, application	High	High: fake-rental deposit scams
Automotive	Region within driving distance × make and model	Days to weeks	Lead (message, call, test drive, reservation)	High: dealer feeds to every portal, as the Bundeskartellamt found in 2026	Moderate: buyers call dealers directly
Jobs	Commute area or remote × occupation	Days	Application started or completed	High: postings aggregated and scraped	High: applications move to employers' own systems

Two patterns stand out. **Vertical portals live with high supply multi-homing and compete on audience and tools**, which is why their dominance metrics (share of time, exclusive audience) matter so much. **Horizontal goods marketplaces compete on cell density and speed**, which is why publication time and alert latency matter more there than anywhere else. Real-estate rentals in tight markets combine both: supply multi-homes, demand is intense and the fraud risk is highest. The Barcelona fake-rental ring that Hearthly faced in July 2027, in which 26 accounts posted 180 fake listings priced about 35% below market, exploited exactly that combination.

Failure Modes

Vanity metrics on the wall. The most common failure is a dashboard of inputs: live listings, sessions, page views, app downloads. These rise with ghost inventory, duplicates, bot traffic and paid boosts, so they reward teams for making liquidity worse. Tradeyard Portugal's growing user count was the comfortable number that hid its thin cells. This is **Vanity Liquidity** in its purest form.

National averages for local markets. A country-level contact rate blends dense and empty cells. Decisions taken on it, such as national campaigns, uniform pricing and one-size ranking, help the cells that need help least.

Measuring from the wrong start event. Time to first contact measured from submission hides how much of the wait is the platform's own pipeline. The fix is a "searchable" event emitted by the publication pipeline, which in turn requires that pipeline to know when a listing is actually visible in search.

Ghosts counted as supply. Expired and sold listings left visible make categories look fuller than they are, consume alerts and contacts and train buyers to ignore results. This is **Ghost Inventory**, and its cost appears in liquidity metrics only if the ghost share is measured in every channel: search, alerts, digests, recommendations, landing pages and partner feeds.

Duplicates counted as depth. In real estate and dealer-heavy categories, multi-agency and multi-feed listings inflate supply and split demand. Hearthly's 1.7 listings per property meant its supply metrics overstated depth by 70%. That is **Seeing Double**, and it is invisible until the unit of supply becomes the real-world entity.

Liquidity metrics that nobody owns. When "contact" means one thing in the product analytics, another in the professional seller's reports and a third in the sales team's renewal deck, every discussion about liquidity becomes a discussion about definitions. Governed metric definitions are an architecture asset, not a reporting detail.

Optimizing one side. Ranking tuned purely for buyer clicks can concentrate exposure on a few professional sellers until private sellers stop listing. OLX's check that unique contacted sellers rose after its ranking change is a simple guard against this. Ranking for Two Sides: Relevance, Freshness and Paid Visibility develops the two-sided objective.

The Chief Architect's View

Put the metric tree in the architecture repository, not only in the data team's. The liquidity tree defines the business reason for most performance and freshness targets. When it is versioned alongside the decision records and the service-level objectives, every target can be traced to a metric and every metric to an owner. At Veldmark, the

architecture advice forum adopted a rule that any new latency or freshness objective must name the liquidity metric it protects.

Govern the definitions at group level, measure at cell level. The definitions of contact, lead, sold, searchable, ghost and duplicate should be one set for the whole group, owned by the data platform with the Group CPO as the business owner. The cell keys (place and category) should come from the shared location and taxonomy services, so that a cell means the same thing in every brand. Data Platform for Marketplaces: Events, Lakehouse and Data Products owns governed metrics; Category Trees That Can Change: Designing Marketplace Taxonomies and Where Is It? Location, Geocoding and Privacy own the keys.

Fund the requirements in proportion to the cells they serve. A 90-second publication target is worth a great deal in Tradeyard NL's dense goods cells and much less in Hearthly's property cells, where time windows are weeks. Group platforms should support different service levels per vertical and category instead of imposing the strictest one everywhere at the strictest one's cost.

Treat multi-homing as a given, and invest in what it cannot copy. Professionals will keep publishing everywhere. The durable advantages are the ones that depend on the platform's own systems: freshness, deduplication, data quality, ranking, alerts, trust and professional tooling. That is also why neutrality commitments such as Keybridge's matter: a group that misuses data from its multi-homing tools destroys the trust that makes professionals use them.

Watch the counter-metrics as closely as the metrics. Ghost share, duplicate ratio and unsafe contact rate are the early warnings. A cell whose contact rate rises while its unsafe contact rate rises faster is being exploited, not succeeding.

Questions to Ask

1. What is our cell definition for each vertical and category, and who decided it?
2. What share of our cells are liquid by our own definition, and what share of listings and searches do they hold?
3. Is time to first contact measured from submission or from the moment a listing is searchable, and does the publication pipeline emit that event?
4. How do we know a listing is sold? What share of sold outcomes are seller-reported, feed-reported and inferred?
5. Is ghost share measured in alerts, digests, recommendations and partner feeds, or only in search?
6. Do we count supply in listings or in real-world entities, and what is our duplicate ratio per vertical?
7. Which of our latency, freshness and capacity targets can be traced to a liquidity metric, and which cannot?

8. Where do our professional sellers multi-home, and what do we offer them that a competitor receiving the same feed cannot?
9. Which contact channels do we allow per category, and what is the unsafe contact rate for each?
10. Is there one definition of "contact" across product analytics, professional reports and sales materials?

From Liquidity to History

Liquidity explains why listing marketplaces become local monopolies, why their groups are portfolios of country leaders and why the same group can run a thriving market in one country and a thin one next door. It also explains the shape of the industry: newspapers lost their classified columns to sites that were more liquid, horizontal sites lost categories to vertical portals with deeper supply, and groups formed by acquiring liquid local leaders rather than building new ones. From Newspaper Columns to Portfolios: How the Classifieds Industry Evolved tells that story with dates and shows what each phase left behind in the systems that Chief Architects inherit.

Sources and Evidence

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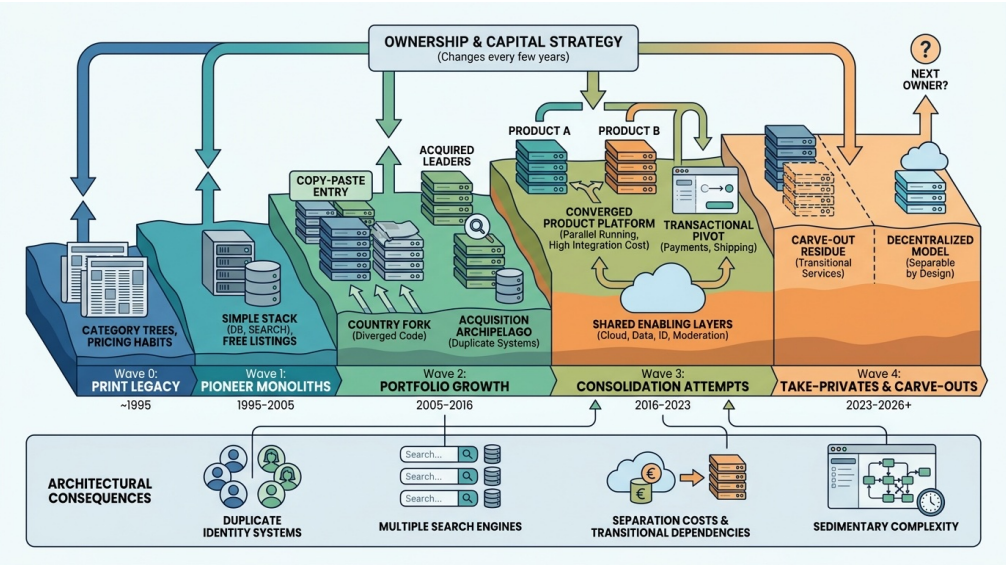
Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

3. From Newspaper Columns to Portfolios: How the Classifieds Industry Evolved



IN THIS CHAPTER, YOU WILL: Follow the classifieds industry from newspaper columns to the portfolio groups of 2026 through a dated, sourced timeline. You will see what each wave of strategy left in the systems (country copies, acquired stacks, shared platforms, rebrands and carve-outs) and learn to read a group's architecture as the sediment of its history.

WHY THE BUSINESS CARES: Marketplace groups are bought, merged, split and sold more often than their systems are replaced. The history explains why synergy cases routinely overestimate what a merger saves, why break-ups cost more than their spreadsheets assumed, and why the owner's capital strategy should shape the platform roadmap.

WHY ARCHITECTS SHOULD CARE: Most of what a chief architect inherits was not designed; it was deposited by a strategy that has since

changed. Knowing which wave created a system tells you what assumptions it carries, who will defend it and how expensive it will be to separate, share or retire.

KEY POINTS:

- **The industry moved in five waves, and each wave left systems behind.** After print columns came the online pioneers, portfolio building, consolidation and private-equity break-ups, each with a characteristic architecture: minimal free local sites, country copies of one codebase and acquired national stacks kept "as a separate business", shared group platforms, and finally carve-outs that had to undo the sharing.
- **Network effects are local, so groups are portfolios of local number-ones.** Leadership is won per country and vertical, which is why acquired national leaders mostly survived while many greenfield group brands closed. The architecture follows the same logic: strong local products sit on top of whatever the group has managed to share underneath.
- **Capital strategy sets platform topology, and it changes faster than platforms do.** Adevinta converged in 2023 and decentralized in 2025; AVIV integrated in 2021 and rowed back in 2025. A shared platform built for synergies becomes a separation cost when the owner wants to sell, so separability belongs in every group platform from the first design review.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Copy-Paste Country** — The copy-paste market-entry model of the 2000s launched each country on its own copy of the codebase, and those copies diverged into separate systems that every later improvement has to touch several times.
- **Deal-Shaped Architecture** — Portfolio groups bought national leaders and promised to run them as separate businesses, so the group still pays for several search engines, identity systems and data platforms years after the deal.
- **The Forced Fit** — When consolidation programs tried to converge whole product platforms across countries, the platform ran into the local differences that had made each marketplace win, and the record shows several groups reversing course within two to four years.

Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics showed that leadership is won country by country, so marketplace groups are portfolios of local leaders. This chapter shows what building those portfolios did to the systems

a Chief Architect inherits, and answers the question every review of such a landscape reaches: *why is it like this?* Why does a group run three search engines, four identity systems and a monolith from 2006? The honest answer is rarely "because someone chose it". It is usually "because a strategy that made sense at the time put it there, and the strategy changed before the system did". **A marketplace group's architecture is the sediment of its corporate history**.

The chapter traces the classifieds industry from newspaper columns through the first free websites, the country-by-country land-grabs of Schibsted, eBay and Naspers, the vertical portals, the consolidation of 2020–2023 and the private-equity take-privates and break-ups of 2024–2026. It owns two terms: the **industry timeline** used throughout the book and the **portfolio model**, the way multi-country groups are built from local marketplaces. It does not repeat the taxonomy of marketplace models (Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy) or the economics of liquidity, and it tells the Adevinta story only in outline; Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform tells it in full. Later chapters build on it: Map What the Marketplace Does Before What Runs It maps what Veldmark's brands duplicate, and After the Deal: Consolidating Platforms After Acquisitions and One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture turn its separability rule into a method.

For each wave, the chapter gives the dated public record, then asks what the wave did to the systems. **Every company statement below carries its source and year**, and where the record is thin or old, the chapter says so. Architectural readings of real companies are labeled as inferences. The worked example is Veldmark, whose fictional history was written to mirror the industry's, so that its 2027 landscape can be read the same way.

Setting the Scene at Veldmark: Twenty-Three Years in One Diagram

In her first week as Group Chief Architect, in February 2027, Ingrid Halvorsen asked each brand's engineering lead for one slide: every production system, with the year it was built and the reason it exists. Veldmark is the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build): four brands, eight countries, **€1.31 billion** of revenue in 2026 and about **1,150 engineers**. The slides came back with dates from 2004 to 2027.

Ingrid laid them out by year rather than by brand, and the landscape changed shape. The systems clustered into five layers, and each layer matched a period of Veldmark's corporate history:

Period	What Veldmark did	What it left in the systems
2004–2008	Tradeyard launched in the Netherlands (2004); **Yardstick**, the PHP monolith on MySQL, rewritten in 2006; Motorvane launched in Germany (2008) as a separate automotive brand	A monolith in a co-located Amsterdam data center that still runs Tradeyard and blocks 140 engineers
2009–2014	Tradeyard expanded by copying itself: Belgium (2009) and Portugal (2014) launched on their own copies of Yardstick; Poland (2012) came through the purchase of a local site	Four country category trees; Solr for three countries and Elasticsearch for Poland, which kept its own search after the acquisition
2015–2022	Motorvane added the Netherlands, Poland and Austria on one Kotlin service platform on AWS; Tradeyard built about 40 services on AWS around Yardstick from 2019	Two cloud estates that share nothing; Tradeyard Poland's cars section competing with Motorvane Poland for the same dealers
2023–2025	Acquisitions: Hearthly France (2023), Keybridge (2024), Casavista in Spain (2025, rebranded Hearthly.es in 2026); a private-equity investor took majority ownership in 2025	A third cloud (Hearthly on Google Cloud), a fourth (Keybridge on Azure), a third search engine (OpenSearch), a third data warehouse
2027–	The **Common Ground** program: **€140 million** over 2027–2030 to share platforms beneath the brands	The decisions this book follows

The board had approved Common Ground in January 2027 on a slide that said "four of everything". Ingrid's diagram said something more useful: **each "of everything" was a fossil from a different strategy**. The four identity systems were not four mistakes. They were the residue of organic launches, a country acquisition, an automotive build and two recent purchases, each of which was right, or at least defensible, when it was made.

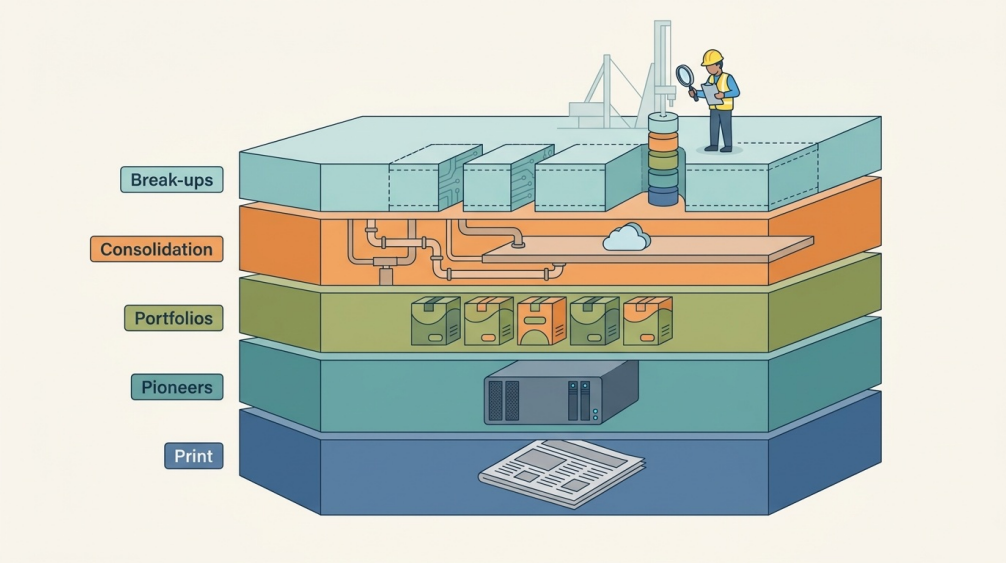


Figure 1: *A group's architecture is sediment: each wave of strategy, from print columns to break-ups, left systems behind that outlived the strategy that deposited them.*

Two questions followed. Mateus Carvalho, the Group CTO, asked which of these layers the industry's larger groups had managed to consolidate, and at what cost. Henrik de Boer, the CEO, asked a sharper one: *if our owner sells a brand in 2030, which of the platforms we are about to build will we have to take apart?* The rest of this chapter is the industry's answer to both.

Concepts: Waves, Portfolios and Sediment

- **Classifieds**: marketplaces whose job ends when a buyer contacts a seller (defined fully in Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy). In print, a classified was a short paid notice in a newspaper column; online, the listing became free and the monetization moved to visibility, professional subscriptions and advertising.
- **Local number-one**: the marketplace with the largest share of audience and supply in one country and one vertical. Because network effects are local (Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics), most of the industry's value sits in local number-ones.
- **Portfolio model**: a group strategy that owns many local marketplaces, usually one leader per country and vertical, and shares some capabilities beneath their brands. The group's value is the sum of local leaderships plus whatever the shared layer saves or enables. This chapter owns the term.
- **Copy-paste market entry**: launching a marketplace in a new country by deploying a copy of an existing marketplace's codebase and operating model, then localizing it. Fast to start, expensive later, because the copies diverge.

- **Shared platform**: capabilities built once and run for several marketplaces, from infrastructure and data to whole product stacks. One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture owns the design.
- **Convergence**: moving several marketplaces' *product* platforms (listing, search, transactions, apps) onto one platform, as distinct from sharing *enabling* layers such as cloud or data.
- **Take-private**: the acquisition of a listed company by a private-equity buyer, followed by delisting.
- **Carve-out** and **transitional services agreement (TSA)**: separating a business from its group for sale, and the contract under which the seller keeps providing shared services (hosting, software, operations) to the buyer for a period.
- **Architectural sediment**: this book's name for the systems each strategy leaves behind after the strategy itself has changed. Sediment is not a defect. It becomes a problem when nobody remembers which strategy deposited it and therefore what assumptions it carries.

The chapter divides the history into five waves. The boundaries are approximate and overlap, because different groups reached each wave at different times.

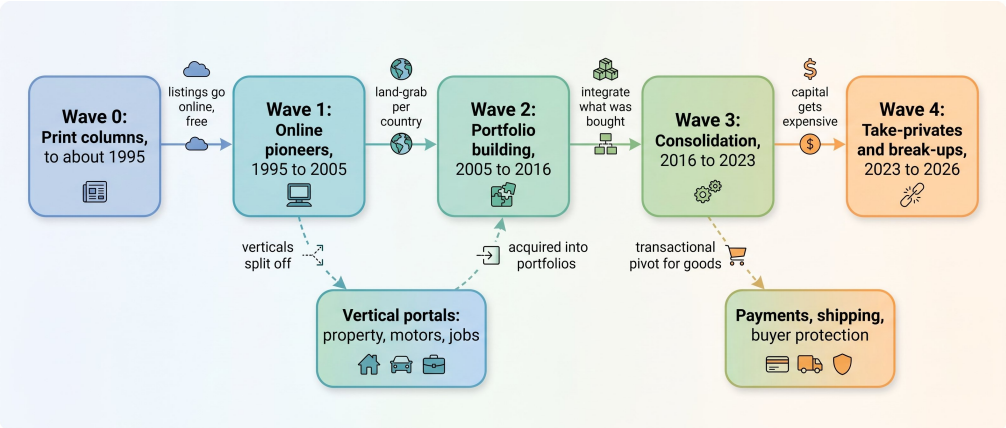


Figure 2: *Five waves of classifieds strategy. Vertical portals grew alongside the horizontals and were absorbed into portfolios; the transactional pivot runs through the consolidation wave. Each arrow is a strategic change that left the previous wave's systems in place.*

The Timeline in Dated Events

The table is the book's reference timeline. It keeps to events with public sources; the sections that follow add the architectural reading.

Date	Event	Source
1977	*Thames Valley Trader* founded in the UK; renamed Auto Trader in 1988	Wikipedia, Auto Trader Group (2026)
1995–1996	Craig Newmark's e-mail list for Bay Area events becomes craigslist; web-based from 1996, a company from 1999, other US and Canadian cities from 2000	Wikipedia, Craigslist (2026)
1996	mobile.de founded in Hamburg; 50,000 listings and 1,200 dealers by 1998	mobile.de newsroom timeline (n.d.)
1998	ImmobilienScout24 and AutoScout24 (as MasterCar.de) founded in Germany	Wikipedia, Scout24 and AutoScout24 (2026)
1999	Marktplaats founded in the Netherlands	eBay 8-K exhibit (2004)
2000	Rightmove founded by four UK estate-agency and financial groups; Gumtree founded in London	Wikipedia, Rightmove (2026); eCG brand page (2014 snapshot)
2004	eBay buys mobile.de (about €121 million, closed April 2004) and Marktplaats (about €225 million, November 2004), the latter to be run "as a separate business"	eBay SEC filings (2004)
2005	eBay launches Kijiji as a greenfield brand and buys Gumtree and LoQUo	eCG brand pages (2014 snapshots)

Date	Event	Source
2006	Schibsted founds leboncoin (France) and, with Styria, willhaben (Austria); OLX founded; Zillow launches; Rightmove lists in London	Adevinta AR 2023; Wikipedia (2026)
2008	eBay buys dba.dk and Bilbasen (about US$390 million); its classifieds unit has "five major brands" in more than 20 countries	eBay (2008-10-06)
2009	eBay Kleinanzeigen launches in Germany, built "to process 10K new ads per day"	Berlin Tech Blog (2019)
2010	Naspers takes a majority stake in OLX	Wikipedia, OLX (2026)
2013	Final print edition of Auto Trader UK, after 36 years	Wikipedia, Auto Trader Group (2026)
2014	Naspers raises its OLX stake to 95% and rebrands acquired local sites to OLX across Eastern Europe	Wikipedia, OLX (2026)
2015	Auto Trader lists in London; Immowelt and Immonet combine under Axel Springer; Kleinanzeigen introduces Kafka; Marktplaats replaces eBay's Voyager search with Elasticsearch	Wikipedia (2026); Immowelt (2015); Berlin Tech Blog (2019); Marktplaats Technology (2015/2017)
2016	Naspers consolidates classifieds under OLX Group; Schibsted presents "one global product and tech organization"; Facebook launches Marketplace	Wikipedia (2026); investor-material summary (2016); Meta (2016)
2017	Marktplaats launches transactional services	Adevinta AR 2023
Feb–Apr 2019	Schibsted demerges its non-Nordic marketplaces as Adevinta, listed in Oslo	Schibsted/Vend (2019-02-25)

Date	Event	Source
2019–2020	OLX buys control of Frontier Car Group (OLX Autos); OLX merges its Middle East and South Asia businesses into EMPG (39% stake) and combines letgo US with OfferUp (40% stake); Hellman & Friedman buys AutoScout24	TechCrunch (2021); OLX Group (2020); Wamda (2020); Wikipedia (2026)
25 Jun 2021	Adevinta completes its acquisition of eBay Classifieds Group; UK competition remedies force the sale of Gumtree UK, Motors.co.uk and Shpock	Adevinta (2021)
2022	Prosus sells Avito (RUB 151 billion); Adevinta sells Gumtree Australia and its Mexican brands	Naspers (2022-10-14); Adevinta (2022-08-25)
2023	eBay Kleinanzeigen becomes Kleinanzeigen; Adevinta reorganizes by vertical and plans "a unified global platform" for its generalists; Prosus exits OLX Autos	Adevinta (2023-05-16); AR 2023; News24 (2023-03-02)
May 2024	Permira- and Blackstone-led consortium completes the take-private of Adevinta	Adevinta (2024-05-29)
Apr 2025	AVIV and Stepstone separate from Axel Springer under KKR and CPP Investments	KKR (2025-04-29)
2025–2026	Adevinta moves to "a decentralised, standalone operating model" and sells its Spanish business to EQT (completed 27 Feb 2026); CoStar completes the purchase of Domain; OLX buys La Centrale (€1.1 billion) and agrees to sell its Central Asian sites	Adevinta Strategic Report 2025 (2026); CoStar 10-K (2026); OLX Group (2025, 2026)

The timeline shows two patterns before any architecture. First, **the same assets change hands repeatedly**: Gumtree has had at least three owners, mobile.de three, AutoScout24 two. Second, **the waves are short compared with the systems**. Kleinanzeigen's original platform, built in 2009, served through eBay's ownership, Adevinta's listing, its take-private and its decentralization before Kleinanzeigen began migrating users off it in 2025 (Berlin Tech Blog, 2025). A system routinely outlives two or three owners.

Wave 0 and Wave 1: From Paid Columns to Free Pages

Print economics

For most of the twentieth century, classifieds were a newspaper's most profitable pages: small paid notices for jobs, cars, homes and goods, sold by the line, with the newspaper as the only local marketplace. The listing *was* the product, and the reader paid for the paper that carried it. Specialist print titles grew alongside: Auto Trader began in 1977 as *Thames Valley Trader*, a magazine of car photographs, and published its final print edition only in June 2013, "after thirty six years" (Wikipedia, Auto Trader Group, 2026).

The architectural legacy of print is not code but **vocabulary and pricing habits**. Category columns became category trees. The idea that a professional pays per listing, or per week of visibility, survives in insertion fees and bump products. And the newspaper's habit of treating each city as its own market survives in the per-country, per-city structure of almost every classifieds platform.

Free, local and minimal

The pioneers inverted the print model: listings became free for private sellers, and money came from a few paid categories. Craigslist began in 1995 as Craig Newmark's e-mail list for San Francisco events, moved to the web in 1996, incorporated in 1999 and expanded to other cities from 2000; its revenue came mainly from paid job ads in selected US cities, with $25 job posts in New York and Los Angeles from August 2004 (Wikipedia, Craigslist, 2026, widely reported).

Free listings pushed down the newspapers' classified prices. Robert Seamans and Feng Zhu studied craigslist's entry into US local markets in the 2000s and found that newspapers relying heavily on classifieds saw **classified-ad rates fall by 20.7%**, with smaller effects on circulation and display-ad rates; they estimated that classified-ad buyers saved about **$5 billion** between 2000 and 2007 (NYU Stern, 2013, on a paper then forthcoming in *Management Science*).

In Europe the same years produced the national leaders that still dominate: mobile.de (1996), ImmobilienScout24 and AutoScout24 (1998), Marktplaats (1999), Rightmove and

Gumtree (2000). Some were start-ups; others were founded by the incumbents the internet threatened: Rightmove's founding shareholders were Countrywide, Halifax, Royal & Sun Alliance and Connells, estate-agency and financial groups that would otherwise have been its customers (Wikipedia, Rightmove, 2026). That origin matters for architecture, because **a portal founded by its suppliers inherits a feed-first relationship with them**: agents' systems push listings in, and the portal's job is to publish them well.

What Wave 1 left in the systems

- **Minimal, monolithic stacks** built for modest scale. Kleinanzeigen's 2009 architecture was "a plain old database backend", "a search backend based on Apache Solr™ (with a Master/Slave setup)" and web servers, built "to process 10K new ads per day" (Berlin Tech Blog, 2019). By 2019 the same marketplace had 34 million live ads and up to a million new ads a day. The original design survived a hundredfold growth because it was simple; it also became the system that had to be strangled later.
- **Free listings as the default for private sellers**, which shifts the economic burden onto visibility products, professional subscriptions and advertising, and onto moderation, because free listings attract abuse. How Listing Marketplaces Make Money and Trust and Safety as a Platform pick up both threads.
- **Categories as the unit of the business**. Andrew Parker's 2010 post on "the unbundling of craigslist" showed how start-ups carved niches out of craigslist's categories, and observed that "a team of thirty-or-so people manages a top 10 ranked English-speaking website" (Parker, 2010). Category-level competition is why taxonomy and vertical-specific schemas later became strategic (Category Trees That Can Change: Designing Marketplace Taxonomies).

Wave 2: The Portfolio Builders

Between roughly 2005 and 2016, three groups turned local classifieds into international portfolios: Schibsted, eBay and Naspers (later Prosus) with OLX. They used different entry methods, and each method left a different architecture.

Schibsted: launch, copy, localize

Schibsted, a Norwegian media group, moved its growth focus from newspapers to online classifieds in the 2000s, anchored on FINN.no in Norway and Blocket in Sweden. Its international expansion is widely described as built on the Blocket platform; a 2012 Nieman Lab article is quoted as saying Schibsted ran classifieds in 28 countries on it. That article could not be retrieved for this book, so the claim is reported, not verified.

The standard account of the playbook (launch or buy a generalist site per country, run it on a shared code base with free private listings, monetize later through verticals and advertising) is industry lore without a primary Schibsted source. Dated facts are clearer: Schibsted founded leboncoin on 28 April 2006 and willhaben in 2006 as a 50/50 joint venture with Styria (Adevinta Annual Report 2023, published 2024).

The record does show that **the federation stayed local in product and technology for a long time**, and that the tension with central scale was visible early. A Schibsted investor presentation in 2016 carried the heading "one global product and tech organization" (widely reported). In 2019, Adevinta's first annual report as a listed company named the trade-off directly: "one of our historic success formulas has been local autonomy and decentralised capacity in product and technology", while data and analytics would be centralized because "investment requirements are large and the technology is uniform" (Adevinta Annual Report 2019, published 2020).

The federation also produced group platforms from local inventions. FIAAS, the deployment abstraction later used across Adevinta's marketplaces, was built at FINN.no and open-sourced in early 2019; FINN.no reported 350 microservices and about 1,200 deployments a week at KubeCon Europe in 2017 (both as documented in Adevinta's 2020 talk write-up and the 2017 slides). Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform follows that thread.

> **INFERENCE, NOT DOCUMENTED:** A copy-paste entry model produces country codebases that start identical and diverge with every local change. After a few years, the group owns several related but incompatible systems: the same data model with different extensions, the same category idea with different trees, the same search engine at different versions. This is the mechanism behind **The Copy-Paste Country**, and it is consistent with the later history, in which the group's central scale had to be built as *new* shared components rather than by merging the country copies.

eBay: buy the leader, keep the brand

eBay assembled its classifieds portfolio mainly by buying national leaders and keeping their brands. It bought mobile.de in 2004 for about €121 million and Marktplaats the same year for about €225 million, stating that Marktplaats would be run "as a separate business" (eBay 8-K exhibit, 2004). It added Gumtree and LoQUo in 2005, dba.dk and Bilbasen in 2008 for about US$390 million, 2dehands/2ememain in 2013 and Vivanuncios in 2015. Where it had no leader, it launched a group brand: Kijiji in 2005 in Canada, Italy and Taiwan, later in other countries, and eBay Kleinanzeigen in Germany in 2009 (eBay, 2008; eCG brand pages, 2014 snapshots).

The bought leaders mostly survived and grew; many greenfield launches were closed, renamed or sold: Kijiji Hong Kong and Taiwan closed in 2015, Kijiji China became

Baixing, Kijiji India became Quikr, and eBay's Mexican brand iBazar was retired when Vivanuncios was bought (eBay, 2015; Wikipedia, Kijiji, 2026). The exception that proves the rule is eBay Kleinanzeigen, a greenfield brand that became Germany's leader, partly because it absorbed an existing Kijiji audience and launched with about 700,000 live ads.

Technically, eCG was a federation with group functions, not a single platform. Group job advertisements from 2009–2011 speak of a "sister Classifieds platform" and "a variety of eCommerce platforms that form the portfolio of the group", with reuse encouraged rather than mandated (eCG careers pages, 2009–2011 snapshots). The clearest documented shared element is the inheritance from the parent. Marktplaats ran **Voyager**, a search engine "developed internally by eBay in the USA", from around 2008; by 2015 eBay "has moved on from Voyager several years ago" and it "has almost become a black box for us". eCG companies jointly evaluated replacements and chose Elasticsearch; Marktplaats migrated itself in three months with two or three developers, behind unchanged interfaces, with zero downtime (Marktplaats Technology, 2015, republished 2017).

Two lessons travel. **Coordinate the technology choice across brands, but let each brand migrate itself**. And **a parent's platform can become orphaned legacy faster than local code**, because its owner moves on and local knowledge decays.

Naspers and OLX: the global roll-up

OLX was founded in 2006 as a classifieds alternative outside the United States. Naspers took a majority stake in 2010 and 95% in 2014, the year it rebranded acquired local sites to OLX across Romania, Bulgaria, Kazakhstan, Belarus, Hungary, Poland and Indonesia; in 2016 it consolidated its classifieds under OLX Group in Amsterdam (Wikipedia, OLX, 2026, widely reported). The vertical brands kept their own names: Otomoto and Otodom in Poland, Standvirtual and Imovirtual in Portugal, Autovit and Storia in Romania. In 2018 the group described itself as "a network of leading classifieds platforms in 45 markets" (OLX Tech, 2018).

Rebranding to one name implies a stronger push toward shared product than eBay's keep-the-brand strategy. The public engineering record, however, documents shared practices and some shared code more than one global platform: one frontend repository serving three brands in 2026, a centralized learning-to-rank repository with per-brand configuration in 2025, and the standardization of user identification "across 71 services" at OLX Motors in 2024–2025 (OLX Tech, 2026). That last item is a direct trace of sediment: user identity had been per platform, and consolidation had to fix it first.

The vertical portals grow up

While the horizontals built portfolios, the verticals became the industry's most profitable businesses. Vertical portals (property, motors, jobs) monetize professionals through

subscriptions and depth products, and the leaders earn very high margins: Rightmove reported a **70%** underlying operating margin on £425.1 million of revenue for 2025, and Autotrader a 70% operating margin for the year to March 2026 (Rightmove, 2026; Autotrader, 2026). Both report that most of their traffic arrives directly rather than through paid channels.

Verticals were also the assets portfolios most wanted to own. eBay bought mobile.de; Axel Springer assembled the property portals that became AVIV, combining Immowelt and Immonet in 2015 and buying Meilleurs Agents for €200 million in 2019 (Immowelt, 2015; FrenchWeb, 2019); OLX built and bought motors and property brands per country. Their architecture differs from the horizontals' because their supply arrives through professional feeds, their items have authoritative identities (a vehicle identification number, an address) and their data products, such as price evaluations and valuations, are part of what they sell (Same Platform, Different Physics: How Listing Verticals Differ).

What Wave 2 left in the systems

Entry method	Example	Typical architecture left behind
Copy the codebase into a new country	Schibsted federation (industry account)	Diverged country copies; per-country category trees and data models; the **Copy-Paste Country**
Buy the national leader, keep it separate	eBay with Marktplaats, mobile.de	Self-sufficient national stacks with their own search, identity, data and billing; the **Deal-Shaped Architecture**, accepted on purpose
Launch a group brand on group technology	Kijiji, eBay Kleinanzeigen	Platforms built by a group team; fragile where the brand did not reach leadership
Buy and rebrand to one name	OLX in Eastern Europe	Brand unity ahead of platform unity; identity and data consolidation work deferred

Entry method	Example	Typical architecture left behind
Build or buy verticals per country	mobile.de, Otomoto, Immowelt	Feed ingestion, vertical schemas and valuation data per country, deeply local by necessity

Each method was a reasonable way to win local leadership quickly, and local leadership was what created the value, so the table is not a ranking. **The architecture cost was deferred, not avoided.**

Wave 3: Consolidation

From around 2016, the portfolio builders turned to integrating what they owned. Three forces drove it: the cost of running many stacks, the rising expectations of users (apps, personalization, payments, AI-assisted search) that each local team could not meet alone, and deals that combined portfolios of comparable size.

The Adevinta–eBay Classifieds deal

The largest was Adevinta's acquisition of eBay Classifieds Group, announced on 21 July 2020 for US$2.5 billion in cash plus about 540 million Adevinta shares, which left eBay with about 44% of Adevinta's capital. It completed on 25 June 2021, with a claimed footprint of number-one positions in 16 countries and targeted run-rate cost synergies of **€130–165 million after three years** (Adevinta, 2021).

Competition law shaped the result before any integration began. The UK Competition and Markets Authority found that Shpock and Gumtree competed closely in UK generalist classifieds; the undertakings required the sale of Gumtree UK with Motors.co.uk and of Shpock, and "Adevinta and eBay have offered to provide transitional services" to the buyers (CMA, 2021). Within about eighteen months Adevinta also sold its Australian, Mexican and South African horizontals, focusing on "five core European markets: Benelux, France, Germany, Italy and Spain" (Adevinta, 2022).

Integration then proceeded layer by layer. Adevinta's 2023 annual report describes consolidating infrastructure "to a single public cloud vendor for our EU5 businesses", plans to "close three of four data centre footprints", and a history of consolidating "enabling capabilities such as cloud and data, and overarching elements such as chat and moderation". The integration of eBay Classifieds cost €69 million in 2023, and the rebrand of eBay Kleinanzeigen to Kleinanzeigen another €15 million (Adevinta Annual Report 2023, published 2024). The rebrand alone shows how far sediment reaches: the old domain was to "remain accessible for several more years" (Adevinta, 2023).

Sharing the enabling layers worked and paid, at least while the portfolio held together. Adevinta's internal Kubernetes platform served "more than 4,000 developers" in 2023 and ran "over 2,000 Kubernetes nodes and 30 clusters across 25 marketplaces" in 2024; experimentation went from roughly ten tools across 22 marketplaces to one (Adevinta tech blog, 2023; 2024; 2024).

In 2023 Adevinta went further and set out to achieve **convergence of the product platform itself**. It decided to stop "individually addressing local market customer needs — such as the full transactional journey, AI-assisted search and modern approaches to first-party advertising", and "Kleinanzeigen, leboncoin and global teams embarked on transforming the two platforms into a unified global platform to support Adevinta's generalist marketplaces" (Adevinta Annual Report 2023). The convergence spanned two countries, languages and legal regimes. In May 2024 the project was "progressing well and according to plan"; no later public source confirms that it was delivered (Adevinta, 2024).

Regional mergers and joint ventures

OLX consolidated differently, by merging regional businesses into larger local players and keeping minority stakes. In June 2020 it contributed dubizzle and its Middle East and South Asia operations to Emerging Markets Property Group for a 39% stake; in March 2020 it combined letgo's US business with OfferUp for 40%; in 2020 its Brazilian joint venture with Adevinta bought Grupo ZAP for about R$2.9 billion (Wamda, 2020; OLX Group, 2020). Architecturally, a merger into another operator is the cleanest exit: the merged business runs on the partner's platform, and the group keeps equity, not systems.

Axel Springer's property business took the integrate-first path. When AVIV appointed a chief technology officer with multi-country platform experience in 2021, its CEO said "we are aligning our tech roadmap across the organization to create a European real estate marketplace" (Online Marketplaces, 2021).

The transactional pivot runs through the wave

In the same years the horizontals moved toward payments and shipping for goods, and **payments became the main product capability the consolidators wanted to build once**. Marktplaats launched transactional services in 2017 and shipping in 2019; leboncoin launched a transactional model in some generalist categories in 2019; eBay Kleinanzeigen launched "Sicher bezahlen" for all users in October 2020; Adevinta's group transactional revenue reached €103 million in 2023, about 6% of the total (Adevinta Annual Reports 2019 and 2023; onlinemarketing.de, 2020). OLX described its Pay&Ship flow in 2023, and Prosus recorded "the de-prioritisation of pay-and-ship in Europe" two years later (OLX Tech, 2023; Prosus FY25 results, 2025). Recommerce specialists built the model from the start: Vinted reported **€10.8 billion** of gross merchandise value and **€1.1 billion** of revenue for 2025 on "vertically-integrated shipping and payment

infrastructure" (Vinted, 2026). Beyond the Contact: Payments, Shipping and Buyer Protection owns the pivot.

What Wave 3 left in the systems

- **Shared enabling platforms** (cloud landing zones, Kubernetes, data, experimentation, media, moderation services) with real fleet economics, and with the coupling that comes with them: one upgrade, many tenants.
- **Partially converged product platforms**, whose state at the end of the wave is often not public.
- **Transitional-services dependencies** wherever a brand was sold while running on group components. The CMA remedy depended on them; so did the sale of Gumtree Australia, which Adevinta promised to support "during the transition".
- **Rebrand residue**: redirect estates, token-based design systems and identity migrations. mobile.de's Android team reported in 2025 that a rebrand "took only two weeks to implement once the base refactoring was done", after removing "technical debt from 8 years ago" (Berlin Tech Blog, 2025).

Wave 4: Take-Privates, Focus and Break-Ups

From 2022 the cost of capital rose, and the industry's owners changed their minds about breadth.

Focus and retreat at OLX. Prosus sold Avito in Russia for RUB 151 billion in October 2022 and announced its exit from OLX Autos, its inventory-holding car-trading business, in March 2023, after the business had reached $1.6 billion of revenue in the year to March 2022, just over half of classifieds revenue, while losing money (Naspers, 2022; News24, 2023). India and Indonesia were sold in 2023, Uzbekistan and Kazakhstan agreed for sale in 2025. By the year to March 2026, OLX operated in seven markets, earned US$992 million of revenue at a 48% adjusted EBITDA margin, and had bought La Centrale in France for €1.1 billion (OLX Group, 2026; Online Marketplaces, 2026). From 45 markets in 2018 to seven in 2026 is the starkest public example of a group concluding that **local leadership in a few markets beats global breadth**.

The Adevinta take-private and break-up. A consortium led by Permira and Blackstone offered NOK 115 per share in November 2023 and completed the take-private in May 2024. In July 2024 work began on a new operating model "empowering local marketplaces". By 2025 the group had "evolved from a centralised structure to a decentralised, standalone operating model… This transition has included the ramp-down of central functions, the transfer of key capabilities, the localisation of intellectual property, and the realignment of reporting structures" (Adevinta Strategic Report 2025, published 2026). It sold its willhaben stake in 2025 and its Spanish business to EQT,

completed on 27 February 2026; EQT sold the property portals Fotocasa and Habitaclia on to Scout24 (Scout24, 2025). The same report notes that cloud emissions "more than doubled", mainly because decentralization "required parallel running of central and local services during a transitional period". **Undoing a shared platform costs real money, and for a while you pay for both.**

The outcome of mobile.de's sale process is not public as of October 2026. The Financial Times reported in October 2025 that Adevinta was exploring an IPO or a sale at up to about €10 billion (as relayed by PYMNTS, 2025). In August 2026 mobile.de's own press release still called it an Adevinta subsidiary, while the German competition authority's October 2026 clearance of its purchase of Carwow's German business named "Oak Germany Buyer GmbH" as mobile.de's parent (Bundeskartellamt, 2026).

The AVIV reversal. AVIV and Stepstone became separate joint ventures majority-owned by KKR and CPP Investments on 29 April 2025, with Axel Springer keeping about 10% (KKR, 2025). In a December 2025 interview, the interviewer (not AVIV) said that AVIV had rowed "back on its plans for a fully integrated platform for all its European portals", and Immowelt's new head explained that "AVIV has strengthened the autonomy of its local businesses", using group functions "where they create clear advantages… sometimes local and sometimes cross-group" (Online Marketplaces, 2025). A sale of Yad2, AVIV's Israeli marketplace, to Apax was signed the same month. What the integrated platform consisted of, and how far it got, is not public.

Platform buyers keep buying. The consolidation logic did not disappear; it moved to owners who integrate *into* a platform rather than building one across peers. CoStar completed its purchase of Domain in Australia in August 2025 and said it was "ready to apply that same proven playbook in Australia" (CoStar, 2025). AutoScout24, owned by Hellman & Friedman since 2020, migrated "110 AWS accounts across 4 business units" of Trader in Canada into its own AWS organization in 2025 (AutoScout24, 2025).

> **INFERENCE, NOT DOCUMENTED:** For owners pursuing break-up value, a shared product platform is a liability at exit: it creates transitional services, intellectual-property questions and double running costs. The same shared platform that a synergy case valued in 2021 became a separation cost in 2025. The documented pattern does not say centralization was wrong; it says **platform topology has to match the capital strategy**, and that the capital strategy can change within two or three years.

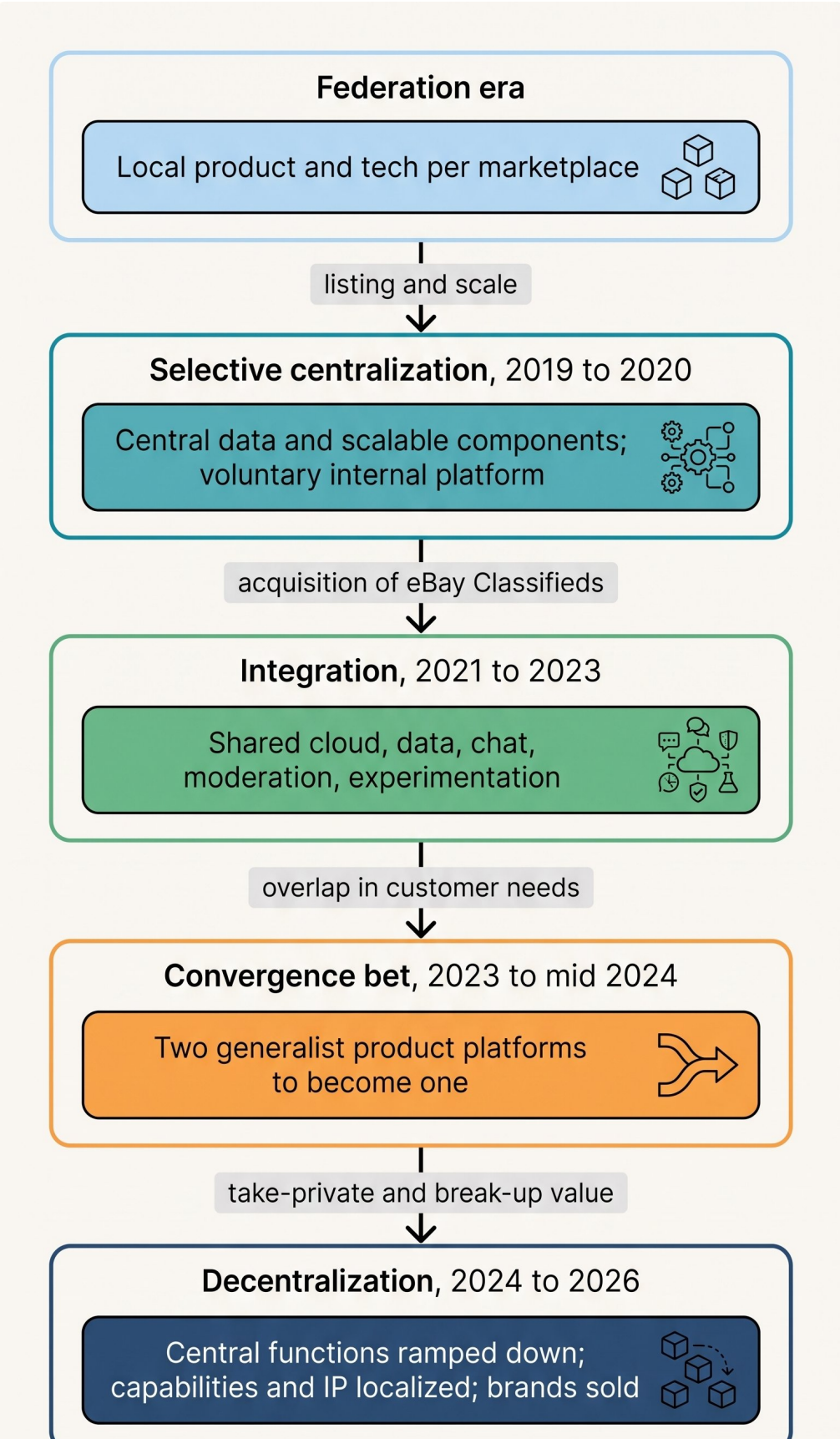

Federation era
Local product and tech per marketplace
listing and scale
Selective centralization, 2019 to 2020
Central data and scalable components; voluntary internal platform
acquisition of eBay Classifieds
Integration, 2021 to 2023
Shared cloud, data, chat, moderation, experimentation
overlap in customer needs
Convergence bet, 2023 to mid 2024
Two generalist product platforms to become one
take-private and break-up value
Decentralization, 2024 to 2026
Central functions ramped down; capabilities and IP localized; brands sold

Figure 3: *The centralize–decentralize swing in Adevinta's own words, 2019–2026. Each step was justified by the ownership and strategy of its moment; the systems built in one step became the inheritance, or the separation cost, of the next.*

Reading a Group's Architecture as Sediment

The waves give an architect a practical tool: for each major system, ask **which wave deposited it, and under what assumption about ownership and sharing**. The answers predict behavior better than the technology does.

Layer of sediment	Assumption it carries	Typical symptom in 2027	What to do with it
Pioneer monolith (Wave 1)	One country, one team, modest scale	Single deployable; slow releases; schema nobody fully understands	Strangle by capability; never "rewrite and switch"
Country copies (Wave 2, copy-paste)	Countries are similar enough to share a codebase once, then diverge	Same concept implemented several ways; fixes built per country	Converge reference data first (taxonomy, attributes), code later
Acquired national stacks (Wave 2, buy-and-keep)	The acquisition stays a separate business	Duplicate search, identity, data, billing; local pride	Share beneath the product only where the layer is uniform and expensive
Inherited parent technology	The parent keeps investing	Orphaned components; knowledge in another company	Replace early, coordinated across brands, migrated per brand
Shared enabling platforms (Wave 3)	The portfolio stays together	Upgrade blast radius; tenants depending on incidental behavior	Treat as products with tenancy, cost allocation and exit paths

Layer of sediment	Assumption it carries	Typical symptom in 2027	What to do with it
Converged product platforms (Wave 3)	Markets' needs overlap enough to share product	Local workarounds; feature requests queuing at the center	Re-examine per capability; keep local edges local
Carve-out residue (Wave 4)	A separation will happen once, quickly	Transitional services that last years; parallel running costs	Design separability in advance, not at signing

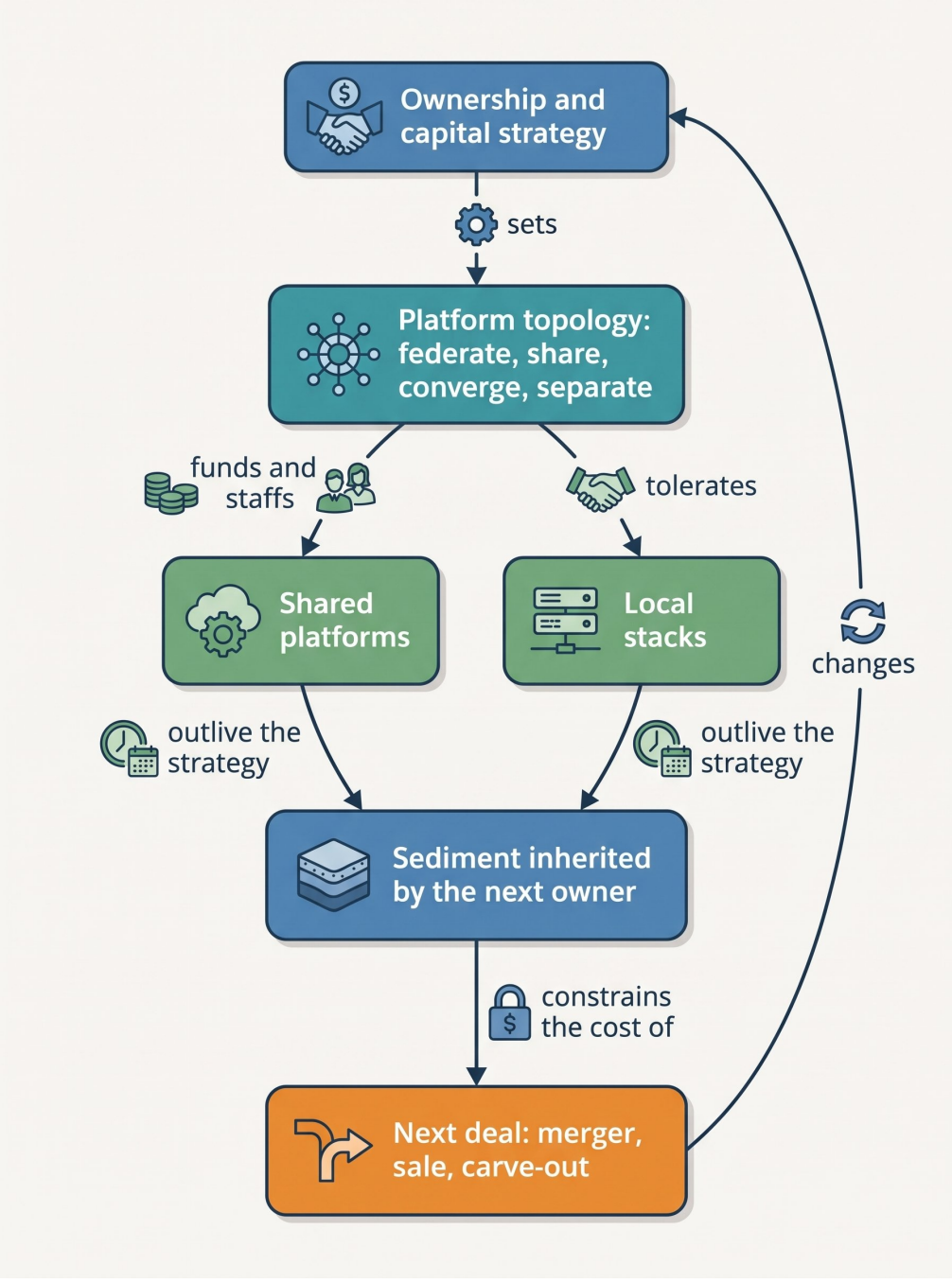


Figure 4: *The loop behind the timeline. Ownership sets platform topology; platforms and local stacks outlive the strategy that created them; that sediment then sets the cost of the next deal, which changes the ownership again. The architect's leverage is in making the sediment cheap to separate.*

In Practice: Veldmark's History, Read as Sediment

Veldmark's history is fictional, but it was written to follow the industry's waves, and Ingrid's February 2027 diagram can be read with the table above.

Wave 1: the pioneer monolith. Tradeyard launched in the Netherlands in 2004, when the founding family's regional publishing business moved its classified pages online. Yardstick, the 2006 PHP rewrite on MySQL, was built for one country and a few hundred thousand listings. In 2026 Tradeyard Netherlands alone carried **6.1 million** live listings, and Yardstick remained a single deployable for **140 engineers**, released twice a week, in a co-located Amsterdam data center. Its assumption, one country and one team, is the reason the Common Ground plan strangles it capability by capability (insertion flow out in Q2 2028, search in Q4 2028, decommissioned at the end of 2029) instead of rewriting it.

Wave 2: copies and an acquisition. Belgium (2009) and Portugal (2014) launched on copies of Yardstick, each with its own database and deployment. Between 2016 and 2018 the copies were folded back into one Yardstick codebase, still with a database and deployment per country; the merge is where Yardstick's roughly 2,600 country branches come from (see One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture). Poland (2012) came through the purchase of a local classifieds site that was rebranded Tradeyard and moved onto Yardstick's code, but kept its own Elasticsearch search because the migration was never finished. The result is in the 2027 numbers: four country category trees with **1,420**, **1,380**, **980** and **760** leaf categories, **3,900** distinct attribute keys of which 41% are used on fewer than 0.1% of listings, and about **six weeks** to add a category. That is **The Copy-Paste Country** in reference data, which is why the group taxonomy and attribute service (Category Trees That Can Change: Designing Marketplace Taxonomies) comes before any attempt to share Tradeyard's code across countries.

Wave 2, vertical: Motorvane. Motorvane launched in Germany in 2008 as a separate automotive brand rather than a Tradeyard category, because its supply was dealers with feeds, not private sellers with photographs. Its 2017–2020 rebuild as about 120 Kotlin services on AWS, and its expansion to the Netherlands, Poland and Austria on that one platform, is the vertical-portal pattern: deeply specific ingestion and data, shared across countries within the vertical. The sediment shows in Poland, where Tradeyard's cars section and Motorvane Poland were selling to the same dealers under two different models by 2027 (Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy describes how the overlap was resolved).

Wave 3 and Wave 4 at once: acquisitions under a new owner. Veldmark bought Hearthly France in 2023, Keybridge in 2024 and Casavista in Spain in 2025, and a private-equity investor became majority owner in 2025. Each acquisition brought its own cloud, search engine and data warehouse. Unlike the industry's consolidators of 2021, Veldmark started its consolidation program *after* its change of owner, which makes

Henrik's question, "what will we have to take apart?", a design input rather than an afterthought.

Ingrid's analysis of the inventory produced a table she showed the board in March 2027:

Capability	Copies in 2027	Wave that created the copies	Common Ground decision	Separability rule
Identity	4	Every wave	Veldmark ID, one login with brand profiles (approved June 2027, rollout 2028)	Per-brand profiles and tenant-scoped data, so a brand can be cut out with its users (Who Is on the Other Side? Identity and Verification)
Search	3 engines	Wave 1 (Solr), Wave 2 acquisition (Elasticsearch, PL), Wave 4 acquisition (OpenSearch)	Shared OpenSearch platform for Tradeyard and Hearthly by end of 2028; Motorvane stays on Elasticsearch until 2029	One cluster set per brand, indexes rebuildable from the event stream
Taxonomy and attributes	4 Tradeyard trees plus vertical schemas	Wave 2 copies	Group taxonomy and attribute service (October 2027)	Per-brand trees on one service; schemas exportable

Capability	Copies in 2027	Wave that created the copies	Common Ground decision	Separability rule
Events	4 technologies	Every wave	Kafka on Confluent Cloud, listing event contract v1 (July 2027)	Topics keyed by brand; consumers registered
Data warehouses	3	Waves 2–4	Group data platform	Per-brand accounts and data products with contracts
Product catalog and entitlements	4	Every wave	One service by Q2 2028	Catalog content owned by each brand
Keybridge platform	1 (Azure)	Wave 4	Stays separate and neutral	Already separate by design

The last column is the lesson of Wave 4 applied in advance. Every shared platform in Common Ground is specified with **tenant-scoped data**, **per-brand accounts and cost centres**, **exportable contracts and a documented exit path**, because the owner that approved the program is, by its nature, a future seller. The run-cost target of **€38 million per year** by 2030 is treated as a business case that must survive the sale of any one brand, not as a synergy that assumes the portfolio is permanent.

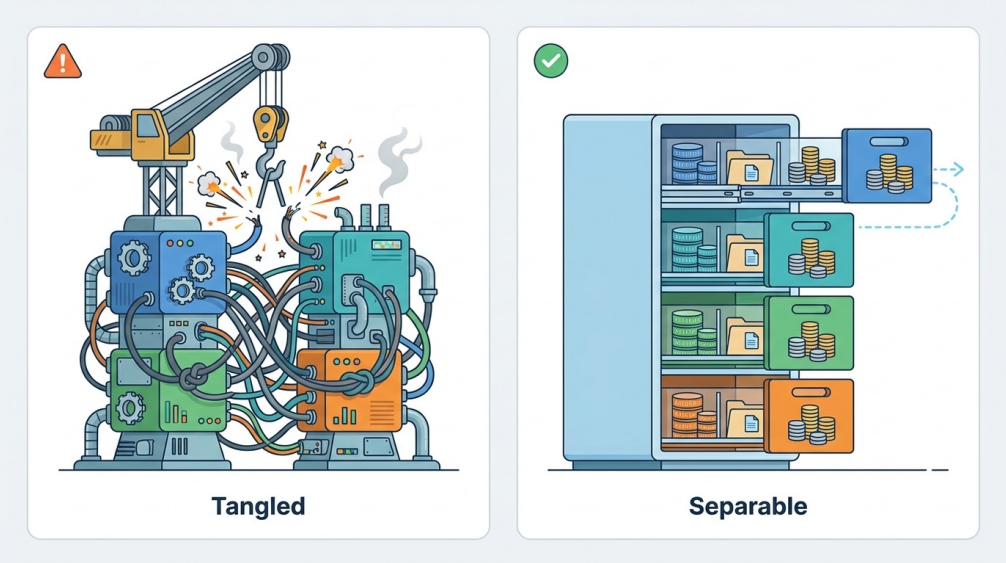


Figure 5: *Because the owner that funds a shared platform is a future seller, every Common Ground platform carries tenant-scoped data, per-brand accounts and a documented exit path, so a brand can leave like a drawer, not a demolition.*

Trade-offs: Growth Strategies and Their Architecture Bills

Each wave's growth strategy sent an architecture bill that arrived later.

Strategy	Speed to local leadership	Architecture left behind	When it is the right choice	When it backfires
Greenfield launch on a copy of an existing codebase	Fast to start, slow to win	Diverging country copies	Similar markets; a team that can keep copies aligned; a plan to converge reference data early	When the copies diverge for years and every improvement is built several times

Strategy	Speed to local leadership	Architecture left behind	When it is the right choice	When it backfires
Greenfield launch on a group platform	Slow to start	One platform, group-owned	The group platform is mature and markets are similar	When the brand does not reach leadership, the platform's cost has no market under it (Kijiji outside Canada)
Buy the local leader, keep it separate	Immediate	A full national stack per acquisition	Strong leaders, very different markets, short holding horizon	When the group wants scale benefits it never designed for
Buy and integrate onto a group platform	Immediate, then a long migration	Converged platforms and transitional dependencies	A buyer with a proven platform and a long horizon (CoStar with Domain, AutoScout24 with Trader)	When local differences were the reason the target won
Regional merger into another operator	Immediate, with diluted control	Equity, not systems	Markets where the group is number two	When control matters more than exposure

Strategy	Speed to local leadership	Architecture left behind	When it is the right choice	When it backfires
Vertical spin-out from a horizontal	Slow	Separate vertical stack, possible overlap with the horizontal's category	Categories with professional supply and data products	When the horizontal's category and the vertical compete for the same customers
Convergence of whole product platforms	Slow, large	One product platform for several brands	Same model, same supply type, same legal regime, stable ownership	When ownership or strategy changes before convergence completes

A second trade-off sits underneath: **how much to share, given the capital strategy**.

Topology	What is shared	Best fit for	Cost at exit
Federation	Practices, standards, some tools	Short holding horizons; very different markets	Low
Shared enabling layers	Cloud, data platform, CI/CD, media, moderation tooling	Most groups, most of the time	Moderate, if tenancy and accounts are clean
Shared capability services	Identity, messaging, search platform, catalog and entitlements, event backbone	Groups with a long horizon or a credible exit path per brand	Moderate to high; depends on separability by design
Converged product platform	Listing, search, apps, transactions	One model across similar markets, stable owner	High: transitional services, IP splits, parallel running

Adevinta's 2019 rule, "centralise where investment requirements are large and the technology is uniform", remains the best single default in the public record. The swings that followed suggest a second rule: **share whatever you can separate cheaply, and be explicit about what you cannot**.

Vertical Differences

The waves hit verticals differently, and that is visible in their architectures today.

Dimension	General classifieds	Real estate	Automotive	Jobs and recommerce
Pioneer origin	Start-ups and newspaper spin-offs; free private listings	Often founded by or with agents; feed-first	Print magazines and start-ups; dealer-driven	Job boards from print; recommerce born transactional
How portfolios grew	Copy-paste launches and rebranded acquisitions	Country-by-country acquisitions of leaders	Acquisitions of national leaders	Separate groups (Stepstone) or recommerce specialists (Vinted)
Consolidation outcome	Shared enabling layers; product convergence attempted and reversed	Integration attempted and rowed back (AVIV); integration into a platform buyer (CoStar–Domain)	Strong national stacks; selected cross-border platforms (AutoScout24)	Vinted merged its per-country "portals" into one marketplace with per-user consent (2022)

Dimension	General classifieds	Real estate	Automotive	Jobs and recom-merce
Why local wins	Language, consumer habits, local trust	Agent software ecosystems, disclosures, address systems, contract norms	Dealer systems, vehicle data, financing partners	Labour law and taxonomies (jobs); carriers and payment methods (recom-merce)
Typical sediment	Pioneer monoliths and country forks	Feed adapters per country and portal-owned agent software	Feed importers, vehicle decoders, price models per country	Order, payment and logistics state; legal-entity-per-country data

Real estate and recommerce sit at opposite ends. In real estate, **local differences are structural**: each country has its own professional software ecosystem and mandatory disclosures, which is the most plausible reading of why AVIV's integrated platform was scaled back. Vinted shows the opposite case: a recommerce marketplace whose model is the same everywhere could merge its country "portals", but even it had to migrate per user and obtain consent, because each country had been a separate legal entity with separate databases (Vinted Engineering, 2022). Same Platform, Different Physics: How Listing Verticals Differ develops the comparison.

Failure Modes

Misreading the history produces seven recurring mistakes.

Copying the last wave's answer. A new chief architect who arrives during a consolidation wave proposes convergence; one who arrives during a break-up proposes autonomy. Both are copying a fashion rather than reading their own group's ownership, markets and sediment.

The synergy case that counts platforms as savings. Deal models often count the retirement of duplicate platforms in year two or three. The public record shows long tails: rebrands that keep old domains for years, integration costs that recur (€69 million at Adevinta in 2023), transitional services for divested brands. A synergy that depends on platform convergence should carry the convergence's risk, not a straight-line saving.

Building shared platforms with no exit path. A shared platform without tenant-scoped data, per-brand accounts and cost allocation turns every later sale into a carve-out project. Adevinta's report of parallel running during decentralization is the visible cost; the invisible one is the deal that does not happen because separation is too expensive.

Treating acquired stacks as temporary. "We will migrate them next year" is the sentence that creates **Deal-Shaped Architecture**. Acquired stacks that are not on a dated, funded plan become permanent, and their teams build around them.

Forcing convergence where local differences are the product. Real-estate ingestion, dealer feeds, local payment methods and regulatory disclosures are where local marketplaces win. A converged platform that flattens them invites **The Forced Fit**: local teams route around it, and the group pays for both.

Forgetting inherited dependencies. Components inherited from a former parent, a former sister brand or a TSA keep running quietly until their owner stops supporting them. Marktplaats' Voyager is the documented example.

Rebrands priced as marketing projects. A brand change touches domains, search-engine equity, identity, design systems, app stores and payment descriptors. mobile.de's two-week app rebrand was possible only after a refactoring that removed eight years of debt.

The Chief Architect's View

At group scale, history is an input to four standing decisions.

Know the owner's horizon, and design to it. A family owner, a listed company and a private-equity fund imply different holding periods and different exit routes. The architecture function should ask for the capital strategy explicitly, in writing, and should revisit platform topology when it changes. This is not disloyal; it is the information without which a platform roadmap is a guess. At Veldmark, a decision record from March 2027, Ingrid's second month, states the assumption plainly: any brand may be sold within five years.

Make separability a non-functional requirement. Every shared platform should be specified with tenant-scoped data, per-brand cloud accounts and cost centres, exportable contracts and a tested exit path. A rule I use: **no shared platform goes live without a written answer to "how would we hand this brand's slice to a buyer in six months?"** If the answer is a transitional services agreement, the agreement's scope belongs in the design.

Separate the enabling layers from the product edge. The public record is consistent: sharing infrastructure, data platforms, media, moderation tooling and experimentation paid; converging whole product platforms across countries was slow and frequently

reversed. Decide per capability, using Adevinta's 2019 rule as the default and the separability rule as the constraint. After the Deal: Consolidating Platforms After Acquisitions and One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture turn this into a method.

Keep a sediment register. For each major system: the wave that created it, the assumption it carries, its owner, its consumers and its fate (keep, share, converge, retire, separate). It is a short table, reviewed with the architecture roadmap. It turns "why is it like this?" from folklore into a decision record, and it tells the next owner what they are buying.

What to watch: the share of engineering spent on duplicated capabilities; the number of transitional or cross-brand dependencies a sale would break; run costs per brand that can be stated without allocation arguments; and, after each deal, whether the platform plan was updated within a quarter.

Questions to Ask

Take these into a strategy review or a due-diligence meeting:

1. For each major system, which strategy created it, and does that strategy still hold?
2. What is the owner's expected holding period, and which exit routes (sale of the group, sale of a brand, listing) does the platform roadmap need to support?
3. If one brand were sold next year, which shared services would it lose, and how long would a transitional services agreement need to run?
4. Which of our shared platforms are enabling layers, and which are product platforms? Is the distinction deliberate?
5. Which acquired stacks are on a dated, funded plan, and which are "temporary" without one?
6. Are any components inherited from a former parent or sister company, and who still maintains their knowledge?
7. Does our synergy case count platform retirements as savings, and does it carry the risk of the convergence they depend on?
8. Where are local differences the reason our marketplaces win, and does our platform plan protect them?
9. What did the last rebrand or migration actually cost, including redirects, identity and design-system work, and is that cost in the next plan?

From History to Capabilities

The history explains how the systems got here; it does not say what they should become. To decide what to share, converge or keep local, a group first needs a description of what

its marketplaces *do* that is independent of the systems that happen to do it today. Map What the Marketplace Does Before What Runs It builds that description: a capability map for listing marketplaces, the value streams that cross it, and a heat map that shows where Veldmark's four brands duplicate one another and where their differences are the point.

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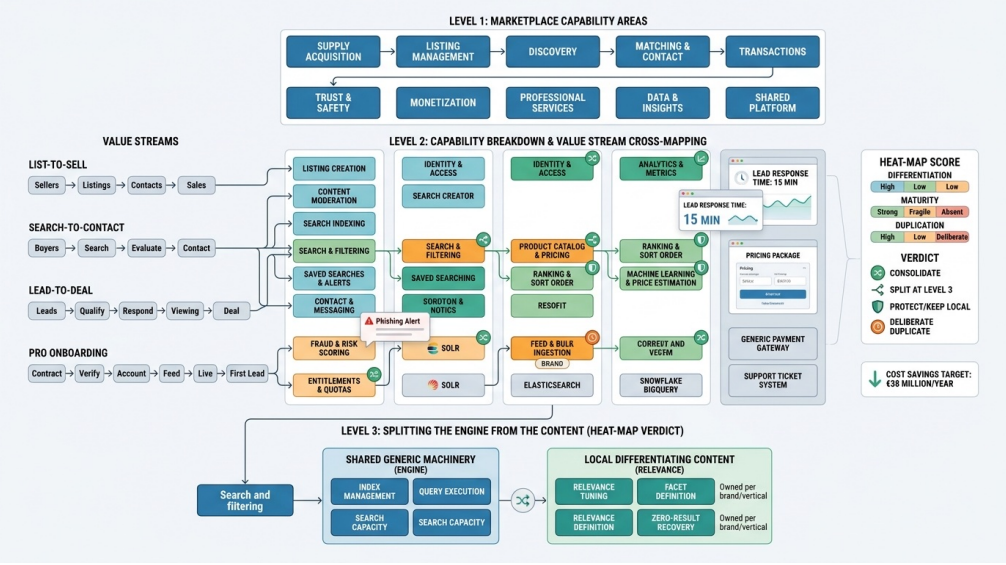
Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

4. Map What the Marketplace Does Before What Runs It



IN THIS CHAPTER, YOU WILL: Build a three-level capability map for a listing marketplace and cross-map it to four value streams. You will then heat-map it for differentiation, maturity and duplication across brands, and follow Veldmark as it uses the map to find five overlaps worth consolidating, one commercial overlap that no platform can fix, and one duplication it must keep on purpose.

WHY THE BUSINESS CARES: A group CFO or an investor asks a simple question: what do we pay for several times, and what makes us money? A capability map answers it in business words that stay true through reorganizations and replatformings. A systems inventory cannot answer it, because it names systems, not what they do for the business.

WHY ARCHITECTS SHOULD CARE: Every later decision in this book needs a map to stand on: where the bounded contexts go, what becomes a shared platform, what stays local and what to consolidate after an acquisition. Without the map, each of those debates turns into a contest between system owners.

KEY POINTS:

- **A capability is what the marketplace does, not how it does it or who does it.** Capabilities outlive systems, teams and org charts. That is why they are the right unit for comparing four brands that share almost no code. A map that names systems or departments is an inventory in disguise.
- **Capabilities matter only through the value streams they serve.** List-to-sell, search-to-contact, lead-to-deal and pro onboarding are the streams that turn supply into liquidity. Cross-mapping streams to capabilities shows which capability is the bottleneck for an outcome the business measures, and the bottleneck is where investment should go.
- **A heat map decides with three separate scores: differentiation, maturity and duplication.** Duplication alone proves nothing. A duplicated capability that differentiates a vertical should stay local. A generic one that is immature in four places is the consolidation case. Most real answers come from splitting a capability at level 3, where the engine is shared and the content stays local.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Deal-Shaped Architecture** — without a shared capability vocabulary, nobody can even list which capabilities the group pays for several times, let alone decide which to consolidate.
- **The Forced Fit** — a heat map that scores only duplication pushes differentiating capabilities onto a central platform that does not fit them, and local teams route around it.
- **Vanity Liquidity** — capabilities measured by their own activity (listings ingested, searches served) instead of by the value-stream outcomes they serve (contacts, sales) get optimized for the wrong thing.

From Newspaper Columns to Portfolios: How the Classifieds Industry Evolved showed how marketplace groups came to run several brands on stacks from different waves. Before such a group can decide what to share, it needs the third piece of business knowledge this part supplies: a description of **what the marketplace does that would still be true after the next reorganization and the next replatforming**. That description is a capability map. Badly made maps deserve their reputation as enterprise-architecture paperwork. A well-made one is the most useful single page a Chief Architect owns, because it is the only artifact on which a GM, a CFO, a platform lead and an acquirer's integration team can point to the same box and mean the same thing.

This chapter defines four terms that are routinely confused (capability, value stream, product and system), builds a three-level map with ten top-level areas, and cross-maps

it to four value streams. It then heat-maps the map for **differentiation**, **maturity and duplication** and reads the result without the reflex that every duplicate should be merged. It takes the business models of Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy and the liquidity metrics of Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics as given. It does not decompose the software: Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models turns this map into boundaries of models and teams, Same Platform, Different Physics: How Listing Verticals Differ tests what genuinely differs between verticals, and After the Deal: Consolidating Platforms After Acquisitions applies the map after acquisitions.

The running example is Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build). In March 2027 its new Group Chief Architect uses a capability map to test a €140 million program that the board approved before she arrived.

Setting the Scene at Veldmark: What Do We Pay for Four Times?

It is Tuesday 9 March 2027. Ingrid Halvorsen has been Veldmark's Group Chief Architect for five weeks. The board approved the **Common Ground** program in January: €140 million over 2027–2030 to build shared platform capabilities beneath the group's four brands, retire Tradeyard's PHP monolith Yardstick by the end of 2029, and cut run costs by **€38 million a year** by 2030. The program document lists what to share: a group identity, trust and safety, media, a search platform, messaging, a product catalog and entitlements, an event backbone, a data platform, and a taxonomy and attribute service.

Henrik de Boer, the Group CEO, asks Ingrid the question the private-equity owner asked him: *what exactly do we pay for four times, and which of those things make us money?* The program list does not answer it: it names platforms to build, not what the business does or how often it does it. Mateus Carvalho, the Group CTO, has a systems inventory of 412 applications across the four brands. It answers a different question: it says that Motorvane runs about 120 Kotlin services and Hearthly runs Go and TypeScript on Google Cloud. It does not say whether "Motorvane dealer packages" and "Hearthly agency subscriptions" are the same capability.

Two days earlier, on Sunday evening, a private seller in Rotterdam listed a city bike for €240 on Tradeyard. Within half an hour the listing had been published (six minutes after submission), matched 312 saved searches, drawn its first contact and attracted a phishing message that pretended to be a Safe Deal payment link. Ingrid uses that listing in her first workshop. She asks the brand CTOs which of their systems the bike passed through, and then which *capabilities* it needed. The first answer differs for every brand. The second, once the group agrees on the words, is nearly identical for all four.

Systems differ by brand; capabilities mostly do not. That is the chapter's argument in one example. Comparing brands, finding overlap and deciding what to share all have to happen at the level where the brands are comparable.

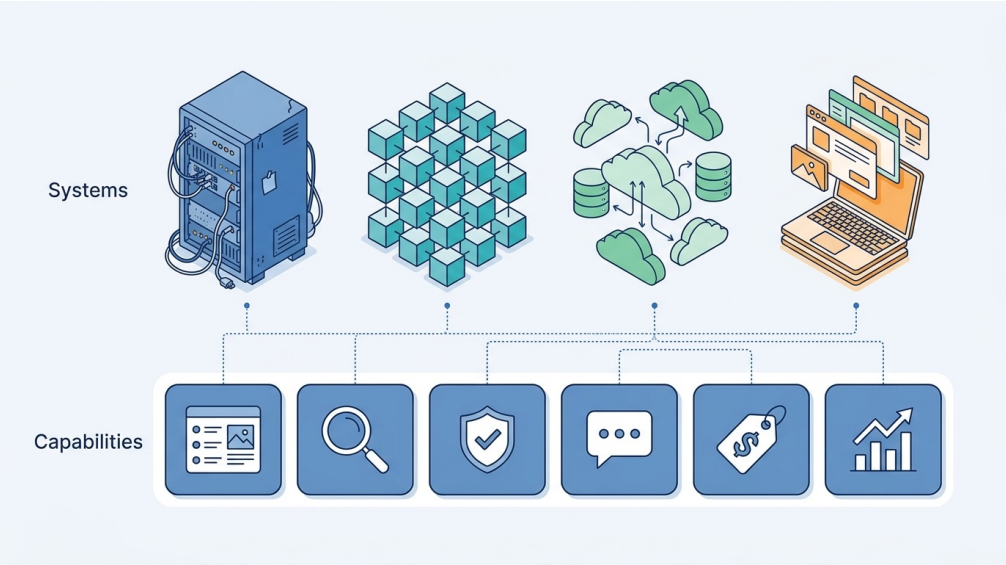


Figure 1: *Systems differ by brand; capabilities mostly do not, which is why comparing brands and deciding what to share has to happen at the capability level.*

Capability, Value Stream, Product, System

Four terms do most of the work, and confusing them produces most of the bad maps.

A **business capability** is what the business does, described independently of how, where and by whom. The Open Group's TOGAF standard defines it, in paraphrase, as a particular ability that a business may possess or exchange to achieve a specific purpose, and devotes a whole series guide to it (TOGAF Series Guide: Business Capabilities, Version 2, 2022). For a marketplace, *Content moderation*, *Saved searches and alerts* and *Entitlements and quotas* are capabilities. "The Solr cluster", "the Trust team" and "Yardstick's ad-insertion module" are not.

A **value stream** is the end-to-end sequence of stages that delivers an outcome to a customer or stakeholder. The Object Management Group's business-architecture work says that "the Value Stream view defines the end-to-end set of activities that deliver value to external and internal stakeholders" (OMG Business Architecture Working Group, *Business Architecture Overview*, 2012, as quoted in Wikipedia: Business architecture). In business architecture, value streams are commonly cross-mapped to the capabilities that enable each stage (Wikipedia: Value stream). In a marketplace the main value

streams are the journeys that create liquidity: a seller lists and sells, a buyer searches and contacts.

A **product** is a packaged offer that a customer uses or buys: "Tradeyard for iOS", "Motorvane Dealer Hub", the "Top listing" bump. A product composes many capabilities, and a capability serves many products.

A **system** (or application) is software that implements part of one or more capabilities: Yardstick, the Motorvane listing service, Hearthly's OpenSearch cluster.

Term	Answers the question	Lifetime	Marketplace example	Owned by
Business capability	*What* must the business be able to do?	Decades; changes when the business model changes	Fraud and risk scoring	A business owner, plus an architecture steward for the map
Value stream	*How does value reach* a customer, stage by stage?	Years; changes with the customer journey	Search-to-contact	A product or journey leader
Product	*What do we offer or sell?*	Months to years	Motorvane "Premium" dealer package	A product manager
System	*What software runs it?*	3–15 years	Yardstick, a Solr 8 cluster	An engineering team
Process	*In what steps does work get done?*	Months; changes with tooling	The manual review of new agencies at Hearthly	An operations lead

Three properties make a capability map useful, and each is easy to break.

1. **Stable.** If the map changes when the org chart changes, it is an org chart. If it changes when a system is replaced, it is an inventory. A good test: the capability *Duplicate and entity resolution* existed when Hearthly's duplicates were removed by hand, and it will still exist when a model does the work.
2. **Implementation-free.** No system names, vendor names, team names or channel

names in capability labels. "Salesforce CRM" is a system. "Professional account management" is a capability.

3. **Comparable**. The same capability must mean the same thing in every brand, at least at levels 1 and 2. This is what allows a heat map across brands at all. Vertical vocabulary belongs in the descriptions and in level 3, not in the names of the top two levels.

Only a capability map can answer Ingrid's question. A systems inventory counts 412 applications and cannot say how many of them do the same job. In 2014 James Lewis and Martin Fowler made the same move for software: they described microservices as "organized around business capability", in contrast to teams split by technology layer, and they tied the idea to Conway's law (Lewis and Fowler, "Microservices", 2014). Melvin Conway's original observation was that organizations are "constrained to produce designs which are copies of the communication structures of these organizations" (Conway, "How Do Committees Invent?", *Datamation*, 1968). A capability map is the business-side half of that bargain: it names the units that teams and services should align to before anyone draws a team or a service.

A Three-Level Capability Map for Listing Marketplaces

The map below has ten level-1 areas. They follow the flow of a listing marketplace: supply comes in, becomes a listing, is discovered, is matched with demand and sometimes transacted, while trust, monetization, professional services, data and a shared platform cut across all of it.

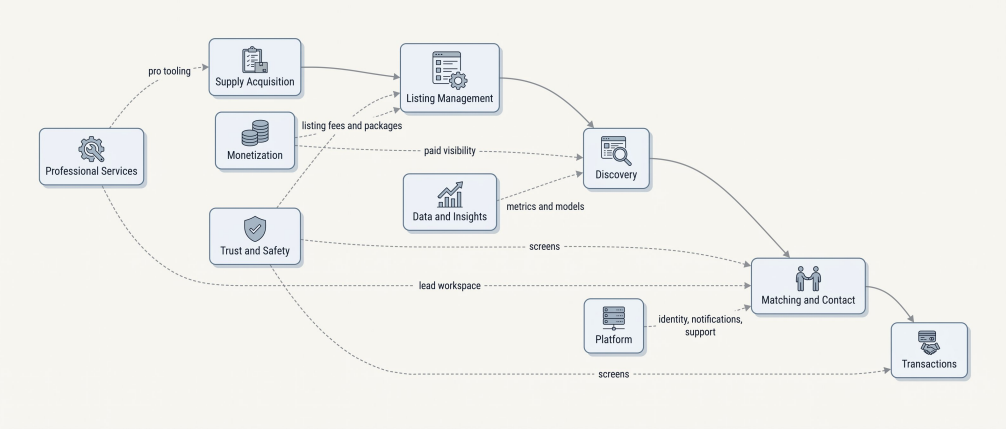


Figure 2: *The ten level-1 capability areas of a listing marketplace. Five follow the liquidity flow from supply to transaction; five cut across it. Dotted edges show where*

the cross-cutting areas act on the flow; Platform and Data serve every area, and only a few of their edges are drawn.

Level 2 breaks each area into capabilities that a business owner would recognize and that can be compared across brands. Veldmark's group map has **54** level-2 capabilities.

Level 1	Level 2 capabilities	What the area is for
Supply Acquisition	Private seller acquisition; Professional sales and account acquisition; Partner and integration management; Feed and bulk ingestion; Valuation-led supply capture	Getting sellers, professionals and their inventory onto the marketplace
Listing Management	Listing creation and editing; Listing lifecycle and publication; Taxonomy and attribute management; Media management; Location and geo reference; Normalization and enrichment; Duplicate and entity resolution	Turning raw supply into consistent, findable, correct listings
Discovery	Search and filtering; Query understanding; Ranking and sort order; Recommendations; Map and location search; Saved searches and alerts; SEO and landing pages	Matching buyer intent to inventory

Level 1	Level 2 capabilities	What the area is for
Matching and Contact	Contact and messaging; Lead management and routing; Call and contact tracking; Viewing and test-drive scheduling; Reviews and ratings	Turning interest into a conversation or a qualified lead
Trust and Safety	Policy management; Content moderation; Fraud and risk scoring; Identity and business verification; Notice, appeals and case management; Regulatory reporting	Keeping listings, users, messages and transactions safe and lawful
Transactions	Checkout and protected payment; Shipping and handoff; Order and dispute management; Payouts and reconciliation	Completing a sale on the platform where the model calls for it
Monetization	Product catalog and pricing; Entitlements and quotas; Orders, contracts and billing; Paid visibility delivery; Advertising sales and delivery	Charging for what the marketplace provides, and proving delivery
Professional Services	Pro inventory workspace; Pro performance and ROI reporting; Pro CRM and lead workspace; Market insights for professionals; Professional software (SaaS)	Serving dealers, agencies and merchants as businesses
Data and Insights	Behavioral tracking and consent; Analytics and governed metrics; Experimentation; Machine learning and price estimation; Data products and licensing	Knowing what happens and deciding with evidence

Level 1	Level 2 capabilities	What the area is for
Platform	Identity and access; Notifications; Customer support and back office; Privacy and data-subject rights; Developer platform and runtime	Shared foundations that every customer-facing capability needs

Four placement choices will be challenged, so the map records the reason for each.

- **Taxonomy and attributes sit under Listing Management, not Discovery.** Search uses them heavily, but the reason they exist is to define what a listing is. Placing them under Discovery leads to the common mistake of a taxonomy owned by the search team and shaped by facets. Category Trees That Can Change: Designing Marketplace Taxonomies and Attributes Without Chaos: Schemas, Dictionaries and Validation own the design.
- **Identity verification sits under Trust and Safety; identity and access sits under Platform.** Logging in is a foundation. Deciding how much to trust a person or a business is a risk decision with policy behind it. The two reference models described below make the same split: authentication as a platform service, verification as a trust capability that authentication escalates to. Who Is on the Other Side? Identity and Verification owns the detail.
- **Professional Services is a level-1 area of its own.** In verticals the professional seller is the paying customer, and the tooling they use is a product line with its own economics. Folding it into Supply Acquisition hides a revenue line, and folding it into Monetization hides a product. Serving the Pros: Dealer, Agent and Merchant Tooling covers its architecture.
- **Transactions is a level-1 area even where it is small.** Most of a classifieds group's value still settles off-platform, but the transactional pivot is a recurring strategic question. Keeping the area on the map, scored as immature or absent where it is absent, keeps the question visible.

Level 3: where the real decisions are made

Level 3 is where a capability is split into the parts that behave differently. Veldmark's map has **212** level-3 capabilities. Most are written only for the areas under active decision. Mapping everything to level 3 on day one is wasted effort: the map goes stale before it is used.

Two examples show why level 3 matters.

Level 2	Level 3 capabilities	Why the split matters
Search and filtering	Index management and near-real-time indexing; Query execution and faceting; Filter and facet definition per category; Search relevance tuning; Zero-result recovery; Search capacity and caching	The first two and the last are generic engine capabilities. Facet definition and relevance tuning are vertical knowledge. A shared search *platform* can own the engine while each brand owns its relevance
Entitlements and quotas	Package and entitlement definition; Grant and consumption ledger; Enforcement at insertion; Enforcement in ranking and display; Delivery evidence and audit	The ledger and enforcement points are generic and must be correct. Package definitions are commercial strategy that differs by brand and country

The pattern recurs across the map: **the machinery is generic and the content is specific**. Search engine versus relevance. Entitlement ledger versus package design. Moderation pipeline versus policy. Taxonomy service versus category trees. A heat map at level 2 often says "duplicated and differentiating", which leads nowhere. At level 3 the same capability usually splits cleanly into a shared part and a local part. Most of Veldmark's eventual answers came from that split.

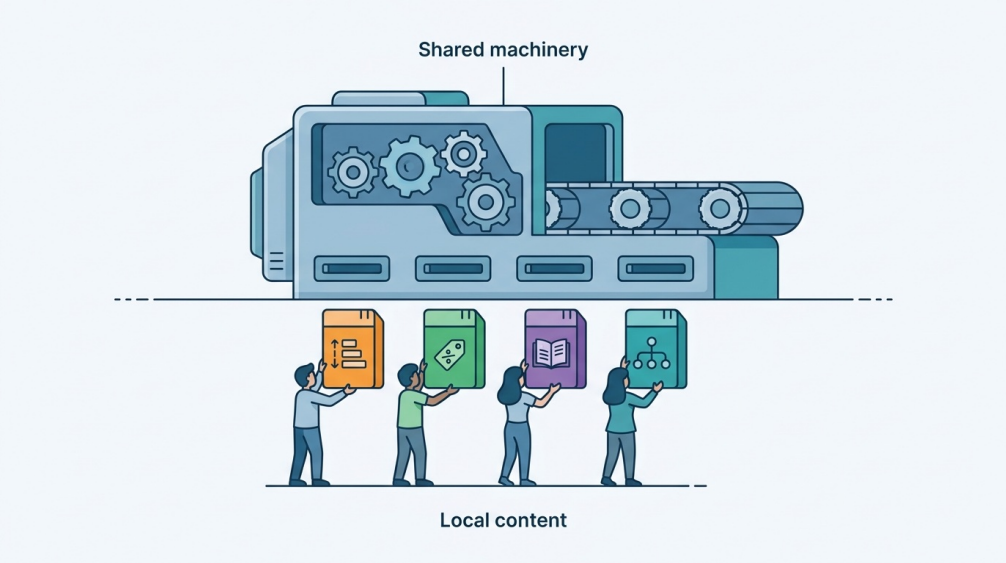


Figure 3: *At level 3 a duplicated capability usually splits cleanly: the machinery (search engine, entitlement ledger, moderation pipeline, taxonomy service) is shared, while relevance, packages, policy and category trees stay local.*

One skeleton, many verticals

One map serves every vertical, and reference modeling shows why. I maintain two generic reference models built with the Productscape method, one for a horizontal general-listings marketplace and one for a real-estate portal. They describe no real company. They share an identical set of **88 capabilities** in the same five root groups. The real-estate model's "Estimation Funnel" and the general model's "Seller Listing Funnel" occupy the same slot, as do "Property Valuation Tool" and "Pricing Guidance Engine". About two-thirds of the 88 have identical meaning in both. Roughly a dozen keep their responsibility under vertical vocabulary (agency for storefront, agent for seller). Only **nine** are genuinely different capabilities in the same slot: protected payment versus mortgage financing, a listing funnel versus a valuation-led lead funnel, shipping preferences versus rental management, and a few others.

The lesson reaches beyond those two models: **a marketplace capability map generalizes across verticals by renaming and re-weighting, not by adding new top-level structure.** That is why Veldmark keeps one group map for four brands. Vertical differences show up in three places: in level-3 capabilities that exist only in one vertical (*Mandate management* for agencies, *Vehicle identification and history* for motors), in the heat-map scores, and in the descriptions. Same Platform, Different Physics: How Listing Verticals Differ owns the question of what genuinely differs between verticals.

Value Streams: Where Capabilities Meet Outcomes

A capability map becomes useful only when it is cross-mapped to the value streams that produce the outcomes the business measures. For a listing marketplace those outcomes are liquidity: listings that sell, searches that end in contact, leads that become deals (Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics defines the metrics). Four value streams cover most of a group's value.

Value stream	Stages	Outcome metric	Main customer
List-to-sell	Start listing → describe and price → publish → get contacts → agree → hand off → mark sold	Listing-to-sale rate; time to sale; time from submission to searchable	Private seller
Search-to-contact	Express intent → refine → evaluate listing → contact → (save and return)	Search-to-contact rate; time to first contact; alert-to-contact rate	Buyer or seeker
Lead-to-deal	Receive lead → qualify → respond → viewing or test drive → offer → deal → attribution	Lead response time; lead-to-viewing; attributed deals per paying professional	Dealer or agency
Pro onboarding	Contract → verify business → set up account and users → connect inventory feed → first listings live → first leads	Days from contract to first live listing; feed success rate; first-90-day retention	Dealer or agency

The cross-map names the capabilities each stage needs. Figure 4 draws list-to-sell for the Rotterdam city bike, with the timings from that Sunday evening.

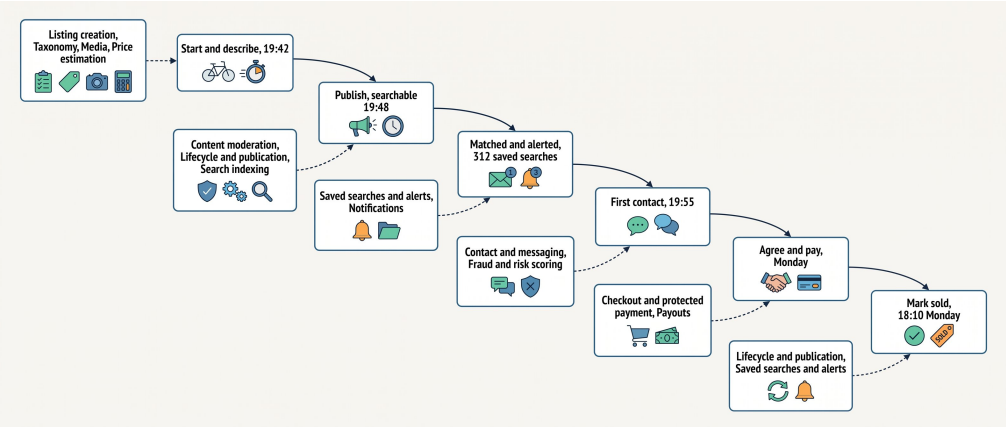


Figure 4: *List-to-sell for the €240 Rotterdam city bike, cross-mapped to the level-2 capabilities each stage needs. The six-minute publication delay sits in two capabilities, moderation and indexing. The Ghost Inventory failure at the end, a sold bike featured in the next morning's alert digest, sits in the hand-off between lifecycle and alerts.*

The cross-map turns a vague complaint into a located one: each of the bike's three problems sits in a specific capability.

1. **Six minutes from submission to searchable.** The stream-level metric belongs to *Listing lifecycle and publication*, but the delay is produced by *Content moderation* (the review queue) and *Search and filtering* (Solr batch commits). The 90-second target for 2028 is therefore an investment in two capabilities, not in "publishing". From Submitted to Searchable: The Publication Pipeline owns the design.
2. **A phishing message delivered at 20:11.** The domain in the fake Safe Deal link had been registered that day. The gap is in *Fraud and risk scoring* applied to *Contact and messaging*, not in moderation of listings, which had done its job.
3. **A sold bike in the next morning's alert digest.** The bike was marked sold at 18:10 on Monday and removed from search at 18:11, yet the next morning's alert digest still featured it. The stream crosses from *Listing lifecycle and publication* to *Saved searches and alerts* without a contract for unpublish events, so the digest never consumed them. That is a capability interface problem, which Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models and Every Change Is an Event: Event-Driven Listing Architecture take up.

The second use of the cross-map is to find **shared bottlenecks**. When the same capability appears in the bottleneck stage of several value streams, it is a high-leverage investment. At Veldmark, *Fraud and risk scoring* sits in the critical stage of three streams: messaging in list-to-sell and search-to-contact, rental listings in search-to-contact at Hearthly, and business verification in pro onboarding. That is a stronger

argument for a shared trust-and-safety platform than "we have four stacks".

Capability (level 2)	List-to-sell	Search-to-contact	Lead-to-deal	Pro onboarding
Listing lifecycle and publication	**Bottleneck** (6 min)	Freshness of results	Feed removal lag	First listings live
Saved searches and alerts	Speed of first contact	**Return visits**	Price-drop alerts	—
Fraud and risk scoring	Message scams	**Fake listings**	Fake leads	**Business verification**
Lead management and routing	—	—	**Response time**	First leads
Feed and bulk ingestion	—	—	Price and status sync	**Feed connection**
Entitlements and quotas	Paid bumps	Paid placement in results	Package limits	**Package activation**
Analytics and governed metrics	Sale capture	Contact definition	**Attribution**	Onboarding funnel

Bold cells mark where Veldmark's 2027 evidence placed the binding constraint of each stream. The table is a working tool, not a deliverable. It is redrawn when a constraint moves.

Two cautions keep value streams honest.

- **Measure stream outcomes, not capability activity.** "Listings ingested per day" is a capability activity metric. "Share of dealer listings searchable within ten minutes of a feed update" is a stream outcome. A capability team that reports only its own activity will optimize the wrong thing. That is the Vanity Liquidity pattern applied inside the architecture.
- **Map jobs to streams, then streams to capabilities.** One generated capability model I reviewed had mapped customer jobs directly to capabilities, skipping the value streams. The result was a model in which every capability appeared to serve every customer. The value stream is the layer that explains *why* a capability matters for an outcome.

Heat-Mapping: Differentiation, Maturity, Duplication

A heat map scores each capability on several dimensions and colors the map by the scores, so that patterns show on one page. A multi-brand marketplace group needs three dimensions, scored separately.

Differentiation asks how much this capability makes customers choose us over a competitor, in this vertical. It is scored by product and commercial leaders, not by engineers. The DDD Crew's core domain charts make the same separation: business differentiation is "provided by product managers or business stakeholders", while engineers judge model complexity (DDD Crew, Core Domain Charts, n.d.). Ask: how hard would it be for a competitor to match us here, and would customers notice if we were merely average?

Maturity asks how well the capability works today, against what the business needs. It is scored with evidence, on a five-point scale, per implementation.

Score	Meaning	Evidence that justifies it
1	Absent or manual	No system; spreadsheet or human process
2	Fragile	Works, but incidents, backlogs or known gaps against current volume
3	Adequate	Meets today's needs; hard to change
4	Strong	Meets needs, measured, changes at reasonable cost
5	Leading	Measurably better than competitors; others copy it

Duplication asks how many separate implementations of the capability the group runs, and whether they do the same job. It is a count, not a judgement, and it is the easiest score to collect. That is why it dominates badly made heat maps.

The three scores combine into a small number of verdicts.

Differentiation	Maturity across implementations	Duplication	Typical verdict
Low	Low, uneven	High	**Consolidate onto a shared platform.** The classic synergy case

Differentiation	Maturity across implementations	Duplication	Typical verdict
Low	High in one brand	High	**Promote the strongest implementation** to group use, if it can be made multi-tenant
Low	Any	One	**Buy or keep**; consider a commodity product
High	Any	High	**Split at level 3**: share the machinery, keep the differentiating content local
High	Low	One	**Invest locally**; this is where the brand competes
High	High	One	**Protect**; do not disturb with a platform program
Any	Any	Deliberate	**Keep separate on purpose**, when a legal, commercial or neutrality reason requires it

The last row is the one maps forget. Some duplication is a design decision. Veldmark's case is Keybridge, the agent CRM it acquired, which publishes agency listings to many portals, including Hearthly's competitors. In March 2027 the group decided that Keybridge stays a separate, neutral SaaS, and that Hearthly may not use Keybridge agency data for ranking or sales targeting. Keybridge's listing distribution and Hearthly's feed ingestion are therefore *duplicated on purpose*. A heat map that scored them for consolidation would have recommended destroying the trust that Keybridge's business depends on.

A heat map is only as good as its evidence. Three rules keep it honest.

1. **Score with named evidence and a date**. "Maturity 2: 138,000 listings a week go to human review, and the queue p95 is four minutes" is evidence. "Maturity 2: the team feels it is weak" is not. A generated reference model I reviewed showed the same failure: KPI values with no provenance, numbers that look measured and are not.

2. **Score per implementation, then aggregate.** "Trust and safety: maturity 3" averages a strong Tradeyard NL implementation with an absent Keybridge one and hides both.
3. **Separate the scorers.** Product leaders score differentiation, engineering leads score maturity, architecture counts duplication. When one group scores everything, the scores converge on that group's interests.

In Practice: What the Public Record Shows

No marketplace group has published its capability map. The public record does show the reasoning that capability maps are used for, and its limits.

Adevinta: from duplicated tools to shared enabling capabilities. In a 2024 engineering post, Adevinta described the state of experimentation in 2021, when it ran 22 marketplaces: "Every team had its own tool for A/B testing, resulting in inefficiencies, redundancy and escalating costs." A cross-functional working group used a scoring matrix with a three-to-five-year horizon and consolidated about ten tools towards one in-house platform (Adevinta, "From ten to one", 2024). Experimentation is a textbook low-differentiation, high-duplication capability, and the consolidation took years even so. The group's 2023 annual report says its central technology focus had historically been "consolidating enabling capabilities such as cloud and data, and overarching elements such as chat and moderation" (Adevinta Annual Report 2023, 2024). Those are exactly the capabilities a heat map tends to score as generic: infrastructure, data, messaging and moderation.

Adevinta: overlap of needs is not the same as a shared product. The same 2023 report describes a bolder step. Citing "a pronounced overlap in customer needs across all of Adevinta's markets", the group decided to avoid duplicating investments in the full transactional journey, AI-assisted search and first-party advertising, and to transform the Kleinanzeigen and leboncoin platforms "into a unified global platform" for its generalist marketplaces (Adevinta Annual Report 2023, 2024). After the 2024 take-private, the 2025 strategic report of its reporting entity describes the opposite direction: "a decentralised, standalone operating model", with "the ramp-down of central functions, the transfer of key capabilities, the localisation of intellectual property" (Aurelia UK Feederco Ltd, Strategic Report 2025, published 2026). Whether the unified platform shipped is not publicly documented. The documented sequence carries a lesson for heat maps: an overlap score measures similarity of needs, not the cost of sharing, and the owners' capital strategy can turn a shared platform from a synergy into a separation cost. Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform tells the full story; After the Deal: Consolidating Platforms After Acquisitions draws the consolidation lessons.

Rightmove and Autotrader: monetization capabilities as revenue levers. Two UK portals publish unusually precise revenue bridges, and both show that product and

pricing capabilities are differentiating, even though the machinery behind them is generic. Rightmove's results for 2025 attribute 62% of the growth in agency average revenue per advertiser (ARPA) to product rather than price or volume (Rightmove plc, 2025 results, 2026). Autotrader's FY26 results describe an "annual 1 April 2025 pricing and product event, which combined a like-for-like price increase with additional products" (Auto Trader Group plc, FY26 results, 2026). For a capability map, the inference is that *package design and pricing* is a differentiating level-3 capability, while *entitlement ledger and enforcement* is generic. That is the split Veldmark uses below. How Listing Marketplaces Make Money and Packages, Bumps and Quotas: Products, Pricing and Entitlements own the detail.

Domain-first decomposition at Vinted. In a 2026 engineering post, Vinted described the first step of leaving its monolith: "Our first step wasn't to break the monolith apart. It was to understand it." Using domain-driven design, it "identified almost 300" domains over "at least two years", and that understanding "guided how we restructured teams" (Vinted Engineering, 2026). Vinted's domains are closer to bounded contexts than to business capabilities, but the order is the point: describe what the business does before deciding what runs it. Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models picks up the story.

Veldmark Worked Example: Five Overlaps, One Commercial Clash, One Deliberate Duplicate

Ingrid runs the mapping in three weeks of March 2027.

1. **Week 1: draft the map**. The group architecture team drafts levels 1 and 2 from the four brands' product documentation and the two reference models, without looking at the systems inventory. Brand CTOs review the draft for one question only: does every capability mean the same thing in your brand? Seven labels change; none of the ten level-1 areas does.
2. **Week 2: map systems to capabilities**. Each brand maps its applications to level-2 capabilities. Of the 412 applications, 371 map cleanly. The other 41 are either platform plumbing (build tools, monitoring) or Yardstick, which maps to 23 capabilities on its own.
3. **Week 3: score and read**. Noor Hassan, the Group Chief Product Officer, and the brand product leads score differentiation. Engineering leads score maturity per implementation with dated evidence. Architecture counts duplication. Olivier Petit, VP Monetization, adds revenue attribution for the monetization area.

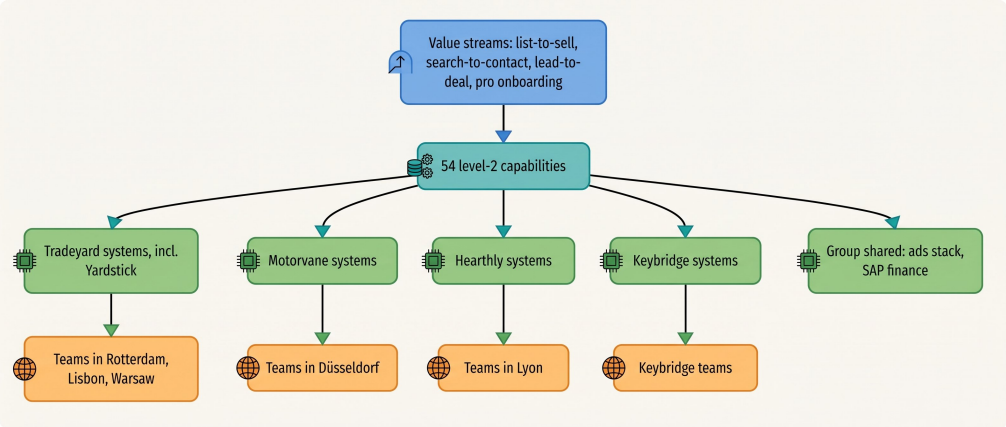


Figure 5: *The layering Veldmark used: value streams explain why capabilities matter, capabilities are the comparable unit, and systems and teams are mapped to capabilities, never the other way round. Only the capability layer is shared vocabulary across all four brands.*

The heat map

The table shows the rows that drove decisions. Maturity is listed per implementation (T = Tradeyard, M = Motorvane, H = Hearthly, K = Keybridge).

Level-2 capability	Implementations	Differentiation	Maturity (evidence, March 2027)	Verdict
Identity and access	4	Low	T 3, M 3, H 2, K 3; no single login across brands	**Overlap 1: consolidate** (Veldmark ID)
Content moderation; Fraud and risk scoring	4 stacks (T NL/BE in-house, T PL/PT outsourced queues, M, H)	Low as machinery, high as policy	T NL/BE 3; T PL/PT 2; M 2 (dealer KYB only at contract); H 2	**Overlap 2: split** at level 3; share the pipeline and scoring, keep policy config-urable per country

Level-2 capability	Implementations	Differentiation	Maturity (evidence, March 2027)	Verdict
Search and filtering	3 engines (Solr 8, Elastic-search, OpenSearch)	Engine low; relevance high	T 2 (6-minute indexing, January brown-out); M 4; H 3	**Overlap 3: split**; shared engine and indexing, brand-owned relevance
Product catalog and pricing; Entitlements and quotas	4 catalogs	Package design high; ledger low	T 2; M 3; H 3; K 3	**Overlap 4: split**; one catalog model and entitlement service, brand-owned packages and prices
Analytics and governed metrics	3 warehouses (Snowflake, BigQuery, Azure Synapse)	Low	All 3; "contact" defined three different ways	**Overlap 5: consolidate** definitions first, platforms later
Media management	3	Low	T 3; M 3; H 3 (1.8 billion stored images)	Consolidate after the five overlaps; cost-driven
Taxonomy and attribute management	3, plus four Tradeyard country trees	Content high; service low	T 1–2 (six weeks per new category); M 4; H 3	Split: shared service, brand-owned trees (already in Common Ground)
Ranking and sort order	3	High	T 2; M 4; H 3	**Keep local** on the shared engine

Level-2 capability	Implementations	Differentiation	Maturity (evidence, March 2027)	Verdict
Machine learning and price estimation	2 models (Motorvane market value, Valuo)	High	M 4; H 4	Keep models local; share the ML platform
Professional sales and account acquisition, Poland	2 (Tradeyard PL cars, Motorvane PL)	High	T PL 2; M PL 3	**Commercial overlap**: a portfolio decision, not a platform one
Feed and bulk ingestion; listing distribution	2 (Hearthly ingestion, Keybridge multi-diffusion)	High	H 3; K 4	**Deliberate duplicate**: keep separate (neutrality decision, March 2027)

Reading the map

The five overlaps match the "four of everything" problem that the board already knew about: four identity systems, four trust-and-safety stacks, three search engines, four product catalogs, three data warehouses. The map's contribution was not to discover them. It was to change the answer for three of them and to add two findings that no platform program would have produced.

- **Two overlaps were straight consolidations**. Identity and access scored low on differentiation everywhere and immature in Hearthly. The brands do not compete on login. Veldmark ID, approved in June 2027, follows. For analytics, the map showed that the costly duplication was not the three warehouses but the three definitions of a *contact*. The verdict was to govern the metric first and move platforms later, which Data Platform for Marketplaces: Events, Lakehouse and Data Products develops.
- **Three overlaps were splits, not merges**. Trust and safety, search, and the product catalog each scored "duplicated and differentiating" at level 2. At level 3 they split into generic machinery and local content. The search verdict is what later shaped the September 2027 decision: a shared OpenSearch platform for Tradeyard

and Hearthly, while Motorvane, whose search scored 4 and whose relevance is its own, stays on Elasticsearch until a 2029 review. The catalog verdict became "share the model, not the packages": one group catalog and entitlement service, with Motorvane's dealer packages and Hearthly's agency subscriptions defined by the brands.

- **The commercial overlap**. Tradeyard Poland's cars section and Motorvane Poland sell to the same dealers, with two sales teams, two package price lists and two feeds from the same dealer management systems. The heat map flagged it because the same level-2 capability appeared twice in one country, with high differentiation in both. No shared platform resolves it, because the conflict is between two go-to-market strategies. Ingrid's recommendation to the group leadership team was to treat it as a portfolio decision for Henrik de Boer and Katarzyna Nowak, Tradeyard Poland's general manager, with a technical option attached: one dealer feed endpoint feeding both brands, whichever brand ends up owning the relationship.
- **The deliberate duplicate**. Keybridge's distribution stays separate, as described above. The map records it as a duplicate with a reason and a review date. Otherwise the next cost-cutting exercise would find it again and try to merge it.

The map also changed Common Ground's **sequence**. The program's list put media high. The heat map scored media as low differentiation but adequate maturity in all three implementations: expensive, not broken. Trust and safety scored immature in three places and sat in the bottleneck stage of three value streams. The leadership team moved the shared trust-and-safety platform forward. It went live for Tradeyard NL and BE in November 2027, and media consolidation followed the five overlaps.

The output Ingrid presented on 31 March 2027 was one page: the ten level-1 areas, the 54 level-2 capabilities colored by verdict, and seven annotated findings. It did not contain a target architecture. That came later, in Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models, and the map is what made the later document arguable.

Trade-offs: Ways to Build a Capability Map

There are four common starting points, and the choice shapes what the map can and cannot answer.

Approach	How it works	Strengths	Weaknesses	When it is right
Reference-model first	Start from an industry or generic reference map, then adapt	Fast; complete; vocabulary not owned by any brand	Imports someone else's assumptions; generic text hides vertical specifics	A group with several brands and no shared vocabulary; the first draft after an acquisition
Systems-inventory first	List applications and infer capabilities from what they do	Grounded in what exists; easy to collect	Encodes today's system boundaries; misses what no system does; duplicates hide behind different names	Rarely as a starting point; useful as the week-2 mapping step
Value-stream first	Walk the customer journeys and name the capabilities each stage needs	Ties every capability to an outcome; finds gaps	Misses back-office and platform capabilities; can drift into process mapping	A single marketplace optimizing liquidity
Org-first	Map what each team or department does	Easy buy-in; ownership is obvious	Produces the org chart; breaks at the next reorganization	Almost never. Map ownership to capabilities afterwards instead

Veldmark combined the first and third: a reference model for the skeleton, value streams to test and weight it, and the systems inventory only as a mapping step.

Granularity is the second trade-off. Level 1 is for boards and investors. Level 2 is for portfolio decisions and heat maps. Level 3 is for platform scoping and bounded contexts. Level 4 usually means process, not capability, and belongs to the teams. The stopping rule: **decompose a capability one level further only when the parts would**

get different verdicts. Search splits at level 3 because the engine and the relevance get different verdicts. *Notifications* does not, because all its parts would get the same one.

Vertical Differences

The skeleton is shared, but each vertical weights it differently and adds a few level-3 capabilities of its own.

Dimension	General classifieds	Real estate	Automotive	Jobs and recommerce
Most differentiating areas	Discovery (local freshness), Trust and Safety (scams), Transactions where shipping applies	Professional Services (agency tooling), Data and Insights (valuation), Supply Acquisition (mandates)	Professional Services (dealer tooling), Data and Insights (price evaluation), Feed ingestion	Jobs: matching and applications. Recommerce: Transactions and shipping
Vertical-only level-3 capabilities	Shipping label creation; pickup scheduling	Mandate management; viewing scheduling; energy and legal disclosures; property reference	Vehicle identification and history; finance and insurance referral; test-drive booking	CV and application handling; catalog matching for recommerce
Supply Acquisition shape	Self-serve funnel at volume	Agencies must win mandates first; owner valuation captures supply	Dealer sales and feed onboarding	Employer sales; for recommerce, self-serve
Transactions	Protected payment in shippable goods	Off-platform (notary, lease)	Mostly off-platform; reservations and deposits emerging	Jobs none; recommerce central

Dimension	General classifieds	Real estate	Automotive	Jobs and recom-merce
Where the money is mapped	Paid visibility, transac-tional fees	Agency subscrip-tions, valuation leads, new-build campaigns	Dealer packages, depth products, finance referrals	Pay-per-post or per-application; buyer fees

At Veldmark this explains two of the heat-map outcomes. Valuo and Motorvane's market-value model sit in the same level-2 capability, *Machine learning and price estimation*. They are differentiating in both verticals and built on different data, so they stay local on a shared ML platform. And Hearthly's *Feed and bulk ingestion* scores higher on differentiation than Tradeyard's, because 84% of Hearthly's listings arrive through feeds from about 160 software vendors, compared with a mostly private, self-serve supply at Tradeyard.

Failure Modes

The inventory in disguise. Capability names that are system names ("Solr search", "Salesforce CRM") make the map a duplicate of the inventory and useless for comparison. The test: could the capability survive the replacement of every system that implements it today?

The org chart in disguise. A map with a box per department changes at every reorganization, and every owner defends their box. Veldmark's map has no team names in it. Ownership is a separate table that maps capabilities to owners, and it is expected to change.

Duplication as the only score. A heat map colored only by number of implementations recommends merging everything that is duplicated, including the differentiating capabilities. That is how The Forced Fit begins. Score differentiation separately, by the people who own the commercial outcome.

Mapping everything to level 4. A 600-box map takes six months, is out of date on delivery, and nobody reads it. Map level 3 only where a decision is pending.

Generated or uniform dependencies. Capability tools make it easy to draw dependencies between capabilities. When every capability depends on three others with boilerplate descriptions, as in one generated reference model I reviewed, the graph hides the real dependencies. Keep only the relationships that shape ownership, data movement, reliability, release sequencing or incident response.

Synthetic precision. Maturity scores and KPIs with decimals and no evidence look measured and invite false confidence. Every score needs a source and a date.

A workshop, not a practice. A map produced once for a program kick-off is stale within a year. The map earns its cost only if it is used in the decisions that follow: acquisitions, platform scoping and budget. That means an owner, a version and a review cycle.

Forgetting the deliberate duplicates. If the reason for keeping a duplicate is not written on the map, the next efficiency program will rediscover it and try to merge it.

The Chief Architect's View

At group scale the capability map is less a model than a governance instrument. Five decisions follow from that.

1. **Own the vocabulary.** The Chief Architect owns the level-1 and level-2 names and definitions and changes them only through a decision record. Brands own the level-3 detail in their areas and their scores. The map is published, versioned and short; Tool 1 in the Field Kit: Templates for the Marketplace Architect gives a template for a capability-map row and its heat-map scores.
2. **Separate the scorers and publish the evidence.** Product scores differentiation, engineering scores maturity, architecture counts duplication, finance attributes cost and revenue. A disputed score is a useful conversation. A score nobody can trace is a political one.
3. **Make "share the machinery, keep the content" the default hypothesis** for any duplicated capability, and require evidence to move away from it in either direction: towards full consolidation or towards full local ownership.
4. **Tie the map to money.** Every level-2 capability should eventually carry its run cost and, where it can be attributed, its revenue. Veldmark's €38 million run-cost target became credible only when it could be allocated to capabilities: identity, trust and safety, search, catalog and data, plus media later.
5. **Use the map at every acquisition.** An acquired marketplace is mapped onto the group's capabilities in its first month, before any integration plan. This is the cheapest defence against Deal-Shaped Architecture, and it is the starting point of After the Deal: Consolidating Platforms After Acquisitions.

What to watch over time: the number of deliberate duplicates (it should be small and each should have a reason), the share of capabilities with an owner and dated evidence, and whether teams and bounded contexts are actually aligning to the map. If after two years team boundaries still cut across level-2 capabilities, the map is decorating a wall. Who Owns What: Organizing Marketplace Engineering covers the team side.

Questions to Ask

1. Could every capability name on our map survive the replacement of every system that implements it today?
2. Do our level-1 and level-2 capabilities mean the same thing in every brand and vertical? Where they do not, is the difference real or only vocabulary?
3. For each value stream, which capability holds the binding constraint, and what is our evidence?
4. Who scored differentiation, who scored maturity, and can we trace each score to a dated source?
5. For each duplicated capability, have we tried splitting it at level 3 into shared machinery and local content?
6. Which duplicates do we keep on purpose, and is the reason written on the map with a review date?
7. Which overlaps are commercial (two brands selling to the same customer) and so need a portfolio decision rather than a platform?
8. What does each level-2 capability cost to run, and which ones can we attribute revenue to?
9. When we acquire the next marketplace, how quickly can we map it onto our capabilities?

From Capabilities to Contexts

The capability map tells Veldmark *what* the group does and where it does it several times. It does not say where one model of a listing should end and another begin, which team should own which part, or how the parts should talk to each other. That is a decomposition question, and it needs a different tool: domain-driven design's bounded contexts and context maps, applied in Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models.

Sources and Evidence

Company engineering and investor sources

- Adevinta Annual Report 2023, Adevinta ASA (filed as an eBay exhibit), 2024 — consolidation of enabling capabilities; the "pronounced overlap in customer needs" rationale for a unified generalist platform.
- Strategic Report 2025, Aurelia UK Feederco Ltd (Adevinta's reporting entity), 2026 — the move to "a decentralised, standalone operating model".

- "From ten to one: streamlining experimentation tooling at Adevinta", Adevinta Tech Blog, 2024 — duplicated A/B tools across 22 marketplaces and the consolidation decision.
- Rightmove plc 2025 results announcement, Rightmove, 2026 — 62% of agency ARPA growth from product.
- Auto Trader Group plc FY26 results, Autotrader, 2026 — the annual 1 April pricing and product event.
- "Building a global event-driven platform, part 1", Vinted Engineering, 2026 — almost 300 domains identified over at least two years before decomposition.

Standards, regulation and documentation

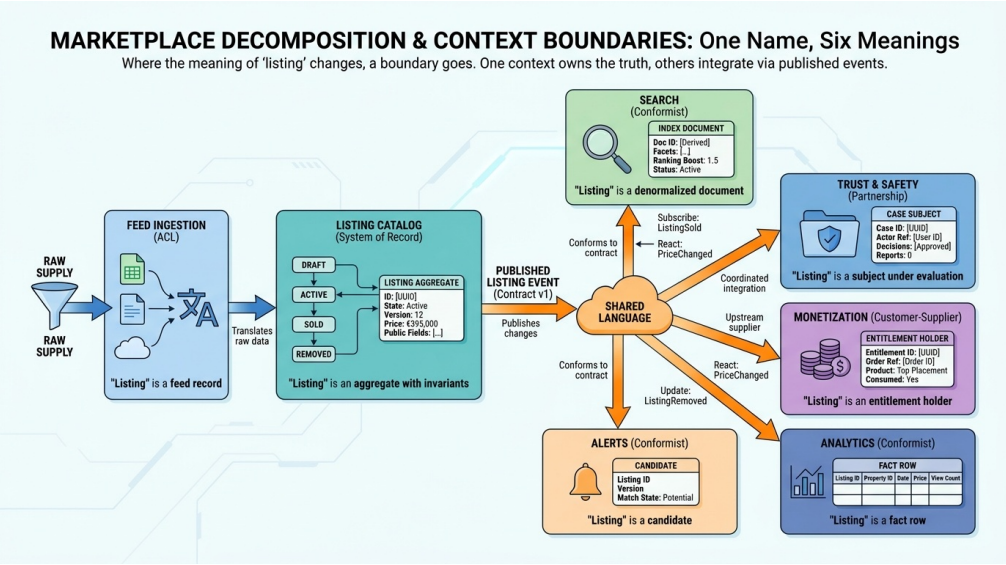
- TOGAF Series Guide: Business Capabilities, Version 2, The Open Group, 2022 — the business capability concept and definition (paraphrased; the guide sits behind a free Open Group login).
- Business architecture and Value stream, Wikipedia, n.d. — summaries of the OMG value stream definition and of capability cross-mapping in business architecture.
- Core Domain Charts, DDD Crew, n.d. — separating business differentiation from model complexity, scored by different people.

Research and practitioner sources

- "How Do Committees Invent?", Melvin E. Conway, *Datamation*, 1968 — Conway's law.
- "Microservices", James Lewis and Martin Fowler, 2014 — services organized around business capabilities.
- Productscape reference models for a general-listings marketplace and a real-estate marketplace, author's working models, 2026 — the shared 88-capability skeleton and the nine vertical-specific slots; generic models, not descriptions of any company.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

5. Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models



IN THIS CHAPTER, YOU WILL: Apply domain-driven design to a listing marketplace: classify subdomains as core, supporting or generic, find the bounded contexts where the word "listing" changes meaning, and draw a context map with explicit relationships. You will see where aggregates set transactional boundaries, why the listing event is the group's most important published language, and how Veldmark draws the Common Ground context map across four brands and a PHP monolith.

WHY THE BUSINESS CARES: Boundaries decide how fast a marketplace can change. A group whose brands share one listing table cannot launch a new paid product, a new vertical or a new country without a cross-team project. Clean boundaries let a brand change its ranking or its packages in a week without asking anyone else.

WHY ARCHITECTS SHOULD CARE: Boundaries are the architecture decision that is most expensive to reverse. They decide which team owns

which truth, what can be shared across brands, where translation happens and what breaks when a model changes. Services, topics and databases follow from them.

KEY POINTS:

- **A boundary goes where the meaning of a word changes**. "Listing" means a feed record in supply, an aggregate with invariants in the catalog, a subject under evaluation in trust, a denormalized document in search, a thing that holds entitlements in monetization and a row of facts in analytics. Each meaning deserves its own model and its own owner. One model for all of them becomes the god listing table.
- **One context owns the listing's truth; every other context owns only its own view**. The listing catalog decides state, version and public fields. Search, alerts, trust and monetization build their own models from the published listing event and never write listing state back. The search index is a read model, never the system of record.
- **Draw the relationships, not just the boxes**. Partner feeds need an anti-corruption layer, search conforms to the published language, monetization is a customer–supplier partner of the catalog, and a neutral SaaS business may need "separate ways". The context map is where a group decides who adapts to whom, and that decision is also an organizational one.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Search as System of Record** — when no context is named as the owner of listing truth, the search index becomes the only complete copy, and nothing else can rebuild it.
- **Billing by Guesswork** — without a monetization context that owns entitlements, "is this listing paid for?" is answered by flags on the listing table, read and written ad hoc by insertion, ranking and rendering code.
- **Ghost Inventory** — when contexts share a database instead of reacting to published lifecycle events, each consumer learns about a sale on its own schedule, and some never learn.

The previous chapter, Map What the Marketplace Does Before What Runs It, described what a marketplace does: 54 level-2 capabilities in ten areas, mapped to the value streams that produce liquidity. A capability map is not a design. It does not say how many models of a listing there should be, which team owns which one, or how they talk. Those are decomposition questions, and **the most useful tool for them is domain-driven design's bounded context**: a boundary inside which one model and one vocabulary hold, owned by one team.

Listing marketplaces are an unusually clear case for that tool, because their central noun is overloaded. Ask six teams what a listing is and you get six correct answers that cannot all be true of one data structure. The seller's draft, the moderator's case subject, the search document, the paid placement, the lead's source and the analyst's fact row are all called "the listing". Teams that share one model of it end up negotiating every change with every other team.

The rest of the book builds inside the boundaries drawn here. The Listing Aggregate: Identity, Lifecycle and Invariants designs the catalog's listing, Every Change Is an Event: Event-Driven Listing Architecture specifies the event contract's envelope and evolution, and Who Owns What: Organizing Marketplace Engineering turns the contexts into teams. This chapter defines the terms and draws the map: subdomains first, then the contexts and the meanings they own, then the context map and its relationships, then aggregates, then the listing event as the shared language. The running example is Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build).

Setting the Scene at Veldmark: Six Answers to "What Is a Listing?"

It is April 2027. Veldmark has just adopted its architecture principles and set up a weekly, open architecture advice forum. Sven Lindqvist, principal engineer for the event platform, has three months to ship **Listing event contract v1**, which goes live on 1 July 2027. Before he can define an event, he needs an answer to a question that sounds trivial. Ingrid Halvorsen, the Group Chief Architect, puts it on the forum's agenda: *what is a listing, and who decides?*

Six people answer, and each is right.

- **Daan Visser**, who owns Tradeyard's PHP monolith Yardstick, says a listing is a row in the `ads` table. The table has 214 columns and is read by 38 Yardstick modules. Paid-visibility flags, moderation status, search boost and the seller's phone number live side by side in it.
- **Felix Brandt**, Motorvane's CTO, says a listing is what the dealer's feed says it is. About 88% of Motorvane's dealer listings arrive through dealer management system (DMS) feeds. The Essen Volkswagen Golf sold on 29 April 2027, and the feed's removal arrived 26 hours later. Felix's question is who is allowed to say that a listing is gone.
- **Aisha Rahman**, Group Head of Trust & Safety, says a listing is a subject under evaluation: content, an actor, a device, a history of reports and decisions. Her model must outlive the listing, because a removal decision must survive the listing's deletion.

- **Tomasz Wiśniewski**, head of the group search platform, says a listing is a flattened document with facets, a ranking timestamp and a boost. He does not want the seller's phone number in it.
- **Olivier Petit**, VP Monetization, says a listing is something that can hold a paid product, such as a bump, a top placement or a package slot. What he cares about is whether what was sold was delivered.
- **Julien Marchand**, Hearthly's CTO, says that in real estate the listing is not even the main thing. The Lyon flat that three agencies listed in May 2027 is one property with three listings, and Hearthly's buyers, valuation model and market reports all care about the property.

The forum's conclusion is the starting point of this chapter: **"listing" is six concepts that share a name and an identifier**. The design question is not what *the* listing model should be. It is which context owns which meaning, which one owns the truth, and how the others learn about changes.

Figure 1: *One name, six models: the monolith, the feed, trust and safety, search, monetization and real estate each see a different 'listing', linked only by a shared identifier.*

Concepts: Subdomains, Contexts and Context Maps

Domain-driven design (DDD) was introduced by Eric Evans in *Domain-Driven Design: Tackling Complexity in the Heart of Software* (2003). Its strategic part deals with large models and many teams, and five of its terms carry this chapter.

A **domain** is the area of business the software serves: here, running a listing marketplace. A **subdomain** is a part of it with its own concerns, such as discovery, trust and safety or billing. Subdomains are found in the business. The capability map's level-1 and level-2 areas are a good first draft of them.

Subdomains differ in strategic value. Evans distinguished the **core domain**, where the business competes and the best people should work, from **generic subdomains** that every business needs and nobody wins with. Vaughn Vernon added the middle category of **supporting subdomains**: specific to the business, necessary, but not where it wins (*Implementing Domain-Driven Design*, 2013). The classification is a business judgement, and it maps onto the differentiation score of the capability heat map.

A **bounded context** is the boundary within which one model and its vocabulary apply consistently. Martin Fowler calls it "a central pattern in Domain-Driven Design" and explains that DDD "deals with large models by dividing them into different Bounded Contexts". He adds that the dominant factor in drawing the boundaries is usually human culture, because the model is the language people use (Fowler, "BoundedContext", 2014). Fowler's example is an electricity utility in which "meter" meant three different things in different departments. "Listing" is the marketplace's meter.

The **ubiquitous language** is the vocabulary used consistently inside one context, in conversation, documents and code. It is ubiquitous *within* the context, not across the company.

A **context map** shows the bounded contexts and the relationships between them: who is upstream, who adapts to whom, and where translation happens. The relationship patterns have standard names, summarized below from the DDD Crew's open reference (DDD Crew, Context Mapping, n.d.).

Relationship	What it means	Typical marketplace use
Partnership	Two teams coordinate planning and manage the integration together	Listing catalog and trust and safety during a pipeline redesign
Shared kernel	A small, explicit part of the model that two teams co-own	Listing identifiers and tenant codes
Customer–supplier	Upstream plans for the downstream team's needs	Listing catalog (supplier) and monetization (customer)
Conformist	Downstream adopts the upstream model as it is	Safe Deal conforming to the payment provider's model

Relationship	What it means	Typical marketplace use
Anti-corruption layer (ACL)	Downstream translates to keep its model clean	Feed ingestion translating 160 vendors' formats; a translator in front of a monolith
Open host service	Upstream offers a well-defined protocol to many consumers	Identity, taxonomy, the partner listing API
Published language	A documented shared language for exchange	The listing event contract; feed standards
Separate ways	No integration; each side solves its own problem	A neutral SaaS business that must not consume the portal's data
Big ball of mud	A part with mixed models and no clear boundary, to be contained	A 20-year-old monolith

Two more terms come from DDD's tactical design. An **aggregate** is, in Fowler's words, "a cluster of domain objects that can be treated as a single unit", and "transactions should not cross aggregate boundaries" (Fowler, "DDD_Aggregate", 2013). A **domain event** is a record that something meaningful happened in a context, published so that other contexts can react.

These concepts are easy to confuse with the previous chapter's. The table separates them.

Concept	Unit of	Decided by	Example
Business capability	What the business does	Business and architecture, via the capability map	Saved searches and alerts
Subdomain	Business area with a strategic value	Business, with architecture	Discovery (core at Tradeyard)
Bounded context	One model and one language	Architecture, with the owning team	Alerts context

Concept	Unit of	Decided by	Example
Service or deployable	Software runtime	The owning team	`alert-matcher`, `digest-builder`
Team	People and ownership	Engineering leadership	Tradeyard Alerts team

The usual healthy shape is **one team owns one or more contexts; one context is owned by exactly one team; a context contains one or more services**. A context split across two teams is a coordination problem disguised as an architecture. Conway's observation that systems copy the communication structures of the organizations that build them (Conway, 1968) cuts both ways here. Context boundaries that ignore team boundaries will not hold, and team boundaries drawn without a context map produce contexts by accident.

Subdomains: Where a Marketplace Competes

Before drawing contexts, classify the subdomains. The classification decides where to invest, what to buy and what to share across brands. It differs by marketplace model and vertical, which is why it must be done per brand even when the map is shared.

Subdomain	General classifieds (Tradeyard)	Automotive (Motorvane)	Real estate (Hearthly)	Why it differs
Discovery: search relevance, ranking, alerts	**Core**	**Core**	**Core**	Every listing marketplace competes on matching; the signals differ
Supply quality: insertion, normalization, entity resolution	Core (C2C insertion funnel)	Supporting (feeds and VIN decoding)	**Core** (duplicates across agencies)	Where supply quality is the visible weakness, it is core

Subdomain	General classifieds (Trade-yard)	Automotive (Motor-vane)	Real estate (Hearthly)	Why it differs
Trust and safety	**Core** (C2C scams)	Supporting (dealer-heavy supply)	Core for rentals (fake-rental fraud)	Private supply and rentals attract fraud
Lead management and pro tooling	Supporting	**Core** (dealers pay for leads)	**Core** (agencies pay for leads)	In verticals the professional is the paying customer
Valuation and price estimation	Supporting (price hints)	Core (market-value model)	**Core** (Valuo is a product)	A valuation that is sold is core
Monetization: packages and pricing	Core (VAS design)	Core (package staircase)	Core (agency subscriptions)	Package design is commercial strategy
Monetization: entitlement ledger, billing	Generic	Generic	Generic	Must be correct; nobody chooses a marketplace for its invoicing
Transactions and payments	Supporting (Safe Deal)	Generic or absent	Absent (deals close at the notary)	The marketplace model decides
Taxonomy and attributes	Supporting	Supporting	Supporting	Necessary and specific, rarely visible as a differentiator

Subdomain	General classifieds (Tradeyard)	Automotive (Motorvane)	Real estate (Hearthly)	Why it differs
Identity, notifications, media storage	Generic	Generic	Generic	Commodity capabilities with marketplace-specific configuration

The classification has two practical consequences.

First, **core subdomains are where brand-specific contexts belong**, even inside a shared platform. Motorvane's ranking and Hearthly's valuation model are core. A group platform may provide the engine underneath them, but the contexts that hold the relevance model and the valuation model belong to the brand. This is the bounded-context version of the capability chapter's rule: share the machinery, keep the content.

Second, **generic subdomains are candidates for buying or for one group context**. Veldmark ID, the group identity approved in June 2027, is a generic subdomain implemented once, as an open host service with brand-specific profiles. Nobody at Veldmark argues that their brand's login is a competitive advantage.

Classification gets political when one group platform serves brands that classify the same subdomain differently. At Tradeyard, trust and safety is core. At Motorvane, whose supply is 91% from about 21,400 dealers verified at contract signature, it is supporting. A group trust-and-safety platform must serve both, which is why Trust and Safety as a Platform makes the pipeline shared and the policies configurable per brand and country.

One Listing, Many Models

The decomposition starts from the contexts in which the listing appears and what each one needs it to be. The table follows the Lyon apartment through ten of them. In May 2027 three agencies list the same 68 m² T3 flat in Lyon 7e. **Agence Rhône Habitat**, a 14-branch network, publishes through Keybridge at €389,000; the other two agencies publish through two different real-estate software vendors at €389,000 and €395,000.

Context	What "listing" means here	Identity	Owns	The Lyon flat in this context
Feed ingestion	A record in a partner's payload, in the partner's vocabulary	Partner ID plus external reference	Feed runs, mapping rules, validation results, source snapshots	Three feed records in two vendor formats plus Keybridge's, each with its own external ID
Listing catalog (system of record)	The seller's offer, as published, with invariants and a lifecycle	Group listing ID	State, version, public fields, references to everything else	Three listings, three owners, three prices
Entity resolution	A member of a cluster that describes one real thing	Canonical property ID	Clusters, links, merge and split lineage	One canonical property with three member listings
Trust and safety	A subject under evaluation, linked to actors, devices and reports	Case ID, subject references	Policies, scores, decisions, cases, appeals	Three subjects, all approved; agencies verified at onboarding
Search	A denormalized, query-optimized document	Document ID derived from listing ID	Index mappings, analyzers, relevance configuration	One collapsed result: "also listed by two other agencies"
Saved searches and alerts	A candidate that may match stored intent	Listing ID plus version	Saved searches, match state, sent-alert log	One alert per saved search, not three

Context	What "listing" means here	Identity	Owns	The Lyon flat in this context
Monetization	A target of paid products and package slots	Entitlement and order IDs referring to the listing	Catalog of products, entitlements, consumption ledger, delivery evidence	Each agency consumes a slot in its own package
Messaging and leads	The source and subject of a conversation or lead	Conversation ID, lead ID	Conversations, leads, routing, response clocks	A buyer's enquiry is routed to the agency whose listing the buyer contacted
Valuation	An observation of an asking price for a property	Property ID plus observation date	Models, estimates, comparables	One property observation, not three; Valuo estimate €372,000–€401,000
Analytics	A fact row with dimensions, and an event subject	Listing ID and property ID	Governed metrics, tracking plan	Three listings, one property, contacts counted once per buyer

Each context owns something no other context should write, and the "Owns" column is where the boundaries become visible. Ingestion owns the partner's raw truth and the mapping. The catalog owns the listing's state. Entity resolution owns cluster identity. Trust owns decisions and evidence. Search owns the index shape. Monetization owns what was sold and delivered. Messaging owns conversations. When two contexts write the same field, one of them is wrong about its boundary.

The same real-world change means different things in different contexts, as the Lyon flat shows. If Rhône Habitat cut its price to €379,000, ingestion would see a changed field in a feed record. The catalog would see a `PriceChanged` transition with

a new version. Search would see a document update, alerts a possible price-drop alert, valuation a new asking-price observation and analytics a row in the price-history fact table. One published event serves all of them, but each context interprets it in its own model.

The generic domain models this book draws on reach the same structure by a different route. Both put the listing in one system-of-record capability with one owner, many readers and a separate search projection. The real-estate model also separates the listing from the property, through a property reference that "connects multiple classifieds, valuations, and location signals to the same underlying property without treating every listing as a separate property". **The listing is not the thing; it is a publication about the thing**. The Listing Aggregate: Identity, Lifecycle and Invariants turns that into an aggregate design, and One Property, Three Listings: Duplicates and Entity Resolution owns the canonical entity.

Architecture: The Marketplace Context Map

The context map below is Veldmark's **Common Ground context map**, as approved by the architecture advice forum in May 2027. It shows the target for 2028, with Yardstick still in place behind a translator.

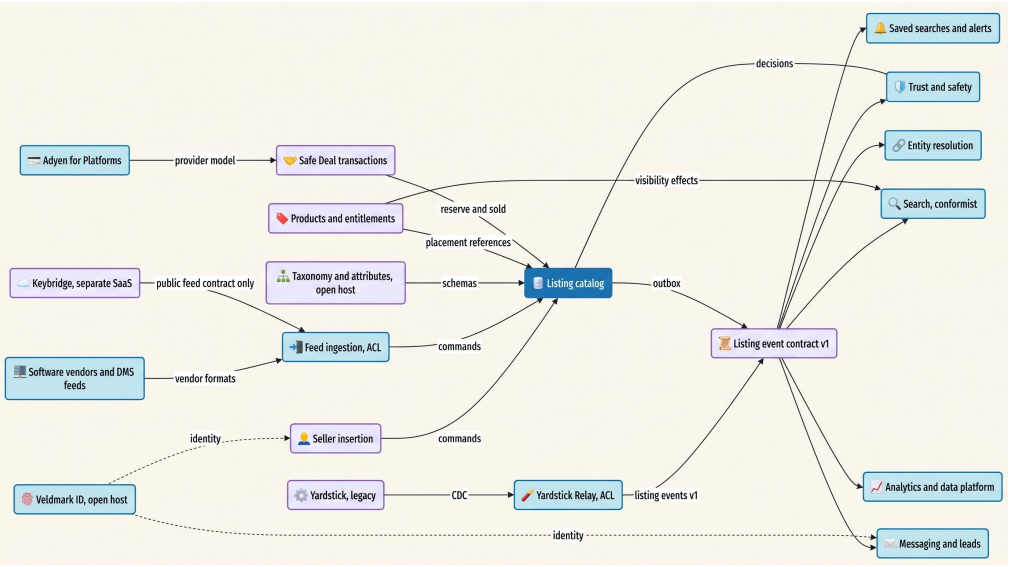


Figure 2: *The Common Ground context map (target 2028). The listing catalog owns listing truth and publishes it as Listing event contract v1. Partner feeds and the Yardstick monolith reach the contract through anti-corruption layers. Search conforms to the contract. Trust and safety, monetization and transactions change listings only through*

commands to the catalog. Keybridge sends listings in through the same public feed contract as any vendor and consumes nothing.

Each relationship on the map is a decision, and each has a reason.

Feed ingestion is an anti-corruption layer. Hearthly receives 84% of its listings from about 160 real-estate software vendors, and Motorvane receives most of its dealer stock from DMS exports. Each source has its own field names, enumerations, photo conventions and ideas of what "deleted" means. If the catalog adopted any vendor's model, every vendor's quirk would leak into search, trust and analytics. Ingestion translates all of them into catalog commands, keeps the source payloads for reconciliation and reports errors back to partners. Many Doors In: Forms, APIs and Bulk Feeds owns the design. The relationship is deliberately one-way: the catalog does not know which vendor a listing came from, except through a provenance reference.

The listing catalog is upstream of almost everything, and it publishes a language rather than exposing a model. Consumers do not read the catalog's tables. They read Listing event contract v1: thirteen event types with the listing's public state, version and cause (see Every Change Is an Event: Event-Driven Listing Architecture). That makes the catalog an **open host service with a published language**. The catalog can refactor its storage without telling anyone, as long as the contract holds.

Search is a conformist to the published language. The search context adopts the contract's vocabulary for listing state and public fields, and adds its own model on top: analyzers, facets, boosts. It never writes listing state back. If the catalog says a listing is sold, search removes it; search does not decide. That keeps the index a read model and prevents Search as System of Record. When Vinted moved its search engine to Vespa, it put a "search contract" in a Go gateway in front of the engine so that product applications did not change (Vinted Engineering, 2024). That is the same idea from the other side: a stable language at the boundary lets the model behind it change.

Monetization is a customer–supplier partner of the catalog, and the owner of entitlements. Monetization needs the catalog to accept placement references and to publish visibility changes. The catalog needs monetization to tell it what a listing is entitled to. The rule that prevents Billing by Guesswork is that **entitlements live in the monetization context**: what was sold, what was granted, what was consumed and what was delivered. The listing carries only references to active placements. Ranking reads visibility effects that monetization publishes. Packages, Bumps and Quotas: Products, Pricing and Entitlements owns the design.

Trust and safety acts on listings only by command. The trust context keeps its own model of listings as subjects, with actors, devices, reports and evidence. When it decides to remove a listing, it sends an enforcement command to the catalog, which emits `ListingRemoved` with the cause. The decision record stays in trust and safety, where it survives the listing's deletion for appeals and regulatory reporting. The two contexts work as a **partnership** during the 2027 pipeline redesign, because the publication

pipeline's 90-second target cannot be met unless both change together (From Submitted to Searchable: The Publication Pipeline).

Transactions conform to the payment provider. Safe Deal runs on Adyen for Platforms, which is the licensed party. Veldmark's transaction context adopts the provider's model of payments, payouts and accounts rather than inventing its own, and translates only at the edge where an order meets a listing: a reservation, a sale. Conforming is the right choice when the upstream is a regulated provider whose model you cannot change. The cost is portability, which Beyond the Contact: Payments, Shipping and Buyer Protection discusses together with the sibling book's Adyen: One Platform from Checkout to Settlement chapter.

Identity and taxonomy are open host services. Both serve every context with a stable protocol. Taxonomy publishes versioned category trees and attribute schemas, and listings record the taxonomy version they were classified under. Identity provides one login and brand-specific profiles. Neither owns listings, and neither should know about them.

Keybridge goes separate ways. Keybridge publishes agency listings to Hearthly and to Hearthly's competitors. In March 2027 Veldmark committed that Hearthly would not use Keybridge agency data for ranking or sales targeting. On the context map, that commitment is a relationship: Keybridge reaches Hearthly only through the same public feed contract as every other vendor, and it has no subscription to Hearthly's listing events. A context map that showed Keybridge as a partner of the catalog would have contradicted a commercial promise.

Yardstick is a big ball of mud, contained. Tradeyard's monolith still holds the `ads` table and its 214 columns until the insertion flow moves out in Q2 2028. It is not decomposed in place. **Yardstick Relay**, a translator service, reads Yardstick's tables through change data capture (CDC) and emits contract events, so consumers see the same language from Tradeyard as from Motorvane. The rule Veldmark adopted is that CDC is a source, never a contract: only the translator reads Yardstick's tables. When the new insertion service ships, it publishes the same contract through its own outbox, and consumers do not notice. Changing the Engine in Flight: Modernizing a Live Marketplace owns the strangler plan.

Aggregates and transactional boundaries

Bounded contexts set the boundaries of models. Aggregates set the boundaries of transactions inside them. Most damage in listing systems comes from transactions that try to span what should be two aggregates, often in two contexts.

Aggregate	Context	Consistency rule inside	What it must not include
Listing	Listing catalog	State transitions, versions, invariants on price, category and required attributes	Entitlement balances, moderation evidence, conversations
Entitlement account	Monetization	Grants, consumption and expiry balance, idempotent per operation	Listing state; it refers to listings by ID
Case	Trust and safety	Decision, evidence, policy version, appeal state	The listing's current content; it keeps a snapshot
Conversation	Messaging and leads	Message order, participants, delivery and safety state	Listing state beyond a reference and a snapshot of title and price
Canonical property	Entity resolution	Cluster membership, lineage	Listing prices and states, which it reads from events
Saved search	Alerts	Query, channels, consent, throttling state	Matched listings, which are a separate log

Consider a bump. A seller buys a "Top listing" for seven days. A naive implementation decrements a counter, sets a flag on the listing and reindexes, all in one database transaction across three tables in two contexts. It works until Yardstick's MySQL is replaced, or until the entitlement service moves to the group platform. The aggregate design splits it.

1. Monetization records the order and grants the entitlement in the entitlement account aggregate, idempotently keyed by the order ID.
2. Monetization sends `ApplyPlacement` to the catalog with the entitlement reference, and publishes `VisibilityChanged` for search.
3. The catalog records the placement reference on the listing and emits a new version.

4. If step 2 or 3 fails, monetization retries with the same key. If the listing has been removed in the meantime, the catalog rejects the command and monetization refunds or re-credits the entitlement.

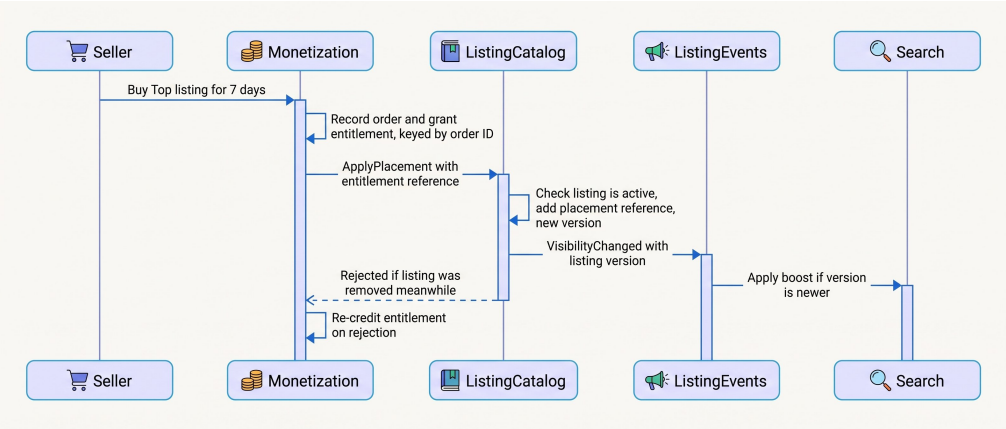


Figure 3: *A bump crosses two aggregates in two contexts without a shared transaction. Each step is atomic inside its own aggregate, the order ID makes retries safe, and a rejection from the catalog is repaired by monetization, which owns the entitlement.*

Each step is consistent within one aggregate. Between them, the system is eventually consistent, and the boundaries say who repairs what. This is the trade an architect makes on purpose: **atomic consistency within an aggregate, explicit eventual consistency between aggregates, and idempotency everywhere**.

The listing event as published language

The listing event contract is the context map's most important artifact: the language through which the catalog tells every other context what happened to a listing. Its design rules follow from the map.

- **Owned by the catalog, designed for consumers.** The catalog team owns the contract, but its first version was designed with the four first consumers: search, alerts, trust and safety, and the warehouse. A published language that only the producer understands is a leaked internal model.
- **Public state, not the catalog's internals.** Events carry the listing's public state, version and cause, and leave out personal contact data and internal performance data. Consumers that need contact details fetch them through an access-controlled interface when they need them.
- **Business transitions, not row changes.** `ListingSold` and `PriceChanged` are domain events. "Row 84711032 updated" is not. A translator such as Yardstick Relay exists to turn the second into the first.

- **Versioned and keyed per listing.** Each event carries a strictly increasing listing version and is keyed by listing ID, so every consumer can apply events in order and ignore stale ones.
- **Group identifiers as a shared kernel.** The listing ID format (brand, country and local ID, for example `vm:tradeyard:nl:84711032`) and the tenant codes are the one piece of model all contexts share. They are small, stable and co-owned, which is what a shared kernel should be.

The Rotterdam city bike shows why the contract matters for boundaries, not only for plumbing. On Monday 8 March 2027 Femke marked her €240 bike sold at 18:10, and search removed it at 18:11. Yet the alert digest featured it the next morning, because the digest builder assembled its candidates from its own copy of the data and did not consume removals. Under the context map, the alerts context subscribes to `ListingSold` and purges its candidate set, and every consumer declares in its contract how fast it removes a listing that has left the active state. Ghost Inventory is a boundary failure before it is a latency failure: **one context learned about the sale and another did not, because they did not share a language**.

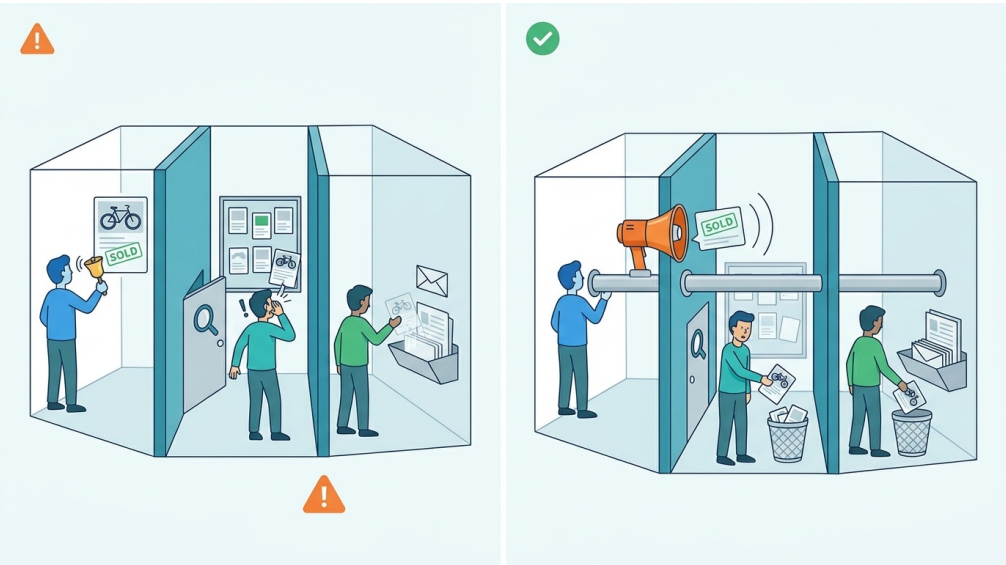


Figure 4: *Ghost Inventory is a boundary failure before it is a latency failure: the alert digest featured the sold bike because it never consumed the sale, and a shared ListingSold event closes that gap.*

In Practice: What the Public Record Shows

Few marketplaces publish their context maps. Several have published enough to show the shape of the decisions.

Vinted: domains before services. In a January 2026 engineering post, Vinted described leaving its Rails monolith. "Our first step wasn't to break the monolith apart. It was to understand it." Domain-driven design gave the team "the vocabulary" to find natural boundaries. "It took us at least two years to understand the domains, as we identified almost 300 of them," and DDD "guided how we restructured teams". Synchronous calls between parts were replaced by business events, so "services could publish state changes and let other domains react whenever they were ready" (Vinted Engineering, 2026). Two lessons transfer: the decomposition came first and was slow, and the integration that made it work was events between domains, not shared tables.

AutoScout24: services aligned to bounded contexts. In a 2018 post, an engineer described AutoScout24's architecture after its migration from a .NET monolith to JVM microservices on AWS: services each focused on a bounded context, mostly synchronous APIs, some Kafka for asynchronous work, and pages composed from fragments served by the services (AutoScout24 Tech Blog, 2018). The migration itself, "Project Tatsu", started in November 2014 with one team, grew by adding teams every two to three months, and followed a "shared nothing" principle (InfoQ, 2016). The evidence is old and describes the state of that time; the current architecture is not public. It is an early documented case of a listing marketplace using bounded contexts as the unit of service ownership.

Kleinanzeigen: what happens when the database is the contract. In a January 2025 post on the Adevinta tech blog, Kleinanzeigen's search team described a pipeline that uses Debezium CDC to stream MySQL changes into Kafka, joins them with SQL against the source database, enriches them over HTTP, and publishes a denormalized per-user object to a compacted Kafka topic as a re-consumable cache. The author notes that Debezium troubleshooting is harder because problems "often occur at the infrastructure level rather than in the application code" (Adevinta Tech Blog, 2025). The pattern is a pragmatic way to get data out of an existing database with configuration rather than code. Its architectural cost, which is an inference rather than something the post states, is that the source schema becomes a de facto public contract for every downstream consumer. That is why Veldmark puts a translator between CDC and the published language.

Adevinta: an anti-corruption layer as a product. In a February 2023 post, Adevinta described a dispute-management back office built as a reusable component "that can be plugged into any marketplace". It ingests per-marketplace events through Kafka, SQS or REST and normalizes them into a common model; its constraint was "We couldn't hardcode anything… or make any assumptions" (Adevinta Tech Blog, 2023). In context-map terms, a group-level context consumes each marketplace's language through its

own translation layer. The design lets the shared context serve several marketplaces whose models differ. Its dependency is that each marketplace must publish good events. Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform discusses how such shared components fared when the group later decentralized.

The reference models: one owner, many readers. The two generic domain models behind this book, one for general listings and one for real estate, assign the listing to a single system-of-record capability, with a search index as a projection and a change event as the interface. In the general model, the listing is owned by one capability and read by eleven others, and two more query it through search. The models are not descriptions of any company, but their structure is the canonical pattern, and their mistakes are instructive. The general model's data catalog assigns credentials, sessions and the audit log to the cookie-consent capability. That is a reminder that ownership assignments must be reviewed as carefully as boundaries.

Veldmark Worked Example: Drawing Common Ground's Boundaries

Ingrid's team draws the map in four working sessions in April and May 2027, using the capability map from March as input.

Step 1: candidate contexts from capabilities. The 54 level-2 capabilities collapse into 19 candidate contexts. Some capabilities share a model and become one context, for example *Saved searches and alerts* with the alert parts of *Notifications*. Others split. *Search and filtering* splits into a group search platform context (engine, indexing, capacity) and brand relevance contexts, because the capability heat map gave them different verdicts.

Step 2: find the language breaks. For each candidate, the team writes down what "listing", "seller" and "price" mean there. Where two candidates use the same words with the same meaning and the same invariants, they merge. Where one word changes meaning, there is a boundary. The exercise produced one surprise. At Motorvane, "price" means the dealer's asking price. At Hearthly France it means the price including agency fees, which is why the three Lyon listings show €389,000, €389,000 and €395,000 for one flat. Valuation needs the net price. The catalog keeps the advertised price and the fee basis as separate fields, and valuation owns the translation.

Step 3: assign truth. For every piece of data, one context is named as owner. The forum records 31 ownership decisions. Five took more than one session:

Data	Candidates	Decision	Why
Listing state (active, sold, removed)	Catalog, search, ingestion	**Catalog**	Search is a read model; ingestion reports what partners say, the catalog decides
Paid visibility	Catalog, monetization, ranking	**Monetization**, with references on the listing	Ends the `top_until` columns in Yardstick's `ads` table
Moderation decision	Catalog, trust and safety	**Trust and safety**, with enforcement state on the listing	Decisions must survive deletion, for appeals and statements of reasons
Canonical property	Catalog, entity resolution, valuation	**Entity resolution**	The property outlives every listing and is read by valuation
Sold price for off-platform sales	Catalog, analytics	**Catalog**, as an optional outcome on `ListingSold`	It is a seller-reported fact about the listing, used by pricing

Step 4: draw relationships and name the translation points. The map in Figure 2 is the result. Four anti-corruption layers are named and owned: feed ingestion (one per brand, sharing a group library), Yardstick Relay, the Hearthly bridge from Google Pub/Sub to the group event backbone on Confluent Cloud, and the transaction context's edge towards Adyen.

Step 5: assign teams. Each context gets exactly one owning team. Thirteen of the 19 contexts were already owned by one team. Four had split ownership, all inside Yardstick. Two had no owner. The two orphans were *removal propagation to alerts and digests*, which went to the Alerts team as part of the alerts context, and *taxonomy*, which went to the new group taxonomy and attribute team whose service went live in October 2027.

The Essen Golf shows the map at work. When the dealer sold the Golf on 29 April 2027, the feed's removal arrived 26 hours later. Under the map, nothing downstream decides that the Golf is gone. Ingestion sees the removal in the next feed run and sends `Withdraw` to the catalog, which emits `ListingRemoved` with the cause "feed

removal". Search, alerts and leads react to that event within their declared deadlines. The 26-hour delay is still there, because it sits in the dealer's DMS. But the map makes it **one problem in one context**: ingestion's freshness contract with the dealer. Before the map, it was a different problem in each of six places.

Trade-offs

Three decisions recur in every decomposition. None has a universal answer.

Granularity of contexts.

Option	Strengths	Weaknesses	When it is right
Few large contexts (5–8 per marketplace)	Fewer integrations; easier to change models inside a context	Contexts grow internal language breaks; teams step on each other	Small marketplaces; early in a migration
Contexts per level-2 capability (15–25)	Aligns with the capability map and team sizes	Needs a mature event platform and contract discipline	A group-scale platform; Veldmark's choice
Many fine-grained domains (100+)	Precise ownership; small blast radius	High coordination and integration cost; contexts become services in disguise	Very large organizations that have done the domain work first, as Vinted's 300 domains suggest

Integration style between contexts.

Style	Strengths	Weaknesses	When it is right
Shared database	Simple at first; strong consistency	Couples every context to one schema; no boundary at all	Never across contexts; acceptable inside one

Style	Strengths	Weaknesses	When it is right
Synchronous APIs	Simple to reason about; immediate answers	Temporal coupling; cascading failures at peak	Queries and commands that need an answer now: identity, entitlement checks at insertion
Published events	Decoupled in time; many consumers; replayable	Eventual consistency; contract governance needed	Listing changes, lifecycle, anything with many consumers
CDC from tables	Gets data out without code changes	Schema becomes the contract; no business intent	A transitional source behind a translator

One group model or one model per context.

Option	Strengths	Weaknesses	When it is right
Enterprise canonical listing model used everywhere	One vocabulary; easy reporting	Becomes the god model; every change needs every team	Only for the small shared kernel: IDs, tenant codes
Per-context models plus a published language	Each context evolves alone; translation is explicit	Requires a governed contract and translation work	The default for a marketplace group
Per-brand models with no shared language	Full local freedom	No shared platform is possible; reporting is reconciliation	Separately owned businesses, or one that may be sold

Vertical Differences

The context map's skeleton is shared across verticals. What changes is which contexts exist, which are core and where the entity sits.

Dimension	General classifieds	Real estate	Automotive	Jobs and recom-merce
Entity behind the listing	Usually none; a product identity only for regulated or catalog goods	**Canonical property**, durable across years and listings	**Vehicle**, identified by VIN	Jobs: vacancy, weakly identified. Recom-merce: catalog product
Extra contexts	Shipping and handoff; protected payment	Mandates and agent authority; viewings; disclosures; valuation	Vehicle data and history; finance and insurance referral; test drives	Applications and CVs; orders, shipping and disputes
Dominant supply language	Insertion funnel commands	Vendor feeds, translated by an ACL	DMS feeds, translated by an ACL	Employer ATS feeds; seller apps
Where the hardest boundary is	Messaging versus trust (scams arrive in conversa-tions)	Listing versus property (duplicates, history)	Listing versus vehicle (relisting, history, valuation)	Jobs: application data with sensitive personal data. Recom-merce: order versus listing
Transactions context	Supporting, where shipping applies	Absent; milestones only	Emerging (reserva-tions, deposits)	Recommerce core; jobs absent

The real-estate column explains why Hearthly's map has two contexts that Tradeyard's does not: entity resolution as a core context, and a mandates context that records which

agency is authorized to market which property. The general-listings resilience analysis points to the same separation for goods in one case: when product-safety rules make the marketplace responsible for recalled items, a product identity must exist separately from the listing so that one safety notice can suspend every affected listing. Same Platform, Different Physics: How Listing Verticals Differ develops the differences further.

Failure Modes

The god listing table. One table holds the seller's content, the moderation state, paid flags, search boosts, contact data and counters, and every module reads and writes it. Yardstick's 214-column `ads` table is the example. Every change needs a cross-team review, and no team can say which columns it owns. The way out is not to split the table first. It is to name the owner of each group of columns, put a translator in front, and move consumers to the published language one at a time.

The shared database between contexts. Two services in different contexts read and write the same schema "for now". The boundary on the diagram does not exist in the system. The symptom is a migration that needs three teams to deploy together.

The canonical model for everything. A group "enterprise data model" defines one listing schema that every context must use. It becomes the god table at group scale, and each brand works around it. Keep the shared kernel small: identifiers, tenant codes and the published event, nothing more.

Contexts drawn by technical layer. "The API context", "the database context" and "the front-end context" are layers, not contexts. They guarantee that every business change crosses every boundary.

The search index as owner. A consumer, usually search, becomes the place where listing state is corrected, because it is the copy people look at. Soon the catalog cannot rebuild the index, and the index is the only complete copy. This is Search as System of Record, and the cure is a named owner of listing truth with a rebuild path that is tested.

Entitlement checks everywhere. Insertion, ranking, rendering and messaging each check package names and flags in their own code. Nobody can say what was sold versus delivered. The cure is a monetization context that owns entitlements and publishes their effects.

Conformist by accident. A context adopts a vendor's or a partner's model without deciding to, because the first integration was the quickest. Five years later the vendor's field names are in the search index. Make conformity a recorded decision, as Veldmark did for the payment provider, and put an anti-corruption layer everywhere else.

Split ownership. One context, two teams. Each team assumes the other handles the edge cases, and the edge cases are where Ghost Inventory lives. The two orphaned

responsibilities that Veldmark found, removal propagation to digests and taxonomy, were both in this category.

Too many contexts too early. Fine-grained contexts before the domain is understood produce distributed monoliths: many services, one change at a time. Vinted's two years of domain work came before the services.

The Chief Architect's View

At group scale, the context map is the architecture's constitution. These are the decisions I would keep at group level, and the ones I would leave to brands.

Decided at group level, by decision record:

1. **The owner of listing truth per brand**, and the rule that every other context's copy is a read model that can be rebuilt.
2. **The published language**: the listing event contract, its compatibility rules and its governance (see Every Change Is an Event: Event-Driven Listing Architecture).
3. **The shared kernel**: identifier formats, tenant codes, the taxonomy reference format. Keep it small, and change it rarely and with every owner present.
4. **The relationships that carry commercial or legal promises**, such as Keybridge's separate-ways relationship. These belong in the same register as the promises themselves.
5. **The rule for anti-corruption layers**: every external or legacy model reaches the published language through a named, owned translator.

Decided by brands, within those rules: the internal models of their core contexts (relevance, valuation, package design), the services inside each context, and the decomposition of their own legacy systems.

Who decides: the architecture advice forum reviews changes to the context map, the shared kernel and the published language; the owning team decides inside its context. Ingrid's rule of thumb is that a change which alters what another team must do needs a decision record, and a change which does not, does not.

What to watch:

- **Contexts with more than one owning team**, and responsibilities with none. Both should be zero, and both drift.
- **Writes into a context by a non-owner**. Any service that writes listing state without going through the catalog's commands is a boundary breach.
- **The number of consumers reading a legacy table directly** instead of the published language. At Veldmark this is the Yardstick migration metric that matters most, because every direct reader is a hidden dependency for the strangler plan.

- **Translation layers that start to make business decisions.** An ACL that begins to decide what a listing's state is has become a second catalog.
- **Separability.** When a business might be sold or carved out, the contexts it depends on, and the data they hold for it, must be separable. Adevinta's 2025 report of "parallel running of central and local services during a transitional period" as it decentralized shows the cost when that is not designed in (see Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform).

One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture applies the same map to tenancy across countries and brands, and Who Owns What: Organizing Marketplace Engineering turns it into team design.

Questions to Ask

1. For each brand, which context owns listing truth, and can every other copy, including the search index, be rebuilt from it?
2. Where does the meaning of "listing", "seller" and "price" change in our system, and is there a boundary at each change?
3. Which subdomains are core for each brand, and are those contexts owned by the brand even when the engine underneath is shared?
4. What is our published language between contexts, who owns it, and which consumers still read tables instead?
5. Which relationships are anti-corruption layers, which are conformist, and was each choice made on purpose?
6. Who owns entitlements, and does any code outside that context check package names or paid flags?
7. Does any context write listing state without going through the owner's commands?
8. Does any context have two owning teams, or any responsibility, such as removal propagation, have none?
9. Which relationships on the map carry a commercial or legal promise, and are they enforced by access control?
10. If a brand had to be carved out next year, which shared contexts would it have to leave, and how much of its data would they hold?

From One Map to Four Verticals

The context map says where the models are and how they talk. It deliberately treats the four brands alike, because most of the map is shared. The next chapter, Same Platform, Different Physics: How Listing Verticals Differ, asks the opposite question: what genuinely differs between general classifieds, real estate, automotive and jobs, and therefore which contexts must never be forced into one shape.

Sources and Evidence

Company engineering and investor sources

- "Building a global event-driven platform, part 1", Vinted Engineering, 2026 — DDD used to identify almost 300 domains over at least two years; teams restructured; business events between domains.
- "Goodbye Elasticsearch, hello Vespa", Vinted Engineering, 2024 — a search contract in a gateway that let the engine change without changing product applications.
- An engineer's account of a first year at AutoScout24, AutoScout24 Tech Blog, 2018 — microservices each focused on a bounded context; describes the state at that time.
- News report on AutoScout24's move to microservices, InfoQ, 2016 — Project Tatsu, its incremental team growth and "shared nothing" principle.
- "Make data migration easy with Debezium and Apache Kafka", Adevinta Tech Blog (Kleinanzeigen), 2025 — CDC from MySQL into a compacted topic as a denormalized cache.
- "From hello world to a dispute management back office", Adevinta Tech Blog, 2023 — a cross-marketplace component that normalizes each marketplace's events into a common model.

Standards, regulation and documentation

- Context Mapping, DDD Crew, n.d. — names and definitions of the context-map relationship patterns.
- Debezium outbox event router, Debezium documentation, n.d. — keying events by aggregate ID to preserve per-aggregate order.

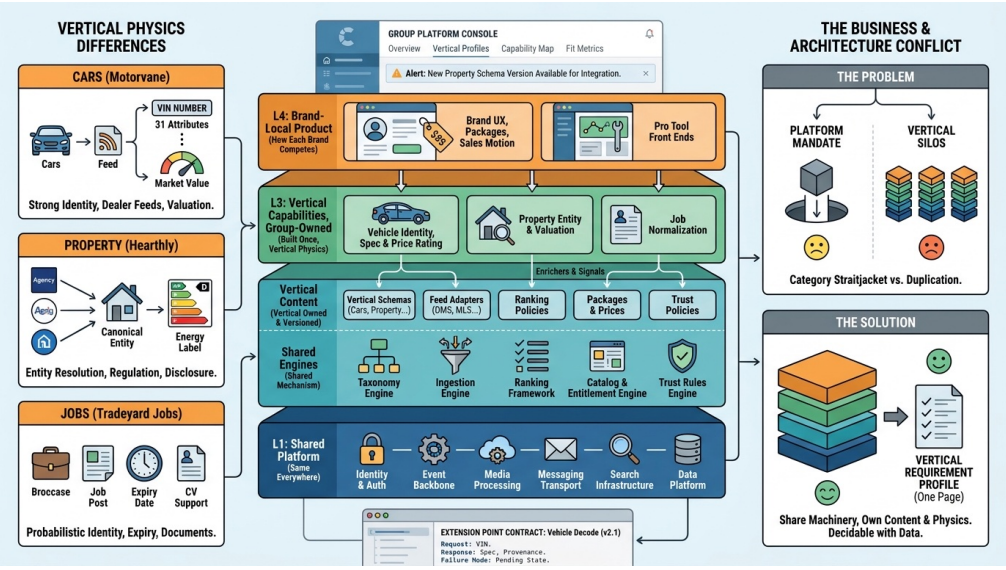
Research and practitioner sources

- Eric Evans, *Domain-Driven Design: Tackling Complexity in the Heart of Software*, Addison-Wesley, 2003 — bounded context, context map, core domain, generic subdomains.
- Vaughn Vernon, *Implementing Domain-Driven Design*, Addison-Wesley, 2013 — supporting subdomains and practical context mapping.
- "BoundedContext", Martin Fowler, 2014 — the pattern, language as the dominant boundary factor, the "meter" example.
- "DDD_Aggregate", Martin Fowler, 2013 — aggregates as units of consistency; transactions do not cross them.

- "How Do Committees Invent?", Melvin E. Conway, *Datamation*, 1968 — the link between communication structures and system design.
- Productscape reference models for a general-listings marketplace and a real-estate marketplace, author's working models, 2026 — single listing owner with search as a projection; property reference separate from listings; generic models, not descriptions of any company.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

6. Same Platform, Different Physics: How Listing Verticals Differ



IN THIS CHAPTER, YOU WILL: Compare general classifieds, real estate, automotive, jobs and recommerce on the dimensions that change architecture, and learn to write a vertical requirement profile for each. You will then use a four-layer share test to decide, capability by capability, what a multi-vertical group should build once and what each vertical must own.

WHY THE BUSINESS CARES: A group that runs several verticals is paid to share. The synergy case assumes one listing platform, one search platform and one trust stack. When sharing reaches the features that win a vertical locally, those features get worse, professional customers notice, and the savings arrive as lost revenue.

WHY ARCHITECTS SHOULD CARE: "Is this shared or vertical?" is the question behind almost every Common Ground-style program. It is usually answered by organizational politics or by the most confident engineer in the room. A written profile per vertical and an explicit share test turn it into a decision that can be reviewed, revisited and defended.

KEY POINTS:

- **Verticals differ in physics, not just in categories.** Who supplies the inventory, whether the item has an identity outside the listing, how long a listing lives, how many attributes it carries, what a lead is worth and where the money moves are different in each vertical. Each of these differences changes ingestion, data models, search, trust and monetization.
- **Share the machinery, own the content, and give the vertical capabilities a single owner.** Identity, events, media processing, messaging transport and search infrastructure can serve every vertical. Taxonomies, schemas, ranking policies and packages are vertical content running on shared engines. Vehicle identity, property entities and job normalization are vertical capabilities that the group should build once, owned by the vertical that knows them best.
- **The vertical requirement profile is the artifact that makes this decidable.** One page per vertical, filled in with numbers, lets a group see which differences are real and which are habit. The profile must be revisited when a vertical changes its business model, because a transactional pivot or a new lead model changes the profile more than any technology does.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Category Straitjacket** — treating real estate, cars and jobs as subtrees of a horizontal category tree forces every vertical requirement through one generic model that changes slowly.
- **The Forced Fit** — a group platform designed around one vertical's physics is imposed on the others, and their teams route around it.
- **The Attribute Junkyard** — a shared attribute store without vertical dictionaries accumulates thousands of keys whose meaning differs per vertical, so filters and feeds disagree.

Part I has so far treated the verticals alike. Map What the Marketplace Does Before What Runs It shows that a classifieds capability map generalizes across verticals through renaming and re-weighting, not through new top-level structure, and Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models draws one context map for all four Veldmark brands. This chapter asks where that sameness stops. The question matters because a group's synergy case assumes one listing platform can carry a used sofa, a family car, a three-room apartment and a warehouse job, with a different category tree for each. Many groups have tried, and the public record shows how often they had to undo it.

The difference is in the physics: the forces that act on each capability. A car arrives through a dealer's machine feed with a 17-character identity that decodes into 31 attributes. A sofa arrives through a phone camera with a title and a price. An apartment arrives three times, from three agencies, and must be recognized as one property. A job arrives from an applicant tracking system and expires on a date that a search engine enforces. These forces decide where ingestion load comes from, what the data model must hold, how search ranks, what fraud looks like and how money is made.

This chapter defines those forces, compares the main verticals on them, and turns the comparison into a decision method. It owns two terms. A **vertical requirement profile** is a one-page description of a vertical's physics, written with numbers, that architecture decisions can be checked against. A **shared-versus-specific decision** is the classification of a capability into one of four layers, from fully shared platform to brand-local product, with stated reasons. The method is applied to Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), whose three listing brands span four verticals. Part II starts from the result: The Listing Aggregate: Identity, Lifecycle and Invariants and the chapters after it design a listing model with room for the item identities, external systems of record and vertical schemas this chapter finds.

Setting the Scene at Veldmark: One Platform for Four Verticals?

It is mid-March 2027. The board approved Common Ground in January: **€140 million** over four years to give Veldmark's brands shared platform capabilities under the principle "share platforms, not products". Ingrid Halvorsen, the Group Chief Architect, joined in February. She has called a two-day scoping workshop in Rotterdam because the program's capability list (Veldmark ID, trust and safety, media, search platform, messaging, product catalog and entitlements, event backbone, data platform, taxonomy and attribute service) says nothing about which parts of each capability are shared.

Three positions arrive before the workshop starts.

- **Felix Brandt**, **Motorvane's CTO**, argues that Motorvane needs almost nothing from a group listing platform. 91% of its 1.45 million listings come from 21,400 dealers, 88% of those through dealer management system (DMS) feeds, and every car is enriched from vehicle data the horizontal brands do not have. "A shared listing service built for Tradeyard's sofas would be a regression for us."
- **Julien Marchand**, **Hearthly's CTO**, argues the opposite for one capability. Hearthly France's 1.9 million listings describe about 1.12 million distinct properties. He needs a property entity that outlives every listing, and he would like the group to build it once rather than see Tradeyard Portugal's small property section invent a second one.

- **Katarzyna Nowak**, **Tradeyard Poland's general manager**, has the hardest case. Tradeyard Poland runs general goods, a cars section that competes with Motorvane Poland for the same dealers, a property section and a jobs section, all on the same horizontal stack. She wants to know whether her cars section should keep its own vehicle data, borrow Motorvane's or be handed to Motorvane. She also wants to grow jobs, and nobody at Veldmark has built a jobs platform.

Ingrid does not open with the capability list. She asks each brand to bring one real listing and to describe it with the same twelve questions: who supplied it, through which channel, whether the item exists outside the listing, how long the listing lived, how many structured attributes it carried, how many images, how precise its location is, what the first contact was worth to the seller, who paid Veldmark and for what, where the money of the sale moved, which regulations shaped the listing, and what the typical fraud looks like. The answers become the first vertical requirement profiles, and the rest of this chapter is what she does with them.

Concepts: What "Vertical" Means Architecturally

Horizontal and vertical marketplaces

A **horizontal marketplace** lists many kinds of goods and services under one brand and one category tree; a **vertical marketplace** specializes in one kind of item, such as property, vehicles or jobs, with data, tools and monetization built for it. Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy owns the business definitions. This chapter cares about one consequence: a horizontal marketplace usually contains small vertical *sections* (cars, property, jobs), and those sections inherit the horizontal platform's physics even when their own physics differ. Tradeyard Poland's cars section is a vertical living inside a horizontal stack.

The vertical requirement profile

A vertical requirement profile describes a vertical by the forces that act on its systems, not by its features. I use fourteen dimensions, grouped by the part of the architecture they stress.

Group	Dimension	The question it answers	What it stresses
Supply	**Supply mix**	What share of new listings comes from private sellers, professionals and partners?	Identity depth, account hierarchy, monetization base
	Ingestion path	Form, API, bulk feed, or replication from an external system of record?	Ingestion engine, idempotency, reconciliation
	External system of record	Does the authoritative listing live outside the marketplace?	Who wins conflicts; deletes; freshness
Item	**Item identity**	Does the item have an identity outside the listing (VIN, parcel, address, product code)?	Entity resolution, enrichment, history
	Listing lifetime	How long does a listing stay live, and what ends it?	Freshness SLAs, expiry jobs, Ghost Inventory risk
	Attribute density	How many structured attributes does a typical listing carry, and how many are mandatory?	Schema service, validation, facets
	Media profile	How many images, which other media, uploaded or pulled?	Media pipeline cost and design
	Location semantics	What does location mean, and how precise may it be shown?	Geo model, privacy, map search
Demand	**Decision horizon**	Minutes, days or months from first search to decision?	Alert latency vs decision-support depth

Group	Dimension	The question it answers	What it stresses
	Lead value	What is one contact worth to the seller?	Lead objects, attribution, dispute processes
Money	**Monetization unit**	Who pays, per what unit?	Catalog, entitlements, billing
	Transaction locus	Where does the money of the sale move: off-platform, through a partner, through the marketplace?	Payments, ledger, buyer protection
Rules	**Regulatory load**	Which rules turn into mandatory fields, display duties or prohibited content?	Schema, rendering, compliance evidence
	Fraud typology	What does the typical scam look like?	Risk signals, moderation, messaging safety

Table 1: *The fourteen dimensions of a vertical requirement profile, grouped by the part of the architecture they stress.*

The profile's value is comparative: two verticals that differ on two dimensions can share far more than two that differ on ten. It is filled in with numbers where numbers exist ("88% of dealer listings via DMS feeds", "1.7 listings per property") and with a short sentence where they do not.

The shared-versus-specific decision

A **shared-versus-specific decision** places a capability, or a part of one, in one of four layers.

1. **Shared platform**. The capability means the same thing in every vertical and changes at the same rate. One implementation, one owner, configuration only. Examples: authentication, the event backbone, image storage and resizing.
2. **Shared engine, vertical content**. The *mechanism* is the same, but the *content* is vertical and owned by the vertical. One implementation runs many vertical

configurations. Examples: a taxonomy and attribute service running separate vertical schemas; an ingestion engine running vertical adapters; a ranking framework running vertical policies.

3. **Vertical capability, group-owned**. The capability exists only because of one vertical's physics, but more than one brand needs it. Build it once, owned by the vertical team that understands it best, and offer it to the others. Examples: vehicle identity and specification decoding; a property entity and valuation model.
4. **Brand-local product**. The capability is how a brand competes: its user experience, packages and prices, sales motion and much of its professional tooling front end. Shared code here is a liability.

The decision is per capability *part*, not per capability. "Search" is not one decision. The engine and its operations may be layer 1, the index mappings and ranking policies layer 2, a vehicle "price rating" signal layer 3, and the result page layer 4.

How the Verticals Differ

Table 2 compares five verticals on the fourteen dimensions. Each cell names the typical case; real marketplaces vary, and documented examples follow.

Dimension	General classifieds	Real estate	Automotive	Jobs	Recommerce
Supply mix	Mostly private sellers	Mostly agencies; some private owners and landlords	Mostly dealers	Employers, agencies and aggregated career sites	Mostly private sellers at very high volume; a pro channel
Ingestion path	App or web form	Feeds and APIs from agent software; MLS replication in the US	DMS feeds through integration vendors; per-ad APIs	Applicant tracking system (ATS) feeds, career-site crawling, employer self-service	App form with photo-first flows

Dimension	General classifieds	Real estate	Automotive	Jobs	Recommerce
External system of record	None; the ad is the record	Often: the agency CRM, or the MLS	Often: the dealer's DMS	Often: the employer's ATS	None
Item identity	Weak; typed by the seller	Persistent property: address, parcel, building unit	Strong: VIN, registration plate, type-approval codes	Probabilistic: title mapped to an occupation taxonomy	Weak; brand, size and image similarity
Listing lifetime	Days to weeks; ends when sold	Weeks to months; the property outlives it	Weeks; ends at sale, then a feed delete	Fixed validity window; expiry is enforced externally	Days to weeks; ends at checkout
Attribute density	Low and uneven	High: rooms, surface, energy class, charges, fees	High and authoritative: decoded spec, options	Medium: role, contract type, salary range, location, remote	Medium: brand, size, condition
Media	A few phone photos	Many photos pulled by URL, floor plans, tours	Many photos, often from the dealer's photographer	Few images; documents (CVs) on the demand side	Several photos per item
Location semantics	Approximate pickup area; privacy matters	Exact address is the product, if allowed	The dealer's site	Workplace, region or "remote"	Shipping origin; pickup point
Decision horizon	Hours	Weeks to months	Days to weeks	Days	Minutes to days

Dimension	General classifieds	Real estate	Automotive	Jobs	Recommerce
Lead value	Low; first buyer wins	Very high; an instruction or a let is worth thousands	High; a qualified lead is a likely sale	Medium; an application, often billed per click or application	No lead; a checkout
Monetization unit	Paid visibility, insertion fees, display ads	Per-branch subscriptions, per-listing depth, seller leads	Per-dealer packages with slots, prominence, leads, finance referrals	Posting packages or pay per click or started application	Buyer fees per transaction, bumps
Transaction locus	Off-platform, increasingly on-platform	Always off-platform, at a notary or lease signing	Mostly at the dealer; online reservation is growing	Off-platform, in the employer's ATS	On-platform: payment, shipping, disputes
Regulatory load	Platform law, prohibited items, trader status	Energy disclosures, fees, fair housing, licensing	Emissions labeling, consumer guarantees, finance disclosure	Anti-discrimination, pay transparency, CV data protection	Trader status, payments, product safety
Typical fraud	Fake payment links, courier scams	Fake rentals and deposit theft	Vehicle deposit scams, clocked mileage	Fake jobs harvesting personal data or fees	Payment and shipping fraud, counterfeits

Table 2: *Five verticals on the fourteen profile dimensions. The rows that differ most, item identity, external system of record and transaction locus, are the ones that decide*

data models and system boundaries.

Shared platforms most often break on three rows: item identity, external system of record and transaction locus.

Item identity: is the listing the thing, or an event on the thing?

In automotive and real estate, the item has an identity the listing merely references, and that changes the data model. In general classifieds, **the ad is the item**: a used sofa has no identity outside its listing, and when the listing is deleted nothing remains to link to.

- **Vehicles carry machine-readable identity**. The US National Highway Traffic Safety Administration's vPIC service decodes a 17-character VIN into manufacturer data and accepts batches of up to 50 VINs; its databases can be downloaded and run in-house (NHTSA vPIC, version 4.10, accessed October 2026). In Germany, mobile.de's public Seller API carries the type-approval codes `hsn` and `tsn` and a Schwacke make-and-model code on each ad (mobile.de Seller API, n.d., accessed 2026). Autotrader UK's developer portal publishes a taxonomy of vehicle types, makes, models, generations and derivatives, and keys valuations on a derivative identifier plus first-registration date and mileage (Autotrader Connect, n.d., accessed 2026).
- **Properties outlive their listings**. In Great Britain, a Unique Property Reference Number identifies every addressable location, and Ordnance Survey publishes about 40 million of them as open data (OS Open UPRN, n.d.). Zillow's FY2025 annual report describes a "living database" of about 173 million US homes, with user updates to more than 46 million property records (Zillow Group 10-K, filed February 2026). A property model of this kind treats listings as events on properties.
- **Jobs have only probabilistic identity**. A vacancy has no registry. Its identity comes from normalizing a free-text title into an occupation taxonomy such as the EU's ESCO, which in version 1.2.1 has 3,039 occupations and 13,939 skills in 28 languages (European Commission, ESCO, updated December 2025), or the US O*NET-SOC taxonomy of 1,016 occupational titles (O*NET Resource Center, n.d.).
- **Recommerce sits between**. Some used goods can be matched to a product catalog by barcode or model number; most cannot. eBay's Product-Based Shopping Experience tried to make catalog association mandatory from 2018; eBay's developer documentation now states that the mandate for all categories and brands was rolled back, while product and instance attributes stayed separate (eBay PBSE playbook, n.d., accessed 2026).

Where the item has an identity, model identity, specification and listing as three separate things. One physical car produces many listings over its life, on many platforms; one apartment is listed for rent, sold and re-listed by different agencies. Price

history, duplicate detection, valuations and fraud checks all hang off the identity, not the listing. The Listing Aggregate: Identity, Lifecycle and Invariants and One Property, Three Listings: Duplicates and Entity Resolution own the models; the point here is that a horizontal listing service with no place for an external identity cannot serve those verticals well, however good its category tree.

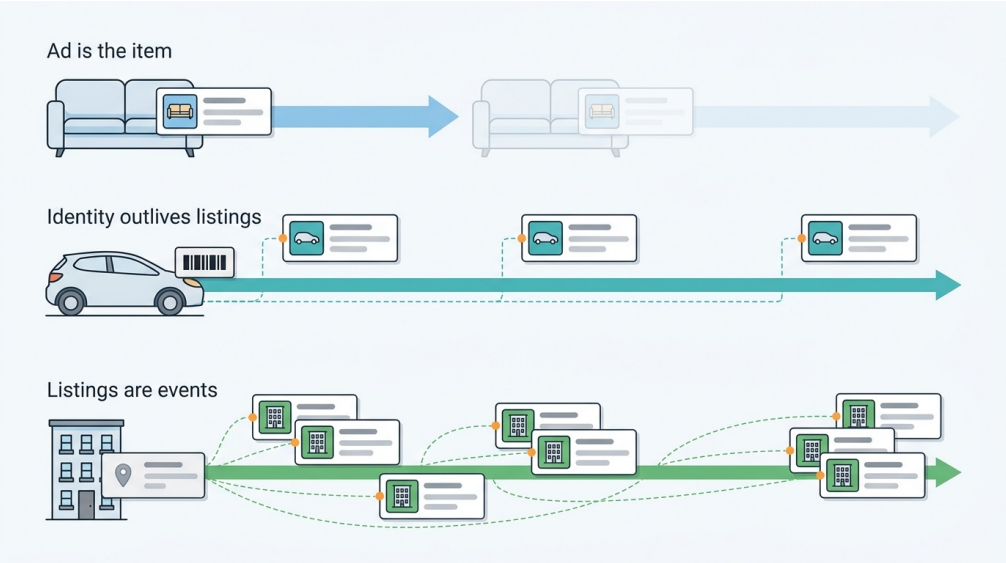


Figure 1: *In general classifieds the ad is the item; for vehicles and property the item has an identity that outlives many listings, so identity, specification and listing must be modeled as separate things.*

External system of record: who wins a conflict?

A private seller on a horizontal marketplace edits the listing on the marketplace, so the marketplace is the system of record. A dealer edits the car in its DMS; an agency edits the property in its CRM; in the US, the listing is created in a multiple listing service (MLS) that the portal receives licensed copies from. **In professional verticals, the marketplace holds a copy of a record it does not own.**

That reverses several defaults. Deletes arrive late or not at all: Rightmove's Real Time Datafeed specification gives integrators a `GetBranchPropertyList` method that returns a snapshot of a branch's properties so they can check that nothing is "still live on Rightmove which should have been removed" (Rightmove RTDF specification, v1.4.1, updated November 2023). RESO, the US real-estate standards body, describes replication as first pulling a full dataset, then requesting recent changes (RESO Web API, n.d.). The ingestion engine has to support push, pull, full snapshots and deltas, and reconcile them; Many Doors In: Forms, APIs and Bulk Feeds owns that design.

A platform designed for form-based insertion, where the marketplace's own write is always the truth, has none of this.

Transaction locus: where does the money move?

Verticals range from no platform money flow to a full order platform. The productscape reference model for real estate records the money of the property sale as "typically settled off-platform at notary or lease signing", with no platform flow at all. Recommerce is the opposite. Vinted describes its marketplace as sitting "in an ecosystem of vertically-integrated shipping and payment infrastructure", with more than 500,000 pick-up and drop-off points in Europe and its own carrier in five countries (Vinted 2025 results, April 2026). Automotive is moving between the two: Autotrader UK reports more than 6,700 retailers on its online reservation product, Deal Builder, and 137,000 full reservations in its year to March 2026 (Autotrader FY2026 results, May 2026).

Where money moves, the listing becomes an order: payment intents, shipping labels, tracking events, disputes and a ledger. Beyond the Contact: Payments, Shipping and Buyer Protection owns that pivot. **Transaction locus is the profile dimension most likely to change during a vertical's life**, and with it the whole right-hand half of the architecture.

In Practice: What the Public Record Shows

Automotive: enriched, not authored

The documented automotive marketplaces describe a listing that is **enriched from authoritative data rather than typed in**, and a supply chain of professional feeds.

- **Supply through integrators**. mobile.de grants Seller API access to *Transfer Service Providers*, the DMS and inventory-tool vendors that upload for dealers, and to self-uploading dealers. It asks clients not to send parallel requests for the same ad and makes create idempotent through a request identifier (mobile.de Seller API, n.d., accessed 2026).
- **Valuation as a listing attribute**. The same API returns a price rating with five labels from "very good" to "high" price and the euro boundaries of each band, or a "no rating" state with a reason; mobile.de recalculates it when images arrive. CarGurus rates eligible used listings in five bands based on its Instant Market Value, recalculated daily from "tens of millions of data points" (CarGurus 10-K, filed February 2026; CarGurus help center, updated May 2026).
- **Regulation as schema**. Germany's passenger-car labeling ordinance (Pkw-EnVKV) requires dealers advertising new cars to show energy consumption, CO2 emissions and CO2 class as soon as engine data appears, and presumes a car is

new if first registered at most eight months ago or driven 1,000 km or less (Pkw-EnVKV, as amended March 2026). "New" is therefore a computed classification, not the dealer's category choice.

Real estate: a property-first, regulation-heavy schema

Real-estate portals publish the clearest evidence that a vertical's schema is **driven by law and by an external software ecosystem**.

- **Mandatory disclosures**. In France, property ads must show the energy and climate classes from the energy performance diagnostic (DPE), the estimated annual energy costs, and a warning for F and G homes; G-rated homes may not be let since January 2025 (Service-Public.fr, DPE, page verified January 2026). The EU's recast Energy Performance of Buildings Directive, (EU) 2024/1275, had to be transposed by 29 May 2026 and moves certificates to a common template by the same date (Directive (EU) 2024/1275, Articles 19 and 35; European Commission, EPBD, n.d.).
- **Schema versions track regulation**. Rightmove's feed specification added shared ownership, ground rent, service charge and council-tax band in April 2022, and utilities, flood risk and accessibility in November 2023 (Rightmove RTDF specification, v1.4.1). The fields arrived on regulators' timelines, not product roadmaps.
- **Rich structure at scale**. Zillow's engineers described nearly 50 filters and more than 200 attributes per property in a 2020 post on guided search (Zillow Tech Hub, via Medium RSS, September 2020); the current state is not public.
- **Country-specific integration standards**. Germany's OpenImmo standard has run as XML exchanged over FTP since 2001; its REST and JSON successor, not compatible with the XML version, was expected as a final release at the end of 2026 (OpenImmo e.V., n.d., accessed October 2026). France, the UK and the US each have their own ecosystem.

The evidence also shows that real estate resists multi-country sharing. AVIV Group's CEO spoke in 2021 of "aligning our tech roadmap across the organization" (Online Marketplaces, August 2021). In December 2025 an interviewer noted that AVIV had rowed "back on its plans for a fully integrated platform for all its European portals", and the new head of Immowelt said AVIV "has strengthened the autonomy of its local businesses" while keeping "group-level functions where they create clear advantages" (Online Marketplaces, December 2025). What the integrated platform contained is not public.

Jobs: normalization, documents and an external expiry clock

Jobs marketplaces combine four properties no other vertical has together.

- **Probabilistic normalization on the critical path**. In a January 2024 post, OLX described extracting job roles "buried within the titles and descriptions" with an LLM service subscribed to ad events, then adding them to the search index; about 2,000 new or updated ads a day produced about 4,000 API requests (OLX tech blog, January 2024).
- **Documents, not photos**. InfoJobs, then part of Adevinta, uses the group's shared media service, which for InfoJobs "stores, transforms and retrieves mostly documents (résumés, CVs, cover letters)", converting uploads to PDF with LibreOffice (Adevinta tech blog, January 2025). A shared media service built for images had to learn document semantics.
- **Expiry enforced by a search engine**. Google's job-posting structured data requires `validThrough` where a job expires and warns that expired jobs that stay marked up "may trigger a manual action" (Google Search Central, updated September 2026). Expiry becomes a distribution SLA.
- **Performance pricing**. Indeed bills employers on "job seeker interactions like clicks or started applications" (Indeed pricing, n.d., accessed 2026), which makes a jobs platform resemble an advertising system with budgets and pacing.

Recommerce: an order platform with a search engine attached

In recommerce, the hard problems are order, payment and logistics state, and search over an enormous low-structure catalog, not taxonomies or feed contracts. Vinted's engineering record shows both. Its search moved to Vespa between May 2023 and April 2024 and serves about a billion active items, with 4.64 seconds at the 99th percentile for an item update to reach the index (Vinted Engineering, September 2024). Its January 2026 post describes moving from synchronous calls to business events with saga-style orchestration across "almost 300" domains (Vinted Engineering, January 2026).

Groups organized by vertical, and back again

Adevinta's annual report for 2023 described an organization by verticals (Mobility, Re-Commerce, Real Estate) and a plan to turn the leboncoin and Kleinanzeigen platforms into "a unified global platform" for its generalist marketplaces (Adevinta Annual Report 2023, April 2024). Its 2025 strategic report describes the move to "a decentralised, standalone operating model" with "the localisation of intellectual property" (Adevinta Strategic Report 2025, May 2026). The record does not say how much of the unified platform shipped. It does show that **the unit of sharing (country, vertical, group) is**

a strategic choice that ownership can reverse, which One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture and After the Deal: Consolidating Platforms After Acquisitions take up.

A smaller documented case shows the share test working at feature level. Kleinanzeigen's photo-based listing assistant, reported in November 2025, covered nearly all categories but excluded vehicles and intangibles such as jobs and services (mobiFlip, November 2025). The interpretation is mine: **a shared capability can declare the verticals it does not serve**, which is better than serving them badly.

Architecture: A Platform With Vertical Seams

The four layers of the share test become an architecture when each layer has a clear owner and the layers meet at defined extension points. Figure 2 shows the shape I recommend for a multi-vertical group.

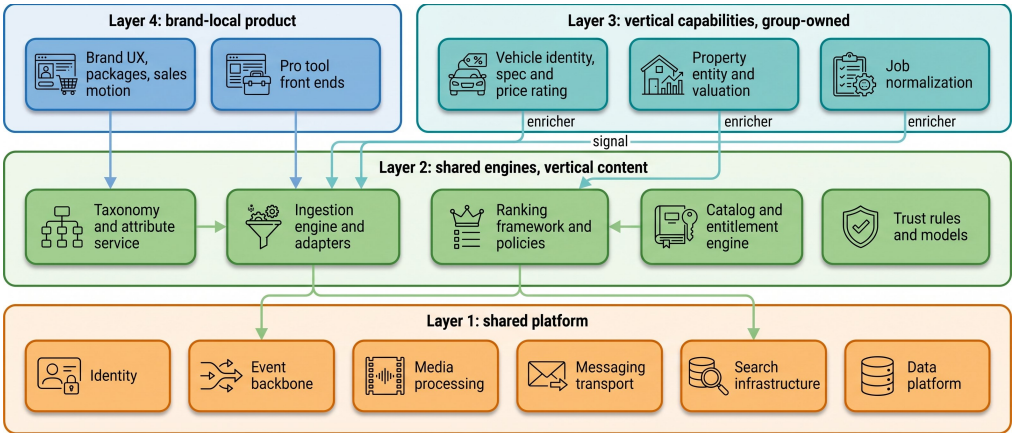


Figure 2: *Four layers with different owners. Layer 1 is the same everywhere. Layer 2 runs vertical content on shared engines. Layer 3 holds capabilities that exist because of one vertical's physics but are built once. Layer 4 is how each brand competes. Vertical capabilities plug into shared engines as enrichers and signals, never as forks of the engine.*

Layer 1: what is the same everywhere

The candidates for a fully shared platform are capabilities whose semantics do not depend on what is listed: authentication and accounts at the identity layer (the brand profile on top is not shared; see Who Is on the Other Side? Identity and Verification), the event backbone and its schema registry (Every Change Is an Event: Event-Driven Listing Architecture), media storage, transformation and delivery (Photos Sell: Media Pipelines

for Listings), messaging transport and storage (From Interest to Contact: Messaging and Lead Management), the operation of search clusters (Search Architecture for Listings: Indices, Freshness and Scale), the experimentation platform and the data platform.

Even here the profiles add requirements the first vertical would not have written. InfoJobs needed document conversion and sensitive storage; real estate needs pull-by-URL image ingestion with change detection: Rightmove's feed fetches up to 999 media items per property and does not re-fetch images whose media update date has not changed (Rightmove RTDF specification, v1.4.1). **A layer-1 capability is shared only if its requirements were collected from every vertical profile**, not from the vertical that built it first.

Layer 2: shared engines with vertical content

Layer 2 is where most of the value of a group platform sits, and where most of the damage is done when the boundary is wrong.

Engine	What is shared	Vertical content	Who owns the content
Taxonomy and attribute service	Versioned trees, attribute dictionary mechanics, validation, migration tooling	Category trees, vertical schemas, mandatory regulatory fields, enumerations	The vertical's catalog or data team
Ingestion engine	Transports, idempotency, snapshot and delta reconciliation, error reports, quotas	Format adapters (OpenImmo, DMS exports, ATS feeds), field mappings, vertical validation rules	The vertical's supply team
Ranking framework	Retrieval, feature logging, model serving, experimenta-tion hooks, paid-label plumbing	Ranking policies, vertical features (price rating, distance, freshness), paid-placement rules	The vertical's search team

Engine	What is shared	Vertical content	Who owns the content
Catalog and entitlement engine	Products, price lists, entitlements, quota checks, audit trail	Packages, slot rules, prices per country and seller type	The brand's monetization team
Trust rules and models	Case management, enforcement actions, review queues, model hosting	Policies, vertical rules (rental-deposit patterns, VIN mismatches), thresholds	The vertical's trust team under group policy

Table 3: *Layer-2 engines: the mechanism is shared, the content is a vertical asset with its own owner and release cycle.*

The test of a good layer-2 boundary is that **a vertical can change its content without a release of the engine**. If adding a mandatory energy-cost field for French listings needs a code change in the shared attribute service, the content has leaked into the engine, and the Category Straitjacket follows.

OLX documents layer-2 thinking: a configuration framework for ranking models that balance the visibility of paid ads across countries and verticals, including Motors and Real Estate (OLX tech blog, June 2025), so one ranking machinery is configured per vertical. Attributes Without Chaos: Schemas, Dictionaries and Validation and Category Trees That Can Change: Designing Marketplace Taxonomies show how the taxonomy engine keeps vertical schemas separate without separate services.

Layer 3: vertical capabilities with one owner

Some capabilities exist only because of a vertical's physics, yet several brands need them. Veldmark's Tradeyard Poland cars section needs vehicle identity as much as Motorvane does, and Tradeyard Portugal's property section needs to recognize the same apartment listed twice. The wrong answers are to build them twice, or to generalize them into a layer-2 engine that nobody understands.

The right answer is a **vertical capability**, **group-owned**: one implementation, owned by the vertical team with the deepest knowledge, offered to other brands through a published contract. The vehicle capability takes a VIN or registration and returns a specification with provenance and version; the property capability takes an address and listing facts and returns a canonical property identifier and its history. Both plug into the shared ingestion engine as enrichers and into the ranking framework as signal providers, as Figure 2 shows.

The owner has obligations a brand team does not have: an internal service-level objective, versioned contracts, a roadmap that includes consuming brands, and licensing. Vehicle and property data often come from licensed providers whose terms may limit redistribution, which is a reason to keep the capability small and behind one contract.

Layer 4: what the brand keeps

Brand experience, packages and prices, sales tooling and the professional front ends stay with the brand. Serving the Pros: Dealer, Agent and Merchant Tooling argues why the pro front end belongs here even when its back end is shared. The public record supports keeping pricing local in particular: package design and prices differ by country and vertical in every portal annual report cited in How Listing Marketplaces Make Money.

Extension points along the listing flow

Figure 3 shows where the vertical seams sit on the path of one listing. The flow is shared. At each stage, the vertical contributes content or a capability through a declared extension point.

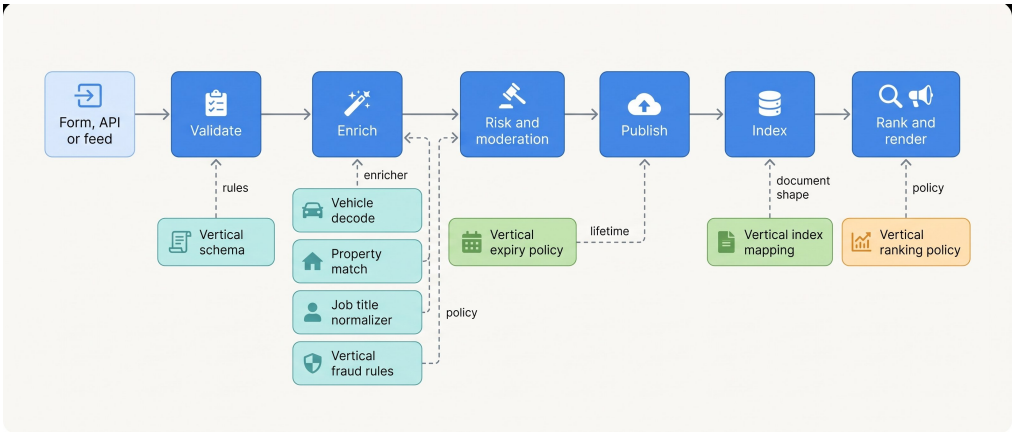


Figure 3: *One listing flow, eight vertical seams. Solid arrows are the shared publication path that From Submitted to Searchable: The Publication Pipeline owns. Dotted arrows are vertical contributions: rules, enrichers, policies and mappings, each versioned and owned by the vertical. A vertical that needs a new seam asks for one; it does not copy the pipeline.*

Two design rules keep the seams honest. First, **each extension point has a contract and a failure mode**. A vehicle decoder that times out must leave the listing publishable with a "decode pending" state, because a dealer feed of 200 cars cannot wait for one provider. Second, **the number of seams is governed**. Every new extension point is a

permanent cost to the engine team, so new seams go through the same review as new shared capabilities.

Veldmark Worked Example: Three Listings, Three Physics

The workshop drafts its profiles from recent listings. Over the spring, Ingrid's team checks them against the three listings this book follows through Parts II and III (Table 4).

Profile dimension	Rotterdam city bike (Tradeyard NL)	Essen Volkswagen Golf (Motorvane DE)	Lyon apartment (Hearthly FR)
Supplier	Femke, a private seller	Autohaus Ruhrblick, a dealer	Three agencies, including Agence Rhône Habitat (14 branches)
Ingestion path	App form, Sunday 7 March 2027, 19:42 CET	DMS feed, Wednesday 14 April 2027	Two real-estate software vendors and Keybridge, May 2027
System of record	Tradeyard	The dealer's DMS	Each agency's software
Item identity	None outside the listing	VIN, decoded into 31 attributes	One canonical property behind three listings
Price	€240, sold for €225	€21,900, cut to €21,400	€389,000
Price guidance	None	Market value €22,800, "good price"	Valuo range €372,000–€401,000
First contact	13 minutes after listing (19:55)	Part of 23 raw leads in 10 days (19 after merging repeat contacts)	Not tracked as one number; leads go to three agencies

Profile dimension	Rotterdam city bike (Tradeyard NL)	Essen Volkswagen Golf (Motorvane DE)	Lyon apartment (Hearthly FR)
Lifetime	About 22.5 hours to sale	15 days to sale; feed delete 26 hours later	Weeks to months
Regulatory field	Trader status (private)	Trader status and the B2C conformity guarantee; no CO2 label, because Pkw-EnVKV exempts used cars	DPE energy class D
Money of the sale	Through Safe Deal "pay at pickup"; buyer fee €11.95	At the dealer	At a notary
Characteristic failure	Phishing link at 20:11; alert digest after sale	Late feed delete (Ghost Inventory)	Duplicates across agencies (Seeing Double)

Table 4: *The same twelve questions, three different answers. The bike's physics are speed and safety, the car's are identity and feeds, the apartment's are entity resolution and disclosure.*

The table confirms what Felix and Julien argued. The bike's critical numbers are minutes: 6 minutes to searchable, 13 minutes to first contact, a phishing message 29 minutes after listing. The Golf's critical numbers are attributes and hours: 31 decoded attributes, a market-value estimate, a 26-hour-late delete. The apartment's critical number is a ratio: Hearthly France's 1.9 million listings describe about 1.12 million properties, **1.7 listings per property**. No single listing model optimizes for all three without making explicit room for identity, external systems of record and vertical schemas.

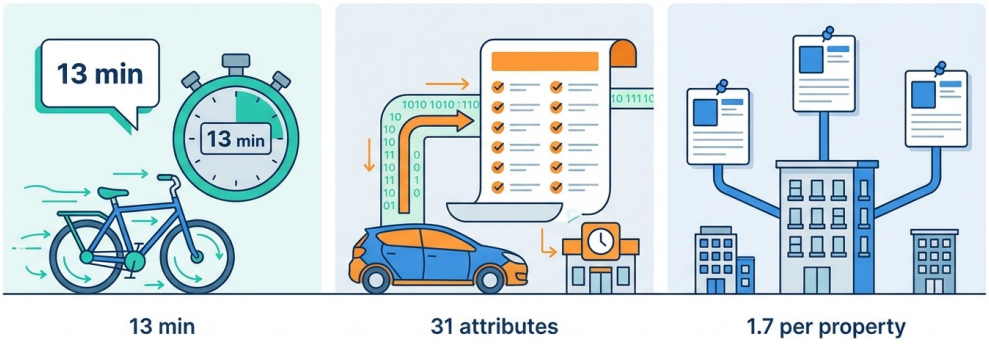


Figure 4: *Three listings, three physics: the bike's critical numbers are minutes, the Golf's are decoded attributes and feed hours, and the Lyon flat's is a ratio of 1.7 listings per property.*

The group's 2026 figures from the ledger add two profile numbers for platform design: implied listing lifetime and revenue per live listing.

Brand	Live listings	New listings per week	Implied average lifetime	2026 revenue	Revenue per live listing per year
Tradeyard	13.2M	2.3M	about 5.7 weeks	€410M	about €31
Motorvane	1.45M	210,000	about 6.9 weeks	€360M	about €248
Hearthly	2.6M	240,000	about 10.8 weeks	€395M	about €152

Table 5: *Implied lifetime is live listings divided by weekly new listings, a steady-state approximation. Revenue per listing differs by a factor of eight between Tradeyard and Motorvane.*

Two consequences follow. **Write load and value are inversely distributed**. Tradeyard produces about 84% of the group's 2.75 million weekly new listings, but each earns a fraction of a Motorvane car, so a pipeline tuned for Tradeyard's volume must not treat a dealer's listing as cheap. And **cost models must be read against the profile**: a media

service charged per image makes Hearthly's 1.8 billion stored images look like a problem, when each Hearthly listing earns about five times what a Tradeyard listing earns.

Katarzyna's sections

Tradeyard Poland is the case where profiles matter most, because it runs four verticals on one horizontal stack. The workshop fills in profiles for its cars and jobs sections, and the numbers are local to this chapter: the cars section carries about **410,000** live listings, about 40% from dealers, and the jobs section about **140,000** live job ads, most posted directly by small employers.

- **Cars section**. Its profile matches Motorvane's on identity, attribute density and fraud (deposit scams on vehicles), and differs on supply mix (more private sellers) and monetization (insertion fees, not dealer packages). The workshop's conclusion is architectural, not commercial: the cars section should consume Motorvane's vehicle identity and price-rating capability as a layer-3 service, through the shared ingestion engine, while its packages and prices stay with Tradeyard Poland. Whether the two brands should keep competing for the same Polish dealers is a portfolio question, left to After the Deal: Consolidating Platforms After Acquisitions.
- **Jobs section**. Its profile differs from everything else at Veldmark: probabilistic identity (job titles in Polish), documents on the demand side once applications move on-platform, an external expiry clock enforced by Google, and anti-discrimination and pay-transparency rules. The workshop concludes that jobs does not justify a vertical stack at 140,000 listings. It needs layer-2 content (a jobs schema mapped to ESCO occupations, an expiry policy, jobs-specific trust rules), and one layer-1 change: document support in the media service before CV upload is offered. If Tradeyard Poland ever charges per application, the profile's monetization row changes and the decision is revisited.

The workshop's classification

The workshop ends with a first shared-versus-specific table for the Common Ground capability list. Table 6 is the version the architecture advice forum ratifies in April 2027.

Capability part	Layer	Owner	Reason
Login, sessions, account security	1	Group identity team (Veldmark ID)	Same semantics everywhere
Event backbone and schema registry	1	Platform engineering	Same semantics; already decided for July 2027
Image storage, processing, delivery; document support	1	Media platform	Same mechanism; requirements from all profiles
Messaging transport and storage	1	Messaging platform	Same mechanism; safety rules are layer 2
Taxonomy and attribute engine	2	Group taxonomy team	Same mechanism; trees and schemas owned per vertical
Vertical schemas (goods, vehicles, property, jobs)	2 content	Each vertical's data team	Vertical semantics and regulation
Ingestion engine	2	Group supply platform	Same transports and reconciliation
Feed adapters (DMS formats, French and Spanish property formats)	2 content	Motorvane, Hearthly	Ecosystem-specific
Ranking framework	2	Group search platform	Same machinery
Ranking policies	2 content	Each brand's search team	How each brand competes
Vehicle identity, specification and price rating	3	Motorvane	Vehicle physics; consumed by Tradeyard Poland
Property entity and valuation (Valuo)	3	Hearthly	Property physics; consumed by Tradeyard Portugal's property section later
Packages, prices, pro front ends	4	Each brand	How each brand sells

Table 6: *Veldmark's first shared-versus-specific classification. Two vertical capabilities*

get group owners; four engines get vertical content owners. Keybridge stays outside the table: under the March 2027 neutrality decision it is a separate SaaS business, not a Hearthly capability.

Felix accepts the table on one condition: the shared ingestion engine must meet Motorvane's dealer-feed requirements before Motorvane moves onto it, measured by the same feed success rate and delete latency Motorvane reports today. Ingrid makes that condition a stage gate. It is the defense against the Forced Fit: **a vertical moves onto a shared engine when the engine meets the vertical's profile, not when the program plan says so**.

Trade-offs: How Much to Share

Groups choose among four broad options for supporting several verticals. The share test is a way of mixing them per capability; Table 7 compares them as pure strategies.

Option	What it means	Strengths	Weaknesses	When it is right
A. One horizontal platform with vertical categories	Real estate, cars and jobs are subtrees of one category tree on one listing model	Lowest cost; one codebase; fast for small sections	No place for external identity, external systems of record or vertical regulation; the Category Straitjacket	Small vertical sections inside a horizontal brand, where the vertical is not a competitive business
B. Shared core with vertical extension points	Layers 1 and 2 shared; vertical content and layer-3 capabilities owned by verticals	Shares machinery without forcing physics; vertical teams own what differentiates them	Extension-point governance; engine teams must serve several masters; slower first delivery	Groups with several material verticals and a stable ownership horizon

Option	What it means	Strengths	Weaknesses	When it is right
C. Separate vertical stacks	Each vertical brand runs its own platform end to end	Full fit to physics; independent pace; easy to sell a business	Pays for every capability several times; inconsistent trust and identity; Deal-Shaped Architecture	Verticals with very different physics and no shared customers, or businesses likely to be sold
D. A vertical platform reused by horizontal sections	The strongest vertical brand's stack serves the same vertical inside horizontal brands	Best vertical capability everywhere; one vertical owner	Couples the horizontal brand to another brand's roadmap; commercial conflicts between brands	Where a horizontal brand's section and a vertical brand compete for the same professionals and data

Table 7: *Four strategies. Most groups end up with B for the platform and D for one or two vertical capabilities; pure A and pure C are each right only under narrow conditions.*

My judgement: **choose option B as the default, use D only for layer-3 capabilities, and keep C available for any business the owners might sell.** Adevinta's 2023–2025 record is the reminder for the last clause. A shared platform that was a synergy in one ownership era became a separation cost in the next. Option B survives that change better than a fully merged product platform, because layer-1 and layer-2 services can be designed to be separable: tenant-scoped data, per-tenant accounts and clean ownership of vertical content.

Option B is not free. Without engine teams who treat vertical owners as customers and say no to forks, it decays into A (the engine absorbs vertical content) or into C (each vertical forks the engine).

Vertical Differences in Who Decides

The profile also differs on a dimension the earlier tables leave implicit: who has the power to change the vertical's requirements. Architecture has to be ready for changes

from those parties, on their timelines.

Vertical	Party that can change requirements	Documented example	Architectural readiness
General classifieds	Regulators; fraudsters	EU trader traceability duties (DSA Article 30) from February 2024	Trader status and evidence as account data; fast rule deployment
Real estate	Regulators; standards bodies; agents' associations and MLSs	NAR's 2025 listing options allowing delayed IDX marketing; French DPE changes in 2026 and 2027	Effective-dated attributes; visibility flags per listing; versioned schemas
Automotive	Regulators; data providers; dealers as paying customers	Pkw-EnVKV amendments; Autotrader's Deal Builder backlash in late 2025	Configurable, reversible platform states on dealer inventory; licensed-data abstraction
Jobs	Search engines; regulators	Google's job-posting expiry rules (updated September 2026)	Expiry as a first-class event; indexing integration
Recommerce	Payment and logistics partners; tax authorities	Vinted Pay's own e-money licence and wallet rollout (2023–2026)	Order and ledger boundaries designed for a change of payment partner

Table 8: *Each vertical has outside parties who can rewrite its requirements. The architecture should make their likely changes cheap.*

The automotive row needs one comment. After retailer pushback, Autotrader paused the automatic rollout of Deal Builder and introduced "request a reservation" (Autotrader FY2026 results, May 2026). In a dealer vertical, **the supplier is also the paying**

customer, so per-dealer configuration and reversible platform states are part of the profile.

Failure Modes

Profiling the vertical that shouted loudest. The first shared engine is usually built for the largest brand. At Veldmark that would be Tradeyard, with 84% of insertion volume. The engine then encodes form-based insertion, the marketplace as system of record and weak item identity, and every other vertical becomes an exception. The remedy is to collect requirements from every profile before the engine's contracts are fixed.

Vertical exceptionalism. The opposite failure: every vertical declares itself unique, every capability lands in layer 3 or 4, and the group pays for everything several times. The profile is the defense here too. If two verticals' profiles match on the dimensions a capability depends on, the capability is shared, whatever the teams prefer.

Content leaking into the engine. A shared attribute service gains a hard-coded "energy class" field because French real estate needed it first; Spanish and German rules then need different ones. Every regulatory change now needs an engine release. This is how the Category Straitjacket forms inside a well-intentioned platform.

One attribute store, many meanings. A shared store without vertical dictionaries ends with "condition" meaning a goods grade, a vehicle damage category and a property renovation state, under the same key. Facets, feeds and models then disagree about what a listing is. Tradeyard's own figures show the start of this: **3,900** distinct attribute keys, 41% of them used on fewer than 0.1% of listings. Attributes Without Chaos: Schemas, Dictionaries and Validation owns the cleanup.

Ignoring the external system of record. A horizontal pipeline that treats the marketplace's write as the truth will keep a sold car live until the dealer's feed delete arrives, as with the Essen Golf's 26-hour gap. Professional verticals need reconciliation against the source system, which the horizontal pipeline never needed.

Missing a regulatory field as a feature gap. A property listing without a DPE class is not incomplete; in France it is unlawful to advertise it that way, with fines up to €3,000 for an individual and €15,000 for a legal entity (Service-Public.fr, DPE). If mandatory regulatory fields sit in the same validation tier as optional ones, compliance defects pass as data-quality warnings.

The Chief Architect's View

At group scale, the vertical question is a standing governance question, not a one-time design decision. The Chief Architect's job is to make it cheap to ask and hard to dodge.

What to decide.

- **One profile per vertical, per brand where the brand differs**. Tradeyard's goods, Tradeyard Poland's cars and jobs, Motorvane, Hearthly France and Hearthly Spain each get a profile. It is one page, numbers first, and lives in the architecture repository next to decision records; Tool 2 in the Field Kit: Templates for the Marketplace Architect is the template.
- **The layer of every Common Ground capability part**, with the reason, the owner and the condition for moving a vertical onto it. Table 6 is the format.
- **The owners of layer-3 capabilities**, with their obligations: an internal service-level objective, versioned contracts, consumers on the roadmap, licensing managed centrally.
- **The list of extension points**, governed like an API: additions reviewed, deprecations announced, each with a contract and a failure mode.

Who decides. The vertical owner decides its content and argues its profile. The engine owner decides the mechanism. The architecture advice forum decides the layer when they disagree, and records the decision. The group CTO decides when the forum's decision changes funding.

What to watch.

- **Fit metrics per vertical on shared engines**. Feed success rate and delete latency for Motorvane on the ingestion engine; duplicate rate for Hearthly; publication time for Tradeyard. A shared engine that is cheaper but worse on a vertical's critical metric is a Forced Fit in progress.
- **Exception counts**. The number of vertical-specific branches in shared engine code, and the number of requests for new extension points. Rising numbers mean the boundary is wrong.
- **Profile drift**. Any change in a vertical's business model, such as a transactional pivot, a move to pay-per-lead or an acquisition, triggers a profile review before the next architecture decision.
- **Separability**. For every shared capability, whether a brand could be carved out with its data and content in months rather than years. Ownership horizons in this industry are shorter than platform lifetimes.

In the long term, **vertical boundaries move**. Horizontal marketplaces spin off motors, property and jobs brands; vertical portals add transactions and software; groups organize by vertical and then by country. Layers that match today's organization will be wrong after the next reorganization. Layers that match the physics survive it, because the physics of a car, an apartment and a job change more slowly than org charts.

Questions to Ask

1. Do we have a written requirement profile for every vertical and every brand section that runs on the platform, with numbers rather than adjectives?

2. For each shared capability, which vertical's profile did its requirements come from, and which profiles were never consulted?
3. Where does the item have an identity outside the listing, and does our data model keep identity, specification and listing separate there?
4. For which verticals is our marketplace not the system of record, and how do we reconcile against the source, including deletes?
5. Can a vertical change its schema, adapters or ranking policy without a release of the shared engine?
6. Which capabilities exist only because of one vertical's physics, who owns each of them, and which other brands consume them?
7. What metric tells us that a shared engine serves each vertical at least as well as the system it replaced, and who reviews it?
8. Which regulatory fields are mandatory in each vertical and country, and are they validated as blocking errors or merely as warnings?
9. If one vertical business were sold next year, which shared capabilities would have to be separated, and how long would it take?
10. Which vertical is closest to changing its business model, and what in its profile would change first?

From Physics to Models

This chapter closes Part I with the business physics that every later design must respect: what a marketplace is, how liquidity works, how the industry formed, what capabilities exist, how they divide into contexts and how verticals differ. Part II — LISTING: Modeling the Thing Being Sold turns to the thing being sold. Its first chapter, The Listing Aggregate: Identity, Lifecycle and Invariants, defines what a listing is, what must always be true about it, and how it changes over time. It also shows how a well-designed aggregate leaves room for the vehicle identities, property entities and external systems of record that this chapter has shown to be unavoidable.

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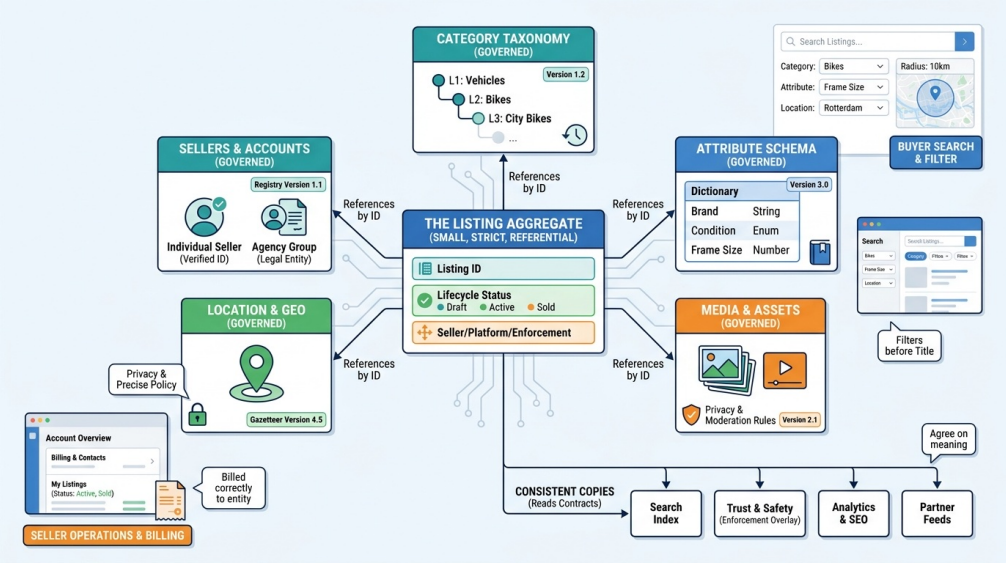
Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

Part II — LISTING: Modeling the Thing Being Sold



IN THIS PART, YOU WILL: Design the listing domain model: the listing aggregate and its lifecycle, category taxonomies, attribute schemas, media, location, and the sellers and organizations that supply listings.

WHY THE BUSINESS CARES: Buyers filter on category, attributes and place before they read a title, and professional sellers are billed through the account model. Weak models mean listings nobody finds, sold items that keep drawing contacts and invoices sent to the wrong business.

WHY ARCHITECTS SHOULD CARE: Search, alerts, trust, monetization, analytics, SEO and partner feeds all copy the listing. The shape chosen here decides whether those copies agree.

KEY POINTS:

- **The listing is a small aggregate that holds references, not a wide table.** It owns identity, invariants and a lifecycle with separate seller, platform and enforcement dimensions, and references categories, attributes, media, places and accounts by ID.
- **Each referenced subdomain is governed, versioned reference data.** Taxonomies, attribute dictionaries, gazetteers and media assets change on their own schedules and need owners, compatibility rules and migrations, not admin screens and hard-coded IDs.
- **Several of these models carry privacy and legal weight.** Photo metadata, a seller's precise location and trader status are personal or regulated data, so their boundaries belong in the data model, not in the user interface.

FAILURE PATTERNS THIS PART ADDRESSES:

- **The Category Straitjacket** — Treats the category tree as versioned data, so a change is a migration, not a release.
- **The Attribute Junkyard** — Gives every attribute one definition that forms, feeds, filters and rules compile from.
- **Ghost Inventory** — Makes every lifecycle transition an event with a deadline, so no copy outlives the listing.

Part I drew the domain boundaries and decided who owns the listing's truth; Part II designs **the thing being sold** inside them. Most marketplaces modeled the listing once, at launch, and have extended it ever since. Fifteen years later the `ads` table has two hundred columns, category IDs live in the code, and nobody can say which copy of a listing is the true one.

This part argues that a listing is **an offer about a thing, not the thing itself**. The aggregate stays small and strict. Around it sit five subdomains, each with its own owner and rate of change: categories, attributes, media, location and the parties who list. Give each one a contract, and ingestion, search, trust and billing read the same meaning instead of guessing it.

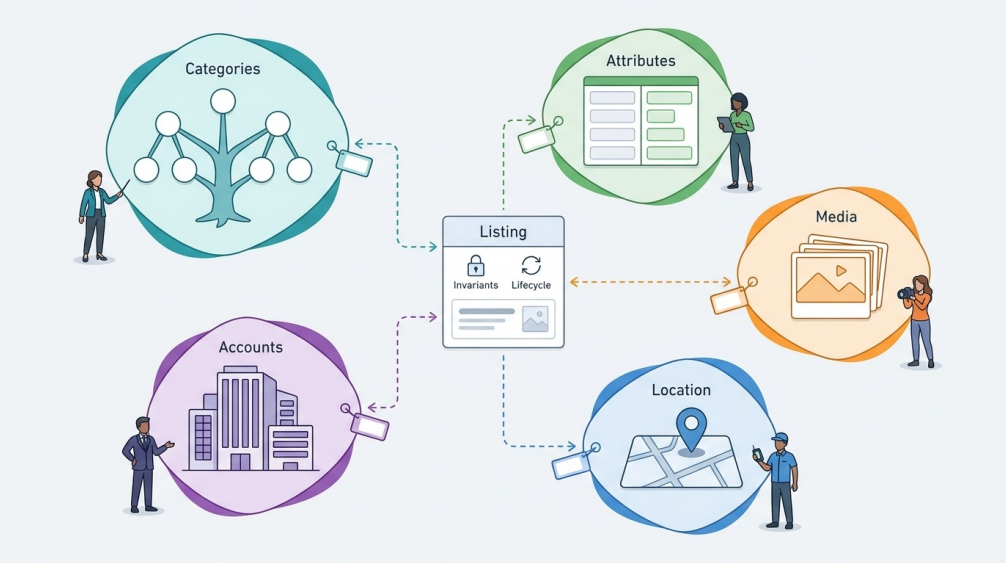


Figure 1: *The listing stays a small, strict aggregate; categories, attributes, media, location and accounts are separate governed subdomains it references by ID.*

Chapters in this part

Chapter	The question it answers	Read it when…
The Listing Aggregate: Identity, Lifecycle and Invariants	What is a listing, what must always be true about it, and how does it change?	Your status field has twenty values, or sold listings still get contacts.
Category Trees That Can Change: Designing Marketplace Taxonomies	How do you design a category tree that can change without breaking everything?	Adding or splitting a category takes weeks and a cross-team release.
Attributes Without Chaos: Schemas, Dictionaries and Validation	How should attributes be modeled so they can be validated, searched and evolved?	Filters are empty, or one fact is stored under many keys.
Photos Sell: Media Pipelines for Listings	How do you ingest, store, moderate and serve billions of images?	The media bill outgrows live listings, or stolen photos go unnoticed.

Chapter	The question it answers	Read it when…
Where Is It? Location, Geocoding and Privacy	How do you model location so search, maps and privacy all work?	Distances leak addresses, or the map and the results disagree.
Private Sellers, Dealers and Agencies: Modeling Who Lists	How should you model the people and organizations that supply listings?	Agencies, dealer groups or franchises do not fit your user table.

How this part connects

Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models in Part I showed where the listing means different things. This part designs the Supply-side core those meanings depend on. Part III — SUPPLY: Getting Inventory In puts the model to work: forms and feeds, enrichment, duplicates and publication. Later parts read these models as contracts. Search indexes the attribute schemas, trust and safety writes the enforcement overlay, and entitlements attach to accounts.

The Veldmark thread

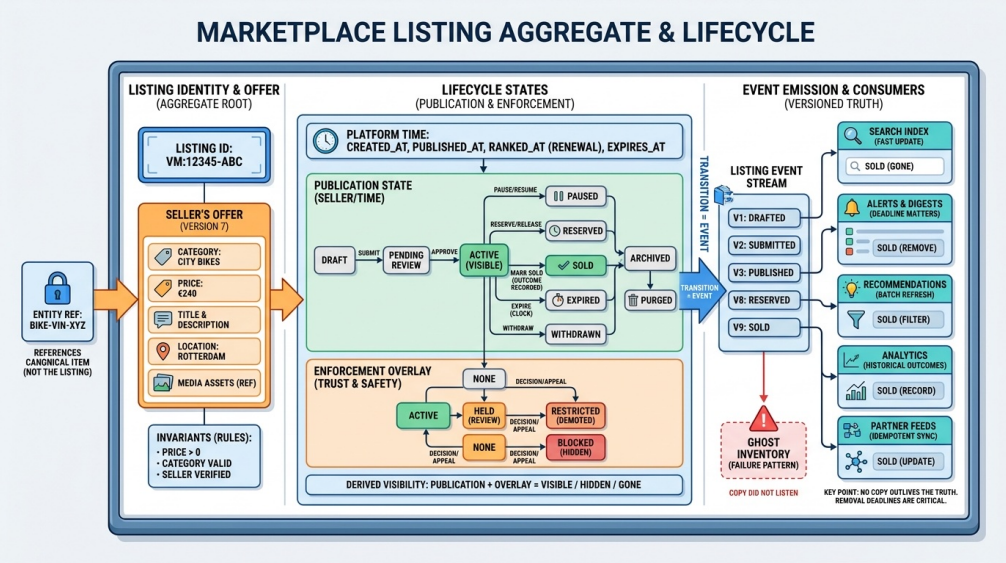
Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), starts 2027 with four Tradeyard country trees and 3,900 attribute keys, 41% of them used on fewer than 0.1% of listings. Ingrid Halvorsen's team decides:

- a **common listing lifecycle** with an enforcement overlay owned by trust and safety, shipped in Listing event contract v1 on 1 July 2027;
- **Veldmark Place**, a gazetteer and precision-policy service (May 2027), so private sellers' precise points never reach a client;
- a **group business party registry** for verified legal-entity facts (August 2027);
- the **group taxonomy and attribute service**, live for Tradeyard in October 2027, and the **bikes split** on Monday 1 November 2027;
- a **group media decision** (November 2027), with a shared perceptual-hash index from Q1 2028.

Femke's €240 Rotterdam city bike, the Lyon apartment and Agence Rhône Habitat's fourteen branches carry the thread.

By the end of Part II you should be able to say who owns each part of a listing, how it is versioned and which systems copy it. Begin with The Listing Aggregate: Identity, Lifecycle and Invariants.

7. The Listing Aggregate: Identity, Lifecycle and Invariants



IN THIS CHAPTER, YOU WILL: Define the listing as an aggregate, with an identity, an owner, a small set of invariants and an explicit lifecycle. You will see how versions, price history, availability and multi-unit offers fit into it, and you will follow one €240 city bike through every state and event of its two-day life.

WHY THE BUSINESS CARES: Every buyer contact on a listing that is already sold, every bump charged on a listing that is already blocked and every "new" listing that is really an old one reposted costs trust or money. The marketplace general manager feels it as wasted contacts, refunds, support tickets and buyers who stop believing the results.

WHY ARCHITECTS SHOULD CARE: The listing is the object that every other context copies: search, alerts, recommendations, trust, monetization, analytics, SEO and partner feeds. If its states, versions and events are vague, every copy invents its own interpretation, and the platform cannot say which listings are actually for sale.

KEY POINTS:

- **A listing is an offer, not the thing.** It is a seller's time-bounded public statement about an item, a vehicle, a property or a job, at a price and under conditions. Keep the listing separate from the underlying entity, from what the seller has paid for and from the moderation case, and reference them by ID instead of copying them in.
- **The lifecycle has three owners, so model two dimensions.** Seller intent (draft, active, paused, sold, withdrawn) and platform time (expiry, renewal, archiving) share a publication state; enforcement (held, blocked, demoted) is a separate overlay, because it changes for different reasons and has its own appeals. One overloaded status field is how listings end up both "sold" and "promoted".
- **Every transition is an event with a deadline.** The listing service owns the aggregate and publishes a versioned event for each state change. Each consumer has a contract for how quickly it must react, and removal deadlines matter more than insertion ones. Ghost inventory is what happens when one copy does not listen.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Ghost Inventory** — a listing marked sold in its owning service stays in a downstream copy that never processed the transition.
- **Search as System of Record** — when nobody owns the aggregate, the search index becomes the only place where a listing's current state can be read.
- **Billing by Guesswork** — paid flags stored as listing fields drift from what the seller bought and from the listing's actual state.

Every listing marketplace has a table called something like `ads`, `items` or `listings`, and most have had it since the first release. It starts with a title, a price, a category and a status column. Fifteen years later it has two hundred columns, a status field with twenty values, three flags that mean "promoted" in different ways and a `deleted` column that nobody trusts. **The listing is the oldest and most copied object in the platform, and usually the least designed.**

This chapter treats the listing as what domain-driven design calls an **aggregate**: a cluster of data with one root, one owner and a set of rules that must hold after every change, updated only through that owner. Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models explained why the listing means different things in Supply, Search, Trust and Monetization, and Same Platform, Different Physics: How Listing Verticals Differ showed that cars and apartments have identities that outlive any listing. This chapter designs the Supply-side core that all of them depend on: identity, invariants, lifecycle, versions and the events that leave the aggregate. Categories,

attributes, media, location and seller accounts each get their own chapter in Part II; here they appear only as references the aggregate holds.

The test of a listing model is simple. **At any moment, can the platform say whether a listing is for sale, who can see it, who can contact it and why?** Most platforms can answer for the owning database and fail for the copies. The states and events defined here are what later chapters use to answer for every copy: From Submitted to Searchable: The Publication Pipeline measures freshness against them, search and Tell Me When It Appears: Saved Searches and Alerts react to them, and Every Change Is an Event: Event-Driven Listing Architecture and System of Record and Read Models: Listing Data Management carry them to every copy.

Setting the Scene at Veldmark: Four Answers to One Question

Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), has a listing whose state its systems could not agree on. On **Sunday 7 March 2027 at 19:42 CET**, Femke, a private seller in Rotterdam, listed a used city bike on Tradeyard Netherlands for **€240**. It sold on **Monday 8 March** for **€225** through Tradeyard Safe Deal's "pay at pickup". Femke marked it sold at **18:10**; it left search at **18:11**. On Tuesday morning, an alert digest assembled the evening before still told 214 buyers about it.

Ingrid Halvorsen, who joined as Group Chief Architect in February 2027, asks a question in her second month that sounds pedantic: *what was the state of Femke's bike at 08:00 on Tuesday?* Daan Visser, Tradeyard's engineering director, pulls the answers from four systems.

System	What it said at 08:00 on Tuesday 9 March	Why
Yardstick (PHP monolith, MySQL)	`status = 4` (sold), `is_active = 0`, `top_until` still set to Wednesday	The bike never had a paid product; `top_until` was a default copied from a template
Solr (search)	Not present	Removed at 18:11 by the sold-listing hook

System	What it said at 08:00 on Tuesday 9 March	Why
Alert digest service	Included in 214 digests sent at 07:00	The digest was assembled at 17:30 Monday from a MySQL replica snapshot taken before the sale, and nothing re-checked it before sending
Snowflake (analytics)	"Expired"	The nightly extract had no sold state for Safe Deal pickups, and its fallback rule mapped "inactive and no end reason" to expired
Safe Deal order service	Completed, fee **€11.95** charged	Correct

Only the order service and Yardstick agreed, and only on the most basic fact. Search was right for the wrong reason: a hook in Yardstick deleted the document, which is not the same as the search platform knowing the listing was sold. Analytics counted a successful sale as a failure. The alert service sent a sold bike to buyers it was supposed to help.

Figure 1: *At 08:00 on Tuesday 9 March, Femke's sold bike had four different states in four systems, and the alert digest had already sent it to 214 buyers.*

Sven Lindqvist, the principal engineer for the event platform, is preparing the group's first **Listing event contract**, due to go live in July 2027 on the new Kafka backbone. Ingrid's conclusion is that the contract cannot be written until the group agrees what a listing *is*: what identifies it, what states it can be in, which rules always hold and which changes are events. The rest of this chapter is that agreement.

What a Listing Is, and What It Is Not

A **listing** is a seller's public, time-bounded offer about a specific thing: an item to sell, a vehicle, a property to sell or let, a job to fill. It carries the seller's description of the thing, a price or price state, a location, media and contact options, and it has a lifecycle that ends. The productscape model this book draws on (see Map What the Marketplace Does Before What Runs It) defines it as "a seller-created marketplace offer with lifecycle, visibility, and transaction state". That definition is right, but it does not separate the listing from its neighbors.

Concept	What it is	Identity	Lifetime	Example
Listing	One seller's offer, as published	Platform listing ID	Days to months	Femke's €240 bike ad
Underlying entity	The physical or legal thing the offer is about	VIN, address or parcel, none for most goods	Years	A specific apartment in Lyon; a specific Golf
Product	A catalog type of which the item is an instance	GTIN, catalog ID	Years	"iPhone 14, 128 GB"
Offer (commercial)	Terms under which the thing is available	Often the listing itself	As listing	Price, availability, delivery, financing
Placement	Paid visibility applied to a listing	Entitlement or order ID	Hours to weeks	A 7-day "top listing"

Concept	What it is	Identity	Lifetime	Example
Case	A trust-and-safety record about a listing or account	Case ID	Months to years	The report that removed a fake rental

The distinction between listing and entity is the one that marketplaces discover late. One apartment produces many listings: several agencies at once, then a new listing each time it is sold or let again. One car produces a dealer listing, a private listing three years later and a listing on every competing portal. eBay's attribute model formalizes part of this: each attribute ("aspect") is marked as applying to the **product** or to the **individual item**, so a listing can inherit product facts from a catalog while condition stays on the listing (eBay Taxonomy API specification, n.d.). eBay also shows the limit of pushing the idea too far: its Product-Based Shopping Experience required catalog association for listings from 2018, and eBay's own playbook now states that the mandate for all categories and brands was rolled back (eBay PBSE playbook, n.d., covering 2018–2020).

The rule I use: **the listing references the entity; it never is the entity**. The listing holds an optional `entity_ref` (a canonical property, a vehicle identity, a catalog product) that One Property, Three Listings: Duplicates and Entity Resolution maintains. Prices, descriptions and states belong to the listing. Facts about the thing that outlive any listing, such as a property's surface area from a registry or a vehicle's first registration, belong to the entity, with provenance.

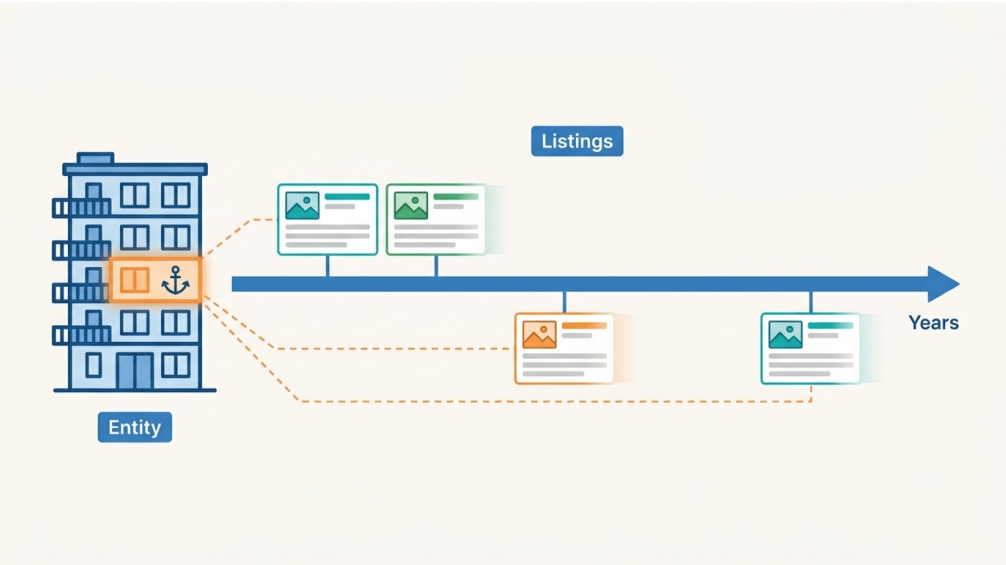


Figure 2: *The listing references the entity and never is the entity: one apartment outlives many listings, each a separate offer with its own lifecycle.*

Inside the aggregate, and outside it

The test for what belongs inside is transactional: a field belongs inside if a change to it must be validated together with the listing's other rules in one write. Otherwise it is a reference.

Inside the listing aggregate	**Referenced by ID, owned elsewhere**	**Owner of the reference**
Listing ID, external keys, version	Seller account and organization	Private Sellers, Dealers and Agencies: Modeling Who Lists
Category ID **and the taxonomy version** it was classified under	The category tree itself	Category Trees That Can Change: Designing Marketplace Taxonomies
Attribute values, with provenance per value	Attribute definitions and schemas	Attributes Without Chaos: Schemas, Dictionaries and Validation
Title, description, language	Media assets and renditions (the listing holds ordered media IDs)	Photos Sell: Media Pipelines for Listings
Price, currency, price state	Price estimate or valuation	Models in Production: ML Platforms and Use Cases
Location reference and display precision	Geocode, administrative areas	Where Is It? Location, Geocoding and Privacy
Contact options chosen by the seller	Conversations and leads	From Interest to Contact: Messaging and Lead Management
Lifecycle states and timestamps	Moderation decisions and cases	Trust and Safety as a Platform
Ranking timestamp (last renewal)	Paid products, entitlements and placements	Packages, Bumps and Quotas: Products, Pricing and Entitlements

Inside the listing aggregate	Referenced by ID, owned elsewhere	Owner of the reference
Availability, quantity, unit structure	Orders and payments	Beyond the Contact: Payments, Shipping and Buyer Protection

Two rows matter most. **Paid visibility is a reference, not a flag.** A `top_until` column on the listing is a copy of a fact owned by the entitlement service, and copies drift: Femke's bike carried a default `top_until` it never paid for. The listing may cache a projection of active placements for display, but the entitlement service decides what was sold, and ranking reads placements, not listing columns (see Ranking for Two Sides: Relevance, Freshness and Paid Visibility). **Moderation decisions are also references.** The listing knows that it is held or blocked and by which decision; the decision itself, with its evidence, reviewer and policy version, lives in the trust-and-safety case store, because it must survive the listing's deletion.

Identity: One ID, Several Keys

A listing needs one **internal identity** that never changes and never carries meaning. Everything else is a key that maps to it.

Key	Purpose	Rules
Listing ID	The aggregate's identity everywhere inside the platform	Globally unique across brands and countries; never reused; never encodes anything that can change, such as category, owner or status. Veldmark's group ID namespaces brand and country (`vm:<brand>:<country>:<id>`), which a listing never changes
Public ID	Shown in URLs and support conversations	Short, stable, may be numeric for legacy reasons; maps one-to-one to the listing ID

Key	Purpose	Rules
External key	The partner's or dealer's own reference	Unique per `(integration, external key)`; the idempotency key for feeds
Request key	Deduplicates a create request from a flaky mobile connection	Short-lived; scoped to the client
Entity reference	Links to the canonical property, vehicle or product	Optional, many-to-one, may change as resolution improves

External keys are where most identity bugs live, and real feed contracts show the pattern. Rightmove's Real Time Datafeed uses the agent's own `Agent_Ref` as the listing key and offers a `GetBranchPropertyList` call that returns a snapshot of a branch's properties, so that a software provider can check that no listing is "still live on Rightmove which should have been removed" (Rightmove RTDF specification v1.4.1, 2023). mobile.de's Seller API supports an insertion request ID that makes create calls repeatable and tells clients not to send parallel requests for the same ad (mobile.de Seller-API, n.d.). Both are external-key designs: the partner's identifier is honored at the boundary, and the platform's identifier is used inside. Many Doors In: Forms, APIs and Bulk Feeds owns the feed side of this.

The hard identity question is **relisting**. When a sold or expired listing comes back, is it the same listing or a new one?

Situation	Same listing ID	New listing ID	My default
Seller renews an expired listing within the renewal window	Keeps URL, history and favorites	Loses them	**Same ID**, new version, state back to active
Seller sold, the sale fell through, item is back	Honest history, "back on the market" signal	Looks fresh	**Same ID**, transition sold → active, recorded

Situation	Same listing ID	New listing ID	My default
Seller deletes and reposts to look new	Not what the seller asked for	What the seller asked for	**New ID**, but linked by One Property, Three Listings: Duplicates and Entity Resolution to the old one, so ranking and duplicate rules can see the repost
A property is put on the market again a year later	Mixes two sale campaigns	Clean campaign metrics	**New ID**, same entity reference
Dealer re-imports the same vehicle after a feed outage	Keeps continuity	Creates duplicates	**Same ID**, matched by external key

The pattern is consistent: **the same campaign keeps its ID; a new campaign gets a new ID tied to the same entity**. Freshness gaming by delete-and-repost is then a duplicate-detection and ranking problem, not an identity one, which is where it belongs.

Invariants: What Must Always Be True

An **invariant** is a rule that holds after every successful command on the aggregate. Invariants are the reason to have an aggregate at all: if a rule can be checked only across many listings or many services, it is a policy, not an invariant, and it is enforced asynchronously. Mixing the two is the commonest modeling error, because it leads either to distributed transactions or to rules that silently stop being checked.

Rule	Invariant or policy	Where it is enforced	When
A listing has exactly one owning account (organization for professionals)	Invariant	Listing service	Every command
A listing is classified under a **leaf** category valid in the taxonomy version it records	Invariant	Listing service, against cached taxonomy	Create, recategorize
Required attributes for the category are present and valid	Invariant **at submission**, not in draft	Listing service, against the category schema	Submit, edit of an active listing
Price is a positive amount in the market currency, or an explicit state (free, on request, negotiable)	Invariant	Listing service	Every price change
`created_at` never changes; the ranking timestamp moves only through renewal or a paid bump	Invariant	Listing service	Every command
An active listing has passed the moderation gate required for its category and risk level	Invariant on the transition to active	Listing service, using a decision reference	Publish, material edit

Rule	Invariant or policy	Where it is enforced	When
A listing under an enforcement block cannot be made visible by its seller	Invariant	Listing service	Every seller command
The seller's trader status is declared before a listing is published	Invariant (regulatory)	Listing service, from the account	Publish
A listing does not exceed the seller's paid listing slots	**Policy**	Entitlement service, checked at publish with a reservation	Publish; reconciled later
The same item is not listed twice by the same seller	**Policy**	Duplicate detection, asynchronous	After publish
A sold listing receives no new paid placements	**Policy** with an invariant guard	Entitlement service refuses; listing rejects the reference	Placement purchase

The trader-status row is regulatory. The Omnibus amendments to the EU Consumer Rights Directive require marketplaces to tell consumers whether the other party is a trader, based on that party's declaration, and the Digital Services Act adds trader-traceability duties for online marketplaces (see Rules of the Marketplace: DSA, DAC7 and Vertical Regulation). A listing published without the declaration is a compliance defect, so it belongs among the invariants on publish.

Two design rules follow from the table.

Validate by state, not once. A draft must accept incomplete data, or the insertion funnel dies. eBay's Inventory API shows the split: an offer becomes a live listing only when it is published, and some fields are required only at publish time (eBay Inventory API overview, n.d.). The same listing schema therefore has a lenient mode for drafts and a strict mode for the transition to pending review or active.

Enforce on create and on revise. eBay's aspect metadata warns that a seller "will be blocked from listing or revising an item without" required aspects (eBay Taxonomy API specification, n.d.). That is how a marketplace backfills a newly required attribute over time without a mass migration: each edit must satisfy the current rules. The price is friction for sellers who want to change only the price, so I exempt price-only and pause-only commands from new attribute requirements and apply them on every other edit.

The Lifecycle: Two Dimensions, Not One Status

A **listing lifecycle** is the set of states a listing can be in, the transitions between them, who may trigger each transition, and what each state means for visibility, contact and retention. The usual implementation is a single `status` column. It fails because three different parties change a listing's state for unrelated reasons.

- **The seller** creates, submits, pauses, resumes, reserves, marks sold and withdraws.
- **The platform clock** expires, renews, archives and purges.
- **Trust and safety** holds, blocks, restricts and restores, with appeals.

When these share one column, every combination needs its own value ("paused_and_-blocked", "expired_under_review"), or one owner overwrites another's state. The classic bug: a seller "resumes" a listing that trust and safety blocked, because the resume command set `status = active`. The fix is structural. I model the lifecycle as **a publication state owned by the listing, plus an enforcement overlay owned by trust and safety**, and derive visibility from both. Seller and clock transitions share the publication state, because both decide whether the offer is open. Enforcement gets its own field, because it changes for other reasons and has its own appeals.

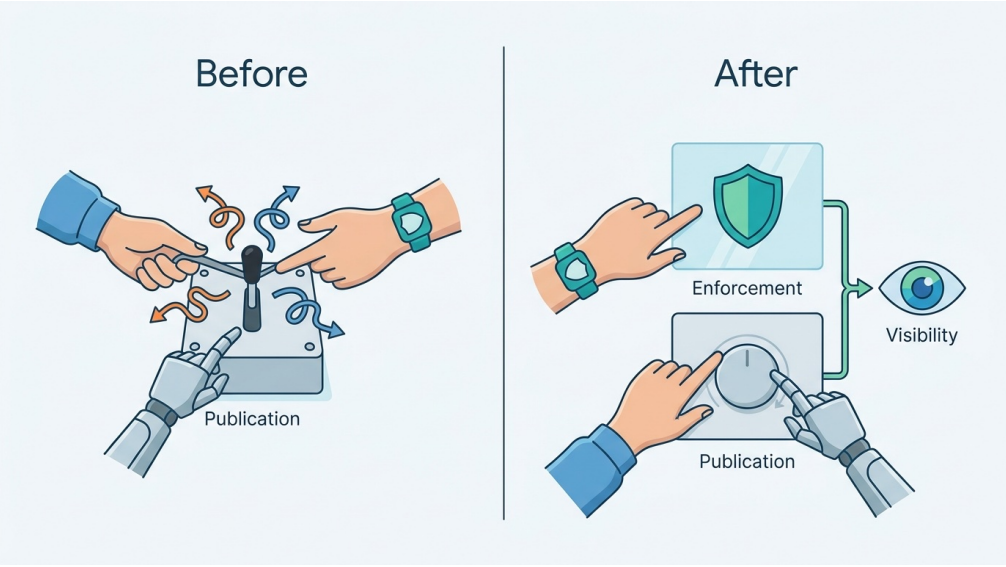


Figure 3: *Seller, platform clock and trust and safety change a listing for unrelated reasons, so the lifecycle is a publication state plus an enforcement overlay, with visibility derived from both.*

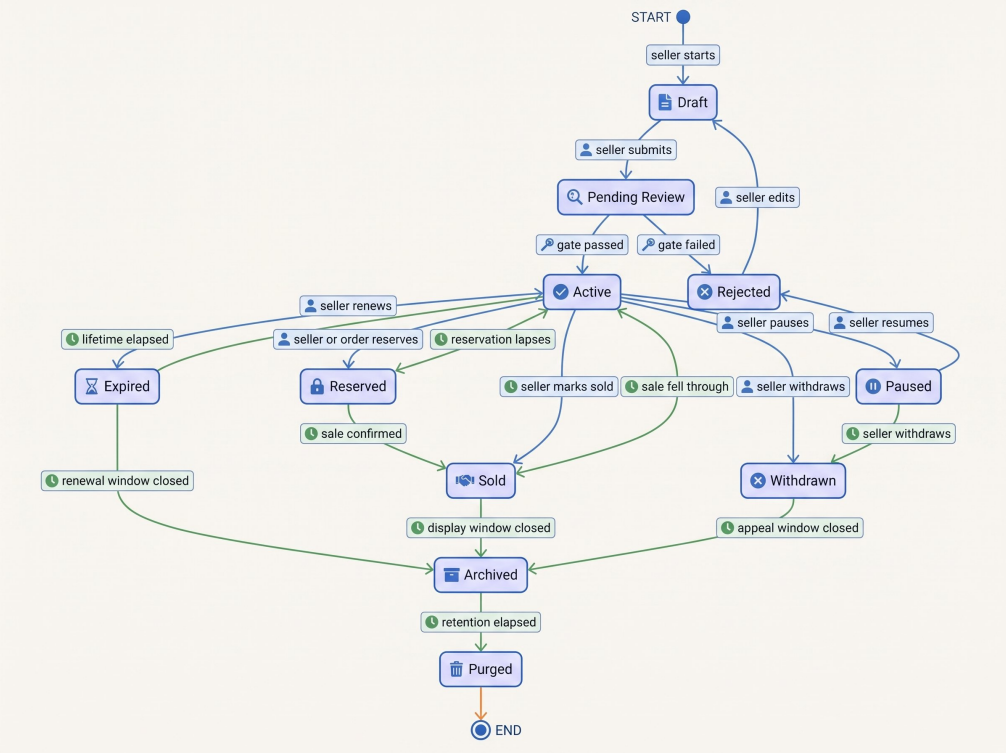


Figure 4: *The publication lifecycle owned by the listing service. Seller transitions and clock transitions are distinct; renewal returns the same listing to active; nothing is deleted until retention ends. Enforcement is deliberately absent: it is an overlay, shown in the next table.*

The enforcement overlay has its own small state set: **none**, **held** (awaiting review, not visible), **restricted** (visible but demoted, or hidden in some countries), **blocked** (not visible, seller cannot restore) and **restored** (after appeal). It is orthogonal to publication. A listing can be active and blocked, which means invisible; it can be sold and blocked, which matters for buyer-protection claims. The overlay matters legally: under the Digital Services Act a decision to remove, disable or *demote* content requires a statement of reasons to the affected user (Regulation (EU) 2022/2065, Art. 17, 2022), so a "restricted" state that ranking applies silently is still an enforcement action with a record. Trust and Safety as a Platform owns the decisions; the listing holds the overlay state and the decision ID.

Visibility, contactability and alerting are then **derived**, never stored independently:

Publication state	Searchable	Detail page	Contactable	Alertable and recommendable	URL behavior
Draft, pending review, rejected	No	Owner only	No	No	404 to the public
Active (overlay none or restored)	Yes	Yes	Yes	Yes	200
Active (overlay restricted)	Demoted or hidden per country	Yes	Yes	No	200
Active (overlay held or blocked)	No	No	No	No	404, or 410 when blocked
Paused	No	Owner, plus buyers with an open conversation	Existing conversations only	No	404 to new visitors
Reserved	Yes, marked reserved, or no, per vertical	Yes	Existing conversations, plus waitlist per vertical	No	200
Sold	No	Yes, marked sold, for a display window	No	No	200 with related live inventory, then 410

Publication state	Searchable	Detail page	Contactable	Alertable and recommendable	URL behavior
Expired	No	Owner; public "no longer available" page	No	No	200 or 410 per SEO policy
Withdrawn	No	No	No	No	410
Archived	No	No	No	No	410

The URL column belongs to Found by Google: Web, App and SEO Architecture for Listings, but its inputs come from here: an expired listing with lasting search demand can keep a 200 page that says it is unavailable and shows live alternatives, while a listing removed for fraud should return 410. That decision is only possible if the lifecycle distinguishes *why* the listing left.

Renewal, bumps and the timestamps that make recency honest

Ranking and analytics depend on three timestamps, and confusing them breaks both.

Timestamp	Meaning	Who moves it
`created_at`	When the listing first existed	Nobody, ever
`published_at`	When it first became active (first publication)	Listing service, once
`ranked_at` (renewal timestamp)	The recency the ranking uses	Renewal, paid bump, relist within the same campaign

Bumps and renewals are not state transitions. They move `ranked_at` and nothing else. mobile.de's Seller API documents this cleanly: an ad carries `creationDate`, `modificationDate` and a `renewalDate`, the date the ad "was last marked as new, which makes it appear new in search" (mobile.de Seller-API, n.d.). Vinted documents its Bump as a paid feature that lasts three or seven days, "or until the item sells" (Vinted Help, "What is an item Bump?", n.d.) — a time-boxed placement with its own end

condition, not a new listing. If a bump resets `created_at`, recency ranking becomes a paid lottery and every "time to sell" metric in analytics becomes meaningless.

Expiry is a data-quality control, not just a revenue lever. Downstream channels enforce it whether or not the marketplace does: Google Merchant Center expires products 30 days after their last update unless they are refreshed (Google Merchant Center Help, n.d.), and Google's job-posting guidance says expired jobs must be removed and that failure to do so "may result in a manual action" (Google Search Central, JobPosting, updated 2026). A lifetime per category, with a renewal path that asks the seller to confirm availability, is the cheapest ghost-inventory control there is. From Submitted to Searchable: The Publication Pipeline owns the expiry jobs and their SLAs.

Outcomes, not just end states

"Sold" is not one fact. The productscape model notes that "capturing whether it sold on-platform, off-platform, or was withdrawn improves future supply and pricing signals". I give every terminal transition an **outcome record**: sold on platform (with the order reference and final price), sold elsewhere, no longer available, sold through this marketplace's contact but paid off-platform, or unknown (expired without an answer). The outcome is what Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics measures as sell-through, and what valuation and price-guidance models train on. Without it, Snowflake's fallback rule turned Femke's sale into an expiry.

Versions, Change History and Price History

A **listing version** is an immutable snapshot of the aggregate's seller-visible content after one accepted change, numbered with a monotonically increasing integer per listing. Versions do four jobs.

1. **Ordering.** Downstream consumers apply changes only if the version is newer than what they hold; the version number becomes the external version for the search index (see Search Architecture for Listings: Indices, Freshness and Scale) and the ordering key for events.
2. **Concurrency.** Commands carry the version they were based on. A dealer feed and the dealer's staff editing the same listing in the pro tool cannot silently overwrite each other; the second writer gets a conflict and re-reads. This is optimistic concurrency, and it is what mobile.de's "no parallel requests on the same ad" rule approximates.
3. **Evidence.** Trust and safety must be able to show what the listing said when it was reported, approved or removed. A statement of reasons that cites content the listing no longer contains is useless in an appeal.

4. **Re-moderation.** A change is **material** if it alters what was approved: category, price by more than a threshold, images, title, or contact details in the description. Material changes send an active listing back through the moderation gate; immaterial ones (a typo, a pickup note) do not. The productscape model records the classic abuse this guards against: material edits "used for bait-and-switch must be caught". A scammer lists an ordinary phone, passes moderation, then edits in a different price and a payment link.

How to store history

Approach	How it works	Strengths	Weaknesses	Right when
State plus history table plus outbox	Current row is the truth; each accepted change writes a version row and an outbox event in the same transaction	Familiar; cheap reads; history is queryable; events are reliable	History schema drifts from current schema; two shapes to maintain	Most listing services; my default
Event-sourced aggregate	The event log is the truth; current state is a fold over events	Perfect audit; easy replay; versions are free	Upcasting old events on every schema change; rebuilding projections; hard GDPR erasure of personal data inside events	Small, high-stakes aggregates such as orders; rarely worth it for a 200-field listing

Approach	How it works	Strengths	Weaknesses	Right when
CRUD with change data capture (CDC)	Writes go to a table; a CDC tool streams row changes	No change to the legacy code; good as a migration step	Row changes are not domain events; the table schema becomes a public contract; intent ("sold" vs "edited") is lost	Strangling a monolith, as a first stage

The outbox pattern writes the event into an outbox table in the same database transaction as the state change, and a relay publishes it; Debezium's outbox router keys messages by the aggregate ID so that per-listing order is preserved (Debezium documentation, Outbox Event Router, n.d.). Every Change Is an Event: Event-Driven Listing Architecture owns the mechanics; the point here is that **the aggregate decides what an event means**, and CDC on a legacy table cannot, which is why I treat raw CDC as a stage on the way to an outbox, not as the end state.

Price history is a regulated series

Price deserves its own history beyond versions, for three reasons. Buyers react to it: watchers get price-drop alerts (see Tell Me When It Appears: Saved Searches and Alerts). Models learn from it: asking-price trajectories and the gap to the final price feed valuation and price guidance. And regulation reads it: the EU Price Indication Directive, as amended by the Omnibus Directive and applied from 28 May 2022, requires any announcement of a price reduction by a trader to indicate the prior price (European Commission, Price Indication Directive, n.d.). The prior price is defined in Article 6a as the lowest price the trader applied during a period of at least 30 days before the reduction; check the national transposition for the exact rule in each market. A dealer listing that shows "price reduced" must therefore be backed by a price series the platform can query, per listing and per seller, with timestamps.

I store a **price event series** per listing: amount, currency, price state, effective time, source (seller, feed, automatic repricing rule) and the version that carried it. "Price reduced" badges, prior-price display and price-drop alerts all read from it. They do not compare two snapshots.

Facts that change without the listing changing

Some attributes change meaning without anyone editing the listing. France's energy performance diagnosis (DPE), which sale and rental ads must display, is recalculated with new primary-energy factors: the official guidance describes a factor change on 1 January 2026 and a further change on 1 January 2027 (Service-Public.fr, DPE, 2026). A flat's class can move from E to D with no change to the flat. This is the case for **bitemporal** thinking: record when a fact was true (valid time) separately from when the platform learned it (recorded time). I do not make the whole listing bitemporal; I make the few regulated, recalculable attributes carry an effective date and a source, and I keep the version history to answer "what did the ad show on 3 May?".

Availability, Units and Multi-Item Listings

Not every listing offers one thing that sells once. The aggregate must say how many things it offers and when they are available, because search filters, alerts and the sold transition all depend on it.

Shape	Example	Model	Sold transition
Single unique item	A used bike, a used car, an apartment for sale	One listing, quantity 1	One sale ends it
Fungible quantity	A shop's 12 identical chairs	One listing with `quantity_available`	Each sale decrements; zero means sold out, not sold
Variants	Shoes in sizes 38–44	Parent listing with variant children, each with quantity	Per variant
Program with units	A new-build development with 40 apartments in four types	Development (parent) → unit types → units, each unit listable	Per unit; the program ends when all units are reserved or sold
Rental with availability window	A flat available from 1 September; a holiday let	One listing with `available_from` and optional calendar	"Let agreed" ends the campaign; the flat returns as a new campaign

Shape	Example	Model	Sold transition
Multiple openings	A job with five vacancies	One listing with `openings`	Filled when all are filled, or when the employer closes it

Two rules keep this manageable. **The parent is a listing only if buyers can contact it.** A new-build program page is a listing (it collects leads); a variant family is usually not (buyers buy a variant). And **"sold out" and "sold" are different states**: a shop listing at zero stock may be restocked, so it should pause rather than end, while a unique item that sells is finished. Real-estate data standards show how rich this vocabulary gets. The RESO Data Dictionary, the US real-estate data standard, defines eleven `StandardStatus` values, including *Coming Soon*, *Active Under Contract* (an offer accepted but still on the market), *Pending*, *Hold*, *Withdrawn* and *Closed* (RESO Data Dictionary 2.0, StandardStatus, revised 2023). A portal does not need all eleven, but it needs a mapping from each feed's states to its own, and that mapping is part of the aggregate's contract, not an ingestion detail.

Architecture: One Owner, Many Copies

The listing aggregate is owned by one service per marketplace: the **listing service** (in productscape terms, Classifieds Management). It accepts commands, enforces invariants, writes versions and emits events. Everything else is a reader or a contributor of references.

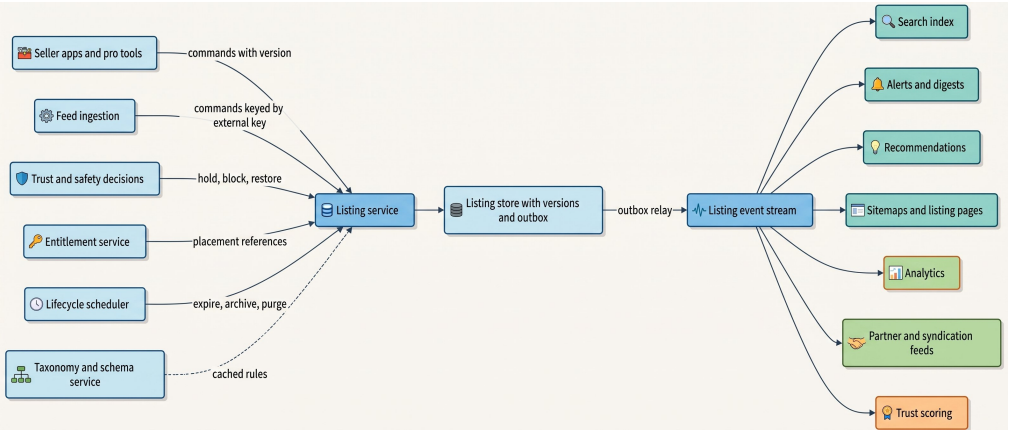


Figure 5: *One owner, many copies. Every write enters through the listing service, including platform-clock and enforcement transitions; every copy is rebuilt from the*

event stream. No consumer writes listing state back, and the search index can be rebuilt from the store at any time.

The command interface is small and explicit. I prefer intent-named commands to a generic `PATCH`, because intent is what the events must carry.

Command	Allowed from	Resulting event	Consumers that must react, and how fast
`CreateDraft`, `EditDraft`	Draft	`ListingDrafted` (internal)	None outside Supply
`Submit`	Draft, rejected	`ListingSubmitted`	Moderation and fraud scoring, seconds
`Approve`, `Reject` (from the gate)	Pending review	`ListingPublished` or `ListingRejected`	Search, alerts, sitemap: inside the publication SLA
`Edit`	Active, paused	`ListingRevised` with changed fields and materiality	Search, price-drop alerts, re-moderation if material
`Pause`, `Resume`	Active, paused	`ListingPaused`, `ListingResumed`	Search and alerts: removal within the removal SLA
`Reserve`, `ReleaseReservation`	Active, reserved	`ListingReserved`, `ReservationReleased`	Search badge, messaging
`MarkSold` (with outcome)	Active, reserved	`ListingSold`	Search, alerts, recommendations, digests: removal SLA; analytics: daily
`Renew`	Active, expired	`ListingRenewed` (`ranked_at` moved)	Search ranking fields
`Expire` (clock)	Active	`ListingExpired`	Same as sold
`Withdraw`	Active, paused	`ListingWithdrawn`	Same as sold
`ApplyEnforcement`, `LiftEnforcement`	Any	`ListingRestricted`, `ListingBlocked`, `ListingRestored`	Search, alerts, messaging: minutes, faster for fraud

Command	Allowed from	Resulting event	Consumers that must react, and how fast
`Archive`, `Purge` (clock)	Terminal states	`ListingArchived`, `ListingPurged`	Every copy, including media and analytics, within the erasure deadline

The table names events by meaning. The published contract carries twelve domain event types: drafts and internal moderation steps stay inside Supply, and related transitions share one type distinguished by a `cause` field (see Every Change Is an Event: Event-Driven Listing Architecture). Three properties make the contract work.

Events carry state, not only notice. Each event carries the new version, the publication state, the overlay and the changed fields, so a consumer that missed an event can still converge on the latest one. A consumer receiving version 7 when it holds version 5 applies version 7; a consumer receiving 5 when it holds 7 ignores it.

Every consumer declares its removal deadline. Insertion latency is visible to sellers, who check that their listing appears. Removal latency is invisible to everyone except the buyer who contacts a sold item, which is why it degrades unnoticed. I make each consumer of the listing stream state, in its contract, how fast it removes a listing that leaves the active state, and I monitor that **sold-to-gone** latency per consumer. From Submitted to Searchable: The Publication Pipeline owns the end-to-end freshness SLAs and unpublish propagation; this chapter defines the events they are measured on.

Consumers that cache must re-check at send time. A digest, an e-mail campaign or a recommendation batch that assembles candidates hours before sending must verify each listing's current state at send time, or subscribe to removals and purge its candidate set. This is the single rule that would have kept Femke's bike out of Tuesday's digests.

In Practice: What Public Evidence Shows

No classifieds company has published its full listing lifecycle or event model. The evidence comes from partner APIs, help pages, standards and investor reports, which reveal contracts rather than internals. Besides the mobile.de, Vinted, eBay and Google sources already cited, three are worth tabulating.

Source	What it documents	Lesson for the aggregate
Rightmove RTDF v1.4.1 (2023)	`Agent_Ref` as key; snapshot reconciliation to find listings "still live … which should have been removed"	Lost deletes are normal; reconcile
RESO Data Dictionary 2.0 (revised 2023)	Eleven standard statuses, including *Active Under Contract*, *Coming Soon* and *Hold*	Vertical lifecycles are richer than goods lifecycles; map them explicitly
Auto Trader plc FY26 results (2026)	Retailer backlash to the Deal Builder reservation rollout in late 2025; Auto Trader paused auto-rollout and introduced "request a reservation"	A platform-controlled state on a dealer's stock changes the dealer's operating model

The Auto Trader case shows that **who may trigger a transition is part of the business model**; that lesson is my interpretation rather than Auto Trader's statement. In its results for the year to March 2026, Auto Trader describes pausing the automatic rollout of Deal Builder, holding listening sessions and introducing "request a reservation" after retailer pushback (Auto Trader Group plc, FY26 results, 2026). Trade press reported that the original flow let consumers place a paid reservation and that the revised one leaves the reservation to the retailer's discretion (Car Dealer Magazine, 2025). In a dealer-supplied vertical, a "reserved" state that the platform sets on the dealer's inventory is a change to the dealer's sales process. Transition permissions belong in the lifecycle specification, per vertical and per seller type, and should be configurable without code.

The Rotterdam city bike, transition by transition

Here is Femke's bike under the Common Ground lifecycle that Veldmark adopted in 2027, replayed against the March events. The times and version numbers match the event log in Every Change Is an Event: Event-Driven Listing Architecture; versions 4 to 7, Femke's Sunday-night edits and a photo reorder, are omitted.

Time (CET)	Command	Version	Publication state / overlay	Event	What should happen downstream
Sun 19:31	`CreateDraft` from the app	1	Draft / none	`ListingDrafted`	Nothing public
Sun 19:42	`Submit` (category Bikes, the pre-2027-split leaf, €240, approximate Rotterdam location, six photos)	2	Pending review / none	`ListingSubmitted`	Automated moderation and risk scoring
Sun 19:48	`Approve` (automated)	3	Active / none	`ListingPublished`	Searchable; matched against **312** saved searches
Sun 19:55	—	3	Active	—	First contact; **14** conversations within 24 hours

Time (CET)	Command	Version	Publication state / overlay	Event	What should happen downstream
Sun 20:11–20:26	—	3	Active	—	A buyer account sends a fake Safe Deal payment link; Femke reports at 20:19; the *buyer's account* is suspended at 20:26. The listing is untouched: the enforcement targets an account, not this listing
Mon 14:02	`Reserve` by the Safe Deal order (pay at pickup)	8	Reserved / none	`ListingReserved`	Delisted from search; other conversations told it is reserved

Time (CET)	Command	Version	Publication state / overlay	Event	What should happen downstream
Mon 18:10	`MarkSold` (outcome: sold on platform, €225, order reference)	9	Sold / none	`ListingSold`	Removed from search within the removal SLA (it was gone at 18:11); **purged from digest candidate sets**; analytics records a sale; detail page shows “sold” with related bikes

Time (CET)	Command	Version	Publication state / overlay	Event	What should happen downstream
Sep 2027 (request Mon 6 Sep)	`Purge`, on Femke's erasure request	—	Purged	`ListingPurged`	Every registered copy erases her personal data, including the public sold page that was still live; the trust-and-safety case and Safe Deal records stay under their own retention (see System of Record and Read Models: Listing Data Management)

Three observations from the replay. The phishing episode, which dominates Femke's evening, **never touches the listing aggregate**, and it should not: it is an account and messaging case. A model that "blocked the listing" to stop the scam would have punished the victim. The reservation is set by the order, not by Femke, which is allowed for private Safe Deal sales and would be configurable for a dealer. And the sale price of €225 differs from the asking price of €240. It is recorded in the outcome, not written back as a price change, because writing it back would trigger a "price reduced" signal and price-drop alerts on an item that is no longer for sale.

The 07:00 digest on Tuesday is the Ghost Inventory failure in miniature. Under the new contract the digest service subscribes to `ListingSold`, `ListingExpired`, `ListingWithdrawn` and `ListingBlocked`, removes those listings from assembled digests, and re-checks state through the listing read API in the last minute before sending. Veldmark's March 2027 audit found the problem was not rare: in one week, about **41,000** Tradeyard NL listings that had ended in Yardstick were still present in at least one downstream copy 24 hours later. Most were in the digest cache and the recommendations candidate store, neither of which listened to Yardstick's removal hook.

Trade-offs

Beyond the storage choice above, four decisions shape every listing model.

Decision	Option A	Option B	When A is right	When B is right
Status model	One status field	Publication state plus enforcement overlay (and derived visibility)	A single-country, single-vertical start with in-house moderation and one team	Any marketplace with independent trust and safety, appeals, or regulation that treats demotion as a restriction, which in the EU is all of them

Decision	Option A	Option B	When A is right	When B is right
Delete semantics	Hard delete on seller request	Soft delete, archive, then purge on a retention schedule	Never for published listings	Always: appeals, disputes, fraud investigation and DSA complaint windows need the content; purge it when they close
Aggregate scope	Wide: attributes, media metadata, placements and stats in the listing row	Narrow: core plus references; stats, placements and media owned elsewhere	Early product, when one team owns everything	Once separate teams own media, monetization and analytics; the wide row becomes a shared database by another name
Ownership across brands	One group listing service for every brand	One listing service per marketplace, implementing one group lifecycle contract	Brands with the same vertical and similar rules	Brands in different verticals with different transitions and permissions

The retention row has a regulatory floor and ceiling. The Digital Services Act gives users at least six months to complain about moderation decisions, so the content behind a decision must survive that window; GDPR's storage-limitation principle requires personal data to be deleted when it is no longer needed. Tiered retention per data class resolves the tension: content snapshots for the appeal window plus a buffer, aggregates for longer, personal data purged on schedule (see Rules of the Marketplace:

DSA, DAC7 and Vertical Regulation and System of Record and Read Models: Listing Data Management, which owns erasure across copies).

Vertical Differences

The aggregate's shape is common across verticals; its lifecycle, units and invariants are not. Same Platform, Different Physics: How Listing Verticals Differ compares the verticals broadly; this table covers what changes in the listing model.

Dimension	General classifieds	Real estate	Automotive	Jobs	Recommerce (shipped goods)
Typical lifetime	Days to weeks; fixed lifetime with renewal	Weeks to months; ends with sale or let	Weeks; dealer feeds keep listings live until removed	Fixed validity (`validThrough`)	Until sold; long tail of stale items
States beyond the core	Reserved	Coming soon, under offer, sale agreed, let agreed	Reserved (platform- or dealer-set), deposit taken	Paused, filled	In checkout, shipped, returned
Who triggers transitions	The private seller, mostly	Agency software via feed	Dealer management system via feed; platform for reservations only with consent	Employer or applicant-tracking system	Seller and the order flow
Units	Single item, some quantities	Programs with units; rentals with availability dates	Single vehicle; fleets as many listings	Multiple openings	Quantities and variants

Dimension	General classifieds	Real estate	Automotive	Jobs	Recommerce (shipped goods)
Entity behind the listing	Usually none	Property (address, parcel)	Vehicle (VIN)	Employer and occupation	Catalog product when identifiable
Price semantics	Asking price, free, negotiable	Price with fees; rent plus charges; regulated disclosures	Price with prior-price rules for traders; finance quotes	Salary range, often optional	Fixed price, offers
Strictest invariant on publish	Category and prohibited-item rules	Mandatory disclosures (energy class, fees)	Vehicle identity and mileage	Mandatory fields for job search engines; no discriminatory content	Shipping and payment eligibility
Main ghost-inventory source	Sellers who never mark sold	Feeds that send no delete	Feed delay after a sale	Expired jobs not removed	Items sold elsewhere

At Veldmark the table translates into configuration, not forks. Tradeyard, Motorvane and Hearthly share the publication states and the overlay. Hearthly enables *under offer* and *let agreed* as sub-states of reserved and sold; Motorvane allows a platform-set reservation only for dealers who opt in. Each brand configures lifetimes per category and transition permissions per seller type. The Essen Golf from Motorvane shows the automotive ghost source: it sold on 29 April 2027, but the dealer feed's removal arrived 26 hours later, so the listing stayed active and contactable a full day after the car was gone. The aggregate cannot fix a late feed; it can make the gap measurable, by recording when the feed said sold against when leads stopped converting. Many Doors In: Forms, APIs and Bulk Feeds owns the feed reconciliation.

Failure Modes

The overloaded status column. One enum that mixes seller intent, platform state, moderation and monetization ("top", "highlighted", "pending_payment") grows until nobody can say which values are visible. Symptom: queries with long `status IN (...)` lists that differ between services.

Consumers that never hear the end. A copy that subscribes to publications but not to removals, or that caches candidates for hours, keeps sold listings alive. This is Ghost Inventory, and it lives in digests, recommendation candidate stores, partner syndication feeds and SEO caches more often than in search.

Bumps that reset age. If renewal or a paid bump writes `created_at`, recency ranking rewards the most frequent payer, "new today" filters fill with old items and time-to-sell analytics collapse.

Hard deletes. A seller deletes a listing that was reported as a scam an hour earlier. Without a soft-delete and retention design, the evidence for the statement of reasons, the appeal and any police request is gone.

Edits that bypass the gate. If only `Submit` triggers moderation, a scammer edits an approved listing into a different one. Materiality rules and re-scoring on edit close the gap; leaving them out is Moderation as an Afterthought at the aggregate level.

Expiry jobs that fall behind. A clock-driven transition implemented as a nightly batch job can fall days behind after an outage, and nobody notices because nothing alerts on listings that should have expired. Monitor the backlog of overdue transitions as a first-class metric.

Purges that miss copies. Erasure that deletes the row in the listing store but not the search document, the image renditions, the analytics extract and the partner feed is not erasure. The `ListingPurged` event exists to give every copy the same deadline.

The Chief Architect's View

Decide the lifecycle contract at group level; let brands own the services. At Veldmark, the decision record adopted at the architecture advice forum ahead of the contract's July 2027 go-live says that the Listing event contract v1 fixes the publication states, the enforcement overlay, the event names, the version semantics and the outcome codes for every brand. Each marketplace may keep its own listing service, which is the only realistic path while Yardstick still runs Tradeyard, but it must emit the contract. This is "share platforms, not products" applied to the most central object: the shared thing is the language, not the code.

Make ownership explicit for every dimension. The listing team owns publication states and invariants. Trust and safety owns the overlay and its decisions. Monetization

owns placements and entitlements. Data owns the outcome taxonomy's analytical definitions. Every disputed field gets one owner, recorded in a decision record, because the listing is where unowned fields accumulate.

Govern changes to the state vocabulary like an API change. Adding a state such as *under offer* touches search filters, alert rules, analytics, SEO and every partner feed mapping. Treat it as a versioned contract change with a deprecation window, not as a new enum value.

Watch four fitness functions.

Metric	Definition	Veldmark 2027 baseline	Target
Sold-to-gone latency	p95 time from `ListingSold` (or expired, withdrawn, blocked) to removal in every registered consumer	Unknown for most consumers; about 1 minute for search	Under 10 seconds for search; under 2 minutes for alerts and recommendations; under 15 minutes for partner feeds (the From Submitted to Searchable: The Publication Pipeline SLA card)
Lingering copies	Listings ended in the owning service but present in any copy after 24 hours	About 41,000 per week (Tradeyard NL)	Near zero, with each case explained
Overdue transitions	Listings past their lifetime still active	Not measured	Zero beyond one scheduler cycle
Outcome coverage	Share of ended listings with a known outcome	Not measured	Above 60% for private goods; above 90% for feed-driven verticals

Resist the god listing. Every team will want one more column on the listing because it is the easiest place to put data. Each column added for a reader's convenience is a copy

of someone else's fact. The aggregate should get narrower over the years, not wider; projections are where readers get the shapes they want.

Questions to Ask

1. Which service owns the listing aggregate, and can every other copy, including search, be rebuilt from it?
2. Can a seller command ever undo a trust-and-safety decision? Show the code path that prevents it.
3. Which timestamps exist, who may move each one, and does any bump or renewal change `created_at`?
4. For each consumer of listing events, what is its declared removal deadline, and where is the dashboard that measures it?
5. Which consumers assemble candidates in advance (digests, recommendations, feeds), and do they re-check state before sending?
6. Which edits are material, and do material edits re-enter moderation?
7. How is relisting handled: same ID, new ID, and how is delete-and-repost detected?
8. Where is price history stored, and can the platform show a trader's prior price for any reduction it displays?
9. What is the retention schedule per data class for ended listings, and which event tells every copy to purge?
10. How are feed-specific states, such as RESO statuses or an agency software's codes, mapped to the platform's lifecycle, and who owns that mapping?

From Aggregate to Category

The aggregate holds a category ID and a taxonomy version, and treats both as given. They are not. The category decides which attributes are required, which moderation rules apply, what a listing costs and where it appears, and the tree that defines categories has to change while millions of listings sit in it. Category Trees That Can Change: Designing Marketplace Taxonomies designs that tree.

Sources and Evidence

Company engineering and investor sources

- Seller-API documentation — mobile.de, n.d. Ad timestamps including `renewalDate`, full-document updates, renewal endpoint, insertion request IDs and the rule against parallel requests per ad.

- What is an item Bump? — Vinted Help, n.d. Bump duration and the "until the item sells" end condition.
- Inventory API overview — eBay Developers, n.d. Offer-to-listing publication and publish-time required fields.
- Taxonomy API specification v1.1.1 — eBay Developers, n.d. Required aspects enforced on listing and revising; product versus item aspects.
- Product-Based Shopping Experience playbook — eBay Developers, n.d. (covers 2018–2020). Catalog-association mandate and its rollback.
- Real Time Datafeed Specification v1.4.1 — Rightmove, 2023. `Agent_Ref` keys and snapshot reconciliation for lost removals.
- Full-year results FY26 — Auto Trader Group plc, 2026. Deal Builder reservations, retailer backlash and the "request a reservation" change.
- Auto Trader tweaks Deal Builder product, making reservations optional — Car Dealer Magazine, 2025. Trade-press account of the reservation changes.

Standards, regulation and documentation

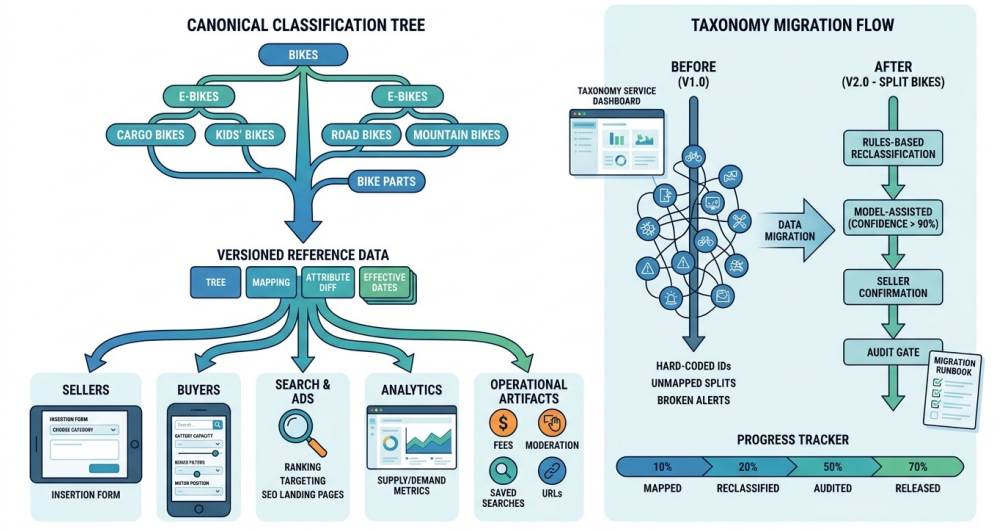
- Regulation (EU) 2022/2065 (Digital Services Act) — European Union, 2022. Statements of reasons for removal and demotion (Art. 17); complaint handling (Art. 20); trader traceability (Art. 30).
- Price Indication Directive — European Commission, n.d. Article 6a rules on announced price reductions, applying from 28 May 2022, and the Commission's 2021 guidance.
- StandardStatus field — RESO Data Dictionary 2.0, revised 2023. The eleven standard listing statuses in US real estate.
- Diagnostic de performance énergétique (DPE) — Service-Public.fr, 2026. Mandatory DPE display in ads and the 2026 and 2027 calculation changes.
- Product expiration — Google Merchant Center Help, n.d. 30-day expiry after the last update.
- Job posting structured data — Google Search Central, updated 2026. Removal of expired jobs and manual actions.
- Outbox Event Router — Debezium, n.d. Outbox table structure and aggregate-keyed messages.

Research and practitioner sources

- Productscape general-listings and real-estate domain models (author's sibling repository; digest in this book's research notes), 2026. Listing definitions, stream pain points (bait-and-switch edits, zombie listings, outcome capture) and the residuality finding that the listing must be separated from the thing listed.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

8. Category Trees That Can Change: Designing Marketplace Taxonomies



IN THIS CHAPTER, YOU WILL: Learn what a marketplace category actually controls, how to structure a taxonomy (trees, polyhierarchy, leaves, facets, country and navigation views), how to publish it as versioned reference data with mappings to external taxonomies, and how to run a category migration. You will follow Veldmark's split of "Bikes" into four leaves across four country trees.

WHY THE BUSINESS CARES: Categories decide what sellers are asked, what buyers can filter, what listings cost, which ads can be sold and which landing pages rank in Google. A marketplace that needs six weeks to add a category cannot follow demand, and one that changes categories carelessly loses alerts, organic traffic and comparable metrics overnight.

WHY ARCHITECTS SHOULD CARE: Category IDs end up hard-coded in fee rules, moderation rules, search mappings, SEO tables, dashboards and partner feeds. Unless the taxonomy is owned, versioned and distributed as data, every change becomes a coordinated release across a dozen systems.

KEY POINTS:

- **The category is a schema key, not a label.** It selects the attribute schema, validation, fees, moderation policy, facets, ranking and ad targeting for every listing in it. A taxonomy change is therefore a schema migration, and it should be designed, reviewed and released like one.
- **Separate classification from navigation and from countries.** One canonical classification tree, with stable IDs and leaf-only listing, is the source of truth. Browse trees, country views, localized labels, URL slugs and mappings to Google, schema.org or vehicle and job taxonomies are derived from it and can change without touching listings.
- **Ship taxonomy changes as data with a migration plan.** Each release carries the new tree, a mapping from every retired node, an attribute diff and effective dates. Merges map deterministically, splits need reclassification, and saved searches, URLs, partner feeds and analytics are migrated deliberately, not left to break.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Category Straitjacket** — Tradeyard's new categories took six weeks because category IDs were embedded in code, search schemas and SEO tables.
- **The Attribute Junkyard** — without category-scoped schemas, attributes accumulate as free-form keys that filters, facets and feeds interpret differently.
- **The Copy-Paste Country** — four Tradeyard country trees that cannot share models, filters or features are a fork in reference data.

Ask a product manager what a category is and you will hear "where the listing goes". Ask the systems and you get a different answer. The category decides which fields the insertion form shows, which are required, what an insertion fee costs, which moderation rules and models apply, which facets search offers, which ranking model scores the results, which ads can be targeted and which landing pages exist. **The category is the most widely shared key in a listing platform, and the one most often treated as content.**

That double nature is the source of the trouble. Categories look like editorial content, so they are edited like content: renamed in an admin tool, rearranged for a merchandising campaign, created by copying a neighbor. But they behave like schema, so every edit ripples into validation, pricing, search mappings, URLs, saved searches and dashboards. The result, after a decade, is the **Category Straitjacket**: a tree so entangled with code that nobody dares change it, next to an "Other" bucket that keeps growing because the tree no longer fits the supply.

Part II keeps the listing small and has it reference governed subdomains by ID. The category is the first of them and the one that selects the contract for the rest. The Listing Aggregate: Identity, Lifecycle and Invariants established that a listing records its category *and the taxonomy version it was classified under*, and took both as given. This chapter designs the tree and its versions: what a category controls, how to separate classification from navigation and countries, how to release the tree as versioned reference data, and how to run splits and merges as migrations. Attributes Without Chaos: Schemas, Dictionaries and Validation then designs the schemas that each category selects.

Setting the Scene at Veldmark: Six Weeks for a New Leaf

In September 2027, Noor Hassan, Veldmark's Group Chief Product Officer, wants Tradeyard to split bikes. Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), sells more used bikes on Tradeyard than almost anything else. E-bikes have become a large share of supply and an even larger share of value, but buyers cannot filter by battery capacity or motor position, and cargo bikes and kids' bikes sit in the same generic leaf as city bikes. Veldmark Ads has an e-bike retail-media campaign it cannot target. The request is four leaves under Bikes: **city bikes**, **e-bikes**, **cargo bikes and kids' bikes**.

Daan Visser, Tradeyard's engineering director, explains why the answer has always been "six weeks". Tradeyard runs four country category trees, built when each country launched: **1,420 leaf categories in the Netherlands**, **1,380 in Belgium**, **980 in Poland and 760 in Portugal**. A new category means:

- adding it in Yardstick's admin tool for each country, where it gets a different numeric ID per country;
- copying fee rules (Poland charges insertion fees in some categories) and moderation rules, which reference category IDs in PHP configuration;
- adding fields to the Solr schema for NL, BE and PT and to the Elasticsearch mapping for PL, and reindexing;
- creating SEO landing pages by hand in a table keyed by URL slug;
- asking the analytics team to update dashboards that filter by category *name*;
- telling the professional sellers whose feeds send category codes, some of which map by name.

Bike data is spread across **19 country-specific leaves** holding about **640,000** live listings: about 380,000 in the Netherlands, 120,000 in Belgium, 95,000 in Poland and 45,000 in Portugal. About **410,000** of them sit in generic "bikes" leaves that the split must divide; the rest are already in sport-bike and parts leaves that map one to one.

Ingrid Halvorsen, the Group Chief Architect, does not want to solve bikes. She wants the next request to take two days, which is the target Common Ground set for new categories. The group taxonomy and attribute service, one of Common Ground's shared capabilities, goes live in October 2027 with Tradeyard as its first consumer. The bikes split, live on **Monday 1 November 2027**, is its first real migration. Tomasz Wiśniewski, who heads the group search platform, and Priya Raman, Head of Data and ML, join her to design it. Katarzyna Nowak, Tradeyard Poland's general manager, joins with a condition: Poland's tree must stay recognizably Polish.

What a Category Controls

A **category taxonomy** is a controlled hierarchy of classes into which every listing is placed, together with the rules that hang off each class. A **category node** is one class in it; a **leaf** is a node with no children; a **path** is the chain from the root to a node. The taxonomy is load-bearing because every system in the table below keys on it.

What depends on the category	Example	Owning chapter
Attribute schema and validation	E-bikes require battery capacity; apartments require floor area and energy class	Attributes Without Chaos: Schemas, Dictionaries and Validation
Insertion form and its order	Vehicle forms start from make and model; goods forms start from photos	Many Doors In: Forms, APIs and Bulk Feeds
Fees and paid products	Insertion fees in Tradeyard Poland's vehicle and property categories; bump prices per category	Packages, Bumps and Quotas: Products, Pricing and Entitlements
Moderation policy	Prohibited items, pre-moderation for high-risk categories	Trust and Safety as a Platform
Search facets and sort	Frame size for bikes, mileage for cars	Search Architecture for Listings: Indices, Freshness and Scale
Ranking model and features	Separate models or features per vertical	Ranking for Two Sides: Relevance, Freshness and Paid Visibility

What depends on the category	Example	Owning chapter
Ad targeting and retail media	Segment "e-bikes, Netherlands" sold to manufacturers	Selling Attention: Display, Native and Retail-Media Advertising
SEO landing pages	"E-bikes in Utrecht"	Found by Google: Web, App and SEO Architecture for Listings
Saved searches and alerts	A stored query scoped to a category	Tell Me When It Appears: Saved Searches and Alerts
Analytics and market insights	Supply, demand and sell-through per category	Data Platform for Marketplaces: Events, Lakehouse and Data Products
Legal disclosures	Energy class in property ads; vehicle mileage rules	Rules of the Marketplace: DSA, DAC7 and Vertical Regulation

The public evidence agrees. eBay states that "every eBay listing must be listed in an eBay leaf category" and returns required, recommended and optional attributes per leaf (eBay Listing Metadata Guide, n.d.). Amazon retrieves attribute requirements per product type and marketplace (Amazon SP-API Product Type Definitions guide, updated 2026). Google needs a product's category to apply category-specific requirements (Google Merchant Center Help, n.d.). In each case **the category selects the contract** the listing must meet.

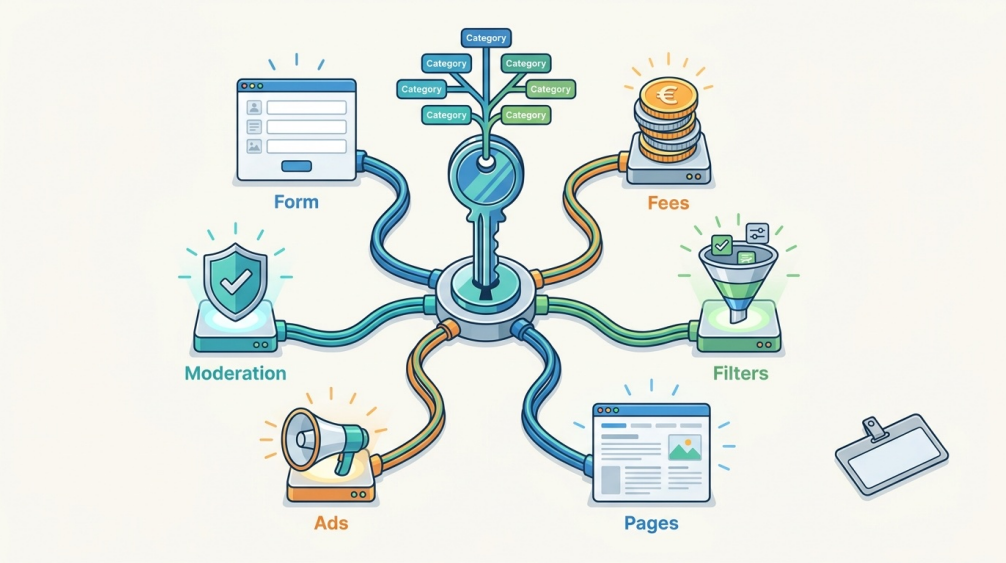


Figure 1: *A category is not a label but the key that selects the form, fees, moderation, filters, ad segments and landing pages a listing gets.*

Four Structural Choices

Four structural decisions shape every taxonomy. Each has a default that works for most marketplaces and conditions under which it does not.

Strict tree, polyhierarchy or facets

A **strict tree** gives each node exactly one parent. A **polyhierarchy** lets a node have several parents: "cargo bikes" under both Bikes and Family. **Faceted classification** avoids deep trees by classifying along independent dimensions (type, condition, material) instead.

Structure	How it works	Strengths	Weaknesses	Right when
Strict tree, leaf-only listing	One parent per node; listings only in leaves	One schema per leaf; unambiguous fees and policies; simple analytics	Items that belong in two places need a navigation alias	The classification tree of almost every listing marketplace

Structure	How it works	Strengths	Weaknesses	Right when
Polyhierarchy in classification	Nodes with several parents	Natural for browsing	Which parent's schema, fee and policy apply? Analytics double-count	Rarely in classification; acceptable in navigation
Shallow tree plus facets	Few categories; type and other dimensions as attributes	Fewer migrations; flexible filters	Schemas must vary by attribute value; weaker category-level policies	Real estate, where "property type" behaves like a facet; fashion sizes and colors

My default is a **strict classification tree with leaf-only listing**, plus polyhierarchy only in the *navigation* layer described below. Leaf-only listing matters because a listing in an interior node escapes the leaf's schema, fees and policies, and interior nodes then fill with under-described supply. The cost is that sellers must choose a leaf, which is why category prediction (later in this chapter) is part of the design, not an add-on.

Classification is not navigation

Amazon's public documentation describes two separate structures: **browse nodes**, the customer-facing navigation graph, and **product types**, which define attribute schemas. New items are placed "in the appropriate place(s) within the graph", plural, while the product type decides what must be filled in (Amazon SP-API, browse nodes and item type keywords, updated 2026). The lesson, which is my inference from that design, is that **classification for schema selection and classification for browsing are different concerns**. A marketplace that fuses them has to change its schemas every time merchandising wants a new browse path, such as "Back to school" or "Moving house".

I separate three layers:

1. **The classification tree** — canonical, strict, leaf-only, with stable IDs. It selects schemas, policies, fees and models. It changes slowly and only through releases.
2. **Navigation trees** — per brand and country, derived from the classification tree, with aliases, polyhierarchy and curated groupings. They change freely.

3. **Labels and slugs** — localized names and URL segments per country and language, properties of a node, never its identity.

Breadth, depth and the "Other" leaf

There is no correct number of leaves; there is a correct process for changing them. Two signals tell you the tree no longer fits the supply. The first is the **share of new listings in "Other" leaves**, which grows when sellers cannot find their item. The second is **attribute sparsity inside a leaf**: when one leaf holds items so different that its schema is either too thin to be useful or full of attributes that apply to a fraction of listings. Veldmark's 3,900 distinct attribute keys, 41% of them used on fewer than 0.1% of listings, are partly a taxonomy symptom. The generic bike leaves carried battery fields that applied to a third of their listings and wheel-size fields that applied to kids' bikes only.

A leaf is justified when it **changes a contract**: a different schema, fee, policy, facet set or ad segment. A distinction that changes none of those is an attribute value or a navigation grouping, not a leaf.

One tree or many: localization

Multi-country marketplaces choose between country trees, one global tree, or a canonical tree with country views.

Model	How it works	Strengths	Weaknesses
Country trees	Each country owns an independent tree	Full local fit; no coordination	Nothing shared: models, filters, ad segments and analytics must be mapped per country; the Copy-Paste Country in reference data

Model	How it works	Strengths	Weaknesses
One global tree, translated	Same nodes everywhere; labels translated	Maximum sharing	Forces local markets into one structure; local categories (a Portuguese tile category, a Polish agricultural one) have no home
Canonical tree with country views	One canonical set of nodes with stable IDs; each country enables a subset, with its own labels, slugs and navigation	Shared semantics where they exist, local fit where they do not	Needs governance: who may add a canonical node, and when a country-only node is allowed

I recommend the canonical tree with country views, because it meets Katarzyna's condition. Each country view decides **which canonical nodes exist** for it, **what they are called** and **how they are browsed**. A node can be country-specific when the distinction exists in only one market, but it is still a canonical node with a canonical ID, so that ML models, ads and analytics can treat it consistently.

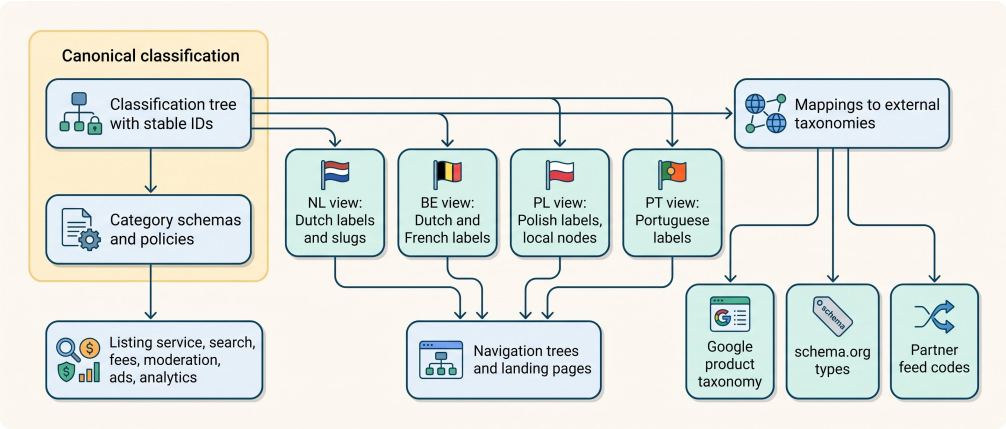


Figure 2: *Three layers. The canonical classification tree selects schemas and policies; country views, navigation trees and external mappings are derived from it. Labels, slugs, browse paths and partner codes can change without touching a single listing.*

The Taxonomy as Versioned Reference Data

Reference data is data that other data points at to give it meaning, changes rarely, and must be identical wherever it is used: currencies, country codes, categories. Treating the taxonomy as versioned reference data means four things.

Stable, meaningless IDs. A node's ID never changes and never encodes its position, name or country. Display names and URL slugs are separate localized properties. When a node moves, its ID stays. When its meaning changes enough that old listings would be misclassified under the new meaning, it is **retired and replaced** by a new node, not edited.

Immutable releases. The tree is published as numbered snapshots. Shopify's open-source Standard Product Taxonomy shows the convention: calendar-versioned releases such as `2024-07`, at most one stable release per quarter (Shopify product-taxonomy repository, current version 2026-08). eBay attaches a `categoryTreeVersion` to each marketplace tree and tells integrators to cache it and compare: "If a new version is detected, call getCategoryTree or getCategorySubTree to fetch the updated categories" (eBay Listing Metadata Guide, n.d.).

Every release ships its migration. A release is not only a new tree. It needs four artifacts:

Artifact	Content	Public precedent
New tree	Nodes, parents, country presence, labels, slugs	eBay category trees; Shopify releases
From⟶to mapping	For every retired node: its successor, or "no successor"	eBay `getExpiredCategories` returns `fromCategoryId`/`toCategoryId` pairs for combined and split categories
Attribute diff	New, changed and removed attributes per leaf, with requiredness	eBay's Taxonomy SDK uses bulk aspect files "to surface changes (new, modified, and removed entities)"
Effective dates	When each new requirement becomes mandatory	eBay's `expectedRequiredByDate`, "only an approximate date"

The eBay precedents come from its Taxonomy API specification (eBay Taxonomy API specification v1.1.1, n.d.). Shopify goes further and publishes **mappings between its**

own versions, such as 2022-02 to 2024-07, and to other taxonomies, including Google's (Shopify product-taxonomy integrations, n.d.). That makes the migration a data product that every consumer can apply in the same way.

Consumers pin versions. A listing records the version it was classified under. A search mapping, a fee table, an ML model and a partner feed each declare which version they understand. During a migration, two versions are live, and the taxonomy service answers "what is node X in version Y?" for both. Google shows the receiving side of this: it accepts categories from older taxonomy versions and translates them to the latest (Google Merchant Center Help, n.d.).

The taxonomy service

At Veldmark the group taxonomy and attribute service is a small product with four interfaces.

Interface	Consumers	Behavior
Authoring and review	Category managers per brand and country; the taxonomy team	Drafts of a release; change requests with impact analysis (listings, saved searches, URLs, fees, models affected); approval workflow
Release publication	All consumers	`TaxonomyReleased` event on the group backbone, carrying the release ID and links to the four artifacts
Read API and snapshots	Listing service, search indexer, insertion forms, pro APIs	Full snapshot per release; node lookup by ID and version; mapping lookup; served from cache with long TTLs
Embedded client	Services on the hot path	A library that loads the snapshot into memory, so validating a listing never calls the network

A listing service that calls the taxonomy service for every insertion has made the

taxonomy a runtime dependency of supply. A snapshot cached in memory and refreshed on `TaxonomyReleased` makes it a dependency only of change.

Mapping to External Taxonomies

A marketplace never owns the only taxonomy its data must fit. It must emit categories to Google and social catalogs, structured data for search engines, codes for partners, and it must ingest categories from dealer systems, agency software and feeds. The rule I use: **external taxonomies are mapping targets, never the internal model.**

External taxonomy	What it is	How marketplaces use it	Caution
Google product taxonomy	Tree of >-separated paths with numeric IDs	Emitting product feeds for Shopping ads	The published file's header still reads version 2021-09-21 (Google taxonomy file); Google now assigns categories automatically and treats `google_product_category` as an override in specific cases (Google Merchant Center Help, n.d.)
schema.org	Web structured-data vocabulary (`Product`, `Vehicle`, `Accommodation`, `RealEstateListing`, `JobPosting`, `Offer`)	Generating structured data at the page edge	An output format, not a category tree; `RealEstateListing` is still in the "new" area (schema.org, V30.1, 2026)

External taxonomy	What it is	How marketplaces use it	Caution
Shopify Standard Product Taxonomy	Open-source categories, attributes and values	A reference for goods categories and a source of mappings	Built for new goods; thin on used and local categories
GS1 Global Product Classification	Four levels down to a "brick", with eight-digit codes	New-goods catalogs with barcodes	Rarely useful as the primary tree for private classifieds
Vehicle taxonomies	Make → model → generation → derivative, licensed from data providers or built by portals	The backbone of automotive classification	Changes monthly as new models appear; license terms may restrict redistribution
Occupation taxonomies (ESCO, O*NET)	Standard occupation and skill classifications	Classifying job listings	Mapping from free-text titles is probabilistic

External structures change under you, as two cases show. First, Google added vehicle-listing structured-data documentation in October 2023, added phase-out notices in June 2025 and removed the documentation in September 2025 because these types "are no longer shown in Google Search results" (Google Search Central, vehicle listing structured data changelog, 2025). A portal that had shaped its vehicle model around Google's schema would have had to reshape it twice in two years.

Second, vehicle taxonomies change with the market. Auto Trader UK's developer documentation describes its taxonomy as vehicle types, makes, models, generations and derivatives, and its release notes record model splits that "will impact derivatives" (Auto Trader developer portal, n.d.). In automotive, the category tree beneath "Cars" is effectively a licensed or curated taxonomy that the marketplace maps onto, and its changes are migrations like any other.

For jobs, the European Commission's ESCO classification lists 3,039 occupations and 13,939 skills in 28 languages in version 1.2.1 (ESCO, "What is ESCO", updated 2025). A jobs marketplace usually keeps a shallow internal category tree for browsing and fees and maps listings to occupation codes as an enrichment, because a job title is not reliably classifiable by the employer.

Mappings live in the taxonomy service as versioned tables: `(internal node,`

`internal version) → (external code, external version, mapping type)`, where the type is *exact*, *broader* or *narrower*. A narrower mapping (many internal nodes to one Google category) loses nothing on output; a broader one (one internal node to several partner codes) needs an attribute rule to decide. Inbound mappings, from a dealer system's codes to internal nodes, are owned by Many Doors In: Forms, APIs and Bulk Feeds but stored here, so a taxonomy release can update them in the same change.

Category Migrations: Splits, Merges, Moves and Retirements

A **category migration** is any taxonomy change that alters which node an existing listing, saved search, URL, rule or metric points at. Changing a label is not a migration. Everything else is.

Operation	Listings	Saved searches and alerts	URLs and SEO	Fees, policies and models	Analytics	Partner feeds
Rename (label or slug only)	Unchanged	Unchanged	New slug; 301 from the old slug	Unchanged	Unchanged (IDs stable)	Unchanged
Move (new parent)	Unchanged (ID stable)	Unchanged	Path-based URLs change; 301	Inherited rules may change: review	Roll-ups change; restate history	Unchanged
Merge (A and B into C)	Deterministic: all to C	Rewrite to C	301 from A and B to C	C's rules apply; check fee changes	Mapping view keeps A+B history comparable	Accept old codes, map to C

Operation	Listings	Saved searches and alerts	URLs and SEO	Fees, policies and models	Analytics	Partner feeds
Split (A into C, D, E)	**Reclassification needed**	Expand to C+D+E, or ask the user	Keep A as parent hub or 301 to the best child	New rules per child	History of A cannot be split exactly; estimate or restate	Old code maps to a default child until partners update
Retire (no successor)	Listings must move or end	Notify users	410 or 301 to parent	Remove rules	Freeze history	Reject old code with a clear error
Add (new leaf)	Optional reclassification from parent or "Other"	None	New landing pages if demand exists	New rules	New series	New code

Merges are deterministic; splits are lossy. eBay's expired-category mapping returns one target per retired category and explicitly covers "combined categories and split categories" only; for a split, a single from⟶to pair can name only a default target (eBay Taxonomy API specification, n.d.). Every listing in a split category must be reclassified, and reclassification at scale is a machine-learning problem.

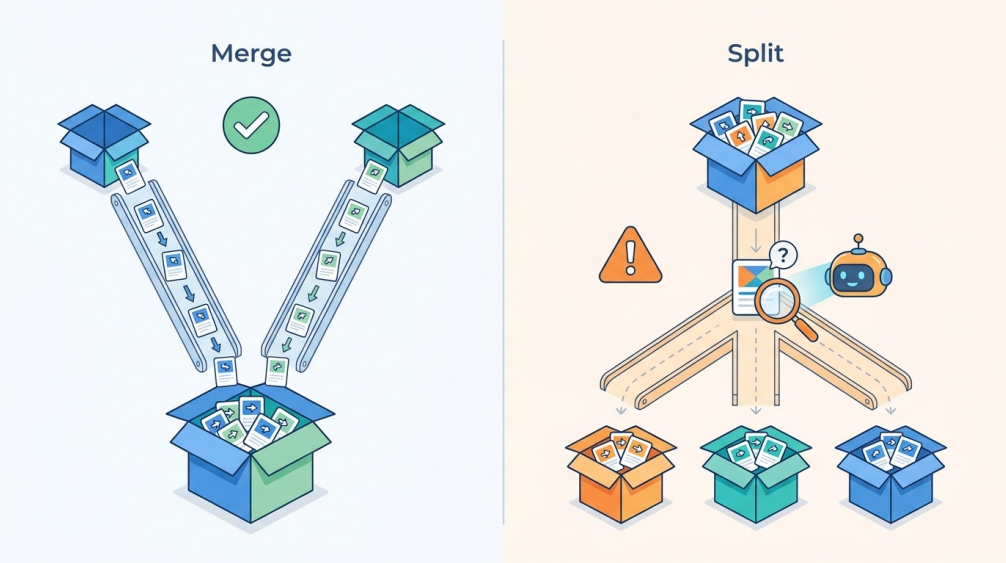


Figure 3: *Merges are deterministic, because every listing has one destination; splits are lossy, because every listing in the split category must be reclassified.*

Reclassification at scale: the Mercari pattern

The best public case of a mass reclassification is Mercari's. In a 2024 engineering post, Mercari described renewing its category structure, which greatly increased the number of categories and required reassigning existing items — more than three billion of them — without labeled data for the new structure (Mercari Engineering, "LLM-based Approach to Large-scale Item Category Classification", 2024). Classifying every item with GPT-3.5 Turbo was estimated at about USD 1 million and about 1.9 years. Mercari instead used the large model to label a sample of several million items, choosing among about ten generated candidate categories per item, and then propagated those labels to the full corpus with a k-nearest-neighbor classifier over embeddings of name, description, metadata and the item's *original* category. An open-source multilingual embedding model matched the commercial embeddings on Mercari's internal evaluation. Mercari published no accuracy figure; the post says only that kNN "performed satisfactorily".

The pattern generalizes: **a large model as labeler, a cheap model as propagator, the old category as a feature**. The old category is the most informative input, because a split only has to choose among the old node's children. Two additions make it safe in a live marketplace: a confidence threshold below which listings are not moved silently, and a human audit sample whose accuracy gates the run.

The migration runbook

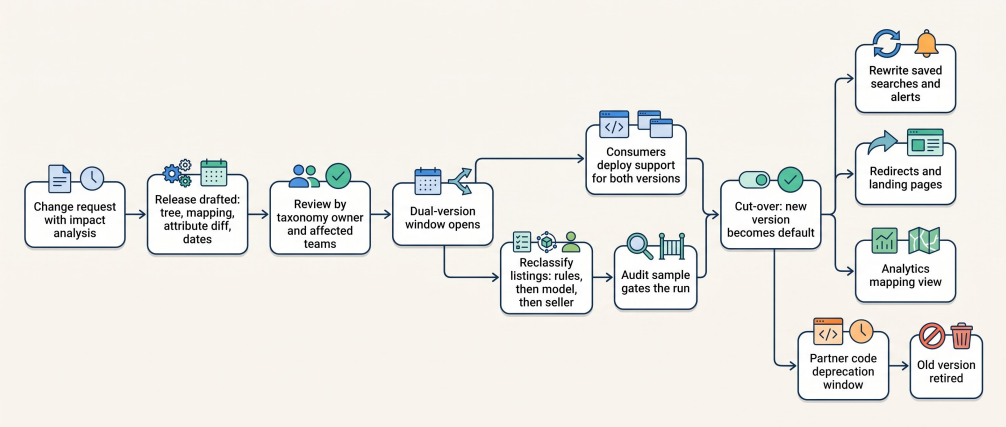


Figure 4: *A category migration as a release with a dual-version window. Consumers learn both versions before cut-over; listings move by rules first, then by model above a confidence threshold, then by asking sellers; saved searches, URLs, analytics and partner codes each have their own step.*

Seven steps, each with a concern that teams commonly forget.

1. **Version the taxonomy**. Immutable snapshots; listings record the version they were classified under.
2. **Publish the mapping**. Merges deterministic; splits model-assisted; retirements explicit.
3. **Open a dual-version window**. Search, APIs and fee rules accept both old and new IDs until cut-over. Partner feeds keep sending old codes for months, so the inbound mapping outlives the window.
4. **Reclassify**. Rules first (existing attribute values, unambiguous keywords), then a model above a confidence threshold, then the seller for the remainder, prompted at their next edit. The remainder sits in a default child with a low-confidence flag rather than in a dead node.
5. **Backfill attributes**. New mandatory attributes start as recommended with an expected-required date and are enforced only on create and revise, following eBay's pattern. A split that makes battery capacity mandatory for 200,000 existing e-bike listings on day one would block every one of their sellers from changing a price.
6. **Migrate URLs and saved searches**. Saved searches are user data: rewrite them, never drop them. Use permanent server-side redirects for retired category pages; Google recommends a permanent redirect "whenever possible" when a URL changes (Google Search Central, redirects, updated 2026). New landing pages appear only where there is enough inventory; Google's faceted-navigation guidance points the same way, recommending 404 for filter combinations with no results (Google Search Central, faceted navigation, n.d.).

7. **Keep analytics comparable**. A mapping view in the warehouse expresses old categories in new terms (or the reverse), so year-on-year category metrics survive. Analysts are the consumers most often forgotten.

Category Prediction

Category prediction suggests or assigns a listing's category from its title, description, photos or structured data. It matters twice: at insertion, where choosing the right leaf decides everything that follows, and in migrations, where it reclassifies existing listings.

Public evidence is mostly product-level. eBay's 2023 "magical listing" announcement described AI drafting a listing from a photo, including "detailed category/sub-category" (eBay, "Magical listing tool", 2023). German trade press reported in November 2025 that Kleinanzeigen's app suggests a title, description and category from a photo across most categories, excluding complex-input categories such as vehicles and intangibles such as jobs and services, and that sellers can accept suggestions fully, partially or not at all (mobiFlip, 2025). I found no first-party engineering description of either system.

Four design rules follow, and they are my recommendations rather than any company's documented practice.

- **Suggest the top three; let the seller choose**. A wrong silent assignment puts the listing under the wrong schema, fee and policy. A seller who confirms a suggestion has also confirmed the contract.
- **Make the model taxonomy-version-aware**. A model trained on version 2027.10 cannot predict leaves that exist only in 2027.11. Either retrain per release or, better, pass candidate categories as input, as Mercari did with generated candidates and as its 2024 attribute-extraction work did by passing attribute keys at request time (Mercari Engineering, "Fine-Tuning an LLM to Extract Dynamically Specified Attributes", 2024).
- **Measure acceptance and correction, not only accuracy**. The useful labels are the seller's choice among suggestions and later recategorizations by moderators or the seller.
- **Treat category manipulation as a fraud signal**. Sellers who choose a cheaper or less-moderated category for a high-risk item are a known abuse. A large disagreement between the predicted and chosen category on a high-risk leaf should raise the listing's risk score (see Scams, Fakes and Account Takeovers: Fraud Prevention for Listings).

In Practice: The Bikes Split at Veldmark

The bikes split was decided on **4 October 2027**, released as group taxonomy version **2027.11** and made effective on **Monday 1 November 2027**. It was the first release to go through the new taxonomy service, and it doubled as the first harmonization step for the four Tradeyard trees.

The tree before and after

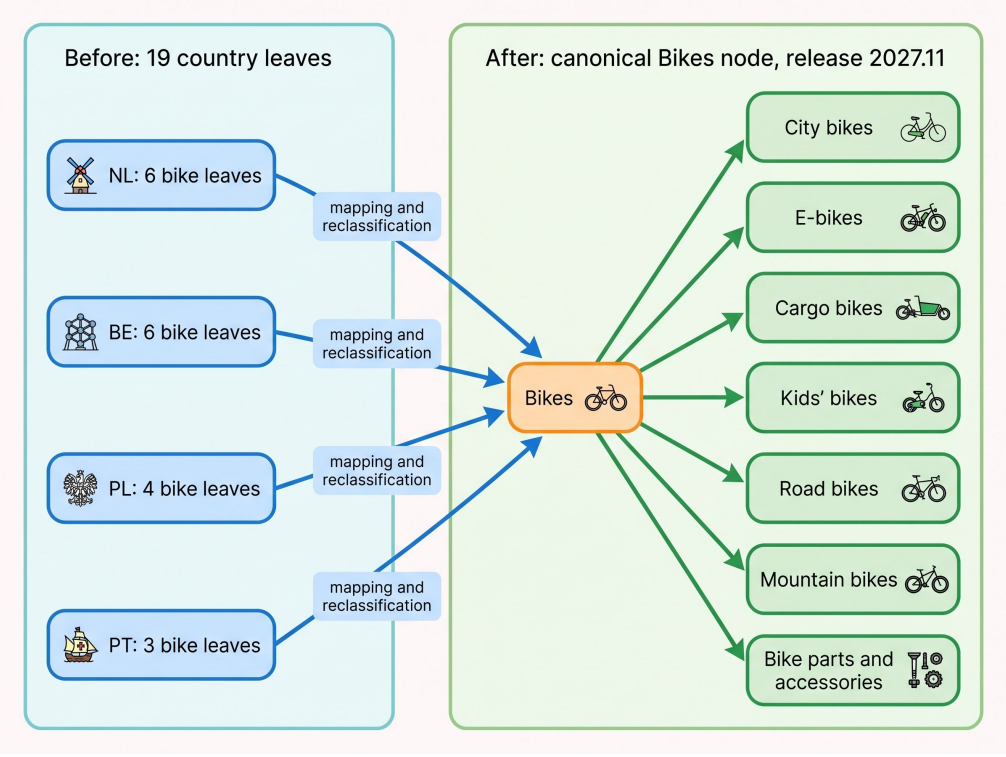


Figure 5: *The bikes split. Nineteen country-specific leaves become one canonical Bikes node with seven leaves present in all four country views. Road, mountain and parts leaves map one to one; the generic leaves are divided into city, e-bikes, cargo and kids' bikes.*

Every country view enables all seven leaves, with local labels: *Stadsfietsen* and *E-bikes* in the Netherlands, *Vélos de ville* alongside the Dutch labels in Belgium, *Rowery miejskie* in Poland, *Bicicletas de cidade* in Portugal. Polish navigation keeps bikes under its existing "Sport i hobby" browse node, so Poland's buyers see their familiar structure while the classification underneath is shared.

Reclassifying 410,000 listings

The 230,000 listings in sport-bike and parts leaves mapped one to one. The 410,000 in generic leaves went through three tiers.

Tier	Method	Listings	Share
Rules	Existing attribute values (NL and BE listings with "electric: yes"), unambiguous terms in titles ("bakfiets", "rower dziecięcy"), kids' wheel sizes of 20 inches or less	About 189,000	46%
Model	Text-and-image classifier trained on 60,000 listings labeled by a large language model and checked by moderators; moved only above 0.92 confidence	About 205,000	50%
Seller	Placed in City bikes with a low-confidence flag; seller asked to confirm at next edit or renewal	About 16,000	4%

A moderator audit of 3,000 automatically moved listings found **96.8%** correctly classified; the run had been gated at 95%. The errors clustered in one place the rules had not anticipated: e-cargo bikes, which the model sent to E-bikes and the category managers wanted in Cargo bikes. The fix was a rule (cargo wins over electric) and a new attribute, `electric assist`, on Cargo bikes, which is the general lesson: **when two leaves both claim an item, decide which distinction is the category and which is an attribute**.

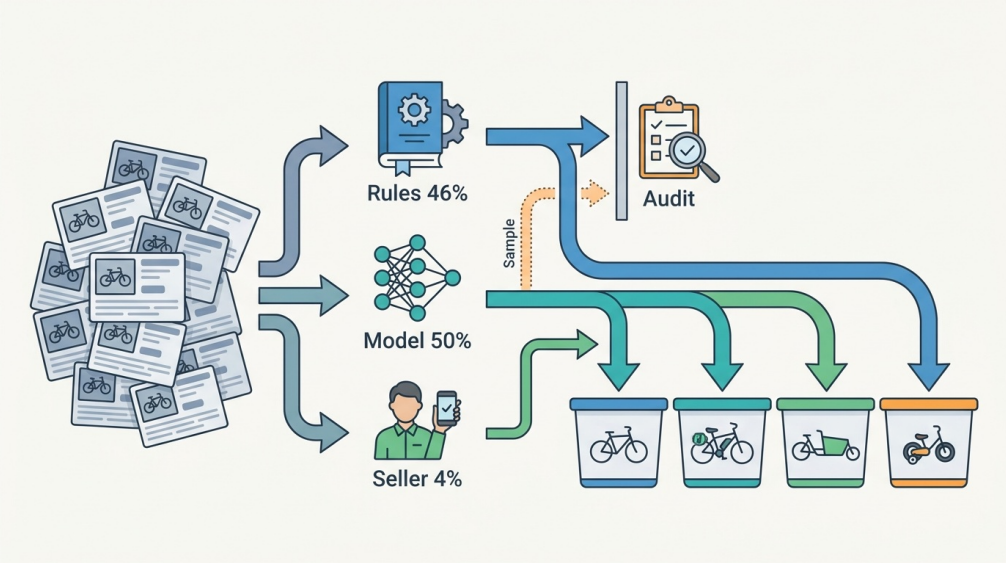


Figure 6: *Veldmark moved 410,000 generic bike listings by rules (46%), a model above 0.92 confidence (50%) and seller confirmation (4%), with a moderator audit gating the automated moves.*

Femke's city bike from March, long since sold and archived, illustrates the version rule from The Listing Aggregate: Identity, Lifecycle and Invariants: its archived record still says it was classified under the old Dutch generic leaf in the old version, and the analytics mapping view reports it as a City bikes sale. Nobody rewrote history; the mapping translates it.

Everything else that moved

Concern	What Veldmark did	Number
Saved searches	Searches on a generic bike leaf were rewritten to the union of the four new leaves, preserving their results; users were offered a one-tap narrowing	About 1.3 million rewritten

Concern	What Veldmark did	Number
New attributes	E-bikes: battery capacity (Wh), motor position; Cargo bikes: load type, electric assist; recommended now, required on create and revise from **1 March 2028**	5 new attributes
Landing pages	Bikes hub kept at its URL; new child pages created only for city × leaf combinations with at least 30 live listings	About 2,900 new pages
Redirects	Retired country-leaf URLs redirected permanently to their canonical successors	About 2,300 URLs
Fees and products	Bump prices per leaf inherited from the old generic leaves; no fee change at cut-over	—
Search	One new mapping version per engine with the new leaves and attributes; old IDs accepted until 31 January 2028	Solr and Elasticsearch both
Analytics	Warehouse mapping view from old leaves to new, with model-estimated splits for history before November	—
Pro feeds	Shops and bike dealers sending old codes mapped to City bikes by default, with a feed warning; codes accepted until 30 April 2028	410 professional feeds
Veldmark Ads	E-bike segment available from 1 November	—

The elapsed time from decision to cut-over was four weeks, but the *taxonomy* change took two days of work in the service; the rest was reclassification, its audit, the saved-

search rewrite and SEO review. Ingrid's measure of success is the second request. In December 2027, Tradeyard Netherlands adds a "Fatbikes" leaf under Bikes, with no split of existing listings, in **two days**.

Harmonizing four trees

The harmonization program applies the bikes method to the rest of the tree. The canonical tree's first release, 2027.10, contains **1,640** leaf nodes; each country tree became a view over it, with country-only nodes allowed where the distinction is local. Areas are harmonized in order of value: first where shared ML models, ad segments or Motorvane and Hearthly integration benefit most (vehicles, electronics, bikes, furniture), last where categories are small and local. The rule for a country-only node is simple: it must change a contract (schema, fee, policy or ad segment) that no canonical node provides.

Trade-offs

Decision	Options	Choose the first when	Choose the second when
Reclassify a split	Silent model reclassification vs ask every seller	Confidence is high, the audit gate passes and the schema difference is small	The categories carry different fees or policies, so a wrong move changes what the seller pays or what rules apply
Saved searches after a split	Expand to the union of children vs ask the user to choose	Always as the default: it preserves results	Never as the only step; asking afterwards improves alert relevance
Old category URL	Keep as a parent hub vs redirect to a child	The parent still exists and has inventory	The node is retired or its inventory moved almost entirely to one child

Decision	Options	Choose the first when	Choose the second when
Taxonomy ownership	Central taxonomy team vs each brand owns its tree	Brands share verticals, models and ad segments	Brands are in unrelated verticals with no shared consumers
External taxonomy as backbone	Adopt a licensed or public taxonomy vs map to it	The vertical has an authoritative, licensed structure (vehicles) and you accept its change cadence	The external taxonomy was built for another purpose (Google Shopping, new-goods catalogs)
Category vs attribute	New leaf vs new attribute value	The distinction changes schema, fees, policy or ad segment	It changes only filtering or display

Vertical Differences

Dimension	General classifieds	Real estate	Automotive	Jobs
Tree shape	Wide and deep; thousands of leaves	Shallow: transaction type (sale, rent, new build) by property type	Shallow top, deep vehicle taxonomy beneath (make, model, generation, derivative)	Shallow browse tree plus occupation codes
Who chooses the category	Private seller, helped by prediction	Agency software via feed; category follows legal property type	Derived from the vehicle identity (VIN or registration decoding)	Employer, then normalized by enrichment

Dimension	General classifieds	Real estate	Automotive	Jobs
Change cadence	Quarterly or as demand shifts	Rare; driven by regulation and new property products	Monthly, as manufacturers release models and derivatives	Yearly for codes; titles change constantly
External anchor	Google and Shopify taxonomies for output	National data standards and disclosure rules	Licensed vehicle data providers	ESCO, O*NET
Typical migration	Split of a popular generic leaf	New property type or regulatory category	Model split or rename in the vehicle taxonomy	Code version upgrade
Main risk	"Other" growth and misclassification	Disclosure rules tied to the wrong type	Valuation and price-rating models keyed on derivatives that changed	Probabilistic mapping errors that hide jobs from search

At Veldmark, the three verticals share the taxonomy *service* and its release process, not the tree contents. Motorvane's vehicle classification is a mapping over a licensed vehicle taxonomy, released when the provider releases; Hearthly's property types follow French and Spanish practice and change almost never; Tradeyard's goods tree is the one that changes most and benefits most from shared tooling. Tradeyard Poland's cars section and Motorvane Poland, which compete for the same dealers, now classify cars against the same vehicle taxonomy, which makes their overlap measurable for the first time.

Failure Modes

Category IDs in code. `if ($categoryId == 1234)` in a fee rule or a moderation rule is the root of the Category Straitjacket. Every rule keyed on a category should read the category's properties from the taxonomy service instead ("requires pre-moderation", "insertion fee band B").

Slugs or names as identity. Dashboards that filter by category name and URLs that

resolve by slug break on the first rename. Identity is the ID; everything else is a property.

Deleting nodes. A deleted node leaves orphaned listings, broken URLs, dead saved searches and holes in analytics. Retire nodes; never delete them.

Splits without saved-search migration. Alerts on the old node silently match nothing, and users conclude the marketplace has no supply. They do not complain; they leave.

Mandatory attributes on day one. A new required field enforced on existing listings blocks sellers from routine edits and pushes them to delete and relist, which damages freshness metrics and creates duplicates.

Unmapped partner codes. A dealer feed that sends a retired code fails validation, and the listings it carries go stale or disappear. Inbound mappings must outlive the dual-version window.

Analytics discontinuity. After a split, the old category's series stops and four new ones start. Without a mapping view, every year-on-year report becomes wrong, and someone eventually concludes the category collapsed.

Ranking and price models keyed on retired nodes. A model that uses category as a feature, or is trained per category, degrades silently after a migration. Model owners must be on the release's consumer list and retrain or remap before cut-over.

The ever-growing "Other". When the tree does not fit the supply, sellers choose "Other", and listings there have no useful schema. Watch the share; it is the earliest signal that the tree needs a release.

The Chief Architect's View

Run the taxonomy as a product with an owner. At Veldmark, a small taxonomy team inside Common Ground owns the canonical tree, the service and the release process; brand and country category managers own their views and propose canonical changes. The team's customers are the listing services, search, trust and safety, monetization, ads, data and SEO, and its roadmap is driven by their change requests.

Classify changes by blast radius. Not every change needs a committee.

Change class	Examples	Who decides	Lead time target
Cosmetic	Label, slug, navigation grouping, landing-page curation	Country category manager	Same day

Change class	Examples	Who decides	Lead time target
Additive	New leaf with no reclassification; new optional attribute	Taxonomy team review	Two days
Structural	Split, merge, move with reclassification; new required attribute	Taxonomy team plus affected owners; decision record	Two to four weeks, dominated by migration
Cross-group	Canonical changes affecting several brands; adopting an external taxonomy	Architecture advice forum	As needed

Keep releases regular and boring. A monthly release train with a published calendar lets consumers plan. Emergency releases exist for legal changes and are rare.

Measure the taxonomy's fitness. Five metrics tell you whether the tree serves the marketplace.

Metric	What it reveals	Veldmark direction
Lead time for a new category	Whether the straitjacket is gone	Six weeks (2027 baseline) to two days
Share of new listings in "Other" leaves	Fit between tree and supply	Down, per country
Category correction rate	Share of listings recategorized by sellers, moderators or models after publication	Down, per leaf
Prediction acceptance	Share of insertions where the seller chose a top-three suggestion	Up
Consumers on the current release	Whether downstream systems keep up	All within one release

Decide what is shared before you share it. The canonical tree is a tool for shared semantics, not a mandate for identical products. Country views exist so that Poland

can stay Polish. The architectural decision is that **identity and contracts are shared**, **presentation is local**, which is the "share platforms, not products" principle applied to reference data.

Questions to Ask

1. Where is the canonical list of categories, who owns it, and can you produce the tree as it was on a given date?
2. Do category IDs appear in code, configuration or dashboards, and how many places would a split touch?
3. Are navigation trees and URL slugs separate from the classification tree?
4. What does a taxonomy release contain, and do consumers receive the from→to mapping, the attribute diff and effective dates as data?
5. How are splits reclassified, what confidence threshold applies, and what audit gates the run?
6. What happens to saved searches, alerts and landing pages when a category is split or retired?
7. How long are old category codes accepted from partner feeds, and what error do partners get afterwards?
8. Which ML models use category as a feature or are trained per category, and are they on the release's consumer list?
9. How do country trees relate to each other, and can a shared model or ad segment address "e-bikes" in every market?
10. What share of new listings lands in "Other", and who acts when it rises?

From Categories to Attributes

Categories decide which schema applies; they do not define the schemas. The battery capacity that e-bikes now require, the energy class on every French apartment and the 31 attributes a VIN decode fills for the Essen Golf all need a dictionary, types, units, enumerations and governance of their own. Attributes Without Chaos: Schemas, Dictionaries and Validation designs them.

Sources and Evidence

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- RealEstateListing, Vehicle, Offer — schema.org, V30.1, 2026. Structured-data types as output formats.
- What is ESCO — European Commission, updated 2025. Occupation and skill counts in v1.2.1.
- Redirects and Google Search — Google Search Central, updated 2026. Permanent server-side redirects for changed URLs.
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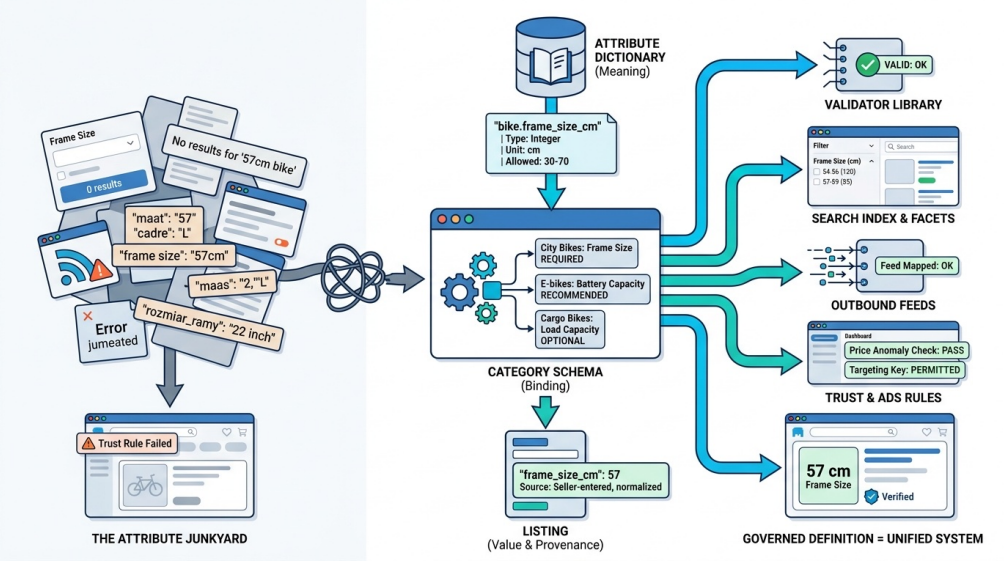
Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

9. Attributes Without Chaos: Schemas, Dictionaries and Validation



IN THIS CHAPTER, YOU WILL: Learn how to separate an attribute's meaning (the dictionary) from where it is required (the category schema) and from the value a seller gives (the listing). You will compare the storage options, design a schema service that compiles one definition into forms, validators, index mappings and feed mappings, and follow Veldmark as it turns 3,900 attribute keys into one dictionary for three verticals.

WHY THE BUSINESS CARES: Buyers filter on attributes before they read a single title. When the "frame size" filter is empty on most bike listings, buyers do not find the bikes, so sellers sell less and pay for visibility that does not convert. When a French listing lacks its energy class, the problem is no longer conversion: it is a legal defect in every ad.

WHY ARCHITECTS SHOULD CARE: Attributes are the most widely shared data in a listing system. Forms, feeds, validation, search, ranking,

trust rules, ads targeting, SEO pages and analytics all read them. An attribute without a governed definition gets a different meaning in each of those systems. A change without a compatibility rule breaks them one at a time, often weeks later.

KEY POINTS:

- **Separate three things that most systems blur: the definition, the binding and the value.** The attribute dictionary says what an attribute means, its type, unit and allowed values. The category schema says where it is required, recommended or optional. The listing holds the value, with its provenance. Each layer has a different owner and changes at a different rate.
- **Compile the schema once and distribute the results.** The same versioned category schema should generate the insertion form, the validator used by every ingestion channel, the search mapping, the facet configuration and the outbound feed mappings. Systems that each interpret attributes on their own drift apart, and that drift is the Attribute Junkyard.
- **Evolve attributes the way you evolve an event schema: with compatibility rules.** Adding an optional attribute is safe. Making an attribute required needs advance notice, an effective date and enforcement on create and revise. Changing an attribute's meaning, type or unit means creating a new attribute, never editing the old one.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Attribute Junkyard** — Without a dictionary, every country, team and feed adds its own keys, and the same fact ends up stored under many names in many formats.
- **The Category Straitjacket** — When attributes are columns or code, every new category or attribute needs a release and a migration, so taxonomy change stalls.
- **Feed Roulette** — Partner feeds that are validated against nothing, or against rules that differ from the web form, deliver attributes that no filter or facet can use.

A listing title says "Gazelle Chamonix, nice condition." A buyer searching for a city bike with a 57 cm frame, three gears and a front basket cannot use that title. They use **attributes**: the structured, typed facts about the item that drive filters, facets, sorting, ranking, validation and the feeds that carry the listing elsewhere. In most listing marketplaces, attributes do more work than any other part of the listing, and they get less design attention than almost any other part.

The neglect has a pattern. Attributes start as a few columns; a second vertical adds a key-value table; countries and feeds add their own keys. Years later the platform has thousands of keys, many of them duplicates in different languages, and nobody can say which systems depend on which. That is **The Attribute Junkyard**, and its cost shows up far from its cause: in empty facets, in duplicate detection that cannot compare listings, and in trust rules that never fire because the attribute they check is spelled four ways.

Part II models the listing as a small aggregate that references governed subdomains, and attributes are the subdomain that the most systems read. This chapter gives them a **contract**: the attribute dictionary, the category schema, the storage model, the validation and the governance that keep one definition true across every system that reads it. It builds on The Listing Aggregate: Identity, Lifecycle and Invariants, which holds attributes as a validated value collection, and on Category Trees That Can Change: Designing Marketplace Taxonomies, which owns the category tree each schema binds to. Later chapters depend on it: Many Doors In: Forms, APIs and Bulk Feeds validates feeds against it, From Raw Supply to Usable Listings: Normalization and Enrichment cleans values toward it, and Search Architecture for Listings: Indices, Freshness and Scale turns it into filters and facets. Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), provides the worked example.

Setting the Scene at Veldmark: Eleven Ways to Say "Frame Size"

It is late August 2027. Veldmark's group taxonomy and attribute service is due to go live for Tradeyard in October, and the bikes split (city bikes, e-bikes, cargo bikes and kids' bikes as separate leaf categories) is planned for **Monday 1 November 2027**. Ingrid Halvorsen, the Group Chief Architect, has asked Daan Visser, who owns Tradeyard's Yardstick monolith, and Tomasz Wiśniewski, who runs the group search platform, a simple question: what will the new "frame size" facet on the city-bikes page show?

The answer comes back as a spreadsheet. Tradeyard runs four country category trees (NL with 1,420 leaf categories, BE 1,380, PL 980, PT 760) and stores **3,900 distinct attribute keys**. **41% of those keys are used on fewer than 0.1% of listings**. Frame size alone appears under eleven keys: `framemaat`, `frame_size`, `frame size`, `maat`, `rozmiar_ramy`, `rozmiar ramy`, `tamanho_quadro`, `tamanho do quadro`, `cadre`, `size` and `fs`. The values are worse than the keys: `57`, `57 cm`, `57cm`, `22"`, `22 inch`, `L`, `heren 57`, `M/L`. Only about **23%** of live Dutch bike listings carry a usable frame-size value at all.

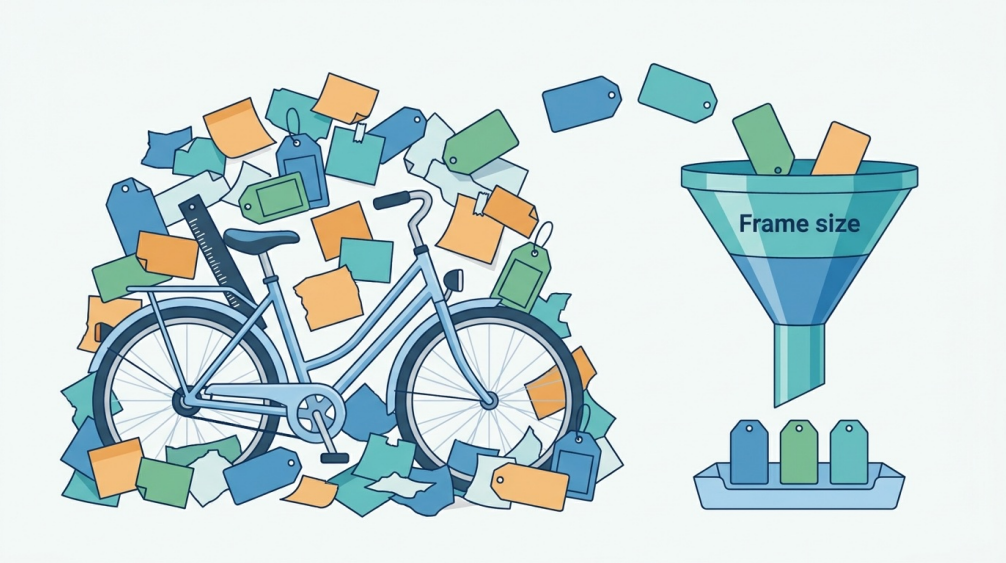


Figure 1: *Frame size lived under eleven keys with values such as 57, 57 cm, 22 inch and L, so only about 23% of live Dutch bike listings reached the frame-size filter.*

Femke's city bike, the Rotterdam listing this book follows, is typical. She listed it for **€240** on **Sunday 7 March 2027** and typed `57` into a free-text "maat" field. Is that 57 centimeters, the frame measured center-to-top, or a size from the brand's chart? The system does not know, so the bike appears in no frame-size filter at all.

The problem is not only Tradeyard's. Motorvane's typed vehicle model, which VIN decoding fills with **31 attributes** on a listing such as the Essen Volkswagen Golf, uses names and enumerations that differ from those of Tradeyard Poland's competing car section. Hearthly holds mandatory French energy-performance fields, such as the energy class **D** of the 68 m² Lyon apartment, in documents structured differently in France and in the Spanish platform it acquired in 2025.

Ingrid's question to the architecture advice forum is the question of this chapter: **can one attribute dictionary and one schema service serve general classifieds, automotive and real estate, without forcing Motorvane and Hearthly into Tradeyard's shape, and without creating a central team that every attribute change has to queue for?**

Three Layers: Definition, Binding and Value

Most attribute problems start with one confusion: treating an attribute's meaning, the categories it applies to, and the value on a listing as a single thing. They are three layers, and each has its own owner and lifecycle.

- An **attribute definition** says what a fact *means*: its stable key, its data type, its canonical unit, its allowed values, its cardinality and its labels in each language.

"Frame size: the length of the seat tube in centimeters, integer, 30–70" is a definition. It does not say where the attribute is used.

- An **attribute dictionary** is the governed collection of definitions. It is the single place where a platform says what each of its attributes means. eBay calls its attributes *aspects*; Amazon calls its attribute contracts *product type definitions*. The idea is the same.
- A **category schema** binds definitions to a category: which attributes a listing in this category must, should or may carry, which subset of allowed values applies, which validation rules apply, and in which order and groups the form shows them. "City bikes: frame size required, gear count recommended, basket optional" is a category schema. Schemas are versioned; each listing records the version it was validated against.
- An **attribute value** is what one listing holds: the value, its unit as entered, the normalized value, and its **provenance** (seller-entered, extracted by a model, decoded from a registry, supplied by a partner feed). The pipeline that produces and records provenance belongs to From Raw Supply to Usable Listings: Normalization and Enrichment; this chapter only needs it to exist.
- **Reference data** is a controlled vocabulary that is larger and more structured than an enumeration: vehicle makes, models, generations and derivatives; property types; energy classes; brands. It has its own identifiers, hierarchy and suppliers, and often its own license.

Layer	Question it answers	Typical owner	Changes	Example
Attribute definition	What does this fact mean, and how is it represented?	Attribute steward for the domain	Rarely; a change of meaning creates a new attribute	`bike.frame_size_cm`, integer, cm, 30–70
Category schema	Where is it required, recommended or optional?	Category owner, with the steward	With taxonomy releases and product needs	City bikes v4: frame size required
Reference data	Which values exist, with which IDs?	Data team or licensed supplier	Continuously (new car models, new brands)	Make → model → generation → derivative

Layer	Question it answers	Typical owner	Changes	Example
Attribute value	What is true of this item?	The seller, the feed or the enrichment pipeline, recorded with provenance	With every edit or re-enrichment	57, entered as "57", normalized 57 cm, seller-provided

The separation pays off three ways. **Meaning becomes reusable**: frame size is defined once and bound to every bike leaf. **Change becomes local**: making it required for city bikes touches no definition and no stored value. And **ownership becomes possible**, because each layer has someone who can be asked about it. Within the The Listing Aggregate: Identity, Lifecycle and Invariants, attributes are a value collection validated against the category schema version current at each write.

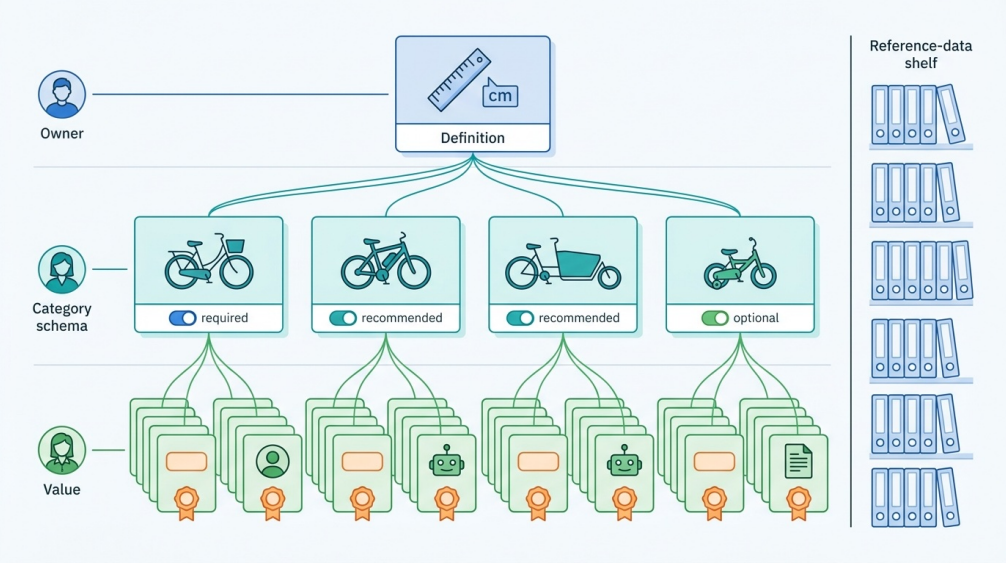


Figure 2: *An attribute's meaning, the categories that bind it and the value on a listing are three layers with their own owners, so frame size is defined once and reused by every bike leaf.*

The Attribute Dictionary Entry

A dictionary entry is a small contract, and two kinds of evidence say which fields it needs: the public dictionaries that marketplaces expose to their sellers, and the systems that read attributes downstream.

The most complete public example is eBay's Taxonomy API. Its `AspectConstraint` object records, per leaf category, requiredness, whether the value must be selected from a list or may be free text (`aspectMode`), data type and format, cardinality, whether the aspect describes the catalog product or the individual item (`aspectApplicableTo`), a maximum length, and an approximate date after which an aspect becomes required (`expectedRequiredByDate`) (eBay Taxonomy API specification, v1.1.1, n.d.). The specification warns that required aspects also report as recommended, "but they are actually required and a seller will be blocked from listing or revising an item without these aspects."

Amazon's Selling Partner API publishes its attribute requirements as JSON Schema documents, built on **JSON Schema 2019-09** with a small Amazon vocabulary on top: `editable` (whether an existing item's value can change), `maxUtf8ByteLength` (length in bytes rather than characters), `selectors` (which properties make an array element unique), and display labels for enumerations (Amazon SP-API, Product Type Definitions use-case guide, updated 2026). Definitions reflect the latest requirements unless a client pins a version.

Combining those public models with what downstream systems need gives the entry below. It is a recommendation, not a description of any one company's system.

Field	Purpose	Notes
Key	Stable, language-neutral identifier	Never a display label; never reused after retirement
Labels and help text per locale	What sellers and buyers see	Localization is a property of the entry, not its identity
Description	The precise meaning, including how it is measured	"Seat-tube length, center of bottom bracket to top of tube"
Data type	integer, decimal, boolean, date, enumeration, reference, text, range	eBay distinguishes numeric ranges as their own type

Field	Purpose	Notes
Canonical unit and accepted input units	Comparability	cm for frame size; inches accepted and converted
Allowed values	Range, pattern, enumeration or reference-data set	Closed enumerations power facets
Mode	Selection only, or free text with suggestions	eBay's `aspectMode` is the public precedent
Cardinality	Single or multiple values, with a maximum	Bike "accessories" is multi-valued
Scope	Product attribute or instance attribute	Color of a phone model vs scratches on this phone
Editability	Whether the value can change after publication	A VIN should not change; a price can
Purposes	Facet, filter, sort, ranking feature, trust rule, ads targeting, feed, SEO	Drives which compiled artifacts include it
Sensitivity	Personal data, special-category risk, legally mandated	Steers retention, display and ads use
Mappings	External standards per version	OpenImmo, RESO, mobile.de reference tables, schema.org, Google Merchant
Owner and status	Accountability and lifecycle	Proposed, active, deprecated, retired

Two of these fields are the ones most often missing: purposes and scope.

Purposes make the dependency graph visible: the impact of a change is known before it is made. Without them, the only way to find dependents is to break something.

Scope separates what a catalog product can supply from what only the seller knows, and eBay's catalog history shows why it matters. Between 2018 and 2020 eBay ran its Product-Based Shopping Experience, which required many listings to be associated with a catalog product; the playbook now states that the mandate for all categories and brands was rolled back, and that catalog matching remains recommended rather than required (eBay PBSE playbook, n.d.). The durable result is the split between **product aspects**, which a matched catalog product can supply, and **instance aspects**, such as condition, that only the seller knows. A used-goods marketplace that does not mark the difference

cannot prefill product facts from a match without overwriting the seller's facts.

Veldmark's Tradeyard stewards wrote the frame-size entry below. It is fictional; its shape is what carries over.

Field	`bike.frame_size_cm`
Labels	NL "Framemaat", FR (BE) "Taille du cadre", PL "Rozmiar ramy", PT "Tamanho do quadro", EN "Frame size"
Meaning	Seat-tube length in centimeters, as printed by the manufacturer or measured center-to-top
Type, unit	Integer; canonical unit cm; inches accepted and converted (× 2.54, rounded)
Allowed values	30–70
Mode, cardinality	Selection from a list of integers, single value
Scope, editability	Instance attribute (a product match may suggest it); editable
Purposes	Facet (bucketed), filter, outbound feed, duplicate-detection feature
Related entry	`bike.frame_size_class` (XS–XL), a separate enumeration, because letter sizes are brand-specific and cannot be converted to centimeters reliably
Owner, status	Tradeyard goods stewardship; active from schema release 2027.10

The related entry is the part a quick design gets wrong. A size "L" from one brand is 54 cm and from another 58 cm. Converting letter sizes to centimeters would create false precision. Keeping two attributes, and letting the facet offer both, is less elegant and more honest.

Category Schemas and How They Evolve

A category schema is a list of bindings, each of which says: this attribute, in this category, with this requiredness, this value subset, these extra rules, in this position on the form. Three design choices shape how schemas behave over time.

Inheritance down the tree. Attributes bound at "Bikes" (frame material, condition, brand) are inherited by every bike leaf; each leaf adds its own (battery capacity for e-bikes, load capacity for cargo bikes, wheel size in inches for kids' bikes). A leaf may tighten an inherited binding, from optional to recommended or from a full enumeration to a subset, but should not loosen one. Loosening in a child makes "every bike has a condition" false, and that kind of invariant is what search and analytics rely on.

Requiredness with three levels, not two. *Required* blocks publication. *Recommended* is shown prominently, counted in quality scores and requested by nudges, but never blocks. *Optional* is available. eBay's model shows why the middle level exists: a recommended aspect with an `expectedRequiredByDate` gives sellers and integrators a runway before enforcement. Autotrader's developer portal makes the same commitment from the API side. It announces breaking changes at least 30 days ahead and lists adding mandatory parameters and new validation errors as breaking, while adding properties is not (Autotrader Connect developer documentation, n.d.).

Immutable versions. A schema version, once published, never changes. Every listing records the version it was last validated against, which lets the platform answer "was this listing valid when it was published?" and lets old listings stay readable when the schema moves on. The category tree has its own versioning, owned by Category Trees That Can Change: Designing Marketplace Taxonomies; a taxonomy release usually ships with a set of schema versions, but the two are separate artifacts.

The rules for evolving schemas are the same compatibility logic that event platforms use. Confluent's Schema Registry defines **backward** compatibility (consumers on the new schema can read data written with the previous one), **forward** compatibility (consumers on the previous schema can read data written with the new one), **full** compatibility (both), transitive variants that check against all earlier versions, and **none**; the default is backward (Confluent, Schema Evolution and Compatibility, n.d.).

An attribute schema has two kinds of consumer to protect, and a change can be safe for one and breaking for the other: **producers** (sellers, forms and partner feeds that write listings) and **readers** (search, trust, ads, feeds out, analytics). The table applies the logic to the changes that actually happen.

Change	Producers	Readers	Rule I use
Add an optional or recommended attribute	Safe	Safe if readers ignore unknown attributes	Ship any time; readers must tolerate unknown keys
Add a value to a closed enumeration	Safe	Risky: readers with closed switch statements, facet configurations and feed mappings may reject or drop it	Publish the value, then wait for mapped readers to confirm before allowing it in forms

Change	Producers	Readers	Rule I use
Make an attribute required	**Breaking** for every producer that does not send it	Safe	Recommended first, with an effective date at least one feed cycle away; enforce on create and revise, never retroactively
Narrow a range or remove an enumeration value	**Breaking** for producers	Safe	Treat as "make required": runway, then enforce on write
Rename a label	Safe	Safe	A label is not the key; change freely
Change type, unit or meaning	Breaking for both	Breaking for both	**Create a new attribute**; migrate values; deprecate the old one
Retire an attribute	Producers must stop sending it	Readers lose it	Deprecate, stop accepting on write, keep reading old values until no reader uses it

The last two rows are where most damage happens. A team that changes `mileage` from kilometers to miles "for the UK launch" in place, or turns a free-text field into an enumeration under the same key, corrupts every historical value and every reader at once. **A change of meaning is a new attribute**. The cost is a migration; the alternative is silent corruption.

Enforcing new requirements on *create and revise* rather than on all stored listings is what makes backfill gradual. eBay's warning that sellers are blocked "from listing or revising" without required aspects is the public precedent. Listings that predate the requirement stay published, keep their old schema version, and are brought up to date the next time their seller or feed touches them, or by an enrichment run that fills the value with model provenance.

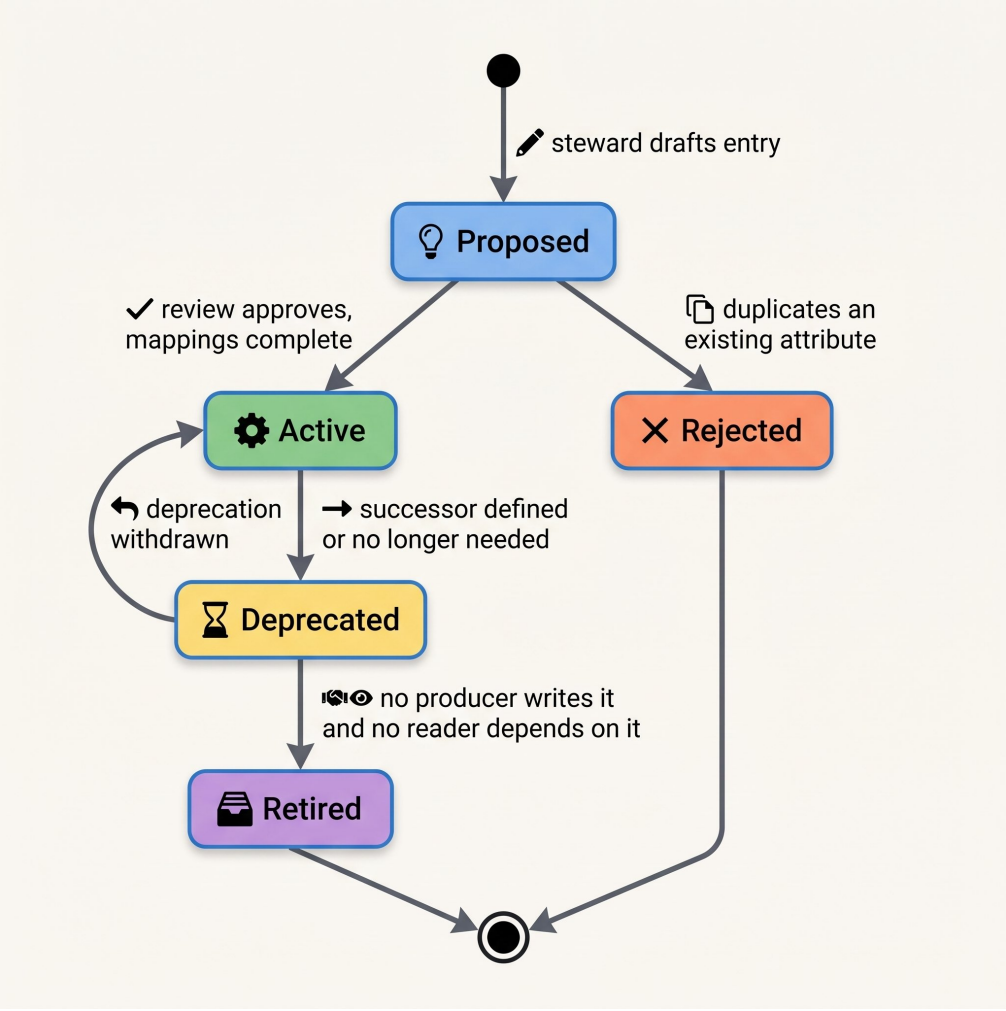


Figure 3: *The lifecycle of an attribute definition. An active attribute never changes its meaning; deprecation stops new writes while readers continue, and retirement waits until the dependency list in the entry's purposes is empty. A retired key is never reused.*

Storage Models: Wide Tables, EAV, Documents and Typed Schemas

How attributes are stored is the question architects reach for first, and it matters less than the dictionary. Still, the choice constrains validation, query patterns and the cost of change.

Wide tables put every attribute in a column. **Entity–attribute–value (EAV)** stores rows of `(listing_id, attribute_id, value)`; it needs no schema changes but many

self-joins, and it is where junkyards grow fastest, because creating a new attribute is an `INSERT` that no reviewer sees. **Documents** store each listing's attributes as one JSON document, validated against the category schema before write. **Typed per-vertical models** give a vertical its own types or tables, such as a vehicle with engine, transmission and emissions as typed structures.

A document store needs a schema as much as any other. PostgreSQL's `jsonb` with GIN indexes supports containment queries, and its `jsonb_path_ops` operator class produces "usually much smaller" indexes for `@>` queries but cannot answer key-existence queries. The documentation warns that any update "acquires a row-level lock on the entire row" and advises keeping documents to a manageable size with "a somewhat fixed structure" (PostgreSQL documentation, JSON Types, current). That advice is a polite way of saying that a document store without a schema is EAV with better syntax.

Model	Strengths	Weaknesses	Right when
Wide table	Simple; typed; fast	Does not scale past a few hundred attributes; sparse	A single vertical with a small, stable attribute set
EAV	Extensible without DDL; metadata in tables	Joins; weak typing; invisible growth	Legacy; or the storage layer under a strict dictionary that controls what may be created
Validated document	One read per listing; flexible; schema enforced in the application	Whole-row updates; indexing trade-offs; needs a validator everywhere it is written	Long-tail categories in horizontal marketplaces
Typed vertical model	Strong validation; domain logic; analytics-friendly	Code and deploys to evolve; does not scale to thousands of leaf categories	Automotive, real estate and jobs, where domain logic is deep

Model	Strengths	Weaknesses	Right when
Hybrid (typed core, validated documents for the long tail, typed vertical models where justified)	Each part fits its job	Two storage patterns to run; the dictionary must cover both	**Most multi-vertical groups**

The hybrid is where most large marketplaces end up, and it is my recommendation for a group: **typed core fields** on the listing aggregate (price, location, category, seller, status, timestamps), **schema-validated documents** for long-tail attributes, and **typed vertical models** for the two or three verticals with deep domain logic. The rule that matters is that the typed vertical model is *registered in the dictionary*. Motorvane can keep its Kotlin vehicle types, but `vehicle.mileage_km` must still be a dictionary entry with a meaning, a unit and mappings, so that search, ads, analytics and Tradeyard Poland's car section all mean the same thing by it.

The store is not the facet engine

Whatever the primary store, **faceted queries run on the search index, not on the system of record**. Vinted's 2023 account of faceted search over almost 600 million items describes facets served by Elasticsearch terms aggregations, with multi-select facets requiring a query per facet, facets split from item search at a cost of about 20% more cluster CPU to make item search about 71 ms faster, and brand cardinality cut from about 2,000,000 values to about 40,000 verified brands, which lowered p99 latency by about 15 ms (Vinted Engineering, "Faceted search using Elasticsearch", 2023). The brand curation is an attribute-dictionary decision with a measured search effect: a free-text value became a reference entity with a verified flag.

The search index has its own version of the storage dilemma. Elasticsearch's documentation explains that `nested` fields index each object as a hidden Lucene document, call nested queries "typically expensive", and suggest the `flattened` type for large sets of arbitrary key-value pairs, which in turn gives up typed range queries (Elastic, nested field type, current). The usual compromise mirrors the storage hybrid: typed fields for the high-traffic filters of each vertical, and a generic structure for the long tail. That only works if something decides which attributes deserve typed fields. In the design below, the purposes field of the dictionary entry makes that decision, and Search Architecture for Listings: Indices, Freshness and Scale owns the index design that consumes it.

Architecture: A Schema Service That Compiles

Treat schemas as **source code that compiles into artifacts**, not as a service that every system calls at request time to ask "is this value valid?" or "which attributes does this category have?" A request-time service puts a network hop on the insertion path, the ingestion path and the indexing path, and it turns a governance tool into an availability dependency.

The components:

1. **Dictionary store**. Definitions, reference-data sets, category bindings and mappings, all versioned. The authoring side is a workflow tool with review, not a database console.
2. **Schema compiler**. On each release, it produces, per category and schema version: a **JSON Schema document** (standard JSON Schema 2020-12 plus a small marketplace vocabulary for purposes, editability, scope and display hints, the same approach as Amazon's); a **form definition** for web and apps; a **search mapping and facet configuration** fragment; **outbound feed mappings**; and **inbound feed mappings** from partner standards.
3. **Validator library**. One validation implementation, embedded in every service that writes listings (insertion flow, partner API, bulk feed importer, back-office edits), loading compiled schemas from a local cache. The same code produces the same error codes everywhere.
4. **Reference-data service**. Makes, models, brands, property types and energy classes, with IDs and aliases, serving lookups and autocomplete, and fed by licensed suppliers where needed.
5. **Release events**. A `SchemaReleased` event on the event backbone tells consumers that new compiled artifacts exist. Consumers pull them and apply them on their own schedule within an agreed window.
6. **Quality metrics**. Coverage, validity and long-tail ratios per attribute and category, computed from listing events, feeding the stewards' dashboards.

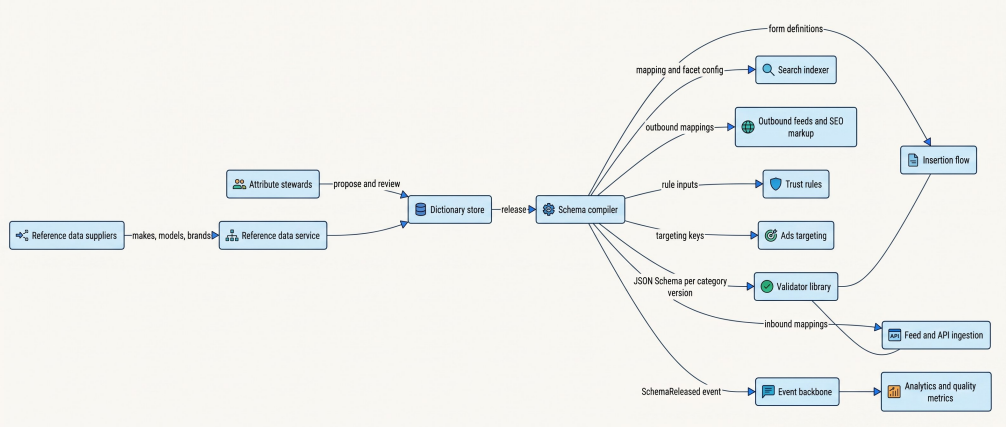


Figure 4: *One definition, many compiled artifacts. Stewards change the dictionary; the compiler turns each release into the validator, forms, mappings and configurations that other systems load locally. No system on the listing write path calls the schema service at request time.*

Three properties of this design matter more than its components.

Every channel validates with the same code. The most common source of Feed Roulette is that the web form enforces rules the feed importer does not, or the other way round. With one validator library and one compiled schema, a dealer feed and a private seller's app submission get the same answer and the same error code. The error code is part of the contract: Many Doors In: Forms, APIs and Bulk Feeds owns how errors are reported back to partners, and it needs stable, documented codes to do that.

Validation has two outcomes, not one. A required attribute that is missing blocks the write. A recommended attribute that is missing, or a value outside a soft range (a city bike priced at €9,000, a flat of 6 m²), is accepted and flagged, so that quality scoring, nudges or trust rules can act. Blocking everything suspicious hurts supply; accepting everything feeds the junkyard. The schema says which rule is hard and which is soft.

Readers tolerate the unknown. A reader that fails on an unfamiliar attribute or enumeration value turns every additive change into a coordinated release. Readers should ignore attributes they do not know and map unknown enumeration values to a safe default ("other") while logging them. This is the reader-side half of backward compatibility, and it is a coding standard, not a schema feature.

Veldmark's **Listing event contract v1** (live **1 July 2027**) carries attributes as a typed map, not as event fields. One event field per attribute would make every new attribute an event-schema change; the map takes its meaning from the dictionary, at the cost of weaker compile-time typing. Each map entry carries key, normalized value, unit and schema version; Every Change Is an Event: Event-Driven Listing Architecture covers the event side.

Reference Data: Makes, Models, Property Types and Energy Classes

Enumerations work for a dozen values. They do not work for vehicle derivatives, brands or property types, which have thousands of values, hierarchies, aliases and suppliers. Those are **reference data**, and they need their own service and governance.

Automotive reference data is the deepest case. Autotrader UK exposes a vehicle taxonomy that descends from vehicle types to makes, models, generations and derivatives; a generation combines body type and years, and its valuation and metrics calls are keyed on the derivative ID plus first-registration date and odometer reading (Autotrader Connect, n.d.). mobile.de's Seller-API serves reference-data endpoints such as gearboxes, colors, makes and models, with valid values depending on the seller's site, and its ads carry the German type-approval codes `hsn` and `tsn` and a `schwackeCode`, which the documentation describes as the classification of makes and models given by Eurotax/Schwacke (mobile.de Seller-API documentation, n.d.). In the United States, the National Highway Traffic Safety Administration's (NHTSA) vPIC API decodes VINs from manufacturer submissions, supports partial VINs and batches of up to 50 VINs, and recommends always sending the model year (NHTSA vPIC API, version 4.10, 2026). European marketplaces typically rely on licensed commercial providers for decoding and specification data.

Three design rules follow, and they apply beyond cars.

1. **Keep identity, specification and listing apart**. The VIN identifies a vehicle; the taxonomy derivative and its options describe its specification; the listing is one seller's offer of it at a price. One vehicle produces many listings over its life. One Property, Three Listings: Duplicates and Entity Resolution relies on that separation.
2. **Make the internal taxonomy a mapping layer over licensed ones**. Suppliers change their trees: Autotrader's release notes record model splits that affect derivatives. A marketplace that uses a supplier's IDs as its own inherits every supplier change as a migration. Map supplier IDs to internal IDs, version the mapping, and check the license before redistributing supplier data through public APIs.
3. **Record decode results with provenance and version**. When the decoder or the supplier's data changes, the platform must know which values came from which version, so it can re-run decoding without overwriting what the dealer corrected.

The Essen Golf shows why the third rule matters. Autohaus Ruhrblick's DMS (dealer management system) feed sends the listing on **Wednesday 14 April 2027**, and VIN decoding fills **31 attributes**. The 1.5 TSI engine was sold in more than one power output, and the decoder returns two candidates, 96 kW and 110 kW, while the dealer's feed

states 96 kW. The dictionary marks `vehicle.power_kw` as a field where the decode is a suggestion and a consistent dealer statement wins, with both kept. The market-value model that rates the car a "good price" against €22,800 reads the resolved value; had decode and dealer disagreed silently, it would have rated the car against the wrong derivative.

Real-estate reference data is less about a supplier tree and more about regulation. In France, sale and rental ads must show both the energy class and the climate class from the energy-performance diagnosis (DPE, A to G), homes rated F or G must carry the mention "logement à consommation énergétique excessive", and ads must show estimated annual energy costs with the reference year of prices; fines reach €3,000 for individuals and €15,000 for legal entities. G-rated homes have been barred from new rental leases since 1 January 2025, with F following in 2028 and E in 2034. From 1 January 2026 the primary-energy factor for electricity dropped from 2.3 to 1.9, which can move a home's class without any change to the home (Service-Public.fr, DPE, 2026).

For the dictionary, this makes DPE class, climate class and energy-cost range **required** attributes in French residential schemas; the class carries the **DPE date and calculation method** as companion attributes, because the same home's class depends on when it was calculated; and rental eligibility is a **derived** attribute computed from class and date, not a seller input. The Lyon apartment's class D is unremarkable today and could matter for a future rental listing of the same flat.

The German equivalent for cars is the passenger-car energy-labeling ordinance (Pkw-EnVKV). Dealers advertising new cars must state energy consumption, CO2 emissions and the CO2 class as soon as motorization data is first shown, with a presumption that a car is "new" if first registered at most eight months ago or with 1,000 km or less (Pkw-EnVKV, amended 2026). "New" is therefore a **computed regulatory classification** derived from registration date and mileage, not a category the dealer picks, and a car can cross the line while it is listed.

Brands sit between the two. Vinted's reduction from about two million brand strings to about 40,000 verified brands is the best public example of turning free text into reference data. The pattern is an entity with a canonical name, aliases, a verified flag and a parent (a sub-brand), with unverified values kept on the listing but excluded from facets.

Normalization of Units and Values

How values are cleaned belongs to From Raw Supply to Usable Listings: Normalization and Enrichment; the dictionary defines the target. These are the rules I would put in a group's attribute standard.

- **One canonical unit per attribute**: centimeters, kilometers, square meters, kilowatt-hours, minor currency units. Conversions are defined in the dictionary, not in each service.

- **Keep the value as entered** beside the normalized value, for display, audit and seller disputes. The seller typed `22"`; the listing shows `22"` to buyers who expect inches and `56 cm` in the filter.
- **Normalize on write**, so that search, ranking, trust rules and duplicate detection compare like with like. Normalizing on read in every consumer guarantees disagreement.
- **Make special states explicit**: "price on request", "negotiable", "free", "unknown", "not applicable". A magic number (price 0, mileage 999,999) is a junkyard waiting to happen.
- **Never convert what cannot be converted**: letter sizes, brand-specific sizes and condition grades from different schemes stay separate attributes.

Using standard codes inside the dictionary, such as schema.org's UN/CEFACT unit codes (`KMT` for kilometers) and ISO 4217 currencies, costs nothing and saves a mapping layer later (schema.org Vehicle, V30.1, 2026).

Attributes for Search, Trust, Ads and Feeds

Different readers need different things from the same attribute, and the purposes field in each entry records which reader needs what, so one definition can serve all of them.

Purpose	What the reader needs	Dictionary requirement	Example
Facet	Low cardinality, closed values, pre-bucketed ranges	Enumeration or reference set; bucket boundaries defined centrally	Frame size in 2 cm buckets; verified brands only
Filter and sort	Typed, normalized, comparable values	Canonical unit; numeric type	Mileage in km; living area in m^2
Ranking feature	Coverage and stable meaning over time	Version history; provenance for model trust	Photo count, attribute completeness, price versus market

Purpose	What the reader needs	Dictionary requirement	Example
Trust rule	Values a rule can reason about, including absences	Hard and soft ranges; mandatory compliance flags	Price far below market for make, model and year; missing DPE
Ads targeting	Keys that advertisers buy against, under consent rules	Sensitivity and permitted-use flags	Make and model for car-maker campaigns
Outbound feeds and SEO	Mapping to the receiving schema	Versioned mappings per channel	Google Merchant fields; schema.org markup generated at the edge
Analytics	Comparable values across schema versions	Mapping views across versions	Year-over-year e-bike share after the bikes split

Three of these rows carry lessons the table cannot show.

Trust rules depend on attribute quality more than on rule cleverness. A vehicle price-anomaly rule compares price with the market for make, model, year and mileage; if model is free text, the rule is noisy or silent. The Barcelona fake rentals in Scams, Fakes and Account Takeovers: Fraud Prevention for Listings were priced about 35% below market, an anomaly computable only because district, surface and rooms were structured. Trust and Safety as a Platform owns the rules; this chapter owns their inputs.

Ads targeting needs a sensitivity flag. A real-estate listing's accessibility features are harmless as a filter and problematic as a targeting key, because of what they reveal about the viewer. The dictionary records which attributes the ads stack may use; Selling Attention: Display, Native and Retail-Media Advertising owns targeting and consent.

Outbound schemas are outputs, not models. Google added vehicle-listing structured-data documentation in October 2023, added phase-out notices in June 2025 and removed the documentation in September 2025 because the types "are no longer shown in Google Search results" (Google Search Central, vehicle listing structured data changelog, 2025). A marketplace that had modeled its vehicle attributes on that schema would have built on sand. Generating markup from the dictionary through a versioned mapping, as Found by Google: Web, App and SEO Architecture for Listings recommends, makes such changes a mapping edit.

Mapping to feed standards

Inbound standards set the vocabulary partners send, and each comes with its own size and change rate.

- **RESO Data Dictionary** (US real estate): Data Dictionary 2.0 is the ratified version, with more than 1,700 fields and 3,100 lookups; RESO says adoption of the full dictionary "is not expected for most organizations and is not required for certification" (RESO Data Dictionary, n.d.).
- **OpenImmo** (German-speaking markets): the XML version defines about 250 features in XML Schema, typically exchanged as a ZIP with media over FTP; OpenImmo 2.0, a REST and JSON API, is not compatible with the XML version (OpenImmo, n.d.).
- **Rightmove's Real Time Datafeed** shows how a portal's inbound schema tracks regulation: version 1.4.0 (April 2022) added shared ownership, ground rent, service charge and council-tax band, and version 1.4.1 (November 2023) added utilities, parking, restrictions, flood risk and accessibility (Rightmove Real Time Datafeed specification v1.4.1, 2023).

All three show that a mapping is a **versioned artifact with an owner**, not a transformation hidden in an importer: it states the standard and version, the direction, the transform, what is lost, and what happens to unmapped values (rejected, stored as raw text, or queued for a steward). Hearthly's France team maintains connectors to about 160 real-estate software vendors; when each connector has its own private mapping of "heating type", there are 160 meanings of heating type. Many Doors In: Forms, APIs and Bulk Feeds owns the feed contracts; the dictionary owns what their fields mean once inside.

In Practice: Public Examples and Veldmark's Cleanup

The public record on attribute design is thin compared with search or payments, but three groups of examples teach most of what is needed.

eBay and Amazon publish their dictionaries as data. Beyond the constraint fields described earlier, eBay's bulk `fetchItemAspects` file covers all leaf categories in a marketplace, can exceed 100 MB compressed, and is used by its open-source Taxonomy SDK to surface new, modified and removed entities between versions. Amazon's JSON Schema contracts let partners validate locally with off-the-shelf tools, and errors caught before submission never become support tickets.

Mercari and Etsy show the schema as an input to models. In a 2024 engineering post, Mercari described a fine-tuned small language model that extracts attributes whose keys

are supplied at request time, because required attributes vary by category and change over time, so schema changes ship without retraining (Mercari Engineering, 2024). Etsy reported in 2025, according to a secondary summary (the original post was not accessible for this book), that LLM extraction raised complete attribute coverage in target categories from 31% to 91% (ZenML summary, 2025). **Enrichment can fill a dictionary, but it cannot replace one**; extraction without a target schema builds a second junkyard, faster.

Autotrader's validation errors show regulation and policy encoded as schema rules. Its documented API responses include a rejection for a price below a minimum and a rejection for vehicles the UK's DVLA classifies as category A or B write-offs, which "are not permitted for sale" (Autotrader Connect, n.d.). A validation rule with a stable error code is a policy that every channel enforces the same way.

Veldmark: cleaning the junkyard and building one dictionary

Veldmark's attribute work ran from June to October 2027, under Common Ground's taxonomy and attribute workstream. The numbers below are Veldmark's, and fictional.

Step 1: profile what exists. The team extracted every attribute key and value distribution from Yardstick's EAV tables (about 410 million rows, including expired listings), the Polish Elasticsearch mappings and the Portuguese forms. For each of Tradeyard's 3,900 keys they recorded usage per country and category, value types observed, and every reader they could find in code, feeds and dashboards.

Step 2: cluster and decide. Each key was assigned one of three fates.

Fate	Keys	What happened
Mapped to a canonical dictionary attribute	2,310	Merged into **1,240** canonical attributes; synonyms and translations became labels and aliases
Retired	990	Unused, garbage or duplicate of the title; values kept in the listing history, not migrated
Kept as country extensions	600	Genuinely local (Polish vehicle registration fields, Portuguese energy fields) in a country namespace
Total	**3,900**	

Frame size's eleven keys became `bike.frame_size_cm` and `bike.frame_size_class`. The largest single source of duplicates was not language but history: keys created for one campaign or one partner feed and never cleaned up.

Step 3: namespaces, not one flat list. The dictionary has three tiers. The **group core** holds attributes with one meaning everywhere (condition, color, brand, mileage, living area, energy class). **Vertical namespaces** (`vehicle.*`, `property.*`, `goods.*`) hold attributes whose meaning is shared within a vertical across brands. **Country extensions** hold the rest. Promotion from extension to vertical or core needs a steward's review and a migration plan. The tiers answer Ingrid's question about centralization: the group standardizes meaning where it is shared and leaves the rest local, but everything is in one dictionary, so everything has a definition.

Step 4: the bikes split. The four new leaves inherit eight attributes from "Bikes" and add their own. E-bikes add `bike.motor_position` (front hub, rear hub, mid-drive), `bike.battery_capacity_wh` and `bike.assist_speed_kmh`. Battery capacity was published as **recommended** with an expected-required date of **1 March 2028**, four months after the split, because Tradeyard's professional bike dealers send feeds that needed time to add it. The rollout followed the compatibility rules in two releases.

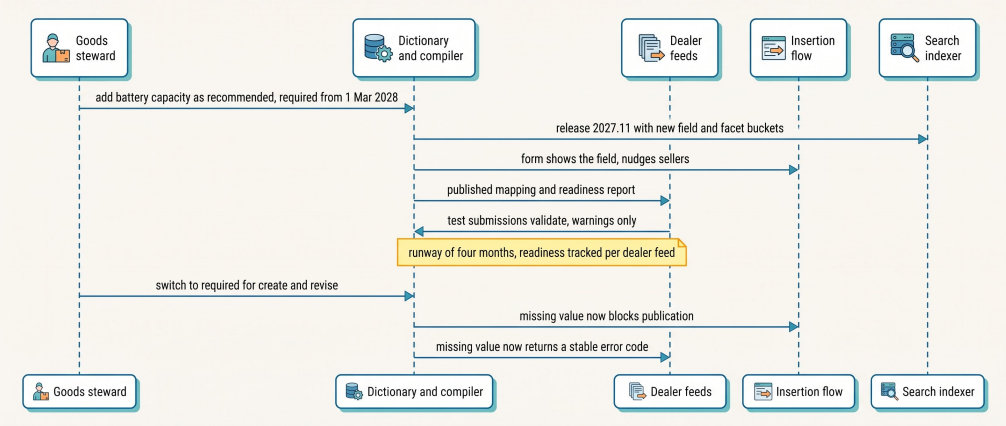


Figure 5: *Making an attribute required is a producer-breaking change, so it ships in two releases: first recommended with a published effective date, then enforced on create and revise once the readiness report shows the feeds are ready. Existing listings are not invalidated.*

Listings that existed before 1 November were re-classified by a model, with low-confidence cases left in a transitional category until their sellers confirmed at the next edit; Category Trees That Can Change: Designing Marketplace Taxonomies tells that part.

Step 5: backfill without blocking. Existing values were normalized and confirmed later, not demanded before the listing could stay live. Femke's bike shows how. Her `maat: 57` value sits in the city-bike context, and the normalizer maps it to 57 cm with medium confidence, provenance "seller-entered, unit inferred". It now appears in the frame-size facet, and the seller app asks her to confirm the unit the next time she opens

a listing. Together with the extraction pilot described in From Raw Supply to Usable Listings: Normalization and Enrichment, which reads frame sizes from descriptions and photos, the share of live Dutch bike listings with a usable frame-size value rose from **23%** to **71%** by the end of September 2027.

Step 6: Motorvane and Hearthly join without moving. Motorvane kept its typed vehicle model and registered its 212 vehicle attributes in the `vehicle.*` namespace, with mappings to its reference-data supplier and to Tradeyard Poland's car fields, which were then retired in favor of the vehicle namespace. Hearthly registered its French and Spanish property attributes in `property.*`, with DPE fields as required companions in French residential schemas and the Spanish energy certificate as a separate attribute, not a translation of the DPE, because the two schemes are calculated differently. Neither brand moved its storage. Both now validate with the compiled schemas. On the search side, the first group schema version gave typed index fields only to canonical attributes with facet, filter or sort purposes, which covered about 610 of Tradeyard's original keys; everything else sits in a flattened field (Search Architecture for Listings: Indices, Freshness and Scale).

Step 7: governance with a clock. Common Ground set a target of two days for a new category, against six weeks before, and the same target for attributes: a steward-reviewed new attribute is live in forms, validators and search within **two working days**. Stewardship is federated: Tradeyard goods, Motorvane vehicles and Hearthly property each have a named steward, and a small group platform team, under Yara Mendes's platform engineering organization, runs the service and the compiler. A new core attribute, or a promotion to core, needs a decision record in the architecture advice forum.

The cleanup's most useful by-product was the reader inventory: 31 of the 990 retired keys were still read by something, mostly old reports and one partner export, and each reader was migrated first. Without the inventory, those 31 would have been discovered as incidents.

Trade-offs

Each choice in this chapter has a cost. The table compares the main alternatives.

Decision	Option A	Option B	Choose A when	Choose B when
Storage	Typed per-vertical models	Validated documents	The vertical has deep domain logic and a stable attribute set (cars, property, jobs)	The long tail of categories changes often (general goods)
Dictionary scope	One group dictionary with namespaces	Separate dictionaries per brand	Brands share search, trust, ads or data platforms, or will	Brands are truly independent and will be sold or run apart
Validation	Strict: reject on violation	Lenient: accept and flag	Legally mandated fields; values that break downstream systems	Recommended fields; soft ranges; anything that would suppress supply
Enumerations	Closed (selection only)	Open (free text with suggestions)	Facets and filters depend on it; the value set is known	The value set is open-ended (models of rare items) and curation can follow
Schema evolution	Central release train	Continuous per-attribute releases	Many consumers with manual mappings; regulated changes	Mature compiled artifacts and tolerant readers

The **validation** row is a liquidity decision: every hard rule loses some supply, so the right set of hard rules is the smallest one that protects legal compliance and downstream integrity.

Vertical Differences

Attributes are where the verticals differ most visibly, which is why one dictionary with namespaces works better than either one flat schema or separate systems.

Dimension	General classifieds	Real estate	Automotive	Jobs	Recommerce
Attributes per listing	Few (5–15), varying widely by category	Many (40–100+), dense	Many, mostly decoded	Moderate, with free-text sections	Few, product-oriented
Source of truth	Seller's form	Agency software and feeds	Registry decode plus dealer DMS	Employer or applicant-tracking system	Seller, plus optional catalog match
Reference data	Brands, sizes	Property types, energy classes, administrative areas	Makes, models, derivatives, options (licensed)	Occupation and skills taxonomies (ESCO, O*NET)	Brands, product catalogs
Regulatory attributes	Few (product safety, prohibited items)	Many: energy performance, fees, disclosures	Energy and CO2 labeling for new cars; write-off status	Pay transparency, non-discrimination	Product safety, trader status
Main junkyard risk	Long-tail keys per country and category	Per-vendor mappings of the same field	Supplier taxonomy changes and decode conflicts	Free-text titles and salary formats	Unverified brand strings

Dimension	General classifieds	Real estate	Automotive	Jobs	Recommerce
Typical storage	Validated documents	Typed property model	Typed vehicle model	Typed model plus text	Documents plus catalog references

In jobs, ESCO v1.2.1 lists 3,039 occupations and 13,939 skills in 28 languages (European Commission, 2025); mapping a free-text title to an occupation is probabilistic, so occupation carries a confidence score and an override. Same Platform, Different Physics: How Listing Verticals Differ compares the verticals more broadly.

Failure Modes

The failures below are the ones I would look for first in a design review.

1. **Keys created without review**. EAV or schemaless documents let any team or feed importer create a key. Within a few years the dictionary is whatever the data says. The control is that unknown keys are rejected on write or quarantined, never silently stored.
2. **Meaning changed in place**. A unit, type or definition changes under the same key, and every historical value becomes ambiguous. The control is the rule that a change of meaning is a new attribute, enforced by the compiler.
3. **Required without runway**. A new mandatory attribute is enforced on the day it is announced, partner feeds start failing, and supply drops until someone disables the rule. The control is recommended-first with an effective date, and a dashboard showing which producers are ready.
4. **Validation only in the user interface**. The web form enforces rules; the API and the bulk importer do not. Feeds then deliver what the form forbids. The control is one validator library on every write path.
5. **Facet values drifting from the dictionary**. The search index stores raw values, the facet shows `heren 57` and `57cm` as separate options, and buyers stop using the filter. The control is facets built only from normalized values and reference IDs.
6. **Readers that reject the unknown**. A consumer crashes on a new enumeration value, and the dictionary team becomes a release coordinator. The control is tolerant readers and contract tests run against each release.
7. **Sensitive attributes in the wrong place**. An attribute that reveals health, religion or ethnicity, directly or by proxy, ends up as an ads targeting key or a ranking

feature. The control is a sensitivity flag with permitted uses, reviewed by the data protection lead.

8. **The dictionary team as a bottleneck**. Central review becomes a six-week queue, teams route around it, and the junkyard regrows in country extensions. The control is federated stewardship, a time target and self-service for extensions.

The Chief Architect's View

At group scale, attributes are a governance problem with an architecture attached. The decisions that belong to the Chief Architect, and the forum, are few but durable.

What is shared. I would share the dictionary, the schema compiler, the validator library and the reference-data service across the whole group, because they cost little to share and every shared downstream platform (search, trust, ads, data) depends on them. I would not force a shared storage model. Motorvane's typed vehicle model and a validated document store for Tradeyard's long tail are both legitimate; the dictionary is what makes them speak the same language.

Who decides. Stewards own definitions in their namespace. Category owners own bindings. The platform team owns the service and the compiler, not the content. The architecture advice forum decides on the group core: what enters it, what leaves it, and the compatibility policy. Lucía Ortega's data protection and regulatory function reviews sensitivity flags and regulatory attributes, because the DPE, energy labeling and trader-status fields are legal obligations, not product preferences.

What to measure. Five numbers tell whether the dictionary is working: **coverage** of required and recommended attributes per category; **validity** (share of values that pass the dictionary without a soft flag); the **long-tail ratio** (share of keys used on fewer than 0.1% of listings, which was 41% at Tradeyard and should fall and stay low); **time to a new attribute** (target two working days); and **facet coverage** for the top filters per category. If the long-tail ratio climbs again, the junkyard is regrowing somewhere, usually in a country extension or a new feed.

What to watch over the long term. Three pressures will test the design. **Model-generated attributes** will outgrow seller-entered ones, so stewards must define what a model may fill without confirmation (AI Across the Listing Lifecycle). **Regulation** keeps adding mandatory attributes and calculation changes like the 2026 DPE factor, so the dictionary needs effective dating, not just versions. And **acquisitions** bring new dictionaries: mapping an acquired brand's attributes into a namespace is cheap, immediately useful and the right first step in After the Deal: Consolidating Platforms After Acquisitions.

Questions to Ask

1. Where is the single definition of each attribute's meaning, type, unit and allowed values, and who owns it?
2. Can a new attribute be added, and an attribute made required, without a code release? How long does each take today?
3. Do the web form, the partner API and the bulk feed importer use the same validator and return the same error codes?
4. What happens when a reader receives an attribute or enumeration value it does not know?
5. Which attributes are required by law in each country and vertical, and how does the platform prove every live listing carries them?
6. For each attribute used as a facet, a ranking feature, a trust rule or an ads key, is that use recorded, so that the impact of a change is known before it is made?
7. How are licensed reference data (vehicle taxonomies, brand lists) mapped to internal IDs, and what happens when the supplier splits a model?
8. What share of attribute keys is used on fewer than 0.1% of listings, and is that share rising or falling?
9. Is the source of each value (seller, feed, registry decode, model) recorded, and which source wins in a conflict?
10. Which attributes are sensitive, and what prevents them from becoming targeting keys or ranking features?

From Attributes to Pictures

The attribute dictionary describes what a listing says in words and numbers. Buyers, however, decide in pictures first, and media is a second, much larger body of listing data with its own pipeline, storage economics, privacy risks and fraud signals. Photos Sell: Media Pipelines for Listings takes it up.

Sources and Evidence

Company engineering and investor sources

- Taxonomy API specification (OpenAPI v1.1.1) — eBay, n.d. — `AspectConstraint` fields, enforcement on listing and revising, `expectedRequiredByDate`, bulk aspect files and SDK diffs.
- Product-Based Shopping Experience playbook — eBay, n.d. (covers 2018–2020) — catalog mandate and its rollback; product versus instance aspects.

- Product Type Definitions API use-case guide — Amazon, updated 2026 — JSON Schema 2019-09 attribute contracts with a custom vocabulary and version pinning.
- Faceted search using Elasticsearch — Vinted Engineering, 2023 — facets on the search index, CPU and latency trade-offs, brand curation from about 2 million to 40,000 values.
- Fine-Tuning an LLM to Extract Dynamically Specified Attributes — Mercari Engineering, 2024 — attribute keys supplied at request time.
- Autotrader Connect developer documentation — Autotrader, n.d. — vehicle taxonomy depth, breaking-change policy, validation errors for minimum price and write-off categories.
- Seller-API documentation — mobile.de, n.d. — reference-data endpoints, `hsn`/`tsn` and Schwacke codes.
- Real Time Datafeed specification v1.4.1 — Rightmove, 2023 — inbound schema versions adding regulated property fields.

Standards, regulation and documentation

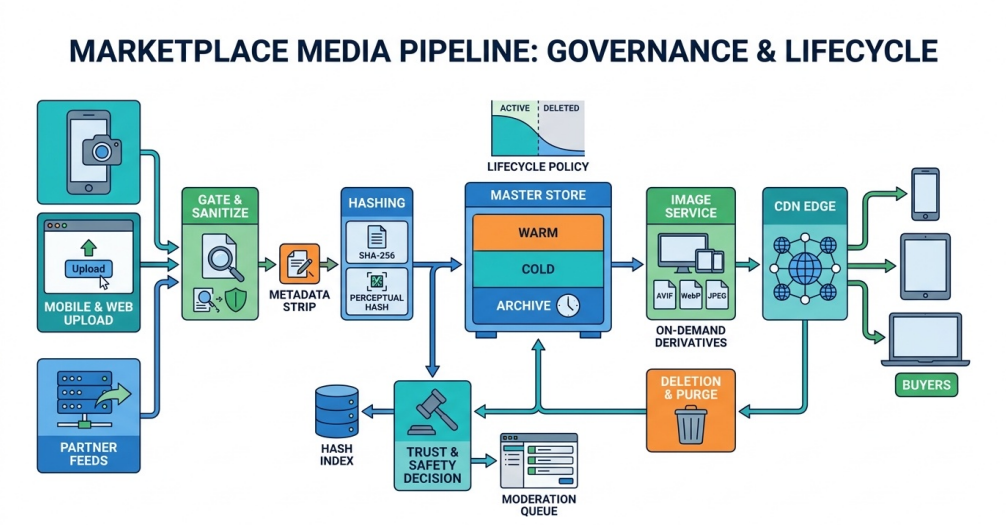
- Schema Evolution and Compatibility — Confluent, n.d. — backward, forward, full and transitive compatibility; backward as default.
- JSON Types — PostgreSQL Global Development Group, current — `jsonb`, GIN operator classes, row-level locking on update.
- JSON Schema Draft 2020-12 — json-schema.org, 2022 — the current JSON Schema release.
- Nested field type — Elastic, current — cost of nested queries; `flattened` as an alternative.
- vPIC API — NHTSA, version 4.10, 2026 — VIN decoding from manufacturer submissions, partial VINs, batches of 50.
- Diagnostic de performance énergétique (DPE) — Service-Public.fr, 2026 — mandatory ad disclosures, fines, rental bans, 2026 primary-energy factor change.
- Pkw-EnVKV — German Federal Ministry of Justice, amended 2026 — energy and CO_2 labeling in new-car advertising; "new" presumption.
- RESO Data Dictionary — RESO, n.d. — Data Dictionary 2.0, 1,700+ fields and 3,100 lookups.
- OpenImmo — OpenImmo e.V., n.d. — XML standard with about 250 features; incompatible 2.0 REST API.
- Vehicle listing structured data (changelog) — Google Search Central, 2025 — documentation added 2023, removed 2025.
- Product data specification — Google Merchant Center, n.d. — price with ISO 4217 currency.
- Vehicle — schema.org, V30.1, 2026 — `QuantitativeValue` with UN/CEFACT unit codes.
- What is ESCO — European Commission, 2025 — v1.2.1 occupation and skill counts.

Research and practitioner sources

- LLM-powered product attribute extraction (summary of Etsy's post) — ZenML LLMOps database, 2025 — Etsy's reported coverage rise from 31% to 91%; secondary source, original not accessible.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

10. Photos Sell: Media Pipelines for Listings



IN THIS CHAPTER, YOU WILL: Follow a listing photo from a phone, a browser or a partner's server through validation, sanitization, hashing, classification, derivative generation, delivery and deletion. You will see where media costs come from, how perceptual hashing catches stolen photos, and how Veldmark plans to cut Hearthly's media bill while closing the gap that let a fake-rental ring operate for days.

WHY THE BUSINESS CARES: Photos are the first thing a buyer judges and the main thing a fraudster steals. Listings with good photos get more contacts, galleries are most of a marketplace's delivered bytes, and stored images become one of the larger lines on the infrastructure bill, growing faster than live listings.

WHY ARCHITECTS SHOULD CARE: The media pipeline touches security (malicious files), privacy (location in photo metadata, faces, number plates), trust (stolen and duplicate photos), search quality (cover photos, quality

scores), feeds (media pulled by URL) and cost (storage tiers and CDN delivery). Decisions made for one of these, such as storing every rendition forever, quietly set the cost and risk of all the others.

KEY POINTS:

- **A listing references media IDs, not URLs, and media has its own lifecycle.** Give every asset a content-derived identity, a sanitized master, a moderation state and a retention class. Then CDN changes, format changes and deletions are media-service operations, not listing migrations.
- **Generate derivatives on demand at the edge; store originals deliberately.** Pre-generating every size and format multiplies storage and freezes yesterday's formats. An image service behind a CDN produces AVIF, WebP or JPEG at the size each device needs, while lifecycle rules move originals of ended listings to colder storage and delete them when their retention ends.
- **Hashes are trust infrastructure, not housekeeping.** Exact hashes remove duplicate uploads; perceptual hashes and embeddings find stolen photos across sellers, listings and brands. A match that cannot hold a listing, carries no seller context or covers only one brand lets fraud run for days.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Seeing Double** — Re-pulled feed images, reposted listings and multi-agency listings fill storage and search with copies that only hashing can recognize.
- **Moderation as an Afterthought** — Image matches that only create queue items while the listing goes live let stolen photos, contact details and prohibited content reach buyers first.
- **Ghost Inventory** — Photos of removed or sold listings stay reachable on the CDN and in caches long after the listing is gone, which is also a privacy problem.

Listing marketplaces are, measured in bytes, image businesses. A text listing is a few kilobytes; its photos are several megabytes, multiplied by every derivative size, every format and every year the platform keeps them. Buyers scroll galleries before they read descriptions, and professional sellers judge a portal partly by how well it presents their photography. **Media is the largest body of listing data, and usually the least governed.**

The media pipeline is the set of systems that accepts photos, videos, floor plans and documents, makes them safe, understands them, stores them and serves them. In most

marketplaces it grew in stages: an upload endpoint, a resizing job, a CDN in front, then classifiers and duplicate detection bolted on as fraud and quality problems appeared. Each stage was reasonable. Together they often produce a system that stores far more than it needs, serves formats a decade old, and checks for stolen photos too late to stop the fraud that uses them.

Part II models the listing as a small aggregate that references governed subdomains by ID, and media is the largest of them. This chapter designs its pipeline as one system with one contract: media IDs, sanitized masters and a `MediaReady` event that feeds, trust and safety, ranking and image models all depend on. It builds on the listing aggregate in The Listing Aggregate: Identity, Lifecycle and Invariants and the attribute contract in Attributes Without Chaos: Schemas, Dictionaries and Validation, hands moderation policy to Trust and Safety as a Platform and duplicate clustering to One Property, Three Listings: Duplicates and Entity Resolution, and follows Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), through Hearthly's media bill and the fake-rental ring that its image checks caught too late.

Setting the Scene at Veldmark: 1.8 Billion Images and a Fake-Rental Ring

It is September 2027. Julien Marchand, Hearthly's CTO, brings two documents to Ingrid Halvorsen, the Group Chief Architect.

The first is the media bill. Veldmark stores about **6.4 billion images** across its brands and adds about **9.5 million a day**. Hearthly, with 2.6 million live listings, stores **1.8 billion** of them, on Google Cloud Storage behind Cloud CDN. Hearthly's media run cost in 2026 was about **€6.2 million**: about €2.6 million for storage, €2.8 million for CDN delivery and cache fill, and €0.8 million for processing and fetching images from partner feeds. The storage line grew 19% in 2026 while Hearthly's live listings grew 4%.

Three causes are visible in the numbers. First, about 84% of Hearthly's listings arrive through feeds from about 160 real-estate software vendors, and many connectors re-send every photo URL on every update without a change marker; Hearthly re-downloads the photos and stores each as a new object. A sample showed that about **38%** of stored originals are byte-for-byte copies of another stored image. Second, the French platform pre-generates **nine JPEG renditions** of every image at upload and keeps them as long as the original. Third, nothing is ever deleted: about **96%** of stored images belong to listings that are no longer live.

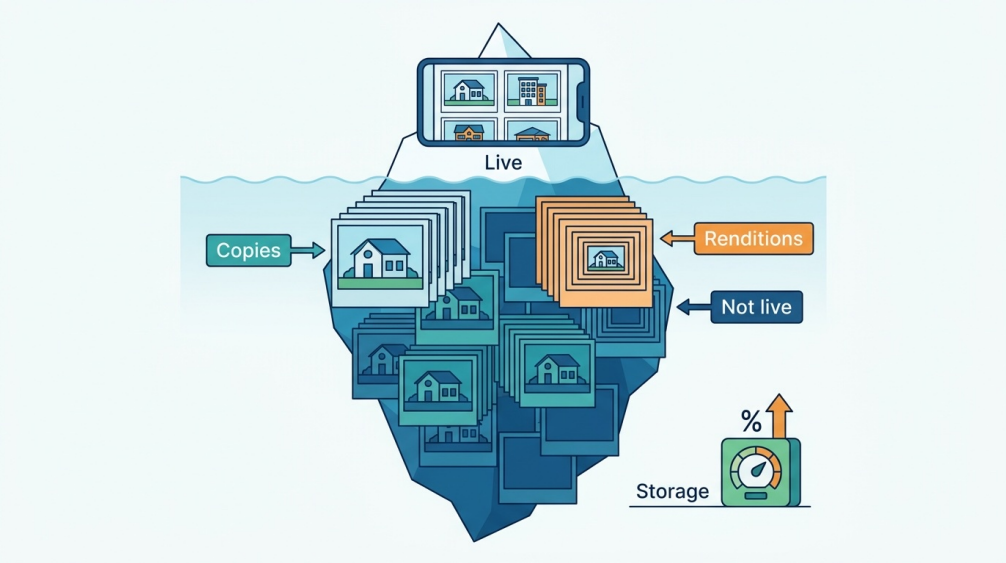


Figure 1: *About 96% of Hearthly's 1.8 billion stored images belong to listings that are no longer live, and about 38% of originals are byte-for-byte copies of another stored image.*

The second document is the review of the **Barcelona fake-rental ring**. In **July 2027**, **26** accounts posted **180** fake rental listings in Eixample on Hearthly.es, using photos stolen from real listings, priced about **35%** below market, and asking for deposits by bank transfer outside the platform. Perceptual-hash matches against existing listings, combined with price anomalies, exposed the ring, but only after **63** people had reported losses totaling about **€71,000**. The media part of the review is uncomfortable because detection worked: Hearthly Spain's hash index matched the first stolen photos within minutes of upload. The match did not hold the listing; it created a "possible duplicate" item in a general moderation queue while the listing went live. The hash service returned a distance and nothing else, so reviewers could not tell a stolen photo from a second agency marketing the same flat. And the index held only Hearthly.es images, so photos copied from other portals or from Veldmark's other brands were never compared. Scams, Fakes and Account Takeovers: Fraud Prevention for Listings tells the decision side of the story.

Ingrid's question to the architecture advice forum is the chapter's question: **what does one media pipeline for the group look like, such that it stores less, serves faster, protects sellers' privacy and catches stolen photos before buyers see them?**

Concepts: Assets, Masters, Derivatives and Hashes

Several costly mistakes in media systems come from treating different things as one.

- A **media asset** is one piece of listing media: a photo, a video, a floor plan, a 3D

tour reference or a document such as an energy certificate or a CV. It has an identity, an owner, a type, a moderation state, rights metadata and a retention class, independent of any URL.

- The **original** is the file as received. The **master** is the sanitized version the platform keeps: re-encoded, orientation applied, metadata stripped. I recommend storing the master and, in most cases, not keeping the original at all.
- A **derivative** (or rendition) is a version of the master at a given size, crop, format and quality, possibly with a watermark. Derivatives are disposable: they can always be regenerated from the master.
- An **image service** generates derivatives on request from a URL that encodes the transformation, typically behind a CDN that caches the results. The open-source imgproxy and thumbor are common examples; imgproxy is built on libvips and uses signed URLs to prevent abuse (imgproxy, n.d.; thumbor, n.d.).
- **Content addressing** derives an asset's storage key from a cryptographic hash of its bytes, such as SHA-256. Identical uploads collapse to one stored object, and a derivative URL built from the hash never changes, so it can be cached indefinitely.
- A **perceptual hash** is a short fingerprint computed from an image's visual content, so that re-compressed, resized or lightly edited copies produce similar fingerprints. Two images are compared by the Hamming distance between their hashes: the number of differing bits. Meta's PDQ, open-sourced in 2019, is the most widely used public algorithm for this (Meta, open-source photo and video matching, 2019).
- The **media pipeline** is the end-to-end system: upload or fetch, validation, sanitization, hashing, classification and moderation, master storage, derivative delivery, lifecycle and deletion.

A listing stores media IDs and an order, never URLs. This is the most important design rule. If listings hold URLs, every CDN change, domain change, format change or watermark change becomes a migration of millions of listing records, and deleting a photo means finding every place a URL was copied. If listings hold IDs, those are media-service operations. The same rule appears in reference designs and in public partner APIs: mobile.de's Seller-API has clients pre-upload images, receive references, and then reference those in the ad (mobile.de Seller-API, n.d.).

Architecture: The Upload-to-Edge Pipeline

Media arrives through three doors, and the pipeline behind them should be one.

Mobile and web uploads from private sellers and professionals. In the common pattern, the client asks the media service for a short-lived, pre-signed upload URL to object storage, uploads directly, and receives a media ID. The upload completes asynchronously, and the media service processes the file when storage reports it. This keeps large files away from application servers and lets uploads resume on poor mobile

connections. Resumable uploads have their own traps: OLX described in 2026 how users with several browser tabs open each tried to resume the same large upload, duplicating objects in storage, and how it used the Web Locks API so that only one tab proceeds (OLX Engineering, Web Locks API, 2026). Server-side content addressing is the backstop: a duplicate upload costs bandwidth, not storage.

Partner feeds from dealer management systems and real-estate software. Some standards bundle media with the data; OpenImmo's XML exchange typically ships images zipped with the XML file over FTP (OpenImmo, n.d.). Others have the portal pull media by URL. Rightmove's Real Time Datafeed accepts up to 999 media items per property and 20 MB per image, deletes an image when its URL is removed from an update, and, when the sender supplies a media update date, "will not re-retrieve images on the same URLs if they have the same media update date", which the specification says "relieves pressure on the client's image servers" (Rightmove Real Time Datafeed specification v1.4.1, 2023). That one field is the difference between a feed that fetches changed images and one that fetches everything every time. Hearthly's re-download problem is the cost of not requiring it, or of not using HTTP conditional requests and content hashes when the sender does not provide it.

Back-office and generated media: moderator edits, virtual staging, AI-generated listing videos, floor plans redrawn by a vendor. These enter through the same API with their provenance recorded.

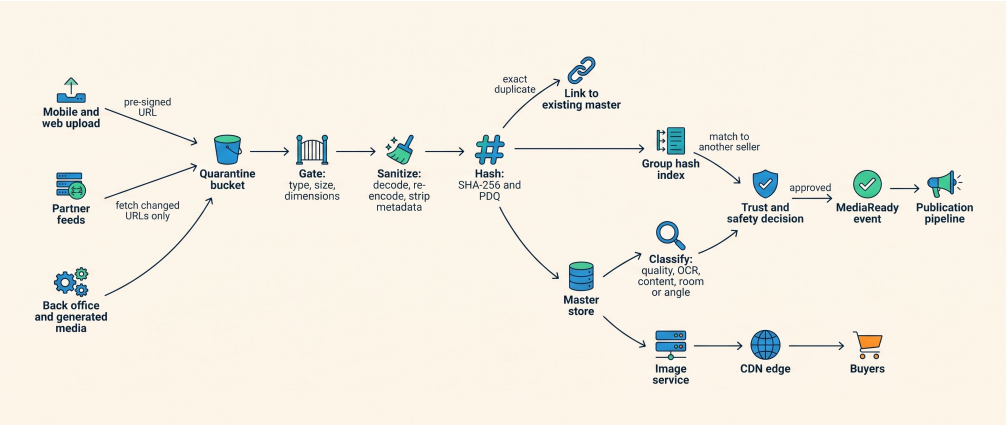


Figure 2: *One pipeline behind three doors. Files land in quarantine, are gated, re-encoded and hashed before anything else sees them, and reach buyers only as derivatives generated by the image service from a sanitized master. The hash index and classifiers feed the trust decision that the publication pipeline waits for.*

The stages, and the evidence behind each:

Stage	What happens	Blocks publication?	Notes and evidence
1. Quarantine	File lands in a bucket nothing else reads	—	Nothing downstream touches unvalidated bytes
2. Gate	Check type by content (magic bytes), size, and dimensions *before* full decoding	Yes	imgproxy checks image type before downloading fully and real dimensions before decoding, to guard against "image bombs"
3. Sanitize	Decode, apply orientation, re-encode to a master format; strip EXIF, XMP and trailing bytes	Yes	Re-encoding discards payloads hidden in metadata or after the image data
4. Malware scan	For documents (PDF floor plans, energy certificates, CVs)	Yes, for documents	ClamAV describes itself as "not a traditional anti-virus or endpoint security suite"; use it as one layer
5. Hash	SHA-256 for exact duplicates; PDQ for near-duplicates	Exact: no; perceptual: by policy	Exact match links to the existing master
6. Classify	Quality, OCR, prohibited content, category hints, room type or vehicle angle	By policy	Feeds moderation, ranking and gallery order

Stage	What happens	Blocks publication?	Notes and evidence
7. Decide	Trust rules combine hash matches, classifier output and listing context	Yes, for blocking outcomes	Owned by Trust and Safety as a Platform
8. Deliver	Image service generates derivatives; CDN caches them	—	Immutable, content-addressed URLs
9. Delete	Purge derivatives from CDN, delete master by retention class, keep hashes by policy	—	Retention classes below

Two public lessons shape the gate and the document path. First, in a 2025 post, Adevinta's InfoJobs team described its shared media service, YAMS, returning HTTP 500 errors for PDFs produced by a regional public employment service; each failing file was two complete PDFs joined together, with a duplicate header and stale cross-reference offsets, and the failures had been classified as internal faults instead of file-format errors (Adevinta Tech Blog, "Engineers to the rescue", 2025). The general rule: **a malformed file is a client error with an actionable message**, and where the source is a third party the seller cannot control, the pipeline should offer repair. Second, in jobs and real estate the document path carries personal data. CVs, energy certificates and battery-health certificates for electric cars, which mobile.de accepts as PDF, JPEG or PNG up to 10 MB, are media assets with personal data and their own retention rules (mobile.de Seller-API, n.d.).

What blocks publication, and what does not

Which checks block publication is a product decision. If every check is synchronous, a seller with twenty photos waits; if every check is asynchronous, stolen photos go live. The split I use:

- **Synchronous with the upload** (seconds): gate, sanitize, exact hash, a fast perceptual-hash lookup against the hash index, and a fast prohibited-content classifier. These are cheap and decisive.

- **Before the listing becomes searchable** (within the publication budget): the full trust decision for listings in high-risk segments (rentals, high-value vehicles, new accounts), combining hash matches with price anomalies and account signals. From Submitted to Searchable: The Publication Pipeline owns the budget; Veldmark's target for Tradeyard is 90 seconds at p95.
- **After publication** (minutes to hours): quality scoring, room and angle classification, gallery reordering suggestions, embedding-based similarity for heavily edited copies, and re-scoring when other listings arrive.

The Barcelona ring sat in the second row's gap: the comparison ran within minutes, but its result was a queue item, not an input to a decision that could hold the listing.

Privacy: What Photos Reveal

A listing photo carries more personal data than its seller realizes, and the platform is the one accountable for publishing it.

Metadata. EXIF GPS tags can record latitude, longitude, altitude, timestamp and the direction the camera faced (ExifTool GPS tags, n.d.). A private seller who photographs a bike in their hallway publishes their home's coordinates unless the pipeline strips them. When Femke listed her **€240** city bike on Tradeyard NL on **Sunday 7 March 2027**, her phone embedded the coordinates of her Rotterdam home in all six photos. Tradeyard strips EXIF before generating derivatives, so buyers never saw them, but its image service kept the unmodified originals, with GPS intact, for the lifetime of the account. One misconfigured bucket would expose them. Stripping must happen before the master is written, and the raw original should not be kept unless a documented purpose requires it, with restricted access and a short retention. Location privacy for the listing itself, the approximate pin, is Where Is It? Location, Geocoding and Privacy's subject; the photo is a second path to the same data.

Figure 3: *A phone photo can carry the seller's home coordinates, so metadata must be stripped before the master is written, and the raw original should not be kept.*

Content. Photos show faces (sellers, tenants, children), number plates, documents on desks, family photos on walls and the layout of an occupied home. A common practice is to detect and blur number plates and faces in vehicle and property photos, at least for private sellers, and to flag interiors of occupied rentals for tenant consent. The detection is a classifier; the policy, including when blurring is mandatory, belongs to the data protection lead.

Generated and edited images. Virtual staging, sky replacement and AI-generated listing videos bring marking and disclosure duties. Article 50 of the EU AI Act, which applies from **2 August 2026** under the regulation as adopted in 2024 (check for later amendments to its timeline), requires providers of systems that generate synthetic images to mark outputs in a machine-readable way, and deployers of systems that generate or manipulate images resembling existing places or objects in a way that could falsely appear authentic to disclose it (EU AI Act, Article 50, 2024). Whether a virtually staged room falls under the deployer duty is a question for counsel; the architectural consequence holds either way. **Provenance is a media attribute**: the pipeline must record whether an image is a photograph, an edited photograph or generated, and the listing must be able to display that.

Deletion. When a listing is removed for fraud, sold or deleted by the seller, its derivatives must stop being served. That means purging CDN caches, not only deleting the master: a content-addressed URL cached for a year at the edge outlives a deleted object for exactly that year. This is the media form of **Ghost Inventory**, and it is also a privacy failure when the seller exercised a right to erasure. Retention of hashes after deletion is a separate decision, covered below.

Derivatives, Formats and the CDN

Whether to store renditions or generate them decides a large part of the media bill.

Pre-generated renditions are computed at upload in a fixed set of sizes and formats and stored beside the master. They are simple to serve and need no compute at request time. Their costs grow with every image ever stored, they lock in the formats chosen at the time, and adding a new size or format means reprocessing the entire archive.

On-demand derivatives are generated by an image service when first requested and then cached by the CDN. OLX described this design in 2020: a CDN in front serving "around 80%" of traffic from cache, with the remaining requests, "millions of requests a day" and about 30,000 per second at peak on the busiest cluster, reaching an on-demand image processor that resizes, rotates, watermarks, re-encodes and compresses. OLX replaced a thumbor-based processor, which needed "~6000 pods" at peak, with **Dali**, a Rust service that first used OpenCV and then libvips, and that handled about twice thumbor's load per instance (OLX Engineering, "Presenting Dali", 2020). The post is from 2020, and OLX's current system is not public. The design is still the reference pattern.

Modern formats make on-demand generation more valuable. As of October 2026, AVIF is supported by about 96% of browsers in use globally and WebP by about 97% (Can I use: AVIF; Can I use: WebP, accessed 2026-10-19). An image service can choose the format per request, from the browser's `Accept` header or through `<picture>` sources, with JPEG as the fallback. A platform that pre-generated JPEGs in 2018 serves those JPEGs until it reprocesses its archive.

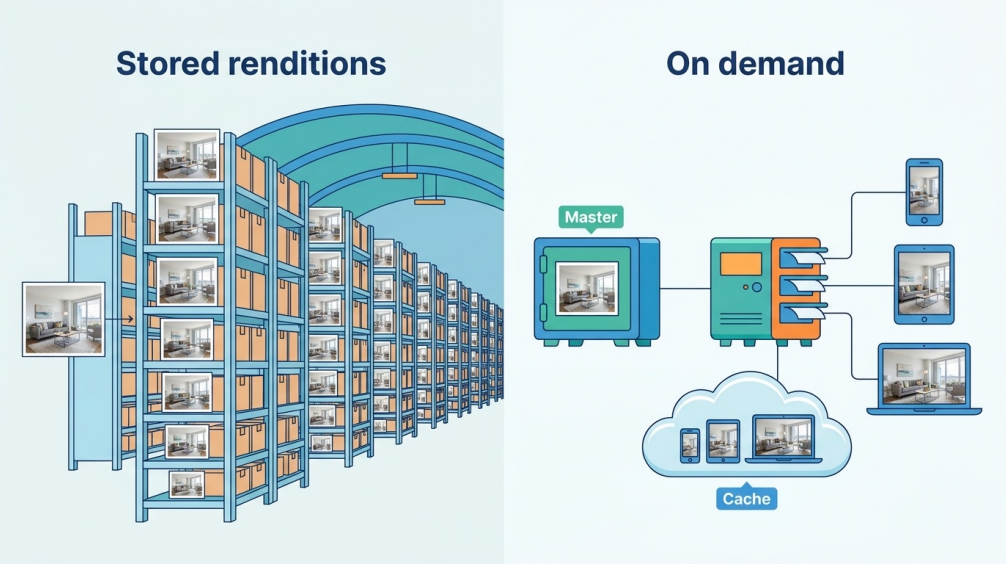


Figure 4: *Pre-generated renditions grow with every image ever stored and lock in old formats; one master plus on-demand derivatives behind a CDN serves modern formats without storing them.*

Three rules keep on-demand generation affordable and safe.

1. **Bound the variant space.** Allow a fixed set of widths, crops, qualities and formats, not arbitrary parameters. Every distinct URL is a cache entry and a potential cache miss; unbounded parameters are both a cost problem and a denial-of-service vector.
2. **Sign transformation URLs.** Only the platform can mint a derivative URL, so outsiders cannot ask the image service to render arbitrary sizes. imgproxy's signed URLs are the public example.
3. **Make URLs immutable.** A derivative URL built from the master's content hash and the transformation never changes meaning, so it can carry a long cache lifetime. Changing an image means a new hash and a new URL, not a cache purge. Deletion is the exception and needs an explicit purge.

Outbound channels set floors on derivatives. Google Merchant Center will enforce a minimum of 500 × 500 pixels for product images from **31 January 2027** (Google Merchant Center product data specification, n.d.). Any marketplace that syndicates listings to shopping channels needs to know, per listing, whether it has an image that meets the floor; that is a derived media attribute, and its absence is a feed error, not a surprise.

Storage Tiers and the Media Lifecycle

Most stored listing images are never viewed again after their listing ends. Cloud object stores price that pattern explicitly. Google Cloud Storage offers Standard, Nearline, Coldline and Archive classes with minimum storage durations of none, 30, 90 and 365 days, retrieval fees for the colder classes, and an Autoclass option that manages transitions automatically (Google Cloud Storage classes, updated 2026-10-13). Amazon S3's Standard-IA has a 30-day minimum and Glacier Instant Retrieval a 90-day minimum, both with millisecond access and per-GB retrieval charges; S3 Intelligent-Tiering moves objects not accessed for 30 consecutive days to an infrequent-access tier and after 90 days to an archive instant-access tier (Amazon S3 storage classes, n.d.).

The minimum durations matter for media more than for most data. An image deleted from Archive after two months is billed for twelve. The lifecycle policy must therefore follow the **listing's** lifecycle, which only the platform knows, rather than access patterns alone.

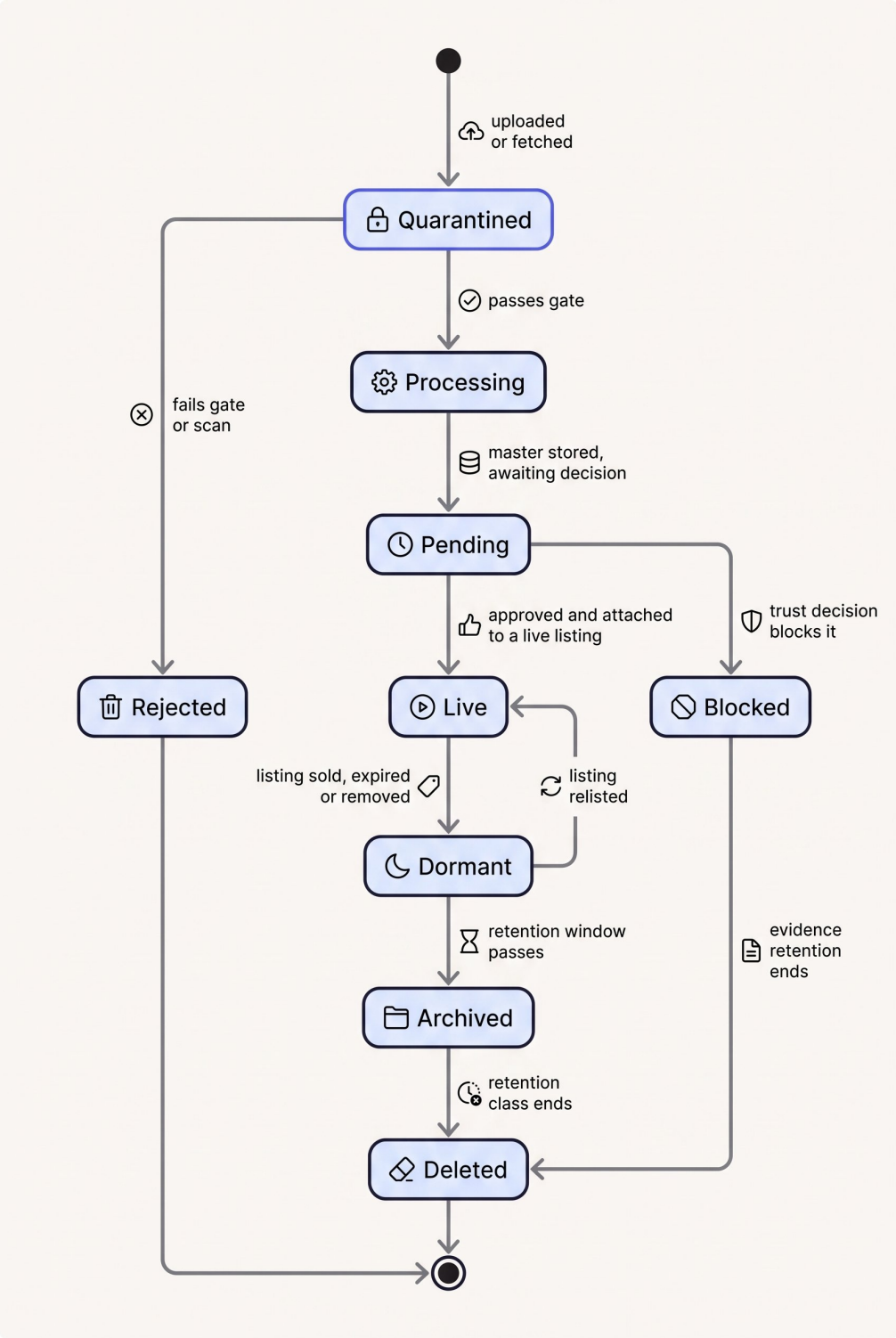

uploaded
or fetched
Quarantined
passes gate
Processing
fails gate
or scan
master stored,
awaiting decision
Pending
approved and attached
to a live listing
trust decision
blocks it
Rejected
Live
Blocked
listing sold, expired
or removed
listing
relisted
Dormant
retention window
passes
evidence
retention
ends
Archived
retention
class ends
Deleted

Figure 5: *A media asset's lifecycle follows its listing. Dormant media stays in warm storage for relisting and disputes; archived media moves to a cold class; deletion purges derivatives from the CDN and removes the master, while hashes may be kept under their own documented retention.*

Retention classes make the lifecycle explicit and defensible. A workable set:

Class	Applies to	Master retention after the listing ends	Hash retention	Rationale
Standard listing media	Photos of ordinary listings	12–24 months, warm then cold	Longer, documented	Relisting, disputes, fraud investigations
Removed for fraud	Media of listings removed by trust and safety	For the appeal window and investigations	Kept on block lists	Evidence; future matching
Erasure requested	Media covered by a deletion request	Delete promptly, purge CDN	Only if a separate legal basis applies	Data-protection rights
Documents	CVs, certificates, identity-bearing files	Shortest workable period	Usually none	High personal-data content
Professional archives	Agency or dealer media under contract	As the contract specifies	As policy	Contractual rights

The hash-retention column is where privacy and fraud prevention meet. A perceptual hash of a removed scam listing's photos is valuable for years, because the same stolen photos return. A hash is still derived from personal data in some cases, so its retention needs a documented purpose and legal basis; fraud prevention is the usual one. Rules of the Marketplace: DSA, DAC7 and Vertical Regulation covers the legal framework and System of Record and Read Models: Listing Data Management the platform-wide retention machinery.

Perceptual Hashing: Finding the Same Photo Twice

Duplicate and stolen photos are two problems with opposite policies, and the hashing layer must serve both.

- **The same seller reusing photos** (reposting a listing, or a dealer listing two similar cars with the same studio shots) is a spam or ranking-abuse problem at worst, and often legitimate.
- **Different sellers using the same photos** is a strong fraud signal in rentals and high-value goods, where scammers copy real listings, and a normal event in real-estate sales, where several agencies market the same property, sometimes with the same photographer's pictures.

The layered design that public evidence supports has four levels.

1. **Exact match** by SHA-256. Free, certain, defeated by any re-encoding.
2. **Perceptual hash** by PDQ or similar. PDQ's documentation recommends discarding hashes with a quality score of 49 or less, because flat or featureless images produce unreliable hashes, and starting with a match threshold of Hamming distance 31 or less, tuned on one's own data; the algorithm can produce additional hashes for rotations and flips, but "does not guarantee exact rotational invariance" (PDQ documentation, ThreatExchange repository, n.d.). Perceptual hashes survive re-compression and resizing and are weaker against crops, overlays and mirroring.
3. **Embedding similarity** from an image model, with approximate nearest-neighbor search, for copies that have been cropped, watermarked, color-shifted or deliberately perturbed. OLX has published work on detecting image duplicates with perceptual hashes on Elasticsearch, and on a triplet-loss network trained to catch fraudsters who augment images to evade hashing; those posts could not be accessed directly for this book and are known through search summaries (OLX Engineering, n.d.).
4. **Human review** for the gray zone between thresholds.

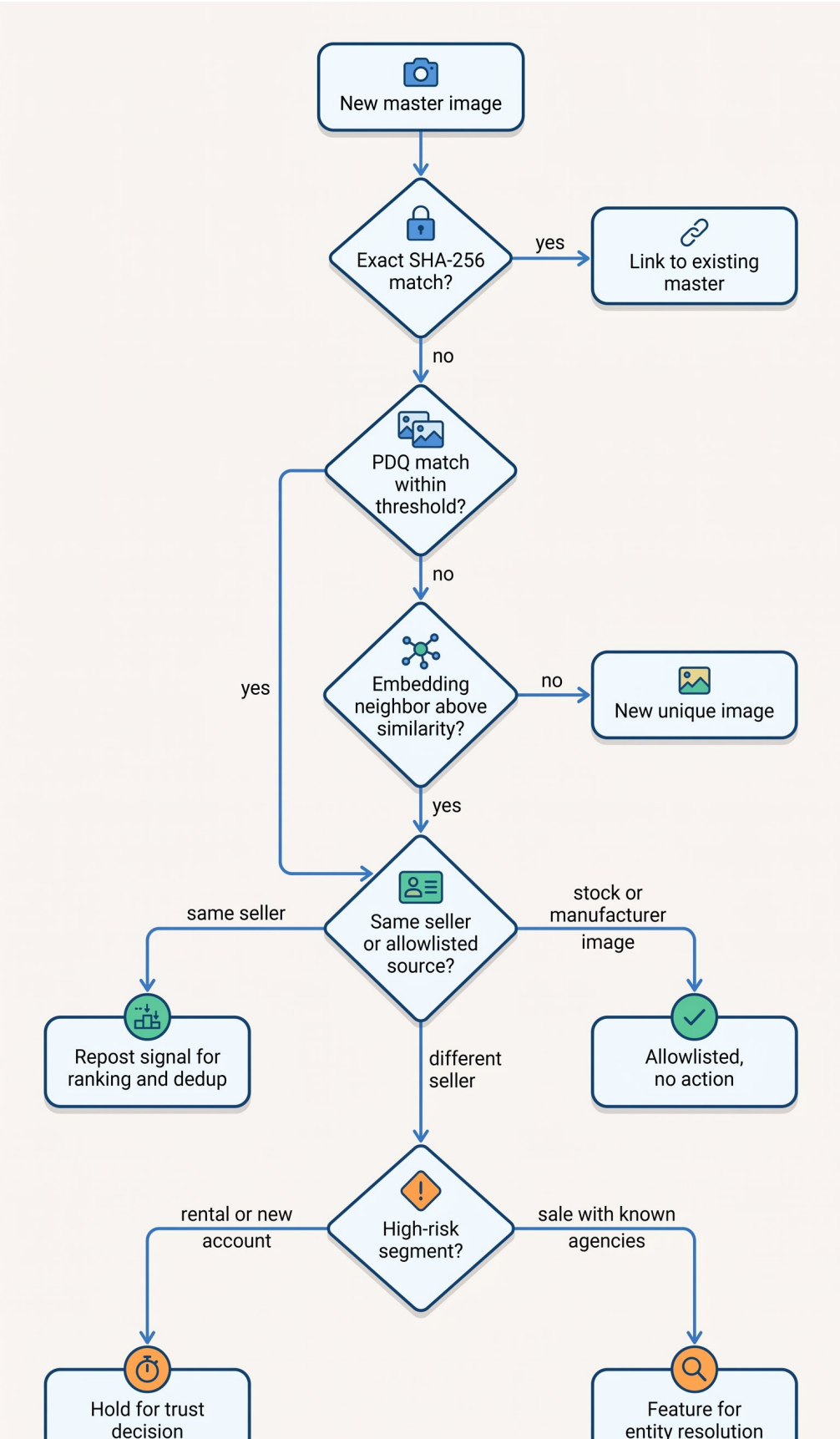

New master image
Exact SHA-256 match?
yes
Link to existing master
no
PDQ match within threshold?
yes
no
Embedding neighbor above similarity?
no
New unique image
yes
Same seller or allowlisted source?
same seller
stock or manufacturer image
Repost signal for ranking and dedup
Allowlisted, no action
different seller
High-risk segment?
rental or new account
sale with known agencies
Hold for trust decision
Feature for entity resolution

Figure 6: *Layered duplicate detection. Cheap checks run first; the same match leads to different actions depending on who the sellers are and which segment the listing is in. A copied photo is a fraud signal for a new account's rental and a clustering feature for three agencies selling the same flat.*

The allowlist in Figure 6 is not optional. Manufacturer press images in new-car listings, developer renderings in new-build real estate and catalog photos in recommerce legitimately appear on thousands of listings. Without an allowlist, the detector floods moderators and is switched off.

The hash index is also the clearest case for **sharing across brands**. A photo stolen from a Hearthly France listing and reused on Hearthly.es, or a car photo moved from Motorvane to Tradeyard Poland's car section, is invisible to a per-brand index. The index is small (a PDQ hash is a few dozen bytes) and the matching service is generic, while the decisions stay with each brand's trust rules. The Lyon apartment shows the other side: when **three agencies** listed the same 68 m² flat in **May 2027**, two of them used the same set of photographer's images, and the perceptual-hash matches were among the strongest signals One Property, Three Listings: Duplicates and Entity Resolution used to cluster the listings into one canonical property. The same signal that blocks a scam supports a legitimate merge, which is why the hashing service reports matches and leaves the decision to the consumer.

Understanding Images: Quality, Order, Text and Content

Beyond safety, image models score quality, order galleries, read text in photos and fill attributes, and each changes how a listing performs.

Quality scoring predicts whether a photo is sharp, well exposed and appealing. Google's NIMA model, published in 2017, predicts a distribution of human ratings for technical and aesthetic quality and is the standard public reference for such scores (Google Research, "Introducing NIMA", 2017). OLX described in 2020 seller recommendations based on photo count, blur and darkness (OLX, "Data Science at OLX", 2020). Quality scores feed ranking features, seller nudges and cover-photo selection; Ranking for Two Sides: Relevance, Freshness and Paid Visibility decides how much weight they deserve.

Ordering and cover selection decide what a buyer sees first. Autotrader UK reported in its FY2026 results that its Co-Driver tools, which generate descriptions and tag, order and optimize images, had made 700,000 "smart image re-orders" and generated 1.9 million descriptions (Autotrader Group FY26 results, 2026). In automotive the classifier labels angles (front three-quarter, interior, dashboard with odometer); in real estate it labels rooms (living room, kitchen, façade, floor plan). Ordering is a suggestion to professional sellers, who often care about their gallery, and a default for private sellers.

Text in images is both a moderation signal and a search signal. Adevinta's 2023 engineering post on its OCR service described detecting insults, hidden messages, URLs, e-mail addresses and phone numbers in ad images, with volume growing from more than 100 million to 330 million requests a month, accuracy on an internal benchmark rising from 0.4 to 0.8 after a move to PaddleOCR models, and a cost about twelve times lower than an external OCR API (Adevinta Tech Blog, "Text in Image 2.0", 2023). Phone numbers written on a photo are the classic way to route buyers off-platform, the first step of many scams described in From Interest to Contact: Messaging and Lead Management.

Content and attribute signals. Classifiers detect prohibited items, nudity and weapons, predict categories and fill attributes. Adevinta described in 2023 a shared team running a text detector, a car classifier, a background remover and an image classifier for its marketplaces on shared GPU serving (Adevinta Tech Blog, GPU serving, 2023). mobile.de's documentation shows images feeding valuation itself: its price rating is recalculated "including the image data" when images arrive after the ad, and mobile.de recommends uploading images first (mobile.de Seller-API, n.d.). So a `MediaReady` event must reach the models that depend on images, and listing-level outputs must be recomputed when it does. Models in Production: ML Platforms and Use Cases owns the serving infrastructure.

Video, 3D Tours and Floor Plans

Richer media shows up mostly in real estate and automotive in the public record, and it behaves differently from photos.

- **Video** needs transcoding to adaptive streaming formats (HLS or DASH), poster frames, frame and audio moderation, and much more storage and egress per asset. Meta's catalog format accepts up to 20 video URLs per item (Meta catalog product fields, n.d.). An OLX news item from February 2026 describes AI-powered listing videos; its content was not accessible for this book.
- **3D tours and panoramas** are usually produced and hosted by specialist providers and embedded. CoStar's FY2025 annual report records the completion of its acquisition of Matterport, a 3D-capture company, on 28 February 2025 (CoStar Group FY2025 Form 10-K, 2026), and Zillow's FY2025 annual report describes its Showcase tier with interactive floor plans and 3D touring (Zillow Group FY2025 Form 10-K, 2026). mobile.de's Seller-API lists interior and exterior "Auto Panorama" images among its resources. For most marketplaces, a tour is an external media object with its own lifecycle, consent and moderation, referenced by the listing like any other asset.
- **Floor plans** are images that carry data. Zillow's ZInD dataset, presented in 2021, contains 67,448 panoramas and 2,737 floor plans of 1,575 homes for academic use (Zillow ZInD, n.d.); it is evidence of research investment, not a description

of Zillow's production pipeline. A floor plan can yield room counts and surface areas, which become attribute values with model provenance, checked against the seller's stated living area under the rules in Attributes Without Chaos: Schemas, Dictionaries and Validation.

The architectural rule for all three: **one media model, several processing profiles**. A video is a media asset with a transcoding profile, a tour is an asset of type "external embed" with a provider and a consent record, and a floor plan is an image with an extraction profile. A separate system for each type multiplies deletion, moderation and privacy work.

In Practice: Public Examples and Veldmark's Plan

Public evidence on listing media pipelines is uneven but consistent. **Adevinta** runs a shared media service, YAMS ("Your Adevinta Media Service"), which a 2024 engineering post described as a service that "all of the marketplaces in Adevinta rely on" to deliver "more than 800 million images, documents and videos daily" (Adevinta Tech Blog, logging at scale, 2024); a further post, not accessible for this book and known through a search summary, reportedly put its storage at more than 3 petabytes, with on-the-fly resizing, cropping, watermarking and format conversion. Given the break-up of Adevinta's portfolio from 2025, whether YAMS still serves all of its former marketplaces is not public. **OLX** documented the CDN-plus-on-demand-processor pattern and its scale in 2020. **Rightmove** documented a pull-by-URL feed contract with change detection. **mobile.de** documented pre-uploaded image references, per-seller image limits and images as valuation inputs. Together they describe the same architecture: IDs not URLs, sanitized masters, on-demand derivatives, change-aware feed fetching and models downstream of a media-ready event.

> **INFERENCE, NOT DOCUMENTED:** None of these companies has published its full storage tiering or retention policy for listing images. The lifecycle and retention classes in this chapter are a reference design, not a description of any of their systems.

Veldmark: Hearthly's media plan in numbers

Veldmark's forum approved a group media decision in **November 2027**, and Hearthly is its first large adopter. All figures are fictional.

One contract, staged infrastructure. The group media platform defines one media API (upload, fetch-by-URL, attach, order, delete), one asset model with retention classes, one derivative URL scheme and one `MediaReady` event on the group event backbone. The

platform runs on AWS, the group's primary cloud. Hearthly stays on Google Cloud through 2028 under the ledger's cloud decisions, so it runs the same media service code against Google Cloud Storage, and its storage moves only when Hearthly's broader platform does. The **group perceptual-hash index** is the first shared component, live for all brands in **Q1 2028**, because it needs no storage migration at all: brands send hashes, not images.

Hearthly's four changes.

1. **Content addressing and change-aware fetching**. Originals become masters keyed by SHA-256, and feed fetchers send conditional requests and compare hashes before storing anything. Connectors that support a media change date, as Rightmove's contract does, must send it from the 2028 feed contract version onward. The 38% of byte-identical copies, about **680 million** objects, are collapsed into references.
2. **No stored renditions**. The nine pre-generated JPEG renditions are replaced by an image service behind Cloud CDN, serving AVIF or WebP with JPEG fallback in six bounded widths. In a September 2027 test on Hearthly France, bytes delivered per gallery view fell by about **38%** at equal visual quality, which is what moves the CDN line.
3. **Lifecycle by listing state**. Masters of listings ended more than 90 days ago move to Coldline; after 24 months they move to Archive and are deleted when their retention class ends. Twenty-four months reflects how often the same property is relisted by the same agency in France. Masters of fraud removals and documents follow their own classes.
4. **Hash check before searchable, for rentals**. On Hearthly.es and Hearthly France, rental listings with a PDQ match to a different seller's image, or to a hash on the fraud block list, are held for the trust decision within the publication budget instead of producing a queue item while the listing goes live. Each match now carries the relationship between the two sellers (same account, same agency group, known co-listing agency or unrelated), which is what reviewers lacked in July. The index covers all of Veldmark's brands from Q1 2028.

The cost effect, as planned for the end of 2028:

Line	2026 actual	End-2028 plan	Main driver
Stored data	about 5.4 PB	about 2.3 PB	Dedup, no stored renditions, deletion by retention class

Line	2026 actual	End-2028 plan	Main driver
Storage cost	€2.6M	€1.1M	Less data; most of the rest in Coldline and Archive
CDN delivery and cache fill	€2.8M	€1.9M	Modern formats, bounded sizes, higher cache hit ratio
Processing and feed fetching	€0.8M	€0.6M	Fewer re-downloads; on-demand compute partly offsets
Total media run cost	**€6.2M**	**€3.6M**	About **€2.6M a year** toward Common Ground's €38M run-cost target

The processing line falls least, because on-demand derivative generation costs compute that pre-generation did not. That is the honest trade: Hearthly pays a little more compute per request to stop paying storage for nine renditions of 1.8 billion images forever.

The plan also closes Tradeyard's privacy finding: from the November 2027 decision, no brand stores raw originals with metadata. Tradeyard's existing originals are being rewritten as sanitized masters, a backfill over about 2.9 billion images that runs into 2028.

Trade-offs

Decision	Option A	Option B	Choose A when	Choose B when
Derivatives	Pre-generated and stored	Generated on demand, cached at the CDN	Very small catalogs; no compute available at the edge	Large archives; formats and sizes still evolving (almost always)

Decision	Option A	Option B	Choose A when	Choose B when
Original files	Keep raw originals	Keep only sanitized masters	A documented need (legal evidence, professional photography contracts), with restricted access	Default; removes metadata risk
Feed media	Pull by URL	Push with the data (ZIP, multipart upload)	Senders host images reliably and supply change markers	Senders cannot host images; batch exchange is the norm (OpenImmo-style)
Perceptual check timing	Before searchable	After publication	High-risk segments (rentals, new accounts, high-value goods)	Low-risk segments where publication speed matters more
Hash index scope	Group-wide	Per brand	Brands share fraud patterns or supply (almost always within a group)	Legal separation is required, as for a neutral SaaS such as Keybridge
Media service ownership	One group platform	One per brand	Shared cloud and team capacity; similar media types	Brands on different clouds with no near-term convergence (then share the contract first)

Decision	Option A	Option B	Choose A when	Choose B when
Rich media	Host and process in-house	Embed from specialist providers	Video at scale is core to the product	3D tours and specialist formats; low volume per listing

The **hash index scope** row needs one qualification at Veldmark. Keybridge, the neutral agent SaaS, publishes agency listings to Hearthly's competitors, and the March 2027 decision bars using Keybridge agency data for Hearthly's ranking or sales targeting. Hashes of Keybridge-handled media therefore do not enter the group index; the listings Keybridge sends to Hearthly are hashed by Hearthly on arrival, like any other feed.

Vertical Differences

Dimension	General classifieds	Real estate	Automotive	Jobs	Recommerce
Media per listing	Few phone photos	Many professional photos (up to 999 per listing in Rightmove's feed), floor plans, tours	Many dealer photos, panoramas, sometimes video	Logos, rarely photos; CVs and documents from applicants	Several phone photos, often standardized
Main source	Direct upload	Feeds, pulled by URL or zipped	DMS feeds and dealer apps	Employer systems; applicant uploads	Direct upload

Dimension	General classifieds	Real estate	Automotive	Jobs	Recommerce
Main fraud use	Stolen photos for fake goods	Stolen photos for fake rentals	Stolen photos for deposit scams on non-existent cars	Fake employer branding; malicious documents	Stolen photos; counterfeit goods
Privacy risk	EXIF location of homes; faces	Occupied interiors; tenants; address	Number plates; dealer staff	CVs: personal and sensitive data	Faces, homes
Image models	Category and attribute hints, prohibited items	Room type, quality, floorplan extraction	Angle, damage, odometer reading	Document parsing	Category, brand, condition
Legitimate duplicates	Rare	Common (multi-agency listings)	Some (manufacturer images, same dealer group)	Common (agencies reposting the same job)	Catalog photos

The jobs column is the reminder that a shared media service must handle documents, not only images, as Adevinta's InfoJobs case showed. Same Platform, Different Physics: How Listing Verticals Differ has the wider comparison.

Failure Modes

1. **URLs in listings.** Listings store CDN URLs; a CDN migration or domain change becomes a data migration across every listing and every partner that copied them.
2. **Storing everything forever.** No retention classes, nine renditions per image, and every feed re-download stored as a new object. Storage grows with time instead of with live inventory.
3. **Raw originals with metadata.** Derivatives are stripped, but the stored originals keep GPS coordinates and are one misconfigured bucket away from a breach.

4. **Matches that decide nothing**. Perceptual-hash matches that only create queue items, carry no seller context or cover a single brand let stolen-photo fraud run until victims report it.
5. **No allowlist**. Manufacturer, developer and stock images flood the duplicate detector, moderators stop trusting it, and it is turned off.
6. **Deletion without purge**. The master is deleted but cached derivatives keep serving for months; removed fraud listings and erased personal photos remain reachable.
7. **Unbounded transformations**. An image service that renders any size or quality on request becomes both a cost leak and an attack surface.
8. **Malformed files as server errors**. Bad uploads produce 500 errors, alerts and retries instead of a clear error the seller or partner can act on.
9. **Models that never hear about images**. Valuations, quality scores and category predictions computed before photos arrive are never recomputed, because no event tells them the media is ready.

The Chief Architect's View

Media is one of the easiest capabilities to share in a marketplace group and one of the most expensive to leave unshared. The decisions I would take to the forum:

Share the contract first, the infrastructure second. A common asset model, media API, derivative URL scheme and `MediaReady` event cost little and make every later consolidation cheaper. Storage and image services can follow cloud decisions at their own pace. The hash index and classifiers can be shared from day one, because they process fingerprints and model calls, not storage.

Set retention as policy, with owners. Retention classes need sign-off from the data protection lead, Lucía Ortega at Veldmark, and from trust and safety, Aisha Rahman, because each wants something different from the same bytes. Write the classes down, implement them as lifecycle rules, and audit them.

Watch five numbers. **Stored bytes per live listing** (it should be flat, not rising); **bytes delivered per gallery view**; **CDN cache hit ratio**; **time from upload to `MediaReady`** at p95; and **share of image matches against unrelated sellers that reach a decision before the listing is contactable**. The last one is the number the Barcelona review was really about.

Plan for generated media. Edited and generated listing images are already arriving: staged rooms, cleaned backgrounds, videos generated from photos. The pipeline must record provenance per asset, carry it to the listing page and to outbound feeds, and keep the original photograph where a buyer might reasonably need to see it. AI Across the Listing Lifecycle takes up the product side.

Keep neutrality boundaries in the media layer too. Shared fraud infrastructure is a strong case for sharing, but where a group has committed to separation, as Veldmark has for Keybridge, the boundary must be enforced in the hash index and the classifiers, not only in ranking.

Questions to Ask

1. Do listings store media IDs or URLs, and what would a CDN or domain change cost today?
2. Where is metadata stripped, and do any stored files still contain GPS coordinates or other EXIF data?
3. How many stored bytes per live listing does the platform hold, and how has that changed over the last two years?
4. Are derivatives generated on demand or stored? Which formats are served, and to what share of traffic?
5. Do feed fetchers detect unchanged media, or re-download and re-store everything on each update?
6. When are perceptual-hash checks run relative to publication, and against which brands' images?
7. What retention class applies to the media of a sold listing, a fraud removal, an erasure request and an applicant's CV, and who signed off?
8. When a listing is removed, how long do its images remain reachable on the CDN?
9. Which models depend on images, and are they recomputed when media arrives or changes?
10. How does the platform record and display whether an image is a photograph, an edited photograph or generated?

From Pictures to Place

Photos show what an item is; the listing also has to say where it is. Location drives search, maps, trust signals and privacy risks of its own, and photos are only one way to leak it. Where Is It? Location, Geocoding and Privacy takes up geocoding, precision and the privacy of a seller's address.

Sources and Evidence

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- Zillow Group Form 10-K for FY2025 — Zillow Group, 2026 — Showcase, interactive floor plans and 3D touring.
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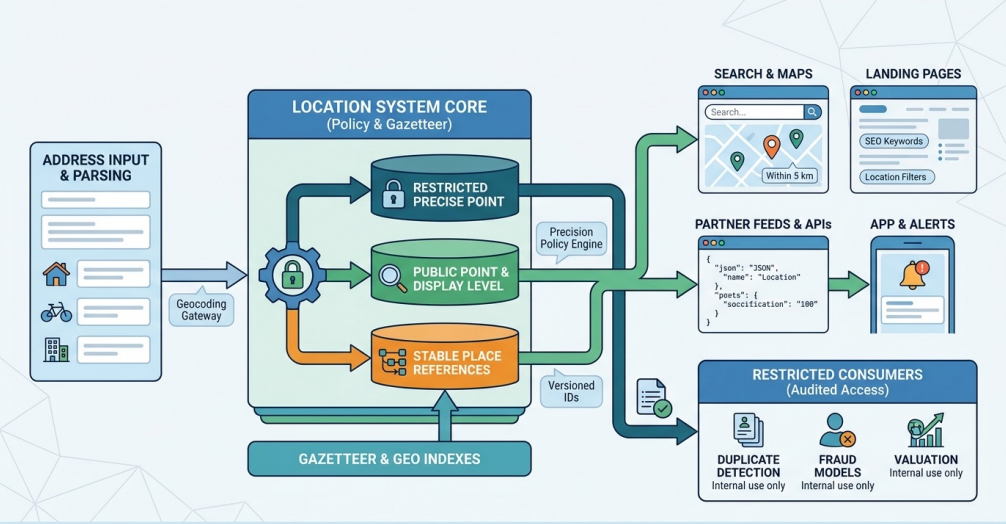
- Amazon S3 storage classes — Amazon Web Services, n.d. — minimum durations and Intelligent-Tiering transitions.
- Product data specification — Google Merchant Center, n.d. — 500 × 500 pixel image minimum from 31 January 2027.
- Catalog product fields reference — Meta, n.d. — up to 20 videos per item.
- OpenImmo — OpenImmo e.V., n.d. — media zipped with XML for exchange.
- EU AI Act, Article 50 — Regulation (EU) 2024/1689, unofficial reproduction, 2024 — marking of synthetic images and deep-fake disclosure; applies from 2 August 2026.

Research and practitioner sources

- Introducing NIMA: Neural Image Assessment — Google Research, 2017 — image quality scoring as a rating distribution.
- OLX Engineering posts on image duplicate detection and triplet-loss fraud detection — OLX, n.d. — known through search summaries only; the originals were not accessible for this book.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

11. Where Is It? Location, Geocoding and Privacy



Key Principle: Store three location representations, enforce privacy and licenses in the data model, not the UI.

IN THIS CHAPTER, YOU WILL: Follow a Rotterdam bike and a Lyon apartment from a typed address to a searchable, mappable and privacy-safe location. Along the way you will see how addresses differ by country, what geocoders and their licences allow, how place hierarchies and geo indexes work, and how a precision policy decides what each audience may see.

WHY THE BUSINESS CARES: In most listing verticals, location is the first filter a buyer applies. If the location is wrong, a listing is invisible. If the location is too precise, a private seller can be found at home by a stranger. If it relies on a geocoder's results the licence did not allow the marketplace to store, a supplier can send a legal letter that forces a rebuild.

WHY ARCHITECTS SHOULD CARE: Location looks like one attribute but is really three: a precise point, a public point and a set of place references. Each one has its own consumers, accuracy, licence terms and privacy class. A design that stores one field and lets every consumer read it will leak data, break licences and produce search results that disagree with the map.

KEY POINTS:

- **Store three representations, not one**. Keep a restricted precise point, a public point derived from it deterministically by a precision policy, and a list of stable place IDs from a versioned gazetteer. Search radius, maps, landing pages and analytics each read the representation meant for them.
- **Privacy is enforced on the server and by policy**. The precise point must never reach a client, a partner feed, structured data or an analytics event. Distances shown to buyers have to be computed from the public point, otherwise an attacker can work out the seller's address by measuring distances from several places.
- **Geocoding is a licensing decision before it is a technical one**. Some providers restrict storage or which maps their results may appear on, some charge extra for permanent storage, and open data can carry share-alike terms. Pick providers per country and per use, record the terms with every coordinate, and keep the gazetteer under the group's control.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Attribute Junkyard** — free-text location strings ("R'dam centrum", "near the station") stand in for structured place references, so filters, alerts and landing pages each guess differently where a listing is.
- **The Copy-Paste Country** — every country builds its own address model, geocoder integration and privacy rule, so a cross-border search, a group map tile or a privacy fix has to be built four times.
- **Seeing Double** — when locations are imprecise or inconsistent, the same property listed by three agencies cannot be grouped, because location is the first key every duplicate detector relies on.

A marketplace answers "where is it?" millions of times a day, and its wrong answers look like search bugs. A buyer searching "within 10 km of Utrecht" gets a city bike that is 14 km away, because the seller's postcode was resolved to the centroid of a different municipality. A landing page for "apartments in Lyon" leaves out flats in Villeurbanne, which buyers think of as Lyon but which is a separate commune. A private seller's home address can be worked out from distances in the app. None of these is a search-ranking problem. Each comes from how location was modeled.

Part II models the listing as a small aggregate that references governed subdomains by ID (The Listing Aggregate: Identity, Lifecycle and Invariants). Location is the subdomain whose model decides whether a private seller's home stays private. **Location is not an attribute but a small subsystem**: country-specific address capture, geocoding, a gazetteer of places, geo indexes, a precision policy and a set of quality and fraud signals.

Its gazetteer is reference data governed like the category tree in Category Trees That Can Change: Designing Marketplace Taxonomies, and it relies on Photos Sell: Media Pipelines for Listings to strip coordinates from photos. Later chapters build on it: it feeds search (Search Architecture for Listings: Indices, Freshness and Scale), map discovery (Searching by Place: Map, Polygon and Commute Search), alerts (Tell Me When It Appears: Saved Searches and Alerts), duplicate detection (One Property, Three Listings: Duplicates and Entity Resolution), fraud detection (Scams, Fakes and Account Takeovers: Fraud Prevention for Listings) and SEO landing pages (Found by Google: Web, App and SEO Architecture for Listings). It is also one of the few listing fields that is personal data almost every time a private seller is involved.

The chapter uses Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), as its running example. Veldmark runs four brands in eight countries, and its listing brands hold five different ideas of what a location is.

Setting the Scene at Veldmark: Five Answers to "Where Is It?"

In late March 2027, Lucía Ortega, Veldmark's Group Data Protection Officer, brings Ingrid Halvorsen, the new Group Chief Architect, a complaint from a Tradeyard Netherlands seller. A man had turned up at her door in Rotterdam about a sofa she had listed. She had never given him her address. Tradeyard showed only "3014 Rotterdam" on the listing page, so how had he found her?

Daan Visser's team reconstructs it in two days. The Tradeyard app sorted results by distance and showed distances to one decimal place ("2.3 km"). The distance was computed from the **centroid of the seller's full six-character postcode** (for example 3014 AB), which in a dense Dutch street covers a few dozen front doors. That centroid was also returned, with six decimal places, in the search API's JSON payload, which the app did not display but which anyone could read. With the payload alone, the man had a point within a few tens of meters of the seller's door. Without it, he could have searched from three places and intersected the three distance circles. The page showed a district; the system behind it handed out a doorstep.

Figure 1: *The page showed a district, but distances to one decimal place from a postcode centroid let three searches intersect at the seller's front door.*

Ingrid asks each brand how it handles location. The answers show why Common Ground needs one location model:

Brand	What is captured	What is stored	What is shown publicly	Geocoder
Tradeyard NL/BE	Postcode from the account profile; optional street	Postcode centroid in Yardstick's MySQL database and in Solr	Postcode district and city on the page; the precise centroid in the API	In-house postcode table built from the national address registers
Tradeyard PL/PT	Free-text city plus optional postcode	Coordinates from a commercial geocoder, kept indefinitely	City; map pin at the stored coordinates on an open-data map	Commercial API on a standard plan

Brand	What is captured	What is stored	What is shown publicly	Geocoder
Motorvane	Dealer branch address from the contract; postcode for private sellers	Exact dealer address; postcode centroid for private sellers	Dealer address in full; city for private sellers	A second commercial API
Hearthly FR	Full address from agency feeds	Exact point from the French national address base	The agency chooses: exact, street, or district	Self-hosted geocoder on the national address base
Hearthly ES (former Casavista)	Full address, including private landlords' rentals	Exact point	**Exact pin for every listing, including private landlords**	Third commercial API

Each of these rows was reasonable when it was built. Together they mean that Veldmark cannot answer a regulator who asks what location data it publishes about private individuals. It cannot say whether its Polish and Portuguese coordinates were licensed for storage. It cannot draw one group map, run one radius search across Motorvane's German and Austrian sites, or compare rents per neighborhood across Hearthly's two countries. The rest of this chapter is the model Ingrid's team proposes.

Concepts: What "Location" Means for a Listing

Four kinds of location

A listing can carry several locations, and confusing them is the first modeling mistake.

Kind	Definition	Example	Who needs it
Item location	Where the thing being sold physically is	The Lyon apartment; the Essen Golf on Autohaus Ruhrblick's lot	Search, maps, duplicate detection, valuation
Seller location	Where the seller is based	A dealer's head office; a private seller's home	Trust, regulatory disclosure, contact routing
Handover location	Where buyer and seller meet or where the item is collected	"Pickup near Rotterdam Centraal"	Buyer convenience; safety advice
Service area	The region a seller serves, for services and jobs	A removals company covering South Holland; a remote-friendly job	Search for services and jobs, where a point is the wrong shape

For goods, item and seller location usually coincide, which is why many systems store one field. They part ways often enough to matter: a dealer group with ten branches, a seller who keeps the item at a parent's house, a property listed by an agency in another city, and every shipped item. When Tradeyard Safe Deal ships a parcel (see Beyond the Contact: Payments, Shipping and Buyer Protection), the item location stops being a hard filter and becomes a delivery-cost input. A listing flagged "ships nationwide" should match a radius search it would otherwise fall outside.

Addresses are country-shaped

An **address model** is the structure a marketplace uses to capture and store a postal address. There is no universal one. The postcode alone shows the range across Veldmark's countries:

Country	Postcode shape	How specific it is	Practical consequence
Netherlands	Four digits plus two letters ("3014 AB")	Six-character code covers part of a street; the four-digit part a neighborhood	Postcode plus house number identifies an address; the four-digit part is a natural public district
Belgium	Four digits	A municipality or part of one	Bilingual place names: one postcode, two official names in Brussels
Germany, Austria	Five digits (DE), four (AT)	Delivery districts; not administrative units	A postcode can cross municipal borders, so postcode and municipality must be stored separately
France	Five digits	Delivery offices; one postcode can cover several communes, and Lyon, Paris and Marseille split by arrondissement	Postcode alone cannot answer "which commune"
Spain, Portugal, Poland	Five digits (ES), "1234-567" (PT), "12-345" (PL)	Varies from district to street segment	Validation rules per country; formats must be data, not code

Industry practice is to parse free-text addresses into components before matching. The open-source **libpostal** library does this with a statistical parser trained on OpenStreetMap and other open address data. Its README reports 99.45% full-parse accuracy on held-out data and normalization in "60+ languages" (libpostal, n.d.). Its model is about 1.8 GB, which in practice means it runs as a shared service, not inside every application.

Geocoding and its outputs

Geocoding turns an address or place name into coordinates. **Reverse geocoding** turns coordinates into an address or place. Both are inexact, and a geocoder's result should

always come with three things the listing keeps:

1. **Match level** — what the coordinate actually represents: a rooftop or building entrance, an interpolated position along a street, a postcode centroid, a locality centroid. A "point" that is the centroid of Rotterdam is a very different fact from a point at a front door, even though both are two numbers.
2. **Confidence** — the geocoder's own score, which drives whether the result is accepted, shown to the seller for confirmation, or sent to review.
3. **Provenance** — which provider, which data version, under which licence, on which date. This is what lets a team answer a licensing audit or re-geocode everything from one provider when a contract ends. From Raw Supply to Usable Listings: Normalization and Enrichment owns provenance for derived attributes in general; location is the case where the licence makes it mandatory.

Places, gazetteers and hierarchies

A **gazetteer** is a reference dataset of named places (countries, regions, municipalities, districts, neighborhoods) with stable identifiers, names in several languages, geometries and parent-child relations. Two public models show the design space:

- **OpenStreetMap's `admin_level`** runs from 2 to 10. Level 2 "is almost always a de-facto independent country", but higher levels "vary in meaning between countries": a city can sit anywhere from level 4 (Berlin) to level 10 (OpenStreetMap Wiki, n.d.). A marketplace cannot use the levels as a cross-country hierarchy without a mapping per country.
- **Who's On First** gives each place a stable 64-bit ID and placetypes from continent down to neighbourhood, macrohood and microhood. It supports **multiple hierarchies** for places with more than one parent, and it advises querying by ID, never by name (Who's On First, n.d.).

Marketplaces need both multiple parents and places no administration defines. A place can have several parents: Lyon's 7th arrondissement is in the commune of Lyon, in the Métropole de Lyon and in the Rhône department. And buyers use names that do not match any administrative unit ("Randstad", "Costa del Sol", "Eixample"). The gazetteer therefore holds two kinds of place: **administrative places** imported from official sources, and **marketplace places** curated by the marketplace for how buyers search, each with a geometry and links to the administrative places it overlaps.

Listings reference places by ID, and every listing carries the IDs of all its ancestors. "All listings in Rotterdam" then becomes a term filter on a place ID, not a polygon query, and renaming a place changes no listing.

Location precision

Location precision is the level of detail at which a location is known, stored or shown. A shared scale makes policies writable. Veldmark adopts six levels:

Level	Name	Example (Tradeyard NL)	Example (Hearthly FR)
P0	Country	Netherlands	France
P1	Region	South Holland	Auvergne-Rhône-Alpes
P2	Locality	Rotterdam	Lyon
P3	District	Postcode area 3014; a neighborhood	Lyon 7e arrondissement
P4	Street or block	A street segment	The street name, without house number
P5	Exact	Address with house number; rooftop point	Address, building and unit

Precision applies three times over: **capture precision** (what the seller or feed gave), **stored precision** (what the platform keeps after geocoding) and **display precision** (what a given audience may see). The design mistake in the Rotterdam case was to let display precision follow stored precision by default, with the page template as the only place that blurred it.

Geo indexes

A **geo index** is a data structure that makes spatial queries cheap: "points within this circle", "points inside this polygon", "counts per map tile". Four families matter in listing systems:

- **The search engine's native types**. Elasticsearch and OpenSearch index `geo_point` and `geo_shape` fields in BKD trees. Shapes are tessellated into triangles, the precision is about 1e-7 degrees, and tessellation cost grows with vertex count (Elastic documentation, n.d.). This is what radius and polygon queries should use at query time.
- **Geohash**. A base-32 string built by interleaving latitude and longitude bits. Shared prefixes mean nearby cells, so prefixes work as keys in stores without spatial indexes. Two points a meter apart on either side of a cell edge can have completely different prefixes, so proximity queries must always include neighboring cells.
- **H3**. Uber's hexagonal hierarchical grid, published in 2018, has 16 resolutions, each about one-seventh the area of the one above. Hexagons were chosen because a

hexagon's center is the same distance from all six neighbors (Uber, 2018). Children are only approximately contained in their parents. Average cells range from 36.13 km² at resolution 6 to 0.105 km² at resolution 9 (H3 documentation, n.d.).

- **S2.** Google's library projects a cube onto the sphere and orders cells along a Hilbert curve. It has levels 0 to 30 and 64-bit cell IDs that preserve locality, and a region coverer approximates any shape with a small set of cells (S2 Geometry, n.d.). Airbnb's 2026 paper on embedding-based retrieval uses S2 cells as a listing-location feature (Airbnb, 2026).

The four are complementary: **the engine's native types serve queries, H3 serves aggregation and privacy, and S2 or geohash serve key-value lookups**. The trade-offs section compares them job by job.

Architecture: Three Representations and the Services Behind Them

The location value object

Inside the listing aggregate (The Listing Aggregate: Identity, Lifecycle and Invariants), location is a value object with three parts that are written together but read by different consumers.

Part	Contents	Classification	Readers
Precise location	Normalized address components; precise point; match level; confidence; provider and data version; capture channel	Restricted personal data for private sellers; confidential for professionals	Geo service, duplicate detection, fraud models, valuation; never clients or partners
Public location	Public point; display precision level; display label ("3014 Rotterdam"); radius of uncertainty for maps	Public	Search documents, listing pages, maps, apps, partner feeds, structured data

Part	Contents	Classification	Readers
Place references	Place ID at the finest level allowed for display, plus all ancestor IDs, with the gazetteer version	Public	Search filters, landing pages, alerts, analytics, ad targeting

Each part can be recomputed from the one above it: the public point from the precise point and the policy, the place references from the public or precise point and the gazetteer. That is what makes policy changes and boundary reforms backfills rather than migrations.

A private seller's precise listing location is personal data; the gazetteer is not. The GDPR is explicit that **location data is one of the identifiers that make a person identifiable** (GDPR Art. 4(1), unofficial reproduction). The real-estate Productscape domain model behind this book's research classifies its geo reference data as internal and non-personal, but the listing that carries a private owner's location as "possible-personal". Its general-listings model treats the listing as non-personal, which the model's own review notes is wrong for private sellers, whose listings routinely embed location and contact data (see the research digest).

Components

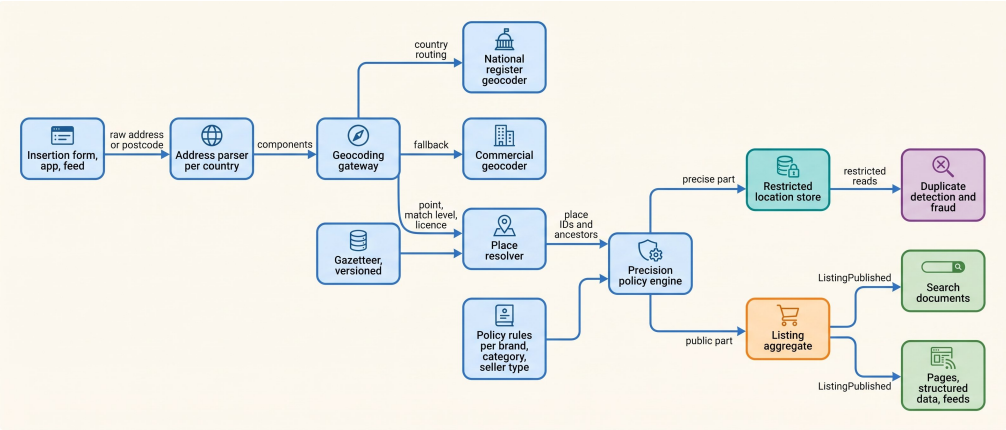


Figure 2: *Location enters as text and leaves as two different products. Everything to the right of the precision policy engine sees only what the policy allows, while duplicate detection and fraud models read the restricted store through an audited path.*

The components and their responsibilities:

1. **Address capture per country**. The form, app and feed mappers know the country's address shape, validate postcodes and offer autocomplete. Address formats are configuration in the gazetteer, not code in the form. That is the same configuration-over-code argument One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture makes for the platform as a whole.
2. **Geocoding gateway**. One internal API in front of all providers. It routes by country and use, caches within what each licence allows, records provenance and match level, and enforces per-provider quotas. No product service calls a geocoding provider directly; that rule alone makes a provider switch a configuration change.
3. **Gazetteer service**. Owns place IDs, names in every language, aliases, geometries, hierarchies and versions. It is reference data in the same sense as the category tree in Category Trees That Can Change: Designing Marketplace Taxonomies and needs the same governance: versioned releases, change notes, mappings when places split or merge.
4. **Place resolver**. Given a point, returns the place IDs that contain it and their ancestors, using the gazetteer version stamped on the result.
5. **Precision policy engine**. Given the precise location, the brand, category, seller type and the seller's own choice, it produces the public part. It is a pure function with versioned rules, which makes it testable and auditable.
6. **Restricted location store**. Holds the precise part, separated from the listing's public read models and with access logged. System of Record and Read Models: Listing Data Management owns the general model of read models and erasure; for location, the separation is the control.

Deriving the public point

The precision policy can derive the public point in two sound ways:

- **Snap to a cell**. Replace the precise point with the center of the H3 cell containing it, at a resolution chosen for the display level. At resolution 7 the average cell has an edge of 1.41 km; at resolution 8, 0.53 km (H3 documentation, n.d.). Every listing in the same cell gets the same public point, which is honest and simple, but in sparse rural areas a cell may contain a single house.
- **Seeded offset**. Move the point by a random distance and bearing within a radius, using a seed derived from the listing ID. **The offset must be deterministic**. If it is drawn again on every render, every edit or every relisting, an attacker who collects enough versions can average them back to the true point.

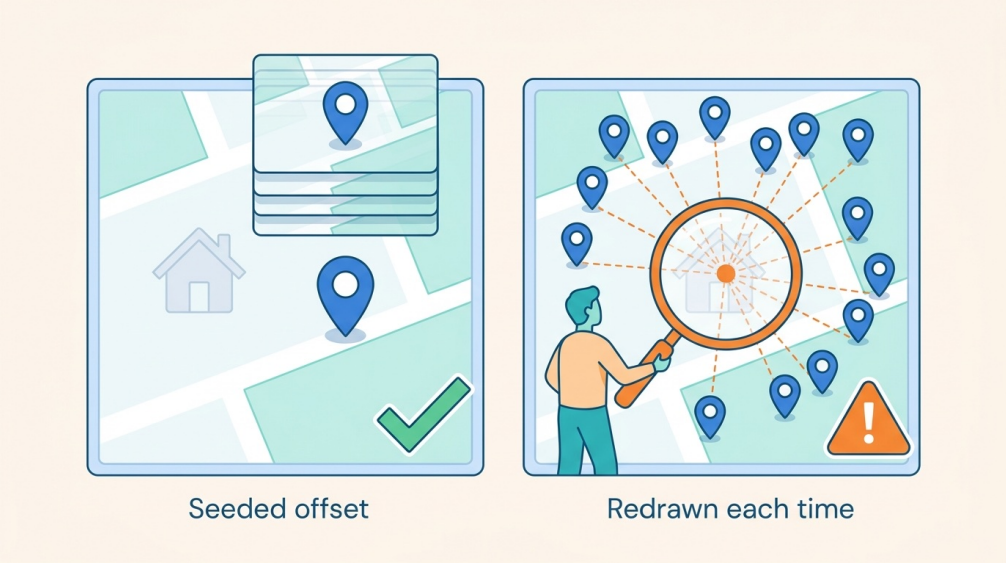


Figure 3: *A public offset must be seeded by the listing ID: if it is drawn again on every render or edit, enough versions average back to the true point.*

Two refinements apply to both. First, the public point must stay inside the public place: a snapped or offset point for a Rotterdam listing that lands in the river or in the next municipality creates support tickets and wrong landing pages. Second, in sparse areas the policy should coarsen automatically until the public area contains a minimum number of addresses. The national address registers make that count possible.

The best-documented public example of display obfuscation is Airbnb's. By default, its map shows a radius "within about a half mile (less than 1 kilometer) of the street address", and "guests don't receive the address until their reservation is confirmed" (Airbnb, 2025). The principle carries over to classifieds directly: the exact address is released when there is a reason to release it, to a person the platform knows, not when the listing is published.

The rule for distances

Distances leak locations, as the Rotterdam case showed. A distance from a searcher's chosen point to a listing's precise point, shown with enough precision, turns every search into a measurement, and three measurements locate the seller. **Distances shown to users, and distance sorting, must be computed from the public point**, and displayed values should be rounded to the precision of the public location (whole kilometers for district-level listings). Radius filters can still use the public point; the fuzziness at the edge of the circle is the same size as the fuzziness of the location, which is acceptable.

Internal consumers are different. Duplicate detection for real estate needs the precise point, since two listings of the same flat should agree within meters (One Property, Three Listings: Duplicates and Entity Resolution uses H3 resolution 9 cells as a blocking key).

Fraud models need it to compare a listing's location with the account's history. These reads go to the restricted store and are logged.

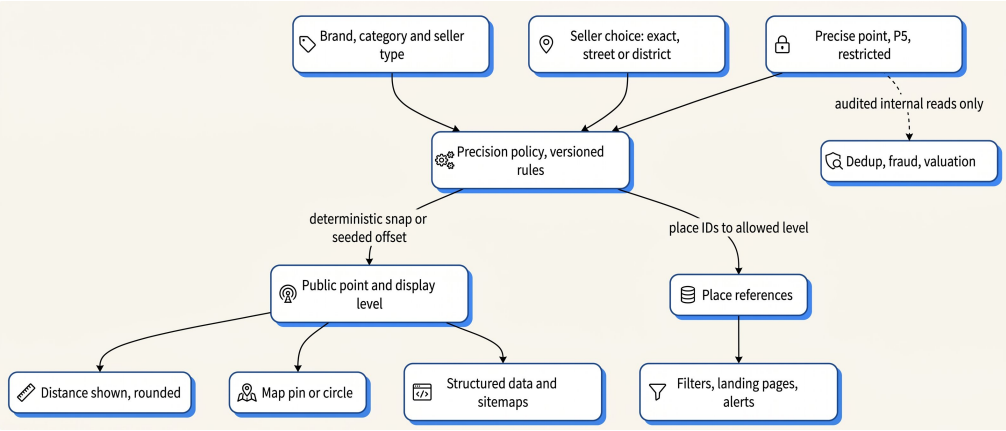


Figure 4: *One precise point, many audiences. Public consumers see a derived point and place references; only internal models with a documented purpose read the precise point, and every distance a buyer sees comes from the public side.*

Location in the listing event and the search document

The listing event contract (Every Change Is an Event: Event-Driven Listing Architecture) carries the public part and the place references, not the precise part. Consumers that need the precise point — duplicate detection, fraud scoring, valuation — call the restricted store with their own credentials. The cost is an extra call for those consumers. The gain is that the event stream, the data lake and every downstream copy stay free of precise private locations. That matters, because System of Record and Read Models: Listing Data Management found seven copies of Femke's bike listing alone, and erasing a precise point from seven copies is much harder than never having written it there.

The search document (Search Architecture for Listings: Indices, Freshness and Scale) then holds:

- a `geo_point` for the public point, used by radius filters, distance sorting and map clustering;
- a keyword array of all place IDs and ancestors, used for "in Lyon" filters and for landing pages;
- one or two H3 cell IDs at the resolutions used for aggregation and heat maps;
- the display label and display level, so the result card does not need a second lookup.

For real estate, the search document may also hold a precise `geo_point` that is never returned in responses but is used for polygon queries when the agency has chosen exact

display. Searching by Place: Map, Polygon and Commute Search owns the details of polygon, isochrone and clustering queries; this chapter's contribution is the rule about which point those queries may use.

Radius, Polygon and Place: Getting the Semantics Right

Each kind of location query hides a decision that produces a visible bug if left implicit.

"Within 10 km of Utrecht" — from where? The radius has to start from a point. If the buyer typed a city, the choice is between the city's centroid, its main station, or a population-weighted center. For a small town the choice makes little difference. For Rotterdam it matters: the municipality includes Hoek van Holland on the coast, about 30 km from the city center, so the geometric centroid of the municipality can fall somewhere unhelpful. Veldmark's gazetteer stores an explicit **reference point** for every place used as a search origin, curated for the largest cities and computed for the rest.

"In Lyon" — which Lyon? The commune of Lyon has nine arrondissements. Villeurbanne, which most buyers consider part of Lyon, is a separate commune inside the Métropole de Lyon. "Apartments in Lyon" can mean the commune, the metropolis or a buyer's mental map. The answer is not to pick one silently but to offer both as places: an administrative place ("Lyon, commune") and a marketplace place ("Lyon and surroundings") with a curated geometry. The query-understanding layer decides which one a typed query means (What Did They Mean? Query Understanding and Autocomplete); the gazetteer makes both possible.

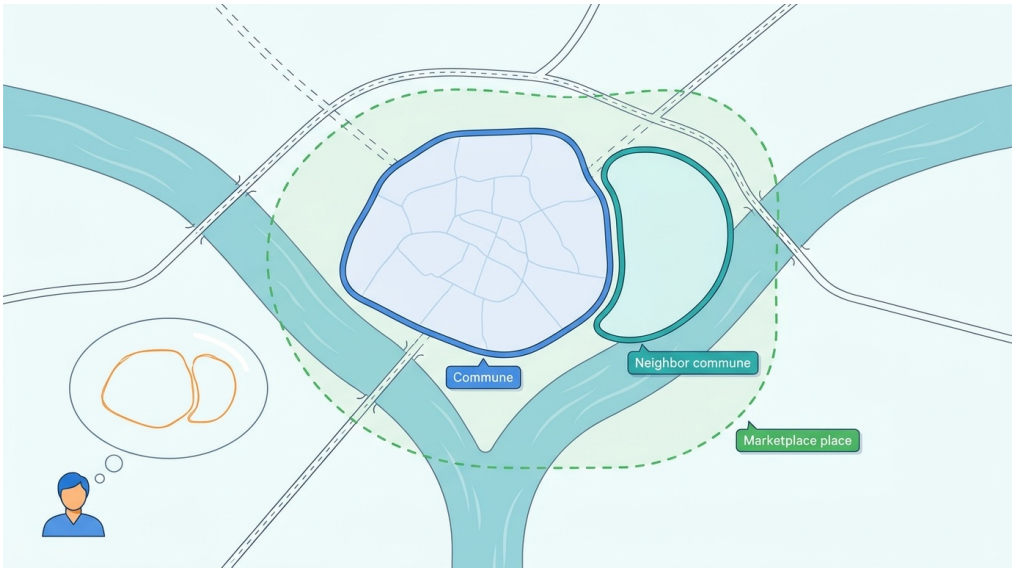


Figure 5: *"In Lyon" can mean the commune or the wider area buyers have in mind, so the gazetteer holds administrative places and curated marketplace places, each with its own ID.*

Brussels in two languages. The Brussels-Capital Region has 19 municipalities with official French and Dutch names. A Tradeyard Belgium buyer may type "Elsene" or "Ixelles" for the same place. Names belong to place IDs, not the reverse, and alias lists cover common spellings and English names ("The Hague", "Den Haag", "'s-Gravenhage").

Polygon or place? A drawn polygon and an administrative place answer different questions. A polygon is a geometric filter on points. An administrative place is a containment fact that was computed once, when the listing was published, against a specific gazetteer version. They can disagree at boundaries, for example when a listing's public point was snapped across a municipal line. The rule Veldmark adopts is that **place filters use stored place IDs and polygons use public points**, and the listing page always names the place that was stored, so what the buyer sees is consistent with the filter that found it.

Cross-border radius. Motorvane operates in Germany, the Netherlands, Poland and Austria. A buyer in Aachen searching "within 50 km" is, geographically, searching parts of three countries. Whether the result should include Motorvane's Dutch listings is a product decision, but it is only possible if the three countries' listings share one coordinate system, one gazetteer and one index or a federated query. A **country fork in the location model rules out cross-border search** before anyone has decided whether it is wanted.

Travel time instead of distance. For housing and jobs, buyers care more about commute time than distance. Rightmove's "My Places" feature computes travel times

from listings to a user's saved places using the third-party TravelTime API. The vendor's case study says Rightmove chose a fixed annual licence with unlimited requests partly to avoid the cost growth of pay-as-you-go pricing (TravelTime, n.d.; a vendor case study). The lesson for this chapter is that **the routing provider's commercial model is an architectural constraint**, in the same way as the geocoder's licence. Searching by Place: Map, Polygon and Commute Search covers isochrone queries in depth.

Geocoding Providers and Licences

What the terms say

Licences for geocoding and map data differ in ways that decide what a marketplace can store. The terms below were checked on the providers' pages in October 2026; providers change them, so treat this as a snapshot.

Option	What the terms say	Consequence for a listing system
Google Geocoding API	Caching and storage are generally restricted, but place IDs may be stored indefinitely. Results displayed on a map must be shown on a Google Map (Google, last updated 2026-10-16)	A permanent table of listing coordinates, shown on a non-Google map, does not fit these terms. Store place IDs and re-geocode as the terms allow, or use another provider for stored coordinates
Mapbox Geocoding	Default "temporary" results "are not allowed to be cached"; `permanent=true` allows storing results indefinitely and requires a payment method or contract (Mapbox, n.d.)	Storage rights are a separately priced product. Budget for permanent geocoding if coordinates live in the listing database

Option	What the terms say	Consequence for a listing system
OpenStreetMap Nominatim, public instance	"An absolute maximum of 1 request per second"; bulk geocoding discouraged; results must be cached by the user (OSM Foundation, n.d.)	Unusable for production ingestion. Self-host Nominatim or another geocoder, or buy a hosted OSM-based service
OpenStreetMap data	Open Database License: attribution required; a database built on the data may be distributed "only under the same license" (OpenStreetMap, n.d.)	Get legal advice on whether an enriched place or address database counts as a derivative database before mixing it with proprietary data
Pelias	Open-source geocoder on Elasticsearch; imports OSM, OpenAddresses, Who's On First and GeoNames; MIT licensed (Pelias, n.d.)	A natural self-hosted option for a marketplace that already runs Elasticsearch or OpenSearch, and a source of place hierarchies
National address registers	The Dutch BAG is open data under the Creative Commons Public Domain Mark, with municipalities as the source holders (Kadaster, n.d.). France's Base Adresse Nationale is published under Licence Ouverte 2.0 (data.gouv.fr, updated 2026-10-19). Great Britain's OS Open UPRN publishes coordinates for about 40 million addressable locations, free to use (Ordnance Survey, n.d.)	Where a good open register exists, it is usually the best primary geocoding source: authoritative, storable and free. Coverage and licence vary by country

The general rule is that **the provider is chosen per country and per use, and the licence is a property of every stored coordinate**. Geocoding a listing for storage and indexing is one use. Autocomplete in the insertion form is another, with different volume and different terms; public Nominatim, for example, explicitly prohibits client-side autocomplete. Displaying a base map is a third, often from a different provider again.

Building versus buying

Most marketplace groups end up with a mix:

- **National registers, self-hosted**, where they exist and are open: the Netherlands and France in Veldmark's footprint. They give authoritative, storable rooftop points for addresses and stable address identifiers that also help duplicate detection.
- **A commercial geocoder with storage rights** for countries without a usable open register and for messy free-text input.
- **An open-data gazetteer** (built from OSM, Who's On First or national boundary datasets) for administrative places, extended with the marketplace's own curated places. The legal review of share-alike terms happens once, at the gazetteer boundary.

The volumes that drive the cost are usually smaller than people fear. Most private listings are geocoded from a postcode, which is a lookup in a local table, not a provider call. Full address geocoding is concentrated in real estate and dealer feeds, where addresses change rarely and results can be stored.

In Practice at Veldmark

The Rotterdam city bike

Femke lists her €240 city bike on Tradeyard Netherlands on Sunday 7 March 2027 at 19:42. Her account profile holds her six-character postcode in Rotterdam. Under the policy Ingrid's team adopts later that year, the bike's location would be processed like this:

1. **Capture**. The insertion form pre-fills the postcode from her profile and asks whether the bike is there. She confirms. Her photos were taken in her hallway, and their EXIF metadata carries GPS coordinates within a few meters of her door. The media pipeline strips EXIF on upload (Photos Sell: Media Pipelines for Listings); this chapter's privacy rule would be pointless without it, since EXIF carries `GPSLatitude` and `GPSLongitude` tags (ExifTool, n.d.).

2. **Geocode**. The postcode resolves, through the gateway, to the centroid of that six-character postcode from the national address register. Match level: postcode centroid. Stored precision: close to P5 in a dense street, which is exactly why it is restricted.
3. **Place references**. The resolver returns the four-digit postcode area 3014, the district, the municipality of Rotterdam, the province of South Holland and the Netherlands, all stamped with the gazetteer version.
4. **Public location**. The Tradeyard private-seller rule sets display level P3. The public point is the center of the H3 resolution-7 cell that contains her postcode centroid, constrained to stay inside postcode area 3014. The label reads "3014 Rotterdam".
5. **Indexing**. The search document receives the public point, the place IDs and the label. The event carries the same.
6. **Search**. A buyer 2.6 km away sees "about 3 km" and a shaded district on the map. Of the 312 saved searches the bike matches, those with a radius are evaluated against the public point (Tell Me When It Appears: Saved Searches and Alerts).

Nothing in steps 4 to 6 can reveal more than the district. The precise centroid exists only in the restricted store. The 20:11 fake payment link that reaches Femke that evening is a messaging and fraud problem, not a location one (From Interest to Contact: Messaging and Lead Management); but a scammer who wanted to turn up at her door instead would now find nothing more precise than a 5 km² cell.

The Lyon apartment

The three-room, 68 m² flat in Lyon 7e arrives in May 2027 from three agencies through two software vendors and Keybridge. The feeds carry full addresses with differences in spelling and one with the unit number.

1. **Parse and normalize**. The parser splits street, number, postcode and commune; normalization expands abbreviations ("av." to "avenue") so the three addresses compare equal.
2. **Geocode**. The French national address base gives a rooftop point for the building and a stable address identifier. All three listings resolve to the same building within a few meters.
3. **Place references**. Lyon 7e arrondissement, commune of Lyon, Métropole de Lyon, Rhône, Auvergne-Rhône-Alpes, France, plus Hearthly's curated places that contain the point.
4. **Precision**. Each agency chooses its display level in its account settings (Private Sellers, Dealers and Agencies: Modeling Who Lists). Agence Rhône Habitat shows the street without a house number (P4); the other two show the arrondissement (P3). The listing page for the clustered property uses the coarsest choice among the agencies whose listings are shown together, so one agency's discretion is not overridden by another's openness.

5. **Duplicate detection.** One Property, Three Listings: Duplicates and Entity Resolution reads the precise points from the restricted store, blocks on H3 resolution-9 cells and address identifier, and clusters the three listings into one canonical property.
6. **Valuation.** Valuo uses the precise point and the property's attributes to estimate €372,000 to €401,000, drawing on geolocated public transaction data for comparables.

Agencies hide the street for privacy and for business. The owner may still live there, and an agency that publishes the exact address invites buyers to approach the owner directly, bypassing the agency. Whatever the reason, the portal's job is to honor the choice consistently, including in the map, the structured data and the alert e-mail.

The Barcelona fake-rental ring

In July 2027, 26 accounts post 180 fake rental listings in Eixample with photos stolen from real listings, priced about 35% below market and asking for deposits by bank transfer. The ring is detected through perceptual-hash matches against existing listings and price anomalies (Scams, Fakes and Account Takeovers: Fraud Prevention for Listings). Location is part of both signals:

- **Price anomaly is a location feature.** "35% below market" means below the median asking rent for comparable flats in the same small area. Hearthly computes that median per H3 resolution-8 cell (average area 0.737 km²) and property type. Without a consistent grid, "the market" would be the whole of Barcelona, and the anomaly would disappear in the average.
- **Stolen photos come with a location.** Many of the stolen photos belonged to listings in other districts. A perceptual-hash match between two listings whose precise points are kilometers apart is a much stronger fraud signal than a match between two listings in the same building, which may be the legitimate case of two agencies listing the same flat.

After the incident, Aisha Rahman's team adds two more location signals for new accounts: a listing location far from every location the account has shown in its device and network history, and a burst of listings concentrated in one high-demand district. Neither signal is decisive alone, but both are cheap and work well in combination.

Location quality, measured

Ingrid's team defines a small set of location quality metrics per brand and country, published weekly:

Metric	Definition	Tradeyard NL, Q3 2027	Hearthly FR, Q3 2027
Match-level mix	Share of listings geocoded at rooftop, street, postcode, locality level	97% postcode centroid; 2% street; 1% locality	88% rooftop; 9% street; 3% locality
Low-confidence rate	Share of new listings whose geocode fell below the acceptance threshold	0.4%	2.1%
Coordinate defects	Points at (0, 0), outside the country, or with latitude and longitude swapped	31 listings	112 listings
Place drift	Listings whose stored place IDs disagree with the current gazetteer version	0 after the January backfill	4,800 pending a commune merger update
Precise-point exposure	Precise points found in any public payload by the weekly scan	0 (target)	0 (target)

These figures are Veldmark's illustrative baselines. Swapped coordinates are a perennial bug because GeoJSON and WKT order coordinates as longitude, latitude, while most people say "latitude, longitude" (Elastic documentation, n.d.). A Lisbon listing with swapped coordinates lands in the Indian Ocean. The quality check is trivial; the damage when nobody runs it is not.

Trade-offs

Where to store location

Option	How it works	Strengths	Weaknesses	When it is right
Single point per listing	One latitude-longitude field, read by everyone	Simple; fast to build	Privacy depends on every consumer blurring correctly; no audit; licence unclear	A prototype, or a vertical where every listing is a business at a public address
Precise plus public point	Restricted precise point; derived public point	Privacy enforced once; distances safe	No place hierarchy, so "in Rotterdam" needs polygons	Small single-country marketplaces
Three representations	Precise, public, and place references with ancestors	Privacy, consistent filters, cheap landing pages, backfillable	A gazetteer to run and govern; more fields per event	Multi-country groups and any marketplace with SEO landing pages
Property-first (real estate)	Location belongs to a persistent property entity; listings reference it	Location agreed once for every listing of a property; supports valuation and history	Requires entity resolution and a property store	Real-estate portals at scale (One Property, Three Listings: Duplicates and Entity Resolution)

Which geo index for which job

Job	Best fit	Why	Avoid
Radius and polygon filters at query time	Engine-native `geo_point` and `geo_shape`	Exact geometry; maintained by the engine	Geohash prefix filters, which miss neighbors at cell edges
Map clustering by tile	Engine tile aggregations	Matches map zoom levels	Precomputed clusters that go stale as inventory churns
Heat maps, local price statistics, privacy cells	H3	Uniform neighbors and areas; the same cell everywhere	Administrative areas of very different sizes
Key-value lookups and range scans	S2 or geohash	Locality-preserving 64-bit or string keys	Engine queries for simple lookups
"All listings in place X"	Place-ID term filter	Cheap; stable across geometry changes	Polygon queries on large administrative areas
Duplicate blocking (real estate)	H3 fine resolution plus address identifier	Candidate pairs limited to nearby listings	Exact coordinate equality, which misses geocoder noise

Public-point methods

Method	Privacy	Map quality	Main risk
Show place label only, no pin	Strong	Poor; buyers cannot judge distance	Lower engagement on map-led verticals
Snap to H3 cell center	Good in dense areas; weak where a cell holds one house	Pins stack on cell centers	Sparse-area exposure unless the policy coarsens

Method	Privacy	Map quality	Main risk
Deterministic seeded offset	Good if seeded per listing and bounded to the place	Natural-looking spread	Re-randomized offsets can be averaged out
Circle of uncertainty	Good	Honest, but large circles look vague	Circles drawn around the precise point give the center away

The last row is a classic mistake: a "privacy circle" centered on the real address reveals the address as its center. The circle must be centered on the public point.

Vertical Differences

Dimension	General classifieds	Real estate	Automotive	Jobs and services
What is located	The item, usually at the seller's home	A persistent property	A vehicle on a dealer lot or at a private home	A workplace, a service area, or "remote"
Typical capture	Postcode from the profile	Full address from agency software	Dealer branch address from the contract; postcode for private sellers	Office address; region; remote flag
Default public precision	District (P3) for private sellers	Agency's choice; district for private landlords	Exact for dealers (a business address); locality or district for private sellers	Locality for workplaces; region for service areas

Dimension	General classifieds	Real estate	Automotive	Jobs and services
Role of location in search	Radius filter, often secondary to category	**The primary filter**; maps, polygons, commute	Radius matters, but buyers travel far for the right car	Commute; remote and hybrid flags change the shape
Location in duplicate detection	Weak	**Central**: same building, same unit	Weak; the VIN identifies the car (From Raw Supply to Usable Listings: Normalization and Enrichment)	Moderate: same employer, same office
Regulatory angle	GDPR for private sellers	GDPR; mandatory disclosures linked to the property, such as France's energy class	Trader address display for dealers (Rules of the Marketplace: DSA, DAC7 and Vertical Regulation)	Equality law on job ads; remote-work tax questions
Shipping	Changes radius semantics for shipped goods	Not applicable	Occasional delivery services	Not applicable

Real estate is the vertical where location carries the most weight. The property outlives every listing of it, so its location is resolved once against the property, not per listing (Same Platform, Different Physics: How Listing Verticals Differ). General classifieds is the vertical where location is most often personal data. Automotive sits in between: dealer addresses are public business information, and the car's identity comes from the VIN, not from where it stands.

Failure Modes

1. **The precise point leaks through a side door**. The page shows a district, but the precise point is in the search API response, the JSON-LD structured data, an image's EXIF, a sitemap, an analytics event, a partner feed or an alert e-mail's static map. The control is architectural: the public read models never receive the precise point, and a weekly scan of public payloads looks for coordinates that match restricted ones.
2. **Distances give away the address**. Precise distances from a searcher's point to a seller's precise point allow trilateration. Compute from the public point; round to the display precision.
3. **Jitter that is not deterministic**. A random offset recomputed on each request or edit is averaged out by anyone who collects enough samples.
4. **Licence debt**. Coordinates from a provider whose terms forbid storage, or whose results may appear only on its own maps, sit in the listing database for years. Discovering this during a provider negotiation or an acquisition's due diligence is expensive. Record provider and licence per coordinate from day one.
5. **Name-based place matching**. Systems that join on place names break on bilingual names, renames and homonyms; there are several Bergens in Veldmark's markets alone. Join on IDs.
6. **Stale place references after boundary reforms**. Municipal mergers and boundary changes happen every year in several European countries. If place IDs were computed once and never recomputed, landing pages and filters quietly diverge from reality. Recompute ancestors as a backfill each time the gazetteer version changes, the same way Category Trees That Can Change: Designing Marketplace Taxonomies handles category migrations.
7. **Centroid fallbacks treated as precise**. A listing geocoded to "Rotterdam" lands on the city centroid. If the match level is not stored, that listing appears in every "near the center" search and distorts local price statistics. Low-precision listings should be excluded from fine-grained statistics and shown as area matches.
8. **Polygon queries that explode**. A hand-drawn polygon with thousands of vertices, or an administrative boundary at full resolution, makes every query slow, because shape tessellation cost grows with vertex count. Simplify shapes before querying and use place-ID filters for administrative areas.
9. **One country's model imposed on another**. A Dutch-style "postcode plus house number" form fails in Portugal and Poland, and a French "commune" field means little in the Netherlands. Address formats belong in configuration per country.

The Chief Architect's View

In most organizations nobody owns location end to end. Search teams treat it as an index field, product teams as a form field, legal teams as a privacy question, and finance

as a line on a provider invoice. At group scale it needs one owner and four explicit decisions.

Decision 1: one gazetteer and geocoding gateway for the group. Place IDs are shared language across search, alerts, landing pages, ads targeting, analytics and fraud. If each brand keeps its own, the group cannot aggregate, cannot run cross-border search and pays for several provider contracts. Veldmark's decision record, approved at the architecture advice forum in May 2027, puts a **Veldmark Place** service (gazetteer, geocoding gateway, place resolver and precision policy engine) inside the Common Ground taxonomy and attribute platform, under Yara Mendes's platform engineering organization. It goes live for Tradeyard alongside the taxonomy service in October 2027, with Hearthly and Motorvane following in 2028.

Decision 2: the precision policy is group policy with brand parameters. The rule that precise private locations never leave the restricted store is not negotiable per brand. The display levels per category and seller type are brand parameters, owned by brand product leads and reviewed by Lucía Ortega's privacy team. Changes to the policy are versioned and backfilled like any other reference data. Hearthly Spain's exact pins for private landlords are the first thing the policy fixes, in its first release.

Decision 3: providers are chosen per country and use, behind the gateway. National open registers where they exist; one commercial provider with storage rights for the rest; a self-hosted open geocoder for bulk work. The gateway's licence register is reviewed when contracts renew and during every acquisition's due diligence. When Veldmark bought Casavista in 2025, nobody asked what its coordinates were licensed for; the next deal team will.

Decision 4: location quality is a published metric. The match-level mix, coordinate defects, place drift and precise-point exposure are reported weekly per brand and country, next to the publication-time and duplicate-rate metrics in From Submitted to Searchable: The Publication Pipeline and One Property, Three Listings: Duplicates and Entity Resolution. Precise-point exposure has a target of zero and is treated as a privacy incident when it is not.

What to watch over the long term: open address registers improving in more countries, which can retire commercial contracts; the share of shipped transactions, which changes what location means for general goods; regulators' interest in how platforms protect private sellers' locations; and AI search agents that scrape listing pages and analytics endpoints at scale, which make every leaked coordinate easier to harvest (AI Across the Listing Lifecycle).

The design is too much for a single-country marketplace where nearly every listing comes from businesses at public addresses: it can live with a precise point and a place hierarchy, and skip the policy engine. The moment private sellers appear, it cannot.

Questions to Ask

1. For each brand, what is the most precise location of a private seller that any public payload contains — page, API, structured data, sitemap, feed, analytics, alert e-mail?
2. Are distances shown to buyers computed from the precise point or the public point, and how are they rounded?
3. Is the public point derived deterministically, and does it stay inside the public place?
4. For every stored coordinate, can we name the provider, the data version and the licence terms that allow us to keep and display it?
5. Do listings reference places by stable ID with all ancestors, or by name? What happens to landing pages and filters when a municipality merges?
6. Where does "in [city]" mean the administrative unit and where a marketplace-curated area, and who decides?
7. Which internal consumers read precise locations, for which documented purpose, and is each read logged?
8. Do we store the geocoder's match level and confidence, and do low-precision listings stay out of fine-grained price statistics?
9. Could a buyer search across our country sites near a border today, and if not, is that a product decision or an accident of the data model?
10. What is our weekly figure for coordinate defects, and who looks at it?

From Place to Seller

Location tells the marketplace where a listing is; it does not say who is selling it. That question — private sellers, dealers, agencies with fourteen branches, and the accounts, roles and integrations behind them — is the subject of Private Sellers, Dealers and Agencies: Modeling Who Lists.

Sources and Evidence

Company engineering and investor sources

- Setting expectations with an accurate location — Airbnb, 2025. Default approximate-location radius and address release after booking.
- Embedding-based retrieval at Airbnb — Airbnb, 2026. S2 cells as a listing-location feature; the prevalence of map-based filters.

- H3: Uber's Hexagonal Hierarchical Spatial Index — Uber, 2018. Design and rationale of H3.
- Rightmove property POI search case study — TravelTime, n.d. (vendor case study). Travel-time search and the fixed-licence pricing choice.

Standards, regulation and documentation

- Article 4 GDPR — Definitions — unofficial reproduction of Regulation (EU) 2016/679, n.d. Location data as an identifier of a natural person.
- Geocoding API policies — Google, last updated 2026. Storage restrictions, place-ID exception and display on Google Maps.
- Geocoding API — Mapbox, n.d. Temporary versus permanent geocoding.
- Nominatim usage policy — OpenStreetMap Foundation, n.d. Rate limit, caching and autocomplete rules for the public instance.
- Copyright and License — OpenStreetMap, n.d. Open Database License attribution and share-alike terms.
- Key:admin_level — OpenStreetMap Wiki, n.d. Administrative levels and their variation by country.
- Placetypes — Who's On First, n.d. Stable place IDs, placetypes and multiple hierarchies.
- H3 resolution table — H3 project, n.d. Average cell areas and edge lengths.
- S2 cell hierarchy — S2 Geometry, n.d. Cell levels, Hilbert ordering and region covering.
- Geo-shape field type — Elastic, n.d. BKD indexing, tessellation cost and coordinate order.
- Basisregistratie Adressen en Gebouwen (BAG) — Kadaster, n.d. Dutch address and building register as open data maintained by municipalities.
- Base Adresse Nationale — data.gouv.fr, updated 2026. French national address base under Licence Ouverte 2.0.
- OS Open UPRN — Ordnance Survey, n.d. Open coordinates for about 40 million addressable locations in Great Britain.
- GPS tags — ExifTool, n.d. GPS metadata carried in image EXIF.

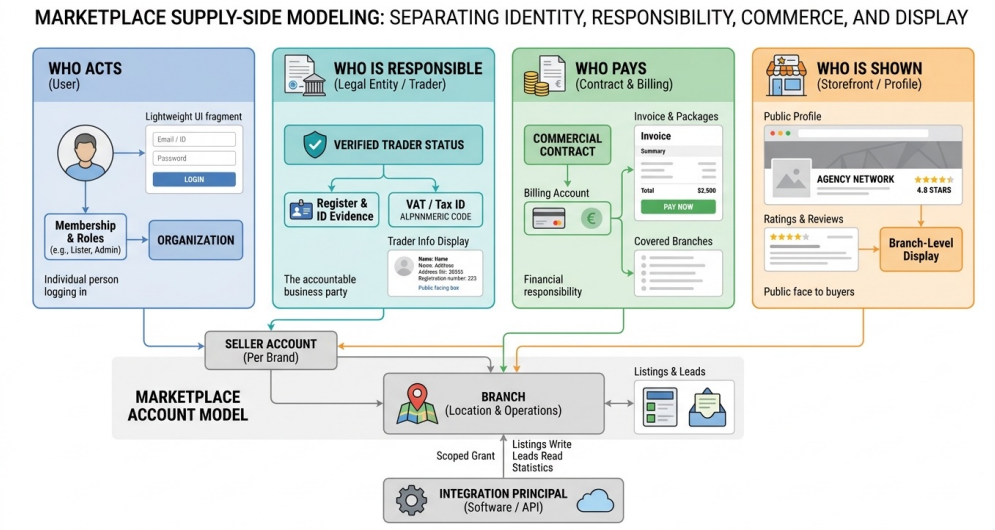
Research and practitioner sources

- libpostal — OpenVenues, n.d. Statistical address parsing and normalization; reported accuracy and model size.
- Pelias — Pelias project, n.d. Open-source geocoder on Elasticsearch with open data importers.

- *Productscape domain models for general listings and real-estate marketplaces* — the author's research digest, 2026. Geo reference and property reference data assets; personal-data classification of listings.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

12. Private Sellers, Dealers and Agencies: Modeling Who Lists



IN THIS CHAPTER, YOU WILL: Model the supply side of a marketplace, from a private seller with one bike to an agency network with fourteen branches, five of them franchised. You will see how organizations, branches, users, roles and integrations fit together, why trader status has to be a governed attribute, and how a multi-brand group can share identity without merging every account.

WHY THE BUSINESS CARES: Professional sellers pay most of the bills in vertical marketplaces, and they buy as organizations: a dealer group with several sites, an agency network with franchisees, a head office that pays for some branches but not others. An account model that cannot represent them produces wrong invoices, misrouted leads and reviews attached to the wrong business, and it fails the trader rules in EU consumer and platform law.

WHY ARCHITECTS SHOULD CARE: The seller account is referenced by almost every other context: listings, leads, entitlements, billing, trust,

regulation and analytics. Its identifiers end up in feeds, invoices and regulatory reports. Getting its shape wrong is expensive to fix, because every one of those contexts has copied it.

KEY POINTS:

- **Who acts, who is responsible, who pays and who is shown are four separate questions.** A user logs in and acts; a legal entity is the trader and is responsible; a billing account pays; a storefront or branch is what buyers see. Model each one explicitly. In simple cases they collapse into a single record; in an agency network with franchisees, none of them coincide.
- **Trader status is a governed attribute, not a checkbox.** It decides which consumer law applies, what the listing must disclose, what must be verified under the EU Digital Services Act and, once payments run through the platform, what is reported to tax authorities. Store the declaration, the evidence, the verification result and the history, and run a process for private sellers who behave like traders.
- **Organizations own listings; integrations are principals; groups share identity, not accounts.** Staff come and go, so listings belong to the organization and are assigned to branches. A software vendor posting for 500 agencies is one client with 500 scoped grants. Across a group, share the login and the verified legal-entity facts, and keep brand accounts separate, with purpose limits on how they are linked.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Billing by Guesswork** — when the account hierarchy does not say who bought a package for which branches, every service that checks entitlements makes up its own answer.
- **Deal-Shaped Architecture** — each acquired brand brings its own seller accounts, and the group cannot even count how many businesses it serves.
- **Feed Roulette** — integrations authenticated with shared passwords and no scoped grants cannot be told apart, throttled or revoked one dealer at a time.

Part II has modeled the listing as an aggregate with its category, attributes, media and location; the last governed subdomain it references is the seller account, and "seller" means something different in every vertical. On a general classifieds site it usually means a person with a phone number. On a real-estate portal it means an agency branch, staffed by agents, owned by a company that may be one franchisee in a national network, publishing through a software vendor's feed. On an automotive marketplace it means a

dealer group with several sites, a DMS (dealer management system) vendor that uploads the stock, a head office that negotiates the package, and salespeople who answer the leads.

Most marketplaces start with the first meaning and extend it one exception at a time, until the "user" table has a company name column, a parent-user ID for branches, a "pro" flag, a VAT number field and a shared login for the DMS vendor. **The account model is the place where marketplaces most often confuse identity with commerce**. This chapter builds a model that keeps them separate from the start and still collapses to "one person selling one bike". Later chapters build on it: entitlements attach to its accounts and branches (Packages, Bumps and Quotas: Products, Pricing and Entitlements), leads route through its branches (From Interest to Contact: Messaging and Lead Management), and partner feeds authenticate as its integration principals (Many Doors In: Forms, APIs and Bulk Feeds).

The chapter works through the model with Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build): Femke, a private seller on Tradeyard; Autohaus Ruhrblick, a dealer on Motorvane; and Agence Rhône Habitat, an agency network that lists on Hearthly and runs its business on Keybridge.

Setting the Scene at Veldmark: Fourteen Branches, Three Wrong Answers

In May 2027, Agence Rhône Habitat publishes the three-room flat in Lyon 7e that later becomes this book's example of entity resolution. Agence Rhône Habitat (fictional) is a network of **14 branches** around Lyon. Nine are operated directly by the parent company; **five are franchisees**, separate companies that use the network's brand under contract. The agency manages its listings in Keybridge, Veldmark's agent CRM, which publishes them to Hearthly and to two competing portals.

The same week, the network's head of digital sends Claire Fontaine, Hearthly's CEO, a letter listing three problems:

1. **The wrong trader**. On Hearthly, listings from the five franchise branches show the parent company's name, registration number and address in the trader-information panel. Legally, each franchisee is the trader for its own listings.
2. **Reviews in the wrong place**. Buyer reviews of one franchisee, including some angry ones, appear on the profile of the whole network and on every branch page.
3. **An agent who left**. An agent who left the Villeurbanne branch in February still "owns" 212 listings on Hearthly. New leads for those listings go to her personal e-mail address, which she had used to create her Hearthly login in 2023.

A fourth problem is one Hearthly finds itself: the five franchisees' subscriptions are invoiced to the parent company, which has started refusing to pay them.

Julien Marchand, Hearthly's CTO, traces all four to one design decision. Hearthly's account model has a single entity, the **advertiser**, which is simultaneously the login, the legal identity, the billing party, the public profile and the owner of listings. Branches are advertisers with a parent-advertiser pointer. Users are e-mail addresses attached to an advertiser. Feeds authenticate as the advertiser. The model was built for single-office agencies and has been stretched to cover networks, franchises and staff turnover.

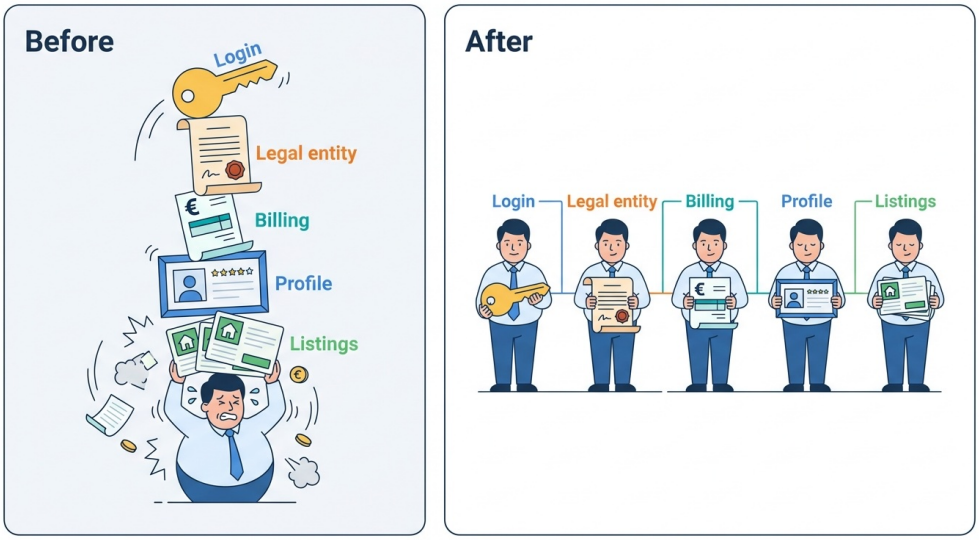


Figure 1: *Hearthly's advertiser was at once login, legal identity, billing party, public profile and listing owner; separating those roles fixes the wrong trader, misplaced reviews, orphaned listings and wrong invoices.*

Ingrid Halvorsen, Veldmark's Group Chief Architect, sees the same shape in every brand. Tradeyard has user accounts with a "business" flag. Motorvane has dealer accounts keyed by its customer number, with sites added as sub-accounts and DMS vendors logging in with dealers' credentials. Keybridge, a SaaS product, has a proper tenant and user model, but with no link to anything else in the group, by design. With Veldmark ID due for approval in June 2027 and a group product catalog due in 2028 (Packages, Bumps and Quotas: Products, Pricing and Entitlements), the account model has to be settled first, because both depend on it.

Concepts: The Vocabulary of Supply

Private and professional sellers

A **private seller** is a person selling their own goods outside any trade or business. A **professional seller** (dealer, agency, merchant, developer, employer) sells in the course of a business. The distinction matters commercially, because professionals pay subscriptions and need tooling (Serving the Pros: Dealer, Agent and Merchant Tooling). It matters even more legally.

In EU law the term is **trader**. The Digital Services Act (DSA) imposes **trader traceability** on online platforms that allow consumers to conclude distance contracts with traders. Before a trader can use the service, the platform must obtain "the name, address, telephone number and email address of the trader", a copy of an identification document or electronic identification, "the payment account details of the trader", the trade register and registration number where applicable, and a self-certification that the trader only offers compliant products or services. The platform must "make best efforts to assess whether the information … is reliable and complete", keep the information securely "for a period of six months after the end of the contractual relationship with the trader concerned" and then delete it. It must show the name and contact details, the registration and the self-certification to buyers "at least" where the product or service is presented (Regulation (EU) 2022/2065, Art. 30, applying since 17 February 2024).

Whether a classifieds site that only connects buyers and sellers lets consumers "conclude distance contracts" is arguable; the safe reading, used throughout this book, is that its trader listings are in scope. Rules of the Marketplace: DSA, DAC7 and Vertical Regulation owns the legal analysis. This chapter's concern is what the obligations do to the data model:

Regulation	What depends on trader status	Data the account model must hold
DSA Art. 30 (trader traceability)	Onboarding gate, verification, display on listings, retention and deletion	Legal name, address, contacts, ID evidence, payment account, register number, self-certification, verification status and dates, end of relationship

Regulation	What depends on trader status	Data the account model must hold
EU consumer law (Modernisation Directive, applying from 28 May 2022)	Marketplaces must tell consumers whether the counterparty is a trader, based on the counterparty's own declaration, and that consumer rights do not apply when it is not (European Commission, n.d.)	Seller's declaration, its date, and what was shown to the buyer
EU Platform-to-Business Regulation	Business users get ranking transparency and complaint handling	Which accounts are business users
EU DAC7 tax reporting	Once the platform knows the consideration paid to sellers, it reports sellers above the threshold to tax authorities. Goods sellers with fewer than 30 sales and €2,000 or less in the period are excluded (Belastingdienst, n.d.)	Name, address, date of birth or registration, tax identifiers, financial account, quarterly consideration and fees, per seller and Member State
US INFORM Consumers Act (for comparison)	High-volume third-party sellers on marketplaces that process their payments must be verified, and larger ones disclosed (FTC, 2023)	Verified identity, bank account, tax ID, annual re-certification

One verified record serves many laws, each with its own view of it. The obligations also grow with the business model: a pure lead-generation classifieds site sits mostly outside DAC7 and INFORM, but **launching in-platform payments brings seller reporting with it**, because the platform then knows what each seller was paid. Tradeyard Safe Deal is that change (Beyond the Contact: Payments, Shipping and Buyer Protection).

Trader status as a governed attribute

Trader status is the platform's record of whether a seller account acts as a trader. It has four parts:

1. **Declaration** — what the seller said, when, in which flow, with which wording on screen.
2. **Evidence** — register extracts, VAT numbers, ID documents, eID results, payment-account checks.
3. **Verification result** — the platform's assessment, by which method, at which date, with an expiry or re-check date.
4. **Platform assessment** — whether the platform's own signals agree with the declaration.

The fourth part is where marketplaces most often have nothing. A private seller who lists 400 phone cases a year in their original packaging is very likely a trader, whatever they declared. The account model needs a **reclassification process**: signals (listing volume, new goods in volume, repeated identical listings, business-like descriptions, high Safe Deal turnover) open a case; the seller is asked to confirm or explain; the outcome changes the status, the display and the obligations, and is recorded. The signals belong to trust and safety (Trust and Safety as a Platform); the status, the case outcome and its history belong to the account.

Organizations, branches and networks

An **organization** is a legal entity that sells: a company, a sole trader's business, a public body. It is the trader, signs contracts and is the subject of business verification (Who Is on the Other Side? Identity and Verification covers KYB, know-your-business checks). Industry practice and published reference models add three more levels:

- **Branch** (also site, location, rooftop, office): a physical place of business with its own address, opening hours, staff, inventory and lead routing. Branches usually belong to one organization.
- **Network** (franchise, brand group, dealer group): an umbrella across *several* organizations that share a brand. A franchise network is not a legal parent of its franchisees, which is why a model that makes branches children of one legal entity fails.
- **Storefront** or **profile**: the public identity shown to buyers (name, logo, description, ratings). It may correspond to a branch, an organization or a network brand.

Public portal interfaces use the same levels. Rightmove's Real Time Datafeed specification gives each software provider a **Network ID** and each agent office a **Branch ID**

with sales and lettings channels; a call "will be rejected if sending a channel to which the branch does not have access", and the feed provider must be authorized for each branch (Rightmove RTDF specification v1.4.1, 2023).

The choice of commercial unit (branch, forecourt, dealer, organization) is a modeling decision that shows up in investor reporting. Rightmove's commercial unit is the branch: its average revenue per advertiser (ARPA) is defined as revenue from agency and new-homes advertisers divided by the number of advertisers, and it reported 16,385 agency branches at the end of 2025 (Rightmove results announcement, 2026). Autotrader UK reports its unit as retailer forecourts (13,942 on average in FY26) with an average revenue per retailer of £2,995 a month (Autotrader FY26 results, 2026).

Users, roles and memberships

A **user** is a person who logs in. In a group with a shared login, the user is a Veldmark ID identity (Who Is on the Other Side? Identity and Verification owns authentication). A user is not a seller. A user has **memberships**: a relation to an organization, scoped to the whole organization or to specific branches, carrying roles such as administrator, lister, lead handler, finance or read-only. A private seller is a user whose only membership is to their own personal seller account.

Two rules follow from separating users from organizations:

1. **Listings are owned by the seller account and assigned to a branch; users are only the last editor.** When the Villeurbanne agent leaves, her membership ends; the 212 listings stay with the branch and are reassigned, and her lead routing ends with her membership.
2. **People can belong to several organizations.** An agent who moves between agencies, a consultant who manages listings for three dealers, an owner of two franchises: one user, several memberships, no duplicate logins.

Integration principals

An **integration principal** is a software client that acts on behalf of sellers: a real-estate CRM, a DMS vendor, a feed aggregator, a listing tool. It is not a user and not a seller.

mobile.de's public Seller API keeps integration principals separate from sellers. Integration is through **Transfer Service Providers** (the DMS and inventory-tool vendors that upload for dealers) or through self-uploading dealers. Providers serving more than ten customers must supply contractual and technical contacts, their software, a VAT ID and customer numbers before access is granted, and every ad must name the dealer's customer number. API credentials are independent of sellers, and an API user sees only the sellers linked to it (mobile.de Seller API, n.d.).

OAuth makes the same separation. The OAuth 2.0 Security Best Current Practice (RFC 9700, January 2025) requires exact-match redirect URIs, recommends PKCE for all clients, forbids the implicit grant, says the resource-owner password grant "MUST NOT be used", and recommends sender-constrained, audience-restricted tokens (IETF RFC 9700, 2025). In listing terms: a third-party tool acting for one agency uses the authorization-code flow, with the agency's administrator consenting to scopes; a contracted feed vendor uses client credentials with grants bound to specific organizations and branches. **A vendor serving 500 dealers is one client with 500 grants, not 500 shared passwords.**

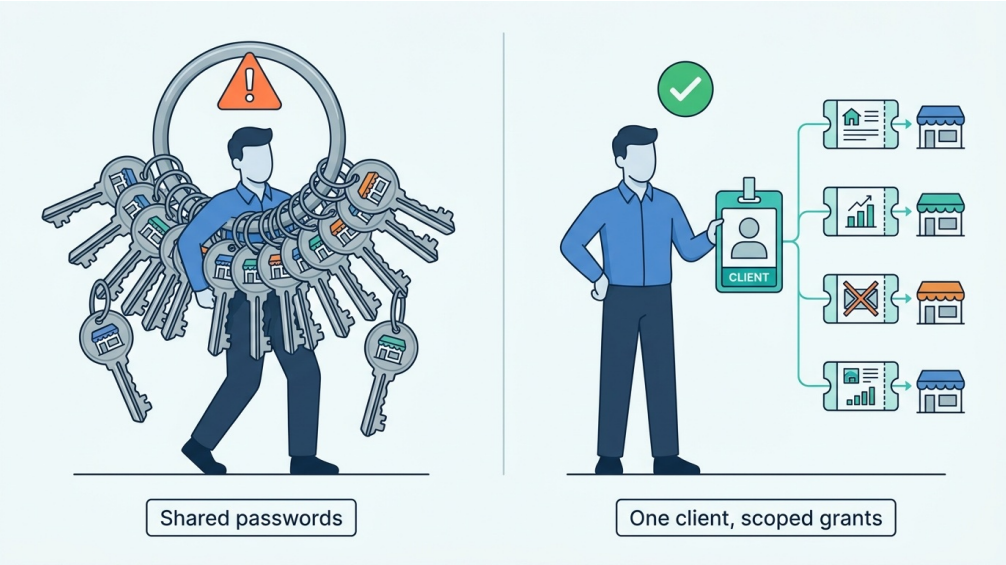


Figure 2: *An integration vendor serving many dealers should be one client with a scoped grant per dealer, never a collection of shared seller passwords.*

Architecture: The Seller Account Model

One entity per question

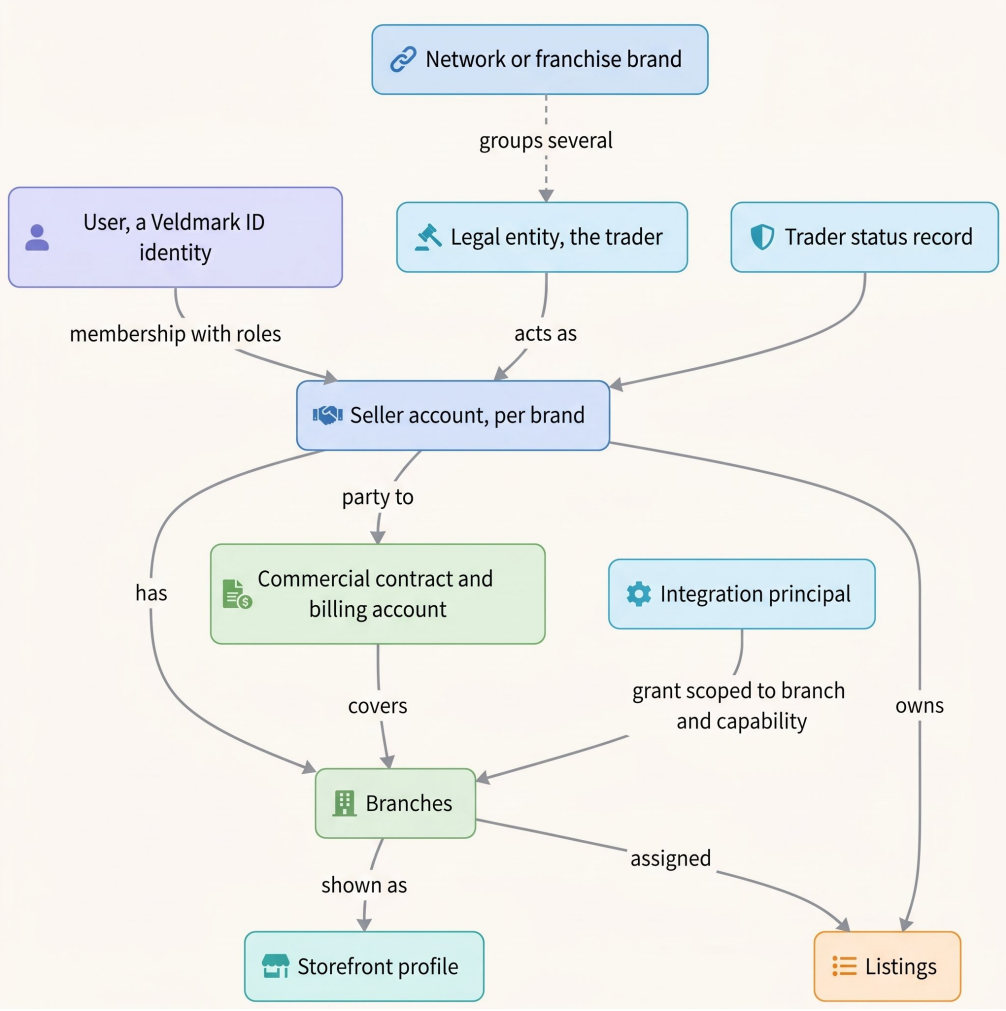


Figure 3: *Eight entities, each answering one question. Users act, legal entities are responsible, contracts pay, and storefronts are shown; seller accounts tie them to one brand, branches tie them to a place, networks group traders under one umbrella, and integrations get scoped grants instead of logins.*

Each entity answers one question and lives in one place:

Entity	Answers	Key attributes	Lives in
User	Who acts?	Identity ID, verified contact points, authentication methods	Group identity (Who Is on the Other Side? Identity and Verification)
Legal entity	Who is responsible?	Legal name, register and number, VAT or tax IDs, registered address, verification evidence and status	Group business party registry
Seller account	Who sells on this brand?	Brand, seller type (private or professional), trader status record, lifecycle state, contact preferences, link to a legal entity or to a private user	Brand account service
Branch	Where, and who handles it?	Address and location (Where Is It? Location, Geocoding and Privacy), opening hours, channels (sale, rent), lead routing rules, assigned staff	Brand account service
Commercial contract and billing account	Who pays, for what?	Contracting party, invoiced party, packages, covered branches, price list, payment terms	Product catalog, entitlements and billing (Packages, Bumps and Quotas: Products, Pricing and Entitlements)

Entity	Answers	Key attributes	Lives in
Storefront	Who is shown?	Display name, logo, description, ratings, the trader information that must be displayed	Brand account service, rendered by the brand
Network	Which brand umbrella?	Network name, member legal entities, shared display assets	Brand account service
Integration principal	Which software acts?	Client ID, vendor legal entity, grants per seller account, branch and scope, quotas	Group integration platform (Many Doors In: Forms, APIs and Bulk Feeds)

In the simple cases, most of this collapses. Femke is a user with one private seller account; there is no legal entity, branch, contract or storefront beyond her public profile. A single-office agency is one legal entity, one seller account, one branch, one contract and one storefront. The model costs nothing in the simple case and is essential in the complex one.

Agence Rhône Habitat, modeled correctly

Element	Before (Hearthly 2027)	After
Legal entities	One advertiser record for the network	Six: the parent company and five franchisees, each with its register number and its own verification
Seller accounts on Hearthly	1 parent advertiser and 13 child advertisers	Six seller accounts, one per legal entity

Element	Before (Hearthly 2027)	After
Branches	14 child advertisers	14 branches: 9 under the parent's account, 1 each under the franchisees'
Network	Implicit in the parent pointer	One network, "Agence Rhône Habitat", grouping the six legal entities for display
Contracts and billing	One contract, invoiced to the parent	Six contracts: the parent's covers its 9 branches; each franchisee pays for its own
Trader information on listings	Parent's details on every listing	The legal entity that owns the listing's seller account
Reviews	Pooled at network level	Attached to the branch, shown at branch level; network page shows a clearly labeled aggregate
Users	61 e-mail logins, 7 of them former staff	Users with memberships scoped to branches; leavers' memberships ended by the agency's administrators
Integration	Keybridge authenticated as the parent advertiser	Keybridge is an integration principal with grants for the 14 branches, scoped to listings and leads

A franchise network is a display and commercial concept, not a legal hierarchy, and the trader rules follow the legal hierarchy. The table is Veldmark's illustration, but any franchise network has the same shape.

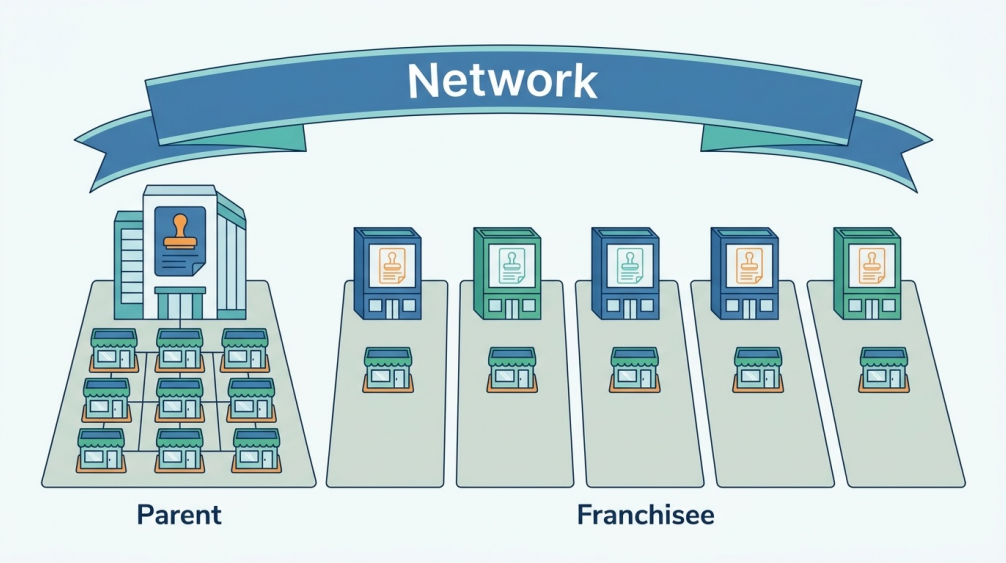


Figure 4: *Agence Rhône Habitat is one network over six legal entities: the parent runs nine branches, and each of the five franchisees is its own trader with one branch.*

Account lifecycle

A **seller account lifecycle** is the set of states an account passes through, from registration to erasure. It differs for private and professional accounts mainly in the onboarding gate.

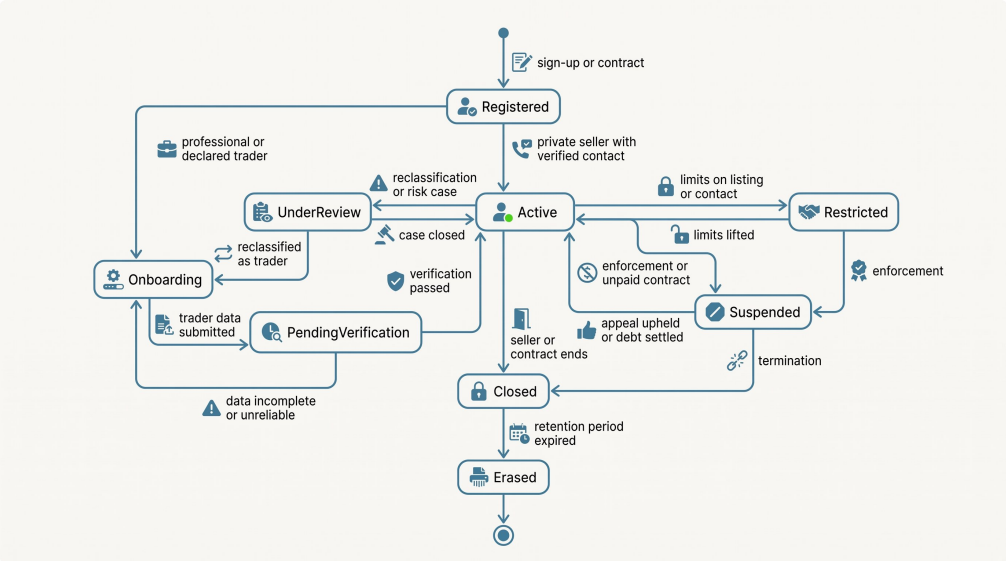


Figure 5: *A seller account's lifecycle. Professional accounts pass a verification gate before their listings go live, private accounts can be reclassified into it, and closing an account starts retention clocks rather than deleting data at once.*

Three transitions need explicit design:

- **Reclassification** (Active to UnderReview to Onboarding) is how a private account becomes a trader. Its listings may stay visible during the case, but new listings beyond a volume limit wait until verification completes.
- **Suspension** must propagate. A suspended account's listings leave search, its integration grants stop working, its scheduled bumps stop and its pending leads are handled. From Submitted to Searchable: The Publication Pipeline owns unpublish propagation; the account service owns the event that triggers it. Every restriction is also a moderation decision that needs a statement of reasons under the DSA (Trust and Safety as a Platform).
- **Closed to Erased** is not one clock but several. DSA trader data is kept for six months after the relationship ends and then deleted. Invoices follow accounting retention rules, which are much longer. Fraud block lists may keep hashed identifiers longer, with a documented justification. The account service has to know which data class follows which clock, and System of Record and Read Models: Listing Data Management has to propagate erasure to every copy.

Contact preferences and lead routing

Contact options belong to the branch and the seller account, not to the listing alone: which channels are open (chat, masked phone, e-mail form, viewing or test-drive booking), opening hours, whether phone numbers are shown or masked, and who receives a lead. A dealer's lead for a car on its Bochum site goes to the Bochum sales team's queue, not to whoever uploaded the car. From Interest to Contact: Messaging and Lead Management owns lead objects and routing; the account model provides the routing rules and their owners.

For private sellers, contact preferences are also safety controls. A new private account whose phone number is hidden until it has some history, and whose messages go through on-platform chat, is harder to use for the off-platform scams described in Scams, Fakes and Account Takeovers: Fraud Prevention for Listings.

Reputation and ratings

Where ratings attach is a modeling decision with legal weight:

Attach ratings to	Works for	Breaks when
The user	Private sellers and buyers	A professional's staff change; the rating follows a person, not the business
The seller account	Single-entity businesses	A franchise network pools ratings across separate traders
The branch	Agencies and dealer groups with several sites	Buyers want a single network view (offer it as a labeled aggregate)
The individual agent	Real estate, where buyers choose agents	The agent leaves; ratings must stay with the branch or move with the agent by an explicit rule

EU consumer law, as amended by the 2019 Modernisation Directive, requires a business that gives access to consumer reviews to say whether and how it checks that they come from people who actually used or bought the product (Article 7(6) of the Unfair Commercial Practices Directive, inserted by Directive (EU) 2019/2161). A marketplace that shows ratings of sellers should therefore record, with each rating, what made the reviewer eligible: a completed Safe Deal transaction, a lead followed by a confirmed viewing, or nothing at all.

In Practice: Documented Patterns and Three Veldmark Accounts

Documented patterns at real marketplaces

The public record on seller models is thin; most of it is in partner API documentation and investor reports, not engineering blogs.

- **mobile.de** separates the seller from the commercial account (a seller ID and a customer number), documents a "multi-locations" resource for dealers with several sites, and treats transfer service providers as separate principals linked to the sellers they serve (mobile.de Seller API, n.d.). This is the clearest public example of integration principals with scoped access.
- **Rightmove's** feed specification models a network (software provider) by branch (office) by channel (sales or lettings) tenancy, authenticated with mutual TLS client certificates issued by Rightmove (Rightmove RTDF specification v1.4.1, 2023). Its

leads and performance calls are also scoped by branch. Rightmove's customer checks and its requirement that agents belong to a redress scheme act as know-your-business controls: Rightmove says it carries out checks on all its customers before they can advertise (Rightmove guide, n.d.).

- **CarGurus** defines a paying dealer as "a dealer account with an active, paid subscription at the end of a defined period" and reported 34,409 paying dealers at the end of 2025 (CarGurus Form 10-K, 2026). The definition is a reminder that the commercial unit is an account with a contract, not a legal entity.
- **The Bundeskartellamt's clearance on 8 October 2026** of mobile.de's acquisition of Carwow's German business noted that dealers and consumers often use several platforms in parallel and that many platforms let dealers export their data to other platforms; its main stated reasons for clearance were Carwow's new-car focus and competition from AutoScout24 (Bundeskartellamt, 2026). Dealers keep their data portable across marketplaces, and the integration-principal model is how a marketplace accepts that.

INFERENCE, NOT DOCUMENTED: None of these sources describes the internal account schema of the companies concerned. The seller account model in this chapter is a reference architecture derived from their public interfaces and from regulatory requirements, not a description of any company's systems.

Femke, a private seller

Femke's Tradeyard account in March 2027 is a user record with an e-mail address, a verified phone number and a postcode. Under the target model:

- She is a user with one membership: her own private seller account on Tradeyard Netherlands. When Veldmark ID rolls out in 2028, the same identity gives her a separate Motorvane profile if she ever sells a car, but no Motorvane seller account until she creates one.
- Her trader status is "private, self-declared" with the date and the wording shown in the insertion flow. Her listings show "Private seller", which is the consumer-law disclosure that EU law asks marketplaces to make.
- Her sale of the city bike for €225 through Safe Deal "pay at pickup" counts towards DAC7. She makes four Safe Deal sales in 2027 for €610 in total, well under both thresholds, so Tradeyard collects nothing beyond what Safe Deal itself needs.
- Her public profile shows her first name, her district (not her postcode; see Where Is It? Location, Geocoding and Privacy), her member-since date and ratings from completed Safe Deal transactions, each marked as verified by a transaction.

The accounts Tradeyard's reclassification job flags only look like Femke's. In 2026, about **18,000** Tradeyard Netherlands accounts declared as private each published more than 100 listings; a sample review by Aisha Rahman's trust team found that most were resellers (an illustrative Veldmark figure). Each of them is a trader the platform has not verified and whose buyers have been told, wrongly, that consumer rights do not apply. The reclassification process is the fix, and it is also a revenue opportunity: reclassified traders are the obvious prospects for Tradeyard's professional subscriptions.

Autohaus Ruhrblick, a dealer

Autohaus Ruhrblick (fictional) is one legal entity with two sites, Essen and Bochum, and a DMS from a vendor that serves about 1,900 of Motorvane's dealers. In the target model:

- **Legal entity**: Autohaus Ruhrblick GmbH, verified against the commercial register when it signed its contract in 2019, which until Motorvane's June 2027 re-check pilot was the only check Motorvane ran on dealers (Who Is on the Other Side? Identity and Verification). In the target model the registry re-verifies it on a schedule (every 90 days for business claims), including its VAT ID against the EU's VAT information exchange system.
- **Seller account**: one Motorvane Germany account, professional, trader status verified.
- **Branches**: Essen and Bochum, each with its address, opening hours and lead routing: e-mail leads to each site's sales inbox, calls to its tracked number, test-drive bookings to its calendar.
- **Contract**: one dealer package (Motorvane's Pro 150: 150 listing slots for €1,450 a month) covering both branches, with the slots shared between them (Packages, Bumps and Quotas: Products, Pricing and Entitlements).
- **Integration principal**: the DMS vendor's client, with a grant for both branches, scoped to listings write and statistics read. The vendor cannot read leads, because Autohaus Ruhrblick has not consented to that scope.

When the dealer lists the 2021 Volkswagen Golf on Wednesday 14 April 2027, the listing is owned by the seller account, assigned to the Essen branch, and its provenance records the DMS vendor's client as the writer. The 23 raw leads in ten days (19 after deduplicating repeat contacts from the same buyers) go to the Essen queues. The trader-information panel on the listing shows the GmbH's name, address, register number and self-certification, which is what DSA Art. 30(7) asks for. When the feed's removal arrives 26 hours after the sale on 29 April, the delay is a problem for Many Doors In: Forms, APIs and Bulk Feeds, not for the account model, but the account model is what lets Motorvane measure feed latency per integration principal and per dealer and raise it with the right vendor.

Overlap across brands: Poland

Tradeyard Poland's cars section and Motorvane Poland compete for the same dealers, one of the eight problems Veldmark starts 2027 with. Nobody knows how many dealers are customers of both, because the two brands' accounts share no identifiers. Matching the two brands' professional accounts on the Polish tax identification number (NIP) in a one-off analysis finds about **2,900 dealers with active accounts on both**, about 40% of them on paid packages with both brands (an illustrative Veldmark figure).

The number is a strategic fact, not just a data-quality one: it is the population for any decision to move car supply from Tradeyard to Motorvane in Poland, and it is why the group needs a registry of legal entities that both brands can read. It is also a warning about what not to do with the registry, which is the subject of the next section.

Group-Wide Identity Versus Per-Brand Accounts

A multi-brand group has three realistic options for who-lists data:

Option	What is shared	Strengths	Weaknesses	When it is right
Separate accounts per brand	Nothing	Brand autonomy; no cross-brand data-protection questions; easy to sell a brand	Duplicate logins; no group view of customers; KYB done several times; bans evaded by moving brands	Groups that are portfolios held for sale, not platforms
Shared identity and party registry, brand accounts	The login (group identity) and verified legal-entity facts	One login; KYB once; group customer view possible; brands keep their commercial relationship	Purpose limits and access control on the links between brand accounts must be designed and enforced	Groups building a shared platform beneath separate brands, like Veldmark's "share platforms, not products"

Option	What is shared	Strengths	Weaknesses	When it is right
One group account across brands	Everything: one customer account, one contract, one profile	One commercial relation-ship; cross-brand bundles	Brands lose control of their customer relation-ships; data-protection and competition questions; hard to divest	Groups that run one brand in several verticals

Veldmark chooses the middle option, and the Keybridge case shows why the purpose limits matter. In March 2027, Veldmark committed that **no Keybridge agency data is used by Hearthly for ranking or sales targeting**, with access controls and an annual independent review. Agence Rhône Habitat is both a Keybridge customer and a Hearthly advertiser. A shared registry of legal entities will link the two accounts, because the same company is the trader in both. The question is what that link may be used for.

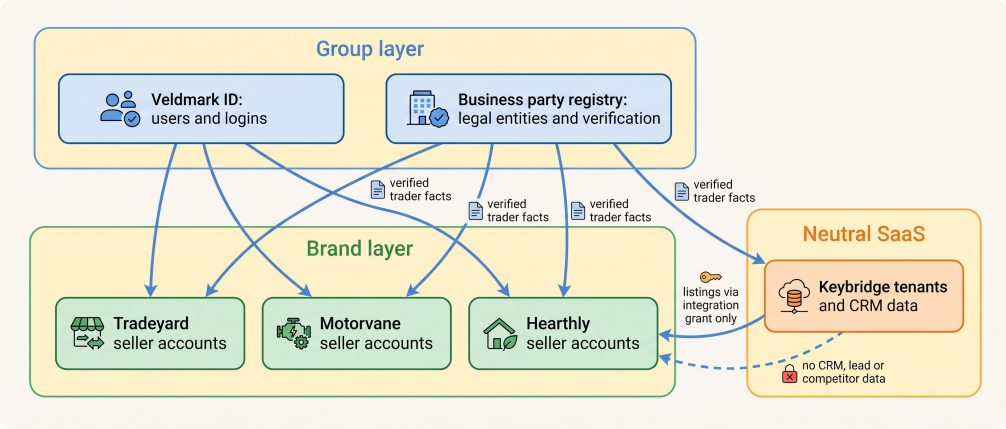


Figure 6: *Share identity and verified legal facts, not commercial data. The registry gives every brand the same verified trader record; Keybridge reaches Hearthly only as an integration principal with scoped grants, and its CRM data never crosses the dashed line.*

The rules Veldmark writes into the decision record:

1. **The registry holds public-register and verification facts only**: legal name,

register number, tax IDs, registered address, verification status and evidence references. It holds no packages, prices, revenues, leads or listing counts.

2. **Brand accounts link to registry entries, not to each other**. A query "which brands is this company a customer of?" is a privileged report, available to group finance and to approved strategy analyses such as the Poland overlap, not an API that brand sales tools can call.
3. **Keybridge reads and writes verification facts but nothing else**. Its CRM data, its customers' leads from other portals and its customers' listings on competing portals stay in Keybridge. Keybridge acts on Hearthly only as an integration principal, with the same grants any other software vendor would get.
4. **Every cross-brand use of the registry is logged** and included in the annual independent review of the Keybridge commitment.

The data-protection side is owned by Lucía Ortega's team: for private users, Veldmark ID links profiles across brands only when the user uses more than one brand, and brand-specific data stays with the brand. The competition side is for counsel, but the architecture should make the commitment checkable.

Trade-offs

Modeling private sellers

Option	How it works	Strengths	Weaknesses
User with a seller flag	The user record is the seller	Simplest; fits a C2C-only site	Breaks at the first professional; reclassification means a migration
Degenerate organization of one	Every seller is an organization; private sellers have one member and no legal-entity data	One model for every seller	Private sellers carry fields that make no sense for them; risk of collecting business data from consumers

Option	How it works	Strengths	Weaknesses
Seller account linked to either a user or a legal entity	The seller account is the common entity; it references a private user or a legal entity	Clean privacy boundary; reclassification is a relink plus verification	Two link types to handle in every consumer of the account

I recommend the third option. It keeps private sellers' data minimal, makes reclassification an explicit operation and lets the trader rules attach to legal entities, where they belong.

Who owns a listing

Owner	Works for	Breaks when
The user who created it	Private sellers	Staff leave; listings orphaned or follow the person
The seller account	Most businesses	Multi-site businesses need branch-level routing, statistics and entitlements
The seller account, assigned to a branch	Agencies, dealer groups, networks	Nothing structural; requires branch maintenance by the seller

Integration authentication

Option	Strengths	Weaknesses
Seller's own credentials shared with the vendor	No work for the platform	Cannot scope, rotate, throttle or revoke per vendor; violates RFC 9700's ban on the password grant in spirit and in letter

Option	Strengths	Weaknesses
Static API key per vendor	Simple	Usually all-or-nothing access; keys leak and live forever
Mutual TLS certificate per vendor with branch authorization	Strong, used by Rightmove's feed	Certificate lifecycle to manage; coarse scopes
OAuth clients with grants per seller account, branch and scope	Fine-grained, revocable, auditable; seller consent possible	More moving parts; needs a consent and grant-management user interface

Vertical Differences

Dimension	General classifieds	Real estate	Automotive	Jobs
Typical supply	Mostly private sellers (78% of new Tradeyard listings); small traders	Agencies and developers; private landlords in rentals	Dealers dominate; private sellers a minority	Employers and staffing agencies
Commercial unit	Account, sometimes none	Branch (office)	Dealer or forecourt	Employer account; sometimes per job slot
Hierarchy depth	Shallow	Deep: network, franchisee, branch, agent	Group, legal entity, site	Group, company, recruiter
Who acts	The seller	Agents and back-office staff, mostly via software	Sales staff; DMS uploads	Recruiters; applicant-tracking systems

Dimension	General classifieds	Real estate	Automotive	Jobs
Integration principals	Few	About 160 vendors for Hearthly; agent CRMs	DMS vendors; 88% of Motorvane dealer listings via DMS feeds	Applicant-tracking systems; job aggregators
Individual professional identity	Rare	The agent matters to buyers; licensing regimes vary by country	Salesperson rarely shown	The recruiter is often shown
Trader questions	Private-versus-trader reclassification is the main issue	Agency KYB; private landlords as possible traders	Dealer KYB; private sellers who trade cars	Employer verification; fake job ads

Real estate is the vertical where the hierarchy is deepest and the individual agent is a visible, rated person, often under a professional licensing regime in the country concerned. Automotive is where integration principals matter most, because most stock arrives from DMS vendors. General classifieds is where the trader boundary is hardest to police, because the population is huge and the signals are behavioral.

Failure Modes

1. **Listings owned by users.** Staff turnover orphans listings or routes leads to people who left, as with the Villeurbanne agent. Ownership belongs to the seller account; users are editors.
2. **The trader flag as a checkbox.** A boolean set once at sign-up, with no evidence, no date, no wording record and no reclassification, cannot answer a regulator and lets resellers pose as private sellers.
3. **The legal hierarchy assumed to equal the display hierarchy.** Franchisees displayed under the network's legal identity give buyers the wrong trader and the wrong contact for consumer claims.
4. **Billing hierarchy assumed to equal the legal hierarchy.** Head offices pay for some branches and not others; contracts must name the branches they cover.

5. **Shared credentials for integrations**. Vendors logging in as dealers cannot be told apart, throttled or revoked one dealer at a time, and every password reset breaks a feed.
6. **Suspension that does not propagate**. A suspended account's listings stay searchable, its feeds keep writing and its scheduled bumps keep running: Ghost Inventory with an enforcement problem attached.
7. **One retention clock**. Deleting all of a closed account's data at once loses invoices that must be kept; keeping all of it forever breaks the DSA's six-month deletion rule for trader data.
8. **Merging across brands by e-mail address**. Matching accounts on an e-mail address across brands links people who never asked to be linked, and in Veldmark's case could breach the Keybridge commitment. Link through the identity the user chose to use, and through verified legal entities with purpose limits.
9. **Pooled reputation**. Ratings aggregated across separate traders make good franchisees pay for bad ones and mislead buyers about who they are dealing with.

The Chief Architect's View

The seller account is a supporting domain in the language of Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models: no buyer chooses a marketplace for its account model. But almost every core domain depends on it, so a group should standardize it early, before the product catalog, entitlements and messaging copy their own versions.

Decision 1: one supply-side model for all brands. Users, legal entities, seller accounts, branches, networks, contracts, storefronts and integration principals, with the meanings in this chapter. Each brand implements it in its own account service on its own timeline, but the identifiers and events (SellerAccountCreated, TraderStatusChanged, BranchAdded, MembershipEnded, AccountSuspended) follow one group contract. The product catalog and entitlement service due in Q2 2028 attaches entitlements to seller accounts and branches as defined here; without that, Packages, Bumps and Quotas: Products, Pricing and Entitlements would be building on four different meanings of "customer".

Decision 2: a group business party registry, launched with Veldmark ID. Approved at the architecture advice forum in August 2027, owned by the identity platform team, live with Veldmark ID in 2028. It holds verified legal-entity facts and evidence references, serves DSA trader traceability, consumer-law declarations, P2B business users and DAC7 for every brand, and runs re-verification on a schedule. The purpose rules above are part of the decision record.

Decision 3: trader status is shared between accounts and trust. The account service owns the status and its history; the trust-and-safety platform owns the reclassification

signals and cases. Aisha Rahman and the account platform's product lead jointly own the policy, with Lucía Ortega's team signing off on what is displayed to buyers.

Decision 4: integrations get grants, not passwords. Every brand moves feed and API partners to client credentials or certificates with grants per seller account and branch, starting with Motorvane's DMS vendors in 2028. This is also what makes per-vendor feed quality measurable (Many Doors In: Forms, APIs and Bulk Feeds).

What to watch: the share of professional listings whose seller has a verified trader record (target 100%); reclassification cases opened and resolved per month; orphaned listings (owned by an account with no active member); integration principals still using shared credentials; and the volume of cross-brand registry queries, which should stay small and fully logged.

The condition under which this is too much: a single-brand, C2C-only marketplace without payments can live with users and a trader declaration for a long time. It stops being enough the day the first dealer asks for a second login.

Questions to Ask

1. For any listing, can we say which user last changed it, which legal entity is the trader, which contract pays for its visibility and which branch handles its leads?
2. What happens to a professional's listings and leads when the employee who created them leaves?
3. Where is trader status stored, with what evidence, and how is a private seller who behaves like a trader detected and reclassified?
4. Does the trader information on each listing come from the legal entity that is actually responsible, including for franchisees?
5. How do integration partners authenticate, and can we revoke one partner's access to one dealer without affecting anyone else?
6. Which data classes of a closed account follow which retention clock, and who deletes what when?
7. Where do ratings attach, and what makes a rating eligible to be shown as verified?
8. Across our brands, can we count how many businesses we serve, and who is allowed to ask that question?
9. If we launched in-platform payments in a new country tomorrow, could we produce the seller data that tax reporting would require?
10. Which other services hold their own copy of the account hierarchy, and which of them would disagree with the account service today?

From the Listing to Its Supply

With the listing, its taxonomy, attributes, media, location and seller modeled, Part II's account of the thing being sold is complete. Part III — SUPPLY: Getting Inventory In turns to how that supply actually arrives, through forms, APIs and bulk feeds from thousands of partners, starting with Many Doors In: Forms, APIs and Bulk Feeds.

Sources and Evidence

Company engineering and investor sources

- Seller API manual — mobile.de, n.d. Transfer service providers, seller and customer identifiers, multi-locations, linked-seller access.
- Real Time Datafeed Specification v1.4.1 — Rightmove, 2023. Network, branch and channel tenancy; mutual TLS; branch-scoped leads and performance.
- Results announcement for the year to 31 December 2025 — Rightmove, 2026. ARPA definition per advertiser; 16,385 agency branches.
- Rightmove guide on customer checks — Rightmove, n.d. Pre-advertising customer checks and redress-scheme membership for agents.
- Full-year results FY26 — Autotrader Group, 2026. Retailer forecourts and average revenue per retailer.
- Form 10-K for 2025 — CarGurus, 2026. Paying-dealer definition and count.
- Bundeskartellamt clears mobile.de's acquisition of carwow.de — Bundeskartellamt, 2026. Dealer multi-homing and data export as constraints on market power.

Standards, regulation and documentation

- Regulation (EU) 2022/2065 (Digital Services Act), Article 30 — European Union, 2022. Trader traceability: data to collect, best-efforts verification, six-month retention and display; quotations checked against the Official Journal text.
- Directive (EU) 2019/2161 (Modernisation or Omnibus Directive) — European Union, 2019. Trader declaration on online marketplaces (new Art. 6a of the Consumer Rights Directive) and review-verification information (new Art. 7(6) of the Unfair Commercial Practices Directive).
- Consumer Rights Directive — European Commission, n.d. Modernisation Directive amendments applying from 28 May 2022, including obligations of online marketplaces.
- Information for platform operators (DAC7) — Belastingdienst, n.d. Reportable activities, seller data and the de minimis rule.

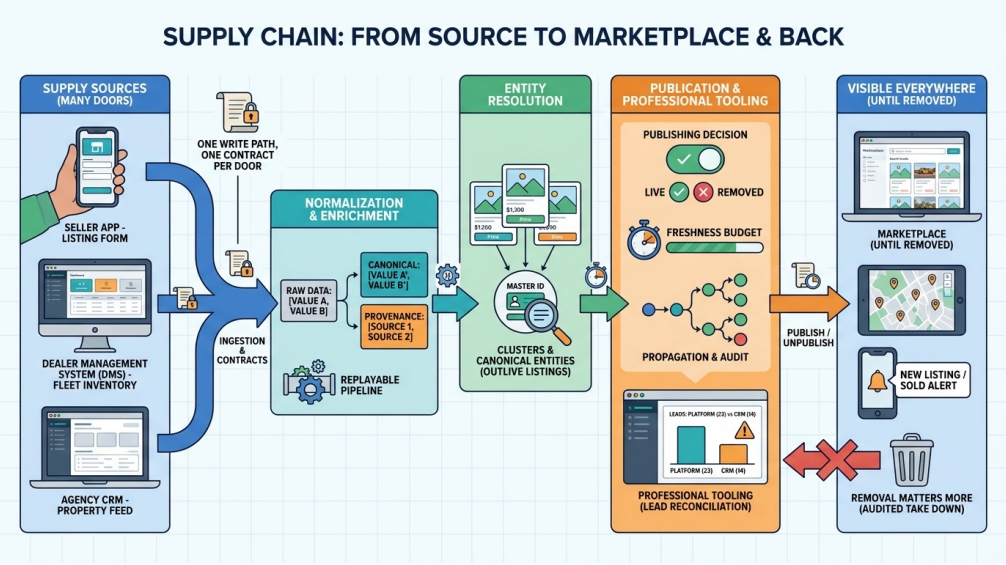
- Informing Businesses about the INFORM Consumers Act — US Federal Trade Commission, 2023. High-volume seller verification and disclosure.
- RFC 9700: Best Current Practice for OAuth 2.0 Security — IETF, 2025. Grant types, PKCE, sender-constrained tokens.

Research and practitioner sources

- *Productscape domain models for general listings and real-estate marketplaces* — the author's research digest, 2026. Seller entity, storefront and intermediary models; professional personas and integration partners.
- The book's research briefs on listing data, trust and regulation, real-estate portals and monetization, 2026 — summaries of the primary sources above and of EU consumer-law provisions on trader declarations and reviews.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

Part III — SUPPLY: Getting Inventory In



IN THIS PART, YOU WILL: Follow supply from the moment it reaches the marketplace until it is visible everywhere, and later removed everywhere: ingestion and feed contracts, normalization and enrichment, entity resolution, publication and professional tooling.

WHY THE BUSINESS CARES: In vertical marketplaces most inventory comes from professionals through machine feeds, and they pay for the leads it produces. A lost feed, a ghost listing or a lead count that does not match the dealer's CRM costs renewals, not just data quality.

WHY ARCHITECTS SHOULD CARE: Supply is where outside systems write into yours. Every weakness here (an unstated contract, an unrecorded

source, an unregistered copy) becomes a search, trust or billing problem downstream.

KEY POINTS:

- **Many doors, one write path, one contract per door.** Forms, APIs and feeds should end in the same idempotent listing command. Each channel needs a contract that says what absence means, who owns each field and how both sides reconcile, because partner state and platform state will drift.
- **Keep what was said, record where every value came from.** Normalize to canonical values next to the original, attach provenance to every value, and run enrichment as a versioned, replayable pipeline. Resolve duplicates to entities that outlive their listings, with cluster identifiers governed as a product contract.
- **Publishing is a decision followed by a propagation, and removal matters more.** A short synchronous path decides whether a listing may go live. Events carry it everywhere within a budget, and the same mechanism, audited per consumer, takes it down again.

FAILURE PATTERNS THIS PART ADDRESSES:

- **Feed Roulette** — Gives every channel a contract, validation, error feedback and reconciliation.
- **Seeing Double** — Separates abuse from legitimate multi-listing and clusters the second for presentation.
- **Ghost Inventory** — Makes removal reach every copy, with evidence per consumer.

Part II — LISTING: Modeling the Thing Being Sold modeled the listing: a small aggregate referencing governed categories, attributes, media, places and accounts. This part puts that model under load. **Supply arrives through many doors, at different rhythms, from systems the marketplace does not control**, and each door tends to grow its own idea of what a listing is.

The argument is that supply is **one chain, not five features**. A record enters through a channel under a contract, is normalized and enriched with recorded provenance, is recognized when it describes something already known, and is published and later unpublished within a stated freshness target. The professionals who supply most of the inventory then check the result against their own systems.

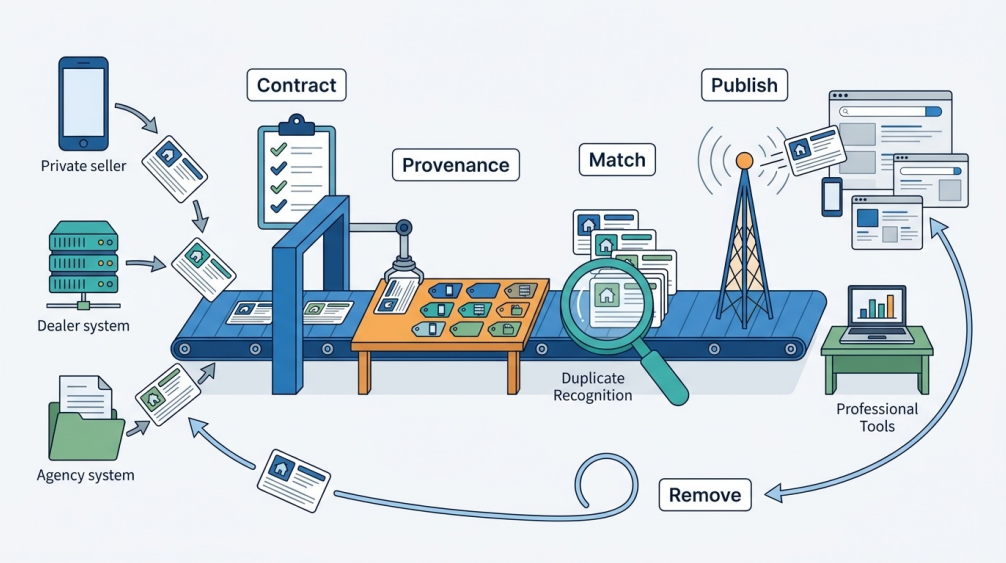


Figure 1: *Supply is one chain, not five features: every door feeds the same contracted path, and the same path takes listings down again.*

Chapters in this part

Chapter	The question it answers	Read it when…
Many Doors In: Forms, APIs and Bulk Feeds	How should a marketplace accept supply from private sellers, professionals and partners at scale?	A partner's bad file can unpublish thousands of listings overnight.
From Raw Supply to Usable Listings: Normalization and Enrichment	How do you turn heterogeneous input into consistent, enriched listings?	Nobody can tell which values a seller typed and which a model inferred.
One Property, Three Listings: Duplicates and Entity Resolution	How do you recognize that several listings describe the same thing, and what do you do about it?	One flat appears three times, or reposts crowd out new supply.
From Submitted to Searchable: The Publication Pipeline	What happens between "publish" and searchable, and how fast must it be?	Publication takes minutes, or sold items still appear in alerts.

Chapter	The question it answers	Read it when…
Serving the Pros: Dealer, Agent and Merchant Tooling	What do professional sellers need, and how should that tooling be built?	Dealers dispute your lead counts, or the group sells software to professionals.

How this part connects

This part consumes Part II's contracts. Ingestion validates against the compiled Attributes Without Chaos: Schemas, Dictionaries and Validation, writes commands to the The Listing Aggregate: Identity, Lifecycle and Invariants and attaches records to Private Sellers, Dealers and Agencies: Modeling Who Lists. The liquidity metrics of Part I — MARKETPLACE: The Business Behind the Listings set its targets: time to first contact sets the publication budget, and ghost rates set the removal guarantee. Part IV — DISCOVERY: Matching Intent to Inventory then reads the result: search, alerts and recommendations are only as fresh, deduplicated and trustworthy as the supply this part delivers. Trust and safety (Part V — TRUST: Keeping the Marketplace Safe) owns the moderation gate inside the publication pipeline.

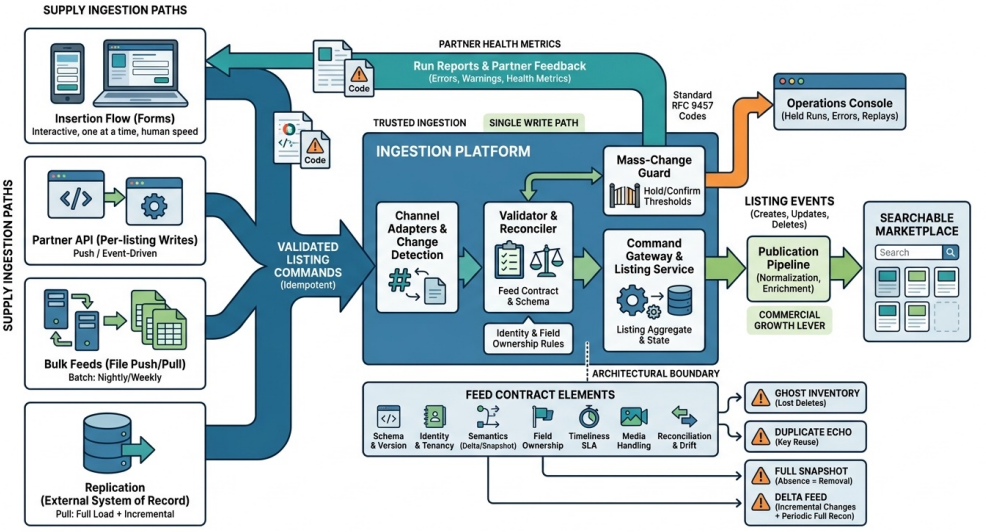
The Veldmark thread

At Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), supply problems arrive as incidents before they arrive as designs. In this part Ingrid Halvorsen's team:

- Responds to the **8 March 2027** Motorvane incident, in which truncated snapshot files unpublished **38,400** listings, by adopting **Supply Contract v1** in September 2027: one envelope, one error catalog, mandatory mass-change guards and brand-local adapters.
- Traces the **Essen Volkswagen Golf**'s 31 decoded attributes value by value, and gives every value a recorded source.
- Clusters the **Lyon apartment**'s three agency listings into one canonical property, using only what each listing carries as published to Hearthly, never Keybridge's own data.
- Writes Tradeyard's **90-second** p95 publication target as an SLA card with a budget per stage, after finding that the 6-minute baseline ignored the 6% of listings held for human review.
- Reconciles the Golf's **23** Dealer Hub leads against the **14** in the dealer's CRM, and keeps Keybridge's cross-portal report out of Hearthly's reach.

Begin with Many Doors In: Forms, APIs and Bulk Feeds.

13. Many Doors In: Forms, APIs and Bulk Feeds



IN THIS CHAPTER, YOU WILL: Learn the four doors through which supply enters a listing marketplace (the insertion flow, partner APIs, bulk feeds and replication from an external system of record) and the contract that should govern each. You will follow a dealer's Volkswagen Golf through Motorvane's feed pipeline, see why a sold car stayed online for 26 hours, and design the identity, reconciliation, error-reporting and back-pressure rules that make ingestion trustworthy.

WHY THE BUSINESS CARES: In automotive and real estate, most inventory never touches the marketplace's own forms. It arrives by machine from dealer and agency software, and a broken feed is a dealer with an empty shop window and a reason to cancel. For private sellers, the insertion flow is the most expensive funnel the marketplace runs: every seller who gives up halfway is supply that a competitor gets.

WHY ARCHITECTS SHOULD CARE: Ingestion is where the platform accepts other people's version of the truth. Without explicit identity, full-versus-delta semantics, ownership of fields and reconciliation, the platform's

listings drift from reality one silent difference at a time, and nobody can tell which copy is right.

KEY POINTS:

- **Many doors, one write path.** Forms, partner APIs, file feeds and replication differ in transport and rhythm, but each should end in the same idempotent command on the listing service, validated by the same compiled schema, keyed by the partner's own reference. Separate ingestion stacks per channel are how one listing ends up with three meanings.
- **A feed contract is more than a schema.** It states what a record means when it is absent (full snapshot) or when it says nothing about a field (delta), which side owns each field, how fast removals must arrive, how errors are reported back and how both sides reconcile. Most ingestion incidents are a contract question nobody asked.
- **Assume the partner's state and yours will drift, and build the reconciliation in.** Lost deletes, truncated snapshots and duplicate sends are normal operating conditions, not exceptions. Mass-delete guards, periodic snapshot comparison and per-partner health metrics are what turn Feed Roulette into an operated supply chain.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Feed Roulette** — Partner feeds accepted without contracts, validation or feedback make every nightly import a surprise for the dealer and for the platform.
- **Ghost Inventory** — Removals that arrive late, or never, through feeds keep sold cars and let flats searchable and contactable long after they are gone.
- **Seeing Double** — One seller's stock arriving through two channels, or one property through three software vendors, multiplies listings unless ingestion recognizes the same partner record.

A marketplace's supply arrives through more doors than its product team usually counts. A private seller photographs a city bike in the app. A dealer's management system pushes 200 cars every evening. An agency's CRM sends a flat to fifteen portals at once. A developer uploads a spreadsheet of new-build units. A partner API creates listings for a fleet of rental vans. Each door was built at a different time, by a different team, for a different customer, and **each tends to grow its own idea of what a listing is**.

Ingestion is the first link in Part III's supply chain: the boundary where the listing model of Part II — LISTING: Modeling the Thing Being Sold meets systems the marketplace does not control, and where outside supply becomes listing commands. Every later step,

from normalization to search, works only on what this boundary accepts. This chapter owns three terms. An **ingestion channel** is a distinct path by which listings enter, defined by who uses it, how data moves and what rhythm it has. A **feed contract** is the agreement that governs a channel: schema, identity, semantics, ownership, errors, limits and reconciliation. A **delta feed** sends only changes, as opposed to a **full snapshot** that sends a partner's complete current inventory every time.

Five neighboring chapters own what ingestion does not. The Listing Aggregate: Identity, Lifecycle and Invariants owns the listing's identity, invariants and lifecycle; ingestion writes to it through commands. Attributes Without Chaos: Schemas, Dictionaries and Validation owns the compiled schemas that ingestion validates against. From Raw Supply to Usable Listings: Normalization and Enrichment owns what happens to a record's values after it is accepted, and From Submitted to Searchable: The Publication Pipeline owns the path from accepted to searchable. Serving the Pros: Dealer, Agent and Merchant Tooling owns the dealer and agent products built on top of these channels. Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), supplies the worked examples.

Setting the Scene at Veldmark: The Morning 38,400 Cars Disappeared

On **Monday 8 March 2027** at 02:40 CET, a mid-sized dealer-management software vendor that Motorvane registers as a transfer service provider (the industry term for a vendor that uploads stock on dealers' behalf) sent its nightly full-stock files for **286 dealers**. A storage fault on the vendor's side had cut most files short. Motorvane's importer did what it had been built to do: it compared each file with the dealer's live stock and treated every missing vehicle as sold. By 02:55, **38,400 listings** were unpublished, saved-search alerts had stopped matching them, and paid "top" placements on about 1,900 of them had lapsed. Dealers opened their lead inboxes that morning to silence. Motorvane's dealer support took more than 600 calls before the vendor resent complete files at 09:40 and the stock reappeared, with new listing IDs for about a fifth of it, because the importer had already purged the external-key mapping for those vehicles.

Felix Brandt, Motorvane's CTO, brought the incident to the architecture advice forum, where Ingrid Halvorsen, Veldmark's Group Chief Architect since February, had asked each brand to describe how supply comes in. Julien Marchand, Hearthly's CTO, described the opposite problem from the same root. Of Hearthly's 2.6 million live listings, **84% arrive through feeds from about 160 real-estate software vendors**. Some of those vendors send changes but never deletes, and Hearthly's France team estimated, from a sample of contact attempts that agencies answered with "already sold", that a few percent of live listings at any time were ghosts. Daan Visser, who owns Tradeyard's Yardstick monolith, added a third: Tradeyard's professional sellers upload

CSV files through a 2014-era importer that validates differently from the web form, so the same listing passes one door and fails the other.

Ingrid's question to the forum was not "which ingestion technology should we use?" It was: **for each door, what is the contract, who owns the truth, and how would we know if our copy were wrong?** None of the three brands could answer all three parts for all of its channels. The rest of this chapter is the answer Veldmark worked out over the following months.

Concepts: Doors, Contracts and the Truth Outside

Four kinds of ingestion channel

Channel	Who uses it	How data moves	Rhythm	Where the truth lives
Insertion flow (web and app forms)	Private sellers; small professionals	Interactive, one listing at a time, with drafts	Human; seconds to minutes per step	**The marketplace**
Partner API (per-listing writes)	Software vendors and larger professionals; self-uploading dealers	Push, synchronous request and response per listing or small batch	Near-real-time, event-driven	The partner's system, mirrored per call
Bulk feed (file push or scheduled pull)	Dealer and agency software; developers; merchants with catalogs	Push to an endpoint or file drop, or the marketplace fetches a partner URL on a schedule	Batch: hourly, nightly, weekly	**The partner's system**, sent as snapshot or delta

Channel	Who uses it	How data moves	Rhythm	Where the truth lives
Replication from an external system of record	Portals down-stream of multiple-listing services (MLS) or cooperative databases	Pull: full load, then incremental changes since a high-water mark	Continuous or frequent polling	**The external system**, licensed to the portal

Where the truth lives decides the design. When the marketplace holds the truth, as with a private seller's bike, ingestion is mostly a user-experience and validation problem. When the truth lives in a dealer's dealer management system (DMS) or an agency's CRM, the marketplace holds a copy, and **the central engineering problem becomes keeping the copy aligned with a source it does not control**. In the United States the listing record is created in an MLS and portals receive licensed copies, so the portal is downstream of a system of record that is not even the seller's own software (see Zillow, Rightmove, Idealista and REA: What Property Portals Teach).

Scraping is not a fifth channel in this book. Some aggregators have built supply by crawling other sites, at the cost of legal, licensing and quality problems; a marketplace group that wants supply from partners should offer them a contract instead. Syndication, where the marketplace sends its listings out to partners and search engines, is the mirror image of ingestion and is covered in Every Change Is an Event: Event-Driven Listing Architecture and Found by Google: Web, App and SEO Architecture for Listings.

Full snapshots and delta feeds

A **full snapshot** feed sends the partner's complete current inventory each time. Its semantics are simple and dangerous: **absence means removal**. The platform diffs the snapshot against its own state, creates what is new, updates what changed and removes what is missing. A snapshot is self-healing, because any earlier mistake is corrected by the next complete file, but a truncated file is indistinguishable from a dealer who sold everything.

A **delta feed** sends only changes: creates, updates and explicit deletes. It is efficient and fast, but it is not self-healing. A lost delete is never re-sent, a delete for a listing the platform never received is an error that nobody notices, and over months the two sides drift.

Neither is sufficient alone: the dependable pattern, visible in the public contracts discussed below, is **incremental changes for speed plus periodic full reconciliation for correctness**. Rightmove's Real Time Datafeed is incremental by design but provides

a `GetBranchPropertyList` call that returns "a snapshot of a branch's properties", which a software provider can use to check that properties were created as expected, are up to date, and that there "aren't any properties still live on Rightmove which should have been removed" (Rightmove RTDF specification v1.4.1, last updated 2023). The RESO Web API, the US real-estate standard, describes the replication pattern as first pulling a full dataset, then requesting recent changes (RESO Web API, n.d.).

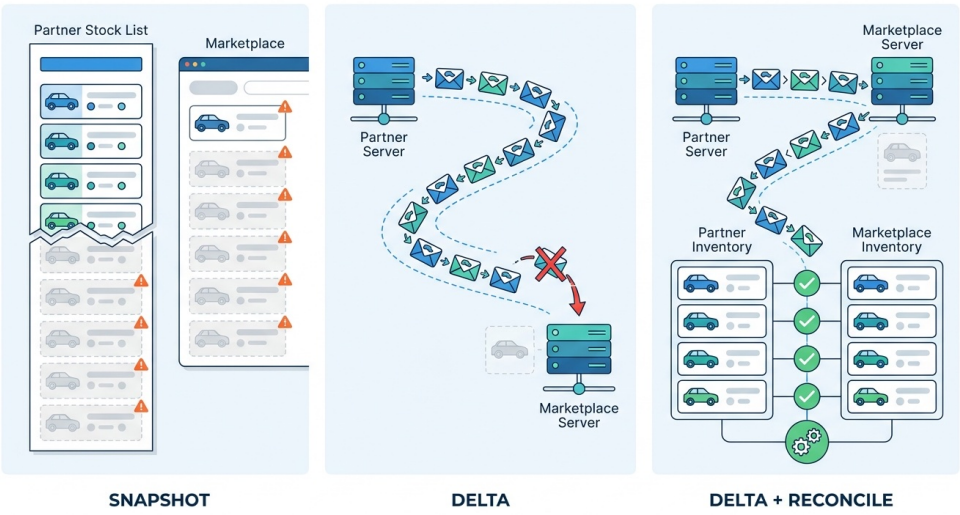


Figure 1: *A truncated snapshot looks like a dealer who sold everything, and a lost delete leaves a ghost; incremental changes plus periodic full reconciliation guard against both.*

The feed contract

A **feed contract** is everything both sides must agree on for a channel to be operable. Schema validity is the part everyone remembers; it is perhaps a third of the contract.

Clause	Question it answers	What goes wrong without it
Scope and tenancy	Which sellers, branches, channels and categories may this integration write?	A vendor writes stock into the wrong dealer account; a lettings branch publishes into sales

Clause	Question it answers	What goes wrong without it
Identity	Which field is the partner's stable key for a listing, and is it unique per seller or per integration?	Re-sent records create duplicates; key reuse overwrites another car
Semantics	Is this a snapshot or a delta? What does absence mean? What does an empty field mean: "unknown", "remove the value" or "unchanged"?	A truncated file deletes stock; a blank field wipes a value the seller set in the pro tool
Field ownership	For each field, does the feed, the seller in the marketplace's tools, or the platform own it?	The nightly feed reverts a price the dealer cut by hand at noon
Schema and version	Which compiled category schema validates the record, and how are versions pinned and retired?	A new mandatory field breaks every vendor on the same night
Timeliness	How fast must creates, updates and especially removals arrive?	Sold cars stay listed for a day (Ghost Inventory)
Media	Pushed bytes or pulled URLs; change detection; limits	Images re-downloaded every night; broken images on live listings
Errors and feedback	How does the partner learn what failed, why and how to fix it?	Silent rejects; support tickets as the error channel
Limits	Rate limits, batch sizes, concurrency per listing, quotas per seller	One large feed starves others; parallel writes corrupt a listing
Reconciliation	How do both sides verify they agree, and how often?	Drift accumulates unnoticed
Change management	How much notice before breaking changes; what counts as breaking	Partners learn of changes from failures

The contract is a product artifact as much as a technical one. Partners judge a

marketplace by how predictable its feed is, and the Productscape model this book started from lists "partner integration reliability", certification lead time and "validation error recurrence" among the integration partner's key measures. In feed-driven verticals that matters commercially: Germany's competition authority, clearing a mobile.de acquisition in October 2026, noted that dealers multi-home and that many platforms offer dealers data export to other platforms (Bundeskartellamt, 2026). **Inventory in automotive is rarely exclusive, so the feed is where a marketplace competes for the dealer's attention.**

The Public Contracts: What Real Marketplaces Publish

Few marketplaces publish their internal pipelines, but several publish their partner contracts in detail, and those contracts are good evidence of what the operators learned. All the documentation below is public; where it is undated, that is noted.

Contract	Vertical and market	Transport and format	Identity	Notable rules
Rightmove Real Time Datafeed (v1.4.1, 2023)	Real estate, UK	HTTPS JSON calls validated against published schemas; mutual TLS with a Rightmove-issued client certificate	The agent's `Agent_Ref` per branch and channel	Network (software provider) and branch tenancy; incremental sends plus `GetBranchPropertyList` snapshot reconciliation; `RemoveProperty` archives rather than deletes; media pulled by URL; success, warning and error responses

Contract	Vertical and market	Transport and format	Identity	Notable rules
mobile.de Seller-API and XML upload (n.d.)	Automotive, Germany	REST with JSON or XML vendor media types; zipped XML upload with images	`internalNumber` or `seller-inventory-key` from the dealer; platform `mobileAdId`	Access only for transfer software providers and self-uploading dealers; idempotent create via an insertion request ID; full-document PUT; no parallel requests for the same ad
Autotrader UK developer platform (n.d.)	Automotive, UK	REST APIs (Stock, Taxonomy, Vehicles, Images and others) with token authentication; Stock Notifications webhooks	Retailer stock records	Validation rules encode policy (minimum price, write-off categories); breaking changes announced at least 30 days ahead
OpenImmo (n.d.)	Real estate, German-speaking markets	XML Schema of about 250 features; images zipped with the XML, typically sent over FTP; OpenImmo 2.0 is REST and JSON	Object identifiers from the sender's CRM	Association-run standard since 2001–2002; 2.0 is not compatible with the XML version

Contract	Vertical and market	Transport and format	Identity	Notable rules
RESO Data Dictionary and Web API (n.d.)	Real estate, US	OData V4 Web API; Data Dictionary 2.0 with more than 1,700 fields and 3,100 lookups	Listing keys from the MLS	Full load then recent changes; RETS deprecated; per-MLS credentials and license rules
eBay Sell Feed and Inventory APIs (n.d.)	General e-commerce	Asynchronous upload tasks with status and downloadable result files; schedules	Seller-defined SKU "unique across the seller's inventory"	Inventory item, offer and publish are separate steps

Four lessons recur across them.

The partner's key is the listing's identity at the boundary. Rightmove states it directly: it uses `Agent_Ref` to decide whether a property is new, and an update must "provide the same Agent_Ref". mobile.de's bulk format carries the dealer's own `seller-inventory-key` and requires transfer service providers to name the dealer's customer number on every ad (mobile.de upload interface XML, n.d.). The platform's own listing ID stays inside; the external key, scoped to the integration and seller, is what makes a re-sent record an update instead of a duplicate. The Listing Aggregate: Identity, Lifecycle and Invariants records it as the listing's external key.

Creates must be safe to retry. mobile.de's Seller-API accepts an `X-Mobile-Insertion-Request-Id` header on create; a retry with the same value returns HTTP 303 pointing to the ad already created rather than creating a second one, and the documentation says GET, PUT and DELETE are idempotent while POST is not (mobile.de Seller-API, n.d.). It also tells clients not to send multiple requests for the same ad at the same time, which is a per-listing serialization rule written into the contract.

Tenancy is part of the contract. Rightmove gives each software provider a network ID and each office a branch ID with sales and lettings channels, and rejects a call that sends "a channel to which the branch does not have access". mobile.de grants API access only to transfer service providers and self-uploading dealers, and the user sees only sellers linked to its account. Both separate the *integration principal* (the vendor) from the *seller* it acts for, which is the model Private Sellers, Dealers and Agencies: Modeling Who Lists recommends.

Validation rules are where policy lives. Autotrader's documented error responses include a price rejected for not meeting a minimum of 75 and a refusal of vehicles that the UK's Driver and Vehicle Licensing Agency (DVLA) classifies as category A or B write-offs (Autotrader developer portal, n.d.). Rightmove's schema versions track regulation: version 1.4.0 (April 2022) added ground rent, service charge and council-tax band, and 1.4.1 (November 2023) added utilities, parking, flood risk and accessibility fields. The feed contract is therefore a compliance surface: a field the law requires in a property ad must be required in the feed, or the platform publishes non-compliant ads at machine speed (see Rules of the Marketplace: DSA, DAC7 and Vertical Regulation).

Architecture: Many Doors, One Write Path

The channel determines the transport and the rhythm, never the meaning. Every channel ends in the same small set of commands on the listing service, validated by the same compiled schema, and every channel reports outcomes through the same error catalog.

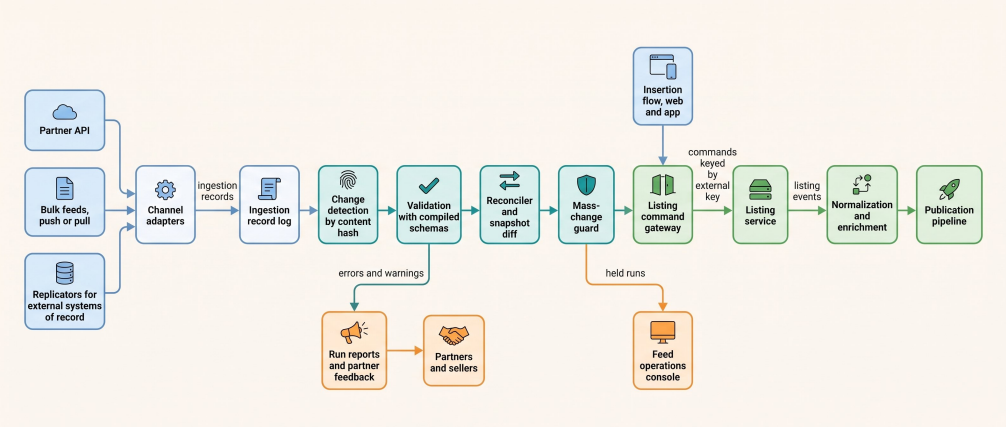


Figure 2: *Many doors, one write path. Adapters turn each transport into the same ingestion record; change detection, validation, reconciliation and a mass-change guard sit between the partner and the listing service; errors flow back to partners through one report format. The insertion flow writes through the same command gateway and validator.*

Components and responsibilities

Component	Responsibility	Must not
Channel adapters	Terminate transports (HTTPS calls, SFTP and HTTPS file drops, scheduled fetches, replication clients); authenticate the integration; parse formats (OpenImmo XML, vendor CSV, JSON); emit one **ingestion record** per partner listing	Apply business rules or write listings directly
Ingestion record log	Durable, append-only log of every received record with partner, seller, channel, external key, payload, received time and run ID	Be pruned before the reconciliation and dispute windows end
Change detection	Hash the semantically relevant payload per record and skip records identical to the last accepted version	Skip records whose schema version or mapping changed
Validator	Validate against the compiled category schema from Attributes Without Chaos: Schemas, Dictionaries and Validation; separate blocking errors from warnings	Implement its own rules that differ from the insertion flow
Reconciler	Interpret snapshot or delta semantics; compute creates, updates and removals per seller; apply field-ownership rules	Decide removals for a seller whose snapshot is suspect

Component	Responsibility	Must not
Mass-change guard	Hold runs whose effect exceeds thresholds (removals, price changes, category changes) for review or confirmation	Silently drop held changes
Command gateway	Translate accepted changes into intent-named commands (`Submit`, `Edit`, `MarkSold`, `Withdraw`) with idempotency keys and expected versions	Bypass the listing service's invariants
Run reports	Produce per-run and per-record outcomes with stable error codes, delivered by API, file and notification	Report only totals
Feed operations console	Show partner and seller health, held runs, error trends; allow controlled replays	Become the only place errors are visible

The **ingestion record** is the key internal abstraction. It is channel-neutral: `(integration_id, seller_id, external_key, run_id, sequence, operation, payload, received_at, partner_updated_at, schema_version)`. A record from Rightmove-style JSON, an OpenImmo ZIP and a CSV row from a Tradeyard pro seller look identical after the adapter. That is what lets one reconciler, one validator and one report format serve every partner, and it is what Veldmark lacked: three brands had built importers whose first step was "parse the file" and whose last step was "write to the database", with business rules scattered in between.

Identity and idempotency

Ingestion has three kinds of key, and confusing them causes most duplicate and overwrite bugs.

Key	Scope	Purpose	Example
External key	Per integration and seller	Identifies a partner's listing across all sends; maps to the platform listing ID	`Agent_Ref`; a dealer's stock number
Request idempotency key	Per request, short-lived	Makes a retried create or command safe	mobile.de's insertion request ID; the key semantics Stripe documents, with keys up to 255 characters, results stored once execution begins, keys pruned after at least 24 hours and a mismatch error if the parameters differ (Stripe idempotent requests, n.d.)
Record sequence	Per external key	Orders updates for the same listing	A partner `updated_at` timestamp, or a sequence number per send

Two design rules follow. First, **the mapping from external key to listing ID is listing data, not importer state**. It belongs in the listing aggregate (as its external key) and survives removals, so a car that reappears after a feed outage gets its old listing ID, URL, favorites and price history back. Motorvane's March incident turned into a duplicate problem because the importer purged the mapping when it unpublished the cars. Second, **there is no standard HTTP idempotency header to point partners to**: the IETF draft for an `Idempotency-Key` header reached version 07 in October 2025 and expired without becoming an RFC (IETF Datatracker, 2025). Define the header in your own contract and document its semantics.

Ordering and concurrency

Two updates to the same listing can arrive out of order, from retries, parallel workers or two channels. Three rules keep them in order:

- **Serialize per listing**. Partition ingestion records by `(seller, external key)` so that one worker applies one listing's changes in order, the internal equivalent of mobile.de's "do not send multiple requests for the same ad at the same time".
- **Apply only newer partner versions**. If a record carries a partner timestamp or sequence older than the last applied one, record it and skip it. Rightmove's `Update_Date` plays this role and also drives its reconciliation.
- **Use expected versions on commands**. The command gateway passes the listing version it read; the listing service rejects the command if another writer changed the listing in between, and the reconciler re-reads and re-applies field-ownership rules.

Field ownership: who may change what

Feed-driven listings have three writers: the partner's feed, the seller using the marketplace's own pro tool, and the platform (moderation, enrichment, entitlements, lifecycle clock). The contract must say who owns each field, or the last writer wins and the dealer's noon price cut is reverted by the evening snapshot.

Field group	Default owner for feed listings	Feed may overwrite?	Notes
Identity facts (VIN, address, property type)	Feed	Yes, with re-moderation if material	Changes to identity facts on a live listing are a classic bait-and-switch signal
Price, availability, description, media	Feed	Yes	Unless the seller has set a **lock** in the pro tool, which the contract must document

Field group	Default owner for feed listings	Feed may overwrite?	Notes
Visibility products (top, bumps)	Platform entitlements, ordered via feed or tool	Only via product calls	Rightmove applies premium and featured products through separate feed calls that fail if credits are insufficient; paid state is never a field in the listing payload
Moderation state	Platform	Never	A feed cannot un-block a listing
Enriched and derived values	Platform, with provenance	Never directly; the feed changes inputs	See From Raw Supply to Usable Listings: Normalization and Enrichment
Lifecycle clock (expiry, renewal)	Platform	Indirectly: a fresh send can count as confirmation	Policy per vertical

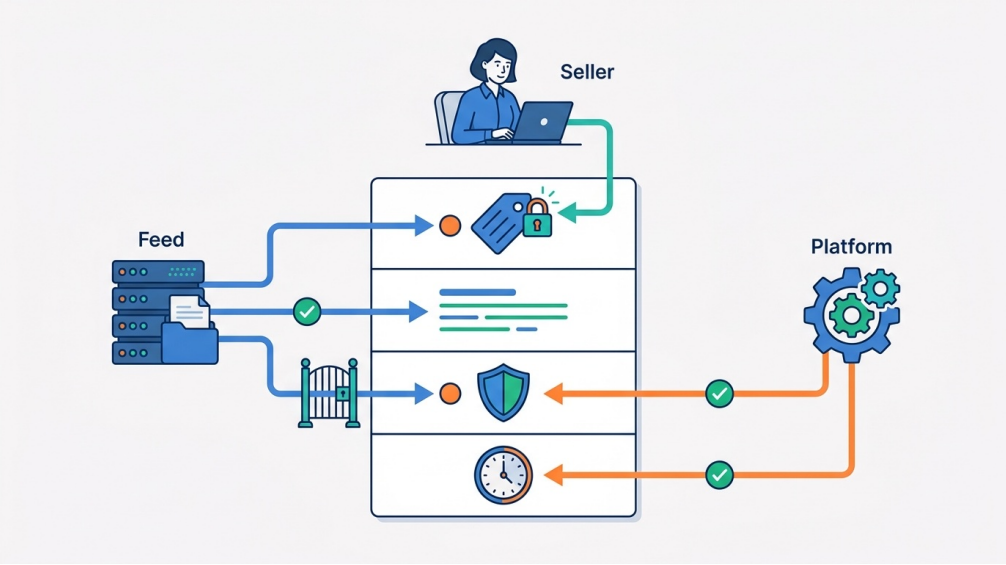


Figure 3: *Each field group has one owner: the feed updates what it owns, a seller's lock holds the noon price cut against the evening snapshot, and moderation state never comes from a feed.*

Reconciliation and the mass-change guard

The reconciler turns a run into changes, and the guard sits where a snapshot run becomes removals. For a delta run the reconciler applies operations. For a snapshot run it computes, per seller, the set difference between the snapshot and the seller's live feed-owned listings.

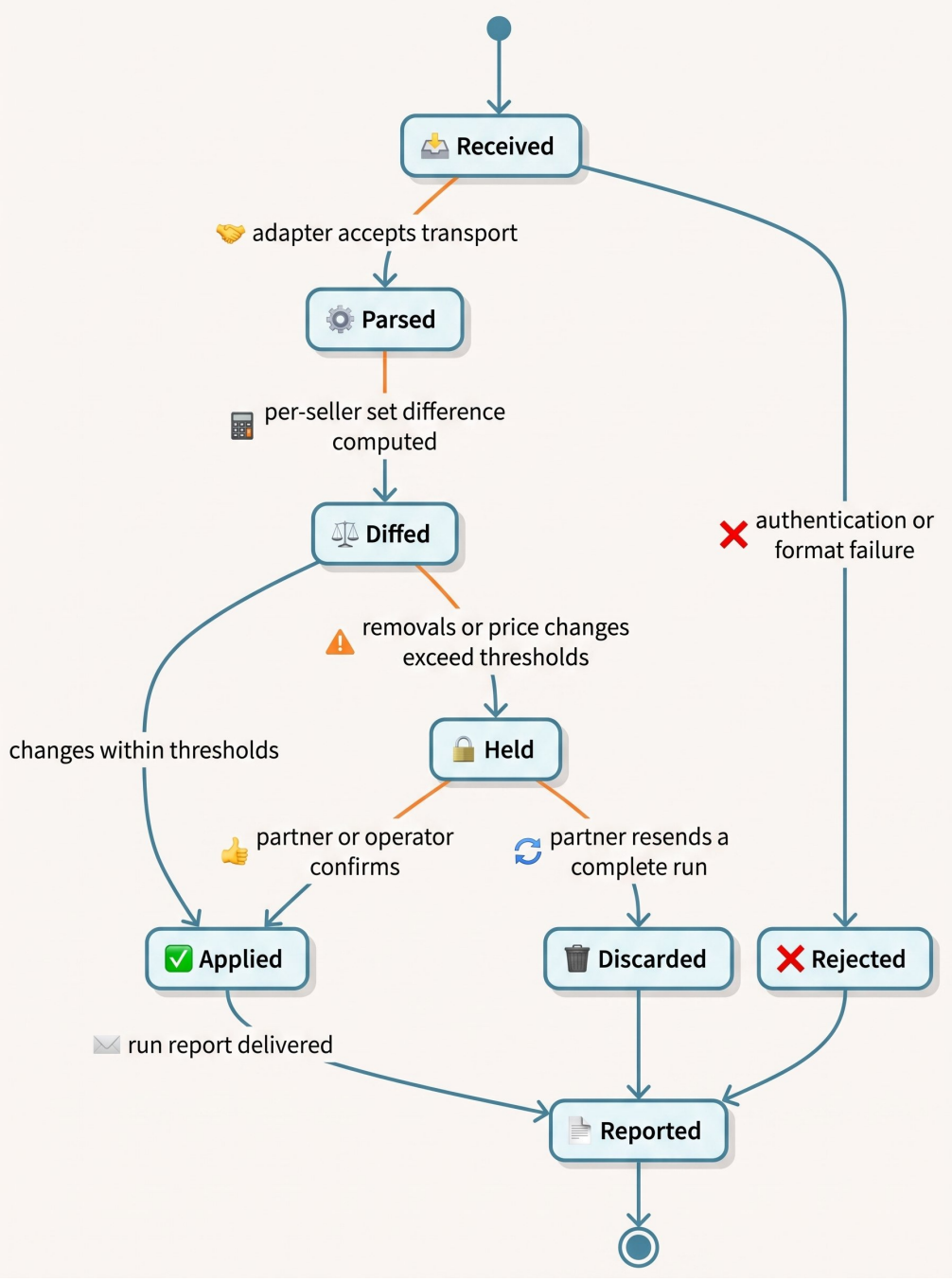


Figure 4: *The life of a snapshot run. A run that would remove or reprice more of a seller's stock than its thresholds allow is held, not applied; the partner learns about it through the same report as any other outcome.*

A **mass-change guard** holds a snapshot run for a seller when its effect exceeds a

threshold, for example removal of more than a set share of that seller's live stock, or more than a set count, or a change to a large share of prices. The thresholds are per seller size, because a small dealer selling three of its eight cars on a Saturday is normal and an agency losing 60% of its stock overnight is not. Held runs are confirmed by the partner through an API call or by an operator, or superseded by a complete resend. The guard is cheap, and it is the single control that would have turned Motorvane's 38,400 removals into 286 held runs and one phone call to the vendor.

Delta channels need the reverse control: **scheduled reconciliation**. At least weekly, and daily for high-value verticals, the platform requests or receives a full list of the partner's live external keys and compares it with its own. Listings live on the platform but absent from the partner's list become removal candidates, through the same guard. Listings present at the partner but missing on the platform are lost creates and go back into the pipeline.

Errors and feedback to partners

Partners experience a marketplace through its error reports. A good report is per run and per record, uses stable machine-readable codes, points to the offending field, and tells the partner what to do.

- **Three outcomes, not two.** Rightmove's responses distinguish success, *warning* ("the call has been successful, but one of the business rules has failed") and error, and makes warnings visible to the agent in an error report in its agent back office (Rightmove RTDF, 2023). The warning tier is what lets a platform accept a listing missing a recommended attribute while still telling the sender.
- **Standard envelopes.** RFC 9457, "Problem Details for HTTP APIs" (July 2023, obsoleting RFC 7807), defines `type`, `title`, `status`, `detail` and `instance`; its example validation extension carries an `errors` array with JSON Pointers to the failing fields (RFC 9457, 2023). mobile.de returns HTTP 400 with an `errors` array of keys and arguments. Either works; what matters is that the codes are stable and documented, and that the same code comes from the insertion flow, the API and the file importer, because they share one validator.
- **Asynchronous results for asynchronous channels.** For file feeds, eBay's Sell Feed API models each upload as a task with an ID, a status and a downloadable result file (eBay Sell Feed API, n.d.). The equivalent for every bulk channel is a run ID returned at upload and a run report retrievable by API, sent by webhook or e-mail, and visible to the seller in the pro tool.
- **Malformed input is the sender's error.** When Adevinta's media service for InfoJobs returned HTTP 500 for PDFs produced by a regional employment service, each failing file turned out to be two complete PDFs joined together; the failures had been classified as internal faults rather than file-format errors (Adevinta Tech Blog, 2025). An ingestion boundary that answers bad input with a 5xx hides a partner's bug inside the platform's own error budget.

Back-pressure, quotas and fairness

Feed traffic is spiky in a way user traffic is not: dealer software exports at the same evening hour, and one dealer group's 200,000-row file can arrive at the same time as 3,000 single-dealer files.

- **Per-partner queues and concurrency limits** keep one large partner from starving the rest. Work is scheduled by partner and seller, not first come, first served.
- **Rate limits with honest signals**. API channels return HTTP 429 with a retry hint when a client exceeds its limits; Autotrader documents that inbound connections are subject to "dynamic reputation scoring". File channels accept the upload and queue it, returning the run ID immediately.
- **Quotas tied to entitlements**. Professional packages often include a number of listing slots. Whether a feed record beyond the seller's quota is rejected, accepted as a draft or accepted and billed is a commercial rule owned by Packages, Bumps and Quotas: Products, Pricing and Entitlements; ingestion enforces it by asking the entitlement service, never by keeping its own count. When an importer applies one quota rule and the billing export counts feed rows by another, dealers are billed for cars that were never published; Motorvane found exactly that in 2027 (see Packages, Bumps and Quotas: Products, Pricing and Entitlements).
- **Priority by effect, not by arrival**. Removals and price changes are processed before creates and description edits, because a late removal is a buyer contacting a sold car and a late description edit is invisible.

Media through the feed

Media dominates feed bytes. Rightmove pulls media by URL rather than accepting pushed files, allows up to 999 media items and 20 MB per image, removes images whose URLs disappear from an update, and with a `Media_Update_Date` "will not re-retrieve images on the same URLs if they have the same media update date", which "relieves pressure on the client's image servers". mobile.de pre-uploads images and references them by ID in the ad. Either way the ingestion layer only references media; fetching, scanning, hashing and resizing belong to Photos Sell: Media Pipelines for Listings. The contract decision that ingestion must make is whether a listing may go live before its images are processed, and the answer differs by vertical.

The insertion flow is a channel too

Product teams usually own the insertion flow as a user-experience problem, and it is one: a **conversion funnel** from "start listing" to "published", where every step that asks a casual seller for something they do not have loses supply. The Productscape

model's seller journey names the typical pains: forced account creation before the seller sees value, mobile uploads over weak connections, and a moderation rejection "with an unclear reason" that leaves the casual seller stuck.

Architecturally, it is a channel like the others:

- **Drafts are listing state**. A draft saved server-side, keyed to the account and resumable on another device, is a listing in the draft state of The Listing Aggregate: Identity, Lifecycle and Invariants, not browser storage. Resumable media uploads need coordination even inside one browser; OLX described in 2026 how several open tabs each tried to resume the same large upload and duplicated objects in storage, which it solved with the Web Locks API (OLX Engineering, 2026).
- **Category first, then the schema**. The category selects the attribute schema, fees and moderation rules, so the flow predicts the category early and loads the compiled form for it.
- **The same validator**. The form renders the compiled schema and submits to the same command gateway. A rule enforced in the form but not in the feed, or the reverse, is a defect.
- **Assistance is a suggestion layer**. Photo-to-listing features have moved from experiment to mainstream: eBay announced a photo-based "magical listing" tool in September 2023 (eBay, 2023), and German trade press reported in November 2025 that Kleinanzeigen suggests title, description and category from a photo for almost all categories, excluding vehicles and intangibles such as jobs and services (mobiFlip, 2025). The assistant fills the draft; the seller confirms; provenance records which values came from a model. AI Across the Listing Lifecycle owns the assistant and From Raw Supply to Usable Listings: Normalization and Enrichment owns the provenance.

The funnel metric that matters is not form completion but **published-and-contacted**: a flow that publishes listings faster by asking for less, and so produces listings nobody can find with filters, has moved the problem downstream. Tradeyard measures start-to-publish rate and the share of new listings that receive a first contact within 48 hours, side by side.

In Practice: The Essen Golf Comes In Through a Feed

Motorvane is the clearest case, because most of its supply arrives by machine. Of its **1.45 million** live listings, 91% come from **21,400 dealers**, and **88% of dealer listings arrive through DMS feeds**: about **1.16 million** listings maintained by machine. Most of Motorvane's transfer service providers send a nightly full snapshot per dealer; a minority send deltas through the Seller API during the day. New dealer listings run at about

210,000 a week, about 30,000 a day, but the nightly snapshots carry well over a million records, because every car in stock is re-sent every night.

That ratio gives the first design fact: **most feed records are unchanged**. After the Common Ground review, Motorvane added content-hash change detection in front of validation: on a typical weekday, about **91% of snapshot records** are byte-identical in their relevant fields to the last accepted version and are dropped before validation, normalization, enrichment and indexing. The remaining roughly 100,000 records a day are creates, price changes, description edits and removals, and they are what the rest of the pipeline pays for.

Wednesday 14 April 2027: create

Autohaus Ruhrblick in Essen (fictional) uses a DMS whose vendor sends a full snapshot every day at 17:00. The **2021 Volkswagen Golf 1.5 TSI**, **48,000 km**, **€21,900**, entered the dealer's stock that morning.

Time (CEST)	Step	What happens
17:00	Snapshot arrives	148 vehicles for Ruhrblick, within its 150-slot package; run ID issued; adapter emits 148 ingestion records
17:01	Change detection	145 records unchanged and skipped; 2 price changes; 1 new external key, the Golf's stock number
17:01	Validation	The Golf passes the used-car schema: VIN present, first registration, mileage, price with VAT status, required emissions fields
17:01	Reconciliation	No removals; guard not triggered; a `Submit` command with the external key as idempotency key
17:02	Listing service	Listing created, version 1, external key mapped; `ListingSubmitted` emitted

Time (CEST)	Step	What happens
17:02–17:04	Normalization and enrichment	VIN decoding fills **31 attributes**; market-value estimate **€22,800**, rated "good price" (see From Raw Supply to Usable Listings: Normalization and Enrichment)
17:05	Publication	Searchable; run report shows 1 created, 2 updated, 0 errors, 1 warning (a recommended equipment field missing on another car)

Over the next ten days the Golf draws **23 raw leads**: 14 e-mails, 7 calls and 2 test-drive bookings, or 19 once repeat contacts from the same buyers are removed (see Serving the Pros: Dealer, Agent and Merchant Tooling).

Thursday 22 April 2027: price change

The dealer cuts the price to **€21,400** in the DMS at 11:20. The change reaches Motorvane in the 17:00 snapshot, nearly six hours later, and triggers price-drop alerts to **1,180 watchers** shortly after 17:00. Six hours is acceptable for a price cut; the next step shows when it is not.

Thursday 29 April 2027: sold, and still listed for 26 hours

The Golf is sold at 15:10 on Thursday 29 April. Ruhrblick's sales staff record the sale in the DMS the next morning. The car is absent from Friday's 17:00 snapshot, and the reconciler removes it at 17:10 on **Friday 30 April**, **26 hours after the sale**. In between, the listing stayed searchable, appeared in a morning alert digest, and drew three more e-mail enquiries for a car that no longer existed. None of the components misbehaved. The ghost was in the contract: a daily snapshot is a 24-hour removal SLA by construction, and nobody had chosen that SLA.

The fix Motorvane adopted is a **hybrid contract**. Vendors may keep sending nightly snapshots for completeness, but status changes (sold, reserved, withdrawn) must arrive as deltas through the Seller API within 15 minutes of being recorded in the DMS. Vendors that cannot meet it are labeled in the dealer's pro tool, so the dealer can see what their software choice costs them. The contract cannot fix the gap between the handshake on the forecourt and the DMS entry the next morning; Serving the Pros: Dealer, Agent and Merchant Tooling addresses that with a one-tap "sold" action in the dealer app that writes directly to Motorvane and is honored by the next snapshot.

Hearthly: 160 vendors, one contract

Hearthly's problem is breadth. About **160 software vendors** send listings for **36,500 agencies** across France and Spain, in formats that grew per vendor over years. The French market has historically worked through per-portal export connectors in the agency software, and Spain's former Casavista platform brought its own importer in 2025. Hearthly's ingestion team found three classes of vendor: about 40 that send complete, timely data; about 90 that send acceptable data with recurring warnings; and about 30 that cause most incidents, including the vendors that never send deletes.

Two of those vendors and Keybridge supplied the three listings of the Lyon apartment in May 2027. Keybridge turns neutrality into an ingestion requirement. It is a Veldmark company that publishes agency listings to Hearthly and to Hearthly's competitors, and the March 2027 decision keeps it a neutral SaaS. For ingestion, neutrality has a concrete meaning: **Keybridge integrates with Hearthly through the same public contract, credentials model, limits and error catalog as every other vendor**, with no private endpoint, no earlier access to schema changes and no field that other vendors cannot send. Anything else would be visible to the agencies that use competing software, and would confirm their suspicion.

Trade-offs

Decision	Option A	Option B	When A is right	When B is right
Transport for professional supply	**Per-listing API push**	**File feed** (push or pull)	Partners with modern software and event-driven stock changes; removals must be fast	Long-tail vendors, legacy CRMs, large catalogs; partners who can only export files

Decision	Option A	Option B	When A is right	When B is right
Feed semantics	**Full snapshot**	**Delta**	Partners with unreliable change tracking; small inventories; when self-healing matters more than latency	Large inventories, frequent changes, low latency targets; requires scheduled reconciliation
Acceptance	**Synchronous validation** at the API	**Accept and process asynchronously**, report per run	Interactive tools and single-listing APIs where the caller can fix and retry	Bulk files; spiky loads; enrichment-heavy validation
Media	**Pull by URL** with change detection	**Push bytes** or pre-upload	Partners with stable image hosting; large media counts (real estate)	Mobile and small partners without hosting; when the platform needs the bytes immediately

Decision	Option A	Option B	When A is right	When B is right
Partner authentication	**Mutual TLS client certificates**	**OAuth 2.0 tokens**	Few large B2B integrations, strong non-repudiation (Rightmove's model)	Many third-party tools acting for sellers with consent and revocation; follow RFC 9700's current best practice, which forbids the password grant and requires Proof Key for Code Exchange (PKCE) for public clients (RFC 9700, 2025)
Standards	**Adopt an industry standard** (OpenImmo, RESO)	**Own contract** with adapters	The ecosystem already speaks it; you are one receiver among many	No standard exists or it lags the regulation you must meet; support standards as adapters into your contract
Ownership of ingestion across brands	**One group ingestion platform**	**Shared contract and libraries, local adapters**	Brands share partner ecosystems and formats	Partner ecosystems differ by vertical and country, as at Veldmark

Standards change slowly, which is why the standards row favors adapters. OpenImmo's XML version dates from 2001, and its REST and JSON successor, not compatible with the

XML version, was expected to reach final release only at the end of 2026 (OpenImmo, n.d.). Rightmove dropped XML in 2019 and moved to a certificate-authenticated JSON API. **A receiver must plan to run adapters for two generations of a standard for years**, which is one more reason the standard should be an adapter into the platform's own contract, not the contract itself.

Vertical Differences

Dimension	General classifieds	Real estate	Automotive	Jobs
Dominant channel	Insertion flow (Tradeyard: 78% of new listings from private sellers)	Feeds from agency software (Hearthly: 84% of listings)	DMS feeds through transfer service providers (Motorvane: 88% of dealer listings)	Employer and applicant-tracking-system feeds, plus self-service
Where the truth lives	The marketplace	Agency CRM, or an MLS in the US	Dealer DMS	Employer's applicant tracking system
Typical semantics	Interactive commands	Mixed; snapshots common in file-based ecosystems	Nightly snapshots common; deltas through APIs	Deltas plus expiry dates
Identity anchor	Platform listing ID	Agency reference per branch; property address	Dealer stock number; VIN	Employer requisition ID
Hardest problem	Funnel conversion and abuse at insertion	Breadth of vendors; mandatory disclosures; license and embargo rules	Removal latency; feed truncation; paid-slot accounting	Expired jobs; duplicate postings across job boards

Dimension	General classifieds	Real estate	Automotive	Jobs
Removal pressure	Seller must remember to mark sold	Under-offer and let states lag	Sold cars draw leads within hours	Search engines penalize stale job markup
Regulatory content in the contract	Trader declarations under the EU Digital Services Act	Energy class and costs, fees, local disclosures	Emissions and consumption labeling, VAT status	Salary transparency, anti-discrimination rules

Real estate adds a visibility dimension general classifieds lack. In the US, the National Association of Realtors' multiple-listing-options policy, effective 25 March 2025, allows "delayed marketing exempt" listings whose public marketing through IDX and syndication is postponed for a period set by the local MLS (NAR, n.d.). An ingestion pipeline downstream of such sources must carry **embargo and visibility flags as data on each record** and honor them, as a compliance control rather than a ranking signal.

Jobs add expiry as a hard requirement: Google's guidance warns that failure to remove expired job postings "may result in a manual action" (Google Search Central, updated 2026), so a jobs feed that cannot express closure is a search-engine risk.

In the EU, trader traceability adds a rule to every feed channel. The EU Digital Services Act, applicable since 17 February 2024, requires marketplaces to collect and make best efforts to verify trader information before traders can use the service (Article 30) (Regulation (EU) 2022/2065, 2022). For feed channels this means the integration must not let a vendor create listings for a trader whose traceability data is incomplete.

Failure Modes

Absence treated as truth without a guard. The Motorvane incident in one sentence: a truncated snapshot looks exactly like a dealer who sold everything. Every snapshot channel needs a mass-change guard, and its thresholds need an owner who reviews them as the partner base changes.

Deltas without reconciliation. A delta channel without periodic full comparison accumulates ghosts at the rate partners lose deletes. The drift is invisible until someone samples it, which is why the reconciliation report, not the error log, is the primary health signal for delta partners.

External-key mapping held by the importer. When the mapping from partner key to listing ID lives in importer tables and is cleaned up on removal, a re-sent listing becomes

a new listing: new URL, lost favorites, lost price history and a duplicate if the old one was not fully removed. Keep the mapping on the listing and retire it only when the listing is purged.

Two channels, one seller, no recognition. A dealer moves from one DMS vendor to another and both send stock for a week, or a dealer sends the same car through a vendor and through the pro tool. Ingestion should recognize the same seller and the same strong key (VIN, agency reference) across channels and **merge at the boundary**, keeping one listing with two sources. One Property, Three Listings: Duplicates and Entity Resolution handles duplicates across sellers; within one seller, the merge belongs here.

Different validation behind different doors. Tradeyard's CSV importer accepted listings that the web form would reject and rejected some the form accepted. Sellers learned to use whichever door let their listing through, and the attribute data told two stories. One validator library, compiled from one schema, ends this.

Errors only in support tickets. If partners learn about rejections from sellers who cannot find their listings, the support desk becomes the error channel and every fix costs a conversation. Machine-readable run reports, available to both the vendor and the seller, cost less than one support agent.

Partner bugs counted against the platform. Malformed input answered with 5xx, retries amplified into storms, and one partner's backlog delaying everyone's removals. Classify errors at the boundary, isolate partners in queues, and track each partner's error budget separately from the platform's.

Breaking changes by surprise. A new mandatory field enforced on a Tuesday breaks every vendor on Tuesday night. Publish the requirement as recommended first, with an effective date, the way eBay publishes an `expectedRequiredByDate` for item aspects (see Attributes Without Chaos: Schemas, Dictionaries and Validation), and announce breaking changes with a fixed notice period; Autotrader's documented minimum is 30 days and lists what counts as breaking.

Regulatory fields optional in the feed. If the feed contract treats a legally required disclosure as optional, the platform publishes non-compliant ads at feed speed. Required-by-law fields must be blocking in the validator for the categories and countries where they apply, with the rule versioned by effective date.

The Chief Architect's View

At group scale, ingestion is less a single system than a **set of contracts and the discipline to operate them.** The decisions Veldmark made, and the reasoning, generalize.

1. **Share the contract, not necessarily the importer.** Ingestion is not one of the Common Ground shared platforms, and Ingrid did not make it one. Motorvane's partners are German, Dutch, Polish and Austrian DMS vendors; Hearthly's are French

and Spanish agency-software vendors; Tradeyard's are mostly its own pro sellers. A single group importer would serve three ecosystems that share almost nothing. What Veldmark shares is the **Supply Contract v1**, adopted at the architecture advice forum in September 2027: one ingestion-record envelope, one error catalog in RFC 9457 format with stable codes, the field-ownership defaults above, the requirement that every snapshot channel run behind a mass-change guard and every delta channel reconcile at least weekly, and a shared validator library that loads schemas from the group taxonomy and attribute service. Adapters stay local. This is the book's "share platforms, not products" principle applied at the boundary.

2. **Make removal latency a contracted number**. Every channel's contract states a removal SLA, and the platform measures it from the partner's own timestamp to the listing leaving search. Motorvane's hybrid contract (snapshots for completeness, deltas for status within 15 minutes) is the pattern. From Submitted to Searchable: The Publication Pipeline owns the freshness SLAs end to end; ingestion owns the part that happens before the listing service.

3. **Treat partner health as a commercial metric**. For each integration and each large seller, publish feed success rate, warning rate, time from partner change to platform, reconciliation drift and held runs. Give the same numbers to the partner and to the seller. The Productscape model's insight that **feed health is a commercial growth lever** is right: dealers blame the marketplace for empty lead inboxes, whoever caused them.

4. **Decide who owns each door**. Veldmark assigns each channel an owning team per brand, a contract owner and a partner-operations function that onboards and certifies vendors. Certification is a test suite that a vendor runs against a sandbox before production credentials are issued; mobile.de's requirement that transfer service providers serving more than ten customers supply contractual and technical contacts, software details, VAT ID and customer numbers before access (mobile.de Seller-API manual, n.d.) is a public example of gating access on accountability.

5. **Govern changes like an API program**. Version the contract, pin vendors to versions, announce deprecations with a notice period, and keep a public change log. The cost of a breaking change is paid by 160 vendors and 36,500 agencies, not by the team that made it.

6. **Keep the evidence**. The ingestion record log is evidence: what the partner sent, when, and what the platform did with it. It answers dealer disputes ("you deleted my cars"), billing disputes over listing slots, and regulators' questions about who published a listing. Retain it for the dispute and audit windows, with personal data minimized.

What to watch: the share of live listings past their channel's freshness SLA; held runs per week and how many were real; reconciliation drift per delta partner; the 20 partners with the most warnings; and sold-to-gone latency for feed listings by vendor. A sudden change in duplicates or ghost reports upstream of search is almost always a feed change first.

Questions to Ask

1. For each ingestion channel, where does the truth about a listing live, and how would we detect that our copy has drifted from it?
2. What does absence mean in each feed, and what stops a truncated snapshot from removing a seller's stock?
3. Where is the mapping from the partner's key to our listing ID stored, and does it survive a removal and a re-send?
4. Do the insertion flow, the partner API and the file importer run the same validator against the same compiled schema, and return the same error codes?
5. Which fields can a feed overwrite, which can the seller lock in our tools, and which can only the platform change?
6. What is the contracted removal latency per channel, and what is the measured sold-to-gone latency per vendor?
7. How does a vendor learn, without calling support, that a record failed or was held, and how does the seller learn it?
8. How is one partner isolated from another under load, and are removals prioritized over creates?
9. Which legally required disclosures are blocking in the feed contract, per country and category, and who updates them when the law changes?
10. Does any integration, including one owned by our own group, have access or terms that other partners do not?

From Accepted to Usable

The records that pass this boundary are accepted, but they are not yet usable: prices with thousands separators in the wrong locale, makes spelled five ways, addresses without coordinates and vehicles described only by a VIN. The next chapter, From Raw Supply to Usable Listings: Normalization and Enrichment, follows the same Golf from accepted record to consistent, enriched listing, and asks how the platform records which of its values the dealer said and which the platform derived.

Sources and Evidence

Company engineering and investor sources

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by URL and Media_Update_Date, success/warning/error responses, regulation-driven schema versions.
- Seller-API documentation and Seller-API manual — mobile.de, n.d. Transfer service provider access model, insertion request ID returning 303, idempotency of methods, no parallel requests per ad, error format, access requirements for providers with more than ten customers.
- Upload interface XML — mobile.de, n.d. Bulk XML with seller-inventory-key and customer number per ad.
- Autotrader developer portal — Autotrader UK, n.d. Stock API and Stock Notifications webhooks, validation errors encoding policy, 30-day breaking-change policy, dynamic reputation scoring.
- Sell Feed API overview and Inventory API overview — eBay, n.d. Asynchronous upload tasks with result files; SKU identity; separate publish step.
- Bundeskartellamt clears mobile.de's acquisition of Carwow's German business — Bundeskartellamt, 2026. Dealer multi-homing and data export as competitive constraints.
- "Magical" listing tool — eBay, 2023. Photo-based listing assistance.
- Kleinanzeigen führt KI-gestützte Anzeigenerstellung per Foto ein — mobiFlip (trade press), 2025. Photo-to-listing suggestions and excluded categories.
- Handling concurrency on the Web with Web Locks API — OLX Engineering, 2026. Duplicate resumable uploads across tabs.
- Engineers to the rescue: when your public employment service makes it impossible to apply for jobs — Adevinta Tech Blog, 2025. Malformed input misclassified as server errors.

Standards, regulation and documentation

- OpenImmo — OpenImmo e.V., n.d. XML standard of about 250 features over FTP; incompatible REST/JSON 2.0.
- RESO Data Dictionary and RESO Web API — Real Estate Standards Organization, n.d. Data Dictionary 2.0 scale; OData-based Web API; full-then-changes replication; RETS deprecation.
- Multiple Listing Options for Sellers — National Association of Realtors, n.d. (effective 25 March 2025). Delayed-marketing and office-exclusive listing states.
- RFC 9457: Problem Details for HTTP APIs — IETF, 2023. Error envelope with JSON-pointer validation errors.
- RFC 9700: Best Current Practice for OAuth 2.0 Security — IETF, 2025. PKCE, no password grant, sender-constrained tokens.
- Idempotent requests — Stripe, n.d. Canonical idempotency-key semantics.
- draft-ietf-httpapi-idempotency-key-header — IETF, 2025 (version 07, expired). No standard idempotency header exists.

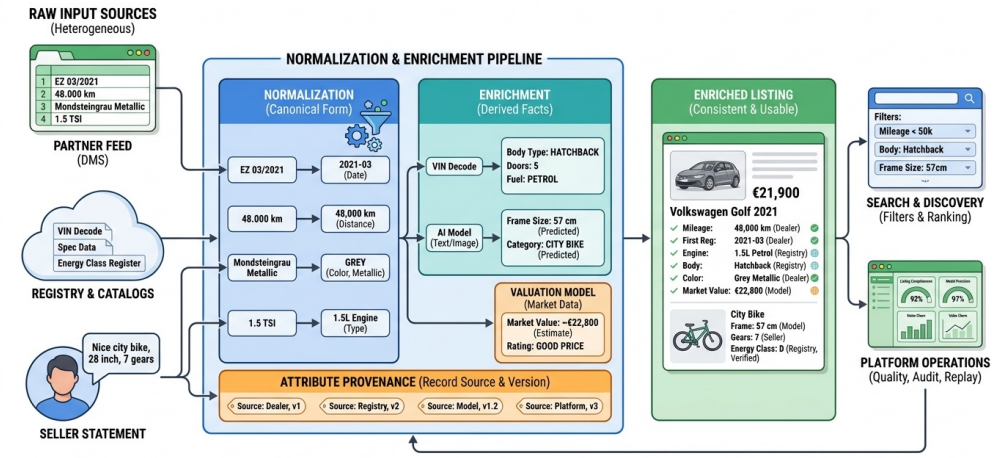
- Job posting structured data — Google Search Central, 2026 (last updated). Expired-job removal and manual actions.
- Regulation (EU) 2022/2065 (Digital Services Act) — European Union, 2022 (applicable from 17 February 2024). Trader traceability (Article 30) and compliance by design (Article 31).

Research and practitioner sources

- Productscape general-listings and real-estate marketplace domain models — the author's generic domain models (not descriptions of any company), digested in this book's research notes, n.d. Integration-partner measures, seller-journey pains and the "feed health as a growth lever" insight.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

14. From Raw Supply to Usable Listings: Normalization and Enrichment



IN THIS CHAPTER, YOU WILL: Learn the difference between normalizing a value and enriching a listing, follow a dealer's Volkswagen Golf from a raw DMS record to a listing with 31 decoded attributes and a market-value rating, and design an enrichment pipeline whose steps are idempotent, versioned and replayable. You will see how attribute provenance settles conflicts between the seller, a registry and a model, how quality scores and human correction loops work, and what ML enrichment costs at marketplace scale.

WHY THE BUSINESS CARES: Buyers find listings through filters, and filters only work on values that are present and consistent. Every listing whose mileage, frame size or energy class is missing or malformed is invisible to the buyers filtering for it, and every wrong derived value, such as a car rated against the wrong engine, is a complaint from a paying professional.

WHY ARCHITECTS SHOULD CARE: Enrichment multiplies the writers of a listing: the seller, the feed, registries, models and moderators. Without provenance and versioning, nobody can tell which value came from where, re-running a better model overwrites what sellers corrected, and the derived data ends up living only in the search index, where it cannot be rebuilt.

KEY POINTS:

- **Normalize to one canonical value, and keep what was said**. Store each attribute in its dictionary's canonical type and unit, next to the original text, unit and locale as received. Normalization must be deterministic and versioned, so that search, ranking and duplicate detection compare like with like and every transformation can be replayed.
- **Every value carries its provenance, and precedence is a policy per attribute**. Seller statements, feed values, registry decodes, model extractions and moderator corrections are different kinds of truth. Decide per attribute which source wins, keep the losers, never let a derived value silently overwrite an asserted one, and verify legally mandated disclosures rather than invent them.
- **Enrichment is a versioned, replayable pipeline with a budget**. Each enricher is an idempotent function of the listing version, the reference data and its own version, and writes back to the listing's system of record. At classifieds volume, model calls are rarely the main cost; re-processing the live stock, evaluating each new version and reviewing the uncertain cases are.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Attribute Junkyard** — Without canonical normalization, the same fact arrives as "Benzin", "Super", "petrol" and "1", and no filter, facet or feed can agree on what a listing is.
- **Set-and-Forget AI** — Extraction and classification models write values into millions of listings with no evaluation gate, no monitoring and no owner, and their errors become the catalog.
- **Search as System of Record** — Enrichment results written only into the search index cannot be rebuilt, audited or corrected, so the index quietly becomes the only place those facts exist.

A listing accepted at the boundary of Many Doors In: Forms, APIs and Bulk Feeds is not yet a usable listing. This chapter is the next link in Part III's supply chain: the step that records where every value came from. A dealer management system (DMS) writes "48.000 km" with a German thousands separator, a Portuguese private seller types

"bicicleta de cidade 28' "' into the title and nothing into the attributes, a French agency's software sends an address without coordinates, and a feed describes a car by its vehicle identification number (VIN) and little else. **Search, ranking, alerts, trust rules and analytics all need the same fact in the same form**, and none of them should have to parse a seller's text to get it.

This chapter owns three terms. **Normalization** turns a received value into the canonical representation the attribute dictionary defines, without changing what it means. **Enrichment** adds facts the seller did not state, derived from reference data, registries, models or other listings. **Attribute provenance** is the record, per value, of where it came from, when, by which process and version, and with what confidence.

The neighbors own the rest. Many Doors In: Forms, APIs and Bulk Feeds owns the channels and contracts that deliver records; Attributes Without Chaos: Schemas, Dictionaries and Validation owns the dictionary and compiled category schemas that define the canonical forms; Where Is It? Location, Geocoding and Privacy owns geocoding and location precision; Photos Sell: Media Pipelines for Listings owns image processing; One Property, Three Listings: Duplicates and Entity Resolution owns duplicates and canonical entities; Models in Production: ML Platforms and Use Cases owns training, serving and model governance. This chapter owns the pipeline that applies all of these to a listing, and the provenance that makes the result trustworthy. One Property, Three Listings: Duplicates and Entity Resolution compares the canonical values it produces, and the search of Part IV — DISCOVERY: Matching Intent to Inventory filters on them. Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), supplies the worked examples.

Setting the Scene at Veldmark: Which of These Values Did Anybody Say?

In April 2027, Noor Hassan, Veldmark's Chief Product Officer, asked a question in a product review that turned into an architecture review. Tradeyard Netherlands' bike filters were part of the case for the November bikes split into city, e-bike, cargo and kids categories. Yet only about **23% of live bike listings** carried a frame size, and fewer than half said whether the bike had electric assist. Priya Raman, Head of Data & ML, proposed running a language model over the live bike listings to extract the missing values. Daan Visser, who owns Tradeyard's Yardstick monolith, pointed out that Yardstick had no place to store a value that the seller had not typed: anything written into the attribute columns would look exactly like a seller statement.

The same week, Felix Brandt brought a Motorvane example. The **2021 Volkswagen Golf 1.5 TSI** from Autohaus Ruhrblick in Essen (fictional) had arrived through its DMS feed on **Wednesday 14 April 2027**. VIN decoding had filled **31 attributes**, and the market-value model had rated the car "good price" against an estimate of **€22,800** for

its **€21,900** asking price. Felix's team had found, in a sample of older listings, cars rated against the wrong engine variant because a decoder value had silently replaced a dealer value. When the decoder supplier later corrected its data, Motorvane could not re-run decoding without also overwriting values dealers had fixed by hand, because nothing recorded which values had come from where.

Ingrid Halvorsen, the Group Chief Architect, put both examples on one slide with one question: **for each value on a listing, do we know who said it, what we did to it, and whether we may change it?** The rest of this chapter is the design that answers yes.

Concepts: Canonical Values, Derived Facts and Their Origins

Normalization and enrichment are different operations

	Normalization	Enrichment
What it does	Converts a stated value into its canonical form	Adds a value nobody stated
Example	"48.000 km" becomes 48,000 km; "Benzin" becomes fuel type PETROL	VIN decodes to body type HATCHBACK; title text yields frame size 57 cm
Changes meaning?	Never; if it cannot convert without guessing, it flags	Yes: it asserts a new fact
Determinism	Deterministic given the normalizer and reference-data versions	Often probabilistic, with a confidence
Provenance	Inherits the source's provenance, with a normalizer version	Its own provenance: registry, model, catalog, other listings
Failure mode	Silent misparse, such as a thousands separator read as a decimal point	Confident, wrong derived value

The distinction matters because the two have different owners, different tests and different rules for being overwritten. A normalized mileage is still the dealer's statement; a decoded body type is the platform's assertion, backed by a registry.

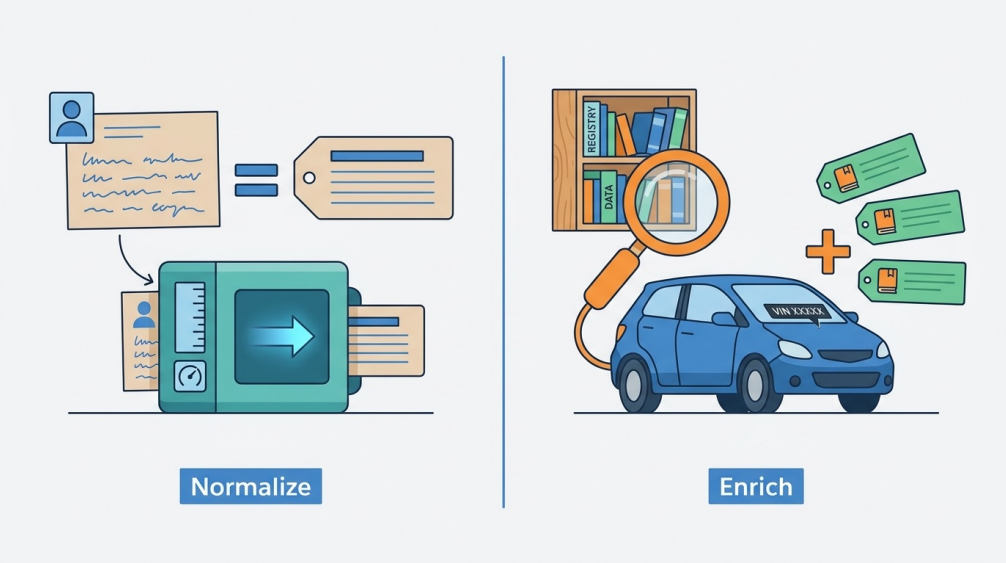


Figure 1: *Normalization rewrites what the seller said into canonical form without changing its meaning; enrichment asserts a new fact, so it carries its own source and its own rules for being overwritten.*

Canonical value and original value

Each attribute value is stored twice: the **canonical value** in the dictionary's type and unit, used by every system that compares, filters or computes, and the **original value** exactly as received, with its unit and the locale in which it was parsed, used for display, audit and re-normalization. Schema.org's `QuantitativeValue` uses UN/CEFACT unit codes such as `KMT` and `SMI` for odometer readings (schema.org Vehicle, V30.1, 2026), and Google's merchant feeds require prices as a number plus an ISO 4217 currency code (Google Merchant Center, n.d.). Using such codes in the dictionary costs nothing and saves a mapping layer later. Conversion for display, such as miles for a British visitor, happens at the edge, never in storage.

Attribute provenance

Provenance is a small record attached to every attribute value. A useful minimum:

```
attribute:        vehicle.power_kw
canonical_value:  96
original:         "96 kW"            (locale de-DE)
source:           partner_feed       (integration TSP-031, run 2027-04-14T17:00)
asserted_by:      seller             (Autohaus Ruhrblick)
process:          normalizer units@2027.03
confidence:       1.0
alternatives:     registry_decode -> [96, 110]   (decoder licensed-vin@2027.04)
status:           resolved, consistent with derivative
observed_at:      2027-04-14T17:01:12+02:00
```

There are seven **source classes**, and they should be a closed vocabulary across the group:

Source class	Meaning	Typical examples	Default trust
`seller`	Stated by the seller in the insertion flow or pro tool	Price, condition, description, mileage	Authoritative for intent and condition; legally the advertiser's statement
`partner_feed`	Stated by the seller through their software	Everything a DMS or agency CRM sends	As `seller`, but subject to mapping errors
`registry`	Decoded from an authoritative or licensed reference source	VIN decode, building identifier, energy certificate register	Authoritative for identity and specification facts
`catalog`	Inherited from a matched catalog product	Phone model specifications	Authoritative for product facts, not instance facts
`model`	Inferred by an ML or rules model, with confidence	Category, frame size, room type, translation	A suggestion until accepted or above a policy threshold
`moderator`	Set or corrected by a human reviewer	Corrected category, removed contact details	Overrides models; may be appealed by the seller

Source class	Meaning	Typical examples	Default trust
`platform`	Computed by the platform from other values	Price per m², quality score, H3 cells	Recomputed whenever inputs change

The listing aggregate holds attribute values with provenance (see The Listing Aggregate: Identity, Lifecycle and Invariants); the dictionary in Attributes Without Chaos: Schemas, Dictionaries and Validation says, per attribute, which source classes are allowed and how conflicts resolve. This chapter's pipeline produces the records.

Normalization: Ten Deterministic Steps

Normalization is a list of small, deterministic rules. I would expect any listing platform operating in several European countries to need the ten below.

Step	What goes wrong without it	Rule
Encoding	Old DMS and CRM exports in Windows-1252 or ISO-8859-1 produce "MÃ¼nchen"; mixed encodings within one file	Decode by declared encoding, detect and repair known mojibake patterns, store UTF-8, reject undecodable records with a clear error
Text	HTML fragments, tracking pixels, repeated whitespace, invisible characters used to evade moderation	Strip markup to an allowed subset, normalize Unicode (NFC), collapse whitespace, keep the original for audit
Numbers and locale	"48.000" read as 48 in an English parser; "1,5" as 15	Parse with the record's declared or inferred locale; for ambiguous values, compare with plausible ranges from the dictionary and flag rather than guess

Step	What goes wrong without it	Rule
Units	Square meters and square feet, cc and liters, kW and PS mixed in one field	Convert to the dictionary's canonical unit; keep unit as received; flag implausible conversions
Enumerations	"Benzin", "Super", "petrol", "gasolina" and "1" for the same fuel	Map through alias tables to dictionary value IDs, per country and per partner; unknown values go to a mapping queue, not into the canonical field
Named entities	Makes, brands and models spelled many ways	Resolve against an entity dictionary with aliases and a verified flag. Vinted reported about **2,000,000** unique brand values and restricted brand facets to about **40,000** verified by its brand specialists, which cut P99 facet latency by about 15 ms (Vinted Engineering, 2023)
Dates and times	"03/2021" versus "2021-03"; first registration as `yyyyMM` text; feeds in local time without offset	ISO 8601 with explicit offsets; mobile.de's API, for example, uses ISO 8601 date-times in the Europe/Berlin offset and `yyyyMM` text for first registration (mobile.de Seller-API, n.d.)

Step	What goes wrong without it	Rule
Prices and currency	"21.900,- €"; prices with and without VAT; "price on request" encoded as 0 or 1	Minor units plus ISO 4217 code; VAT status as a separate attribute; "on request", "negotiable" and "free" as explicit price states, never magic numbers. Store the currency the seller used (złoty at Tradeyard Poland); convert only for display
Addresses	Free-text addresses with abbreviations and typos	Parse into components and generate normalized variants; libpostal, an open-source address parser and normalizer trained on OpenStreetMap and OpenAddresses data, reports 99.45% full-parse accuracy on held-out data and handles abbreviation expansion in "60+ languages" (libpostal, n.d.). Its model is about 1.8 GB, which makes it a service rather than a library in every pod. National reference bases, such as France's open Base Adresse Nationale, published under the Etalab 2.0 open license with a public geocoding service (adresse.data.gouv.fr, n.d.), anchor the result

Step	What goes wrong without it	Rule
Phone numbers and contact data	Numbers in many formats; contact details inside descriptions and images	E.164 for structured contact fields; detection of contact details in free text and images is a trust-and-safety rule (see Trust and Safety as a Platform)

Three design rules apply to every step. **Normalizers are versioned code with test suites**, because a change to an alias table or a locale rule changes millions of canonical values. **Failure to normalize is a flag, not a guess**: an unparseable mileage goes into a quality state and, for feed listings, a warning in the partner's run report (see Many Doors In: Forms, APIs and Bulk Feeds). **Mapping tables are data with owners**: the alias that maps a Polish DMS vendor's fuel code to the dictionary belongs to someone who reviews the unknown-value queue every week.

Enrichment: Facts Nobody Typed

Enrichment does four things: it fills attributes sellers did not provide, it adds facts sellers cannot know, it adds platform-computed signals, and it prepares the listing for other languages and systems.

Enrichment	Typical inputs	Output	Typical source class	Owner of the capability
Registry decoding	VIN; German type-approval codes (HSN/TSN); building or parcel identifiers	Specification attributes, identity links	`registry`	Vertical teams; licensed data

Enrichment	Typical inputs	Output	Typical source class	Owner of the capability
Geocoding and area assignment	Normalized address	Coordinates with precision, administrative area IDs, grid cells	`platform` from `registry` data	Where Is It? Location, Geocoding and Privacy
Property and regulatory data	Address, energy certificate number	Building facts, verification of disclosed energy class, flood zone	`registry`	Real-estate vertical
Category prediction	Title, description, photos	Category with confidence	`model`	ML platform, taxonomy owners
Attribute extraction from text	Title, description, category schema	Missing attributes with confidence	`model`	ML platform, vertical schema owners
Attribute extraction from images	Photos, floor plans	Room types, vehicle angles, text in images, condition cues	`model`	Photos Sell: Media Pipelines for Listings and ML platform
Translation	Title, description	Text in other languages	`model`	Shared language service
Market value and price rating	Attributes, location, comparables	Estimate, band, label	`model`	Models in Production: ML Platforms and Use Cases
Quality and completeness scores	Everything above	Scores and nudges	`platform`	Listing data domain

Enrichment	Typical inputs	Output	Typical source class	Owner of the capability
Embeddings	Text, images	Vectors for retrieval and similarity	`model`	Search and recommen-dations

Registry decoding: when the truth exists elsewhere

Where a registry exists, decoding beats extraction from text. In the US, NHTSA's vPIC service decodes VINs, supports batches of up to 50 VINs, accepts partial VINs with * for unknown characters, and publishes standalone databases for download, so a marketplace can run decoding in-house rather than depend on a public endpoint (NHTSA vPIC API, n.d.). European marketplaces typically license specification data from commercial providers; mobile.de's Seller-API carries German type-approval codes `hsn` and `tsn` and a `schwackeCode`, a make-and-model classification "given by Eurotax/Schwacke" (mobile.de Seller-API manual, n.d.). Autotrader UK's developer platform exposes a taxonomy of vehicle types, makes, models, generations and derivatives, and keys valuation calls on a derivative ID plus first registration date and mileage (Autotrader developer portal, n.d.).

Decoders teach the pipeline two lessons. They return **families as well as exact answers**: an engine sold in more than one power output may decode to a set of candidates, and the dealer's statement resolves it. And decoded data **changes with the supplier's releases**, so each decoded value must record the decoder and data version that produced it, or a correction from the supplier cannot be applied without trampling dealer edits.

Real estate has registries too, though fewer of them are identifiers for units. France publishes its transaction register, *Demandes de valeurs foncières* (DVF), as open data from the tax administration, covering the last five years and updated twice a year, in April and October (data.gouv.fr, last updated 2026). ADEME, the French environment agency, publishes the energy-performance diagnostics (DPE) of existing dwellings made since July 2021 as an open dataset with an API, updated weekly, about 15.8 million records as of October 2026, and warns that the data is raw diagnostician input that ADEME does not correct (ADEME open data, 2026). That last warning is the general point about registries: **authoritative does not mean clean**, and the provenance record should say which register and which extract a value came from.

Model-based enrichment: the public evidence

Marketplaces have published more about model-based enrichment since 2023 than about almost any other part of their listing systems, and the evidence points one way: **use**

models to propose values against a schema that is passed in, validate the output structurally, gate releases on measured precision, and keep the big model offline and the small model online.

- **Mercari, title suggestions (2023).** Mercari described an offline step in which GPT-4 extracted the attributes of effective titles, and an online step in which GPT-3.5 Turbo detected those attributes in new listings and suggested titles. Malformed outputs (varied casing, apologetic refusals, extra lines) grew with volume during rollout and required pre- and post-processing; Japanese text could cost more than twice the tokens of English (Mercari Engineering, 2023).
- **Mercari, dynamic attributes (2024).** Because required attributes vary by category and change over time, Mercari supplied the attribute keys *at request time* to a fine-tuned 2-billion-parameter model (gemma-2b-it, QLoRA-tuned, quantized to 4 bits), about 95% smaller than the base model, slightly better than gpt-3.5-turbo-0125 on BLEU, and estimated at more than 14 times cheaper; fine-tuning gave more control over hallucination than prompting alone (Mercari Engineering, 2024).
- **Mercari, re-categorization (2024).** When Mercari renewed its category structure, classifying all existing items with GPT-3.5 Turbo was estimated at about USD 1 million and 1.9 years. Mercari instead labeled a sample of several million items with the LLM and propagated labels to the rest with a k-nearest-neighbor classifier over embeddings (Mercari Engineering, 2024). Category Trees That Can Change: Designing Marketplace Taxonomies uses this case for migrations; here it is a cost pattern.
- **Etsy, attribute extraction (2025).** A third-party summary of Etsy's engineering post, whose original blocked automated access, reports parallel prompts per attribute, validation of structured output, business rules, evaluation against vendor-annotated ground truth and expert-reviewed "silver labels", production runs halted when precision or recall deviates from a baseline, and complete attribute coverage in target categories rising from **31% to 91%** (ZenML LLMOps Database summary of Etsy Code as Craft, 2025). Treat the figures as reported, not verified.
- **Adevinta, text in images (2023).** Adevinta's optical character recognition (OCR) service detects insults, hidden messages, URLs, e-mail addresses and phone numbers in ad images. After moving to PaddleOCR on SageMaker, volume grew from more than 100 million to 330 million requests a month; accuracy on an internal benchmark rose from 0.4 to 0.8, latency improved 7.5 times, and the service was about 12 times cheaper than an external OCR API (Adevinta Tech Blog, 2023).
- **mobile.de, ordering of enrichments (n.d.).** mobile.de's price rating uses images: "If images are uploaded later … the price rating will be recalculated including the image data", and the documentation recommends uploading images first (mobile.de Seller-API manual, n.d.). Enrichers depend on each other, and the pipeline must re-run downstream enrichers when an upstream input arrives late.
- **Google, receiver-side classification (n.d.).** Google Merchant Center now assigns a product category to every product automatically, and the merchant-supplied

category is an override used in a few cases (Google Merchant Center, n.d.). Large receivers of supply increasingly classify items themselves rather than trust the sender.

Architecture: A Replayable Pipeline That Writes Back

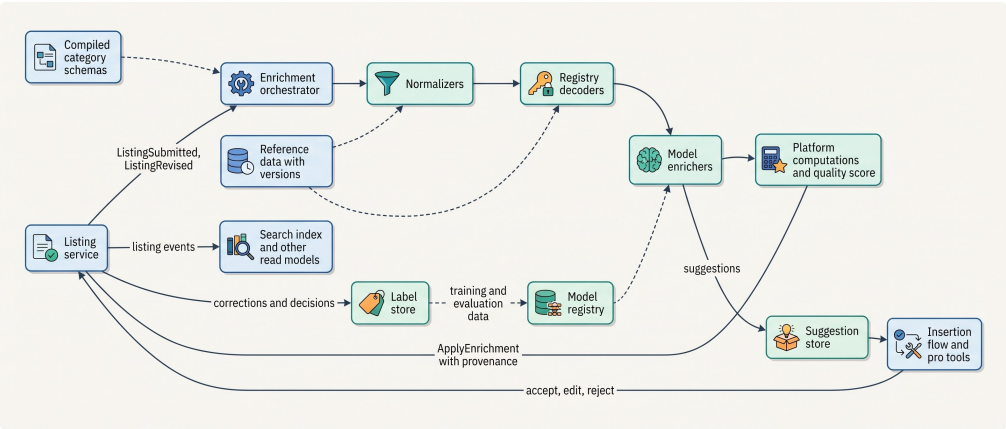


Figure 2: *The enrichment pipeline as a loop around the listing service. Enrichers read listing events and versioned reference data, and write results back as commands with provenance; read models such as the search index are rebuilt from the listing, never written directly. Suggestions go to sellers, and their decisions become labels.*

The rules that make it replayable

1. **Enrichers write back to the system of record.** Every enrichment result enters the listing through a command, `ApplyEnrichment`, carrying the values, provenance and the listing version it was computed from. The listing service accepts it only if that version is still current, or if the attribute's inputs have not changed since. The search index receives enriched values through listing events like any other value. An enrichment written only into the index is the start of **Search as System of Record**: when the index is rebuilt, the value is gone, and nobody can audit or correct it.

2. **Each enrichment step is a pure function with an idempotency key.** The key is `(listing ID, input fingerprint, enricher ID, enricher version, reference-data version)`. The input fingerprint covers only the fields the enricher reads, so a description edit re-runs text extraction but not VIN decoding. Running the same step twice produces the same output and no new version.

3. **Enrichers declare dependencies.** The orchestrator runs them as a small directed graph: normalization before decoding, decoding before valuation, images before the price rating that reads them. When an input changes, it re-runs the affected subgraph, which is how the mobile.de rule "recalculate when images arrive" generalizes.

4. **Derived values never overwrite asserted ones silently.** The precedence policy in the dictionary decides what is displayed and used, but the listing keeps every source's value. When a seller later edits an attribute, the seller's value becomes the resolved value and the model's suggestion is retired, not deleted, so the disagreement remains available as a training label.

5. **Some steps block publication; most do not.**

Step	Blocking before publication?	Why	Degraded mode
Normalization of required attributes	**Yes**	Invalid canonical values break filters and invariants	Reject or hold with a clear error
Category validation or prediction for form listings	**Yes**, when the seller did not choose a leaf	The category selects schema, fees and moderation rules	Ask the seller to confirm
Registry decode for vehicles	Usually yes in automotive	Specification attributes drive filters and valuation	Publish with seller values, decode later, mark as pending
Geocoding	Yes when location precision drives display rules	Privacy rules depend on precision (see Where Is It? Location, Geocoding and Privacy)	Fall back to postcode or city precision
Attribute extraction	**No**	Suggestions improve the listing over minutes	Listing is published with stated values only
Translation, embeddings, quality score	**No**	Used by search and recommendations later	Read models update when results arrive

Step	Blocking before publication?	Why	Degraded mode
Market value and price rating	No	Display and ranking feature	Show no rating, the "no rating" state mobile.de also exposes

The blocking set must fit inside the publication budget owned by From Submitted to Searchable: The Publication Pipeline; Tradeyard's target there is 90 seconds at p95, and a synchronous call to a language model on the insertion path consumes a large share of it.

Versioned enrichers and re-processing

An enricher changes for three reasons: its code or model changes, its reference data changes (a decoder release, a new alias table, a boundary reform), or the schema it fills changes (a new attribute, a category split). Each change raises a question that most pipelines answer by accident: **which existing listings should be re-processed, in what order, and with what protection?**

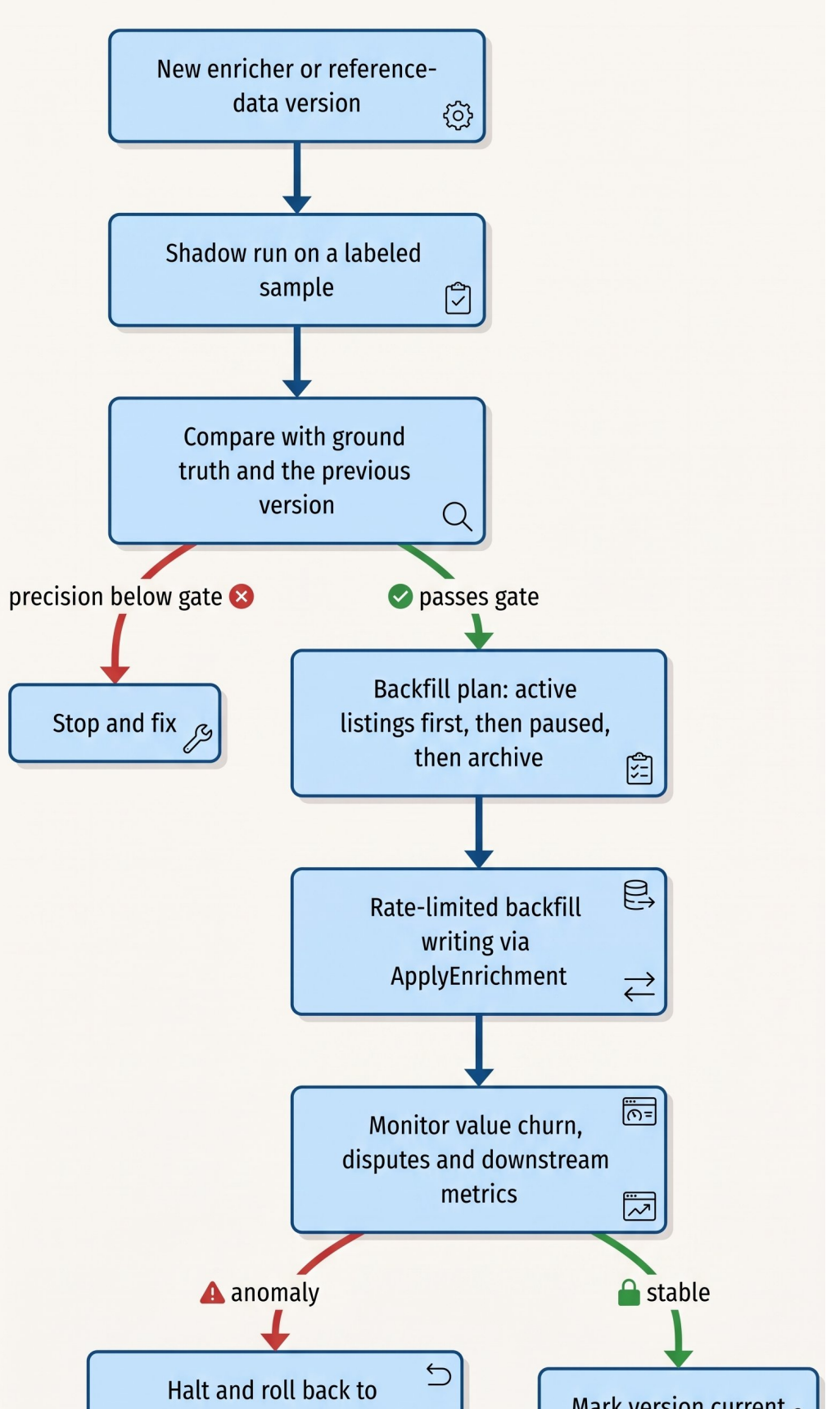

New enricher or reference-data version
Shadow run on a labeled sample
Compare with ground truth and the previous version
precision below gate
passes gate
Stop and fix
Backfill plan: active listings first, then paused, then archive
Rate-limited backfill writing via ApplyEnrichment
Monitor value churn, disputes and downstream metrics
anomaly
stable
Halt and roll back to
Mark version current

Figure 3: *Releasing an enricher like a schema migration. A new version is shadow-tested against labels, backfilled by priority through the same write path, and monitored for unexpected churn; because every value records the version that produced it, a bad release can be rolled back value by value.*

Etsy's reported practice of halting production runs when precision or recall drifts from a baseline is the public precedent for the gate. Two details make it work in practice. **Value churn is a metric**: if a new version changes 4% of values where the evaluation predicted 1%, something is different between the sample and production. And **rollback is a data operation**: because provenance records the enricher version per value, the pipeline can restore the previous version's values for every listing it touched, without touching values that sellers or moderators set in between.

Quality scores

A **listing quality score** summarizes how usable a listing is for discovery and trust. I would build it from four measurable dimensions, defined per category schema:

Dimension	Definition	Example signal
Completeness	Share of required and recommended attributes with a resolved value	23 of 26 recommended used-car attributes present
Validity	Values conform to type, range and enumeration	Mileage within the plausible range for the age
Consistency	Values agree with each other and with registries	Stated power matches the decoded derivative; price per m² plausible for the district
Confidence	Share of resolved values backed by assertion or high-confidence derivation	Seller-confirmed versus model-only values

The quality score is an input to seller nudges ("add the frame size to appear in 3 more filters"), to moderation sampling, and to ranking as one feature among many (see Ranking for Two Sides: Relevance, Freshness and Paid Visibility). It should not be confused with a relevance score, and it should not reward listings for values that a model guessed: a listing whose completeness comes entirely from unconfirmed suggestions is not as good as one the seller described.

Human correction loops

Every human decision about a value is a label, and the pipeline should collect them deliberately.

- **Seller decisions on suggestions**. Kleinanzeigen's photo-based assistant, as reported in 2025, lets sellers accept suggestions fully, partially or not at all (mobiFlip, 2025). Each choice is a labeled example, provided the interface records which suggestion was shown and what the seller did with it.
- **Seller edits after publication**. A seller who changes a model-filled value is correcting the model.
- **Moderator corrections**. A moderator who moves a listing to another category or removes a value sets `moderator` provenance; the correction is both an enforcement record (see Trust and Safety as a Platform) and a label.
- **Buyer reports**. "Wrong mileage" or "not an e-bike" reports, routed to the listing's quality state, are noisy labels worth sampling.
- **Professional disputes**. A dealer who disputes a decoded value through the pro tool is often right about their own car; the dispute should override the decode for that listing and feed a report to the decoder's supplier.

Label quality is the hidden risk. Sellers accept suggestions they did not read, and moderators under time pressure correct the most visible errors and leave the rest. A small, regularly refreshed **gold sample** labeled by trained reviewers is what calibrates everything else.

In Practice: The Essen Golf, Value by Value

The Golf's record arrived in Ruhrblick's 17:00 snapshot on 14 April 2027 and was created at 17:02 (see Many Doors In: Forms, APIs and Bulk Feeds). The table follows a selection of its values from the feed to the listing. Motorvane's decoder is a licensed vehicle-specification service; its supplier is not named here.

Attribute	Raw from the DMS feed	Canonical value	Source class	What the pipeline did
Make	“VW”	Volkswagen (make ID)	`partner_feed`	Alias mapping; confirmed by decode
Model and generation	“Golf VIII”	Golf, generation 8	`partner_feed`, `registry`	Alias mapping; generation confirmed by decode
Engine and trim	“1.5 TSI Life”	Engine 1.5 TSI; trim Life	`partner_feed`	Decode confirms engine family; trim not decodable, dealer value kept
Power	“96 kW”	96 kW	`partner_feed`	Decode returned two candidates, 96 and 110 kW; dealer value consistent with one, so resolved to the dealer’s
First registration	“EZ 03/2021”	2021-03	`partner_feed`	Date normalization
Mileage	“48.000 km”	48,000 km	`partner_feed`	German-locale parse; plausibility check for a 2021 car passed
Price	“21.900,- €”	2,190,000 minor units, EUR	`partner_feed`	Locale parse; VAT-deductible flag from a separate feed field

Attribute	Raw from the DMS feed	Canonical value	Source class	What the pipeline did
Exterior color	"Mondsteingrau Metallic"	Base color GREY; metallic; manufacturer name kept	`partner_feed`	Alias table plus the original string for display
Equipment	"Navi, ACC, Sitzhzg."	Navigation, adaptive cruise control, heated seats	`partner_feed`	Abbreviation dictionary for the vendor's export
Body type, doors, fuel, displacement, transmission and 21 more	absent	26 further specification values	`registry`	VIN decode; together with the values the decoder confirmed above, **31 attributes**
Location	Branch ID	Essen; coordinates, area IDs and grid cells from the branch record	`platform`	Professional seller, exact location allowed
Market value	—	**€22,800**; label "good price"	`model`	Valuation enricher after decode; re-run on 22 April when the price fell to €21,400

Before enrichment the listing met all 14 required attributes of the used-car schema and 9 of 26 recommended ones; afterwards it met 23 of 26. The search filters for body type, transmission and fuel, used by most Motorvane buyers, now include it; before decoding, a buyer filtering on "automatic" or "manual" would never have seen it.

The power value shows the failure that provenance prevents. The decoder could not tell the two power outputs of the 1.5 TSI apart from the VIN, and an earlier version of

Motorvane's pipeline had taken the first candidate. The valuation model then compared some cars with the wrong derivative, and dealers saw ratings that did not match their cars. With both values stored and a policy that the dealer's statement wins when it is consistent with the decoded family, the rating reads the right derivative, and when the supplier publishes a more precise decode, the pipeline can update the cars where no dealer value exists and leave the rest alone.

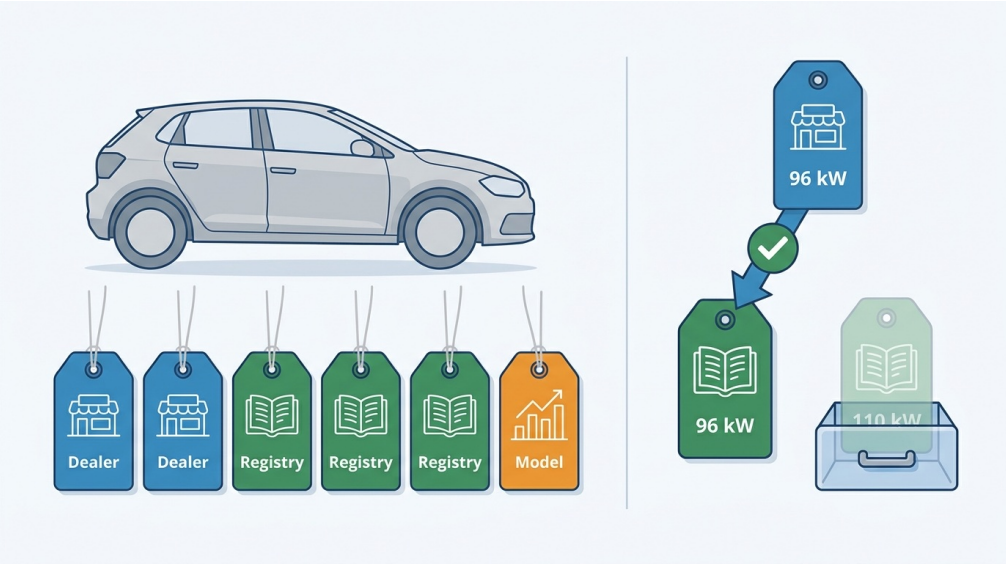


Figure 4: *Every value on the Golf carries its source; because both decoder candidates are stored, the dealer's 96 kW wins where it is consistent, and a later, more precise decode can update only cars with no dealer value.*

Tradeyard: extracting bike attributes before the split

Priya's team ran the bike-extraction pilot on Tradeyard Netherlands from June to September 2027, ahead of the 1 November bikes split. The design followed the evidence above:

- A fine-tuned small language model, self-hosted, receives the title, description and the **bike schema's attribute keys at request time**, the Mercari pattern, so the November schema change did not require retraining. A vision classifier reads the first three photos for bike type and electric assist.
- Outputs are validated against the compiled schema; values below a confidence threshold are discarded.
- For new listings, extracted values are shown in the insertion flow as **suggestions** for the seller to confirm. For live listings, values above a higher threshold are

applied with `model` provenance and shown to buyers; the seller is asked to confirm them at the next edit.
- A gold sample of 2,000 listings, labeled by reviewers, gates each release: the pilot's model reached 93% precision on frame size and 97% on electric assist.

Frame-size coverage on live bike listings rose from **23% to 71%**; sellers accepted **64%** of the suggestions shown to them in the insertion flow and edited another 11%. The values made the November split possible: the e-bike flag and bike type became inputs to the re-categorization job owned by Category Trees That Can Change: Designing Marketplace Taxonomies.

Hearthly: the Lyon apartment's disclosures

The **68 m² three-room flat in Lyon 7e**, **offered at €389,000 with energy class D**, arrived in May 2027 from three agencies through two software vendors and Keybridge. Enrichment did three things before One Property, Three Listings: Duplicates and Entity Resolution clustered the three listings into one property. It normalized the three addresses against the Base Adresse Nationale, which produced one address for all three despite two spellings of the street. It assigned area IDs and grid cells for the district. And it **verified**, **rather than derived**, the energy class: the DPE class is a legally required disclosure that the advertiser must state, so enrichment compares the stated class with the energy-certificate register where a certificate number is supplied, and flags a mismatch to the agency rather than replacing the value. Valuo, Hearthly's valuation model, estimated **€372,000–€401,000**; it trains on DVF transaction prices, not on asking prices (see Models in Production: ML Platforms and Use Cases).

The rule generalizes: **enrichment may fill attributes that serve discovery, but must not invent values that the law requires the advertiser to state**. Energy class, fees and other mandated disclosures are verified, flagged and reported back; they are never silently supplied by the platform (see Rules of the Marketplace: DSA, DAC7 and Vertical Regulation).

Tradeyard Belgium: translation as enrichment

Tradeyard Belgium serves Dutch- and French-speaking buyers on one marketplace. Machine translation of titles and descriptions is an enrichment like any other: the translation is stored with `model` provenance and the model version, indexed for cross-language search, shown to buyers with an "automatically translated" label and a link to the original, and re-generated when the seller edits the text. Attribute values need no translation, because they are dictionary IDs with localized labels; that is one more benefit of normalizing enumerations rather than storing words.

The Cost of Enrichment at Scale

At classifieds volume, calls times price per call is usually the smallest number on the enrichment bill. A rough Veldmark model, with all prices illustrative rather than any provider's quotes:

Cost driver	Volume	Rough cost	Comment
Text extraction on new and materially edited Tradeyard listings	2.3M new listings a week plus about 0.9M material edits	At about 2,000 tokens per call and an illustrative €0.40 per thousand calls for a small model, roughly €1,300 a week	Cheap per listing; cheaper still self-hosted
One backfill of all Tradeyard live listings	13.2M listings	Roughly €5,000 per pass in model cost, plus days of pipeline capacity	Every enricher version, schema change or category split can require one
Evaluation per release	Gold samples per category and country	Reviewer time: a 2,000-listing sample at about a minute per listing is about 33 hours	Recurs with every model and schema release
Human review of uncertain values	1% of 3.2M weekly runs at 30 seconds each	About 270 reviewer-hours a week, around seven full-time people	Usually the largest variable cost
Image enrichment	About 9.5 million new images a day group-wide	Part of the group's image inference spend of about €3.1M a year (see Models in Production: ML Platforms and Use Cases)	GPU-bound; cost scales with images, not listings

Three conclusions follow. **Per-call model cost is rarely the constraint** once small or self-hosted models are used; Mercari's USD 1 million estimate came from brute-

forcing a large API model over more than 3 billion items, and its answer was a cheaper architecture, not a smaller ambition. **Re-processing is the multiplier**: an organization that changes its schema monthly and backfills each time pays the backfill cost twelve times a year, which argues for prioritized backfills (active listings first) and for fingerprints that skip unaffected listings. **People are the main cost**: evaluation and gray-zone review scale with the number of models, categories and countries, so the governance model decides the budget more than the inference price.

Trade-offs

Decision	Option A	Option B	When A is right	When B is right
Where enrichment runs	**At write time**, results stored on the listing	**At read or index time**, computed on the fly	Values used by several consumers, audited or shown to sellers	Cheap, consumer-specific transformations, such as display formatting
Where derived values live	**Listing system of record**, with provenance	**Search index only**	Always for values that anyone may need to audit, correct or rebuild	Index-specific fields such as analyzers and n-grams, which are derivable from stored values
Model type	**Rules and dictionaries**	**Learned models** (small fine-tuned or large general)	Closed vocabularies, registry mappings, units	Free text and images; long-tail categories; use the schema as input

Decision	Option A	Option B	When A is right	When B is right
Application	**Suggest** to the seller	**Auto-apply** with `model` provenance	New listings, where the seller is present; values buyers rely on for decisions	Live stock where sellers are absent; high-precision values; always visible as derived
Timing	**Synchronous**, blocking publication	**Asynchronous**, after publication	Values that drive validation, privacy or fees	Everything else
Ownership across the group	**Shared enrichment services**	**Vertical enrichers**	Generic capabilities: geocoding, translation, image classification, a model gateway	Domain logic: VIN decoding, energy-certificate checks, valuation
Conflict policy	**Asserted value wins**	**Registry wins**	Condition, intent, equipment the seller added, legal disclosures	Identity and factory-specification facts, unless the seller's value is consistent with the registry family

Vertical Differences

Dimension	General classifieds	Real estate	Automotive	Jobs
Main normalization problem	Free text and casual input; many languages	Addresses, surfaces, fees and agency-software codes	Locale-specific numbers and dates from DMS exports; option codes	Titles, locations, salary formats
Strongest enrichment	Category prediction and attribute extraction from text and photos	Address normalization, geocoding, building and transaction data	Registry decoding to a licensed taxonomy	Mapping to occupation taxonomies, skills extraction, salary normalization
Authoritative external data	Rare (brand dictionaries, product catalogs for some goods)	Address bases, transaction registers, energy-certificate registers	VIN and type-approval decoders, licensed specification data	Occupation taxonomies
Values that must not be derived	Few beyond trader declarations	Energy class, fees and other mandatory disclosures	Emissions and consumption disclosures where required, vehicle history claims	Pay ranges where pay-transparency law requires the employer's figure
Biggest model risk	Wrong category hides the listing	Wrong unit match inside a building; biased valuations	Wrong derivative drives a wrong price rating	Wrong occupation code hides the job from matching
Seller presence for confirmation	High at insertion	Low: agency software sends and leaves	Low: DMS sends nightly	Medium

The last row, seller presence, decides whether suggestions can work. In general classifieds the seller is in the insertion flow, so suggestions work. In feed-driven verticals nobody is there to confirm a suggestion, so either the value is applied with derived provenance and shown as such, or it is reported back to the partner as a recommended improvement in the run report.

Failure Modes

Derived values indistinguishable from asserted ones. Yardstick's attribute columns could not tell a model's guess from a seller's statement. Once mixed, the two cannot be separated, and every later decision, including what to show, what to re-run and whose value to trust in a dispute, is made blind.

Re-running enrichment over seller corrections. A new decoder release or model version that writes every value it computes overwrites the corrections sellers and moderators made. Precedence rules and provenance per value prevent it; a backfill job that bypasses the listing service does not.

Enrichment only in the index. Category predictions, extracted attributes or quality scores written directly into search documents vanish at the next reindex, cannot be shown to sellers, and cannot be audited. This is Search as System of Record arriving through the side door.

Silent misparses. A thousands separator read as a decimal point turns 48,000 km into 48 km, and, without a plausibility rule, the car ranks as nearly new. Every numeric normalizer needs a plausibility check from the dictionary.

Models without gates. An extraction model released without a labeled sample, a precision threshold and churn monitoring writes its error rate into millions of listings. That is Set-and-Forget AI, and the remedy is the release process in Figure 3 plus a named owner.

Unknown values forced into known ones. Alias tables that map unrecognized feed values to the nearest known enumeration, rather than to an unknown state with a mapping queue, convert a mapping gap into wrong data.

Enrichment that changes legal disclosures. A platform that "corrects" an agency's stated energy class, or fills it from a model, publishes a disclosure the advertiser did not make. Verify and flag; do not replace.

Cost surprises from re-processing. A category split, a schema change and a model upgrade in one quarter, each followed by a full backfill, can exhaust a quarter's inference and review budget. Plan backfills like migrations, with priority and fingerprints.

The Chief Architect's View

1. **Make provenance a group standard before it is a feature**. Veldmark adopted one provenance vocabulary (the seven source classes above, plus process, version, confidence and time) in October 2027, alongside the group taxonomy and attribute service. Every brand's listing store must be able to hold it, and every new enricher must write it. It is cheaper to require this before the pipelines multiply than to retrofit it into four listing stores.

2. **Share generic enrichers, keep domain enrichers local**. The shared set at Veldmark is small: address normalization and geocoding, translation, image classification and text-in-image detection, and a model gateway with structured-output validation, all built on the group's ML platform. VIN decoding stays with Motorvane, energy-certificate checks and Valuo with Hearthly, because their reference data, licenses and failure modes are vertical. This is the same line as for ingestion: share capabilities whose inputs and outputs are generic, keep those whose meaning is domain-specific.

3. **Put every enricher under model governance, including rules**. Each enricher has an owner, a version, an evaluation set, a release gate and a dashboard of coverage, precision on the gold sample, value churn and dispute rate. Rule-based normalizers are included; an alias table changed by mistake damages as many listings as a model.

4. **Decide the precedence policy per attribute, in the dictionary**. Who wins between seller, feed, registry and model is a product and legal decision, not an engineering default. Put it in the attribute dictionary of Attributes Without Chaos: Schemas, Dictionaries and Validation, where stewards own it, and make the pipeline enforce it.

5. **Budget re-processing explicitly**. Each quarter's plan should list the expected backfills (schema changes, category migrations, model releases) and their costs in compute and review time. The bikes split was planned with its backfill; the decoder correction at Motorvane was not, and waited three months for capacity.

6. **Treat enrichment as a supply-side liquidity lever**. Completeness is findability: a listing missing the attribute a buyer filters on does not exist for that buyer. Measure enrichment by its effect on filtered-search coverage and contact rates, not by attributes filled (see Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics).

What to watch: coverage of filterable attributes per category and country; the share of resolved values that are model-only; gold-sample precision per enricher; value churn per release; seller rejection rates of suggestions; dealer and agency disputes of derived values; and the unknown-value queue for each mapping table.

Questions to Ask

1. For any attribute value on a listing, can we say who stated or derived it, by which process and version, and when?
2. Do normalizers keep the original value, unit and locale, and do numeric values pass plausibility checks?
3. Where do enrichment results live, and could we rebuild the search index without losing any of them?
4. What is the precedence policy between seller, feed, registry and model for each important attribute, and who owns it?
5. Which enrichment steps block publication, and do they fit the publication budget?
6. When an enricher or its reference data changes, how do we decide which listings to re-process, and how do we roll back a bad release?
7. Which legally required disclosures do we verify, and is there any path by which the platform supplies them instead of the advertiser?
8. Which human decisions become labels, and how is label quality checked?
9. What did the last backfill cost in compute and reviewer time, and how many are planned this quarter?
10. Which enrichers are shared across brands, which are local, and is that line drawn by domain meaning or by history?

From Clean Records to Duplicates

A normalized, enriched listing can now be compared with other listings, which is what makes the next problem visible: several of them describe the same flat or the same car. One Property, Three Listings: Duplicates and Entity Resolution takes the canonical addresses, decoded VINs and image hashes produced here and asks when several listings are one thing, and what each side of the market should see when they are.

Sources and Evidence

Company engineering and investor sources

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- Text in Image 2.0: improving OCR service with PaddleOCR — Adevinta, 2023. OCR volumes, accuracy, latency and cost versus an external API.
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- Autotrader developer portal — Autotrader UK, n.d. Vehicle taxonomy to derivative level; valuations keyed on derivative, registration and mileage.
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Standards, regulation and documentation

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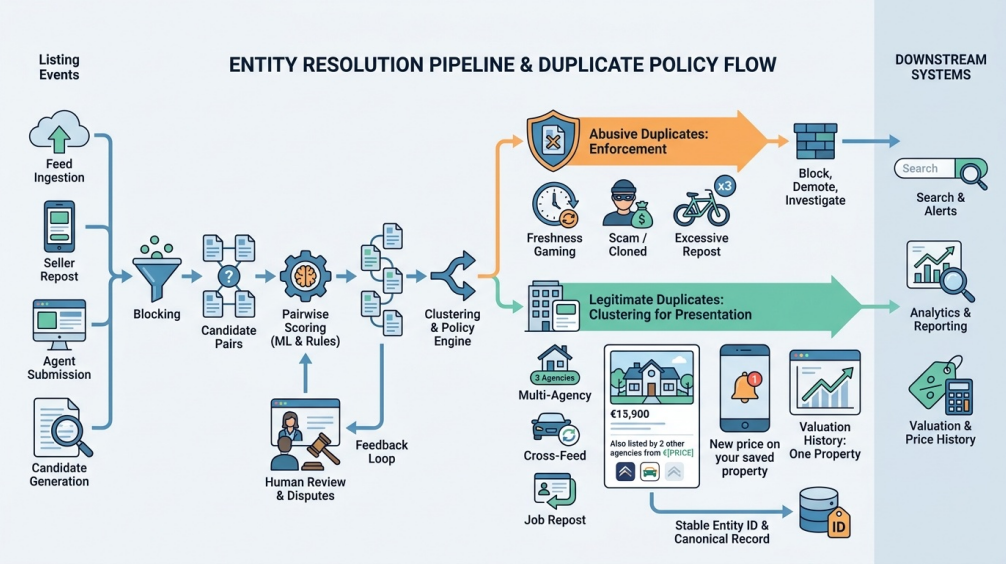
Research and practitioner sources

- libpostal — OpenVenues, n.d. Open-source address parsing and normalization; reported accuracy, language coverage and model size.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

15. One Property, Three Listings: Duplicates and Entity Resolution



IN THIS CHAPTER, YOU WILL: Learn the five kinds of duplicate listing, the real-world identities that listings can be resolved to, and the pipeline that turns candidate pairs into stable clusters and canonical entities. You will also see how merge policies decide what buyers, sellers and downstream systems see, using the Lyon apartment that three agencies listed on Hearthly.

WHY THE BUSINESS CARES: Duplicates make a marketplace look bigger than it is and work worse than it should. Buyers scroll past the same flat three times, agencies argue about whose card was shown, alerts fire twice, and the inventory numbers used to sell subscriptions overstate the real supply by about 70%.

WHY ARCHITECTS SHOULD CARE: Entity resolution creates an identifier that search, alerts, valuation, analytics, fraud detection and pro reporting will all depend on. If that identifier is unstable, lives only in the search index, or is produced by a model nobody evaluates, every one of those systems inherits the error.

KEY POINTS:

- **"Duplicate" names two problems with opposite answers.** Reposts, freshness gaming and stolen listings are abuse, and the action is enforcement. Several agencies listing the same flat, or one dealer feeding the same car through two channels, is legitimate, and the action is clustering for presentation. One detector can serve both, but the policies, owners and appeals must stay separate.
- **Resolve listings to an entity that outlives them.** A property or a vehicle has an identity that persists across listings, agencies and years. Model it as its own record with a stable identifier, owned by a named domain, and keep the listing as an offer that refers to it. Goods mostly have no such identity, so do not invent one.
- **The cluster identifier is a product contract, not a model output.** Blocking, pairwise scoring and clustering produce candidate links. A governed process turns them into stable cluster IDs with merge and split lineage, measured precision and recall, a correction path for sellers, and a policy for which listing represents the cluster. That policy must never be for sale silently.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Seeing Double** — reposts, multi-feed copies and multi-agency listings crowd result pages and alerts, and nobody owns the decision of what to show once.
- **Set-and-Forget AI** — a pairwise matcher merges and hides listings without labeled evaluation, drift monitoring or an owner who can explain a wrong merge to an agency.
- **Ghost Inventory** — a cluster keeps a property "on the market" after its last live listing has gone, because the cluster's state was never tied to the lifecycle of its members.

Part III follows supply from the seller to the result page. From Raw Supply to Usable Listings: Normalization and Enrichment left every listing with a canonical address, a decoded VIN where one exists, and image hashes, so listings can now be compared, and the comparison shows that many of them describe the same thing. This chapter's online checks run inside From Submitted to Searchable: The Publication Pipeline, and search, alerts and valuation read the cluster identifiers it creates.

The duplicates come in several kinds. A private seller reposts a bike every two days because new listings rank higher. A dealer's management system and a listing aggregator both send the same car. Three agencies hold mandates on one flat and each publishes it. A fraudster copies a real rental listing, photos and all, and prices it 35% lower. On a result page each looks the same: one thing appears more than once. **They**

are different problems, and treating them as one is the most common mistake in this area.

This chapter covers two capabilities. **Duplicate detection** finds listings that are copies or near-copies of each other. **Entity resolution** (ER), sometimes called record linkage, decides which listings refer to the same real-world thing and gives that thing an identity. A **canonical entity** is the resulting record: one property, one vehicle, sometimes one product, with a stable identifier and attributes reconciled from its listings. In real estate, the canonical entity is the **canonical property**. This chapter owns all three terms.

Neighboring chapters own adjacent parts. Many Doors In: Forms, APIs and Bulk Feeds owns the feeds that bring the copies in. From Raw Supply to Usable Listings: Normalization and Enrichment owns the address, VIN and attribute normalization that matching depends on. Photos Sell: Media Pipelines for Listings owns perceptual hashing as an image capability. Scams, Fakes and Account Takeovers: Fraud Prevention for Listings owns what happens once a duplicate turns out to be a scam. This chapter owns the matching, the clusters, the canonical entity and the policies that decide what each side of the market sees.

Setting the Scene at Veldmark: The Lyon Apartment, Three Times

In May 2027, Hearthly France receives three listings for a three-room (T3) **68 m²** flat in Lyon's 7th arrondissement, energy class (DPE, the French energy performance diagnostic) **D**. Hearthly is the real-estate portal of Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build). **Agence Rhône Habitat**, a network of **14 branches**, publishes it through Keybridge, the group's agent CRM. Two other agencies hold mandates too, and their transaction software sends the flat through two different real-estate software vendors. Rhône Habitat lists it at **€389,000**. The other agencies list it at €389,000 and €395,000, because French sale prices are usually shown fees included, and the agencies charge different fees.

Before 2027, a buyer searching "T3 Lyon 7e under €400,000" saw three cards with three photo sets, two prices and three agency names. A buyer who saved that search received three alerts. Hearthly's market report counted three listings. The price history behind Valuo, Hearthly's valuation product, showed three properties for one sale. Agencies complained twice: first that "their" flat was shown below a competitor's copy, and then that a competitor's lower-fee copy undercut them on the same page.

The flat is not unusual. Across Hearthly France, **1.9 million** live listings describe about **1.12 million** distinct properties, an average of **1.7** listings per property. Claire Fontaine, Hearthly's CEO, puts the problem to Ingrid Halvorsen, Veldmark's Group

Chief Architect, in two sentences: "Buyers think we have more flats than we do, and agencies think we pick favorites. Fix the first without making the second worse."

Julien Marchand, Hearthly's CTO, has a nightly deduplication job that already groups some listings by exact address. Ingrid asks four questions about it. *What is the thing being duplicated? Who decides that two listings are the same, and how often is that decision wrong? What does each audience see once they are grouped? And who can dispute a grouping?* The rest of this chapter answers those questions.

Five Kinds of Duplicate

The word "duplicate" covers five situations that differ in who creates them, why, and what the marketplace should do.

Kind	Typical example	Who creates it, and why	Typical signals	Right action	Owning context
Same-seller repost	The same bike posted three times in a week	A seller wanting more exposure, or one who forgot the first listing	Same account or device, near-identical text and photos, same category and area	Block the new one, or replace the old one and keep its history	Trust and safety; listing policy
Relisting to game freshness	A listing deleted and recreated every few days to reach the top of a recency sort	A seller substituting a free repost for a paid bump	Same as above, plus a delete-and-recreate pattern in time	Enforce: keep the original creation date, limit reposts, offer a paid bump instead	Trust and safety with monetization

Kind	Typical example	Who creates it, and why	Typical signals	Right action	Owning context
Cross-feed duplicate	One dealer's car arriving through its DMS (dealer management system) feed and an aggregator feed	An integration configuration: two channels for one seller	Same organization, same VIN or partner reference, different channel IDs	Merge silently at ingestion: one listing, two sources	Supply and ingestion
Multi-agency listing	The Lyon flat, listed by three agencies	Legitimate: several professionals hold mandates on one property	Same address or parcel, matching surface and rooms, different organizations, overlapping photos	Cluster for presentation: show the property once, list every agency	Listing and discovery
Stolen or cloned listing	A rental in Barcelona copied from a real listing, priced far below market	Fraud	Different seller, matching photos (perceptual hash), price anomaly, new account	Enforce: hold or remove, investigate the account network	Trust and safety (Scams, Fakes and Account Takeovers: Fraud Prevention for Listings)

Two more cases look like duplicates but are not. **The same entity with a different offer**, such as one flat listed both for sale and for rent, or a car listed again by a new

owner two years later, is a new listing on the same entity. Price history and valuation need the link, but search should not hide either listing. **Similar but distinct entities**, such as two identical flats on different floors of a new building or two white 2021 Golfs at one dealer, are the main source of wrong merges.

The chapter's first rule follows: **detection can be shared**; **decisions cannot**. The same blocking and similarity machinery finds a same-seller repost, a multi-agency cluster and a stolen copy. What separates them is the relationship between the sellers and the evidence of intent. The design note in Veldmark's research puts it plainly: *same seller, same photos* is spam or ranking abuse; *different seller, same photos* is a fraud signal or a legitimate multi-agency listing, and seller, device and organization keys are what tell them apart. The OLX duplicate pipeline described publicly in 2019, discussed below, used exactly those keys to generate candidates.

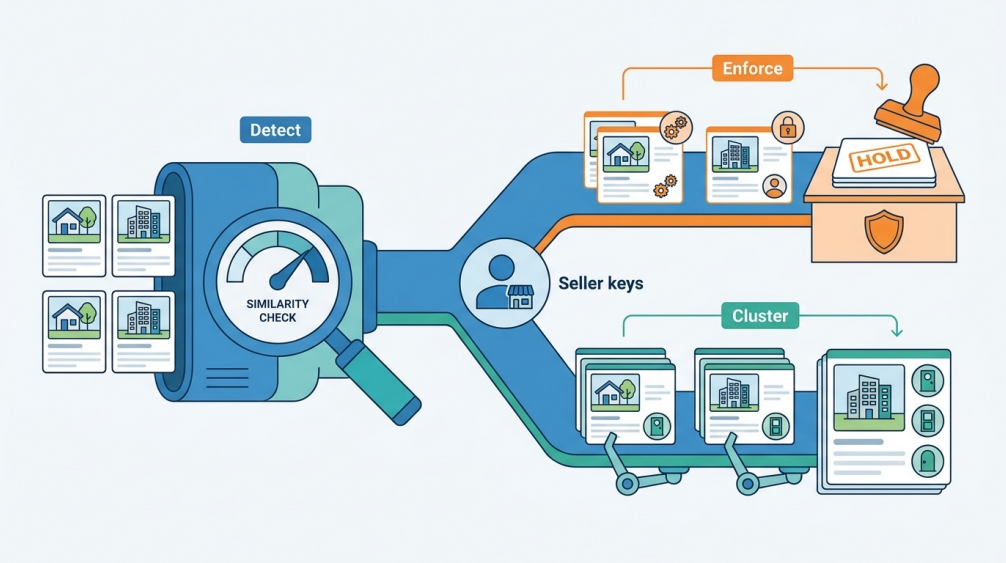


Figure 1: *The same detection machinery finds reposts, stolen copies and multi-agency listings; the relationship between the sellers decides whether the result is enforcement or presentation.*

The two families of action also sit under different legal regimes. Removing or demoting a listing because it is an abusive duplicate is a content-moderation restriction. Under the EU Digital Services Act (DSA), the affected seller is owed a statement of reasons, and demotion counts as a restriction (Art. 17; see Rules of the Marketplace: DSA, DAC7 and Vertical Regulation). Grouping three agencies' listings under one property card is a presentation choice. It has no DSA consequences, but the rules for choosing the card a buyer sees first are a ranking parameter, and those rules have transparency obligations (see Ranking for Two Sides: Relevance, Freshness and Paid Visibility). An architecture that runs both through one "dedup" flag cannot meet either set of obligations.

What Is the Thing? Real-World Identity by Vertical

Entity resolution needs an answer to a prior question: *same what?* The answer differs by vertical, and it decides how far resolution can go.

Vertical	Real-world entity	Identifiers available	How reliable	What ER can achieve
Real estate	Property (dwelling unit), inside a building, on a parcel	National address bases, building registries, cadastral parcels; in Great Britain the Unique Property Reference Number (UPRN); in the US county assessor parcel numbers	Good for the building, weak for the unit inside it; listings often omit or blur the address	A durable **canonical property** with price and listing history
Automotive	Vehicle	The 17-character vehicle identification number (VIN), plus registration plates in some countries	Strong when present; dealers often send it, private sellers often do not	Exact merging of cross-feed duplicates; vehicle history across owners
General goods (used, unique)	The physical item	None	No identifier; the listing *is* the item	Near-duplicate detection only, mostly for abuse

Vertical	Real-world entity	Identifiers available	How reliable	What ER can achieve
Catalog goods (new, recom-merce)	The product model, not the item	Global Trade Item Number (GTIN) or EAN barcodes, manufac-turer part numbers	Good for products; says nothing about the individual item	Product matching for attributes and price comparison, not deduplication
Jobs	The vacancy	Employer reference numbers, sometimes	Weak; the same vacancy is reposted by agencies and job boards	Clustering of reposted vacancies

Real estate is where the entity matters most, because **the property outlives every listing of it**. The same flat is listed for sale, sold, let, relisted by a different agency, and valued off-market. Public reference data exists, but each country's is different. In Great Britain, Ordnance Survey's open UPRN product publishes coordinates for about 40 million "addressable locations", each with "a unique numeric identifier for every addressable location in Great Britain" (Ordnance Survey, n.d.). France has the Base Adresse Nationale, an open national address base updated twice a week (adresse.data.gouv.fr, n.d.), and the Référentiel National des Bâtiments (RNB), which gives each building "un identifiant unique et pérenne", the ID-RNB, designed as a pivot for linking building data between administrations and private parties (RNB, n.d.). In the US, county assessor parcel numbers vary by county, and data aggregators sell normalized parcel IDs. None of these identifies a **unit inside a building**. In a Lyon apartment block, the address and the ID-RNB narrow the candidates to one building; floor, surface, room count and photos must do the rest.

Zillow is the documented example at scale of a property-first model. Its annual report for fiscal 2025 describes a "living database" of about 173 million US homes, with user-generated updates to more than 46 million property records, which powers its Zestimate valuation and its indices (Zillow Group 10-K, filed 2026-02-11). The filing describes the database, not how listings are matched to it, and the internal matching method is not public.

Vehicles are easier when the VIN is there. A VIN identifies one vehicle across feeds, dealers and owners, so two listings with the same VIN at the same time are a cross-feed duplicate or a dealer group listing the car at two sites, and one listing with a VIN seen two years earlier is the same car with a new seller. The work is in the listings without a

VIN, and in protecting the VIN itself, which some markets treat as personal data when linked to an owner. Enrichment from the VIN is owned by From Raw Supply to Usable Listings: Normalization and Enrichment.

General goods have no entity to resolve to. **Do not invent one**. A marketplace that tries to attach every used item to a product catalog meets the problem eBay met when it mandated catalog association for listings in 2018–2019 and later relaxed it (see Attributes Without Chaos: Schemas, Dictionaries and Validation). For goods, duplicate detection protects the result page from abuse; it does not build an entity.

The Matching Pipeline

The reference architecture below is an inference from public algorithms and the few public marketplace designs, not a description of any company's system. Its stages are standard in the record-linkage literature; what is specific to listings is the mix of signals and the speed at which clusters change.

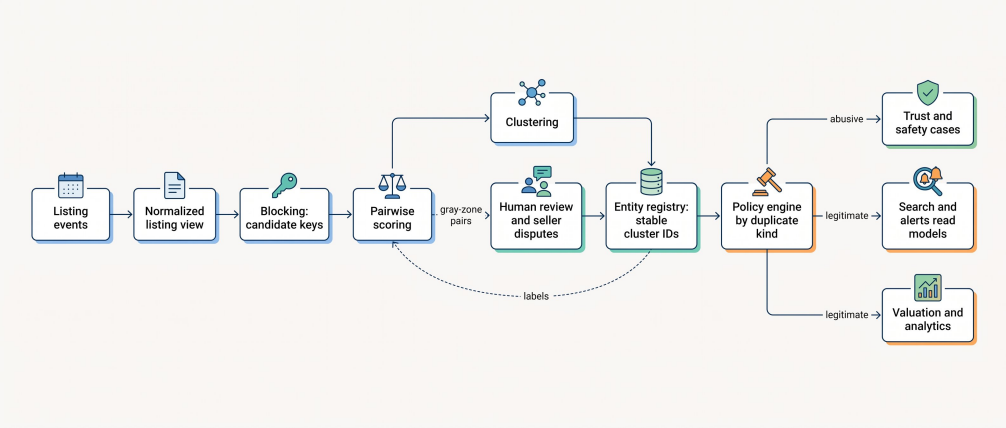


Figure 2: *The matching pipeline. Blocking, scoring and clustering propose links; the entity registry owns the identifiers; a policy engine turns one cluster into different actions for different duplicate kinds; and human decisions feed back into both the registry and the model.*

Blocking: finding candidates cheaply

Comparing every listing with every other listing is impossible: Hearthly France alone has 1.9 million live listings, which is about 1.8 trillion pairs. **Blocking** generates a small set of candidate pairs using cheap keys with high recall, so that the expensive scoring runs only on plausible pairs. A listing usually gets several blocking keys, and any shared key makes a pair a candidate:

- **Category and place**: property type and an H3 hexagonal cell at fine resolution (H3 is Uber's open hierarchical geo index; see Where Is It? Location, Geocoding and Privacy), plus a rounded price band.
- **Normalized address** where one exists: the address normalized by the enrichment stage, and the building ID once the address is linked to it.
- **Hard identifiers**: VIN, partner listing reference, the agency's own mandate number when a feed carries it.
- **Text fingerprints**: MinHash locality-sensitive hashing over shingles of title and description. MinHash estimates the Jaccard similarity of two sets; it goes back to Andrei Broder's 1997 work on document resemblance, first used for duplicate web pages. SimHash, published by Manku, Jain and Das Sarma at Google in 2007 (WWW 2007), does the same job with a single fingerprint compared by Hamming distance.
- **Image fingerprints**: bands of a perceptual hash for each photo. Meta's PDQ, released as open source on 1 August 2019 (Meta, 2019), is a common choice; its documentation suggests starting with a match threshold of Hamming distance 31 or less and discarding low-quality hashes, then tuning on your own data (PDQ documentation, n.d.). Photos Sell: Media Pipelines for Listings computes these hashes for every image.
- **Account keys**: seller, organization, device and network identifiers, which separate same-seller reposts from cross-seller matches.

Blocking is a recall decision. A pair that shares no key is never compared, so a missing key is a silent miss. The usual failure is a key that seems safe but splits true matches: blocking on exact address misses the agency that writes "rue de Marseille" without a number, and blocking on the H3 cell misses listings near a cell boundary unless neighboring cells are included.

Pairwise scoring: is this pair the same thing?

Each candidate pair gets a score from field-level comparisons: text similarity, image distance, geographic distance, surface and room agreement, price gap, same organization, same device, time between postings. Two families of scorer are in common use.

Probabilistic record linkage in the Fellegi–Sunter tradition weighs how much each field's agreement or disagreement says about a match, and can estimate those weights without labeled data. Splink, an open-source library from the UK Ministry of Justice, implements it; its documentation says "no training data is required", that it links "a million records on a laptop in around a minute", and that Spark suits 100 million records or more (Splink, n.d.). It is a good first scorer because its weights are explainable to a reviewer.

Supervised pairwise classifiers, often gradient-boosted trees, learn from pairs that people have labeled as same or different. The best public classifieds-specific design is

still OLX's, presented by Alexey Grigorev at PyCon DE in 2019: candidate generation by category, city, seller, IP and device, and image hashes, then a classifier trained on human-labeled pairs using price difference, location distance, title similarity and image similarity. For images it layered MD5, then dHash, then MobileNet embeddings reduced with PCA and indexed with random-projection locality-sensitive hashing in Elasticsearch, fed by an S3 upload event, AWS Lambda and a queue, with a target of about 10 million listings a day. The talk also noted that cropping, rotating and flipping break the hashes (PyCon DE, 2019). The design is seven years old and OLX's current system is not public, but the shape has aged well.

Scores must be **calibrated** so that thresholds mean something, and the score bands map to actions: above an upper threshold, link automatically; below a lower one, do nothing; between them, send the pair to human review. The gray zone is where the money is spent, so its width is a budget decision.

Clustering: from pairs to groups

Pairs are not clusters. If listing A matches B and B matches C, the obvious step is to put A, B and C together, which is **connected components** clustering. It is simple and it fails in a predictable way: **chaining**. One weak link between two otherwise separate groups merges them, and in dense housing a chain of plausible links can merge a whole building. Safer methods require a cluster to be internally consistent: every member must match most other members (correlation clustering, or a minimum average link score), and a cluster above a size limit is split or sent to review. In real estate, a cluster with more than a handful of members is almost always wrong, because a property rarely has more than four or five agencies at once.

The entity registry: stable identifiers and lineage

The entity registry keeps cluster identifiers stable. Clustering produces new groups each time it runs, but consumers need **identifiers that stay the same** when it reruns, so that a buyer's saved property, a price-history chart and a valuation series do not jump to a new ID every night. The registry assigns each cluster a durable ID, keeps the ID when membership changes slightly, and records **merge and split events** with lineage when two clusters become one or one becomes two. A merge keeps the older ID and retires the other with a pointer; a split keeps the ID on the larger part. Consumers subscribe to these events like any other domain event (see Every Change Is an Event: Event-Driven Listing Architecture).

The registry is also where the **canonical entity** lives: the canonical property's address, building, unit features and identifiers, with per-attribute provenance. Survivorship rules decide which value wins when members disagree. Surface might come from the energy diagnostic where present, else the most common value among professional listings; the address comes from the most precise geocode. The listing keeps the seller's own values.

The canonical property does not overwrite the listing, because the listing is the seller's statement and carries legal weight in some markets. Valuation, price history and analytics read from the canonical record; the listing page shows what the seller published.

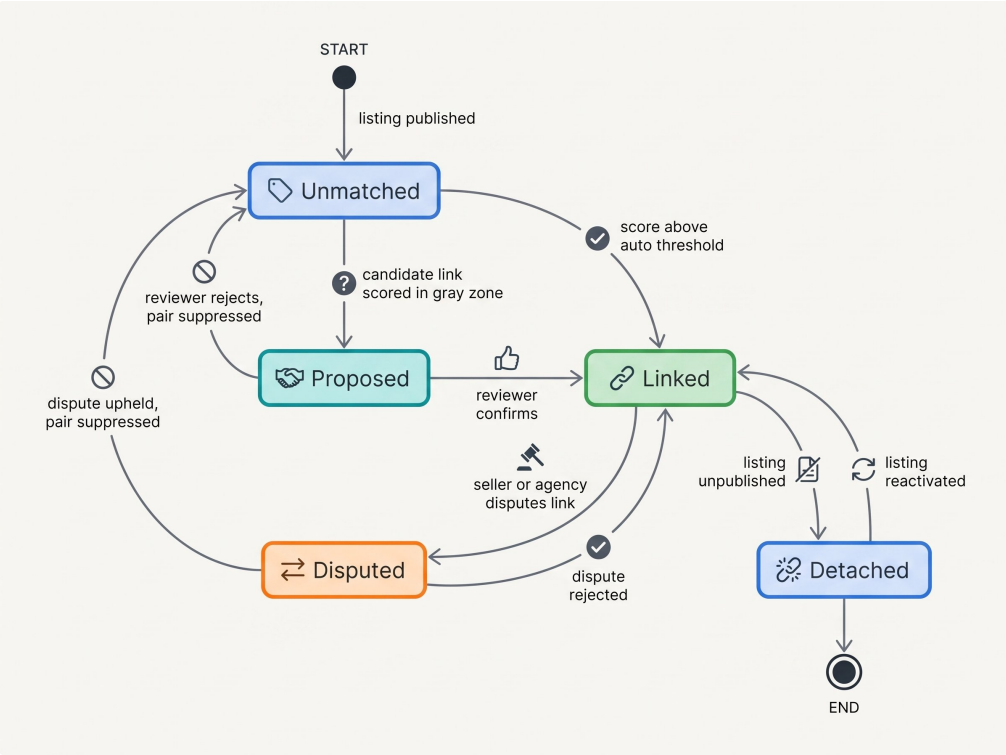


Figure 3: *A listing's membership in a cluster has its own lifecycle. Human decisions are durable: a rejected or disputed pair is suppressed so that the next model run cannot re-create it, and unpublished listings detach so the cluster's state follows its live members.*

A human "not the same" decision must be stored as a constraint (Figure 3), not just as a label for the next training run, or the nightly job will reunite the pair and the agency that complained will complain again.

Membership also follows the listing lifecycle (see The Listing Aggregate: Identity, Lifecycle and Invariants). When the last live listing in a cluster is unpublished, the cluster is no longer "on the market", even though the property entity persists. A cluster that stays active after its members have gone is Ghost Inventory in a new form: an alert subscriber or a market report sees a property for sale that nobody is selling.

Online and batch

Resolution runs at two speeds. **Online** checks at publication time are cheap and local: exact and perceptual image hashes against recent listings, same-seller text near-duplicates, VIN and partner-reference lookups. They catch reposts and cloned listings before they go live, and they attach new listings to existing clusters when a strong key matches. **Batch** resolution, typically nightly, runs global blocking and clustering across all listings, finds cross-seller and cross-feed clusters, and repairs clusters as listings change. The online path must finish inside the publication budget set in From Submitted to Searchable: The Publication Pipeline; anything that does not fit runs after publication and adjusts the presentation a few minutes later.

Merge Policies: What Each Audience Sees

A cluster changes something different for each audience, and choosing what it changes is product policy expressed in architecture. The most common mistake is to apply one policy, usually "hide the others", everywhere.

Audience or system	Policy options	Recommended default for legitimate clusters
Buyer, search results	Show all; collapse to one card with "also listed by"; hide the others	Collapse to one property card, with every agency reachable from it
Buyer, listing page	Show only this listing; show "also listed by" with other agencies and prices	Show the other listings, including prices, with clear agency names
Buyer, saved-search alerts	Alert per listing; alert per property	One alert per property; a new agency on a known property is an update, not a new match
Buyer, price-drop alerts	Per listing; per property	Per property, on the lowest live asking price, with the agency named

Audience or system	Policy options	Recommended default for legitimate clusters
Agency or dealer	Not shown; shown own clusters only	Show each agency the clusters its own listings belong to, never other agencies' private data
Valuation and price history	Per listing; per property	Per property, using one asking-price series per listing to avoid triple counting
Inventory and market metrics	Count listings; count properties	Report both; use properties for "supply" and listings for "advertising volume"
Lead routing	To the card shown	To the agency the buyer chose; never re-route a lead to another agency

Collapsing in search

The usual implementation stores the cluster ID on each listing's search document and asks the engine to return one document per cluster. Elasticsearch calls this **field collapsing**: the `collapse` parameter keeps only the top-sorted document for each value of a single-valued keyword or numeric field, and optional `inner_hits` return the other members. Its documentation also lists the costs an architect should know before relying on it: the total hit count still counts all matching documents, not the number of groups; aggregations are not collapsed; each `inner_hits` request adds a query per collapsed hit; and collapsing cannot be combined with scroll (Elastic, n.d.). So the result list shows 1,000 properties while the facet counts and the "2,400 results" header still count listings. **Either the counts are computed on a property-level document, or the page shows numbers that disagree with the list**. A common answer is a second, property-level index for counting and map pins, built from the same events; Search Architecture for Listings: Indices, Freshness and Scale covers the index design.

Choosing the representative

When a cluster collapses to one card, one listing's photos, price and agency appear first. This is the most politically sensitive decision in the chapter. The options are the most complete listing, the earliest published, the lowest price, the highest quality

score, rotation among agencies, or the listing that pays for visibility. Each one rewards a different behavior. "Lowest price" rewards the agency with the lowest fee. "Earliest" rewards the agency that publishes first. "Quality score" rewards good photos and complete attributes, which is what buyers want. "Paid" silently turns deduplication into an auction.

My rule is that **the representative rule is a published ranking parameter**, decided by product and legal together, and never changed by an engineer tuning a model. If paid placement affects it, the card says so. The EU Platform-to-Business Regulation and the DSA both require the main ranking parameters to be described to business users and consumers (Rules of the Marketplace: DSA, DAC7 and Vertical Regulation). A representative rule is one of them. **Multiplier Creep often starts in a cluster rule that nobody thinks of as ranking**.

A documented example: Idealista's duplicate control

The clearest public product example comes from Idealista, the Spanish, Italian and Portuguese property portal. In April 2021 it launched a duplicate-control tool for home seekers that groups the ads for one property published by several agencies or individuals, *including on other portals*, so that users can compare them in one place. Users see the number of duplicates, price changes, removed ads and new duplicates, and get push notifications when a price changes, an ad is removed or a new duplicate appears. Agents see duplicates only of their own properties, and only once a listing has been published. The tool is updated daily (Online Marketplaces, 2021-04-11). The report describes the product, not the matching system, and the current state of the tool is not public. Three design choices are worth noting: the cluster is presented as **buyer information** rather than hidden, the **agent view is limited to their own properties**, and the cadence is daily, which suggests batch resolution.

In Practice: Resolving the Lyon Apartment

Here is how the Lyon flat moves through the pipeline Hearthly built in 2027. The design is Veldmark's, and the numbers are illustrative.

The three listings. The listings arrive over four days in May 2027:

	Listing A	Listing B	Listing C
Agency	Agence Rhône Habitat (branch Lyon Jean-Macé)	A second agency, independent	A third agency, part of a franchise
Channel	Keybridge	Software vendor 1	Software vendor 2
Asking price, fees included	€389,000	€389,000	€395,000
Surface and rooms	68 m², T3	68 m², 3 rooms	67.8 m², 3 rooms, 2 bedrooms
Address in feed	Full street and number	Street only	Street and number, different spelling
DPE	D	D	D
Photos	14	11	9, of which 6 match listing A's photos (PDQ distance below 20)

Blocking. All three share the building: listings A and C geocode to the same address, and the address links to one building ID. Listing B has no street number, but it shares the H3 cell, the property type and the price band, and five of its photos fall into the same perceptual-hash bands as listing A's. Blocking produces three candidate pairs and, in the same building and cell, 41 pairs with other listings, mostly other flats in the same block.

Scoring. Pair A–C scores 0.97: same address, same DPE class, surface within 0.5%, six shared photos, different organizations. Pair A–B scores 0.88: no street number, but matching surface, rooms and DPE, five shared photos taken from the same viewpoints. Pair B–C scores 0.84. The 41 other pairs score below 0.3. Hearthly's thresholds put automatic linking at 0.92 and the gray zone between 0.70 and 0.92.

Clustering and review. A–C links automatically. A–B and B–C go to the review queue. A reviewer sees the shared photos and the same balcony, confirms both within the hour, and the three listings become one cluster with one canonical property ID. The canonical record takes its address from listing A, its surface from the DPE certificate data, and its building ID from the address link.

Presentation. Search shows one card. The representative rule Hearthly published in 2027 picks the listing with the highest completeness and media-quality score, here listing A, and shows "also listed by 2 other agencies, from €389,000". All three agencies are one

click away, and a lead goes to whichever agency the buyer contacts. The price-drop alert subscribes to the cluster, so when listing C drops to €389,000 it is reported as "now €389,000 at a third agency", not as a new listing. Valuo values the property once, with its estimate of **€372,000–€401,000**, and the property's price history records three asking-price series against one property instead of three properties.

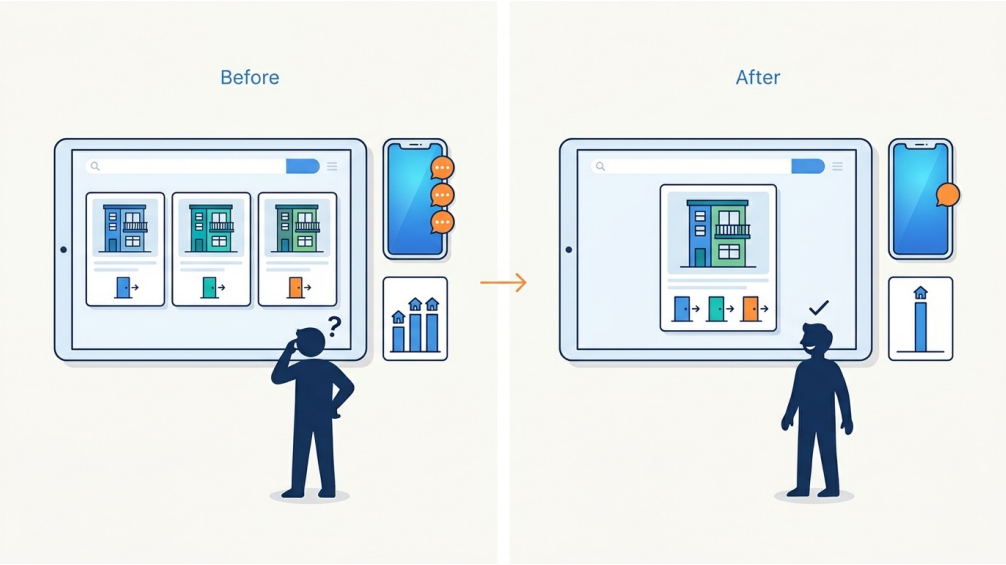


Figure 4: *Resolved, the Lyon flat is one property card with every agency one click away, one alert per property and one property in Valuo's price history instead of three.*

The Keybridge constraint. Veldmark decided in March 2027 that Hearthly may not use Keybridge agency data for ranking or sales targeting (see Meet Veldmark: Four Brands, One Platform to Build). Keybridge holds far more about the Lyon flat than Hearthly does, including the mandate and the seller's identity. Entity resolution at Hearthly may use **only what each listing carries as published to Hearthly**, exactly as it would for a listing from any other software vendor. That rule is enforced in the data path: the matcher reads Hearthly's listing read model, never Keybridge's database, and the access-control review checks it every year. Without it, a better match rate would cost Veldmark the trust of every agency that uses Keybridge to publish to Hearthly's competitors.

What changed. After the first quarter of the new pipeline, Ingrid's review uses four numbers, all illustrative. Search pages in Lyon show 31% fewer cards for the same queries. Saved-search alerts for Lyon sales fall by 24% with no fall in click-through, which means the removed alerts were noise. Precision measured on 4,000 labeled pairs is 98.6% and recall 91%. Agency disputes run at about 60 a month across France, two thirds of them for flats in the same building that the matcher wrongly merged. Those disputes are the most valuable labels the team has.

The same machinery pointed at abuse

The same detectors run on Tradeyard, Veldmark's general classifieds brand, for a different purpose. A same-seller repost of a listing already live in the same category is blocked at publication, with a message that offers a paid bump instead. A delete-and-recreate within days keeps the original creation timestamp, so the new listing gains no freshness advantage (see The Listing Aggregate: Identity, Lifecycle and Invariants on why bumps and renewals update a ranking timestamp, never the creation time). This is a monetization and trust decision as much as a technical one: **a marketplace that sells bumps and tolerates free reposts is competing with itself**.

The registry must record why listings were linked, because a stolen copy must never be shown to buyers as "also listed by". At Hearthly Spain, the same kind of perceptual-hash matching flagged the Barcelona fake-rental ring in July 2027. Twenty-six accounts posted 180 fake rental listings in Eixample using photos stolen from real listings and priced about 35% below market. The match between a new, unrelated account and photos already attached to another agency's live listing, combined with the price anomaly, is the fraud signal. The hash matches fired from the first day, but they went to a different queue from the price flags, and eleven photo-match items were closed as legitimate multi-agency duplicates before an analyst linked the cluster. The policy engine sends such clusters to trust and safety, not to the presentation layer. The case and its cost are in Scams, Fakes and Account Takeovers: Fraud Prevention for Listings.

Trade-offs

Decision	Option	When it is right	What it costs
Where duplicates are resolved	At ingestion, before publication	Abuse checks, strong keys (VIN, partner reference)	Latency in the publication path; must be cheap and local
	Batch, after publication	Cross-seller clusters, global repair	Duplicates visible for hours
	At query time only (collapse on a similarity key)	Prototypes	No stable entity; counts and alerts stay wrong

Decision	Option	When it is right	What it costs
Scorer	Probabilistic linkage (Fellegi–Sunter, Splink)	No labels yet; explainability needed	Weaker on images and free text
	Supervised classifier	Labeled pairs available; mixed signals	Needs a labeling program and monitoring
	Embedding similarity	Augmented images, multilingual text	Hard to explain; expensive at scale
Clustering	Connected components	Small, sparse data; strong keys	Chaining into giant clusters
	Consistency-constrained clustering with size limits	Dense housing, many weak signals	More complex; needs tuning
Presentation	Hide duplicates	Abusive duplicates only	Agencies lose visibility; disputes
	Collapse with "also listed by"	Legitimate multi-agency clusters	Count and facet inconsistencies to solve
	Show all with a badge	Early stage, low confidence	Buyers still see the echo
Entity ownership	Shared group registry	Several brands in one vertical and country	Coordination and legal review across brands
	Per-brand registry	Brands in different verticals	Duplicated machinery

The trade-off that decides most of the others is **where the cluster ID is stored**. If it exists only as a field on search documents, the index becomes the only place that knows which listings belong together, and nobody can rebuild it from anything else. That is **Search as System of Record** in miniature. The registry is the system of record for clusters; search, alerts and analytics hold copies.

Vertical Differences

Dimension	General classifieds	Real estate	Automotive	Jobs
Dominant duplicate kind	Same-seller reposts, freshness gaming, cloned scam listings	Multi-agency listings, cross-feed copies	Cross-feed copies, dealer groups listing at several sites	Reposts by recruiters and job boards
Resolvable entity	None for unique goods; product for catalog goods	Property, building, parcel	Vehicle by VIN	Vacancy, weakly
Strongest signals	Account, device, image hashes, text	Address and building ID, surface, rooms, energy class, photos	VIN, partner reference, make, model, mileage	Employer, title, location, text
Main action	Enforcement	Clustering for presentation	Silent merge at ingestion	Clustering, enforce-ment for spam
Typical scale of clusters	Many small groups	Two to five members	Two members	Two to dozens
Specific risk	Stock and manufac-turer photos flooding the detector	Over-merging units in one building	Missing VINs on private listings	Merging distinct vacancies with the same title
Who disputes	Rarely anyone	Agencies, strongly	Dealers, occasionally	Employers and agencies

In general classifieds, a perceptual-hash detector without an allowlist of manufacturer images floods moderators with false matches, because **legitimate stock photos** are common in some categories, such as new phones or furniture.

In automotive, the VIN makes resolution easy, which tempts teams to skip the gray zone entirely; the cost shows up when a dealer group lists one car at two sites with different prices and the merge picks the wrong one as representative.

Failure Modes

1. **One "duplicate" flag for everything**. Abusive and legitimate duplicates share a boolean, so agencies' legitimate listings get demoted like spam, or scam copies get shown as "also listed by". Keep the duplicate kind on every link.
2. **Giant clusters**. Connected components chains weak links until a 40-flat building becomes one "property". Cap cluster size, require internal consistency, and alert when a cluster grows past a threshold.
3. **Unstable cluster IDs**. The nightly job renumbers clusters, so favorites, alert subscriptions and price histories break or jump between properties. The registry must keep IDs stable and publish merges and splits as events.
4. **Human decisions forgotten**. A reviewer separates two flats; the next batch run merges them again. Store reviewer and dispute outcomes as constraints the clustering must respect.
5. **Counts that disagree with the page**. Collapsed results with listing-level facet counts tell buyers "2,400 results" above 1,000 cards. Decide what a result is and count that.
6. **The representative rule as a hidden auction**. An engineer adds paid status to the representative score. Agencies notice within a week. Treat the rule as a published ranking parameter with an owner.
7. **Clusters that outlive their listings**. The last live member is sold and unpublished, but the cluster still drives alerts and market reports. Tie the cluster's market state to the lifecycle of its members.
8. **The unmeasured matcher**. Thresholds were set once, the model was never re-evaluated, and a new software vendor's address format halves recall without anyone noticing. Keep a labeled evaluation set, refreshed every quarter, and monitor match rates per source.
9. **Using data you are not allowed to use**. Matching across brands, or using a CRM's private data, links records the group has no legal basis or business permission to link. Make the allowed inputs part of the matcher's contract and review them.
10. **Keeping hashes forever**. Perceptual hashes of removed listings are valuable against fraud and are also personal data in some contexts. Retain them for a documented purpose and period, owned with trust and safety.

The Chief Architect's View

At group scale, entity resolution raises five decisions.

1. Share the machinery, not necessarily the entities. Blocking, scoring, clustering, the registry pattern, review tooling and evaluation are the same problem in every brand, and Common Ground should build them once. Entities are vertical: Hearthly's canonical property, Motorvane's vehicle and Tradeyard's abuse clusters have different identifiers, policies and owners. The platform provides the engine; each vertical domain owns its entity model and its policies.

2. Name the owner of each entity. The canonical property belongs to Hearthly's listing data domain, because valuation, search and pro reporting depend on it. Abuse clusters belong to the trust-and-safety platform under Aisha Rahman, because their actions are enforcement with appeals. Cross-feed merges belong to ingestion. Each owner sets thresholds, answers disputes and signs off on model changes. **A matcher without an owner is Set-and-Forget AI, even if it is very accurate.**

3. Decide the legal boundary of linking. Linking listings across brands, such as a Tradeyard Portugal flat and a Hearthly listing, or using Keybridge data at Hearthly, is a data-protection and competition question before it is a technical one. Lucía Ortega, the Group Data Protection Officer, decides with the brand owners which sources may be linked, and the decision is written into the matcher's input contract.

4. Govern the representative rule as ranking. Product, legal and the search platform decide it together, publish it, and change it through the same experiment and review process as any ranking change (see Ranking for Two Sides: Relevance, Freshness and Paid Visibility and Testing a Two-Sided Market: Experimentation Platforms).

5. Watch a small set of numbers. Listings per entity by market (1.7 at Hearthly France is the baseline); precision and recall on a labeled sample; gray-zone volume and review cost; cluster churn per night; dispute rate and overturn rate by agency; and the share of live listings with a resolved entity. A sudden change in any of them is usually a feed or enrichment change upstream, which is why the dashboard belongs next to the feed-health dashboard of Many Doors In: Forms, APIs and Bulk Feeds.

Questions to Ask

1. For each duplicate kind we see, what is the action, who owns it, and does the seller get a statement of reasons when the action is a restriction?
2. What real-world entity are we resolving to in this vertical, and which identifiers do we actually receive, at what fill rate?
3. Where is the cluster ID stored as the system of record, and can we rebuild every downstream copy from it?
4. Are cluster IDs stable across runs, and are merges and splits published as events with lineage?
5. How do reviewer decisions and seller disputes constrain the next clustering run?
6. What are precision and recall on a labeled sample from the last quarter, per market and per source?

7. What stops a chain of weak links from merging a whole building or a dealer's whole stock?
8. What is the published rule for the representative listing, and does payment affect it?
9. Do result counts, facets, map pins and alerts count the same thing as the result list?
10. Which data sources is the matcher allowed to read, who approved that list, and how is it enforced?

From Clean Supply to Live Supply

Entity resolution turns many copies into one view of what is really for sale. It works only if it fits inside the path every listing takes from submission to visibility, with cheap checks before publication and the heavier clustering after. That path, its stages, its latency budget and the way removals propagate to every copy, is the subject of From Submitted to Searchable: The Publication Pipeline.

Sources and Evidence

Company engineering and investor sources

- Fighting fraud: finding duplicates at scale — Alexey Grigorev (OLX), PyCon DE, 2019. Candidate generation, pairwise classifier and layered image hashing for duplicate listings; the hashing limits.
- Zillow Group Form 10-K for fiscal 2025 — Zillow Group, filed 2026. The "living database" of about 173 million homes and 46 million user-updated property records.
- Idealista launches new tool to help organize duplicate listings — Online Marketplaces, 2021. Idealista's duplicate-control tool: cross-portal grouping, notifications, agent visibility rules, daily updates.

Standards, regulation and documentation

- OS Open UPRN — Ordnance Survey, n.d. Unique identifiers and coordinates for about 40 million addressable locations in Great Britain.
- Base Adresse Nationale — French government, n.d. National open address base and its update cadence.
- Référentiel National des Bâtiments — RNB team, n.d. The ID-RNB as a unique, durable building identifier and open data.

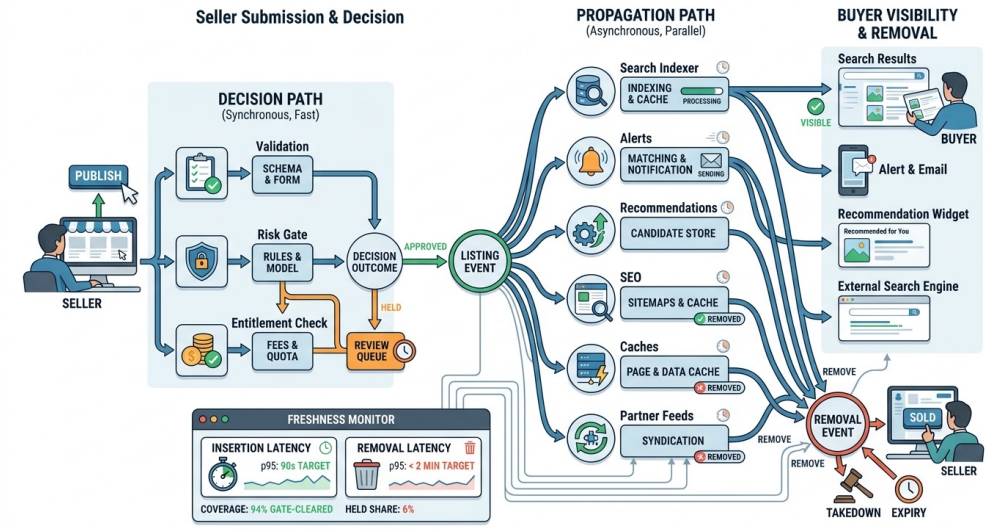
- Collapse search results — Elastic, n.d. Field collapsing semantics and limits: total hits, aggregations, inner hits, scroll.
- PDQ and TMK+PDQF open-source release — Meta, 2019. Release of the PDQ perceptual hash.
- PDQ documentation — Meta ThreatExchange, n.d. Quality and Hamming-distance starting thresholds.
- Regulation (EU) 2022/2065 (Digital Services Act), Art. 17 — European Union, 2022. Statements of reasons for restrictions, including demotion; detailed in Rules of the Marketplace: DSA, DAC7 and Vertical Regulation.

Research and practitioner sources

- Detecting near-duplicates for web crawling — Manku, Jain and Das Sarma, Google, WWW 2007. SimHash fingerprints for near-duplicate detection.
- A. Broder, "On the resemblance and containment of documents" — 1997. Origin of MinHash for estimating document similarity.
- Splink — UK Ministry of Justice, n.d. Open-source Fellegi–Sunter record linkage without training data; scale guidance.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

16. From Submitted to Searchable: The Publication Pipeline



IN THIS CHAPTER, YOU WILL: Follow a listing through every stage between "publish" and "findable": validation, enrichment, the moderation gate, entitlement checks, indexing, cache invalidation and alerts. You will learn which stages belong in the synchronous path, how to set and measure a freshness SLA per vertical, how ordering and idempotency keep removed listings from coming back, and how to make unpublishing reach every copy. The example is Tradeyard's move from a 6-minute p95 to a 90-second target.

WHY THE BUSINESS CARES: Speed to first contact is a liquidity lever. A private seller who checks for her listing and cannot find it posts it again or leaves, and a buyer alerted to a bike that sold yesterday learns to ignore alerts. Publication latency and removal latency both show up in contact rates, support tickets and seller retention.

WHY ARCHITECTS SHOULD CARE: The publication pipeline crosses more bounded contexts than any other flow in a marketplace: listings, media,

trust and safety, monetization, search, alerts, SEO and analytics. Without one owner and one end-to-end measure, each team optimizes its own stage and nobody can say why a listing took six minutes or why a sold one is still visible.

KEY POINTS:

- **Publishing is a decision followed by a propagation.** Keep the synchronous path to what is needed to decide whether a listing may go live: validation, the risk gate and the entitlement check, within about two seconds. Everything else (indexing, alerts, SEO, recommendations) is asynchronous propagation driven by one listing event, with a latency budget per stage.
- **A freshness SLA is a contract with three parts.** It says which listings it covers, which clock it measures (platform latency from receipt to visible, or source latency from the real-world change), and what happens to the listings that fall outside it, such as the ones held for human review. A single p95 that hides those choices is not an SLA.
- **Unpublishing is harder than publishing and matters more.** A removal must reach every copy: search, alerts and digests, recommendation candidates, caches, sitemaps, ads and partner feeds. Version-gated events, tombstones, send-time re-checks and a per-consumer propagation monitor are what make removal reliable. Ghost Inventory lives in the copies nobody registered.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Ghost Inventory** — sold and removed listings survive in digests, candidate stores, caches and partner feeds because removal was designed for the search index only.
- **Moderation as an Afterthought** — a moderation queue added serially in front of the indexer makes every listing wait for the slowest check, instead of tiering checks by risk.
- **Search as System of Record** — "published" is defined as "present in Solr", so nobody can tell whether a listing missing from search was never approved, failed to index or was removed.

The publication pipeline is the path that turns an accepted listing into visible supply, and back again. It is the last link in Part III's supply chain: Many Doors In: Forms, APIs and Bulk Feeds, From Raw Supply to Usable Listings: Normalization and Enrichment and One Property, Three Listings: Duplicates and Entity Resolution bring a record in, clean it and match it, but none of that reaches a buyer until the listing appears in search, an alert, a search-engine result or a recommendation. On the way the listing

passes through a dozen systems owned by half a dozen teams. Each adds latency, each can fail, and each holds a copy that must later be removed. Part I's time to first contact sets the publication budget, and its ghost rates set the removal guarantee; the search, alerts and recommendations of Part IV — DISCOVERY: Matching Intent to Inventory are only as fresh as this pipeline makes them.

This chapter owns three terms. The **publication pipeline** is the sequence of stages, synchronous and asynchronous, between a seller's (or feed's) submission and the listing's visibility on every surface, and the reverse sequence for removal. A **freshness SLA** (service-level agreement) is the target, per vertical and per channel, for how quickly a change becomes visible and how quickly an ended listing disappears. **Unpublish propagation** is the mechanism that carries the end of a listing, whether sold, expired, withdrawn or taken down, to every system that holds a copy, with evidence that each one did.

Neighboring chapters own the parts. The Listing Aggregate: Identity, Lifecycle and Invariants owns the lifecycle states and the events each transition emits. Trust and Safety as a Platform owns the moderation gate inside this pipeline and the enforcement decisions that trigger takedowns. Every Change Is an Event: Event-Driven Listing Architecture owns the event contract, the outbox and the per-event-family objectives. Search Architecture for Listings: Indices, Freshness and Scale owns indexing and its share of the budget. Tell Me When It Appears: Saved Searches and Alerts owns matching and fan-out. This chapter owns the whole path, the budget that ties the stages together and the measurement that proves it works.

Setting the Scene at Veldmark: Where Did Six Minutes Go?

On Sunday **7 March 2027** at **19:42 CET**, Femke, a private seller in Rotterdam, publishes a used city bike for **€240** on Tradeyard Netherlands. Tradeyard is the general classifieds brand of Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build). Automated moderation clears the listing at **19:44**. It becomes searchable at **19:48**. It matches **312** saved searches, the first buyer writes at **19:55**, and fourteen conversations follow within 24 hours. The bike sells on Monday for **€225** through Tradeyard Safe Deal. Tradeyard marks it sold at **18:10** and removes it from search at **18:11**. On Tuesday at 07:00, it appears in **214** alert digests, which were assembled at 17:30 on Monday from a database replica snapshot and never re-checked.

Ingrid Halvorsen, Veldmark's Group Chief Architect, puts the timeline on a slide in her first month. Six minutes from submit to searchable is Tradeyard's **p95** (the time within which 95% of listings become searchable). Common Ground, Veldmark's €140 million platform program, has set a target of **90 seconds** p95 by the end of 2028. Daan

Visser, who owns Yardstick, Tradeyard's PHP monolith, explains where the time goes: a moderation consumer that pulls listings from RabbitMQ in batches, a batch indexer that collects changes and pushes them to Solr on a timer, and Solr commits that make new documents visible on another timer.

Ingrid's first question is not how to make each stage faster. It is *which listings the six minutes describes*. The answer changes the problem. Tradeyard's dashboard measured only listings that the automated gate cleared. **94%** of new listings are cleared that way; the other **6%**, about **138,000 a week**, go to human review and become visible after minutes or hours. Nobody had a number for them. Her second question is about the other direction: *how long does it take for a sold listing to disappear from everything, not just from search?* Nobody had that number either.

Concepts: Stages, Clocks and Moderation Modes

Eight stages from submission to full visibility

A listing passes through eight stages between submission and full visibility. The order below is logical; the architecture section decides which run in sequence and which in parallel.

Stage	What it decides or produces	Typical owner	Sync or async
1. Validation	Is the listing well-formed for its category schema: required attributes, value ranges, price, location?	Listing context, using the attribute service (Attributes Without Chaos: Schemas, Dictionaries and Validation)	Synchronous
2. Media readiness	Are the required images uploaded, scanned and resized, and are the derivatives available?	Media pipeline (Photos Sell: Media Pipelines for Listings)	Usually started before submission; checked synchronously

Stage	What it decides or produces	Typical owner	Sync or async
3. Risk gate	Allow, hold for review, or reject, based on rules, block lists and a light model	Trust and safety	Synchronous, a few hundred milliseconds
4. Entitlement and pricing check	May this seller publish here: listing slots, insertion fees, quota?	Monetization (Packages, Bumps and Quotas: Products, Pricing and Entitlements)	Synchronous
5. Enrichment	Category prediction, attribute extraction, geocoding precision, duplicate checks	Supply and data (From Raw Supply to Usable Listings: Normalization and Enrichment, One Property, Three Listings: Duplicates and Entity Resolution)	Mostly asynchronous
6. Asynchronous risk checks	Image models, duplicate and stolen-photo search, price anomaly, account graph	Trust and safety	Asynchronous; may gate visibility for high-risk tiers
7. Indexing and cache invalidation	Search document built, index refreshed, cached result pages and counts updated	Search platform	Asynchronous
8. Notification and distribution	Saved-search matching, alerts, sitemaps, recommendations, partner syndication	Discovery, SEO, partnerships	Asynchronous

The first four stages answer one question: **may this listing go live?** The last four answer another: **where must it now appear?** Mixing the two is the root of most slow pipelines. A pipeline that indexes inside the request makes the seller wait for search; a pipeline that sends the listing to a moderation queue *after* the indexer has seen it shows unmoderated content.

Two clocks

A freshness SLA must say which clock it measures.

- **Platform latency** runs from the moment the marketplace receives a change (a form submission, an API call, a feed record) to the moment the change is visible. The marketplace controls it fully.
- **Source latency** runs from the moment something changes in the real world (a car is sold at the dealership, a flat is let) to the moment the marketplace receives it. The marketplace controls it only indirectly, through feed contracts, seller prompts and reconciliation.

For private sellers who use the marketplace's own app, the two are nearly the same. For dealers and agencies they are not. When Motorvane's dealer Autohaus Ruhrblick sold the Essen Volkswagen Golf on Thursday 29 April 2027, the removal arrived through the next day's feed snapshot, **26 hours** after the sale (see Many Doors In: Forms, APIs and Bulk Feeds). Motorvane's own platform latency on that removal was ten minutes. Both numbers are true, and only the second is the publication pipeline's to fix. The first is a feed-contract and reconciliation problem. **An SLA that does not separate the clocks blames the wrong team**.

Pre-moderation and post-moderation

Pre-moderation checks a listing before it is visible; **post-moderation** publishes first and checks afterwards. Large classifieds platforms commonly run both at once, choosing per listing by category risk, seller trust and the gate's risk score. Trust and Safety as a Platform owns the moderation spectrum and its legal context. Here the point is structural: **the moderation mode decides which stages sit on the critical path to visibility**. A verified dealer's car can go live on the gate's verdict, with image and duplicate checks running afterwards. A new account's first rental listing in a city with a known fake-rental problem should wait for those checks. The pipeline must support both paths with the same components, and the SLA must report them separately.

The pipeline's actual mode must match what the marketplace tells users. In December 2015 a Paris court found leboncoin guilty of misleading commercial practices because it claimed every ad passed a validation process when not all were reviewed before going

live, while confirming that, as a hosting provider, it had no duty to check every ad in advance (Clubic, 2015).

A material edit is a new submission: changes to price, payment terms, contact details, photos or category must re-enter the gate, not bypass it on the way to the indexer. In a December 2024 undercover test by the UK consumer organization Which?, Gumtree blocked a first scam-style ad, but edited versions passed review "in minutes" (Which?, 2024).

Architecture: A Decision Path and a Propagation Path

The reference architecture below is Veldmark's target design and an inference from public building blocks; no classifieds operator has published its complete publication pipeline.

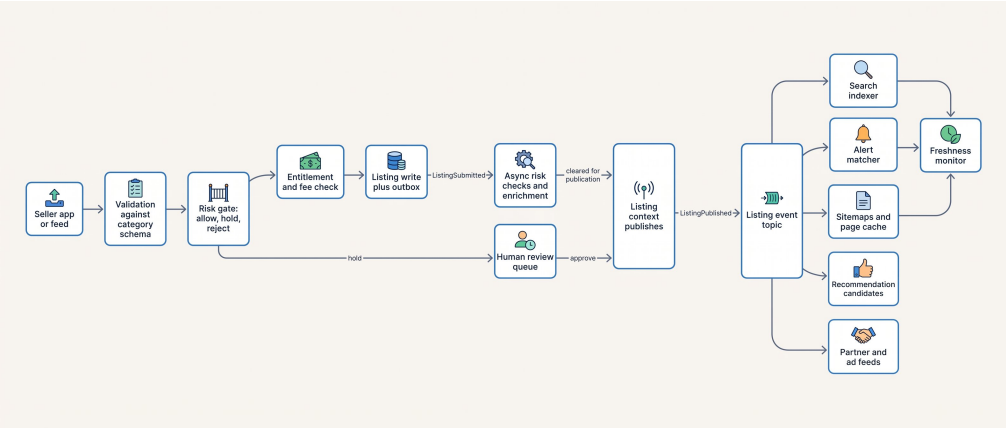


Figure 1: *The decision path (left) ends when the listing context records the decision and publishes one event; the propagation path (right) fans that event out to every consumer, each of which reports to a freshness monitor. The human-review path rejoins at the same publication step, so a held listing propagates exactly like an automatically cleared one.*

The synchronous path: decide, and stop

The synchronous path runs while the seller waits. Its job is to give a definite answer: *published*, *published shortly*, *needs your input* or *rejected, here is why*. It should do four things and nothing else.

1. **Validate** the listing against the versioned category schema. Blocking errors return to the seller with field-level messages; warnings are recorded as quality signals.

2. **Confirm media readiness**. The best way to keep image processing off the critical path is to start it during drafting: photos upload, are scanned, stripped of location metadata and resized while the seller is still writing the description. At submission the pipeline checks that the required derivatives exist.
3. **Run the risk gate**. Rules, block-list lookups and a light model return allow, hold or reject within a few hundred milliseconds.
4. **Check entitlements**. Does a professional seller have a free listing slot? Is an insertion fee due in this category and country, as for vehicles and property at Tradeyard Poland? A missing entitlement is a decision outcome ("buy a slot to publish"), not a later failure.

Then the listing context writes the listing and its event in one transaction, using the transactional outbox (see Every Change Is an Event: Event-Driven Listing Architecture), and returns. **Everything after the write is propagation**. The seller sees "your listing is being published" with a promise the platform can keep, such as "usually within two minutes".

Indexing, alert matching, sitemap updates, recommendation refreshes and partner syndication must never be called from the request. Each of them can be slow or down, and none of them decides whether the listing may exist.

The asynchronous path: one event, many consumers

The asynchronous path begins with the listing event, and it should fan that event out rather than pass the listing from stage to stage.

In the **sequential design**, each stage calls the next: the moderation worker calls the enrichment worker, which calls the indexer, which calls the alert service. Latency is the sum of all stages, a slow stage blocks everything behind it, and adding a consumer means changing the stage before it. Yardstick's pipeline was a version of this, with timers instead of calls.

In the **fan-out design**, the listing context publishes `ListingPublished` once, and every consumer subscribes. Indexing, alerts, SEO and recommendations run in parallel. Latency to each surface is the decision time plus that consumer's own processing. A new consumer subscribes without touching anyone else.

Asynchronous checks that may still change the decision, such as image models, duplicate search and price-anomaly scoring, sit between `ListingSubmitted` and `Listing-Published` for the risk tiers that require them, and *after* `ListingPublished` for the tiers that go live on the gate. In the second case they can still demote, hold or remove a live listing, through the enforcement path of Trust and Safety as a Platform.

Ordering, idempotency and the zombie listing

Fan-out is only safe if every consumer applies events correctly when they arrive twice, late or out of order. Three rules from Every Change Is an Event: Event-Driven Listing Architecture apply directly.

- **Per-listing order.** Key events by listing ID so that one listing's events stay in order on one partition.
- **Version-gated apply.** Every event carries the listing's aggregate version; a consumer applies an event only if its version is higher than the one it holds, and ignores it otherwise. Vinted's 2021 indexing pipeline used the Kafka offset as the Elasticsearch document version with external versioning for the same reason, and documented that increasing topic partitions broke the ordering assumption (Vinted Engineering, 2021).
- **Tombstones, not deletions.** When a listing ends, consumers record the end with its version rather than simply deleting their copy. Otherwise a late `ListingUpdated` for an older version finds no record, treats the listing as new, and recreates it.

These rules prevent the **zombie listing**, as many teams call it. A reindex job or a replayed backlog writes a listing's old "active" state after its "sold" state has been applied, and the sold listing comes back to life in search. It is the most common way that a backfill creates Ghost Inventory, and it is invisible unless something compares every copy with the system of record.

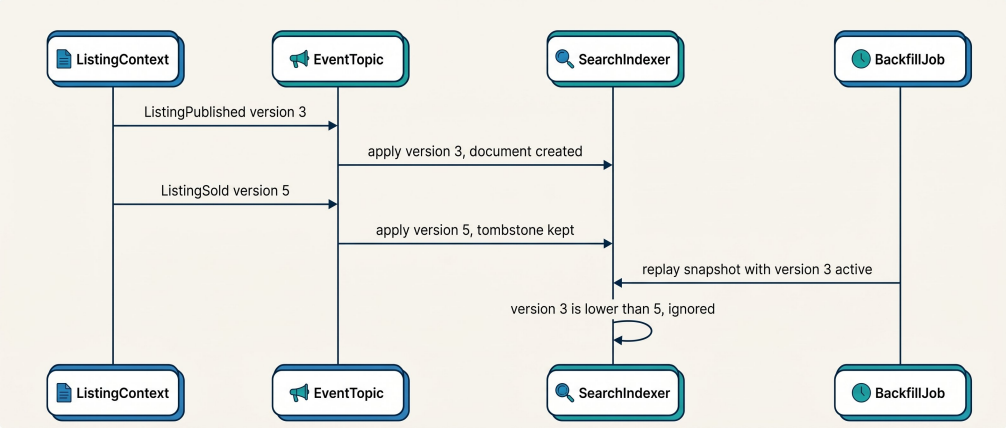


Figure 2: *Why consumers keep tombstones and gate on versions. Without the tombstone, the backfill's version 3 would find no document and recreate the sold listing; with it, the stale write is ignored.*

Expiry and renewal

The **lifecycle scheduler**, which ends the listings no seller ends, is part of the pipeline and has its own failure modes. Listings expire when their lifetime elapses, and professional listings expire when a feed stops confirming them.

- **Batch expiry at midnight** creates a removal storm: hundreds of thousands of `ListingExpired` events at once, competing with live traffic for the indexer and the alert matcher. Spread expiry times across the day, or rate-limit the scheduler's output.
- **A scheduler that falls behind** after an outage leaves listings live past their expiry, and nothing alerts on a transition that did not happen. Monitor the backlog of overdue transitions as a first-class metric.
- **Renewal is not republication**. A renewal or a paid bump moves a ranking timestamp; it must not reset the creation date, re-trigger "new listing" alerts or re-enter the gate unless content changed. mobile.de's Seller API, for example, keeps a separate `renewalDate`, "the date the ad was last marked as new, which makes it appear new in search", next to the creation and modification dates (mobile.de Seller-API, n.d.).

Downstream channels enforce expiry whether the marketplace does or not. Google Merchant Center expires products 30 days after their last update unless they are refreshed (Google, n.d.), and Google's job-posting guidance warns that failing to remove expired jobs "may result in a manual action" (Google Search Central, updated 2026). Lifecycle rules and their durations belong to The Listing Aggregate: Identity, Lifecycle and Invariants; the pipeline's job is to execute them on time and propagate their results.

Unpublish Propagation

Removal is the mirror image of publication, but it touches more systems and its failures cost more. It touches more systems because a listing that has been live for a week has been copied into places a new listing has not yet reached: digests, recommendation candidate stores, partner feeds, ad audiences, cached pages. A new listing that becomes visible two minutes late costs a little liquidity. A fraudulent listing that stays visible after a takedown sends buyers to a scammer under the marketplace's own brand.

Every copy, registered

The first requirement is a **consumer registry**: a list of every system that holds a copy of listing data, the events it consumes, its removal objective and its owner. Veldmark's March 2027 audit found that about **41,000** Tradeyard NL listings that had ended in

Yardstick were still present in at least one downstream copy 24 hours later, mostly in the alert-digest cache and the recommendation candidate store, neither of which listened to Yardstick's removal hook (see The Listing Aggregate: Identity, Lifecycle and Invariants). Neither consumer was in any list. **You cannot propagate a removal to a copy you do not know exists.**

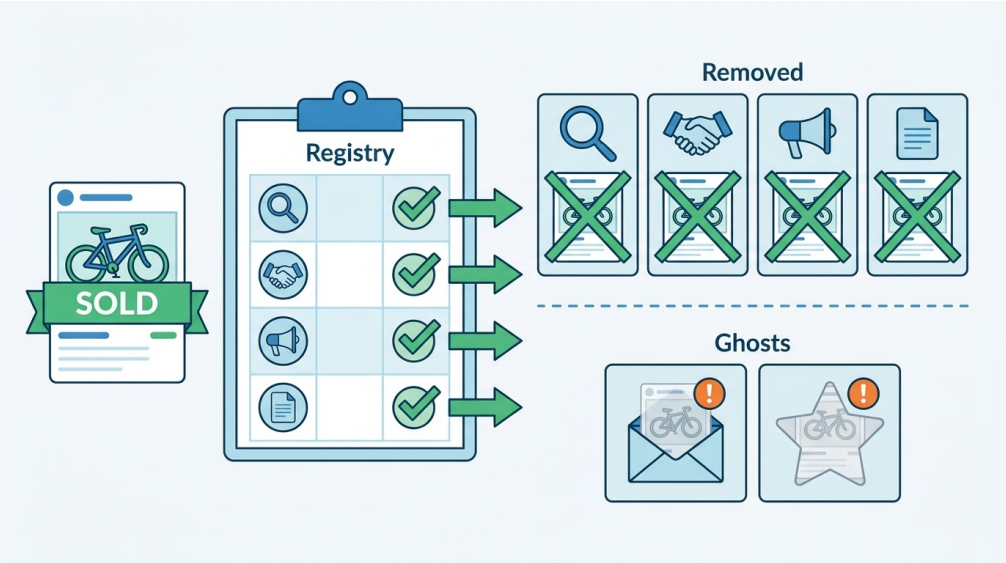


Figure 3: *Removal reaches only the copies the registry knows about; in Veldmark's March 2027 audit, about 41,000 ended Tradeyard listings lingered, mostly in the alert-digest cache and recommendation store that no list recorded.*

Consumer	What it holds	Removal action	Removal objective (Veldmark target)
Search index	Listing document	Delete or hide	p95 10 seconds, p99 30 seconds
Result-page and count caches	Cached pages, facet counts	Invalidate affected keys, or rely on a short time to live (TTL)	Within the cache TTL, at most 20 seconds

Consumer	What it holds	Removal action	Removal objective (Veldmark target)
Alert matcher and digests	Pending and assembled notifications	Drop queued alerts; purge from assembled digests; re-check at send time	p95 2 minutes, and never sent after removal
Recommendation candidates	Candidate stores, similar-listing lists	Evict	p95 2 minutes
Listing page and SEO	Rendered page, sitemap entry	Serve the "sold" or "gone" page per policy; drop from sitemap	p95 2 minutes; sitemap next generation
Messaging	Conversations about the listing	Mark listing ended; warn on takedown	p95 2 minutes
Advertising	Sponsored placements, retargeting	Stop serving and billing	p95 2 minutes
Partner and syndication feeds	Copies at partners and comparison sites	Send deletion in the next delta	Within 15 minutes
Analytics	Event history	Record the end; keep the evidence	Daily

The objectives in the table are the targets Veldmark set in 2027; the search and event figures follow Search Architecture for Listings: Indices, Freshness and Scale and Every Change Is an Event: Event-Driven Listing Architecture. How each consumer renders an ended listing, for example a "sold" page with related listings versus a 410 response for a fraud takedown, is decided by Found by Google: Web, App and SEO Architecture for Listings from the reason the listing ended. The pipeline's job is to carry that reason.

Re-check at send time

Consumers that assemble work in advance must re-check each listing's current state immediately before sending, through the listing read API or a compacted state topic. A digest built at 17:30 and sent at 07:00, a recommendation batch computed

overnight, a marketing e-mail scheduled for the morning: each holds a list of listings that may have ended in between. Subscribing to removal events and purging assembled lists is necessary but not enough, because a purge can fail or arrive late. Femke's bike was in 214 Tuesday digests because the digest service did neither. The re-check costs one batched lookup per digest; the alternative costs the trust of 214 buyers.

Takedowns ride the same rails, faster

A takedown for fraud or policy violation is an enforcement decision owned by trust and safety, but its propagation should not be a separate mechanism. The decision service emits its decision, the listing context applies it and emits the resulting listing event, and every registered consumer removes the listing exactly as it would for a sale, on a **priority lane** with a tighter objective (see Trust and Safety as a Platform). Two separate removal mechanisms, one for sales and one for takedowns, guarantee that one of them falls behind the consumer registry. The only differences for takedowns are priority, the 410 page rather than a "sold" page, the removal of pending messages, and the statement of reasons the Digital Services Act requires for the restriction (Rules of the Marketplace: DSA, DAC7 and Vertical Regulation).

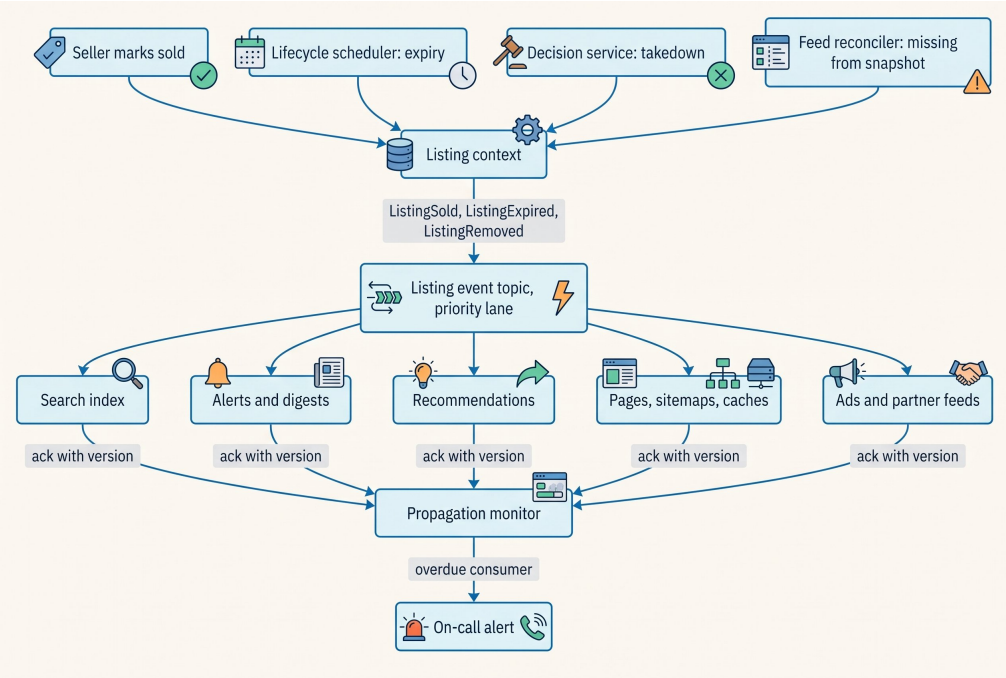


Figure 4: *Four causes, one removal path. Sales, expiry, takedowns and feed reconciliation all end in the listing context, which emits one removal event; each registered consumer acknowledges with the version it applied, and the propagation monitor alerts on any consumer that misses its objective.*

Observability: Proving the Pipeline Works

A pipeline that spans eight teams needs measurement that no single team can game. Five instruments are worth building.

1. **The listing journey**. Every event carries the time of its cause, and every consumer records when it applied the event. Joined by listing ID, these give a per-listing timeline: submitted, gate decided, cleared, published, indexed, alert-matched, removed, removed from each consumer. The p95 of each interval is the stage budget's actual value.
2. **Synthetic canaries**. A synthetic listing is created, published and removed every minute in each tenant (country and brand), in a hidden category, and its end-to-end times are measured from outside the pipeline. Veldmark's event platform runs this for search removal (see Every Change Is an Event: Event-Driven Listing Architecture); the publication pipeline extends it to insertion and to every surface.
3. **Stuck-listing detection**. Listings that sit in a state longer than its objective: submitted but not decided after five minutes, cleared but not published after one, held for review past the review objective, published but absent from the index after two. Each is a queue on the operations dashboard, not a log line.
4. **Copy reconciliation**. An hourly comparison between the system of record and each major copy: active in the record but missing from the index (lost publications), ended in the record but present in a copy (Ghost Inventory). The comparison works only because the record, not the index, defines what is published. If the index is the only place that knows, the pipeline has become **Search as System of Record**, and none of these questions can be answered.
5. **The two SLAs on one page**. Insertion latency and removal latency, per brand, per channel and per moderation path, side by side. Removal latency is often more important than insertion latency, and it is rarely measured separately.

In Practice: How Others Got Faster, and Tradeyard's Budget

Shape first, engine second

The best public record of a classifieds-scale freshness program is Vinted's. In 2023, its engineers described moving item indexing from Sidekiq delta jobs, under which item visibility could take up to about seven minutes, to change data capture feeding Apache Flink jobs that build read models in Redis, from which search documents are assembled; visibility fell to about one minute (Vinted Engineering, 2023). In 2024, after moving search to Vespa, they reported a 5-second visibility interval and a p99 of 4.64 seconds to

index a single item after upload (Vinted Engineering, 2024). Two lessons transfer. The first improvement came from **changing the pipeline's shape**, from timed batches to event-driven streams, not from tuning the engine. The second came from the engine's own visibility model, which is why the engine's refresh semantics belong in the budget.

Each engine documents when new documents become visible, and at what cost. In Elasticsearch, a refresh makes buffered documents searchable by writing a new segment; the default is every second, but only for indices that received a search request in the last 30 seconds, and a refresh is "much lighter" than a full commit (Elastic, n.d.). In Solr 8, documents become visible through a soft commit or a hard commit that opens a new searcher; soft commits give faster visibility without guaranteeing durability (Apache Solr Reference Guide 8.11, n.d.). When Solr opens a new searcher, the old one keeps serving while the new one auto-warms its caches from the old one's contents (Apache Solr Reference Guide 8.11, n.d.). That warming cost is why teams running Solr commonly set soft-commit intervals in tens of seconds rather than one second, and why commit frequency is a capacity decision as much as a freshness one.

Tradeyard's baseline

Ingrid's team reconstructed the six minutes from Yardstick's logs, for listings cleared by the automated gate in a week of March 2027. The stage times do not add up to the end-to-end p95, because the slow cases in each stage are rarely the same listings, but they show where the time goes. The figures are illustrative.

Stage (Yardstick, March 2027)	Median	p95	Why
Submit, validation, write, RabbitMQ publish	2 s	4 s	Synchronous PHP request
Thumbnail generation before moderation	18 s	70 s	Started at submission, not during drafting
Moderation consumer (batch pull, rules, models)	50 s	140 s	Pulls batches every 60 seconds; models run after rules
Batch indexer pick-up and push to Solr	70 s	150 s	Runs every 2 minutes; long runs delay the next
Solr soft commit and searcher warming	40 s	85 s	60-second soft commits; warming at peak
End to end, submit to searchable	**3 min 10 s**	**6 min**	

Femke's bike is a typical case: cleared at 19:44, it waited for the indexer's next run and the next soft commit, and was searchable at 19:48.

Three findings came out of the reconstruction. **Every timer stacks.** A 60-second batch pull, a 2-minute indexer cycle and a 60-second commit interval each add half their interval on average and nearly all of it in the worst case, before any work is done. **Images were processed at the wrong moment**, after submission, when they could have been processed while the seller was typing. And **the held 6% had no measure at all**: the review queue's own dashboard reported backlog size, not time to publication.

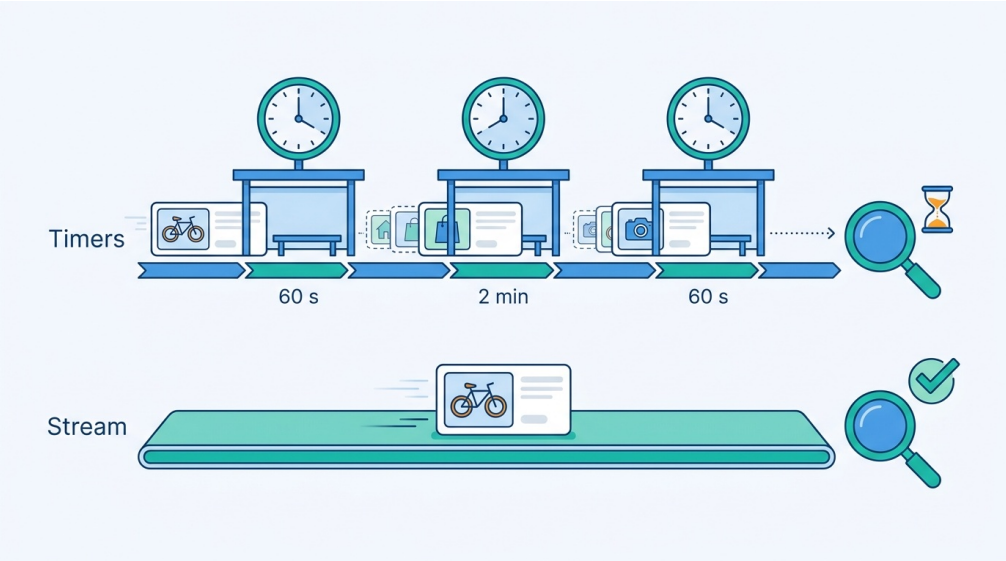


Figure 5: *Every timer stacks: a 60-second pull, a 2-minute indexer cycle and a 60-second commit each add waiting before any work is done, which is why event-driven stages win most of the gain.*

The 90-second budget

Ingrid's team wrote the Common Ground target, a p95 of 90 seconds by the end of 2028, as a contract with three parts, recorded in a publication-pipeline SLA card (a template is in the Field Kit: Templates for the Marketplace Architect):

- **Coverage.** The 90-second target applies to listings cleared by the automated gate, which was 94% of Tradeyard's new listings in 2026. Listings held for human review have their own objective: p90 under 30 minutes from submission to decision during staffed hours. The share of listings held is published next to both numbers, so that the fast path cannot be met by holding more listings.
- **Clock.** Platform latency, from receipt of the submission to visible in search, for listings submitted through Tradeyard's own apps and web. Professional feeds have their own SLA, measured from receipt of the feed record.

- **Removal**. Search removal p95 10 seconds; every registered consumer p95 2 minutes; partner feeds within 15 minutes; no alert, digest or recommendation ever sent for a listing that ended before the send.

The budget allocates the 90 seconds to stages. Media and the asynchronous risk checks run in parallel, so only the longer of the two counts.

Stage (target design, 2028)	p95 budget	How it is met
Synchronous path: validation, gate, entitlement, write	2 s	Gate under 300 ms; images already processed during drafting
Outbox to listing event topic	2 s	Event platform objective
Asynchronous checks and media confirmation (parallel)	30 s	Image models and duplicate search only for tiers that need them before publication
Publication decision and `ListingPublished`	1 s	Listing context applies the clearance
Indexing, refresh and visibility	15 s	Event-driven indexer; OpenSearch refresh every 5 seconds (Search Architecture for Listings: Indices, Freshness and Scale)
Result-page and count caches	20 s	First results page cached at most 20 seconds
Slack	**20 s**	Absorbs peaks such as Sunday evenings
Total	**90 s**	

The plan delivers it in two steps, because the platform changes underneath. In the second half of 2027, while Tradeyard still runs Solr, the team replaces the timed batch indexer with an event-driven indexer fed from the listing event contract (live since 1 July 2027), starts image processing during drafting, and shortens the moderation consumer's pull to a stream. The p95 for gate-cleared listings falls to about **2 minutes** by the end of 2027, on Solr, without touching the engine. The rest comes with the insertion flow's move out of Yardstick in the second quarter of 2028 and the search migration to OpenSearch in the fourth quarter. **The pipeline's shape gives most of the gain**; **the engine gives the last part**.

The bike's sale tests the removal side. Under the new contract, `ListingSold` at 18:10 reaches the search index within seconds, as it already did. It also reaches the digest service, which purges the bike from the assembled digests, and the 07:00 send re-checks

every listing's state in its final minute. The bike appears in no Tuesday digest. The propagation monitor shows the recommendation candidate store acknowledging at 18:11 and the partner syndication feed at 18:19.

Freshness SLAs by Vertical

A group does not need one SLA. It needs one SLA *format*, with values per vertical and channel.

	General classifieds (Trade-yard)	**Automotive (Motor-vane)**	**Real estate (Hearthly)**	**Jobs**
Main supply channel	Private sellers in the app	Dealer DMS feeds (88% of dealer listings)	Agency software feeds (84% of listings)	Employer and agency feeds, job boards
What sellers notice	"Is my listing live?" within a minute or two	Inventory in sync with the DMS	Listing live before a competitor portal	Posting live on the advertised date
Insertion objective (platform latency)	90 s p95 for gate-cleared listings	Minutes from feed receipt	Minutes from feed receipt	Minutes; scheduled starts honored
Dominant removal risk	Sellers who do not mark sold	Feeds that send the sale late or not at all	Let or sold flats that agencies do not withdraw; duplicates by other agencies	Filled vacancies left open
Source-latency control	Prompts and "still available?" checks	Snapshot reconcilia-tion, dealer contract terms	Reconciliation, agency reporting, duplicate clusters	Validity dates, employer confirma-tions

	General classifieds (Tradeyard)	Automotive (Motorvane)	Real estate (Hearthly)	Jobs
Moderation path	Risk-tiered; most listings on the gate	Gate for known dealers; private sellers tiered	Gate for agencies; private landlords and rentals in high-risk cities wait for image checks	Gate; scam-heavy job categories wait
Special constraint	Peak load on Sunday evenings at Tradeyard	Bulk price changes from dealers	Embargo and visibility rules from source licenses	Search engines penalize expired postings

In real estate, **a listing that waits for a duplicate check before going live** is slower, but it lets entity resolution attach the listing to an existing property cluster before buyers see it (see One Property, Three Listings: Duplicates and Entity Resolution). Hearthly accepted a slower insertion objective for that reason. For rentals in cities with known fake-rental activity, the image and duplicate checks are pre-publication; the Barcelona ring of July 2027, which used photos stolen from real listings, is the case behind that rule (Scams, Fakes and Account Takeovers: Fraud Prevention for Listings).

Trade-offs

Decision	Option	When it is right	What it costs
Shape of the asynchronous path	Sequential stages	Small marketplace, few consumers	Latency adds up; one slow stage blocks all; adding consumers is invasive

Decision	Option	When it is right	What it costs
	Fan-out from one event	Many consumers, several teams	Needs an event contract, idempotent consumers and a propagation monitor
Moderation mode	Pre-moderation for all	High-risk verticals, small volume	Slow publication; reviewer cost scales with volume
	Risk-tiered pre and post	Large classifieds	Complexity; two SLAs to report; policy must be stated honestly
	Post-moderation for all	Low-risk, trusted sellers only	Exposure window for fraud
Visibility trigger	Index on submission, filter by status	Very low latency needs	Unmoderated content one filter bug away from buyers
	Index only after the publication decision	Default	Moderation latency is on the path
Engine visibility	Frequent refresh or soft commit	Freshness-critical tenants	Indexing and cache-warming cost; segment merges
	Longer intervals	Cost-sensitive, low-churn indices	Stale results
Removal mechanism	Separate takedown path	Never	Drifts from the consumer registry
	Same path, priority lane	Default	Priority handling in every consumer
Digest freshness	Purge on removal events	Necessary	Missed or late purges
	Plus re-check at send time	Default	One batched lookup per send

The trade-off that decides the others is **where "published" is defined**. If it is defined by the listing context's state and event, the pipeline can be measured, reconciled and rebuilt. If it is defined as "the indexer has seen it", every question about a missing or lingering listing becomes an investigation.

Failure Modes

1. **The moderation queue in series**. A queue added in front of the indexer makes every listing wait for the slowest check, including listings whose risk does not justify it. Tier the checks; put only the necessary ones on the critical path.
2. **Timers that stack**. Every batch interval adds up to its full length in the worst case. Replace timed batches with event-driven consumers; where an engine needs an interval, budget it explicitly.
3. **Indexing before the decision**. A listing indexed at submission and hidden by a status filter is one filter bug away from showing unmoderated content. Index after the publication decision.
4. **Edits that skip the gate**. Material edits to price, payment terms, contact details or photos go straight to the indexer. Treat them as new submissions.
5. **One p95 for everything**. A p95 over gate-cleared listings hides the held ones; a p95 over all listings is dominated by the review queue. Report both paths and the share held.
6. **Removal measured only in search**. The index is fast, and the digest, the recommendation store and the partner feed are not. Measure removal per registered consumer.
7. **Zombies from replays**. A backfill or reindex writes an old active state over a sold listing. Version-gate every consumer and keep tombstones.
8. **Expiry storms and expiry lag**. A midnight batch overloads consumers; a lagging scheduler leaves expired listings live. Spread expiry and alert on overdue transitions.
9. **Unregistered copies**. A team adds a cache, an export or a model's candidate set without subscribing to removals. Make registration in the consumer registry a condition of access to the listing event topic.
10. **Renewal as republication**. Renewals and bumps reset the creation date or fire "new listing" alerts. Keep renewal a ranking-timestamp change.

The Chief Architect's View

One owner for the whole path. Each stage has an owner; the path needs one too. At Veldmark it is the listing platform team under Yara Mendes's platform engineering organization, which owns the SLA card, the budget, the consumer registry and the

freshness monitor. It does not own the stages. It owns the numbers that show whether the stages add up.

The SLA card as a decision record. Coverage, clock, insertion objective, review objective, removal objectives per consumer, and the stage budget are written on one card per brand and channel, decided at the architecture advice forum, and changed only through it. When a team wants to add a pre-publication check, the question is which stage's budget pays for it. When product wants to promise sellers "live in one minute", the card shows what that would cost.

Registration as the price of access. Any system that consumes listing events must register: what it holds, what it does on removal, its removal objective and its owner. The event platform enforces this at subscription time. It is the cheapest Ghost Inventory control available.

Honesty about the mode. The legal and product teams must be able to state, from the pipeline's configuration, what share of listings is checked before publication, by what, in which categories. The leboncoin judgment shows that a marketing claim about moderation is a factual claim. The pipeline should be able to produce the evidence for it.

What to watch. Insertion latency per brand, channel and moderation path; the share of listings held; review-path latency; removal latency per consumer and the worst consumer; the overdue-transition backlog; copy-reconciliation mismatches per day; and synthetic-canary results at the Sunday-evening peak. A number that improves while the held share rises is not an improvement.

Questions to Ask

1. Which stages are on the synchronous path, and could any of them move out without changing the publication decision?
2. Which listings does our freshness SLA cover, which clock does it measure, and what is the objective for listings held for review?
3. What is the stage budget, and which stage's budget pays for the next check someone wants to add?
4. Where is "published" defined: in the listing record and its event, or in the index?
5. Which edits re-enter the risk gate, and who decided that list?
6. How do consumers handle duplicate, late and out-of-order events, and what stops a replay from resurrecting a sold listing?
7. Which systems hold copies of listing data, and is every one of them in the consumer registry with a removal objective and an owner?
8. Which consumers assemble work in advance, and do they re-check each listing's state at send time?

9. Do takedowns use the same removal path as sales, on a priority lane, and how quickly does the last consumer acknowledge?
10. What does the synthetic canary show at our weekly peak, for insertion and for removal?

From Pipeline to People

The publication pipeline treats every seller the same way, but the sellers who supply most of a vertical marketplace's inventory are not private individuals with one bike. They are dealers and agencies who manage hundreds of listings, track their performance and expect tooling to match. What they need, and how to build it without compromising the neutrality of the marketplace, is the subject of Serving the Pros: Dealer, Agent and Merchant Tooling.

Sources and Evidence

Company engineering and investor sources

- Vinted Search Scaling Chapter 1: Indexing — Vinted Engineering, 2021. Kafka offsets as external document versions; partition changes breaking order.
- Vinted Search Scaling Chapter 7: Rebuilding search indexing pipeline — Vinted Engineering, 2023. From delta jobs (up to about 7 minutes) to CDC and Flink (about 1 minute).
- Vinted Search Scaling Chapter 8: Goodbye Elasticsearch, Hello Vespa Search Engine — Vinted Engineering, 2024. 5-second visibility interval and 4.64-second p99 end to end.
- Seller-API documentation — mobile.de, n.d. Separate creation, modification and renewal dates.
- Leboncoin a prétendu à tort qu'il relisait toutes les annonces — Clubic, 2015. The court finding on leboncoin's moderation claim and hosting status.

Standards, regulation and documentation

- Near real-time search — Elastic, n.d. Refresh semantics and the search-idle default.
- UpdateHandlers in SolrConfig — Apache Solr Reference Guide 8.11, n.d. Hard and soft commits and visibility.
- Query settings in SolrConfig — Apache Solr Reference Guide 8.11, n.d. Searcher auto-warming from the previous searcher's caches.

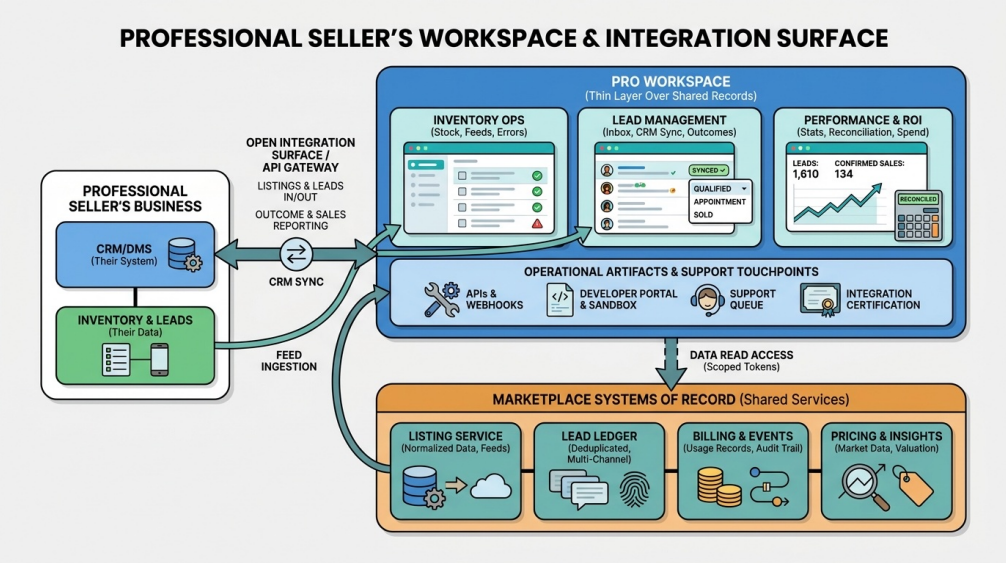
- Product expiration — Google Merchant Center Help, n.d. Products expire 30 days after the last update.
- Job posting structured data — Google Search Central, updated 2026. Removal of expired jobs and manual actions.
- Regulation (EU) 2022/2065 (Digital Services Act), Art. 17 — European Union, 2022. Statements of reasons for restrictions, including takedowns; detailed in Rules of the Marketplace: DSA, DAC7 and Vertical Regulation.

Research and practitioner sources

- Scam ads slipping through the net too easily on second-hand marketplaces — Which?, 2024. Edited scam ads passing review “in minutes” after a first block.

Veldmark is fictional; architectural recommendations and labeled inferences are the author’s judgement.

17. Serving the Pros: Dealer, Agent and Merchant Tooling



IN THIS CHAPTER, YOU WILL: Learn what professional sellers need from a marketplace (inventory operations, performance and ROI reporting, lead management, pricing and market insight, team permissions and integrations) and how to architect it as a workspace over shared systems of record. You will also see why software sold to professionals, such as an agent CRM, creates a conflict of interest with the portal, and how to design that conflict out.

WHY THE BUSINESS CARES: Professionals pay most of the bills in automotive and real estate. They renew when they can prove that the marketplace produced sales, and they downgrade quietly when they cannot. Pro tooling is where that proof is made or lost, and where a portal either becomes part of the professional's daily workflow or stays a cost line.

WHY ARCHITECTS SHOULD CARE: Pro tools read from every domain at once: listings, leads, statistics, entitlements, prices and accounts. Built

carelessly, they become a second system of record with numbers that disagree with billing and with the professional's own CRM. Built well, they are a thin workspace and an API surface that partners can build on.

KEY POINTS:

- **Professionals buy outcomes and judge them with their own systems.** A dealer or agency compares the marketplace's lead count with its CRM and its sales. Statistics, lead exports and invoices must come from the same events, carry the same identifiers and survive that comparison, or the renewal conversation is lost before it starts.
- **Pro tooling is a workspace, not a system of record.** Inventory, leads, entitlements and accounts belong to their domains. The pro workspace composes them for one organization, with permissions that follow its hierarchy, and exposes the same capabilities through APIs, because many professionals will use them from partner software rather than from the portal.
- **Software sold to professionals must be neutral by architecture.** When the group also sells the CRM that publishes to competing portals, neutrality has to be enforced by separate tenancy, API parity, data boundaries and audit evidence. In the EU, the Platform-to-Business Regulation already requires portals to describe their data access and any differentiated treatment of their own services.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Vanity Liquidity** — pro dashboards that report views and impressions instead of contacts, leads and sales sell exposure the professional cannot convert, and renewals fail.
- **Feed Roulette** — professional feeds without visible validation and error reports leave dealers and agents to discover missing stock from their customers.
- **Billing by Guesswork** — pro features gated by package checks scattered across the tooling make it impossible to say what a professional bought and what they received.

A private seller visits a marketplace a few times a year. A dealer or an estate agent lives in it. Their sales staff open the lead inbox before their e-mail, their managers read the weekly statistics before the board meeting, and their integration partner pushes inventory every few minutes. For these customers the marketplace is not a website. It is **one of the systems their business runs on**, next to their dealer management system (DMS) or agency CRM, and they judge it the way they judge their other systems: by whether its numbers reconcile.

This chapter closes Part III's supply chain at the point where the professionals check it. Many Doors In: Forms, APIs and Bulk Feeds and From Submitted to Searchable: The Publication Pipeline brought their stock in; pro tooling is where they see what that stock produced, and where the marketplace's liquidity becomes visible to the people who pay for it. A Motorvane dealer pays for listing slots and prominence; what they want to know is how many qualified buyers each car produced and how many of those bought. A Hearthly agency pays a subscription; what it wants to know is whether the portal brought it sellers and tenants.

This chapter owns three terms. **Pro tooling** is the set of products a marketplace gives professional sellers to run their business on it: inventory and visibility operations, performance and ROI reporting, lead management, pricing and market insight, team and account administration, and the APIs behind all of them. **Lead management for pros** is the professional's side of a lead: receiving, assigning, answering, qualifying and recording the outcome of buyer contacts, inside the portal or in the professional's own CRM. **Pro SaaS** (software as a service) is software sold to professionals as a product in its own right, such as an agent CRM or a dealer website builder, usable whether or not the professional lists on the group's portals. The lead itself, conversations and lead routing belong to From Interest to Contact: Messaging and Lead Management; packages and entitlements to Packages, Bumps and Quotas: Products, Pricing and Entitlements; the organization model to Private Sellers, Dealers and Agencies: Modeling Who Lists; feeds to Many Doors In: Forms, APIs and Bulk Feeds.

Setting the Scene at Veldmark: Two Arguments in One Week

It is the second week of June 2027 at Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build). Two arguments about professional tooling reach Ingrid Halvorsen, the Group Chief Architect, in the same week.

The first comes from Essen. Autohaus Ruhrblick, the dealer that sold the 2021 Volkswagen Golf through Motorvane in April, is up for renewal. It pays **€1,450 a month** for Motorvane's "Pro 150" package with 150 listing slots, above Motorvane's average of about €927 per dealer (see Packages, Bumps and Quotas: Products, Pricing and Entitlements and How Listing Marketplaces Make Money). Its sales director has compared the Golf's numbers. Motorvane's dealer portal, **Dealer Hub**, shows **23 leads** in the first ten days. The dealer's CRM received **14**. The sales director's conclusion is that Motorvane counts generously and the package is worth less than its price; the proposal on the table is a downgrade to a smaller package. Felix Brandt, Motorvane's CTO, wants funding to rebuild Dealer Hub, which runs on a reporting database fed nightly from three lead systems.

The second comes from Lyon. Keybridge, the agent CRM that Veldmark acquired in 2024, publishes listings for its 9,800 agency customers to many portals, including Hearthly's competitors. One of those agencies is Agence Rhône Habitat, a 14-branch network whose Lyon branch listed the Lyon apartment in May. Emma Laurent, Keybridge's founder and general manager, wants to launch a "portal performance" report that shows each agency how many leads each portal produced per listing. Claire Fontaine, Hearthly's CEO, asks a reasonable-sounding question at the next leadership meeting: could Hearthly see the aggregated comparison, to learn where it underperforms?

The March 2027 decision on Keybridge is clear on principle: **no Keybridge agency data is used by Hearthly for ranking or sales targeting**, with access controls and an annual independent review. It says nothing about how. Ingrid's task is to answer both arguments with one design: what professionals need, how the tooling should be built, and how a group that sells software to professionals keeps that software neutral.

Concepts: What Professionals Need

The professional's jobs

The productscape reference model of a general listings marketplace, a generic model and not a description of any company, gives its professional-merchant persona three jobs: operate inventory and visibility at scale, convert demand into qualified transactions, and prove value to justify continued spend. Its fears are equally specific: "paying for exposure without measurable return" and a feed sync that "has silently dropped half their stock". The real-estate model adds a job that has no equivalent for dealers: an agency must first **win mandates** from owners before it has anything to list, so prospecting and valuation tools are part of its toolkit.

Table 1 maps those jobs to capabilities and to documented products.

Professional job	Capability	Documented example
Operate inventory at scale	Portfolio view, bulk edit, feed health and error reports, listing quality checks	mobile.de's Seller API exposes listing quality checks and statistics (mobile.de Seller API, n.d., accessed 2026)

Professional job	Capability	Documented example
Create listings efficiently	Assisted listing creation from photos, decoded data and generated text	Autotrader's Co-Driver "anecdotally" cut listing time from 28 to 5 minutes; 86% of retailers had used a Co-Driver tool (Autotrader FY2026 results, May 2026)
Control visibility	Buy and apply paid products per listing or per branch	Rightmove's feed applies premium and featured listings through the same API as property data (Rightmove RTDF, v1.4.1, November 2023)
Convert demand	Lead inbox, assignment, reply assistance, call handling, CRM export	mobile.de's Lead Manager drafts 45% of written replies with AI and offers a Lead API into existing CRMs (mobile.de newsroom, July 2026)
Price to sell	Price ratings, days-to-sell forecasts, price simulation	mobile.de's coming "Preis-Planer" simulates price changes in 2% steps against days on market, price rating and search position (mobile.de newsroom, July 2026)

Professional job	Capability	Documented example
Win supply (real estate)	Valuation tools, seller-lead detection, prospecting maps	AVIV's Ultimmo offer, as described in 2022 (the current offer may have changed), combined an estimation tool, seller-prospect detection among buyer contacts and a prospecting map (MySweetImmo, November 2022)
Prove value	Performance statistics, ROI reports, account management	Rightmove's feed offers branch and property performance calls (Rightmove RTDF); CarGurus' account managers provide "ROI tracking" (CarGurus 10-K, filed February 2026)
Run a team	Users, roles, branches, lead ownership	mobile.de lists "multi-locations" among its Seller API resources (mobile.de Seller API)

Table 1: *Eight professional jobs and the capabilities that serve them. Every row has a public example, and every row reads from more than one domain.*

Pro tooling is a composition problem: no row of the table lives in one service. A statistics page reads listings, events, leads and entitlements; a lead inbox reads conversations, call records, accounts and the listing.

Outcome, not exposure

The professional's real question is causal: *what did the marketplace produce that I would not have had otherwise?* Two pieces of public evidence show how hard that question is.

- **Self-selection distorts package dashboards**. In an August 2026 post, OLX's engineers describe how professional sellers choose a package when they publish, and naive metrics made premium-package ads "appear to perform worse" than

cheaper ones. After propensity-score matching, moving from the cheapest to the most expensive package was associated with **97.4%** more conversions, and a comparison with a randomized test found the matched estimates consistently higher than the experimental ones, a sign of remaining confounding (OLX tech blog, August 2026).

- **Incumbents report outcomes, not views.** Rightmove's 2025 results claim its products deliver "7 out of 10 vendor instructions within resale, and 8 out of 10 tenants for lettings properties" (Rightmove FY2025 results, February 2026). Autotrader reports that full reservations through Deal Builder "continue to be the very best enquiries in terms of conversion to sale" (Autotrader FY2026 results, May 2026).

The architecture therefore needs a **performance funnel per listing, branch and organization**, from impressions through detail views and contacts to deduplicated leads and, where the data exists, outcomes. Views and impressions belong in the funnel as diagnostics, not as the headline. A dashboard whose top number is views is the Vanity Liquidity pattern served to a paying customer.

The lead from the professional's side

From Interest to Contact: Messaging and Lead Management defines a **lead** as a buyer contact with commercial intent attached to a listing and a seller. From the professional's side, a lead has a lifecycle the marketplace often cannot see: it is assigned to a salesperson, answered, qualified, turned into an appointment and won or lost, usually in the professional's CRM. Figure 1 shows that lifecycle as the pro workspace should model it.

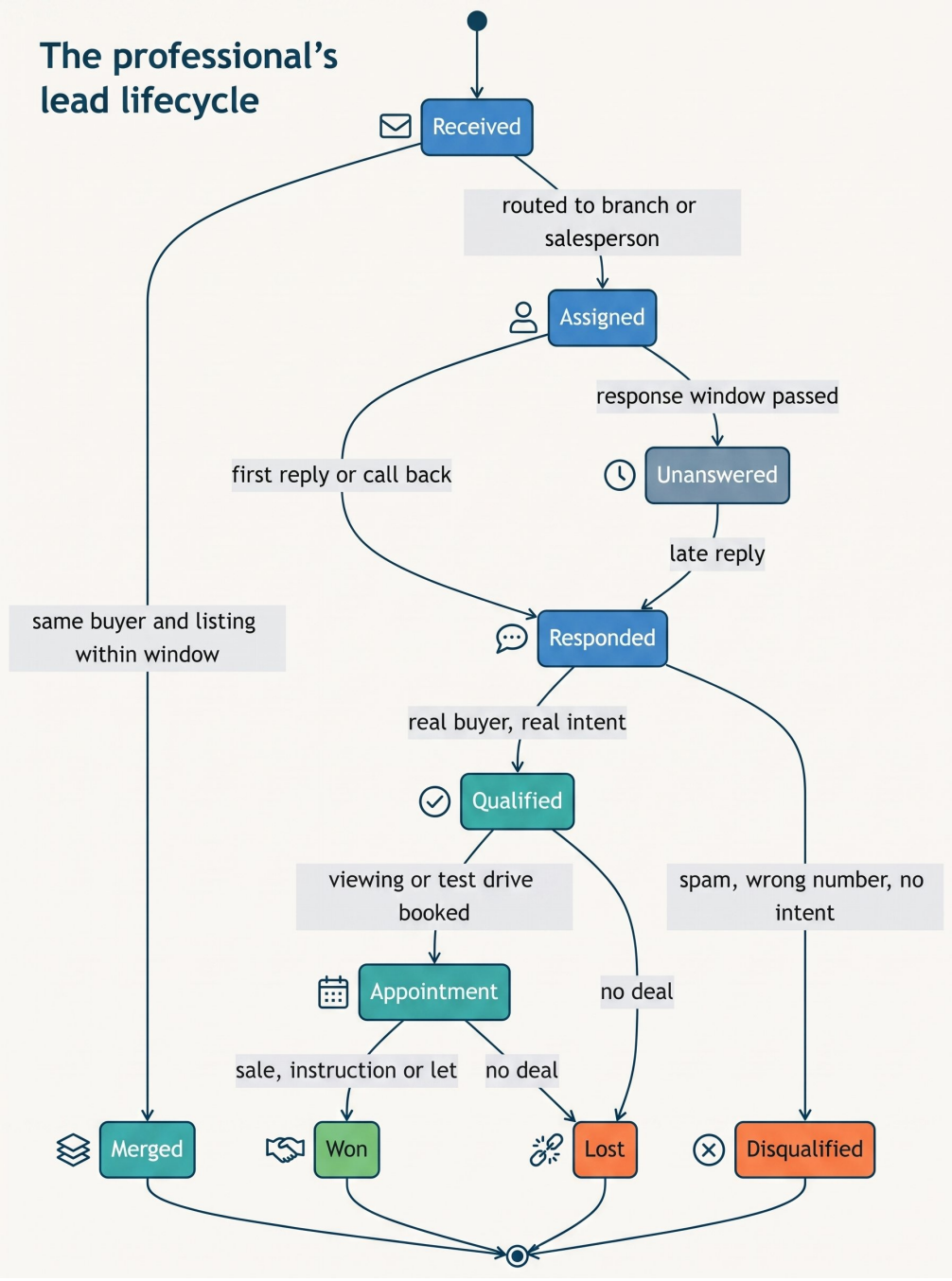


Figure 1: *The professional's lead lifecycle. The marketplace records Received, Merged, Assigned and Responded itself; the later states usually happen in the professional's CRM and come back only through an integration. ROI reporting is only as good as that return path.*

Response time and **outcome** are the two measurements in the figure that matter most. Response time is the one the marketplace can see and the professional can improve; mobile.de's 2026 lead-performance report tracks exactly response time, response rate and processing status. Outcome is the one the professional sees and the marketplace needs; without it, ROI is an estimate.

Pro SaaS and the neutrality problem

Many marketplace groups also sell software to professionals. Scout24's "Transaction Enablement" line, covering the Propstack CRM and the Sprengnetter and bulwiengesa data and valuation businesses, earned €106.6 million in 2025 (Scout24 preliminary FY2025 results, February 2026). Zillow's FY2025 annual report lists its agent CRM Follow Up Boss, which added AI message suggestions, call summaries and "intelligent lead routing" (Zillow 10-K, filed February 2026). In US automotive, Cox Automotive's brands include the Autotrader.com marketplace alongside vAuto inventory software, the Dealertrack DMS and the VinSolutions CRM (Cox Automotive, n.d.). Cars.com runs dealer websites for about 7,700 dealers through Dealer Inspire and D2C Media (Cars.com 10-K, filed February 2026).

Owning that software brings an advantage and a conflict. The advantage: the software sits where listings originate, and switching CRMs is painful. The conflict: **the CRM that creates a professional's listings knows what the professional publishes to every competitor**, how each portal performs and which sellers are about to come to market. If the portal can see that, competing portals and the professionals themselves have reason to distrust both products.

Architecture: A Workspace Over Systems of Record

Three properties of the pro workspace

The reference architecture I recommend (Figure 2) has three properties: the pro workspace owns no business records, every capability is available through the same API the partners use, and statistics are computed from the same event stream as billing.

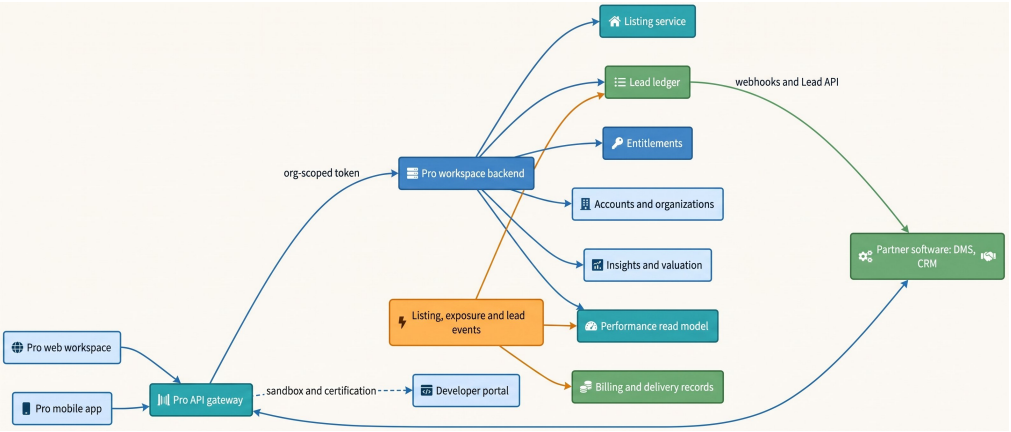


Figure 2: *The pro workspace composes domain services for one organization. The portal UI, the app and partner software reach the same capabilities through one gateway with organization-scoped tokens. Statistics, the lead ledger and billing all consume one event stream, so their numbers can be reconciled.*

The pieces, and why each sits where it does:

- **Pro API gateway**. One entry point for the portal's own front ends and for partners. Tokens are scoped to an organization and, where needed, a branch and a capability (listings write, leads read). The productscape model gives the partner surface its own gateway, sandbox and developer portal with "certification and SLA controls", which matches what the public portals document.
- **Pro workspace backend**. A backend-for-frontend that composes listing, lead, entitlement, account, insight and statistics data for one organization. It caches, but it does not own; if it disappears, no record is lost. This is the defense against the second system of record.
- **Lead ledger**. One store of deduplicated leads across channels (messages, tracked calls, bookings, reservations), with a stable lead identifier and a buyer-deduplication key. How Listing Marketplaces Make Money proposes it at Motorvane as the basis for any per-lead billing; here it is also the basis for the inbox, the statistics and the CRM export.
- **Performance read model**. Aggregates exposure, views, contacts and leads per listing, branch and organization from events. It is a read model in the sense of System of Record and Read Models: Listing Data Management, rebuildable from the stream.
- **Insights and valuation**. Price ratings, days-to-sell forecasts and demand data, served from the vertical capabilities described in Same Platform, Different Physics: How Listing Verticals Differ (vehicle price rating at Motorvane, Valuo at Hearthly).

Inventory operations: show the professional what the pipeline did

Most professional inventory arrives by feed, so the most valuable inventory tool is often not an editor but a **mirror of the ingestion pipeline**: which records arrived, which were rejected and why, which are waiting for images, which are live and which were removed. The productscape model's partner persona fears "opaque moderation or validation failures"; mobile.de's API returns errors as stable keys with arguments; Autotrader's developer portal documents validation errors such as a vehicle rejected because the UK Driver and Vehicle Licensing Agency (DVLA) classifies it as a category A or B write-off (Autotrader Connect, n.d.). Many Doors In: Forms, APIs and Bulk Feeds owns the feed contract and error reports. The pro tooling's job is to make that report readable by a sales manager, not only by an integration engineer, because the manager is the one who notices that a car is missing. Feed Roulette is a tooling failure as much as an ingestion failure.

Bulk operations follow the same rule. A bulk price change of 200 cars is a batch of commands against the listing service, executed asynchronously, with a job record the professional can inspect. It is not a direct write by the workspace into a listing table.

Performance and ROI: one event stream, three consumers

The professional, the sales team and billing must see the same numbers. That is only possible if **statistics, leads and delivery records are derived from the same events with the same identifiers**: listing ID, organization and branch, lead ID, buyer-deduplication key and the paid product that was active at the time. Packages, Bumps and Quotas: Products, Pricing and Entitlements requires delivery records ("what was sold versus delivered") for every paid product; the pro statistics page should be a view over those same records, so that a "Top listing" bought on Monday shows the impressions it actually produced.

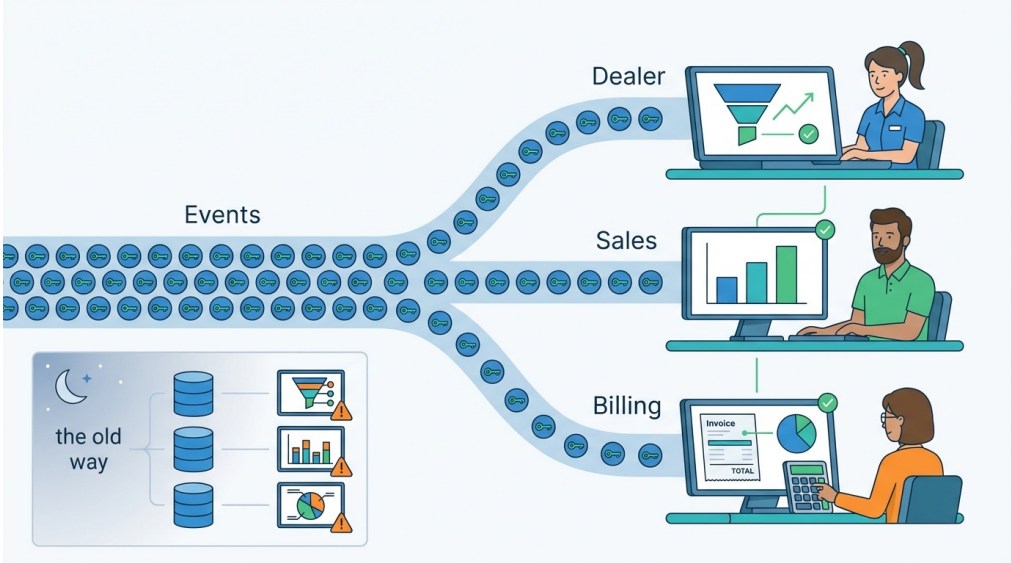


Figure 3: *Statistics, leads and delivery records derived from one event stream with the same identifiers let the professional, the sales team and billing see the same numbers.*

Three design rules follow.

1. **Count leads after deduplication, and show the raw contacts beside them.** Hiding the deduplication makes the marketplace look generous; showing both makes the rule auditable.
2. **Separate observed facts from estimates.** Views and leads are observed. "Sales attributed to Motorvane" is an estimate, unless the professional's system confirms the sale. Label estimates as such, and show the method.
3. **Do not compare packages naively.** As the OLX case shows, professionals self-select into packages. A tool that tells a dealer "premium listings perform worse" because premium was bought for hard-to-sell cars will cost the marketplace revenue and credibility. Comparative claims need matching or experiments; Testing a Two-Sided Market: Experimentation Platforms owns the methods.

Lead management: inbox inside, export outside

Professionals split into two groups: those who work leads in the portal's inbox and those who work them in their own CRM. The architecture must serve both with the same lead object.

- **Inside**, the inbox is a view over the lead ledger and the conversation store, with assignment rules that follow the organization hierarchy, reply assistance and response-time tracking. mobile.de's 2026 update adds AI reply suggestions, a

WhatsApp channel tied to the listing and an AI phone assistant; in the pilot, 91% of answered calls came outside opening hours and one in three became a qualified lead (mobile.de newsroom, July 2026).

- **Outside**, a Lead API and webhooks deliver every lead, from every channel, with its stable identifier, to the professional's CRM. mobile.de describes its Lead API as transferring leads "directly and securely into existing CRM systems". Autotrader publishes Messages, Calls and Deals APIs for the same purpose (Autotrader Connect).
- **Back**, an outcome channel lets the CRM or DMS report qualification, appointments and sales. For vehicles, the VIN makes this unusually reliable: a sale recorded in the dealer's DMS can be matched to the listing and its leads without asking anyone to fill in a form. For property, the return path depends on the agency's software and on consent.

Lead data is personal data about the buyer. When the lead is exported, the professional becomes responsible for it in its own systems; the marketplace must know which leads were exported to which integration and under which terms, so that erasure requests and complaints can be handled. Autotrader's "Buying Signals" model, which scores enquiry intent from "verified but anonymised sales transactions and consumer interactions" (Autotrader FY2026 results), shows the direction: the lead object will carry model outputs, which then become part of what is exported and must be explainable.

Pricing and market insight: a control loop, not a widget

Price tools turn the marketplace's market data into pricing advice for each professional. CarGurus tells dealers the smallest price cut needed to reach the next deal-rating band ("Next Best Deal Rating") and sells PriceVantage, ML pricing software that uses "VIN-level activity, turn-time predictions, lead potential, market days' supply" (CarGurus 10-K, filed February 2026). Autotrader's API computes a "price adjusted days to sell" for a proposed price (Autotrader Connect). mobile.de's Preis-Planer links price steps explicitly to search position.

Two architectural consequences follow; both are my inference from the published designs. First, **these tools create a feedback loop**: when dealers price to band boundaries, the market distribution the model learns from shifts, so the valuation owner must monitor clustering at band edges. Second, **the insight is computed from all professionals' data and shown to each one**. That is legitimate and valuable, but it is also exactly the kind of aggregated data whose access the EU's Platform-to-Business Regulation asks portals to describe (see below). The insight service should know, for every output, which population it was computed from and which access rule applies.

Team, branches and permissions

Professional accounts are hierarchies: a legal entity, brands, branches, users and integrations. Private Sellers, Dealers and Agencies: Modeling Who Lists owns the model; the pro tooling is its main consumer and enforces three rules: **listings and leads belong to the organization**, **not to the salesperson**, so a departing employee's inbox is reassigned rather than lost; **permissions are scoped to branch and capability**, so the Lyon branch of Agence Rhône Habitat cannot reprice the Villeurbanne branch's listings; and **integrations are principals**, with their own scopes and audit trail. The OAuth 2.0 Security Best Current Practice (RFC 9700, January 2025) gives the baseline for partner tokens: authorization code with Proof Key for Code Exchange (PKCE) for third-party tools acting for a professional, no password grant, and sender-constrained or rotated refresh tokens (RFC 9700, January 2025).

The partner API as a product

For many professionals the marketplace's own UI is secondary. Autotrader reports that its data is "integrated with over 220 technology partners", with an average of **155 million** calls a month on its technology and data services, up from 91 million a year earlier (Autotrader FY2026 results, May 2026). Its developer portal announces breaking changes at least 30 days ahead and defines what counts as breaking (Autotrader Connect). Rightmove's feed gives each software provider a network identifier and each agent office a branch identifier, and authenticates with Rightmove-issued client certificates (Rightmove RTDF).

My judgement is that **the pro API is a product with its own customers**: software vendors. It needs versioning and deprecation rules, a sandbox, certification, per-integration rate limits and its own reliability objectives. The Bundeskartellamt's October 2026 decision clearing mobile.de's acquisition of Carwow's German business noted that dealers multi-home and that "many platforms offer dealers data export to other platforms" (Bundeskartellamt, October 2026). In feed-driven verticals inventory is not exclusive, so the quality of the integration surface is part of what makes a professional choose one portal's workflow over another's.

Neutrality by Design: When the Group Sells the CRM

What the law already asks

The EU's Platform-to-Business (P2B) Regulation, (EU) 2019/1150, has applied to online intermediation services since 12 July 2020. Two of its articles bear directly on a group that sells pro SaaS. **Article 7** requires a description of "any differentiated treatment which they give, or might give" to goods or services offered by the provider itself or

"any business users which that provider controls" compared with other business users, covering access to data, ranking, remuneration and "technical interfaces" (Regulation (EU) 2019/1150, Art. 7). **Article 9** requires a description of the technical and contractual access to data that business users and consumers provide or generate, including whether the provider has access, whether business users have access "including in aggregated form", and whether data is provided to third parties (Regulation (EU) 2019/1150, Art. 9). The European Commission proposed repealing the regulation in November 2025, citing overlap with the Digital Services Act and Digital Markets Act; as of October 2026 that is a proposal (European Commission, updated July 2026).

Whatever happens to the regulation, its articles describe the questions a neutral design must answer. **Can the portal see the CRM's data? Does the CRM get better interfaces to the portal than its competitors? Does the portal's ranking treat the CRM's listings differently?** A design that cannot answer each with evidence is not neutral, whatever its policy says.

The four controls

Neutrality is enforceable when it rests on four architectural controls, each of which produces evidence.

Control	What it means	Evidence it produces
Separate tenancy	The SaaS runs in its own accounts, identity realm and data stores; no portal service has credentials to them	Infrastructure inventory; access-policy exports; quarterly access reviews
API parity	The SaaS publishes to the group's portal through the same public feed API, with the same fields, limits and service levels as any other vendor	Contract tests run against the public API; integration configuration diff; support-ticket SLAs by vendor

Control	What it means	Evidence it produces
Data boundary	Data flows from the SaaS to the portal only as the listing payloads the agency chose to publish; data from the portal to the SaaS only through the public lead and performance APIs available to every vendor	Data-flow inventory; data-catalog tags on every dataset; egress logs
Ranking blindness	Portal ranking, sales and pricing systems carry no feature that identifies the publishing software	Feature inventories of ranking and sales models; model cards; review of sales-targeting queries

Table 2: *Four controls that turn a neutrality promise into checkable architecture. Each produces evidence an independent reviewer can test.*

The controls cost something. API parity means the group's own CRM cannot ship a portal feature its competitors cannot also use, which slows integration work the product team would like to do. Separate tenancy means duplicate infrastructure and no shared customer database. Those costs are the price of selling software to customers who also buy from the portal's competitors.

Veldmark Worked Example: The Golf's 23 Leads and Keybridge's Report

Reconciling the Golf

Felix's team reconstructs the Golf's ten days from the three lead systems, as the lead-ledger design in How Listing Marketplaces Make Money proposed. Table 3 shows the result.

Channel	Contacts in Dealer Hub	After deduplication (one buyer, one listing, 14 days)	Reached the dealer's CRM	Why the CRM count differs
E-mail and messages	14	14	14	Exported by the nightly e-mail lead export
Tracked phone calls	7	4 (three callers had already e-mailed)	0	Calls rang the showroom through a tracked number and were never exported
Test-drive bookings	2	1 (one booker had already e-mailed)	0	Bookings stayed in Dealer Hub's calendar
Total	**23**	**19**	**14**	

Table 3: *Three counts of the same demand. Dealer Hub reported raw contacts; the dealer's CRM saw only e-mails. Neither number was the number the dealer needed.*

Both sides were wrong, in different directions. Dealer Hub's 23 counted the same buyers more than once. The dealer's 14 missed every phone and booking lead. The honest number was **19 unique buyers**, and one more fact closes the loop: the car's VIN appears in the record of the 29 April sale that Autohaus Ruhrblick's DMS exported the next morning, and the buyer's e-mail address matches one of the 19. The sale can be attributed to a Motorvane lead with evidence, not estimated.

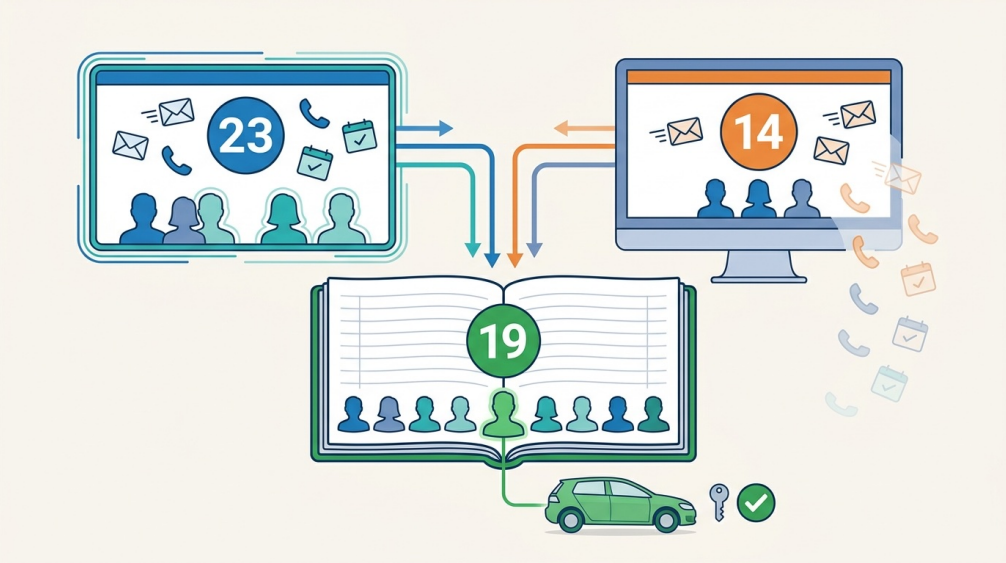


Figure 4: *Dealer Hub's 23 counted some buyers twice and the dealer's 14 missed every call and booking; the honest number was 19 unique buyers, and a VIN match tied the sale to one of them.*

Felix's rebuild becomes four specific changes rather than a new portal.

1. **Dealer Hub's lead view becomes a read model of the lead ledger**, showing raw contacts and deduplicated leads side by side, with the rule stated.
2. **The Lead API and webhooks export all channels**, not only e-mail, each lead with its stable ID, channel and deduplication key. Calls carry duration and a recording consent flag; bookings carry the slot.
3. **A sale-matching feed** lets dealers' DMSs, or their integration vendors, report sales by VIN. Matching is opt-in per dealer and reported back in Dealer Hub as "confirmed sales from Motorvane leads".
4. **Statistics move from the nightly reporting database to the event stream**, so the statistics page, the lead ledger and the package delivery records agree within minutes.

The renewal conversation changes with the numbers. Over the same quarter, Autohaus Ruhrblick's 150 slots produced 1,610 deduplicated leads and, among dealers that opt in to sale matching, about one confirmed sale for every twelve leads; the downgrade is withdrawn. The numbers are illustrative, but the mechanism is not: **the professional renewed because both systems finally told the same story**.

Keybridge's portal performance report

Emma's report is useful to agencies and legitimate for Keybridge. The question is what data it uses and who sees the result. Figure 5 shows the boundary Ingrid's team designs.

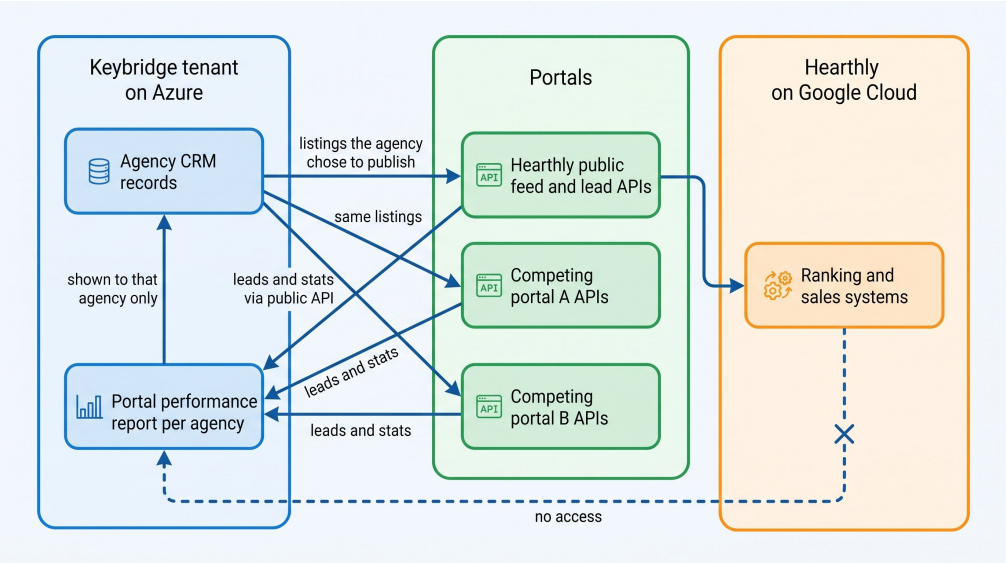


Figure 5: *Keybridge's neutrality boundary. Keybridge publishes to Hearthly through the same public API as any other vendor and receives Hearthly's leads and statistics through the same public APIs as everyone else. The portal performance report is computed inside Keybridge, per agency, and shown only to that agency. No Hearthly system can read Keybridge's data.*

The design answers Claire's question: **no**. Hearthly cannot see the cross-portal comparison, aggregated or not, because it is computed from data agencies entrusted to Keybridge, including data from Hearthly's competitors. Hearthly can learn where it underperforms the way any portal can: from its own data, from agency feedback and from public market research. The decision record notes the alternative Claire raised, an "anonymized benchmark" sold to Hearthly, and rejects it. Even anonymized, it would be Keybridge agency data used for Hearthly's sales targeting, which the March decision forbids, and it would undermine Keybridge's position with every agency that multi-lists.

Three implementation details make the boundary auditable.

- **Keybridge is treated as one of Hearthly's roughly 160 software vendors.** It uses the same feed API, the same error reports and the same support queue. A contract test suite, run monthly against the public API with Keybridge's credentials and with a test vendor's credentials, proves that both receive the same fields and limits.
- **Hearthly's data catalog tags every dataset with its origin.** Listings published through Keybridge are tagged as such for operational support only. The ranking feature store and sales tools are barred, by policy checked in their deployment pipelines, from reading that tag.

- **The annual independent review receives a standard evidence pack**: access-policy exports for both clouds, egress logs between Keybridge and Hearthly, the contract-test results, the data-flow inventory and the feature inventories of Hearthly's ranking and sales models. Lucía Ortega, the group's data-protection officer and regulatory lead, owns the pack and the P2B descriptions it supports.

The separate clouds (Keybridge on Azure, Hearthly on Google Cloud) help, but they are not the control. If Veldmark later moves both to AWS, the controls must survive the move, which is why they are written as policies and tests rather than as a consequence of hosting.

Trade-offs: Build, Integrate or Own

A group decides, per capability, whether to build pro tools itself, to let partners provide them through APIs, or to own a SaaS product that professionals use beyond the portal. Table 4 compares the three.

Option	What it means	Strengths	Weaknesses	When it is right
Build in the portal	Inbox, statistics, bulk tools and insights in the portal's own workspace	Direct control of the experience; tightly tied to the marketplace's data; differentiates the portal	Competes with professionals' existing software; must match CRM features to be used	Capabilities only the portal can provide: its statistics, its price ratings, its paid products, its leads
Integrate through APIs	Expose capabilities to the CRMs, DMSs and websites professionals already use	Meets professionals where they work; partners carry feature breadth; scales across verticals	Partners own the daily relationship; the portal's UI is used less; API product costs	Professionals with established software ecosystems, which is most dealers and agencies

Option	What it means	Strengths	Weaknesses	When it is right
Own pro SaaS	Sell a CRM, DMS or website product to profession-als, including for competing portals	Revenue line; presence at the origin of listings; switching costs	Neutrality obligations and their costs; separate product and sales motion; distrust if handled badly	Markets where the group can run the SaaS as a neutral, separately governed business

Table 4: *Three ways to serve professionals. Most groups need the first for portal-specific capabilities and the second for everything else; the third is a separate business with its own rules.*

The rule I use is: **build what only the portal knows, expose everything through the API, and own SaaS only if you are prepared to run it as if a competitor owned the portal**. The middle row is the one most often under-funded. A portal that invests in its own inbox but not in its Lead API ends up with a feature most professionals do not use and a CRM integration they complain about.

Owning SaaS also changes the economics of a group exit. After the Deal: Consolidating Platforms After Acquisitions and Adevinta's 2024–2026 decentralization (Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform) show that businesses sold separately must be separable; a pro SaaS product built with separate tenancy is easier to sell, or to keep, than one entangled with the portal.

Vertical Differences

Professional tooling varies more across verticals than almost any other capability, because professionals' businesses differ.

Vertical	The professional	System they already run	What they need most from the portal	Typical tension
Automotive	Dealer with sales staff, often several sites	DMS and dealer CRM, often from the same vendor	Feed reliability, lead export across channels, price ratings and days-to-sell, sale matching	Platform-controlled states on dealer stock, as with Autotrader's Deal Builder rollout pause in late 2025
Real estate	Agency or network with branches and agents	Agency CRM or transaction software that multi-lists to portals	Seller leads and valuation tools, branch statistics, viewing requests, mandate support	The portal or its parent owning the CRM; agencies withholding supply
General classifieds	Shop or small business selling goods	Shop system, sometimes none	Bulk listing, storefront page, simple statistics, paid visibility	Trader obligations: DSA traceability for every professional account
Jobs	Employer or recruitment agency	Applicant tracking system	Applicant flow into the ATS, budget and pacing controls, candidate matching	Performance billing disputes over clicks and applications

Vertical	The professional	System they already run	What they need most from the portal	Typical tension
Recommerce	Professional reseller	Inventory spreadsheet or reseller tool	Pro channel with trader status, shipping and payout tools	Separating traders from private sellers; Vinted allows commercial selling only through Vinted Pro

Table 5: *The professional and their existing system decide what the portal must provide. In automotive and real estate, the integration surface matters as much as the portal's own tools.*

In real estate and recommerce, pro tooling does more than save the professional time. In **real estate**, supply acquisition is part of the professional's job, so valuation and seller-lead tools are pro tooling: Rightmove describes its Opportunity Manager as predicting "potential vendors from their usage of the Rightmove platform" (Rightmove FY2025 results). In **recommerce**, Vinted gives the reason it routes commercial sellers into a separate Pro channel: "businesses have legal obligations that private sellers don't" (Vinted newsroom, September 2026). Pro tooling there is first a compliance boundary and only then a productivity product.

Failure Modes

The second system of record. The pro portal grows its own listing tables, lead tables and statistics database, filled by nightly jobs. Within a year its numbers differ from billing, from the listing service and from the professional's CRM, and nobody can say which is right. Dealer Hub's nightly reporting database was this pattern in early form.

Statistics that flatter. Raw contacts reported as leads, views as the headline, estimates shown as facts. The professional's CRM eventually disagrees, and the portal's credibility goes with it. Vanity Liquidity, sold to the customer who pays.

Leads that stop at the portal's edge. E-mail leads are exported; calls, bookings and chat are not. The professional undercounts the portal's value and then downgrades, as Autohaus Ruhrblick nearly did.

Feature gating in the front end. "Premium statistics" or "AI replies" are switched on by checks of the package name scattered through the workspace code. A package redesign

then breaks features in unpredictable ways, and no one can prove which customers received what. Entitlement checks belong in Packages, Bumps and Quotas: Products, Pricing and Entitlements, read through one interface.

An API treated as a side project. Breaking changes without notice, no sandbox, error messages only engineers can read. Partners pass the pain to professionals, and professionals blame the portal. The productscape partner persona's fear is exactly this: "breaking changes disrupting merchant inventory".

Neutrality by promise. The group's CRM gets an early-access endpoint "temporarily", a portal analyst gets read access to a CRM dataset "for a one-off analysis", and the ranking model gains a feature that correlates with the publishing software. Each step is small; together they are the end of neutrality, and in the EU they are also differentiated treatment that should have been described.

Rolling out platform-controlled states without consent. A "reserved" flag, an automatic price suggestion or an AI agent that answers buyers on the professional's behalf changes how the professional runs their business. Autotrader's own account of pausing Deal Builder's auto-rollout, holding listening sessions and introducing "request a reservation" is the documented reminder (Autotrader FY2026 results). Per-organization configuration and reversibility belong in the design.

The Chief Architect's View

At group level, pro tooling raises three decisions that brand teams cannot make alone.

What to decide.

- **Which pro capabilities are shared engines and which are brand front ends**. In the language of Same Platform, Different Physics: How Listing Verticals Differ, the lead ledger, the performance read model, the pro API gateway and the developer portal are layer-2 engines; Dealer Hub and Hearthly's agency workspace are layer-4 products. A group that shares the front end forces one vertical's workflow on another's professionals.
- **The pro API as a governed product**: versioning and deprecation policy, certification, rate limits, service levels and an owner with a roadmap that includes partners.
- **The neutrality regime for any pro SaaS the group owns**: the four controls, the evidence pack, the reviewer and the escalation path when a brand asks for an exception.

Who decides. Brand product teams decide their workspaces. The group platform owns the shared engines and the API product. The architecture advice forum decides when a brand asks to bypass an engine or to read data across the neutrality boundary, and its

decision records are part of the review evidence. Group legal and the data-protection officer own the P2B descriptions of data access and differentiated treatment.

What to watch.

- **Reconciliation gaps.** For a sample of professionals, the difference between portal-reported leads and leads received by their CRM. A gap above a few percent is a defect, not a disagreement.
- **Lead export coverage.** The share of leads, by channel, that reached an integration. Channels below full coverage are where the portal's value disappears.
- **Response time and outcome return.** Median first response per professional, and the share of leads with a reported outcome. The first improves liquidity; the second makes ROI measurable.
- **API health as partners see it.** Error rates by error key, breaking changes announced on time, certification lead time.
- **Neutrality incidents.** Every access request that crosses the SaaS boundary, granted or refused, with the reason.

Professionals increasingly reach marketplaces through software and, from 2025–2026, through AI agents. As of 2026, Autotrader and OLX have shipped apps inside ChatGPT, Rightmove has submitted one, and mobile.de's call assistant answers buyers for dealers. **The pro API will be called by agents acting for professionals as well as by CRMs**, which raises the stakes on scoped authorization, explainable outputs and audit trails. AI Across the Listing Lifecycle takes this up.

Questions to Ask

1. For a sample of our largest professional customers, do our lead counts match the leads their CRM received, by channel? If not, where do they diverge?
2. Does every lead channel (messages, calls, bookings, chat, reservations) reach the Lead API and webhooks with a stable identifier and a deduplication key?
3. Are our pro statistics, our lead ledger and our paid-product delivery records computed from the same events?
4. Which numbers on our pro dashboards are observed and which are estimates, and does the dashboard say so?
5. Does our pro workspace own any business record that does not exist in a domain service?
6. Can a professional's software do everything our own pro UI can do, through a documented, versioned API?
7. How do we tell partners about breaking changes, and how long is the notice?
8. Where are package and entitlement checks performed in the pro tooling, and could we list every feature a given customer is entitled to?

9. If we own software sold to professionals, which data can the portal see, which interfaces does that software get that competitors do not, and what evidence would prove our answers to an independent reviewer?
10. Which platform-controlled states or automated actions do our tools apply to professionals' inventory or leads, and can each professional switch them off?

From Supply to Discovery

This chapter closes Part III. The supply chain is now complete: listings come in through forms and feeds, are normalized and enriched, are recognized when they describe the same thing, are published within a freshness target, and are managed by the professionals who supply most of the value. Part IV — DISCOVERY: Matching Intent to Inventory turns to the other side of the market: matching buyers' intent to that inventory. Its first chapter, Search Architecture for Listings: Indices, Freshness and Scale, asks how to build search for millions of short-lived listings with rich filters, and why the index that serves them must never become the system of record.

Sources and Evidence

Company engineering and investor sources

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- Cox Automotive — Cox Automotive, n.d. Marketplace alongside inventory, DMS and CRM software.
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- Rightmove FY2025 results — Rightmove plc, 2026. Outcome claims and Opportunity Manager.
- Zillow Group Form 10-K for FY2025 — Zillow Group, 2026. Follow Up Boss capabilities.
- Scout24 preliminary FY2025 results — Scout24 SE, 2026. Transaction Enablement revenue including the Propstack CRM.
- Pros de l'immo : AVIV Group lance Ultimmo — MySweetImmo, 2022 (trade press). Agent offer with estimation and prospecting tools.
- Correlation lied to us: rethinking product impact with causal inference — OLX Tech, 2026. Self-selection into packages and matched estimates.
- Keeping second-hand at the heart of Vinted — Vinted Group, 2026. Commercial selling only through Vinted Pro.

Standards, regulation and documentation

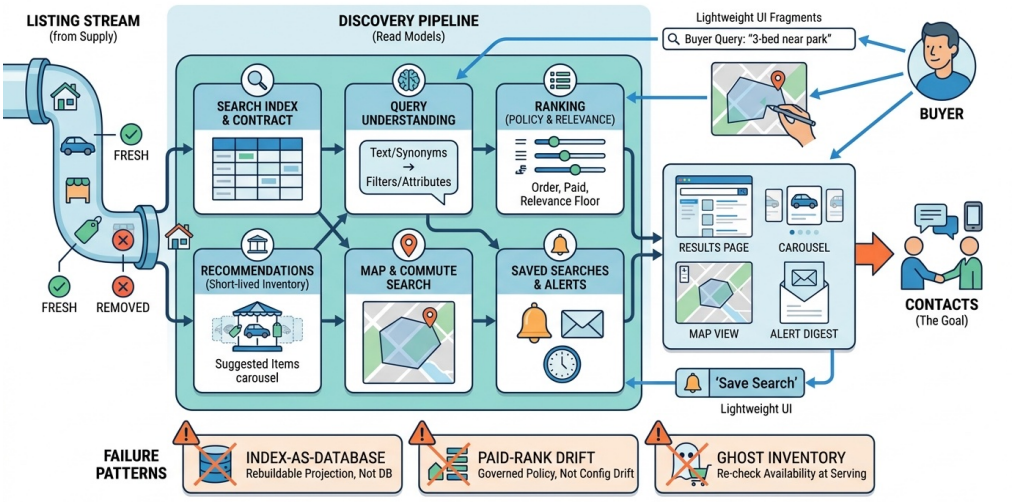
- Regulation (EU) 2019/1150, Article 7 — EU text as adopted, 2019. Differentiated treatment of the provider's own services.
- Regulation (EU) 2019/1150, Article 9 — EU text as adopted, 2019. Description of data access, including aggregated data.
- Platform-to-business trading practices — European Commission, 2026. Application since July 2020 and the November 2025 repeal proposal.
- RFC 9700: Best Current Practice for OAuth 2.0 Security — IETF, 2025. Baseline for partner tokens.
- Freigabe: Übernahme Carwow durch mobile.de — Bundeskartellamt, 2026. Dealer multi-homing and data export as competitive constraints.

Research and practitioner sources

- Productscape domain models for a general listings marketplace and a real-estate marketplace (author's sibling repository, 2026). Professional and integration-partner personas, jobs, fears and partner integration surfaces. Generic models, not descriptions of any company.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

Part IV — DISCOVERY: Matching Intent to Inventory



DISCOVERY: One Pipeline, Six Surfaces. Judged by Freshness & Contacts.

IN THIS PART, YOU WILL: Build the discovery layer: the search index and its contract, query understanding, two-sided ranking, recommendations for short-lived inventory, map and commute search, and saved searches with alerts.

WHY THE BUSINESS CARES: Nearly every contact starts on a results page, a carousel, a map or an alert. Discovery decides which listings get seen and whether paid visibility is worth buying.

WHY ARCHITECTS SHOULD CARE: Discovery runs the busiest read paths in the company, the ones most tempted to become a database, and couples taxonomy, events, ML models and monetization on every query.

KEY POINTS:

- **Every discovery surface is a read model of the listing stream**. Search and recommendation indices, map tiles and alert matchers must be rebuildable from it, and must honor removals faster than insertions.
- **Structure beats cleverness in listing discovery**. Most queries are filters in disguise. The largest gains come from compiling words, places and saved searches into the marketplace's own categories and attributes; embeddings and language models are layers over that core.
- **Order is a policy with owners**. Ranking, recommendations and alerts decide which sellers get exposure. Separate the relevance model from a versioned ranking policy that bounds paid visibility, and judge changes on contacts, not clicks.

FAILURE PATTERNS THIS PART ADDRESSES:

- **Search as System of Record** — Treats every index as a read model that regular drills prove can be rebuilt.
- **Multiplier Creep** — Moves boosts out of engine configuration into a governed policy with a relevance floor.
- **Ghost Inventory** — Re-checks availability at serving and send time on carousels, map tiles and digests.

Part III ended with supply that is clean, deduplicated and searchable within a freshness target. Part IV turns to **the other side of the market**: the buyer who types three words, draws a shape on a map or saves a search and waits. Listing discovery looks like web search but behaves like a high-write operational system: documents live for days, most queries contain no text, and it is judged by how fast a new listing appears and a sold one disappears.

The argument of this part is that discovery is **one pipeline with six surfaces**, not six products. Each reads the same stream and taxonomy, and each must answer three questions. Is this listing still available? Who decided its position? Did it lead to a contact?

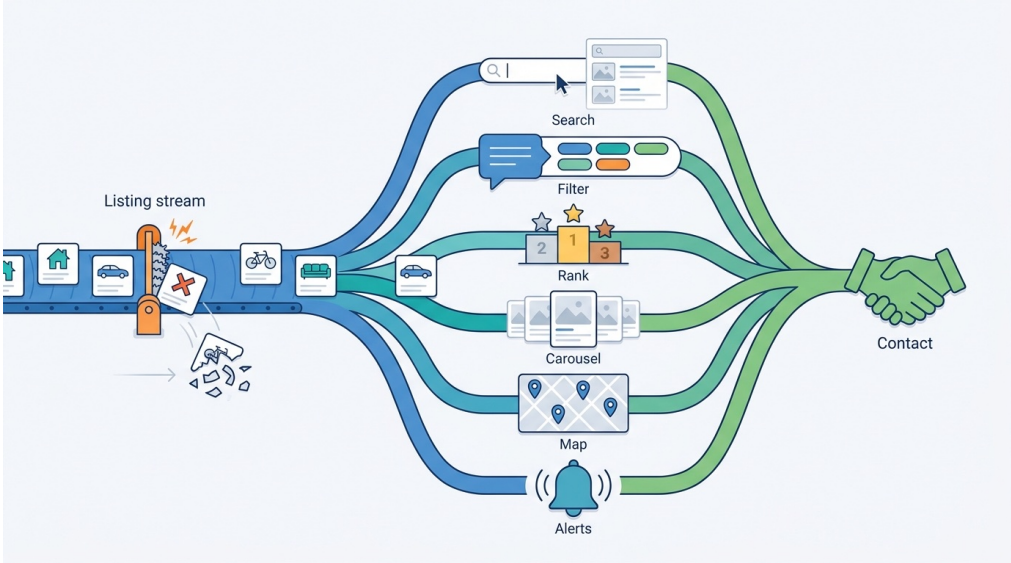


Figure 1: *Discovery is one pipeline with six surfaces: each reads the same listing stream, must forget a sold listing quickly, and is judged by the contacts it produces.*

Chapters in this part

Chapter	The question it answers	Read it when…
Search Architecture for Listings: Indices, Freshness and Scale	How do you build search for millions of short-lived listings with rich filters?	You are choosing an engine, planning a reindex or sizing for the evening peak.
What Did They Mean? Query Understanding and Autocomplete	How do you turn short, ambiguous queries into the right structured search?	Zero-result rates are high, or synonyms live in scattered config files.
Ranking for Two Sides: Relevance, Freshness and Paid Visibility	How should results be ordered so buyers find the right listing and sellers get fair exposure?	Paid listings crowd the first page, or nobody can explain position 3.
Recommendations for Inventory That Disappears	How do you recommend items that may be sold tomorrow?	Carousels show sold items, or new listings never appear in them.

Chapter	The question it answers	Read it when…
Searching by Place: Map, Polygon and Commute Search	How do you build discovery where place is the main filter?	Map traffic dominates, or polygon queries break the latency budget.
Tell Me When It Appears: Saved Searches and Alerts	How do you notify millions of users within minutes when a match appears?	Alerts run on a cron job, or digests advertise listings already gone.

How this part connects

Discovery reads the categories, attributes and location model from Part II — LISTING: Modeling the Thing Being Sold, and the publication pipeline and entity clusters from Part III — SUPPLY: Getting Inventory In. Three later parts constrain or supply it. Part V — TRUST: Keeping the Marketplace Safe puts its risk gate in front of search and alerts. Part VI — MONEY: Conversion and Monetization sells the paid placements that ranking must blend and account for. Part VII — PLATFORM: The Distributed System Underneath supplies the event backbone, experimentation and ML platform these chapters assume.

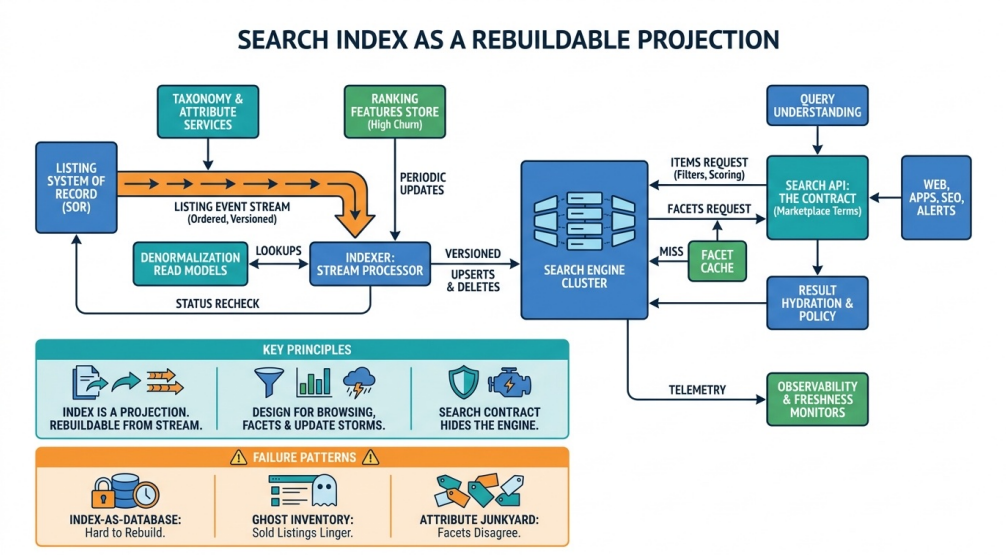
The Veldmark thread

Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), runs three search engines. On Sunday 17 January 2027 Tradeyard NL search browned out at the evening peak, and about 9% of searches failed for 48 minutes. Over 2027, Ingrid Halvorsen, Tomasz Wiśniewski's search platform team and the brands decide on:

- **Ranking policy v1** (June 2027), which allows at most four paid listings in the first 20 positions, admits top listings only above a relevance floor and goes live for Tradeyard NL and BE on 1 December 2027;
- a **rebuilt Motorvane similar-vehicles carousel**, which cuts new listings with no carousel impression from 41% to 8%;
- a **shared search platform on OpenSearch** (14 September 2027), with every application moving to Listing Search API v1 first and Motorvane staying on Elasticsearch until a 2029 review;
- a **geo query compiler** for Hearthly's draw-your-area search;
- a **group saved-search service** targeting five minutes p95 from searchable to alert for 46 million saved searches.

By the end of Part IV you should be able to say, for each surface, how fast it forgets a sold listing and who owns the order it shows. Begin with Search Architecture for Listings: Indices, Freshness and Scale.

18. Search Architecture for Listings: Indices, Freshness and Scale



IN THIS CHAPTER, YOU WILL: Learn how to design the search layer of a listing marketplace: the listing document, index topology, indexing pipeline, freshness and removal latency, facets, caching, reindexing, and capacity planning for evening peaks. Along the way you will follow Veldmark's decision to move from three search engines to one search platform.

WHY THE BUSINESS CARES: Most buyer sessions pass through the results page, so search latency, freshness and availability decide whether supply turns into contacts. A listing that appears six minutes late, or a sold car that stays visible for a day, costs both sides of the market.

WHY ARCHITECTS SHOULD CARE: The search index is the busiest read model in the company and the one most tempted to become a database. Choices about document shape, topology and update paths are expensive to reverse once ranking, facets, alerts and SEO pages depend on them.

KEY POINTS:

- **The index is a read model, not a record**. Build every search document from the listing event stream and the listing system of record, version it with a monotonic sequence, and prove regularly that you can rebuild the whole index from those sources. If you cannot, the index has quietly become the database.
- **Design for browse, facets and update storms, not for text relevance alone**. Most listing traffic is category, location and price filters with counts. Most writes change price, bump time, status or counters rather than text. Keep volatile fields off the expensive update path, put facets on their own path, and measure removal latency separately from insertion latency.
- **Put a contract in front of the engine**. A stable search API owned by a platform team, with engine adapters behind it, lets a group run two engines during a migration, change engines later and keep ranking, synonyms and schemas under marketplace control. Engine choice matters less than this seam.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Search as System of Record** — Search documents that hold data found nowhere else, or that can only be rebuilt by a week-long batch job, turn every mapping change into a risky migration.
- **Ghost Inventory** — Removal is the update teams measure least, so sold and fraudulent listings linger in results, facets and counts.
- **The Attribute Junkyard** — Thousands of loosely defined attribute keys mapped straight into the index explode the mapping and make facets disagree with filters.

Part III ended with a listing that From Submitted to Searchable: The Publication Pipeline declares searchable within a freshness target. Part IV asks how buyers find it, and the search index is where finding begins: it is the first and busiest of the discovery read models, and the rest of Part IV runs on the document, the contract and the indexing path this chapter defines. What Did They Mean? Query Understanding and Autocomplete turns text into queries against that contract, Ranking for Two Sides: Relevance, Freshness and Paid Visibility orders what it retrieves, and Searching by Place: Map, Polygon and Commute Search and Tell Me When It Appears: Saved Searches and Alerts query the same index by area and in reverse.

A listing search engine looks like web search and behaves like a high-write operational store. Its documents live for days or weeks, not years. Most of its queries contain no text at all: a buyer picks a category, a city and a price band and scrolls. Its documents change constantly, mostly in fields unrelated to text: a price cut, a paid bump, a “reserved” flag, a seller suspension that fans out to thousands of listings. And it is judged on two freshness

numbers, how fast a new listing appears and how fast a sold one disappears, that matter more to sellers and buyers than another point of relevance.

The argument rests on one principle from System of Record and Read Models: Listing Data Management: **the search index is a read model**. Everything in this chapter, from document versioning to reindexing to capacity planning, follows from treating it as a read model that can be thrown away and rebuilt, and from refusing to let it become the only place where something is true. The chapter covers the **structure** of that layer: the search document, index topology, the write and read paths, reindexing and capacity. Engine choice comes last, because it matters less than the seam in front of it.

Setting the Scene at Veldmark: Three Engines and a Sunday Evening

Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), entered 2027 with three search engines. Tradeyard, the general-classifieds brand, ran **Solr 8**, self-hosted next to its PHP monolith Yardstick, for the Netherlands, Belgium and Portugal, and **Elasticsearch 7** for Poland, a leftover from a 2016 rebuild of the Polish site. Motorvane, the automotive brand, ran **Elasticsearch 8** on Elastic Cloud. Hearthly, the real-estate portal, ran **OpenSearch** that it managed itself on Google Kubernetes Engine. Between them they served 17.25 million live listings, an average of about **9,000 queries per second** and a peak of about **31,000** on Sunday evenings, with Tradeyard carrying about 70% of the load.

The case for change arrived on **Sunday 17 January 2027 at 20:35 CET**. Tradeyard Netherlands' search p99 latency rose from about **400 ms** to about **7 seconds** and stayed there for **48 minutes**; about **9%** of searches failed. The post-incident review, which Tomasz Wiśniewski, then Tradeyard's search lead and later head of the group search platform, wrote with Daan Visser's Yardstick team, found three causes that were harmless on their own. A TV campaign had pushed traffic about 30% above the usual Sunday peak. A full reindex of the Dutch collection, scheduled for Sunday evenings since 2019 because "the office is quiet then", was writing into the live collection and committing every few seconds. And every commit opened a new Solr searcher, which cleared the filter and query-result caches that the Sunday browse traffic depended on. Solr's own documentation is explicit that caches "are cleared after a commit and usually need to be re-populated before their benefit can be seen again" (Apache Solr Reference Guide, n.d.). At peak, warming never caught up.

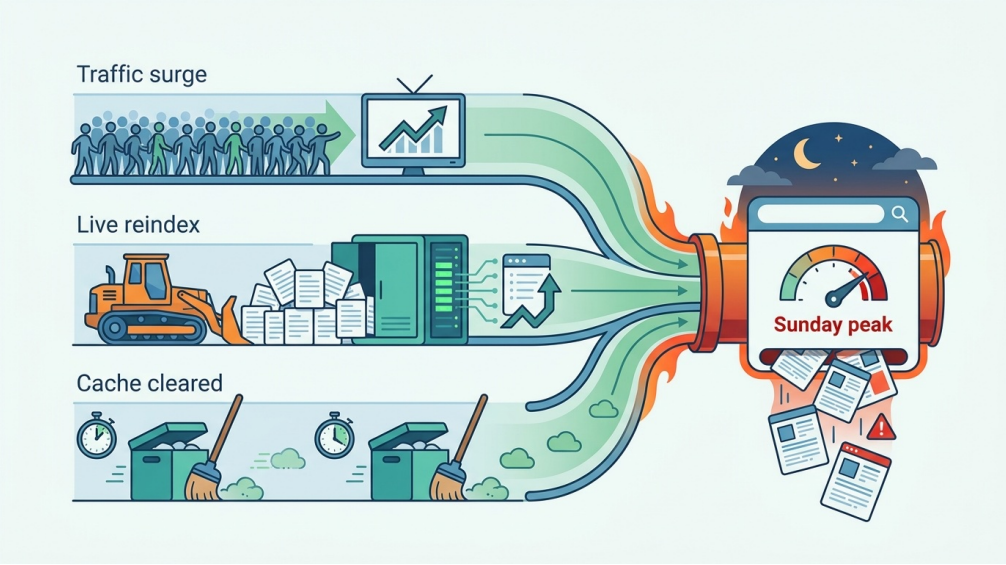


Figure 1: *The Tradeyard NL brownout had three causes that were harmless on their own: traffic 30% above the usual Sunday peak, a reindex writing into the live collection, and commits that kept clearing the caches the browse traffic needed.*

The same review listed a slower problem: a new Tradeyard listing took a p95 of **6 minutes** from submit to searchable. The Rotterdam city bike that Femke listed for €240 on Sunday 7 March 2027 at 19:42 showed it, becoming searchable at 19:48. Part of the delay was the moderation queue, which From Submitted to Searchable: The Publication Pipeline owns. Part was search: Yardstick's indexer ran as a batch job that collected changes and committed them on a timer. Common Ground, Veldmark's €140 million platform program, set a target of **90 seconds** p95 by the end of 2028.

When Ingrid Halvorsen joined as Group Chief Architect in February 2027, the search question in front of her was not "which engine is best?" It was three questions that every multi-brand group meets: *what must be true of any index we run, what should be shared across brands, and what may stay different?* The decision that answered them, taken on **14 September 2027**, is the thread of this chapter.

Concepts: What a Listing Search Layer Is

Search document. The denormalized representation of one listing in the index, assembled from several sources and shaped for queries, not for writes. An **index** (Solr: collection) is a named set of documents with one mapping and one layout of shards and replicas.

Index alias. A stable name that points at one or more physical indices. Clients read and write through the alias, so the physical index behind it can be replaced. In Elasticsearch,

removing and adding an alias target in one request is atomic: the alias "never points to both" targets during the swap (Elastic documentation, n.d.).

Browse query and text query. A browse query is filters with no free text: category, location, price, attributes, sort. A text query adds keywords. Unlike web or retail search, listing search is dominated by browse traffic over short-lived, often unique items written by millions of non-professional sellers. Vinted split "browsing queries (filters without search text)" onto a dedicated index with no analyzed text fields in 2021, because they behaved differently from text search (Vinted Engineering, 2021).

Filtering versus scoring. A filter decides whether a document matches, contributes no score and can be cached as a bit set. Scoring orders the matches. Listing queries are mostly filters, which stay cheap and cacheable until a per-user element enters them.

Facet (aggregation). The counts per value shown next to results: brands, sizes, makes, price bands, number of rooms. Facets are computed over the whole match set, not the page, which often makes them the most expensive part of a results request.

Freshness. Two latencies, not one. **Insertion latency** is the time from a listing becoming publishable to being findable. **Removal latency** is the time from a listing being sold, expired or taken down to disappearing from results, facets, counts and alerts. From Submitted to Searchable: The Publication Pipeline owns the end-to-end freshness service-level agreement (SLA); this chapter owns the index part of it.

Update storm. A burst of document updates triggered by something other than seller edits of content: bulk price changes, seller-level changes that fan out, bump campaigns, counters fed into ranking.

Reindexing. Rebuilding an index's documents, usually into a new physical index with a new mapping, then switching readers to it. It is needed for mapping changes, analyzer changes, shard-count changes and engine upgrades. In Elasticsearch, `number_of_shards` "can only be set at index creation time" (Elastic documentation, n.d.), so even a capacity change can mean a reindex.

Search contract. The API that product applications use to search, expressed in marketplace terms (category, attribute keys, geo, sort keys, ranking profile) rather than engine terms. Engine adapters translate it. Vinted's 2024 account of its Vespa migration credits a "search contract" in a Go gateway with letting the engine change without changing product applications (Vinted Engineering, 2024).

Architecture: The Index as a Rebuildable Read Model

A listing search layer has two paths that meet at the engine: a write path that turns listing changes into documents, and a read path that turns requests into engine queries

and responses. The diagram shows the reference architecture I recommend; it is a design, not a description of any named company's system.

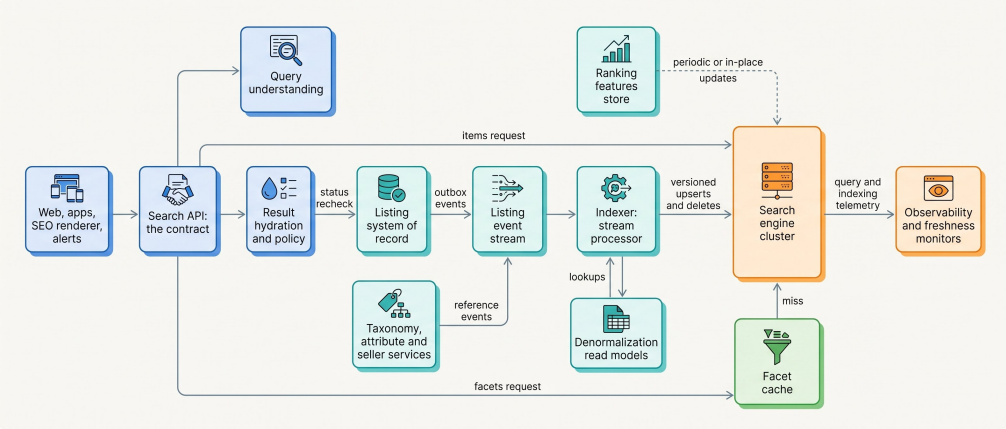


Figure 2: *The search layer as a read model. Documents are built only from the event stream and reference data, the search contract hides the engine from product applications, and facets travel a separate path from results so their cost does not block the result list.*

The table sets what each component owns and what it must not own.

Component	Owns	Must not own
Listing system of record	The listing aggregate and its lifecycle (see The Listing Aggregate: Identity, Lifecycle and Invariants)	Search-specific shaping
Listing event stream	Ordered change events per listing, with a sequence number (see Every Change Is an Event: Event-Driven Listing Architecture)	Business logic of consumers
Indexer	Building documents, coalescing updates, versioning, deletes, backfills	Any data not derivable from its inputs
Read models for denormalization	Fast lookups of seller, category, enrichment and media data	Being a second system of record

Component	Owns	Must not own
Search engine	Matching, filtering, scoring, aggregation, storage of derived documents	Source-of-truth fields; business rules about who may see what
Search API (contract)	Query model in marketplace terms; routing; engine adapters; quotas	Ranking policy decisions owned by marketplaces
Result hydration and policy	Final checks on status and visibility; rendering data	Re-implementing search

Three boundary rules matter more than the components.

First, nothing enters the index that cannot be rebuilt from the stream and reference data. If a field exists only in the index because a script once wrote it there, the index has become a database. Vinted's 2023 rebuild of its indexing pipeline makes the point operationally: after moving to change data capture (CDC) into Kafka and Apache Flink, with denormalization reads from Redis hashes, "reindexing" became publishing messages to a Kafka topic (Vinted Engineering, 2023). A rebuild is then a normal consumer workload, not a special project.

Second, the read path rechecks what is expensive to get wrong. The index always lags the system of record by some seconds. For most fields that is acceptable; for a listing taken down for fraud a second ago, it is not. A cheap status recheck on the 20 to 50 listings of a results page against a fast store, filtering out anything no longer active, closes the gap that no indexer can close. This is a design recommendation, not a documented practice of a specific company, and it costs one batched lookup per page.

Third, the contract speaks marketplace, not engine. Product teams send "category `bikes/e-bikes`, attribute `frame_size` in [M, L], within 15 km of a point, sort by relevance with profile `tradeyard-default`". The adapter turns that into Solr, Elasticsearch, OpenSearch or Vespa syntax. This lets a group run two engines side by side during a migration, and it keeps engine-specific query syntax out of hundreds of call sites.

Designing the Listing Search Document

The standard pattern is one document per listing, denormalized at write time. Elastic's own tuning guidance says to denormalize to avoid joins, noting that "nested can make queries several times slower" and parent-child relations can be much worse (Elastic documentation, n.d.).

The design decision is how to group the fields by source and change rate, not the exact field list.

Field group	Examples	Source	Change rate	Index treatment
Identity and version	listing ID, seller ID, sequence number	System of record	Every change	Keyword; sequence used as external version
Lifecycle	status, published, bumped and expiry timestamps	System of record	Low, but critical	Keyword and date; status is a filter on every query
Taxonomy	category path at each level, taxonomy version	Taxonomy service (see Category Trees That Can Change: Designing Marketplace Taxonomies)	Low; bulk on migrations	Keyword per level
Filterable attributes	frame size, make, model, year, mileage, rooms, energy class	Attribute schema (see Attributes Without Chaos: Schemas, Dictionaries and Validation)	Low	**Typed fields** for attributes used as filters, facets or sort
Long-tail attributes	the thousands of rarely used keys	Attribute schema	Low	One flattened key-value field, searchable by exact value only

Field group	Examples	Source	Change rate	Index treatment
Price	amount, currency, normalized value, price band	System of record	Medium; bulk from dealers	Numeric plus pre-computed band keyword
Location	point, administrative area IDs, precision level	Geocoding (see Where Is It? Location, Geocoding and Privacy)	Low	Geo point plus keyword area IDs
Text	title and description per language	Seller input, translations	Low	Analyzed per language
Ranking features	quality score, seller reputation, engagement aggregates	Feature pipelines	**High**	Kept off the text path; see update storms
Monetization flags	promoted, bump timestamp, product tier	Entitlements (see Packages, Bumps and Quotas: Products, Pricing and Entitlements)	Medium; bursty	Separately updatable fields
Embeddings	image or text vectors	ML enrichment	Low	Vector field, if hybrid retrieval is used (see What Did They Mean? Query Understanding and Autocomplete)

Sparse attributes and the mapping explosion

The hardest part of the document is attributes. Tradeyard has **3,900** distinct attribute keys across its four country taxonomies, and **41%** of them appear on fewer than 0.1% of listings. Mapping each key as its own typed field would collide with the engine's guard rails: Elasticsearch caps mapped fields per index at 1,000 by default, a limit that exists "to prevent mappings and searches from becoming too large" (Elastic documentation, n.d.). Teams hit the limit, raise it, and find that every field also costs memory, merge time and cluster-state size.

There are three designs, each with a known cost.

1. **Typed fields per attribute**. Correct types, range queries and facets. Fails at long-tail scale.
2. **A generic key-value array mapped as nested objects**. Flexible and typed, but each nested object becomes a hidden Lucene document. A listing with 30 attributes becomes 31 documents, and Elastic's documentation calls nested queries "typically expensive" (Elastic documentation, n.d.).
3. **A flattened field** (`flattened` in Elasticsearch, `flat_object` in OpenSearch). Cheap and schema-free, but values are keywords: no numeric ranges, limited aggregation behavior.

The workable design is a **promotion rule**: an attribute becomes a typed field when the category schema marks it as a filter, facet or sort key, and everything else goes into the flattened field. The attribute dictionary in Attributes Without Chaos: Schemas, Dictionaries and Validation is the source of that flag, so promotion is a schema change with a review, not an engineer's guess. Veldmark's group attribute service, live in October 2027, publishes this flag. Of Tradeyard's 3,900 keys, about 610 were promoted to typed fields in the first schema version; the rest stayed searchable only as exact values.

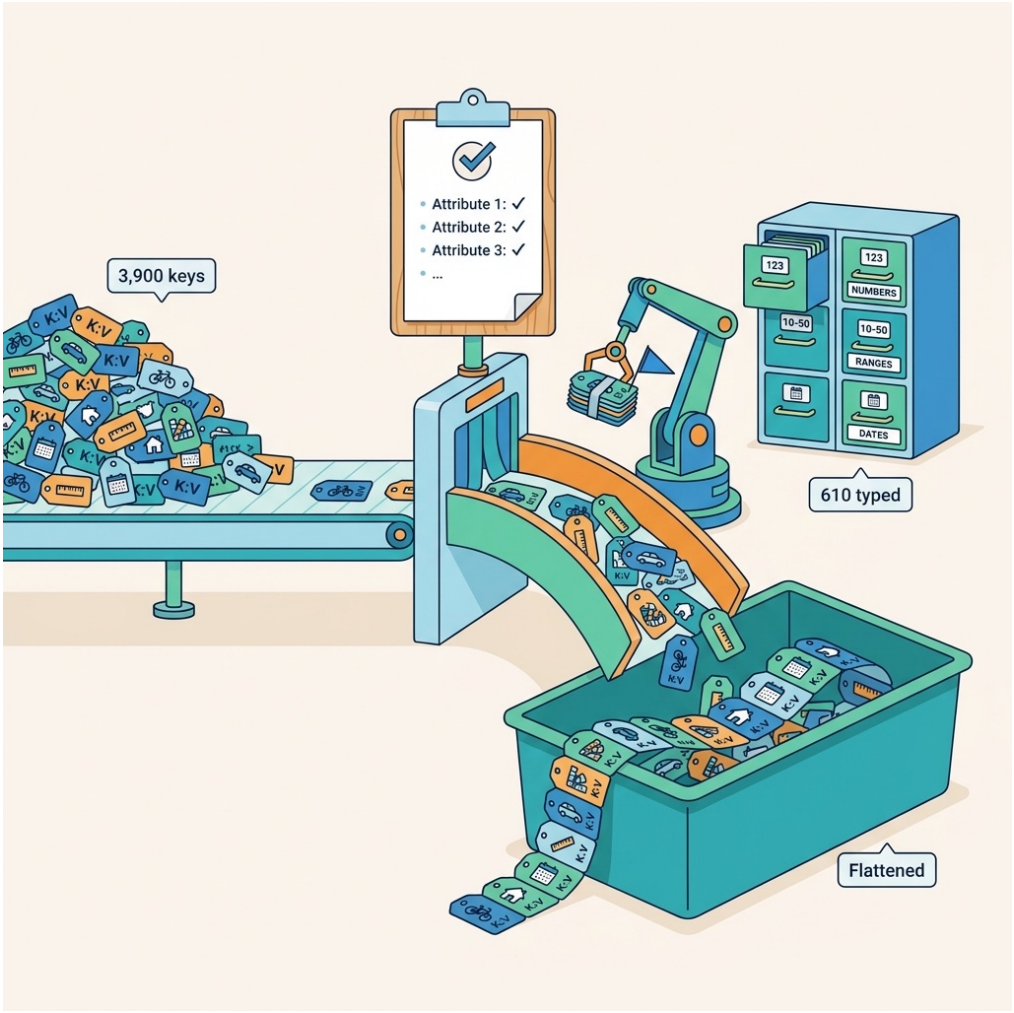


Figure 3: *The promotion rule: only attributes the schema marks as filter, facet or sort keys become typed fields (about 610 of Tradeyard's 3,900), and the long tail stays searchable as exact values in one flattened field.*

Do not let the index absorb the junkyard. If attributes arrive in the index as free strings, facets will show "Gazelle", "gazelle" and "GAZELLE" as three brands. Vinted reported in 2023 that reducing brand cardinality from about 2,000,000 values to 40,000 cut facet p99 by about 15 ms (Vinted Engineering, 2023). Normalization belongs upstream, in From Raw Supply to Usable Listings: Normalization and Enrichment; the index should only see dictionary values.

Index Topology: How to Partition

Index topology decides how many indices exist, what each contains and how they are laid out in shards and clusters. The public record shows most of the options in use, several of them at Vinted at different times.

Partitioning key	Documented example	Strength	Cost
Per country or brand	Vinted's 2026 autocomplete indices are split per country (Vinted Engineering, 2026)	Isolation; analyzers and schemas per market; small blast radius	Cross-border search needs multi-index queries
Per language	Vinted's 2021 Elasticsearch setup used index-per-language	Correct analysis per language	Multilingual countries need several indices or several text fields
Per workload	Vinted's separate count cluster and text-free browse index, 2021	Better caching; noisy workloads isolated	Duplicate storage and indexing
Per time	Vinted's browse index partitioned by creation month, 2021	Recent-first queries touch fewer shards; old partitions cache well	Updates must find the right partition
Per vertical	Common industry practice; schemas differ by vertical	Typed attributes per vertical	Horizontal sites need cross-vertical queries
One global cluster	Vinted's Vespa deployment for all traffic, 2024	Consistency and simpler operations	Larger blast radius

Three questions settle most topology choices. Does any query cross the boundary (cross-border shipping, international property buyers)? Do analyzers differ (language)? Does one partition dominate load (a large home market next to small new ones)? The answer is rarely a single key. For a multi-country group it is typically one index per brand and

country, with per-language text fields where a country is multilingual, on shared clusters sized by load, and separate clusters only where a workload or a market needs isolation.

Belgium shows why a multilingual country needs language fields, not language indices. Tradeyard Belgium serves Dutch and French speakers with one inventory; a listing in Brussels may be written in either language. An index per language would split the inventory and force every query to two indices. One index with `title_nl` and `title_fr` fields, each with its own analyzer, and a language detector at ingest that decides which field receives the text, keeps one inventory and correct analysis.

Shard sizing is a capacity decision, not a default. Vinted's 2017 experience, when uneven placement created hot nodes, is the general lesson: decide shard counts from document counts, update rates and query rates per partition, then revisit them as partitions grow. A shard plan that was right for Tradeyard Portugal's 1.1 million listings is wrong for Tradeyard Netherlands' 6.1 million.

The Write Path: Near-Real-Time Indexing and Update Storms

Freshness mechanics

In Lucene-based engines, a document becomes searchable when a **refresh** (Elasticsearch and OpenSearch) or a **soft commit** (Solr) opens a new view of the index segments. Elasticsearch refreshes every second by default, but only for indices that received a search in the previous 30 seconds (Elastic documentation, n.d.). Solr's guidance is to choose "as long an interval as your application can tolerate" for soft commits, often 15–60 seconds, and to consider disabling caches when the interval is very short (Apache Solr Reference Guide, n.d.).

The trade-off is direct: every refresh creates segments to merge and, in Solr, a new searcher with cold caches. Vinted ran Elasticsearch with a 300-second refresh interval before moving to Vespa, which shows how far a high-traffic site may push the interval to protect query performance. Freshness then has to be bought elsewhere: with an engine that updates in place, or with a read-path recheck for the fields that matter.

Ordering and idempotency

Updates for one listing must be applied in order, and replays must be harmless. The reliable mechanism is **external versioning with a monotonic sequence per listing**: the indexer writes each document with the listing's sequence number as its version, and the engine rejects writes with an older version. Vinted used Kafka offsets for this in 2021, and its experience is the warning that comes with it: the offset is monotonic only within a partition, so changing the partition count broke ordering. A sequence number

owned by the listing aggregate, carried in every event (see Every Change Is an Event: Event-Driven Listing Architecture), survives repartitioning, replays and backfills.

A delete is a write and needs a version too. If the indexer processes a late "price changed" event after a "sold" event without versioning, the sold listing comes back to life. This is one of the most common causes of **Ghost Inventory**, and it is invisible in testing because it needs a race.

Update storms

Listing documents change for reasons unrelated to their text. A dealer group cuts prices on 4,000 cars at 09:00. A seller is suspended and every one of their 900 listings must disappear. A paid bump campaign starts. A seller's rating changes. In Lucene-based engines, Elastic documents that the update API gets the document, applies the change and indexes the result (Elastic documentation, n.d.). Because Lucene segments are immutable, a partial update is a full re-index of that document, plus merge pressure later. Vespa's attribute fields avoid this by updating in memory; its documentation names "real-time inventory and price changes" and feeding click or view counts into ranking as typical uses (Vespa documentation, n.d.).

Whatever the engine, five mitigations recur. They are my design recommendations drawn from this evidence, not a documented recipe from one company.

1. **Keep high-churn ranking features out of the document** in Lucene-based engines. Engagement counters belong in a feature store joined at ranking time, or in a periodic batch refresh, not in per-view document updates.
2. **Coalesce updates per listing** in the indexer with a short window, so ten edits in a minute become one write.
3. **Treat seller-level changes as fan-out jobs** with their own rate limit, except removals, which go first.
4. **Give bulk professional feeds their own lane**, rate-limited separately from private-seller edits, so a dealer's 09:00 price run does not delay a consumer's new listing.
5. **Prioritize removals over everything else**. A removal queue that waits behind 400,000 price updates is how sold listings stay contactable.

Tradeyard's 2027 indexer violated most of these. Yardstick wrote a view counter into the Solr document on every listing view, so of about **9 million** document writes a day, about **6.4 million** were counter updates that never passed through any event stream. They more than tripled the write and merge load for no ranking benefit, because the ranking used a daily aggregate anyway. Removing them was the cheapest performance fix of the year.

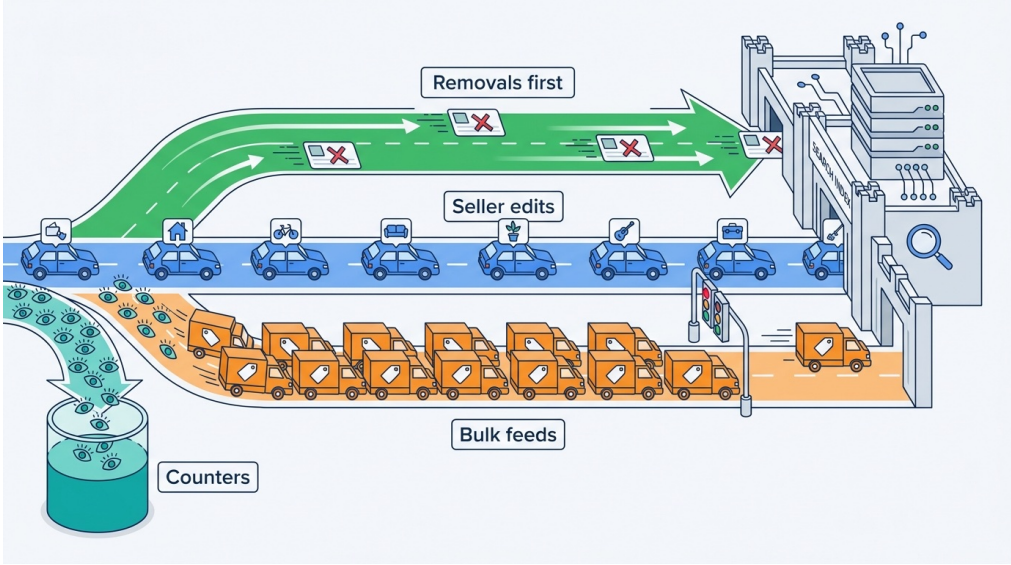


Figure 4: *Update storms are tamed by lanes: removals go first, bulk professional feeds are rate-limited separately from private edits, and high-churn counters stay out of the document altogether, as Tradeyard's 6.4 million daily counter writes showed.*

Measuring freshness

Freshness drifts unless the indexer measures it on every write. It should emit, for every document write, the time between the source event and the moment the engine acknowledged a refresh that includes it, split by operation: insert, update, **removal**. Vinted's figure of 4.64 seconds p99 from Flink to Vespa is the kind of number to know, even if yours is larger. A synthetic probe that publishes a test listing every minute per market, searches for it, then removes it and searches again, catches what metrics miss, such as an alias pointing at a stale index.

The Read Path: Facets, Caching and Latency Budgets

Facets on their own path

Facets are computed over the whole match set, so their cost grows with result counts, not page size. Vinted's 2023 account of faceted search on Elasticsearch is the most concrete public record (Vinted Engineering, 2023): multi-select facets needed one request per facet, each excluding its own filter; splitting facets from item search cost about 20% more CPU but made item search about 71 ms faster; facet responses were cached at about a

50% hit ratio; document collection was capped at 100,000 per shard because counts above 500 displayed as "500+"; and facet latency settled at about 120 ms p99.

Five lessons carry over to any engine. **Approximate counts are a product decision**, and "500+" is acceptable to buyers. **Cap facet cardinality**: show the top 30 brands and a search box, not 40,000 values. **Cache facets on normalized filter keys** and keep per-user elements out of them. **Pre-bucket numeric ranges** at index time. **Serve facets on a separate request**, so the result list renders while counts arrive.

Caching layers

A listing search stack usually has four caches, and each has its own invalidation story.

Cache	Holds	Invalidated by	Risk
Engine filter caches	Bit sets for repeated filters	Refresh or commit (Solr clears caches per new searcher)	Cold caches after frequent commits at peak
Engine query or request caches	Results or aggregations for repeated requests	Refresh	Low hit ratio with short refresh intervals
Application facet and result cache	Responses keyed by normalized query	Time to live	Stale counts; removed listings in cached pages
CDN or page cache	Rendered SEO landing pages	Time to live, purge	Sold listings on landing pages (see Found by Google: Web, App and SEO Architecture for Listings)

The application caches are where freshness and cost collide. A 60-second cache on popular browse pages can absorb a large share of peak load, at the price of showing a sold listing for up to a minute. The status recheck in the hydration step makes that trade acceptable: cached result IDs are cheap to recheck, and cached facet counts that are slightly off are tolerated.

Latency budgets

Every step of a results request spends part of one latency budget. The budget below is Veldmark's p99 target for the shared platform's search API, measured at the API and excluding the client network. It illustrates how to divide a budget; it is not an industry standard.

Step	p99 budget	Notes
Request parsing, routing, quotas	10 ms	In the search API
Query understanding	25 ms	Cached interpretations; see What Did They Mean? Query Understanding and Autocomplete
Items query in the engine	150 ms	Retrieval, filters, first-phase scoring
Re-ranking	40 ms	If a learning-to-rank model runs outside the engine; see Ranking for Two Sides: Relevance, Freshness and Paid Visibility
Hydration and status recheck	25 ms	One batched lookup for the page
Facets	200 ms, in parallel	Separate request; the page renders without waiting
Result list total	**250 ms**	Facets may finish later

The budget also settles arguments: when a team proposes a step that costs 80 ms, such as a cross-encoder re-ranker, it forces the question of what that step displaces.

Zero-Downtime Reindexing

Mappings change with the taxonomy, analyzers with query-understanding work, shard counts with growth, and engines get upgraded; each needs a reindex. The baseline procedure is the alias swap; the diagram shows it with a live update stream, which is where the correctness risk lives.

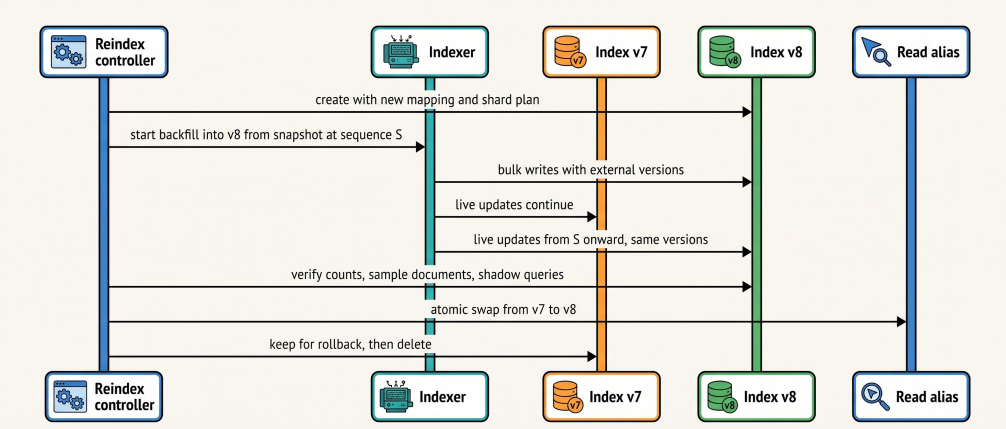


Figure 5: *A reindex with a live stream. Backfill and live updates both write into the new index with the listing's sequence as the version, so the order in which they arrive does not matter. The swap happens only after counts, samples and shadow queries agree.*

Five practices separate a boring reindex from an incident.

1. **Version every write.** With external versions, the backfill can overlap the live stream safely; whichever write is newer wins.
2. **Throttle the backfill and never run it at peak.** Tradeyard's January brown-out was a reindex into the live collection at the weekly peak. A backfill into a new index, throttled and scheduled away from the evening peak, would have been invisible.
3. **Validate before the swap.** Compare document counts per category, sample documents field by field, and run a shadow copy of live queries against both indices, comparing top-k overlap and latency.
4. **Keep the old index for rollback** for a defined window, then delete it, because old indices cost as much as new ones.
5. **Automate it.** Vinted reported in 2021 that, after a year of practice, its alias-based index management became "predictable, automated, and boring" (Vinted Engineering, 2021). Boring is the target.

The ability to reindex from the stream is also the test for **Search as System of Record**. Veldmark's rule, adopted with the 2027 decision, is a quarterly **rebuild drill**: each market's index is rebuilt from the event stream and read models into a fresh index, compared with the live one, and the differences explained. A field that differs and cannot be explained is a field that only exists in the index.

Capacity Planning for Evening Peaks

Listing traffic is peaky and the peaks are predictable. Rightmove reported in 2021 that its busiest minute was "8:48 pm on Wednesday during the final ad break on prime-time

TV" (Google Cloud Press Corner, 2021); the source is more than three years old, and the pattern may have shifted since. Evening leisure browsing and marketing campaigns produce sharp peaks, so capacity planning belongs on the same calendar as the marketing plan. Veldmark's own peak falls on Sunday evenings; that is a property of the fictional group, not a documented industry fact.

Planning from user queries understates engine load. Each results page can generate several engine requests: the items query, one facet request per multi-select facet (as Vinted described), a count, sometimes a sponsored-listings query. Veldmark's own measurement in 2027 was an average of **2.6 engine requests per Tradeyard results page**. At a group peak of 31,000 queries per second with Tradeyard carrying about 70%, Tradeyard alone generates about 21,700 search API requests and about **56,000 engine requests per second** at peak.

Five practices follow; they are simple, if not cheap.

- **Size for the peak plus campaign headroom**, not the average. Veldmark sizes Tradeyard for 1.5 times the trailing peak, because its January campaign added 30% on its own.
- **Replay real traffic in load tests**, with the real mix of browse, text, facets and deep pagination, while the indexer runs at peak write rates. A load test without writes measures a different system.
- **Ban maintenance at peak**. Reindexes, large merges, node rotations and bulk backfills run in a window the platform publishes.
- **Define degradation modes in advance**: drop facets to cached-only, cap pagination depth, serve cached browse pages longer, shed bot traffic first. Sunday Evening Peaks: Reliability, Performance and Cost owns degradation design; the search platform must expose the switches.
- **Track cost per thousand searches** by market. It is the number that makes capacity decisions comparable across brands.

Choosing an Engine: What the Public Record Shows

Most listing marketplaces that have published their search engines run Lucene-based engines: Elasticsearch, OpenSearch or Solr. The best-documented exception is Vinted's move to Vespa. The table shows what is documented, with dates; a company is missing because no primary source was found, not because it runs nothing.

Company	Engine, as documented	Evidence date	What the source supports
Vinted	Elasticsearch from 2015, **Vespa** for item search from November 2023 and facets from April 2024	2024	Engine history, topology, latency and freshness figures (Vinted Engineering, 2024)
leboncoin	Elasticsearch cluster of about 90 nodes plus approximate nearest-neighbor retrieval and a neural re-ranker	2023	Press report of a conference talk (Le Monde Informatique, 2023)
Rightmove	Elasticsearch since 2014, on Elastic Cloud on Google Cloud; percolators for alerts	2025	Vendor case study (Elastic, 2025)
Homegate (SMG, Switzerland)	Elasticsearch; percolator-based alerts	2024	Vendor blog with a guest author (Elastic, 2024)
Airbnb	Apache Lucene-based retrieval with an inverted-file (IVF) vector index used as a filter	2026	Research paper (Abdool et al., 2026)

Licensing no longer separates Elasticsearch from OpenSearch: both are now available under open-source terms with different governance. Elastic moved Elasticsearch away from an open-source license in 2021, which produced the AWS-led OpenSearch fork; in August 2024 Elastic announced it would add the GNU Affero General Public License (AGPL) as a third license option next to its Elastic License v2 and the Server Side Public License (SSPL) (Elastic, 2024). In September 2024 AWS transferred OpenSearch to the newly formed OpenSearch Software Foundation under the Linux Foundation (Linux Foundation, 2024). The practical differences for a marketplace are managed-service options, feature roadmaps (vector, hybrid and learning-to-rank tooling) and the team's

skills.

Vinted's migration: the best-documented case

Vinted's numbered "Search Scaling" series is the most useful public record of a listing search estate evolving over a decade.

- **2017**. One Elasticsearch cluster served ten countries with 46 indices; uneven shard placement caused hot nodes, and re-sharding to about 2.5–3 million documents per shard cut weekly slow queries from about 7 million to under 200,000 (Vinted Engineering, 2017).
- **2021**. Kafka Connect wrote documents to Elasticsearch using the **Kafka offset as the external version**, which made writes order-safe until a partition-count increase broke per-listing ordering (Vinted Engineering, 2021). Browse queries moved to a dedicated text-free index partitioned by creation month, with cache hit ratios above 90% (Vinted Engineering, 2021).
- **2023**. Delta jobs every seven minutes, with week-long backfills, were replaced by CDC, Kafka and Flink; new items became visible "within a minute" (Vinted Engineering, 2023).
- **2024**. Six Elasticsearch clusters of 20 data nodes became one Vespa deployment of 60 content nodes. Visibility of changes fell from 300 to 5 seconds (4.64 s p99 from Flink to searchable), the feed carried about 10,300 update or remove operations per second, peak load was about 20,000 requests per second under 150 ms p99, and ranking depth rose to 200,000 candidates. Vinted contributed Lucene analyzers to Vespa so language analysis carried over (Vinted Engineering, 2024).
- **2025**. The index passed one billion searchable items, up from about 100 million in late 2019 (Vinted Engineering, 2025).

Vinted's before-and-after figures cannot be credited to the engine alone: they combine an engine change with a hardware change (from 0.5 TB SSD to 3 TB NVMe) and a topology change (from six clusters to one). Vespa's own 2024 benchmark against Elasticsearch is a vendor benchmark.

What the series does show is the order in which the hard problems arrived: shard sizing, update ordering, workload separation, freshness, then one engine for lexical, vector and structured retrieval. Facets were the last workload Vinted moved.

Decision criteria

A group should ask which engine fits its workload, team and cloud, not which is fastest in a benchmark. The criteria I use:

Criterion	Why it matters for listings	Lucene-based (Elasticsearch, OpenSearch, Solr)	Vespa
Partial updates of volatile fields	Price, bump time, status and counters dominate writes	A partial update rewrites the whole document into a new segment; frequent counter updates cost like re-indexing	Attribute fields update in place in memory, visible immediately to ranking, grouping and sorting (Vespa documentation, n.d.)
Multi-phase ranking on the content nodes	Learning to rank over deep candidate sets	Rescoring and plugins; ranking often moves to a separate service	Built-in first, second and global phases (Vespa documentation, n.d.)
Facet maturity	Facets are the heaviest browse workload	Very mature aggregations	Grouping language; Vinted moved facets last
Managed-service options	Small platform teams	Several managed offerings per cloud	Vespa Cloud, or self-managed
Talent and ecosystem	Hiring and support	Large	Smaller
Zero-downtime schema change	Mappings evolve with the taxonomy	Reindex plus alias swap	Continuous reindexing without alias switches, as Vinted reports

None of these is decisive alone. A marketplace whose ranking needs deep candidate sets and in-place feature updates has a strong case for Vespa. A group with several small search teams, a primary cloud with a managed Lucene-based service and a workload dominated by browse and facets has a strong case for staying in the Lucene family.

In Practice: Veldmark's September 2027 Search Platform Decision

By summer 2027, Tomasz Wiśniewski led the new group search platform team in Warsaw and Lisbon. His proposal to the architecture advice forum compared four options.

Option	For	Against
A. Keep three engines; share nothing	No migration	Three skill sets, three on-call rotations, three ranking toolchains; Tradeyard's Solr is tied to Yardstick
B. Elasticsearch on Elastic Cloud for all brands	Motorvane already runs it	Hearthly and Tradeyard Poland would migrate anyway; licensing and cost negotiated per deployment
C. **OpenSearch on Amazon OpenSearch Service** for Tradeyard and Hearthly; Motorvane stays on Elasticsearch	AWS is the group's primary cloud; Hearthly already runs OpenSearch; managed service suits a small team; Tradeyard Poland's Elasticsearch 7 mappings port with modest effort	Two engines remain; Hearthly stays on Google Cloud through 2028, so it calls the platform across clouds until it moves
D. Vespa for all brands	In-place attribute updates and phased ranking fit Tradeyard's update storms and future ranking	Largest migration; few people in the group know it; no managed option on the group's cloud contract

The decision on **14 September 2027** was option C, with three conditions that matter more than the engine. First, every product application moves to the **group search contract**, Listing Search API v1, before its engine changes; Motorvane adopts the contract on Elasticsearch, so a 2029 engine review becomes an adapter change rather than a rewrite. Second, documents are built only from the Common Ground listing event contract v1 on Confluent Cloud, live since 1 July 2027, plus the taxonomy and attribute services; no index field without a source. Third, Vespa was evaluated and deferred, with an explicit trigger for reopening it: if Tradeyard's update storms or ranking depth cannot meet their

budgets on OpenSearch, the 2029 review revisits it.

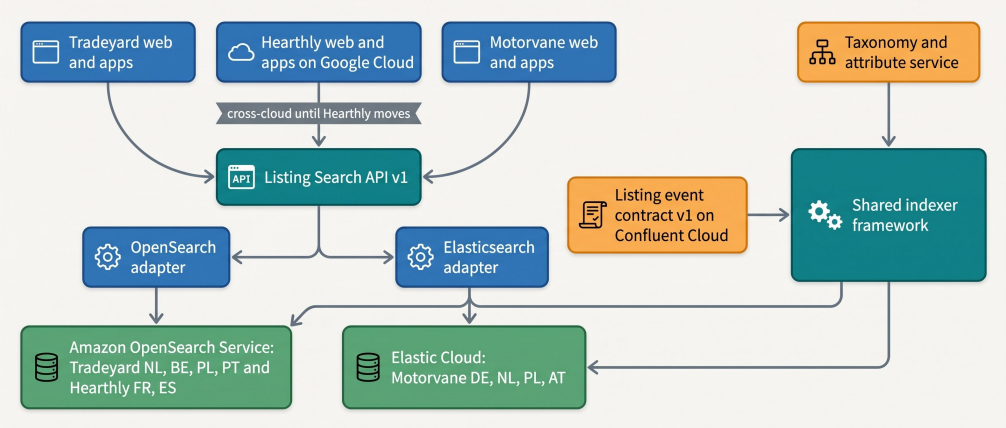


Figure 6: *Veldmark's target search platform after the September 2027 decision. Two engines remain, but every product application calls one contract and every index is fed by one indexer framework from one event contract, so the 2029 engine review becomes an adapter decision.*

The Rotterdam bike shows what the new write path is meant to buy. Femke's listing passed automated moderation at 19:44 in March 2027 and sat in Yardstick's batch indexer until the next commit cycle; it was searchable at 19:48. On the target platform, the `ListingPublished` event reaches the indexer in about 2 seconds, the document builder reads seller, category and media data from read models in about 3 seconds, and the OpenSearch refresh interval is 5 seconds. The index's share of the 90-second publication budget becomes **15 seconds p95**, and the rest belongs to validation and moderation.

Removal gets a tighter target: **10 seconds p95** for sold, removed and taken-down listings, on a priority lane. When the bike sold on 8 March, the removal itself was fast, 18:10 to 18:11; the stale copy that reached buyers the next morning lived in the alert digest, which Tell Me When It Appears: Saved Searches and Alerts owns. That is the point of measuring removal across every copy, not only the index.

The migration order follows risk. Tradeyard Portugal, the smallest market, moves first in the second quarter of 2028, behind the contract, with shadow traffic for four weeks. Tradeyard Poland follows, then Belgium and the Netherlands, timed with Yardstick's search extraction in the fourth quarter of 2028. Hearthly moves last. Analyzer parity is the explicit migration risk: Vinted had to bring its Lucene analyzers into Vespa, and Tradeyard's Solr analyzers for Dutch and French carry fifteen years of tuned synonyms and stopwords. Every analyzer is ported with a test set of real queries whose tokens must match before and after.

Trade-offs

Decision	Option	When it is right	When it is wrong
Engine	Lucene family	Browse and facet heavy; small teams; managed service on your cloud	Deep ranking with frequent in-place feature updates
Engine	Vespa	Large update rates on ranking fields; deep phased ranking; strong in-house search team	Small team; heavy reliance on mature aggregations
Topology	One index per brand and country	Multi-brand groups; isolation; distinct schemas	Heavy cross-border search
Volatile fields	Feature store at ranking time	High churn; Lucene engines	Ranking happens inside the engine and cannot join
Freshness	Short refresh interval	Freshness matters; cache dependence is low	Browse traffic depends on warm caches
Freshness	Longer interval plus read-path recheck	High query volume; removal correctness is what matters	New-listing visibility is the main product promise
Facets	Approximate, cached, separate path	Large marketplaces	Regulated or contractual counts that must be exact

Vertical Differences

Concern	General classifieds	Real estate	Automotive	Jobs
Volume and churn	Highest: millions of listings, many inserts and removals per second at peak	Fewer listings, long-lived, many duplicates across agencies	Mid volume; dealers dominate	Mid volume; fixed expiry dates
Dominant query	Browse plus short text	Location, price, rooms, map	Make, model, year, price, mileage	Text: title, skills, location
Attributes	Thousands of sparse keys	Rich schemas; Zillow cited 200+ attributes per property in 2020 (the current figure is not public) (Zillow Tech Hub, 2020)	Dense, well-defined reference data (makes, models)	Moderate; text matters most
Update storms	Bumps, private edits, suspensions	Agency feed refreshes	Dealer bulk price changes via feeds	Bulk reposting by recruiters
Duplicates	Reposts	Same property from several agencies; collapse at query time (see One Property, Three Listings: Duplicates and Entity Resolution)	Same car on several dealer sites	Same job via several agencies

Concern	General classifieds	Real estate	Automotive	Jobs
Freshness emphasis	Insertion and removal both	Removal of let or sold homes; new listings for alerts	Removal after sale; feeds often lag	Expiry
Geo	Radius around the buyer	Polygon and map dominate (see Searching by Place: Map, Polygon and Commute Search)	Radius with delivery options	Commute

The table explains why one engine can serve several verticals while one document schema cannot. Hearthly's documents carry a property-cluster ID from entity resolution so results can collapse three agency listings of the Lyon apartment into one card; Motorvane's carry decoded vehicle attributes and dealer IDs; Tradeyard's carry the promoted and flattened attribute fields. The platform shares the engine, the indexer framework, the contract and the operations, while each marketplace owns its schema.

Failure Modes

Beyond the failures already described (reindexing into the live index at peak, unversioned deletes, repartitioned offsets used as versions, removals queued behind bulk updates), four more deserve a name.

The index as the only copy. A batch job writes a field, say a moderation score, into the index and nowhere else. Months later a mapping change requires a reindex, and the field cannot be rebuilt. The rebuild drill exists to find these fields while they are few.

Per-user elements in shared filters. A blocked-sellers list added to every query destroys filter and facet cache hit ratios. Vinted accepted slightly inaccurate facet counts to keep a per-user filter out of facet queries.

Analyzer drift during a migration. The new engine tokenizes "e-bike" differently and a share of queries returns different results. Port analyzers with test sets of real queries, as What Did They Mean? Query Understanding and Autocomplete describes.

Deep pagination and scraping. Bots page through results thousands deep, and each deep page costs more than the last. Cap pagination depth for anonymous traffic and serve partners through feeds, not search.

The Chief Architect's View

At group level, search is a platform decision with a product boundary running through it. The platform owns the engine, the clusters, the indexer framework, the contract, capacity, freshness monitoring and cost. Each marketplace owns its document schema (through the attribute and taxonomy services), its analyzers and synonyms, its ranking profiles and its experiments. Write that split down, because every later dispute, about who may change a synonym or who pays for a cluster, will be settled by it.

The decisions I would insist on, in order of importance:

1. **The index is rebuildable, and we prove it quarterly**. This one rule prevents **Search as System of Record** and makes engine changes, mapping changes and disaster recovery the same capability.
2. **Product applications talk to a contract, never to the engine**. It is the cheapest insurance against a future engine change, and the only way to run two engines during a migration.
3. **Removal latency is a named service-level objective (SLO)**, measured per market and per removal reason, with a priority lane. Insertion latency matters to sellers; removal latency is what protects buyers.
4. **The attribute dictionary decides index fields**. No engineer maps a field that the schema does not flag.
5. **Maintenance windows avoid the peak**, and the peak is planned with marketing.

Who decides what: the group search platform team decides the engine, topology and operations, with the architecture advice forum reviewing engine and contract changes. Marketplace search teams decide schemas, analyzers and ranking within the contract. The CTO arbitrates cost allocation, which should follow cost per thousand searches by market.

What to watch over time: freshness percentiles per market and operation, the share of engine load from bots, cost per thousand searches, the number of index fields without a source, cache hit ratios at peak, and the age of the oldest unported analyzer. Revisit the engine decision on a fixed cycle, as Veldmark does in 2029, with the evidence the contract makes cheap to gather: shadow traffic against a candidate engine behind the same API.

Questions to Ask

1. If the search cluster were lost tonight, how long would it take to rebuild every index from the event stream and the system of record, and when did we last do it?
2. Is there any field in the index that exists nowhere else?
3. What are our insertion and removal latencies, at p95, per market and per removal reason, and who is alerted when they drift?
4. Which document updates are counters or other high-churn values, and do they need to be in the document at all?
5. How many engine requests does one results page generate at peak, and which of them can be cached or approximated?
6. What is the version on each document write, and does it survive a change in stream partitioning?
7. Which product applications build engine-specific queries directly instead of calling the search contract?
8. What maintenance can run during the evening peak, and what stops it from doing so?
9. How are new attribute keys promoted to typed index fields, and who approves it?
10. What would we need to see to reopen the engine decision, and how would we test a candidate without changing product code?

From Engine to Intent

A search layer that is fast, fresh and rebuildable still has to understand what people type into it. A buyer in Utrecht who types “ebike” and one who types “elektrische fiets” mean the same thing, and neither has told the engine which category, which attributes or which language they mean. That translation from words to structured queries is the subject of the next chapter, What Did They Mean? Query Understanding and Autocomplete.

Sources and Evidence

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- Near real-time search, Elastic documentation, n.d. — refresh behavior and defaults.
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- Nested field type, Elastic documentation, n.d. — cost of nested objects.
- Tune for search speed, Elastic documentation, n.d. — denormalization guidance.
- Update a document, Elastic documentation, n.d. — partial updates re-index the document.
- Commits and transaction logs and Caches and query warming, Apache Solr Reference Guide, n.d. — soft commits, commit intervals, caches cleared on new searchers.

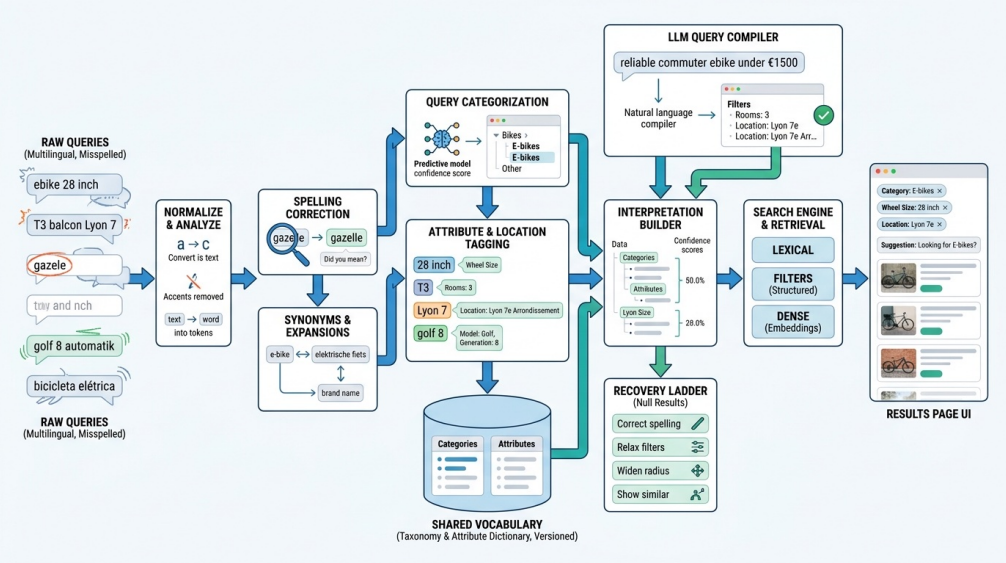
- Partial updates and Phased ranking, Vespa documentation, n.d. — in-place attribute updates; ranking phases.
- Elasticsearch is Open Source. Again!, Elastic (S. Banon), 2024 — license history and the AGPL option.
- Linux Foundation Announces OpenSearch Software Foundation, Linux Foundation, 2024 — OpenSearch governance.

Research and practitioner sources

- Applying Embedding-Based Retrieval to Airbnb Search, Abdool et al. (Airbnb), 2026 — Lucene-based retrieval with an IVF filter.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

19. What Did They Mean? Query Understanding and Autocomplete



IN THIS CHAPTER, YOU WILL: Learn how listing marketplaces turn short, misspelled, multilingual queries into structured searches: language analysis per country, spelling and synonyms, query categorization, attribute extraction, autocomplete, null-result recovery, hybrid retrieval with embeddings, and natural-language search with large language models. You will follow Tradeyard's e-bike queries through the November 2027 bikes split.

WHY THE BUSINESS CARES: A query that returns nothing, or the wrong category, is a buyer who leaves and a seller who gets no contact. Query understanding decides how much of the inventory a buyer can actually reach with the words they use.

WHY ARCHITECTS SHOULD CARE: Query understanding couples search to the taxonomy, the attribute dictionary, the language stack and, increasingly, to ML models and LLM providers. Built as scattered rules and synonyms, it breaks every time a category changes; built as a governed

layer with its own evaluation, it becomes the cheapest relevance lever the marketplace has.

KEY POINTS:

- **In listing search, understanding a query mostly means compiling it into filters.** Buyers type "ebike 28 inch" or "T3 balcon Lyon 7"; the marketplace already has categories and attributes for those words. Mapping words to the taxonomy and attribute dictionary does more for relevance than better text matching, and it must be versioned with them.
- **Every interpretation needs a confidence and a way back.** Apply high-confidence interpretations as filters, weaker ones as boosts, show them as removable chips, and recover from thin results with a defined relaxation order. A wrong filter silently hides inventory; a visible, reversible one costs a tap.
- **Embeddings and LLMs are layers over the structured core, not replacements.** The documented marketplace cases use dense retrieval to fill low-recall searches and LLMs to compile natural language into the existing filter schema, validated before the engine runs. They need the same per-country evaluation, monitoring and owners as any model.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Attribute Junkyard** — Query tagging can only map words to attributes that exist once, with one meaning; free-form attributes leave queries nothing to compile to.
- **The Category Straitjacket** — A query categorizer hard-wired to one version of the category tree turns every taxonomy change into a relevance regression.
- **Set-and-Forget AI** — Query classifiers, spelling models and LLM query compilers decide what buyers see, yet often run without evaluation sets, monitoring or an owner.

A listing query is short, often misspelled, written in whatever language and dialect the buyer speaks, and usually **a request for a filter in disguise**. "ebike gazelle" means category e-bikes, brand Gazelle. "golf 8 automatik" means make Volkswagen, model Golf, generation 8, automatic gearbox. "T3 balcon Lyon 7" means three main rooms, a balcony, Lyon's 7th arrondissement. None of these buyers is asking for text relevance. They are asking the marketplace to recognize its own vocabulary in their words.

Query understanding is the set of steps that turns the raw query string into a structured query: normalized and analyzed text, a predicted category, extracted attribute values and locations, corrections, expansions and, increasingly, a semantic representation.

This is where Part IV's claim that structure beats cleverness becomes a design: the categories and attributes that queries compile to come from the taxonomy and attribute service of Category Trees That Can Change: Designing Marketplace Taxonomies and Attributes Without Chaos: Schemas, Dictionaries and Validation. It sits between the search contract and the engine described in Search Architecture for Listings: Indices, Freshness and Scale, inside a latency budget of about 25 ms at Veldmark. It hands its output to Ranking for Two Sides: Relevance, Freshness and Paid Visibility, which orders what retrieval finds.

This chapter owns three terms: **query understanding**, **query categorization** (predicting the category a query is about) and **hybrid retrieval** (combining lexical, structured and embedding-based retrieval in one result set). It treats autocomplete as part of query understanding, because a good suggestion is a query the system already understands.

Setting the Scene at Veldmark: Eleven Ways to Say E-Bike

Veldmark is the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build). On **Monday 1 November 2027**, Tradeyard's bike categories (six country leaves in the Netherlands, most of them generic) would be replaced by canonical leaves for city bikes, e-bikes, cargo bikes, kids' bikes and more, the first change to run through the new group taxonomy and attribute service (see Category Trees That Can Change: Designing Marketplace Taxonomies). In early October, Priya Raman, Head of Data & ML, and Tomasz Wiśniewski, head of the group search platform, sat down with a month of Tradeyard Netherlands query logs to answer one question: *when the e-bikes category exists, will the people looking for e-bikes land in it?*

The logs were sobering. Tradeyard NL had about **380,000** live bike listings across all bicycle categories, of which about **214,000** would land in the new city-bike leaf and about **38,000** were e-bikes. About a quarter of the 1.9 million weekly bike searches expressed e-bike intent, in at least eleven forms: "ebike", "e-bike", "e bike", "e-fiets", "efiets", "elektrische fiets", "elektrischefiets", "fiets met trapondersteuning", "elektrische damesfiets", brand-plus-word forms like "gazelle elektrisch", and, a trap, "speed pedelec", which in the Netherlands is legally a moped with its own license plate, not an e-bike.

The Solr analyzer that Tradeyard Netherlands had used for years treated them very differently. "e-bike" was split into "e" and "bike", and "bike" matched almost every bike listing whose seller had written the English word, so the query returned about 31,000 results, most of them ordinary bikes. "ebike" was one token found in about 2,100 listings. "elektrische fiets" matched about 18,400 listings, because Dutch sellers write it that way but do not write "elektrischefiets" as one word, which returned 140. The same intent produced four result sets that shared less than half their top 50. Contact rates per search differed by a factor of three across the variants.

Belgium added French ("vélo électrique"), Poland "rower elektryczny" and Portugal "bicicleta elétrica". Each country's analyzer had its own history of hand-maintained synonyms that nobody fully owned. The bikes split would make the problem visible: from 1 November, a query that did not resolve to the e-bikes category would miss a category that held every e-bike.

Concepts: From Query String to Structured Query

Query normalization and analysis. Unicode normalization, lowercasing, accent folding, tokenization, stemming and decompounding, per language. Analysis must agree between the query side and the index side, or matching fails silently.

Spelling correction. Mapping a misspelled token to an intended one, either automatically ("auto-correct") or as a suggestion ("did you mean"), against the marketplace's own vocabulary of brands, models, place names and category words.

Synonyms. Equivalences ("e-bike = elektrische fiets") and one-way expansions ("gazelle → bike brand"), maintained per language and market.

Query categorization. Predicting the category, or a ranked list of categories with probabilities, that a query is about. It is a classification problem whose label set is the category taxonomy.

Query tagging (attribute extraction). Recognizing spans of the query that name attribute values ("gazelle" → brand, "28 inch" → wheel size, "T3" → rooms 3) or places ("Lyon 7" → an arrondissement). It is named-entity recognition against the attribute dictionary in Attributes Without Chaos: Schemas, Dictionaries and Validation and the location model in Where Is It? Location, Geocoding and Privacy.

Interpretation. The structured result of all these steps: categories, filters, boosts, corrected text and, optionally, an embedding, each element with a confidence. One query may have several candidate interpretations.

Null and low-result recovery. What the system does when an interpretation returns nothing or too little: correct, relax, widen, broaden, or offer a saved search.

Hybrid retrieval. Running lexical, structured and dense (embedding-based) retrieval for the same request and fusing the results.

LLM query compiler. A large language model (LLM) that turns a natural-language request into the marketplace's existing structured query, which the classic engine then executes.

Architecture: Shared Vocabulary, Local Policy

Query understanding runs inside the search API, before retrieval, for every request that has text. The diagram shows the reference architecture I recommend. Steps that need

models call them through a cache: the head of the query distribution repeats, so the models do their real work on the tail.

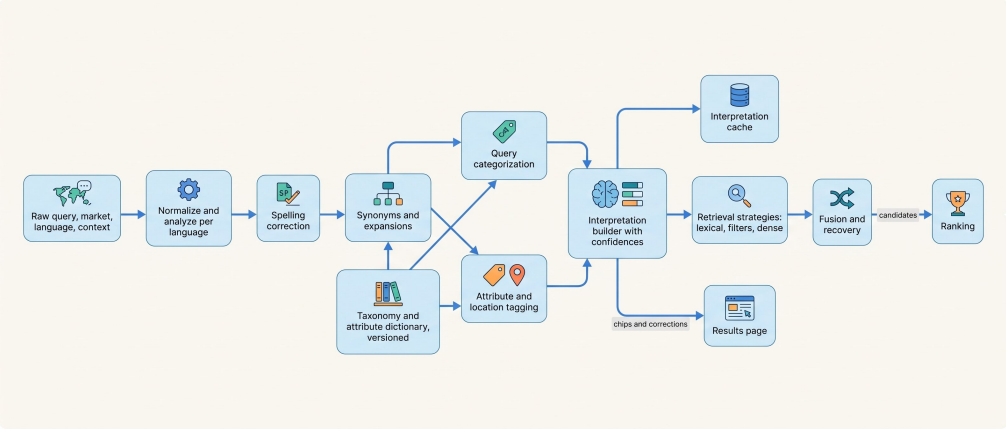


Figure 1: *Query understanding as a compiler. Each step adds structure with a confidence, the taxonomy and attribute dictionary supply the vocabulary, and the interpretation is shown back to the buyer as chips and corrections so that wrong guesses can be undone.*

Each step has its own owner and its own reason to change.

Component	Owned by	Changes when
Analyzers per language	Group search platform, with marketplace input	Engine migration; language added
Spelling vocabulary	Generated from listings and query logs per market	Daily
Synonyms	Marketplace search teams, per market	Continuously, with review
Query categorizer	Data & ML, per brand	Taxonomy version changes; retraining
Attribute and location tagger	Data & ML, built on the dictionary	Dictionary releases
Interpretation policy (filter, boost or suggest)	Marketplace search teams	Experiments
Evaluation sets and harness	Group search platform	Quarterly refresh

The table encodes a principle: **the vocabulary is shared, the policy is local**. The taxonomy and attribute service gives every brand one dictionary of what words can mean. How aggressively a marketplace turns those meanings into filters is a product decision each marketplace owns and tests.

Language Analysis per Country

Analysis is the source of most silent failures in query understanding. Four mechanisms matter for European listing marketplaces.

Tokenization and hyphens. Product and model names are full of hyphens, digits and units: "e-bike", "1.5 TSI", "28-inch", "T3", "iPhone 13". A tokenizer that splits on every hyphen turns "e-bike" into "e" and "bike"; one that never splits leaves "e-bike" unequal to "ebike". The reliable approach is to index and query several forms: the original token, the joined form ("ebike") and the split parts with lower weight. Units and model codes need explicit patterns: a Dutch seller may write "28 inch", "28"" or "28 duim" for the same wheel size.

Accent and character folding. Vinted's 2026 autocomplete indexes both accented and ASCII-folded forms across 24 languages, so that "z" matches "Zara" and "Žalgiris", while "Ž" matches only the accented form (Vinted Engineering, 2026). The asymmetry is deliberate: a buyer who types the accent has asked for the accented form.

Stemming. Language stemmers reduce "fietsen" to "fiets" and "appartements" to "appartement". They help recall and occasionally hurt precision, which is why brand and model names should be protected from stemming with a keyword list.

Decompounding. Dutch and German build compounds: "damesfiets" (women's bike), "kinderfiets", "Anhängerkupplung" (tow bar). A buyer typing "dames fiets" and a seller writing "damesfiets" must meet. Lucene-based engines provide compound-word token filters; Elastic documents that its hyphenation decompounder "was built for Germanic languages" and keeps only sub-words found in a supplied word list, so "Kaffeetasse" yields "Kaffee" and "tasse" (Elastic documentation, n.d.). The word list is the asset: for a marketplace it should come from the category vocabulary and attribute values, not from a general dictionary.

Multilingual markets. Tradeyard Belgium serves Dutch and French speakers with one inventory. Detect the query language (with a fallback that tries both), analyze with the matching analyzer, and search the language-specific text fields described in Search Architecture for Listings: Indices, Freshness and Scale. Category and attribute tagging should work in both languages against the same dictionary, which is one of the strongest reasons to localize the dictionary rather than the index.

Analyzer parity is a migration risk. When Vinted moved from Elasticsearch to Vespa, it contributed Lucene text analyzers to Vespa so that existing language analysis carried over (Vinted Engineering, 2024). Any engine change, and any analyzer change,

should run a test set of real queries per language and compare tokens before and after. Veldmark's set is 5,000 queries per market, sampled across the head, torso and tail of the query distribution.

Spelling and Synonyms

Spelling correction

The simplest correction is fuzzy matching in the engine. Elasticsearch's `AUTO` fuzziness allows no edits for terms of up to two characters, one edit for three to five, and two for longer terms (Elastic documentation, n.d.). It is cheap and useful as a fallback, but it is blind to the marketplace's vocabulary: it will match "golf" to "gold".

A marketplace speller works better because its vocabulary is the marketplace's own. The design I recommend generates a dictionary per market daily from three sources: tokens in live listings weighted by frequency, tokens in successful queries (followed by a click or contact), and the attribute dictionary's values, including brands and models. A correction is applied automatically only when the original returns few results and the corrected form is both close and much more frequent; otherwise it is offered as "did you mean". Brand names and model codes get protection from correction.

Vinted's autocomplete shows that the expensive step is usually not needed. Its fuzzy cascade runs from exact prefix to edit distance 1 to edit distance 2, stopping at ten unique suggestions, and about 62% of requests resolved at exact prefix (Vinted Engineering, 2026).

Synonyms

Synonyms are where hand-maintained search configuration rots. Three rules prevent it.

1. **Apply synonyms at search time, not index time.** Elastic's `synonym_graph` filter is designed for search analyzers, handles multi-word synonyms correctly, and can be made `updateable` so that reloading search analyzers picks up changes (Elastic documentation, n.d.). Changing a synonym should never require a reindex.
2. **Mine candidates from behavior, approve them by hand.** A useful source is reformulation: buyers who search "x", get little, then search "y" and contact. Candidates go to a reviewer per market, because the cost of a wrong synonym is invisible: "speed pedelec = e-bike" would send moped buyers into bicycles and bicycle buyers past the legal difference.

3. **Prefer structure to synonyms where structure exists.** If "e-bike" and "elektrische fiets" both mean the e-bikes category, the better fix is to tag both to the category, not to make them textual synonyms. Synonyms are for text; categories and attributes are for meaning.

Query Categorization and Attribute Extraction

Predicting the category

Query categorization is usually a text classifier trained on query-category pairs taken from behavior: the categories of listings that buyers clicked or contacted after the query. It outputs a distribution over categories, not a single answer. "golf" on Tradeyard is ambiguous between golf equipment and Volkswagen Golfs in its small car section; "golf tas" (golf bag) is not.

Three properties separate a production categorizer from a demo.

- **It is versioned with the taxonomy.** When a category splits, the categorizer's label set changes. Behavioral labels for the new categories do not exist yet. The bridge is the taxonomy service's mapping events, which state how old categories map to new ones (see Category Trees That Can Change: Designing Marketplace Taxonomies), plus the attribute evidence that tells the classifier which old bike listings were e-bikes. Without that bridge, a taxonomy change is a relevance regression: this is how **The Category Straitjacket** reaches search.
- **It is calibrated.** A probability of 0.9 should be right about 90% of the time, because the interpretation policy acts on thresholds.
- **It is monitored like any decision model.** Veldmark's model governance classes search models as "order" models, which decide what buyers see (see Models in Production: ML Platforms and Use Cases). The categorizer needs an owner, a registry entry and monitors on its prediction distribution, which will move sharply on 1 November.

Tagging attributes and places

Attribute tagging finds spans that name values: brands, models, sizes, rooms, energy classes, fuel types. A dictionary matcher over the attribute dictionary's values and aliases gets far, because marketplace vocabularies are bounded; a sequence-labeling model on top handles ambiguity and context. Location tagging is its own problem: "Lyon 7" is an arrondissement, "Den Bosch" is the common name of 's-Hertogenbosch, and "Essen" is a city in Germany and a municipality in Belgium. The location model in Where Is It? Location, Geocoding and Privacy supplies the gazetteer.

The dictionary is the hard dependency. If the attribute service holds one definition of `bike.wheel_size` with its aliases ("28 inch", "28 duim", "28""), the tagger can map the query to a filter. If wheel size exists as four free-text keys entered by different country teams, there is nothing to map to: **The Attribute Junkyard** defeats query understanding before any model runs.

Verticals change what a tag means. On Hearthly France, "T3" means three main rooms (pièces), which is not three bedrooms; on Hearthly Spain, "3 habitaciones" means three bedrooms. On Motorvane, "golf 8" means the eighth generation, not model year 2008 or eight doors. These meanings belong in the dictionary, per market, not in tagger code.

From interpretation to query: filter, boost or suggest

The policy I recommend maps each interpretation's confidence to an action, and shows every applied interpretation to the buyer.

Confidence	Action	What the buyer sees	Risk if wrong
High (for example ≥ 0.85, calibrated)	Apply as a **filter**	A removable chip: "Category: E-bikes ×"	Hides relevant inventory until the chip is removed
Medium	Apply as a **boost**	Results lean toward the interpretation; a suggestion chip	Mild ordering errors
Low	**Suggest** only	"Looking for e-bikes?"	None beyond screen space
Conflicting interpretations	Show **both** as options, or blend	Two category suggestions	Choice cost for the buyer

Two invariants keep the policy honest. First, **adding an interpretation must never increase the result count**. Vinted reported in 2025 that approximate retrieval could make results inconsistent with filters, and set the goal that adding a filter should not increase the number of results (Vinted Engineering, 2025). Second, **an interpretation the buyer removes is a label**: a removed chip is the clearest negative feedback the categorizer will ever get, and it should flow back into training.

Null and Low-Result Recovery

A null-result page is a failed match between demand and supply. Some are genuine (nobody sells a cargo bike within 5 km of a small village tonight); many are artifacts of misspelling, over-eager filters or vocabulary mismatch. The recovery ladder should be explicit, ordered from least to most change to what the buyer asked.

Step	Action	Example
1	Correct spelling	"gazele" to "gazelle"
2	Drop the least-confident machine-added interpretation	Remove an inferred color filter
3	Fall back from all-terms to most-terms text matching	Match three of four words
4	Widen the location radius in steps	10 km to 25 km, labeled
5	Broaden to the parent category	E-bikes to Bikes
6	Fill with dense retrieval	Visually or semantically similar listings, labeled "similar"
7	Offer a saved search	"Tell me when one appears" (see Tell Me When It Appears: Saved Searches and Alerts)

The ladder never removes a filter the buyer chose without saying so. Airbnb has published work on augmenting search with recommendations when searches are too narrow (Airbnb Tech, 2025); the general lesson is that a thin result set is an opportunity to show close alternatives, labeled as such.

Step 7 matters for liquidity: a buyer who saves a search after a null result is standing demand that alerts can serve when supply appears.

Autocomplete and Query Suggestions

Autocomplete is the earliest point at which the marketplace can steer a query toward something it understands. Vinted's April 2026 description is the most complete public account of a marketplace autocomplete system (Vinted Engineering, 2026):

- **Candidates**. About 125 million scored suggestions, regenerated twice a week across more than 50 country-language combinations, built from product metadata (category, brand, color, attributes) and search logs. Query-log candidates were about 2% of the pool but drove about half of suggestion clicks.

- **Ranking**. An offline score based on popularity, sell-through rate and suggestion click-through, normalized per country, language and first letter; then online two-phase ranking in Vespa, with a LightGBM LambdaRank model over 63 features, including per-user features from a feature store, re-ranking the top 20 per node.
- **Matching**. Edge n-grams precomputed at index time cut matching p99 from about 220 ms to 25 ms, with the fuzzy cascade described above.
- **Serving**. Per-country indices in three European data centers; about 4,700 queries per second at 31 ms p99. Cutting the client debounce from 350 ms to 100 ms increased usage.
- **Results**. Suggestion use rose from under 8% to over 17% of search sessions; the learning-to-rank change added about 8% suggestion click-through.

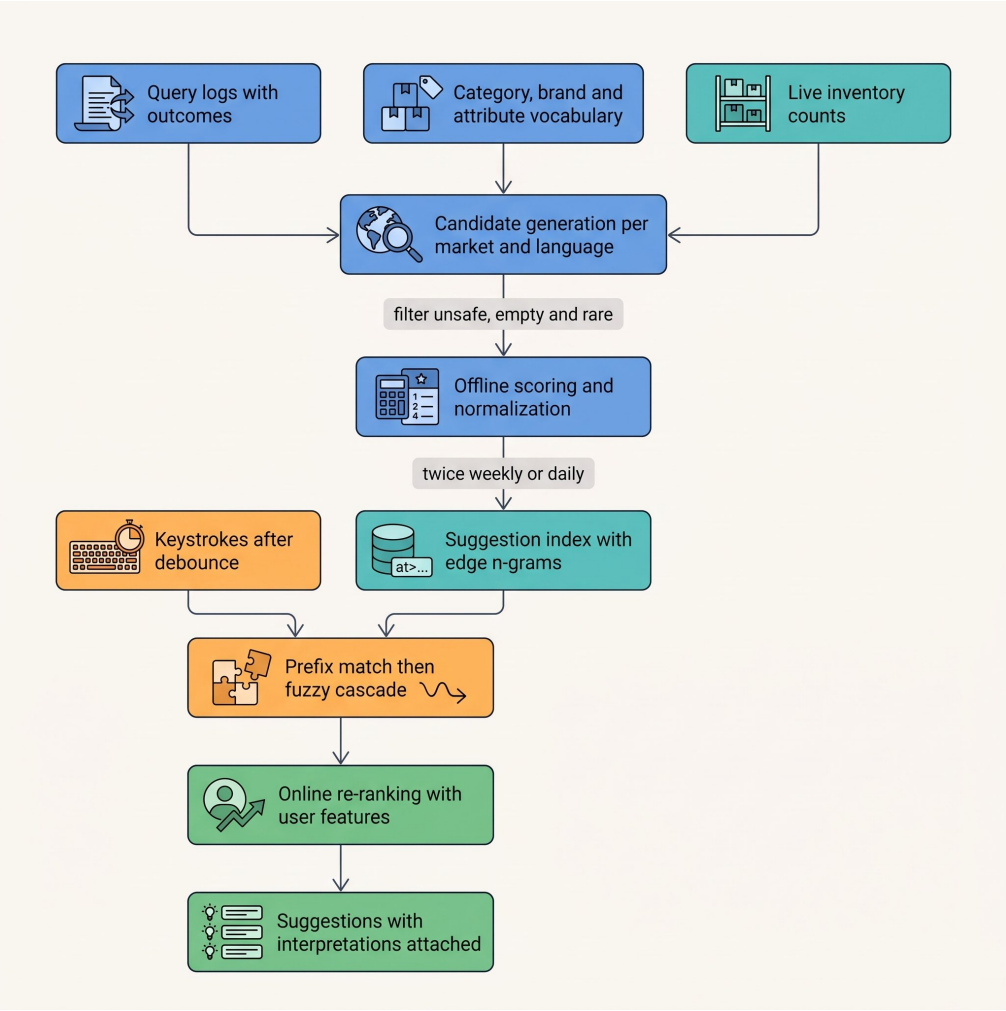


Figure 2: *Autocomplete as two pipelines. An offline job builds and scores candidates from logs, vocabulary and live inventory; an online path matches prefixes and re-ranks per user within a few tens of milliseconds. Attaching an interpretation to each suggestion turns a click into a structured query.*

Three design rules follow for listing marketplaces.

1. **Suggest only what returns inventory**. A suggestion for a query with no live listings in the buyer's market is a null result delivered faster. Candidate generation should join live counts.
2. **Attach interpretations to suggestions**. "e-bike in Fietsen › E-bikes" is better than "e-bike", because the click lands the buyer in the category with no categorizer guess at all.
3. **Filter the candidate pool for safety**. Query logs contain abusive terms, personal data and the probing queries of fraudsters. A suggestion is the marketplace speaking, not the user.

Semantic and Hybrid Retrieval

Lexical matching fails on vocabulary mismatch ("aquamarine blouse" against listings that say "blue top") and on categories where the photo says more than the text. Dense retrieval, which embeds queries and listings in one vector space and retrieves by approximate nearest neighbor (ANN), addresses both. The public record from 2023 to 2026 shows how marketplaces use it.

- **Vinted, November 2025**. Dense retrieval in main search uses a two-tower model: a frozen multilingual CLIP (Contrastive Language-Image Pre-training) query tower with a trained projection head, and an item tower that fuses brand and category embeddings with precomputed CLIP image embeddings. Hierarchical navigable small world (HNSW) graph indices in Vespa are split by blocs of connected markets, with fallback from approximate to exact search when the 500 ms budget is at risk. Dense matches are merged with keyword matches and metadata-filter "interpretations" using **reciprocal rank fusion (RRF) in the global ranking phase**, with a per-request cap on dense-only matches. Filling **low-recall sessions** first improved metrics. Full rollout came after about 50 A/B tests. The image embeddings for all primary photos took about a month to compute and need about 1 TB of memory (Vinted Engineering, 2025).
- **Airbnb, January 2026**. Embedding-based retrieval chose an inverted-file (IVF) index over HNSW because HNSW memory grew under frequent real-time updates and made combining discriminative filters difficult; IVF cluster IDs act as a normal filter in Apache Lucene, at the cost of lower recall. The listing tower runs as a daily batch, so only the query tower runs online (Abdool et al., 2026).

- **leboncoin, 2023.** A press report of a conference talk describes a CamemBERT-based retriever with ANN in Elasticsearch and a neural re-ranker over the top candidates (Le Monde Informatique, 2023); the primary engineering post could not be verified, and the report is about three years old, so the current system may differ.

The two detailed cases reached **different answers to the same problem**, approximate retrieval under heavy filtering with real-time updates, for documented reasons. Neither is generally right; the choice depends on filter selectivity, update rate and the engine already in use. Recommendations for Inventory That Disappears covers the same trade-off for candidate generation.

Fusion

Hybrid retrieval needs a way to merge lexical and dense result lists whose scores are not comparable. Two families are available in the Lucene world.

Method	How it works	Engine support	When to prefer
Reciprocal rank fusion	Sums `1/(k + rank)` across lists; ignores raw scores	Elasticsearch RRF retriever, `rank_constant` default 60 (Elastic documentation, n.d.); OpenSearch score ranker processor, introduced in 2.19, default 60 (OpenSearch documentation, n.d.); Vespa global phase (Vinted)	Default choice; needs no tuning

Method	How it works	Engine support	When to prefer
Score normalization and combination	Rescales each list's scores (min-max, L2, z-score), then combines (arithmetic, geometric or harmonic mean)	OpenSearch normalization processor, introduced in 2.10, `min_max` and `arithmetic_mean` by default (OpenSearch documentation, n.d.)	When weights are tuned per market against judged data

Fusion produces candidates; it does not replace ranking. The fused set goes to Ranking for Two Sides: Relevance, Freshness and Paid Visibility, which applies the learning-to-rank model, freshness and paid-visibility policy.

Where dense retrieval belongs in a listing marketplace

My recommendation follows the evidence: **use dense retrieval to fill recall gaps, not to replace structured retrieval**. Run lexical and structured retrieval first; add dense matches when the session is low-recall, when the query has no strong interpretation, or in image-led categories such as fashion and furniture; cap the share of dense-only results; and make sure dense matches respect every filter the buyer set.

Dense retrieval has real costs: embeddings for every listing, memory for the ANN index, an embedding call per query, and an evaluation burden, because dense results fail in ways lexical results do not. Vinted's tokenizer quirk that flooded a brand-filtered query for "zx750" with about 7,000 extra dense matches is a reminder that the failure modes are new.

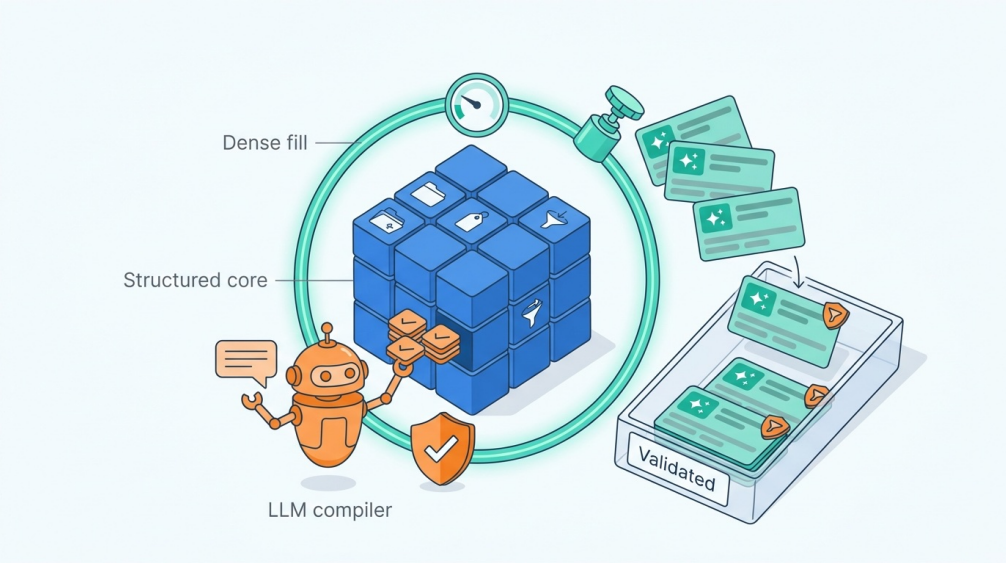


Figure 3: *Embeddings and language models are layers over the structured core: dense retrieval fills low-recall sessions under a cap and every buyer filter, and the LLM compiles language into filters that the attribute dictionary validates.*

Natural-Language and Conversational Search with LLMs

Since 2023, several listing marketplaces have published natural-language search features, and **none of the documented cases replaces retrieval with generation**. They use the LLM to compile language into the existing structured query, or to orchestrate existing search tools.

Marketplace	Feature (date)	What the LLM does	Documented results or limits
coches.net (Adevinta Spain)	Natural-language car search (2024)	A cheap moderation call classifies the query as valid, unethical or invalid; Claude Haiku on Amazon Bedrock converts it into the site's existing filters with few-shot prompts; the frontend validates and applies them	About 6,000 input and 500 output tokens per query, about 20,000 calls a day, about €19 a day; adoption "still limited" (Adevinta Tech Blog, 2024)
leboncoin	Conversational shopping assistant (2024)	GPT-3.5 Turbo extracts preferences as JSON matching the product search API's schema, with category prediction merged into the prompt; JSON mode at temperature 0.1; history capped at 5–10 turns; evaluation by structured diff of expected and generated JSON	OpenAI's Assistants API rejected because messages would be stored on OpenAI's servers (Adevinta Tech Blog, 2024)
leboncoin (cars)	AI search assistant (2024)	Maps needs to keywords, categories and filters	User research: users "are not aware of the benefits" (Adevinta Tech Blog, 2024)

Marketplace	Feature (date)	What the LLM does	Documented results or limits
OLX	OLX Magic agentic assistant (2024)	An agent plans and calls text, visual and web search tools; learning to rank stays underneath	Latency about 5–10 s against 300–500 ms for classic search, per a third-party talk summary (ZenML LLMOps Database, 2024); the talk lists latency, cost and guardrails as challenges (MLOps Community, 2024)
Zillow	Natural-language search (2023; extended 2024; current state not described here)	Commute, affordability, schools and points of interest as criteria; saveable searches	No architecture published (HousingWire, 2024)
Rightmove	"Ask Rightmove" conversational search (2026)	Not described	Early findings (data to 30 June 2026): about a 40% uplift in average time on site and "roughly double the propensity to send leads", per the company's H1 2026 results (Rightmove plc, 2026)

Model names, prices and feature availability in this table change quickly; treat them as dated evidence, not current product facts.

The LLM as query compiler

In the pattern these cases share, the LLM receives the query, the market, and a compact description of the filter schema (the category list and the attribute keys and values relevant to the predicted vertical, drawn from the attribute dictionary), and returns a structured query. A validator checks every category and attribute value against the dictionary and drops or repairs what does not exist. The classic engine runs the query. The buyer sees the result with the compiled filters as chips.

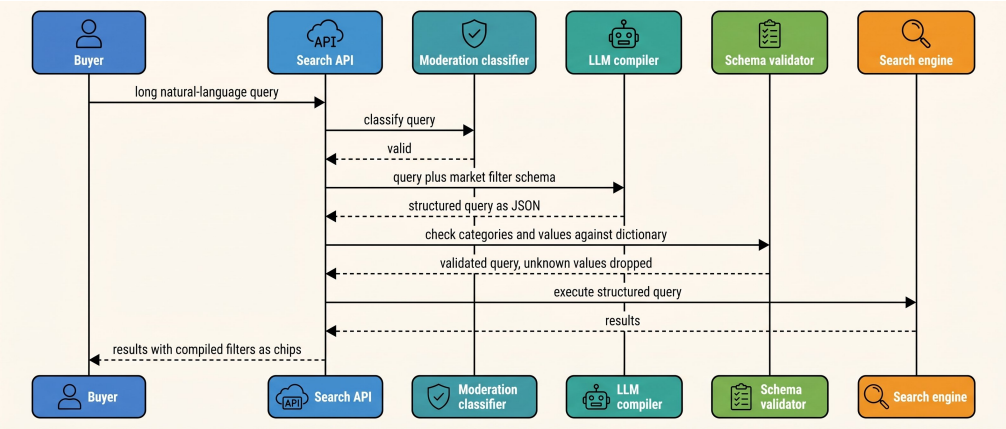


Figure 4: *The LLM compiles, the engine retrieves. Validation against the attribute dictionary keeps hallucinated categories and values out of the query, the engine keeps results grounded in live inventory, and the chips make the compiled interpretation visible and editable.*

The pattern has five properties to defend in a design review. Results are explainable, because the buyer sees the filters. No listing can be hallucinated, because the engine retrieves. Output can be cached by normalized query. Keyword queries never touch the LLM, which bounds cost and latency. And the same structured query can be saved as an alert, which is exactly what Zillow made possible for natural-language searches.

Its limits are latency and data protection. Latency is in hundreds of milliseconds at best, so the compiler should run only for queries that need it, typically long queries or those the classic pipeline cannot interpret with confidence. Query text can be personal data, so prompts and logs fall under the same retention rules as search telemetry, and provider terms decide where text may be processed; leboncoin's rejection of a hosted assistant API over message storage is a documented example of that constraint shaping architecture.

External assistants as a new client

External AI assistants have started to send queries on the buyer's behalf. In two 2026 posts, ImmoScout24 described a Model Context Protocol (MCP) server for its assistant, launched in November 2025, and an app inside ChatGPT over its property data (Scout24 Engineering, January 2026; Scout24 Engineering, March 2026); Rightmove's FY2025 results reported an app submitted to the ChatGPT ecosystem (Rightmove plc, 2026). When an external assistant calls marketplace search, the search contract becomes an external product, with a published filter schema, rate limits, attribution for leads and freshness obligations. The query-understanding layer does not disappear; the external model does part of the compiling, and the marketplace's validator becomes the line of defense. AI Across the Listing Lifecycle covers how Veldmark opens this channel.

Evaluating Query Understanding

Query understanding fails quietly, which makes evaluation the part teams skip. The evaluation harness should be a shared group asset, with market-specific data.

Layer	Offline measure	Online measure
Analysis	Token parity on a fixed query set per language	Not needed if parity holds
Spelling	Precision of automatic corrections on labeled misspellings	Rate of buyers undoing a correction
Categorization	Precision and recall at the filter threshold; calibration	Chip-removal rate; search-to-contact by category
Tagging	Span-level precision and recall per attribute	Chip-removal rate per attribute
Recovery	Share of null results recoverable in replay	Null-result rate; saved searches from null results
Dense retrieval	Recall against a reference, on replayed traffic	A/B tests on low-recall sessions
LLM compiler	Structured diff against expected queries	Adoption, reformulation, contact rate

Two lessons from the public record apply. Airbnb found that offline recall measured on logged data tracked online results poorly, because logs contain only what users were shown; replaying a slice of production traffic to an offline cluster tracked A/B results better (Abdool et al., 2026). And leboncoin's structured-diff evaluation of its JSON compiler, which its authors called deliberately simple, is still better than no evaluation:

comparing expected and generated structured queries is cheap and catches regressions in prompts and model versions.

The query sets must be refreshed on a schedule and after every taxonomy release, because the right answer for "e-bike" changes on the day the e-bikes category appears.

In Practice: Tradeyard's E-Bike Queries Through the Bikes Split

Priya's and Tomasz's plan for 1 November had four parts, each applying a section above.

One vocabulary. The attribute service already held the e-bikes category with its Dutch, French, Polish and Portuguese labels. The team added a query-vocabulary entry: aliases per language ("ebike", "e-bike", "e bike", "e-fiets", "elektrische fiets", "fiets met trapondersteuning", "vélo électrique", "rower elektryczny", "bicicleta elétrica") mapping to the category, and an explicit exclusion for "speed pedelec", which maps to the mopeds category instead. The tagger and the categorizer both read this entry.

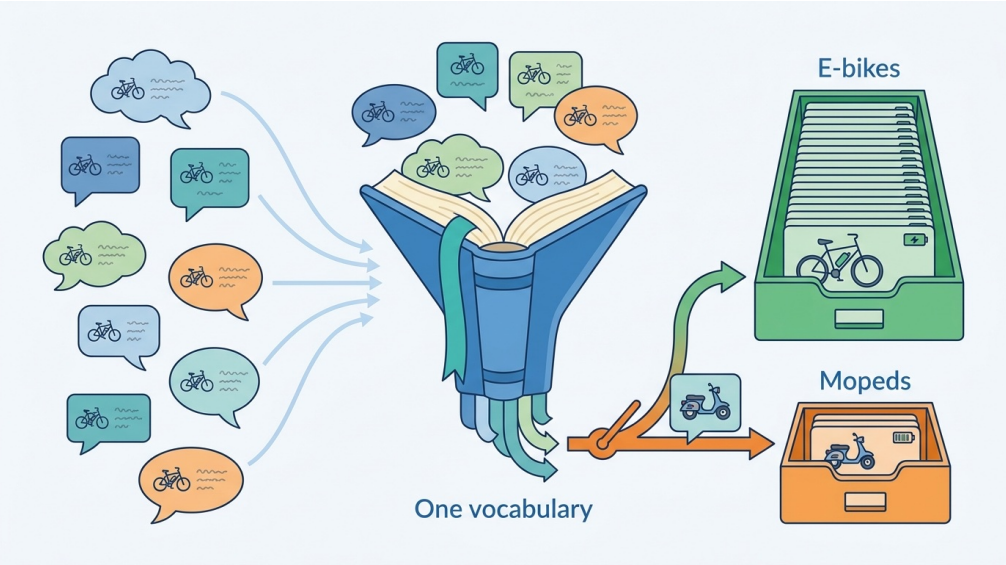


Figure 5: *One vocabulary entry, read by both the tagger and the categorizer, sent 10 of the 11 e-bike query variants to the new e-bikes category and routed speed pedelec, legally a moped, to mopeds.*

Analysis fixes. The Dutch analyzer was changed to index and query the joined and split forms of hyphenated tokens, and the compound word list was generated from category labels and attribute values, so "elektrischefiets" and "damesfiets" decompose. The Solr

analyzer and the target OpenSearch analyzer were checked against the same 5,000-query Dutch test set.

A categorizer that survives the split. Old bike listings were relabeled using the taxonomy service's mapping events and the e-bike attributes from the backfill (motor position, battery capacity), giving the categorizer training labels for the new category before any buyer had used it. The model was entered in the group model inventory as an order-tier model, with a prediction-distribution monitor and Priya's team as owner (the group registry decided on 9 November later made that mandatory for every production model).

Policy and recovery. An e-bike interpretation with calibrated confidence of at least 0.85 is applied as a category filter with a chip; between 0.5 and 0.85 it becomes a boost and a suggestion. If the e-bikes filter returns fewer than 20 results within the buyer's radius, the recovery ladder widens the radius before it broadens to all bikes.

The outcome in the four weeks after the split, against the four weeks before, measured on Tradeyard NL bike searches with e-bike intent:

Measure	Before (October 2027)	After (November 2027)
Query variants landing in e-bike inventory	4 of 11	10 of 11 (speed pedelec correctly excluded)
Overlap of top 50 results across the main variants	under 50%	about 90%
Null or under-five-result rate	6.1%	2.4%
Chip-removal rate on the e-bikes filter	—	3.8%
Search-to-contact rate, e-bike intent	baseline	+11% relative

These are Veldmark illustrations, not industry benchmarks; the shape is what generalizes. The biggest gain came from vocabulary and policy, not from a model. The 3.8% chip-removal rate became a training signal: most removals came from buyers searching "e-bike accu" (battery), who wanted parts, which led to a new interpretation for e-bike parts in the next release.

Two further projects show where Veldmark is going. Hearthly France is testing an LLM compiler for long queries only, about 6% of Hearthly search traffic, with output validated against the French property schema; "T3 balcon Lyon 7 moins de 400k" compiles to rooms 3, balcony, Lyon 7e and a €400,000 price ceiling, and returns the Lyon apartment

at €389,000. Motorvane is extending its tagger with generation codes from its vehicle reference data, so that "golf 8" means the generation of the Essen Volkswagen Golf rather than 2008.

Trade-offs

Decision	Option	When it is right	When it is wrong
Where meaning lives	Synonyms in analyzers	Pure text equivalence; no structure exists	Words that map to categories or attributes
Where meaning lives	Tagging to the dictionary	A governed attribute dictionary exists	Free-form attributes; no dictionary
Interpretation policy	Aggressive filters	Calibrated, well-monitored models; visible chips	Unmonitored models; no chips
Interpretation policy	Boosts and suggestions only	Early models; ambiguous verticals	Filter-heavy verticals where buyers expect structure
Spelling	Engine fuzziness	Fallback; small catalogs	Brand-heavy vocabularies; false matches
Spelling	Marketplace speller	Large vocabularies; many brands and models	No query logs yet
Retrieval	Lexical plus filters only	Structured verticals; tight budgets	Image-led categories; vocabulary mismatch
Retrieval	Hybrid with dense fill	Low-recall sessions; fashion and furniture	Small teams without evaluation capacity
Natural language	LLM query compiler for long queries	Rich filter schemas; long-tail phrasing	Short keyword traffic; strict latency budgets

Decision	Option	When it is right	When it is wrong
Natural language	Agentic assistant	Exploratory, advisory journeys	Main search path; cost per query unbounded

Vertical Differences

Concern	General classifieds	Real estate	Automotive	Jobs
Typical query	Short object words and brands	Place plus rooms, price, features	Make, model, generation, fuel, gearbox	Title, skills, place
Hardest problem	Vocabulary breadth; category ambiguity	Place semantics (districts, neighbor-hoods); room-count conventions per country	Model and trim codes; generations	Title synonyms and seniority
What tagging maps to	Thousands of categories, sparse attributes	Rich property schema	Dense vehicle reference data	Occupation taxonomy
Spelling vocabulary	Brands and colloquial words	Place names	Model codes (“1.5 TSI”, “e-tron”)	Job titles
Natural-language value	Moderate	High: lifestyle and commute criteria	Moderate to high: needs-based car search	High: experience and skills
Dense retrieval value	High in image-led categories	Moderate: photos and descriptions	Low to moderate	High: semantic job matching

Real estate and automotive have the richest structured vocabularies, which makes tagging and LLM compiling most effective there; general classifieds have the broadest vocabulary, which makes categorization and autocomplete the main levers.

Failure Modes

Synonyms as the dumping ground. Every relevance complaint becomes a synonym, until the synonym file is thousands of lines that nobody can explain and every new entry has side effects. Move meaning into the dictionary; keep synonyms for text.

Invisible filters. The system applies a category or attribute filter without showing it. Buyers who were wrong-footed cannot recover, and the team never learns. Every applied interpretation needs a chip.

Taxonomy changes without a categorizer plan. The category tree changes on a Monday; the categorizer still predicts the old categories, or none. Version models with the taxonomy and use mapping events to build labels in advance.

Analyzer drift. An engine upgrade or a well-meant analyzer change splits hyphens differently, and a share of queries quietly changes results. Keep token-parity tests per language in continuous integration.

Unvalidated LLM output. A compiler returns a category or attribute value that does not exist, and the query returns nothing or errors. Validate every output against the dictionary, and log what was dropped.

Autocomplete that suggests nothing. Suggestions built from logs without live counts lead buyers to empty results, or surface offensive or personal strings. Join inventory counts and filter the candidate pool.

Dense results that ignore filters. Approximate retrieval returns listings outside the buyer's filters or inflates counts. Enforce filters on every retrieval path and test the "adding a filter never increases results" invariant.

The Chief Architect's View

Query understanding is a group concern because it is where search meets the group's shared vocabulary. At Veldmark, the decisions I would hold the line on are these.

1. **One vocabulary, many policies**. The taxonomy and attribute service holds what words can mean, in every language, including query aliases and exclusions. Marketplaces own how aggressively they apply interpretations. Marketplaces keep no private vocabularies, and the group does not set their thresholds.

2. **Every model in the query path is an order-tier model**. Categorizers, taggers, spellers, dense retrievers and LLM compilers are registered, owned, evaluated on market-specific query sets and monitored, under the same governance as ranking models (see Models in Production: ML Platforms and Use Cases).
3. **Taxonomy releases include a query-understanding checklist**. No category split ships without updated query aliases, categorizer labels and evaluation sets. This is where The Category Straitjacket is either broken or rebuilt.
4. **LLMs compile; engines retrieve**. Any natural-language feature uses the group's AI gateway (see AI Across the Listing Lifecycle), validates against the dictionary and executes through the search contract. Providers, models and prompts are configuration, so they can change as prices and terms change; Lucía Ortega's data-protection review covers where query text may be processed and how long prompts are kept.
5. **Evaluation is a shared asset**. The group search platform owns the harness and the replay tooling; marketplaces own their query sets and judgments.

What to watch: null-result and low-result rates per market and category, chip-removal rates per interpretation, the share of traffic each layer touches, LLM cost per thousand queries, and the age of each market's evaluation set. A rising chip-removal rate after a taxonomy release is the earliest sign that query understanding and the catalog have drifted apart.

Questions to Ask

1. Which share of our queries is browse, short keyword, long natural language? Do we design for that mix?
2. Where do the meanings of words live: synonym files, analyzer configurations, tagger code or the attribute dictionary?
3. Which interpretations do we apply as filters, at what confidence, and can buyers see and remove them?
4. What happens, step by step, when a query returns nothing?
5. How does our query categorizer learn about a category that was created yesterday?
6. Do we have token-parity tests per language that would catch an analyzer change before buyers do?
7. Do autocomplete suggestions carry interpretations, and do they ever lead to empty results?
8. If we use dense retrieval, can it return a listing that violates a buyer's filter?
9. If we use an LLM, what validates its output, what does it cost per thousand queries, and where is the query text processed and stored?
10. Who owns each model in the query path, and when was its evaluation set last refreshed?

From Retrieval to Order

Query understanding decides what the engine retrieves; it does not decide the order in which buyers see it. That order is a negotiation between relevance, freshness, seller fairness and paid visibility, and it is the subject of the next chapter, Ranking for Two Sides: Relevance, Freshness and Paid Visibility.

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- OLX Magic: transforming e-commerce search experience with AI agents, MLOps Community, 2024 — agentic assistant challenges.
- Building a conversational shopping assistant with multi-modal search and agent architecture, ZenML LLMOps Database (secondary summary of an OLX talk), 2024 — latency comparison and architecture.
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- Rightmove FY2025 results, Rightmove plc, 2026 — app submitted to the ChatGPT ecosystem.
- Shipping the first MCP in HeyImmo at ImmoScout24, Scout24 Engineering, 2026 — MCP server launched November 2025.
- What we learned from launching ImmoScout24 app in ChatGPT, Scout24 Engineering, 2026 — property search app inside ChatGPT.
- Recherche optimisée pour leboncoin grâce aux LLM, Le Monde Informatique, 2023 — leboncoin’s retriever and re-ranker (press report).
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- Academic Publications & Airbnb Tech: 2025 Year in Review, Airbnb Tech, 2025 — augmenting narrow searches with recommendations.

Standards, regulation and documentation

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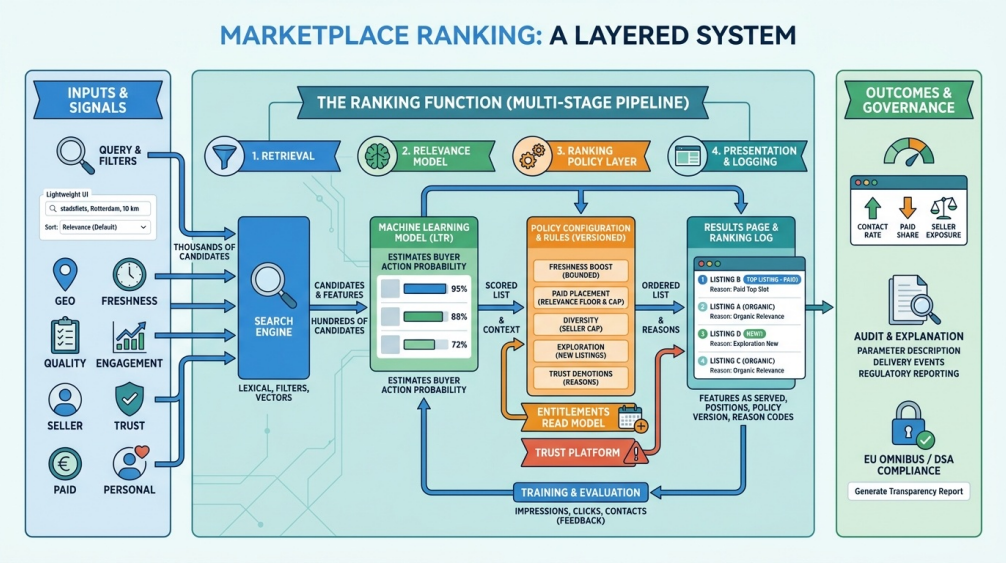
- Hyphenation decompounder token filter, Elastic documentation, n.d. — decompounding for Germanic languages.
- Common options: fuzziness, Elastic documentation, n.d. — `AUTO` edit distances.
- Reciprocal rank fusion, Elastic documentation, n.d. — RRF retriever and its rank constant.
- Score ranker processor and Normalization processor, OpenSearch documentation, n.d. — hybrid fusion options and versions.

Research and practitioner sources

- Applying Embedding-Based Retrieval to Airbnb Search, Abdool et al. (Airbnb), 2026 — IVF versus HNSW under filters; replay-based evaluation.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

20. Ranking for Two Sides: Relevance, Freshness and Paid Visibility



IN THIS CHAPTER, YOU WILL: Take apart the order of a results page: the signals, the multi-stage pipeline, freshness and bumps, learning to rank, position bias, and the blending of paid placements with organic results. You will see how ranking is evaluated and governed, and why the EU treats "relative prominence" as something a marketplace must be able to explain.

WHY THE BUSINESS CARES: The first screen of results is the most valuable real estate the marketplace owns. Buyers spend their attention there, sellers pay to get there, and every ranking change moves contacts, seller retention and paid-visibility revenue at the same time. A general manager who cannot say why a listing sits in position 3 cannot defend the revenue that depends on it, or the buyer trust it consumes.

WHY ARCHITECTS SHOULD CARE: Ranking is where search, monetization, trust and regulation meet in one function that runs on every query. If

boosts live as constants in engine configuration, nobody can reason about the whole. If the ranking policy is a separate, versioned component with logged inputs and outputs, ranking becomes something the group can test, audit and move between search engines.

KEY POINTS:

- **Separate the relevance model from the ranking policy**. The model estimates how likely a buyer is to act on each listing. The policy decides how freshness, paid products, diversity and exposure rules change that order. They need different owners, different change processes and different tests, and mixing them inside engine boost functions is how Multiplier Creep starts.
- **Paid visibility must compete under a relevance floor and be accounted for**. Fixed slots and score multipliers put irrelevant paid listings on top when inventory is sparse. A floor, a cap on paid positions and delivery events that billing can read keep paid products valuable to the sellers who buy them, and keep the order credible to the buyers who read it.
- **Ranking you cannot explain or measure is a liability**. Log the features as served and the positions shown, correct for position bias, judge changes on contacts and seller-side guardrails rather than clicks, and keep a plain-language description of the main parameters and of what payment does to them. EU consumer and platform law already requires that description.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Multiplier Creep** — Boosts for bumps, top listings and pro packages accumulate as multipliers in the ranking until paid listings that do not match the query fill the first screen.
- **Vanity Liquidity** — Ranking tuned on clicks and page views rewards eye-catching listings, while contacts per search and new sellers' first contacts fall.
- **Set-and-Forget AI** — A learning-to-rank model and its feature pipeline decide every listing's exposure, and nobody owns its evaluation, monitoring or the guardrails that keep it fair to both sides.

Every marketplace has an opinion about which listing comes first, whether it admits to one or not. Even "newest first" is a ranking policy: it favors sellers who post often, it makes the bump product possible, and it decides that a city bike listed a minute ago matters more than a better-matched one listed yesterday. **There is no neutral order**;

there are only orders whose effects someone has measured, and orders whose effects nobody has.

Ranking in a listing marketplace is harder than in retail search for three reasons. Inventory is unique and short-lived, so a model has days, not years, to learn about each item. The outcome that matters, a contact that leads to a sale, is rare, delayed and often happens off the platform. And the marketplace serves **two sides at once**: the order that is best for a buyer this minute is not always the order that keeps sellers listing next month, and some sellers have paid to be seen.

Part IV judges every discovery component by how fast and how fairly it turns fresh supply into contacts; ranking is where "fairly" gets decided. This chapter turns the candidates retrieved by Search Architecture for Listings: Indices, Freshness and Scale and interpreted by What Did They Mean? Query Understanding and Autocomplete into an ordered page. It owns **learning to rank** (LTR), **paid-organic blending** and **ranking transparency**. The paid products of Part VI — MONEY: Conversion and Monetization depend on its relevance floor and delivery events, and Recommendations for Inventory That Disappears reuses its machinery without a query. It does not re-teach the product catalog and entitlements behind paid products (Packages, Bumps and Quotas: Products, Pricing and Entitlements), the experimentation methods for two-sided markets (Testing a Two-Sided Market: Experimentation Platforms) or the full regulatory map (Rules of the Marketplace: DSA, DAC7 and Vertical Regulation); it shows where each one enters the ranking function.

Setting the Scene at Veldmark: The "Top Listing" Boost That Ate Relevance

Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), earns **€150 million** a year from Tradeyard's paid visibility products, its value-added services (VAS): bumps, top listings and highlights. In April 2027, two months after she joined as Group Chief Architect, Ingrid Halvorsen asked Priya Raman's data team a narrow question: what does a buyer in Tradeyard Netherlands actually see on the first page?

The answer came back as an archaeology report. Tradeyard's ranking was not designed; it was deposited, layer by layer, in the Solr configuration of Yardstick, the PHP monolith.

Year	Change	Where it lived	Who asked for it
2016	**Top listing** introduced: three reserved slots above the results, rotating among top listings in the category and region	Yardstick results template	Monetization
2019	**Bump** re-dates a listing so it rises in newest-first order	`sort_date` field, overwritten on bump	Monetization
2021	"Relevance" becomes the default sort for text queries; top listings get a **×1.8** score multiplier "so they do not vanish below the slots"	Solr boost function	Search, at Monetization's request
2023	Bumped listings get a recency boost in relevance sort too	Solr boost function	Monetization
2025	Pro "Visibility Plus" subscribers get **×1.3** on all listings	Solr boost function, keyed on a seller flag	Pro sales

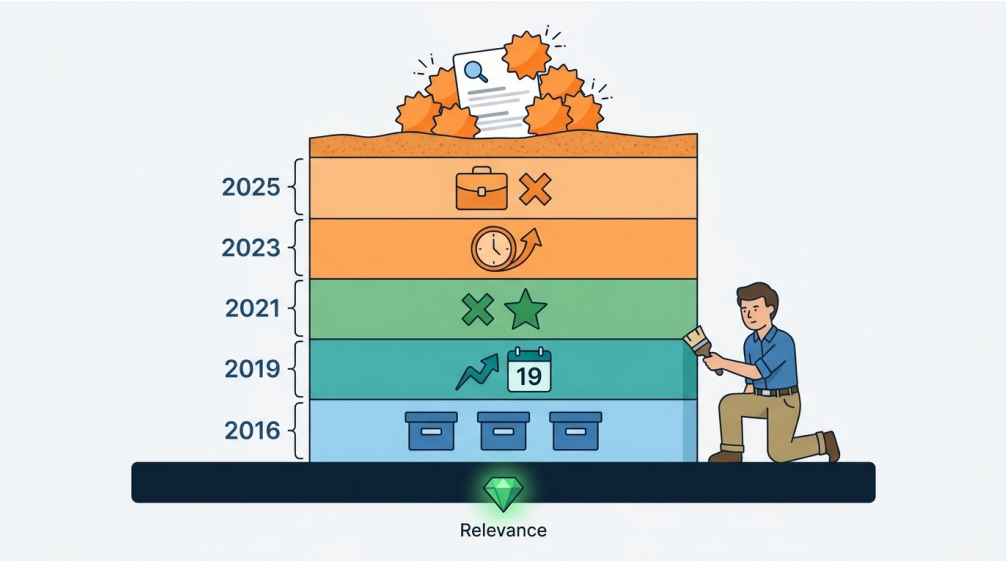


Figure 1: *Tradeyard's ranking was deposited, not designed: five paid-visibility changes, each A/B tested on revenue alone, buried relevance until paid listings held 38% of first-page positions.*

Each change was reasonable when it was made and was A/B tested on revenue per search. None was tested against the others, and none had a relevance guardrail. Priya's sample of Netherlands text queries showed the result (the figures here are illustrative, like everything about Veldmark):

- Paid listings held **38%** of first-page positions in 2027, against **12%** in 2021.
- Almost a third of those paid first-page listings scored below the median relevance of the organic results on the same page.
- The share of search sessions that end with at least one contact fell from **10.6%** in 2021 to **9.4%** in 2026. Over the same period VAS revenue in the Netherlands grew every year.
- The share of new private listings with **no contact within 48 hours** rose from 24% to 33%.

Ingrid made the drift concrete with a familiar listing. On Sunday **7 March 2027**, Femke listed her **€240** city bike in Rotterdam at 19:42; it became searchable at 19:48. A buyer searching "stadsfiets" within ten kilometers of Rotterdam at 19:50 saw Femke's bike at **position 14**. Above it were three top-listing slots, two electric bikes priced around €1,400, a children's bike, and four listings from one bike reseller with Visibility Plus. Femke's first contact, at 19:55, came from one of the **312** saved searches her listing matched (Tell Me When It Appears: Saved Searches and Alerts), not from ranking.

Olivier Petit, VP Monetization, made the fair counterpoint: sellers buy top listings because they work, and the revenue funds the free private listings that make Tradeyard liquid. Tomasz Wiśniewski, head of the new group search platform, added the technical one: Tradeyard would move from Solr to OpenSearch by the end of 2028, and nobody wanted to port five layers of boost functions nobody owned. The question for the architecture advice forum was therefore not "less paid visibility" but "**what should own the order of a results page, and how will we know when it is wrong?**"

What Ranking Is: Prominence, Not a Sort

The most useful definition of ranking comes from EU consumer law. Directive (EU) 2019/2161, the Omnibus Directive, defines **ranking** as "the relative prominence given to products, as presented, organised or communicated by the trader, irrespective of the technological means used for such presentation, organisation or communication" (Directive (EU) 2019/2161, Art. 3, 2019, inserting point (m) into Article 2 of the Unfair Commercial Practices Directive). Three architectural consequences follow.

1. **Ranking is about prominence, not about a sort algorithm**. Reserved slots, carousels, highlighted cards and the first map pins are all ranking, even when no score is involved.
2. **Ranking covers every surface that presents listings in an order**: results pages, category browse, map pins (Searching by Place: Map, Polygon and Commute Search), alerts ordered by match quality and recommendation modules (Recommendations for Inventory That Disappears).
3. "**Irrespective of the technological means**" means a rule in a template is as much ranking as a neural network. The obligations described later in this chapter attach to both.

The chapter uses four more terms.

- **Sort order** is the ordering mode a user can choose: newest, price, distance, or the marketplace's default ("best match", "recommended", "relevance").
- **Relevance model** is the component that estimates, for one query and one listing, how likely the buyer is to act on it: click, save, contact, buy.
- **Ranking policy** is the set of rules applied to the model's output: freshness treatment, paid placements, diversity, per-seller caps, demotions for trust reasons, exploration for new listings.
- **Exposure** is the attention a listing receives, usually modeled as impressions weighted by position. It is what sellers buy, what new listings need, and what a two-sided marketplace has to distribute.

The signals

Signals fall into families with different owners, update rates and risks. The table is the reference I use when a team proposes "just adding a boost".

Signal family	Examples	Typical source	Update rate	Main risk
Query match	Text match (BM25), category and attribute match, query-to-filter agreement	Search index, query understanding	Per query	Keyword stuffing in titles

Signal family	Examples	Typical source	Update rate	Main risk
Geography	Distance, same city or neighborhood, delivery availability	Listing location, user location (Where Is It? Location, Geocoding and Privacy)	Per query	Precision leaks for private sellers
Freshness	Age since publication, time since bump, time since price change	Listing lifecycle (The Listing Aggregate: Identity, Lifecycle and Invariants)	Minutes	Reposting and bump gaming
Listing quality	Photo count and quality, attribute completeness, description length, price plausibility	Enrichment (From Raw Supply to Usable Listings: Normalization and Enrichment)	On change	Rewarding volume over accuracy
Engagement	Click-through rate, saves, contact rate, dwell on the detail page	Behavioral events (Data Platform for Marketplaces: Events, Lakehouse and Data Products)	Minutes to hours	Position bias, feedback loops

Signal family	Examples	Typical source	Update rate	Main risk
Seller	Response rate and time, ratings, verification, tenure, trader status	Accounts, messaging, trust (Private Sellers, Dealers and Agencies: Modeling Who Lists)	Hours to days	Fan-out to every listing of a seller
Trust	Risk score, duplicate cluster, pending reports	Trust and safety (Scams, Fakes and Account Takeovers: Fraud Prevention for Listings, One Property, Three Listings: Duplicates and Entity Resolution)	Seconds to hours	Demotion is a regulated restriction
Paid	Bump timestamp, top-listing entitlement, package tier, sponsored bid	Entitlements (Packages, Bumps and Quotas: Products, Pricing and Entitle-ments)	On purchase and expiry	Multiplier Creep; disclosure duties
Personal	Past searches, viewed categories, price range, saved searches	User profile, consent state	Per session	Consent, filter bubbles, discrimina-tion in housing

Two rules of thumb come from this table. First, **signals that change often should not live in the inverted index**. In segment-based engines such as Elasticsearch, OpenSearch

and Solr, a partial update re-indexes the whole document, so engagement counters and bump timestamps cause update storms; Search Architecture for Listings: Indices, Freshness and Scale covers the mitigation. Second, **every signal in the "paid" and "trust" rows carries a legal duty**, which is a reason to apply them in a layer that can be logged and explained, not inside a score.

Architecture: Model, Policy Layer and Ranking Log

Large marketplaces that publish their designs converge on a **multi-stage pipeline**: each stage scores fewer candidates with a more expensive method. Airbnb's 2026 description of its search stack is typical: retrieval of tens of thousands of homes, a first-stage neural ranker, then a set-wise re-ranker, with each stage scoring about an order of magnitude fewer listings than the last (Abdool et al., 2026). Vespa's documentation names the same phases, first, second and global (Vespa, n.d.). In a 2024 engineering post, Vinted reported that after moving to Vespa it could rank 200,000 candidates per query (Vinted, 2024).

For a listing marketplace I add one stage that the published papers rarely draw separately, because it is where the business rules live: the **ranking policy layer**.

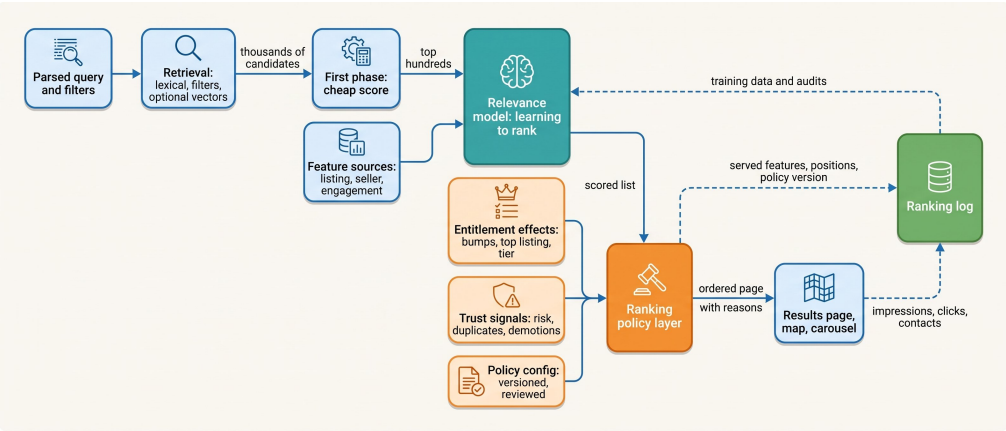


Figure 2: *A ranking function as four stages. The relevance model estimates what the buyer will act on; the policy layer applies freshness, paid placement, diversity and trust rules from versioned configuration; the log records what was served and why, which is what training, evaluation, billing and regulators all need.*

Each component needs a written contract, because Tradeyard's failure was missing boundaries, not bad engineering.

Component	Responsibility	Owns	Must not
Retrieval	Return every listing that satisfies the hard filters and plausibly matches the query	Recall, filter correctness, latency	Apply paid or trust preferences
Relevance model	Estimate the probability of a valuable buyer action per candidate	Model, features, training data, offline metrics	Read entitlements, except as an analysis covariate
Ranking policy	Turn scores into an order under explicit rules: freshness, paid slots and floors, per-seller caps, diversity, exploration, demotions	Policy configuration, its version history, its explanations	Hide rules in model features or engine boost functions
Presentation	Render the order, the labels ("Top listing", "Sponsored") and the "why am I seeing this" entry	Labels, layout	Re-sort the results
Ranking log	Record served features, scores, positions, policy version and reasons per impression	Joinable impression data, retention	Drop the losing candidates' features (needed for training)

The relevance model must not read entitlements. If "has paid" is a model feature, the model learns that paid listings get more contacts, which they do because they are shown higher, and it converts payment into relevance. OLX's 2025 description of its ranking stack is candid about this coupling: it runs a "VAS LTR" model that "balances two objectives: maximizing relevance for users and ensuring visibility for Value Added Services (VAS)", and when OLX added vector retrieval, the keyword-trained model under-scored semantically retrieved results, which "would be particularly problematic for VAS ads retrieved via semantic search", so the model had to be retrained (OLX, 2025). Balancing both objectives in one model is a defensible choice for OLX's scale and team.

The lesson for a group platform is that **a retrieval change became a monetization change**, and only someone who knew the coupling would have seen it coming. A separate policy layer makes the coupling visible.

What one ranked query records

Every ranked page produces data that four other teams depend on. The sequence below shows one text query in the target architecture.

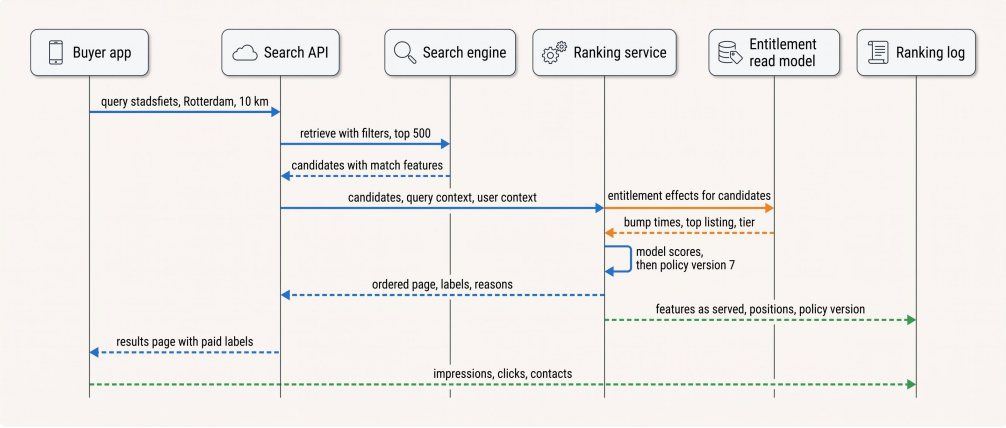


Figure 3: *One ranked query. Entitlement effects are read at ranking time from a fast read model, and the ranking service logs features as served, positions and the policy version. Training, delivery reporting to paying sellers, experiment analysis and regulatory explanations all come from that log.*

Three interface decisions make this work.

1. **Entitlement effects arrive as data, not as code.** The entitlement service publishes effects, such as "top listing active until 14 March 23:59" or "bumped at 19:44", to a read model or onto the listing document by events; Packages, Bumps and Quotas: Products, Pricing and Entitlements owns that design. The ranking service never asks which package a seller has.
2. **The policy is configuration with a version.** Each page records `policy_version`. A change to a slot cap is a reviewed change to a versioned file, with an experiment plan, not an edit to a boost function.
3. **The response carries reasons.** Each result carries a short machine-readable reason code: `organic`, `paid_top_slot`, `bump_recency`, `exploration_new`, `demoted_trust`. The presentation layer uses them for labels, the log for audits, and support for answering a seller's "why is my listing on page three?"

Freshness, Bumps and the Politics of the Default Sort

The **default sort** is the most political decision in a classifieds product, because it decides who wins without any user choosing. Three options dominate.

Default	What it optimizes	Who it favors	Where it fits
Newest first	Novelty; the feeling that the site is alive	Frequent posters, bump buyers, repost abusers	Fast-moving horizontal goods, where many buyers return daily to check what is new
Relevance or "best match"	Expected buyer action per position	Complete, well-priced, well-photographed listings; responsive sellers	Text queries, dense categories, verticals with rich attributes
Hybrid	Relevance with a freshness term and time-boxed paid positions	Depends entirely on the weights	Most mature marketplaces, which is why weights need governance

Newest-first is not naive. For a buyer hunting underpriced bikes, recency is relevance: a good deal listed an hour ago may be gone tonight. That is why many horizontal classifieds keep it as the default for category browsing, and why the **bump**, a paid product that re-dates a listing, is one of the most widely sold paid products. Four rules keep bumps sound.

- **Keep the bump timestamp separate from the creation time.** mobile.de's API, as documented, keeps a `renewalDate` apart from `creationDate` (see The Listing Aggregate: Identity, Lifecycle and Invariants). Overwriting the creation date, as Tradeyard did in 2019, corrupts analytics on listing age and makes recency gameable.
- **A bump is a ranking event, not a lifecycle transition.** It does not change the listing's status; The Listing Aggregate: Identity, Lifecycle and Invariants keeps the two separate.

- **Reposting is a free bump.** If deleting and re-creating a listing resets its age, sellers will do it, and the result is duplicates (One Property, Three Listings: Duplicates and Entity Resolution). The defense is to inherit age, or at least ranking history, from a detected predecessor.
- **Freshness in relevance sort should decay, not step.** A smooth decay, for example an exponential with a half-life tuned per category, avoids the cliff where a listing drops from page one to page five at a fixed age. Half-lives differ by vertical: hours for cheap goods, days for cars, weeks for property.

When a marketplace runs relevance as the default, the bump does not disappear; it becomes a term in the policy. That is exactly where Tradeyard went wrong in 2023: the bump kept its newest-first effect *and* gained a relevance multiplier, so one purchase bought two kinds of prominence without anyone deciding that it should.

Learning to Rank

Learning to rank (LTR) is the use of supervised machine learning to order candidates, trained on logged outcomes. It replaced hand-tuned scoring in large marketplaces because hand-tuned weights stop scaling once there are more than a handful of signals.

Labels: what counts as success

The choice of label is the most consequential design decision, and it is a business decision disguised as a technical one.

Label	Volume	Bias	Fit
Click	High	Rewards attractive thumbnails, clickbait prices, position	Weak alone; useful as an auxiliary task
Save or favorite	Medium	Intent without commitment	Useful for high-consideration verticals
Contact or lead (message, call, form, booking)	Low	Closest observable proxy for a match	The default primary label for classifieds

Label	Volume	Bias	Fit
Transaction or sale	Very low; often unobservable	Cleanest; delayed and off-platform	Possible where payments or shipping run on-platform
Lead quality (reply, viewing, test drive)	Very low	Requires seller-side data	Valuable in pro verticals; feeds two-sided objectives

Most marketplaces end up with a **weighted or multi-task objective**: contacts as the primary label, clicks and saves as auxiliary tasks that help the model learn from sparse data. Airbnb's 2023 "Journey Ranker" is the published reference: it treats intermediate guest actions as milestones and adds host-side outcomes, rejections and cancellations, to the objective (Tan et al., 2023). The classifieds analogue is to count a contact the seller never answers as less than one that gets a reply. That is a **two-sided label**: it rewards listings that produce matches, not just messages.

Models: tree ensembles are still a good default

The published lineage runs from tree ensembles to neural networks, once tree gains plateau. Airbnb described moving from gradient-boosted decision trees (GBDT) to neural networks once GBDT gains plateaued, a journey its authors summed up as "deep learning was steep learning for us" (Haldar et al., 2018). Its later papers added position-bias handling and cold-start treatment (Haldar et al., 2020), similarity-aware diversity (2022) and map-specific ranking (2024). Etsy reported that a unified embedding model for personalized retrieval increased search purchase rate by 5.58% across its A/B tests (Jha et al., 2023).

Closer to classifieds, mobile.de presented learning to rank with XGBoost at Berlin Buzzwords in June 2024, for "millions of searches each day" under a latency target below 30 ms (Berlin Buzzwords, 2024). In 2023 leboncoin engineers described re-ranking Elasticsearch's top candidates with a two-tower model built on DistilBERT, within a budget of a few dozen milliseconds; secondary write-ups of that work report up to 5% more clicks and contacts (ZenML summary, 2023; Le Monde Informatique, 2023). leboncoin's own post could not be read for this book, so treat those figures as reported, not verified.

My recommendation for a mid-sized marketplace is a **GBDT ranker with a LambdaMART-style objective**, trained on contacts with click and save auxiliaries, served in the engine (OpenSearch and Solr both ship LTR plugins; see OpenSearch, n.d.) or in a re-ranking service. Move to neural re-rankers when three conditions hold: tree gains have plateaued in online tests, text and image understanding are the limiting

factor, and the team can run GPU inference within the latency budget. They do not hold for Tradeyard Portugal; they may hold for Tradeyard Netherlands.

Features as served, not as recomputed

LTR needs the feature values **as the ranker saw them**, joined to later outcomes. Recomputing features offline from today's data introduces **training-serving skew**: the model learns from a listing's age, price and seller response rate as they are now, not as they were at the moment of the impression. The fix is architectural: the ranking service logs the feature vector for every candidate it scored, including those shown on later pages or not shown at all, and the training pipeline joins that log to contacts by an impression identifier. This log is large; sampling the non-shown candidates is the usual compromise.

Position bias and new listings

Buyers click what is on top because it is on top. Training on raw clicks therefore teaches the model to reproduce the previous ranking. Joachims, Swaminathan and Schnabel showed in 2017 that propensity weighting, estimating how likely each position is to be examined and weighting clicks by the inverse, yields unbiased learning to rank, and that it works even when queries do not repeat (Joachims et al., 2017). That property matters for classifieds, where the long tail of filter combinations means few exact repeats. Airbnb reported that its position-bias method gave "one of the largest gains" on the inventory its neural ranker had struggled with (Haldar et al., 2020).

Position bias hurts new listings most: they have no engagement history, so a model that leans on engagement ranks them low, so they collect no engagement. In classifieds this is a supply problem, not a model curiosity. A private seller whose first listing gets no contact in its first days often does not list again, and that is **local liquidity** leaving the market (Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics). Two policy tools address it:

- **Exploration slots**. Reserve a small number of positions on early pages for new listings with few impressions, chosen by the model's content-only score. Position-randomized exploration also produces the clean data that propensity estimation needs.
- **Content-based priors**. Give new listings features that do not depend on history: photo quality, attribute completeness, price relative to similar listings. Recommendations for Inventory That Disappears covers the same cold-start problem from the item-similarity side.

Diversity and per-seller caps

Exploration slots and per-seller caps are **exposure** policies: they decide how attention is shared among listings. Singh and Joachims gave this a precise form in 2018: treat exposure as a resource the ranking allocates, and constrain how it is distributed among items while maximizing utility for the user (Singh and Joachims, 2018). Marketplaces rarely implement the formal constraints, but the framing is right.

Listings on a page are also not independent. Airbnb's 2022 work showed that booking probability depends on what else is shown, and that a similarity-aware model reduces redundant results (Haldar et al., 2022). Classifieds have a blunter version: one professional seller with two hundred near-identical listings can flood a query. A **per-seller cap**, for example no more than three listings from one seller in the first twenty positions, is a policy rule, not a model feature. It protects buyers from monotony, protects private sellers from being buried and limits the payoff from bulk posting. It must be tunable per category: a cap that is right for city bikes is wrong for new-car inventory, where one dealer group may legitimately hold most of the relevant stock.

Blending Paid Visibility with Organic Results

Paid visibility is not an embarrassment to hide in the ranking; it is a product that sellers buy, and a buyer-facing claim that the marketplace must keep honest. The design question is how paid placement **competes** with organic results. Four approaches cover the field.

Approach	How it works	Strength	Risk	Evidence
Fixed slots	Positions 1–3, or every Nth position, reserved for paid listings	Simple to sell, deliver and bill	Irrelevant paid listings on top when relevant inventory is sparse	eBay described this problem with its earlier fixed slots (eBay, 2020)

Approach	How it works	Strength	Risk	Evidence
Score multiplier	Paid listings get ×k on the relevance score	Feels proportional	Unbounded: k interacts with every other boost; the result depends on the score distribution of each query	Tradeyard's ×1.8 (illustration)
Separate auction, then merge	Paid candidates ranked by bid × predicted response, merged into floating slots	Clean ad economics	Two models with inconsistent notions of relevance	Common in sponsored-listing products (industry practice)
Unified ranking with a relevance floor	Paid and organic compete in one ordering; paid listings are eligible only above a relevance threshold, with a cap on paid positions	Paid positions vanish when they would hurt the page	Needs per-category tuning and a cannibalization model	eBay's 2020 unified ranker (eBay, 2020)

eBay's 2020 description is the clearest public statement of the unified approach. Promoted and organic listings had previously come from two separate stacks with different algorithms. The unified ranker lets both "compete for top search slots", weighing relevance to the query, the item's and seller's past performance and the ad rate. **No promoted listings appear at the top when few relevant items exist**, or when participation and ad rates are low (eBay, 2020). The source is six years old and eBay's ad products have changed since; the principle has not.

Classifieds add two complications that retail sponsored listings do not have.

1. **Paid products are sold as positions, not as clicks.** A Tradeyard top listing is sold

as "shown in the top slots for seven days", not as a cost per click. A relevance floor therefore changes what the seller bought. The product, its description and its price have to change together with the ranking, and this is why the Monetization team must co-own the policy.

2. **Paid visibility is bought by private sellers**. A floor that suppresses an individual's paid bump on a poorly matching query is correct for buyers, but the seller paid for it. The usual resolution is to sell **guaranteed exposure** (a number of impressions in relevant contexts, with a refund or extension when delivery falls short) rather than an unconditional position.

Delivery accounting

Whatever the blending rule, the system must answer the question that every paying seller eventually asks: **did I get what I paid for?** That requires delivery events emitted by the ranking log (impressions in paid slots, positions, the queries they appeared for) and consumed by the entitlement and billing side, so that "sold" and "delivered" can be reconciled. Packages, Bumps and Quotas: Products, Pricing and Entitlements owns the reconciliation; the ranking function owns emitting the events. Without them, a group cannot defend a paid-visibility dispute, and cannot tell whether a ranking change quietly under-delivered a product it sold.

Measuring whether paid exposure *creates* value is harder than it looks. OLX described in 2026 how premium boost packages for professional car sellers appeared to perform worse than cheaper packages in naive dashboards, because sellers self-select into packages; propensity-score matching, validated against a randomized test, showed uplifts of 17.6% to 97.4% between package levels (OLX, 2026). Seller ROI reporting built on correlation will mislead both the seller and the pricing team.

Deduplication across paid and organic

The same listing should not appear in a paid slot and again in the organic results on the same page. Show it once, in the higher position, and count the impression against the paid product only if the paid slot was the reason it was there. The rule is trivial; implementing it requires the policy layer to see both lists at once, which is another argument against assembling the page in a template.

Transparency: Ranking as a Regulated Function

European law turned ranking explanation from good practice into an obligation. Three instruments apply to a classifieds group operating in the EU, and a fourth applies to the largest platforms. The regulatory chapter, Rules of the Marketplace: DSA, DAC7

and Vertical Regulation, owns the full map; this table shows what each means for the ranking function.

Instrument	What it requires (as relevant to ranking)	Engineering consequence
Platform-to-business (P2B) Regulation (EU) 2019/1150, Art. 5 (applies to business users: dealers, agents, pro sellers)	Terms must set out "the main parameters determining ranking and the reasons for the relative importance of those main parameters"; where ranking can be influenced "against any direct or indirect remuneration", the provider must describe those possibilities and their effects; it is not required "to disclose algorithms" (Regulation (EU) 2019/1150, 2019)	A maintained ranking-parameters description per brand, generated from the policy configuration, that says what each paid product does to ranking
Omnibus Directive (EU) 2019/2161 (consumer side)	Undisclosed paid ranking in search results is blacklisted as always unfair (Annex I point 11a); the main ranking parameters and their relative importance are material information, directly and easily accessible from the results page (Art. 7(4a) of the Unfair Commercial Practices Directive, UCPD) (Directive (EU) 2019/2161, 2019)	A "paid" flag that travels from entitlement to rendering on every surface; a results-page link to the parameters description

Instrument	What it requires (as relevant to ranking)	Engineering consequence
Digital Services Act (EU) 2022/2065, Arts. 26, 27 and 17	Ads labeled in real time with who paid (Art. 26); main recommender parameters in plain language in the terms, and options where offered (Art. 27); a demotion of content is a restriction requiring a statement of reasons (Art. 17)	Label payloads per result; trust-based demotions logged as enforcement actions with reasons (Trust and Safety as a Platform)
DSA Art. 38 (very large online platforms only)	At least one option for each recommender system "which is not based on profiling" (Regulation (EU) 2022/2065, Art. 38, 2022)	A non-personalized ranking variant that can be served on request

Two cautions about this table. First, on 19 November 2025 the Commission proposed repealing the P2B Regulation because of its overlap with the DSA and the Digital Markets Act; as of October 2026 that is a proposal, and the Commission says its non-binding ranking guidelines are proposed to remain in application (European Commission, updated 2026). Second, whether a promoted classifieds listing bought by a private seller counts as an "advertisement" under DSA Art. 26 is a legal judgment, not settled text; the prudent reading treats it as one and labels it.

The DSA point that surprises ranking teams most is Art. 17. **If the ranking quietly pushes down listings that a fraud model suspects, that is a restriction of visibility**, and the affected user is owed a statement of reasons. A "trust demotion" therefore cannot be a hidden multiplier. It must be an enforcement action recorded by the trust platform, and the ranking policy applies it with a reason code. This is one more reason to keep trust rules in the policy layer rather than in the model.

The architectural answer to all four rows is the same: **if the policy is explicit configuration and every result carries reason codes, the parameters description can be generated, the labels are correct by construction, and the audit trail exists.** If the policy is buried in five layers of boost functions, every one of these obligations becomes a manual reconstruction.

Evaluating Ranking Changes

A ranking change is judged three times: offline before an experiment, online in an experiment, and after rollout. Each stage has a known trap.

Offline. Normalized discounted cumulative gain (NDCG) on logged data is the standard metric, and it is a weak predictor in two documented ways. Logs contain only what users were shown, so offline recall on logged data does not track online results well; Airbnb found that replaying a slice of production traffic and measuring against the current ranker's top results tracked its A/B outcomes better (Abdool et al., 2026). And list metrics mislead on maps: Airbnb reported that maps account for roughly 80% of its search interactions and introduced a map-specific NDCG because the list version "fails to model user attention on maps" (Airbnb Tech, 2025). Real-estate portals, where the map is often the main interface, should take that seriously (Searching by Place: Map, Polygon and Commute Search).

Online. Contacts are rare, so A/B tests on contact rate need large samples. Airbnb reported in 2025 that interleaving and counterfactual methods increased experiment sensitivity **by up to 100×**, depending on method and metric (Zhang et al., 2025). Interleaving merges two rankers' results into one page and attributes each contact to the ranker that contributed the listing. It suits ranking-model comparisons; it does not suit policy changes such as slot caps, whose effect is on the whole page.

Interference. Buyers in treatment and control compete for the same listings. A buyer-randomized test of a ranking change that moves attention toward some listings can show a lift that disappears at full rollout, because the "winning" listings took contacts from listings shown to control buyers. Seller-side, two-sided or cluster randomization (by category and region) reduces the bias. Johari and colleagues showed that customer-side randomization is biased when the market is supply-constrained and listing-side randomization when it is demand-constrained (Johari et al., 2020). Testing a Two-Sided Market: Experimentation Platforms owns the designs; the ranking team must know which one a given change needs.

Guardrails

Every ranking experiment should report a fixed set of guardrail metrics beside its primary metric. I use these:

Guardrail	Why	Typical threshold
Search sessions with a contact	The buyer-side outcome	No significant decrease
New listings with a contact within 48 hours	Seller retention and supply	No significant decrease
Seller exposure concentration (share of impressions held by the top 1% of sellers)	Two-sided fairness; flooding	Bounded increase
Paid-product delivery (impressions delivered against impressions sold)	Revenue and dispute risk	Within contract terms
Paid share of first-page positions	Multiplier Creep	Below the policy cap
Contacts on unavailable listings	Ghost Inventory reaching the top	No increase
Latency p99 of the ranking stage	User experience; peak load	Within budget

A change that improves the primary metric and breaches a guardrail is not shipped without an explicit decision by the owners of that guardrail. That rule is what turns "the experiment won" into a group-level governance process.

The Veldmark Remedy: Ranking Policy v1

The architecture advice forum took the Tradeyard case in June 2027. The decision record (an architecture decision record, or ADR) that Ingrid, Tomasz, Olivier and Priya signed had five parts.

1. **One ranking service, engine-agnostic.** A ranking service in the Common Ground platform retrieves the top 500 candidates through Listing Search API v1, the group search contract (Search Architecture for Listings: Indices, Freshness and Scale), from Solr in 2027 and OpenSearch from 2028. It does all second-stage scoring and policy itself, within the 40 ms re-ranking budget that the search platform allots. Boost functions in Solr are frozen and removed one by one. The ranking survives the engine migration because it does not live in the engine.
2. **A relevance model trained on contacts.** A GBDT ranker trained on two-sided labels (contacts weighted up when the seller replies within 24 hours), with click and save auxiliaries, propensity-corrected using a small randomized exploration slice. It has no entitlement features.

3. **Ranking policy v1 as versioned configuration** co-owned by the search platform and Monetization, with changes reviewed in the advice forum:

    - Top listings keep up to three reserved slots, but only for listings whose relevance score clears a **floor**: at least the score of the 20th organic result for that query. Below the floor, the slot stays empty or goes to the next eligible top listing.
    - **At most 4 paid listings in the first 20 positions**, counting slots and bump effects together.
    - Bumps act only on newest-first sort. In relevance sort they add a bounded recency term worth at most four positions, not a multiplier.
    - Visibility Plus becomes a set of seller tools and analytics, not a ranking multiplier.
    - **No more than 3 listings from one seller** in the first 20 positions, configurable per category.
    - **Two exploration positions** in the first 30 for listings under 24 hours old with fewer than 50 impressions.

4. **Delivery and transparency by construction**. Every result carries a reason code; paid-slot impressions are emitted as delivery events to the entitlement service; Tradeyard's ranking-parameters page for buyers and its business-user terms are generated from the policy file and reviewed by Lucía Ortega's regulatory team.
5. **A product change with the ranking change**. Olivier's team repositioned the top listing as **seven days of guaranteed prominence in relevant searches**, with an automatic extension when delivery falls short. The "Spotlight" product that the forum had deferred in May (How Listing Marketplaces Make Money) was withdrawn, because a guaranteed first slot cannot coexist with a relevance floor.

The experiment ran from 6 September to 17 October 2027 in Tradeyard Netherlands, randomized by **category × region cluster** to limit interference. The results, illustrative like all Veldmark figures:

Metric	Control clusters	Treatment clusters	Relative change
Search sessions with a contact	9.4%	9.9%	**+5.3%**
New private listings with a contact within 48 hours	67%	71%	+6.0%

Metric	Control clusters	Treatment clusters	Relative change
Paid share of first-page positions	38%	17%	−55%
Contacts per top listing sold	1.00 (index)	1.21	**+21%**
VAS revenue per search	1.00 (index)	0.975	−2.5%

Fewer paid positions were shown, but each was shown where it fit, so the sellers who paid got more contacts per euro. That is the argument that carried the decision: **the floor made the paid product better, not just the organic results**. Henrik de Boer approved the 2.5% revenue cost per search on the condition that Monetization report delivered impressions and contacts per top listing monthly. Ranking policy v1 went live for Tradeyard Netherlands and Belgium on **Wednesday 1 December 2027**; Poland and Portugal follow with the move to OpenSearch.

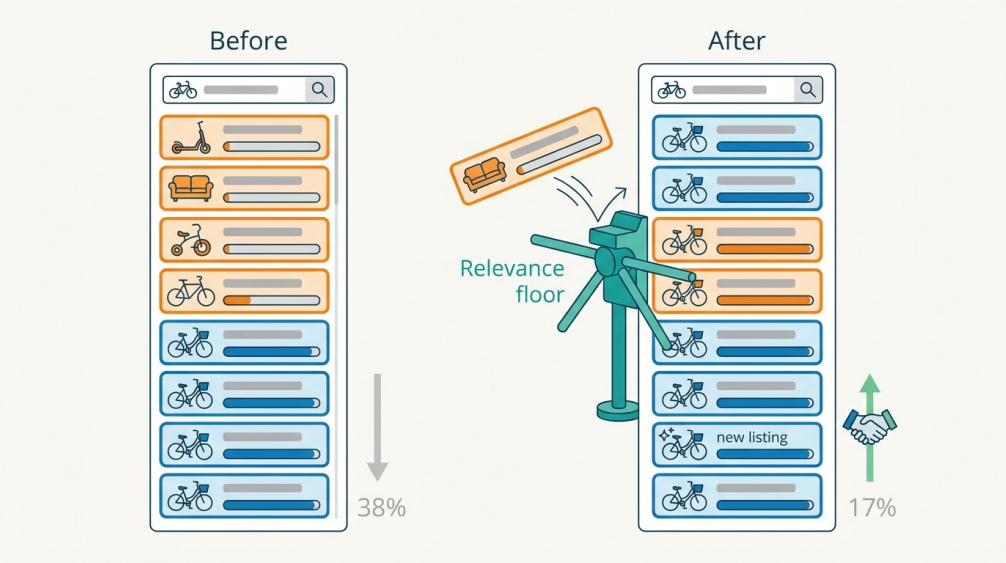


Figure 4: *Ranking policy v1 in Tradeyard NL: with a relevance floor and at most four paid listings in the first 20, the paid share of first-page positions fell from 38% to 17%, while contacts per top listing sold rose by 21%.*

Trade-offs

No choice in this chapter is right everywhere. The table shows when each common option fits.

Decision	Option A	Option B	Choose A when	Choose B when
Default sort	Newest first	Relevance with bounded freshness	Cheap, fast-moving goods; buyers browse daily for deals	Text queries; rich-attribute verticals; dense categories
Where ranking runs	Inside the engine (boost functions, LTR plugin)	Separate ranking service over engine candidates	One engine, one brand, small team, tight latency	Several engines or brands, an engine migration ahead, policy that must be audited
Model family	GBDT (LambdaMART-style)	Neural re-ranker	Tabular features dominate; mid-sized traffic; CPU budget	GBDT has plateaued; text and image understanding limit quality; GPUs available
Paid blending	Fixed slots	Unified ranking with floor and cap	Paid product sold as a position to many small sellers; sparse tooling	Enough relevant inventory; ability to sell guaranteed exposure; delivery accounting in place

Decision	Option A	Option B	Choose A when	Choose B when
Label	Clicks	Contacts, two-sided weighted	Very low traffic; early stage	Enough contact volume to train; lead data available
Personalization	None (contextual only)	Personalized features	Housing in jurisdictions with discrimination exposure; low consent rates	Stable consent; repeat buyers; clear lift without fairness harm
Experiment unit	Buyer	Category × region cluster, or two-sided	Model comparisons via interleaving; small attention shifts	Policy changes that move attention between sellers

Vertical Differences

Ranking physics differ more by vertical than by country.

Dimension	General classifieds (Tradeyard)	Automotive (Motorvane)	Real estate (Hearthly)	Jobs
Dominant query shape	Text plus category browse	Filters: make, model, price, mileage, distance	Map and area, then filters	Title and location, then filters

Dimension	General classifieds (Trade-yard)	Automotive (Motor-vane)	Real estate (Hearthly)	Jobs
Freshness half-life	Hours to days	Days	Days to weeks; rentals hours in tight markets	Days; expired roles must disappear
Main supply concern	Private sellers' first contacts	Dealer flooding; one dealer group dominating a region	Same property from several agencies	Volume posters and duplicate postings
Typical paid product	Bump, top listing, highlight	Depth tiers, featured slots in dealer packages	Premium or featured listing tiers, developer campaigns	Sponsored or promoted jobs
Distinctive signal	Price plausibility, photo quality	**Price rating against a market-value model**; dealer response time	Energy class, price per m², agency response	Salary trans-parency, application rate
Distinctive risk	Scam listings at the top	Price ratings perceived as unfair by dealers	Which agency's listing represents a duplicate cluster; Keybridge neutrality	Discrimination exposure; ranking candidates on the other side

Two cells turn a ranking signal into a business dispute. At Motorvane, the market-value model that labels the Essen Golf a "good price" at €21,900 against an estimate of €22,800 is also a ranking signal, which turns a pricing model into an exposure model; dealers will contest it, and Models in Production: ML Platforms and Use Cases covers its governance. At Hearthly, when entity resolution clusters the Lyon apartment's three agency listings into one canonical property, ranking must decide **which agency's listing represents the cluster**. Doing that by subscription tier would be legal, if disclosed, but it would

also be read as Veldmark favoring paying agencies, and it sits uncomfortably close to the March 2027 Keybridge neutrality commitment. The rule I would choose is the most complete and most recently updated listing, with paid tier as a disclosed tie-breaker at most.

Failure Modes

1. **Boost accretion**. Each paid or business rule added as a multiplier, each tested alone on revenue. The sum is untested. Warning sign: nobody can list all the factors that move a listing up, or say which team owns each.
2. **Paid as a model feature**. The model learns that paid listings get contacts and converts payment into relevance. Warning sign: a feature-importance report with "promoted" or "package tier" near the top.
3. **Clicks as the objective**. The ranker optimizes clickable thumbnails and too-good-to-be-true prices, which are also what scammers post. Warning sign: click-through rises while contacts per search fall.
4. **Training on yesterday's ranking**. Without position correction or exploration, the model reproduces its own past, and new listings never break in. Warning sign: the share of new listings on page one falls with each retraining.
5. **Ghost listings at the top**. Sold and removed listings keep their engagement history and rank well until the index catches up (From Submitted to Searchable: The Publication Pipeline). Warning sign: contacts on unavailable listings concentrated in top positions.
6. **Hidden trust demotions**. A fraud score silently multiplied into the ranking; nobody can produce statements of reasons, and appeal reversals do not restore position. Warning sign: trust and ranking teams disagree about whether demotion exists.
7. **Interference-blind experiments**. Buyer-randomized tests of slot or exposure changes that report lifts the rollout never delivers. Warning sign: experiment wins that do not appear in the next quarter's market-level metrics.
8. **One policy for every country**. Slot counts and caps tuned on the Netherlands, applied to Portugal, where a category may have twelve relevant listings in a region and three paid slots fill a quarter of the inventory. Warning sign: paid share of first-page positions varying by a factor of three between markets.

The Chief Architect's View

Ranking is one of the few components that every part of the business believes it owns. Search owns relevance, Monetization owns paid products, Trust owns demotions, pro sales owns what packages promise and Legal owns the parameters description. The

Chief Architect's job is to make those claims coexist in a structure rather than in a configuration file nobody can read.

What to decide at group level.

- **The layering.** A group standard that relevance models do not consume entitlements, that business and trust rules live in a versioned policy layer, and that every ranked surface logs features as served, positions, policy version and reason codes. This standard is what makes ranking portable across engines and brands.
- **Ownership of the objective.** A small **ranking council** per brand: the product owner for discovery as chair, with search, monetization, trust and data science. It owns the primary metric, the guardrails and the policy file. Changes to the policy go through the architecture advice forum with a decision record when they affect more than one brand, as Veldmark's April 2027 principles require (Who Owns What: Organizing Marketplace Engineering).
- **What is shared and what is local.** Share the ranking service, the log schema, the evaluation tooling, the guardrail definitions and the transparency generator. Keep models and policy values per brand and often per country, because category density, paid products and buyer behavior differ. **Share the machinery of ranking, not the ranking.**

What to watch.

- Paid share of first-page positions per brand and country, monthly. It is the earliest indicator of Multiplier Creep.
- New-listing first-contact rate. It is the supply-side health of ranking.
- Delivery against sold for every paid visibility product. It is the revenue-risk indicator.
- Model freshness and drift for each production ranker, with a named owner. A ranker that has not been retrained for a year is Set-and-Forget AI whatever its original quality (Models in Production: ML Platforms and Use Cases).
- Regulatory change: the P2B repeal proposal, the Digital Fairness Act preparation, and national enforcement of the Omnibus paid-ranking rules.

The long-term consideration. Ranking is increasingly consumed by machines as well as people: AI assistants and agents call marketplace search and present a shortlist (AI Across the Listing Lifecycle). When a third party re-ranks your results, your paid placements and your labels may not survive the trip. The ranking service's output contract, with reason codes and paid flags, is what lets the group negotiate and enforce how its order is presented elsewhere.

Questions to Ask

1. Can we list every factor that moves a listing up or down on our default sort, with the team that owns each and the date it was last tested?
2. Does any relevance model use paid status, package tier or bump state as a feature? If so, why, and what guardrail watches it?
3. What label does our ranker learn from, and does it reward contacts that sellers answer, or only clicks and messages?
4. When relevant inventory for a query is sparse, what happens to paid slots: do they stay filled, or does a floor empty them?
5. Can we tell a paying seller, from logs, how many impressions in which positions and queries their product delivered last week?
6. How are fraud-based demotions represented, and can we produce a statement of reasons for each one?
7. Is our ranking-parameters description generated from the live policy, and when was it last compared with production?
8. Which randomization unit did our last three ranking experiments use, and was it right for the kind of change tested?
9. What share of first-page positions do new listings receive, and is it rising or falling with each model retraining?
10. If we change search engines next year, which parts of our ranking live in the engine and would have to be rebuilt?

From Ranking to Recommendations

Ranking orders what a buyer asked for. The next chapter, Recommendations for Inventory That Disappears, deals with what the buyer did not ask for: similar listings, home feeds and "you may also like" modules that must find relevant items in inventory that may be gone tomorrow. The same machinery reappears there, candidate generation, re-ranking, exposure and logging, but without a query to anchor relevance, which makes cold start, availability and feedback loops even harder.

Sources and Evidence

Company engineering and investor sources

- eBay Makes Promoted Listings in Search Results More Relevant and Dynamic — eBay Inc., 2020. Unified ranker for promoted and organic listings; no promoted listings at the top when few relevant items exist; earlier fixed-slot problems.

- Hybrid search: where keywords meet vectors, enabling classifieds discovery — OLX Engineering, 2025. "VAS LTR" balancing relevance and paid visibility; retraining needed after vector retrieval.
- Correlation lied to us: rethinking product impact with causal inference — OLX Engineering, 2026. Selection bias in package performance; propensity matching validated by A/B test.
- Vinted Search Scaling Chapter 8: Goodbye Elasticsearch, Hello Vespa — Vinted Engineering, 2024. Ranking depth of 200,000 candidates.
- Better search relevance using Learning to Rank at mobile.de — Berlin Buzzwords, 2024. XGBoost LTR, millions of searches per day, under 30 ms.
- Recherche optimisée pour leboncoin grâce aux LLM — Le Monde Informatique, 2023; and LLM-powered search relevance re-ranking system — ZenML (secondary), 2023. leboncoin two-tower re-ranker; reported gains (not verified against the primary post).
- Academic Publications & Airbnb Tech: 2025 Year in Review — Airbnb Tech, 2025. Maps at about 80% of search interactions; map-specific NDCG.

Standards, regulation and documentation

- Regulation (EU) 2019/1150, Article 5 (Ranking) — Official Journal text via the EU Publications Office, 2019. Main parameters, paid influence, no duty to disclose algorithms.
- Directive (EU) 2019/2161 (Omnibus) — Official Journal text via the EU Publications Office, 2019. Article 3: definition of ranking (UCPD Art. 2(m)), UCPD Art. 7(4a) and Annex I point 11a.
- Platform-to-business trading practices — European Commission, updated 2026. Application since July 2020; repeal proposal of 19 November 2025.
- Regulation (EU) 2022/2065 (Digital Services Act) — Official Journal text via the EU Publications Office, 2022. Art. 17 (statement of reasons, including demotion), Art. 26 (advertising labels and who paid), Art. 27 (recommender parameters) and Art. 38 (non-profiling option for very large platforms).
- Phased ranking — Vespa documentation, n.d. First, second and global ranking phases.
- Learning to Rank plugin — OpenSearch documentation, n.d. Feature definitions, feature logging, model upload.

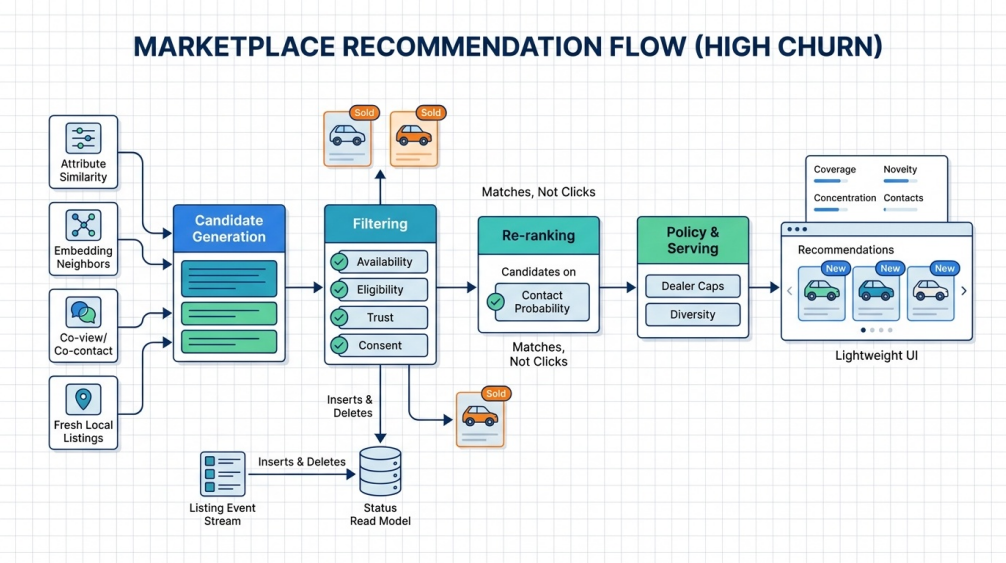
Research and practitioner sources

- Applying Deep Learning to Airbnb Search — Haldar et al., 2018. GBDT plateau and move to neural ranking.

- Improving Deep Learning for Airbnb Search — Haldar et al., 2020. Position bias and cold start.
- Learning To Rank Diversely at Airbnb — Haldar et al., 2022. Listings on a page are not independent.
- Optimizing Airbnb Search Journey with Multi-task Learning — Tan et al., 2023. Milestone labels and host-side outcomes.
- Learning to Rank for Maps at Airbnb — Haldar et al., 2024. List assumptions fail on maps.
- Applying Embedding-Based Retrieval to Airbnb Search — Abdool et al., 2026. Multi-stage pipeline; replayed traffic tracks online results better than logged data.
- Harnessing the Power of Interleaving and Counterfactual Evaluation for Airbnb Search Ranking — Zhang et al., 2025. Up to 100× experiment sensitivity.
- Unified Embedding Based Personalized Retrieval in Etsy Search — Jha et al., 2023. +5.58% search purchase rate.
- Unbiased Learning-to-Rank with Biased Feedback — Joachims, Swaminathan and Schnabel, 2016 (WSDM 2017). Propensity-weighted LTR.
- Fairness of Exposure in Rankings — Singh and Joachims, 2018. Exposure as the quantity to allocate fairly.
- Experimental Design in Two-Sided Platforms — Johari, Li, Liskovich and Weintraub, 2020. Bias of customer-side and listing-side randomization.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

21. Recommendations for Inventory That Disappears



IN THIS CHAPTER, YOU WILL: Map the recommendation surfaces of a listing marketplace, compare content-based, collaborative, session-based and embedding methods under fast inventory churn, and design a pipeline that keeps sold listings out and new ones in: candidate generation, filtering, re-ranking and policy. You will also see how to evaluate recommendations and where consent and regulation constrain them.

WHY THE BUSINESS CARES: Recommendations catch the buyer who did not find what they wanted, or did not know what to search for, and send them to another listing instead of another site. At Motorvane, the similar-vehicles carousel starts 7% of leads in Germany. When they recommend a car that was sold yesterday, they waste the buyer's time and annoy the dealer who has to answer.

WHY ARCHITECTS SHOULD CARE: Most recommender designs assume a stable catalog that a nightly batch job can process. A listing marketplace

replaces a large part of its catalog every week. The architecture must support real-time inserts and deletes, filtered retrieval and content-based cold start. It needs the same event feed and the same availability rules as search, or it will rebuild Ghost Inventory in a new place.

KEY POINTS:

- **Inventory churn is the defining constraint.** A listing may live for days, so a recommender that learns only from interactions knows nothing about new listings and keeps recommending sold ones. Content and image features must carry a listing until behavior arrives. Recommendation indices need real-time inserts, deletes and pre-filtering, not nightly rebuilds.
- **Split the pipeline into candidate generation, filtering and re-ranking, and give each its own job.** Several cheap candidate sources (attribute similarity, embeddings, co-contact) feed one filter that enforces availability, eligibility and consent at serving time. A re-ranker then optimizes for contacts, with a policy that caps dealer concentration and keeps room for new listings.
- **Measure recommendations by matches, not clicks, and design for consent from the start.** Judge them on contacts and on coverage of the inventory, not on carousel click-through. Personalization rests on consent and, for very large platforms, on a non-profiling option, so every surface needs a non-personalized mode that is good enough to stand on its own.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Ghost Inventory** — Recommendation lists computed overnight keep showing sold and removed listings long after search has dropped them.
- **Set-and-Forget AI** — Recommendation models decide which listings get exposure, yet they are rarely evaluated against contacts, coverage or seller concentration, and they often have no owner once the launch A/B test is over.
- **Vanity Liquidity** — Carousels tuned for click-through send buyers to popular, eye-catching listings and leave most of the inventory unseen.

In retail, a recommender learns that people who bought a kettle also bought a toaster, and the lesson stays useful for years because the same kettle is sold every day. In a listing marketplace there is no "same kettle". The Essen Volkswagen Golf with 48,000 kilometers exists once, for fifteen days, and then it is gone. **Every listing is a cold-start problem, and every recommendation is a bet on an item that may disappear before the buyer clicks.**

Churn changes a recommender's architecture more than its algorithms. The published marketplace systems that work have kept their models fairly simple. What they changed is the plumbing: indices that accept inserts and deletes in real time, retrieval that filters while it searches, content and image features that let a listing take part before anyone has viewed it, and serving-time checks that keep sold items out. The failures usually come from the opposite choice, a recommender built as a nightly batch job on top of a catalog that changes by the hour.

This chapter owns **candidate generation**, **similar listings** and **cold start**. Like the rest of Part IV — DISCOVERY: Matching Intent to Inventory, it treats discovery as a read model over the listing stream, judged on contacts. A recommendation module is a ranked list without a query, so it reuses the machinery of Ranking for Two Sides: Relevance, Freshness and Paid Visibility. Its rule that every surface reads the same availability status as search returns in Searching by Place: Map, Polygon and Commute Search and Tell Me When It Appears: Saved Searches and Alerts. It leaves the ML platform underneath, feature stores and model governance, to Models in Production: ML Platforms and Use Cases, and the event stream that keeps everything fresh to Every Change Is an Event: Event-Driven Listing Architecture.

Setting the Scene at Veldmark: The Golf That Kept Being Recommended

Motorvane is the automotive brand of Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build). Every vehicle detail page in Germany carries a carousel of ten "similar vehicles". It was built in 2020 as a nightly Spark job. The job computes item-to-item co-views (pairs of listings viewed in the same session), keeps the top ten neighbors for each listing and writes them to a key-value store that the detail page reads. It was cheap, it raised click-through when it launched, and nobody has looked at it closely since.

In May 2027, Felix Brandt, Motorvane's CTO, asked for a review after a complaint from Autohaus Ruhrblick in Essen. The dealer's **2021 Volkswagen Golf 1.5 TSI**, 48,000 km, arrived through its dealer management system (DMS) feed on Wednesday **14 April 2027** at **€21,900**. It had a good run: **23 leads in 10 days** (raw leads; 19 after deduplicating repeat contacts from the same buyers), a price cut to **€21,400** on 22 April that triggered price-drop alerts to 1,180 watchers, and a sale at 15:10 on Thursday **29 April**. The feed's removal arrived **26 hours** after the sale, at 17:10 on Friday 30 April. Three e-mail enquiries reached the dealer in between, two of them from buyers who had arrived through the carousel. Search dropped the listing within minutes of the removal. The carousel did not: it read lists written by the nightly job, so the sold Golf stayed in other cars' carousels until the job ran again in the early hours of Saturday 1 May.

Figure 1: *Ghost Inventory in the carousel: the Golf's removal arrived 26 hours after the sale, search dropped it within minutes, and the nightly co-view lists kept it in other cars' carousels until the next run.*

Priya Raman's team found that the Golf was not an exception. Their review of the carousel in Germany found (illustrative figures, like all Veldmark numbers):

Finding	Value
Share of detail-page views that arrive from the carousel	**12%**
Share of Motorvane DE leads that start from a carousel click	**7%**
New listings with **no carousel impressions** in their first 72 hours	**41%**
Carousel clicks that land on a sold or removed listing	**3.8%**
Recommendations from the same dealer group as the anchor car	31%
Share of live inventory that appeared in any carousel in a week	46%

The carousel starts 7% of leads, yet 41% of new listings get no impressions from it in their first 72 hours. A co-view recommender can only recommend listings that someone has already viewed together with something else. A car that arrived this morning has no co-views, so it appears in no carousel and gets none of the 12% of traffic the carousel distributes. Dealers pay for packages with listing slots (How Listing Marketplaces Make Money). They notice when a new car gets less attention than an old one, and they do not care that the cause is an algorithm's data requirements.

The question Felix put to Ingrid Halvorsen and Priya was practical: **how should Motorvane recommend cars that may be sold tomorrow and were not there yesterday?**

Recommendation Surfaces

A **recommendation** is a list of listings, searches or sellers that the user did not explicitly request, chosen from what is known about the context, the user or both. Surfaces differ in what anchors them, which decides the method, the freshness requirement and the metric.

Surface	Anchor	Latency budget	Freshness need	Primary outcome
Similar listings on the detail page	One listing	Tens of ms; rendered with the page	Availability within minutes	Contact on the recommended listing
Home feed ("for you", "new for you")	The user's history	100–300 ms	New listings within minutes	Sessions that reach a contact
Continue your search / recently viewed	The user's recent searches and views	Tens of ms	Real time	Return-visit contacts
Recommended searches and category suggestions	The user's history; the query	Tens of ms	Daily is enough	Saved searches created (Tell Me When It Appears: Saved Searches and Alerts)
Zero-result and low-result fallback	The failed query	Within the search budget	Real time	Recovered sessions
"Buyers also contacted" after a contact	The contacted listing	Asynchronous	Hours	Additional contacts for the same buyer
Recommendations e-mails and push	The user's history	Batch	Must be checked again at send time	Opens that lead to contacts

Low-result fallback is where recommendations protect liquidity most directly, and it gets less attention than it deserves. A buyer searching for a cargo bike within five kilometers of a small town may get three results. Similar listings slightly farther away, or in a neighboring category, turn a dead end into a contact. Airbnb describes "augmenting search with recommendations" when searches are too narrow (Airbnb Tech, 2025).

Recommendation e-mails are where Ghost Inventory is most visible: a list computed at 02:00 and opened at 09:00 can be seven hours stale. The Rotterdam city bike that Femke sold on 8 March 2027 still appeared in Tradeyard's alert digest the next morning, and the same failure is waiting in every batch-built recommendation e-mail.

Why Listings Break the Usual Recommender

Retail recommendation assumes a stable catalog, logged-in repeat customers and purchases as labels. A listing marketplace violates all three, and the published marketplace research says so explicitly. FINN.no's 2018 paper on marketplace recommenders (eight years old; the current system is not public), from a system then serving more than a million visitors a day, describes second-hand and unique items, poorly structured item data and interactions that often finish off the platform (Eide and Zhou, 2018). An OLX Jobs study from 2023 notes that classifieds "provide many challenges for recommendation methods, due to the limited information regarding users and items" (Kwieciński et al., 2023).

Property	Retail catalog	Listing marketplace	Consequence
Item lifetime	Years	Days to weeks	Interaction-based item representations are always young or missing
Item multiplicity	One SKU, many buyers	One item, one buyer	"Bought together" barely exists; "viewed together" is the main behavioral signal
Catalog churn	Low	A large part of the catalog is replaced every week	Indices must take inserts and deletes continuously

Property	Retail catalog	Listing marketplace	Consequence
User identity	Often logged in	Often anonymous browsing, logged in only to contact	Session-based methods matter; long user histories are rare
Outcome	Purchase, observed	Contact; sale often unobserved	Labels are sparse and proxy-based
Supply side	Merchant catalog team	Millions of individual sellers and dealers	Exposure from recommendations is a seller-retention lever

The consequence is a design principle: **the recommender must be able to describe a listing it has never seen anyone interact with**. That rules out pure collaborative filtering as the only method, and it puts the listing's own content (category, attributes, price, location, photos and text) at the center of the system.

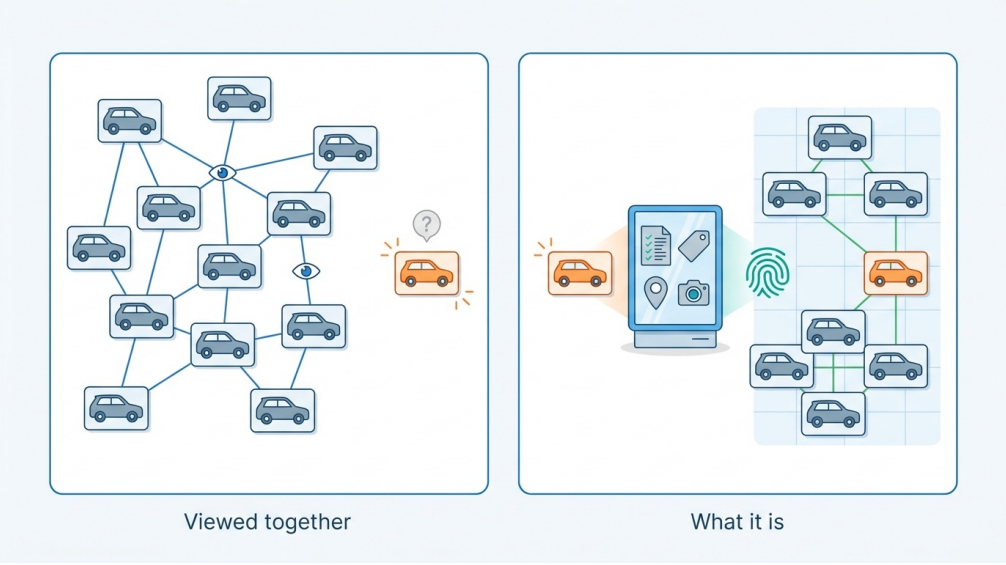


Figure 2: *A co-view recommender can only recommend listings someone has already viewed alongside another, so a new car starts invisible; describing a listing by what it is places it among similar cars from minute one.*

Methods Under Churn

For listings, what matters about each method family is how it treats a new listing and a sold one.

Method	How it works	New listing	Sold listing	Cost	Marketplace evidence
Content-based similarity	Nearest neighbors on attributes, price, location, text, images	Works at once	Disappears if the index is live	Low	Zillow's listing-text embeddings for similar homes (Zillow, 2021)
Item-to-item collaborative filtering (co-view, co-contact)	Listings that appear together in sessions	Absent until it collects co-views	Lingers until the next rebuild	Low; batch	Common baseline (industry practice)
Matrix factorization and graph methods (alternating least squares (ALS), RP3beta, SLIM)	Latent factors or random walks over the user-item graph	Absent	Lingers	Medium; batch	OLX Jobs benchmark: online, RP3beta about 20% stronger than ALS on contacts (Kwieciński et al., 2023)

Method	How it works	New listing	Sold listing	Cost	Marketplace evidence
Session-based sequence models (GRU4Rec and successors)	Predict the next item from the current session	Needs item representation	Lingers in vocabularies	Medium to high	GRU4Rec designed for short sessions where factorization fails (Hidasi et al., 2015)
Learned item embeddings (item2vec-style, two-tower)	Embeddings from sessions, optionally with content towers	Works if a content tower exists	Disappears if the index is live	Medium; approximate-nearest-neighbor (ANN) serving	Airbnb listing embeddings (Grbovic and Cheng, 2018); Vinted two-tower retrieval (Vinted, 2023)
Bandits and ensembles	Choose among several sub-models online	Inherits	Inherits	Medium	FINN.no re-ranking proposals from several sub-models with bandits (Eide and Zhou, 2018)

The documented cases, in date order:

- **Airbnb, 2018.** Grbovic and Cheng's KDD paper introduced listing and user embeddings for real-time personalization in search ranking and in similar-listing recommendations. They reported that these two channels drive 99% of Airbnb's conversions (KDD 2018). The paper, eight years old now, made "embeddings

learned from sessions" the standard pattern for marketplace similarity.

- **Zillow, 2021.** A Zillow engineering post described Word2Vec embeddings of listing descriptions for similar-home recommendations. The team chose Word2Vec after comparing GloVe, BERT and Doc2Vec, trained it on about 10 million descriptions and generated more than 2 million embeddings per model in under ten minutes in a daily Airflow job. Offline gains were largest for vacant land, where structured data is thin (Zillow Tech Hub, 2021). The current state of that system is not public. The lesson still holds: **text carries the listings that attributes describe poorly**.
- **Adevinta, 2022.** A post on complementary-item recommendations described an ANN index built only over items that had been active in the last 90 days and viewed in the last 15 (Adevinta, 2022); the post is four years old and the current implementation is not public. That is a candidate-pool rule that removes much of the dead inventory before any model runs.
- **Vinted, 2023.** Vinted's homepage recommendations had used a read-only Faiss index that "had to be periodically rebuilt" to add new listings and drop sold ones. It also could not pre-filter, so users with restrictive filters could get no recommendations at all. Moving retrieval to Vespa, with real-time updates and pre-filtering, fixed both. In Vinted's 2023 test, exact nearest-neighbor search raised p99 latency by about 40%, from about 50 ms to about 70 ms, without enough benefit, so Vinted kept approximate search with measured recall of about 60–70% (Vinted, 2023).
- **Airbnb, 2026.** In its embedding-based retrieval for search, Airbnb computes the listing tower as a **daily offline batch** and only the query tower online. Serving cost then no longer grows with the number of eligible listings, but listing features can be up to a day old at retrieval time (Abdool et al., 2026). For Airbnb, whose listings persist, that is a sensible trade. For a classifieds site, whose listings are often days old, a day of staleness is a large share of a listing's life.

Vinted and Airbnb made the same trade-off and settled it differently, because their inventory lives for different lengths of time. That is the most useful way to read any published recommender design.

Architecture: Generate, Filter, Re-rank, Apply Policy

Recommendation systems at scale share the multi-stage shape of search ranking. The difference is that there is no query to anchor relevance, so the candidate generation stage carries more of the load. For listings I use four stages with strict responsibilities.

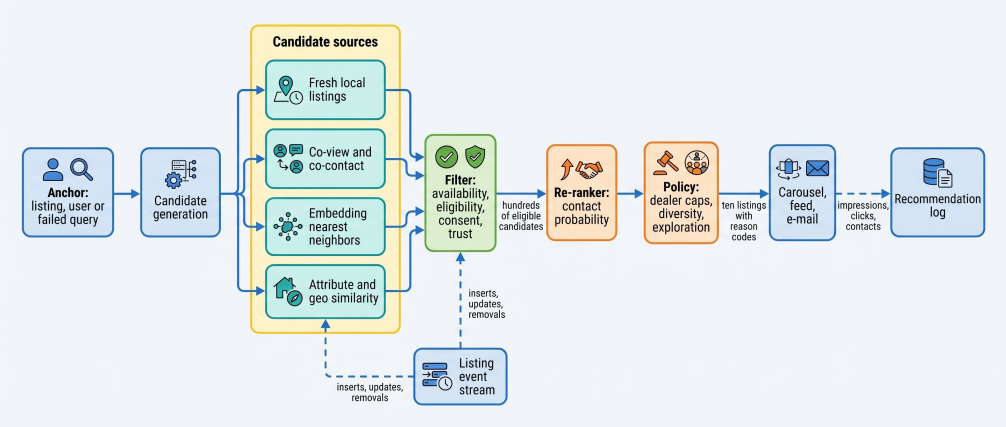


Figure 3: *Recommendations as four stages. Several cheap candidate sources, one of them purely content-based, feed one filter that enforces availability and eligibility at serving time. A re-ranker optimizes for contacts, and a policy caps concentration and reserves room for new listings. The listing event stream keeps both the candidate indices and the filter current.*

Candidate generation

Candidate generation is the stage that selects a few hundred plausible items from the whole live inventory. It is optimized for recall and speed, not precision. Use several sources, each covering a weakness of the others.

1. **Attribute and geography similarity**. A structured query over the search index: same vertical and category, overlapping attribute bands (make and model family, year ±2, mileage band and price band for cars; rooms, area and price per m^2 for property), within a radius. It works from the listing's first minute, because enrichment (From Raw Supply to Usable Listings: Normalization and Enrichment) has already filled the attributes. It is the cold-start backbone.
2. **Embedding nearest neighbors**. Item embeddings from a model trained on sessions and contacts, with a content tower so that new listings get a vector from their attributes, text and primary photo. Serve them from an ANN index that supports filtering during the search. Elasticsearch, for example, documents that in filtered approximate k-nearest-neighbor (kNN) search "the filter is applied during approximate kNN search to ensure that k matching documents are returned", whereas post-filtering "can return fewer than k results" (Elastic, n.d.).
3. **Co-view and co-contact**. Behavioral neighbors from recent sessions. They are strong for listings with history, especially co-contact, because a buyer who contacted two sellers was comparing real alternatives. Recompute them incrementally from the event stream, or at least several times a day, and never treat them as the only source.

4. **Fresh local listings.** The newest listings matching the anchor's category and area, a source that exists to give new supply a chance. It doubles as exploration.

A rule I insist on: **candidate sources return listing identifiers and scores, never rendered content.** The listing card is assembled at serving time from the current listing read model, so price, photos and status are never stale even when a candidate list is.

Filtering

The filter keeps sold, ineligible and untrusted listings off every surface. It is one deliberately simple component, applied to every candidate from every source at serving time.

Filter	Rule	Source of truth
Availability	Only active, published listings; not sold, reserved, expired, paused or removed	Listing status read model fed by unpublish events (From Submitted to Searchable: The Publication Pipeline)
Eligibility	Same country and brand rules; category exclusions (no adult or regulated categories on the home feed)	Policy configuration
Trust	No listings under fraud review or from suspended accounts	Trust platform decisions (Trust and Safety as a Platform)
Duplicates	At most one listing per canonical entity	Entity resolution clusters (One Property, Three Listings: Duplicates and Entity Resolution)
Already seen or contacted	Drop listings the user has contacted; demote ones already viewed	User interaction history
Consent	Personalized candidates only where the user has consented	Consent state on the request

The availability check uses the **same status read model as search**. If search and recommendations disagree about whether a listing exists, buyers will find the gap, as they found it with the Golf.

Re-ranking and policy

The **re-ranker** scores the filtered candidates for the outcome that matters, usually the probability of a contact. It uses features of the anchor, the candidate and the pair: price difference, distance, attribute agreement, the candidate's quality and freshness, and the seller's response rate. A gradient-boosted model is a sound default here for the same reasons given in Ranking for Two Sides: Relevance, Freshness and Paid Visibility. The re-ranker must not use paid status as a feature. Where paid placement in recommendation modules is sold, it belongs in the policy layer, labeled and disclosed, exactly as in search.

The **policy** then applies rules that a model should not learn:

- **Seller and dealer caps**. At most two listings from the same seller or dealer group in a ten-item carousel. That protects buyers from a carousel that is really one dealer's showroom, and protects smaller dealers from being crowded out.
- **Diversity**. Spread across price bands and, where useful, neighboring models, so the carousel offers real alternatives rather than ten near-identical cars.
- **Exploration**. One or two positions for listings with few impressions, chosen from the content-based source.
- **Reason codes**. Each item carries why it was chosen (`similar_attributes`, `viewed_together`, `new_nearby`), for logging, debugging and, where offered, "why am I seeing this".

Keeping Recommendations Alive: Events and Serving-Time Checks

Freshness runs in two directions, and removal matters more than insertion: a sold listing should become ineligible within minutes, as a new listing should become eligible. Both come from the listing event stream that already feeds search (Every Change Is an Event: Event-Driven Listing Architecture). The sequence below replays the Golf's sale in Motorvane's target design.

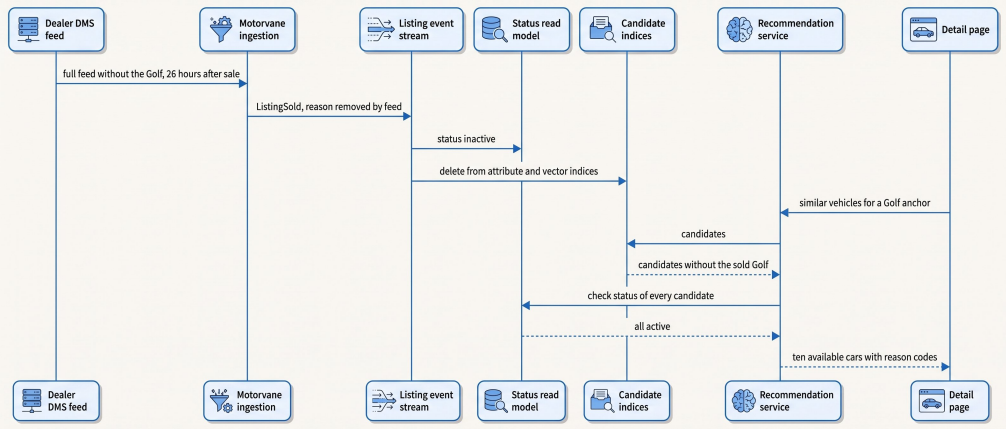


Figure 4: *Two defenses against Ghost Inventory. The unpublish event deletes the listing from the candidate indices, and the serving-time status check catches anything the indices still hold, such as precomputed co-view lists. Neither defense can fix a feed that reports the sale 26 hours late; that is an ingestion problem, owned by the feed contract.*

Three design points follow from the diagram.

1. **Indices that cannot delete are a liability.** Vinted's experience with a periodically rebuilt Faiss index is the documented example. Choose ANN infrastructure that supports real-time upserts and deletes and filtering during the search, or accept that a serving-time filter will discard a growing share of candidates as the index ages.
2. **Precomputed lists are acceptable only behind a live filter.** Co-view neighbors computed every few hours are fine as a candidate source, provided the filter checks every item against current status.
3. **Late feeds are not a recommender problem, but recommenders amplify them.** The Golf's 26-hour removal lag comes from the dealer's DMS export schedule. The fix is in the feed contract and in "sold" signals from Motorvane's own dealer tooling (Many Doors In: Forms, APIs and Bulk Feeds, Serving the Pros: Dealer, Agent and Merchant Tooling). A lead management system that knows the car was sold can unpublish it before the feed does.

E-mail and push recommendations need a **send-time check** as well: build the list in batch if necessary, but check availability again when the message is rendered or sent, and drop items that have become unavailable. For e-mail, rendering the card when the message is opened (a pixel-time or link-time check) is the stronger version. It needs care with e-mail clients that prefetch images.

Cold Start in Three Forms

Cold start is the problem of recommending without the interaction history that collaborative methods need. Marketplaces have three versions of it.

Form	Example	Approach
New listing	The Golf on 14 April, minute one	Content-based candidates; content tower for embeddings; exploration positions; image and text features from enrichment
New or anonymous user	A first visit from a search engine landing page	Contextual recommendations from the landing page and the first query; popular-and-fresh in the user's area; session-based models as the session grows
New market or category	Tradeyard's new cargo-bike leaf from November 2027; a newly launched country	Transfer from neighboring categories and markets; attribute-based similarity until behavior accumulates

For new listings, the published evidence supports **content and image representations**. Airbnb's 2020 work treated cold start for new listings as one of the main ways to improve its deep-learning ranker (Haldar et al., 2020). Its 2026 embedding retrieval reported more bookings for new listings (Abdool et al., 2026). Vinted's 2025 dense retrieval built its item tower from category, brand and precomputed image embeddings, with text features left as future work (Vinted, 2025). The common move is to describe a listing by what it *is*, so the model can place it before anyone has interacted with it.

For anonymous users, the important architectural point is that **the session is the user**. A session-based model, or simply the last few viewed listings used as anchors, turns a logged-out visitor into a personalized one within a few clicks, without a long-term profile. That is also the privacy-friendly option, discussed below.

Feedback Loops and Exposure

A recommender trained on its own output drifts toward what it already recommends. Popular listings collect more impressions, more clicks and more co-views, and so they get recommended more. Three effects are specific to listings.

- **Popularity bias amplifies Ghost Inventory**. The most-viewed listings are often the best deals, and the best deals sell first. A popularity-heavy recommender is therefore biased toward listings close to being sold.
- **Exposure from recommendations is supply-side value**. At Motorvane, a carousel that distributes 12% of detail-page traffic is a marketing channel, and dealers will judge whether it treats them fairly. Coverage, the share of live inventory recommended at least once a week, is a fairness metric as well as a quality metric.
- **Carousels have position bias too**. The first two items get most of the clicks, and the rest of the carousel is seen far less. Log positions, and use the same propensity corrections as in Ranking for Two Sides: Relevance, Freshness and Paid Visibility when training on carousel clicks.

The tools are the same as in ranking: exploration positions, logging of served candidates and positions, propensity-aware training, and policy caps. The difference is emphasis. In search the buyer's query limits how far ranking can drift. In recommendations nothing limits it except the policy.

Evaluating Recommendations

Click-through rate is the most common recommendation metric and one of the least useful. A carousel full of suspiciously cheap cars has excellent click-through.

Offline, measure recall at k and normalized discounted cumulative gain (NDCG) against held-out contacts, not clicks, plus three distribution metrics: **coverage** (the share of inventory recommended), **novelty** (the share of recommended listings under 72 hours old) and **concentration** (the share of impressions going to the top sellers). Replay real traffic where possible. The ANN recall of the serving index is an engineering metric in its own right. Vinted accepted 60–70% recall because exact search cost more latency than it returned in user benefit (Vinted, 2023), and that is the right kind of decision: measured, explicit and revisited.

Online, run A/B tests on the outcomes that matter: contacts per session that started from the surface, contacts on recommended listings, and the guardrails below. Recommendation tests suffer less interference than ranking tests, because they redistribute less attention. They are not immune, though. A carousel change that sends more buyers to

new listings takes attention from older ones, and a seller-level guardrail will catch that. Vinted reported that its dense retrieval in search reached full rollout only after about 50 A/B tests (Vinted, 2025). Iteration at that scale needs an experimentation platform, not a one-off launch test (Testing a Two-Sided Market: Experimentation Platforms).

Guardrail	What it protects
Contacts on unavailable listings from recommendations	Buyers and sellers from Ghost Inventory
New listings with a recommendation impression within 72 hours	New supply
Share of impressions to the top 1% of sellers	Smaller sellers; fairness of a marketing channel
Weekly inventory coverage	The long tail
Recommendation p99 latency	Detail-page performance and search-engine ranking of the page (Found by Google: Web, App and SEO Architecture for Listings)

Privacy, Consent and Regulation

Recommendations are where a marketplace's behavioral data turns into visible decisions about a person, so the legal surface is larger than for search ranking.

- **Consent for tracking and personalization**. Behavioral histories come from tracking, and in the EU, access to device storage for that purpose generally requires consent under the ePrivacy rules. Personalization based on profiles is processing of personal data under the GDPR, with its own legal-basis question. The architectural consequence is that **consent state must travel with the request**, and every surface needs a non-personalized mode: contextual (anchored on the listing or query), local and fresh. It must be good enough that users who decline are not punished with an empty page. Data Platform for Marketplaces: Events, Lakehouse and Data Products owns the tracking plan and consent propagation.
- **DSA transparency**. Article 27 of the Digital Services Act requires online platforms to set out the main parameters of their recommender systems in plain language in their terms, and to describe any options users have to change them. Article 38 requires very large online platforms to offer, for each recommender system, at least one option "which is not based on profiling" (Regulation (EU) 2022/2065, Official Journal, 2022). Few classifieds brands reach the very-large-platform threshold of 45 million average monthly active EU recipients. A non-profiling mode costs little

if it is designed in, and the Article 27 duty applies to every online platform that is not a micro or small enterprise, whatever its reach.

- **Discrimination in housing and jobs**. In the United States, fair-housing rules prohibit advertising that indicates a preference on protected grounds (24 CFR §100.75, n.d.). A personalized recommender for housing listings must therefore be audited for disparate delivery: whether protected groups systematically see different listings. EU anti-discrimination law points the same way for jobs. In these verticals, prefer contextual signals (the listing, the stated search) over inferred demographics, and never infer special-category data.
- **Retention and minimization**. Recommendation features rarely need years of history. Airbnb's 2026 personalization work used long-term sequences of up to 80 events over seven years and short-term sequences of up to 200 views over 21 days (Airbnb Tech, 2026). Its booking cycle is long. A classifieds buyer's intent changes faster, and most of the value is in the last days. Set retention per feature, not per data lake.

Cross-brand data is a separate decision. Veldmark ID, the group identity approved in June 2027 for rollout in 2028, will make it technically possible to recommend Motorvane cars to a Tradeyard user who browsed car accessories. Whether that is allowed depends on consent scope, on the privacy notice of each brand, and, for Keybridge agency data, on the March 2027 neutrality commitment, which rules out using Keybridge data for Hearthly ranking or sales targeting. Lucía Ortega's team owns that answer. The architecture should make cross-brand features **opt-in at the feature level**, with lineage, so that the answer can be enforced.

The Veldmark Rebuild: Similar Vehicles at Motorvane

Felix's team rebuilt the carousel between June and August 2027, inside Motorvane's own stack. Motorvane stays on Elasticsearch 8 until the 2029 search review, so the design uses what that engine provides rather than waiting for the group platform.

1. **Four candidate sources**.

    - An **attribute-similarity query** over the search index (same make and model family, year ±2, mileage within ±25,000 km, price within ±20%, within 100 km and widened when inventory is sparse). It works from the moment vehicle identification number (VIN) decoding has filled the listing's 31 attributes.
    - **Vehicle embeddings** in a filtered kNN index: a two-tower model trained on co-contact sessions, with a content tower over decoded attributes, price, region and the primary photo's image embedding, so new cars get vectors at once.

   - **Co-contact neighbors**, recomputed every two hours from the lead event stream.
   - **Fresh listings** of the same model family nearby.
2. **One serving-time filter** on the shared status read model, plus trust, duplicate and already-contacted rules.
3. **A gradient-boosted decision tree (GBDT) re-ranker** trained on leads from carousel impressions, with position logged and corrected.
4. **Policy**: no more than **two cars from one dealer group** in ten; at least three price bands represented when inventory allows; **one exploration position** for a car under 72 hours old.
5. **Events**: `ListingPublished`, `PriceChanged`, `ListingSold` and `ListingRemoved` from the listing event contract v1 upsert and delete the candidate indices within a minute. A dealer's "mark as sold" in Dealer Hub now produces `ListingSold` at once, without waiting for the next DMS feed.

The rebuilt carousel ran against the old one in an A/B test randomized by user in September 2027. Results, illustrative like all Veldmark figures:

Metric	Old carousel	New carousel	Change
Leads starting from a carousel click (share of Motorvane DE leads)	7.0%	8.1%	**+16% relative**
New listings with no carousel impression in their first 72 hours	41%	**8%**	−33 points
Carousel clicks landing on unavailable listings	3.8%	**0.3%**	−3.5 points
Recommendations from the anchor's dealer group	31%	14%	−17 points
Weekly inventory coverage	46%	79%	+33 points
Carousel click-through rate	1.00 (index)	0.96	**−4%**

The last row is the one that would have killed the project if click-through had been the success metric. The new carousel shows fewer eye-catching popular cars and more new, comparable ones, so it is clicked slightly less and produces more leads. Felix's report to the group put the lesson in one line: "**We were optimizing the carousel for curiosity; the dealers pay us for leads.**"

If the Golf were listed under the new design, it would appear in other cars' carousels within minutes of becoming searchable at 17:05 on 14 April. It would leave every

carousel within a minute of the dealer marking it sold in Dealer Hub. If the dealer does not use Dealer Hub, it would leave within a minute of the status delta that the group's Supply Contract v1 (September 2027) asks DMS vendors to send within 15 minutes of a sale (Many Doors In: Forms, APIs and Bulk Feeds). The 26-hour gap closes at the source, not in the recommender.

Trade-offs

Decision	Option A	Option B	Choose A when	Choose B when
Candidate source mix	Behavioral only (co-view, factorization)	Behavioral plus content-based	Long-lived inventory; dense interaction data	Churning inventory; new listings matter commercially (all classifieds)
Index update model	Batch-built static ANN index	Real-time upsert and delete with filtering	Inventory changes slowly; rebuilds are cheap	Listings live days; filters are restrictive
Item representation freshness	Daily batch embeddings (Airbnb 2026 style)	Embeddings computed on publish	Listings persist for months	Listings are days old; first-day exposure matters
ANN precision	Exact search	Approximate search with measured recall	Small corpus; latency budget allows	Large corpus; recall loss measured and acceptable (Vinted kept 60–70%)
Optimization target	Click-through	Contacts, with coverage and concentration guardrails	Early prototype only	Production

Decision	Option A	Option B	Choose A when	Choose B when
Personalization depth	Contextual and session-based only	Long-term user profiles	Low consent rates; housing and jobs; anonymous traffic	Logged-in repeat users with stable consent and long buying cycles
Ownership	Per-brand recommenders	Shared candidate and serving platform with per-brand models	One brand; very different verticals	Several brands on the same event backbone and status model

Vertical Differences

Dimension	General classifieds (Tradeyard)	Automotive (Motorvane)	Real estate (Hearthly)	Jobs
Main surface	Home feed and similar items	Similar vehicles on the detail page	Similar homes; "new for your search"; map-adjacent suggestions	Similar jobs; jobs for your profile
What "similar" means	Category, price, image similarity, distance	Make, model, year, mileage, price, equipment, distance	Area, price per m², rooms, energy class, neighborhood	Title, skills, seniority, location, pay
Dominant cold-start signal	Photo and title	VIN-decoded attributes	Address, area, floor plan, photos	Title and description text

Dimension	General classifieds (Trade-yard)	Automotive (Motor-vane)	Real estate (Hearthly)	Jobs
Freshness pressure	Very high; items sell in hours	High; days	Very high for rentals; moderate for sales	High near application deadlines
Distinctive risk	Scam listings with attractive prices recommended widely	Dealer concentration; recommending cars from competing dealers on a dealer's own page	Recommending the same property through several agencies; fair-housing exposure	Discrimination in who sees which jobs
Supply-side sensitivity	Private sellers' first contacts	Dealers measure carousel leads and compare packages	Agencies watch exposure across portals	Employers pay per application in some models

The automotive cell "recommending competing dealers on a dealer's own page" is a product decision with commercial weight. Some automotive marketplaces sell "own stock only" recommendations on dealer listings as a package feature. If that is sold, it is a paid effect and belongs in the policy layer with a label, not inside the model.

At Hearthly, the duplicate filter has a fairness side. When the Lyon apartment's three agency listings are clustered into one canonical property, the recommender must choose one listing to show. The rule should match the one ranking uses for the cluster (Ranking for Two Sides: Relevance, Freshness and Paid Visibility), or agencies will see inconsistent treatment across surfaces.

Failure Modes

1. **Nightly lists, hourly inventory**. Recommendations computed once a day and served without an availability check. Warning sign: contacts on sold listings concentrated in recommendation surfaces and e-mails.
2. **Behavior-only candidates**. New listings get no recommendation exposure until they have collected interactions elsewhere. Warning sign: the age distribution of recommended listings is far older than that of live inventory.

3. **Click-through as the goal**. The carousel learns to show the most clickable listings, which include the most suspicious ones. Warning sign: carousel click-through rises while contacts per carousel click fall, or fraud reports on recommended listings rise.
4. **Rendering from the candidate store**. Price and photos shown from a precomputed list rather than the live listing. Warning sign: buyers complain that the price on the card differs from the detail page.
5. **Post-filtering a static ANN index**. Restrictive filters (country, category, price) applied after retrieval leave few or no results. Warning sign: empty or short modules for niche anchors, which Vinted documented before its migration.
6. **Personalization without a fallback**. Users who decline consent see an empty or random feed. Warning sign: a large gap in contact rate between consenting and non-consenting users that comes from the fallback, not from the data.
7. **One dealer's showroom**. No seller caps, so large dealer groups fill every carousel in a region. Warning sign: complaints from small dealers, and top-1% seller share above the policy threshold.
8. **Unowned after launch**. The recommender was shipped by a project team that has moved on. Nobody retrains it, monitors its coverage or notices when an upstream event schema changes. Warning sign: nobody can name the model's owner or its last retraining date.

The Chief Architect's View

Recommendations look like a feature owned by a product squad. At group scale they are a **platform capability with brand-specific models**, and the Chief Architect's work is to draw that line.

What to share.

- **The status read model and the listing event stream**. Search, recommendations, alerts and e-mail must all ask the same question, "is this listing available right now?", and get the same answer. This is the most valuable piece to share, and in Veldmark's case it comes with the Common Ground event backbone.
- **Candidate-serving infrastructure**. An ANN and attribute-retrieval service with real-time upserts, deletes and filtered search, offered to every brand. In Veldmark's plan it rides on the shared OpenSearch platform for Tradeyard and Hearthly from 2028, while Motorvane keeps its Elasticsearch-based design until the 2029 review.
- **The recommendation log schema and evaluation tooling**, including coverage, novelty and concentration metrics computed the same way everywhere.
- **Consent propagation and cross-brand feature lineage**. This is the only way to enforce the answer to "may Motorvane use Tradeyard behavior?"

What to keep local. Models, candidate-source weights and policy values. "Similar" means different things for a city bike, a Golf and a Lyon apartment, and the people who know each vertical should tune it.

Who decides. The discovery product owner of each brand owns the objective and the guardrails, as for ranking. The group data protection officer owns consent scope and cross-brand data use. The architecture advice forum reviews any recommender that uses data from more than one brand, or that sells placement.

What to watch. Contacts on unavailable listings by surface (the Ghost Inventory indicator), new-listing exposure within 72 hours, weekly coverage, seller concentration and the retraining age of every production model. A recommender that nobody has retrained for a year is Set-and-Forget AI, whatever its launch results were (Models in Production: ML Platforms and Use Cases).

The long-term consideration. Recommendations increasingly reach users through channels the marketplace does not control: AI assistants that suggest "similar cars" from a marketplace's API, and agents acting for buyers (AI Across the Listing Lifecycle). A marketplace that exposes similar-listing retrieval as an API product needs the same availability guarantees, rate limits and attribution as its own carousel. Building the pipeline as a service with a contract, rather than as a detail-page feature, keeps that option open.

Questions to Ask

1. What share of our live inventory appears in at least one recommendation module in a week, and how old is the average recommended listing compared with the average live listing?
2. How long after a listing is sold or removed can it still be recommended, on each surface, including e-mail and push?
3. Which candidate source serves a listing in its first hour, before it has any interactions?
4. Does our ANN index support real-time deletes and filtering during the search, or are we post-filtering a periodically rebuilt index?
5. What is our recommenders' primary metric? If it is click-through, what evidence do we have that it tracks contacts?
6. What does a user who declines personalization see, and how does its contact rate compare with the personalized version?
7. Do seller or dealer caps exist on our recommendation surfaces, and who set their values?
8. Are recommendation cards rendered from the live listing read model or from precomputed data?
9. Which data from other brands, if any, feeds our recommenders, and where is the consent and lineage for it recorded?

10. Who owns each production recommender today, and when was each last retrained and evaluated?

From Lists to Maps

Recommendations rank without a query. The next chapter, Searching by Place: Map, Polygon and Commute Search, ranks within a place, where the map is the interface and the geography is the main filter. It deals with polygons, clusters, commute times and the finding that list-based metrics do not describe how people look at a map. The same principles return there: content first, availability always, and measurement on matches rather than clicks.

Sources and Evidence

Company engineering and investor sources

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- Dense Retrieval — Vinted Engineering, 2025. Item tower from categorical and image embeddings; about 50 A/B tests to full rollout.
- How to provide complementary item recommendations in e-commerce marketplaces — Adevinta tech blog, 2022 (possibly superseded). ANN index over items active in the last 90 days and viewed in the last 15.
- Improving Recommendation Quality by Tapping into Listing Text — Zillow Tech Hub, 2021 (read via the publication's RSS feed). Word2Vec listing-text embeddings for similar homes; daily generation; largest gains for vacant land.
- Academic Publications & Airbnb Tech: 2025 Year in Review — Airbnb Tech, 2025. Augmenting narrow searches with recommendations.
- Personalizing Airbnb search by learning from the guest journey — Airbnb Tech, n.d. (2026). Long-term and short-term behavior sequences.

Standards, regulation and documentation

- Filter approximate kNN results — Elastic documentation, n.d. Filtering during approximate kNN versus post-filtering.

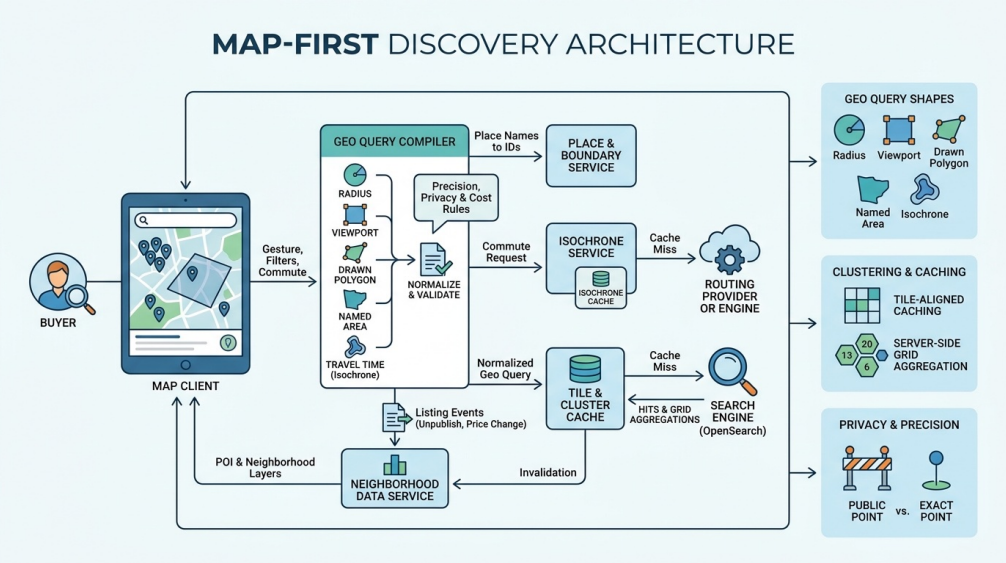
- Regulation (EU) 2022/2065 (Digital Services Act) — Official Journal of the European Union, 2022. Article 27 (recommender transparency for online platforms) and Article 38 (non-profiling option for very large platforms); wording checked against the Official Journal text served by the EU Publications Office.
- 24 CFR §100.75 Discriminatory advertisements — Cornell Legal Information Institute, n.d. US fair-housing advertising rule.

Research and practitioner sources

- Real-time Personalization using Embeddings for Search Ranking at Airbnb — Grbovic and Cheng, KDD 2018. Listing and user embeddings; search ranking and similar listings drive 99% of conversions.
- Improving Deep Learning for Airbnb Search — Haldar et al., 2020. Cold start for new listings.
- Applying Embedding-Based Retrieval to Airbnb Search — Abdool et al., 2026. Daily listing tower and its staleness trade-off; gains for new listings.
- Deep neural network marketplace recommenders in online experiments — Eide and Zhou (FINN.no), 2018. Marketplace-specific challenges; bandit ensembles in production.
- Job recommendations: benchmarking of collaborative filtering methods for classifieds — Kwieciński et al. (OLX), 2023. Offline and online comparison of ALS, RP3beta and others.
- Session-based Recommendations with Recurrent Neural Networks — Hidasi et al., 2015 (ICLR 2016). Session-based models for short, anonymous sessions.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

22. Searching by Place: Map, Polygon and Commute Search



IN THIS CHAPTER, YOU WILL: Learn the five ways a buyer says "here" (radius, viewport, drawn polygon, named area and travel time), how each becomes an index query, how maps cluster and cache results, and how commute search works, following one Lyon apartment search through Hearthly's draw-your-area feature.

WHY THE BUSINESS CARES: In real estate, and increasingly in automotive and local goods, buyers decide on place before anything else. A map that is slow, cluttered with duplicates or wrong at the edges loses the session to a competitor, and a commute filter priced per request can quietly become one of the largest variable costs in the search budget.

WHY ARCHITECTS SHOULD CARE: Map traffic is chattier, harder to cache and more expensive per request than list traffic, and it brings constraints the list never had: polygon complexity, aggregation that can leak private locations, and routing providers whose licenses shape the design.

KEY POINTS:

- **A map is not a list with pins.** Map sessions send many small viewport queries, show clusters and counts instead of a ranked top ten, and need their own ranking and evaluation. Airbnb reports that maps carry about 80% of its search interactions and that list metrics misjudge them.
- **Compile every place expression into a few primitives.** Named areas become term filters on place IDs, drawn shapes become simplified polygons, commutes become isochrone polygons or post-retrieval travel times, and viewports become tile-aligned boxes. The geo query compiler, not the client, owns that translation, so caching, privacy and cost rules apply everywhere.
- **Precision, privacy and routing cost are design inputs, not details.** Every pin and every cluster must be computed from the listing's public point, never from its exact one. One canonical property should show as one pin, not three. And the routing provider's pricing and license decide whether commute search runs per search or per listing.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Seeing Double** — On a map, three agencies' listings of one flat become three stacked pins, and the clutter is visible at a glance in a way a ranked list hides.
- **Ghost Inventory** — Cached map tiles and cluster counts keep showing sold and withdrawn listings long after search has removed them.
- **The Forced Fit** — A shared search platform tuned for list-shaped horizontal classifieds, imposed on a map-first property portal, pushes the portal's teams to build a shadow geo stack.

Part IV argues that most discovery queries are filters in disguise, and that the platform should compile a buyer's words, places and saved searches into its own structure. This chapter does that for place. Search usually treats location as one filter among many, a radius around a postcode applied before price and category; for many listing marketplaces that is backwards. A family looking for a flat starts with *where*: near a school, within half an hour of work, on this side of the river. A car buyer starts with *how far I am willing to drive to see it.* A buyer of a used sofa starts with *how far I can carry it.* **Place is often the first filter and the main one**, and in real estate it is also the main way people browse.

This chapter owns **map search** (discovery where the result set is displayed on, and scoped by, a map), **isochrone search** (filtering by travel time rather than distance) and **geo clustering** (grouping many results into fewer map markers). It builds on Where Is It? Location, Geocoding and Privacy, which owns geocoding, location precision and the geo index; that chapter decides *what point a listing has*, and this one decides *how buyers*

search over those points. It sits on the engine and index design of Search Architecture for Listings: Indices, Freshness and Scale, uses the ranking concepts of Ranking for Two Sides: Relevance, Freshness and Paid Visibility, and turns the canonical properties of One Property, Three Listings: Duplicates and Entity Resolution into map pins. The next chapter, Tell Me When It Appears: Saved Searches and Alerts, turns saved map searches into alerts.

Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), offers the case. Its property portal Hearthly launched a draw-your-area search in France, and by 2027 it has become both the feature agencies praise and the workload the search team fears.

Setting the Scene at Veldmark: The Polygon Problem at Hearthly

It is late September 2027, two weeks after Veldmark decided on **14 September** to move Tradeyard and Hearthly onto a shared search platform on Amazon OpenSearch Service by the end of 2028 (see Search Architecture for Listings: Indices, Freshness and Scale). Julien Marchand, Hearthly's CTO, asks Ingrid Halvorsen, Veldmark's Group Chief Architect, and Tomasz Wiśniewski, who heads the group search platform, for a session about one feature before the migration is planned.

Hearthly's search runs on self-managed OpenSearch on Google Kubernetes Engine. In France it serves **1.9 million** live listings, which entity resolution groups into about **1.12 million** distinct properties (One Property, Three Listings: Duplicates and Entity Resolution), and **0.7 million** in Spain. Julien's numbers for September 2027, local to this chapter and illustrative, are these:

- **61%** of Hearthly France search sessions open the map view at least once; on mobile, 54%.
- **17%** of sessions use *Dessinez votre zone* ("draw your area"), the draw-your-area search launched in France in 2024.
- At the Sunday-evening peak, Hearthly's search API handles about **4,200 requests per second**, of which about **2,700** are map requests: viewport queries, cluster requests and count updates as users pan and zoom.
- Drawn polygons arrive with a median of **146 vertices** and a maximum of more than **2,000**, because the mobile client sends the raw finger trace.
- p99 latency for polygon queries at peak is **1.8 seconds**; for list queries over the same area defined by city, it is **210 milliseconds**.
- About **1.1 million** of Hearthly's 7 million saved searches contain a drawn polygon.

Two complaints arrive with the numbers. Agencies in Lyon, among them **Agence Rhône Habitat** (fictional, 14 branches), complain that the map shows their listing as

one of three stacked pins for the same flat. And product wants commute search, *"30 minutes by public transport to my office"*, for the spring 2028 season, with a routing provider's quote that prices each isochrone request.

Tomasz's worry is that Hearthly's traffic does not fit the shared platform. The shared OpenSearch platform was sized and tuned for Tradeyard, which produces about 70% of group queries and is list-shaped: category, keyword, radius and price, with a ranked page of results. Hearthly is map-shaped. And Hearthly stays on Google Cloud through 2028 while the shared platform runs on AWS, so every pan and zoom could cross clouds.

Ingrid frames three questions for the session. **What query shapes does the map need, and who translates the user's gesture into them? How should results be grouped, cached and ranked on a map? And what does commute search cost, in latency, money and license terms?** The rest of this chapter answers them.

Five Ways to Say "Here"

Buyers express place in five ways. Each has a different meaning, a different index primitive and a different cost. Treating them as one "location filter" is the source of most map-search bugs.

Expression	What the buyer means	Index primitive (Elastic-search/OpenSearch terms)	Cost and cacheability
Radius	"Within 25 km of Essen"	`geo_-distance` filter on a point	Cheap; cacheable if the center is snapped to a grid
Viewport (bounding box)	"What is in the part of the map I am looking at"	`geo_-bounding_-box` filter	Cheap per query, but many queries per session; cacheable only if aligned to tiles
Drawn polygon	"Only this area I traced"	`geo_shape` query with a polygon against a point field	Cost grows with vertex count; rarely cacheable; must be normalized

Expression	What the buyer means	Index primitive (Elastic-search/OpenSearch terms)	Cost and cacheability
Named area	"Lyon 7e", "Eixample", "postcode 3011"	Term filter on a denormalized array of ancestor place IDs	Cheapest and most cacheable; boundaries are reference data
Travel time (isochrone)	"30 minutes by tram from my office at 08:30"	A `geo_shape` filter on an isochrone polygon, or travel times computed per listing after retrieval	One routing call per search or many per page; the provider's pricing dominates

A **bounding box** is the rectangle defined by the map's south-west and north-east corners at the current zoom. An **isochrone** is a polygon enclosing every point reachable from an origin within a time budget, for a given mode of transport and departure time. A **place ID** is a stable identifier for an administrative or named area (country, region, city, district, neighborhood). Where Is It? Location, Geocoding and Privacy recommends storing on each listing the ancestor place IDs, so that "all listings in Lyon 7e" is a term filter, not a polygon query. A **public point** is the coordinate a listing may expose to clients, which for private sellers is deliberately coarser than the true location (also from Where Is It? Location, Geocoding and Privacy).

Elasticsearch documents these primitives, and OpenSearch uses the same names: `geo_point` stores points and `geo_shape` stores polygons, lines and rectangles. Shapes are tessellated into triangles indexed in a BKD tree, at a precision of about 1e-7 degrees, and **tessellation cost scales with the number of vertices** (Elastic, geo-shape field type, n.d.). The `geo_shape` query can run against `geo_point` fields with the spatial relations `intersects`, `within`, `disjoint` and `contains`. It can also reference a shape already indexed in another index by its ID, which is how a platform can store named areas or saved polygons once and refer to them by name (Elastic, geo-shape query, n.d.).

Three semantic traps sit inside the table.

1. **Radius and named area disagree at the edges, and users notice.** A flat 300 meters outside the Lyon 7e boundary is in a 2 km radius around the district center but not in "Lyon 7e". Neither answer is wrong; the product must say which one it applies, and saved searches must store which one the user chose.

2. **A viewport is a display scope, not intent.** A user who zooms out to orient themselves has not asked for all of Lyon. Systems that silently turn the last viewport into a filter on the list view, or into a saved search, produce results the user never meant. The rule I use: the viewport filters the map; it filters the list only when the user has chosen "search this area".
3. **A coarse public point makes "within" probabilistic.** If a private seller's bike in Rotterdam is shown at the center of an H3 cell with an edge of about 1.4 km (resolution 7, see Where Is It? Location, Geocoding and Privacy), a polygon that cuts through that cell includes or excludes the bike based on the cell center, not the bike. For goods that is acceptable. For real estate with exact addresses it is not an issue. For the cases between, such as an agency that hides the street number, the product must decide whether to match on the true point and show the coarse one, which leaks a little information through the result set, or to match on the coarse point and accept edge errors.

The Map Is a Different Product

The strongest public evidence on map search comes from Airbnb, which is not a classifieds site but shares the property: place-first discovery of unique inventory. In a 2024 paper, Airbnb's search team reported that its two result interfaces, a ranked list and price pins on a map, had used the same booking-probability ranking, and that list assumptions did not hold on the map. The team rebuilt the model of user attention for maps (Haldar et al., *Learning to Rank for Maps at Airbnb*, arXiv, 2024). Airbnb's 2025 review of its research says maps account for "roughly 80% of search interactions" and describes a map-specific NDCG (normalized discounted cumulative gain, the standard list-ranking metric), because list NDCG "fails to model user attention on maps" (Airbnb Tech, 2025 year in review). Airbnb's January 2026 paper on embedding-based retrieval adds that nearly all its searches carry filters, "especially map-based geographic filters", and that it uses S2 cells as a location feature (arXiv 2601.06873, 2026).

Property portals have long treated the map as central. Rightmove named "rich, map-centric results" among the three priorities of its 2014 move to Elasticsearch, according to a case study Elastic published in 2025 (Elastic, 2025); the case study does not detail Rightmove's current map stack. Applying Airbnb's ranking findings to property portals is my inference, not something the portals have published. The mechanics carry over, though, and the table shows how they change the architecture.

Property	List view	Map view
Unit of result	A ranked page of 20–50 listings	Every match in the viewport, as pins or clusters
Queries per session	A few, at each filter change	Dozens: every pan, zoom and "search this area"
What must be fast	The first page	Counts and clusters at every zoom level, plus the pins at the deepest zoom
What ranking means	Order of the page	Which pins to show when there are too many, which pin to highlight, and the order of the side list
Attention model	Top positions get most attention, decaying down the page	Attention follows geography and the user's area of interest, not a position
Evaluation	NDCG, click-through by position	Map-specific metrics; contacts per map session; time to shortlist

Two consequences follow. First, **the map is a search client in its own right**, with its own latency budget and query mix. Capacity planning that counts "searches" as list queries will underestimate the map's load. Second, **ranking does not disappear on the map; it moves**. When a viewport over central Lyon holds 2,400 matching flats, the system must decide which 300 to show as individual pins and which to fold into clusters, which pins carry a price label, and what the side list shows. Paid placements raise the same questions on the map that Ranking for Two Sides: Relevance, Freshness and Paid Visibility discusses for lists: a "featured" pin that is larger or always shown is a paid boost, and ranking transparency obligations apply to it.

Architecture: A Geo Query Compiler in Front of the Engine

The design I recommend puts a geo query compiler between the map clients and the engine, and gives each other map concern its own component with a narrow interface. Figure 1 shows them for Hearthly's target state on the shared search platform.

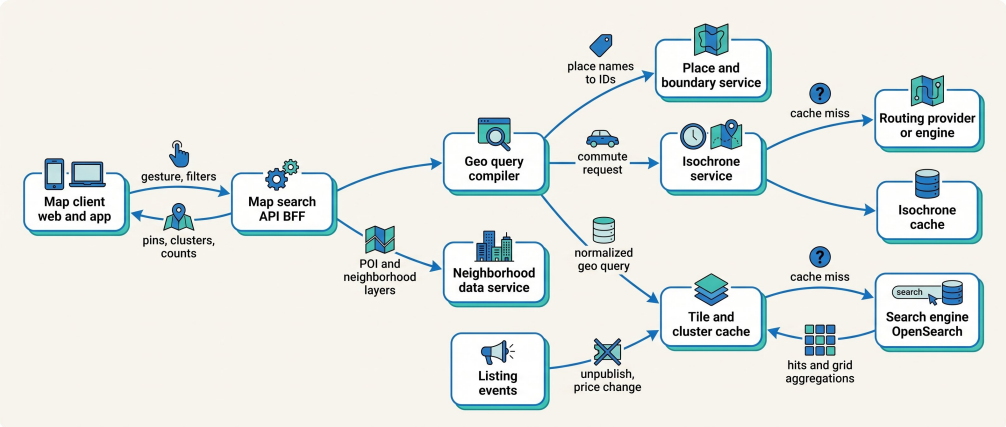


Figure 1: *The geo query compiler turns every place expression into a normalized query before any cache or engine sees it. Commute requests go through an isochrone service with its own cache, and listing events invalidate cached tiles so the map does not show ghost inventory.*

- **Map search API** (a backend-for-frontend, see Found by Google: Web, App and SEO Architecture for Listings). Receives the gesture (viewport, zoom, drawn shape, chosen areas, commute parameters) and the ordinary filters. Returns pins, clusters, counts and the side list in one response shaped for the map client. It owns debouncing rules and response budgets per zoom level.
- **Geo query compiler**. The center of the design. It normalizes every place expression into one of the engine primitives: it resolves names to place IDs, simplifies and validates polygons, snaps viewports to tiles, asks the isochrone service for commute polygons, and enforces the precision rule (match and display fields, see below). Because every query passes through it, it is also where cost limits, vertex caps and cache keys are defined.
- **Place and boundary service**. Reference data owned by the location capability of Where Is It? Location, Geocoding and Privacy: place IDs, hierarchies, boundary polygons and their versions. Boundaries change (municipal mergers, new postcode areas), so saved searches that reference place IDs must be migrated like taxonomy changes (Category Trees That Can Change: Designing Marketplace Taxonomies).
- **Isochrone service**. Turns (origin, mode, time budget, departure time) into a polygon, through a routing provider or a self-hosted engine, with a cache in front. It is a separate service because its cost, latency and license terms differ from everything else in the search path.
- **Tile and cluster cache**. Stores aggregated results per map tile and normalized filter set. It is invalidated by listing events (publication, unpublish, price change) from the listing event stream (Every Change Is an Event: Event-Driven Listing Architecture).
- **Search engine**. Executes geo filters together with the ordinary filters and returns hits and grid aggregations. It holds no geo logic the compiler does not control.

- **Neighborhood data service.** Serves points of interest and area facts (schools, transit stops, noise, flood zones, price statistics) as map layers and, selectively, as filters.

The rule that holds the design together: **clients send intent, not engine queries.** A mobile client that sends a raw 2,000-vertex finger trace straight to the engine, or that constructs its own bounding boxes at arbitrary precision, puts the cost and correctness of the map in the hands of every app release. Mobile releases are slow to reach all users (see Found by Google: Web, App and SEO Architecture for Listings), so a client-side mistake can live for a year. The compiler makes the server the owner.

A pan, a zoom, a count

Figure 2 shows the sequence for a single pan on the map, the most frequent interaction.

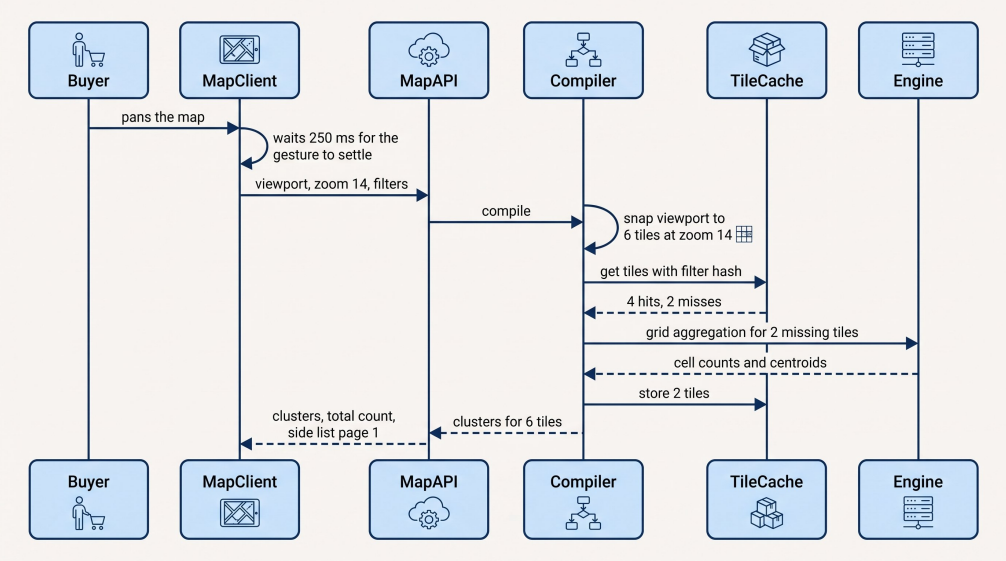


Figure 2: *Snapping the viewport to tiles turns an uncacheable bounding box into a set of reusable cache entries. A pan typically reuses most of the previous tiles, so the engine sees only the edge of the movement.*

Three choices in the figure matter.

1. **Debounce in the client, but enforce budgets on the server.** Clients should wait for a gesture to settle before querying. But the server must still rate-limit per session, because a buggy client release can multiply map traffic overnight.
2. **Snap viewports to tiles.** A raw bounding box is almost never repeated exactly, so it cannot be cached. A viewport rounded to the tiles of a fixed grid at the

current zoom produces cache keys that many users share. Web-map tiles use the standard `zoom/x/y` scheme; Elasticsearch's `geotile_grid` aggregation buckets points into exactly these tiles, at precisions 0 to 29, and returns only non-empty cells. Its documentation warns that the finest precision is smaller than 10 cm × 10 cm and recommends a bounding-box filter to avoid huge numbers of buckets (Elastic, geotile grid aggregation, n.d.).

3. **Separate the count from the pins.** The total count in the header ("2,412 homes in this area") and the cluster counts are the expensive part, much as facet counts are for list search (Search Architecture for Listings: Indices, Freshness and Scale discusses Vinted's 2023 experience with facets). Approximate counts above a threshold ("2,000+") and a separate request path for the side list keep the map responsive.

Clustering: Three Ways to Show Ten Thousand Results

At city zoom levels a map cannot show every match as a pin. **Geo clustering** groups nearby results into a marker with a count, which splits into smaller clusters and finally pins as the user zooms in. There are three places to do it.

Approach	How it works	When it fits	Main risk
Server-side grid aggregation	The engine buckets matching listings into grid cells (`geotile_grid` for square map tiles; `geohex_grid` for H3 hexagons) and returns counts, optionally with centroids and sub-aggregations such as median price	Large result sets; filters change often; the default for real estate and automotive	Engine CPU at peak; grid-shaped clusters look mechanical unless centroids are used

Approach	How it works	When it fits	Main risk
Client-side clustering	The server returns all matching public points in the viewport (capped); the client clusters them, for example with Mapbox's open-source Supercluster library, which builds a hierarchy of clusters per zoom level over a k-d tree	Up to tens of thousands of points; smooth zoom without server round trips	Payload size on mobile; the cap silently drops listings; every point sent is data that can be scraped
Pre-rendered vector tiles	The engine or a tile service returns map tiles with hits and aggregated cells already encoded. Elasticsearch's vector tile search API returns a binary Mapbox vector tile with `hits`, `aggs` (from `geotile` or `geohex` cells) and `meta` layers	Very high read volume with few filter combinations; base layers such as "all homes for sale"	Combinations of filters multiply the tile space; cache invalidation on every listing change

The sources for the table are documentation, all undated: OpenSearch's `geohex_grid` aggregation groups points into H3 cells identified by their H3 index, at precision 0 to 15 (OpenSearch, geohex grid, n.d.). Supercluster describes itself as "a very fast geospatial point clustering library for browsers and Node", built to power clustering in Mapbox GL JS (Mapbox, Supercluster, n.d.). Elasticsearch's `_mvt` endpoint returns the tile layers listed above, with `grid_agg` set to `geotile` or `geohex` and up to 10,000 hits per tile (Elastic, vector tile search API, n.d.). A team on OpenSearch that wants vector

tiles should check the current OpenSearch documentation before assuming the same endpoint exists; the aggregations underneath are available in both engines.

The approaches combine. A common split is server-side aggregation at city and region zoom, client-side clustering of capped point sets at street zoom, and individual pins when the viewport holds fewer than a few hundred matches. H3 hexagons, introduced by Uber in 2018 (Uber Engineering, 2018), make clusters that look less like a chessboard than square tiles do, because every hexagon has six equidistant neighbors; Where Is It? Location, Geocoding and Privacy gives the resolution table.

The privacy trap in aggregation

Aggregations can leak what pins hide. If a private seller's exact point is stored next to the public point, and the cluster centroid or the cell is computed from the exact point, a cluster of one at high zoom places a marker at the seller's front door. The same holds for heatmaps, "price per square meter" grids computed at fine resolution, and the `hits` layer of a vector tile.

The rule is absolute: **every geometry returned to a client is computed from the public point field**, and the exact point field is not readable by the map search API at all. Where Is It? Location, Geocoding and Privacy recommends a deterministic public point per listing, either a snapped cell center or a seeded random offset. Determinism matters again here: if the offset changed with each query, averaging cluster positions over many requests would reveal the true point. For Tradeyard's private sellers the public point is the center of an H3 resolution-7 cell. For Hearthly's agency listings it follows the precision each agency chooses (P3 district, P4 street or P5 exact address), and a property listed by several agencies shows the coarsest of their choices.

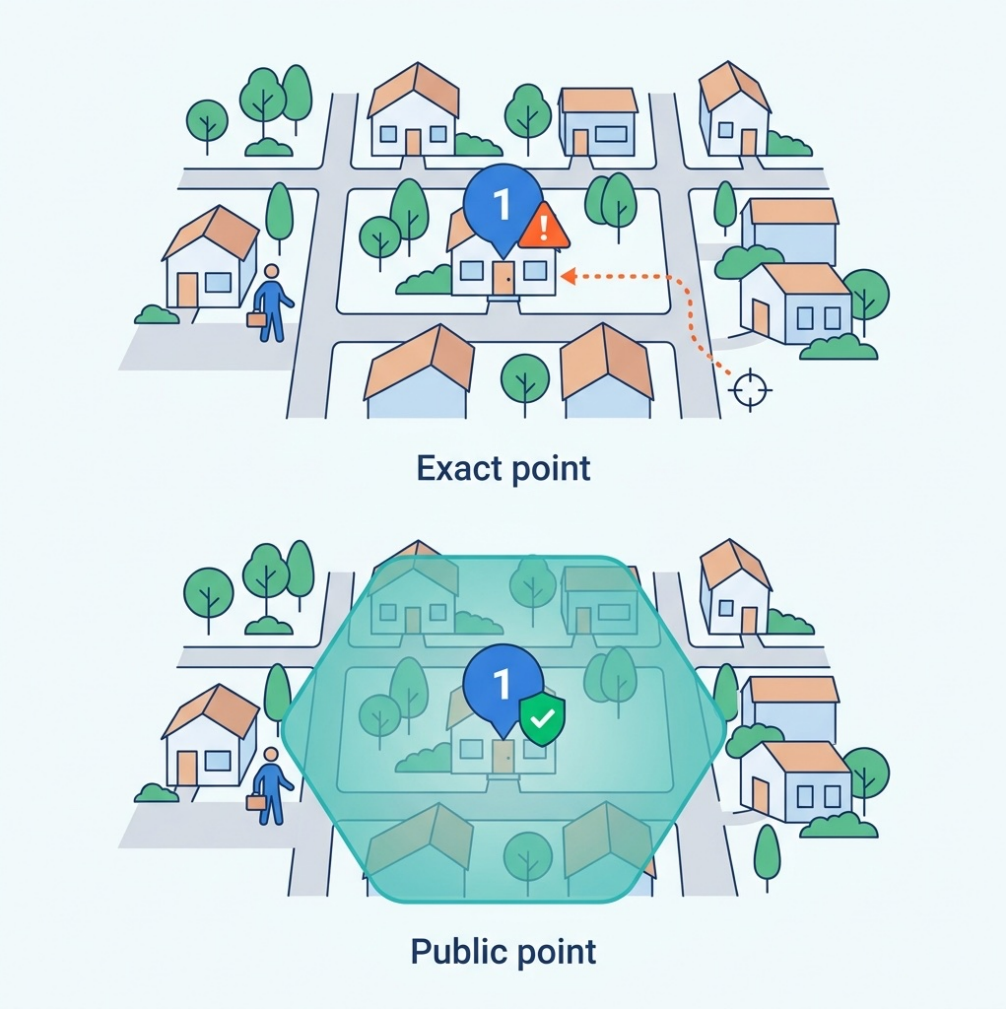


Figure 3: *Aggregations can leak what pins hide: a cluster of one computed from the exact point marks the seller's front door, so every geometry the map returns is computed from the public point.*

Drawn Polygons: Normalize Before You Query

A drawn polygon is the most expressive place query and the most dangerous one. Its cost depends on vertex count, its validity depends on the user's finger, and its meaning depends on rules the user never sees. Large portals offer draw search (Rightmove, Idealista and Zillow among them), but how they implement it is not publicly documented. The natural implementation in Elasticsearch-style engines is a `geo_shape` query with the user's polygon against listing points, and Hearthly uses it.

The geo query compiler should apply six steps to every drawn shape before it reaches

the engine.

1. **Validate and repair**. Close the ring; remove repeated points; fix self-intersections, which a finger trace that crosses itself produces, by splitting into valid polygons or taking the outer boundary; reject zero-area shapes. Engines reject or misinterpret invalid polygons, and the error should be the compiler's, with a clear message to the user, not a 500 from the cluster.
2. **Simplify**. Reduce vertices with a tolerance tied to the zoom level at which the shape was drawn, for example with the Douglas–Peucker algorithm. A trace drawn at city zoom does not need meter-level precision; 15–25 meters is invisible on screen and cuts most vertices.
3. **Cap**. Enforce a maximum vertex count after simplification (Hearthly's target is 64) and a maximum area. A polygon covering half of France is a region search and should be compiled as one.
4. **Snap to known areas when the shape is close to one**. If a drawn shape matches a named area within a tolerance, offer the named area. Named-area filters are cheaper, cacheable and stable when boundaries are versioned.
5. **Store it as an object**. A drawn shape that will be reused, in a saved search or a shared link, gets an ID and is stored once, then referenced (the pre-indexed shape mechanism in the `geo_shape` query, or the platform's own shape store). Saved polygons are the input to alerts in Tell Me When It Appears: Saved Searches and Alerts, so their normalized form is the one to keep.
6. **Pre-filter coarsely**. Add a bounding-box filter or a set of covering cells (H3 or S2 cells that cover the polygon) before the exact polygon test, so that the expensive geometric test runs on few candidates. Whether the engine already does this internally depends on the engine and version; adding it explicitly is cheap.

For Hearthly, steps 2 and 3 alone change the workload. Julien's team tests the Lyon polygons from September: simplification at 20 meters cuts the median from 146 to 31 vertices and the maximum from over 2,000 to the cap of 64. In a replay of a Sunday-evening hour against a staging cluster, polygon query p99 falls from 1.8 seconds to **240 milliseconds**. These are illustrative figures for Veldmark; the direction is what the Elastic documentation predicts.

Figure 4: *Normalize before you query: at Hearthly, simplifying drawn shapes at 20 meters cut the median from 146 to 31 vertices, and polygon p99 at peak fell from 1.8 seconds to 240 milliseconds in replay.*

The remaining semantic choice is **`within` versus `intersects` for points**. For point data the two give the same answer except for points exactly on the boundary, which is rare. The real boundary question is the coarse public point discussed earlier, and the right answer is a product decision that should be written down once, in the compiler, and not rediscovered by each client team.

Commute Search: Isochrones and Travel Times

Commute search lets buyers filter or sort by travel time to a place that matters to them. For property it matters because the cost of a home is partly the cost of getting to work from it. There are two documented forms in the property sector.

- **Per-listing travel times to the user's places.** Rightmove's *My Places* feature calculates travel times from each property to points of interest the user saves (work, school, family), for walking, cycling, driving and public transport, using the third-party TravelTime API. The places persist in the user's session, so travel times appear on every property they view. According to TravelTime's case study (undated; the page carries a 2026 footer and is vendor marketing), Rightmove chose it partly for a **fixed annual license with unlimited requests**, because computing travel times to every place, for every mode, across millions of daily searches "would simply not be possible in a transactional model" (TravelTime, Rightmove case study, n.d.). The case study describes per-listing travel times, not isochrone filtering.

- **Commute as a search criterion.** Zillow added commute time, affordability and nearby points of interest as natural-language search criteria in September 2024, with the example "homes within 30 minutes of downtown Dallas" (HousingWire, 2024-09-04). The article gives no technical detail, so how Zillow computes it is not public.

Behind both forms sit two patterns, which a platform will usually need together.

Pattern	How it works	Cost profile	Accuracy
Isochrone filter	Compute the isochrone polygon for (origin, mode, minutes, departure time) once per search; run it as a `geo_shape` filter like a drawn polygon	One routing call per search; cacheable by quantized origin, mode, budget and time bucket	Approximate at the polygon's edges; the polygon is a simplification of a network
Per-listing travel time	Retrieve listings normally; compute travel times from the top N results to the user's places with a matrix call; display, sort or filter the page	N × places routing computations per page; grows with traffic	Exact per listing for the chosen departure time

The isochrone filter is what makes "show me only homes within 30 minutes" possible over the whole inventory; per-listing travel time is what makes each result card honest. A design that uses only the first shows "28 minutes" on a home that the router would say is 33 minutes away; a design that uses only the second cannot filter, only annotate a page.

Routing providers are an architectural constraint

The commercial model of the routing provider is an architectural input, as the Rightmove case shows. Three kinds of terms matter.

1. **Pricing unit**. Per request, per element of a matrix, or a fixed license. Per-listing travel time on a pay-per-element contract scales with page views, which in a property portal peak at exactly the moments marketing works hardest.
2. **Limits**. Mapbox's Isochrone API, for example, documents (as of 2026) a maximum of 60 minutes per contour, at most four contours per request, profiles for driving (with and without traffic), walking and cycling, a limit of 300 requests per minute, and a requirement that results be displayed on a Mapbox map (Mapbox, Isochrone API, n.d.). The rate limit alone rules it out as an uncached per-search call for a portal at Hearthly's peak, and the display requirement constrains the choice of map renderer.
3. **Storage and display rights**. Whether you may cache results, for how long, and on which maps you may display them. Where Is It? Location, Geocoding and Privacy shows the same issue for geocoding, where storing results is a priced option at some providers.

The alternative is a self-hosted routing engine. Valhalla, an open-source engine, offers an isochrone service that "computes areas that are reachable within specified time intervals from a location", with costing models for car, bicycle, pedestrian and multimodal travel, up to four contours by default, and polygon output (Valhalla documentation, n.d.). Self-hosting moves the cost from license fees to engineering: road and transit data, which for public transport means importing operators' timetables (commonly published in the General Transit Feed Specification, GTFS) and keeping them current, plus capacity for the peaks. For public transport in particular, the departure time changes the answer, so the cache key must include a time bucket and the product must state the assumption ("weekdays, leaving at 08:30").

Caching isochrones

Commute requests repeat often, because people commute to the same few places. The isochrone cache key I recommend is:

- the origin snapped to an H3 resolution-9 cell (about 0.1 km², average edge about 200 m; Where Is It? Location, Geocoding and Privacy), so that two users working in the same building share an entry;
- the mode;
- the time budget, rounded to the options the product offers (15, 20, 30, 45, 60 minutes);
- the departure time bucket (for example weekday morning peak, weekday off-peak, weekend).

Snapping the origin by up to about 200 meters changes the isochrone by a small margin, which is an acceptable trade for a high hit rate; the per-listing travel time on the result

card then corrects the displayed number. Cache entries must expire when the transit timetable or road data changes, and isochrones should be stored in simplified form, through the same normalization as drawn polygons.

Neighborhood Data and Points of Interest

Neighborhood context (schools, transit stops, shops, parks, noise, flood risk and local price statistics) is the part of map search most often under-designed, because it looks like content rather than data.

Four rules apply.

1. **It is reference data with owners, sources and refresh cycles.** School lists, transit stops and flood zones come from public registries or licensed providers, each with its own update cycle and license. OpenStreetMap data, for instance, is under the Open Database License (ODbL), which requires attribution and has share-alike terms for derivative databases (Where Is It? Location, Geocoding and Privacy discusses the legal review this needs). The neighborhood data service should record source, license and as-of date for every layer.
2. **Display is cheap; filtering is not.** "Show schools on the map" is a layer the client fetches per tile. "Only homes within 500 meters of a primary school" is a geo query that must be compiled, typically as a union of buffers precomputed per school or as a precomputed per-listing attribute ("distance to nearest primary school") written at enrichment time (From Raw Supply to Usable Listings: Normalization and Enrichment). Precomputing is usually better: the listing's location rarely changes, and the attribute can be filtered and ranked like any other.
3. **Area statistics must use enough data.** A "median price per m²" for a cell with four sales is noise presented as fact. Aggregate at a resolution where the counts support the number (H3 resolution 8, about 0.74 km² on average, is a common neighborhood-level choice), and suppress cells below a minimum count. For Hearthly, area statistics come from Valuo, Hearthly's automated valuation model, whose prices are learned from France's open transaction data (DVF, *Demandes de valeurs foncières*), never from asking prices (Models in Production: ML Platforms and Use Cases); Hearthly computes its local price medians per H3 resolution-8 cell.
4. **Some neighborhood facts are legally sensitive.** In the United States, fair-housing rules (24 CFR §100.75) prohibit housing advertisements that indicate a preference based on protected characteristics, including through the selection of media or locations (Cornell LII, 24 CFR §100.75). Demographic overlays and "neighborhood character" scores attached to listings are therefore a compliance question for US portals, not a design flourish. Europe has different rules, but the same caution applies to any layer that could be read as steering buyers by who lives in an area; Rules of the Marketplace: DSA, DAC7 and Vertical Regulation owns the details.

Performance and Caching at the Peak

Map traffic peaks with all search traffic. At Veldmark the group peak is about **31,000 queries per second** on Sunday evenings, 20:00–22:00 CET; that peak is Veldmark's illustration, while the documented public example of a sharp, marketing-driven peak is Rightmove's busiest minute, "8:48 pm on Wednesday, during the final ad break on prime-time TV" (Google Cloud press release, 2021; see Sunday Evening Peaks: Reliability, Performance and Cost). Map requests are about 2,700 of Hearthly's 4,200 peak requests per second, because each session sends many of them.

The levers, in the order I would pull them:

1. **Tile-aligned caching of aggregations** with filter-set hashing. Normalize filters (sorted keys, bucketed price ranges) before hashing, as for facets.
2. **Cheaper shapes**: named areas where the user's intent allows, and normalized polygons everywhere else.
3. **Approximate counts** and separate request paths for counts, clusters and the side list.
4. **Zoom-dependent result limits**. At region zoom, clusters only; at street zoom, a capped set of pins. Never return 10,000 points to a phone.
5. **Degradation modes**. Under overload, serve cached clusters even if slightly stale, disable live counts, and fall back from per-listing travel times to isochrone-only. Sunday Evening Peaks: Reliability, Performance and Cost owns the general model; the map needs its own entries in it.

Cache invalidation is where map caching meets **Ghost Inventory**. A cluster tile cached for ten minutes shows a sold flat for ten minutes, and a pin that opens a "listing no longer available" page is the most visible kind of staleness. The listing event stream should invalidate tiles: an unpublish or price-change event identifies the tiles containing the listing's public point at each cached zoom level, and those entries are dropped. With tile keys derived from the public point, this is a direct calculation, not a search. A short time-to-live (one to two minutes) as a backstop limits the damage if an event is missed. From Submitted to Searchable: The Publication Pipeline owns the unpublish propagation SLA; the tile cache is one more consumer that must meet it.

Hearthly's cross-cloud constraint adds one more lever. With the shared search platform on AWS and the Hearthly application on Google Cloud through 2028, Ingrid's recommendation is to run the map search API and the tile cache on the Hearthly side, so that most pans are answered without crossing clouds, and only cache misses go to the engine. The cost of the cross-cloud round trip is then paid only by the edge of each movement.

Worked Example: The Lyon Apartment on the Map

In May 2027, a three-room (T3) **68 m²** flat in Lyon 7e is listed for sale at **€389,000**, energy class (DPE, *diagnostic de performance énergétique*) **D**, by **three agencies**: one listing arrives through each of two real-estate software vendors and one through Keybridge, Veldmark's agency software. One of the agencies is Agence Rhône Habitat. Entity resolution clusters the three listings into one canonical property, and Valuo estimates it at **€372,000–€401,000** (the canonical facts are in Meet Veldmark: Four Brands, One Platform to Build).

Ingrid's team uses the flat as its design walk-through: it replays a buyer's May 2027 evening session as it would run in the target design of Figure 1, with commute search included. Sophie (fictional, she/her) works near Lyon Part-Dieu station and wants a T3 under €400,000 within 30 minutes of the office by public transport.

1. **Commute filter**. Sophie enters her office address and "30 minutes, public transport, weekdays at 08:30". The compiler snaps the origin to its H3 resolution-9 cell and asks the isochrone service. The key is already cached; colleagues of hers asked the same question this week. The isochrone, simplified to 58 vertices, covers most of Lyon 3e, 7e and 8e and parts of Villeurbanne.
2. **Map view**. At city zoom the compiler intersects the isochrone with the viewport, snaps to 9 tiles, and asks for H3 grid aggregations with the filters (sale, T3, up to €400,000). Seven tiles are cache hits. The map shows 14 clusters with 412 matching properties. The count is per **canonical property**, not per listing: the engine collapses listings of the same property (the canonical property ID is in the listing document, see One Property, Three Listings: Duplicates and Entity Resolution), so the three agency listings of the 7e flat count once.
3. **Draw**. Sophie draws around the streets near the Jean-Macé metro stop. Her trace has 181 vertices; the compiler repairs a self-intersection, simplifies it to 27 vertices and intersects it with the isochrone. 63 properties remain.
4. **Pins**. At street zoom the map shows 63 pins. The 7e flat is **one pin**, at the exact address because all three agencies chose to publish it (a clustered property shows the coarsest precision any of its agencies chose). Tapping it shows the three agency offers on one property card, with the price, the DPE class, the Valuo range and a per-listing travel time of **24 minutes**, computed by a matrix call for the pins on screen.
5. **Save**. Sophie saves the search. The saved search stores the normalized polygon by reference, the isochrone parameters (not the polygon, which may change when timetables change) and the filters. From this point, Tell Me When It Appears: Saved Searches and Alerts takes over.

The example shows the value of the compiler as one owner: the canonical-property rule, the public-point rule, the polygon normalization and the isochrone cache are each

enforced once. And it shows the value of entity resolution on the map: without it, the flat would be three stacked pins, and the agency that complained in September would be right.

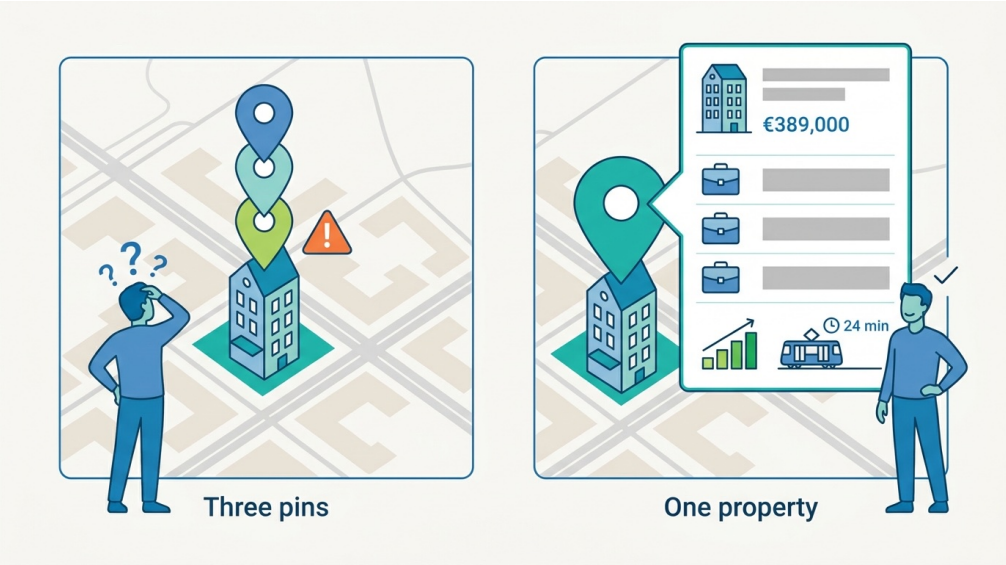


Figure 5: *Entity resolution on the map: the Lyon 7e flat listed by three agencies is one pin and one property card with three offers, instead of the three stacked pins that Agence Rhône Habitat complained about.*

The commute arithmetic, not a feature comparison, decides the routing choice. If 10% of Hearthly's 2,700 peak map requests per second come from commute sessions, and each needs a new isochrone per parameter change plus a travel-time matrix per page of pins, then even with a 70% isochrone cache hit rate the peak load is in the low hundreds of isochrones and tens of thousands of matrix elements per second. On a per-request price list, that cost rises with every marketing campaign; with a fixed license or a self-hosted engine, it is capacity planning. So for the spring 2028 commute launch Hearthly chooses a fixed-license provider for travel-time matrices and a self-hosted engine for cached isochrones (an illustrative Veldmark decision).

Trade-offs

The main design choices, with when each is right:

Decision	Option A	Option B	Choose A when	Choose B when
Where geo logic lives	Clients build engine queries	Server-side geo query compiler	Single web client, small scale, prototype	Several clients, mobile apps, saved searches, any cost or privacy rule
Area filtering	Polygons for everything, including named areas	Place-ID term filters for named areas; polygons only for drawn and commute shapes	Boundaries are not available as reference data	Always, once place hierarchies exist (Where Is It? Location, Geocoding and Privacy)
Clustering	Server-side grid aggregation	Client-side clustering of capped point sets	Large result sets, frequent filter changes	Small result sets, smooth zooming, low scraping risk
Commute	Isochrone filter	Per-listing travel time	The user needs to filter the whole inventory	The user needs exact times on a page of results; usually you need both
Routing	Commercial API, pay per use	Fixed license or self-hosted engine	Low or uncertain volume, fast launch	Volume is high and peaky; display or storage terms constrain the product
Map ranking	Reuse the list ranker for pin selection	Map-specific selection and evaluation	Map is a minor surface	Map carries a large share of interactions, as in real estate

Decision	Option A	Option B	Choose A when	Choose B when
Map unit	One pin per listing	One pin per canonical property or item	Goods, where duplicates are rare and listings are the items	Real estate, multi-dealer automotive inventory, any vertical with multi-listing

The table hides one cross-cutting cost: **every geo feature that becomes a saved search becomes an alert workload**. A drawn polygon in a saved search is percolated against every new listing in the area; an isochrone in a saved search must be recomputed when timetables change. Product decisions on map features are therefore also capacity decisions for Tell Me When It Appears: Saved Searches and Alerts.

Vertical Differences

Dimension	General classifieds (Trade-yard)	Real estate (Hearthly)	Automotive (Motor-vane)	Jobs
Role of place	Pickup distance; local liquidity	**The** primary filter and browse mode	Willingness to travel to a dealer, often 25–200 km	Commute and remote-work options
Typical query	Radius around the user or a postcode	Named areas, drawn polygons, commute	Radius from postcode; sometimes "nationwide with delivery"	City or region; commute radius; "remote" as a non-place

Dimension	General classifieds (Trade-yard)	Real estate (Hearthly)	Automotive (Motor-vane)	Jobs
Point precision	Coarse public point for private sellers	Exact address where the agency allows; hidden street number otherwise	Exact dealer address; coarse for private sellers	Employer office address; sometimes region only
Map view	Secondary, mostly for local goods	Primary; often the majority of sessions	Secondary; dealer locations matter more than car positions	Rare
Duplicates on the map	Reposts, usually filtered by moderation	Multi-agency listings of one property: a pin per canonical property	The same car advertised by a dealer group across branches	The same job through several recruiters
Commute search	Rarely useful	High value	Low; distance to dealer suffices	High value
Neighborhood data	Little	Schools, transit, noise, flood, prices; legally sensitive in some markets	Dealer reviews and opening hours	Transit; employer location

For automotive, the place that matters is the seller's, not the car's, so Motorvane's map is effectively a dealer map, and its main geo query is a radius around the buyer's postcode in German (*Umkreissuche*). Motorvane stays on Elasticsearch until a 2029 review, which is fine: the geo primitives in this chapter exist in both engines. For general classifieds, the strongest geo lever is not the map but **local liquidity**: ranking by distance and recency

together, and expanding the radius when local results are thin, so a buyer in a small town does not see an empty page. Ranking for Two Sides: Relevance, Freshness and Paid Visibility owns the scoring; the compiler owns the radius expansion policy.

Failure Modes

The mistakes I would look for first in a design review of map search:

1. **Raw client geometry**. Clients send unbounded polygons or arbitrary bounding boxes straight to the engine. Symptoms: p99 driven by a few users' shapes; cache hit rates near zero; incident reports that start with "after the app release".
2. **The viewport becomes a filter by accident**. The last map viewport leaks into the list view or into a saved search, and users get alerts for an area they only zoomed into. The fix is an explicit "search this area" state.
3. **Aggregations over exact points**. Cluster centroids, heatmaps or vector-tile hits computed from the private point field. This is a privacy incident waiting to be discovered by a journalist or a stalker; the fix is that the map path cannot read the exact field.
4. **Stacked pins (Seeing Double)**. One property, three listings, three pins. Agencies and buyers both notice, and it undermines trust in every count on the map.
5. **Stale tiles (Ghost Inventory)**. Long-lived tile caches without event-based invalidation keep sold listings on the map, and the pin opens a dead page.
6. **Isochrones without stated assumptions**. A transit commute computed for Sunday at noon, shown to a buyer who commutes on weekdays at 08:30. The departure time and mode must be visible and stored with saved searches.
7. **Per-request routing at peak**. A routing contract priced per call, used uncached in the search path, turns a TV campaign into an invoice. The symptom is a monthly bill nobody forecast.
8. **Boundaries as hard-coded polygons**. District polygons copied into the codebase or the index without versions. When the municipality redraws a boundary, saved searches and SEO landing pages silently change meaning.
9. **List metrics for the map**. Evaluating pin selection and map ranking with list NDCG and position-based click models, which Airbnb's published work shows mislead on maps.

The Chief Architect's View

At group level, map search is a test of the principle Veldmark adopted with Common Ground: **share platforms, not products**. The geo primitives, the geo query compiler, the place and boundary service, the isochrone service and the tile cache are platform

capabilities: they are the same problems for every brand, and building them four times is waste. The map product itself, with its gestures, layers, pin design and the decision whether the map or the list is the default, belongs to each brand. Hearthly's map-first experience and Tradeyard's radius-first experience should not converge.

The decisions I would put in front of the architecture advice forum:

1. **Own the geo query compiler as part of the search platform**. Tomasz's team owns it; brands configure it (vertex caps, snapping tolerances, precision rules per category). This is the seam that keeps the shared platform from becoming **The Forced Fit**: Hearthly gets the map-specific behavior it needs as configuration of a shared component, not as a fork.
2. **Write the precision rule as a platform invariant**. Which field may be matched, which may be returned, per brand and seller type, enforced in the compiler and the index mapping. Lucía Ortega, the group's Data Protection Officer, signs it off; changes go through a decision record.
3. **Treat the canonical property as the map unit for real estate**. This makes entity resolution (One Property, Three Listings: Duplicates and Entity Resolution) a dependency of map search. Its accuracy becomes visible to every user, so its quality metrics belong on the search platform's dashboard, not only the data team's.
4. **Buy or build routing on a volume forecast, not a feature list**. The commercial model, limits and display terms are decided by Ingrid with Mateus Carvalho (CTO) and finance, with a forecast of peak isochrone and matrix volumes. Re-forecast whenever a brand plans a commute-led campaign.
5. **Budget the map separately in capacity planning**. Count map requests, not searches, and size the tile cache and engine for the Sunday-evening peak of map traffic in the map-heavy brands.
6. **Measure the map as a product**. Contacts and saves per map session, time to shortlist, and the share of sessions that end on the map. Owned by the brands' product teams, reviewed with search.

What to watch over the long term: **conversational and natural-language search will express place in sentences** ("near a good school, 30 minutes from Part-Dieu"), as Zillow's 2024 feature already does. That raises the value of the compiler. An LLM that turns sentences into filters, as What Did They Mean? Query Understanding and Autocomplete recommends, should emit place IDs, normalized polygons and isochrone parameters, the same intermediate form as the map client. A platform with one compiler absorbs the new interface; a platform where each client builds its own geo queries has to rebuild them for it.

Questions to Ask

1. Which component translates a user's map gesture into an engine query, and can a client release change the shape or cost of geo queries without a server change?
2. For each brand and seller type, which location field is used for matching and which for display, and can the map path read the exact point at all?
3. Are clusters, heatmaps and area statistics computed from public points, with minimum counts per cell?
4. What are the vertex cap, simplification tolerance and maximum area for drawn polygons, and what happens to a shape that violates them?
5. Is the map unit a listing or a canonical property, and how many duplicate pins does a sample of dense areas show today?
6. How are map tiles and counts invalidated when a listing is sold, removed or repriced, and what is the measured staleness?
7. What does commute search cost per peak hour under the current routing contract, and what are its limits, caching rights and display terms?
8. Do commute results state mode and departure time, and do saved searches store the parameters rather than the polygon?
9. How are map ranking and pin selection evaluated, and are list metrics being used where they do not apply?
10. Which neighborhood layers carry legal or reputational risk in each market, and who approved them?

From Maps to Standing Promises

The map turns place into a query. The next step is to turn a query into a standing promise: when a buyer saves the polygon around Jean-Macé, Hearthly commits to telling her within minutes when a matching flat appears. That promise, and the reverse search and fan-out machinery behind it, is the subject of Tell Me When It Appears: Saved Searches and Alerts.

Sources and Evidence

Company engineering and investor sources

- Learning to Rank for Maps at Airbnb — M. Haldar et al., arXiv, 2024. List-ranking assumptions fail on map interfaces; a rebuilt attention model for maps.
- Airbnb Tech: 2025 year in review of academic publications — Airbnb, 2025. Maps at about 80% of search interactions; map-specific NDCG.

- Embedding-based retrieval at Airbnb — Airbnb, arXiv, 2026. Map-based geographic filters on nearly all searches; S2 cells as a location feature.
- Rightmove case study: property POI search — TravelTime, n.d. (vendor case study). Rightmove's My Places per-listing travel times; fixed annual license with unlimited requests.
- Rightmove Sets up Home With Google Cloud — Google Cloud press release, 2021. Rightmove's busiest minute during a prime-time TV ad break.
- Zillow AI search adds commute time, affordability — HousingWire, 2024. Commute, affordability and points of interest as natural-language criteria.
- H3: Uber's Hexagonal Hierarchical Spatial Index — Uber Engineering, 2018. The H3 grid used for hexagonal clustering and cell keys.

Standards, regulation and documentation

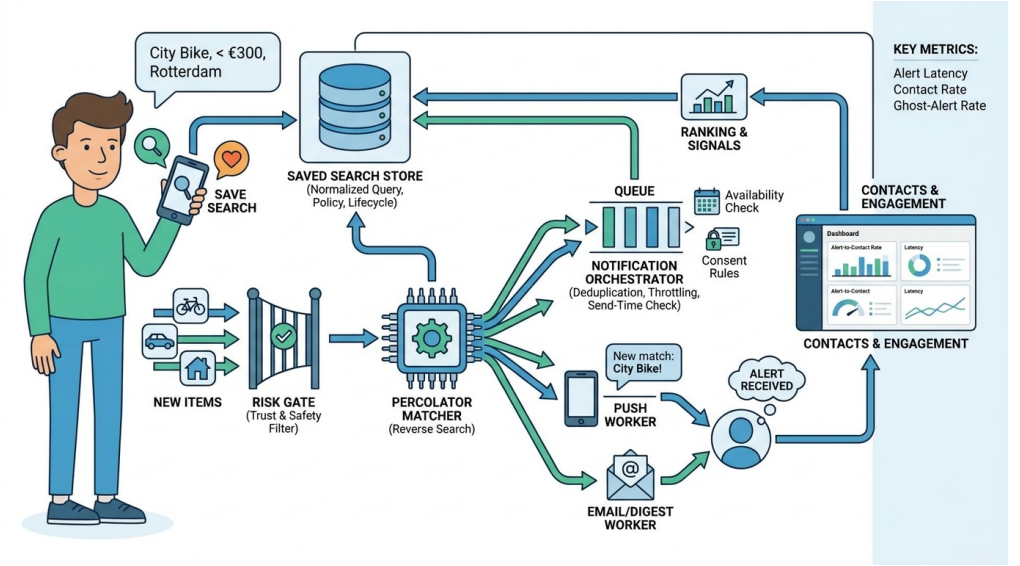
- Geo-shape field type — Elastic, n.d. Tessellation, precision, cost growing with vertex count.
- Geo-shape query — Elastic, n.d. Polygon queries against points; spatial relations; pre-indexed shapes.
- Geotile grid aggregation — Elastic, n.d. Tile-based clustering and the bounding-box advice.
- Vector tile search API — Elastic, n.d. Hits, aggregation and metadata layers in Mapbox vector tiles.
- Geohex grid aggregation — OpenSearch, n.d. H3-cell aggregation at precision 0–15.
- Isochrone API — Mapbox, n.d. Contour, profile, rate and display limits of a commercial isochrone API.
- Isochrone and isodistance service — Valhalla project, n.d. Open-source isochrones with multimodal costing.
- Supercluster — Mapbox, n.d. Client-side hierarchical point clustering.
- 24 CFR §100.75: Discriminatory advertisements, statements and notices — US Code of Federal Regulations via Cornell LII, n.d. Fair-housing limits on housing advertisements.

Research and practitioner sources

- Elastic and Rightmove partnership success story — Elastic, 2025 (vendor material). "Rich, map-centric results" among the priorities of Rightmove's 2014 move to Elasticsearch.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

23. Tell Me When It Appears: Saved Searches and Alerts



IN THIS CHAPTER, YOU WILL: Learn how saved searches become standing queries, how reverse search (percolation) differs from batch re-querying, how matches fan out through push, e-mail and in-app channels with throttles and digests, and how alerts handle price drops, sold listings, taxonomy changes and consent, following Veldmark's move from a ten-minute cron job to a five-minute alert target for 46 million saved searches.

WHY THE BUSINESS CARES: For in-demand local goods, rentals and well-priced cars, the first buyer to make contact usually wins. Alerts decide who that buyer is, bring users back without paid acquisition, and are among the most visible promises a marketplace makes, so a late, noisy or wrong alert costs contacts and erodes trust.

WHY ARCHITECTS SHOULD CARE: Alerts are reverse search plus fan-out: a matching workload that grows with every saved search and every listing change, and a notification workload with hot spots, provider quotas

and legal rules. Both depend on the listing event stream, the taxonomy and the search platform, so alerts expose every inconsistency among them.

KEY POINTS:

- **A saved search is a stored, normalized query, and alerts are reverse search.** Instead of running queries against documents, the platform runs each new or changed listing against millions of stored queries. Percolators (Rightmove and Homegate document them) do this per event; batch re-querying is simpler but only as fast as its interval.
- **Notify on entering the match set, and re-check at send time.** A listing that newly matches a saved search, because it was published, repriced below a limit or moved into an area, deserves an alert; every later edit does not. And anything delivered later than it was computed, such as a digest, must confirm the listing is still available before it goes out.
- **Measure alerts by contacts, not sends.** Latency from searchable to delivered, alert-to-contact rate, unsubscribes and the share of alerts for unavailable listings are the metrics that matter. Alerts are a liquidity lever and a trust surface at the same time, so the risk gate, the consent rules and the frequency caps belong in the design from the start.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Ghost Inventory** — A digest assembled in the afternoon and sent the next morning advertises a bike that sold in between, because the alert system trusted a snapshot.
- **Moderation as an Afterthought** — An alert system that fans out every new listing within minutes also delivers fake listings to the most motivated buyers, unless the risk gate sits in front of it.
- **Vanity Liquidity** — Counting alerts sent rather than contacts made rewards noisy alerts, and users answer noise by unsubscribing.

The earlier chapters of Part IV help a buyer find what is listed at the moment they search. A **saved search** covers what is listed afterwards: it is the same question asked continuously, "tell me when a city bike under €300 appears within 10 km of Rotterdam", "a T3 in this polygon in Lyon under €400,000", "a Golf from 2019 or later with less than 60,000 km". A buyer who saves a search turns a moment of intent into **standing demand**, and the marketplace takes on an obligation: to notify them when supply appears, quickly enough that the notification is worth something.

At Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), the obligation covers **46 million** active saved searches, and the group's brands send about **38 million** alert notifications a day by push

and e-mail. Every new listing, every price cut and every reactivation is a candidate for matching against all of them. **Alerts are where the listing event stream, the taxonomy, the search platform, the trust gate and the notification channels all meet**, and the alert system is only as correct as the weakest of them.

This chapter owns three terms. A **saved search** is a persisted, normalized search query with an owner, a notification policy and a lifecycle. **Percolation** (also called reverse search or prospective search) is matching a document against stored queries rather than a query against stored documents. **Alert fan-out** is the process of turning matches into notifications across channels, with deduplication, throttling and delivery. It builds on Search Architecture for Listings: Indices, Freshness and Scale for the engine, on Every Change Is an Event: Event-Driven Listing Architecture for the event stream, on From Submitted to Searchable: The Publication Pipeline for what "searchable" means and when, and on Searching by Place: Map, Polygon and Commute Search for drawn and commute areas. Ranking for Two Sides: Relevance, Freshness and Paid Visibility and Recommendations for Inventory That Disappears use its engagement signals, and its risk gate is where the trust and safety work of Part V — TRUST: Keeping the Marketplace Safe begins.

Setting the Scene at Veldmark: 312 Saved Searches and a Ten-Minute Cron

On **Sunday 7 March 2027 at 19:42 CET**, Femke, a private seller in Rotterdam, lists a used city bike for **€240** on Tradeyard Netherlands. It becomes searchable at **19:48**, six minutes later, the Tradeyard baseline. It matches **312** saved searches. The first contact arrives at **19:55**, from a buyer who received an alert. Femke has **14** conversations within 24 hours, and on Monday 8 March she sells the bike for **€225** through Safe Deal. Tradeyard marks it sold at **18:10** and search removes it at **18:11**.

On Tuesday morning at **07:00**, **214** alert digest e-mails still feature the bike, and three recipients message Femke to ask whether it is still available. Every Change Is an Event: Event-Driven Listing Architecture traces the mechanism: the digest job assembled its candidate list at **17:30 on Monday** from a read replica of Yardstick, Tradeyard's PHP monolith, stored it, and sent it thirteen and a half hours later without asking whether anything had changed.

When Ingrid Halvorsen, Veldmark's Group Chief Architect, asks how Tradeyard's alerts actually work, Daan Visser, who owns Yardstick, describes two jobs:

- **Instant alerts** run as a Yardstick cron job every **10 minutes**. The job selects listings published since its last run, about 2,300 per run on a Sunday evening in the Netherlands, and matches them against saved searches stored in MySQL, using an index on category and province to pick candidates and PHP code to test the

remaining filters. Matches go to a push queue. Femke's bike was picked up by the 19:50 run, and the 312 pushes were sent between 19:51 and 19:53. That was luck: a listing that becomes searchable at 19:51 waits until 20:00.

- **Digests** are e-mails assembled once a day, in the afternoon, by re-running each saved search that has digest delivery against Solr, filtered to listings published in the past 24 hours. The results are stored and sent the next morning in send-time windows.

Veldmark's 2027 baseline numbers:

Metric	Value
Active saved searches, group	**46 million**: Tradeyard 31 million, Motorvane 8 million, Hearthly 7 million
Alert notifications per day, group	**~38 million**: push about 22 million, e-mail about 14 million (mostly digests), in-app only about 2 million
Tradeyard instant alert latency, searchable to push sent	median **7 minutes**; p95 **23 minutes** on Sunday evenings, when the cron job overruns
Tradeyard digests featuring a listing no longer available at send time	about **4%** of digest items
Saved searches with no opened alert in 12 months	about **38%** of Tradeyard's 31 million

The Common Ground target, agreed with Noor Hassan, the group Chief Product Officer, is that an instant alert reaches the push or e-mail provider within **5 minutes p95 of the listing becoming searchable**, in every brand, by the end of 2028. Together with the 90-second publication target (From Submitted to Searchable: The Publication Pipeline), a buyer would typically hear about a new listing within about six and a half minutes of the seller pressing "publish", if both targets hold.

Ingrid frames the chapter's question for the architecture advice forum: **how do you match every new and changed listing against 46 million stored queries and notify the right people within minutes, without spamming them, alerting them to listings that are gone or fake, or breaking every saved search when the taxonomy changes?**

Concepts: Saved Searches, Match Sets and Alerts

A saved search is a query with a contract

A saved search has four parts, and confusing them causes most alert bugs.

Part	What it holds	Example (Tradeyard NL)
Query	The normalized criteria: category IDs, place IDs or geometry, filters on attributes, keywords	Category *city bikes*; within 10 km of postcode area 3011; price ≤ €300
Owner and identity	Who it belongs to; brand; account; device tokens and addresses by reference	A Tradeyard account; one iOS device; one verified e-mail address
Notification policy	Channel per urgency, frequency, quiet hours, digest time, whether price drops count	Push instantly, e-mail digest daily at 07:00, quiet 23:00–07:00
Lifecycle	Created, last opened, last matched, paused, dormant, expired, migrated; taxonomy and gazetteer versions it was written against	Created January 2026; taxonomy version 2026.4

The **query part must be stored in normalized, versioned form**, not as a URL or an engine query string. A URL is a client concern and changes with every redesign; an engine query is a search-platform concern and changes with every mapping or engine migration. The normalized form is the same intermediate representation that the search API and the geo query compiler of Searching by Place: Map, Polygon and Commute Search use: category and place IDs from the taxonomy and gazetteer, attribute filters keyed by the attribute dictionary (Attributes Without Chaos: Schemas, Dictionaries and Validation), keywords after the analysis of What Did They Mean? Query Understanding and Autocomplete, and geometry stored by reference. From that form, the platform compiles whatever the current matching engine needs.

Figure 1 shows the lifecycle a saved search moves through.

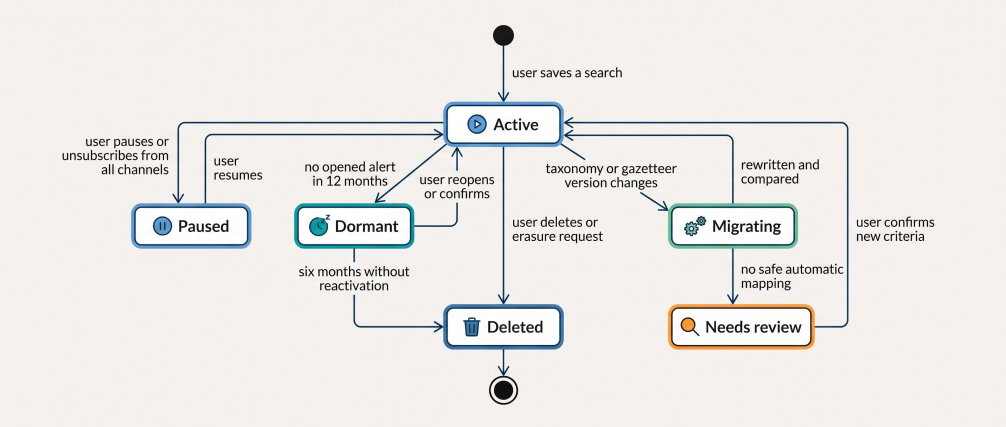


Figure 1: *A saved search is long-lived user data with its own lifecycle. Dormancy and migration are states, not cleanup scripts, so that each one has an owner, a user notice and a measurable backlog.*

A saved search is personal data. The productscape model behind this book names the capability *Saved Search / Alerts* ("persist search intent and alert on new matches by email/push") and records the saved search as personal data, kept "until user deletion or inactivity threshold". That is the right default: a saved search says where someone wants to live, what car they can afford, or what they want to buy, and it is governed by the General Data Protection Regulation (GDPR) like any other personal data.

The match set: notify on entry, retract on exit

For a saved search *S*, its **match set** is the set of live, alertable listings that satisfy *S*'s query right now. Alerts are about changes to the match set, not about listing events as such. A listing event can:

- **Enter** the match set: a new listing is published; a price drops below the saved search's limit; a listing is edited into a matching category or area; a paused listing resumes. These are alert-worthy.
- **Stay** in it: a matching listing gets new photos or a better description. Usually not alert-worthy, with one exception, the price drop for listings the user watches.
- **Leave** it: sold, removed, expired, reserved, repriced above the limit. Never alert-worthy as a match, but always important as a **retraction**: pending notifications and digest entries for that listing must be cancelled.

The design rule follows: **notify on entry**, **retract on exit**, **ignore the rest**. Implementing it requires knowing whether the listing matched *before* the event, which means the matcher needs either the previous state of the listing (event-carried state with the changed fields and their previous values, as the listing event contract of Every Change

Is an Event: Event-Driven Listing Architecture provides) or a **match ledger** that records which (saved search, listing) pairs have already been notified. In practice both are needed: the previous state for the matching logic, and the ledger for deduplication across channels and over time.

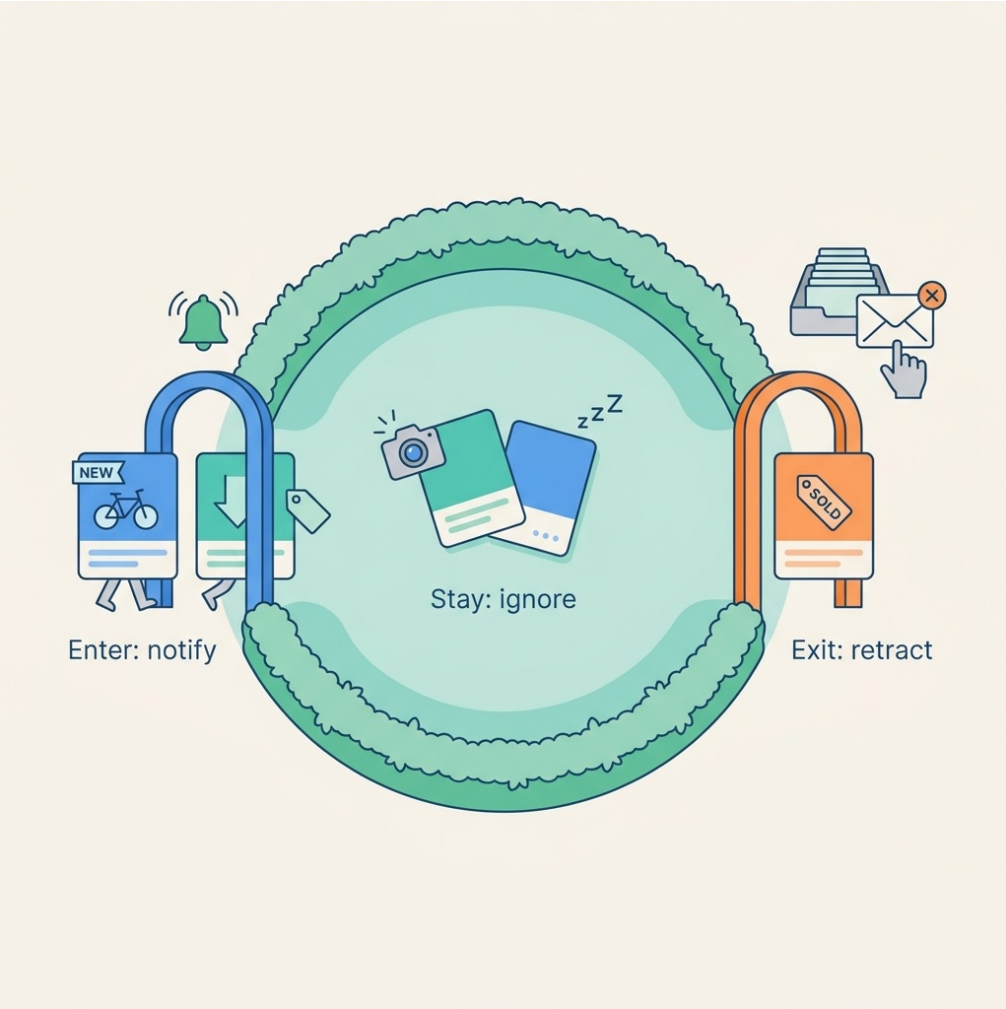


Figure 2: *Notify on entry, retract on exit, ignore the rest: alerts are about changes to a saved search's match set, and a listing that leaves it must cancel every pending notification and digest entry.*

Watchers and price-drop alerts

A **watcher** is a user who has favorited a specific listing. Watching is not a saved search. It is a subscription to one listing, and its alerts are about that listing's changes:

price drops, "reserved", "sold", "back on the market". The Listing Aggregate: Identity, Lifecycle and Invariants keeps a price event series per listing for exactly this purpose. When Autohaus Ruhrblick (fictional), a Motorvane dealer in Essen, cuts the price of a 2021 Volkswagen Golf 1.5 TSI from **€21,900** to **€21,400** on **Thursday 22 April 2027**, one `PriceChanged` event produces two kinds of alerts:

1. **Price-drop alerts to the 1,180 watchers** of that listing.
2. **New-match alerts** to saved searches whose price limit lies between €21,400 and €21,899. For them, the listing *enters* the match set. In this example, about 640 Motorvane saved searches.

Price-drop alerts invite gaming. A dealer who cuts the price by €10 every three days generates an alert every three days. A minimum drop rule, for example at least 2% or a fixed amount by price band, and a cap of one price-drop alert per listing per watcher per week keep the signal meaningful. The Golf's €500 cut is 2.3%, so it passes.

Two Ways to Match: Percolate or Re-query

At Veldmark's scale, instant alerts need percolation; batch re-query is fast enough only for digests.

Batch re-query runs each saved search periodically as an ordinary search against the main index, filtered to listings published or changed since its last run. It reuses the search engine and its analyzers exactly, so a saved search returns what the same search would return on the site. Its cost scales with the number of saved searches, not with the number of listings, and its latency is the batch interval. For Tradeyard's 31 million saved searches, re-running each one every five minutes would mean about 100,000 queries per second, more than three times Veldmark's whole-group search peak of **31,000 queries per second**. Batch re-query is therefore a digest strategy, not an instant-alert strategy, at this scale.

Percolation inverts the problem. Saved searches are indexed as queries; each new or changed listing is submitted as a document; the engine returns the stored queries it matches. Its cost scales with the rate of listing events times the number of candidate queries each event must be tested against, and its latency is per event, so it can be seconds.

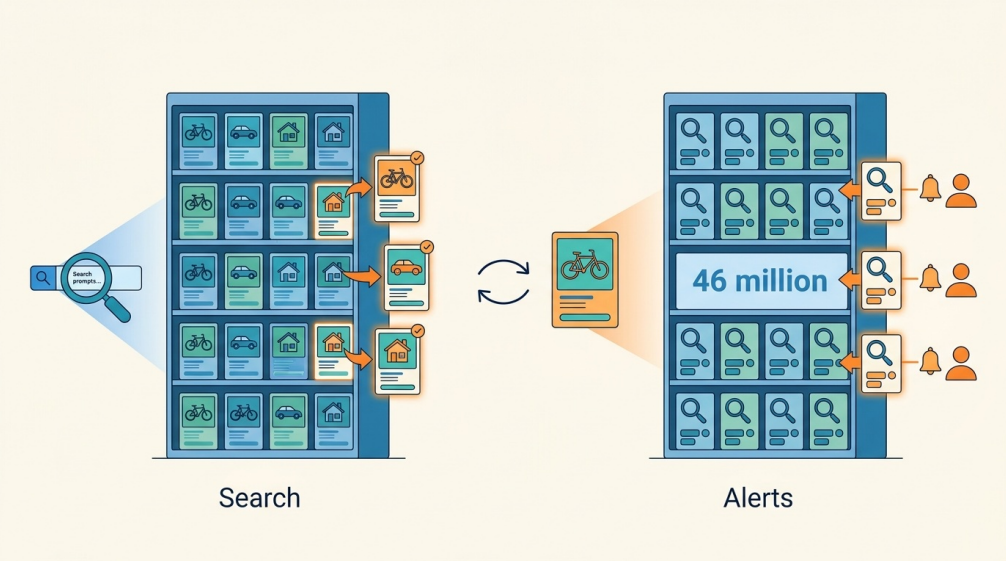


Figure 3: *Alerts are reverse search: instead of one query against many listings, each new or changed listing is matched against the stored queries, 46 million of them across Veldmark.*

Percolation is documented at property portals:

- Rightmove listed "instant property alerts, made possible by Elasticsearch's percolators" as one of three priorities of its 2014 move to Elasticsearch, in a case study Elastic published in 2025 (Elastic, 2025). The case study is vendor material, and describes the 2014 decision; how Rightmove's alerts work today is not public.
- Homegate, the Swiss property portal, described its alerts in a 2024 post on Elastic's blog: an alert is persisted in Homegate's AWS infrastructure and indexed into a "search alert index" in Elasticsearch; when a property is posted, the search alert service "creates a percolate query for the property", finds the matching alerts and notifies users by e-mail or push. Alerts filter on property type, location, price and more and are described as one of Homegate's most popular services (Elastic blog, 2024-04-29).

The engine mechanics are documented by Elastic and apply, with the same field type and query names, to OpenSearch, which supports a `percolator` field type ("specifies to treat this field as a query") and a `percolate` query (OpenSearch documentation, n.d.). The constraints that matter for architecture, from Elastic's percolator documentation (Elastic, n.d.):

1. **Fields a stored query references must exist in the percolator index mapping.** The percolator index is coupled to the listing schema.

2. **Stored queries must be reindexed** into a new index on major upgrades and when percolator improvements arrive, followed by an alias swap. A percolator index is not a store of record; it is a projection of the saved-search store and must be rebuildable from it.
3. **Percolation re-parses each candidate query and re-runs text analysis**, which can dominate the cost. The documented fix is to analyze at index time and match stored tokens with a lightweight analyzer. Wildcard and prefix queries are expensive and should be replaced by subfields such as `edge_ngram`.
4. **Some query types are not supported** (`has_child`, `has_parent`), and data fetched at index time, such as terms lookups or pre-indexed shapes, is fetched once and can go stale.

Point 4 matters for drawn polygons from Searching by Place: Map, Polygon and Commute Search: a percolated saved search that references a stored shape keeps the shape as it was when the saved search was indexed, so a changed shape requires re-indexing the saved search.

Other engines offer the same capability under other names. Lucene's **monitor** module, contributed from the Luwak library in Lucene 8.2 (2019), registers queries and matches a stream of documents against them, using a "presearcher" to select candidate queries before running them (Apache Lucene, monitor package). Vespa's **predicate fields** store boolean constraints in documents and match them against queries carrying attribute values; its documentation names saved searches ("users save searches in the form of predicates, and new items are matched to the saved searches") and says predicate fields suit problems "where practitioners have used Percolator Queries" (Vespa documentation, n.d.).

Candidate selection is where percolation wins or loses

At Veldmark's scale, percolation must select a small candidate set of stored queries for each listing rather than test every listing against every stored query, and the selection key is mostly **category and place**, because almost every saved search in a listing marketplace has both. A plausible decomposition for Tradeyard:

- **Partition** the percolator indices by brand, country and top-level category tree, so a city bike never meets a property search.
- **Index pre-filter keys** with each stored query: the leaf and ancestor category IDs it accepts, the place IDs or coarse geo cells (for example H3 resolution 5 or 6, see Where Is It? Location, Geocoding and Privacy) that cover its area, and a price band. A listing then carries its own keys, and the percolate request filters candidates by those keys before testing the full queries.
- **Push keyword tests last**. Most listing-marketplace saved searches are browse-shaped (category, place, price, attributes) rather than keyword-shaped, which

makes the structured pre-filter selective. Keyword saved searches are analyzed at index time.

For Femke's bike, the NL city-bikes partition held about **1.9 million** saved searches; the place pre-filter for the cells around Rotterdam left about **58,000** candidates; the full test matched **312**. These are illustrative numbers, but the ratios are typical of the problem: the pre-filter does almost all the work.

The structured matcher as an alternative

Some marketplaces skip the engine altogether and build a **structured matcher**: saved searches are decomposed into their structured constraints and stored in an in-memory or key-value index keyed by (category, place cell); each listing looks up the keys it produces and evaluates the remaining numeric and attribute constraints in code. Tradeyard's Yardstick cron job is a primitive version of this in MySQL and PHP. The approach is fast and cheap for browse-shaped saved searches and has no coupling to the search engine's mapping. Its weakness is fidelity: it reimplements the search semantics, and every difference (how "10 km" is measured, how keywords are tokenized, how a missing attribute is treated) becomes an alert that does not match what the user sees on the site, or a listing the site shows that no alert announced.

Architecture: A Saved-Search Store with Rebuildable Percolators

Figure 4 shows the target architecture Ingrid's team proposes for Common Ground.

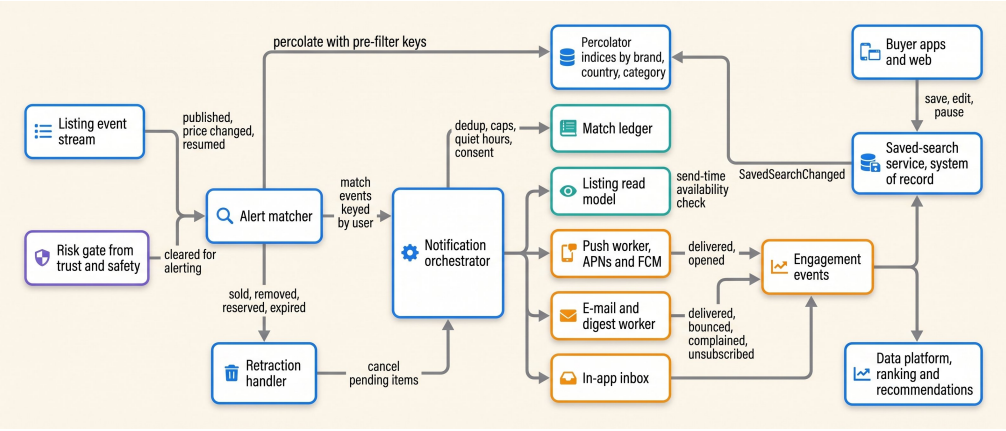


Figure 4: *The saved-search store is the system of record and the percolator indices are rebuildable projections. Listing events enter through a risk gate; matches flow to an*

orchestrator that owns deduplication, policy and a send-time availability check; channel workers send; engagement flows back to saved-search hygiene and to ranking.

The responsibilities:

- **Saved-search service**. The system of record for saved searches: query in normalized form, owner, policy, lifecycle, versions. It emits `SavedSearchChanged` events, from which the percolator indices are built. It owns migrations when the taxonomy or gazetteer changes.
- **Percolator indices**. Projections of the saved-search store on the search platform, partitioned for candidate selection. They can be rebuilt from the store at any time, which is what makes engine upgrades, mapping changes and the move from Solr and Elasticsearch to the shared OpenSearch platform possible. A saved search that exists only in a percolator index is **Search as System of Record** in miniature.
- **Alert matcher**. Consumes alert-relevant listing events (`ListingPublished`, `PriceChanged`, `ListingResumed`, and relevant `ListingUpdated` events that change category, place or matched attributes). It computes the listing's pre-filter keys, percolates, applies the entry rule using the event's previous values, and emits match events keyed by user ID.
- **Retraction handler**. Consumes `ListingSold`, `ListingRemoved`, `ListingReserved`, `ListingExpired` and `ListingPaused`, and cancels pending notifications and digest items for that listing.
- **Risk gate**. Listings above a risk threshold from trust and safety are not alertable until cleared (Trust and Safety as a Platform). The gate is described below.
- **Notification orchestrator**. Groups match events per user, applies deduplication against the match ledger, frequency caps, quiet hours, channel preferences and consent, assembles instant notifications and digests, and performs the **send-time availability check** against the listing read model.
- **Channel workers**. Push (through Apple Push Notification service and Firebase Cloud Messaging), e-mail (through an e-mail service provider or the group's own mail transfer agents), and the in-app inbox. They handle provider quotas, retries and delivery receipts.
- **Engagement events**. Deliveries, opens, clicks, contacts, bounces, complaints and unsubscribes, fed back to saved-search hygiene (dormancy, pausing) and to the data platform for ranking and recommendations (Data Platform for Marketplaces: Events, Lakehouse and Data Products).

The matching approach is a **hybrid**: percolation per event for instant alerts, and for digests a choice between collecting the day's percolation matches (cheap, already computed) and batch re-querying (exact, reuses the main index). Ingrid's recommendation is to build digests from the percolation matches recorded in the match ledger and re-check them at send time, because that keeps one matching semantics for both channels. Batch re-query remains as a reconciliation job that samples saved searches and compares its results with the ledger, to catch matcher bugs.

The bike, replayed on the target design

Figure 5 replays Femke's listing on the Common Ground design.

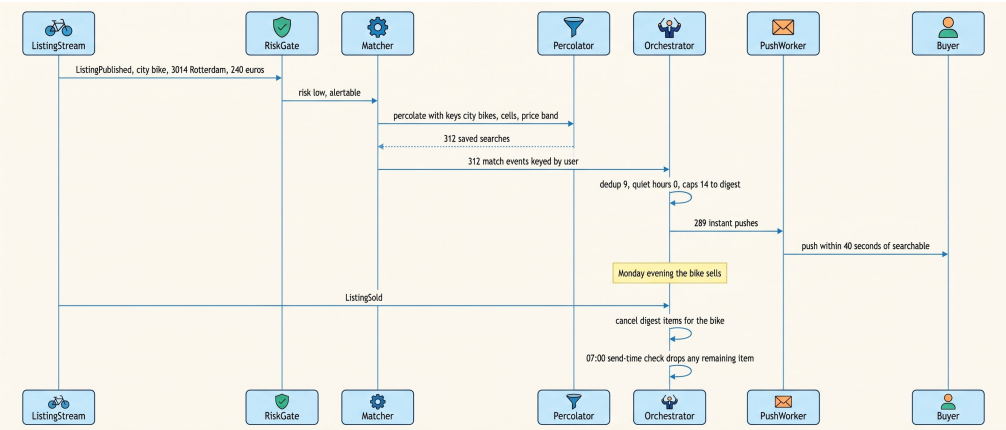


Figure 5: *On the target design the bike's 312 matches become 289 instant pushes within seconds, 14 matches are deferred to digests by frequency caps, and 9 are suppressed as duplicates. The sale on Monday retracts the bike from pending digests, and the send-time check would catch it even if the retraction were missed.*

The match ledger and the caps decide what each of the 312 matches becomes. Nine belong to users with two overlapping saved searches (city bikes near Rotterdam, and all bikes under €300 in South Holland), and the ledger sends one notification per user per listing. Fourteen users had already received their cap of instant alerts for the evening and get the bike in the next digest. The remaining 289 pushes leave within seconds. The numbers in the figure are illustrative.

Fan-out: Hot Listings, Quotas and Bursts

Fan-out must be sized for the tail of the match distribution, not its mean. Most listings match a few dozen saved searches. Some match tens of thousands: a cheap two-room flat in a capital city, an underpriced popular car, a new iPhone at half price. Queues, not synchronous calls, must absorb the burst from such a listing.

Three layers of limits shape the design.

1. **Provider quotas**. Firebase Cloud Messaging documents a default quota for its HTTP v1 API of 600,000 messages per minute per project, with increases on request and at least 15 days' notice for launch or event increases. Per device, it limits collapsible messages to a burst of 20 per app per device, refilling at one message

every three minutes, and caps the rate to a single Android device at 240 messages per minute and 5,000 per hour (Firebase documentation, n.d.). Veldmark's 22 million pushes a day average about 15,000 a minute, far below the project quota, but a Sunday-evening hour with a few hot listings can be an order of magnitude above the average, and a promotional campaign sharing the same project can consume the headroom. Separate projects or quotas per traffic class (alerts versus marketing) keep a campaign from delaying alerts.

2. **User attention**. The real limit is the user. A buyer with six saved searches in dense categories can match fifty listings on a Sunday evening. Frequency caps (for example, at most five instant alerts per user per hour, the rest batched), quiet hours in the user's time zone and collapsing of several matches into one notification ("4 new city bikes near Rotterdam") are product rules enforced in the orchestrator.
3. **Mail reputation**. E-mail providers judge the sender, not the message. Since 1 February 2024, Gmail requires senders of more than 5,000 messages a day to its users to authenticate with SPF (Sender Policy Framework), DKIM (DomainKeys Identified Mail) and DMARC (Domain-based Message Authentication, Reporting and Conformance), to keep spam rates reported in its Postmaster Tools below 0.30%, and to support one-click unsubscribe for marketing and subscribed messages (Google Workspace Admin Help, n.d.). One-click unsubscribe is specified in RFC 8058 (2017) as a `List-Unsubscribe-Post: List-Unsubscribe=One-Click` header, with both unsubscribe headers covered by a valid DKIM signature (IETF RFC 8058, 2017). An alert e-mail that users mark as spam damages the deliverability of every e-mail the brand sends, including password resets and Safe Deal receipts. Separate sending domains or subdomains for alerts and transactional mail limit that blast radius.

At Veldmark's 14 million alert e-mails a day, mail infrastructure is a system with its own engineering, whether it is bought or run. For comparison, leboncoin's engineering blog described in June 2026 moving about **12 million** transactional e-mails a day, the same range, from 20 Postfix servers to PowerMTA on AWS (leboncoin tech blog, 2026-06-03).

The order of a large fan-out is a fairness decision. When 40,000 people are alerted to the same flat, the order in which they are notified decides who gets the viewing. Sending in saved-search creation order privileges old accounts; sending in database order is arbitrary. My recommendation is a randomized order within each fan-out, completed within the latency target, and a policy decision, written down, on whether any paid product may change it. A paid "early alert" product is technically trivial and commercially tempting; it is also a ranking decision in disguise, with the same transparency questions Ranking for Two Sides: Relevance, Freshness and Paid Visibility raises for paid placement.

Channels, Digests and Relevance

Channel	Strengths	Weaknesses	Best for
Push	Seconds to deliver; highest response for urgent matches	Requires app install and permission; on Android 13 (API level 33) and higher, apps must request the `POST_NOTIFICATIONS` runtime permission (Android Developers, n.d.); iOS likewise requires the user's authorization; tokens expire	Instant alerts in competitive categories
E-mail	Universal; rich content; good for many matches at once	Minutes to hours to be read; deliverability rules; spam-folder risk	Digests; users without the app
In-app inbox	No permission needed; no deliverability risk; the record of all alerts	Seen only when the user opens the app	Overflow from caps; history; a fallback when push is disabled
SMS, messaging apps	Very high open rates	Cost per message; consent rules; a channel scammers imitate	Rarely justified for alerts; sometimes for tenants in tight rental markets

A **digest** is a scheduled message that collects matches since the last digest. Digests trade

speed for attention: they suit users who want to browse matches once a day, and they absorb the overflow of frequency caps. Their two requirements follow from the bike: **retraction** of items that left the match set, and the **send-time availability check**, which re-reads each item's state from the listing read model when the digest is rendered for sending. The second is the safety net for the first; together they bring the 4% stale-item rate to near zero.

Alert relevance is not binary. A listing that matches the filters of a saved search can still be a poor alert: an overpriced item, a listing from a seller with a poor response rate, a near-duplicate of something already sent. Ordering the items within a digest by match quality, and suppressing instant alerts for clear duplicates (the same item reposted, or the same property from a second agency, see One Property, Three Listings: Duplicates and Entity Resolution), are where alerts meet Ranking for Two Sides: Relevance, Freshness and Paid Visibility. The simplest useful rule: within a digest, use the same ranking as the site, minus paid boosts, plus a penalty for items the user has already seen.

Alerts as a Liquidity Lever

Alerts are a lever on liquidity in the sense of Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics: they shorten the time from supply appearing to demand responding. The productscape model behind this book records the buyer's view plainly: "Alert speed is the differentiator: for in-demand local goods the first buyer to message usually wins, so minutes matter", and lists among the pains "alerts after the item is sold" and "batching or delayed indexing means the alert arrives hours late". Saving a search "converts a fleeting query into a standing demand signal".

The lever works on both sides of the market. In Femke's case, the first contact arrived seven minutes after the listing was searchable, from an alert. On the seller side, fast first contacts are what make private sellers list again.

The metrics that measure the lever, in the order I would put them on the search platform's dashboard:

Metric	Definition	Why
Alert latency	p50 and p95 from listing searchable to notification accepted by the provider, by brand, channel and hour	The promise; the Common Ground target is 5 minutes p95
Alert-to-contact rate	Contacts made within 48 hours after an alert, per alert sent	The outcome; the defense against **Vanity Liquidity**

Metric	Definition	Why
Saved-search activation rate	Share of search sessions with results that end in a saved search	Demand capture
Ghost-alert rate	Share of alert items whose listing is unavailable when the user opens it	Trust; measures retraction and send-time checks
Unsubscribe and complaint rates	Per saved search and per e-mail sent	Noise; complaints affect deliverability
Dormant share	Saved searches with no opened alert in N months	Hygiene and cost

"Alerts sent" is deliberately absent. A system rewarded for sends will widen matching, relax caps and lower price-drop thresholds, and users will answer with the unsubscribe link.

The Risk Gate: Alerts Amplify Whatever They Carry

An alert system is an amplifier. It takes a new listing and puts it in front of the most motivated buyers within minutes, which is exactly what a scammer wants.

In **July 2027**, **26** accounts post **180** fake rental listings in Barcelona's Eixample district on Hearthly Spain, using photos stolen from real listings, priced about **35%** below market and asking for deposits by bank transfer outside the platform. They are detected through perceptual-hash matches against existing listings and price anomalies, but **63** people report losses totaling about **€71,000** before the takedown. Hearthly's alert logs show that the 180 listings produced about **41,000** alert notifications, most of them pushes to renters with saved searches in Eixample (figures from Hearthly's own logs; Veldmark is fictional). In a tight rental market, the renters most likely to pay a deposit quickly are exactly those with alerts.

The design consequence is a **risk gate in front of the matcher**, owned by trust and safety (Trust and Safety as a Platform, Scams, Fakes and Account Takeovers: Fraud Prevention for Listings):

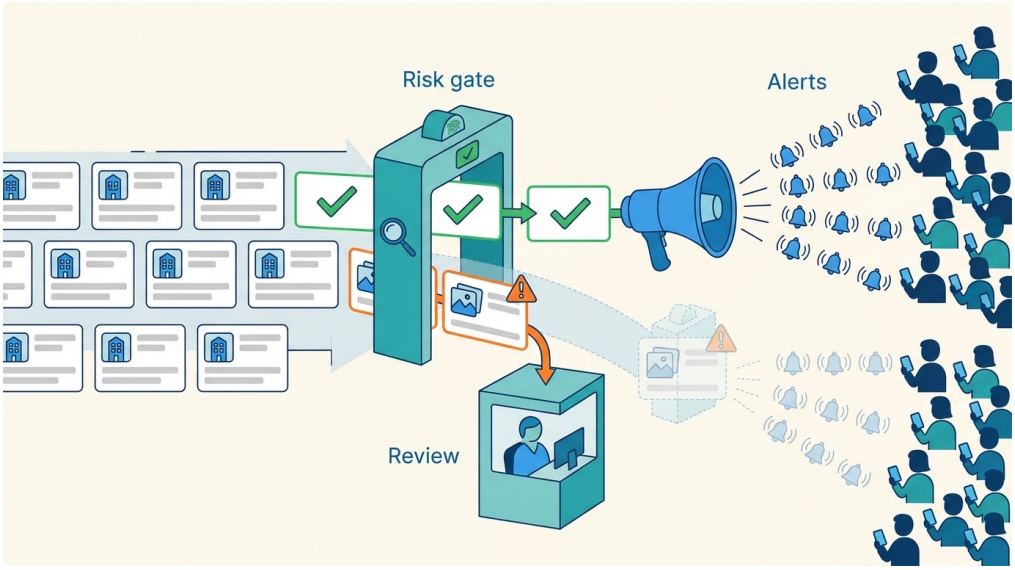


Figure 6: *Alerts amplify whatever they carry: the 180 fake Eixample rentals produced about 41,000 alert notifications, so listings above an alerting threshold stay not alertable until a fast review clears them.*

- Listings with a risk score above an alerting threshold are published (if policy allows) but **not alertable** until a fast review clears them. Alertability is a derived state of the listing, as The Listing Aggregate: Identity, Lifecycle and Invariants recommends for visibility and contactability.
- The alerting threshold is stricter than the publication threshold in categories where alerts carry high stakes: rentals, cars, high-value electronics.
- A takedown event retracts the listing from pending notifications and digests like a sale, and the orchestrator can send a **follow-up safety notice** to users who were alerted to a listing later removed for fraud. That is a policy decision with legal review, not a default.
- A short **alert hold** for new accounts' first listings (minutes, not hours) gives the automated checks of Scams, Fakes and Account Takeovers: Fraud Prevention for Listings time to run before the fan-out.

Every minute of hold costs honest sellers the speed advantage this chapter is about. The hold should be targeted by risk and category, measured, and reviewed with the brands. A general delay would undo the latency target.

Saved Searches Outlive the Taxonomy

Saved searches outlive the reference data they are written against: a category tree, an attribute dictionary, a gazetteer and sometimes drawn shapes and commute parameters,

all of which change. The listing-data brief puts the rule simply: saved searches are user data; migrate them, don't drop them.

Veldmark's test is the **bikes split** that goes live on **Monday 1 November 2027**, when Tradeyard's bike listings move from 19 country leaves into the canonical bike leaves of the new group taxonomy (city, e-bikes, cargo, kids' and others; see Category Trees That Can Change: Designing Marketplace Taxonomies). About **1.3 million** saved searches across Tradeyard's four countries reference the old leaves and must be rewritten. The migration plan:

1. **Version every saved search's taxonomy reference**. Each saved search records the taxonomy version it was written against.
2. **Rewrite deterministically where the mapping is one-to-one or a merge**.
3. **For splits, preserve recall**. A saved search on an old country bike leaf is rewritten to the union of the new leaves that cover it, so it matches everything it matched before. Where keywords make the intent clear ("bakfiets", "e-bike"), the platform *suggests* the narrower category to the user in the next alert, rather than narrowing silently.
4. **Rebuild the percolator indices** from the saved-search store against the new mapping, behind an alias, and run old and new matchers in parallel for a week, comparing match sets.
5. **Never reset the match ledger**. A migrated saved search must not re-announce listings the user has already been told about.

Place references follow the same pattern when municipalities merge or postcode areas change, and drawn polygons when the normalization rules of Searching by Place: Map, Polygon and Commute Search change. Commute saved searches should store their parameters (origin, mode, minutes, departure bucket) and recompute the isochrone when transit timetables change, rather than storing the polygon forever.

Consent, Unsubscribe and Retention

Alerts sit on a legal boundary that differs by country, so this section states principles and points to Rules of the Marketplace: DSA, DAC7 and Vertical Regulation for the details. Lucía Ortega, Veldmark's Data Protection Officer, sets the group policy.

- **What kind of message is an alert?** A notification the user explicitly requested, about listings matching their own criteria, is in Veldmark's policy a service message the user subscribed to, not unsolicited marketing. EU law on electronic marketing (Article 13 of the ePrivacy Directive, 2002/58/EC, as amended in 2009, implemented differently by each member state) generally requires prior consent for direct-marketing e-mail, with a limited exception for existing customers and

similar products that must offer an opt-out in every message (EUR-Lex, Directive 2002/58/EC). The classification is a legal judgement per market, and it changes the moment promotional content is added to an alert: a sponsored listing or a financing offer inserted into a digest can turn a service message into marketing. Veldmark's policy keeps paid content out of alerts unless the user's marketing consent covers it.

- **Unsubscribe must be easy and precise**. One-click unsubscribe (RFC 8058) in every alert e-mail, as the Gmail rules require for subscribed messages; the action should **pause that saved search's e-mail channel**, with a link to manage the others, rather than deleting the account's preferences wholesale or, worse, doing nothing.
- **Push permission is the user's**. The app asks for notification permission when the user saves a search, the moment the value is obvious, not at first launch.
- **Retention**. Saved searches are personal data. Veldmark pauses saved searches with no opened alert for 12 months, tells the user, and deletes them six months later if they are not reactivated; deletion requests remove saved searches, ledger entries and percolator entries together. For Tradeyard, the dormancy rule alone pauses about 38% of saved searches, which also removes about a third of the percolation candidates.
- **Location and inference**. A saved search for a drawn area around a specific street can reveal where someone lives or wants to live; a saved search for accessible housing can reveal health information. Saved searches should not feed advertising segments without consent that covers it, and any special-category inference is off limits for ad targeting (Selling Attention: Display, Native and Retail-Media Advertising).

Worked Example: Sizing the Matcher for Veldmark

Event rate. With the volumes in the scenario ledger, Veldmark's brands produce about 2.75 million new listings a week (Tradeyard 2.3 million, Motorvane 210,000, Hearthly 240,000), an average of about 4.5 per second. With price changes, resumptions and matched-attribute edits, the alert-relevant event rate is, say, three times that, about 14 per second on average. Sunday evenings and dealer feed runs push peaks to perhaps 60–80 events per second. These are modest rates. **Matching is not expensive because of events; it is expensive because of candidates per event**.

Candidates. With brand-country-category partitions and place pre-filters, a typical Tradeyard listing meets tens of thousands of candidate saved searches, as the bike did (58,000). At 80 events per second and 50,000 candidates each, the percolator evaluates up to 4 million stored queries per second at peak, which is a cluster to size and test, not a footnote. Reducing candidates is the cheapest lever: dormancy pausing removes about a third; finer place pre-filters remove more.

Fan-out. 38 million notifications a day average about 440 per second, with Sunday-evening peaks perhaps five times higher and individual hot listings producing bursts of tens of thousands. Queues between the orchestrator and channel workers absorb the bursts; the latency target applies to the queue's drain time.

Ownership of cost. Matching runs on the shared search platform, so its capacity is in Tomasz Wiśniewski's budget, but its demand comes from brand product decisions: a new "alert me for any listing in my city" feature can double candidates per event. Ingrid's rule is that **any change that widens saved-search semantics goes through the same capacity review as a new search feature**.

Transition. The Yardstick cron job keeps sending until Tradeyard's search moves out of Yardstick in the fourth quarter of 2028 (Changing the Engine in Flight: Modernizing a Live Marketplace). In between, Tradeyard runs both: from Q4 2027 the new matcher consumes the listing event stream (live since 1 July 2027) in comparison mode, sending nothing while its match sets are compared with the cron job's; it takes over sending market by market as each Tradeyard country moves to the shared search platform (Portugal first, in Q2 2028), and the cron job stops sending when Belgium and the Netherlands move. Motorvane keeps its alerts on Elasticsearch percolators, consistent with its engine staying until the 2029 review; it adopts the saved-search service and the orchestrator so that notification policy and consent are group-wide.

Trade-offs

Decision	Option A	Option B	Choose A when	Choose B when
Matching for instant alerts	Percolate per event	Batch re-query on an interval	Many saved searches, latency target in minutes	Few saved searches, or only daily digests
Matching engine	Search-engine percolator (Elastic-search, OpenSearch) or Vespa predicates	Custom structured matcher	Semantics must equal site search; keyword saved searches matter	Browse-shaped saved searches dominate; engine coupling is a burden; extreme volume

Decision	Option A	Option B	Choose A when	Choose B when
Digest content	Accumulated percolation matches, re-checked at send	Batch re-query at digest time	One matching semantics for all channels	The percolator is not trusted yet, or digests need ranking the matcher cannot provide
Saved-search storage	Normalized form in a saved-search store; engines hold projections	Engine-specific stored queries as the record	Always, beyond a prototype	Never at scale: it couples user data to the engine version
Alert triggers	Entry into match set, plus watcher events	Every listing change that matches	Users value precision	Never; it produces noise
Fan-out order	Randomized within a fan-out	Fixed (creation or account order)	Competitive categories, fairness matters	Low-competition categories where order is irrelevant
Risk gate	Gate alerts separately from publication	Same gate for both	High-stakes categories, fraud pressure	Low-risk categories with fast moderation

The underlying tension is speed against safety and attention. Every protective mechanism (risk holds, caps, digests) delays some alerts; every speed mechanism (instant push, no holds) increases noise and risk. The architecture should make each mechanism a configurable policy per brand and category, measured against alert-to-contact rate, rather than a hard-coded behavior.

Vertical Differences

Dimension	General classifieds (Trade-yard)	Real estate (Hearthly)	Automotive (Motor-vane)	Jobs
Typical saved search	Category + radius + price; often keywords	Area (named, drawn, commute) + rooms + price; sale or rent	Make, model, year, mileage, price, radius; many attributes	Title keywords, location, contract type
Speed sensitivity	High for desirable local goods	Very high for rentals in tight markets; moderate for sales	Moderate; high for underpriced cars	Low to moderate; daily digests common
Dominant channel	Push	Push for rentals; e-mail digests for sales	Push and e-mail	E-mail
Watcher alerts	Sold and reserved notices	Price drops, "under offer", back on market	**Price drops are central**	Rare
Duplicates in alerts	Reposts	Multi-agency listings: alert once per canonical property	Same car at several dealer branches	Same job via several recruiters
Fraud pressure on alerts	Medium (electronics, bikes)	**High for rentals**	Medium (under-priced cars)	Medium (fake job offers)
Taxonomy change impact	High: large, evolving trees	Low: stable property types, but place boundaries change	Medium: model and trim catalogs change yearly	Medium: job taxonomies change

For real estate, alerts should be per **canonical property**: when three agencies list the Lyon apartment in May 2027, a buyer with a saved search over Lyon 7e should get one

alert for the flat, with the three offers attached, not three alerts. For automotive, the feed lag documented in the Golf example matters: when the car sold on 29 April, the dealer's feed reported the removal 26 hours later, so retractions arrive late through no fault of the alert system, and watchers can receive alerts on a car that is already gone. The send-time check cannot see a sale the system of record does not know about; Many Doors In: Forms, APIs and Bulk Feeds owns the feed contract that closes that gap.

Failure Modes

1. **Snapshot digests (Ghost Inventory).** Digests built from a snapshot and sent hours later, with no retraction and no send-time check. The bike's 214 stale digests are the canonical case.
2. **Saved searches stored as engine queries or URLs.** An engine migration or a URL redesign breaks or silently changes millions of saved searches. The fix is the normalized store and rebuildable projections.
3. **Alerts on every change.** The matcher notifies whenever a matching listing changes, so a dealer's daily feed run re-announces their whole stock. Users unsubscribe; deliverability suffers.
4. **Taxonomy changes that orphan saved searches.** A category split that maps saved searches to nothing, or to the wrong child, so users stop receiving alerts without knowing why.
5. **Alerts as a fraud amplifier (Moderation as an Afterthought).** No risk gate in front of the matcher; fake listings reach the most motivated buyers first.
6. **Shared quotas with marketing.** A promotional push campaign in the same messaging project or sending domain delays or degrades alerts on the evening that matters.
7. **The cron that overruns.** Batch alerting that is fine on Tuesday and late on Sunday, because the batch interval grows with load. Tradeyard's p95 of 23 minutes on Sunday evenings is this.
8. **Unbounded saved-search growth.** No dormancy rule; the candidate set and the cost grow forever, with a third of the work done for people who no longer read the alerts.
9. **Unsubscribe that does not work.** An unsubscribe link that leads to a login page, or that takes days to apply. It is a legal risk and a deliverability risk.
10. **Measuring sends (Vanity Liquidity).** Dashboards that celebrate notification volume while the alert-to-contact rate falls.

The Chief Architect's View

At group level, saved searches and alerts are a natural shared platform, and one of the clearest cases in Common Ground for the principle **share platforms, not products**. The

saved-search store, the matcher, the notification orchestrator, the channel workers and the policy engine for caps, quiet hours and consent solve the same problems for every brand. What alerts say, how they look, which categories get instant push and how digests are composed belong to each brand.

The decisions I would bring to the architecture advice forum:

1. **One saved-search service as system of record, owned by the search platform.** Tomasz's team owns the store, the normalized query format and the percolator projections; brands own the user experience. The normalized format is the same one the search API and the geo query compiler of Searching by Place: Map, Polygon and Commute Search use, so a saved search always means what the same search means on the site.
2. **One notification orchestrator for the group.** Owned by the platform engineering group under Yara Mendes, because it serves alerts, messaging notifications (From Interest to Contact: Messaging and Lead Management) and transactional messages. Caps and quiet hours must count across all of them, or users experience the sum.
3. **Alertability is a derived listing state, governed by trust and safety.** Aisha Rahman's team owns the risk gate and its thresholds per category; changes are decision records.
4. **Consent and channel policy are group policy.** Lucía Ortega owns the classification of alerts per market, the unsubscribe semantics and the retention rule; the orchestrator enforces them.
5. **Alert latency and ghost-alert rate are platform service-level objectives (SLOs).** Reported weekly per brand, next to search latency. The 5-minute p95 target is the promise; the ghost-alert rate is the honesty check.
6. **Taxonomy and gazetteer changes include saved-search migration in their definition of done.** No category split goes live without a migration plan and a parallel-run comparison.
7. **Paid influence on alerts is a product decision with transparency obligations.** Sponsored items in digests or paid early alerts are decided by Noor Hassan and Olivier Petit together with the regulatory lead, not by an engineering default.

What to watch over the longer term: **natural-language saved searches.** A 2023 trade-press report says Zillow's natural-language searches can be saved for alerts (The Title Report, 2023); Zillow has not published how. A large language model (LLM) that turns "a quiet two-bedroom near a good school, under €400,000" into filters, as What Did They Mean? Query Understanding and Autocomplete recommends, must produce the same normalized form, or the saved search will mean one thing on the day it was saved and something else when the model is upgraded. Pinning the compiled query at save time, and re-compiling only with the user's confirmation, keeps alerts stable across model versions.

Questions to Ask

1. Where is the system of record for saved searches, and can the percolator or matching index be rebuilt from it in a day?
2. What is the alert latency, from searchable to provider acceptance, at p50 and p95 on the busiest evening of the week, per brand and channel?
3. Does the matcher notify on entry into the match set only, and how does it know whether a listing matched before the event?
4. How are sold, removed, reserved and taken-down listings retracted from pending notifications and digests, and is there a send-time availability check?
5. Which listings are not alertable, who sets the risk threshold, and how long is the hold for new accounts?
6. What happens to saved searches when a category is split, a boundary changes or a polygon rule changes? Show the last migration's comparison.
7. Are alerts and marketing sent through separate quotas, projects and sending domains, and what are the current spam-complaint and unsubscribe rates?
8. How are users ordered within a large fan-out, and may any paid product change that order?
9. How many saved searches are dormant, what is the retention rule, and does deletion reach the store, the ledger and the percolator?
10. Which metric does the alerts team optimize: sends, opens or contacts?

From Discovery to Trust

Alerts bring buyers back to listings that are fresh and real. Whether those listings *are* real is the work of the next part of the book, Part V — TRUST: Keeping the Marketplace Safe, which begins with how a marketplace organizes trust and safety as a system rather than a queue (Trust and Safety as a Platform).

Sources and Evidence

Company engineering and investor sources

- Elastic is the key to helping Rightmove customers find and buy their dream home — Elastic (vendor case study), 2025. Percolator-based instant property alerts among the priorities of Rightmove's 2014 migration.
- Homegate's fast and modern search experience — Elastic blog with Homegate, 2024. Search alert index, percolate query per new property, e-mail and push notification.

- How we migrated 12 million transactional emails per day from Postfix to PowerMTA — Edouard Perret, leboncoin tech blog, 2026. About 12 million transactional e-mails a day moved from 20 Postfix servers to PowerMTA on AWS.
- Zillow natural-language search can be saved for alerts — The Title Report, 2023 (secondary report). Natural-language saved searches.

Standards, regulation and documentation

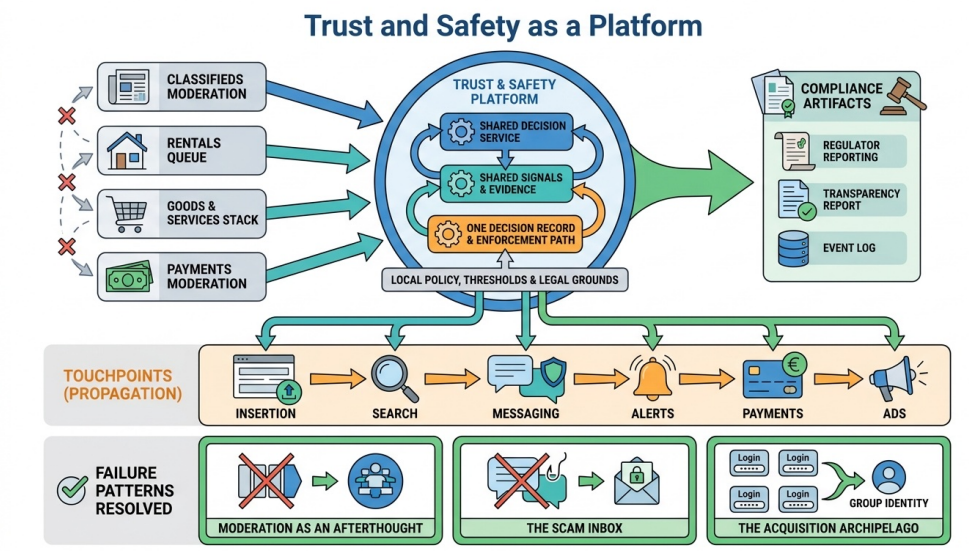
- Percolator field type — Elastic, n.d. Stored-query mechanics, reindexing, analysis cost and limits.
- Percolator field type — OpenSearch, n.d. Percolator field and percolate query in OpenSearch.
- Lucene monitor package — Apache Lucene, n.d. (module added in Lucene 8.2, 2019). Registered queries matched against document streams.
- Predicate fields — Vespa, n.d. Boolean constraints matched against query attributes; saved searches as a named use.
- Throttling and quotas — Firebase Cloud Messaging, n.d. Project quota, per-device limits.
- Notification runtime permission — Android Developers, n.d. `POST_NOTIFICATIONS` from Android 13.
- Email sender guidelines — Google Workspace Admin Help, n.d. Requirements for bulk senders from 1 February 2024.
- RFC 8058: Signaling One-Click Functionality for List Email Headers — IETF, 2017. One-click unsubscribe headers and DKIM coverage.
- Directive 2002/58/EC (ePrivacy Directive) — European Union, 2002, as amended. Article 13 on unsolicited electronic communications.

Research and practitioner sources

- Productscape domain models for general-listings and real-estate marketplaces (author's sibling repository, digest in this book's research notes), n.d. The saved-search capability, its data asset and retention, and the buyer journey quotes on alert speed.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

Part V — TRUST: Keeping the Marketplace Safe



IN THIS PART, YOU WILL: Architect trust and safety as a platform, stop fraud from fake listings to payment-link scams, decide how much to know about users and how to verify it, and turn the Digital Services Act and related rules into system requirements.

WHY THE BUSINESS CARES: A scammed buyer rarely comes back, and a wrongly blocked seller tells other sellers. Since 2024 every EU moderation decision is also a record a regulator can ask about, and fines under the Digital Services Act can reach 6% of worldwide turnover.

WHY ARCHITECTS SHOULD CARE: Trust touches insertion, search, messaging, alerts, payments and ads. Without one decision record, one

propagation path and one identity model, a group cannot say what it removed, why, or whether the removal took effect.

KEY POINTS:

- **Trust and safety is a platform of separate functions joined by one decision record.** Policy, detection, decision, enforcement and review change at different speeds. Share the machinery, keep policy local, and count an enforcement action as done only when every consumer has applied it.
- **Fraud is caught at the setup, by scoring actors and adding friction in proportion to risk.** In classifieds the money usually leaves by bank transfer. Link listings, accounts, devices, payment accounts and photos, and ask for identity at the moment of risk, not at the door.
- **Regulation asks for a few artifacts, built once.** A decision record, a trader record, disclosure surfaces and reproducible reports serve most obligations, and every regulated number should be computable again from the event log.

FAILURE PATTERNS THIS PART ADDRESSES:

- **Moderation as an Afterthought** — Replaces a queue bolted onto insertion with a platform that produces evidence by design.
- **Fast Mail for Fraudsters** — Puts link, pattern and identity controls into the channel where scams close.
- **Deal-Shaped Architecture** — Replaces four ban lists and four logins with shared signals and one group identity.

Part IV built surfaces that put listings in front of buyers within seconds. Fraudsters exploit that speed: a fake rental that reaches saved searches in minutes reaches the most motivated victims first. Part V turns to **the trust that makes contact between strangers possible**.

This part argues that trust is **a platform capability, not a team with a queue**. Most marketplaces build it in the order incidents arrive, and a decade later the group has four moderation stacks that cannot share a signal. These chapters replace that history with one design: shared decision service, signals and evidence; local policy, thresholds and legal grounds.

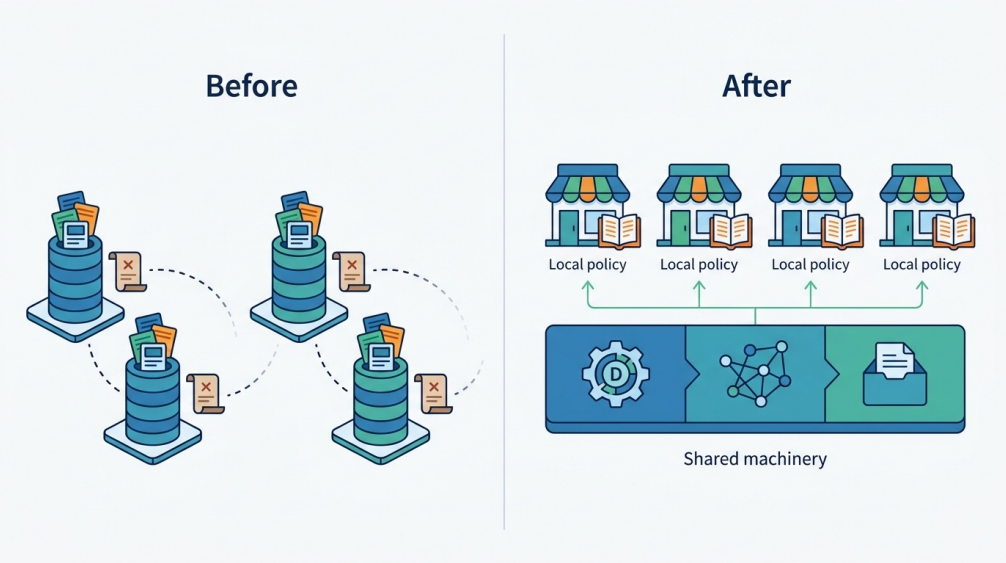


Figure 1: *Part V replaces four moderation stacks that cannot share a signal with one shared decision service, signals and evidence, while each brand keeps its own policy, thresholds and legal grounds.*

Chapters in this part

Chapter	The question it answers	Read it when…
Trust and Safety as a Platform	How should trust and safety be architected across listings, users, messages and transactions?	Each brand runs its own moderation stack, or removals never reach alerts and caches.
Scams, Fakes and Account Takeovers: Fraud Prevention for Listings	How do marketplaces detect and stop fraud, from fake listings to payment scams?	Scams close in your messaging, or fraud signals land in separate queues.
Who Is on the Other Side? Identity and Verification	How much should a marketplace know about its users, and how does it verify it?	You are designing a group login, verified badges or business checks.

Chapter	The question it answers	Read it when…
Rules of the Marketplace: DSA, DAC7 and Vertical Regulation	Which regulations shape the architecture, and how do they become requirements?	A transparency report cannot be reproduced, or a payments feature will move your legal perimeter.

How this part connects

Trust reads the lifecycle and enforcement overlay from Part II — LISTING: Modeling the Thing Being Sold, the publication gate from Part III — SUPPLY: Getting Inventory In and the discovery surfaces from Part IV — DISCOVERY: Matching Intent to Inventory, all of which must obey its decisions. It hands three threads to Part VI — MONEY: Conversion and Monetization. Messaging is where scams close. In-platform payments bring DAC7 seller reporting into scope. Trader records feed billing as well as compliance.

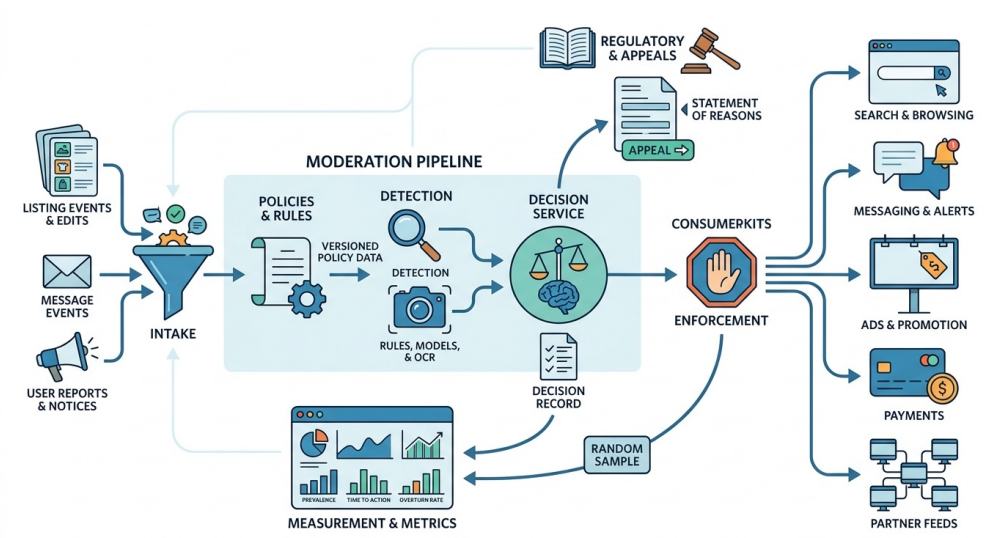
The Veldmark thread

Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), starts 2027 with four trust-and-safety stacks, four identity systems and five sources of statements of reasons. Two 2027 incidents show what that split costs. On 7 March 2027 a fake Safe Deal payment link reached Femke sixteen minutes after her first buyer contact. In July a Barcelona ring of 26 accounts posted 180 fake rentals, and 63 people lost about €71,000. Ingrid Halvorsen's team, with Aisha Rahman and Lucía Ortega, decides:

- **Veldmark ID** (June 2027): one login across brands with brand profiles, designed so that a sold brand can be cut out;
- an **obligation register** that maps the DSA and related rules to controls and evidence queries;
- a **shared trust-and-safety platform**, live for Tradeyard NL and BE in November 2027, with a cross-brand block list of hashed identifiers;
- **first-contact friction** on Tradeyard (December 2027).

By the end of Part V you should be able to say, for any enforcement decision, who made it, under which policy version, where it propagated and how you would prove all three a year later. Begin with Trust and Safety as a Platform.

24. Trust and Safety as a Platform



IN THIS CHAPTER, YOU WILL: Take trust and safety apart into policies, detection, decisions, enforcement, review and measurement. You will learn which parts belong in a shared group platform and which stay local, and how one enforcement decision has to reach every system that shows, alerts on or monetizes a listing.

WHY THE BUSINESS CARES: A buyer who is scammed once rarely comes back, and a seller who is wrongly blocked tells other sellers. Trust is the reason a buyer contacts a stranger, so moderation quality, speed and fairness decide liquidity as directly as search does. Since 2024, the Digital Services Act has also made every moderation decision a record a regulator can ask about.

WHY ARCHITECTS SHOULD CARE: Trust and safety touches every bounded context: insertion, search, messaging, alerts, payments, ads and support. A platform without one decision record and one propagation path cannot say what it removed, why, or whether the removal actually took effect.

KEY POINTS:

- **Trust and safety is five functions, not one queue.** Policy, detection, decision, enforcement and review fail in different ways and change at different speeds. Architect them as separate components joined by one decision record, and treat policies as versioned data rather than as code scattered across services.
- **An enforcement action is only real once every consumer has applied it.** Removing a listing has to reach search, alerts, recommendations, messaging, SEO pages, ads and payments. Emit enforcement as events with the same discipline as listing state changes, and monitor how long propagation takes.
- **Measure harm, not activity.** Queue sizes and removal counts show how busy the team is. Prevalence from random sampling, time to action, precision, and the appeal overturn rate show whether the marketplace is safe and fair. A shared platform pays for itself mostly through shared evidence, tooling and signals, while policy stays local where law and markets differ.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Moderation as an Afterthought** — a queue bolted onto the insertion flow cannot re-score edits, route reports or produce the evidence the DSA requires, and it grows faster than the team that empties it.
- **Ghost Inventory** — a takedown that reaches the listing database but not the alert digest, the recommendation cache or the SEO page leaves fraudulent listings visible after they were "removed".
- **Set-and-Forget AI** — moderation classifiers that auto-decide 94% of listings, with no prevalence sampling or overturn monitoring, look accurate while a new scam pattern goes past them.

Most marketplaces build trust and safety (T&S, the policies, systems and people that keep users safe and the inventory honest) in the order problems appear. A keyword filter arrives after the first wave of spam, a review queue after the first press story, a report button after the first regulator letter. Each piece is attached to whatever system was nearest. Ten years later the group has **four moderation stacks that cannot share a signal**, and nobody can answer a simple question: *what did we remove last month, why, and did it actually disappear?*

Part IV built search, recommendations and alerts to put a listing in front of buyers within seconds, and each of them must also obey a removal. This chapter, the first of Part V, treats trust and safety as a **platform capability** with its own architecture. It defines the five functions and the components that implement them. It follows one enforcement decision through every system that has to obey it, and sets out the metrics that show

whether the platform works. It also argues for a split: share the machinery across brands, keep the policy local.

Fraud is the most expensive harm a classifieds platform faces, and it has its own chapter, Scams, Fakes and Account Takeovers: Fraud Prevention for Listings, which owns risk scoring, fraud signals and step-up friction. Identity levels and verification belong to Who Is on the Other Side? Identity and Verification. The legal detail of the Digital Services Act (DSA) and other regulation belongs to Rules of the Marketplace: DSA, DAC7 and Vertical Regulation. Here we build the platform those chapters plug into. The running example is Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build).

Setting the Scene at Veldmark: Four Stacks and One Phishing Link

On Sunday **7 March 2027** at 20:11 CET, a "buyer" sent Femke, a private seller in Rotterdam, a fake Tradeyard Safe Deal payment link for her **€240** city bike. The message was delivered because the link's domain had been registered that day and no rule looked at domain age. Femke reported it at 20:19. Tradeyard Netherlands' in-house rules engine suspended the account at 20:26, and an analyst wrote a rule that then withheld **37** similar messages to other sellers. By the standards of most classifieds platforms, seven minutes from report to suspension is fast. The details, including the bike's sale the next day, are in From Interest to Contact: Messaging and Lead Management and Scams, Fakes and Account Takeovers: Fraud Prevention for Listings.

Ingrid Halvorsen had joined Veldmark as Group Chief Architect five weeks earlier. On Tuesday **16 March** she asked Aisha Rahman, the Group Head of Trust & Safety, to walk her through what had happened *outside* Tradeyard NL. The answers were less comfortable than the seven minutes:

- The phone number used to register the suspended account had also registered a **Motorvane** private-seller account three weeks earlier. Motorvane could not know. Each brand runs its own identity system, so no shared block list exists.
- Tradeyard **Belgium** shares the Netherlands rules engine, but the new rule took until Wednesday to reach it. Each country's rules are deployed with the country's Yardstick configuration, and Yardstick, Tradeyard's PHP monolith, deploys twice a week.
- Tradeyard **Poland** and **Portugal** use an outsourced moderation queue with its own keyword lists. Neither received the rule.
- **Hearthly** runs in-house rules on Google Cloud. Its Spanish brand, the former Casavista, still uses Casavista's review tool.

Aisha's inventory of the group's T&S landscape fitted on one slide:

Brand	Detection	Human review	Policies	Appeals	DSA statements of reasons
Tradeyard NL/BE	In-house rules engine plus three ML models (listing, image, message)	46 in-house moderators, Lisbon	Wiki pages per country, partly translated	E-mail to support; no link to the original decision	Template in Yardstick; nightly batch to the Transparency Database
Tradeyard PL/PT	Vendor keyword lists; NL image model via an API	112 outsourced moderators, two vendors	Vendor playbooks	Vendor ticket system	Vendor export, monthly
Motorvane	Dealer KYB at contract signature only; private-seller checks	17 moderators, Düsseldorf	Contract terms for dealers	Account managers	Hand-built for private sellers
Hearthly FR/ES	In-house rules; manual review of new agencies	35 moderators, Lyon and Barcelona	Two documents, French and Spanish	Support tool	Separate exporters for FR and ES

That made **210 moderators**, four definitions of a "prohibited item", three appeal processes and five ways of producing a DSA **statement of reasons** (the notice the DSA requires a platform to send a user whose content it restricts). At Tradeyard alone, **94%** of new-listing decisions were automated and about **138,000 listings a week**, 6% of 2.3 million, went to human review.

Ingrid's question for the Common Ground program was not whether to centralize the moderators. It was narrower and harder: **which parts of trust and safety are one**

platform, and which must stay four? The rest of this chapter is the answer the architecture advice forum adopted in mid-2027. The shared T&S platform went live for Tradeyard NL and BE in November 2027.

Concepts: The Five Functions of Trust and Safety

Trust and safety work is easier to architect once it is split into functions that change at different speeds and fail in different ways.

Function	Question it answers	Changes at the speed of	Typical failure
Policy	What is allowed, for whom, where?	Law, market and scam trends: weeks to months	Policies exist only as wiki prose; rules drift from the written policy
Detection	Which items, accounts or messages might violate a policy?	Adversaries: hours to days	Detection runs only at creation; edits and messages are not scored
Decision	What do we do about this item, given policy and evidence?	Policy and risk appetite	Several services decide independently and contradict each other
Enforcement	Did the action take effect everywhere?	System architecture	Removal reaches the database but not search, alerts or caches
Review and redress	Was the decision right, and can the user contest it?	Volume and law	Appeals are support tickets with no link to the decision or evidence

A sixth concern runs across all five: **measurement and transparency**, the ability to say, reproducibly, how much harm exists, how much was caught, how fast, and how often decisions were wrong.

Figure 1: *Trust and safety is five functions that change at different speeds and fail in different ways, joined by one decision record and watched by measurement.*

Five definitions are used throughout the chapter.

- A **policy** is a named, versioned rule of conduct with a scope (category, country, seller type), a rationale and a legal or contractual ground. "No live animals in Tradeyard PT except through registered breeders" is a policy. A keyword list is not; it is one detector for a policy.
- The **moderation pipeline** is the sequence of components that evaluates content and behavior against policies and produces decisions: synchronous gates, asynchronous detectors, human review queues and sampling. This book uses the term for listings, accounts and messages alike. This chapter owns it.
- An **enforcement action** is the concrete restriction a decision imposes: a warning, a reach limit, a demotion, a hold, a removal, a suspension, a payment freeze. Every action has a target (listing, account, message, payment), a scope (country, brand), a duration and a reason. This chapter owns the term.
- A **case** is the unit of human work: one or more items, the evidence, the decisions and the communications that belong together. A fake-rental ring of 26 accounts is one case, not 180 tickets.
- **T&S metrics** are the measures of harm, speed and accuracy defined later in this chapter: prevalence, time to action, precision and recall, and the appeal overturn rate. This chapter owns them.

The functions also separate the **listing view** from the **actor view**. A listing can violate a policy on its own: a weapon, a counterfeit, a discriminatory rental ad. An actor can be a problem even when every listing looks clean: a fraud ring, a takeover account, a reseller

of stolen goods. Mature platforms decide on both, and the decision record links them. The Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models chapter treats Trust as its own bounded context with its own model of the listing. That model is a *subject under evaluation*, not a catalog item.

Policies as versioned reference data

The most underrated design decision in trust and safety is to **treat policies as data**: a registry of policy records that rules, models, reviewer guidelines, user notices and reports all point to. A policy record at Veldmark looks like this:

Field	Example (Tradeyard, payment-link phishing)
Policy ID and version	`POL-FRAUD-PAYLINK` v3, effective 18 March 2027
Scope	Messages and listings; all Tradeyard countries; all account types
Statement	Messages may not contain links or QR codes that imitate Tradeyard payment, shipping or Safe Deal pages
Ground	Contractual (terms of service section 7.2); in NL also an illegal-content ground (fraud)
DSA category	Scams and fraud
Detectors	Domain-age rule, platform-brand-in-URL rule, QR decoder, message model v12
Allowed actions	Withhold message; suspend account; freeze Safe Deal payouts pending review
Reviewer guideline	Link to guideline v3, with five labeled examples
User-facing reason	Template ID and approved text in Dutch, French, Polish, Portuguese
Owner	Group T&S policy lead; local policy owner per country

Three consequences follow. First, a decision can cite **policy ID and version**, so a statement of reasons, an appeal and a transparency report all refer to the same thing. Second, a policy can have **country overlays** (Poland may prohibit something the Netherlands allows) without forking the platform. Third, the policy registry is where legal review happens. Lawyers review policies and user-facing reasons. Engineers review detectors. Neither needs to read the other's code.

The Category Trees That Can Change: Designing Marketplace Taxonomies chapter treats category trees as versioned reference data. The same reasoning applies here, with one addition: **a decision must stay explainable under the policy version that was in force when it was taken**, even after the policy changes.

Architecture: The Moderation Pipeline and Its Components

The reference architecture below is assembled from public building blocks: Meta's Sigma rule engine, described in a 2015 engineering post; Stripe Radar's documented rule operations; OLX's published chat moderation; the DSA's notice, statement-of-reasons and appeal requirements; and the practices recommended in a July 2026 expert report commissioned by Ofcom, the UK online-safety regulator. **It is a reference-architecture inference. No classifieds company has published its T&S architecture in this form.**

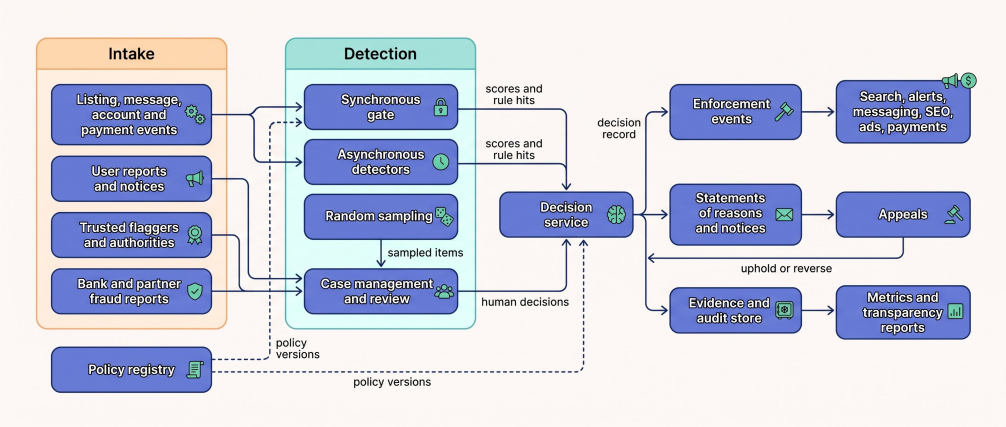


Figure 2: *The trust and safety platform as components. Detection and human review both feed one decision service, which is the single writer of decision records; enforcement, user notices, appeals, evidence and metrics all derive from those records.*

Component	Responsibility	Key interface
Policy registry	Versioned policies, overlays, reviewer guidelines, reason templates	Read API; change events when a policy version takes effect
Synchronous gate	Rules, list lookups and a light model inside the insertion or send path; returns allow, hold or reject	Called by insertion, edit, message send, login; strict latency budget

Component	Responsibility	Key interface
Asynchronous detectors	Image models, duplicate and stolen-photo search, text and LLM classifiers, price anomaly, graph features	Subscribe to listing and message events; publish scores
Random sampling	Draws unbiased samples of live listings and messages for human labeling	Feeds review queues; feeds prevalence metrics
Decision service	Combines rule hits, scores, actor context and human input into an action from the enforcement ladder; writes the decision record	The only component that writes decisions
Case management	Queues, prioritization, reviewer tooling, evidence views, service-level timers, quality sampling	Reviewer UI; case API for support and legal
Enforcement events	Publishes `EnforcementApplied` and `EnforcementLifted` events; tracks acknowledgement by consumers	Event backbone; propagation dashboard
Statement-of-reasons service	Renders user notices, strips personal data, submits to the DSA Transparency Database	Driven by decision records
Appeals	Accepts complaints, routes them to qualified reviewers, reverses or upholds	Writes a new decision linked to the original

Component	Responsibility	Key interface
Evidence and audit store	Immutable record of decisions, rule and model versions, reviewer identity and content snapshots, with retention per data class	Query API for appeals, regulators, law enforcement requests
Metrics	Prevalence, time to action, precision, overturn rates, transparency-report figures	Built from decision records and samples, not from operational logs

The synchronous gate and the asynchronous path

The pipeline has two speeds. The **synchronous gate** sits inside the user's action: publishing a listing, editing it, sending a message, logging in. It has a budget of a few hundred milliseconds at most and can only afford deterministic rules, lookups against block lists (phone numbers, e-mail addresses, IBANs, device identifiers, image hashes) and a cheap model. Its outcomes are *allow*, *hold for review* and *reject*.

The **asynchronous path** runs on events after the action. Image classification, duplicate search, OCR on images (text in an image is a classic way to smuggle a phone number past text filters), graph features linking accounts, price anomaly against a valuation model and LLM classifiers all take longer. They can still demote, hold or remove a listing that is already live.

Which checks run in which path is a **risk decision, not a technical one**. A €15 children's book can go live on the gate's verdict and be re-checked within minutes. A rental flat in a city with a known fake-rental problem should wait for the image and duplicate checks, because the harm of a fake listing being live for ten minutes is larger than the cost of a ten-minute delay. The From Submitted to Searchable: The Publication Pipeline chapter owns the end-to-end publication path and its freshness targets. This chapter owns the moderation gate inside it.

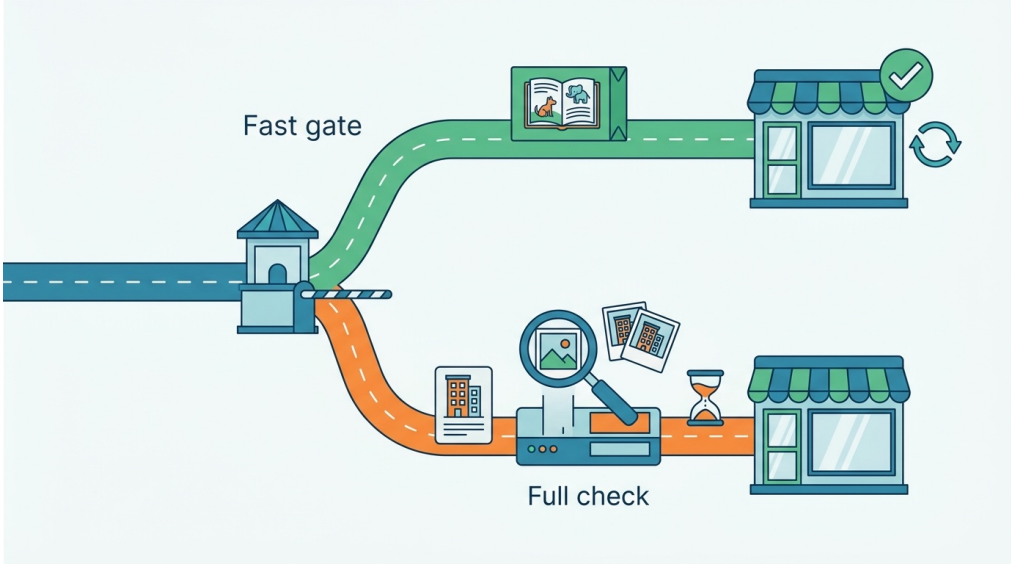


Figure 3: *Which checks run before publication is a risk decision: a €15 book can go live on the gate and be re-checked within minutes, while a rental in a fake-rental hotspot waits for the image and duplicate checks.*

Moderation modes: a spectrum, not a choice

Pre-moderation (checking before publication) and post-moderation (publishing first and checking afterwards) are usually presented as a choice. In practice they are the ends of a spectrum, and large platforms run several points on it at once.

Mode	Time to live	Exposure to harm	Cost	Where it fits
Full human pre-moderation	Minutes to hours	Lowest	Highest	Small sites; high-risk categories such as pets, tickets, rentals
Automated pre-moderation, human review for the grey zone	Seconds; grey zone minutes	Low	Medium	The mainstream default for large classifieds

Mode	Time to live	Exposure to harm	Cost	Where it fits
Publish, then score asynchronously	Instant	A window of exposure	Lower	Low-risk categories and trusted sellers
Reactive only (notice and action)	Instant	Highest	Lowest	Legally minimal; not defensible for fraud at scale

The design that works is **risk-tiered**: the mode depends on the category's risk, the seller's trust level (account age, verification level from Who Is on the Other Side? Identity and Verification, history) and the listing's risk score. A verified dealer's car listing from a DMS feed can go live on the gate. A new account's first rental listing waits for the full check.

Two public cases show why the mode matters.

- **Re-moderate on every material edit**. In a December 2024 undercover test, the UK consumer organization Which? posted fake camera ads on Facebook Marketplace, Gumtree and Nextdoor. Gumtree blocked the first scam-style ad, but **edited versions passed review "in minutes"**, and on Facebook one ad stayed live for almost 48 hours after the price was halved and Apple gift cards demanded (Which?, 2024). The lesson is architectural: a change of price, payment terms, contact details or photos is a new submission, and the pipeline must treat it as one.
- **Say what you actually do**. In December 2015, a Paris court found leboncoin guilty of misleading commercial practices because it claimed all ads passed a validation process when not all did. The court also confirmed leboncoin's status as a hosting provider, which does not oblige it to check every ad before publication (Clubic, 2015). The case is more than ten years old, but the principle has not aged: the moderation mode is a factual claim, and the terms and marketing must describe the mode the architecture actually runs.

Human review: queues, priorities and sampling

Human review exists for three reasons: to decide the grey zone that automation cannot, to produce **labels** that train and evaluate the models, and to satisfy the law where the law requires a human (DSA appeals, discussed below). Each reason calls for a different queue design.

- **Grey-zone queues** should be prioritized by expected harm multiplied by uncertainty, not by arrival time. A suspected fake rental with ten contacts in its first hour outranks a possibly miscategorized sofa.
- **Label queues** need an unbiased sample. If reviewers only see what the models flagged, the labels describe the models' blind spots poorly, because the blind spots never reach a reviewer.
- **Regulated queues** (appeals, trusted-flagger notices, authority orders) carry legal timers and need qualified reviewers.

Stripe's documentation for its Radar fraud product gives a rule that transfers directly: **a review queue that almost always ends in the same outcome should become an automated rule** (Stripe Radar docs, n.d.). The inverse also holds. An automated rule that reviewers would frequently overturn should go back to the queue.

The **random sample** gives the only number in trust and safety that does not depend on what the detectors already catch. The Ofcom-commissioned expert report of July 2026 recommends that larger platforms regularly review randomly sampled ad impressions and the most-viewed ads to estimate the prevalence of violating content (Integrity Institute for Ofcom, 2026). For a classifieds platform the equivalent is a stratified sample of live listings and delivered messages per category and country, reviewed blind.

LLMs in the pipeline

Large language models (LLMs) entered production moderation between 2024 and 2026. The public classifieds evidence is thin but real. In a January 2026 post, OLX described a generative-AI system that detects and blocks suspicious images in sales chats in real time, live in Poland, Portugal, Bulgaria, Romania and other markets (OLX Group, 2026). Llama Guard, published by Meta researchers in December 2023, showed that a policy taxonomy can be supplied to a safety model as a prompt rather than retrained (Inan et al., 2023).

My rule for placing LLMs in a moderation pipeline has four parts:

1. **Grey zone only**. Run LLM classifiers on the uncertain band of a cheap classifier and on items already headed for human review, not on all traffic. At Tradeyard's 2.3 million new listings a week, the cost difference is an order of magnitude.
2. **Policy agility is the main value**. A new scam script can be described in a prompt the day it appears, then distilled into a cheaper classifier once labels accumulate.
3. **Deterministic floors stay**. Hash lists, block lists and hard rules cannot be argued with by a cleverly worded listing. Prompt injection in a listing description is an attack surface.
4. **Drafts, not decisions, where the law needs a human**. An LLM can summarize a case and draft the explanation in a statement of reasons. It cannot decide a DSA complaint, which must not be decided "solely on the basis of automated means".

Enforcement Actions and Their Propagation

The enforcement ladder

Removal is one rung of a ladder of actions, and often not the right one. The Ofcom-commissioned experts describe a graduated ladder, from a nudge through posting limits, payment freezes and re-verification to suspension. They note that payment freezes need a higher evidentiary threshold (Integrity Institute for Ofcom, 2026). A ladder for a classifieds platform:

Rung	Action	Target	Typical trigger	DSA statement of reasons?
1	In-context warning to the counterparty	Conversation	Payment keywords, off-platform move	No restriction on the sender's content
2	Nudge or correction request	Listing	Missing mandatory field, wrong category	No, unless visibility is restricted
3	Friction or re-verification (see Scams, Fakes and Account Takeovers: Fraud Prevention for Listings)	Account	Unusual login, risk score in the grey band	Yes, if service is restricted
4	Reach limit or **demotion**	Listing or account	Low-trust actor, suspected duplicate	**Yes**: demotion is a restriction
5	Hold for review	Listing, message	Gate verdict "hold"	Yes, if it lasts and the user is told
6	Removal or disabling	Listing, message	Clear policy violation	Yes
7	Payment freeze	Payouts	Strong fraud evidence on a transaction	Yes (monetary restriction)

Rung	Action	Target	Typical trigger	DSA statement of reasons?
8	Suspension or termination	Account, and linked accounts	Fraud, repeated violations	Yes

The fifth column surprises many ranking teams: Article 17 of the DSA covers **demotion** as well as removal, so a ranking system that quietly pushes suspected scam listings down the results is taking a moderation decision that needs a statement of reasons. The DSA Transparency Database schema has a value for exactly this case, `DECISION_VISIBILITY_CONTENT_DEMOTED` (European Commission, n.d.). **Ranking teams are in the moderation business**, and actor-trust down-ranking belongs behind the decision service, not inside a ranking feature. Ranking for Two Sides: Relevance, Freshness and Paid Visibility covers the ranking side of this boundary.

One decision, many consumers

A removal is not finished when the listing's status changes. A classifieds platform shows, alerts on and monetizes a listing through many systems, and each holds its own copy of it.

Consumer	What it holds	What enforcement must do there
Listing system of record	Status	Set the restricted state, keep the version history
Search index	Listing document	Delete or hide within seconds
Saved-search alerts and digests	Pending notifications	Drop queued alerts; exclude from digests already built
Recommendations	Candidate caches, embeddings	Evict; block from similar-listing results
Messaging	Conversations about the listing or with the account	Warn counterparties; withhold pending messages; block new contact
SEO pages and sitemaps	Rendered listing page	Return **410** (gone), not a soft 404; drop from sitemaps

Consumer	What it holds	What enforcement must do there
Advertising	Sponsored placement, retargeting audiences	Stop serving; stop billing
Payments	Open Safe Deal transactions, payouts	Hold or freeze per policy
Partner feeds and syndication	Copies sent to partners or price-comparison sites	Send deletion
Analytics	Event history	Mark as enforced; do not delete the evidence

The pattern that works is the one Every Change Is an Event: Event-Driven Listing Architecture sets out for listing state changes: the decision service emits an **`EnforcementApplied`** event with the decision ID, target, action, scope and duration. Each consumer applies it idempotently and acknowledges it. A propagation monitor tracks the time from decision to the last acknowledgement for each consumer. When an appeal reverses the decision, an `EnforcementLifted` event restores the listing **and its ranking state**, because a reinstated listing buried on page nine has not really been reinstated.

The failure mode is **Ghost Inventory** caused by enforcement. Veldmark's ledger records one benign version. Femke's bike was marked sold at 18:10 on Monday 8 March and removed from search at 18:11, yet the evening alert digest still featured it the next morning. A sold bike in a digest wastes a buyer's time. A *fraudulent* listing in a digest, after the platform has decided to remove it, sends buyers to a scammer under the platform's own brand. Unpublish propagation for sold and expired listings is owned by From Submitted to Searchable: The Publication Pipeline. Enforcement should ride on the same mechanism, with a higher priority and a stricter time budget.

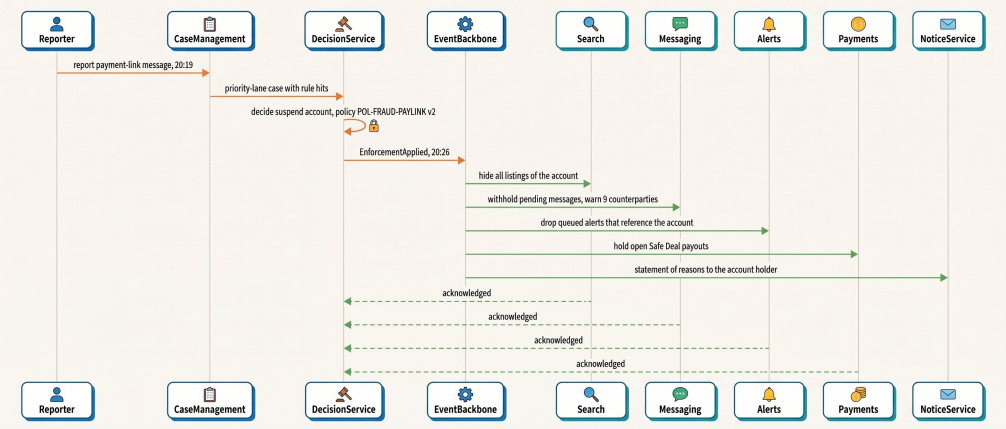


Figure 4: *Enforcement as an event. One decision fans out to every consumer that holds a copy of the account's listings or conversations, and each consumer acknowledges, so the platform can prove when the suspension actually took effect everywhere.*

The "warn 9 counterparties" step in Figure 4 is the target design, not what happened in March 2027. On that Sunday the suspension in Yardstick blocked the account from sending new messages, but the other sellers it had already contacted received no warning. Two of them found out the next day, from Femke's warning in a neighborhood chat group.

Intake, Cases and Appeals

Notice and action

Detection is only one way violations arrive. Users report listings and messages, rights holders report counterfeits, authorities send orders, banks report accounts that received fraud proceeds, and since the DSA, **trusted flaggers** (organizations granted that status by a national Digital Services Coordinator) send notices that must be processed with priority.

Article 16 of the DSA requires an easy, electronic notice mechanism and lists what a notice should contain: an explanation, the exact location (URL), the notifier's name and e-mail address, and a statement of good faith. The platform must confirm receipt, decide "in a timely, diligent, non-arbitrary and objective manner", tell the notifier the outcome and disclose any use of automated means (Regulation (EU) 2022/2065, Art. 16). Article 22 adds the priority lane for trusted flaggers. The Rules of the Marketplace: DSA, DAC7 and Vertical Regulation chapter covers the legal detail. The architectural consequence is that **a notice is an entity with a lifecycle**, not an e-mail in a support inbox:

Notice field	Why the architecture needs it
Notice ID, received timestamp, channel	Time-to-action metrics; confirmation of receipt
Notifier type (user, trusted flagger, rights holder, authority, bank)	Routing and priority; transparency counts by source
Target references (listing, message, account)	Case grouping
Claimed policy or legal ground	Routing to the right reviewer skill
Good-faith statement, notifier contact	Legal completeness; outcome communication
Status and decision link	Proves the notice was decided and how

Case management

Case management is where most T&S platforms either earn or lose their reviewers' productivity. The features that matter are ordinary: grouping related items into one case (all listings of a ring, all reports about one account), showing the evidence the detectors used, showing the actor's history across brands where the law permits, prefilled reason templates from the policy registry, service-level timers, and a second-review sample for quality control.

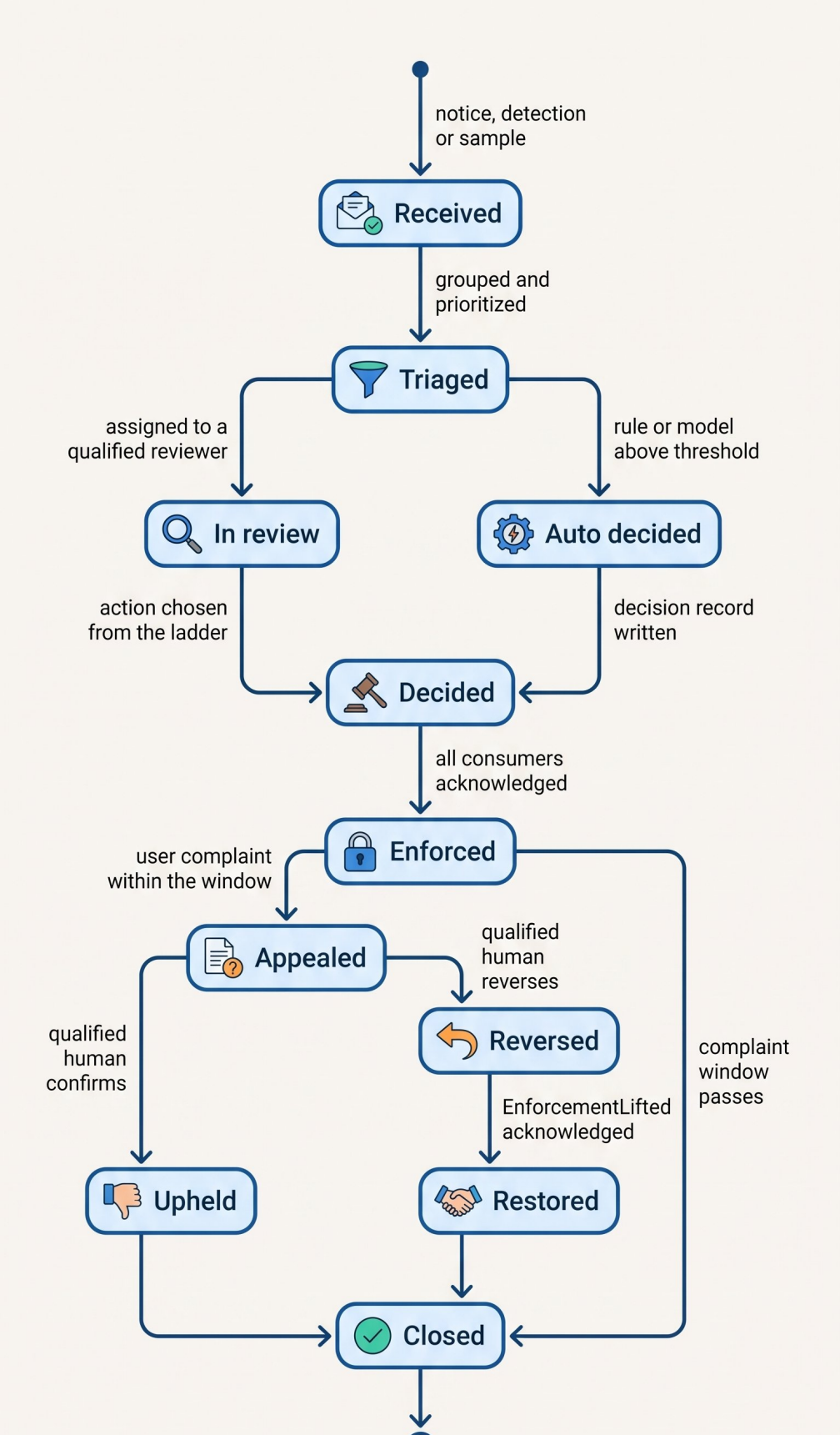

notice, detection or sample
Received
grouped and prioritized
Triaged
assigned to a qualified reviewer
rule or model above threshold
In review
Auto decided
action chosen from the ladder
decision record written
Decided
all consumers acknowledged
Enforced
user complaint within the window
Appealed
qualified human reverses
Reversed
qualified human confirms
complaint window passes
EnforcementLifted acknowledged
Upheld
Restored
Closed

Figure 5: *The lifecycle of a case. "Decided" and "Enforced" are separate states because a decision that has not propagated is not yet in effect; an appeal produces a new decision linked to the original, never an edit of it.*

Appeals and redress

Article 20 of the DSA requires a free, electronic internal complaint system available for **at least six months** after a decision. Complaints must be handled in a timely, non-arbitrary way, unfounded decisions must be reversed, and decisions must be taken under the supervision of qualified staff, not solely by automated means (Regulation (EU) 2022/2065, Art. 20). Users can then escalate to certified out-of-court dispute settlement bodies (Art. 21).

Three architectural rules follow:

1. **An appeal creates a new decision linked to the original.** Never overwrite the first decision. The overturn rate depends on keeping both.
2. **Evidence must outlive the complaint window.** The content snapshot, the rule and model versions and the reviewer identity must be retrievable for at least six months, and longer where out-of-court proceedings or litigation follow. Retention is set per data class, which System of Record and Read Models: Listing Data Management covers alongside erasure.
3. **Reversals are the best labels the platform gets.** A reversed automated decision is a confirmed false positive. Route reversals back to the policy owner and the model owner every week.

Statements of reasons as a by-product, not a project

Every restriction an EU platform imposes on a user's content or account needs a statement of reasons to that user (Art. 17), and every statement must be submitted, without personal data, to the Commission's DSA Transparency Database (Art. 24(5)). For the 180 days to 19 October 2026 it held about **4.4 billion** statements from 375 platforms, with **41%** of decisions fully automated (DSA Transparency Database, accessed 19 October 2026).

At Tradeyard Netherlands, about 1.9% of new listings were removed for policy or fraud reasons in 2026. With roughly 1.05 million new Dutch listings a week, that is about **20,000 statements of reasons a week from listing removals alone**, before account suspensions and message decisions. Nobody writes those by hand. A platform that has a decision record with policy ID, ground, action, scope, automation flags and source can generate the statement mechanically. A platform without one builds five exporters, as Veldmark had. The Rules of the Marketplace: DSA, DAC7 and Vertical Regulation chapter follows Veldmark's statement-of-reasons pipeline in detail. The point here is

that **the decision record is the upstream design**, and the regulatory output is a view of it.

Measuring Trust and Safety

T&S teams are usually measured on activity: items reviewed, listings removed, queue age. Those numbers say how busy the team is. They do not say whether the marketplace is safe, and they can improve while the marketplace gets worse. A model that misses a new scam pattern reduces removals and queue sizes at the same time.

Metric	Definition	What it tells you	How it misleads
Prevalence	Share of live listings (or delivered messages, or contacts) that violate a policy, estimated from a blind random sample	How much harm buyers actually meet	Needs enough samples per stratum; rare harms need large samples
Exposure-weighted prevalence	Prevalence weighted by views or contacts	Harm as experienced, not as stored	Hides harm in long-tail categories
Time to action	Time from first availability (or first report) to enforcement acknowledged by all consumers	Exposure window	Measuring to "decided" instead of "enforced" flatters it
Precision	Share of enforcement actions that were correct, from second review and appeals	Collateral damage to good users	Appeals undercount false positives: most wrongly blocked users give up

Metric	Definition	What it tells you	How it misleads
Recall	Share of violating items the pipeline caught, estimated from the random sample	Detection coverage	Cannot be measured without the sample
Appeal overturn rate	Reversed decisions as a share of appealed decisions, per policy and per detector	Fairness and model drift	Low appeal rates may mean an unreadable appeal flow, not good decisions
Report rate	User reports per 1,000 views or contacts	Visible harm and user trust	Falls when users give up reporting
Reviewer agreement	Agreement between two reviewers on the same item	Guideline clarity	Can be high for unclear policies if reviewers share habits

Two practices make these numbers trustworthy. First, **compute them from decision records and samples, not from operational logs**, so that the same query produces the same figure for the DSA transparency report, the board and the model owner. Second, **report them per policy and per detector**. "Precision 96%" across all policies hides a phishing rule at 99.5% and a counterfeit rule at 70%.

Veldmark's numbers, before and after

Aisha's first measurement exercise, in April 2027, drew a blind sample of 4,000 live Tradeyard NL listings per week, stratified by 20 category groups. It found a prevalence of policy-violating listings of about **0.6%**, against the 0.2% the team had assumed from removal counts. Most of the gap sat in two places: phone-repair "services" posted as goods, and electronics listings with contact details hidden in images, which the text filters could not see.

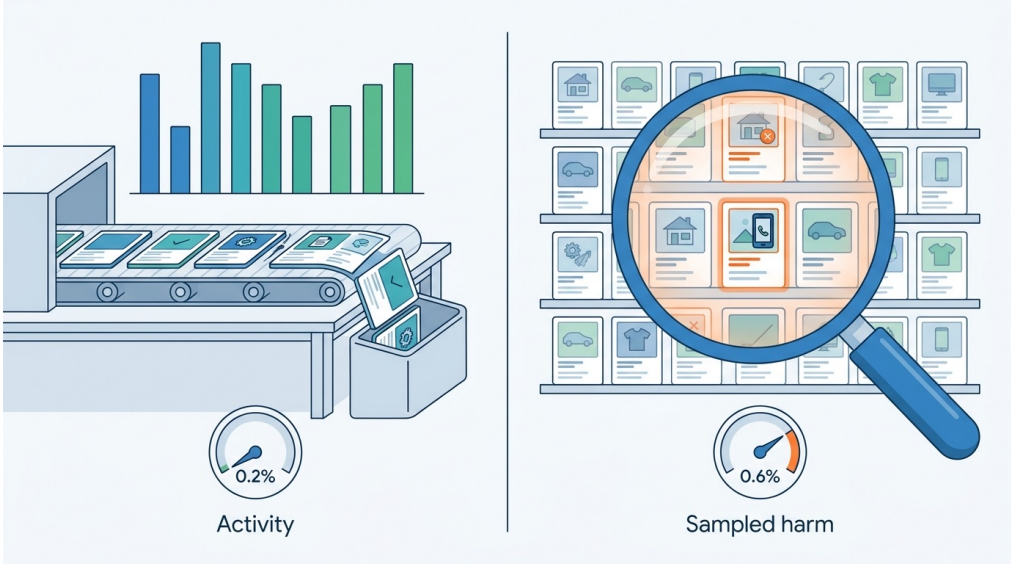


Figure 6: *Removal counts suggested 0.2% of Tradeyard NL's live listings violated a policy; a blind random sample of live listings found about 0.6%, mostly in harms the detectors could not see.*

These weekly queue numbers set the shared platform's design targets:

Quantity (Tradeyard, weekly)	2026 baseline	Design target, end of 2028
New listings	2.3 million	about 2.4 million
Automated decision rate	94%	96%
Listings to human review	about 138,000	about 96,000
Tradeyard moderators (in-house and outsourced)	158	about 120, with 15 more on sampling and appeals
Reviews per moderator per week	about 870	about 800
Median time to action on fraud reports	3 h 10 min	under 30 min
Appeal overturn rate on automated removals	not measured	measured per policy, alert above 10%

The design did not aim to remove moderators. It moved about **15 of them from grey-zone queues to sampling and appeals**, the two activities that produce the labels and the fairness evidence. Raising automation from 94% to 96% at constant precision was expected to come mostly from three sources: re-routing queues that almost always ended in the same outcome to rules, an OCR detector for contact details in images, and actor-level decisions that handle a ring as one case.

In Practice: What Is Publicly Documented

Classifieds companies publish very little about their T&S architecture. What is public is enough to anchor the reference design, but not to describe any company's internals.

- **Shared moderation in a portfolio**. Adevinta's 2023 annual report, published in April 2024, says the central technology team's focus had historically been "on consolidating enabling capabilities such as cloud and data, and overarching elements such as chat and moderation" (Adevinta Annual Report 2023). Moderation was among the first capabilities a multi-country group chose to share. The report does not describe the architecture, and Adevinta's operating model changed after its 2024 take-private.
- **Shared moderation models**. In a March 2023 engineering post, Adevinta described a "Text in Image" OCR service run by a central team across its marketplaces. It detects insults, hidden messages, phone numbers, e-mail addresses and URLs inside ad images. The post reports volume growing from more than 100 million to 330 million requests a month (Adevinta Tech Blog, 2023). A shared detector with one API contract and many tenants is exactly the kind of component a group platform should own.
- **Real-time chat moderation**. OLX's January 2026 post describes real-time moderation of chat messages, a generative-AI image check that blocks suspicious images before delivery, and account bans "as they strike" (OLX Group, 2026). It gives no architecture and no human-review detail.
- **Rules as code, deployed in minutes**. Meta's 2015 post on Sigma describes a rule engine that evaluates policies on every interaction at more than a million requests per second, with policies written in Haskell and hot-swapped into production continuously (Engineering at Meta, 2015). The system is more than ten years old and has surely changed since, but the principle, that **a new rule should reach production in minutes, not with the next monolith deploy**, is exactly what Tradeyard Belgium lacked on 7 March.
- **Rule operations**. Stripe's Radar documentation describes backtesting rules on the past six months of data before activation, a 0% traffic allocation that runs a rule in shadow mode and reports what would have happened, gradual rollout, and an audit trail of who changed each rule (Stripe Radar docs, n.d.). Payments are not classifieds, but rule lifecycle management transfers unchanged.

INFERENCE, NOT DOCUMENTED: Public job postings for Adevinta-group companies in 2025–2026 mention a global fraud detection and moderation component deployed in several countries. That suggests a shared T&S platform serving multiple country marketplaces, but job postings describe intent, not production design. This book does not treat it as a documented architecture.

One Platform for Four Brands: What to Share

Veldmark's principle for Common Ground is **share platforms, not products**. Trust and safety tests it hard, because some of T&S is pure machinery and some of it is local law, local fraud patterns and local language.

Component	Share across the group?	Reasoning
Decision service and decision record	**Share**	One schema for decisions makes appeals, DSA output and metrics uniform
Enforcement events and propagation monitor	**Share**	Rides on the group event backbone; every brand has the same consumer problem
Case management and reviewer tooling	**Share**, with per-brand queues and permissions	Tooling is expensive to build well; reviewers can move between brands
Evidence and audit store	**Share**, partitioned by brand and country	One retention engine, one place for regulators' and police requests
Statement-of-reasons service	**Share**	Same EU law, same database, same schema
Block lists and hash lists (phones, IBANs, devices, image hashes)	**Share where lawful**, with documented purpose and retention	Fraud rings cross brands; the Motorvane phone number is the proof
Rules engine runtime and rule lifecycle (backtest, shadow, rollout, audit)	**Share**	Machinery; rules themselves are per tenant
Policy registry	**Share the registry, localize the content**	Group baseline policies plus country and brand overlays
Image, OCR and text models	**Share base models**, fine-tune or threshold per brand	Language and category mix differ; the infrastructure need not

Component	Share across the group?	Reasoning
Vertical detectors (fake rentals, VIN reuse, discriminatory housing language)	**Local or vertical**	Only meaningful in one vertical
Thresholds and risk appetite	**Local**, within group guardrails	A bike marketplace and a rental portal tolerate different exposure
Reviewers and their language skills	**Local pools**, shared tooling	Policy judgement needs language and market context

The table hides one hard constraint: **sharing signals across brands is a data protection question before it is an engineering one**. Lucía Ortega, Veldmark's Data Protection Officer, approved a shared block list of hashed phone numbers, e-mail addresses, IBANs and device identifiers for accounts terminated for fraud, with a documented fraud-prevention purpose (the GDPR recognizes fraud prevention as a legitimate interest), a two-year retention limit and a data protection impact assessment. She did not approve sharing raw message content or behavioral histories between brands. The architecture enforces that line: the shared store holds hashes and decision references, and per-brand stores hold content.

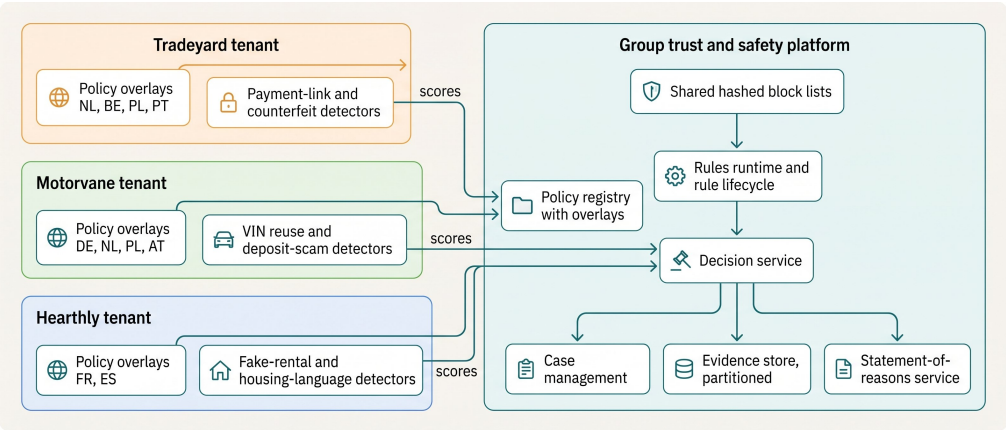


Figure 7: *The tenancy split adopted for Common Ground. The machinery is shared; policy overlays, vertical detectors and thresholds belong to each brand's tenant. Keybridge, a SaaS tool that does not publish listings to consumers itself, is not a tenant.*

Keybridge's absence is deliberate. It sends agencies' listings to many portals, including Hearthly's competitors, and Veldmark committed in March 2027 not to use Keybridge

agency data for Hearthly's advantage. A moderation decision on a Keybridge-originated listing at Hearthly is Hearthly's decision about Hearthly's copy. It is not shared back into Keybridge, and Keybridge does not consume the block lists. The After the Deal: Consolidating Platforms After Acquisitions chapter discusses why neutral SaaS businesses inside a marketplace group need explicit data boundaries.

Trade-offs

Where to build trust and safety

Option	Strengths	Weaknesses	Right when
Per-brand stacks	Local fit; no coordination; fast local decisions	Signals do not cross brands; tooling built four times; DSA output inconsistent	Brands are genuinely unrelated, or the group expects to sell them soon
Shared platform, local policy (Veldmark's choice)	Shared machinery and evidence; signals cross brands where lawful; one regulatory output	Platform team becomes a bottleneck if overlays are hard; migration cost	A group with several brands under the same law and overlapping fraud
Fully central T&S (one policy, one team, one queue)	Consistency; scale economies in review	Local law and language suffer; brands lose control of their customer experience	Brands share one product and one market
Vendor platform (T&S software and outsourced review)	Speed to start; mature tooling; elastic review capacity	Policy knowledge sits with the vendor; signals and labels may not be portable; data protection reviews per vendor	Early-stage marketplaces, or as the review workforce under a group-owned decision record

A vendor can supply review capacity, but not the records the platform learns from.

Outsourced review is normal in the industry, and Veldmark keeps its two review vendors. What it does not outsource is the **decision record**, **the policy registry and the labels**. A platform whose labels live in a vendor's system cannot retrain its own models or switch vendors without losing its history.

Where to put human judgement

Choice	Benefit	Cost
More pre-moderation	Less exposure to harm	Slower publication, which costs liquidity, especially for private sellers
More automation	Speed, consistency, lower cost per decision	False positives at scale; needs sampling and appeals to stay honest
LLMs in the grey zone	Fast policy iteration; better explanations	Cost, latency, adversarial prompts; cannot decide appeals
More sampling	Honest prevalence and recall	Reviewer time spent on mostly clean items

Moderation costs liquidity. Every minute a listing waits is a minute it cannot be found, and the Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics chapter shows why the first hours of a listing's life matter most. The answer is not a single mode but **risk tiers with explicit time budgets per tier**, reviewed with the marketplace general managers who carry the liquidity targets.

Vertical Differences

Dimension	General classifieds	Real estate	Automotive	Jobs
Main harms	Payment-link phishing, advance-payment scams, counterfeit, prohibited items	Fake rentals with stolen photos, deposit fraud, discriminatory ads	Deposit scams, cloned listings, VIN reuse, mileage misstatement	Fake job offers, money-mule recruitment, data harvesting
Supply mix	Mostly private sellers, high volume	Mostly agencies via feeds	Mostly dealers via feeds	Employers and agencies
Natural control point	The listing and the message	The lister's identity (agency checks)	The dealer contract and the vehicle identity	The employer's identity
Pre-moderation tolerance	Low: sellers expect instant publication	Medium: agency feeds tolerate minutes	Medium	Medium
Vertical-specific policy	Product safety, recalls	Energy-class disclosure, fair housing rules	Vehicle disclosure, odometer	Pay transparency, discrimination
Typical detector	Brand in URL, keyword, image OCR	Duplicate and stolen-photo search, rent anomaly	VIN decoding, price vs. market value	Contact-detail extraction, salary anomaly

The table explains why a group platform must allow **vertical detectors and vertical policies** as first-class tenant content. Hearthly's most important detector, stolen-photo matching against existing listings, is close to irrelevant for Tradeyard's second-hand clothes. Motorvane's most important control is not a detector at all. It is dealer verification at onboarding, which Who Is on the Other Side? Identity and Verification covers.

Failure Modes

1. **Moderation only at creation**. Edits, price changes and new photos are not re-scored, so scammers publish a clean listing and change it afterwards. The Which? test is the public proof.
2. **Several deciders**. The insertion service, the ranking service and the messaging service each enforce their own idea of "suspicious". Their decisions contradict each other, and none of them emits a statement of reasons.
3. **Decided but not enforced**. The decision is recorded, but alerts, recommendations, SEO pages or partner feeds keep the listing. Nobody measures propagation, so nobody notices.
4. **Policies in code**. A rule's intent lives only in a regular expression in a service written six years ago. When law changes, nobody can find all the places to change.
5. **Rules that deploy with the monolith**. A rule written at 20:30 on a Sunday reaches the second country on Wednesday. Fraud adapts in hours.
6. **No random sample**. Precision is estimated from appeals and recall not at all. A model that misses a new scam pattern looks better every week.
7. **Appeals as support tickets**. Complaints are not linked to decisions, reversals do not restore ranking, and overturns are never fed back to model owners.
8. **Labels owned by the vendor**. The review vendor's tool holds the history. Switching vendors, or training a new model, starts from zero.
9. **Silent demotion**. Ranking pushes suspected scams down without a decision record, which is both a DSA problem and an unexplainable experience for sellers who are wrongly affected.
10. **Badges that outlive the account's integrity**. A verified badge stays on an account after it is taken over, and makes the takeover more effective. Scams, Fakes and Account Takeovers: Fraud Prevention for Listings covers the detection side.

The Chief Architect's View

Trust and safety is one of the few capabilities where **sharing pays from the first year**, because fraud rings cross brands and the EU legal regime is the same for all of them. It is also one where centralizing too much destroys value, because policy judgement depends on language, market and vertical. The decisions I would make, and who should make them:

Decision	Who decides	What to watch
The decision record schema and the rule that the decision service is the only writer	Chief Architect, with the T&S lead and the DPO	Any service that restricts content without a decision record
The enforcement event contract and propagation targets per consumer	Chief Architect and the event platform owner	Propagation time per consumer, p95 and worst case
Group baseline policies and the overlay mechanism	Group T&S lead, legal, country policy owners	Number of overlays per policy; policies with no detector or no reviewer guideline
What signals may cross brands	DPO, with the T&S lead	Purpose, retention and access logs for shared lists
Risk tiers and time budgets per tier	Marketplace GMs with the T&S lead	Publication time vs. prevalence per tier
Automation thresholds	Policy owner and model owner jointly	Overturn rate and sampled precision per policy
Review vendors and their scope	T&S lead and procurement	Label portability; vendor access to evidence

Three long-term trends belong on the architecture roadmap.

- **T&S is a consumer of everything**. It needs listing events, message events, login events, payment events and media events. A group without a reliable event backbone cannot build a shared T&S platform, which is why Veldmark's backbone (July 2027) came before its shared T&S platform (November 2027).
- **Regulation will keep adding outputs, not inputs**. The DSA, the UK Online Safety Act and successors mostly ask for evidence, notices and reports derived from decisions. A platform built around a clean decision record absorbs new regulation as new views. One built around queues absorbs it as new projects.
- **Adversaries learn the platform**. Detection thresholds, rule logic and model features must not leak through rejection messages. A statement of reasons must give the ground and the facts. It does not need to disclose the threshold.

The metric I would put in front of the board is not removals. It is **sampled prevalence of fraudulent listings, by brand, with its trend**, next to time to action on fraud reports. Those two numbers say whether buyers can trust what they see, and how long they are exposed when they cannot.

Questions to Ask

1. Which component writes moderation decisions, and how many other services restrict content without going through it?
2. When a listing is removed, which consumers apply the removal, how do we know they did, and what is the p95 time until the last one acknowledges?
3. Are edits, price changes, photo changes and contact-detail changes re-scored as new submissions?
4. Where are our policies written down, how are they versioned, and can every decision cite the policy version it applied?
5. What is our prevalence of policy-violating listings from a blind random sample, per category, and how has it moved this quarter?
6. What is the appeal overturn rate per policy and per detector, and who receives the reversals?
7. How long does a new rule take to reach production in every country, and can it run in shadow mode first?
8. Which signals do we share across brands, under what legal basis and retention, and who audits access?
9. Does ranking demote listings for trust reasons, and if so, does each demotion produce a decision record and a statement of reasons?
10. If we changed our review vendor tomorrow, would we keep our labels, guidelines and case history?

From Platform to Adversary

This chapter built the machinery: policies, detection, one decision service, enforcement that reaches every consumer, cases, appeals and metrics. It deliberately treated detection as a slot that detectors plug into. The most demanding detectors are those aimed at people who study the platform and adapt to it within days: phishing crews, fake-rental rings and account-takeover operations. Scams, Fakes and Account Takeovers: Fraud Prevention for Listings takes up that adversary, with risk scoring, the signals that expose organized fraud, step-up friction, and Veldmark's encounter with a fake-rental ring in Barcelona.

Sources and Evidence

Company engineering and investor sources

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Standards, regulation and documentation

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Research and practitioner sources

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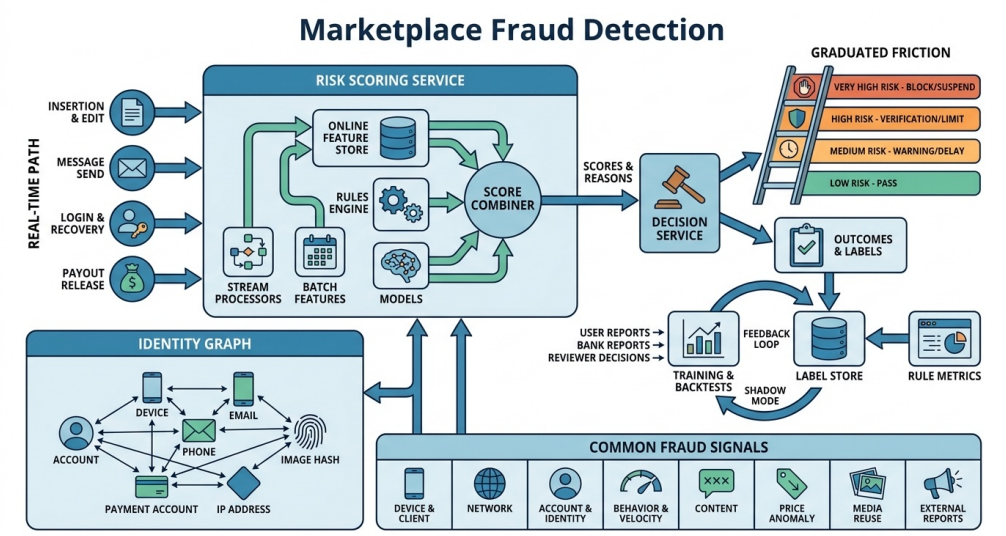
- Llama Guard: LLM-based input-output safeguard for human-AI conversations — Inan et al., arXiv, 2023. Policy taxonomy supplied as a prompt.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

25. Scams, Fakes and Account Takeovers: Fraud Prevention for Listings



IN THIS CHAPTER, YOU WILL: Learn the fraud types that classifieds platforms face and the signals that expose them. You will see how a risk scoring service combines real-time and batch features, rules and models, and how step-up friction, messaging controls and feedback loops turn a score into prevention, using a fake-rental ring in Barcelona.

WHY THE BUSINESS CARES: Most classifieds fraud is a bank transfer the platform never sees and cannot reverse. The victim blames the brand anyway. Fraud losses do not appear on the marketplace's profit and loss statement, but they show up in lost buyers, support costs, press coverage and, increasingly, in demands from banks and regulators that platforms carry part of the cost.

WHY ARCHITECTS SHOULD CARE: Fraud prevention is a real-time decision system with adversarial inputs, delayed and biased labels, and a

latency budget inside the insertion, messaging and login paths. The design of features, identity linking and feedback loops decides whether the platform learns faster than the fraudsters do.

KEY POINTS:

- **Classifieds fraud happens before the payment, so prevention has to happen there too.** In lead-generation marketplaces the money usually moves off the platform by bank transfer. The controls that work sit before it: who can list, what can be said in messages, which links get through, and steering buyers towards a protected payment.
- **Score the actor, not only the item.** A fake rental looks plausible one listing at a time. The fraud becomes visible when listings, accounts, devices, payment accounts and photos are linked into one graph. Signals that are weak on their own (a stolen photo, a low price, a new account) become decisive together. The risk scoring service exists to combine them within the latency of a user action.
- **Answer risk with graduated friction, and learn from every outcome.** Blocking everything suspicious punishes good users. Step-up friction (verification, delays, limits) costs genuine users little and fraudsters a lot. Thresholds should come from an explicit cost model, and labels from victims, banks, reviewers and appeals have to reach the models within days, because organized fraud adapts within days.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Fast Mail for Fraudsters** — payment-link phishing and deposit requests close in the conversation, so messaging without link, domain and pattern controls becomes the platform's main fraud channel.
- **Set-and-Forget AI** — a fraud model with stale labels and no shadow testing goes on approving the scam pattern that replaced the one it was trained on.
- **Seeing Double** — when stolen photos are treated as harmless duplicates, the deduplication pipeline that should expose a fake listing ends up hiding it.

A listing marketplace is a machine for getting strangers to trust each other enough to meet, call or pay. Fraud exploits exactly that. The fraudster borrows the trust the platform has built, uses it to start a conversation, and then moves the victim somewhere the platform cannot see: a bank transfer, a fake payment page, a messaging app. **Most of the loss happens after the victim has left the platform, but the platform is where the scam was set up.**

This chapter is about catching the setup. It owns three terms. A **fraud signal** is an observable property of an item, actor, session or relationship that changes the probability of fraud. A **risk score** is a calibrated estimate of that probability for a specific action, such as publishing a listing, sending a message, logging in or releasing a payout. **Step-up friction** is an additional check imposed in proportion to risk, which genuine users pass easily and fraudsters do not.

This chapter supplies the scores that Part V's shared trust platform acts on. Trust and Safety as a Platform built the decision service that turns a score into enforcement and appeals; fraud scoring is how that service catches fraudsters who exploit the speed Part IV gave new listings. The strongest frictions it recommends are identity checks, which Who Is on the Other Side? Identity and Verification designs next. Payment-side fraud in transactional flows (chargebacks, payout holds) is covered in Beyond the Contact: Payments, Shipping and Buyer Protection. The running example is Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build).

Setting the Scene at Veldmark: The Barcelona Fake-Rental Ring

On Thursday **1 July 2027**, a new account on Hearthly Spain published a rental listing for a renovated two-bedroom flat in Barcelona's Eixample district at about 35% below comparable rents. Hearthly Spain was the former Casavista, acquired in 2025 and rebranded in 2026. Over the next twelve days, **26 accounts** published **180** such listings. Every listing used photos copied from real listings, most of them still live on Hearthly and published by real agencies. Each "landlord" was abroad, could not arrange a viewing and asked for a deposit by bank transfer to secure the flat.

By the time the listings came down on Tuesday **13 July**, **63 people** had reported losses totaling about **€71,000**, roughly €1,130 each, in most cases one month's advertised rent paid as a "deposit". The listings had received about 4,700 contacts.

The detectors did not miss the ring, and that is the uncomfortable part. The ledger records that it was detected through **perceptual-hash matches** against existing listings plus **price anomalies**. A perceptual hash is a fingerprint of an image that survives resizing and recompression (Photos Sell: Media Pipelines for Listings owns the capability). Hearthly Spain's hash index matched the first stolen photos on 1 July, within minutes of upload. Its rent-anomaly check flagged most of the listings as well. The signals existed from the first day. What failed was everything between the signals and a decision.

- **Each signal went to a different place**. The photo match created a "possible duplicate" item in the general moderation queue. The rent anomaly created a separate, low-priority "price check" item. No component combined them.

- **Reviewers saw listings, not actors.** A reviewer looking at one listing whose photos matched an agency's listing saw something that looked like a second agency marketing the same flat, which is common in real estate (One Property, Three Listings: Duplicates and Entity Resolution). Eleven of the first photo-match items were closed as legitimate duplicates.
- **Nobody linked the accounts.** The 26 accounts used five devices and four Spanish bank accounts, which turned out to belong to recruited money mules. Hearthly Spain's stack had no identity graph, and none of the accounts had been asked to prove more than an e-mail address (Who Is on the Other Side? Identity and Verification). Each account looked new and unremarkable.
- **The first victim report, on Friday 9 July, was routed to support** as a dispute between a tenant and a landlord.

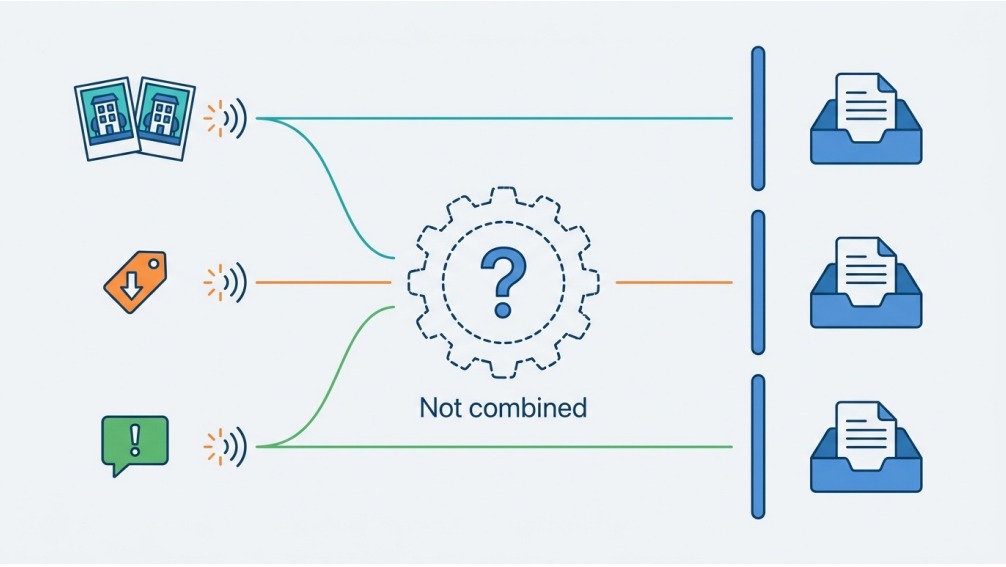


Figure 1: *In Barcelona the photo matches, rent anomalies and the first victim report all existed early, but each went to a different place and no component combined them into one decision.*

On Monday 12 July, a Hearthly analyst in Barcelona noticed that three victim reports named listings whose photos had been flagged. She queried the hash matches, the price flags and the account metadata by hand, and linked all 26 accounts in an afternoon. A combined rule went live on Tuesday morning, and by 16:40 that day all 180 listings had been removed and the accounts suspended. Hearthly Spain was not yet on Veldmark's shared trust-and-safety platform, which went live for Tradeyard Netherlands and Belgium in November 2027.

Ingrid Halvorsen, the Group Chief Architect, asked Aisha Rahman, the Group Head of Trust & Safety, and Priya Raman, Head of Data & ML, one question in the review that

followed: "**We had the signals on day one. What would it have taken to act on day one?**" The rest of this chapter is the answer: a risk scoring service that scores actors as well as items, an identity graph, a combination of rules and models, graduated friction, and feedback loops measured in hours.

The Threat Model: What Classifieds Fraud Looks Like

Why classifieds differ

Most classifieds platforms are lead-generation businesses: buyer and seller meet on the platform, but the money often moves elsewhere (Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy covers the business models). The platform sees the listing and the conversation. It often does not see the payment and cannot reverse it. The dominant fraud is therefore **authorized push payment (APP) fraud**: the victim willingly sends money, usually by bank transfer, to someone who is not who they claim to be.

Purchase scams, the typical classifieds fraud, are the most common kind of APP fraud in UK Finance's data, the most detailed public series. Its 2026 annual fraud report, covering 2025, counts **175,809 purchase-scam cases** in the UK with losses of **£118.1 million**. Purchase scams were **71% of all APP scam cases** and had the lowest average value of the APP types, **£671**. It describes the typical pattern: although "many online platforms offer secure payment options, the criminal will persuade their victim to pay via a bank transfer instead". **78% of purchase-scam losses were returned to victims**, which reflects the UK's mandatory reimbursement regime from October 2024 and explains why banks now ask the government to "mandate seller verification and secure payment protections on online marketplaces" (UK Finance, 2026). As of October 2026 that is a lobbying position, not law.

The architectural consequence is that a classifieds platform's controls are mostly **pre-payment**: who can list, what can be listed, what can be said and linked in messages, and how strongly buyers are steered towards a protected in-platform payment where one exists.

A typology

Fraud type	Mechanism	Strongest signals	Main control point
Advance payment / non-delivery	Fake seller asks for a bank transfer, gift cards or a "friends and family" payment, then disappears	Below-market price, stock photos, new account, payment terms in text	Listing scoring; message scanning; protected payment
Payment-link phishing	Fake *buyer* sends the *seller* a link or QR code to a counterfeit "payment received" or courier page that harvests card or bank data	New domain, platform brand in a foreign URL, QR code in chat	Message gate: link and domain controls
Fake rentals	Copied photos of a real property, low rent, landlord "abroad", deposit before viewing	Stolen photos, rent anomaly, no viewing, unregistered lister	Listing scoring with image reuse; lister verification
Vehicle deposit scams	A car priced to sell fast; deposit to "hold" it, or delivery through a fake escrow or shipping firm	Price vs. market value, VIN or plate reused across listings, external escrow domains in text	Listing scoring; dealer verification; message scanning
Triangulation	Fraudster relists a real seller's item; the victim pays the real seller, believing it is for the fraudster's item; two victims dispute each other	Cloned listing with different contact data	Duplicate detection across sellers

Fraud type	Mechanism	Strongest signals	Main control point
Overpayment	Fake buyer "overpays" with a fake transfer screenshot and asks for the difference back	Payment-proof images, refund keywords	Message image checks; warnings
Account takeover (ATO)	Credentials phished or stuffed; an aged, reputable account then posts scam listings	Session change: new device, network, client; sudden change of behavior	Login risk; step-up authentication
Fake sellers and fake shops	Batches of new accounts or pseudo-dealers with stolen identity or company data	Shared devices, IPs, e-mail patterns, register mismatch	Onboarding risk; identity graph; business verification
Counterfeit and prohibited goods	Branded goods at implausible prices; weapons, drugs, recalled products	Brand × price anomaly, rights-holder notices, image classifiers	Listing scoring (Trust and Safety as a Platform)

Three public observations sharpen the table.

- **Images are a large share of chat fraud**. In a January 2026 post, OLX said fraudulent images make up about **30% of fraud cases** on its Polish marketplace. Its generative-AI chat image system blocks suspicious images in real time and detects fraudulent QR codes in chat with **80% precision**. It is live in Poland, Portugal, Bulgaria, Romania and other markets (OLX Group, 2026).
- **Scammers edit after approval**. In Which?'s December 2024 undercover test, Gumtree blocked a first scam-style ad, but edited versions passed review "in minutes" (Which?, 2024).
- **Fraud is industrial**. Meta reported in November 2024 that it had removed more than two million accounts linked to scam centers in Southeast Asia and elsewhere (Meta, 2024). Classifieds fraud is increasingly run by organized groups that test a platform's defenses systematically and change scripts within days. A rules engine updated twice a week loses that race.

Fraud Signals

A fraud signal is useful only if it is available at decision time, hard for the fraudster to change cheaply, and not too costly for genuine users. The families below are the ones that matter for listings.

Signal family	Examples	Available when	Cost to the fraudster to evade
Device and client	Device identifier, emulator or rooted device, client integrity attestation, app version	Login, insertion, message send	Medium: device farms and emulator kits exist
Network	IP reputation, data-center or proxy IP, IP country vs. market, IP velocity	Every request	Low to medium: residential proxies are cheap
Account and identity	Account age, verification level, e-mail and phone age and type (virtual numbers), recent changes of contact details or password	Every action	Medium: aged accounts are bought, or taken over
Behavior and velocity	Listings per hour, first contacts per hour, time from sign-up to first listing, copy-paste patterns, typing cadence	Rolling windows	Medium: slowing down costs the fraudster throughput

Signal family	Examples	Available when	Cost to the fraudster to evade
Content	Payment terms, off-platform contact details, urgency phrases, text in images (OCR), language mismatch with market	Insertion, edit, message	Low per item, but every evasion is a new pattern to learn
Price	Price vs. market estimate for the category, model, location and condition	Insertion, edit	High: a fake bargain needs a low price to work
Media reuse	Perceptual-hash or embedding match to another seller's photos, to stock photos, to known scam images	After upload, seconds	Medium: cropping and flipping defeat simple hashes, but embeddings catch more
Graph	Shared devices, phones, IBANs, IP ranges, images and text across accounts; distance to known fraud	Near real time if maintained incrementally	High: rings must share infrastructure to be economical
External	Bank fraud reports, phishing-domain feeds, domain registration age, trusted-flagger notices, chargebacks	Minutes to weeks	Not controllable by the fraudster

Two signals deserve special attention because they are hard to fake. **Price** is the bait: a fake bargain has to be a bargain. A valuation or market-price model, built for buyers, doubles as a fraud feature. Motorvane's market-value model, which rates the Essen

Volkswagen Golf at €22,800 against an asking price of €21,900, can flag a 2021 Golf offered at €12,000 just as easily (Models in Production: ML Platforms and Use Cases owns these models). **The graph** is the ring's weak point: organized fraud has to reuse devices, phone numbers, bank accounts and photos to stay profitable, and every reuse is a link.

The Barcelona ring illustrates both. Its prices were about 35% below market, which required structured surface, rooms and district to compute (Attributes Without Chaos: Schemas, Dictionaries and Validation). Its 26 accounts shared five devices and four IBANs. No single account was alarming; the cluster was unmistakable.

Duplicate photos: one detector, two policies

Image reuse is the strongest single signal against fake rentals and cloned listings, and the best public classifieds design for it is still a 2019 conference talk. At PyCon DE 2019, Alexey Grigorev described OLX's duplicate-listing detection for fraud. Candidates were generated by category, city, seller, IP or device and image hashes. A pairwise classifier, trained on human-labeled pairs, compared price difference, location distance, title similarity and image similarity. Images were matched with hashes and then neural embeddings indexed for approximate search in Elasticsearch, at a target of about ten million listings a day. The talk also noted that cropping, rotation and flipping defeat simple hashes (PyCon DE, 2019). The system is seven years old and certainly replaced, but the shape is sound.

A platform needs both a hash for cheap exact and near-exact matches and an embedding for crops and edits. Meta's open-source PDQ perceptual hash documents its own limits: a suggested match threshold, a quality score below which hashes should be discarded, and no guaranteed invariance to rotation or flips (Meta ThreatExchange, n.d.).

The design lesson from Barcelona is about the policy, not the detector. **"Same photos" means opposite things depending on who posted them.** Same seller, same photos is a repost: a spam or ranking problem. Different seller, same photos is either a legitimate multi-agency listing or a stolen one, and the evidence that separates them is the relationship between the sellers: same agency network, same software vendor feed, verified agency accounts, or a new private account with a foreign phone number. The One Property, Three Listings: Duplicates and Entity Resolution chapter makes the same point from the clustering side. A photo match between unrelated sellers must go to fraud scoring with the seller context attached, never to a deduplication queue on its own.

Architecture: The Risk Scoring Service

A **risk scoring service** answers one question for many callers: *how risky is this action, by this actor, right now?* Callers include the insertion and edit flows, the message send

path, login and account recovery, contact-detail reveal, and payout release where the marketplace handles payments. The service does not decide the enforcement action; it returns scores, reasons and recommended friction to the decision service described in Trust and Safety as a Platform.

The design below is a **reference-architecture inference**, built from public adjacent systems (Meta's Sigma rule engine, Stripe Radar's documented rule operations, Uber's 2017 description of a shared feature store, Mercari's 2026 risk-based authentication) and from the evidence above. No classifieds company has published a fraud-scoring architecture in this form.

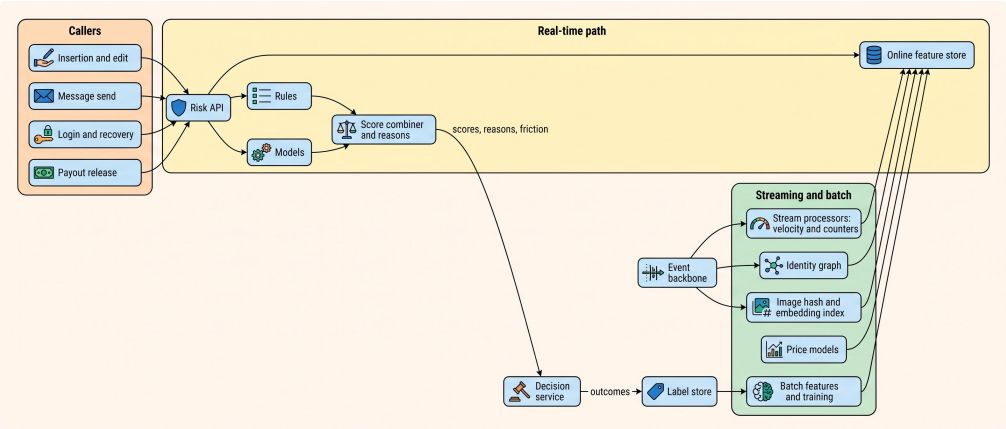


Figure 2: *The risk scoring service. The real-time path reads precomputed features and evaluates rules and models within the caller's latency budget; streaming and batch jobs keep the features, the identity graph and the image index current; decisions and their outcomes return as labels.*

Real-time and batch: what runs where

The real-time path has a budget measured in tens of milliseconds, because it sits inside a user's action. It cannot compute a graph traversal or an image embedding on the request. It **reads** features that streaming and batch jobs have already computed: counters ("first contacts sent by this device in the last hour"), graph features ("distance from this account to the nearest account terminated for fraud"), image matches ("photos of this listing match photos of another seller's live listing") and price features ("asking price relative to estimate").

Work	Where it runs	Latency	Example
Block-list lookups, simple rules	Real-time path	Milliseconds	Phone number on the shared block list; URL domain registered today
Model inference on precomputed features	Real-time path	Tens of milliseconds	Listing risk, message risk, login risk
Velocity counters	Stream processing	Seconds	First contacts per device per hour
Image hashes and embedding matches	Asynchronous after upload	Seconds	Photo match to another seller's listing
Identity graph updates	Stream processing, incremental	Seconds to minutes	New account shares a device with a suspended one
Graph-wide community detection	Batch	Hours	Rings of accounts sharing infrastructure
Model training and backtesting	Batch	Daily to weekly	Retraining on last week's labels

A **feature store**, a shared system that serves the same feature definitions online for scoring and offline for training, is justified once real-time behavioral features drive decisions. Uber's 2017 description of Michelangelo is the standard reference: precomputed online features, the same pipelines for training and serving, and predictions logged and joined with outcomes (Uber Engineering, 2017). No classifieds company has documented one publicly, and many classifieds models run on attributes the listing record already holds. Fraud is the use case that changes the calculation, because its best features are behavioral, cross-entity and time-sensitive. The essential property is **point-in-time correctness**: training must use the feature values as they were when the decision was taken, not as they are now, or the model learns from information it would not have had. Models in Production: ML Platforms and Use Cases owns the feature store as a platform component.

Rules and models together

Rules and models are not alternatives. They do different jobs.

	Rules	Models
Best at	Known patterns, hard policy, fast response to a new script	Combining many weak signals, generalizing to variants
Time to deploy	Minutes, if the rule platform allows it	Days to weeks (labels, training, validation)
Explainability	Direct: "domain registered today"	Needs reason codes from feature attributions
Failure mode	Rot: rules accumulate, overlap and block good users	Drift: the scam changes and the model keeps scoring the old pattern
Adversarial behavior	Easy to probe and evade once discovered	Harder to probe, but learnable from rejections

The operating model that works is the one Stripe documents for Radar: rules written as conditions on attributes, including model scores, **backtested** against historical data before activation, run in **shadow mode** at 0% traffic to report what they would have done, rolled out gradually, and recorded in an audit trail (Stripe Radar docs, n.d.). Meta's 2015 description of Sigma adds the deployment discipline: policies as code, hot-swapped into production continuously (Engineering at Meta, 2015).

The Barcelona combined rule shows the division of labor. It was written in an afternoon:

> photo match to a listing of an **unrelated** seller **and** rent at least 25% below the district estimate **and** account younger than 30 days **and** lister not a verified agency ⟶ hold all the account's listings and open one case for the account cluster.

That rule stopped the ring. A model trained on the ring's 180 listings, and on the 2,000 or so fake-rental listings Hearthly Spain had removed over the previous year, replaced it six weeks later, because the next ring would use 20% discounts and slightly older accounts. **Rules are the fast response**; **models are the durable one**. A platform that only has one of the two either reacts too slowly or overfits to last month.

Scoring actors, not just items

The deepest design choice in the service is the **unit of scoring**. A listing score answers "is this listing fraudulent?" An actor score answers "is this account, or this cluster of accounts, operated by a fraudster?" Organized fraud is an actor problem. A ring's individual listings are designed to pass item-level checks. Its infrastructure is not.

The **identity graph** links accounts through shared attributes: devices, phone numbers, e-mail patterns, payment accounts, IP ranges, image hashes, even near-identical listing text. Its main uses are to:

- **propagate known fraud**: an account two hops from a terminated fraud account starts with a higher risk score;
- **detect rings**: dense clusters of new accounts sharing infrastructure are suspicious even before any report; and
- **scope enforcement**: when one account is confirmed as fraud, the linked accounts are reviewed in the same case, so a ring is taken down at once rather than one listing at a time.

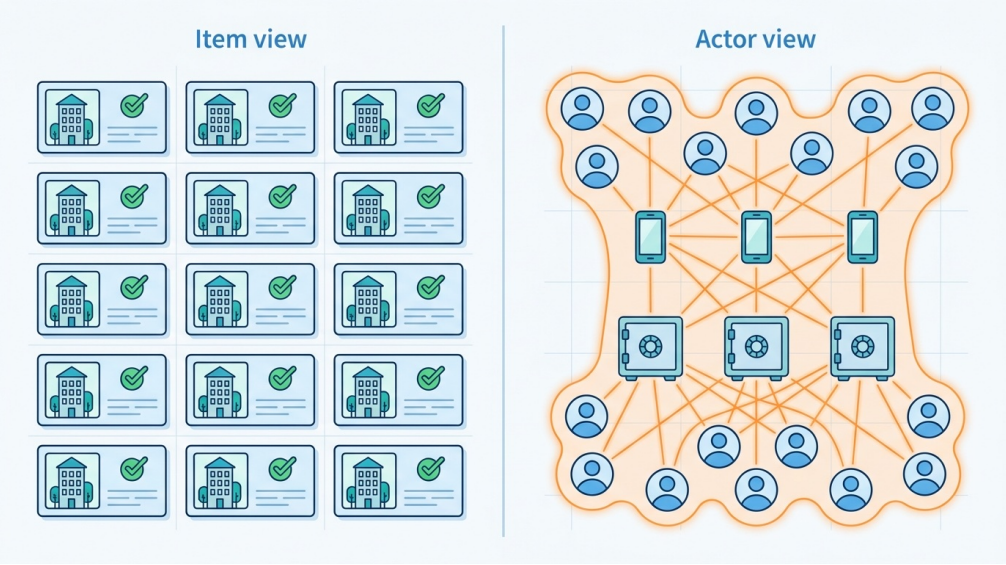


Figure 3: *A ring designed to pass item-level checks shows up in the actor view: the 26 Barcelona accounts shared five devices and four bank accounts, which an identity graph links at creation.*

The graph has two hard constraints. Shared attributes are not proof: families share devices, offices share IP addresses, agencies share photos legitimately. Graph features should raise scores and widen cases, not trigger terminations on their own. And the

graph is personal data. At Veldmark, the shared cross-brand block list holds only hashed identifiers of accounts terminated for fraud, with a two-year retention approved by Lucía Ortega, the Group Data Protection Officer, under a fraud-prevention legitimate interest. Each brand's graph of ordinary accounts stays within that brand.

Step-Up Friction

Blocking is a blunt answer to a probability. Most risky actions are taken by genuine users, and a platform that blocks every account with a score above 0.5 loses customers it needed. **Step-up friction** asks for more evidence instead: something a genuine user can provide quickly and a fraudster cannot, or can only at a cost that ruins the economics of the scam.

Airbnb described the approach publicly in 2018 as **targeted friction**. Machine-learning models score bookings for payment fraud. Instead of blocking above a threshold, Airbnb applies frictions such as micro-authorizations (two small charges the cardholder must identify), 3-D Secure authentication or billing-statement verification. The threshold is chosen by minimizing a loss that weighs the cost of genuine users abandoning the friction against the fraud losses it prevents, and the effect of each friction on good users is measured by A/B testing on low-risk users (InfoQ, 2018, summarizing an Airbnb engineering post). The system is eight years old; the method is the durable part.

A friction ladder for a listing marketplace:

Friction	Defeats	Cost to genuine users	Typical trigger
In-context warning ("never pay a deposit before a viewing")	Naive victims	Almost none	Payment keywords, off-platform request in a message
Delay: listing held for a few minutes or until asynchronous checks finish	Hit-and-run listings	Small	New account in a high-risk category
Limits: listings, first contacts or contact-detail reveals per day	Spam and mass outreach	Small for normal users	Velocity, new account
Phone verification by SMS	Cheap bulk accounts	Small; virtual numbers weaken it	Account risk in the grey band

Friction	Defeats	Cost to genuine users	Typical trigger
Strong authentication (passkey or second factor)	Phishing-based account takeover	Small once set up	Login from a new device; recovery
Identity or business verification (Who Is on the Other Side? Identity and Verification)	Most fake-seller farms	Medium	Rentals, high-value vehicles, payouts
Protected payment only	Advance-payment fraud	Medium: some buyers prefer cash	High-risk category and price band

Two public sources support the ladder's upper rungs. Mercari reported in July 2026 that where passkeys were required, "large-scale phishing financial damages have been effectively zero", and that its risk-based authentication scores session and account posture, using client integrity, network reputation and device signals (Mercari Engineering, 2026). The Ofcom-commissioned expert report of July 2026 recommends "creating delays between creating an account and reaching a large audience" and notes that verification friction "directly disrupts" repeat bad actors (Integrity Institute for Ofcom, 2026).

Choosing a threshold: a worked example

Tradeyard's shared T&S platform introduced a friction in December 2027: a new account that tries to send more than five first-contact messages within an hour must verify a phone number before sending more, if its message-risk score is above a threshold. The question was which threshold. Priya Raman's team ran a two-week shadow evaluation and estimated, per week, about **2,000 fraudulent new accounts** reaching the velocity trigger. Their working assumptions: 85% of fraud accounts fail or abandon the verification; 8% of genuine accounts abandon it; an unstopped fraud account causes about **€180** of expected harm (victim losses, support cost and lost buyers); a lost genuine account costs about **€25** in expected future contribution. All figures are illustrative.

Threshold	Accounts challenged per week	Fraud accounts challenged	Genuine accounts challenged	Fraud stopped	Genuine lost	Weekly cost
No friction	0	0	0	0	0	€360,000
Score ≥ 0.3	60,000	1,800	58,200	1,530	4,656	€201,000
Score ≥ 0.5	22,000	1,500	20,500	1,275	1,640	**€171,500**
Score ≥ 0.7	6,000	1,050	4,950	893	396	€209,160

The weekly cost is (2,000 − fraud stopped) × €180 + genuine lost × €25. The lowest cost is at 0.5, not at the most aggressive threshold and not at the most permissive. Three things matter more than the specific numbers. The **cost of a lost genuine user is a business input**, and the marketplace general manager, not the fraud team, should own it. The 8% abandonment rate is **measured by experiment**, not guessed, because friction design (one tap versus a form) changes it several-fold. And the threshold is **revisited when fraud adapts**: if fraudsters start buying verified phone numbers, the "fraud stopped" column shrinks and the optimum moves.

Fraud Controls in Messaging

Most classifieds scams close in the conversation, so the controls have to act before a message is delivered. Payment-link phishing, deposit requests, fake courier offers and moves to other messaging apps all happen there. From Interest to Contact: Messaging and Lead Management owns the conversation model and its storage; this section covers the fraud controls inside it.

Tradeyard shows the cost of scanning too late. Its users sent about 11 million messages a day in 2026, and NL and BE users filed about **41,000 reports of payment-link phishing** that year. On Sunday 7 March 2027, a fake Safe Deal payment link reached Femke, who was selling a €240 city bike in Rotterdam, because its domain had been registered that day and no rule looked at domain age. The account was suspended seven minutes after Femke reported it, and a new rule then withheld 37 similar messages to other sellers. Seven minutes was fast, but the message should never have been delivered.

A reference design for message-level fraud controls:

1. **Extract** URLs, QR payloads (decoded from images), phone numbers, e-mail addresses, IBANs and messaging-app handles from every message and image.

2. **Check links** against allow and deny lists, a domain-age service and a brand-impersonation check. A domain containing the platform's or its payment product's name that the platform does not own is the strongest phishing tell. Unknown links are rendered non-clickable or behind a warning page.
3. **Withhold before delivery** when the gate is confident: OLX's 2026 description of blocking suspicious chat images before they reach users is the public example.
4. **Warn in context** when it is not. OLX South Africa described in 2018 how payment keywords and attempts to move a conversation to e-mail or WhatsApp triggered safety tips to the counterparty and flagged the account (ITWeb, 2018). The system is long replaced; the warn-in-context pattern is not.
5. **Score the conversation, not only the message.** A sequence of "courier", "safe payment", "deposit" and a link is a pattern no single message shows.
6. **Mask or delay off-platform contact details** for new and unverified accounts, so the scam cannot leave the platform before the platform has had a chance to see it.

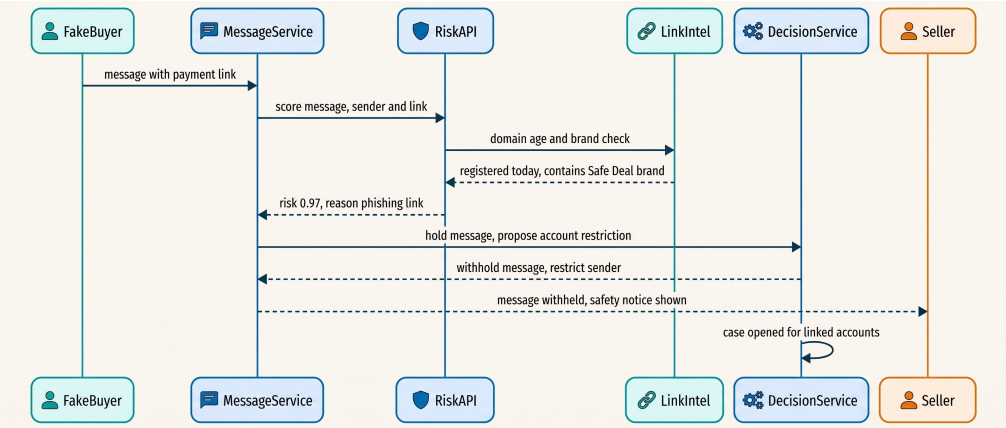


Figure 4: *The 7 March 2027 phishing message as it would be handled by the target design. The message is scored before delivery; the domain-age and brand check is decisive; the decision service withholds the message, restricts the sender and opens a case for linked accounts.*

Scanning private messages for fraud has a legal side. In the EU it engages the GDPR's purpose-limitation and minimization principles and the ePrivacy framework. The DSA also requires platforms to disclose their use of automated means in moderation. Veldmark documents fraud prevention as the purpose, runs a data protection impact assessment, says in its terms that automated tools scan messages for scams, and keeps scanned content only as long as the case needs it. Rules of the Marketplace: DSA, DAC7 and Vertical Regulation covers the legal analysis.

Feedback Loops and Labels

A fraud model is only as current as its labels, and fraud labels are **late**, **biased and incomplete**.

Label source	What it confirms	Delay	Bias
Reviewer decisions	Fraud or not, for queued items	Minutes to hours	Only covers what the system already flagged
User reports	Visible harm	Hours to days	Skews to harms victims notice; many victims never report
Victim loss reports	Confirmed loss	Days	Strongly underreported
Bank and police reports	Confirmed fraud, often with payment details	Weeks	Sparse; arrives after the model has moved on
Chargebacks (where payments run on-platform)	Payment fraud	Weeks to months	Only for on-platform payments
Appeal reversals	Confirmed false positives	Days to weeks	Undercounts: most wrongly blocked users do not appeal
Random-sample review	Unbiased prevalence and recall	Days	Small samples for rare fraud types

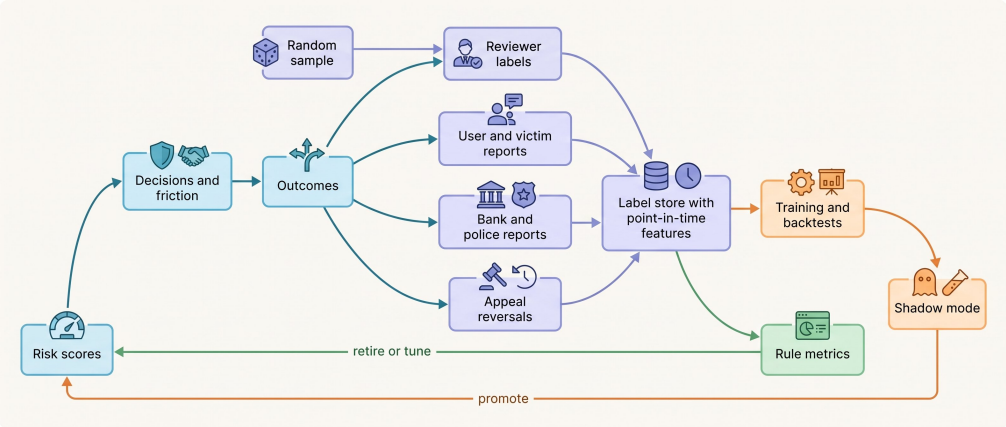


Figure 5: *The fraud feedback loop. Labels arrive from several sources with different delays and biases; the random sample corrects selection bias; new models and rules run in shadow mode before they score live traffic, and rule metrics retire rules that have rotted.*

Four practices keep the loop honest.

- **Correct selection bias with the random sample.** If labels only come from flagged items, the model never learns what it misses. A stratified random sample of live listings and delivered messages, reviewed blind, is the only unbiased source (Trust and Safety as a Platform covers prevalence measurement).
- **Treat late labels as first-class.** A bank report about a transfer three weeks ago must be joined to the listing, the account and the features as they were then. That requires stable identifiers and point-in-time feature logs.
- **Shadow before enforce.** Every new rule and model runs at 0% traffic first and reports what it would have done, as Stripe documents for Radar. The comparison against the current champion is the promotion test.
- **Monitor rules for rot.** Stripe's guidance on rule metrics gives a ready diagnostic: if a rule's blocks rise while confirmed fraud stays flat, the rule may be blocking good users. Every rule needs an owner and an expiry review date.

Adversarial adaptation

Fraudsters probe the gate. Which?'s testers changed devices, Wi-Fi networks and identities between attempts, and edited approved ads into scam ads. Professional fraud groups do the same at scale. Three design rules follow:

1. **Re-score on every material edit** (price, payment terms, contact details, photos), not only at creation.

2. **Do not leak thresholds**. A rejection message, or a statement of reasons under the DSA, must give the ground and the relevant facts. It need not reveal which signal tipped the score or by how much.
3. **Rate-limit retries per identity cluster**, not per account. A ring that is rejected and retries from 26 accounts is one actor learning the system.

Collaboration with Banks and Police

A classifieds platform sees the setup; the bank sees the money. Neither can stop APP fraud well alone.

- **Bank-side checks have limits**. Under the EU Instant Payments Regulation, payment service providers in the euro area have had to offer **Verification of Payee**, a check that the payee name matches the account, before a credit transfer since 9 October 2025 (McCann FitzGerald, 2025). It stops some impersonation, but not a fraudster who uses a money mule's real account in the mule's own name. In Barcelona, the four IBANs belonged to recruited mules, so a name check would have matched.
- **The exchange that matters is identifiers**. Banks can tell a platform which payee accounts received fraud proceeds. Platforms can tell banks which IBANs fraudsters asked victims to pay. Both are fraud signals for the other side. Exchanging them needs a legal basis, a secure channel and agreed formats.
- **Police need evidence packages**. A ring case with linked accounts, listings, messages, payment details and timelines is what an investigation needs. The evidence store from Trust and Safety as a Platform produces it, with a disclosure log.

After Barcelona, Veldmark set up a bank liaison function for Hearthly Spain and Tradeyard, starting with two Spanish and three Dutch banks. It receives reported mule IBANs and adds them, hashed, to the shared block list, so that a message or listing asking for payment to one of them is held before delivery. **Platform-bank exchange is slow to set up and cheap to run.** It is one of the few controls that addresses the money side of a fraud the platform cannot see.

Back to Barcelona: What Day-One Action Would Have Required

Ingrid's question can now be answered component by component.

What happened	Missing capability	Target design
Photo matches queued as "possible duplicates"	Policy separating unrelated-seller matches from multi-agency duplicates	Photo match with an unrelated, non-agency seller is a fraud signal sent to risk scoring with seller context
Price anomaly in a separate low-priority queue	Score combination	One listing risk score combining photo reuse, price, account age and lister verification
26 accounts looked unrelated	Identity graph	Shared devices and IBANs link new accounts at creation; cluster risk raises every member's score
Reviewers judged one listing at a time	Actor-level cases	One case per cluster, with all listings, accounts and evidence
First victim report went to support	Intake routing	Victim reports routed as notices to trust and safety, linked to the listing and the account
Combined rule took 12 days to exist	Rule platform with shadow mode	Analysts write and shadow-test a rule in hours, then promote it

With those components, a plausible timeline is that the first five or six listings go live on 1 July, the cluster forms when the third account links through a shared device that evening, and the remaining listings are held at the gate. That would not have prevented every loss, but most of the €71,000 was paid between 5 and 12 July. Hearthly Spain's move to the shared platform was brought forward to the first quarter of 2028.

Trade-offs

Build, buy or share

Option	Strengths	Weaknesses	Right when
Per-brand in-house scoring	Local fit; teams know their fraud	No cross-brand signals; models built several times; small label sets per brand	One brand, or brands with nothing in common
Shared group risk platform (Veldmark's choice)	Shared graph, block lists and label store; one rule and model lifecycle; larger training sets	Data protection limits on sharing; platform team must support vertical detectors	Several brands under one legal regime with overlapping fraud
Commercial fraud-detection vendor	Fast start; cross-customer intelligence; mature tooling	Labels and features may not be portable; vendor sees platform data; generic models miss vertical fraud	Early-stage platforms, or for specific signals such as device intelligence
Hybrid: vendor signals into an in-house platform	Buys hard-to-build signals (device fingerprinting, phishing-domain feeds) while keeping decisions and labels	Integration and contract work; latency of external calls	Most mature marketplaces

My recommendation for a group is the hybrid: **buy signals, own decisions and labels**. Device intelligence and domain-reputation feeds are expensive to build and benefit from data across many customers. The decision logic, the identity graph and the label store are the platform's memory of its own adversaries, and they should not live in someone else's system.

Friction versus liquidity

Every friction costs liquidity somewhere. A held listing is not searchable; a verification step loses some sellers; a masked phone number slows some genuine contacts. The Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics chapter shows why the first hours of a listing and the speed of first contact matter.

The defensible position is **targeted friction with measured costs**: frictions applied by risk tier, each with an experiment that measures abandonment, and thresholds reviewed with the general managers who own the liquidity targets.

Real-time versus batch detection

Choice	Benefit	Cost
Everything real-time	Earliest possible action	Latency budget limits features; expensive
Real-time gate plus asynchronous scoring	Rich features within minutes	A window of exposure for items published on the gate's verdict
Batch only	Cheap; full graph analysis	Hours or a day of exposure; organized fraud strikes in that window

Most platforms need the middle row, with **risk tiers deciding which items wait for the asynchronous result**. In Barcelona, a rental listing from a new, unverified private account should have waited.

Vertical Differences

Dimension	General classifieds	Real estate	Automotive	Jobs	Recommerce with payments
Dominant fraud	Payment-link phishing, advance payment, counterfeit	Fake rentals and deposit fraud	Deposit scams, cloned listings, fake escrow	Fake job offers, mule recruitment, data harvesting	Shipping and refund fraud, phishing that imitates the platform's payment product

Dimension	General classifieds	Real estate	Automotive	Jobs	Recommerce with payments
Typical loss per victim	Tens to hundreds of euros	One or two months' rent	Hundreds to thousands of euros	Personal data, sometimes "training fees"	Item value
Strongest signal	Domain and link checks, price	Photo reuse, rent anomaly, lister verification	VIN and plate reuse, price vs. market value	Contact details and salary anomaly	Payment and shipping events
Natural friction	Message warnings, limits	Lister verification; viewing before payment	Dealer verification; protected deposits	Employer verification	Protected payment by default
Where the money moves	Off-platform transfer	Off-platform transfer	Off-platform transfer or fake escrow	Usually no money; data is the target	On-platform, through the payment provider

The last row explains the strategic link between fraud and transactions. Moving money onto the platform with buyer protection turns unrecoverable APP fraud into a mediated dispute, which is why UK Finance asks for "secure payment protections on online marketplaces". It also creates a new fraud surface: phishing that impersonates the platform's own payment product, which is exactly what Femke received. Beyond the Contact: Payments, Shipping and Buyer Protection covers the transactional pivot. The fraud consequence is that **launching protected payments changes the threat model, it does not end it**.

Failure Modes

1. **Signals without combination.** Photo matches, price flags and account-age checks each raise their own queue item. No component combines them, so each looks minor. Barcelona is the canonical case.

2. **Item-only scoring**. Rings design listings to pass item checks. Without an identity graph and actor-level cases, enforcement removes listings one by one while the ring posts new ones.
3. **Duplicates and fraud in one queue**. Stolen photos are closed as legitimate duplicates because the reviewer sees two listings of the same flat.
4. **Messages delivered, then scanned**. The phishing link reaches the seller before any check runs. Detection after delivery is forensic, not preventive.
5. **Models trained on last year's fraud**. Labels arrive late, retraining is quarterly, and there is no shadow comparison. The model scores the old pattern well and the new one not at all.
6. **Rules without owners**. Hundreds of rules accumulate. Some block good users, some never fire, and nobody can say which.
7. **Blocking instead of friction**. High-risk-but-genuine users are blocked, appeal rates rise, and the team raises thresholds until fraud gets through.
8. **Verified badges that survive takeover**. An aged, verified account is taken over and its badge makes the scam more convincing. Badges must be suspended on a change of session posture.
9. **Thresholds leaked in rejections**. Detailed rejection reasons teach fraudsters exactly which signal to change.
10. **Late labels dropped**. Bank and police reports arrive weeks later as e-mails and are never joined to the listing and account, so the model never learns from the most reliable labels it has.

The Chief Architect's View

Fraud prevention is where a marketplace group's scale is a genuine advantage, provided the architecture lets it be one. A fraud ring that hits Tradeyard Netherlands will try Tradeyard Belgium next week and possibly Motorvane after that. A shared identity graph, shared block lists and a shared label store let the group learn once. Four separate stacks learn four times, each too slowly.

The decisions I would put on the architecture agenda:

Decision	Who decides	What to watch
A single risk scoring service, with a contract for all callers (insertion, message, login, payout)	Chief Architect, Head of Data & ML, Head of T&S	Callers that implement their own scoring

Decision	Who decides	What to watch
Actor-level scoring and the identity graph, with data protection boundaries	Head of T&S and the DPO	Graph-only terminations; cross-brand data flows outside the approved hashed lists
Rule lifecycle: backtest, shadow, rollout, owner, expiry	Head of T&S	Time from new scam pattern to rule in production; rules with no owner
Friction thresholds and the cost of a lost genuine user	Marketplace GMs with T&S	Abandonment per friction from experiments; fraud stopped per friction
Label pipeline, including bank and police reports	Head of Data & ML	Label delay per source; share of fraud cases with a joined late label
Buy versus build per signal	Chief Architect and procurement	Label and feature portability in vendor contracts
Bank and law-enforcement data exchange	DPO, legal and T&S	Turnaround time; hits on exchanged identifiers

Three numbers belong on the group dashboard: **sampled prevalence of fraudulent listings by brand**, **time from first fraudulent listing to cluster takedown** for confirmed rings, and **fraud losses reported by victims per million contacts**. The second is the one Barcelona exposed: about twelve and a half days, when the signals were there on day one. Under the target design, the number to aim for is hours.

In the long term, fraud and trust will increasingly be **AI against AI**. Generative tools make fake listings, fake documents and convincing messages cheap, and they make fraud scripts easy to translate into every market's language. The same tools make detectors faster to build and policies faster to change. AI Across the Listing Lifecycle covers the broader shift. The architectural defense does not change: a platform that combines signals, scores actors, applies friction in proportion and learns from labels within days will absorb new attack tools. A platform that relies on any single detector, however capable, will not.

Questions to Ask

1. Which actions are scored for fraud risk today (insertion, edit, message, login, recovery, contact reveal, payout), and which are not?

2. Do our photo-reuse matches between unrelated sellers go to fraud scoring with seller context, or to a deduplication queue?
3. Can we link new accounts to known fraud accounts through shared devices, phones, payment accounts or images at the moment of creation?
4. How long does it take from spotting a new scam pattern to having a shadow-tested rule live in every market?
5. Are messages with links, QR codes or payment terms scored before delivery, and is domain age part of the check?
6. How are our friction thresholds set, who owns the cost of a lost genuine user, and when were abandonment rates last measured by experiment?
7. What is the delay from a bank or police report to a joined label in our training data?
8. Which fraud signals do we buy, and do our contracts let us keep the labels and features if we switch vendor?
9. When a verified account shows a change of session posture, does its badge stay visible?
10. For our last confirmed fraud ring, how long did it take from the first fraudulent listing to the takedown of the whole cluster?

From Risk to Identity

Fraud prevention keeps coming back to one question: who is on the other side? A risk score is a probability estimated from indirect evidence, and step-up friction is a request for better evidence. The strongest frictions are identity checks: strong authentication, verified phone numbers, verified identities and verified businesses. They are also the most expensive and the most sensitive. Who Is on the Other Side? Identity and Verification takes up that question: how much a marketplace should know about its users, how it verifies it, and what a verified badge should and should not promise.

Sources and Evidence

Company engineering and investor sources

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- OLX blocks scammers using advanced algorithms — ITWeb, 2018. Warn-in-context on payment keywords and off-platform moves (OLX South Africa).

- Fighting fraud: finding duplicates at scale — Alexey Grigorev, PyCon DE, 2019. OLX duplicate detection: candidate keys, pairwise classifier, hashes and embeddings.
- Bridging the gap between passwords and passkeys with risk-based authentication — Mercari Engineering, 2026. Session and account posture; passkeys and phishing losses.
- Fighting financial fraud with machine learning at Airbnb — InfoQ (Srini Penchikala), 2018, summarizing Airbnb's "targeted friction" post by David Press. Friction instead of blocks; loss-based thresholds; A/B testing frictions.
- Meet Michelangelo: Uber's machine learning platform — Uber Engineering, 2017. Shared feature store with consistent training and serving features.
- Fighting spam with Haskell — Engineering at Meta, 2015. Rules as code, deployed continuously.
- Cracking down on organized crime behind scam centers — Meta, 2024. Scale of organized scam operations.
- Fraud prevention rules (Radar) — Stripe, n.d. (current documentation). Backtesting, shadow mode, rollout, audit trail, rule metrics.

Standards, regulation and documentation

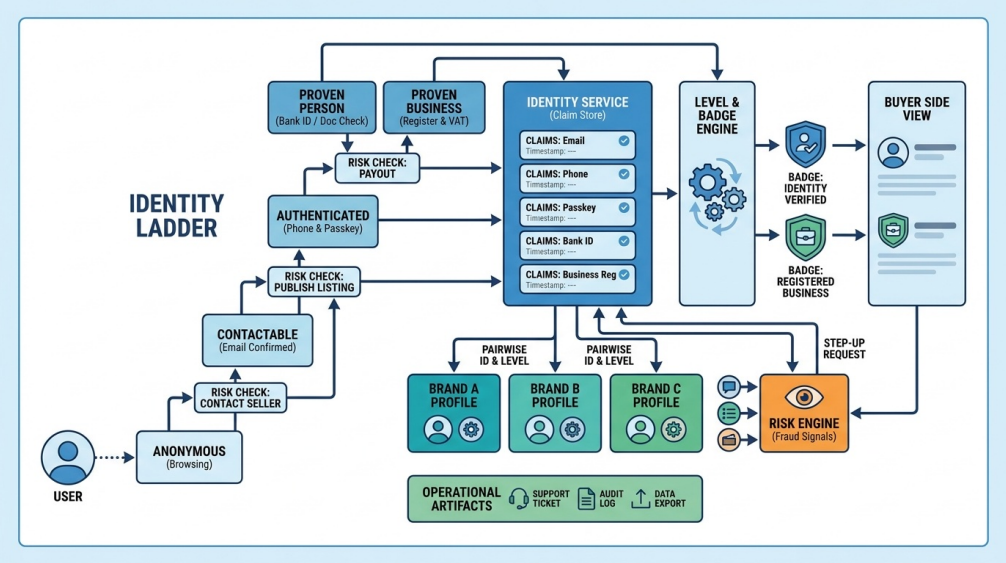
- PDQ perceptual hash — Meta ThreatExchange, n.d. Match thresholds, quality score, invariance limits.
- Are you VoP to go? Prepare for Verification of Payee by October 2025 — McCann FitzGerald, 2025. Verification of Payee obligation for euro-area payment service providers from 9 October 2025.
- Fraudulent advertising and account integrity: expert insights on best practice — Integrity Institute for Ofcom, 2026. Signals, friction between account creation and reach, enforcement ladder.

Research and practitioner sources

- Annual Fraud Report 2026 — UK Finance, 2026. Purchase-scam cases, losses, share of APP fraud, reimbursement, policy asks.
- Scam ads slipping through the net too easily on second-hand marketplaces — Which?, 2024. Undercover test: edited scam ads passing review; evasion by changing devices and identities.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

26. Who Is on the Other Side? Identity and Verification



IN THIS CHAPTER, YOU WILL: Separate the three questions an identity system answers: who someone is, whether the person logging in is still that someone, and whether a business is what it claims to be. You will learn how to turn them into identity levels, verification claims and badges, how to defend accounts against takeover, and how to build one identity service for a group of marketplaces without making the brands inseparable.

WHY THE BUSINESS CARES: Buyers contact strangers, and sellers hand over goods to them. Knowing a little more about who is on the other side is the cheapest trust the marketplace can buy, but every verification step also loses some honest users. The business needs friction placed where the risk is, not spread evenly across everyone.

WHY ARCHITECTS SHOULD CARE: Identity is the one capability every other context depends on: insertion, messaging, payments, trust and safety, regulation and ads. A group that runs one identity system per brand cannot

share a ban, cannot offer one login, and cannot prove which trader it verified. A group that merges them carelessly can no longer sell a brand.

KEY POINTS:

- **Identity is a set of separate claims, not one "verified" flag.** Contactability, authentication strength, proof of person and proof of business answer different questions and fail in different ways. Model each as a claim with a method, a source, a time and an expiry, and derive identity levels from the claims instead of storing a level.
- **Ask for identity at the moment of risk, not at the door.** Browsing and saving searches need no identity. Contacting a seller, publishing a rental, receiving a payout or advertising as a trader each need a specific level. Step-up verification at those moments keeps conversion high and puts the cost where fraud concentrates. Account recovery is part of the same design, because a weak recovery path cancels a strong login.
- **A badge is a promise to the other side, and a group identity is a commitment to the owners.** Show buyers only what was checked ("identity verified with BankID", "registered business"), and revoke the badge when the account's posture changes. Build one group identity service with brand-specific profiles and pairwise identifiers, so a brand can still be separated if it is sold.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Fast Mail for Fraudsters** — when anyone with a fresh e-mail address can message any seller, identity checks never reach the channel where scams close.
- **Deal-Shaped Architecture** — four acquired identity systems mean four logins, four ban lists and four answers to "who is this trader", so a fraudster removed from one brand simply registers on the next.
- **Moderation as an Afterthought** — verification added after a fraud wave, as a one-off document upload, produces badges that nobody can explain and evidence nobody can find.

Part V treats trust as a shared platform, and Scams, Fakes and Account Takeovers: Fraud Prevention for Listings ended on its strongest and most expensive friction: asking users to prove who they are. This chapter decides how much a marketplace should know about its users. Most marketplaces answer that twice, badly. First they ask for almost nothing, because every registration field costs sign-ups. Then, after a fraud wave or a regulator's letter, they ask everyone for an identity document, because the incident made identity feel like the solution. Neither answer holds up. **The right amount of identity depends**

on what the user is about to do. Browsing needs none. Publishing a rental in a city full of fake-rental rings needs a great deal.

The chapter separates *proofing* (establishing who someone is) from *authentication* (establishing that the person logging in is still that someone) and from *business verification* (establishing that an organization exists and is who it says it is). It turns these into identity levels and badges, follows account takeover and recovery, and then takes up the group question: one login for four brands, or four logins forever?

The result is the identity service that neighboring chapters call. Private Sellers, Dealers and Agencies: Modeling Who Lists owns the seller account model, organizations, branches and trader status. Scams, Fakes and Account Takeovers: Fraud Prevention for Listings owns risk scoring, fraud signals and step-up friction as a fraud control. Rules of the Marketplace: DSA, DAC7 and Vertical Regulation owns the legal detail of the Digital Services Act (DSA) and of seller tax reporting, and builds its trader record on the business registry designed here. From Interest to Contact: Messaging and Lead Management gates contact on the identity levels defined here, and Beyond the Contact: Payments, Shipping and Buyer Protection defers to them for payees. The running example is Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build).

Setting the Scene at Veldmark: One Phone Number, Two Brands

On Sunday **7 March 2027**, a "buyer" sent Femke, a private seller in Rotterdam, a fake Tradeyard Safe Deal payment link for her **€240** city bike. The account was suspended at 20:26, seven minutes after Femke reported it (Trust and Safety as a Platform tells that story). When Aisha Rahman, the Group Head of Trust & Safety, traced the account the following week, two facts stood out.

The first was that the scammer's Tradeyard account was **phone-verified**. In Tradeyard's own vocabulary that made it a "verified" account, and the app showed a small green tick next to its name. Phone verification had proved that someone controlled a mobile number for the few minutes it took to type a six-digit code.

The second was that the same number had registered a **Motorvane** private-seller account three weeks earlier. Motorvane could not know, because each Veldmark brand runs its own identity system. Nothing joined them except the fraudster.

Ingrid Halvorsen, the Group Chief Architect, asked Lucía Ortega, the Group Data Protection Officer, and Aisha to describe what each brand knew about its users. The answer took one table:

Brand	Login methods	What "verified" means today	Business verification	Accounts (registered, about)
Tradeyard	Password; Google and Apple sign-in; SMS code for some actions	Phone number confirmed by SMS	Pro sellers: VAT number at subscription, not re-checked	46 million
Motorvane	Password; dealer staff on a separate dealer portal login	E-mail confirmed	Dealers: contract checks at signature only	19 million
Hearthly FR/ES	Password; Hearthly.es still on Casavista's login	E-mail confirmed	New agencies checked manually by sales	15 million
Keybridge	Agency staff accounts, single sign-on for large networks	Not applicable	Agency is the SaaS customer	61,000 staff users

About **80 million** consumer accounts sat in three identity stores, with no shared notion of who a person was. Veldmark's own analysis, matching on verified e-mail addresses, estimated that about **11 million** people held accounts on two or more brands. Three months later, in June 2027, Veldmark approved **Veldmark ID**: one login across brands with brand-specific profiles, rolling out in 2028.

Approving a login was the easy part. The questions Ingrid brought to the architecture advice forum were harder. What does "verified" mean at Veldmark? Which actions need which level? Who keeps the evidence, for how long? And if the private-equity owner sells Hearthly in 2030, can its users leave with it?

Concepts: Identity, Authentication and Business Verification

Five words that teams use interchangeably

Identity discussions go wrong when one word stands for several things. This book uses these definitions:

Term	Meaning	Example at Veldmark
Identity	The person or organization the platform has evidence about	Femke, a natural person resident in Rotterdam
Account	The login: credentials, sessions, recovery methods	Femke's Veldmark ID account
Profile	A brand-specific persona attached to an account: display name, ratings, preferences	"Femke R." on Tradeyard; no Hearthly profile
Seller account	The commercial entity that lists: a private seller or an organization with members (owned by Private Sellers, Dealers and Agencies: Modeling Who Lists)	Autohaus Ruhrblick on Motorvane, with nine staff users
Claim	A statement about an identity, with its evidence: "phone +31… controlled on 2 May 2027, by SMS code"	Femke's phone claim; Ruhrblick's trade-register claim

The distinction between account and identity matters most. One person can hold several accounts, and one shared account can be used by several people (a family, a car dealership's front desk). An identity system that assumes one account per person will either block legitimate sharing or fail to see one fraudster with forty accounts.

Three axes, not one ladder

The clearest vocabulary for this comes from the United States National Institute of Standards and Technology (NIST). Its Digital Identity Guidelines, SP 800-63-4, finalized

in July 2025, separate **identity assurance** (how confident the service is that a person is who they claim to be) from **authenticator assurance** (how confident it is that the person presenting the credential is the account holder), with a third axis for federated logins (NIST, 2025). The guidelines are written for US government systems and oblige no marketplace to follow them, but the separation applies to any platform.

A user can be well proven and weakly authenticated: a dealer who passed a full business check five years ago and still logs in with a reused password. A user can also be strongly authenticated and unproven: an anonymous buyer with a passkey. Fraud exploits both gaps, and **a single "verification level" hides which gap is open**.

For a listing marketplace, three axes are enough:

1. **Proof of person** — what the platform knows about who a natural person is.
2. **Strength of authentication** — how hard it is for someone else to act as the account holder.
3. **Proof of business** — what the platform knows about an organization that lists or advertises.

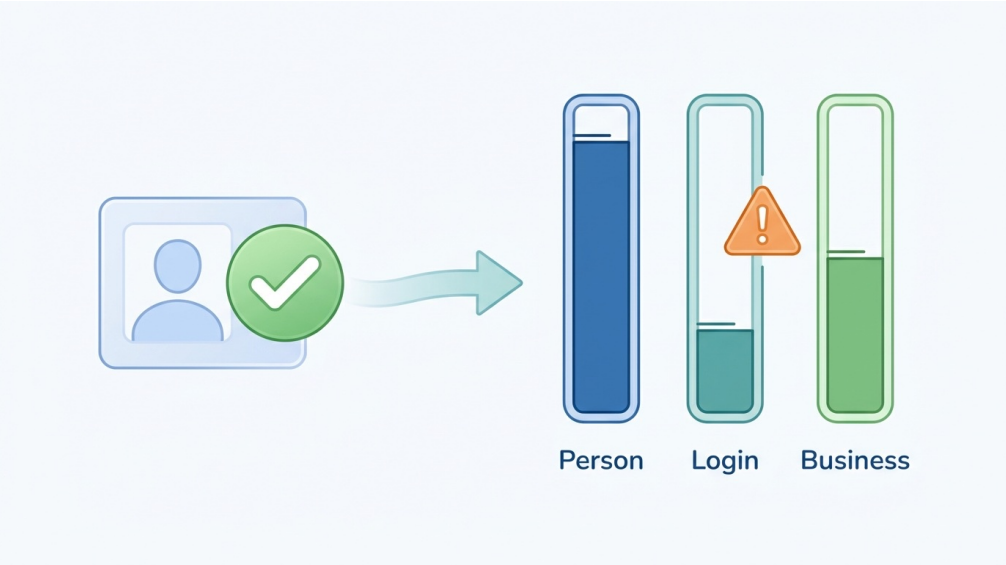


Figure 1: *One "verified" tick hides three separate axes: proof of person, strength of authentication and proof of business, and fraud exploits whichever gap the tick does not show.*

Identity levels

An **identity level** is a named combination of claims that a policy can require. It is a derived value, recomputed when claims are added, expire or are revoked. It is not a

column someone sets. Veldmark adopted six levels for natural persons, plus a separate business ladder.

Level	Name	Required claims	What it defeats	What it does not defeat
L0	Anonymous	None (device-scoped state only)	Nothing; none needed for browsing	—
L1	Contactable	Confirmed e-mail address	Typos, trivial bots	Disposable addresses, account farms
L2	Reachable	L1 plus a confirmed phone number, checked against virtual-number and recycled-number lists	Bulk account creation, because numbers cost money	SIM farms, rented numbers, SIM swap
L3	Protected	L2 plus a phishing-resistant authenticator (a passkey) or a second factor	Credential phishing and stuffing for logins	Social engineering of recovery; a willing fraudster
L4	Identified	L3 plus proof of person: a bank-based eID, a national eID or EU Digital Identity Wallet, or an identity document with a liveness check	Most fake-account farms; repeat offenders under new names	Stolen identities at small scale; money mules using their own names

Level	Name	Required claims	What it defeats	What it does not defeat
L5	Payee-verified	L4 plus a payment account in the person's name, verified by the payment provider	Payouts to accounts in other names	A real person acting as a mule

The business ladder runs in parallel, because an organization is not a person with more documents:

Level	Name	Required claims
B1	Declared trader	The seller declares it acts as a business, and provides name, address, phone, e-mail and trade-register details
B2	Registered business	The trade-register entry and VAT number validated against official sources, and the person onboarding it identified at L4
B3	Accredited professional	B2 plus sector evidence: a dealer contract, a real-estate agent's professional licence where the country has one, or membership of a required redress scheme

B1 and B2 map closely to what the DSA requires of online marketplaces before a trader can offer to EU consumers: name, address, phone and e-mail, a copy of an identification document or an electronic identification, payment account details, the trade-register number, and a self-certification, followed by "best efforts" to check the information against freely accessible official databases (Regulation (EU) 2022/2065, Art. 30). Rules of the Marketplace: DSA, DAC7 and Vertical Regulation covers that obligation as law. Here the point is architectural: **the trader-traceability record is a business identity claim set**, and it belongs in the identity service, not in a compliance spreadsheet.

KYB (know your business) is the name for the B-ladder checks. **KYC** (know your customer) is the payments-industry name for person and business checks that a regulated payment provider must run before it lets a user receive money. The two overlap but have different owners. At Veldmark, Adyen runs KYC for Safe Deal sellers, because Adyen is the licensed party. Adyen's platform documentation says it "must verify the users in your platform" before they can use capabilities such as payouts, and that it re-verifies

when data changes (Adyen, n.d.). Veldmark receives the verification *status* as a claim with Adyen as its source. It does not receive, and should not want, Adyen's evidence.

Verified badges

A **verified badge** is a public statement the marketplace makes to one user about another. That makes it a promise, and a promise has a meaning, a scope and a liability. The rule I use: **a badge must say what was verified**, **by which method**, **and nothing more**. "Identity verified with BankID" is a badge. "Verified" is a liability, because a buyer will read it as "safe to pay", and the platform verified nothing of the kind.

Attackers value badges too. The Ofcom-commissioned expert report on fraudulent advertising and account integrity notes that account takeover is attractive because fraudsters "gain check marks, verification" along with the account (Integrity Institute for Ofcom, 2026). A badge on a taken-over account is worse than no badge. It amplifies the scam it was meant to prevent.

Architecture: The Identity Service and Its Neighbors

Components and responsibilities

The identity capability splits into components that change at different speeds and have different owners. Authentication changes when new authenticator technology appears. Verification changes when a vendor's price or a country's eID scheme changes. Levels and badges change when trust-and-safety policy changes.

Component	Responsibility	Owns	Does not own
Authentication service	Logins, sessions, authenticators (passwords, passkeys, e-mail codes, social login), token issuance to brands through OpenID Connect	Credentials, sessions, authenticator registry	Profiles, risk scores

Component	Responsibility	Owns	Does not own
Account recovery	Regaining access after losing an authenticator; the most attacked flow	Recovery policies, recovery evidence	Normal login
Verification orchestrator	Runs a verification journey with the right provider for the country and level; normalizes results	Verification cases, provider adapters	The provider's raw evidence
Claim store	Every claim with method, source, assurance, time, expiry and status	Claims and their history	Documents and images
Level and badge engine	Derives identity levels from claims; decides which badges an account may show in which brand	Level rules, badge catalog	Fraud decisions
Business registry	B-ladder claims for organizations; register lookups; re-verification schedule; links to Private Sellers, Dealers and Agencies: Modeling Who Lists organizations	Business claims, register snapshots	Contracts and packages

Component	Responsibility	Owns	Does not own
Profile service	Brand-specific profiles and preferences on top of one account	Display names, brand settings	Credentials
Consent and privacy service	Consents, data-subject requests, erasure fan-out	Consent records	Business data

Two boundaries matter most. **Risk scoring lives outside identity.** Scams, Fakes and Account Takeovers: Fraud Prevention for Listings reads identity levels as features and asks the identity service for a step-up. It does not write claims. **Documents live outside the claim store.** The claim says "identity document checked with liveness, provider X, case 8f3…, result pass, name match on the payee account". The image of the passport stays with the provider, for the provider's retention period, or in a sealed evidence vault if the law requires the platform to keep a copy.

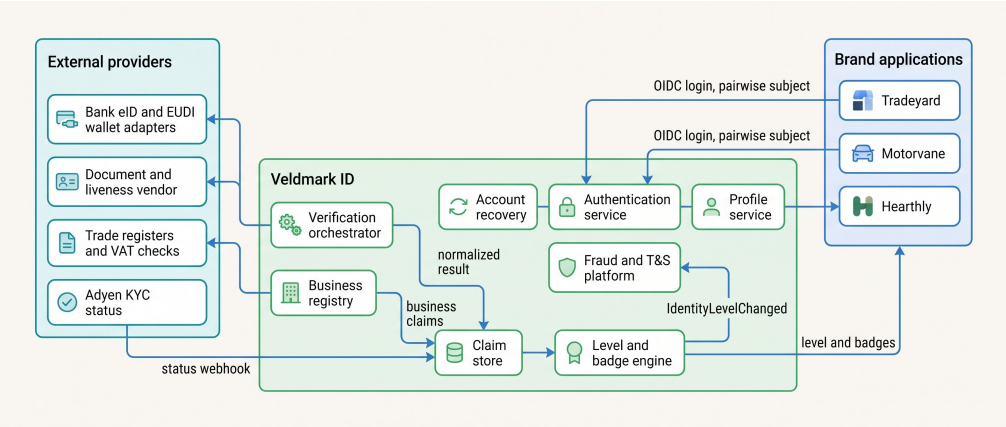


Figure 2: *Veldmark ID's components. Brands authenticate users through one service but receive brand-specific identifiers; verification providers sit behind adapters that write normalized claims; levels and badges are derived from claims and published as events to the fraud and trust-and-safety platform.*

The claim as the contract

The most important interface in the identity service is the shape of a claim, not the login API, because every provider adapter writes one and every consumer reads one.

Veldmark's claim record has these fields:

Field	Example	Why it exists
`claim_type`	`person.identified`, `phone.reachable`, `business.registered`, `payee.verified`	Policies require claim types, not vendors
`subject`	Account ID, or organization ID for business claims	Business claims attach to organizations, not to the staff member who uploaded them
`method`	`bank_eid`, `eudi_wallet`, `document_liveness`, `sms_otp`, `register_lookup`, `psp_kyc`	Different methods carry different assurance
`source`	Provider and case reference	Audit and dispute; no raw evidence
`attributes`	Verified given name, family name, year of birth over 18, country	Only what a policy needs (data minimization)
`assurance`	`low`, `substantial`, `high`	Comparable across providers and countries
`verified_at`, `expires_at`	2 May 2027; 2 May 2030	Claims age; phone numbers get recycled; businesses close
`status`	`valid`, `expired`, `revoked`, `disputed`	Badges must follow status

Two design rules follow. First, **policies reference claim types and assurance, never providers**. "Private-landlord rentals in Spain require `person.identified` at `substantial`" survives a change of document vendor or the arrival of the EU Digital Identity Wallet. "Requires an Onfido check" (or any other named vendor) does not. Second, **claims carry only the attributes a policy needs**. If the only purpose is to show "identity verified", the claim needs the verified name (to match the payee account and the display name policy) and perhaps an adult flag. It does not need the document number, the date of birth or the address. The General Data Protection Regulation (GDPR) requires data to be "limited to what is necessary" and asks for data protection by design (Regulation (EU) 2016/679, Arts. 5(1)(c) and 25). The claim schema is where that principle is enforced.

Seller tax reporting needs more attributes, and they stay out of the general claim. Once Safe Deal makes Tradeyard a reporting platform operator under DAC7, it must collect date of birth and tax identification number for reportable sellers. That collection belongs to a separate, purpose-bound tax profile with its own retention. Rules of the Marketplace: DSA, DAC7 and Vertical Regulation covers the obligation.

The claim lifecycle

Claims expire and get revoked, and the lifecycle has to model every way they end, because a badge that outlives its claim is a false statement. A phone number is reassigned to someone else. A company is dissolved. A passkey is deleted. A payment provider freezes a seller after a fraud report.

A claim's lifecycle. Every transition recomputes the account's identity level and badges; revocation is a first-class state, not a deletion, so the platform can later explain why a badge disappeared.

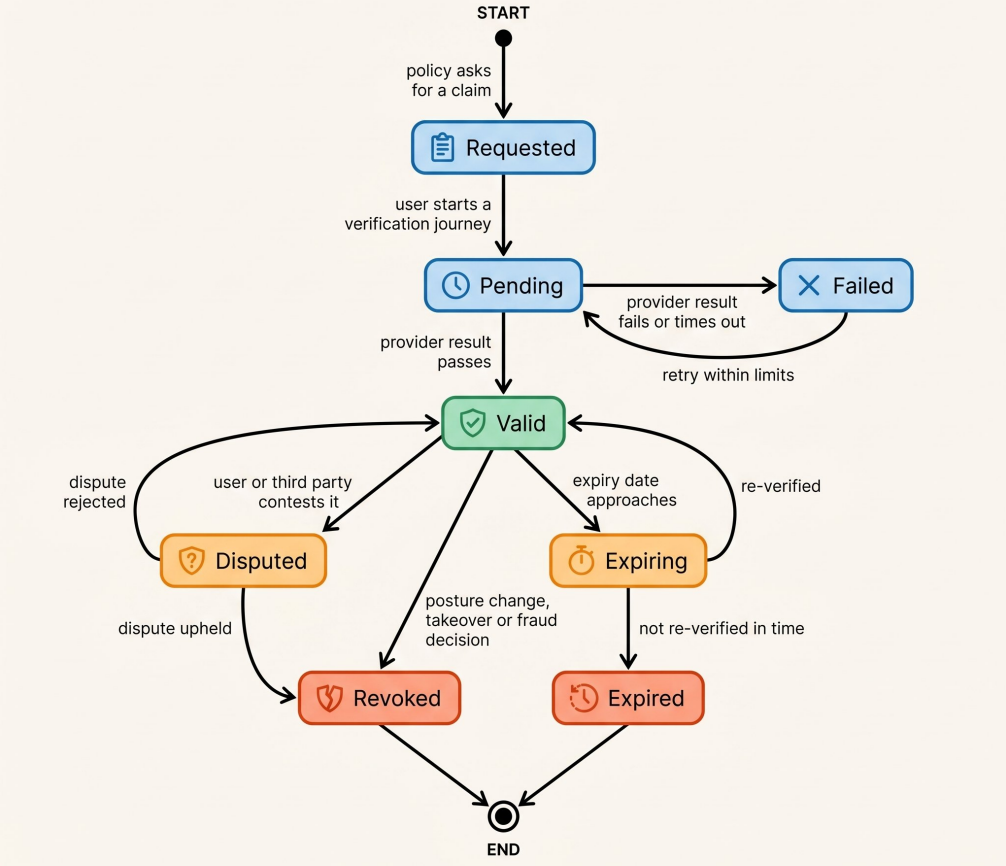


Figure 3: *A claim's lifecycle. Every transition recomputes the account's identity level and badges; revocation is a first-class state, not a deletion, so the platform can later explain why a badge disappeared.*

Authentication and Account Takeover

Authenticators for a consumer marketplace

Most identity incidents begin at authentication, which happens every day for tens of millions of people; proofing happens once in a while. A marketplace needs a small set of authenticators, each with a clear role.

Authenticator	Role	Strength	Weakness
Password	Legacy default	Familiar	Phished, reused, stuffed from other breaches
E-mail one-time code or link	Passwordless default; recovery bridge	No password to steal	Only as strong as the e-mail account
SMS one-time code	Phone confirmation; fallback	Ubiquitous	SIM swap, interception, SMS-pumping fraud against the platform
Social login (Google, Apple)	Low-friction sign-up	Delegates strength to a large provider	Account-merge risks; dependence on a third party
Passkey	Phishing-resistant login	Bound to the site; no shared secret	Device loss; support load; not universal yet
Bank eID (BankID, itsme, iDIN and similar)	Proofing and high-assurance step-up	Bank-grade proofing	Per-country; per-use fees; not every user has one

A **passkey** is, in the FIDO Alliance's words, "an authentication credential based on FIDO standards" that the user unlocks with a biometric, PIN or pattern on their device. Passkeys can be synced across a user's devices through a cloud service or bound to one device, and FIDO describes them as phishing-resistant either way (FIDO Alliance, n.d.). NIST's 2025 guidelines accept syncable authenticators at its middle assurance

level, subject to protections on the sync service, and exclude them from the highest level because syncing conflicts with its non-exportability requirement (NIST SP 800-63B-4, Appendix B). For a marketplace, synced passkeys are the realistic target: strong enough to stop phishing, recoverable enough for consumers.

Mercari, the Japanese second-hand marketplace, documents that required passkeys stop large-scale phishing, at a cost in lost users and support load. In a July 2026 engineering post, Mercari described about **13 million** users with a registered passkey, and "phishing-resistant accounts" that require a passkey, allow login only with it and permit recovery only through high-assurance identity proofing. Where passkeys are required, Mercari reports that "large-scale phishing financial damages have been effectively zero"; where they are optional, losses dropped as adoption grew. Mercari also names the cost: users who cannot register a passkey leave, and users who lose one are locked out and contact support (Mercari Engineering, 2026).

Mercari's migration model is worth copying because it is gradual. Accounts move along four states: password or e-mail link, e-mail link only (password disabled), passkey or e-mail link, and passkey only. **Disabling the password is the first step, not the last.** The e-mail link acts as a bridge, so a user who never adopts a passkey still stops being phishable by password.

FINN, the Norwegian marketplace, moves in the same direction from a different starting point: its help center describes logging in to the shared Vend account with a one-time code sent to the user's e-mail address, and separately encourages two-step authentication because, "as on other digital platforms", some FINN users have had their accounts taken over by others (FINN help center, 2024, updated 2026).

Risk-based authentication

Strong authenticators protect accounts that use them. Risk-based authentication protects the rest. It judges each login and each sensitive action by two kinds of evidence, which Mercari calls **session posture** (signals from the device and network of this session: client integrity, network reputation, device and user-agent properties) and **account posture** (what accumulates over the account's life: authentication capabilities, verification history, behavior). Mercari combines them into one weighted verdict, and weights them differently for different decisions (Mercari Engineering, 2026).

For a listing marketplace, the sensitive actions are predictable, and each deserves its own policy:

Sensitive action	Why attackers want it	Typical control
Login from a new device	Entry point for takeover	Step up to passkey or e-mail code; notify on known devices
Change of e-mail, phone or password	Locks the owner out	Re-authenticate; notify the old contact point; delay the change of the recovery address
Change of payout account	Redirects the victim's sale proceeds	Re-authenticate at L3 or above; hold payouts for a cooling period; notify
Publishing many listings quickly	Uses an aged account's reputation	Velocity limits by level (see Scams, Fakes and Account Takeovers: Fraud Prevention for Listings)
Sending messages with links or contact details	Fast Mail for Fraudsters	Message friction by level (see From Interest to Contact: Messaging and Lead Management)
Enabling a verified badge in a new brand	Borrowed trust	Require fresh authentication and a matching profile name

Recovery is the real login

Every authentication design has a back door: the flow for people who lost their password, their phone or their passkey. Attackers use it because it is weaker. If an account can be recovered by answering a support agent's questions or by receiving an SMS code, the passkey protects nothing.

Mercari's recovery rule is the strictest documented one. Users who cannot be reached by e-mail or phone prove their identity with Japan's My Number Card or with a document and a selfie, and recovery restores a passkey, e-mail address or phone number. "We do not restore passwords," the post says (Mercari Engineering, 2026). The general principle is that **recovery should require at least the assurance the account had**, and that a recovered account should return in a cautious state: badges suspended, payout changes held, until it re-establishes posture.

Group identity programs open a second back door: **pre-hijacking**. Researchers at Microsoft described pre-hijacking attacks, in which an attacker creates an account with

a victim's e-mail address *before* the victim does, and keeps access after the victim later signs up or links a social login. They found at least 35 of 75 popular services vulnerable to one or more of five variants (Sudhodanan and Paverd, USENIX Security 2022). Any program that **merges accounts across brands by matching e-mail addresses** recreates the conditions for this attack. Accounts should be linked only after the user proves control of both, through authentication, not through a string match.

Proofing People: Methods and Their Trade-offs

When a policy asks for L4, the verification orchestrator has to choose a method. The choice depends on country, cost, user experience and the attribute the policy needs.

Method	How it works	Assurance	Coverage	Cost and friction
Bank-based eID (BankID in Norway and Sweden, itsme in Belgium, iDIN in the Netherlands)	The user authenticates with their bank's identity service; the platform receives verified attributes	High	Strong in its country; zero outside	Per-use fee; very low friction for users who have it
EU Digital Identity Wallet	The user presents attributes from a government-backed wallet, with selective disclosure	High, by design	Rolling out from the end of 2026; uneven by country	Fees and relying-party registration rules still settling

Method	How it works	Assurance	Coverage	Cost and friction
Document plus liveness	The user photographs an identity document and takes a selfie video; a vendor checks both	Substantial	Any country with machine-readable documents	Highest friction; vendor fees; demographic accuracy must be tested
Payment-instrument name match	A card or bank account in the user's name is verified by a payment provider	Substantial for payees	Anyone with a bank account	Cheap; proves account control, not innocence
Phone carrier lookup	Checks that a number is real, not virtual, and how long it has been held	Low	Varies by carrier	Cheap; weak against SIM farms

FINN is the clearest documented example of a bank-based eID used for marketplace trust. Its help center, in articles first published in November 2024 and updated in October 2026, says that a user who verifies with BankID gets a symbol on the profile that buyers and sellers see when they make contact, and that "people with this symbol are experienced as safer to trade with". The user can still choose the display name, while FINN knows which BankID verified the account. Only **one** of a person's FINN users can be verified with the same BankID. Verification is mostly voluntary, but it can be required, for example to contact advertisers in some categories or for activity that needs extra security, "to protect you and others against fraud and misuse". Business advertisers do not need to verify (FINN help center, "Why verify my user?", 2024, updated 2026; FINN help center, "How do I become a verified user?", 2024, updated 2026).

Four design choices in that short description are worth copying:

1. **Verified, but pseudonymous**. The platform knows the legal identity; other users see a chosen name and a badge. Privacy and trust are not in conflict.
2. **One verified account per person**. The rule turns a proofing method into a defense against multi-accounting.

3. **Required only where risk concentrates.** Contacting advertisers in some categories, not registering.
4. **A different track for businesses.** Professional advertisers are verified as businesses, not as people.

The **EU Digital Identity Wallet** will change the method mix, but on a slower and narrower path than its publicity suggests. Regulation (EU) 2024/1183 requires each Member State to provide at least one wallet within 24 months of the entry into force of its implementing acts. The first of those acts were published on 4 December 2024, which puts the deadline at the end of 2026, consistent with the Commission's statement that wallets are due by the end of 2026 (Regulation (EU) 2024/1183, Art. 5a; Commission Implementing Regulation (EU) 2024/2979; European Commission, updated 2026). The rules for registering wallet-relying parties apply from 24 December 2026 (Commission Implementing Regulation (EU) 2025/848).

The acceptance obligations for private companies are narrower than many product teams assume. Private relying parties must accept the wallet, on the user's request, only where Union or national law or a contract requires them to use strong user authentication for online identification, and only from 36 months after the implementing acts; small and micro enterprises are exempt. Very large online platforms under the DSA must accept it for user authentication when they require authentication (Regulation (EU) 2024/1183, Art. 5f). As of October 2026 no classifieds, real-estate or automotive marketplace is designated a very large online platform. For most marketplaces, the wallet is therefore **an option to adopt, not an obligation to meet**. It is a good option for exactly the B-ladder and L4 checks: the DSA already allows a trader to identify with "electronic identification" under the eIDAS framework instead of a document copy (Regulation (EU) 2022/2065, Art. 30(1)(b)), and the wallet's selective disclosure lets a platform ask for "over 18" or "name matches" without receiving a birth date.

The architectural conclusion is that **the verification orchestrator must be pluggable by country and method**. A group operating in eight countries will run bank eIDs in some, the wallet in others, document checks as the universal fallback, and will change the mix every year. Each adapter writes the same claim.

In Practice: The Group Identity Question

What a shared identity looks like in public

Schibsted's shared account, the longest-running public example of a marketplace group identity, shows what users feel when the group splits. FINN's help center describes Schibsted account as having been "the shared login solution for all Schibsted services", news media and marketplaces alike. When Schibsted split into two companies, Schibsted Media and Schibsted Marketplaces, the account was separated as well. Users who had

used both media services, such as VG and Aftenposten, and marketplace services, such as FINN, ended up with **two accounts**; privacy choices that had applied to all services at once now had to be made per company; and the marketplace account moved to a new web address, so password managers no longer recognized it and some users were asked to set a new password (FINN help center, "FAQ: Separation of Schibsted Accounts", February 2025, updated 2026).

Schibsted Marketplaces has since been renamed Vend, and FINN describes the Vend account as a shared login and payment solution for Vend services such as FINN.no, Helthjem and Morgenlevering (FINN help center, 2025, updated 2026; FINN help center, "What is a Vend account?", updated 2026).

> **INFERENCE, NOT DOCUMENTED:** The help-center texts describe the user-facing effects, not the internal design. A plausible reading, not a description of Schibsted's or Vend's systems, is that the shared identity carried user data and consents for two businesses in one store, and that separating them required deciding, per user and per data item, which company was the controller.

The lesson for a group architect is not that shared identity is wrong. FINN users kept one login across the marketplace services, and the shared account evidently served Schibsted for years. The lesson is that **a group identity is a bet on the group's shape**. When the shape changes through a demerger, a sale or a carve-out, the identity has to be cut along the new line, and users feel the cut. The Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform case study makes the same point about shared platforms in general: the platform that a synergy case values can become a separation cost when the owners change strategy.

Veldmark ID: designed to be cut

Veldmark is majority-owned by a private-equity investor. Ingrid's design brief for Veldmark ID started from that fact: **assume any brand may be sold, and make the cut cheap**. The architecture advice forum adopted five rules in June 2027.

1. **One account, brand profiles.** The account holds credentials, authenticators, recovery methods and person-level claims. Each brand holds a profile with display name, ratings, favorites and brand settings. Profiles are created on first use of a brand, never in advance.
2. **Pairwise identifiers.** Each brand receives a different, stable subject identifier for the same person through OpenID Connect, so brand databases cannot be joined on a shared user ID. Cross-brand joins happen only inside the identity service and the fraud platform, under documented purposes.

3. **Controller per brand, processor for the group**. Brand data stays the brand's; Veldmark ID's operating entity processes authentication on the brands' behalf. Lucía Ortega's data map records, for every claim type, which entity is the controller.
4. **Separable by export**. A brand can be exported with its users: profiles, brand-scoped claims and a migration package that lets the buyer issue its own logins. Person-level claims a user earned in another brand do not travel unless the user consents.
5. **Keybridge stays outside**. Keybridge is a neutral SaaS business for agencies, and Veldmark committed in March 2027 not to use Keybridge agency data for Hearthly's advantage. Keybridge staff logins remain in Keybridge's own identity system, federated to agency single sign-on, and Veldmark ID never sees them.

Option	What users see	What the group gains	What it costs	When it is right
Per-brand identity (Veldmark in 2026)	A login per brand	Nothing shared; trivial separation	Four ban lists, four authentication stacks, no cross-brand trust	Brands with no shared users, or a group about to sell everything
Shared identity, shared profile	One account, one public persona across brands	Simple; maximal signal sharing	Privacy surprises ("why does the car site know my bike sales?"); expensive separation	Brands sold as one product family
Shared identity, brand profiles, pairwise IDs (Veldmark ID)	One login, separate personas	Shared authentication, recovery, verification and bans	Identity service is critical infrastructure; careful controller mapping	Multi-brand groups that may change shape

Option	What users see	What the group gains	What it costs	When it is right
Federation broker over brand identity providers	Login per brand, linked on request	Gradual adoption without migration	Weakest-link security; four stacks remain	Transition state after an acquisition

Veldmark chose the third option, and planned to pass through the fourth for Hearthly.es, whose Casavista login would be federated in 2028 and migrated in 2029.

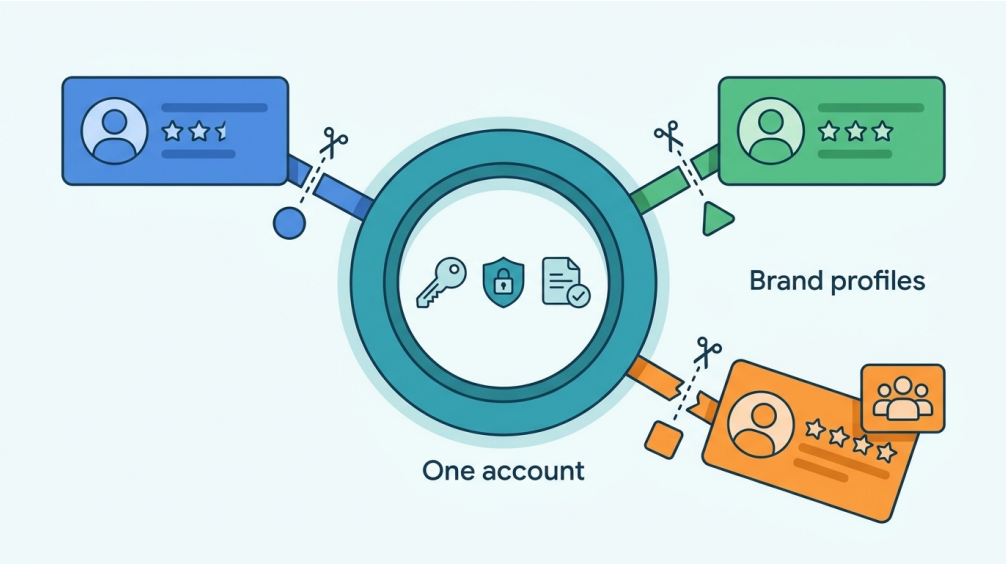


Figure 4: *Veldmark ID shares one account but gives each brand its own profile and its own identifier, so a brand that is sold can be cut out with its users.*

The Barcelona fake-rental ring: a policy worth its cost

The first policy built on the level model came from Hearthly Spain. In **July 2027**, **26** accounts posted **180** fake rental listings in Eixample, using photos stolen from real listings and prices about 35% below market, and asked for deposits by bank transfer outside the platform. **63** people reported losses of about **€71,000** before the listings came down. Scams, Fakes and Account Takeovers: Fraud Prevention for Listings follows the detection: perceptual-hash matches against existing listings and price anomalies. The identity question was simpler and more uncomfortable. None of the 26 accounts had been asked to prove anything beyond an e-mail address, because Hearthly Spain, like Casavista before it, required nothing more of private landlords.

Aisha and Julien Marchand, Hearthly's CTO, proposed a policy for the forum: **private landlords publishing a rental in Spain must reach L4** before the listing goes live, with the name on the identity claim matching the landlord's display name. Agencies were unaffected, because they publish under B2 business claims through feeds. The forum asked for the cost before approving it. The numbers below are Veldmark's planning estimates.

Quantity (Hearthly ES)	Estimate
New rental listings by private landlords per week	about 5,200
Of which by landlords not yet at L4	about 1,900
Verifications started per week, after expected drop-out	about 1,600
Blended cost per verification (document and liveness; Hearthly has no bank eID contract in Spain)	about €1.40
Annual verification cost	about €115,000
Expected drop-out of honest landlords at the verification step	8–12% in the first quarter
Losses reported in one ring, July 2027	about €71,000, plus support and reputation cost

The forum approved the policy in September 2027 with two conditions. Drop-out had to be measured weekly against a control region, and landlords who abandoned verification had to be offered the old path through a partner agency, so that supply did not quietly shrink. The rule went live in **Q4 2027**. It did not stop fraud. It made each fake landlord cost a real, checkable identity, which is what the Barcelona ring had lacked.

Motorvane: from KYB at signature to continuous KYB

Motorvane's 21,400 dealers were checked once, at contract signature, by the sales team. A pilot in June 2027 re-checked every dealer against the German trade register and the EU's VAT number validation service (VIES). It found **212** dealer accounts whose register entry had been deleted or whose VAT number no longer validated; **37** still had live listings. Most were dealerships that had closed or merged, with an employee still logged in. A few were not.

The fix was to make business claims expire. Veldmark's business registry now re-checks register and VAT claims every 90 days, re-verifies the onboarding person's L4 claim when a dealer's administrator changes, and emits a `BusinessClaimRevoked` event that suspends listing publication until the dealer's account manager resolves it. Under the DSA, a marketplace that has reason to believe a trader's information is inaccurate

must ask for correction and, if it is not corrected, swiftly suspend the trader's offers to EU consumers (Regulation (EU) 2022/2065, Art. 30(3)). Continuous KYB is how a platform notices.

Femke's account, rebuilt

Run the March incident again under Veldmark ID's level policy for Tradeyard Netherlands. The scammer's account would still be L2: e-mail confirmed, phone confirmed. Under the new policy, an L2 account younger than 30 days that sends a message containing a link to a domain outside Veldmark's allow list has the message held for a risk check, and the link rendered as text. The domain registered that day fails the check. Femke never sees the link.

Femke's own account shows what a badge should say. When Femke sold the bike through Safe Deal on 8 March, Adyen had already verified her as a payee, because Adyen must do so before it pays anyone out. In 2027 that fact lived only in Adyen and in a Safe Deal table. Under Veldmark ID it arrives as a `payee.verified` claim with Adyen as its source, and Tradeyard shows buyers the badge "Payouts verified by our payment partner". It does not show "Verified". The difference is the point.

Trade-offs

When to ask for identity

Approach	Conversion	Fraud resistance	Data held	When it is right
Identity at registration (everyone proves themselves)	Lowest; every casual seller pays the cost	High, but uniform	Most	Small, high-value, high-risk markets (rentals in a fraud-heavy city; high-value equipment)
Identity by action (step-up at risky moments)	High for the many; friction for the few	High where it matters	Proportional	The default for horizontal and vertical marketplaces

Approach	Conversion	Fraud resistance	Data held	When it is right
Identity by incentive (badges and ranking benefits for verified users)	High	Moderate; fraudsters skip it	Proportional	Complement to step-up; where buyers can choose verified sellers
No identity beyond e-mail	Highest	Low; Fast Mail for Fraudsters	Least	Never at scale for market-places with off-platform payment

Build, buy or federate

Capability	Build	Buy	Comment
Authentication and sessions	Often, on a standard OIDC server	Customer identity platforms	Either works; the group must own the account data model and migration
Passkeys	Use platform standards (WebAuthn)	Included in most identity platforms	Do not build cryptography
Document and liveness checks	Rarely	Usually, from more than one vendor	Keep vendor neutrality through the claim contract
Bank eID and wallet adapters	Thin adapters	Aggregators exist per region	Per-country contracts and fees
Claim store, levels, badges	**Build**	—	This is the group's trust vocabulary; no vendor knows the policies

Capability	Build	Buy	Comment
Business registry	**Build**, with register data feeds	Register data from providers	Joins identity with Private Sellers, Dealers and Agencies: Modeling Who Lists organizations

Vertical Differences

Topic	General classifieds	Real estate	Automotive	Jobs (where relevant)
Who lists	Mostly private sellers (78% of Tradeyard's new listings)	Mostly agencies via feeds (84% at Hearthly); private landlords are the risk	Mostly dealers (91% at Motorvane) via DMS feeds	Employers and recruiters
Main identity risk	Fake buyers and phishing in messaging; account takeover of aged accounts	Fake landlords and stolen photos; fake agencies	Deposit scams by fake private sellers; dealer accounts of closed businesses	Fake employers harvesting candidate data or charging fees
Key check	Phone and passkey for many; L4 for payees and high-risk categories	B2 or B3 for agencies; L4 for private landlords in risky markets	B2 with dealer contract; vehicle registration document for private sellers in high-value bands	B2 for employers; domain-verified e-mail for recruiters

Topic	General classifieds	Real estate	Automotive	Jobs (where relevant)
Badge buyers understand	"Identity verified", "Payouts verified"	"Registered agency", professional licence where it exists	"Approved dealer", "Seller identity verified"	"Verified employer"
Sensitive data on the other side	Low	**High**: tenant application files with income and identity documents (Hearthly's rental services)	Medium: financing applications	**High**: CVs

On a rental portal, identity risk runs in both directions. A fake landlord defrauds tenants, and a fake "tenant" or a careless agency can harvest tenants' identity documents from application files. Hearthly's tenant application files therefore carry their own access rules: the tenant shares a file with a specific listing, the agency sees it for a limited time, and every view is logged. Identity for the person *receiving* sensitive data matters as much as identity for the person posting a listing.

Failure Modes

1. **"Verified" means five things**. A green tick for phone confirmation in one brand, for an ID check in another and for a dealer contract in a third. Buyers cannot tell, and the platform cannot defend any of them.
2. **Badges that outlive their claims**. A dealer's "approved" badge after the business closed; an "identity verified" badge after an account takeover. Badges must follow claim status and posture changes in near real time.
3. **SMS as proof of person**. A phone confirmation proves short-term control of a number. Treating it as identity hands the green tick to SIM farms, as the March 2027 scammer showed.
4. **A strong login with a weak recovery**. Passkeys for login, but support agents restore accounts after a few questions. The attacker uses the support line.
5. **Merging accounts by e-mail string**. A group identity migration that links brand accounts because the e-mail addresses match invites pre-hijacking and silently grants one person's history to another.

6. **Evidence everywhere**. Passport images in support tickets, in a vendor portal, in a data warehouse table "for analysis". Each copy is a breach waiting to happen and a GDPR minimization failure.
7. **Verification without measurement**. No one tracks drop-out at the verification step, fraud rates per level, or false rejections by document type and country. A check that rejects honest users from one country more often than others is a fairness problem as well as a conversion problem.
8. **Vendor lock-in through the data model**. Policies written as "requires vendor X check" and vendor case IDs used as user attributes. Changing vendor becomes a migration.

The Chief Architect's View

Decide the vocabulary centrally, the thresholds locally. The group should own the claim types, assurance levels, identity levels and badge catalog, the way it owns the listing event contract. Brands and countries decide which actions require which level, within group guardrails, because a bike market in Rotterdam and a rental market in Barcelona carry different risks. Aisha's trust-and-safety team proposes level policies; product owns the friction; Lucía signs off the data each policy collects.

Treat identity as tier-zero infrastructure. When Veldmark ID is down, nobody can log in to any brand. It needs the reliability engineering of the search platform, not of an internal tool: multiple availability zones, degraded modes (existing sessions keep working; new logins queue), and a tested regional failover. Sunday Evening Peaks: Reliability, Performance and Cost covers the patterns.

Make separability a requirement, not a hope. Pairwise identifiers, controller mapping and brand export are cheap to build at the start and expensive to retrofit. A private-equity owner will eventually ask what it costs to sell one brand. The identity service should have an answer measured in weeks.

Budget for verification as a variable cost. Document checks and bank eIDs are priced per use. A policy that requires L4 for an action has a cost per listing or per payout that belongs in the business case of that policy, next to the fraud losses it prevents and the honest users it loses.

Watch four numbers. Account-takeover rate per 100,000 active accounts. Share of active accounts at L3 or above (passkey coverage). Drop-out rate at each verification step, by country and method. Fraud reports per 1,000 contacts, by the sender's identity level. If fraud does not fall as levels rise, the levels are measuring the wrong thing.

Keep the EU Digital Identity Wallet on the roadmap, not on the critical path. Build the adapter when wallets are available in a market where Veldmark operates and the fees are known. Do not design a policy that depends on wallet take-up nobody can forecast yet.

Questions to Ask

1. What exactly does each badge in each brand promise, and could a buyer read it as "safe to pay"?
2. Which actions require which identity level, and who approved each requirement and its cost?
3. Are identity levels derived from claims with expiry and revocation, or stored as a flag someone set?
4. What assurance does account recovery require, and is it at least the assurance the account had?
5. What share of active accounts can be phished by password today, and what is the plan to reduce it?
6. Where are identity documents and selfies stored, by whom, for how long, and who can see them?
7. If we change our document-verification vendor, which policies, tables and screens change?
8. How do we link accounts across brands, and could an attacker who registered first with a victim's e-mail inherit the victim's account?
9. Which business claims do we re-check, how often, and what happens to listings when a check fails?
10. If one brand were sold next year, how would its users and their claims leave, and how long would it take?

From Identity to Obligation

Identity answers who is on the other side. Much of what this chapter built, the trader record, the business registry, the evidence trail and the minimization rules, exists because the law now asks marketplaces to know their traders and to explain their decisions. Rules of the Marketplace: DSA, DAC7 and Vertical Regulation takes up those rules directly: the Digital Services Act, seller tax reporting under DAC7, ranking and consumer transparency, and the vertical rules for property and car listings, and shows how to turn them into system requirements rather than annual projects.

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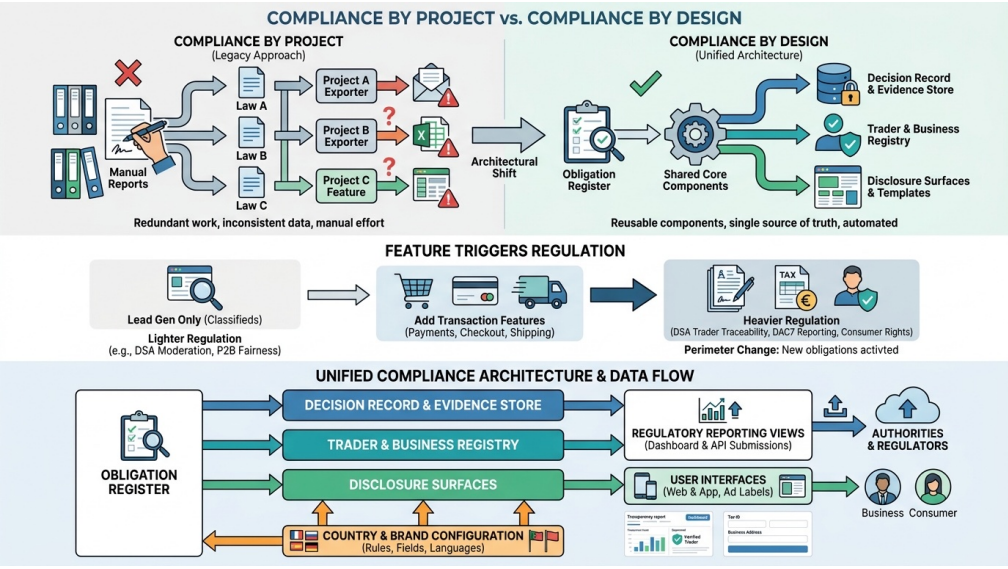
Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

27. Rules of the Marketplace: DSA, DAC7 and Vertical Regulation



IN THIS CHAPTER, YOU WILL: Work out which EU and national rules apply to which parts of a listing marketplace, and why the answer changes when the marketplace adds payments. You will turn the Digital Services Act, the platform-to-business and consumer rules, GDPR, DAC7 seller reporting and vertical disclosure rules into system requirements, and follow Veldmark's statement-of-reasons and DAC7 pipelines from decision to submission.

WHY THE BUSINESS CARES: Fines under the Digital Services Act can reach 6% of worldwide turnover, and consumer-law fines for widespread infringements at least 4% of turnover in the countries concerned. The larger cost is slower: every new rule handled as a one-off project consumes the roadmap, and a feature such as in-platform payment can quietly switch on obligations that nobody budgeted.

WHY ARCHITECTS SHOULD CARE: Regulation now specifies data models, event flows and retention periods: what a decision record must contain,

how long trader data may be kept, what must be shown on a listing page, and which numbers must be reproducible a year later. Those requirements cut across insertion, ranking, ads, messaging, payments and data, and they differ by country.

KEY POINTS:

- **The marketplace's features decide which laws apply**. A pure lead-generation classifieds site sits inside some rules (DSA moderation duties, ranking transparency, platform-to-business fairness) and largely outside others (trader traceability as an online marketplace, DAC7). Adding in-platform payment, checkout or shipping moves it inside. Every transaction feature is also a regulatory decision, and should be reviewed as one.
- **Most obligations ask for the same few artifacts**. One decision record feeds statements of reasons, the Transparency Database, appeals and transparency reports. One trader record serves DSA traceability, platform-to-business duties, consumer disclosures and DAC7. A small set of disclosure surfaces covers ranking, advertising and trader information. Build these once, as components, and treat each law as a view of them.
- **Compliance by design means evidence by design**. Regulators ask what happened, months later, and expect the same answer twice. Keep an obligation register linked to controls and evidence queries, make every regulated number computable from the event log, express country differences as configuration, and retain each data class for exactly as long as the law requires, then delete it.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Moderation as an Afterthought** — when statements of reasons and transparency reports are built as exporters after the fact, each brand produces different numbers and none can be reproduced.
- **Multiplier Creep** — paid bumps and top positions that are not labeled and not described in the ranking disclosures are now a consumer-law infringement as well as a relevance problem.
- **The Copy-Paste Country** — implementing each country's rules in a separate code path multiplies every later change; country differences belong in configuration over shared components.

Until about 2020, regulation reached most classifieds architects as a legal review at the end of a project: a privacy notice, a cookie banner, a clause in the terms. That is no longer possible. The European Union's **Digital Services Act** (DSA), applicable

to all platforms since 17 February 2024, specifies what a moderation decision record must contain, how quickly notices must be handled and what must be submitted to a Commission database. **DAC7**, the EU directive on platform seller reporting, specifies a quarterly per-seller aggregation of payments. Consumer law specifies what a search results page must explain about its ranking. Energy-performance law specifies fields on property listings. **The regulation is now partly written in the vocabulary of systems design**, and the architect who waits for legal review will rebuild the same thing four times.

This chapter maps the regulations that shape listing marketplaces and turns them into requirements. It is not legal advice, and it does not replace counsel; national transposition, guidance and case law change the details. It concentrates on the EU, where Veldmark operates, and treats the UK and US rules more briefly for groups with brands there. Its claim is that the obligations look numerous but ask for a small number of architectural artifacts, and that **compliance by design** (building those artifacts as reusable components with evidence built in) is cheaper than compliance by project.

Part V treats trust as a shared platform, and this chapter explains why the law now shapes that platform. The decision record from Trust and Safety as a Platform and the trader record from Who Is on the Other Side? Identity and Verification are the artifacts most of these rules ask for, and the labeling and tax rules set here bind the payments, paid products and advertising of Part VI. Neighboring chapters own the machinery this chapter regulates. Trust and Safety as a Platform owns the moderation pipeline, the decision service and enforcement propagation. Who Is on the Other Side? Identity and Verification owns identity levels, KYB and the business registry. Beyond the Contact: Payments, Shipping and Buyer Protection owns the payment flows that bring DAC7 into scope, and Ranking for Two Sides: Relevance, Freshness and Paid Visibility and Selling Attention: Display, Native and Retail-Media Advertising own the systems whose disclosures this chapter specifies. The running example is Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build).

Setting the Scene at Veldmark: Five Exporters and One Report

In **February 2027**, Lucía Ortega, Veldmark's Group Data Protection Officer and regulatory lead, had a deadline. Under the Commission's implementing regulation on DSA transparency reports, the first full harmonized reporting period ran from 1 January to 31 December 2026, and reports were due no later than two months after the period ended (Commission Implementing Regulation (EU) 2024/2835). For Veldmark that meant one report per online platform, Tradeyard, Motorvane and Hearthly, on the Commission's template, by the end of February.

Lucía collected the inputs from five places, because each brand and country produced its own statements of reasons: Yardstick's template for Tradeyard Netherlands and Belgium, two vendor exports for Poland and Portugal, Motorvane's hand-built tool and two Hearthly exporters. The numbers did not agree with themselves. Tradeyard Netherlands had submitted about **20,000** statements of reasons a week to the DSA Transparency Database from listing removals alone, yet its moderation log counted removals differently: an edit that was rejected and then accepted counted once in one system and twice in the other. Motorvane counted "dealer listing paused for missing data" as a restriction in one quarter and not in the next. Hearthly Spain had no record of whether its decisions were automated.

Ingrid Halvorsen, who had joined as Group Chief Architect that month, read the draft report and asked two questions. *If the Dutch regulator asks us next year how we got this number, can we produce it again from the same data?* And *what else are we obliged to do that we have not built at all?* Lucía's answer to the first was no. Her answer to the second was a list, which became the obligation register later in this chapter. The shared trust-and-safety platform that went live for Tradeyard Netherlands and Belgium in November 2027 was designed against that list.

Concepts: Who Is the Marketplace in Law?

Legal roles are defined by features

EU digital regulation does not regulate "classifieds sites". It regulates roles, defined by what a service does. One marketplace can hold several roles at once, and each role carries its own obligations.

Role	Defined in	What triggers it	Example at Veldmark
Hosting service	DSA Art. 3	Storing information provided by users	Every brand; Keybridge too, for agency data
Online platform	DSA Art. 3	Hosting that also disseminates user information to the public	Tradeyard, Motorvane, Hearthly

Role	Defined in	What triggers it	Example at Veldmark
Platform allowing consumers to conclude distance contracts with traders	DSA Section 4 (Arts. 29–32)	Consumers can contract with traders through the platform	Tradeyard (pro sellers, Safe Deal); arguably Motorvane and Hearthly for trader listings
Very large online platform	DSA Art. 33	45 million or more average monthly active EU recipients, and designation	None (no classifieds marketplace is designated as of October 2026)
Online intermediation service	P2B Regulation Art. 2	Lets business users offer to consumers "with a view to facilitating the initiating of direct transactions … irrespective of where those transactions are ultimately concluded"	All three consumer brands, for dealers, agencies and pro sellers
Online marketplace	Consumer Rights Directive and UCPD, as amended by the Omnibus Directive	Lets consumers conclude distance contracts with traders or consumers	Tradeyard Safe Deal; contestable for lead-only flows
Reporting platform operator	DAC7 (Directive (EU) 2021/514)	Connects sellers with users for relevant activities, beyond only listing or advertising	Tradeyard, for Safe Deal transactions only
Controller	GDPR	Determining the purposes and means of processing personal data	Each brand's operating entity

Sources: DSA; P2B Regulation (EU) 2019/1150; Omnibus Directive (EU) 2019/2161; DAC7, Council Directive (EU) 2021/514; the list of designated very large online platforms (European Commission, updated 15 October 2026).

The table holds the most important architectural fact in this chapter: some definitions explicitly cover **lead generation**, and others turn on a contract concluded through the platform. The platform-to-business (P2B) Regulation applies "irrespective of where those transactions are ultimately concluded", and the Unfair Commercial Practices Directive's (UCPD) ranking-information rule, inserted by the Omnibus Directive, applies to search results "irrespective of where transactions are ultimately concluded" (Directive (EU) 2019/2161, Art. 3). The definitions that turn on whether **a contract is concluded through the platform** are the DSA's trader-traceability section, the Consumer Rights Directive's marketplace information duties and, through its definition of "platform", DAC7, which excludes software that "exclusively" allows users "to list or advertise" a relevant activity or processes payments without further intervention (Council Directive (EU) 2021/514, Annex V, Section I).

My working rule as an architect is to **assume the contract-based rules apply to every listing published by a trader**. Whether a pure classifieds listing with a contact button "allows consumers to conclude distance contracts with traders" is a question for counsel, but building trader traceability for dealers and agencies costs little and being wrong costs much. The second part of the rule is to **treat every transaction feature as a perimeter change** that triggers a regulatory review before launch. Safe Deal, shipping labels, deposits, rental applications with payment and online car reservation all qualify.

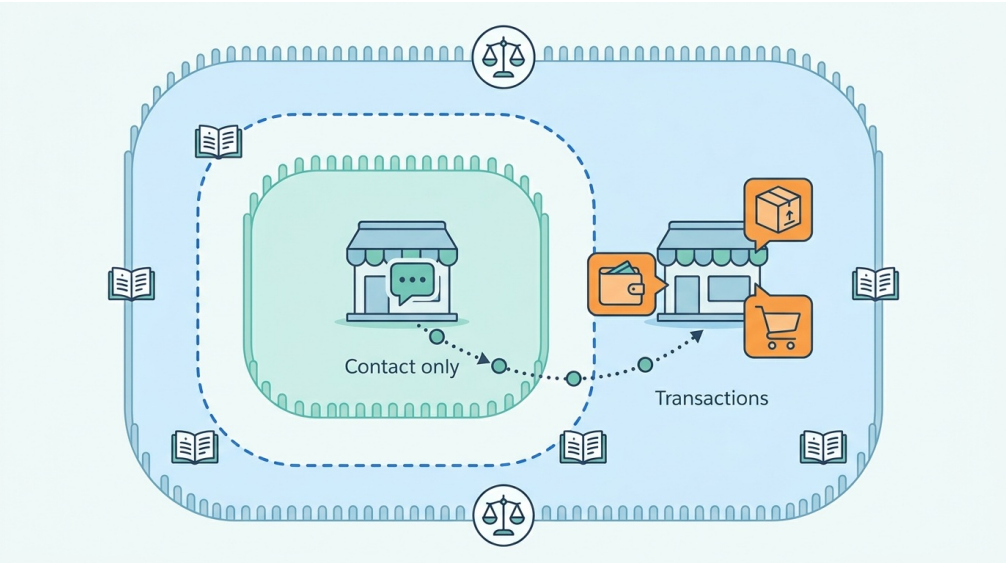


Figure 1: *Lead generation sits inside some rules; adding payment, checkout or shipping moves the marketplace across the perimeter into trader traceability, marketplace duties*

and DAC7, so every transaction feature is also a regulatory decision.

Small-company exemptions do not help a group

The DSA exempts micro and small enterprises from most online-platform duties and from the marketplace section (Arts. 19 and 29), and the P2B Regulation exempts small enterprises from its complaint-handling system (Art. 11(5)). Both refer to Commission Recommendation 2003/361/EC, whose linked-enterprise rules make a small marketplace owned by a large group part of that group. **A newly acquired local marketplace should be assumed to lose any exemption at acquisition**, and its compliance gap belongs in the integration plan.

The obligation, the control and the evidence

Three terms make regulation tractable for architects:

- An **obligation** is a duty stated in a legal source, with an article reference, a jurisdiction and applicability conditions: "DSA Art. 17(1): provide a statement of reasons for any restriction of visibility of a recipient's information, where electronic contact details are known."
- A **control** is the system behavior that meets it: "The decision service emits a statement-of-reasons event for every decision with a visibility, monetary, provision or account restriction."
- **Evidence** is the data that proves the control worked: "For any period, the count of restriction decisions equals the count of statements sent plus the count of documented exceptions, with the query that computes both."

Compliance by design is the practice of building controls into the platform's components, with the evidence query written at the same time as the control. **Compliance by project** builds an exporter to satisfy each new obligation. The first is more expensive for the first law and cheaper for every law after it.

The Digital Services Act as a Set of System Requirements

The DSA is the regulation that most changes listing-system architecture, because it regulates the moderation process itself. The table maps the articles that bind a non-designated online platform with trader listings to the components that implement them. Article text was checked against the Official Journal version of Regulation (EU) 2022/2065.

Article	Obligation in brief	System requirement	Owning component
14 Terms and conditions	Describe content-moderation policies, procedures and tools, "including algorithmic decision-making and human review", in clear language, publicly and in machine-readable form	Policy registry publishes user-facing policy text per brand and country; versioned	Policy registry (Trust and Safety as a Platform)
15 and 24(1) Transparency reports	Annual reports on orders, notices, own-initiative moderation, automated tools with accuracy indicators, complaints, out-of-court disputes and suspensions; Commission templates	All figures computed from decision and notice events, by reporting period, reproducibly	Reporting views over the evidence store

Article	Obligation in brief	System requirement	Owning component
16 Notice and action	Electronic notice mechanism; required notice elements; confirmation of receipt; decision notified with redress options; timely, diligent, non-arbitrary handling; disclose automated means	Notice as an entity with lifecycle and notifier communications	Case management
17 Statement of reasons	Clear, specific reasons to the affected user for removal, disabling, **demotion**, restriction of payments, suspension of service or account; facts, automation, legal or contractual ground, redress	Every restricting decision produces a statement from the decision record	Decision service; statement-of-reasons service

Article	Obligation in brief	System requirement	Owning component
20 and 21 Complaints and disputes	Free, electronic complaints for at least six months after the decision; reversal of unfounded decisions; decided under qualified staff supervision, "not solely on the basis of automated means"; escalation to certified out-of-court bodies	Appeals linked to original decisions; human decision step; evidence kept through the window; case export for dispute bodies	Appeals service
22 Trusted flaggers	Notices from flaggers designated by Digital Services Coordinators get priority	Registry of trusted flaggers from the Commission's database; priority queue; flagger quality metrics	Case management
23 Misuse	Suspend, after a warning, users who frequently post manifestly illegal content, and notifiers who frequently send manifestly unfounded notices	Strike counters per account and notifier; warnings; policy in the terms	Decision service
24(2) Active recipients	Publish average monthly active EU recipients at least every six months	A defined, audited metric from the data platform	Data Platform for Marketplaces: Events, Lakehouse and Data Products

Article	Obligation in brief	System requirement	Owning component
24(5) Transparency Database	Submit statements of reasons "without undue delay", without personal data	PII-free submission pipeline with idempotency	Statement-of-reasons service
25 Interface design	No interfaces that deceive or manipulate users or impair free and informed decisions	Design review for flows such as cancellation, consent and upsell	Product design governance
26 Advertising	Mark each ad in real time; show on whose behalf it is presented and who paid; meaningful information on targeting parameters; no profiling ads based on special-category data	Paid placements, including sponsored listings, carry labels and "why am I seeing this"	Selling Attention: Display, Native and Retail-Media Advertising; listing rendering
27 Recommender transparency	Main parameters of recommender systems and options to change them, in the terms	Ranking and recommendation parameter documentation per surface; user-selectable sort options	Ranking for Two Sides: Relevance, Freshness and Paid Visibility, Recommendations for Inventory That Disappears

Article	Obligation in brief	System requirement	Owning component
30 Trader traceability	Collect and check trader identity, ID or electronic identification, payment account, register number and self-certification before trading; correct or suspend; keep six months after the relationship ends, then delete; display name, address, contact, register and self-certification to users	Business registry with checks and lifecycle deletion; trader panel on listings	Who Is on the Other Side? Identity and Verification
31 Compliance by design	Interface lets traders provide product identification, trader marks, and labeling and product-safety information; best efforts to check before offers go live; reasonable efforts at random checks against official databases	Listing schema fields for safety and labeling; random-check job	Attributes Without Chaos: Schemas, Dictionaries and Validation; trust and safety

Article	Obligation in brief	System requirement	Owning component
32 Right to information	When an illegal product or service was sold, inform consumers who bought it in the last six months, or publish a notice	Record which consumers bought or contacted per listing; notification job	Payments and messaging

Three entries in the table catch ranking, payments and seller-data teams by surprise.

Demotion is a restriction. Article 17 lists "demoting content" among the restrictions that need a statement of reasons, and the Transparency Database has a decision value for it, `DECISION_VISIBILITY_CONTENT_DEMOTED` (DSA Transparency Database API documentation). A ranking feature that quietly pushes suspected scam listings down the results is a moderation decision. Trust and Safety as a Platform puts such demotions behind the decision service for that reason.

Payment holds are restrictions too. "Suspension, termination or other restriction of monetary payments" is in the same list. A Safe Deal payout held by fraud rules is a decision that needs a statement of reasons, unless it falls under a different legal regime the platform can name.

Trader traceability is a data lifecycle, not a form. Article 30 requires the platform to store the trader information "for a period of six months after the end of the contractual relationship" and then to delete it (Regulation (EU) 2022/2065, Art. 30(5)). That is a deletion obligation with a trigger (the relationship ends) and a timer, and it conflicts with any team that keeps dealer documents "just in case".

The statement of reasons as a data contract

The Commission's Transparency Database is the most concrete regulatory interface a classifieds architect will build against. Its API accepts single statements or batches of up to **100** per call, and a batch fails as a whole if any statement fails validation. Required fields include the decision ground, content type, category, content and application dates, the facts relied on, the source of the decision, whether detection and decision were automated, the territorial scope and a **platform unique identifier** (`puid`) of up to 500 characters, and at least one restriction field. The vocabulary is fixed: restriction values such as `DECISION_VISIBILITY_CONTENT_REMOVED` or `DECISION_ACCOUNT_SUSPENDED`; grounds `DECISION_GROUND_ILLEGAL_CONTENT` or `DECISION_GROUND_INCOMPATIBLE_CONTENT`; sources from `SOURCE_ARTICLE_16` to `SOURCE_VOLUNTARY`; automation from

`AUTOMATED_DECISION_FULLY` to `AUTOMATED_DECISION_NOT_AUTOMATED`; and 16 categories, among them `STATEMENT_CATEGORY_SCAMS_AND_FRAUD` and `STATEMENT_CATEGORY_UNSAFE_AND_PROHIBITED_PRODUCTS` (DSA Transparency Database API documentation, n.d.). Submissions must not contain personal data (DSA Art. 24(5)).

The database operates at scale: for the 180 days to 19 October 2026, the database held about **4.4 billion** statements from 375 platforms, 41% of them fully automated decisions (DSA Transparency Database, accessed 19 October 2026).

> **INFERENCE, NOT DOCUMENTED:** The database's schema is, in effect, a regulator-defined moderation event model. A platform whose internal decision record cannot be mapped field by field onto it will end up maintaining a translation layer that loses information. The cheaper design is to adopt the regulatory vocabulary for the fields it covers and extend it for internal needs.

Veldmark's decision record, owned by the shared trust-and-safety platform, maps to the statement as follows:

Decision record field (internal)	Statement field (Transparency Database)	Transformation
`decision_id`	`puid`	Identity; the idempotency key for resubmission
`action` (remove, disable, demote, label, hold payout, suspend account)	`decision_visibility`, `decision_monetary`, `decision_provision`, `decision_account`	Mapping table, one action to one or more values
`policy_id` and `policy_version`	`incompatible_content_ground` or `illegal_content_legal_ground`	Policy registry supplies ground text per language
`reason_facts` (templated, may contain names)	`decision_facts`	PII stripping: names, phone numbers, addresses, IBANs removed or generalized

Decision record field (internal)	Statement field (Transparency Database)	Transformation
`source` (user notice, trusted flagger, detector, sample)	`source_type`	Mapping
`detected_by` and `decided_by`	`automated_detection`, `automated_decision`	Derived from detector and reviewer fields
`territorial_scope`	`territorial_scope`	Country overlay of the policy
`seller_type`	`account_type`	`ACCOUNT_TYPE_BUSINESS` or `ACCOUNT_TYPE_PRIVATE`, from trader status
`content_created_at`, `applied_at`	`content_date`, `application_date`	Dates

Veldmark's statement-of-reasons pipeline

The pipeline that replaced the five exporters has four stages (look up the ground text, send the user's statement and store it as evidence, strip personal data, submit in batches keyed by decision ID), and one rule: **the user's statement and the database submission are two views of one decision, generated from the same record**.

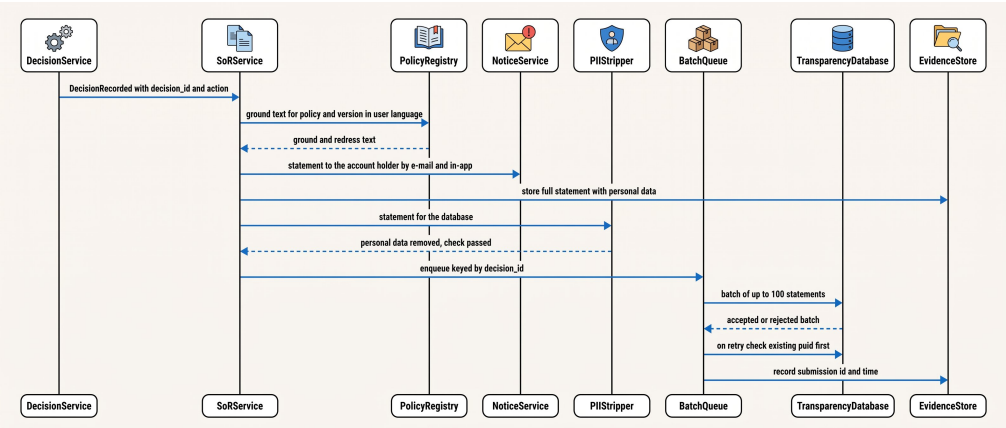


Figure 2: *One decision, two statements. The user receives the full statement with redress options; the Commission's database receives a version stripped of personal data, batched and keyed by the decision ID, so retries never create duplicates and every submission can be traced back to its decision.*

Throughput is not the hard part. In 2027 Veldmark's three consumer platforms generated about **61,000** restricting decisions a week: listing removals and demotions, message withholdings, payout holds and account suspensions, with Tradeyard Netherlands' roughly 20,000 weekly listing removals the largest single source. At 100 statements per batch that is only about 610 API calls a week. Three other things are hard:

1. **Atomic batch failure.** One malformed statement fails 99 good ones. Veldmark validates every statement against the published schema before batching and quarantines failures for repair, so a template bug in one country does not stall the others.
2. **Personal data in free text.** Reviewers' notes and templated facts contain names, phone numbers and addresses. The stripper combines pattern detection for phone numbers, e-mail addresses, IBANs and postcodes with a rule that `decision_facts` may only be assembled from policy-registry templates and structured fields, never from free reviewer text.
3. **Exceptions are decisions too.** The DSA does not require a statement where the platform has no electronic contact details for the user, or for "deceptive high-volume commercial content" (Regulation (EU) 2022/2065, Art. 17(2)). Veldmark's policy registry defines which bot-flood rejections count as the second case, and the decision record stores the exception and its reason, so the count of decisions still reconciles with the count of statements.

One decision shows both views. At 20:26 on Sunday 7 March 2027, Tradeyard Netherlands suspended the account that had sent Femke a fake Safe Deal payment link (the incident is told in Trust and Safety as a Platform). Under the 2027 pipeline, the decision record carries `action = suspend_account`, policy `POL-FRAUD-PAYLINK` version 2, source "user report under the terms", detection not automated, decision partly automated (an analyst confirmed a rule hit), scope NL. The account holder receives a statement in Dutch citing the contractual ground and the complaint route. The database receives `DECISION_ACCOUNT_SUSPENDED`, `DECISION_GROUND_INCOMPATIBLE_CONTENT`, `STATEMENT_CATEGORY_SCAMS_AND_FRAUD`, `SOURCE_TYPE_OTHER_NOTIFICATION`, `AUTOMATED_DECISION_PARTIALLY`, territorial scope `NL`, and facts that describe "a message containing a link to a website imitating the platform's payment service", with no domain name, user name or phone number. The domain, the message and the analyst's identity stay in the evidence store, where an appeal, a police request or a regulator can reach them.

Transparency reports from the event log

The implementing regulation fixes the reporting period for non-designated platforms at the calendar year, requires publication within two months of its end, and requires the reports to stay publicly available for at least five years (Commission Implementing Regulation (EU) 2024/2835, Arts. 2 and 3). Its templates ask, among much else, for counts of notices by category and by trusted flaggers, actions by legal and contractual basis, median times, complaints and reversals, and a qualitative description of automated tools with indicators of their accuracy.

The rule I would apply is that **every number in a regulatory report is a saved query over immutable events**, versioned with the report. Veldmark's report for 2027, published in February 2028, was the first produced this way: the evidence store held decision, notice, complaint and enforcement events with their policy and model versions, and each template cell was generated by a named query whose definition was published internally with the report. The answer to Ingrid's February question changed from "no" to "yes, here is the query and the snapshot it ran on".

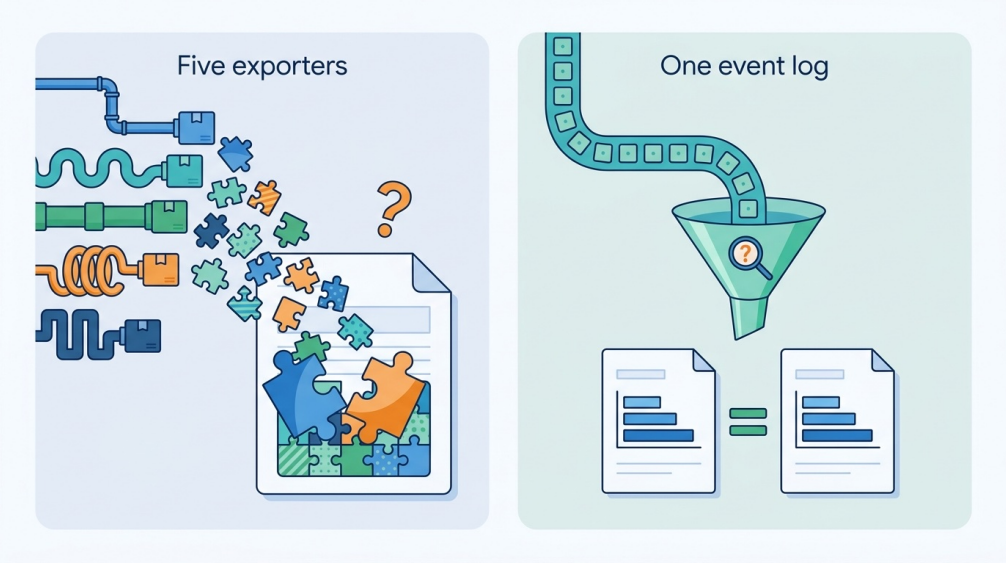


Figure 3: *In February 2027 five exporters produced numbers that did not agree; computed as saved queries over one event log, the 2027 report gives the same answer every time it is run.*

Ranking, Advertising and Consumer Transparency

Four overlapping sources regulate how a marketplace orders and labels what users see. Read together, they ask for three disclosure surfaces.

Source	What it requires	Who it protects
P2B Regulation Art. 5	In the terms for business users: the main ranking parameters and the reasons for their relative importance; where business users can influence ranking against payment, a description of those possibilities and their effects; no obligation to disclose algorithms or information that would enable manipulation	Dealers, agencies, pro sellers
UCPD Art. 7(4a) (Omnibus)	For search results from different traders or consumers, general information on the main ranking parameters and their relative importance, in a section directly reachable from the results page, "irrespective of where transactions are ultimately concluded"	Consumers
UCPD Annex I, point 11a (Omnibus)	Blacklisted: search results that do not clearly disclose paid advertising or payment specifically for higher ranking	Consumers

Source	What it requires	Who it protects
DSA Arts. 26 and 27	Real-time ad labels with advertiser, payer and targeting information; recommender parameters and options in the terms	All users
Consumer Rights Directive Art. 6a (Omnibus)	On online marketplaces, before a consumer is bound: ranking parameters; whether the seller is a trader, based on the seller's declaration; that consumer rights do not apply if not; how obligations are split	Consumers in marketplace contracts

Sources: Regulation (EU) 2019/1150; Directive (EU) 2019/2161; Regulation (EU) 2022/2065.

The three surfaces that satisfy these rules are:

1. **A ranking explanation** per search surface and sort option, in consumer language, reachable from the results page, generated from the same parameter documentation that the ranking team maintains, and a longer business-user version in the professional terms. It must say how paid products (bumps, top positions, depth products) affect order. Ranking for Two Sides: Relevance, Freshness and Paid Visibility owns the parameters; this chapter owns the obligation that they be described.
2. **Labels on every paid placement**: sponsored listings, top-ad slots, highlighted listings that are boosted, and display ads, each with "on whose behalf" and "why am I seeing this". Whether a private seller's €4.99 bump is an "advertisement" in the DSA sense is a question for counsel. Under the UCPD blacklist, an undisclosed paid ranking boost is unlawful either way, so Veldmark labels every paid boost. That is also the strongest structural defense against **Multiplier Creep**: a boost that must be labeled is a boost someone has to justify.
3. **Seller status on every listing**: private or trader, from the seller's declaration, with the trader information panel that DSA Art. 30(7) requires for traders and the consumer-rights notice that CRD Art. 6a requires on marketplaces. One field, set at account level and confirmed per listing, drives all three. Private Sellers, Dealers and Agencies: Modeling Who Lists owns trader status.

The P2B Regulation adds two process rules for business users that belong in the professional-seller platform. Changes to the terms must be notified at least 15 days before they apply, and terminating a business user's whole service requires a statement of reasons at least 30 days in advance, except for repeated infringements or legal obligations (Regulation (EU) 2019/1150, Arts. 3 and 4). For Motorvane, terminating a dealer for unpaid invoices is therefore a workflow with a notice timer, not a switch an account manager flips. In November 2025 the Commission proposed repealing the P2B Regulation, citing its overlap with the DSA and the Digital Markets Act, and says its ranking guidelines would remain in application (European Commission, updated 2 July 2026). Until the repeal is adopted, the rules apply, and the same capabilities are needed for the DSA anyway.

Price reductions and reviews

Two Omnibus rules reach listing features that rarely get legal review.

Price-reduction announcements. The Omnibus Directive inserted into the Price Indication Directive a rule that any announcement of a price reduction must show the prior price, meaning the lowest price the trader applied in a period of at least 30 days before the reduction (Directive (EU) 2019/2161, Art. 2). The obligation is the trader's, and Member States may set different periods. A portal that automatically shows "price reduced" badges on dealer listings is presenting that announcement on the trader's behalf. When the Essen Volkswagen Golf's price fell from **€21,900** to **€21,400** on Thursday 22 April 2027, Motorvane's badge needed the lowest price in the previous 30 days from the listing's price history, not the immediately preceding price. The difference matters only for a dealer who raises a price briefly before "cutting" it, which is exactly the case the rule exists for. The price history in The Listing Aggregate: Identity, Lifecycle and Invariants has to keep every price with its effective time for this reason.

Consumer reviews. Where a trader gives access to consumer reviews, the UCPD treats as material whether and how it ensures that reviews come from consumers who actually used or bought the product, and blacklists claiming that reviews are genuine without "reasonable and proportionate steps" to check, as well as submitting or commissioning fake reviews (Directive (EU) 2019/2161, Art. 3). FINN shows one documented design: ratings between buyer and seller are given after a deal, and both sides' ratings are published at the same time, or after three weeks, so neither party sees the other's rating before giving its own (FINN help center, 2024, updated 2026). Veldmark ties Tradeyard ratings to a completed Safe Deal or a conversation with a confirmed handover, and logs the check as evidence of the steps taken.

GDPR: Retention, Minimization and Address Privacy

For listing systems, three principles of the General Data Protection Regulation (GDPR) produce the most architecture: **data minimization** (personal data "limited to what is

necessary"), **storage limitation**, and **accountability**, the duty to be able to demonstrate compliance (Regulation (EU) 2016/679, Art. 5).

The design that works is **retention per data class**, because other laws require keeping data the GDPR would have the platform delete, while fraud prevention wants to keep it longer still. The GDPR's recitals recognize processing "strictly necessary for the purposes of preventing fraud" as a legitimate interest (recital 47), which supports fraud retention but does not make it unlimited. Each class gets a stated legal basis, a trigger and a timer, enforced by deletion jobs whose runs are themselves logged.

Data class (Veldmark)	Legal driver	Trigger and retention	Notes
DSA trader-traceability data	DSA Art. 30(5)	Six months after the trader relationship ends, then delete	Deletion is mandatory, not optional
Moderation evidence (content snapshot, rule and model versions, reviewer)	DSA Arts. 17, 20 and 21; defense of claims	At least six months after the user is informed (complaint window), extended while an appeal, out-of-court dispute or legal claim is open	Default 12 months at Veldmark
Hashed block-list identifiers for accounts terminated for fraud	GDPR legitimate interest (fraud prevention), DPIA	Two years (Veldmark policy, see Trust and Safety as a Platform)	Hashes and decision references only
DAC7 seller data and reports	DAC7 and national transposition	National record-keeping periods apply; typically years, not months	Separate, purpose-bound tax profile

Data class (Veldmark)	Legal driver	Trigger and retention	Notes
Messages	Contract, fraud prevention	Per brand policy; shorter for content, longer for metadata linked to fraud cases	From Interest to Contact: Messaging and Lead Management

Two GDPR issues are specific to listing marketplaces. **Address privacy**: an exact address on a private seller's or landlord's listing exposes where valuables or empty homes are. Show a postcode area or a jittered map pin for private sellers by default (Where Is It? Location, Geocoding and Privacy owns precision rules). For traders, DSA Art. 30(7) requires the trader's address to be shown, which is uncomfortable for sole traders working from home; the US INFORM Consumers Act has an explicit home-address exception for some sellers, but I found no equivalent in Art. 30. **Automated decisions**: an automated account suspension can be a decision that "similarly significantly affects" a person under GDPR Art. 22, which then requires safeguards including human intervention and the right to contest. The DSA's appeal process provides much of that, and the two should be designed as one mechanism.

DAC7: When Payments Turn a Marketplace into a Tax Reporter

What DAC7 requires

DAC7 extended the EU's directive on administrative cooperation in tax to digital platforms. A **reporting platform operator** must collect information on its sellers, verify it, and report it to the tax authority each year; the authorities exchange it with the sellers' Member States of residence. The key definitions, from Annex V of Council Directive (EU) 2021/514:

- **Relevant activities** are the rental of immovable property (including parking spaces), personal services, the **sale of goods**, and the rental of any mode of transport, carried out for consideration.
- **Consideration** is compensation "net of any fees, commissions or taxes withheld or charged by the Reporting Platform Operator" that is paid or credited to a seller, "the amount of which is known or reasonably knowable by the Platform Operator".
- An **excluded seller** includes one for which the operator facilitated **fewer than 30** sales of goods **and** total consideration of **not more than €2,000** in the period. Both conditions must hold, and the exclusion applies to goods only. Large-scale

property rental entities (more than 2,000 rentals of one property listing) and listed companies are also excluded.
- For each reportable seller the operator reports identity (name, primary address, tax identification number with the issuing Member State, date of birth for individuals, VAT and business registration numbers where they exist), the financial account identifier where available, and **per quarter** the total consideration, the number of relevant activities, and the fees, commissions or taxes withheld or charged.
- Due diligence must be completed by **31 December** of the reportable period, and the report submitted by **31 January** of the following year. The first reportable period was 2023; sellers already registered on 1 January 2023 had until the end of the second period.

The definition of "platform" carries the exclusion that matters most for classifieds: it does not include software that, "without any further intervention in carrying out a Relevant Activity", exclusively processes payments, lets users list or advertise, or redirects users to a platform. A classifieds site that only lists is outside DAC7. A classifieds site that collects the price and pays the seller knows the consideration, and is inside.

Veldmark's DAC7 pipeline

Tradeyard Safe Deal brought Veldmark into DAC7. Motorvane and Hearthly only list and advertise, and Hearthly's rental services (tenant application files) do not collect or pay rent, so neither is a reporting platform operator for its classifieds. Veldmark contracts Safe Deal through Tradeyard B.V., a Dutch entity, for all three Safe Deal countries, so Tradeyard reports to the Dutch tax administration for sellers in the Netherlands, Belgium and Poland, and the authorities exchange the data onward.

The 2026 figures show why DAC7 is mostly a filtering problem. Safe Deal processed **11.6 million** transactions with a GMV of about **€1.02 billion** in 2026. Veldmark's planning estimates for the reportable population were:

Quantity (Safe Deal, 2026)	Estimate
Distinct sellers paid through Safe Deal	about 1.3 million
Sellers with 30 or more sales, or consideration above €2,000	about 21,000
Of which with complete, validated tax identification number and date of birth by 30 November	about 17,500
Reminder cycles needed for the rest	two, plus payout holds for non-responders as the terms allow

The buyer protection fee is charged to the buyer, not withheld from the seller, so it does not reduce consideration. Femke's city bike is a useful edge case. She sold it for **€225** on Monday 8 March 2027, and the buyer paid the **€11.95** fee on top. Her consideration is €225, and with one sale in the year she is an excluded seller. If she sold 30 items through Safe Deal in 2027, even small ones, she would become reportable, and Tradeyard would need her tax identification number before 31 December.

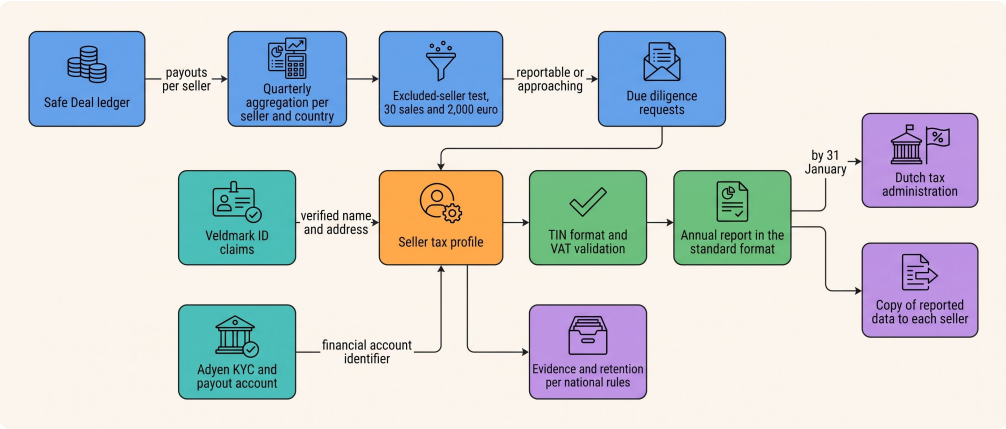


Figure 4: *Veldmark's DAC7 pipeline. The payment ledger provides the quarterly consideration; identity claims and the payment provider supply verified identity and account data; a separate tax profile collects what only tax law needs; the report goes to the tax authority and the same data goes to each seller.*

Three architectural lessons came out of building it. **Aggregate from the ledger, not from orders**, because consideration is what was paid or credited, net of fees, in the quarter it was paid. **Ask early**: the pipeline flags sellers approaching the threshold (20 sales or €1,500) so due diligence starts before the deadline rather than in December. **Keep the tax profile separate**: tax identification numbers and dates of birth are collected for one purpose, under tax law, and must not leak into the general identity claims or the analytics warehouse. Beyond the Contact: Payments, Shipping and Buyer Protection covers the ledger, and RECONCILED's Payments-to-Ledger: Draining the Suspense Swamp covers the accounting integration.

The decision to launch in-platform payments is also a tax-reporting decision, outside the EU as well as inside it: under the US INFORM Consumers Act, only sales made through the marketplace with payment processed by it count toward the high-volume seller threshold (below).

Vertical Rules: Property and Vehicles

Horizontal rules govern the marketplace. Vertical rules govern the listing, and they arrive as fields.

Energy performance in property listings

The recast Energy Performance of Buildings Directive requires Member States to ensure that buildings or units offered for sale or rent have an energy performance certificate, and that its energy performance indicator and class are "stated in online and offline advertisements, including in property search portal websites" (Directive (EU) 2024/1275, Art. 20(4)). Member States had to transpose most of the directive by **29 May 2026**. The recast names property portals explicitly, which removes any argument that the duty belongs only to the advertiser.

France's rules show what national law adds. According to the French government's public-service site, property advertisements must show both the energy class and the climate class of the *diagnostic de performance énergétique* (DPE), must label homes rated F or G as having excessive energy consumption, and must show an estimate of annual energy costs for standard use with the reference year of the energy prices; fines reach €3,000, or €15,000 for a company. Homes rated G may not be newly let since 1 January 2025, with F following in 2028 and E in 2034, and changes to the calculation from 1 January 2026 can move a home's class without any change to the building (Service-Public.fr, 2026).

The **Lyon apartment** shows the consequences for Hearthly. The 68 m² T3 in Lyon 7e, listed in May 2027 at **€389,000** with DPE class **D**, arrived through two software vendors and Keybridge. Each feed carried the energy class; one did not carry the climate class and none carried the estimated annual energy cost in a structured field. Hearthly's ingestion therefore applies four rules:

1. **DPE fields are mandatory structured attributes** for French residential listings: energy class, climate class, energy indicator, estimated annual cost range with its reference year, and certificate date.
2. **A missing field blocks publication with feedback to the feed**, rather than publishing an incomplete ad. In May 2027 about 3.8% of Hearthly France's incoming feed listings failed this check, most of them from two vendors whose schemas predated the cost field.
3. **Values carry effective dates**, because a recalculation can change the class. The listing shows the certificate's current value and keeps the history.
4. **Class constrains category**: a G-rated home cannot be published as a new long-term rental in France, so the validator rejects that combination.

Regulated attributes need an owner who tracks the law, because the schema will change on the regulator's timetable, not the product team's. Attributes Without Chaos: Schemas, Dictionaries and Validation owns the schema mechanics; Many Doors In: Forms, APIs and Bulk Feeds owns feed feedback.

Vehicle listings

Automotive rules are more fragmented, and most apply to dealers rather than to portals. In Germany, the passenger-car energy consumption labeling ordinance (Pkw-EnVKV), which transposes Directive 1999/94/EC, requires manufacturers and dealers who create or use promotional material for **new** passenger cars to state energy consumption, CO2 emissions and the CO2 class, with specific rules for material distributed in electronic form; it was last amended in March 2026 (Pkw-EnVKV, § 5, as amended 2 March 2026). For Motorvane, that means structured fields for new and near-new cars in the dealer feed, validated on ingestion and displayed in the listing, with the same "missing field blocks publication" rule as Hearthly's DPE. The Essen Golf, a 2021 car with 48,000 km, is a used car and outside that rule. Other vehicle duties (mileage, accident history, price inclusiveness) come mostly from national consumer and unfair-competition law and differ by country.

Beyond the EU: UK and US Rules in Brief

The UK and US rules differ from the EU's in structure, but they reward the same components. Veldmark operates only in the EU; many groups do not.

Rule	Scope	Key requirement	Component it reuses
UK Online Safety Act 2023, illegal-content duties	User-to-user services, including classifieds	Fraud is a priority offence (Schedule 7); risk assessments were due by 16 March 2025 and safety measures apply from 17 March 2025	Moderation pipeline, risk assessment, evidence

Rule	Scope	Key requirement	Component it reuses
UK Online Safety Act, fraudulent-advertising duties	Paid-for ads on categorized services	Prevent, minimize and remove fraudulent paid ads; Ofcom's codes consulted in July–October 2026, final statement expected by mid-2027	Ad review; whether paid listing upgrades count is an open question for counsel
UK Digital Markets, Competition and Consumers Act 2024	Consumer reviews, from 6 April 2025	Ban on fake reviews and on hosting reviews without reasonable steps to check they are genuine; the regulator can fine up to 10% of global turnover	Review anchoring and evidence
US INFORM Consumers Act	Online marketplaces with high-volume third-party sellers of new or unused consumer products	200 or more sales and $5,000 or more in 12 months, counting only marketplace-processed sales; collect and verify bank, tax ID and contact information; disclose identity for sellers above $20,000; reporting channel on listings	Business registry, payment ledger

Rule	Scope	Key requirement	Component it reuses
US Fair Housing Act advertising rule (24 CFR § 100.75)	Housing advertisements	No words, photos or targeting that indicate a preference based on protected characteristics	Listing-text classifiers; ad-delivery audits

Sources: Online Safety Act 2023, Schedule 7; Ofcom, illegal harms statement, 2024; Ofcom, fraudulent advertising consultation, 2026; Which?, 2025; US Federal Trade Commission, INFORM guidance, 2023; 24 CFR § 100.75.

INFORM repeats the DAC7 pattern. Because only marketplace-processed sales count, and because most classifieds traffic is used goods, a US lead-generation classifieds site is largely outside it. Payments bring it in.

Compliance as Architecture

Five components that serve every law

The obligations in this chapter form a long list, but they ask for only five systems.

Component	Serves	Key property
Obligation register	Every regulation	Machine-readable: obligation ID, source and article, jurisdiction, applicability conditions, control, owner, evidence query, review date
Decision record and evidence store	DSA Arts. 15–24; GDPR accountability; P2B Art. 4; Online Safety Act	Immutable, versioned, retained per data class
Trader and business registry	DSA Arts. 30–32; P2B business users; CRD Art. 6a seller status; DAC7; INFORM	One verified record, many regulatory views

Component	Serves	Key property
Disclosure surfaces	DSA Arts. 26–27 and 30(7); P2B Art. 5; UCPD Art. 7(4a) and Annex I 11a; CRD Art. 6a; energy and vehicle labeling	Generated from documented parameters and structured fields, never hand-written per page
Regulatory reporting views	DSA transparency reports and active-recipient counts; DAC7 reports	Saved, versioned queries over the event log and ledger; reproducible

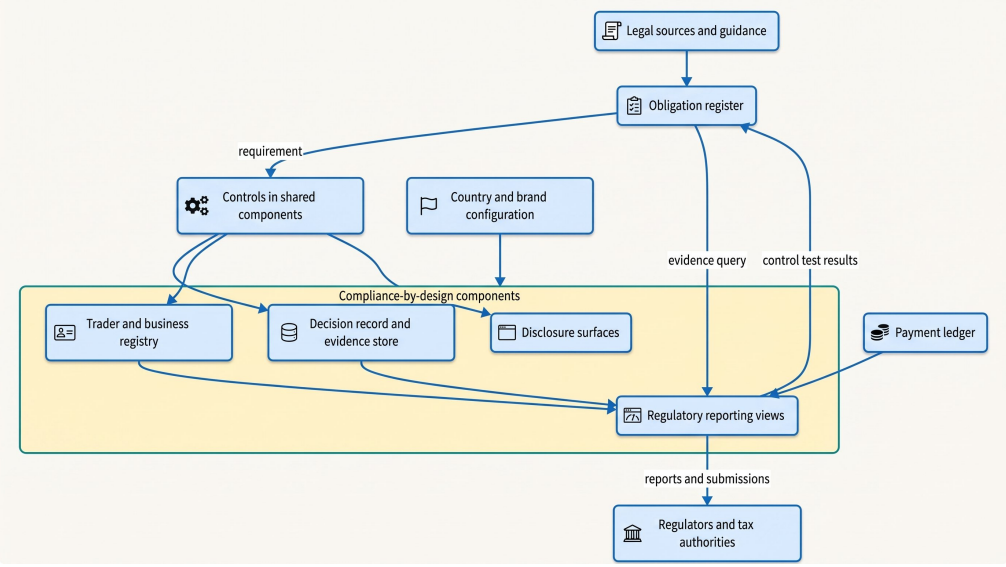


Figure 5: *Compliance by design. Each obligation in the register points to a control in a shared component and to an evidence query; country differences enter as configuration; reports and control tests are computed from the same data, so the register always knows which obligations are demonstrably met.*

Country differences as configuration

The DSA is a regulation and applies directly, but enforcement, penalties and many adjacent rules are national: what counts as illegal content in Poland differs from France, consumer-law transposition differs, DAC7 reporting formats and record-keeping differ, and energy-label rules differ by country. The tempting implementation is a code path per country. That is **The Copy-Paste Country**, and it makes the next change cost eight times as much, once per country.

Veldmark expresses country differences as data. The policy registry holds legal grounds per country; the attribute schema holds mandatory regulated fields per country and category; the disclosure templates hold language and placement per country; the reporting views hold format and authority per country. Engineers change components. Lucía's team, with counsel, changes configuration, through reviewed pull requests against the registry, and the obligation register records which configuration implements which obligation. One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture covers the tenancy model this relies on.

Trade-offs

Approach	Strengths	Weaknesses	When it is right
Compliance by project (one exporter or feature per law)	Fast for the first deadline; no platform dependency	Numbers disagree; every new law repeats the work; evidence is reconstructed	A single small marketplace facing one deadline
Local compliance per brand	Close to local law and language	Four implementations of the same EU regulation; inconsistent decisions across brands	Brands about to be sold, or regulators that differ fundamentally
Shared compliance-by-design components with country configuration	One implementation per obligation type; reproducible evidence; cheaper next law	Up-front investment; a platform team becomes critical to every brand's compliance	Multi-brand, multi-country groups (Veldmark)

Approach	Strengths	Weaknesses	When it is right
Compliance vendor (RegTech) for specific obligations	Speed; maintained templates (for example DAC7 report generation or Transparency Database submission)	Data leaves the platform; the vendor sees only what it is sent; lock-in	A narrow, standardized output with a stable schema

Vertical Differences

Topic	General classifieds	Real estate	Automotive
Main horizontal exposure	DSA moderation volume; trader traceability for pro sellers; DAC7 once payments exist	DSA for agencies; trader panels; P2B for agency subscriptions	DSA and P2B for dealers; dealer termination notice periods
Vertical listing rules	Product safety for goods (DSA Art. 31 random checks)	Energy class and indicator in every ad (EPBD Art. 20(4)); French DPE detail; fair-housing language in the US	CO2 and consumption labeling for new cars (Pkw-EnVKV in Germany); price and mileage rules
DAC7	In scope with in-platform payment for goods	Only if the platform collects rent	Only if the platform processes sale payments

Failure Modes

1. **Exporters instead of records**. Each brand builds a statement-of-reasons exporter from its own logs. The numbers disagree and cannot be regenerated, which is the February 2027 situation at Veldmark.
2. **Silent demotion**. A ranking team down-ranks suspicious listings without a decision record. No statement of reasons is sent, the transparency report undercounts restrictions, and the practice is invisible until a complaint arrives.
3. **Unlabeled boosts**. A "top listing" product that is not visibly marked in results, or a ranking explanation that does not mention paid products. That is both **Multiplier Creep** and a blacklisted practice.
4. **Payments launched without the tax view**. A transaction feature goes live, and the team learns of DAC7 in November, with no tax profile, no seller data and two months to collect it from thousands of sellers.
5. **Trader data kept forever**. Dealer and agency documents retained long after the relationship ended, in breach of DSA Art. 30(5), because nobody owned the deletion trigger.
6. **Law in code**. A country's legal ground, field or label written into a service. Every legal change becomes a release, and every country becomes a fork.
7. **Personal data in the public database**. Free-text reviewer notes copied into `decision_facts` turn a submission into a data breach in a public, searchable database.

The Chief Architect's View

Own the map, not the law. The Chief Architect does not interpret regulation; counsel does. The architect owns the mapping from obligations to components and evidence, and the obligation register is the artifact that makes it visible. At Veldmark, the register has two owners: Lucía Ortega for the legal content and Ingrid's architecture team for the control and evidence columns. Every cross-brand decision record that touches a regulated capability cites the obligations it affects.

Put regulatory review into the product lifecycle at the right moment. The moments that change Veldmark's legal perimeter are predictable: a new transaction feature, a new paid ranking product, a new data-sharing arrangement between brands, an acquisition, a new country. Each triggers a review before design is final, not before launch. Safe Deal's Polish launch in 2026 would have been cheaper if the DAC7 tax profile had been designed with it.

Fund the shared components as platform, not as compliance overhead. The decision record, trader registry, disclosure surfaces and reporting views are also what trust and safety, professional seller tooling, ranking and finance need. Funding them as

"compliance" makes them the first cut in a bad year. Funding them as platform capabilities with compliance as one consumer makes them durable.

Watch the regulatory pipeline. As of October 2026, the Commission has proposed repealing the P2B Regulation, is preparing a Digital Fairness Act on dark patterns, personalization and unfair pricing practices following its 2024 fitness check of consumer law, and EU Digital Identity Wallets are due by the end of 2026 (European Commission, Review of EU consumer law, n.d.). None of these changes the five components. Each changes their configuration. That is the test of whether the architecture is right.

Measure compliance like reliability. Veldmark tracks, per brand and month: the share of restricting decisions with a statement sent and submitted, the median time from decision to submission, the share of trader accounts with complete, current traceability data, the share of regulated listings with all mandatory fields, and the number of obligations in the register whose evidence query passed. A falling number is an incident, with an owner.

Questions to Ask

1. Which of our features let consumers conclude contracts or pay through the platform, and which laws did each feature bring into scope?
2. Does every restricting decision, including demotions and payout holds, produce a decision record and a statement of reasons, and how do we prove the counts match?
3. Could we regenerate last year's transparency report from stored events and get the same numbers?
4. What personal data could reach the Transparency Database through free-text fields, and what prevents it?
5. Where is trader-traceability data stored, who deletes it six months after a relationship ends, and when did that job last run?
6. Do our ranking explanations mention every paid product that affects order, and is every paid boost labeled in results?
7. If we launched in-platform payment in another country next quarter, when would DAC7 due diligence start, and from which ledger would the consideration come?
8. Which regulated listing fields exist per country and vertical, who tracks changes in the law, and what happens to a listing that lacks one?
9. Which country rules are implemented in code rather than configuration, and what would it cost to change each?
10. Is there an obligation register, and for how many obligations can we show a passing evidence query today?

From Trust to Money

Part V has built the trust machinery of a listing marketplace: the shared trust-and-safety platform, fraud prevention, identity and verification, and, in this chapter, the regulation that now specifies much of it. Several of its threads point forward. Payments change the legal perimeter, labels govern paid placements, and trader records feed billing as well as compliance. Part VI — MONEY: Conversion and Monetization turns to how marketplaces convert trust into revenue, starting with the moment a buyer decides to contact a seller, in From Interest to Contact: Messaging and Lead Management.

Sources and Evidence

Company engineering and investor sources

- Vurderinger mellom Kjøper og Selger på FINN.no (Ratings between buyer and seller) — FINN help center, 2024 (updated 2026). Ratings published simultaneously after a deal.

Standards, regulation and documentation

- Regulation (EU) 2022/2065 (Digital Services Act) — Official Journal, 2022. Arts. 3, 14–33, 52 and 93; text checked against the Official Journal version retrieved through the EU Publications Office.
- Commission Implementing Regulation (EU) 2024/2835 — Official Journal, 2024. Transparency-report templates, calendar-year periods, two-month publication, five-year availability.
- DSA Transparency Database and API documentation — European Commission, accessed 19 October 2026 and n.d. Volumes; field vocabulary; batches of 100; `puid` rules.
- Supervision of the designated very large online platforms and search engines — European Commission, updated 15 October 2026. No classifieds marketplace designated.
- Regulation (EU) 2019/1150 (platform-to-business) — Official Journal, 2019. Definitions, Arts. 3, 4, 5, 11; applies from 12 July 2020.
- Platform-to-business trading practices — European Commission, updated 2026. Proposed repeal; guidelines remain.
- Directive (EU) 2019/2161 (Omnibus) — Official Journal, 2019. Price-reduction rule; UCPD Art. 7(4a), 7(6), Annex I points 11a, 23b, 23c; CRD Art. 6a; fines of at least 4%; applies from 28 May 2022.

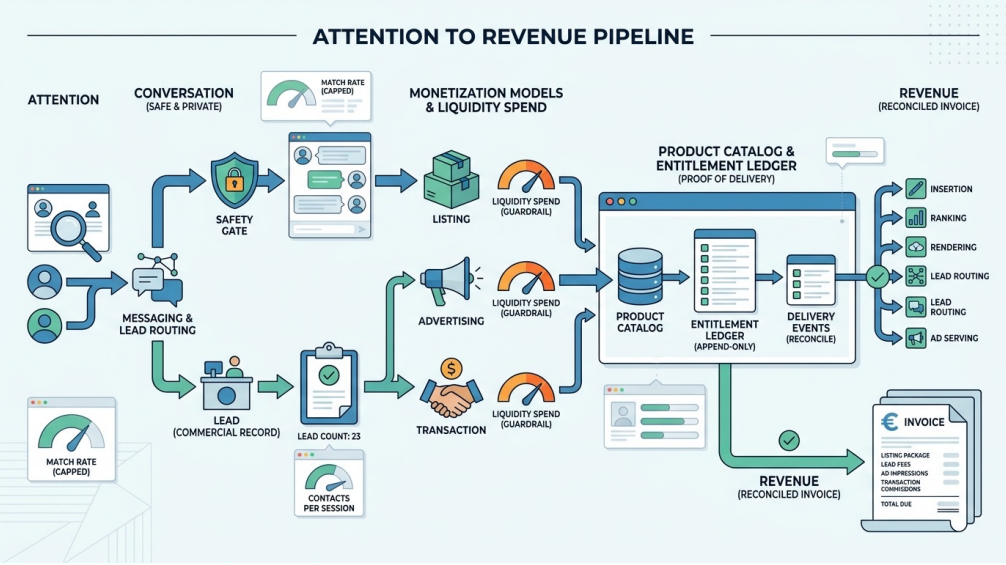
- Council Directive (EU) 2021/514 (DAC7) — Official Journal, 2021. Platform definition and exclusions; relevant activities; consideration; excluded sellers; reporting content and deadlines.
- Regulation (EU) 2016/679 (GDPR) — Official Journal, 2016. Art. 5 principles; Art. 22 automated decisions; recital 47 on fraud prevention.
- Directive (EU) 2024/1275 (Energy Performance of Buildings, recast) — Official Journal, 2024. Art. 20(4) energy class in advertisements including property portals; transposition by 29 May 2026.
- Diagnostic de performance énergétique (DPE) — Service-Public.fr, French government, 2026. DPE advertising content, fines, rental bans, calculation changes.
- Pkw-Energieverbrauchskennzeichnungsverordnung (Pkw-EnVKV) — German Federal Ministry of Justice, as amended 2026. § 5 advertising duties for new passenger cars.
- Online Safety Act 2023, Schedule 7 — legislation.gov.uk, 2023. Fraud as a priority offence.
- Statement: protecting people from illegal harms online — Ofcom, 2024 (updated 2026). Risk-assessment and safety-measure dates.
- Consultation: fraudulent advertising codes of practice — Ofcom, 2026. Paid-ad fraud duties; timeline.
- Informing businesses about the INFORM Consumers Act — US Federal Trade Commission, 2023 (edited 2025). Thresholds, verification and disclosure duties.
- 24 CFR § 100.75, discriminatory advertisements — Cornell Legal Information Institute, n.d. US fair-housing advertising rule.
- Review of EU consumer law — European Commission, n.d. Fitness check and the planned Digital Fairness Act.

Research and practitioner sources

- Fake reviews and sneaky fees banned under new law — Which?, 2025. DMCC Act review rules from 6 April 2025; fines up to 10% of global turnover.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

Part VI — MONEY: Conversion and Monetization



IN THIS PART, YOU WILL: Design the systems that turn attention into contacts and contacts into revenue: messaging and lead routing, payments and buyer protection, monetization models, the product catalog and entitlement ledger, and the advertising stack.

WHY THE BUSINESS CARES: Almost all revenue is decided here. Professionals renew on the leads they can count, and a paid product billed but not delivered becomes a credit note and a lost customer.

WHY ARCHITECTS SHOULD CARE: Every paid product claims a place in ranking, every message passes a safety gate, and every invoice must reconcile with what the platform actually did.

KEY POINTS:

- **Contact is the marketplace's output, and it needs two objects.** A conversation is a private exchange held under privacy and safety rules; a lead is a commercial record delivered, counted and billed. Safety checks sit on the delivery path, because that is where scams close.
- **The unit of charge chooses the architecture, and every model spends liquidity.** Charging per listing, position, account, lead, transaction or audience needs a different system of record, and each needs a guardrail metric owned by someone other than the team that books the revenue.
- **Prove that what was sold was delivered.** Keep catalog, price, entitlement and bill apart, record entitlements as an append-only ledger, and reconcile delivery events from insertion, ranking, rendering, lead routing and ad serving against grants and invoices.

FAILURE PATTERNS THIS PART ADDRESSES:

- **Fast Mail for Fraudsters** — Puts a synchronous safety gate between the sender and the recipient of every message.
- **Multiplier Creep** — Turns boosts, top slots and sponsored listings into explicit, capped effects under one relevance floor.
- **Billing by Guesswork** — Replaces package checks scattered across services with one entitlement ledger and its delivery evidence.

Part V made contact between strangers safe enough to happen. Part VI turns to **what the marketplace earns from it**. A classifieds site usually steps aside before the sale, so it charges for access to attention: the right to list, a better position, a package of slots, a lead, an advertisement.

The argument of this part is that **conversion and monetization are one pipeline with a conflict built in**. The marketplace earns its core revenue when buyers find the right listing quickly. Many of its revenue lines, from paid positions to display ads, earn more when buyers see more pages or when paid listings displace relevant ones. These chapters design each revenue line to sell attention while measuring, and capping, what it costs the match rate.

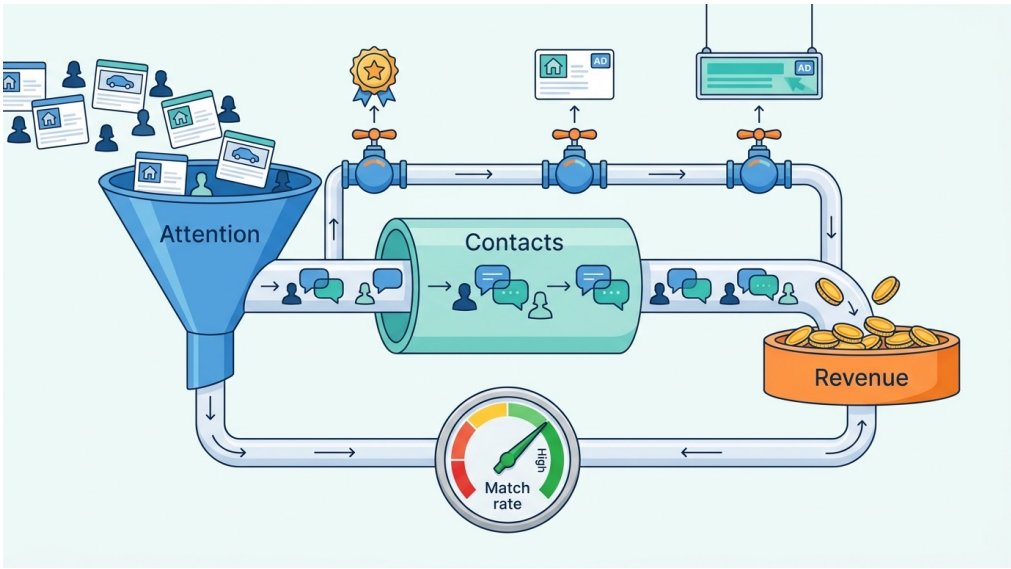


Figure 1: *Part VI treats conversion and monetization as one pipeline: every paid product sells attention, so each one is capped and measured against what it costs the match rate.*

Chapters in this part

Chapter	The question it answers	Read it when…
From Interest to Contact: Messaging and Lead Management	How should buyers and sellers be connected so contacts become deals, without creating a scam channel?	Scams close in your chat, or dealers' lead counts never match yours.
Beyond the Contact: Payments, Shipping and Buyer Protection	When should a classifieds marketplace take part in the transaction, and which systems does it then have to own?	You are adding in-platform payment, shipping or buyer protection, or finance cannot match refunds and payouts to orders at month-end.
How Listing Marketplaces Make Money	Which revenue models fit which marketplace, and what does each require from the systems?	A new revenue model is proposed, or price rises meet professional pushback.

Chapter	The question it answers	Read it when…
Packages, Bumps and Quotas: Products, Pricing and Entitlements	How do you architect what a seller can buy, holds and is billed for?	A price change needs a release, or a dealer disputes an invoice you cannot explain.
Selling Attention: Display, Native and Retail-Media Advertising	How do you run advertising alongside listings without degrading the experience or breaking privacy law?	Ad load, header bidding or consent changes are made without a contacts guardrail.

How this part connects

Part V — TRUST: Keeping the Marketplace Safe handed this part three threads: the message safety gate comes from the trust-and-safety platform, in-platform payments bring DAC7 seller reporting into scope, and trader records feed billing. Paid effects flow into the ranking built in Part IV — DISCOVERY: Matching Intent to Inventory, under its relevance floor. Part VII — PLATFORM: The Distributed System Underneath supplies the plumbing: paid effects and leads travel as events, "contact" becomes a governed metric, and pricing tests get guardrails. For the payments and ledger depth behind Safe Deal, see the sibling book's Payments for Finance Architects: Eleven Reasons the Numbers Disagree.

The Veldmark thread

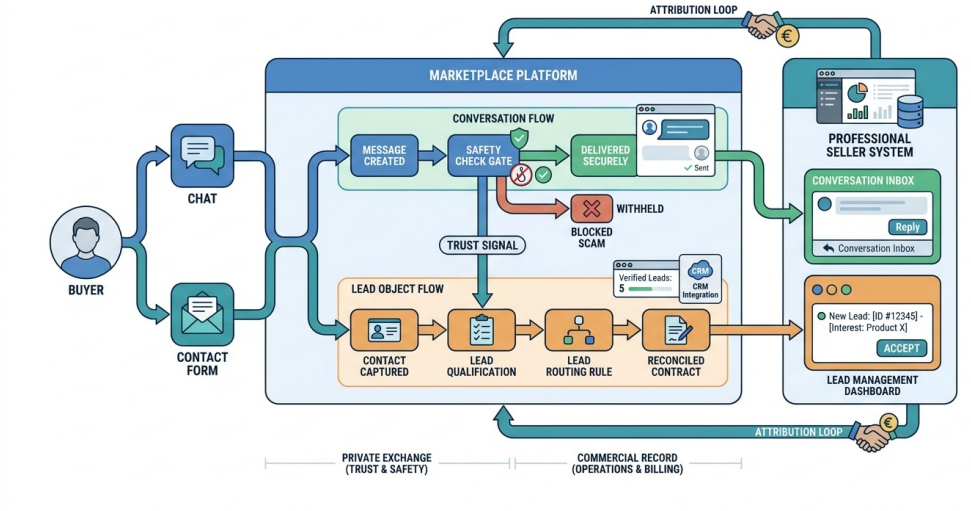
Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), earned €1.31 billion in 2026, sold through four separate product catalogs and a group ad unit. Femke's Rotterdam city bike, sold for €225 through Tradeyard Safe Deal, and the Essen Golf's 23 raw leads (19 once repeat contacts are merged) carry the thread. Ingrid Halvorsen's team, with Olivier Petit, decides:

- **one group conversation service** with a message safety gate (May 2027), live for Tradeyard NL and BE in November 2027, and a lead event contract with a stable lead ID in Q1 2028;
- three **monetization proposals** at the advice forum in May 2027: a Top listing price rise approved, a Motorvane per-lead "lead ledger" counted but not yet billed, and a Hearthly commission pilot for 2028;

- **Veldmark Ads** rules (July 2027): contacts per session as the guardrail for ad changes, and at most three paid positions in the first ten rows on Tradeyard Poland from Q1 2028;
- a **transaction platform** funded before any Safe Deal expansion (September 2027): Safe Deal Portugal approved for Q3 2028 on it, a Motorvane private-car "secure handover" left at design study, and protected rental deposits in Spain declined;
- **Tally**, the group entitlement service, designed in October 2027 after an audit found 1,140 Motorvane dealers overbilled for cars that were never published.

By the end of Part VI you should be able to say, for any paid product, which unit it charges for, what liquidity it spends and how you would prove it was delivered. Begin with From Interest to Contact: Messaging and Lead Management.

28. From Interest to Contact: Messaging and Lead Management



IN THIS CHAPTER, YOU WILL: Take apart the moment a buyer contacts a seller: the contact channels, the conversation service that carries messages, the safety checks that sit on the delivery path, and the lead object that carries a buyer's interest into a dealer's or agency's CRM. You will follow a Rotterdam city bike's fourteen conversations and an Essen Volkswagen Golf's twenty-three leads through each component.

WHY THE BUSINESS CARES: A contact is the first point at which a marketplace produces something a seller can feel. Private sellers judge the platform by whether the messages they receive are real; professionals renew their packages based on how many leads they received, how good those leads were, and whether the portal's count matches their own. Both judgements are made in this part of the system.

WHY ARCHITECTS SHOULD CARE: Messaging is the busiest write path most listing platforms run, the channel where most scams close, and a store

of private correspondence that data-protection law treats with suspicion. Lead delivery is an integration with hundreds of external systems the platform does not control. Getting either wrong shows up as fraud losses, billing disputes and churn, not as an outage.

KEY POINTS:

- **A conversation and a lead are two objects, not one.** The conversation is a private exchange between two parties about a listing, held by the platform and governed by privacy and safety rules. The lead is a commercial record of a buyer's interest, delivered to a professional, counted, billed and later matched to an outcome. Platforms that use the chat store as their lead database, or the CRM e-mail as their only lead record, cannot prove, bill or delete either.
- **Safety checks belong on the delivery path.** Scams close in conversations, often within minutes of first contact. A synchronous gate with a fixed latency budget (links, domain age, payment details, contact details, images and QR codes) decides before the recipient sees the message; slower conversation-level scoring can still withdraw messages and warn recipients afterwards. OLX's 2026 description of blocking fraudulent chat images in real time is the documented direction.
- **Lead routing is a contract with the professional's systems, and lead value is proven downstream.** Leads must reach the right branch or dealer within minutes, exactly once, with consent and an identifier both sides can reconcile. Their value can be measured only with outcome data the portal does not own (a dealer sale, an agency mandate, a closed deal), so the attribution feedback loop has to be designed in from the start.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Fast Mail for Fraudsters** — a messaging service built for delivery speed alone, with safety checks running after delivery or only on reports, becomes the place where payment-link phishing and deposit scams succeed.
- **Vanity Liquidity** — counting messages and form submissions instead of qualified, deduplicated leads that reach a professional's workflow sells exposure that professionals cannot convert, and renewals fail.
- **Ghost Inventory** — contact buttons, saved conversations and lead forms that stay live after a listing is sold keep producing contacts that nobody can satisfy, and charge professionals for them.

Search, ranking and alerts exist to produce one event: a buyer deciding to contact a seller. That event is the marketplace's unit of output. Everything earlier in the funnel

is preparation, and almost everything later (the viewing, the test drive, the payment, the handover) happens partly or wholly outside the platform. **The contact is the last moment the marketplace can see both parties at once**, and the last moment it can protect either of them.

Two very different systems meet at that moment. For private sellers, the contact is a **conversation**: a buyer and a seller exchanging messages about a bike, a sofa or a flat. For professional sellers, the same contact is a **lead**: a commercial record that a dealer or agency expects to receive in its own software, act on within minutes, and count when the renewal invoice arrives. Most platforms start with one of these and bolt the other on. General classifieds start with chat and later discover that dealers want leads in their CRM. Vertical portals start with e-mail leads and later discover that buyers want chat, and that the chat is where the fraud is.

This chapter treats both as first-class, and it opens Part VI because the contact is the first thing the marketplace can count and charge for. The lead it defines is the unit that How Listing Marketplaces Make Money prices and Packages, Bumps and Quotas: Products, Pricing and Entitlements bills; the offer agreed in a conversation is where Beyond the Contact: Payments, Shipping and Buyer Protection begins. It owns the terms **lead**, **conversation** and **lead routing**. It builds on Part V without re-teaching it: fraud detection belongs to Scams, Fakes and Account Takeovers: Fraud Prevention for Listings, and the moderation pipeline and enforcement actions to Trust and Safety as a Platform; this chapter shows where they attach to the contact path. The professional's side of the screen, the lead inbox and CRM tooling, belongs to Serving the Pros: Dealer, Agent and Merchant Tooling.

Setting the Scene at Veldmark: Fourteen Conversations and Twenty-Three Leads

Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), ran four contact systems in early 2027. Two cases from that spring set the agenda for the group's messaging work.

The Rotterdam city bike. On Sunday **7 March 2027** at 19:42 CET, Femke listed a used city bike on Tradeyard Netherlands for **€240**. It became searchable at 19:48 and matched **312** saved searches. The first message arrived at **19:55**. By the next evening Femke had **14** conversations. At **20:11** on Sunday, one "buyer" sent a message with a link to a fake Tradeyard Safe Deal payment page: "I paid via Safe Deal, confirm here to receive the money." The message was delivered. Tradeyard's messaging checks compared links against a deny list, and the link's domain had been registered that afternoon, so it was on no list. Femke recognized the trick and reported the message at **20:19**. The account was suspended at **20:26**, and a rule written that evening withheld **37** similar messages to other sellers. The bike sold on Monday 8 March for **€225** via Safe Deal "pay at pickup".

When Ingrid Halvorsen, the Group Chief Architect, reviewed the case with Aisha Rahman, the Group Head of Trust & Safety, the seven minutes from report to suspension were not the problem. The problem was the sixteen minutes from first contact to phishing link, and the fact that the safety check ran in the wrong place: after the message had been stored and pushed to Femke's phone. The suspended account had already reached nine other sellers before Femke's report, and none of them was warned (the enforcement side is in Trust and Safety as a Platform).

The Essen Volkswagen Golf. On Wednesday **14 April 2027**, the dealer Autohaus Ruhrblick published a 2021 Volkswagen Golf on Motorvane Germany at **€21,900** through its dealer management system (DMS) feed. In ten days the listing produced **23 raw leads**: **14 e-mail enquiries**, **7 phone calls** and **2 test-drive bookings**. The car sold on Thursday 29 April. At the dealer's quarterly review, Autohaus Ruhrblick's sales manager put a different number on the table: the dealership's CRM showed **14** leads from Motorvane for the car. The nine missing were the seven phone calls and the two test-drive bookings. Motorvane recorded calls and bookings in Dealer Hub, its dealer portal, but exported only e-mail enquiries into the dealer's CRM. And the dealer's sales team was sure two of the fourteen were the same person. Neither side was wrong. They were counting different things in different systems, with no shared lead identifier.

The gap mattered beyond one car, because Motorvane was about to charge per lead. It earns **€40 million** a year from leads and financing referrals, and its monetization team was proposing a per-lead charge above a monthly allowance in dealer packages (see How Listing Marketplaces Make Money). A per-lead charge on a number the dealer cannot reconcile is a dispute generator.

Ingrid took two questions to the architecture advice forum in May 2027: **where should safety checks sit in a message's life**, **and what is a lead**, **as an object both Veldmark and a dealer can count?**

Concepts: Contact, Conversation, Lead

The industry uses these terms loosely; this book uses the definitions below.

- A **contact** is any action by which a buyer reaches a seller about a listing: a chat message, an e-mail form, a phone call, a callback request, a booking, an application. A contact is an event.
- A **conversation** is the platform-held thread of messages between a fixed set of participants about one listing (occasionally several). It has participants, messages, attachments, read state and safety state. This chapter owns the term.
- A **message** is one immutable unit of content within a conversation, plus mutable metadata: delivery state, read receipts, safety verdicts, redactions.
- A **lead** is a record that a specific prospective buyer expressed interest in a specific professional's listing or service, created by a defined rule, delivered to

that professional, and tracked to an outcome. A lead has an identifier, a source channel, a timestamp, consented contact data, qualification data, a delivery history and, eventually, a disposition. This chapter owns the term.

- **Lead routing** is the decision of which professional entity (dealer, rooftop, agency branch, individual agent) receives a lead, through which delivery channel, and with what service-level expectation. This chapter owns the term.
- A **connection** is the term some portals use for a counted contact. CarGurus' FY2025 10-K defines connections as "interactions between consumers and dealers on our marketplace through phone calls, email, managed text and chat, and clicks to access the dealer's website or map directions to the dealership" (CarGurus, 2026). Zillow's 10-K describes Premier Agent as the delivery of "validated customer connections, or leads" (Zillow, 2026). The published definitions differ: a click on directions counts as a connection at one portal and would not be a lead at most others. Always ask which definition a number uses.
- **Disintermediation** is buyers and sellers moving off the platform once they have met, to avoid fees or friction. It is defined in Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics. For classifieds, some disintermediation is the business model; for transactional and pay-per-lead models, it is revenue leakage.

The central distinction is between the conversation and the lead: they often begin with the same click, but almost every property differs.

Property	Conversation	Lead
Purpose	Let two parties talk safely	Give a professional a buyer to act on, and count it
Primary owner of the data	The platform (controller for the messaging service)	Shared: the platform creates it; the professional receives a copy and becomes responsible for its use (legal view needed per market)
Lifecycle	Open, active, dormant, closed, deleted	Created, delivered, acknowledged, responded, qualified, outcome, disputed
Identity of the record	Thread ID, stable for the conversation's life	Lead ID, stable across every system that receives it

Property	Conversation	Lead
What counts	Messages delivered, reply rate, time to first reply	Leads delivered, deduplicated, qualified, converted
Retention driver	Data minimization; safety evidence	Contract, billing, attribution, the professional's own obligations
Main risk	Fraud, harassment, privacy	Disputes, double counting, misrouting, consent
Monetization link	Indirect (liquidity, trust)	Direct (pay per lead, share of voice, pay at close, ROI for renewals)

A lead is created from a contact by a rule, and that rule is a business decision encoded in software. Typical rules: the first message from a buyer in a new conversation with a professional creates a lead, later messages do not; a phone call creates a lead only if it connects and lasts longer than a threshold; a booking always creates a lead; repeated contacts from the same buyer about the same listing within a window update the existing lead instead of creating a new one. Each rule changes the number a professional will see on an invoice or ROI report, so the rules need an owner and a version, like a pricing rule.

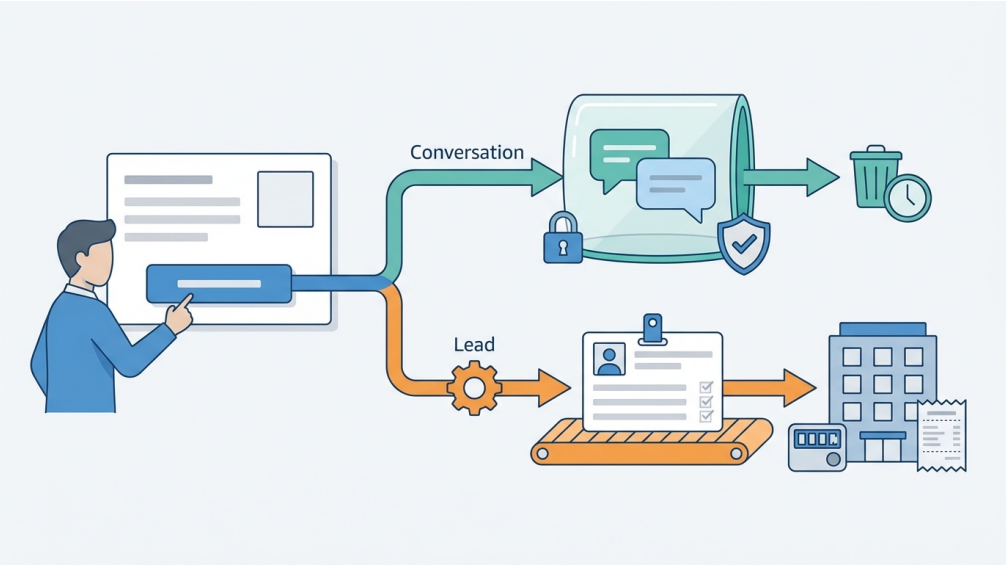


Figure 1: *One click, two objects: the conversation is private correspondence held under*

safety and privacy rules, while the lead is a commercial record, created by a versioned rule, that the professional receives and that is counted and billed.

Contact Channels: What the Platform Sees

Each way a buyer may reach a seller gives the platform a different view of the exchange, a different lever for safety, and a different quality of attribution.

Channel	What the platform sees	Safety leverage	Attribution quality	Typical fit
In-app chat (platform-held conversation)	Every message, attachment and reply time	Highest: can check, warn, withhold and withdraw messages	High: every contact and reply is an event	General classifieds; private sellers; increasingly professionals
E-mail form or e-mail relay	The first message; replies only if relayed through platform addresses	Medium on the first message; low afterwards unless relayed	Medium: first contact is counted, follow-up is not	Real estate and automotive enquiries to professionals
Visible phone number	Only that the number was revealed	Low: can rate-limit reveals	Low: a reveal is not a call	Professionals, legacy listings
Masked or tracked phone number	That a call happened, its duration, sometimes a recording	Medium: can block numbers, cap calls from new accounts	High for calls; the strongest signal of intent in automotive	Dealers, agencies; private sellers who want privacy
Callback or contact request	The request and consented contact data	Medium	High	Professionals with sales teams

Channel	What the platform sees	Safety leverage	Attribution quality	Typical fit
Booking (viewing, test drive, appointment)	The slot, the parties, confirmations and no-shows	Medium: identity and history of both sides	Very high: a booking is close to a deal	Real estate, automotive, local services
Offer or reservation in conversation	Price agreement, sometimes a deposit or payment	High, if the payment stays on-platform	Very high	Recommerce with payments; automotive reservations
Application (rental dossier, job application)	Applicant documents and profile	High for documents; fraud via fake landlords persists	High	Rentals, jobs

Three observations follow from the table.

First, **every channel the platform cannot see is a channel it cannot protect**. Fraud prevention in classifieds is mostly pre-payment: the platform usually does not see the payment, so the conversation is its last chance to intervene (the threat model is in Scams, Fakes and Account Takeovers: Fraud Prevention for Listings). Moving contact off the platform, through visible phone numbers or unrelayed e-mail, removes that chance.

Second, **channels with more friction produce better leads**. A booking or reservation carries more intent than a chat message. Autotrader's FY26 results report **137,000** deals with a full reservation placed through its Deal Builder product (49,000 the year before), which it calls "the very best enquiries in terms of conversion to sale"; it states that Deal Builder enquiries "convert at double the rate into sales" (Autotrader, 2026). The same document shows the cost of pushing a channel too fast: Autotrader attributed part of a fall in retailer numbers to "concerns with the speed and nature of our Deal Builder product roll-out".

Third, **channels evolve away from what the platform cannot attribute**. Rightmove's Real Time Datafeed specification, the API through which UK agency software exchanges data with Rightmove, records in its change log that version 1.3.6 (April 2021) removed the `GetBranchPhoneLeads` and `GetBrandPhoneLeads` calls, leaving e-mail lead exports (Rightmove RTDF v1.4.1, 2023). The specification does not say why; the general point is that whatever a portal cannot deliver into the professional's system, it reports only in its own dashboards, where it is easy to dispute.

Architecture: The Contact Path

The contact path has eight components of its own, listed in the table below. Figure 2 shows the reference architecture Veldmark adopted; it is not a description of any real company's systems.

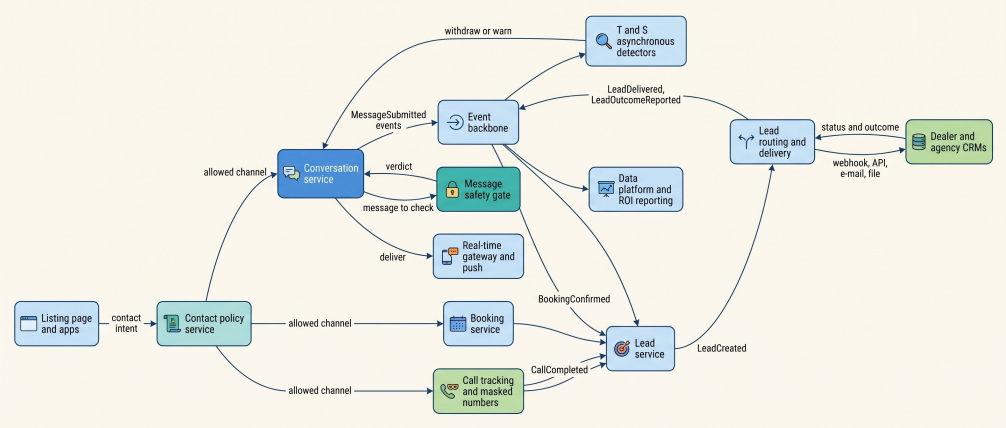


Figure 2: *The contact path at Veldmark. Contact policy decides who may reach whom and how; the conversation service owns messages and calls the safety gate before delivery; the lead service derives leads from conversations, calls and bookings; routing delivers leads to professionals' systems and brings their outcomes back. Only the conversation service stores message content.*

What each component owns, and what it must not own:

Component	Responsibility	Owns	Does not own
Contact policy service	Decides whether this user may contact this seller about this listing now, through which channels, and with what friction (verification step, rate limit, masked number)	Contact eligibility rules per country, category and seller type; rate limits	Identity levels (Who Is on the Other Side? Identity and Verification); risk scores (Scams, Fakes and Account Takeovers: Fraud Prevention for Listings)

Component	Responsibility	Owns	Does not own
Conversation service	Creates conversations, accepts and stores messages, orders them, tracks delivery and read state, enforces retention	Message content, threads, participants, attachments references	Push infrastructure; moderation policy
Message safety gate	Synchronous verdict per message within a latency budget	The call contract and the verdict vocabulary	Policies and detectors (the T&S platform)
Real-time gateway	Holds connections to online clients; fans out to devices; falls back to push and e-mail	Connection state, device registry	Message storage
Call tracking	Allocates tracked numbers, connects calls, records duration and outcome	Number pools, call records	Lead rules
Booking service	Slots, confirmations, reminders, no-shows	Appointments	Calendars inside dealer or agency software
Lead service	Applies lead-creation rules, deduplicates, qualifies and enriches leads	The lead object, its state machine and its store (the lead ledger in Serving the Pros: Dealer, Agent and Merchant Tooling)	Billing (Packages, Bumps and Quotas: Products, Pricing and Entitlements)

Component	Responsibility	Owns	Does not own
Lead routing and delivery	Chooses the recipient and delivers through the professional's channel; ingests acknowledge-ments and outcomes	Routing rules, delivery adapters, delivery log	The professional's CRM

Contact policy is where friction for new accounts lives. Experts interviewed for an Ofcom-commissioned report in 2026 recommend "creating delays between creating an account and reaching a large audience" (Integrity Institute for Ofcom, 2026); for messaging, that means capping how many sellers a new account may contact per hour and masking contact details until the account has history.

Two boundaries cause most of the trouble when they are blurred.

The listing is referenced, not copied. A conversation refers to a listing by its identifier and keeps a small snapshot (title, price, first image) for display, because the listing will change and eventually disappear. When the listing is sold, removed or expired, the listing events from Every Change Is an Event: Event-Driven Listing Architecture must reach the contact path: the contact button must close, open conversations must show the new state, and the lead service must stop creating leads. The city bike's "marked sold at 18:10" on Monday should close new contact within seconds. A platform that checks listing state only when rendering the listing page will keep accepting messages from the conversation screen, where the button is cached: Ghost Inventory in the inbox.

Only one component stores message content. Leads carry the buyer's first message and consented contact details, because the professional needs them. They do not carry the whole conversation, and analytics events carry neither. Every additional copy of message text is a copy that has to be found, secured and deleted. The data platform should receive message *events* (sent, delivered, replied, withheld, reported) with identifiers and timings, not content; Data Platform for Marketplaces: Events, Lakehouse and Data Products owns the tracking plan.

Storing conversations

A conversation store has a simple access pattern and a demanding volume. Users read their own inbox, newest first, and open one thread at a time. Writes are small and frequent; messages are immutable in content but their metadata changes (delivered, read, withheld, redacted).

Even "immutable" messages carry metadata that changes, and that shapes the store. In a **2021** engineering post, OLX described storing chat on Cassandra: "At OLX chat we

need to save the user chats for a long time." Because users "can log in from multiple frontend clients", history must be held on the server, and the data is partitioned by user, since "each user only needs his data". The post describes separate tables for chats and for threads ("where we store the state of a conversation"), and explains why the team moved from a time-window compaction strategy to leveled compaction: messages are not immutable, because "additional states have to be associated with the messages", so updates spread across files and slowed reads (OLX, 2021). The post is more than three years old and the current state of OLX chat is not public, but the lesson about mutable metadata on "immutable" messages is general.

Outside marketplaces, Discord's **2023** account of storing "trillions of messages" is the most detailed public description of the problem at scale. Messages are partitioned by channel plus a fixed time bucket, identifiers are time-sortable, and the hard problem was **hot partitions**: busy channels concentrating reads on a few nodes. Discord added a data-service layer that coalesces identical concurrent requests and routes by channel, and moved from 177 Cassandra nodes to 72 ScyllaDB nodes, with p99 historical-fetch latency falling from 40–125 ms to 15 ms (Discord, 2023).

For a listing marketplace the hot partition is not a busy thread; threads are short. It is the **busy inbox**: a dealer with hundreds of open conversations, an agency branch receiving every enquiry for a popular new-build development, a private seller whose underpriced phone attracts eighty messages in an hour. The reference design I recommend:

- Partition **messages by conversation**, with a time-sortable message ID, so a thread is one partition read.
- Keep a separate **inbox index per participant** (conversation ID, last message time, unread count, listing snapshot), partitioned by participant and bucketed by time, so a dealer's inbox is a bounded read rather than a scan.
- Treat **professional inboxes as a different product**: paginated, filterable by listing and status, and fed to the lead service rather than read by people one thread at a time.
- Give every client-submitted message an **idempotency key**, because mobile clients retry; a duplicated "Is it still available?" is harmless, a duplicated offer is not.
- Store **attachments** through the media pipeline (Photos Sell: Media Pipelines for Listings) as references, never inline, so they get the same scanning, resizing and deletion as listing images.

Delivering in real time

Delivery has three tiers: an open connection to a client that is online (typically WebSockets through a gateway), a push notification to devices that are not, and an e-mail fallback for users who have neither the app nor an open session. The conversation

service should not know which tier was used; it records delivery state as reported by the gateway.

Delivery speed earns money. A buyer who gets an answer in minutes is far more likely to stay than one who gets it the next day; a seller's response time is one of the most useful quality signals a marketplace has, and several portals expose it as a badge or ranking input (see Ranking for Two Sides: Relevance, Freshness and Paid Visibility). But **speed and safety pull against each other on exactly one step: the moment between "stored" and "shown"**. Figure 3 shows how Veldmark resolved it.

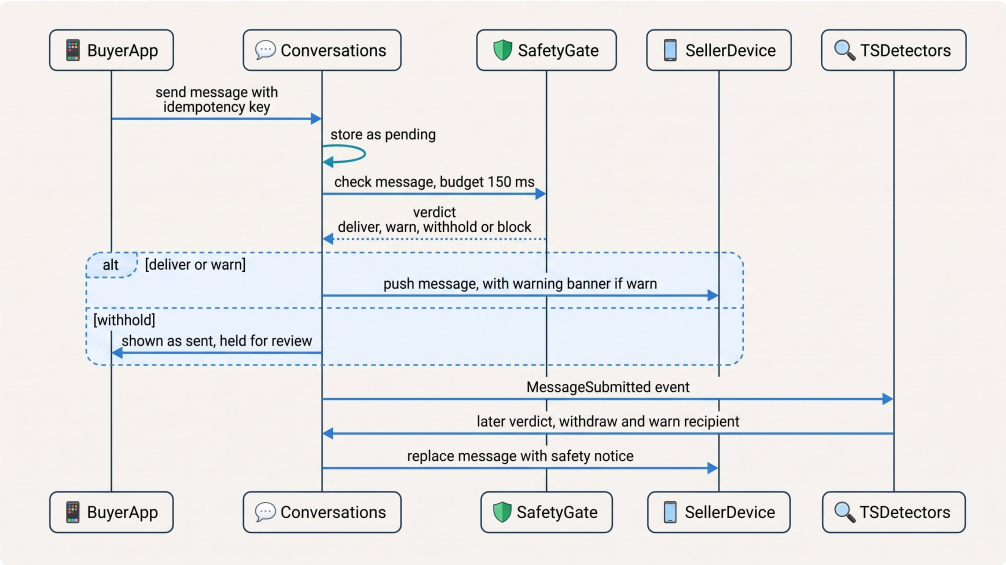


Figure 3: *A message's life with the safety gate on the delivery path. The synchronous check decides before the recipient sees anything; asynchronous detectors can still withdraw a delivered message and warn the recipient. The sender of a withheld message is not told it was withheld, so the check does not become a free testing tool for fraudsters.*

Safety on the Delivery Path

The moderation pipeline, with its synchronous gate and asynchronous detectors, is described in Trust and Safety as a Platform, and the fraud signals in Scams, Fakes and Account Takeovers: Fraud Prevention for Listings; this section covers only what is specific to messages.

Why the gate must sit before delivery. The scam scripts that matter in classifieds close in the conversation. A fake buyer sends a seller a link to a counterfeit "payment received" page that harvests card details; a fake seller asks for a deposit by bank transfer;

a fake landlord asks for money before a viewing. Once a phishing link is on the seller's screen, the platform depends on the seller's judgement.

Large classifieds operators are moving the decision earlier. In a **2026** post, OLX described a "Chat Images GenAI" system that detects and blocks suspicious images in chat in real time, live in Poland, Portugal, Bulgaria, Romania and other countries; it reports that in Poland "fraudulent images make up about 30% of fraud cases", and that its detection of fraudulent QR codes in chat, which can lead to card-harvesting pages, runs at "80% precision" (OLX Group, 2026). An earlier pattern, reported for OLX South Africa in **2018**, was **warn-in-context**: payment keywords and attempts to move a conversation to e-mail or WhatsApp triggered safety tips, and users chatting with a flagged account received an alert (ITWeb, 2018); that system is eight years old and certainly replaced, but the pattern is current. Kleinanzeigen's safety page warns users that "Kleinanzeigen" messages inside the chat can be written by the chat partner (Kleinanzeigen, n.d.), which is why a platform's own brand inside a message is itself a risk signal.

What the gate can afford. The gate runs on every message, so it must be deterministic, cheap and bounded: Veldmark set a budget of **150 ms at p99**, with a fail-open rule for low-risk conversations and fail-closed for messages containing links or payment details from accounts less than 24 hours old. The checks that fit the budget:

Check	What it catches	Typical verdict
URL extraction, allow and deny lists	Known phishing and payment-lookalike domains	Block or withhold
Domain age and registration data	Phishing domains registered hours before use (the city bike case)	Withhold if young and the message mentions payment or the platform brand
Platform brand in a link or text from a counterparty	Impersonation of Safe Deal, shipping or support	Withhold or warn
IBAN, phone, e-mail and messaging-app handles	Moving payment or conversation off-platform	Warn the recipient; mask for new accounts
Image hashes and QR decoding	Known fake payment screenshots; QR codes pointing to phishing pages	Block or withhold
Velocity per sender	One account sending the same text to many sellers	Throttle, withhold

Check	What it catches	Typical verdict
Counterparty and listing risk	Messages from high-risk accounts, or about high-risk listings	Warn, withhold

Slower checks, such as conversation-level text models, large language model (LLM) classifiers on the uncertain band, graph features linking accounts, and image models, run **asynchronously** on message events. Their outputs can withdraw a delivered message, replace it with a safety notice, warn every counterparty of the sender, and feed account-level enforcement. Using LLMs only on the uncertain band is a cost decision: at Veldmark's volume of about **14 million messages a day**, sending every message to an LLM would cost more than the fraud it prevents, while sending the 1–2% the cheap classifier is unsure about is affordable.

The verdict vocabulary matters. "Block" and "allow" are not enough. Veldmark's gate returns four verdicts: *deliver*; *deliver with warning* (the recipient sees a banner such as "Tradeyard never asks you to confirm a payment through a link"); *withhold* (stored, not delivered, queued for review or automatic release, and shown to the sender as sent); *block* (rejected, with an enforcement action on the account). Withholding silently is deliberate: a sender told that a message was blocked learns which words to change. It is also an enforcement decision that, under the EU Digital Services Act (DSA), may require a statement of reasons to the affected user; Rules of the Marketplace: DSA, DAC7 and Vertical Regulation covers when.

Privacy constrains the design. Automated scanning of private messages engages the General Data Protection Regulation (GDPR): purpose limitation, data minimization and transparency. Many platforms rely on legitimate interest for fraud prevention, which needs a documented balancing test and usually a data-protection impact assessment, and the terms of service should say that automated tools check messages. Whether a marketplace's chat is also an "interpersonal communications service" under the European Electronic Communications Code, and therefore subject to the confidentiality rules of the ePrivacy framework, is a legal question for each platform. The Code's definition excludes services where communication is "merely a minor ancillary feature that is intrinsically linked to another service" (Directive (EU) 2018/1972, Art. 2(5)), and a marketplace whose chat exists only to discuss listings has a reasonable argument that it is ancillary. That is a reading to confirm with counsel, not a settled fact. **Get a written legal position before building message scanning, not after**, because the answer changes what you may scan, store and keep.

Retention: What to Keep, and for How Long

Messages are the most sensitive data most listing platforms hold that users did not mean to publish: addresses, phone numbers, bank details, arguments, sometimes photos of identity documents. Retention should therefore be set per data class.

Data class	Example	Retention driver	Veldmark rule (2027)
Message content	Text and attachment references	Usefulness to participants; minimization	Deleted **24 months** after the conversation's last message, or on account deletion
Message metadata	Sender, timestamps, delivery and read state	Abuse investigation, metrics	Pseudonymized after 24 months; aggregates kept
Safety evidence	Withheld or reported messages, verdicts, reviewer notes	Enforcement, appeals, DSA records, law enforcement requests	Kept with the T&S case under the case's retention policy
Lead record (platform copy)	Lead ID, channel, timestamps, recipient, disposition	Billing, disputes, attribution	Kept for the contract term plus the dispute window; contact details removed after 13 months
Lead copy delivered to the professional	Name, contact details, first message	The professional's own obligations	Outside Veldmark's control once delivered; minimized at source

The last row is the reason **lead payloads must be minimized at design time**. Once a buyer's details are in a dealer's CRM, the platform cannot delete them. Rightmove's lead exports illustrate the platform side of the bargain: each brand-level call returns "a maximum of 24 hours within the last 28 days", so leads older than 28 days cannot be retrieved through the API at all (Rightmove RTDF v1.4.1, 2023). A bounded export

window keeps the portal from becoming an indefinite lead archive and pushes the long-term record into the professional's system, where it belongs.

Leads: Creation, Qualification and Lifecycle

The lead service turns contacts into leads. It subscribes to conversation, call and booking events and applies the lead-creation rules, deduplication and enrichment.

Creation and deduplication. At Veldmark the rule is: a lead is created by the first buyer message in a new conversation with a professional, by a tracked call that connects and lasts at least 30 seconds, by a confirmed booking, or by a callback request. Further contacts from the same buyer about the same listing within 30 days attach to the existing lead and update its channel history. "One buyer, three channels, one lead" is the principle, because it is the one a professional will accept; anything else invites the argument Autohaus Ruhrblick started.

Qualification and enrichment. A lead's value depends on what the professional learns from it. The published examples all add structured information at the moment of contact:

- Rightmove says its upgraded **Enhanced Leads** for lettings agents "enriches tenant enquiries with additional, pre-qualification information", is "integrated into partners' CRMs, delivering lead data in real time", is used by "50% of all letting lead senders", and that "over 80% of lead senders are signed in when submitting a lead". The same 2026 report says 60% of consumers attach a property to their Mortgage in Principle (Rightmove, 2026).
- CarGurus' FY2025 10-K describes **Digital Deal**, which captures "purchase intent signals such as trade-in interest and financing preferences", and **Pre-Qualified Leads** from automotive lending partners (CarGurus, 2026).
- Autotrader launched **Buying Signals** in January 2026, "a proprietary machine learning model trained on verified but anonymised sales transactions and consumer interactions". It had featured on more than 800,000 enquiries, and "early results indicate that leads flagged as high-intent convert at twice the average rate" (Autotrader, 2026).

The pattern is consistent: **a signed-in buyer, a structured enquiry and a model score make a lead worth more than a free-text e-mail**, and portals use them both to raise lead value and to justify package prices. The architectural requirement is that qualification data travels in the lead contract as typed fields (financing interest, trade-in, move-in date, budget, verified identity level), not as prose appended to the message, so CRMs can act on it. Lead scoring is a model like any other and needs the ownership and monitoring described in Models in Production: ML Platforms and Use Cases.

The lead lifecycle. A lead needs states that both sides recognize, because each state is a potential dispute.

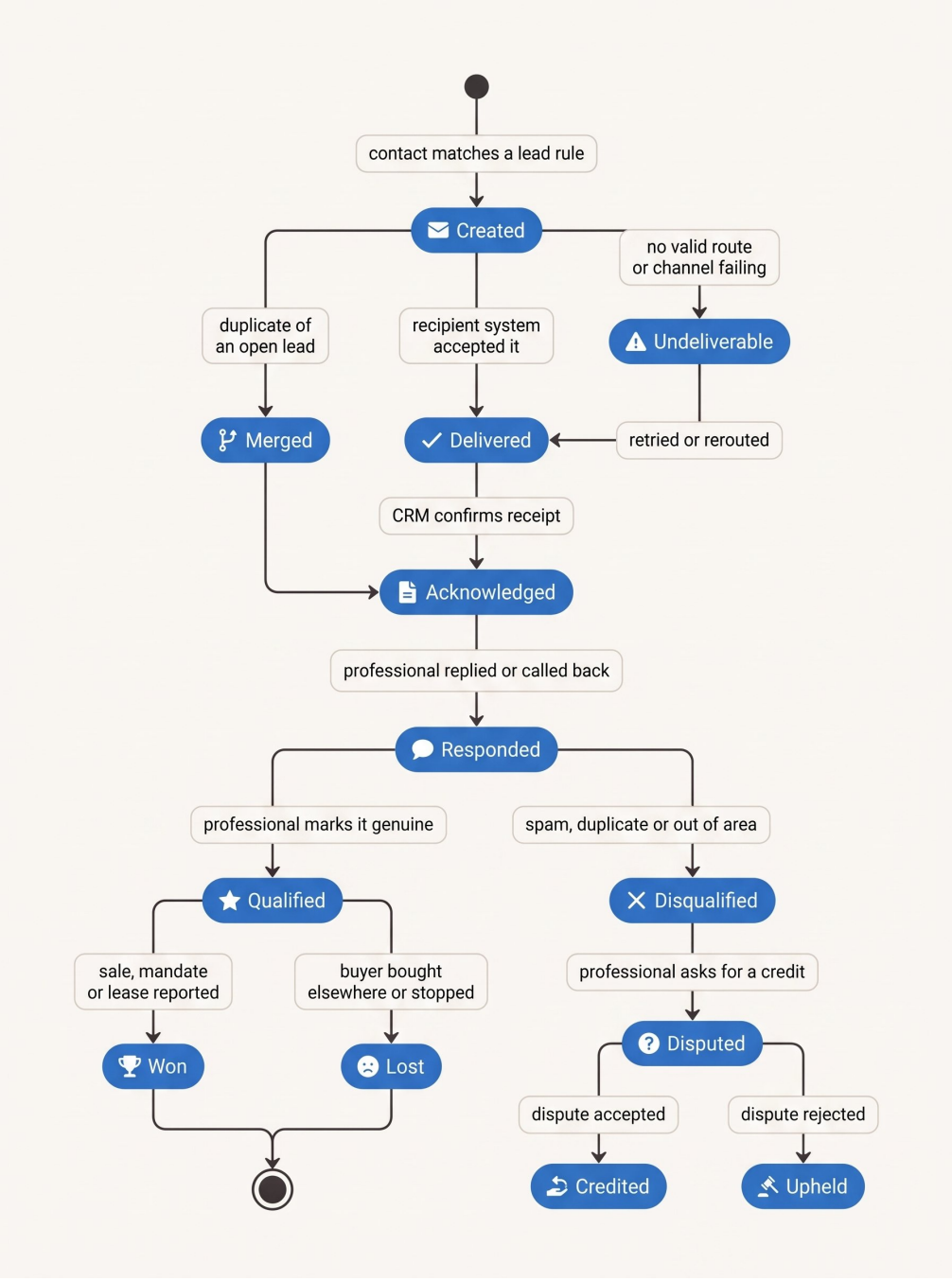


Figure 4: *The lead lifecycle Veldmark adopted. The portal controls the states up to Delivered; everything after Acknowledged depends on data from the professional's systems. Billing rules and ROI reports must say which state they count.*

Two things in Figure 4 are easy to miss. First, **"delivered" means the recipient's system**

accepted it, not that Veldmark sent an e-mail. Second, the states after Acknowledged are reported by the professional, so they are only as good as the integration. A portal that cannot receive outcomes cannot measure lead value except through surveys and renewals.

Lead Routing and Delivery

Routing rules. Most leads have an obvious recipient: the professional who owns the listing. CoStar describes Homes.com as working on a "your-listing your lead basis", a deliberate contrast with models that resell buyer interest to other agents (CoStar, 2026). Other models route by commercial rules. Zillow distributes Premier Agent connections "in proportion to their share of voice, or a Premier Agent partner's share of total advertising purchased in a particular zip code", and "do[es] not promise any minimum or maximum share of connections" (Zillow, 2026). Within a dealer group or agency network, routing usually goes one level further: to the rooftop or branch that holds the vehicle, then to a salesperson by rotation, territory or availability. Zillow says its Follow Up Boss CRM has added "intelligent lead routing" (Zillow, 2026), which places part of the routing decision inside the professional's own software.

The routing decision should be a separate, versioned rule set with its inputs logged, for three reasons. It is commercially sensitive (who gets the buyer), it is subject to transparency obligations where paid placement affects which professional a buyer reaches (see Rules of the Marketplace: DSA, DAC7 and Vertical Regulation), and it is the first thing examined in a dispute.

Delivery channels. Professionals receive leads in systems the portal does not control: dealer management systems, automotive CRMs, agency transaction software, applicant-tracking systems for jobs. The delivery adapter is the anti-corruption layer between the lead contract and each of them.

Delivery channel	How it works	Strengths	Weaknesses	When to use
E-mail (human-readable, sometimes parseable)	Formatted e-mail to a branch address, often parsed by the CRM	Universal; no integration	No acknowl-edgement; parsing breaks; no outcomes; easy to lose	Fallback only

Delivery channel	How it works	Strengths	Weaknesses	When to use
Structured e-mail or file (an industry XML or CSV convention)	Machine-readable payload by e-mail or file drop; in the US, dealer CRMs commonly accept the ADF (Auto-lead Data Format) XML convention	Widely supported in automotive	One-way; versioning by convention	Long tail of dealer CRMs
Pull API	The professional's software polls the portal for new leads	Simple for the portal; the client controls load	Latency; at-least-once delivery that the client must deduplicate	Many small software vendors
Push webhook or API	The portal posts each lead to the recipient's endpoint, retries until acknowledged	Seconds of latency; acknowledgement; idempotent with a lead ID	Every recipient must run an endpoint; retries and dead letters to manage	Large CRMs and dealer groups
Portal pro inbox	Leads shown in the portal's own professional tools	Full control; rich features	One more inbox; duplicates the CRM	Small professionals without a CRM

A pull contract shifts the exactly-once problem to the client. Rightmove's published specification, the clearest public example, advises software providers to call the brand or branch lead exports "at regular intervals (maximum suggested frequency is every 15 minutes) to build up a local database", and to overlap export windows, for example "every 15 minutes, and always request the last 20 minutes' worth of data", because the response's `Replication_Lag` field reports how long data takes to replicate "between

the three Rightmove datacentres", and data within that lag may be missing (Rightmove RTDF v1.4.1, 2023). Overlapping windows mean every lead arrives at least once and often twice, so **the receiving system must deduplicate on the lead identifier**. A push contract keeps the problem at the portal, which must retry until the recipient acknowledges and must send the same lead ID every time.

Push delivery through a lead API is where the larger portals are heading: in a July 2026 platform update, mobile.de described its Lead API as transferring leads "directly and securely into existing CRM systems" (mobile.de, 2026).

My recommendation for a multi-brand group is to define **one lead contract** (a versioned event and API schema, with a stable lead ID, source channel, timestamps, listing and seller references, consented contact data, typed qualification fields and the routing decision), and to implement each delivery channel as an adapter on top of it. Veldmark's Golf disagreement disappears when the calls go through the same contract as the e-mails, and when the dealer's CRM acknowledges each lead ID: both sides then count the same list.

Attribution, ROI and Disintermediation

A professional pays for outcomes and is sold contacts. **Attribution** is the work of connecting the two: matching a lead to a sale, mandate, lease or hire, and reporting the result in a way both sides accept.

The portal rarely owns the outcome. A dealer's sale is recorded in the DMS; an agency's mandate and completion in its transaction software; a property sale, in some countries, in public records months later. The only reliable attribution comes from the professional's systems sending outcomes back (CRM status updates, DMS sales matched to the vehicle identification number), or, for pay-at-close models, from transaction matching over long periods. Zillow's pay-at-close program charges a fee "when a real estate transaction is closed with one of the leads, generally within two years", and treats the fee as variable consideration estimated from "the conversion rate of a lead to a real estate transaction" (Zillow, 2026). That model only works with an outcome pipeline; How Listing Marketplaces Make Money discusses the economics.

Lead counts are also an operational signal: **a broken contact button looks like a quiet day unless someone watches lead counts per platform, channel and country against a forecast**. Adevinta's Benelux analysts described in **2024** an anomaly-detection tool, Artemis, that monitors more than 140 product metrics and caught a drop in iOS leads of about 100,000 a day, valued at about €41,000 a day, within a day (Adevinta, 2024).

Disintermediation is the other side of attribution. Buyers and sellers who meet on the platform can continue by phone, WhatsApp or in person. For a classifieds business that charges for visibility, this is acceptable: the value was the match. For pay-per-lead and transactional models it is leakage, and the temptation is to fight it by hiding contact

details and blocking phone numbers in messages. That works up to the point where it damages the experience that created the match.

The defensible position, in my view, is to **keep users on-platform with value rather than walls**: protected payment (Beyond the Contact: Payments, Shipping and Buyer Protection), booking and reservation flows, verified identity, and a conversation history users trust, with contact details masked only where safety requires it (new accounts, high-risk categories). Marktplaats' buyer-protection product, launched in 2022, can be used only when the deal is arranged through in-platform messaging (AVROTROS Radar, 2022); Kleinanzeigen's "Sicher bezahlen" starts from an agreement in chat (Kleinanzeigen, n.d.). Both tie the protection to the conversation, which gives users a reason to stay in it.

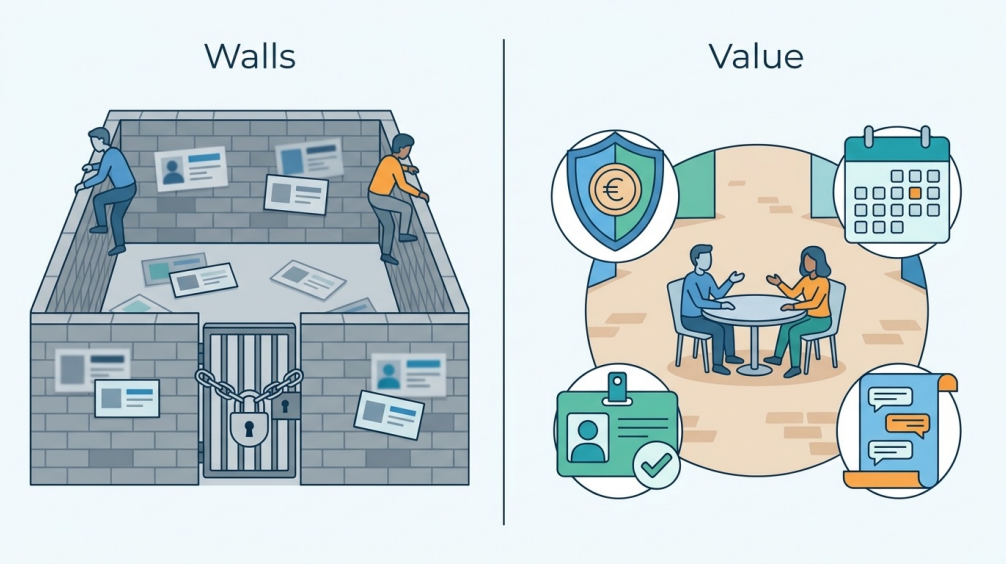


Figure 5: *Hiding contact details and blocking numbers works only until it damages the match; protected payment, booking, verified identity and a trusted conversation history give users a reason to stay on-platform.*

In Practice: The Public Record

Marketplaces publish little about messaging and lead architecture, and most of it is cited above (OLX's 2021 chat storage and 2026 chat-image blocking, Rightmove's lead-export contract, Autotrader's, CarGurus' and Zillow's lead products). Two further documents place contact systems in a group's strategy and in its legal exposure.

Adevinta's 2023 annual report names chat among the capabilities the group consolidated first: "Historically, the team's focus was on consolidating enabling capabilities such

as cloud and data, and overarching elements such as chat and moderation" (Adevinta, 2024). No architecture post on Adevinta's shared messaging service has been published, so how it was built is not public. The strategic reading is still useful: **messaging is generic enough to share across countries and brands, and close enough to trust and safety that sharing it pays twice.**

CarGurus' 10-K adds a reminder that channels carry their own law. Because its platforms "enable us, dealers, and users to send and receive text messages and other mobile phone communications", CarGurus lists compliance with the US Telephone Consumer Protection Act, which restricts calls and texts without prior express consent, as a risk (CarGurus, 2026). A contact channel is also a consent record.

Veldmark Worked Example: One Messaging Platform, Four Brands

The architecture advice forum adopted the reference design in Figure 2 in **May 2027** as part of the Common Ground program. The baseline it replaced:

Brand	Contact channels (2027)	Message store	Safety	Lead delivery
Tradeyard (NL, BE, PL, PT)	Chat; phone reveal for profession-als	Yardstick module on MySQL; WebSocket gateway on AWS since 2021	Deny-list check at send; NL message model asyn-chronous	Pro inbox in Yardstick; e-mail
Motorvane	E-mail form; tracked calls; test-drive booking; chat since 2024	Kotlin service on AWS	Keyword rules	E-mail export to dealer CRMs; calls and bookings only in Dealer Hub

Brand	Contact channels (2027)	Message store	Safety	Lead delivery
Hearthly (FR, ES)	Contact form; agency phone; viewing requests	Forms only, no chat	Manual review of new agencies	E-mail; exports to real-estate software vendors, including Keybridge
Keybridge	Receives leads from many portals for its agencies	—	—	Inbound side of the problem

Volumes. About **14 million messages a day** group-wide, **11 million** of them on Tradeyard. The average is about 160 messages a second; Veldmark planned for about **600 a second** at the Sunday-evening peak, a little above search's peak-to-average ratio of about 3.4. At an average of about 1 KB per stored message including metadata, Tradeyard alone writes about 11 GB of message data a day, about 4 TB a year before replication, which is modest. The capacity problem is not storage. It is the safety gate (one synchronous call per message, so about 600 calls a second at peak inside a 150 ms budget) and the inbox fan-out for professional accounts.

The city bike, replayed. Under the new design, the 20:11 message would have gone through the gate with three signals: a link to a domain registered that day, the words "Safe Deal" in the text, and a sender account created on Saturday. The domain-age rule alone triggers *withhold*. The message is stored, not delivered, and Femke never sees it. The same rule withholds the account's messages to the nine other sellers it had contacted that evening, the asynchronous message model scores the account's history, the account is suspended, and each of those sellers receives a safety notice in the affected conversation instead of the link. The **37** messages withheld on 7 March by a hand-written rule are withheld by the standing domain-age rule within seconds of being sent, not after an analyst writes a rule. Femke's 13 genuine conversations continue unaffected, and the sale happens on Monday as before.

The Golf, replayed. Under the lead contract, each of the 23 contacts produces either a new lead or an update. The **14 e-mail enquiries** come from 13 buyers, so they make 13 leads; one buyer wrote twice and later booked a test drive, and all three of that buyer's contacts sit on one lead. Of the **2 bookings**, that one merges and the other is a new buyer: 1 lead. The **7 calls** go through tracked numbers and all last longer than 30 seconds, but two come from buyers who had already e-mailed: 5 new leads. That gives **19 leads** from the 23 contacts that Motorvane's 2027 reporting counted as leads; the other four were

repeat contacts from the same buyers. The 19 are delivered by webhook to Autohaus Ruhrblick's CRM with acknowledgements. Under the old delivery, the dealer saw 14 records, including the duplicate e-mail, and Motorvane reported 23; under the contract, both sides see the same 19 lead IDs, and the dealer's CRM reports the sale on 29 April against one of them. The proposed per-lead charge can now be priced on a number the dealer can audit.

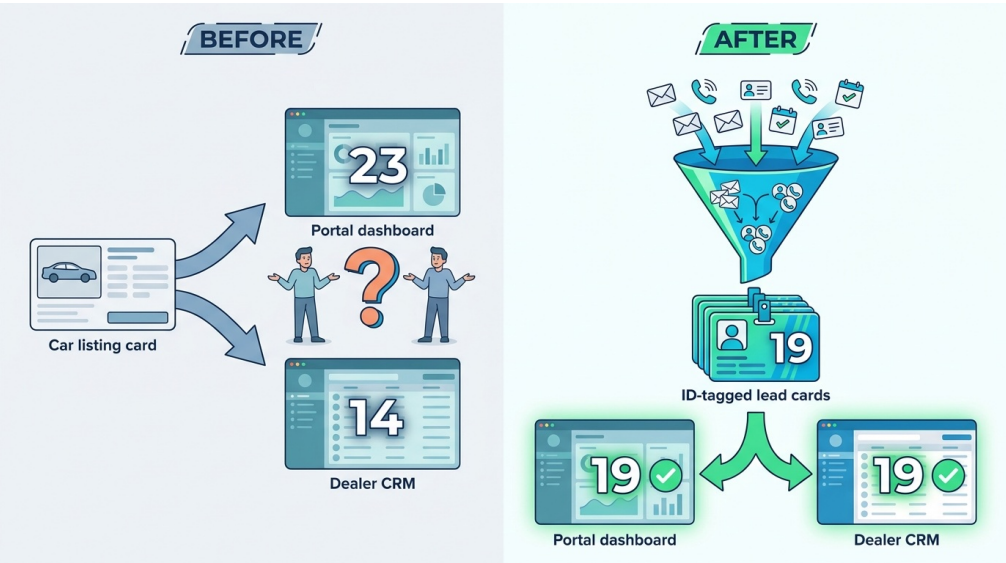


Figure 6: *Under the lead contract, the Golf's 23 contacts merge into 19 leads with stable IDs, so Motorvane and Autohaus Ruhrblick count the same records instead of arguing over 23 against 14.*

The decisions. The forum recorded four decisions, each with a decision record:

1. One **group conversation service** for all brands, with brand-specific user interfaces, delivered in stages: Tradeyard NL and BE first, with the shared trust-and-safety platform, from **November 2027**; Motorvane and Hearthly in 2028. Hearthly gains chat for the first time in that step.
2. The **message safety gate** is a T&S platform service with a contract owned by Messaging: four verdicts, 150 ms p99, fail-closed for links and payment details from new accounts.
3. A **Lead event contract v1** on the group event backbone, with a stable lead ID, live from **Q1 2028**; every delivery channel becomes an adapter; e-mail-only delivery is retired for professionals with a CRM integration by end of 2028.
4. **Keybridge stays on the receiving side only.** Leads from Hearthly reach Keybridge agencies through the same lead contract and delivery adapters as leads to agencies using competing software, and Hearthly does not read lead outcomes

from Keybridge beyond what any software vendor's integration returns. This applies the March 2027 neutrality decision to leads.

Trade-offs

Decision	Option A	Option B	When A is right	When B is right
Build or buy messaging	In-house conversation service	Commercial chat API or SaaS	Messaging volume is large, safety must sit inside the delivery path, and retention rules are strict	Small or new marketplace; chat is secondary to forms; vendor offers synchronous moderation hooks
Safety placement	Synchronous gate plus asynchronous withdrawal	Asynchronous only	Payment scams and phishing are material; peak load is predictable	Low-fraud vertical (most B2B leads); latency budget is very tight
Contact details	Masked by default (relay e-mail, tracked numbers)	Shown after a click	Private sellers; high-fraud categories; pay-per-lead models that need counting	Established professionals who insist on direct contact; low-fraud categories
Lead delivery	Push with acknowledgement	Pull API	Large CRMs and dealer groups; billing depends on counts	Long tail of small software vendors; portal cannot run per-recipient retries

Decision	Option A	Option B	When A is right	When B is right
Lead definition	Strict (deduplicated, qualified)	Generous (every contact counts)	Pay per lead; dealer disputes are costly	Pure subscription; volume reporting supports renewals and nobody is billed per lead
Group platform	One conversation service, brand UIs	One per brand	Shared T&S, shared identity, overlapping users across brands	Brands with no shared users and very different contact flows, or a brand about to be sold

Masking protects private sellers and makes counting possible, but professionals often experience it as the portal standing between them and their customer. CarGurus' free tier offers "capped anonymous leads and connections" (CarGurus, 2026), which shows the other use of masking: as a paid entitlement, where full contact details are part of what the package buys. If masking is an entitlement, it is enforced in the contact policy service from the entitlement service's read model, not in the conversation code (Packages, Bumps and Quotas: Products, Pricing and Entitlements).

Vertical Differences

Aspect	General classifieds	Real estate	Automotive	Jobs	Recommerce with payments
Dominant channel	Chat between private parties	Enquiry form, phone, viewing request	E-mail enquiry, phone, test drive, reservation	Application	Chat with offer and checkout

Aspect	General classifieds	Real estate	Automotive	Jobs	Recommerce with payments
Who receives	Private seller	Agency branch, agent; developer sales office	Dealer rooftop, salesperson	Employer, recruiter, applicant-tracking system	Private seller
Lead object needed?	Rarely, except for professional sellers	Yes, central	Yes, central	Yes (application)	No; the order replaces it
Main fraud in contacts	Payment-link phishing, fake buyers	Fake landlords and deposits (the Barcelona ring in Scams, Fakes and Account Takeovers: Fraud Prevention for Listings)	Deposit scams, fake escrow	Fake job offers, data harvesting	Phishing impersonating the platform's payment
Qualification data	Little	Budget, financing, move-in date, rental dossier	Financing, trade-in, delivery preference	CV, skills, right to work	Not needed

Aspect	General classifieds	Real estate	Automotive	Jobs	Recommerce with payments
Routing complexity	None	Several agencies per property (the Lyon apartment); multi-branch networks	Dealer groups, rooftops, sales teams	Recruiter versus employer	None
Outcome data	Sold flag, payment if on-platform	Mandate, offer, completion; public records months later	DMS sale, often within days	Hire, rarely reported	Order completed

In **real estate**, **the lead goes to the agency whose listing the buyer contacted**, even when entity resolution has clustered several agencies' listings of one property into one canonical record (the Lyon apartment, listed by three agencies, in One Property, Three Listings: Duplicates and Entity Resolution). The buyer contacts a listing, not the cluster, even if the canonical view showed another agency's photos first. Routing a lead to a paying agency instead of the one the buyer chose would be a ranking decision disguised as routing, and a fairness problem with every agency involved.

In **automotive**, outcome data is closest to the portal: a dealer sale appears in the DMS and in the feed's removal within days (the Golf's removal arrived 26 hours after the sale), which makes automotive the vertical where lead attribution can be most precise, if the dealer agrees to share it.

Failure Modes

- **Safety after delivery**. Checks that run on stored messages or only on reports. The fraud succeeds before the check runs. Symptom: report-to-action times look good while losses rise.
- **The chat store as lead database**. Lead counts computed by querying conversations. Every change to chat (merging threads, a new message type) changes billing numbers retroactively.

- **E-mail as the lead contract**. Leads delivered only as e-mails parsed by CRMs. No acknowledgement, no stable ID, no outcomes; every renewal starts with an argument about counts.
- **Duplicate leads from retries and overlaps**. Pull exports with overlapping windows, or push retries without a stable lead ID, create duplicates in the recipient's CRM, and the portal is blamed for "fake" leads.
- **Contacts on gone listings**. Conversation screens and cached contact buttons that keep accepting messages after a sale; lead forms that still route to a dealer for a car sold yesterday. Ghost Inventory, now billed.
- **Copies of message content everywhere**. Message text in analytics events, search logs, support tools and data-warehouse tables. Deletion requests become impossible to honor, and a breach exposes private correspondence.
- **Routing rules hidden in code**. Who gets a lead decided by conditions scattered across services. Nobody can explain a misrouted lead, and paid routing cannot be disclosed accurately.
- **Lead volume treated as success**. Products optimized for more contact submissions (pre-filled messages, one-tap "Is it still available?") that raise counts and lower the professional's reply rate. Vanity Liquidity in the lead funnel.

The Chief Architect's View

At group scale, messaging and leads raise five decisions that belong to the Chief Architect's agenda rather than to a single team.

Share the conversation platform early. Messaging is generic across brands and countries, it is where trust and safety pays off most, and it is a natural client of the group identity (Who Is on the Other Side? Identity and Verification). Adevinta's record suggests the same priority. The brand experience (wording, layout, quick replies, offer flows) stays local; the conversation model, storage, delivery, retention and safety gate are shared. This follows Veldmark's principle of sharing platforms, not products.

Make the lead contract a governed interface. The lead contract is as important as the listing event contract, because money depends on it. It needs a schema registry entry, a versioning policy, an owner in the platform organization, and a change process that includes monetization, the sales organization and the largest CRM partners. Changing the lead-creation rule is a pricing change and should be reviewed as one.

Assign ownership explicitly. At Veldmark: Messaging owns the conversation service, the gate contract and retention; Trust & Safety owns policies, detectors and verdict logic; Monetization owns which lead states are billable; the Data Protection Officer, Lucía Ortega, signs off the scanning and retention design; the Professional Tools teams own the pro inbox and CRM integrations. The architecture advice forum reviews any change that crosses two of these.

Measure the path, not the volume. The metrics I would put on the group dashboard: time to first reply and reply rate (buyer experience); scam reports per 10,000 conversations and the share of withheld messages later released on review (safety and its precision); gate p99 and fail-open rate (operability); lead delivery latency p95 and acknowledgement rate; the **reconciliation gap** between portal and CRM lead counts per professional (trust in the numbers); and lead-to-outcome rate where outcome data exists. Data Platform for Marketplaces: Events, Lakehouse and Data Products should make these governed metrics with one definition across brands.

Plan for AI on both sides of the conversation. Professionals already use AI to answer enquiries; Zillow reports "AI-supported smart message suggestions" in its CRM (Zillow, 2026). Buyers will increasingly reach listings through assistants that send enquiries on their behalf. The conversation model should be able to mark machine-generated messages and agent-initiated contacts, the lead contract should record them, and lead-creation rules should decide whether an assistant's enquiry is a lead. AI Across the Listing Lifecycle takes this further.

Questions to Ask

1. Where in a message's life does the first safety check run: before delivery, after storage, or only on report? What is its latency budget and what happens when it times out?
2. What verdicts can the gate return, and what does the sender see for each?
3. How many systems hold message content, and can you delete a user's messages from all of them within the deadline your privacy policy promises?
4. What is your written legal position on scanning messages, and when was it last reviewed?
5. What exact rule turns a contact into a lead, who owns that rule, and when did it last change?
6. If a dealer and the portal count leads for the same listing, do they get the same number? If not, can you explain the difference lead by lead?
7. How is a lead delivered to a professional's system, how do you know it arrived, and what happens when the recipient is down for six hours?
8. Who decides which professional receives a lead, are the inputs logged, and is any part of that decision paid?
9. When a listing is sold or removed, how many seconds pass before contact is closed in every channel, including open conversation screens and tracked numbers?
10. Which outcome data (sales, mandates, hires) flows back from professionals, and which ROI claims depend on it?

From Contact to Transaction

A contact that turns into an agreement leaves the marketplace with a choice: step aside and let the parties pay each other, or carry the deal through payment, shipping and buyer protection. Tradeyard's Safe Deal began in the conversation, with an offer accepted in chat, and the phishing link that reached Femke imitated exactly that flow. The next chapter, Beyond the Contact: Payments, Shipping and Buyer Protection, follows the money: when a classifieds marketplace should move into transactions, and what it takes architecturally.

Sources and Evidence

Company engineering and investor sources

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Standards, regulation and documentation

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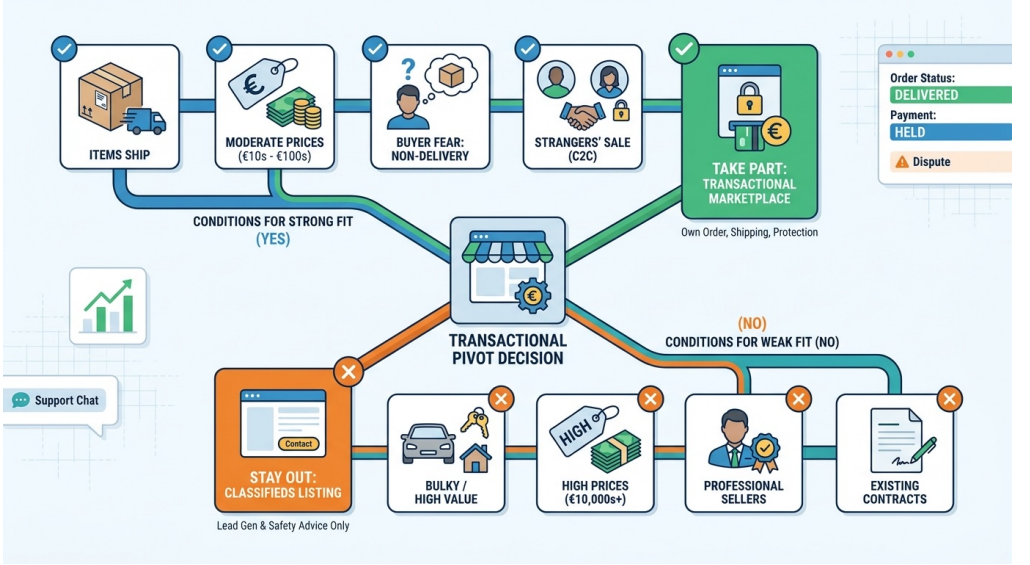
- Regulation (EU) 2022/2065, Digital Services Act, 2022 — statements of reasons for restrictions, including of messages.

Research and practitioner sources

- Fraudulent advertising and account integrity: expert insights on best practice, Integrity Institute for Ofcom, 2026 — friction for new accounts and account-integrity signals relevant to contact policy.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

29. Beyond the Contact: Payments, Shipping and Buyer Protection



IN THIS CHAPTER, YOU WILL: Learn when a classifieds marketplace should take part in the transaction, and how to architect it when it does. You will follow two Tradeyard Safe Deal orders through their whole lives, compare the payment models that marketplace payment products offer, design the order, shipping and dispute contexts, and see which regulatory switches the decision turns on.

WHY THE BUSINESS CARES: In-platform payment with buyer protection makes a marketplace safer, adds a transactional revenue line and makes sales observable for the first time. It also dilutes margin, adds payment, shipping and support costs, puts buyer-protection losses on the profit and loss statement and brings tax reporting with it. A general manager should know the full bill before the first checkout screen ships.

WHY ARCHITECTS SHOULD CARE: A transactional pivot looks like a checkout to product teams and like a ledger, a dispute desk, a carrier

integration and a tax report to everyone else. The order, not the listing, becomes the system of record for the sale. Getting its state machine, its money boundaries and its reconciliation right decides whether the pivot is a trust gain or an operational liability.

KEY POINTS:

- **The transactional pivot is a category decision, not a brand decision.** It works where items ship, prices are moderate and the buyer's main fear is non-delivery. leboncoin, Kleinanzeigen and Marktplaats added payments for goods, not for cars or homes; OLX scaled its pay-and-ship flow back. Gate it per country and category, in configuration, and design it to be switched off again.
- **Let a licensed partner hold the money; own everything around it.** Most classifieds groups use a payment provider's marketplace product (Adyen, Stripe, Online Payment Platform), so the provider is the licensed party that holds funds and verifies payees. The marketplace still owns the order state machine, fee pricing, shipping, release rules, disputes, buyer-protection losses, reconciliation and DAC7 reporting.
- **Buyer protection is a liability with a product attached.** It converts unrecoverable scams into mediated disputes, which is its main value, but it needs evidence-based release rules, a case-management desk, a loss provision and a ledger that can explain every payout. Ship those with the checkout, or the pivot produces unreconciled money and disputes handled by e-mail.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Fast Mail for Fraudsters** — Off-platform payment is where classifieds fraud lands, and a protected-payment brand becomes the scammers' favorite disguise, as the fake Safe Deal link sent to Femke shows.
- **Ghost Inventory** — An item sold through an order but still searchable, alertable or payable lets a second buyer pay for something that is gone, so the order must own the listing's availability.
- **The Forced Fit** — A checkout designed for shippable goods and imposed on car dealers or property agencies fits none of their workflows, and the professionals route around it.

For twenty years, the classifieds bargain was simple: the platform finds the buyer, and the buyer and seller sort out the money themselves. That bargain kept classifieds cheap to run and free of payment regulation. It also left the most dangerous moment of the transaction, **the moment money changes hands, entirely outside the platform's control**, which is why advance-payment scams and fake payment links became the

industry's characteristic fraud (Scams, Fakes and Account Takeovers: Fraud Prevention for Listings).

From about 2017 the large European horizontals began to move into that moment for some categories. They added in-platform payment, shipping labels and buyer protection, so that a buyer in Utrecht can pay a seller in Groningen for a phone without either of them trusting the other. This chapter owns three terms: the **transactional pivot** (adding payment, delivery and protection to a classifieds marketplace), **marketplace payments** (the payment architecture in which a platform collects from buyers and pays out to many sellers) and **buyer protection** (the platform's promise to refund the buyer when a covered problem occurs).

This chapter starts where From Interest to Contact: Messaging and Lead Management ends: with an offer accepted in chat, which the marketplace can leave to the parties or carry through payment and delivery. The question is architectural and strategic: **when should a listing marketplace take part in the transaction, and which systems does it then have to own?** The answer becomes the transactional revenue line that How Listing Marketplaces Make Money compares with the other ways a marketplace earns. The chapter does not re-teach payments. The sibling book RECONCILED covers the payment lifecycle in Payments for Finance Architects: Eleven Reasons the Numbers Disagree, the two main providers in Adyen: One Platform from Checkout to Settlement and Stripe: Every Movement Is a Balance Transaction, and the reconciliation design in Payments-to-Ledger: Draining the Suspense Swamp and Never Update, Only Append: Designing Custom Ledgers.

Setting the Scene at Veldmark: Three Proposals and a Clearing Account

On Wednesday 22 September 2027, the architecture advice forum of Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), had three transaction proposals on its agenda, and Ingrid Halvorsen, the Group Chief Architect, had asked for them to be heard together.

Tradeyard Safe Deal was the reason they existed. Launched in the Netherlands in 2024, Belgium in 2025 and Poland in 2026, it lets private buyers pay in the app and either receive the item by parcel or collect it with "pay at pickup". Payments run on Adyen for Platforms, and Adyen, not Veldmark, is the licensed party. In 2026 Safe Deal processed **11.6 million** orders worth about **€1.02 billion** of gross merchandise value (GMV) and earned **€75 million**: about €59 million of buyer protection fees at **5% of the item price plus €0.70**, and about €16 million of margin on shipping labels. It was Tradeyard's third revenue line, behind visibility products and professional subscriptions.

The three proposals all wanted more of it:

1. **Tradeyard Portugal** (Daan Visser, Tradeyard engineering director): launch Safe Deal in Portugal, the one Tradeyard country without it. The liquidity analysis in Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics had shown that shipping widens a goods market from a district to the whole country.
2. **Motorvane** (Felix Brandt, Motorvane CTO): a "secure payment" for private-to-private car sales in Germany, where buyers and sellers meet strangers with tens of thousands of euros at stake.
3. **Hearthly Spain** (Julien Marchand, Hearthly CTO): a protected rental deposit. In July 2027, a ring of **26** accounts had posted **180** fake rental listings in Barcelona and collected about **€71,000** from **63** people as deposits paid by bank transfer outside the platform (Scams, Fakes and Account Takeovers: Fraud Prevention for Listings tells that story).

Ingrid added a fourth item, which nobody had proposed. At the December 2026 month-end, Veldmark's group finance team had found about **€2.3 million** of Safe Deal items in a clearing account that could not be matched to orders: refunds issued after payouts, label charges netted from settlements, partial refunds agreed in chat. The amount was a fraction of GMV, but nobody could say what it consisted of without a week of spreadsheet work. Her point to the forum was that **Safe Deal had shipped its checkout before it had shipped its ledger**, and that every proposal on the agenda would make that worse unless the transaction platform was designed properly first.

The rest of this chapter is the framework the forum used to decide.

Concepts: What Changes When the Platform Takes Part in the Sale

A classifieds marketplace's job ends at the contact (Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy). A transactional marketplace's job ends when payment and delivery are complete. The transactional pivot moves a category from the first to the second, and brings terms the classifieds side of the organization has usually never needed.

- **Order**: the record of an agreed sale between a buyer and a seller for a listing, with a price, fees, a fulfilment method and a state. It is created when the buyer commits, not when the listing is published.
- **Marketplace payments**: a payment model in which the platform collects one payment from a buyer and pays out to the seller, net of the platform's fees, often after holding the funds for a period. In the EU, the platform either holds a payment licence itself or uses a licensed provider whose marketplace product does the collecting, holding and paying out.

- **Payment service provider (PSP)**: a licensed company that processes payments for merchants; its **marketplace product** (Adyen for Platforms, Stripe Connect, Online Payment Platform) adds seller onboarding, split payments, held balances and payouts.
- **Payee verification (KYC)**: the identity and account checks a licensed provider must run before paying money to someone. For private sellers it is light; for businesses it is know-your-business (KYB). Who Is on the Other Side? Identity and Verification owns the levels.
- **Hold and release**: keeping the buyer's payment in a held balance until a release event (buyer confirmation, a delivery scan plus a waiting period, a handover code), then making it available for payout. This is often called "escrow" informally; legally it is usually not an escrow arrangement, which is why the provider's terms, not the marketing name, define it.
- **Payout**: the transfer of released funds from the seller's balance to their bank account.
- **Merchant of record**: the party whose name is on the buyer's payment statement and which carries the payment-side obligations, refunds and chargebacks (Payments for Finance Architects: Eleven Reasons the Numbers Disagree defines it fully).
- **Buyer protection**: the platform's contractual promise to refund the buyer if a covered problem occurs, such as non-delivery, an item significantly not as described, a counterfeit, a wrong or incomplete item. Covered cases, evidence rules and time windows are product decisions with financial consequences.
- **Dispute** versus **chargeback**: a dispute is opened with the platform under its buyer protection; a chargeback is opened by the buyer's card issuer under card-scheme rules. Both can occur for the same order, and they follow different clocks.
- **Pick-up and drop-off point (PUDO)**: a shop or locker where parcels are handed in and collected; the default shipping method for low-value consumer parcels in much of Europe.

Holding the money and taking the risk are separate: the PSP holds the funds, but a buyer-protection refund that the seller cannot cover is paid by the marketplace. And **the order is a separate aggregate from the listing**: a listing can have many conversations and, for multi-quantity sellers, many orders; an order belongs to one buyer, one seller and one agreed price. The Listing Aggregate: Identity, Lifecycle and Invariants keeps availability on the listing and leaves everything about money to the order.

When to Take Part in the Transaction

The documented record suggests that the pivot pays where four conditions hold together.

Condition	Why it matters	Strong fit	Weak fit
The item ships	Shipping widens the market and creates the delivery event that protection is built on	Phones, clothes, books, small electronics, toys	Sofas, bikes in some markets, cars, homes
The price is moderate	A percentage fee is acceptable; losses per dispute are bounded; payment methods have no limits	Roughly tens to a few hundred euros	Tens of thousands of euros, where a percentage fee is absurd and a single loss is large
The buyer's main fear is non-delivery or misdescription	Buyer protection answers exactly that fear	Distance purchases between private people	Professional sellers with their own guarantees and legal duties
The sale is otherwise completed between strangers	The platform adds trust that does not already exist	C2C goods	Dealer and agency sales, which already have contracts, invoices and showrooms

Category gating is what the European horizontals did. leboncoin "launched a transactional business model in some of the generalist categories" in 2019, according to Adevinta's 2019 annual report. Marktplaats launched transactional services in 2017, shipping in 2019 and buyer protection after that. eBay Kleinanzeigen launched "Sicher bezahlen" for all users in October 2020, initially only for toys, phones, and PCs and notebooks (onlinemarketing.de, 2020). mobile.de, in the same group, earned essentially no transactional revenue in 2023 (Adevinta Annual Report 2023, published 2024).

The economics are documented too. Adevinta's transactional revenue reached **€103 million** in 2023, about 6% of group revenue, and in France the EBITDA margin fell two points "mainly reflecting the evolution of the business mix with an increasing share of transaction services". The same report notes rising "direct costs from transactional services (such as payment and shipping service provider costs)" (Adevinta Annual Report 2023). OLX described its Pay&Ship flow in 2023, "OLX holds the money of the transaction and transfers it to the seller when the package is delivered", and Prosus then recorded "the de-prioritisation of pay-and-ship in Europe" in the second half of the year

to March 2025, while OLX concentrated on motors, real estate and jobs (OLX Tech, 2023; Prosus FY25 results, 2025).

The trust case is the strongest argument. In a December 2024 investigation, the UK consumer organization Which? found that eBay, Vinted, Depop, Shpock, Preloved and Amazon Marketplace offered some buyer protection while Facebook Marketplace, Gumtree and Nextdoor did not, and that a quarter of Facebook Marketplace and Gumtree buyers believed theirs did (Which?, 2024). In-platform payment turns an authorized push-payment fraud, which the victim may never recover, into a dispute the platform can resolve. That is why it belongs as much to Trust and Safety as a Platform as to monetization.

My rule of thumb: **take part in the transaction where the platform's presence removes a fear that stops sales, and stay out where professionals already carry that fear for the buyer.**

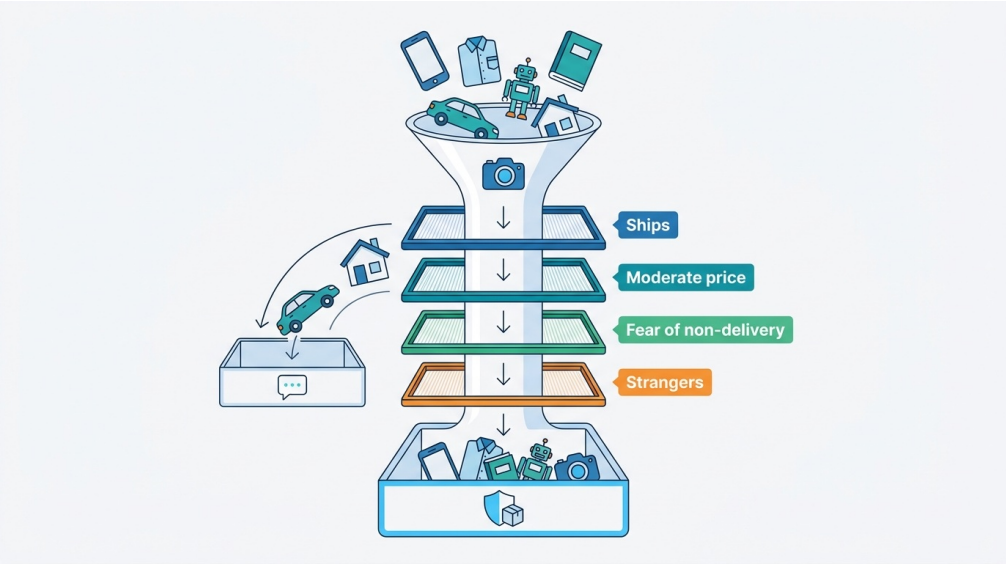


Figure 1: *The transactional pivot is a per-category decision: it pays where the item ships, the price is moderate, the buyer fears non-delivery and the sale is otherwise between strangers, and it stays out where those conditions fail.*

Payment Models: Who Holds the Money

A marketplace that wants to collect from buyers and pay sellers in the EU has three basic options. In practice, classifieds groups have used the first.

Use a licensed provider's marketplace product

The provider is the licensed payment institution or e-money institution. It onboards and verifies sellers, collects the buyer's payment, keeps funds in held balances, splits fees and pays out. Adyen describes its platform product as letting a platform "onboard users of your platform, process payments for them, and pay out their funds", with Adyen verifying users before payouts, the option to "split the payments between one or more users, deduct costs as needed, and hold funds until payout", and payouts on a schedule the platform controls (Adyen documentation, n.d.). Stripe Connect offers three charge types with different money flows: **direct charges** on the seller's connected account, **destination charges** on the platform's account with an immediate transfer to the seller, and **separate charges and transfers**, which let the platform charge first and transfer later or to several sellers. For destination charges and separate charges and transfers, "refunds and chargebacks reduce your platform's balance", and the platform must recover them from the seller by reversing the transfer (Stripe documentation, n.d.). Stripe: Every Movement Is a Balance Transaction and Adyen: One Platform from Checkout to Settlement in the sibling book describe both products in depth.

Specialist marketplace providers serve the same need. Online Payment Platform, a Dutch payment institution, carried Marktplaats' first escrow-style flow in 2017 and eBay Kleinanzeigen's in 2020; Worldline's 2022 announcement of a 40% stake named Marktplaats, eBay Kleinanzeigen and Gumtree among its clients (Worldline, 2022; t3n, 2023). Kleinanzeigen's own help page named "unserem Zahlungsdienstleister Adyen" when consulted in October 2026 (Kleinanzeigen, n.d.); when and why the provider changed is not public. leboncoin has acted as a trusted third party for consumer-goods transactions since 2018 with Adyen as its payment partner (Wikipedia, leboncoin, 2026, widely reported).

Hold a licence yourself

A marketplace can become an e-money or payment institution and hold users' funds itself. Vinted Pay holds an e-money licence from the Bank of Lithuania (2023) and began moving members onto its own wallet in January 2026, "expected to reduce payment-related costs and dependencies", while continuing to work with other providers (Vinted, 2026). A licence brings safeguarding of client funds, capital requirements, supervision and a double-entry ledger the marketplace must run itself (Never Update, Only Append: Designing Custom Ledgers). It makes sense when payment volume is large, payments are the core of the product and the group wants to own the economics; for a classifieds group whose transactions are a minority of revenue, it rarely does.

Stay out of the flow of funds

The marketplace can avoid payments entirely and rely on contact plus safety advice, or pass buyers to an external payment option. This keeps it out of payment regulation and

DAC7, at the cost of the trust and observability gains.

Why "we just collect and pass it on" does not work. The EU's Second Payment Services Directive excludes payment transactions made "through a commercial agent authorised via an agreement" to negotiate or conclude the sale on behalf of only the payer or only the payee (Directive (EU) 2015/2366, Art. 3(b)). A marketplace that collects money on behalf of both buyers and sellers does not fit that exclusion, which is the practical reason most marketplaces use a licensed provider. The precise legal position depends on the contracts and the jurisdiction; treat this as a question for counsel, not a design default.

Comparing the models

Model	Who holds funds	Who verifies payees	Who carries chargebacks and refunds first	Marketplace owns	Fits
Provider marketplace product, platform as merchant (Adyen for Platforms, Stripe destination or separate charges)	The provider, in the platform's or sellers' held balances	The provider	The platform's balance; recovery from the seller by debit or transfer reversal	Order, fees, release rules, disputes, losses, reconciliation	C2C goods at scale; one brand on the statement

Model	Who holds funds	Who verifies payees	Who carries chargebacks and refunds first	Marketplace owns	Fits
Provider marketplace product, seller as merchant (Stripe direct charges, on-behalf-of models)	The provider, in the seller's account	The provider	The seller's balance	Order, fees; less of the risk	Professional sellers with their own customer relationship
Specialist marketplace payment institution (OPP)	The institution, in intermediary accounts	The institution	Contract-dependent	Order, fees, disputes in cooperation with the institution	Mid-size marketplaces wanting a turnkey escrow flow
Own licence (Vinted Pay)	The marketplace, in safeguarded accounts	The marketplace	The marketplace	Everything, including safeguarding and the ledger	Transaction-first marketplaces at very large volume
Off-platform	The parties	Nobody	The victim	Safety advice only	Lead-generation and high-value categories

Veldmark chose the first row. Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models records the consequence: the Safe Deal context **conforms** to Adyen's model of accounts, balances and payouts and translates only at its edge, where an order meets a listing.

Architecture: The Transaction Platform

A transactional pivot adds six bounded contexts beside the listing core. None of them belongs inside the listing service.

Context	Owns	Key events	Must not own
Order	The agreed sale: parties, listing, price, fees, fulfilment method, state, timers	`OrderPlaced`, `OrderPaid`, `OrderShipped`, `OrderDelivered`, `OrderCompleted`, `OrderCancelled`	Listing content; money balances
Payment edge	The translation between orders and the provider: payment sessions, captures, refunds, account holders, payout status	`PaymentAuthorised`, `PaymentCaptured`, `RefundIssued`, `PayoutSettled`	Business rules about when to release
Fees and pricing	Buyer fee, seller fee, shipping prices, promotions, rounding, VAT treatment of fees	`FeeQuoted`, `FeeCharged`	Visibility products (Packages, Bumps and Quotas: Products, Pricing and Entitlements)
Shipping	Carrier contracts, label purchase, PUDO selection, tracking events, address validation	`LabelCreated`, `ParcelHandedIn`, `ParcelDelivered`, `ParcelReturned`	The release decision

Context	Owns	Key events	Must not own
Buyer protection and disputes	Coverage rules, dispute cases, evidence, decisions, refunds funded by the platform, loss provision	`DisputeOpened`, `DisputeDecided`, `ProtectionPaid`	Account sanctions (Trust and Safety as a Platform)
Transaction ledger and reporting	The double-entry record of every amount owed and moved; reconciliation with provider reports; per-seller aggregates for tax	`JournalPosted`, `ReconciliationBreak`, `SellerPeriodAggregated`	Payment execution

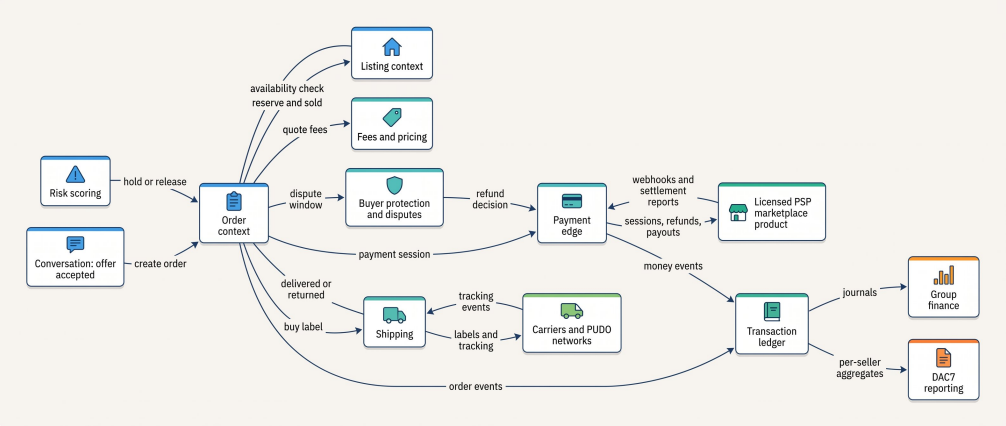


Figure 2: *The transaction platform beside the listing core. The order context owns the sale and its timers; the payment edge translates to the licensed provider; shipping and disputes feed the order's release decision; the ledger receives both order and money events, so finance and tax reporting are built from the same record.*

The order owns availability

The most common integration bug in a transactional pivot is the double sale: two buyers pay for the same one-off item because the listing still looked available. The order context must therefore **reserve the listing atomically when payment is committed**,

and the listing must treat a reservation from an order as authoritative. At Tradeyard, a Safe Deal order emits `ListingReserved` on the listing event stream and the listing transitions to reserved; search shows a badge and alerts stop notifying (Every Change Is an Event: Event-Driven Listing Architecture). A reservation needs a timeout: an order that is never paid, never shipped or never collected must release the listing again, or the reservation becomes Ghost Inventory in reverse, a sellable item nobody can buy.

When the order completes, the listing becomes sold through the same stream, and every copy of the listing must react: search, saved-search alerts, digests, recommendations and sitemaps (Tell Me When It Appears: Saved Searches and Alerts, From Submitted to Searchable: The Publication Pipeline). Femke's bike is the canonical warning. It sold through Safe Deal at 18:10 on Monday 8 March 2027 and was gone from search a minute later, but the next morning's alert digest, assembled from an older snapshot, still featured it to **214** users.

The order state machine

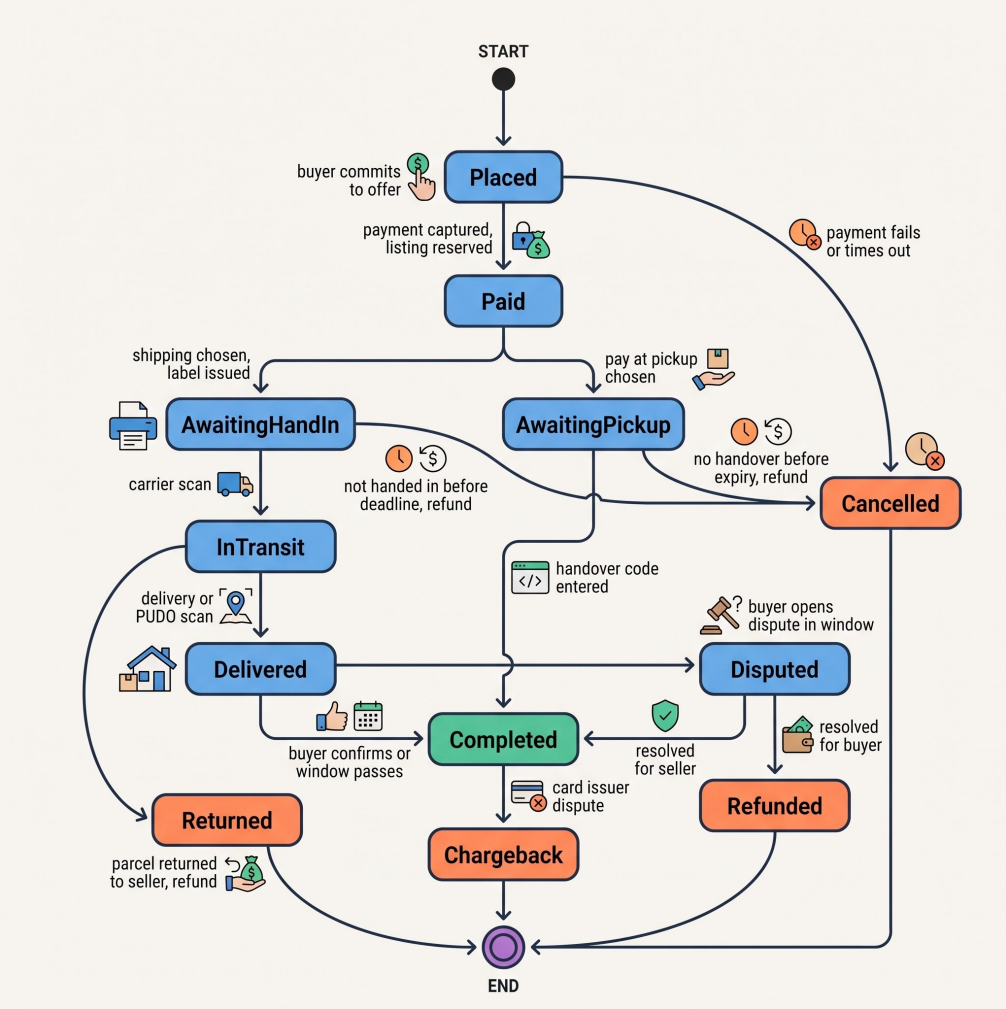


Figure 3: *A protected-payment order for goods. Every waiting state has a deadline, every exit to cancellation or refund returns money and releases the listing, and a completed order can still be reopened by a card chargeback weeks later, which is why the ledger must handle states after payout.*

Four design rules follow from the diagram.

1. **Every waiting state has a timer and an owner.** Unshipped orders, uncollected pickups and silent buyers are the normal case, not the exception. Kleinanzeigen documents one such rule: if the buyer does not confirm receipt, "wird der Betrag nach 14 Tagen automatisch an den Verkäufer überwiesen" (the amount is paid to the seller automatically after 14 days) (Kleinanzeigen, n.d.). Marktplaats' successor to its first escrow product, as described in 2022, paid the seller on day 8 if there were no complaints (AVROTROS Radar, 2022).

2. **Release is a decision, not a side effect.** The order context decides when funds become payable, using the delivery event, the dispute window, the buyer's confirmation and the risk score. The payment edge only executes. A fraud rule must be able to hold a payout, and that hold is itself a restricting decision that needs a statement of reasons under the Digital Services Act (Rules of the Marketplace: DSA, DAC7 and Vertical Regulation).
3. **Money states and order states are reconciled, not assumed equal.** "Paid" in the order means the provider confirmed capture; it does not mean cash has arrived. Provider webhooks arrive asynchronously, at least once and out of order (Adyen: One Platform from Checkout to Settlement), so the payment edge must be idempotent and the ledger must match order events to provider records.
4. **States after completion are first-class.** Chargebacks, late refunds and returned parcels arrive after the seller has been paid. The design needs a policy for recovering money from sellers (debit their balance, net against future payouts, absorb it) and a ledger that records each case.

Shipping as an integration surface

Shipping turns a marketplace into a logistics integrator. Kleinanzeigen's protected-payment flow offers DHL parcels in four sizes and Hermes in four (a small packet and three parcel sizes), handed in at shops, lockers or to the carrier (Kleinanzeigen, n.d.). Vinted reports more than **500,000** pick-up and drop-off points across Europe and runs its own carrier, Vinted Go, in five countries (Vinted, 2026). Adevinta's 2025 report discusses shipment-driven emissions for five of its marketplaces, a reminder that shipping volume is now material enough to appear in sustainability reporting (Adevinta Strategic Report 2025, published 2026).

The shipping context needs one adapter per carrier, a normalized tracking-event model (handed in, in transit, at PUDO, delivered, returned, lost), label pricing by size and route, address validation and a policy for lost parcels. Tracking events are the evidence that buyer protection rests on, so their timeliness is a reliability requirement: a delivery scan that arrives a day late delays a payout by a day.

Shipping also changes discovery. An item that ships nationwide is no longer constrained by a radius search; distance becomes a delivery-cost input rather than a filter (Where Is It? Location, Geocoding and Privacy), and the price a buyer sees in search may need to include the fee and shipping. Vinted's price list describes its buyer-paid fee as varying with "item characteristics, order value, order type" (Vinted UK price list, updated October 2026), and a fee that varies per item has to be computed on the search and listing read path, not only at checkout.

Buyer protection and dispute management

A dispute desk is a case-management system with money attached. Kleinanzeigen's coverage, as published in 2026, lists five cases: item not shipped, significantly different from the description, not authentic, a different item, incomplete (Kleinanzeigen, n.d.). Adevinta built a dispute-management back office as a reusable component "that can be plugged into any marketplace", which ingested per-marketplace transaction events through Kafka, SQS or REST and normalized them into a common model; its designers noted, "We couldn't hardcode anything… or make any assumptions" (Adevinta tech blog, 2023). That is the right shape for a shared capability across brands: an anti-corruption layer per marketplace and a canonical case model.

A protection promise is also a liability that must be provisioned, measured per category and priced into the fee. eBay's 2025 annual report describes its transaction losses as consisting "primarily of losses resulting from our buyer protection programs, chargebacks" (eBay Form 10-K for 2025, filed 2026).

Reconciliation and the ledger

Every amount in an order has to be explainable later: what the buyer paid, which part was the item, the fee and the label, what the provider charged, what was refunded, what was paid out and when, and what the platform absorbed. Provider settlement reports are net and batched, use the provider's vocabulary and arrive after the order events. RECONCILED's Payments-to-Ledger: Draining the Suspense Swamp designs the normalization layer, clearing accounts and three-way match between commerce events, provider records and bank movements; Never Update, Only Append: Designing Custom Ledgers designs the ledger itself. A marketplace adds two requirements on top: **per-seller balances**, because payouts are owed to many parties, and **per-seller, per-period aggregates**, because tax reporting is computed from them.

The regulatory switches

Moving money on-platform turns on obligations that a classifieds site largely avoids.

- **DAC7 seller reporting.** EU platform operators must collect, verify and report seller identity, financial account identifiers and the total consideration per quarter for sellers of goods above the excluded-seller test: fewer than 30 sales and no more than €2,000 of consideration in the period are excluded (Belastingdienst, n.d.). A lead-only site largely falls outside DAC7; in-platform payment makes the consideration known. **The decision to launch payments is also a tax-reporting decision**, and the ledger should aggregate per seller, per quarter and per Member State from the first day. Rules of the Marketplace: DSA, DAC7 and Vertical Regulation owns the detail.

- **Consumer and platform law**. A platform that lets consumers conclude distance contracts with traders takes on trader-traceability duties under the Digital Services Act and information duties under EU consumer law (Rules of the Marketplace: DSA, DAC7 and Vertical Regulation, Private Sellers, Dealers and Agencies: Modeling Who Lists). Professional sellers using a protected-payment flow must be identified as traders.
- **Payment regulation**, through the provider. The provider's onboarding, limits and monitoring become constraints on the product: which sellers can be paid, how much and how fast.

In Practice: Two Safe Deal Orders

The Rotterdam city bike: pay at pickup

Femke listed her city bike on Tradeyard Netherlands for **€240** on Sunday 7 March 2027 at 19:42. Early on Monday afternoon she agreed **€225** with a buyer in chat. The order shows how "pay at pickup" works: the buyer pays in the app before the meeting, the money is held, and the seller receives it at the handover.

Time (Mon 8 Mar 2027)	Step	Context	Money
14:01	Buyer accepts Femke's €225 offer in chat; order placed with "pay at pickup"	Order	Fee quoted: 5% × €225 = €11.25, plus €0.70 = **€11.95**
14:02	Buyer pays €236.95 by iDEAL; provider confirms; listing reserved	Payment edge, Listing	€236.95 captured into Femke's held balance and Tradeyard's fee balance
14:02–18:05	Femke's other open conversations are told the bike is reserved; alerts stop	Listing, Messaging	None

Time (Mon 8 Mar 2027)	Step	Context	Money
18:08	At the handover, Femke enters the buyer's six-digit handover code	Order	€225 released to Femke's available balance; €11.95 recognized as Tradeyard revenue
18:10	Order completed; listing marked sold	Order, Listing	None
Tue 9 Mar	Payout to Femke's bank account	Payment edge	€225 paid out

The handover code is the release event: by giving it, the buyer confirms they have seen and accepted the bike, so pay-at-pickup protection covers payment security and the seller not turning up, not "not as described" after handover. Femke had already been verified as a payee by Adyen before her first Safe Deal payout, which is why Veldmark ID will carry that fact as a `payee.verified` claim (Who Is on the Other Side? Identity and Verification). The €11.95 fee is charged to the buyer on top of the price, so Femke's DAC7 consideration is €225; with four Safe Deal sales in 2027 totaling €610, she is an excluded seller.

A protected-payment brand is the scammer's favorite disguise, as Femke's Sunday evening showed. At 20:11 on Sunday a "buyer" had sent her a link to a fake Safe Deal payment page, on a domain registered that day. Tradeyard users in the Netherlands and Belgium reported about **41,000** payment-link phishing attempts in 2026. The countermeasure is not in the payment platform, which never saw the fake page; it is in the message safety gate that holds links imitating Safe Deal (From Interest to Contact: Messaging and Lead Management). A transactional pivot that does not budget for this is building **Fast Mail for Fraudsters** with its own logo on it.

A shipped order with a dispute

On Tuesday 13 April 2027, a buyer in Antwerp agrees to buy a used camera lens from a private seller in Ghent on Tradeyard Belgium for **€60**, shipped by parcel.

Date	Step	Money and evidence
Tue 13 Apr	Order placed and paid by card	Buyer pays €60 + fee €3.70 (5% × €60 + €0.70) + label €6.95 = **€70.65**; listing reserved
Wed 14 Apr	Seller prints the label and hands the parcel in at a PUDO point	`ParcelHandedIn`; the seller's deadline was three days
Fri 16 Apr	Parcel delivered to a PUDO point near the buyer and collected	`ParcelDelivered`; the two-day dispute window starts
Sat 17 Apr	Buyer opens a dispute: a scratch on the front element not shown in the photos	Funds stay held; the case gets the listing photos, chat history and the buyer's photos
Mon 19 Apr	Seller offers a €15 partial refund in the dispute flow; buyer accepts	`DisputeDecided`; €15 refunded to the buyer's card
Tue 20 Apr	€45 released and paid out to the seller	The fee is not refunded on a partial refund; the label margin (€6.95 charged, about €5.20 carrier cost) stays with Tradeyard

Every step produced an event, and every amount has a ledger line: €70.65 captured; €3.70 fee revenue; €6.95 label revenue against a carrier cost; €15 refund; €45 seller payout; the provider's processing fee, absorbed by Tradeyard. The December 2026 clearing balance existed because, before 2027, partial refunds were agreed in chat and executed manually by support, and nothing linked them to the order.

Figure 4: *The December 2026 clearing balance existed because refunds, label charges and partial refunds were handled outside the order; in the target design every amount is an event with a ledger line, traceable from order to bank movement.*

Safe Deal in 2026, in numbers

Measure (2026)	Value
Orders	11.6 million (NL, BE, PL)
GMV	about €1.02 billion; average item price about €88
Revenue	€75 million: buyer fees about €59 million, shipping-label margin about €16 million
Shipped share	about 78% of orders (about 9 million labels); the rest "pay at pickup"
Distinct sellers paid	about 1.3 million, of whom about 21,000 above the DAC7 excluded-seller test
Disputes opened	about 1.3% of orders (about 151,000); about 70% resolved by agreement or return
Buyer-protection losses funded by Tradeyard	about €4.1 million, 0.4% of GMV
Card chargebacks	under 0.1% of orders

Measure (2026)	Value
Unmatched items in clearing at the December month-end	about €2.3 million

Safe Deal is a good business: €75 million of revenue on €1.02 billion of GMV is a 7.4% take rate, and losses are an order of magnitude below revenue. But it was **run as a feature on top of Yardstick's world view** until 2027: its dispute desk worked from e-mail templates, partial refunds were manual and the finance view was reconstructed at month-end.

The Architecture Advice Forum's Decisions

On 22 September 2027 the forum applied the four conditions and the architecture above to each proposal.

Safe Deal Portugal: approved for Q3 2028, on the target platform rather than as a fourth copy. The conditions all hold for Portuguese goods. The launch adds local payment methods and two national carrier and PUDO networks through new shipping adapters, runs under the same Tradeyard B.V. contract with Adyen, and reports Portuguese sellers under DAC7 through the existing Dutch filing. It waits for the platform work below, because launching a fourth country on the 2024 design would multiply the clearing problem.

Motorvane private car payment: not escrow; a narrow pilot. A 5% fee on the Essen Golf's **€21,900** would be about **€1,095**, which no buyer would pay, and a single misdirected payment of that size is a material loss. Most Motorvane listings come from dealers, who have their own payment processes and legal duties. The forum approved only a design study for a "secure handover" for private-to-private sales in Germany: funds held for at most 72 hours around a scheduled handover, a flat buyer fee, release by handover code, and payee verification at proof-of-person level. It is to be decided in Q2 2028 with Adyen's limits and Motorvane's fraud data in hand. Until then, Motorvane invests in the warning and verification controls that Scams, Fakes and Account Takeovers: Fraud Prevention for Listings describes.

Hearthly Spain protected deposits: declined. A rental deposit is a large payment to a landlord under a lease, governed by its own legal rules, and holding it would put Hearthly into the rental contract itself, a different marketplace model with its own licensing and consumer-law questions. The Barcelona ring is better answered upstream: the private-landlord rule that requires proof of person with name match before publishing rentals in Spain, live in Q4 2027 (Who Is on the Other Side? Identity and Verification), plus warnings in messaging when a counterparty asks for a deposit before a viewing. **This was the forum's clearest case of declining The Forced Fit**: a goods checkout is not a rental product.

The transaction platform: funded first. Before Portugal, Safe Deal moves to the six-context design: an order service that owns availability and timers, a shipping context with normalized tracking events, a dispute case service built on the shared trust-and-safety case management, and a transaction ledger that receives order and money events and feeds group finance. The forum set three objectives for the end of 2028: every Safe Deal amount traceable from order to bank movement; unmatched clearing items under **€100,000** at any month-end; and DAC7 aggregates computed from the ledger rather than assembled from exports.

Trade-offs

Decision	Option A	Option B	Prefer A when	Prefer B when
Who holds funds	Licensed provider's marketplace product	Own e-money or payment licence	Transactions are a minority of revenue; the group wants speed and less regulation	Payments are the core product and volume justifies safeguarding, capital and a full ledger
Merchant of record	Platform	Seller	Private sellers; one brand and one buyer protection	Professional sellers with their own customer relationship and refunds
Release rule	Buyer confirmation or a fixed window after delivery	Fixed days after payment	Reliable tracking events exist	Tracking is weak; simpler but slower and riskier for buyers
Fee payer	Buyer pays a protection fee	Seller pays a commission	Supply growth matters most (free to sell)	Professional sellers or very low price points

Decision	Option A	Option B	Prefer A when	Prefer B when
Fee shape	Percentage plus fixed amount	Flat fee	Moderate prices	High-value items (cars) or tiny items
Shipping	Integrated carriers and labels	Seller arranges shipping	Parcels are standard and tracking drives release	Bulky or specialist items
Scope of a shared platform	One transaction platform for all goods brands and countries	Per-brand transaction stacks	Same model, same provider, same legal regime	Different providers, carriers and consumer rules that make sharing an abstraction tax
Pivot depth	Optional protected payment beside contact	Payment-first, contact second	Classifieds liquidity is the asset	The marketplace is recommerce from the start (Vinted)

Buyer-paid fees keep selling free, which protects supply, and put the price at the moment of highest intent; Vinted describes its mandatory fee, long known as the Buyer Protection fee and called "the Vinted fee" on its UK price list as of October 2026, as paid at checkout and "usually 3% to 8% + £0.3 to £0.8" in the UK (Vinted UK price list, 2026). Marktplaats' 2017 product charged the buyer 2% plus €0.40 (RetailTrends, 2017), its 2022 successor 5% (AVROTROS Radar, 2022).

Sharing the transaction platform across countries is attractive, but it is the layer most exposed to local differences: payment methods, carriers, consumer law and dispute norms. Adevinta's 2023 plan named "the full transactional journey" among the investments it would stop duplicating; under its 2025 decentralization, each market's provider contracts, carriers and consumer-law regimes argue for local ownership again (Adevinta Annual Report 2023; Strategic Report 2025).

> **INFERENCE, NOT DOCUMENTED:** The public descriptions of Kleinanzeigen's and Marktplaats' flows fit a common pattern: a licensed

provider holds funds; the marketplace owns the offer and acceptance in chat, the buyer-fee pricing, label generation and tracking, the release timer and dispute handling. It is a plausible reference architecture, not a description of either company's systems.

Vertical Differences

Dimension	General classifieds goods	Real estate	Automotive	Recommerce and jobs
Is the pivot common?	Yes, category-gated (leboncoin, Kleinanzeigen, Markt-plaats)	Rare; transactions settle with notaries, agents and lease contracts	Reservations and deal steps, rarely full payment	Recommerce is payment-first; jobs have no payment between parties
What gets paid on platform	Item, buyer fee, shipping	At most reservation or application fees	Reservation fees, deposits through dealers, financing referrals	Item, fees, shipping, wallet balances
Who holds funds	Licensed provider	Notaries, agents, landlords	Dealers and their banks; finance partners	Provider or own licence (Vinted Pay)
Main fear protection answers	Non-delivery, misdescription	Fake listings and deposit fraud, answered by verification	Deposit scams, hidden damage, answered by history checks and verification	Non-delivery, counterfeits

Dimension	General classifieds goods	Real estate	Automotive	Recommerce and jobs
Typical failure	Ledger missing; Fast Mail for Fraudsters with the platform's brand	Platform holding money it is not allowed to hold	Workflow imposed on dealers (the reservation backlash)	Operational load of shipping and disputes

Automotive shows the professional side of the trade-off. Autotrader's Deal Builder moved reservations, part-exchange, finance and delivery steps online; the company attributed part of a fall in retailer forecourts in the second half of its 2026 financial year to "concerns with the speed and nature of our Deal Builder product roll-out", and responded by pausing auto-rollout and, as reported in December 2025, introducing a "request a reservation" option that does not require buyer payment (Autotrader, 2026; Car Dealer Magazine, 2025). **Transactional state placed on a professional's inventory must be opt-in, configurable and reversible**. Same Platform, Different Physics: How Listing Verticals Differ compares the verticals in full.

Failure Modes

The checkout ships before the ledger. The order flow goes live, payouts work, and finance discovers at the first month-end that refunds, partial refunds, label charges and chargebacks cannot be matched. Veldmark's €2.3 million clearing balance is the mild version. The fix is to make the ledger and reconciliation part of the launch criteria, not a later phase.

Release by timer only. Funds released a fixed number of days after payment, regardless of delivery evidence, pay scammers who never ship and make disputes after release unrecoverable. Release should depend on the delivery event and the dispute window, with a fraud hold able to stop it.

The double sale. The listing stays available while an order is in progress, a second buyer pays, and one of them must be refunded. The order must reserve the listing atomically and the reservation must expire.

Sold, but still everywhere. A completed order that does not reach every copy of the listing creates **Ghost Inventory**: alerts and digests promote an item that is gone, and a contact form invites buyers to ask about it. The bike in the Tuesday digest is the example.

Disputes by e-mail. Without a case system, evidence lives in mailboxes, decisions are

inconsistent, decision deadlines are missed and there is no data to price protection. A dispute desk needs cases, evidence, decision codes and service levels.

The protected-payment brand turned against users. Phishing pages that imitate the platform's own payment flow follow every launch. Messaging must treat links and payment requests that imitate the brand as high-risk, and users must learn that Safe Deal payments happen only inside the app.

Tax reporting discovered late. A platform that discovers DAC7 after its first year of payments has to collect tax identification numbers from thousands of sellers retroactively, with payout holds for non-responders. The data collection belongs in seller onboarding thresholds.

Provider lock-in by accident. Seller verification data, card tokens and payout accounts held only in one provider's systems make a later change of provider a migration of every seller. The marketplace should hold its own references and the verification outcome, even when the provider holds the documents (Adyen: One Platform from Checkout to Settlement discusses the dependency).

Imposing the goods checkout on other verticals. A checkout built for parcels is extended to cars, homes or rentals because it exists. This is **The Forced Fit** in its transactional form.

The Chief Architect's View

At group scale, transactions raise four standing decisions.

Decide the pivot per profile, and record it. Whether a country and category has protected payment, which fulfilment methods and which coverage belongs in the marketplace profile register (Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy), as configuration. The general manager owns the commercial decision; finance and legal sign off on the provider, the loss provision and the reporting obligations; architecture owns the cost of change and the platform it runs on. A rule I use: **no category gets protected payment without a named owner for disputes, reconciliation and DAC7.**

Share the transaction platform among the profiles that use it, and only them. At Veldmark, Safe Deal is the transaction platform for goods in Tradeyard's countries; Motorvane and Hearthly use the shared layers beneath it (identity, messaging, trust and safety, the event backbone) but not the checkout. Within the platform, keep country-specific parts (payment methods, carriers, consumer-law texts, dispute norms) as adapters and configuration, and keep the order model, the ledger and the dispute case model shared.

Treat the provider as a strategic dependency. Conforming to the provider's model is right for speed, and it ties the platform to that provider's accounts, onboarding and

reporting. Keep the order, the ledger and the seller's verification outcome in Veldmark's own systems, keep provider identifiers as references, and review the dependency at each contract renewal. Separability matters here too: a brand sold with its transaction business needs its seller balances, open orders and ledger history to go with it (From Newspaper Columns to Portfolios: How the Classifieds Industry Evolved).

Measure trust and money together. The metrics that matter are the share of eligible sales that go through protected payment, dispute rate and outcome by category, buyer-protection losses as a share of GMV, payout delay from delivery, reconciliation breaks at month-end and payment-phishing reports. A rising protected share with falling phishing losses is the pivot working; a rising take rate with rising unmatched items is the pivot borrowing from the future.

Questions to Ask

1. For which countries and categories do all four conditions hold (ships, moderate price, fear of non-delivery, strangers), and is the gate configuration or code?
2. Which licensed entity holds the buyer's money, who is merchant of record, and who loses money first when a refund cannot be recovered from the seller?
3. What exactly releases funds to the seller, and can a fraud rule hold a payout with a recorded reason?
4. How does an order reserve the listing, when does a reservation expire, and how quickly does a sale reach alerts, digests and recommendations?
5. Can we trace every amount in an order, including partial refunds, labels, provider fees and chargebacks, from the order to the bank movement?
6. Where do disputes live, what evidence do they use, and what is the loss provision per category?
7. Are DAC7 aggregates computed from the ledger, and do we collect tax identifiers before sellers cross the thresholds?
8. Which data would we lose if we changed payment provider next year: tokens, seller verification, payout accounts, transaction history?
9. What stops scammers imitating our protected-payment brand in messages and e-mail?
10. If we had to switch protected payment off in one category, what happens to open orders, held funds and sellers' expectations?

From Transactions to Monetization

Protected payment adds a revenue line, but it is one of several ways a listing marketplace makes money, and usually not the largest. Visibility products, professional subscriptions,

insertion fees, lead products and advertising each ask for different systems and each spend liquidity differently. How Listing Marketplaces Make Money catalogs the models with their published KPIs and shows what each one requires from the architecture.

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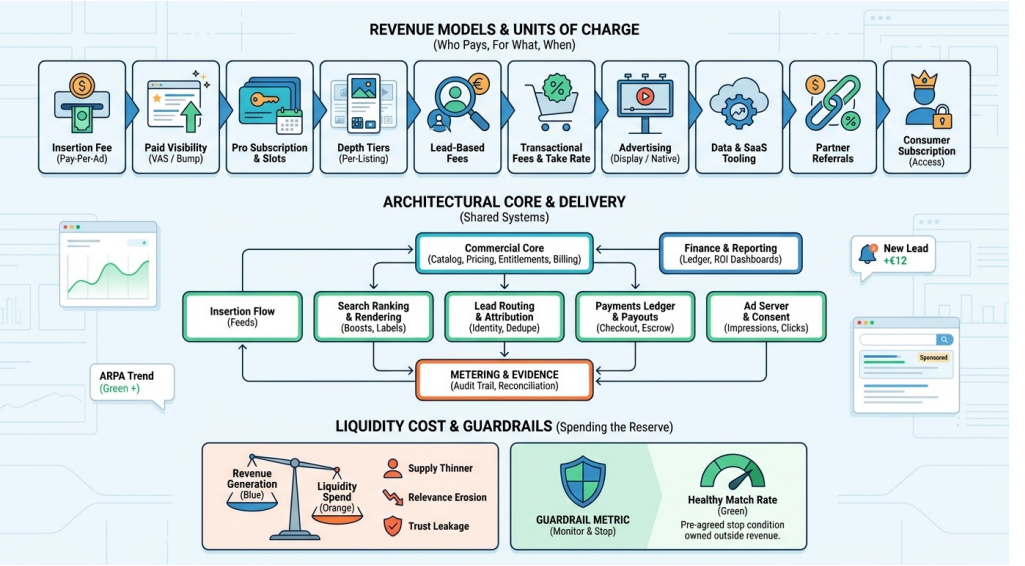
Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

30. How Listing Marketplaces Make Money



IN THIS CHAPTER, YOU WILL: Learn the ten revenue models listing marketplaces use, from insertion fees and paid bumps to dealer packages, pay-at-close leads and take rates. You will read the KPIs that public portals publish and see what each model demands from the systems. Then you will judge three monetization proposals at Veldmark by what they would cost in architecture and in liquidity.

WHY THE BUSINESS CARES: Revenue growth at a mature portal comes mostly from changing what is sold and how it is priced, not from more customers. A monetization model that the systems cannot measure, enforce or bill correctly leaks revenue, invites disputes and, at worst, drives away the supply the marketplace lives on.

WHY ARCHITECTS SHOULD CARE: Every monetization model needs specific systems: per-listing orders, contract and quota management, lead attribution, a checkout and ledger, or an ad stack. Choosing the model chooses

the architecture. Every paid product also touches ranking, so monetization changes are search changes, and search changes are monetization changes.

KEY POINTS:

- **The unit of charge is the architectural decision**. A marketplace can charge per listing, per position, per account, per lead, per transaction or per audience. Each unit needs a different system of record, such as listing orders, contracts and quotas, lead attribution, a payments ledger or an ad server. The KPI a portal publishes (ARPA, ARPR, yield per listing, take rate) shows which unit it chose.
- **Monetization spends liquidity**. Insertion fees reduce supply, paid positions displace relevance, and lead fees push professionals to dispute leads or to take buyers off the platform. Each model needs a guardrail metric that measures what it costs the match rate, owned by someone other than the team that books the revenue.
- **Pricing power is real but conditional**. Leading portals grow revenue per customer through annual price-and-product events while the customer count stays flat. Concentrated professional customers can still push back, as Autotrader's 2026 results show. The systems must let the business change prices and packages as data, measure the effect causally and roll back.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Vanity Liquidity** — revenue plans built on listing counts and page views, instead of contacts and sales, sell professionals exposure they cannot convert. Renewal then fails.
- **Multiplier Creep** — every visibility product is a claim on ranking. Selling more of them, with no guardrail, quietly lowers the relevance the products depend on.
- **Billing by Guesswork** — a new revenue model added as conditions in existing services leaves nobody able to say what a seller bought and what they received.

A listing marketplace connects buyers and sellers and then usually steps aside. The sale happens on a doorstep, at a dealership or at a notary's office. A buyer who meets the seller can complete the deal without the platform. **Classifieds therefore rarely charge for the transaction**. **They charge for access to attention**: the right to list, a better position, a monthly package of slots, a delivered lead, a placed advertisement. Over the last decade, many have also started charging for the transaction itself.

Every one of these is a **monetization model**. In this book, a monetization model is the combination of *who pays*, *for which unit*, *at which moment in the funnel*, and *on what*

pricing basis. The definition is deliberately operational, because each part becomes a system. *Who pays* sets the identity and billing depth. *The unit* sets the system of record for what was sold. *The moment* sets which flow carries the checkout. *The pricing basis* sets what must be metered.

Part VI follows a contact to revenue. From Interest to Contact: Messaging and Lead Management turned contacts into leads, and Beyond the Contact: Payments, Shipping and Buyer Protection carried some of them through payment. This chapter asks what the marketplace charges for along that path. It catalogs the models with public evidence and the KPIs portals publish, then shows the systems each model requires and the liquidity each one spends. Packages, Bumps and Quotas: Products, Pricing and Entitlements, the next chapter, designs the catalog, pricing and entitlement systems that implement them. Selling Attention: Display, Native and Retail-Media Advertising owns display and retail-media advertising, and Beyond the Contact: Payments, Shipping and Buyer Protection owns the transactional pivot. Both appear here only as models.

Setting the Scene at Veldmark: Where Does the Next Euro Come From?

It is mid-May 2027 at Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build). Veldmark's private-equity owner has asked the group to show how revenue grows from **€1.31 billion** in 2026 without cutting into the Common Ground budget. Olivier Petit, VP Monetization, brings three proposals to the architecture advice forum, the weekly open review that Ingrid Halvorsen, the Group Chief Architect, set up in April.

1. **Tradeyard Netherlands: a stronger visibility ladder**. Raise the price of "Top listing", which rotates eligible listings through three reserved slots above the results, and add a new private-seller product, "Spotlight", that guarantees the first of those slots for 48 hours.
2. **Motorvane Germany: a per-lead component**. Keep the dealer packages, with listing slots, but bill every qualified lead above a monthly allowance included in the package.
3. **Hearthly France: a commission share on Valuo seller leads**. Homeowners who request a Valuo estimate are routed to partner agencies. Hearthly would take a share of the agency fee when the property sells, instead of a flat monthly fee for the leads.

Olivier presents the proposals as pricing decisions. Ingrid asks the forum to treat them as architecture decisions and puts four questions to each one. *What unit do we charge for? Which system will hold the truth about what we sold? What will it cost us in liquidity? Who will notice first if it goes wrong?* The rest of this chapter supplies what she needs to answer them. The three proposals come back at the end as a worked example.

Concepts: What a Listing Marketplace Sells

The price structure: who pays

Two-sided markets have a **price structure**, not just a price level. Rochet and Tirole's 2003 paper framed a platform's problem as getting "both sides on board" and studied how platforms divide prices between the two sides. The usual reading of that work is that platforms subsidize the side whose participation creates the most value for the other side (Rochet and Tirole, 2003). Classifieds apply this almost literally: buyers are free, private sellers are free or nearly free because their listings are the depth that attracts buyers, and professionals pay. Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics owns the network-effect argument. Here the consequence is enough: **the free side is not a cost to be minimized but the source of the value the paying side buys**.

Charging the demand side is the exception, and it works only where supply is scarce. Scout24's ImmoScout24 sells consumer "Plus" subscriptions to renters and buyers. In 2025 its private-segment subscription revenue grew 18.8% to €107.2 million, with an average of 506,937 customers. The growth came partly from monetized credit checks inside Plus memberships for renters (Scout24 preliminary FY2025 release, 2026, in German). Demand there is far larger than supply, as in German urban rental markets, and buyers pay for an advantage over other buyers.

What the professional buys: probability of a match

A dealer paying a monthly package is not buying web pages. They are buying a **probability of a match**: views that become contacts, contacts that become test drives, and test drives that become sales. Every paid product is a claim on that probability, and the claim is only as good as the organic liquidity beneath it: a top position is worth something only while enough buyers look at the top of the results. This is why I treat every monetization model as a way of **spending liquidity**. The question is never only "will sellers pay?" but also "what does this charge do to the match rate that makes them willing to pay?"

The unit of charge and the denominator

The most useful single fact about a monetization model is its **unit of charge**: the thing whose quantity sets the bill. Public portals reveal their unit through the KPI they report, because the KPI's denominator is the unit.

Company	Headline unit KPI	Value	Period	Unit of account
Rightmove (UK property)	ARPA, average revenue per advertiser	£1,621 per month	Calendar 2025	Advertiser: agency branch or development
Autotrader (UK motors)	ARPR, average revenue per retailer	£2,995 per month	Year to 31 March 2026	Retailer forecourt
CarGurus (US motors)	QARSD, quarterly average revenue per subscribing dealer	US$6,616 per quarter (US market $7,938)	Q4 2025	Paying dealer
REA Group (Australia property)	Residential Buy yield growth	+13%	Year to 30 June 2026	Listing
Hemnet (Sweden property)	ARPL, average revenue per listing	SEK 9,095 (+12.4%)	Q2 2026	Listing
Zillow (US property)	Residential revenue per visit	US$0.178	Calendar 2025	Consumer visit
eBay	Take rate	13.94%	Calendar 2025	Gross merchandise volume (GMV)

Sources: Rightmove RNS 2026-02-27; Autotrader FY26 results 2026-05-21; CarGurus and Zillow 10-K FY2025 (2026); REA FY26 release 2026-08-06; Hemnet investor page, accessed 2026-10-19; eBay 10-K FY2025 (2026). Full references in Sources and Evidence.

Compare the denominators, not the values: the currencies and periods differ, and the denominators show four different businesses. Per-professional subscription portals (Rightmove, Autotrader, CarGurus) report revenue per account. Seller-pays portals (REA, Hemnet), where the property's seller funds the listing tier, usually through the agent, report revenue per listing. Lead-generation portals (Zillow) report revenue per unit of audience. Transactional marketplaces (eBay) report a take rate on GMV. a16z's "13 Metrics for Marketplace Companies" draws the same line. GMV is "the total sales dollar volume of merchandise transacting through the marketplace", and revenue is "the portion of GMV that the marketplace 'takes'" (a16z, 2020). Classifieds have no

observable GMV, which is why they report ARPA and its relatives.

Three KPI definitions recur in this part of the book. **ARPA** (average revenue per advertiser, or per account) is Rightmove's term, defined as "revenue from Agency and New Homes advertisers in a given month divided by the total number of advertisers during the month, measured as a monthly average over the year" (Rightmove, 2026). **ARPR** is Autotrader's equivalent per retailer forecourt; £2,995 × 13,942 forecourts × 12 months ≈ £501 million, Autotrader's reported retailer revenue, so it counts retailer revenue only. **Take rate** is eBay's "net revenues divided by GMV".

Value-added services

VAS (value-added services) is the industry's name for paid options bought on top of a basic listing: bumps that re-date a listing, top or featured positions, highlights and galleries, longer durations. OLX's engineering team uses the term directly: its ranking stack includes a "VAS LTR" model that balances "maximizing relevance for users and ensuring visibility for Value Added Services (VAS), ads that have paid for additional exposure" (OLX Tech, 2025). Here VAS means options that change how a listing is presented or ranked; subscriptions that bundle VAS remain a separate model.

The Ten Models

Listing marketplaces use ten recurring revenue models. Most groups run several at once. Each brand usually has one center of gravity and three or four supporting models. The table summarizes them; the sections that follow cover what each needs from the systems.

#	Model	Who pays	Unit of charge	Documented example (dated)
1	Insertion fee (pay-per-ad)	Private or small seller	Listing, at publication	Scout24 pay-per-ad +10.9% (FY2025); Autotrader private-seller revenue £23.6m (FY26)

#	Model	Who pays	Unit of charge	Documented example (dated)
2	Paid visibility (VAS)	Private or pro seller	Listing × time window	Vinted Bump for 3 or 7 days (price list, 2026); OLX packages by boosts per 30 days (2026)
3	Pro subscription with quotas	Professional	Account × month, with slots	Rightmove packages and "package commitment" (2025); Autotrader "package staircase" (FY26); CarGurus listing tiers (FY2025)
4	Depth products (per-listing tiers)	Seller, often via the agent	Listing × tier	REA Premiere+ and "depth penetration" (FY26); Hemnet ARPL (2026)
5	Lead-based fees	Professional	Lead, share of leads, or closed deal	Zillow share of voice and Zillow Preferred (FY2025); AVIV France commission share (2023)
6	Transactional fees and take rate	Buyer, seller or both	Order or GMV	eBay take rate 13.94% (2025); Vinted buyer fee (2026); Adevinta transactional €103m (2023)

#	Model	Who pays	Unit of charge	Documented example (dated)
7	Advertising	Brands, agencies, programmatic buyers; sellers for sponsored listings	Impression, click, sale	Adevinta advertising €312m of €1,826m (2023); eBay advertising $1,993m (2025)
8	Data and SaaS	Professionals, partners	Seat, account or dataset	Scout24 Transaction Enablement €106.6m (FY2025); Zillow Follow Up Boss (FY2025)
9	Financing and insurance referrals	Partners (lenders, insurers, brokers)	Referral, application or funded contract	Rightmove Mortgages £6.8m (2025); CarGurus financing-partner revenue (FY2025)
10	Consumer subscriptions	Buyers or renters	Consumer × month	ImmoScout24 Plus, 506,937 average customers (FY2025)

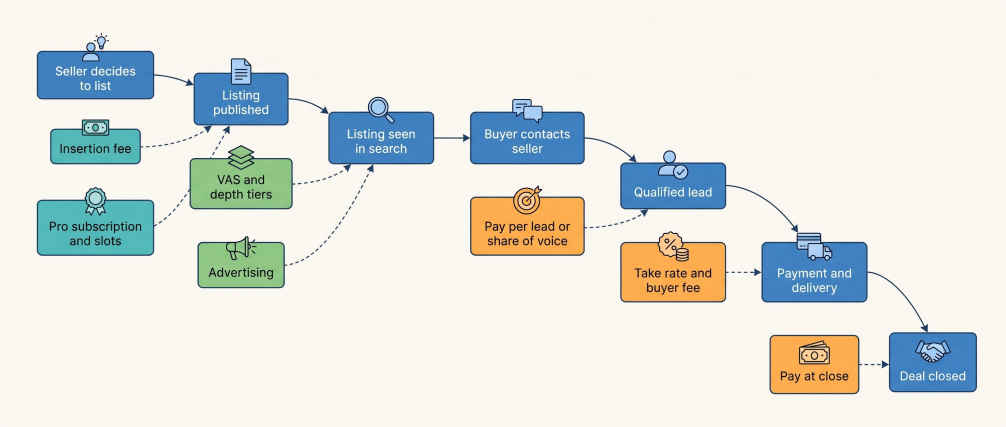


Figure 1: *Each model charges at a different point in the funnel. The further down the funnel the charge sits, the more of the funnel the marketplace must observe, so the more systems it must build: lead attribution after contact, a payments ledger after the lead, outcome matching after the payment.*

1. Insertion fees: charging for the right to list

An **insertion fee** charges a seller to publish a listing. It is usually limited to categories where items are valuable or listings are few: vehicles, property, sometimes jobs. Often it applies only above a free quota. Scout24 reported pay-per-ad listings by private landlords and sellers growing 10.9% in 2025. Autotrader earned £23.6 million in FY26 from individual sellers who "pay to advertise their vehicle". That was 11% less than the year before, which is a reminder that private supply in a paid model is price-sensitive (Autotrader FY26, 2026).

What it requires. Pricing at listing creation, keyed by category, region and seller type. Payment before publication, so the insertion flow becomes a checkout. A rule for moderation: a fee paid for a listing that moderation then rejects must be refunded or credited, automatically. Listing durations become a commercial term, so expiry and renewal (The Listing Aggregate: Identity, Lifecycle and Invariants) turn into billable events.

What it spends. Supply. Every fee removes the marginal listing: the cheap item, the hesitant seller, the "let's see if anyone wants it". In categories where buyers judge the marketplace by depth, that is a direct liquidity cost. The model fits where each listing is valuable to the seller and where a small fee filters out low-intent and fraudulent supply. That last benefit is real: an insertion fee is also a trust control, because a paid listing is a more expensive attack.

2. Paid visibility: bumps, tops and highlights

Paid visibility is the classic horizontal model: free listing, paid prominence. It comes in three mechanics:

- A **bump** (also called refresh or push to top) re-dates the listing so it reappears near the top of date-sorted results. Vinted sells bumps for 3 or 7 days, priced by "duration, item price and … territorial scope" (Vinted price list, updated 2026-10-08).
- A **top or featured slot** reserves positions above or within the organic results. CarGurus sells "Highlight", a fixed monthly fee for "increased vehicle visibility slots" (CarGurus 10-K FY2025, 2026).
- A **highlight**, **gallery or showcase** changes how the listing card looks: color, a larger card, more photos.

What it requires. A scheduler for time-boxed effects, such as running a bump on day 3 and day 6 of a seven-day product. A separately updatable ranking timestamp in the search index: bumps must not overwrite the listing's creation date, a point The Listing Aggregate: Identity, Lifecycle and Invariants makes about mobile.de's separate `renewalDate`. Slot allocation and rotation when more listings have bought "top" than there are top positions. A **paid label** on every promoted result. Under EU consumer law, presenting search results "without clearly disclosing any paid advertisement or payment specifically for achieving higher ranking" is blacklisted as always unfair (Directive (EU) 2019/2161, Annex I point 11a; Rules of the Marketplace: DSA, DAC7 and Vertical Regulation owns the detail). And a measurement of what the paid position actually delivered.

What it spends. Relevance. Every top slot shows a paid listing instead of the most relevant one. The OLX VAS LTR model exists because these two objectives conflict. OLX also documents what happens when the search stack changes underneath a paid product. When it added vector retrieval, the keyword-trained ranking model underscored semantic results, which "would be particularly problematic for VAS ads retrieved via semantic search", and the model had to be retrained (OLX Tech, 2025). **A search change is a monetization change**, and Ranking for Two Sides: Relevance, Freshness and Paid Visibility owns how paid and organic results should be blended.

3. Professional subscriptions with quotas

The vertical portals' center of gravity is the **professional subscription**: a monthly contract that grants a number of listing slots, a number of boosts, access to tools and statistics, and a level of prominence. The contract is usually arranged in tiers, which Autotrader calls a "package staircase".

Published filings show the dominant professional shape: a commitment plus extras. Rightmove's 2025 ARPA grew £97 to £1,621. On the agency side, 62% of the increase came from product rather than price. Its top estate-agency package, Optimiser Edge, reached 35% of independent agents. "More than 50% of independent agents purchased incremental product above their package commitment" (Rightmove, 2026). That last figure describes **a subscription with a commitment plus metered or à-la-carte extras**. CarGurus documents the contract terms: paid listing packages on "monthly, quarterly, semiannual, or annual" terms that "generally auto-renew on a monthly basis and are cancellable by dealers with 30 days"' notice. It also runs a free tier, "Restricted Listings", with "capped anonymous leads and connections" (CarGurus, 2026).

Autotrader's FY26 results give the best public decomposition of how a subscription business grows. ARPR rose £141 to £2,995, made up of three levers (Autotrader, 2026):

Lever	FY26	FY25	What drives it
Price	+£117	+£78	The "annual 1 April 2025 pricing and product event" combining a like-for-like price increase with additional products
Stock	−£48	−£22	Retailers advertising fewer vehicles, with paid stock falling after November 2025
Product	+£72	+£77	"Inclusion of Co-Driver within retailer packages" and more paying new-car retailers, "offset by lower average package penetration"

The stock lever is the architectural lesson. **A dealer can cut spend without churning by advertising fewer cars**, because the package price depends on the number of paid slots. Slot design is therefore a revenue-elasticity decision, and it is encoded in the entitlement service that Packages, Bumps and Quotas: Products, Pricing and Entitlements designs. The product lever is the other lesson. Autotrader bundles its generative-AI listing tool, Co-Driver, *inside* packages instead of selling it separately, which turns a feature flag into a pricing argument.

What it requires. Contract and subscription management: terms, renewals, notice periods, upgrades and proration. An entitlement and quota service that every insertion path checks, including bulk feeds, where a dealer management system can push 300 cars into a 250-slot package overnight. Postpaid invoicing with VAT, credit notes and dunning. An account hierarchy that matches how the customer buys: group, branch and rooftop (Private Sellers, Dealers and Agencies: Modeling Who Lists). ROI reporting, because renewal is negotiated on it. CarGurus describes account managers providing "inventory merchandising, ROI tracking, and profit growth strategies", so ROI data is a sales tool, not just a product feature.

What it spends. Less liquidity than the other models, because the charge sits on the

paying side and does not filter what buyers see. The cost appears elsewhere: in the **power of concentrated customers**, discussed below, and in the temptation to sell prominence inside every package until the packages compete with one another for the same top positions.

4. Depth products: charging per listing tier

In seller-pays property markets, revenue is **listings × yield**. The seller funds a listing tier through the agent, and yield is the mix of depth tiers times their prices, plus add-ons. REA Group reported Residential revenue of A$1,290 million (+12%) for the year to June 2026. Buy revenue grew through "a 13% increase in yield, with national listings flat", helped by a 7% average Premiere+ price rise and "increased depth penetration" (REA Group FY26, 2026). Sweden's Hemnet shows the exposure of the same model. Its Q2 2026 average revenue per listing rose 12.4% to SEK 9,095 while net sales fell 23.1%. On 29 September 2026 it announced "free entry-level formats" to gather all publicly available listings (Hemnet investor page, accessed 2026-10-19). The cause of the volume loss is not covered by the sources used here, so I draw only the structural point: **a per-listing model is exposed to supply leaving**, and its most direct defense is to make entry free.

What it requires. Per-listing orders and upgrades in mid-life (a listing moved from standard to premium on day 10), with the tier stored as a reference on the listing. Agent-mediated billing, often with the agent rebilling the vendor. Delivery evidence per listing, because the person who paid, the vendor, is not the person who sees the dashboard, the agent.

5. Lead-based fees: pay per lead, share of voice, pay at close

Lead-generation portals charge for the contact itself. Three variants have public documentation, all from Zillow's FY2025 10-K (Zillow, 2026):

- **Cost per lead**. Zillow Rentals prices advertising on a "cost per lead, lease, listing or impression basis or for a fixed fee for certain advertising packages".
- **Share of voice**. Under Premier Agent's market-based pricing, "connections are distributed to Premier Agent partners in proportion to their share of voice, or a Premier Agent partner's share of total advertising purchased in a particular zip code".
- **Pay at close**. Under Zillow Preferred (formerly Flex), "agent partners are provided with leads and pay a performance advertising fee when a real estate transaction is closed with one of the leads, generally within two years". The accounting consequence is stated directly: "the transaction price represents variable consideration", estimated from assumptions including "the conversion rate of a lead to a real estate transaction".

In Europe, AVIV France "monetises seller leads through a commission-share model", reported in 2023 (OnlineMarketplaces, 2023).

What it requires. These models put the most new systems behind the revenue line. Cost per lead needs a precise definition of a **billable lead**, plus deduplication (one buyer, three channels, one lead), validity rules, and a dispute and credit process. Share of voice needs an **allocation engine** that routes each connection in proportion to spend per geography, and forecasts the supply of connections per zip code so that sales teams do not sell what will not be delivered. Pay at close needs an **outcome pipeline**: lead identity, assignment, synchronization with the professional's CRM, matching to transactions months later (public records, agent reports, notary data), revenue estimation and true-ups. From Interest to Contact: Messaging and Lead Management owns the lead object and routing. The monetization point is that **the revenue is only as trustworthy as the attribution chain beneath it**.

What it spends. Trust between the portal and its professionals, and sometimes the buyer's experience. Pay-per-lead gives the portal an incentive to manufacture contacts. It gives the professional an incentive to dispute them and to move buyers off the platform before a contact can be counted.

6. Transactional fees and take rates

When the marketplace takes the payment, it can charge on the transaction: a seller commission, a buyer fee, or a margin on shipping. eBay's take rate was 13.94% on US$79.6 billion of GMV in 2025. eBay attributes the growth to first-party advertising and its UK shipping program, "partially offset by lower fees in connection with our U.K. consumer-to-consumer initiative" (eBay 10-K FY2025, 2026). Vinted charges no seller commission and puts a mandatory fee on the buyer at checkout. Long called the "Buyer Protection fee", it appears as "The Vinted fee" on the UK price list as of October 2026. In the UK it is "usually 3% to 8% + £0.3 to £0.8"; in the US "usually 5% + $0.70" (Vinted UK price list, updated 2026-10-08).

Classifieds groups adding this model report modest totals. Adevinta's 2023 transactional revenue was €103 million of €1,826 million (about 6%). In France it was €73 million of €548 million, and Adevinta said the French EBITDA margin fell two points "mainly reflecting the evolution of the business mix with an increasing share of transaction services and a decreasing share of highly profitable advertising revenues" (Adevinta Annual Report 2023, 2024).

What it requires. Checkout, a payment service provider's marketplace product, pay-outs, disputes, buyer protection, an order state machine and a financial ledger. Beyond the Contact: Payments, Shipping and Buyer Protection owns these, and RECONCILED's Payments-to-Ledger: Draining the Suspense Swamp owns the ledger integration. Two monetization points belong here. First, **a buyer fee shown on every listing card puts pricing on the read path**. Vinted's fee varies by "item characteristics, order value, order

type". If the price breakdown appears in search results, the pricing service must answer at search scale, not checkout scale. Second, **buyer protection is a liability as well as a fee**. eBay books losses "resulting from our buyer protection programs, chargebacks" as transaction losses, so the fee must be priced against a provisioned loss, not just against competitors.

What it spends. Margin and simplicity, as the Adevinta figure shows. The trust gain can raise liquidity in categories where buyers fear scams. Payment friction can lower it where a cash handover is the norm.

7. Advertising

Display, native and sponsored-listing advertising monetize the audience rather than the sellers. Adevinta's 2023 advertising revenue was €312 million, about 17% of the total. In its European Markets segment, the former eBay Classifieds horizontals, it was about 27% (€204 million of €744 million). Verticals earn far less: Autotrader's Manufacturer & Agency revenue was £15.2 million in FY26. **Sponsored listings** are a special case: sellers pay to promote their own listings inside the marketplace's ranking, as with eBay's Promoted Listings. They run in the search stack, not in the ad server. Selling Attention: Display, Native and Retail-Media Advertising owns the ad stack, consent and measurement. For this chapter, the point is that advertising is the only model whose revenue depends on traffic volume rather than on matches, so it is the model most exposed to Vanity Liquidity.

8. Data and SaaS

Professionals buy software and data from the portals that serve them: CRMs, valuation data, market analytics. Scout24's "Transaction Enablement" line, covering its Propstack CRM and the Sprengnetter and bulwiengesa data and valuation businesses, earned €106.6 million in 2025 (+17.5%). Zillow owns the agent CRM Follow Up Boss. These lines bring their own billing, often their own acquired stacks, and a governance problem: the portal sells the tool that creates listings for its competitors. Serving the Pros: Dealer, Agent and Merchant Tooling owns the neutrality question; Veldmark's Keybridge is the running example.

9. Financing and insurance referrals

High-intent moments (financing a car, a mortgage in principle, insurance for a new home, broadband for a new rental) can be sold to partners. Rightmove's Mortgages line earned £6.8 million in 2025 (+46%), and Rental Services ancillaries (utilities, broadband, insurance) £2.6 million. Autotrader reported a decline in revenue from its insurance partner in FY26. These models need partner APIs, a consented lead handoff, and **commission reconciliation against the partner's reporting**. That is a small finance integration with a recurring month-end, and it is often forgotten until the first dispute.

10. Consumer subscriptions

Consumer subscriptions let the demand side pay for an advantage when supply is scarce, as ImmoScout24 Plus shows above. The systems are consumer entitlements that gate search features (early access, alerts first), application profiles and credit checks. The model also brings a fairness question that the regulators of rental markets may ask: if paying renters see listings first, what do non-paying renters see?

Architecture: The Systems Each Model Requires

The ten models draw on one shared set of monetization capabilities, each model on a different subset. Mapping one onto the other is the first task for an architect asked to "add a revenue stream".

Capability	Listing fee	VAS	Pro subscription	Depth tier	Lead fees	Transaction	Advertising	Data and SaaS	Referrals	Consumer subscription
Product catalog and pricing	●	●	●	●	●	●	○	●	○	●
Consumer checkout (prepaid)	●	●		○		●				●
Contracts and postpaid invoicing	○	●	●	●			●	●	●	
Entitlement and quota service	●	●	●	●				●		●

Capability	Listing fee	VAS	Pro subscription	Depth tier	Lead fees	Transaction take	Advertising	Data and SaaS	Referrals	Consumer subscription
Ranking and rendering hooks	●	●	●	○	○	●				○
Delivery metering	●	●	●	●		●	○			
Lead identity and attribution		○		●					●	
Payments ledger and payouts					●					
Ad server and consent						●				
Outcome matching and true-ups				● (pay at close)	○				●	

● *central to the model;* ○ *needed in some variants. Author's synthesis from the published product mechanics above; no company has published its monetization architecture in detail.*

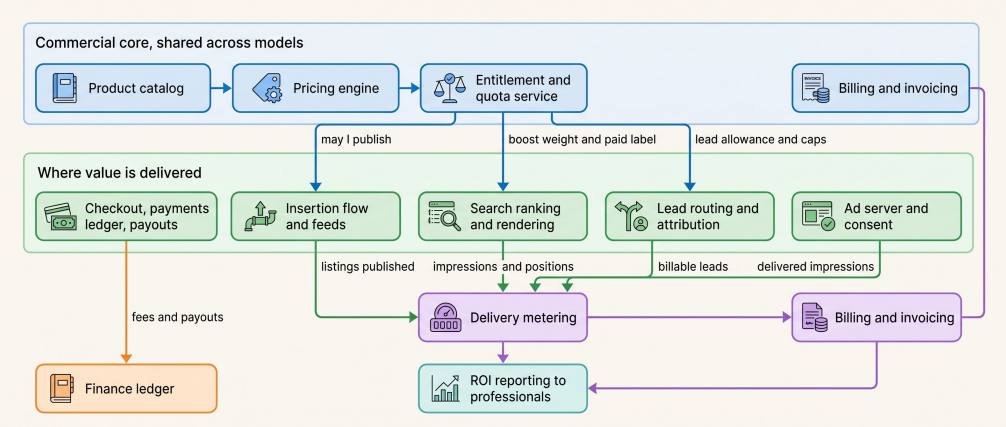


Figure 2: *A reference monetization architecture. The catalog, pricing, entitlement and billing core is shared across models. Each model adds or emphasizes one delivery system. The delivery systems emit metering events, so that what was billed can be reconciled with what was delivered. This is the author's reference inference, not a description of any company.*

Three boundaries in Figure 2 matter more than the boxes.

The listing holds references, not commercial truth. A listing knows that it carries a "Top listing" until Sunday at 20:00. It does not know the price paid, the contract it belongs to or whether the invoice was paid. The Listing Aggregate: Identity, Lifecycle and Invariants keeps monetization flags as references. Commercial truth lives in the entitlement ledger and in billing. This boundary lets a refund, a contract change or a moderation takedown change the listing's paid state without the listing service understanding contracts.

Delivery systems emit evidence. The search stack must report which paid listings appeared in which positions. The lead system must report which leads were billable and which were filtered. The insertion flow must report which listings consumed which slots. Without this evidence, nobody can answer the question every professional will eventually ask: *what did I get for my money?* Packages, Bumps and Quotas: Products, Pricing and Entitlements turns this into the "sold versus delivered" audit.

Ranking is a consumer of monetization, not its owner. Paid products change ranking inputs: a boost weight, a pinned position, a fresher timestamp. The rules for how much a paid input may move the result order belong to the ranking team's guardrails, set jointly with monetization, not to whichever team sold the product last. Without that rule, Multiplier Creep follows.

In Practice: Pricing Power and Its Limits

Growth through price and product events

Mature portals grow revenue per customer far faster than their customer count. Rightmove's total membership grew 1% in 2025 while ARPA grew 6%. Its 2026 guidance is about 1% membership growth and ARPA growth of £110–120. Autotrader's forecourt count fell 0.5% while ARPR grew 5%. CarGurus attributes its 2025 growth to "signing on new dealers at market rates, and revenue expansion driven by subscription tier upgrades, broader adoption of add-on products, and like-for-like price increases". The common mechanism is the **annual price-and-product event**. Once a year, often on a fixed date, prices rise and packages are reshaped with new products bundled in. Autotrader names it: its "annual 1 April 2025 pricing and product event".

For the systems, this means **a price change is a recurring, dated release that must be shipped as data**. The new price list, the new package contents and the migration of every contract to its new package must be prepared in advance, take effect at midnight in each market's time zone, and be reversible if the commercial team backs down after customer pushback. A catalog that needs code changes to add a product, or a pricing table edited in place without effective dates, turns the most important commercial event of the year into a deployment risk. Packages, Bumps and Quotas: Products, Pricing and Entitlements designs the alternative.

When pricing power meets concentrated customers

Pricing power weakens where professional customers are concentrated. Gurley's 2012 framework for evaluating marketplaces lists fragmentation as a factor: "High buyer and supplier fragmentation is a huge positive" (Gurley, 2012). Vertical portals depend on professionals who are less fragmented than private sellers: dealer groups, agency networks, franchises. Their bargaining power shows up in the published record. In its FY26 results, Autotrader reported that average forecourts were 236 (1.7%) lower in the second half. Management attributed the fall to "more difficult cost-related trading conditions and concerns with the speed and nature of our Deal Builder product roll-out". In the same document, Autotrader reports a share of time on site eleven times that of its nearest competitor. **Market leadership does not make a portal immune to customer revolt when it changes the commercial model**.

The architectural reading is about staging. A change to *what professionals are sold* needs the same rollout tools as a risky code change: cohorts, feature flags tied to contracts, a feedback loop from account managers, and a measured rollback path. Pricing and packaging are part of the deployment pipeline, whether or not the organization treats them that way.

Measuring whether a paid product works

A naive dashboard can make the best paid products look the worst, as a 2026 OLX post on causal inference shows (OLX Tech, 2026). At OLX, professional sellers choose a package when they publish, and packages differ by the number of boosts per 30 days. Naive dashboards showed premium-package listings "appeared to perform worse than ads using cheaper packages". This was selection bias: sellers with hard-to-sell cars bought more boosts. After propensity score matching on car attributes, the estimated uplift was +17.6% from Package 1 to Package 2 and +97.4% from Package 1 to Package 4, and the results were validated against a randomized A/B test.

Two consequences follow for architecture. First, **the ROI dashboard that sales teams use for renewals must not be a naive comparison**. If it is, the marketplace tells its best customers that their premium spend is wasted. Second, measuring paid products causally requires experiment assignment, exposure logging and matched controls, which is the platform that Testing a Two-Sided Market: Experimentation Platforms describes. A monetization model whose value cannot be measured honestly will eventually be renegotiated on the professional's own numbers.

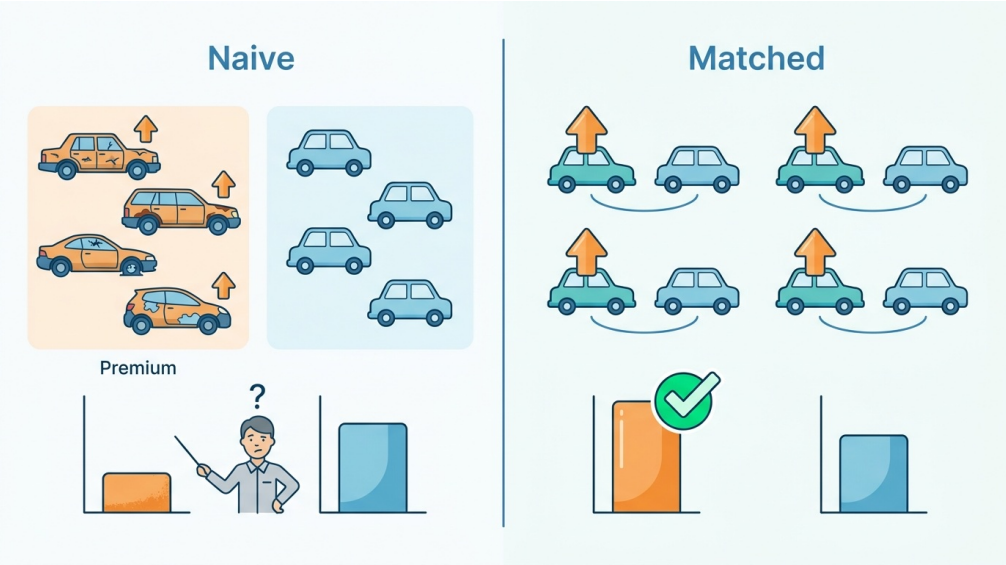


Figure 3: *Sellers with hard-to-sell cars buy more boosts, so a naive dashboard shows premium packages underperforming; comparing matched, like-for-like listings reveals the uplift, which is why ROI reporting needs experiment and matching infrastructure.*

Monetization Versus Liquidity

Every model spends the same reserve, organic liquidity, so every model needs a guardrail. The table below states what each one spends and the guardrail metric I would attach to it. The guardrail should be owned by someone outside the team that books the revenue: in practice, the marketplace's general manager and the liquidity metrics owner (Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics).

Model	What it spends	Guardrail metric	Typical failure
Insertion fee	Supply depth, especially the long tail	New listings per category and region versus the period before the change; search result depth; zero-result rate	Fee extended to a category where depth is what buyers come for
VAS and top slots	Organic relevance	Contact rate on organic results; contact rate per search session; share of first-page results that are paid	More top products sold than the result page can absorb
Pro subscription with slots	Inventory completeness when dealers trim paid stock	Share of a dealer's known stock that is listed; paid stock trend	Price rise answered by silent stock reduction
Depth tier	Supply volume when sellers choose other channels	Listings versus market transactions (share of market supply)	Yield grows while the share of supply quietly falls
Lead fees	Professional trust; buyer experience	Dispute rate on billed leads; response time; share of leads that end off-platform	Portal rewarded for contact volume, professional for disputing

Model	What it spends	Guardrail metric	Typical failure
Transactional fees	Conversion at checkout; margin	Checkout conversion; share of in-platform deals; buyer-protection loss rate	Fee pushes buyers back to off-platform cash deals
Advertising	Page speed and attention	Search-to-contact rate; Core Web Vitals; ad load per page	Ad load raised to hit a revenue target while contacts fall
Consumer subscription	Fairness of access for non-payers	Contact share by non-paying users; complaints	Paying users crowd out the rest of the demand side

Author's recommendations; metric definitions belong in the governed metrics layer described in Data Platform for Marketplaces: Events, Lakehouse and Data Products.

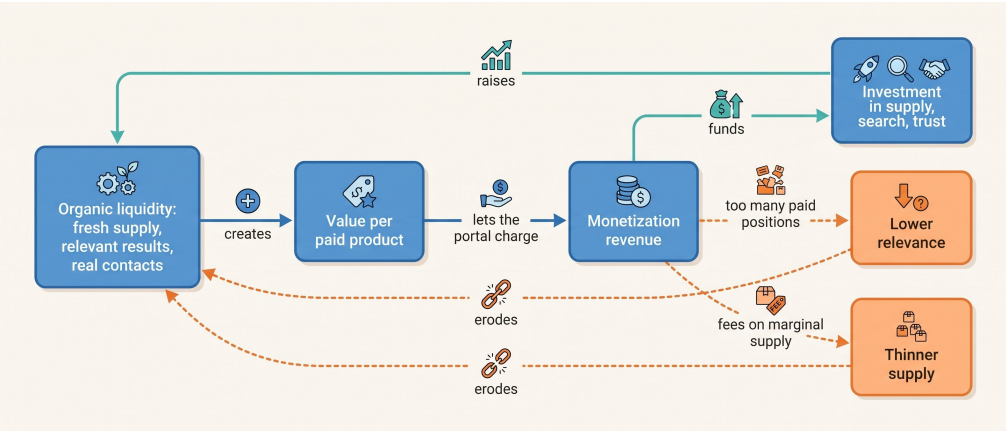


Figure 4: *The monetization–liquidity loop. Revenue funds the investment that raises liquidity, and liquidity sets what paid products are worth. Overcharging the same reserve, through too many paid positions or fees on marginal supply, erodes it from the other side.*

Veldmark Worked Example: Revenue Mix and Three Proposals

What Veldmark sells today

Veldmark's 2026 revenue of €1.31 billion splits across the brands and models as the ledger records. The derived unit figures in the last column are this chapter's arithmetic, not separately reported KPIs.

Brand	2026 revenue	Model mix	Center of gravity	Derived unit view
Tradeyard (horizontal: NL, BE, PL, PT)	€410M	VAS €150M; pro subscriptions €135M; Safe Deal fees €75M; insertion fees (PL vehicles and property) €50M	Paid visibility on free private listings	VAS ≈ €1.25 per new listing (€150M ÷ about 120 million new listings a year); Safe Deal take rate ≈ 7.4% (€75M ÷ about €1.02B GMV on 11.6M transactions)
Motorvane (automotive: DE, NL, PL, AT)	€360M	Dealer subscriptions €238M; depth/VAS €52M; leads and financing referrals €40M; private-seller fees €30M	Dealer packages with slots	Dealer subscription ARPA ≈ €927 per dealer per month (€238M ÷ 21,400 dealers ÷ 12)

Brand	2026 revenue	Model mix	Center of gravity	Derived unit view
Hearthly (real estate: FR, ES)	€395M	Agency subscriptions €270M; new-build campaigns €55M; Valuo data and leads €40M; rental services €30M	Agency subscriptions	Agency ARPA ≈ €616 per agency per month (€270M ÷ 36,500 agencies ÷ 12)
Keybridge (agent CRM SaaS: FR, BE, ES)	€45M	SaaS subscriptions	SaaS	≈ €383 per agency customer per month (€45M ÷ 9,800 ÷ 12)
Veldmark Ads (all brands)	€100M	Display, native, sponsored listings, retail media	Advertising	About 7.6% of group revenue

Veldmark already runs eight of the ten models; only consumer subscriptions are missing. Data and SaaS appear twice, as Valuo and Keybridge. The models are also implemented four times. Each brand has its own product catalog, its own way of enforcing what a seller bought, and its own billing integration with group finance's SAP S/4HANA. That is the "four of everything" problem from Meet Veldmark: Four Brands, One Platform to Build, applied to money. It is why one group product catalog and entitlement service by Q2 2028 is a Common Ground goal.

The ARPA column shows where Motorvane's revenue is exposed: the slot. Motorvane's dealers pay about €927 a month and carry about 62 listings each on average (1.45 million listings × 91% from dealers ÷ 21,400 dealers). The slot is therefore worth about €15 a month per car. That is the unit Motorvane's dealers will cut first when trading gets hard, exactly as Autotrader's stock lever shows. Any proposal that changes Motorvane's packages must model the stock lever, not just the price.

Proposal 1: Tradeyard's stronger visibility ladder

Olivier's case: Tradeyard NL's "Top listing" is underpriced compared with what the category's top positions deliver, and "Spotlight" would add a premium step.

Ingrid's forum answers the four questions:

- **Unit**: listing × 48-hour window × category. **System of truth**: the existing VAS entitlements in Yardstick, the PHP monolith (see Meet Veldmark: Four Brands, One Platform to Build). That is the weak point: Tradeyard's top listings are enforced partly by the reserved slots and partly by a score multiplier in Yardstick's Solr boost function, and nobody owns a report of how many top positions were promised against how many existed per category and hour.
- **Liquidity cost**: high. "Spotlight" takes the most-viewed paid slot in categories where Tradeyard's own data shows most contacts come from the first screen. The forum asks for the guardrail before launch: organic contact rate per search session in the affected categories, with a stop condition agreed in advance.
- **Who notices first**: buyers, through worse first pages, and then private sellers who bought "Top listing" and found themselves rotated out by "Spotlights". The ranking review Ingrid had asked Priya Raman to run in April was already showing that the existing boost was eating relevance (Ranking for Two Sides: Relevance, Freshness and Paid Visibility tells that story). Proposal 1 would repeat it at a higher price.

Decision: **deferred** until the ranking review reports and the group entitlement service exposes paid effects as explicit fields on the listing document. That lets ranking cap paid positions per page and lets the delivery report show what each seller received. A price rise on "Top listing" alone is approved, because it changes no system and no position count. A month later, the ranking policy adopted in June withdrew "Spotlight" altogether (Ranking for Two Sides: Relevance, Freshness and Paid Visibility).

Proposal 2: Motorvane's per-lead component

Olivier's case: dealers already measure Motorvane by leads, so bill the leads. The package would include an allowance, and every qualified lead above it would carry a fee (the forum works with an illustrative €12).

The forum works it through with the Essen Volkswagen Golf. Autohaus Ruhrblick's €21,900 Golf drew **23 raw leads in 10 days**: 14 e-mails, 7 calls and 2 test-drive bookings. Felix Brandt, Motorvane's CTO, has the lead logs checked. One buyer e-mailed twice and later booked a test drive, and two callers had already e-mailed. Under a strict "one buyer, one listing, one lead per 14 days" rule, the 23 raw leads are **19 billable leads**, the same 19 that From Interest to Contact: Messaging and Lead Management counts under the group lead contract. If all of them fell above the allowance, at €12 that one car would cost €228 in lead fees over ten days, against about €15 a month in slot value. The arithmetic shows why the model is attractive and why dealers will fight over every lead definition.

- **Unit**: qualified lead. **System of truth**: none exists. Motorvane counts contacts in three systems (messaging, call tracking, test-drive booking) with three buyer identities, and Veldmark ID, the group login, only arrives in 2028.

- **New systems**: lead identity and deduplication across channels; validity rules (spam, competitor checks, repeated contacts); a dispute and credit process for dealers; metering into billing; and a monthly statement a dealer can reconcile with their own CRM (From Interest to Contact: Messaging and Lead Management).
- **Liquidity cost**: moderate but real. Dealers on a per-lead fee have reasons to hide phone numbers in descriptions and to answer e-mails slowly. The guardrail is the share of contacts that reach the dealer through tracked channels, plus dealer response time.

Decision: **approved for design, not launch**. The first step is a "lead ledger" that counts billable leads under the proposed rule for six months *without billing them*, published to dealers in their statistics. If the ledger and the dealers' CRMs agree within a tolerance Motorvane sets, billing can follow. If they do not, Motorvane will have learned this before invoicing anyone.

Proposal 3: Hearthly's commission share on Valuo leads

Olivier's case: Valuo, Hearthly's automated valuation model, produces homeowners who want an estimate. They are the most valuable leads in property, and a commission share aligns Hearthly with the agency's success.

The forum uses the Lyon apartment. The 68 m² T3 in Lyon 7e was valued by Valuo at **€372,000–€401,000** and listed this month at **€389,000** by three agencies. Suppose its owner had come through a Valuo estimate routed to Agence Rhône Habitat, and the agency's fee was 4% of the price. A 20% commission share would earn Hearthly about €3,100. That would be paid when the sale completes at the notary, typically months after the lead, on a price nobody at Hearthly observes directly.

- **Unit**: completed sale attributed to a routed lead. **System of truth**: the agency's mandate and the notary's deed, both outside Hearthly. Matching requires agency reporting, mandate tracking and a reconciliation against France's public transaction data (DVF), which arrives with a lag.
- **New systems**: lead-to-mandate tracking; outcome matching over months; revenue estimation for unbilled outcomes and true-ups when they resolve; contract terms for disputes; and finance treatment of **variable consideration**, which is the same accounting Zillow describes for Zillow Preferred.
- **Governance cost**: the Keybridge decision of March 2027 forbids Hearthly from using Keybridge agency data for sales targeting. Outcome matching must therefore not draw on Keybridge mandate records, even though they would be the easiest source. Lucía Ortega, the group's data-protection officer and regulatory lead, adds a consent requirement: the homeowner must agree to being routed and to the outcome being tracked.

Decision: **piloted in one region (Lyon metropolitan area) for 2028, alongside the existing flat-fee Valuo lead product**, with finance's agreement on the estimation method before the first invoice. Ingrid's note for the decision record: the model is commercially sound, but it turns Hearthly into a business whose revenue depends on a reconciliation process with partners. That process must be designed before launch, not after the first disputed invoice.

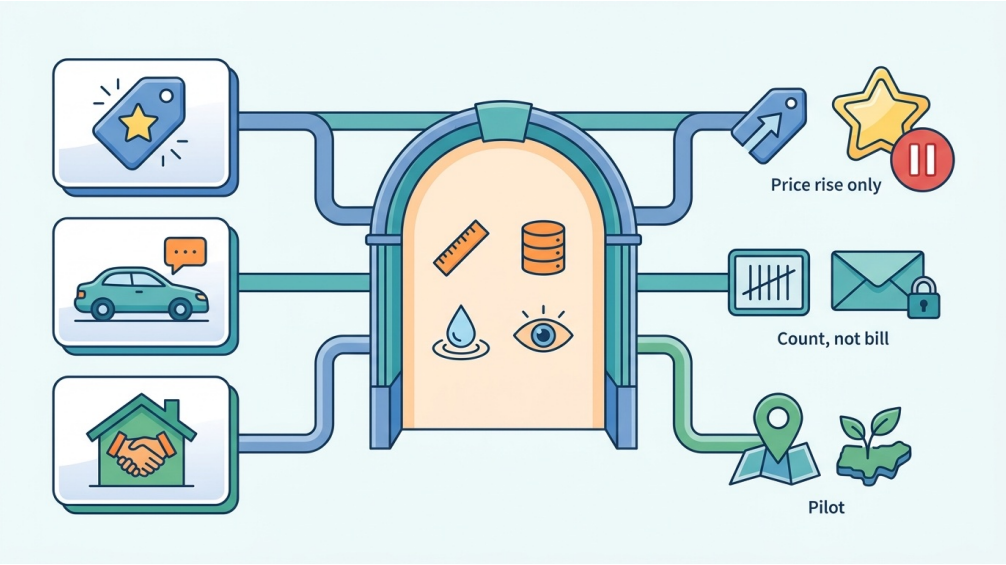


Figure 5: *The forum put the same four questions to each proposal and reached three different verdicts: a Top listing price rise without Spotlight, a lead ledger that counts before it bills, and a one-region commission pilot.*

Trade-offs: Choosing a Model

Model	Revenue predictability	Liquidity risk	New system complexity	Regulatory exposure	Right when
Insertion fee	Medium (volume-driven)	High in depth-driven categories	Low to medium	Low	Each listing is valuable to its seller; the fee also deters fraud; buyers are not driven by long-tail depth
VAS	Medium; seasonal	Medium to high (relevance)	Medium (scheduler, slots, labels, delivery evidence)	High: paid-ranking disclosure (Omnibus, P2B, DSA)	Large free private supply and a default sort that a paid product can improve on
Pro subscription with slots	High (contracts)	Low to medium	Medium to high (contracts, quotas, invoicing)	Medium: P2B terms for paid ranking effects	Professional supply dominates and pros list continuously
Depth tier	Medium (listing volume)	Medium (supply leakage)	Medium (per-listing orders, agent billing)	Medium	Seller pays per listing, through the agent, and a premium tier is valued

Model	Revenue predictability	Liquidity risk	New system complexity	Regulatory exposure	Right when
Pay per lead	Medium	Medium (trust, off-platform leakage)	High (identity, deduplication, disputes)	Medium: consent for contact	Lead quality is measurable and professionals already think in cost per lead
Share of voice	High	Low to medium	High (allocation, forecasting)	Medium	Many professionals compete for leads in each geography
Pay at close	Low (variable consideration, lag)	Low on supply; high on partner trust	Very high (outcome matching, true-ups)	Medium to high	Transactions are large, outcomes are observable, and the portal dominates the lead source
Take rate or buyer fee	Medium to high	Medium (checkout friction)	Very high (payments, ledger, disputes, logistics)	High: payment licensing via partners, DAC7	Shippable goods or high-fraud categories where protection is worth paying for

Model	Revenue predictability	Liquidity risk	New system complexity	Regulatory exposure	Right when
Advertising	Medium; cyclical	Medium (page speed, attention)	High, but usually a shared group stack	High: consent, DSA advertising transparency	Large audiences with commercial intent, mainly horizontals
Consumer subscription	High	Medium (fairness, two-tier demand)	Medium	Medium	Supply is scarce and buyers pay for advantage

Model complexity compounds when models share a page, a judgement the table cannot show. A single Motorvane search result page already combines dealer package prominence, depth products, a sponsored listing and display ads. Each of these was sold by a different team, and each makes a claim on the same few positions. A group that adds models without a shared rule for how paid claims compete for positions ends up with Multiplier Creep, even when each model is sound on its own.

Vertical Differences

Dimension	General classifieds	Real estate	Automotive	Jobs	Recommerce
Main payer	Private sellers (VAS), pros, advertisers	Agencies; vendors via agents in seller-pays markets; developers	Dealers	Employers and recruiters	Buyers (fee), sellers (bumps)
Center of gravity	VAS plus advertising; transactional fees rising	Agency subscriptions or per-listing depth; seller leads	Dealer packages with slots; depth; finance referrals	Pay per posting or per application; employer subscriptions	Buyer fee and shipping margin
Typical unit KPI	Revenue per listing or per paying user; take rate where transactional	ARPA (Rightmove) or yield per listing (REA, Hemnet)	ARPR (Autotrader), QARSD (CarGurus)	Revenue per posting or per employer	Take rate
Lead value	Low per contact	Very high (a seller lead can be worth a commission)	High (a sold car funds the dealer's margin)	Medium to high per qualified applicant	Not applicable: the transaction is the unit

Dimension	General classifieds	Real estate	Automotive	Jobs	Recommerce
Distinctive system need	Bump scheduler, slot rotation, high-volume small payments	Valuation-driven seller leads, mandate tracking, agent rebilling	Feed-driven slot enforcement, stock-level pricing, finance partner APIs	Application tracking, posting duration, multi-posting	Checkout, logistics, buyer protection, provisioning
Distinctive risk	Multiplier Creep; scam listings that buy visibility	Agency concentration; seller-pays volume leakage	Dealer-group bargaining power; stock trimming	Pay-per-application gaming	Margin dilution; buyer-protection losses

For groups, the table has one practical consequence: **the monetization model is the least shareable part of a marketplace**, even when the systems underneath can be shared. AVIV's experience in property points the same way: generic infrastructure can be shared, while ingestion, disclosures, monetization and pro tools are local (One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture discusses the evidence). Veldmark's principle, "share platforms, not products", applies to money with special force. Share the catalog model, the entitlement service, the paid-label pipeline and the metering. Leave the packages, prices and the choice of model to each brand.

Failure Modes

Selling exposure that cannot convert (Vanity Liquidity). A package priced on "views" or "impressions" sells a quantity that has no reliable link to the professional's outcomes. When renewal comes, the professional brings their own numbers: contacts and sales, from their own CRM. A portal that reports only views loses that negotiation. Report contacts and leads, define them in the governed metrics layer, and make sure billing and the ROI dashboard use the same definitions.

Paid positions without a cap (Multiplier Creep). Each paid product is reasonable alone. Together they fill the first page. Without a per-page budget for paid positions,

enforced by ranking and reported to monetization, drift is gradual and invisible in revenue until contact rates fall.

Revenue models added as code (Billing by Guesswork). The fastest way to launch a new product is a condition in the insertion service, another in the search query and a third in the rendering template. After five years and four brands, nobody can say what a seller is entitled to or prove that it was delivered. Packages, Bumps and Quotas: Products, Pricing and Entitlements owns the remedy.

ROI dashboards that mislead, and price events shipped as code. Both are covered above: a naive package comparison makes sales teams discount the products that work best, and a price event that needs a deployment makes the commercial calendar hostage to the release train.

Lead fees without a dispute process. Pay-per-lead launched before a lead definition, a deduplication rule and a credit process exist will be defined by the first hundred disputes, one account manager at a time.

Outcome-based revenue without reconciliation. A commission share or pay-at-close fee whose outcomes are reported only by the professional is a trust arrangement, not a billing system. Finance will treat it as variable consideration and ask how it is estimated. The answer must exist before launch.

Monetization that rewards the scammer. Paid visibility can be bought with a stolen card by an account that listed a fake rental an hour ago. Run every paid boost through the same risk checks as the listing itself, and refund rather than deliver when moderation removes a boosted listing (Trust and Safety as a Platform).

The Chief Architect's View

A Chief Architect does not choose prices. The decisions that are theirs concern the systems that make every model measurable, enforceable and reversible:

1. **Separate model choice from model implementation.** Brands choose their models and prices. The group owns a shared catalog model, entitlement service, paid-label pipeline and metering standard, so that a new model is a configuration of shared capabilities, not a fifth implementation. At Veldmark this is the Common Ground goal of one product catalog and entitlement service by Q2 2028.
2. **Make every paid effect explicit and observable.** Every paid product that changes ranking or rendering must appear as a named field with an expiry on the listing's read model, travel to the search index through events (Every Change Is an Event: Event-Driven Listing Architecture), and produce delivery evidence. This is a compliance requirement (paid-ranking disclosure, P2B terms on ranking effects) as much as a revenue-assurance one.

3. **Attach a liquidity guardrail to every monetization change.** No new model, price event or package change goes live without a named guardrail metric, a pre-agreed stop condition and an owner outside the revenue team. Treat ranking changes as monetization changes in the same review.
4. **Require a "model readiness card" before launch.** For each new model: the unit of charge, the system of truth for what was sold, the delivery evidence, the dispute process, the accounting treatment agreed with finance, the regulatory review, and the rollback path. A model that cannot fill in the card is not ready, whatever its business case says. The entitlement matrix and the decision record in Field Kit: Templates for the Marketplace Architect carry most of these fields.
5. **Fund causal measurement.** Package ROI, price elasticity and the value of paid positions should be measured with experiments or matched comparisons (Testing a Two-Sided Market: Experimentation Platforms). The dashboards used in renewal negotiations should come from the same pipeline.

What to watch over the long term: the share of revenue that depends on outcomes measured outside the platform (pay at close, commission shares, referral commissions), because it is revenue the marketplace cannot fully verify; the trend in professionals' paid stock against their known stock, the silent downgrade; the share of first-page positions that are paid; and the cost of the annual price event in engineering hours. If that cost is growing, the catalog is turning into code.

Questions to Ask

1. What is the unit of charge for this model, and which system is the single source of truth for how many units were sold?
2. Where does the evidence that the product was delivered come from, and can it be reconciled with the invoice line by line?
3. Which liquidity metric does this model spend, who owns the guardrail on it, and what is the pre-agreed stop condition?
4. How many paid claims compete for the first page of results today, and who decides how they are capped and ordered?
5. Can next year's price-and-product event be prepared as data, take effect at a set date in every market, and be rolled back without a deployment?
6. For lead-based fees: what exactly is a billable lead, how are duplicates across channels handled, and what is the dispute process?
7. For outcome-based fees: where does the outcome data come from, how long is the lag, and how does finance estimate revenue before the outcome is known?
8. How is the value of paid products measured: by naive comparison, matched comparison or experiment? Which method feeds the ROI dashboard used in renewals?

9. Which regulatory obligations does this model trigger (paid-ranking labels, P2B terms on ranking effects, DSA advertising transparency, payment licensing), and where in the systems is each one met?
10. If a brand wanted to adopt this model next year, how much of it could it configure from shared group capabilities, and how much would it have to build?

From Models to Machinery

Choosing a monetization model is choosing which unit to count, which system holds the count and how much liquidity to spend. The ten models in this chapter share one commercial core: a catalog of what can be sold, a pricing engine for what it costs this seller today, an entitlement service for what they hold now, and the metering and billing that connect delivery to invoices. Veldmark's three proposals all ran into the same missing piece. None of its brands could say, from one system, what a seller had bought and what they had received. Packages, Bumps and Quotas: Products, Pricing and Entitlements designs that core: product catalog, pricing engine, entitlements and quotas, enforcement points, and the audit trail that proves what was sold was delivered.

Sources and Evidence

Company engineering and investor sources

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- Scout24 preliminary results for 2025 (IR News, German) — Scout24 SE, 2026. Plus consumer subscriptions, pay-per-ad growth, professional subscriptions, Transaction Enablement revenue.
- Zillow Group Form 10-K for fiscal 2025 — Zillow Group, 2026. Share-of-voice pricing per zip code, Zillow Preferred pay-at-close fees as variable consideration, Rentals pricing bases, revenue per visit, Follow Up Boss.
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- Adevinta ASA Annual Report 2023 (Exhibit 99.1 to eBay 10-K/A) — Adevinta, 2024. Revenue by type and segment; transactional revenue and its effect on the French margin.
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Standards, regulation and documentation

- Directive (EU) 2019/2161 — European Union, *Official Journal*, 2019 (wording checked against the Publications Office text, October 2026). Annex I point 11a on undisclosed paid ranking; Article 7(4a) on ranking parameters.
- Regulation (EU) 2019/1150 (P2B) — European Union, *Official Journal*, 2019 (Article 5 checked against the Publications Office text, October 2026). Disclosure of how remuneration affects ranking for business users.

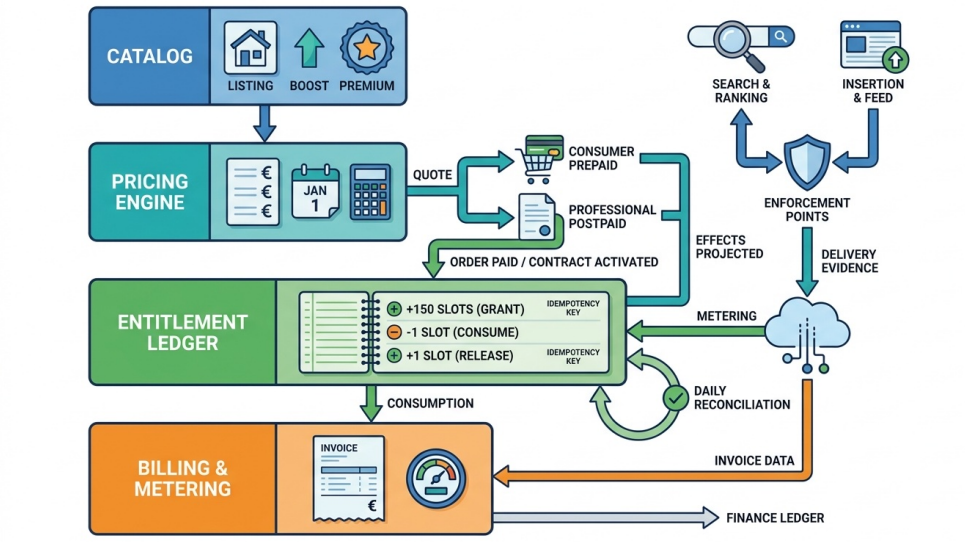
Research and practitioner sources

- Platform Competition in Two-Sided Markets — Jean-Charles Rochet and Jean Tirole, *Journal of the European Economic Association*, 2003. Price structure across the two sides; this chapter relies on the abstract and the standard reading of the paper.
- All Markets Are Not Created Equal: 10 Factors To Consider When Evaluating Digital Marketplaces — Bill Gurley, Above the Crowd, 2012. Fragmentation and payment flow as factors in marketplace economics.

- 13 Metrics for Marketplace Companies — Jeff Jordan, Li Jin, D'Arcy Coolican and Andrew Chen, Andreessen Horowitz, 2020. GMV versus revenue and take rate; match rate.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

31. Packages, Bumps and Quotas: Products, Pricing and Entitlements



IN THIS CHAPTER, YOU WILL: Design the commercial core behind a marketplace's paid products: the product catalog, the pricing engine, orders and contracts, the entitlement and quota ledger, the enforcement points that read it, and the metering and billing that close the loop. You will follow one dealer's disputed invoice at Veldmark to the design that would have prevented it.

WHY THE BUSINESS CARES: Every paid product is a promise to a seller. When the systems that sell, enforce and bill that promise disagree, the marketplace overcharges some customers, under-delivers to others, and cannot change prices or packages without an engineering project.

WHY ARCHITECTS SHOULD CARE: Entitlements are read on the hottest paths in the platform: insertion, every ranked query and every rendered card. They are written by orders, contracts, moderation and schedulers, and they must reconcile with invoices to the cent. Few services are simultaneously

this latency-sensitive, this financially consequential and this easy to scatter across a codebase.

KEY POINTS:

- **Keep catalog, price, entitlement and bill apart.** The catalog says what can be sold, the pricing engine what it costs this seller now, the entitlement ledger what this account or listing holds now, and billing what is owed. Merging any two produces Billing by Guesswork and turns every annual price event into a code release.
- **An entitlement is a ledger, not a counter.** Grants, consumptions, releases, expiries and refunds are append-only entries with idempotency keys. Enforcement points read derived *effects* (a slot, a boost weight, a label, an expiry) from fast read models fed by events, never the contract or the price.
- **Prove that what was sold was delivered.** Insertion, ranking, rendering and lead routing emit delivery events, and a daily reconciliation compares them with grants and invoice lines. This is revenue assurance, a dispute answer and the evidence for paid-ranking disclosure, all from one pipeline.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Billing by Guesswork** — package checks written as conditions in the insertion service, the search query and the card template mean that nobody can say what a seller holds, or prove what they received.
- **Multiplier Creep** — when paid effects are not explicit, capped fields, every new product adds another unbounded multiplier to ranking.
- **Deal-Shaped Architecture** — four acquired brands with four catalogs, four ways of counting slots and four billing integrations pay four times for one capability and still cannot sell a cross-brand package.

How Listing Marketplaces Make Money showed that every revenue model needs the same commercial core: something that defines what can be sold, something that prices it, something that records what a seller holds, and something that bills for it. This chapter designs that core, and with it Part VI's test for every paid product: proof that what was sold was delivered. **A seller who paid for a "Top listing" and never appeared on top, or a dealer billed for cars the platform refused to publish, is a customer the marketplace has wronged.** At scale, these wrongs are rarely bad intent. They come from architecture: the same commercial fact counted in several places, by several teams, with several definitions. The effects this core produces are what Ranking for Two Sides: Relevance, Freshness and Paid Visibility blends into the result order, and what Selling

Attention: Display, Native and Retail-Media Advertising must share with sponsored listings.

The chapter owns four terms. A **product catalog** is the versioned definition of everything sellable: products, packages, add-ons, and the effects each one grants. A **pricing engine** computes the price of a catalog item for a specific buyer, context and date. An **entitlement** is a right, held by an account or a listing for a period, to receive a defined effect, such as "may publish up to 150 vehicles" or "appears in a top slot until Sunday 23:59". A **quota** is a countable entitlement: slots, boosts, leads, credits.

No marketplace whose sources this book uses has published its catalog, pricing or entitlement architecture in detail. The design here is therefore a **reference-architecture inference**, anchored to published product behavior: Rightmove's commitments plus extras, Autotrader's annual price event and stock lever, OLX's boosts per package, CarGurus' capped free tier, Vinted's fee on every card. It also draws on public documentation from billing vendors. It is not a description of any company's systems.

Setting the Scene at Veldmark: Thirteen Cars and €117

It is late September 2027 at Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build). Autohaus Ruhrblick, the Essen dealer whose Volkswagen Golf sold in April, disputes its August invoice from Motorvane. The dealer is on Motorvane's "Pro 150" package: **150 listing slots for €1,450 a month**, with any car above 150 billed at **€9 per car per month** as overage. In August, the dealer management system (DMS) feed delivered **163** cars. The invoice charged overage for 13 cars, **€117**. The dealer's sales manager checked Motorvane's own site. Only 150 Ruhrblick cars were live. The 13 extra cars had never been published.

Felix Brandt, Motorvane's CTO, has the case traced. It turns out to be structural, not a one-off. Motorvane counts a dealer's listings in two places. The **feed importer** enforces the package quota at import time and silently skips cars above the limit. The **billing export**, a monthly job written years earlier for finance, counts the cars *in the feed file*. Each counter is correct by its own definition. Together they bill dealers for cars the platform refused to show. An audit over the previous 14 months finds **1,140 dealers** overbilled by a total of about **€212,000**. Motorvane credits every one of them, and the account managers spend October apologizing.

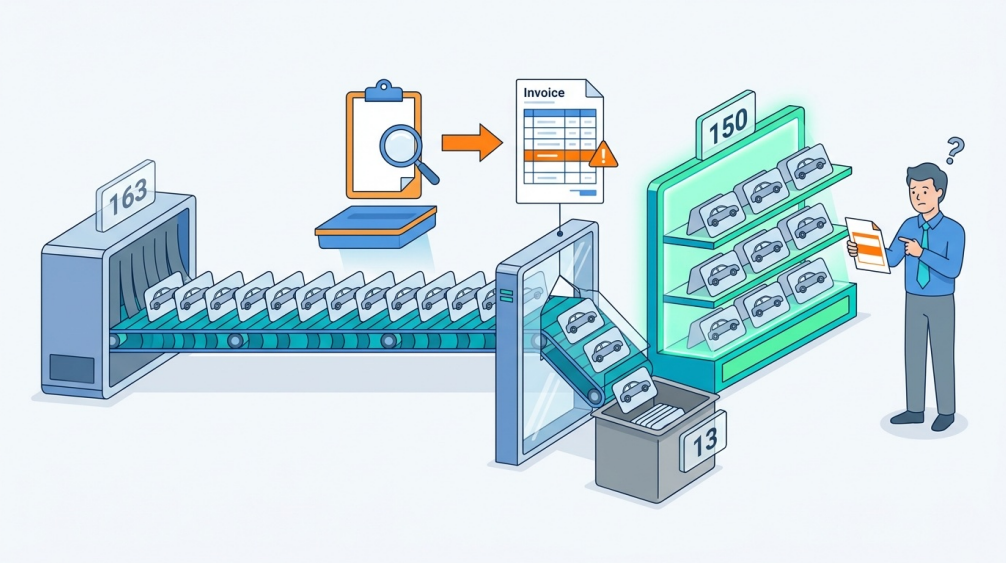


Figure 1: *Two counters, each correct by its own definition: the feed importer published 150 of Ruhrblick's 163 cars and skipped 13, while the billing export counted the feed file and charged €117 overage for cars that were never live.*

Ingrid Halvorsen, the Group Chief Architect, brings the case to the architecture advice forum alongside a second agenda item. Olivier Petit, VP Monetization, wants a group-wide price-and-product event on **1 January 2028**. That means new Tradeyard VAS prices in four countries, a reshaped Motorvane package staircase and a new Hearthly agency tier. Each brand estimates the work in engineering weeks, not in configuration days. Tradeyard's VAS products are hard-coded in Yardstick, its PHP monolith. Motorvane's packages live in a Kotlin service and in the sales CRM. Hearthly's agency tiers live in its Go contract service. Keybridge bills its SaaS customers from its own .NET stack.

Ingrid's framing for the forum: *the Ruhrblick invoice and the price-event estimate are the same problem.* Veldmark has no single place that says what can be sold, what a seller holds and what was delivered. Common Ground's goal of one product catalog and entitlement service by Q2 2028 has to answer both.

Concepts: Four Things That Must Stay Apart

The chapter's core decision is to keep four concerns apart, a separation most marketplaces reach only after the spaghetti:

Concern	Question it answers	Changes when	Owner	Must not contain
Product catalog	What can be sold, and what does each item grant?	A product is launched, retired or repackaged	Monetization product team, per brand, in a shared model	Prices for specific customers; who bought what
Pricing engine	What does this item cost this buyer, in this context, on this date?	Price events, negotiated contracts, promotions, experiments	Monetization and commercial finance	Entitlement state; delivery
Entitlement ledger	What does this account or listing hold right now, and what has it used?	Orders, contract changes, consumption, expiry, moderation, refunds	Monetization platform team	Prices; invoice logic
Billing	What is owed, by whom, on which invoice?	Usage, billing periods, credits, dunning	Finance systems	Enforcement decisions

A fifth element connects them: **delivery evidence**, the record of what the platform actually did, such as listings published under a slot, impressions in a paid position, boosts executed or leads routed. Delivery evidence belongs to none of the four. It is emitted by the systems that deliver value and consumed by the ledger, billing and reporting.

The subsections below define the first three concerns. Billing is a finance system this chapter connects to rather than designs; its interface is in "Metering, billing and revenue recognition" under Architecture.

The catalog: products, packages and effects

A catalog has five object types. The names vary between marketplaces; the shape recurs.

- **Product**. One sellable thing with a defined effect: "Bump", "Top listing (7 days)", "Highlight", "Extra slot", "Lead allowance +50".
- **Package**. A bundle of products and quotas sold as one item, usually on a contract: "Pro 150" = 150 slots + 20 boosts per month + statistics + Co-Driver-style tooling.

Autotrader's "package staircase" and Rightmove's package tiers are packages in this sense (How Listing Marketplaces Make Money gives the evidence).

- **Add-on**. A product sold on top of a package, possibly metered: Rightmove's "more than 50% of independent agents purchased incremental product above their package commitment" describes add-ons.
- **Effect**. What the item does at an enforcement point. This is the object most catalogs forget. "Top listing" is not a flag. It is a set of effects: eligibility for reserved slots, a "Top" label, a duration, and a rule for what happens when the listing is sold or removed.
- **Product version**. Every change to a product's effects creates a new version. Existing entitlements keep the version they were sold under unless a migration says otherwise. Without versions, changing a product silently changes what past buyers bought.

The catalog is **reference data shipped as releases**, like the category taxonomy in Category Trees That Can Change: Designing Marketplace Taxonomies: versioned, reviewed, effective-dated, and published as events that downstream services cache.

The pricing engine: dimensions and dates

A price is a function, not a field. In a listing marketplace the inputs are at least:

Dimension	Example	Why it matters
Country and currency	Tradeyard PL in PLN, NL in EUR	Local price levels and VAT
Category or vertical	Bump in Electronics vs Bump in Cars	Value of visibility differs by category
Region	Lyon metropolitan area vs rural Rhône	Lead value and competition differ
Seller type	Private, professional, franchise member	Different products and price lists
Contract	List price vs negotiated contract price	Pros negotiate; the contract overrides the list
Item value	Vinted's bump price varies with item price	Formula pricing
Channel	Self-serve vs sales-assisted	Different discounts and approval rules
Date	Price list effective from 1 January 2028	Annual price events

Dimension	Example	Why it matters
Experiment arm	Price test cohort B	Price experiments (Testing a Two-Sided Market: Experimentation Platforms)
Promotion	First bump free for new sellers	Bounded, auditable discounts

Vinted's public price list shows formula pricing in practice: its buyer fee varies by "item characteristics, order value, order type (a single item versus a bundle)", and its bump fee by duration, item price and "territorial scope (local or international)" (Vinted, updated 2026-10-08).

Two design rules follow. First, **prices are effective-dated price lists, not editable values**. The 1 January price event is a new price-list version with an effective date, prepared and reviewed weeks earlier, and reversible by deactivating that version. Autotrader's "annual 1 April 2025 pricing and product event" (Autotrader, 2026) is the kind of event this rule exists for. Second, **every price quote carries the version of every input that produced it**: catalog version, price-list version, contract, promotion and experiment arm. When a seller disputes a price six months later, the quote can be reproduced exactly.

Entitlements, quotas and effects

An entitlement has a **holder** (an account, a branch, or a listing), a **product version**, a **validity window**, an optional **quantity**, and a **source**: the order, contract or promotion that created it. A quota is an entitlement with a quantity that consumption reduces. An **effect** is the projection of active entitlements that an enforcement point needs: "account 88412 may hold 150 active vehicle listings and currently holds 150", or "listing 5531 is top-eligible in category 'city bikes', Rotterdam, until 14 March 23:59".

Separating entitlement from effect keeps the hot paths fast. Ranking does not need to know that a top position came from a promotion or a contract. It needs a boolean, a category, a window and a cap. **Enforcement points read effects. Only the entitlement service reads entitlements.**

Architecture: The Commercial Core

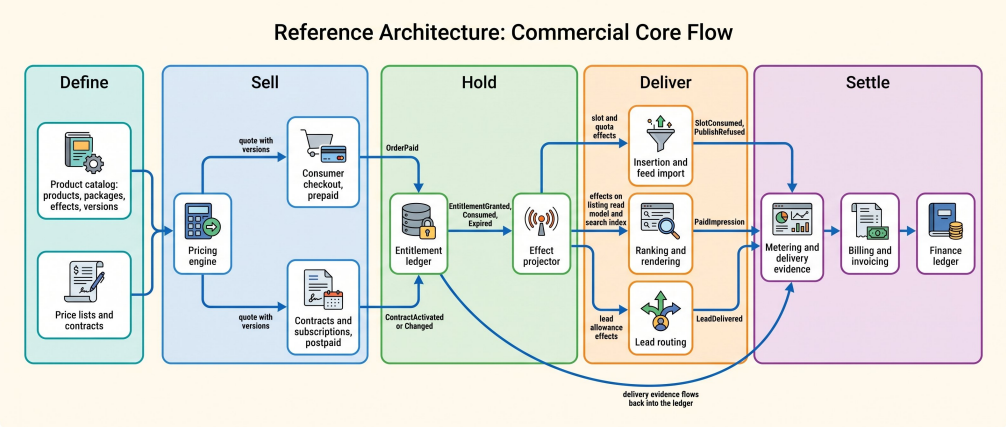


Figure 2: *The commercial core as five stages. Catalog and price lists define, the pricing engine and the two selling paths sell, the entitlement ledger holds, the delivery systems deliver, and metering and billing settle. Delivery evidence flows back into the ledger and billing so that the invoice is computed from what happened, not from what was ordered. Reference architecture, not a description of any company.*

Two selling paths, one ledger

Marketplaces sell through two paths that share almost nothing operationally:

- **Consumer purchases** are prepaid, small, instant and card- or wallet-paid: a €3.99 bump bought on a phone on Sunday evening. They need a checkout, a payment service provider, receipts, and automatic refunds when moderation rejects the listing. Vinted documents that a bump bought while a listing is in review is not processed or charged; that is the kind of rule this path must encode.
- **Professional contracts** are postpaid, monthly, negotiated and invoiced: "Pro 150" for 24 months at a 6% discount, with overage. They need contract management (term, renewal, notice period, upgrades and proration), a sales CRM, VAT invoicing, credit notes and dunning. CarGurus documents the shape: packages sold on "monthly, quarterly, semiannual, or annual" terms that "generally auto-renew on a monthly basis and are cancellable by dealers with 30 days"' notice (CarGurus 10-K FY2025, 2026).

The two paths should share the catalog, the pricing engine and the entitlement ledger, and nothing else. The payment path for consumers and the order-to-cash path for professionals differ in regulation, tooling and failure modes. A group that tries to force one billing system onto both usually ends up with neither working well. What must be common is that a bump bought by a private seller and a bump included in a dealer's package are the same product version with the same effects, recorded in the same ledger.

The entitlement ledger

The ledger is **append-only**. Every change is an entry, never an update of a counter.

Entry type	Created by	Example	Idempotency key
`Grant`	Order paid; contract activated or renewed; promotion	+150 vehicle slots, account 88412, 1–30 Sept	Contract ID + billing period
`Consume`	Insertion; scheduler executing a boost; lead router	−1 slot for listing 5531; −1 boost	Listing ID + slot period; boost schedule ID
`Release`	Listing sold, expired or deleted	+1 slot returned	Listing ID + lifecycle event ID
`Expire`	Validity window ends	Unused 4 boosts lapse on 30 Sept	Entitlement ID + window end
`Suspend` / `Resume`	Moderation hold, unpaid invoice	Top-listing effect paused during review	Case ID
`Revoke` / `Refund`	Moderation removal; cancellation; dispute	Bump revoked, refund issued	Case or order ID
`Adjust`	Make-good, goodwill, correction	+3 days of top listing after under-delivery	Reconciliation run ID + line

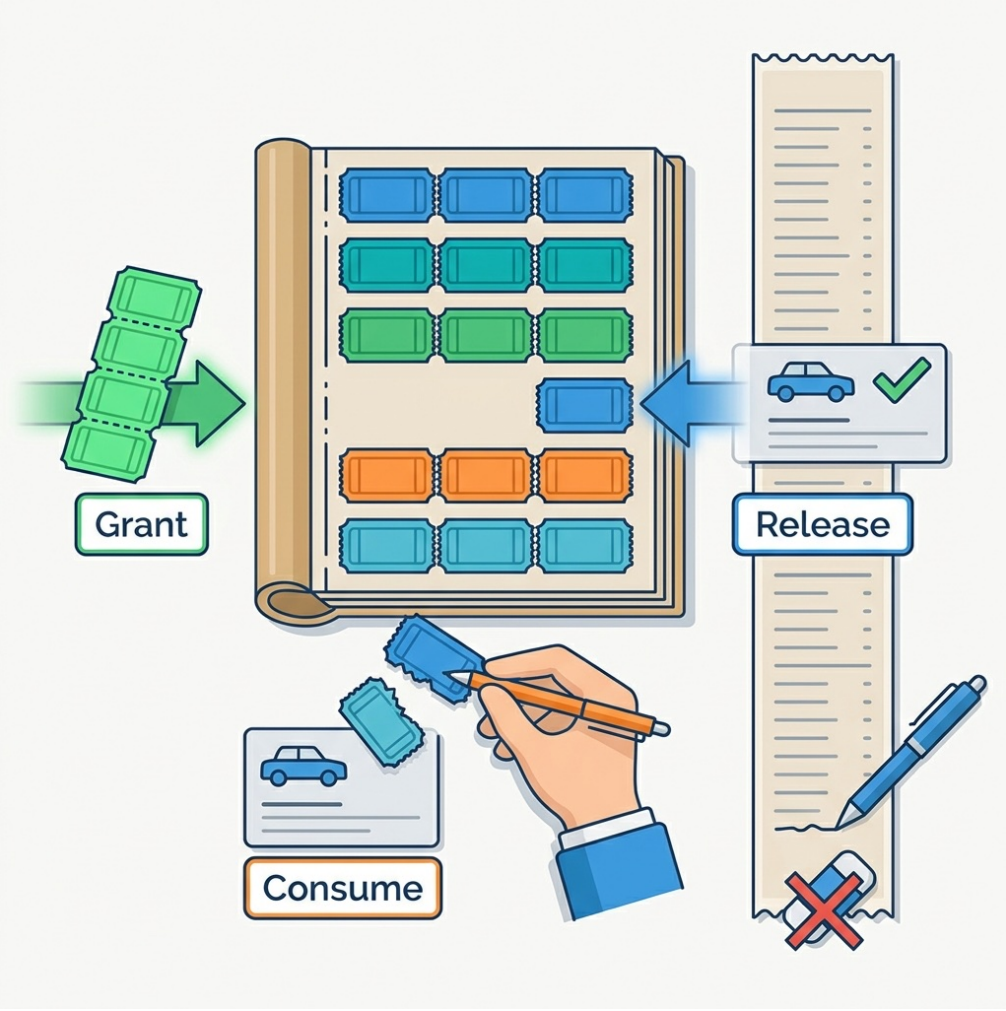


Figure 3: *The entitlement ledger works like a ticket book with a journal: every grant, consumption and release is a new entry, and what a seller holds is the sum of entries, never a counter that someone overwrites.*

The current balance of any entitlement is the sum of its entries. Snapshots and read models make that fast, but the entries are the truth. Three properties follow:

1. **Idempotency**. Every producer supplies a key derived from its own business event. A scheduler that retries a boost after a timeout produces the same key, and the ledger records it once. Stripe's API documentation states the general principle: an idempotency key lets a client "safely repeat the request without risk of creating a second object or performing the update twice" (Stripe, n.d.). In a ledger, the same rule turns a revenue leak (the boost executed twice) and a trust problem (the boost never executed) into detectable states.

2. **Explainability.** "Why can't I publish my 151st car?" has an answer: the grants, consumptions and releases for that account, each traceable to a business event.
3. **Reconciliation.** Billing and the ledger can be compared entry by entry. The Ruhrblick invoice would have been computed from `Consume` and `PublishRefused` entries, not from a count of a feed file.

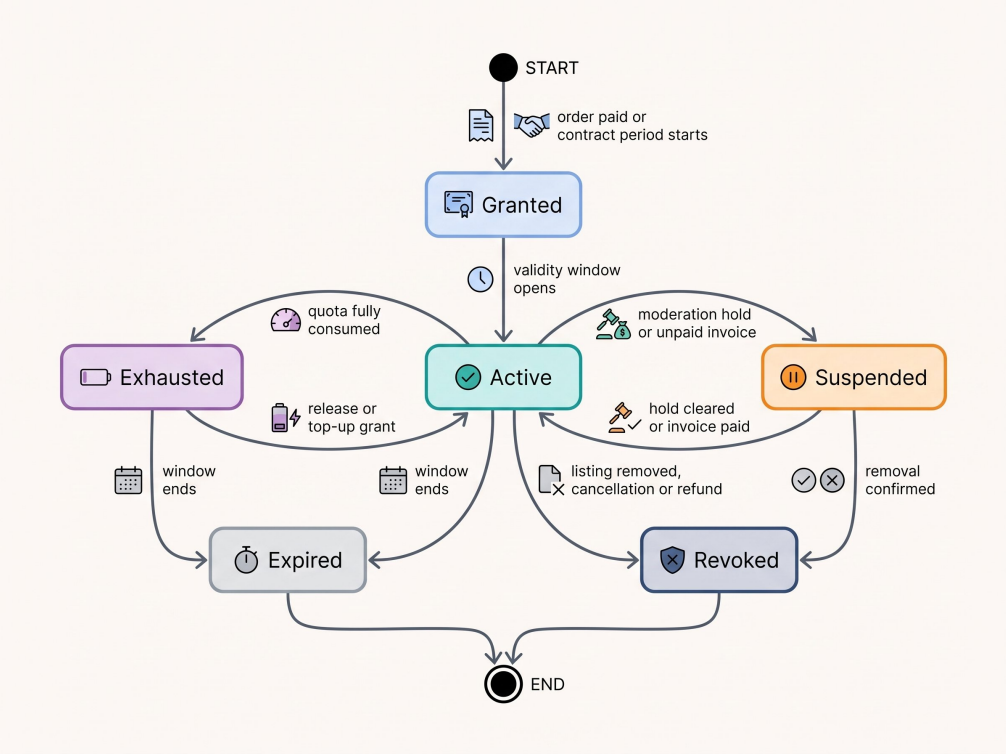


Figure 4: *The lifecycle of an entitlement. Every transition is a ledger entry caused by a business event. Suspended and Revoked are the transitions most often forgotten, and they are where moderation, payments and monetization must agree.*

Effects: projecting entitlements onto the hot paths

Enforcement points need answers in milliseconds, so the design **projects effects onto the read models that the hot paths already use**, through events. At Veldmark's Sunday peak of **31,000 queries per second**, a ranking stage that re-ranks, say, 400 candidates per query (an illustrative figure) would need more than **12 million** entitlement answers per second if it asked the ledger live. No ledger should serve that.

The **effect projector** consumes ledger events and publishes effect changes. In Veldmark's design these are `ListingPaidEffectChanged` for per-listing effects and

`AccountQuotaChanged` for account quotas, carried on the group event backbone (Every Change Is an Event: Event-Driven Listing Architecture). Per-listing effects become explicit fields on the listing document in the search index:

Field	Example	Used by
`paid.top_eligible_until`	2028-03-14T23:59:00+01:00	Ranking: reserved-slot eligibility
`paid.top_scope`	category=city-bikes; region=NL-ZH	Ranking: where the slot applies
`paid.bump_time`	2028-03-08T19:44:00+01:00	Ranking: freshness sort, kept separate from `created_at`
`paid.highlight`	true	Rendering: card style
`paid.label`	"Top"	Rendering and disclosure: the paid label
`paid.product_version`	top-listing-7d@v4	Delivery reporting; disclosure; audits

Ranking for Two Sides: Relevance, Freshness and Paid Visibility owns how these fields are used. Its group standard is that relevance models never read them as features, that a relevance floor applies to reserved slots, and that ranking logs every paid impression. This chapter owns producing the fields correctly and removing them promptly.

A paid effect that outlives its listing is Ghost Inventory with a price tag. If a sold bike keeps its top slot because the `ListingSold` event reached search but not the effect projector, a buyer sees a sold item in a paid position. The fix is that the projector consumes listing lifecycle events too, and that sold, removed or blocked listings carry no paid fields, whatever the ledger says about remaining days.

Two consistency rules apply to every effect. **Effects tolerate bounded staleness; the ledger does not.** A top-listing effect that appears 20 seconds after payment is acceptable. A slot granted twice is not. And **the index is never the record**: if the search index is rebuilt (Search Architecture for Listings: Indices, Freshness and Scale), effects are re-projected from the ledger, never copied from the old index. That rule is what keeps the paid fields from drifting into Search as System of Record.

Enforcement points

Enforcement point	Question	Reads	Latency budget	On ledger unavailability
Insertion (private)	May this listing be published, and does it need a fee?	Account quota effect; price quote	Inside the insertion flow, about 100 ms	**Fail open** for free listings; fail closed for paid publication until payment is confirmed
Feed import (professional)	Which of these 163 cars fit the package?	Account quota effect; dealer priority rule	Batch, minutes	**Queue and retry**; never skip silently; report to the feed
Ranking	Is this listing eligible for a paid position or a bump?	Paid fields on the listing document	Inside the query, sub-millisecond per candidate	Serve the last projected effects; log as stale
Rendering	Which label, badge or card style?	Paid fields on the read model	Page render	Render the label if the effect is present; never render a paid style without a label
Lead routing	Is this lead inside the allowance, masked for the free tier, or billable?	Account lead-allowance effect	Seconds	Route the lead; meter it later; never drop it
Pro tools	May this user use the statistics, Co-Driver-style tools, or bulk edit?	Account feature effects	Interactive	Fail open for paid accounts with a recent effect; log

Enforcement point	Question	Reads	Latency budget	On ledger unavailability
Scheduler	Execute boost 3 of 4 for listing 5531	Ledger, with idempotency key	Asynchronous	Retry with the same key

The right-hand column is a commercial decision written as engineering. **Failing closed protects revenue and damages supply. Failing open protects supply and costs revenue that reconciliation can recover later.** For a liquidity business I default to failing open on anything a paying professional already holds and failing closed only on new consumer charges. That choice must be documented per enforcement point, because during an outage nobody has time to debate it.

Quotas meet feeds

Feeds break naive quota designs. A DMS export contains the dealer's whole stock, not the cars the dealer wants on Motorvane. With 88% of Motorvane's dealer listings arriving by feed (Many Doors In: Forms, APIs and Bulk Feeds), the quota rule *is* the publication rule for most of the supply. There are four possible policies when a feed exceeds the package:

Policy	Behavior	Good for	Risk
Reject the excess, newest first	Publish the first 150 by feed order	Simple	Feed order is arbitrary; the best cars may be skipped
Dealer priority	Dealer marks which cars to publish; the rest wait	Dealer control	Needs a field in the feed contract and tooling
Publish all, bill overage	Every car goes live; overage is metered and billed	Liquidity; revenue	Surprise invoices unless the dealer consented to overage
Publish all, prompt to upgrade	Publish, then prompt, then enforce after a grace period	Upselling	Grace-period rules become a second, implicit contract

Any of the four can be right, but three rules are not optional. The policy must be **a term of the contract stored with the entitlement**, not a behavior of the importer. Every refused car must produce a `PublishRefused` entry and appear in the **feed report back to the dealer**, as the feed contract in Many Doors In: Forms, APIs and Bulk Feeds requires. And billing must count **published listings**, **from the ledger**, never rows in a file.

Metering, billing and revenue recognition

Metering turns delivery evidence into billable quantities: overage cars per month, leads above the allowance, impressions on a cost-per-impression product. Billing combines contract lines (the €1,450 package fee), metered lines (overage), one-off purchases, credits and taxes into invoices, and hands them to the finance ledger. At Veldmark that ledger is group finance's SAP S/4HANA.

Three finance requirements shape the design, and finance must agree them before launch:

- **Recognition over the delivery period.** Adevinta's 2023 accounts state that listing fees are recognized over the period in which the listings are displayed (Adevinta Annual Report 2023, 2024). A seven-day top listing bought on 28 December is therefore partly next year's revenue. The ledger's validity windows are the evidence for that split.
- **Variable consideration** for outcome-based fees, such as pay-at-close leads (How Listing Marketplaces Make Money): revenue is estimated and trued up when outcomes arrive.
- **Credit notes, not edits.** A wrong invoice is corrected by a credit note that references it. The Ruhrblick correction was 1,140 credit notes, each traceable to the ledger entries that justified it.

RECONCILED's Never Update, Only Append: Designing Custom Ledgers covers the accounting design. The architectural point here is that **billing consumes the entitlement ledger and metering, and never recomputes entitlements on its own**. The Ruhrblick defect was exactly such a recomputation.

Sold versus delivered

A daily **delivery reconciliation** closes the loop. It compares three numbers for each paid product and holder: what was **granted** (ledger), what was **delivered** (delivery evidence: slots used, paid impressions, boosts executed, leads routed), and what was **billed** (invoice lines).

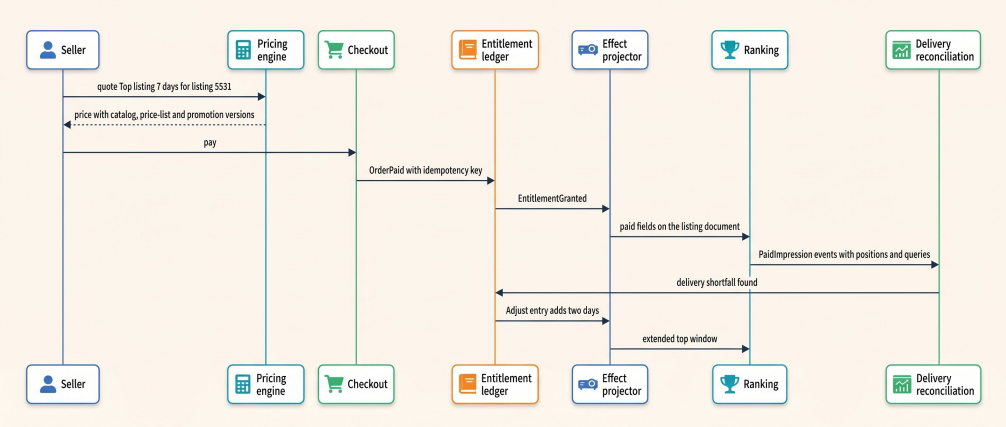


Figure 5: *A top listing from quote to make-good. The quote records the versions that produced the price, the grant is idempotent, effects reach ranking by event, and delivery evidence comes back from ranking. When reconciliation finds a shortfall, the remedy is a ledger entry, not a manual credit.*

Only this loop can keep the promise of Tradeyard's top listing. The ranking decision of June 2027 that Ranking for Two Sides: Relevance, Freshness and Paid Visibility describes repositioned Tradeyard's top listing as **seven days of guaranteed prominence in relevant searches, with an automatic extension when delivery falls short**. Reconciliation also produces two other outputs:

- **The seller-facing delivery report**: "Your top listing appeared 1,840 times in top slots for 'stadsfiets Rotterdam' and related searches." This is the answer to *did I get what I paid for?*, and the input to the ROI reporting that Serving the Pros: Dealer, Agent and Merchant Tooling presents.
- **Disclosure evidence**. EU rules require paid ranking to be disclosed to consumers (Directive (EU) 2019/2161, Annex I point 11a) and the effect of remuneration on ranking to be described to business users (Regulation (EU) 2019/1150, Article 5). The DSA adds advertising transparency for online platforms (Rules of the Marketplace: DSA, DAC7 and Vertical Regulation owns all three). A `paid.label` field that travels from the catalog to the rendered card, and a log that shows which paid effects shaped which results, are how an architecture meets these duties by construction.

In Practice: What the Public Record Shows

No portal in this book's sources publishes its entitlement architecture. The public record does show the behaviors that architecture must support:

Published behavior	Source (dated)	What it requires of the core
"More than 50% of independent agents purchased incremental product above their package commitment"	Rightmove FY2025 results, 2026	Commitments plus metered or à-la-carte extras on one invoice
Annual "1 April 2025 pricing and product event"; a "package staircase" being reshaped in FY27	Autotrader FY26 results, 2026	Effective-dated price lists and package versions; contract migration as data
Stock lever: −£48 per retailer per month from fewer paid cars	Autotrader FY26 results, 2026	Slot quotas measured precisely; downgrade visibility for account managers
Co-Driver included "within retailer packages"	Autotrader FY26 results, 2026	Feature entitlements bundled into packages, not separate SKUs
Packages defined by boosts per 30 days (Package 1 = 1 boost … Package 4 = 4 boosts)	OLX Tech, 2026	Quota of scheduled boosts; idempotent execution; delivery evidence for causal analysis
Free tier with "capped anonymous leads and connections"	CarGurus 10-K FY2025, 2026	Lead-allowance effects at the lead-routing enforcement point
Bump for 3 or 7 days; buyer fee varying by order value and bundle	Vinted price lists, updated 2026-10-08	Time-boxed per-listing effects; formula pricing on the read path
A ranking stack balancing relevance and "visibility for Value Added Services"	OLX Tech, 2025	Explicit, versioned paid fields that ranking can read and cap

Billing vendors document part of the pattern. Stripe's Entitlements feature maps product "features" to customers and fires a webhook, `entitlements.active_entitlement_summary.updated`, when a customer's entitlements change. Stripe recommends that integrators "persist these entitlements internally for faster resolution" (Stripe Entitlements, n.d.). This confirms two parts of the design: entitlements derive from what was bought, and hot paths read a local projection. It also shows the gap a marketplace must fill itself. A SaaS billing entitlement is a **feature switch per customer**.

A marketplace also needs **counted quotas, per-listing and time-boxed effects, scopes (category and region), and delivery evidence**, which subscription billing products are not designed to hold.

That gap shapes the build-or-buy decision. I would buy invoicing, tax, dunning and payment collection. I would build the catalog model, the entitlement ledger and the effect projector, because they encode the marketplace's own products and sit on its hottest paths. Some groups buy configure-price-quote (CPQ) tools for sales-assisted pro contracts. That works if the CPQ tool writes contracts into the shared ledger rather than becoming a second source of entitlements.

Veldmark Worked Example: One Catalog for Four Brands

The architecture advice forum's design

After the Ruhrblick audit, the architecture advice forum agrees a design in October 2027. Yara Mendes, Head of Platform Engineering, owns the platform team that builds it. Its core is the group entitlement service, nicknamed **Tally**. The decisions:

1. **Shared model, local content**. One catalog *model* (products, packages, add-ons, effects, versions) for all brands. Each brand owns its catalog *content* and price lists in its own namespace. This is "share platforms, not products" applied to money: Tradeyard's bumps and Motorvane's packages share a schema, not a price.
2. **One ledger, brand-partitioned**. Tally holds every grant and consumption, partitioned by brand and country, with the event backbone carrying effects to each brand's read models: Kafka for Tradeyard and Motorvane, and a bridge to Google Pub/Sub for Hearthly while it stays on Google Cloud.
3. **Effects on the listing event contract**. `ListingPaidEffectChanged` is added to the Listing event contract as a minor, backward-compatible version. The shared OpenSearch platform indexes the `paid.*` fields. Motorvane's Elasticsearch indexes the same fields.
4. **Billing stays where it is, but consumes Tally**. Brands keep their invoicing integrations into SAP S/4HANA for now. They must compute metered lines from Tally's metering output, not from their own counts.
5. **Keybridge stays out until 2029**. Keybridge, the neutral agent-CRM SaaS, keeps its own billing through 2028. Its customers include Hearthly's competitors, and the March 2027 neutrality decision makes a shared commercial ledger a question for the annual independent review, not for this program.

The numbers that sized it

Flow	Volume (2026 baseline)	Design consequence
Tradeyard VAS purchases	About **38 million** a year, average just under €4 (≈ €150M)	About 104,000 grants a day, peaking on Sunday evenings; the consumer checkout path must scale with listing insertion
Tradeyard new listings	2.3 million a week	Each one is a quota check for professional accounts and a fee check in PL vehicles and property
Motorvane dealer accounts	21,400 dealers; 91% of listings; 88% of dealer listings by feed	Quota enforcement is mostly batch, at feed import; overage policy per contract
Hearthly agencies	36,500 agencies, many in networks (Agence Rhône Habitat: 14 branches)	Entitlements held at network or branch level; slot pools shared across branches
Search peak	31,000 queries/s	Effects must be on the listing document, never looked up per candidate
Group invoicing	About 58,000 professional invoices a month (dealers and agencies), plus Tradeyard pro sellers	Metering output must close before the monthly billing run, with a reconciliation report finance can sign

Replaying the Ruhrblick month

Under Tally, Ruhrblick's August invoice would carry no overage line. The feed delivers 163 cars. The contract stored with the entitlement says *dealer priority, then newest first, no overage without consent.* Tally records 150 `Consume` entries and 13 `PublishRefused` entries. The feed report shows the dealer which 13 cars waited and why. The dealer management system can then mark priority cars in the next feed. Billing reads 150 published cars and bills €1,450, with no overage line. If Ruhrblick had

consented to overage, the 13 cars would have gone live, Tally would have recorded 163 consumptions, and the invoice would have shown €117 that the dealer could reconcile car by car against its own stock list.

Replaying a price event

The 1 January 2028 price event becomes a catalog and price-list release. Tradeyard publishes new VAS price lists for four countries, effective at 00:00 local time. Motorvane publishes `pro-150@v5` and its other package versions, plus a contract migration plan that moves each dealer to the new version at their next renewal date, not on 1 January. That is a commercial choice the data model now allows. Hearthly adds a tier. Each release is reviewed in a pull request, dry-run against last month's orders to show its revenue and invoice effects per customer, and can be deactivated version by version. The estimate drops from engineering weeks to configuration days for every brand whose products have migrated, and that condition is the catch: Tradeyard's VAS products move into Tally only when the insertion flow leaves Yardstick in **Q2 2028**, so the January 2028 event still runs through Yardstick for Tradeyard, one last time.

Migration order

The forum chooses the order by where the pain is. **Motorvane first (Q1 2028)**: dealer quotas and overage, where the overbilling happened. **Tradeyard second (Q2 2028)**: VAS products, timed with the insertion flow's move out of Yardstick. **Hearthly third (second half of 2028)**: agency tiers and new-build campaigns. Each migration runs Tally in **shadow mode** for one full billing cycle first, computing entitlements and invoices alongside the legacy system and reporting every difference. Ingrid's rule for cutover: no brand moves until a shadow month reconciles to zero unexplained differences. A difference is acceptable only if it is explained and the legacy system is the one in error.

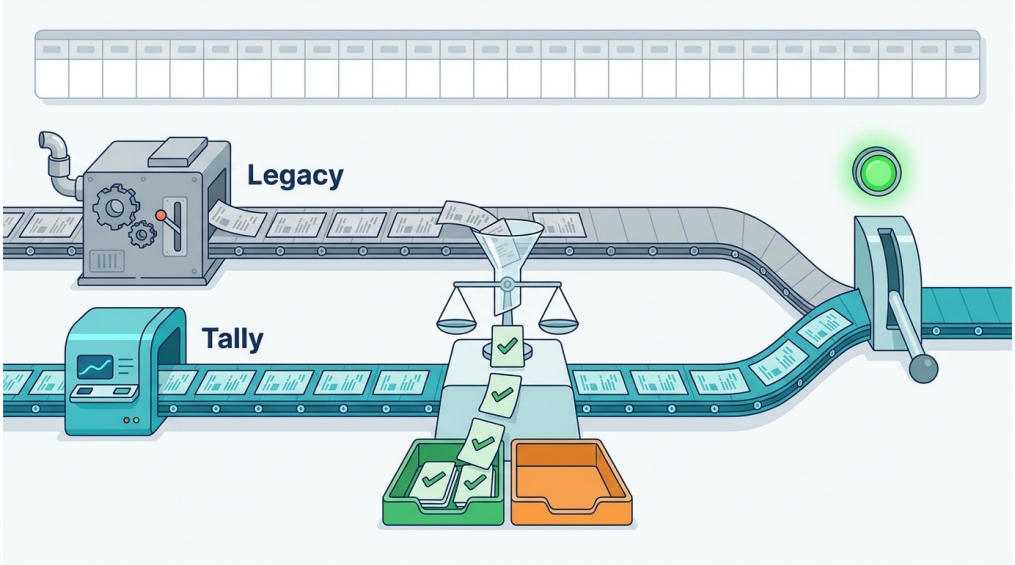


Figure 6: *Each brand runs Tally in shadow mode for one full billing cycle beside its legacy system, and cuts over only when every difference is explained and the legacy system is the one in error.*

Trade-offs

Decision	Option A	Option B	Option C	When each is right
Where entitlements live	Columns on the listing and account tables	A separate entitlement service with counters	**An append-only entitlement ledger with projected effects**	A: one brand, a few products, no disputes. B: moderate scale, few audits. C: multiple brands, contracts, quotas, regulators and disputes

Decision	Option A	Option B	Option C	When each is right
How hot paths read paid state	Query the entitlement service per request	Cache in each consumer	**Project effects onto the listing document by events**	Query: low-volume tools. Cache: account-level checks. Projection: ranking and rendering
Catalog scope in a group	Per-brand catalogs	One central catalog with central content	**Shared model, brand-owned content**	Per-brand: no shared products and no consolidation plans. Central: one brand family with shared go-to-market. Shared model: Veldmark's "share platforms, not products"
Pricing	Prices as product attributes	Price lists per country	**Pricing engine over effective-dated price lists, contracts and formulas**	Attributes: a prototype. Price lists: one country, list prices only. Engine: negotiated contracts, price events, experiments

Decision	Option A	Option B	Option C	When each is right
Billing	Build	**Buy invoicing, tax and collection; build catalog and ledger**	Buy everything, including entitlements	Build: only with unusual billing needs and a large finance-systems team. Buy all: SaaS-like products with feature switches only
Quota enforcement under failure	Fail closed everywhere	Fail open everywhere	**Per enforcement point, documented**	Closed: new consumer charges. Open: held professional entitlements. Never: a single global default

Vertical Differences

Dimension	General classifieds	Real estate	Automotive	Jobs
Typical products	Bump, top listing, highlight; pro shop packages	Agency subscriptions by tier; per-listing depth in seller-pays markets; new-build campaigns	Dealer packages with slots; depth per car; overage	Posting credits; featured jobs; employer subscriptions
Entitlement holder	Mostly the listing (consumer VAS); account for pros	Agency network, branch or listing	Dealer account or rooftop	Employer account; posting
Dominant selling path	Consumer checkout, prepaid	Contracts, postpaid; agent rebills the vendor where sellers pay	Contracts, postpaid	Mixed
Hardest problem	Volume of small purchases; refunds on moderation	Hierarchies: slot pools across branches, network-level contracts	Feeds against quotas; stock-level elasticity	Duration and reposting rules; credits shared across recruiters
Delivery evidence that matters	Paid impressions and position	Leads per listing and branch; campaign impressions	Leads per car; paid stock versus known stock	Applications per posting

Real estate's hardest problem is hierarchy. Agence Rhône Habitat's 14 branches may buy one network contract with a pooled quota, branch-level sub-allocations and one invoice. Others buy per branch. The ledger must let a grant be held at network level and consumed at branch level, and must let Private Sellers, Dealers and Agencies: Modeling Who Lists' organization hierarchy change, for example when a branch leaves the network, without rewriting history.

Failure Modes

Two counters for one fact. The Ruhrblick defect: billing and enforcement count the same thing differently. Any metered line on an invoice must come from the ledger's consumption entries.

Paid fields that outlive the listing. A sold, expired or removed listing keeps its top-slot effect because the projector did not consume the lifecycle event. It is visible to buyers and to regulators, and it is a refund liability.

Product codes in ranking code. `if package == 'GOLD' and country == 'PT'` in a search query. Every new product then needs a search release, and nobody can list all paid effects. Ranking reads `paid.*` fields only.

Price lists edited in place. A price change with no effective date and no version cannot be reproduced for a dispute or rolled back after customer pushback.

Boosts executed twice, or never. A scheduler retry without an idempotency key double-consumes a quota. A failed job with no alert leaves a paid boost undelivered. Both are ledger states that reconciliation should flag within a day.

Moderation and monetization disagree. A listing removed for fraud keeps its paid effect, or is refunded twice. A fake listing that bought a boost with a stolen card is the case to design for (Scams, Fakes and Account Takeovers: Fraud Prevention for Listings). Moderation outcomes must produce `Suspend` or `Revoke` entries, and refunds must be idempotent per case.

Time zones and windows. "Seven days" means 168 hours from purchase in one brand and until 23:59 local time on day seven in another. Define windows in the product version, store them as instants, and render them in the seller's time zone.

Promotions without bounds. Stackable discounts and free products with no budget, no expiry and no owner. Every promotion is a catalog object with a cap, a window and a cost center.

A global fail-closed default. An entitlement-service outage that stops every dealer feed in the group turns a monetization incident into a supply incident. Document the failure behavior per enforcement point and test it.

The Chief Architect's View

What to decide centrally. The catalog model, the entitlement ledger, the effect schema on the listing document and the delivery-evidence events. These are group standards, because they are what let ranking, search, finance and regulators treat four brands as one platform. **What to leave local**. Products, packages, prices, contract terms and the

choice of monetization model. A group that centralizes prices will soon find that the Polish car market and the French property market do not share a price level.

Who owns what. The monetization platform team owns the catalog model, the pricing engine and the ledger. Each brand's monetization team owns its catalog content and price lists, released through review like code. Ranking owns how effects shape the result order, under the policy that Ranking for Two Sides: Relevance, Freshness and Paid Visibility sets out. Finance owns invoices, revenue recognition and the sign-off of each month's reconciliation. Trust and safety owns the suspend and revoke rules. Writing this down in an **entitlement matrix** (product × enforcement point × effect × delivery evidence × owner, Tool 9 in Field Kit: Templates for the Marketplace Architect) is the most useful hour a launch team can spend.

Product	Enforcement points	Effect	Delivery evidence	On listing removal
Tradeyard Top listing 7d	Ranking, rendering	Reserved-slot eligibility above relevance floor; "Top" label	Paid impressions by position and query	Effect removed; no refund after sale; refund if removed by moderation within 24 h
Tradeyard Bump	Ranking (sort), scheduler	`paid.bump_time` updated	`BoostExecuted`	Pending boosts revoked
Motorvane Pro 150	Feed import, insertion, pro tools	150 slots; 20 boosts a month; statistics	`SlotConsumed`, `PublishRefused`, `BoostExecuted`	Slot released
Motorvane lead allowance (pilot)	Lead routing	First N leads included; later leads metered, not billed during the pilot	`LeadDelivered` with deduplication key	Not applicable
Hearthly network tier	Insertion, rendering, pro tools	Pooled slots across branches; agency badge	Slot use per branch; leads per branch	Slot returned to the pool

What to watch. The daily reconciliation's unexplained difference, by brand, in euros.

The time from a catalog change to its effect in production, a measure of whether products are really data. The share of revenue computed from ledger entries rather than legacy counts, which tracks the migration. Effect-projection lag at the Sunday peak. Credits issued for under-delivery, which should rise briefly after launch, when the system starts finding what was always there, and then fall.

The long-term consideration. Every acquisition brings a catalog, an entitlement scheme and a billing integration. A group that has Tally can price the integration of an acquired marketplace's monetization as a content migration plus a shadow month. A group without it starts again each time, which is how Deal-Shaped Architecture forms (After the Deal: Consolidating Platforms After Acquisitions).

Questions to Ask

1. Where is the single source of truth for what this seller holds right now, and can it explain every unit by the business event that caused it?
2. Does billing compute metered lines from entitlement consumption, or does it count something else, such as feed rows, active flags or page views?
3. Can next year's price event be released as a dated, versioned price list, previewed per customer and rolled back without a deployment?
4. Which paid effects appear on the listing document in the search index, and what removes them when the listing is sold, expired or taken down?
5. For each enforcement point, what happens when the entitlement service is unavailable, and who decided?
6. What happens when a feed exceeds a package, and is that policy a term of the contract or a behavior of the importer?
7. Does every quota consumption carry an idempotency key derived from the business event, and what detects a boost executed twice or never?
8. How does a seller see what they received for a paid product, and does that report reconcile with the invoice?
9. Which product version was each existing customer sold, and what happens to them when the product changes?
10. When moderation removes a paid listing, which ledger entries are written, and is the refund idempotent?

From Products to Attention

The commercial core in this chapter turns a marketplace's monetization models into things that can be sold, held, enforced, delivered and billed, and it can prove each step. One family of paid products still needs its own machinery. Display, native and retail-media advertising sells the audience rather than the listing. It runs through ad servers,

header bidding and consent signals, and its buyers are brands and agencies rather than sellers. Selling Attention: Display, Native and Retail-Media Advertising takes up that stack, and asks how a marketplace sells attention without degrading the experience or breaking privacy law.

Sources and Evidence

Company engineering and investor sources

- Rightmove plc full-year results 2025 (RNS) — Rightmove plc, 2026. Package tiers and spending above package commitments.
- Full year results for the year ended 31 March 2026 — Autotrader Group plc, 2026. Annual price-and-product event, package staircase, stock lever, Co-Driver bundled in packages.
- CarGurus Form 10-K for fiscal 2025 — CarGurus, 2026. Contract terms and auto-renewal, capped free tier, Highlight slots.
- Adevinta ASA Annual Report 2023 (Exhibit 99.1 to eBay 10-K/A) — Adevinta, 2024. Recognition of listing fees over the display period.
- Price List (US) — Vinted, updated 2026-10-08. Formula pricing of bumps and the buyer fee.
- Correlation Lied to Us: Rethinking Product Impact with Causal Inference — Enderson Santos, OLX Tech, 2026. Packages defined by boosts per 30 days.
- Hybrid Search — Where Keywords Meet Vectors, Enabling Classifieds Discovery — OLX Tech, 2025. Ranking that balances relevance with visibility for VAS.

Standards, regulation and documentation

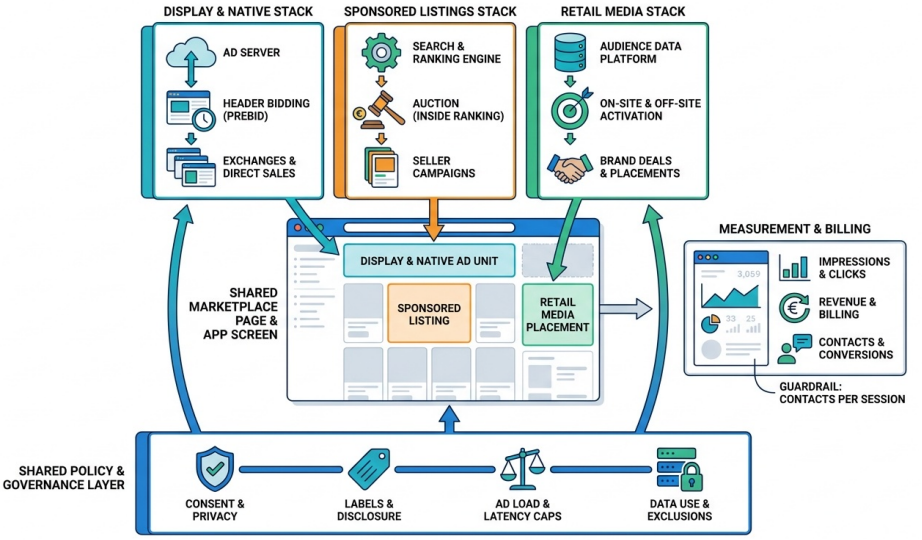
- Entitlements — Stripe documentation, n.d. (read 2026-10-19). Features mapped to products, the entitlement-summary webhook, and the recommendation to persist entitlements locally.
- Idempotent requests — Stripe API reference, n.d. (read 2026-10-19). Idempotency keys for safe retries.
- Directive (EU) 2019/2161 — European Union, *Official Journal*, 2019 (Annex point 11a checked against the Publications Office text, October 2026). Disclosure of paid ranking to consumers.
- Regulation (EU) 2019/1150 (P2B) — European Union, *Official Journal*, 2019 (Article 5 checked against the Publications Office text, October 2026). Description of how remuneration affects ranking for business users.

Research and practitioner sources

- 13 Metrics for Marketplace Companies — Jeff Jordan, Li Jin, D'Arcy Coolican and Andrew Chen, Andreessen Horowitz, 2020. Background on marketplace revenue measures used in the preceding chapter.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

32. Selling Attention: Display, Native and Retail-Media Advertising



IN THIS CHAPTER, YOU WILL: Take apart the three advertising businesses that share a listing marketplace's pages: display and native advertising sold through an ad server and header bidding, sponsored listings sold inside the search stack, and retail media sold to brands such as car makers, banks and insurers. You will see how consent, latency and first-party data shape each one, and follow Veldmark Ads through three decisions in 2027.

WHY THE BUSINESS CARES: Advertising can be a fifth of a horizontal marketplace's revenue and is almost pure margin when it works. It is also the one revenue line that earns more when users see more pages rather than when they find what they want, so it is the line most likely to quietly undermine the marketplace's main business.

WHY ARCHITECTS SHOULD CARE: The ad stack runs third-party code on the most important pages, adds auctions to the critical rendering path, and depends on a consent signal that must reach every request. Sponsored

listings put a second auction inside ranking. Audience products turn the marketplace's behavioral data into something sold to other companies. Each of these is an architecture decision with legal consequences.

KEY POINTS:

- **Three ad businesses share one page but not one stack**. Display and native ads run through an ad server and header bidding; sponsored listings run inside search and ranking; retail media sells audiences and placements to brands across both. They have different buyers, auctions, latency budgets and rules. Build them as three products under one shared policy layer for labels, ad load, consent and data use.
- **Consent and latency set the ceiling on display revenue, so they are architecture decisions**. The consent string must reach every bid request and every audience pipeline, and the header-bidding timeout trades bid density against page speed on pages that also have to win organic search. Changes to the consent dialog or the auction timeout are revenue changes and need the same experiments and guardrails as ranking changes.
- **First-party intent data is the marketplace's advantage and its main liability**. Searches, saved searches and enquiries are unusually strong signals of purchase intent. Use them through consented, purpose-bound segments, never with special-category data or for profiling minors, and never with data the group holds in a neutral role, such as a CRM it runs for agencies.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Multiplier Creep** — sponsored listings and native units added to result pages by an advertising team, outside the ranking guardrails, push relevant organic listings below the fold one placement at a time.
- **Vanity Liquidity** — advertising revenue grows with page views and ad load, so an ads target can reward more pages per session even when buyers need more pages because they are not finding what they want.
- **Billing by Guesswork** — a seller who buys a "Top listing" from the brand and a sponsored campaign from the ad unit holds two claims on the same positions, sold, priced and billed by different systems, and nobody can say which one was delivered.

Part VI has followed a contact to revenue: From Interest to Contact: Messaging and Lead Management turned contacts into leads, How Listing Marketplaces Make Money chose what to charge for, and Packages, Bumps and Quotas: Products, Pricing and Entitlements proved that what was sold was delivered. Advertising is the part's last revenue line, and

the only one paid for attention rather than for matches. The audience has commercial intent: people about to buy a car, rent a flat or furnish a home. Car makers, banks, insurers, energy suppliers and furniture retailers pay to reach it without selling anything on the marketplace. Selling that access is the oldest secondary business in classifieds. **A marketplace earns its core revenue when buyers find the right listing quickly; it earns advertising revenue when they look at more pages**. The architecture has to hold those two in balance, and most of the hard decisions in this chapter are about where the balance is enforced.

The term "advertising" hides three different businesses. **Display and native advertising** sells impressions on the marketplace's pages to advertisers through an ad server, programmatic exchanges and a direct sales team. **Sponsored listings** let sellers pay to promote their own listings inside search results; they are ads, legally, but they run in the search stack. **Retail media** sells a brand's campaign against the marketplace's own audiences and placements, on site and sometimes off site, and is closer to a data product than to a banner. This chapter owns the terms **ad stack**, **first-party audience** and **sponsored listing**. It does not re-teach how paid placements blend into ranking, which Ranking for Two Sides: Relevance, Freshness and Paid Visibility owns, or how visibility products are sold and entitled, which Packages, Bumps and Quotas: Products, Pricing and Entitlements owns. Revenue models and their KPIs are in How Listing Marketplaces Make Money. The chapter's two platform demands, a consent signal on every event and contacts as the guardrail for every ad change, return in Part VII in Data Platform for Marketplaces: Events, Lakehouse and Data Products and Testing a Two-Sided Market: Experimentation Platforms.

Setting the Scene at Veldmark: One Ad Unit, Three Arguments

Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), runs one shared advertising unit, **Veldmark Ads**, which sells across all brands and earned **€100 million** in 2026. It is one of the group's few shared services in 2027: an ad stack built on Google Ad Manager with Prebid header bidding. Its 2026 revenue split, by product and by the brand whose pages carried the inventory:

Product	2026 revenue	Buyers	Where it runs
Programmatic display (open auction and private deals)	€41M	Agencies and their trading desks through supply-side platforms	Ad server with Prebid

Product	2026 revenue	Buyers	Where it runs
Direct-sold display and native	€26M	Advertisers and agencies, sold by the Veldmark Ads sales team	Ad server
Sponsored listings	€19M	Professional sellers on Tradeyard Poland and Motorvane, self-serve	Search result pages, served by each brand's search stack
Retail media	€14M	Car makers, banks and insurers	Ad server placements plus audience deals
Total	**€100M**		

By inventory, Tradeyard pages earned **€58 million**, Motorvane **€24 million** and Hearthly **€18 million**. Keybridge carries no advertising.

In June 2027 three disagreements reached Ingrid Halvorsen, the Group Chief Architect, in the same week.

Page speed versus bids. In the first quarter of 2027, Veldmark Ads raised the Prebid auction timeout on Tradeyard Netherlands' listing and search pages from 1,000 ms to 1,500 ms, to let slower bidders compete. Programmatic revenue per thousand page views rose about 4%. The web platform team reported that the 75th-percentile largest contentful paint on listing pages had worsened from 2.4 to 2.9 seconds, on a site where 46% of web sessions start from organic search. Nobody had measured what the change did to contacts.

A retail-media audience nobody had reviewed. A car maker's agency asked Motorvane for an "in-market for an electric SUV" audience, to be activated on Motorvane pages and off site. The Veldmark Ads team built it from Motorvane search and saved-search behavior. Lucía Ortega, the Group Data Protection Officer, asked three questions nobody could answer: which consent purposes covered the segment, whether dealer-side data had been used, and how the segment would be prevented from leaving Veldmark.

Two products for one position. On Tradeyard Poland, professional sellers could buy a brand-sold "Top listing" (fixed price, seven days) and, since 2026, a Veldmark Ads "Sponsored" campaign (cost per click, self-serve). Both claimed the first rows of the results. A seller with both paid twice for overlapping exposure, on two invoices from two systems. The relevance floor adopted for top listings in Ranking for Two Sides: Relevance, Freshness and Paid Visibility did not apply to sponsored units, because they were inserted by a separate service.

Ingrid's question for the architecture advice forum was not whether Veldmark should sell advertising; at €100 million it clearly should. It was **which parts of advertising are one platform, which slots and data each part may use, and who decides when they conflict**. The rest of this chapter is the answer.

Concepts: The Vocabulary of the Ad Stack

Two distinctions shape everything after this section. **Display and native ads are sold to advertisers who may have no listings; sponsored listings are sold to sellers about their listings**. The first is a media business; the second is a visibility product that happens to be priced by auction. And **a first-party audience is a data product**, so it needs the purpose limitation, access control and ownership of any other data product (Data Platform for Marketplaces: Events, Lakehouse and Data Products), not only a campaign setup. The table defines the ad-industry terms the chapter uses, most of them unfamiliar to marketplace engineers.

Term	Definition
Ad stack	The set of systems that decide, serve, measure and bill advertising on a marketplace's surfaces: consent, ad server, header bidding, supply-side connections, audience services, creative rendering, measurement and policy. This chapter owns the term.
Ad server	The publisher's decision engine. It holds campaigns ("line items") with targeting, priorities and prices, and for each ad slot decides which campaign or programmatic bid wins. Google Ad Manager is the common publisher ad server.
Supply-side platform (SSP)	An exchange that sells the publisher's impressions to many buyers in real-time auctions.
Demand-side platform (DSP)	The buyer's system, used by agencies and advertisers to bid in those auctions.

Term	Definition
Header bidding	Running an auction among several SSPs before the ad server decides, so programmatic demand competes with direct campaigns. **Prebid** is the open-source implementation most publishers use, in the browser (Prebid.js), on a server (Prebid Server) and in apps (Prebid Mobile).
Private marketplace (PMP) deal	A programmatic deal offered to selected buyers, often with audience or placement packaging, at a floor price.
Native ad	An ad rendered in the visual format of the surrounding content, for example a card in the listing grid.
Sponsored listing	A seller's own listing promoted into prominent positions in search results or recommendations in exchange for payment per click, per sale or per day, served by the marketplace's search stack and labeled as paid. This chapter owns the term.
Retail media	Advertising sold to brands against a retailer's or marketplace's own audiences and placements, often with closed-loop measurement against on-platform outcomes.
First-party audience	A segment of users defined from the marketplace's own data (searches, saved searches, viewed listings, enquiries), built under the consent and purposes users granted, and activated for advertising on or off the platform. This chapter owns the term.
Consent management platform (CMP)	The component that asks users for consent and records it; in Europe usually integrated with the IAB Transparency and Consent Framework (TCF), which encodes choices in a **TC string** passed to vendors.

Term	Definition
Ad load	The share of a page's or feed's positions given to advertising.
CPM, eCPM, CPC	Price per thousand impressions; effective revenue per thousand impressions after all sources; price per click.
Viewability	Whether an ad had a chance to be seen. The Media Rating Council's desktop guideline counts a display impression as viewable when at least 50% of its pixels are in view on an in-focus tab for at least one continuous second (MRC, 2015).

Architecture: Three Stacks, One Policy Layer

Veldmark's reference architecture keeps three decision paths apart and puts one policy service over all of them (Figure 1). It is a reference design, not a description of any real company's systems.

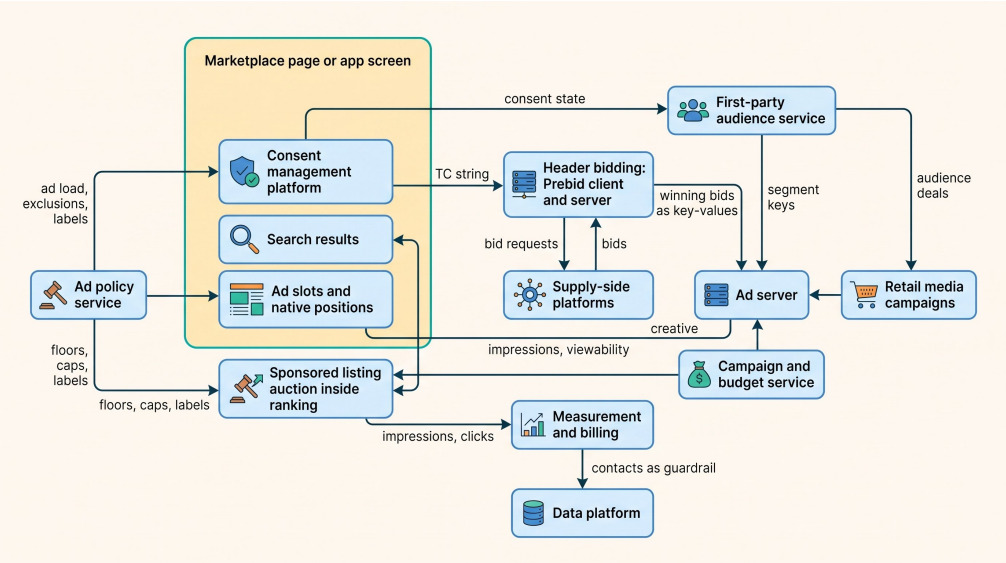


Figure 1: *Veldmark's ad stack. Display and native ads are decided by the ad server after header bidding; sponsored listings are decided inside the search stack; retail media uses both, plus the audience service. One policy service sets ad load, labels, exclusions and*

floors for every path, and measurement reports contacts next to revenue.

Component	Responsibility	Owner at Veldmark
Consent management platform	Collects and stores consent per user, device, purpose and vendor; exposes the TC string and a server-side consent record	Group privacy, with Veldmark Ads as main consumer
Header bidding (Prebid.js, Prebid Server, Prebid Mobile)	Runs the programmatic auction within a timeout; passes bids to the ad server	Veldmark Ads
Ad server	Decides each slot among direct campaigns, deals and bids; renders creatives; reports delivery	Veldmark Ads
First-party audience service	Builds consented segments from marketplace data; exposes segment keys to the ad server and audience deals; enforces exclusions	Veldmark Ads, with Data & ML; DPO approves purposes
Sponsored listing auction	Selects and orders paid listings within the result set, under relevance floors and slot caps	Each brand's search team, on the group search platform
Campaign and budget service	Sponsored-listing campaigns, bids, budgets, pacing, invalid-click filtering, billing events	Veldmark Ads
Ad policy service	Ad load per surface, label rules, category and competitor exclusions, frequency caps, floors for paid positions	Group Chief Product Officer's organization; configuration, not code

Component	Responsibility	Owner at Veldmark
Measurement and billing	Impressions, viewability, clicks, attributed contacts; invoices and revenue recognition	Veldmark Ads with group finance

The display path

A display impression on a listing page is a short distributed transaction with a hard deadline. The page loads; the CMP has (or obtains) the user's consent; header bidding sends bid requests to several SSPs with the consent string; bids return within the timeout; the winning bids are passed to the ad server as key-values; the ad server compares them with direct campaigns and deals and returns a creative; the creative renders and reports an impression and, later, viewability.

The timeout trades bids against speed, and Prebid's documentation says so plainly. "Any demand partner who hasn't responded with a bid within the time allotted will not be included in the auction"; fewer bidders speed up processing but reduce the chance of the best bid, and finding the balance "might require some time and experimentation". Server-side header bidding through Prebid Server is "required in environments that don't have HTML header sections in which to put JavaScript", which is the case for mobile apps and AMP, and Prebid Mobile works "in conjunction with Prebid Server" (Prebid, n.d.). For a listing marketplace, where a large share of traffic is in apps, the server-side path is not optional.

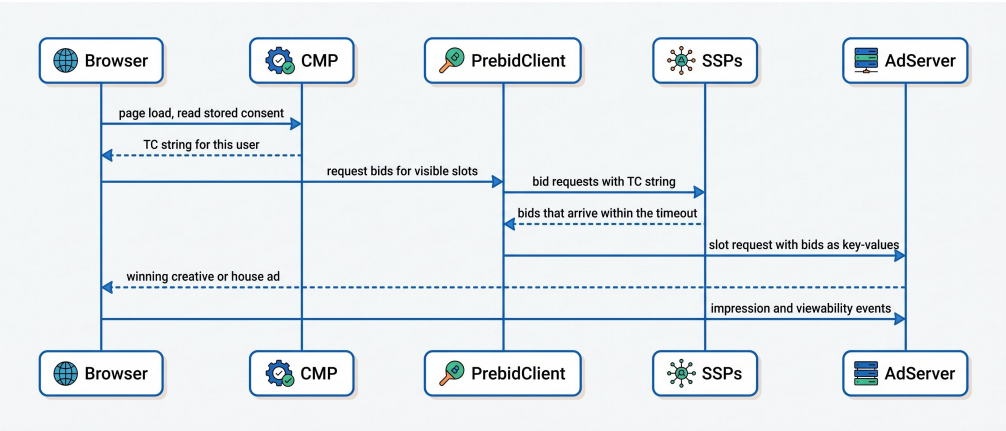


Figure 2: *One display impression. The timeout bounds the bid step; bids that arrive late are dropped; the ad server decides between programmatic bids and direct campaigns.*

Every request downstream of the CMP carries the consent string, and slots below the fold are requested only when they approach the viewport.

Five rules keep the auction off the page's critical path; Found by Google: Web, App and SEO Architecture for Listings owns Core Web Vitals themselves.

- **Never block the listing content on the auction**. The listing's title, price, photos and contact button render first; ad slots fill asynchronously into reserved space, so a late ad does not move content.
- **Reserve slot dimensions** so that a filled or empty ad never shifts the layout.
- **Lazy-load below-the-fold slots**, and request bids only for slots that will be seen.
- **Move bidders server-side** where their latency is high or their scripts are heavy, accepting that some buyers bid less without browser-side identifiers.
- **Treat the timeout as a product parameter with an owner and an experiment**, measured on revenue and on contacts, not on revenue alone.

Supply-chain hygiene is part of the stack. The IAB Tech Lab's ads.txt ("Authorized Digital Sellers") lets a publisher declare publicly which companies may sell its inventory, "making it harder for bad actors to profit from selling counterfeit inventory"; app-ads.txt extends this to apps, and sellers.json makes seller relationships transparent (IAB Tech Lab, n.d.). These files are small, but they are configuration that must be kept in step with every SSP contract, per domain and per app, for every brand.

The ad server is a dependency to manage. Google Ad Manager is the default choice for most publishers, and it is a reasonable one. It is also a concentration risk: in April 2025 a US federal court held that Google "violated antitrust law by monopolizing open-web digital advertising markets" (US Department of Justice, 2025), and the remedies that follow may change how the products are offered. An architect does not need to predict the outcome. They need line items, targeting keys, creatives and reporting defined in a group-owned model, so that the ad server is replaceable in a quarter rather than a year.

Native placements in the listing grid

Native ads, such as a bank's mortgage card between the fifth and sixth listing or a car maker's model card in a results grid, sit inside the result list. Two rules follow. They must be labeled as advertising: the EU Digital Services Act (DSA) requires that recipients can identify, "in a clear, concise and unambiguous manner and in real time", "that the information is an advertisement, including through prominent markings", on whose behalf it is presented and who paid for it (DSA, Art. 26(1)). And their positions count against the same ad-load budget as sponsored listings, because to the buyer they are all positions not given to the most relevant listing.

The sponsored-listing path

Sponsored listings are the advertising product closest to the core marketplace. eBay's FY2025 10-K says it generates "a meaningful amount of our revenue from our Promoted Listings (a first-party advertising offering) and, to a lesser extent, third-party advertising"; eBay's advertising revenue was **US$1,993 million** in 2025, against US$79.6 billion of GMV (eBay, 2026). For classifieds, OLX has described its learning-to-rank stack running a "VAS LTR" model that balances "maximizing relevance for users and ensuring visibility for Value Added Services (VAS), ads that have paid for additional exposure", and the retraining needed when vector retrieval changed which paid listings were retrieved (OLX, 2025). Paid visibility lives inside ranking, whatever team sells it.

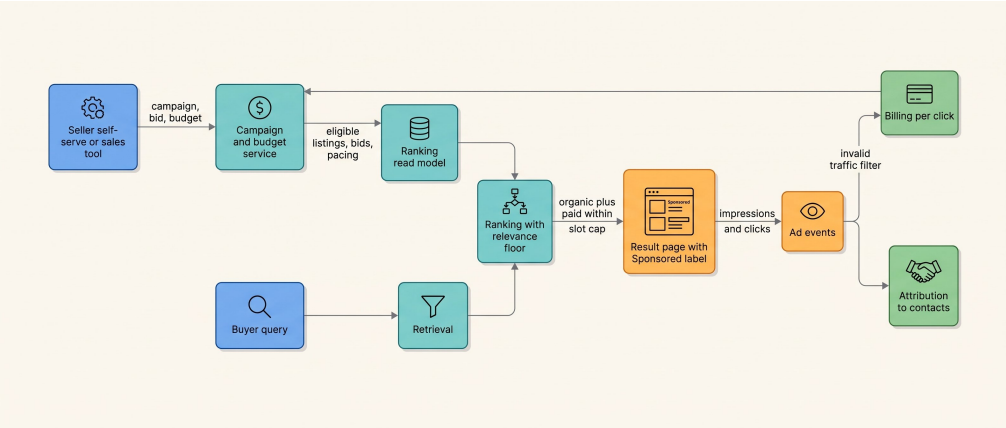


Figure 3: *Sponsored listings at Veldmark. Campaigns and budgets live in the ad unit's service; the decision about which paid listings appear, where, and only above a relevance floor, is made inside ranking from a read model; billing and attribution consume the impression and click events.*

Five design choices keep paid positions inside the marketplace's rules:

- **The auction runs inside ranking, not after it**. A separate service that inserts sponsored units into a finished result page cannot apply the relevance floor, slot caps or deduplication against organic results; Ranking for Two Sides: Relevance, Freshness and Paid Visibility owns how paid and organic candidates are blended. The campaign service publishes eligibility, bids and remaining budget as a read model; ranking reads it the way it reads entitlement effects.
- **Budgets need pacing**. A campaign with €300 a day should not spend it in the first hour of a Sunday peak. Pacing throttles eligibility over the day and must tolerate the bounded staleness of a read model; overspend of a few percent is normally credited, not prevented with a synchronous budget check on every query.

- **Invalid traffic is filtered before billing.** Bots, scrapers and accidental double clicks generate clicks; a click billed to a seller must survive a filter whose rules are stable and documented, and disputes need the event trail.
- **Attribution uses contacts, not only clicks.** The marketplace can measure what most advertising cannot: whether a click led to a message, a call or a booking (From Interest to Contact: Messaging and Lead Management). Reporting contacts per euro to sellers is both the best sales argument for sponsored listings and the best guardrail against selling clicks that do not convert.
- **Sponsored listings and visibility products are one product family.** If the brand sells fixed-price top listings and the ad unit sells auctioned sponsored listings for the same positions, the catalog, the entitlement effects and the slot allocation must be one model (Packages, Bumps and Quotas: Products, Pricing and Entitlements), whoever invoices.

The regulatory floor is unusually clear. The EU's Unfair Commercial Practices Directive, as amended by Directive (EU) 2019/2161, blacklists "providing search results in response to a consumer's online search query without clearly disclosing any paid advertisement or payment specifically for achieving higher ranking of products within the search results" (Annex I, point 11a) (Directive (EU) 2019/2161). For business users, Regulation (EU) 2019/1150 requires online intermediation services to describe, in their terms, any possibility to influence ranking "against any direct or indirect remuneration paid by business users" and "the effects of such remuneration on ranking" (Regulation (EU) 2019/1150, Art. 5). A sponsored listing therefore needs a machine-readable paid flag from campaign to rendering, and a product description that tells sellers what the auction does. Rules of the Marketplace: DSA, DAC7 and Vertical Regulation covers the obligations in full.

The retail-media path and the audience service

Retail media sells brands three things: placements (a car maker's sponsorship of a model's search pages; a bank's mortgage calculator on property listings), audiences (people in market for a product, defined from marketplace behavior), and measurement against outcomes the marketplace can see. CarGurus' FY2025 10-K describes the audience product directly: an "Audience Targeting" add-on that retargets off site using CarGurus audience data, targeted by "geography, search history, CarGurus website activity", alongside advertising sold to manufacturers, other brands and "non-endemic programmatic advertisers" (CarGurus, 2026). Autotrader reports its manufacturer and agency advertising separately: **£15.2 million** in FY26, up 14%, and expected to grow in FY27 (Autotrader, 2026). Rightmove reports about **10 million** consumers subscribed to its marketing CRM (Rightmove, 2026), a first-party channel in its own right.

The **first-party audience service** is where the privacy risk concentrates. Its design rules at Veldmark:

1. **Segments are defined as data, reviewed and versioned.** A segment definition names its sources (search events, saved searches, viewed listings, enquiries), its look-back window, its minimum size, its allowed purposes and its allowed activations (on site, deal ID, clean room, off site). The DPO approves new sources and purposes; the Ads team composes segments from approved building blocks.
2. **Consent is checked at build time and at activation.** A user enters a segment only with consent for the relevant purposes recorded on their events (Data Platform for Marketplaces: Events, Lakehouse and Data Products), and the activation re-checks the current consent, because users withdraw it.
3. **Excluded data never enters.** No special categories of personal data: the DSA forbids presenting advertisements "based on profiling … using special categories of personal data" (Art. 26(3)), and forbids profiling-based ads to recipients a platform knows "with reasonable certainty" to be minors (Art. 28(2)) (DSA). Searches for listings that reveal health (mobility aids, medical equipment) or religion are excluded at the source category, not filtered later. Data held in a neutral or processor role, such as Keybridge's agency CRM data, is out of scope by policy and by access control.
4. **Raw data does not leave.** Off-site activation uses segment identifiers matched in a clean room or through the buyer's platform under contract, with minimum audience sizes, never exported user lists. A segment that leaves the building cannot be recalled.
5. **Sellers' interests are protected.** A car maker's audience built from Motorvane searches is, from a dealer's point of view, Motorvane selling its dealers' buyers to a competitor (new-car leasing against used-car stock). The policy service can exclude retail-media placements from dealer listing pages, or limit them to manufacturer messages that support dealers. That is a commercial decision; the architecture must make it enforceable.

Consent: The Signal That Sets the Revenue

In the EU, reading or storing identifiers on a device requires consent under Article 5(3) of the ePrivacy Directive unless strictly necessary, as Data Platform for Marketplaces: Events, Lakehouse and Data Products explains. For advertising, the industry mechanism is the IAB Europe Transparency and Consent Framework. IAB Europe lists its versions from v1.1 (2018) to **v2.3**, **published in April 2025**, and states that "TCF participants have until 28 February 2026 to adopt TCF v2.3"; v2.3 makes the "Disclosed Vendors" segment of the TC string mandatory (IAB Europe, n.d.). Google requires publishers serving personalized ads in the European Economic Area and the UK to use "a certified CMP integrated with the TCF", a requirement in force since **16 January 2024**; traffic without it may still receive non-personalized or limited ads (Google Ad Manager Help, n.d.).

Three architectural consequences follow.

Consent is per user, per device, per purpose and per vendor, and it changes. The CMP's record must be available server-side, not only in the browser, because Prebid Server, the audience service and app backends all need it. Every bid request carries the TC string; every audience build and activation checks the purpose; every withdrawal propagates within a stated time.

The consent dialog is a revenue component. The share of users who consent to advertising purposes determines how much inventory can be sold with personalized targeting. Changing the dialog's wording, layout or timing changes revenue, and is also the most regulator-visible design decision in the stack. It belongs under experimentation with legal review, not under a growth team's unilateral control.

"Consent or pay" is not a safe default. The European Data Protection Board's Opinion 08/2024 on consent-or-pay models, adopted on 17 April 2024 for large online platforms, concluded that "in most cases, it will not be possible for them to comply with the requirements for valid consent" with a binary choice between behavioral advertising and a fee, and that controllers should consider a free alternative "without behavioural advertising" (EDPB, 2024). No classifieds marketplace appeared on the European Commission's list of designated very large online platforms when it was checked in October 2026 (European Commission, updated 15 October 2026), but the opinion sets the direction. The more robust design is **consent or contextual**: users who decline see ads targeted by page context (category, location, price band), which a listing marketplace can do unusually well because the page itself states the intent.

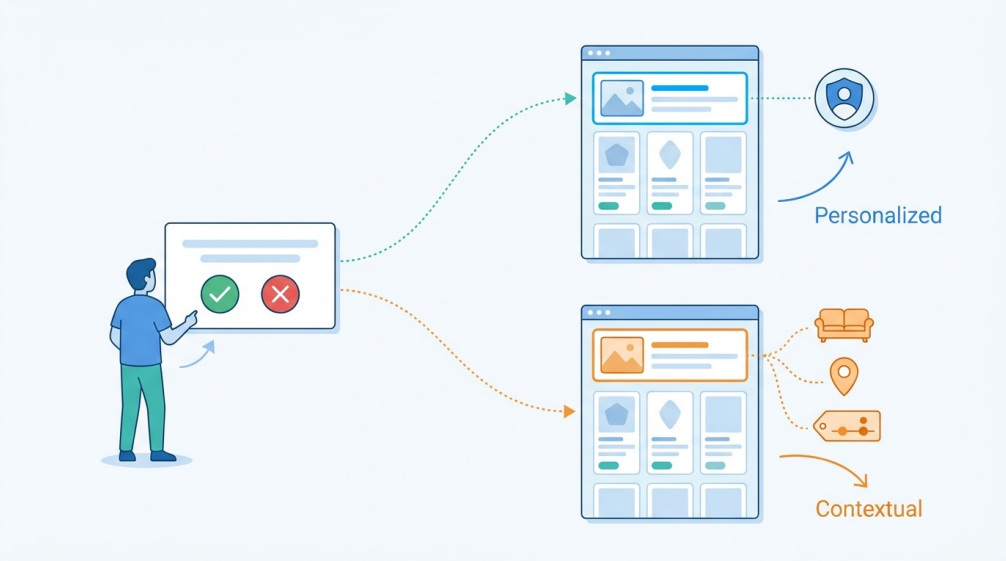


Figure 4: *Consent or contextual: users who decline personalized advertising still see*

relevant ads targeted by what the page itself states, its category, location and price band, which listing marketplaces can do unusually well.

A marketplace should depend as little as possible on third-party identifiers, because browser identity plans keep changing. In October 2025 Google announced that Chrome would "maintain our current approach to offering users third-party cookie choice", and that it would retire several Privacy Sandbox technologies, including Topics and Protected Audience, citing "low levels of adoption" (Google Privacy Sandbox, 2025). Signed-in users, first-party identifiers and contextual signals are under the marketplace's control; the rest of the ecosystem's identity plumbing is not.

In Practice: What the Public Record Shows

Marketplaces publish advertising revenue more readily than advertising architecture, but the figures show how large and cyclical the business is and where its decisions run.

Company and source	What is documented	What it shows
Adevinta, annual report 2023 (published 2024)	Advertising revenue €312M of €1,826M (about 17%); €204M of €744M (about 27%) in European Markets, the former eBay Classifieds horizontals; France's advertising revenue "down 7% … due to the impact of reduced activity from media agencies and programmatic"	Advertising is large on horizontals and cyclical
Adevinta, annual report 2023	Business-continuity planning covers "programmatic suppliers or supply side platforms" as suppliers key to operations; advertising revenue is "recognised as the ads are displayed"	SSPs are critical dependencies; delivery data drives revenue recognition

Company and source	What is documented	What it shows
Adevinta, annual report 2023	The convergence plan for its generalist platforms named "modern approaches to first-party advertising and personalised marketing" among investments not to duplicate	First-party advertising as a shared group capability
eBay, 10-K FY2025 (2026)	Advertising US$1,993M; Promoted Listings described as "a first-party advertising offering" producing "a meaningful amount" of revenue	Sponsored listings can dwarf display at transactional marketplaces
CarGurus, 10-K FY2025 (2026)	Advertising to manufacturers and non-endemic programmatic buyers; Audience Targeting off site	The automotive retail-media and audience model
Autotrader, FY26 results (2026)	Manufacturer & Agency revenue £15.2M, +14%, next to £585.3M Autotrader revenue	On a vertical leader, display is small next to professional revenue
OLX, tech blog (2025)	"VAS LTR" balancing relevance and paid visibility; retraining after vector retrieval	Paid visibility is a ranking concern

Display advertising follows the advertising market, not the marketplace's liquidity. Adevinta's numbers show both the size and the fragility of the business: European Markets earned more than a quarter of its revenue from advertising, while the French business reported a 7% decline driven by agencies and programmatic demand (Adevinta, 2024). Adevinta has been private since 2024, so no later revenue mix is public. A group should therefore never plan its platform investments on the assumption that display revenue grows with audience.

Veldmark Worked Example: Three Decisions in 2027

The architecture advice forum took three decisions in July 2027, each with a decision record.

Decision 1: the auction timeout is set with contacts as a guardrail

Priya Raman's team ran a four-week experiment on Tradeyard Netherlands listing and search pages, randomized by user, with three arms: the 1,500 ms timeout, 900 ms, and 900 ms with four of the nine bidders moved to Prebid Server and below-the-fold slots lazy-loaded.

Metric (Tradeyard NL web, 4 weeks)	1,500 ms	900 ms	900 ms, server-side and lazy-load
Programmatic revenue per 1,000 page views	100 (index)	96.8	98.1
Largest contentful paint, p75, listing pages	2.9 s	2.5 s	2.3 s
Search-to-contact rate	100 (index)	100.5	100.9
Contacts per 1,000 sessions	100 (index)	100.6	101.0

The third arm lost 1.9% of programmatic revenue per page view and gained about 1% in contacts. At Tradeyard Netherlands' scale, a 1% change in contacts is worth more to the marketplace than 1.9% of the programmatic revenue earned on its web pages, because contacts drive renewals, VAS purchases and Safe Deal transactions. The forum adopted the third arm, and a rule: **any change to ad timeouts, slot counts or ad load ships behind an experiment with contacts per session as a guardrail metric** (Testing a Two-Sided Market: Experimentation Platforms owns guardrails). Revenue that requires fewer contacts needs a decision by the CPO, Noor Hassan, not an Ads configuration change.

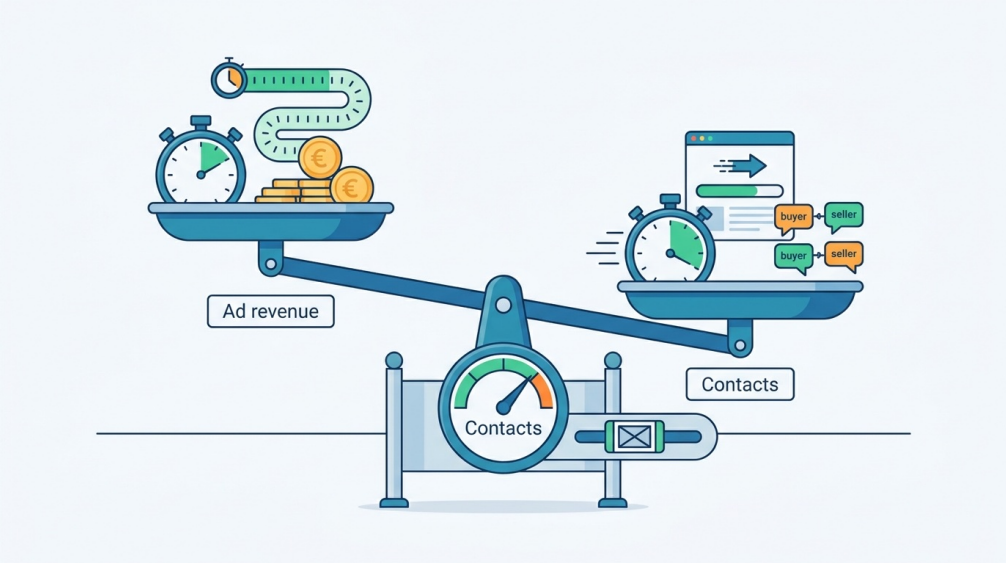


Figure 5: *The shorter auction cost 1.9% of programmatic revenue per page view and gained about 1% in contacts, which is worth more to the marketplace; ad timeouts, slot counts and ad load now ship behind an experiment with contacts per session as the guardrail.*

Decision 2: audiences are built from approved building blocks

The car maker's "electric SUV" audience was rebuilt under the audience-service rules. Its sources were limited to Motorvane search and saved-search events for electric SUVs in the previous 30 days, from signed-in or consented users with the relevant TCF purposes; no dealer-side data, no Keybridge or Valuo data, no listing-contact content. Minimum segment size per activation was set at 1,000 users. Activation was allowed on Motorvane pages and through private deals; off-site activation only through a clean room with aggregate reporting. The segment contained about 310,000 users in Germany when first built, and fell to about 240,000 after the consent check removed users whose purposes did not cover it, a 23% difference that the original segment had silently ignored.

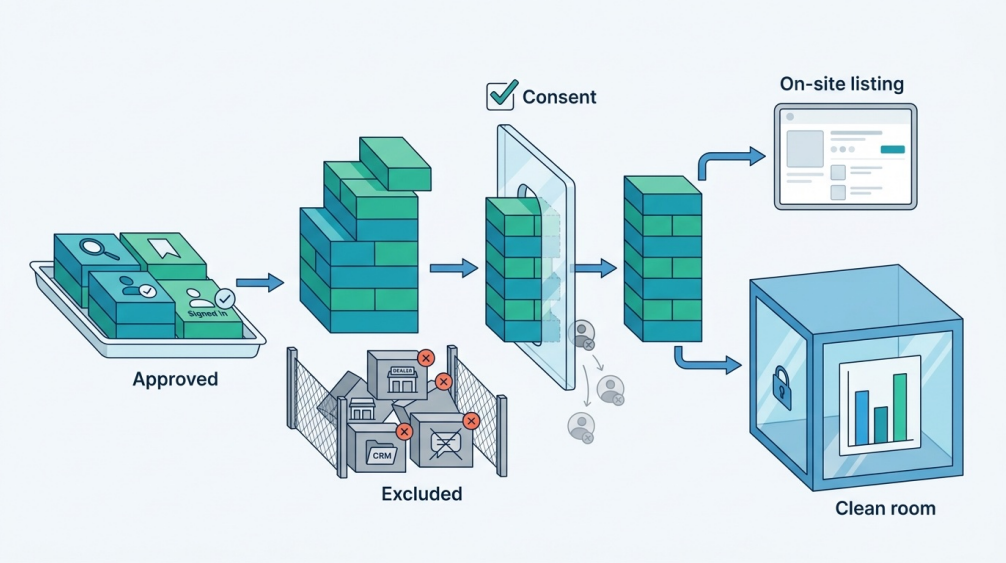


Figure 6: *Audiences are built from approved building blocks only: dealer, Keybridge, Valuo and message data stay out, the consent check shrank the electric SUV segment from about 310,000 to about 240,000 users, and off-site activation runs only through a clean room.*

The forum also accepted Felix Brandt's condition for Motorvane: **retail-media placements for car makers do not appear on dealer listing detail pages**, only on search, editorial and model pages, and dealers can see which placements ran next to their stock. Motorvane dealers pay **€238 million** a year in subscriptions; a €14 million retail-media line is not worth their trust.

Decision 3: one family of paid positions

From the first quarter of 2028, sponsored listings and top listings on Tradeyard Poland become one product family in the group product catalog (Packages, Bumps and Quotas: Products, Pricing and Entitlements): fixed-price "Top listing" and auctioned "Sponsored" are two pricing modes for the same slot type. Ranking reads both from one read model; both are subject to the relevance floor; the result page carries at most three paid positions in the first ten rows, whichever mode bought them. Veldmark Ads keeps the auction, budgets and self-serve tool; the brand keeps the product description and the slot rules; one invoice line per seller per period shows both. Sponsored-listing revenue on Tradeyard Poland fell about 6% in the first months after the floor, while contacts per paid position rose; the same pattern Ranking for Two Sides: Relevance, Freshness and Paid Visibility reports for top listings.

Trade-offs

Decision	Option A	Option B	When A is right	When B is right
Ad server	Market-standard ad server (e.g., Google Ad Manager)	Alternative or in-house decisioning	Broad demand, standard direct sales, small ad-tech team	Very large first-party ad business (sponsored listings, retail media) where the core decision is not a display auction
Header bidding	Client-side (Prebid.js)	Server-side (Prebid Server)	Web pages where buyers bid more with browser identifiers	Apps, AMP, heavy bidders, strict page-speed targets
Sponsored listings owner	Search team serves; ad unit sells	Ad unit serves its own units	Paid positions must obey relevance floors and slot caps	Placements outside ranking, such as a separate carousel clearly marked as ads
Audience activation	On site and private deals only	Off-site activation through clean rooms or partners	Strict privacy posture; strong own inventory	Brands demand reach beyond the marketplace and contracts and clean rooms are mature

Decision	Option A	Option B	When A is right	When B is right
Users who decline consent	Contextual advertising	Consent or pay	Almost always, for market-places	Only with a free non-behavioral alternative and legal review
Ad load	Low, guarded by contacts	High, guarded by revenue	Marketplaces whose core revenue depends on matches	Content sites without a matching business; rarely right for listings
Sales	In-house sales team	Agency or ad network representa-tion	Large, multi-brand inventory; retail-media relation-ships	Small brands or new countries

Building an in-house display ad server is rarely justified for a marketplace, because the demand sits in the programmatic ecosystem. But the most valuable advertising at transactional and large classifieds marketplaces is sponsored listings and retail media, which are decided in the marketplace's own systems anyway. **Own the decisions that use your listings and your audiences; rent the plumbing for generic display.**

Vertical Differences

Aspect	General classifieds	Real estate	Automotive	Jobs	Recommerce
Share of revenue from ads	Highest; a fifth to a quarter at some horizontals	Low to moderate	Low; manufacturer advertising is small next to dealer revenue	Moderate (employer branding)	Sponsored listings can be large
Main display and retail-media buyers	Consumer brands, retailers, telecoms	Banks and mortgage brokers, insurers, utilities, movers, developers	Car makers, finance and insurance, leasing	Employers, recruiters	Brands, retailers
Sponsored listings	Common, sold to pros and private sellers	Usually depth products and developer campaigns instead	Dealer depth products; sponsored placements	Sponsored or promoted jobs	Core product
Sensitive data risk	Category searches revealing health, religion	Financial situation, household composition	Financing interest	Health, ethnicity, union membership in CVs and searches	Low to moderate
Main conflict	Ad load versus browsing	Bank ads versus agents' own mortgage partners	Car-maker campaigns versus dealer stock	Employer branding versus job-seeker privacy	Promoted items versus relevance

In real estate, a property search reveals a household's budget, location preferences

and sometimes life events. The Veldmark rule for Hearthly was that mortgage and insurance advertisers buy contextual placements (on listings in a price band and region) and audiences built only from consented search behavior, and **never segments derived from Valuo valuation requests**, which are homeowners asking about their own property for a different purpose.

Failure Modes

- **The ad unit's own ranking**. Sponsored units inserted after ranking by a separate service, outside relevance floors and slot caps. Multiplier Creep by a second door.
- **Revenue-only ad experiments**. Timeout, slot and ad-load changes evaluated on revenue per page view. Contacts fall slowly and nobody connects the two.
- **Consent that stops at the browser**. The TC string reaches Prebid.js but not Prebid Server, the audience pipeline or the apps' backends. Personalized ads reach users who declined.
- **Audiences without owners**. Segments built by campaign managers from raw event tables, with no purpose review, no minimum size and no record of where they were activated.
- **Exported user lists**. Hashed e-mail lists sent to advertisers "for matching". They cannot be recalled, and hashing does not make them anonymous.
- **Advertising that competes with sellers**. Retail-media placements on professional sellers' own listing pages, promoting alternatives to their stock. Renewals suffer before revenue reports show anything.
- **Third-party scripts with full page access**. Ad and measurement scripts that read the page, including listing contact details or user data, and send it to vendors. The ad stack needs the same content-security discipline as any other third-party code.
- **Stale ads.txt and sellers.json**. Authorized-seller files that lag behind SSP contracts. Buyers stop bidding on inventory they cannot verify.
- **Ads on gone listings**. Retail-media and native placements booked against listing pages that are sold or removed, and pages kept alive for advertising inventory. Ghost Inventory monetized.

The Chief Architect's View

At group scale, advertising raises five questions that belong on the Chief Architect's agenda.

What is shared, and what is not. The ad server, header bidding, CMP integration, ads.txt management, audience service and measurement are generic and belong to one group stack; Veldmark already had this right, and it is the reason advertising was one

of its few shared services in 2027. Sponsored-listing serving is not generic: it belongs inside each brand's ranking on the shared search platform. The product family of paid positions belongs in the group product catalog. This is "share platforms, not products" applied to advertising.

Who owns the page. A listing page has one owner: the brand's product team. Advertising rents positions on it under a policy the product organization sets: maximum ad load per surface, label rules, exclusions, and the guardrail metrics that any change must pass. At Veldmark the policy is configuration in the ad policy service, owned by Noor Hassan's organization, with changes reviewed in the architecture advice forum when they cross brands.

What data advertising may use. The audience service is a data product with purposes approved by the DPO, Lucía Ortega, and with hard exclusions: special categories, minors, data held for professionals in a neutral role (Keybridge), data collected for another purpose (Valuo). Every segment has an owner, a definition, a size and an activation log.

How dependent to be. A marketplace whose display business runs entirely on one vendor's ad server and one browser's identity choices has accepted two dependencies it does not control. Keep the campaign, targeting and reporting model group-owned, keep identity first-party, and review the exit plan when the legal and market landscape changes, as it did in 2025.

How to measure. Advertising reports revenue, eCPM, fill rate and viewability. The group dashboard should show them next to contacts per session, search-to-contact rate and page speed for the same surfaces, so that nobody can raise one set without seeing the other. Advertising is the only revenue line that does not depend on matches, and the measurement must put the matches back in.

Questions to Ask

1. Which advertising products run on your result pages, which systems decide each one, and does every paid position, whoever sold it, obey the same relevance floor and slot cap?
2. What is your header-bidding timeout, who owns it, and when was it last tested against contacts rather than revenue alone?
3. Does the consent state reach server-side bidding, app backends and the audience pipeline, and how quickly does a withdrawal take effect everywhere?
4. What do users who decline consent see, and has the alternative been reviewed against the EDPB's 2024 opinion?
5. Who approves a new audience segment, what sources may it use, what is the minimum size, and where has it been activated?
6. Which data is excluded from advertising by policy, and is that enforced by access control or only by convention?

7. Can a professional seller see which advertisements ran next to their listings, and are any of them promoting a competitor?
8. If your ad server vendor changed terms or products next year, how long would migration take, and which parts of your campaign model would you lose?
9. Are ads.txt, app-ads.txt and sellers.json updated as part of every SSP contract change?
10. Is every advertisement, sponsored listing and native unit labeled with who it is presented for and who paid, as the DSA requires?

From Revenue to the Platform Underneath

Advertising closes Part VI: the marketplace now has contacts, transactions, products and attention to sell, and each depends on systems that run beneath every brand. The ad stack needed a consent signal on every event, a read model of paid positions inside ranking, and measurement that joins impressions to contacts. Those are platform problems. Part VII — PLATFORM: The Distributed System Underneath turns to that platform, starting with the event backbone in Every Change Is an Event: Event-Driven Listing Architecture.

Sources and Evidence

Company engineering and investor sources

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Standards, regulation and documentation

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- Directive (EU) 2019/2161, *Official Journal*, 2019 (Annex point 11a checked against the Publications Office text, October 2026) — blacklisted undisclosed paid ranking.
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- Prebid overview, Prebid.org, n.d. — header bidding, timeouts, Prebid Server and Mobile.
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Research and practitioner sources

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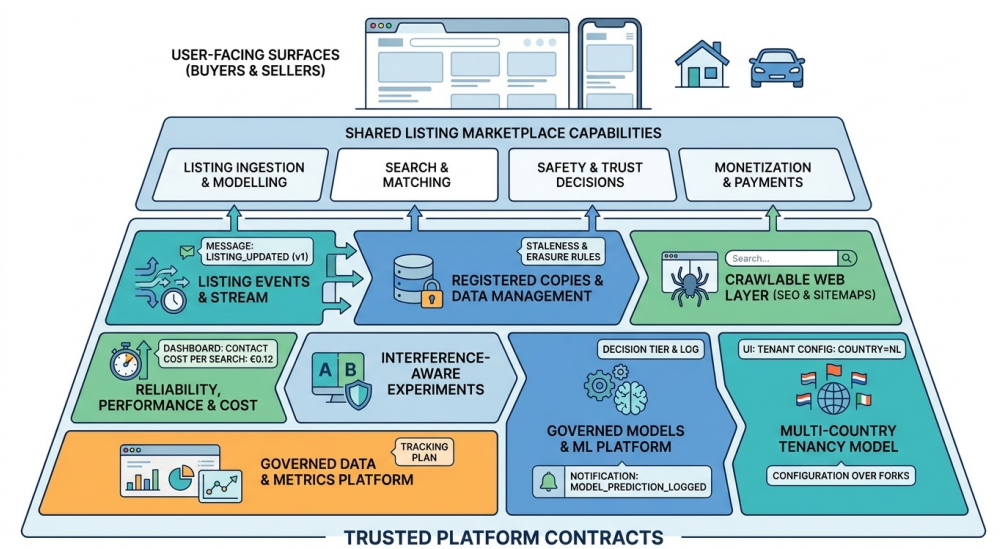
Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

Part VII — PLATFORM: The Distributed System Underneath



IN THIS PART, YOU WILL: Design the shared platform underneath the listing marketplace: listing events, data management and read models, web and SEO architecture, the data platform, experimentation, machine learning, multi-country tenancy, and reliability, performance and cost.

WHY THE BUSINESS CARES: A sold car that still gets alerts, dead pages in Google's index, a price test with the wrong sign and a search outage at the peak all cost contacts. Users see each as a product failure.

WHY ARCHITECTS SHOULD CARE: One listing write has ten or twenty readers, and every brand and country wants its own variant. The platform

decides whether those readers agree, whether a market can change without a fork, and whether a number or a model can be trusted.

KEY POINTS:

- **The platform is a set of published contracts, not a set of products.** One owned listing event stream, a register of every copy with its staleness budget and rebuild path, and governed metrics let many teams build on shared data without coupling to each other's internals.
- **Two-sided markets break the defaults.** Buyers and sellers affect each other, so experiments need interference-aware designs and guardrails for the side not being tested, and models need governance by the decision they make, not by their algorithm.
- **Share beneath the product, and measure by journey.** Make the tenant first-class and market differences configuration, with forks only on an expiry date. Measure reliability as journey objectives with error budgets, and cost per search, per live listing and per contact.

FAILURE PATTERNS THIS PART ADDRESSES:

- **Search as System of Record** — Makes the search index one registered read model that can be rebuilt from the event stream.
- **The Copy-Paste Country** — Replaces copied codebases with a tenancy model, configuration over code and owned extension points.
- **Set-and-Forget AI** — Gives every production model an owner, a decision tier, a prediction log and monitoring.

Parts II to VI designed what a marketplace does: model listings, take supply in, match it to demand, keep it safe and make money from it. Part VII turns to **the distributed system that all of those capabilities stand on**. Most groups never designed this layer; it grew from polling jobs, nightly dumps and copied country codebases.

The argument of this part is that **a shared platform earns trust through contracts, not mandates**. Events, copies, URLs, metrics, experiments, models, tenants and service levels each need a written promise, an owner and a way to check the promise is kept. Then the brands above can move at their own pace without breaking each other.

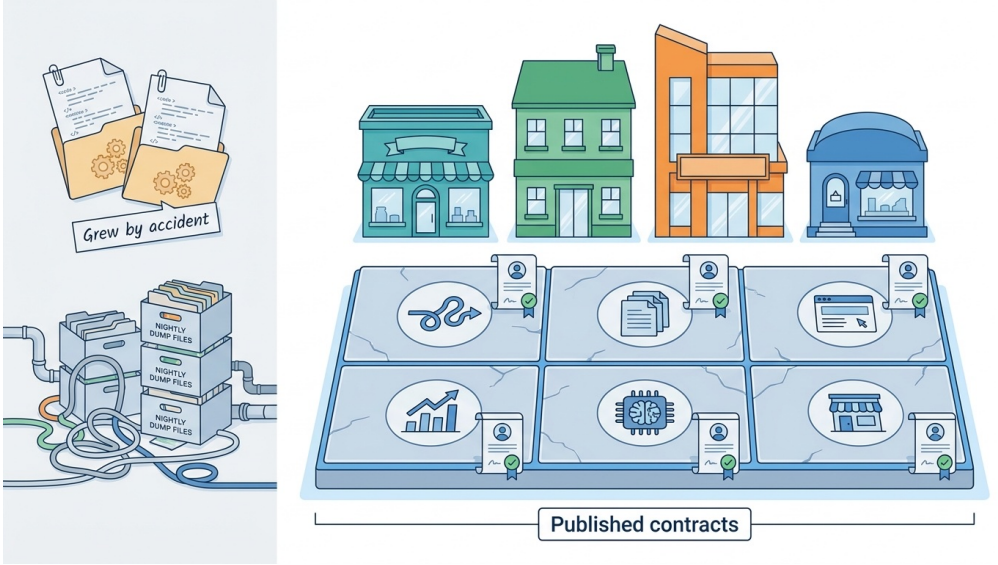


Figure 1: *Part VII rebuilds the layer that grew from polling jobs, nightly dumps and copied codebases as a set of published contracts, each with an owner and a check, so the brands above can move at their own pace.*

Chapters in this part

Chapter	The question it answers	Read it when…
Every Change Is an Event: Event-Driven Listing Architecture	How should listing changes reach every downstream system?	Removals arrive late, or a new consumer takes a quarter to build.
System of Record and Read Models: Listing Data Management	Where does listing data live, and how do the copies stay consistent?	Nobody can list every copy of a listing, or an erasure request takes weeks.
Found by Google: Web, App and SEO Architecture for Listings	How should web and app layers serve both users and crawlers?	Sold listings stay indexed, or a redesign cost you organic traffic.
Data Platform for Marketplaces: Events, Lakehouse and Data Products	How should a marketplace group collect, govern and use its data?	Each brand counts “contact” differently, or impressions lack their ranking context.

Chapter	The question it answers	Read it when…
Testing a Two-Sided Market: Experimentation Platforms	How do you run trustworthy experiments when buyers and sellers affect each other?	A winning price or ranking test disappoints once it ships to everyone.
Models in Production: ML Platforms and Use Cases	Which models matter, and what platform runs them responsibly?	Nobody can list your production models, or an estimate is about to drive offers.
One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture	How do you share a platform across countries and brands without one product?	Every rule is built once per country, or teams route around the platform.
Sunday Evening Peaks: Reliability, Performance and Cost	How do you keep a listing platform fast, available and affordable at scale?	Search fails at the weekly peak, or nobody knows the cost of a search.

How this part connects

Every earlier part rests on this layer. Part VI — MONEY: Conversion and Monetization assumed it for paid effects, leads, contact counts and pricing tests. The lifecycle from Part II — LISTING: Modeling the Thing Being Sold becomes the event contract, discovery in Part IV — DISCOVERY: Matching Intent to Inventory reads the stream, and trust decisions in Part V — TRUST: Keeping the Marketplace Safe must propagate to every copy. Part VIII — LEADERSHIP: The Chief Architect's Agenda takes the platform to the Chief Architect's agenda: consolidating acquired stacks onto it, organizing teams around it and retiring the monolith while it runs.

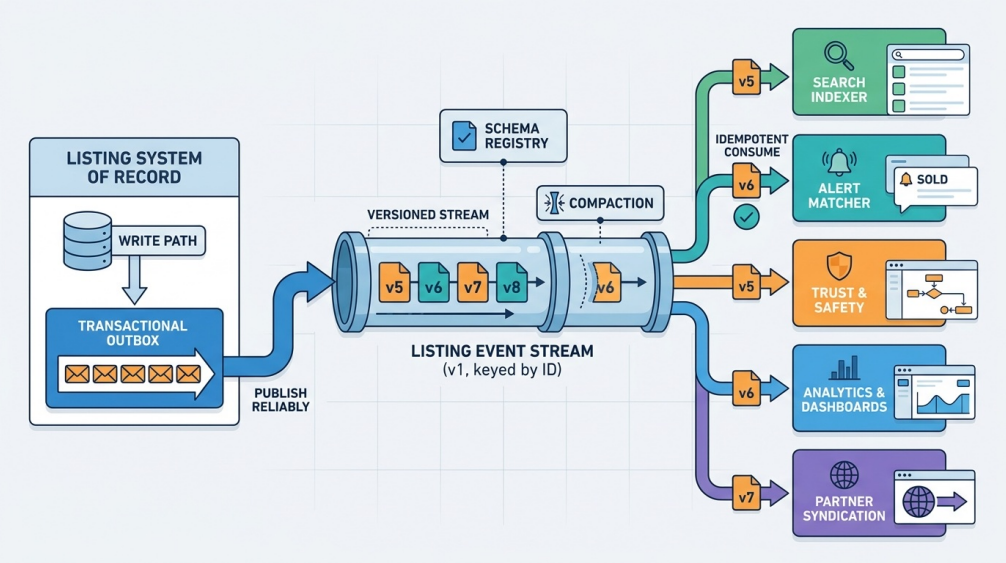
The Veldmark thread

Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), starts 2027 with four messaging technologies, three warehouses, 146 production models on four training stacks and a PHP monolith full of country branches. On Sunday 17 January 2027 a TV campaign and a scheduled reindex browned out Tradeyard NL search for 48 minutes. Ingrid Halvorsen's team, with Sven Lindqvist, Priya Raman and Yara Mendes, decides:

- **journey service level objectives** with error budgets and a search degradation ladder (April 2027);
- **server-rendered pages and event-driven sitemaps** for Tradeyard's 40 million indexed URLs (May 2027), and **Tenancy model v1**: ten brand × country tenants (9 June 2027);
- **Experimentation standard v1** (30 June 2027), after a bump-price test that looked like a win, and **Listing event contract v1** on Kafka (Confluent Cloud), live on 1 July 2027;
- an **Iceberg-based data platform** with tracking plan v1 (September 2027), a **group retention schedule** and the consumer registry extended to every listing copy (October 2027), and a **group model registry** mandatory for every production model (9 November 2027).

By the end of Part VII you should be able to say, for any listing fact, where it is true, which copies hold it and how you would rebuild or erase them. Begin with Every Change Is an Event: Event-Driven Listing Architecture.

33. Every Change Is an Event: Event-Driven Listing Architecture



IN THIS CHAPTER, YOU WILL: Learn how to design the listing event stream that feeds search, alerts, trust and safety, analytics, SEO and partners. The chapter covers what a listing event is, how to publish it reliably with a transactional outbox or change data capture, how to keep it ordered and idempotent, how to evolve its schema, and how to replay it. It follows Veldmark's Common Ground listing event contract v1 from design to go-live.

WHY THE BUSINESS CARES: A listing that is sold but still alerted, a price cut that never reaches the watchers, or a scam ad that stays visible after takedown each cost contacts, trust and money. Every one of them is a change that did not reach a downstream copy in time.

WHY ARCHITECTS SHOULD CARE: In a listing marketplace one write has ten or twenty readers. The way changes travel from the system of record to those readers decides publication latency, removal latency, the cost of reindexing, and whether a new consumer takes a week or a quarter to build.

KEY POINTS:

- **The listing event is a published contract, not an integration detail.** One owner emits it, for every lifecycle transition, with a stable identity, a per-listing version and a schema that consumers can rely on for years. Consumers that read the owner's database instead are coupling themselves to its internals.
- **Reliable publication and safe consumption are two separate halves.** The producer must never commit a change without its event (outbox or change data capture). The consumer must tolerate duplicates, gaps and replays (idempotency by event ID or listing version). Neither half alone gives correct downstream state.
- **Carry enough state to heal, and keep the history replayable.** Events that carry the listing's current public state let a consumer recover from a lost or skipped message with the next one. A compacted state topic plus a disciplined backfill path turns reindexing and new consumers from projects into routine operations.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Ghost Inventory** — removals that travel by polling, nightly dumps or best-effort messages arrive late or never, so sold listings stay searchable and alertable.
- **Search as System of Record** — when the search index is the only complete projection of listing changes, nothing else can rebuild it; a replayable event stream is the alternative source.
- **Deal-Shaped Architecture** — four brands with four messaging technologies and four event vocabularies mean every group capability integrates four times.

A listing is written by one party and read by many. The seller or a dealer's feed changes the price once; search must re-index it, saved-search alerts must decide whether to notify watchers, the market-value model must re-score it, the trust-and-safety system must check whether the change is suspicious, the sitemap must update `lastmod`, the warehouse must record it, and a partner syndication feed may need to carry it. **The architecture question is not how to store the listing but how to move its changes**, because that movement is where freshness, removal latency and consistency are won or lost.

Most marketplaces start with the cheapest answers: downstream systems poll the listing database, read a replica, receive a nightly dump, or get a message that the application sends after it commits — when it remembers to. Each answer works until the number of readers grows. Then removals arrive late, a reindex takes a weekend, and nobody can say which copy of a listing is right. **An event-driven listing architecture replaces**

many private integrations with one published stream, owned by the listing's system of record and designed as carefully as an external API.

This chapter opens Part VII — PLATFORM: The Distributed System Underneath, whose argument is that a shared platform earns trust through contracts, not mandates. The listing event is the first of those contracts and the one the rest of the part reads from. It turns the lifecycle and invariants of The Listing Aggregate: Identity, Lifecycle and Invariants and the stages of From Submitted to Searchable: The Publication Pipeline into a published stream, the stream that search, alerts and trust and safety in Parts IV and V already assumed. The chapter owns the **listing event**, the **transactional outbox** and **schema evolution** for listing events. Where the copies live, how consistent they must be, and how erasure reaches them belong to the next chapter, System of Record and Read Models: Listing Data Management. How behavioral tracking events reach the data platform belongs to Data Platform for Marketplaces: Events, Lakehouse and Data Products.

Setting the Scene at Veldmark: The Digest That Did Not Know the Bike Was Sold

On Tuesday 9 March 2027, Ingrid Halvorsen, Veldmark's Group Chief Architect (Veldmark is the fictional marketplace group this book follows; see Meet Veldmark: Four Brands, One Platform to Build), reads an incident note from Tradeyard Netherlands. Femke's €240 Rotterdam city bike, listed on Sunday evening, sold on Monday for €225 through Safe Deal. Tradeyard marked it sold at **18:10** and Solr removed it from search at **18:11**. Yet the next morning's alert digest e-mail still featured the bike to people with matching saved searches. Three of them messaged Femke to ask whether it was still available.

Sven Lindqvist, principal engineer for the group event platform, traces the path. Tradeyard's alert digest job assembled its candidate list on Monday afternoon from a read replica of Yardstick's MySQL database, the PHP monolith at the heart of Tradeyard. It stored the list and sent it at 07:00 on Tuesday. Nothing told the digest that the bike had sold in between. The search indexer heard about the sale because Yardstick sends a RabbitMQ message after the sold flag is committed, but the digest job was never subscribed to that message, and nobody had a list of who was.

Sven's inventory of Tradeyard NL alone finds **eleven** downstream readers of listing changes. Four consume RabbitMQ messages that Yardstick emits from application code. Three poll the MySQL replica on schedules between one and fifteen minutes. Two receive nightly CSV exports. One, the Snowflake warehouse load, reads a daily table dump. One, the new trust-and-safety scoring service on AWS, reads Kafka topics that a Yardstick plug-in fills on a best-effort basis. Across the group the picture multiplies: Motorvane publishes to Kafka on Amazon MSK, Hearthly to Google Pub/Sub, Keybridge to Azure Service Bus, each with its own event names and payloads.

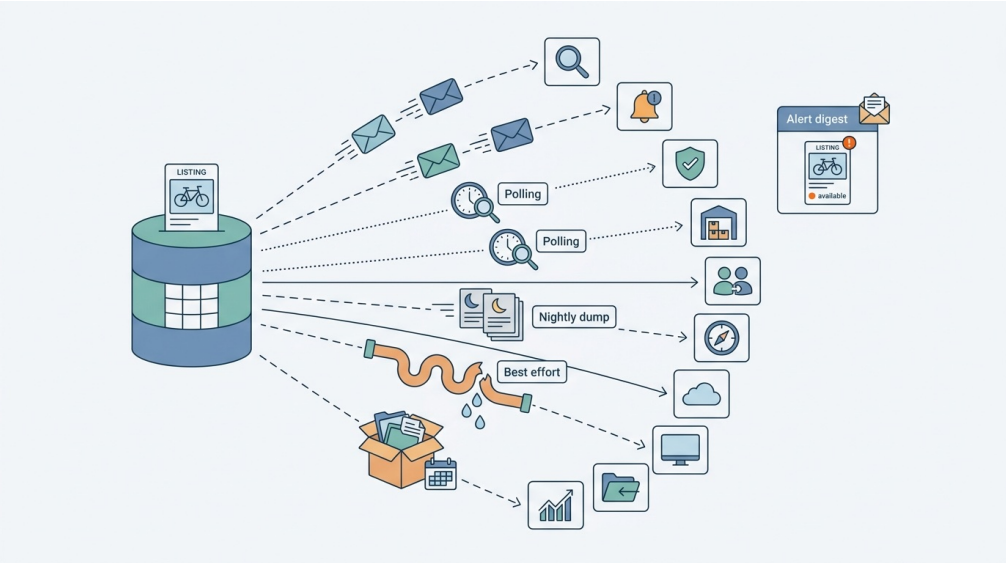


Figure 1: *Before the contract, Tradeyard NL alone had eleven readers of listing changes on five different mechanisms, and nobody had the list, so the digest never heard that the bike had sold.*

Ingrid's question to the architecture advice forum is the chapter's question: **how should listing changes flow to search, trust, alerts, analytics and partners, across four brands, so that every reader sees every change once, in order, and in time?** The Common Ground program's answer, live on 1 July 2027, is the group event backbone on Kafka (Confluent Cloud on AWS) and the **Listing event contract v1**.

Concepts: What a Listing Event Is

Domain events, change events and messages

Three things are often called "events", and the difference matters for ownership.

Term	Definition	Who decides its shape	Example
Domain event	A statement, in business language, that something happened to an aggregate: past tense, immutable, meaningful to other contexts	The owning domain, as a published contract	`ListingSold`, `PriceChanged`
Change event (change data capture, CDC)	A row-level record of an insert, update or delete in a database table, read from the database's transaction log	The database schema, which was designed for the owner's internal needs	`UPDATE ads SET status=4 WHERE id=…`
Command or message	A request for someone to do something; may be refused	The sender and receiver jointly	`ReindexListing`, `SendAlert`

A **listing event**, in this book, is a domain event about one listing aggregate, emitted by the listing's owning context whenever the aggregate passes an invariant-preserving transition: submitted, published, updated, price changed, paused, reserved, sold, expired, removed. The aggregate and its transitions are defined in The Listing Aggregate: Identity, Lifecycle and Invariants; this chapter treats the event as the **published language** of that aggregate (the term Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models owns): the vocabulary other contexts use to learn about listings without knowing how the listing context stores them.

Change data capture is a mechanism, not a contract. It is often the right way to *get* events out of a legacy system, but a raw table change exposes the producer's schema to every consumer. Renaming a column in Yardstick should be Daan Visser's team's business; if eleven consumers parse the binlog of the `ads` table, it is everybody's.

Notification, state transfer, sourcing

Martin Fowler's 2017 essay "What do you mean by 'Event-Driven'?" separates four patterns that hide behind the label. Three matter here.

- **Event notification.** The event says only that something changed ("listing 123

updated") and consumers call back for details. It is small and keeps the producer's schema private, but every consumer becomes a reader of the producer's API at the moment of the event, and Fowler notes that a logical flow running across many notifications becomes hard to see.
- **Event-carried state transfer**. The event carries the state consumers need, so they can maintain their own copy without calling back. The cost, in Fowler's words, is that "there's lots of data schlepped around and lots of copies".
- **Event sourcing**. The events *are* the system of record and the current state is derived from them. For listings, this is rarely worth its cost; the listing context usually keeps a relational record and publishes events from it.

Listing events should carry state. A popular listing change has many readers at the same moment — a price cut fans out to search, alerts, valuation and analytics within seconds — and a callback storm on the listing service at the Sunday-evening peak is exactly the load nobody wants. The real design question is *which* state, which the envelope section answers.

Ordering, delivery and idempotency

Three properties decide whether consumers end up correct.

- **Ordering**. Kafka preserves order within a partition, so keying listing events by listing ID gives per-listing order. The Debezium outbox router follows the same convention, using the aggregate ID as the message key "which preserves per-aggregate order" (Debezium documentation, n.d.). There is no useful global order across listings, and consumers should not need one.
- **Delivery**. Practical pipelines deliver **at least once**: a producer retry or a consumer restart can deliver the same event twice. End-to-end exactly-once across a database, a broker and a search index is not on offer in general, whatever individual components promise.
- **Idempotency**. A consumer is **idempotent** if processing an event twice has the same effect as processing it once. For listing events the robust rule is **version-gated apply**: each event carries the listing's aggregate version, and a consumer applies an event only if its version is newer than the one it has stored.

Architecture: The Listing Event Stream

Components and responsibilities

The architecture has five parts, and each has one owner.

1. **The listing context** (per brand, until shared) owns the listing aggregate and is the only producer of listing events. Trust and safety does not emit `ListingRemoved`; it emits an enforcement decision, and the listing context applies it and emits the resulting listing event. This keeps every listing event consistent with the record.
2. **The publication mechanism** guarantees that a committed change produces its event: a transactional outbox in services that can change their write path, change data capture with a translation layer in systems that cannot.
3. **The event backbone** — Kafka, in Veldmark's case — stores events durably, partitions them by listing ID and retains them long enough for replay.
4. **The schema registry** stores every version of the event schema and rejects incompatible changes at publish time.
5. **Consumers** — search indexer, alert matcher, trust-and-safety scoring, sitemap generator, warehouse loader, partner syndication, pro-seller statistics — each own their read model and their consumer offsets.

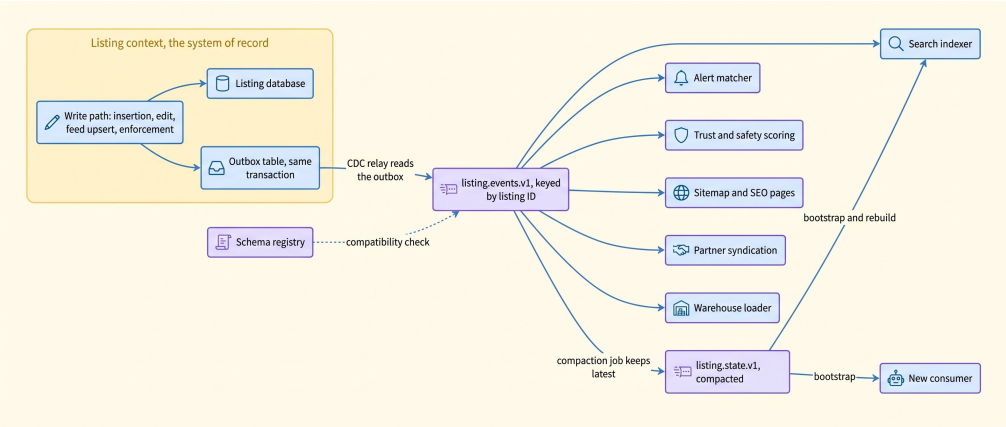


Figure 2: *One producer, one contract, many consumers. The outbox makes publication part of the write; the domain-event topic carries changes; the compacted state topic lets any consumer, including one built next year, start from the current state without asking the producer for a dump.*

The transactional outbox

The **transactional outbox** solves the dual-write problem. If the listing service commits to its database and then publishes to Kafka, a crash between the two steps leaves a listing sold in the database and active everywhere else — the bike problem in miniature. If it publishes first and then commits, a failed commit leaves consumers believing in a change that never happened.

The outbox writes the event into an `outbox` table **in the same database transaction** as the state change. A separate relay — commonly change data capture on the outbox table — publishes outbox rows to the broker. The Debezium outbox event router documents the expected columns: a unique `id` usable for consumer de-duplication, an `aggregatetype` that selects the topic, an `aggregateid` used as the message key, a `type` and a `payload`; its stated purpose is to avoid inconsistencies between "a service's internal state" and the events other services consume (Debezium, n.d.).

The outbox has three properties that matter for listings:

- **The producer chooses the event shape**. The payload is written by domain code, so it can say `ListingSold` with a sale channel and a price, instead of exposing `status=4`.
- **Publication is at least once**. The relay can publish a row twice after a restart; consumers must de-duplicate.
- **Order follows commit order per listing**, provided the relay preserves table order and keys by listing ID.

The cost is a write-path change in every producer, and an extra table to clean up. That is why legacy systems usually start somewhere else.

Change data capture as the strangler's first tool

A monolith like Yardstick cannot be given an outbox in a week; its write paths are scattered across controllers, cron jobs and admin tools. **Table-level change data capture** reads the database's transaction log and turns every row change into an event without touching application code. In a 2025 Adevinta engineering post, Kleinanzeigen's search team described using Debezium on existing Kafka Connect clusters to stream MySQL changes, joining them into a denormalized per-user object published to a compacted topic, at up to 50,000 messages per second during full re-consumption; the authors found Debezium powerful but harder to troubleshoot because issues "often arise at the infrastructure level rather than in the application code" (Adevinta Tech Blog, 2025).

Raw CDC events are not domain events. The pattern that keeps them from leaking is a **translation layer** — an anti-corruption layer, in domain-driven design terms — that consumes the raw change stream, joins what it needs, infers the business transition and emits the contract event. The rule I use: **CDC is allowed as a source, never as a contract**. Only the translator reads the monolith's tables; everyone else reads the translator's output. When the write path moves out of the monolith, the new service publishes the same contract through its own outbox, and consumers do not notice the switch. CDC and outbox are therefore stages, not alternatives, and Changing the Engine in Flight: Modernizing a Live Marketplace uses this sequence for Yardstick's strangler plan.

Mechanism	What it guarantees	What it costs	When it is right
Publish after commit, in application code	Nothing: events are lost on crashes and on code paths that forget to publish	Cheap to start	Never for removals; tolerable for best-effort analytics
Polling the database for changed rows	Eventually sees state, but misses intermediate transitions and deletes unless soft-deleted	Load on the database; latency of the poll interval	Small consumers, early stage
Table-level CDC, consumed directly	Every committed row change, in commit order	Producer schema becomes a public contract; joins in every consumer	Short-lived migration tactic only
CDC plus translator	Every committed change, translated to the contract	One translator to own and operate; inference of intent from rows is imperfect	Legacy producers during a strangler migration
Transactional outbox	Every committed change, in the producer's own vocabulary	Write-path change in the producer	The target state for every listing producer

The event envelope

Every listing event at Veldmark, whatever its type, shares one envelope. The fields are chosen for what consumers need to be correct, not for what is easy to emit.

Field	Meaning	Why consumers need it
`event_id`	Unique ID, generated in the outbox row	De-duplication of redeliveries
`event_type`	`ListingPublished`, `PriceChanged`, …	Routing and filtering
`listing_id`	Group listing ID, e.g. `vm:tradeyard:nl:8471103`	Message key; per-listing order
`listing_version`	Aggregate version after this change, monotonically increasing	**Version-gated apply**; detecting gaps
`occurred_at` / `published_at`	When the change committed; when the event reached the topic	Freshness service level objectives (SLOs); ordering diagnostics
`tenant`	Brand, country, vertical	Multi-brand routing, data residency, per-tenant SLOs
`cause`	Seller action, feed import, moderation decision, expiry job, enforcement, erasure	Analytics and trust need to know *why*, not only *what*
`correlation_id`	Links to the feed run, insertion session or enforcement case	Tracing a change across systems
`schema_version`	Registered schema ID	Deserialization
`state`	The listing's current public state after the change	State transfer; self-healing consumers
`change`	Type-specific detail: old and new price, changed attribute keys, removal reason	Consumers that care about the delta, such as price-drop alerts

Two choices in the envelope deserve justification.

Both delta and state. A `PriceChanged` event carries the old and new price in `change` and the full public listing state in `state`. The alert matcher needs the delta to decide whether a price-drop alert fires; the search indexer needs the state and can ignore the delta. Carrying both makes the event larger — about 4 KB on average at Veldmark, without media, which reference image IDs only — but it means a consumer that missed `ListingUpdated` version 7 is repaired by `PriceChanged` version 8. **Self-healing beats perfect delivery**.

Public state only. The `state` block contains what a buyer can see plus the attributes search and trust need. It does not contain the seller's phone number, e-mail, exact address of a private seller, or internal performance data. The generic real-estate domain model this book uses draws the same line for its listing entity: "public fields can be published; draft, owner, and performance data remain internal and role-controlled". Keeping personal data out of the stream shrinks the erasure problem that System of Record and Read Models: Listing Data Management has to solve. Consumers that need contact data fetch it through an access-controlled API at the moment of use.

The event catalog

Contract v1 defines thirteen listing event types; pause and resume share a row below. The list is short on purpose; every new type is a cost for every consumer.

Event type	Emitted when	Main consumers
`ListingSubmitted`	Seller or feed submits a new listing or a re-moderated edit	Trust and safety scoring, category prediction
`ListingModerated`	Moderation approves, rejects or requests changes	Listing context itself, seller notifications, transparency reporting
`ListingPublished`	Listing becomes visible to buyers	Search, alerts, sitemap, recommendations, warehouse
`ListingUpdated`	Attributes, text or media change without price change	Search (partial update), trust (re-scoring on risky edits)
`PriceChanged`	Asking price changes	Search, price-drop alerts, valuation and price-rating models
`VisibilityChanged`	A paid visibility product starts or ends (bump, top listing)	Ranking, pro statistics; billing is owned by Packages, Bumps and Quotas: Products, Pricing and Entitlements
`ListingPaused` / `ListingResumed`	Seller pauses or resumes	Search, alerts
`ListingReserved`	Seller or Safe Deal reserves the item	Search (label), alerts (suppress)

Event type	Emitted when	Main consumers
`ListingSold`	Seller marks sold, or a transaction completes	Search removal, alerts, SEO page state, warehouse, recommendations
`ListingExpired`	Lifetime ends without renewal	Search removal, sitemap removal, seller reminder
`ListingRemoved`	Seller deletes, or the platform removes for policy or fraud	Search removal, alerts, SEO page status, transparency reporting
`ListingErased`	Personal data purged after retention or an erasure request	Every consumer holding a copy (see System of Record and Read Models: Listing Data Management)

Two exclusions are deliberate. **Bumps are not lifecycle transitions**: a bump changes a ranking timestamp, not the state of the listing. It is a `VisibilityChanged` event that leaves `created_at` untouched, as mobile.de's API separates `renewalDate` from `creationDate` (mobile.de Seller-API, n.d.). And **views, clicks and contacts are not listing events**: they are behavioral tracking events with their own taxonomy, volume and consent rules, owned by Data Platform for Marketplaces: Events, Lakehouse and Data Products. Mixing the two would put billions of tracking events on the stream that removals depend on.

Topics and retention

Veldmark runs two topics for the contract, both partitioned by listing ID.

- **`listing.events.v1`** — every domain event, retained for **14 days**. Consumers that fall behind by less than two weeks catch up from their offsets.
- **`listing.state.v1`** — a **compacted** topic holding the latest `state` per listing. Kafka's log compaction keeps at least the last record for each key and never reorders records; a record with a null value (a *tombstone*) marks the key as deleted, and tombstones are themselves removed after `delete.retention.ms`, 24 hours by default (Confluent documentation, n.d.). A new consumer reads the state topic from the beginning to bootstrap and then switches to the events topic.

Closed listings stay on the state topic in their final state (sold, expired, removed) for 30 days, so consumers learn about the closure, and are then tombstoned. Erased listings are tombstoned immediately. A consumer that is offline longer than the tombstone

retention can miss a deletion — which is why the state topic is not a substitute for periodic reconciliation, discussed under failure modes.

In Practice: Public Evidence and Contract v1

What classifieds companies have documented

Public evidence on listing event models is thin; no classifieds company in the research for this book published its listing event names or topic layout. What is documented is the plumbing around them.

- **Adevinta's data platform**. In a 2022 engineering post, Adevinta called Kafka "the backbone of the data platform", carrying nearly a billion user, behavioral, content and transaction events a day, with topics in both Avro (Confluent wire format) and JSON; the team built a deserializer that infers the format from the first byte rather than enforcing one format (Adevinta Tech Blog, 2022; older evidence, and the current platform may differ). The lesson for a group is that mixed formats accumulate unless a contract forbids them, and the tooling cost of tolerating them is real.
- **Kleinanzeigen's CDC pipeline** (2025, above) shows table-level CDC feeding a compacted topic as a re-consumable cache.
- **Vinted's search indexing**. In a 2023 post, Vinted described rebuilding its indexing on Apache Flink fed by CDC: Flink jobs build read models in Redis and emit change events from which downstream jobs build Elasticsearch documents. Reindexing works by publishing batch events to a Kafka topic that merges with organic changes, and item visibility latency fell from up to seven minutes to about one (Vinted Engineering, 2023).
- **Vinted's global platform**. In two 2026 posts, Vinted described replacing synchronous calls between almost 300 domains with business events — services "publish state changes and let other domains react whenever they were ready" — and using saga-style orchestration with compensations for multi-step workflows instead of distributed transactions (Vinted Engineering, part 1, 2026). Part 2 states that teams must handle events that "arrive late or in a different order than expected" (part 2, 2026).
- **Adevinta Spain's data contracts**. A 2024 post describes immutable JSON data contracts in GitHub — schema reference with X.Y versioning, Kafka landing topic, a list of personal-data fields, service levels — from which ingestion jobs are generated; failed events go to a quarantine table, and the time from contract promotion to usable data fell from about five days to about thirty minutes (Adevinta Tech Blog, 2024).
- **AutoScout24, 2018**. A 2018 post described mostly synchronous microservices, with "some" using Kafka for asynchronous work (AutoScout24 Tech Blog, 2018).

It is old evidence and the current state is not public, but it is a useful reminder that event-driven listing flows are a choice, not a default.

Veldmark: building Listing event contract v1

Sven Lindqvist's team designs the contract between March and June 2027, with one representative from each brand and from the four first consumers: search, alerts, trust and safety, and the warehouse. Four decisions shape it.

One contract, three publication paths. Each brand reaches the contract differently in 2027, and the design accepts that.

Brand	Listing system of record (2027)	Publication path to v1	Target
Tradeyard	Yardstick, PHP on MySQL	Debezium CDC on Yardstick's listing tables into a translator service, **Yardstick Relay**, which emits v1 events	Outbox in the new insertion service when it moves out of Yardstick (Q2 2028)
Motorvane	Kotlin listing service	Transactional outbox, relayed by Debezium	Already target
Hearthly	Go listing service on Google Cloud	Outbox to Google Pub/Sub, then a bridge to Confluent Cloud that preserves the listing key	Direct publication if Hearthly moves to AWS
Keybridge	Not a marketplace; a SaaS publishing agency listings to many portals	Publishes to Hearthly through the same feed interface as any software vendor; **does not consume** Hearthly listing events	Unchanged: the March 2027 neutrality decision applies

Keybridge gets no subscription to Hearthly's listing stream. Keybridge sends listings to Hearthly's competitors, so a subscription would undermine the neutrality commitment

Veldmark made to agencies in March 2027. The event backbone has access control per topic, and the architecture decision record says so explicitly.

Feed noise is suppressed at the source. About 88% of Motorvane's dealer listings arrive through dealer management system (DMS) feeds, many of them daily full snapshots. In 2027 that is about **1.16 million** listings re-sent every day, of which about 91% are unchanged; only about 100,000 records a day carry a real change. Motorvane's ingestion computes a content hash per listing and emits nothing when it matches the last accepted version; the ingestion chapter, Many Doors In: Forms, APIs and Bulk Feeds, owns the mechanism. Without the hash, the stream would carry about a million no-op `ListingUpdated` events a day and every consumer would pay for them.

Versions are mandatory. Every producer must emit a strictly increasing `listing_version`. For Yardstick Relay this means reading the `updated_seq` column the Tradeyard team added to the listing table in April 2027, because MySQL timestamps had one-second resolution and edits within a second were common in bulk tools.

Avro, full transitive compatibility. The contract uses Avro schemas in Confluent Schema Registry with the **FULL_TRANSITIVE** compatibility mode on `listing.events.v1`, explained in the schema section below.

At go-live on **1 July 2027**, the contract has 23 registered consumer groups. By the end of 2027 it carries about **4.2 million listing events a day**, with a peak of about **1,100 events per second** on Sunday evenings, and a publication-latency objective of **p95 under two seconds** from commit to topic. Tradeyard produces about 62% of the events, because private listings change state often and live briefly.

The bike, replayed on the contract

The Rotterdam bike is a March 2027 story, before the contract existed. Sven's team uses it as the acceptance test for v1: replay Femke's listing through the new design and check that every consumer would have behaved correctly.

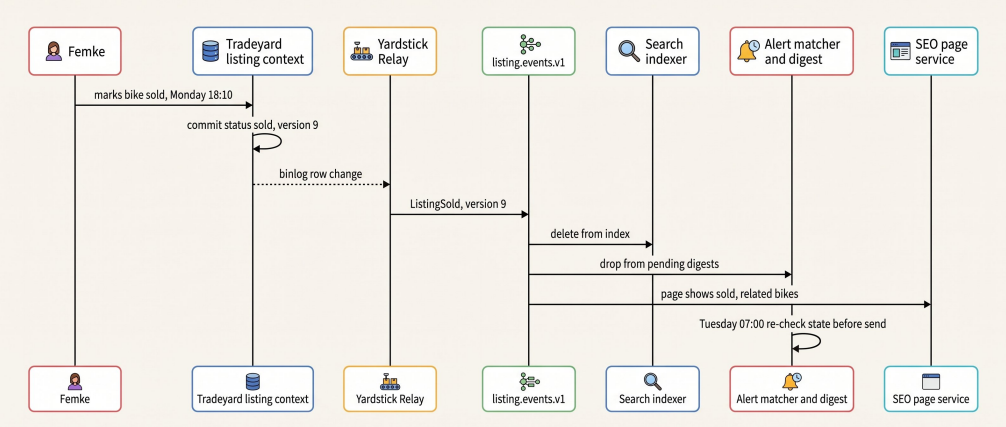


Figure 3: *The bike's sale as a single event with three consumers. The digest is fixed twice over: it removes the bike when* `ListingSold` *arrives, and it re-checks listing state at send time, because a derived artifact that lives for thirteen hours must not trust a snapshot.*

Time (CET)	Event	Version	What consumers do
Sun 7 Mar 19:42	`ListingSubmitted`	1	Trust scoring runs the fast path; category confirmed as city bikes
Sun 19:47	`ListingModerated` (approved)	2	Listing context transitions to active
Sun 19:48	`ListingPublished`	3	Search indexes; alert matcher finds **312** matching saved searches; sitemap adds the URL
Sun 20:26	—	—	The phishing account is suspended; that is an event on the *account*, not on Femke's listing
Mon 8 Mar 14:02	`ListingReserved` (Safe Deal)	8	Search shows "reserved"; alerts suppress new notifications

Time (CET)	Event	Version	What consumers do
Mon 18:10	`ListingSold`, price €225, channel Safe Deal	9	Search removes; digest drops the bike; SEO page switches to a sold state
Mon +30 days	Tombstone on `listing.state.v1`	—	Consumers that track closures purge their state

Versions 4 to 7 are Femke's edits and a photo reorder on Sunday night. The table also shows what the event stream does **not** do: it does not detect the phishing message, which is a messaging event handled in From Interest to Contact: Messaging and Lead Management and Scams, Fakes and Account Takeovers: Fraud Prevention for Listings, and it does not make the sold page an SEO decision, which Found by Google: Web, App and SEO Architecture for Listings owns. It makes sure every one of those systems hears the same facts in the same order.

The Essen Golf gives the contract its second test, on Motorvane. When Autohaus Ruhrblick cuts the price from €21,900 to €21,400 on Thursday 22 April 2027, one `PriceChanged` event reaches the alert matcher, which sends price-drop alerts to **1,180** watchers, and the market-value model, which re-scores the listing. When the car sells on 29 April, though, the dealer's feed reports the removal **26 hours** later. No event architecture can publish a change the system of record has not received. **The stream makes downstream copies as fresh as the source, not fresher**; upstream lateness is a feed-contract problem, handled by reconciliation and staleness controls in Many Doors In: Forms, APIs and Bulk Feeds and From Submitted to Searchable: The Publication Pipeline.

Schema Evolution

A listing event schema lives for years and has consumers its producers have never met. Changing it safely is the hardest operational discipline in this chapter.

Compatibility modes

Confluent Schema Registry enforces compatibility between schema versions. BACKWARD — the default — means consumers using the new schema can read data written with the previous one, and it is checked only against the latest version; the transitive variants check against all previous versions; FORWARD means old consumers can read new data; FULL means both (Confluent documentation, n.d.). For Protobuf, the

documentation recommends BACKWARD_TRANSITIVE because adding new message types breaks forward compatibility.

A multi-brand listing stream needs **full transitive** compatibility, because consumers and producers upgrade on independent schedules and consumers replay old events. Under it, any consumer, on any schema version, can read any event on the topic. In practice, full compatibility in Avro allows adding and removing **optional fields with defaults** and little else. That constraint is the point.

The rules Veldmark publishes

Change	Allowed in v1?	How
Add an optional field with a default	Yes	Register the new schema; consumers ignore it until they upgrade
Add a new event type	Yes, with notice	Announced four weeks ahead in the event catalog; consumers must ignore unknown types
Add a value to an enumeration (e.g. a removal reason)	Only if consumers declare a default branch	Enumerations in v1 are strings validated against a published list, not Avro enums, to avoid this trap
Rename a field	No	Add the new field, populate both, deprecate the old one, remove it in v2
Change a field's meaning or unit	**Never** in place	A new field with a new name; meaning changes are the breaks that registries cannot detect
Remove a required field, change a type	No	New major version

The fifth row, a change of meaning, is the dangerous one, because a registry checks syntax, not semantics. If Hearthly starts sending `surface_m2` as habitable surface instead of total surface, every compatibility check passes and every valuation is wrong. **Semantic changes need human review**, and Veldmark routes every schema change on

the contract through the architecture advice forum's event-catalog review, which takes one week.

Breaking changes: v2 by parallel run

A breaking change means a new topic, `listing.events.v2`, and a period in which the producer publishes both. Consumers migrate one by one; the v1 topic is retired only when the consumer registry shows no active v1 consumer groups. The cost — double publication, two sets of tests, months of coordination — is deliberately high, so that v2 happens rarely and for good reasons. Veldmark's expectation, recorded in the decision record, is that v1 will last at least until Yardstick is decommissioned at the end of 2029.

Client analytics events are a harder schema problem than service events, because they ship in app binaries that cannot be recalled; Vinted documented generating tracking code from schemas and repairing malformed web events on the server (Vinted Engineering, 2019). That is one more reason to keep behavioral tracking off the listing stream and in Data Platform for Marketplaces: Events, Lakehouse and Data Products.

Replay, Backfills and Dead Letters

Replay is a feature you design

A listing stream earns most of its value on bad days. A search mapping changes and the index must be rebuilt. A new recommendation service needs all live listings. An enrichment bug wrote wrong attributes to 300,000 listings and they must be corrected everywhere. Each of these is a **replay**: re-delivering state to consumers that already processed it.

Three replay sources exist, and they serve different purposes.

Source	Use	Limit
Events topic from an older offset	A consumer that fell behind or deployed a bug and must reprocess the last days	14-day retention at Veldmark
Compacted state topic from the beginning	Bootstrapping a new consumer; rebuilding a read model	Latest state only; no history of intermediate changes

Source	Use	Limit
Backfill from the system of record	Corrections after a data fix; re-publishing after a schema addition	Load on the producer; must not starve live traffic

Backfills are where designs fail, because a backfilled event can overwrite a newer live one. If a correction of 300,000 listings is published into the same partitions as live changes, the alert matcher may process a backfilled `ListingUpdated` after a live `ListingSold` and resurrect a sold listing. Version-gated apply prevents that: the backfilled event carries the listing's *current* version, which is not newer than the `ListingSold` the consumer already applied, so it is ignored.

Vinted documented merging batch reindex events with organic changes on one topic (Vinted Engineering, 2023); the post does not say how stale batch events are kept from overwriting newer state, and my inference is that any design of this kind needs a version or timestamp comparison in the consumer to be safe.

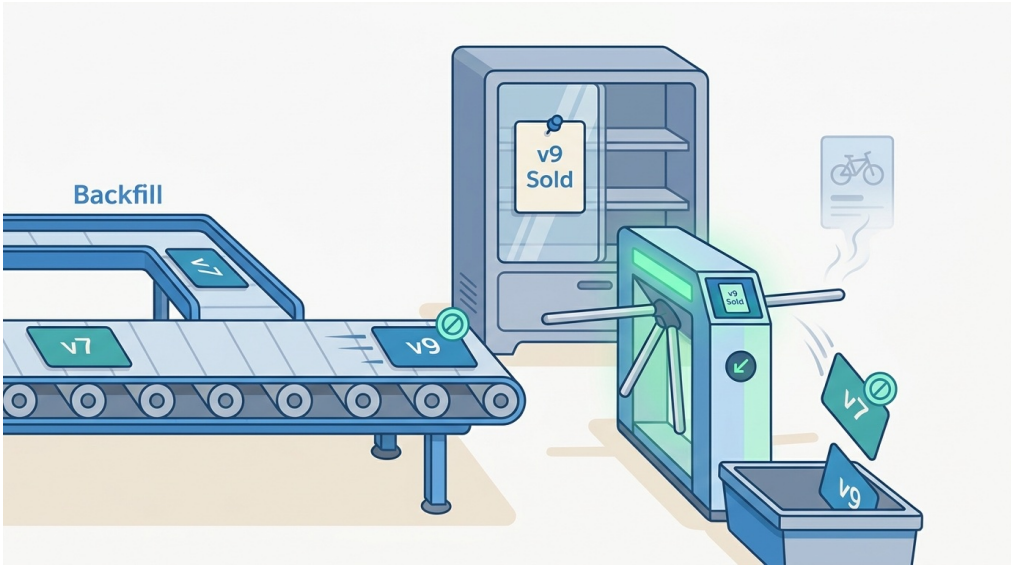


Figure 4: *Version-gated apply: a consumer applies an event only if its listing version is newer than the one it stores, so redeliveries, replays and backfills cannot bring a sold listing back to life.*

Veldmark publishes backfills to a separate topic, `listing.backfill.v1`, with the same schema and a `cause` of `backfill`. Consumers read it at lower priority and apply the same version gate. The separation lets the alert matcher, which must not send alerts for old listings, ignore backfills entirely, and lets the search team throttle a full

rebuild without slowing live publication.

Dead letters without losing order

A **dead-letter queue** (DLQ) holds events a consumer could not process — malformed payloads, failed lookups, bugs. For listing events, a DLQ carries a trap: if version 5 of a listing goes to the DLQ and version 6 is processed, the consumer's state is now ahead of the event it failed on. When someone later reprocesses version 5 from the DLQ without a version check, it overwrites the newer state.

Two rules handle this. First, **because events carry full state, skipping one is usually safe**: version 6 already contains everything version 5 did. The DLQ becomes a diagnostic tool, not a queue to drain. Second, for consumers that need every delta — the price-drop alert matcher is one — a failed event **parks the listing**: further events for that listing ID wait until the failure is resolved, while other listings continue. Parking is per key, so one broken listing does not stop the partition.

Removals are a separate service level

Removal events need a tighter service level than publication events, because a late removal costs more than a late publication. A new listing that becomes searchable in two minutes instead of ninety seconds is a small loss; a scam listing removed at 20:26 that stays searchable until 21:00 is a real one. Veldmark sets separate objectives per event family: publication events p95 under two seconds to topic; **removal events (`ListingSold`, `ListingRemoved`, `ListingErased`) p99 under five seconds to topic and under thirty seconds to search deletion**, alarmed per consumer. The search index's removal latency is measured end to end by a synthetic listing that is created and removed every minute in each tenant.

Trade-offs: Choosing the Shape of the Stream

Decision	Option A	Option B	When A is right	When B is right
Event content	Thin notification, consumers call back	**State plus delta**	Payload contains sensitive data; few consumers; low change rate	Many consumers, peak fan-out, need for self-healing (the listing case)
Granularity	One generic `ListingChanged` with a changed-fields list	**Typed domain events**	Very early, single consumer; mostly state replication	Consumers react to specific transitions (sold, price drop); transparency and analytics need intent
Publication	**Outbox in the producer**	CDC plus translator	Producer can change its write path	Legacy producer; strangler migration; temporary by design
Topic layout	One topic per event type	**One topic per aggregate, keyed by ID**	Event types with very different volumes and consumers	Per-listing ordering across types is required (it nearly always is)
Backfills	Same topic as live events	**Separate backfill topic**	All consumers version-gate and none must ignore old data	Some consumers must not react to old changes; rebuilds need throttling

Decision	Option A	Option B	When A is right	When B is right
Schema format	JSON Schema	**Avro or Protobuf with registry**	Partner-facing payloads, human readability, external webhooks	Internal, high-volume, long-lived contract with enforced compatibility
Broker	One group broker	One per brand, bridged	Group shares platforms (Common Ground's premise)	Brands are run as separate businesses, likely to be sold, or on different clouds for a long time

Two of these deserve a note. **Typed events versus one generic event** turns on whether consumers need to know *why* the listing changed. Search does not; alerts, trust and safety, regulatory reporting and analytics do. A generic event with a `cause` field and a changed-keys list is a reasonable first step; typed events are where the contract ends up.

One broker versus bridged brokers is a portfolio decision as much as a technical one. Veldmark chose one backbone because Common Ground's principle is to share platforms beneath the brands. Hearthly's bridge from Pub/Sub is the documented exception, with a review date when Hearthly's cloud choice is revisited after 2028. A group that expects to sell a brand should think twice before entangling its events with the rest, which After the Deal: Consolidating Platforms After Acquisitions discusses.

Vertical Differences

Aspect	General classifieds	Real estate	Automotive	Jobs
Dominant source of events	Private sellers' edits and short lifetimes; many sold events	Feeds from software vendors; long lifetimes; frequent price changes	DMS full-snapshot feeds; many no-op updates without hashing	Employer and applicant-tracking feeds; hard expiry dates
Volume per live listing	High: listings live days to weeks	Moderate	Low per listing after hash suppression, spiky after feed runs	Low
Event that matters most	`ListingSold` and `ListingRemoved` (ghosts and scams)	`PriceChanged` and the cluster event from entity resolution	`PriceChanged` and the feed-driven removal	`ListingExpired`; search engines penalize stale job postings
Extra event types often needed	Reservation and transaction events (Safe Deal)	`PropertyClustered` from One Property, Three Listings: Duplicates and Entity Resolution; mandate or exclusivity changes	Vehicle identity events (VIN decoded, history checked)	Application events, owned by a separate context
Main consumers beyond search	Alerts, trust and safety	Valuation model, agency statistics, duplicate cluster views	Market-value model, dealer statistics, financing partners	Search engine job indexing, applicant notifications

For a jobs board, the latency from `ListingExpired` to the page change is an SEO risk, not a nicety. Google's structured-data guidance for job postings says expired jobs must be removed by setting `validThrough` in the past, returning 404 or 410, or removing the markup, and that "failure to take timely action on expired jobs may result in a manual

action" (Google Search Central, updated 2026).

In real estate, the entity-resolution cluster is itself a stream: when three agencies list the Lyon apartment, consumers need both the three listing events and the event that says they describe one property.

Failure Modes

- **The forgotten publisher**. Events are emitted from application code after commit, and one code path — an admin tool, a bulk-expiry cron job, a moderator's console — does not emit them. Removals from that path never reach search. The outbox or CDC exists to make this impossible; a nightly reconciliation that compares the system of record with each read model catches what slips through.
- **Consumers on the producer's tables**. A team that cannot wait for the contract reads the monolith's replica or its raw CDC stream. It works, until the producer changes a column. Every such consumer is a hidden veto on the producer's schema; the consumer registry should list them, and the architecture forum should set a date for each to move.
- **Ordering broken by re-keying**. A consumer republishes events keyed by seller ID or category, or a bridge loses the key. Per-listing order is gone, and a `ListingSold` can overtake the `PriceChanged` before it. Hearthly's Pub/Sub bridge has a contract test for exactly this.
- **The unversioned upsert**. A consumer writes `state` into its store without comparing versions. Redeliveries and backfills now resurrect old states, including sold listings — Ghost Inventory produced by the event pipeline itself.
- **Semantic drift**. A field keeps its name and changes its meaning. Registries do not catch it; reviews and contract tests with real examples do.
- **Snapshot consumers**. A consumer reads the stream correctly but builds a long-lived artifact — a digest, a partner export, a newsletter — from a snapshot and sends it hours later. The bike digest was this. **Anything that is delivered later than it is computed must re-check availability at delivery time**.
- **Tombstones missed**. A consumer is down for longer than the tombstone retention and never learns that a listing was erased. Weekly reconciliation against the state topic's key set and the system of record closes the gap.
- **Personal data on the stream**. Seller phone numbers or exact addresses enter `state` "because one consumer needs them". From then on, every consumer, every retained offset and every backup holds personal data subject to erasure. Keep it out; fetch at use.
- **The stream as a dumping ground**. Views, impressions and experiment exposures are added to the listing topic because the infrastructure is there. Volumes rise by two orders of magnitude and removal latency degrades at the peak. Separate streams for separate purposes.

The Chief Architect's View

At group scale the listing event contract is one of a handful of artifacts that outlive systems. Yardstick will be gone by the end of 2029; Solr will be replaced by OpenSearch; the alert matcher will be rewritten. The contract, if designed well, survives all of it. That makes the decisions around it governance decisions, not implementation details.

What to decide centrally. The envelope, the identity scheme for listings across brands, the event catalog, the compatibility rules, the service levels for removal events, the access policy per topic, and the rule that only the owning context emits listing events. These are cheap to decide once and very expensive to decide four times.

What to leave local. Each brand's internal events, its choice of outbox or CDC during migration, its consumers' read models, and brand-specific event types that do not belong in the group contract (Motorvane's financing-partner events, Hearthly's mandate events). The group contract should be the narrow waist, not the whole hourglass.

Who decides. At Veldmark the event platform team, led by Sven Lindqvist, owns the backbone and the catalog; the listing context in each brand owns its events' content; the architecture advice forum reviews every contract change; and Lucía Ortega, the group data protection officer, signs off on any field that could carry personal data. Consumers register before they get access, which gives the group the list of readers that the bike incident showed nobody had.

What to watch. Publication latency per producer; removal latency end to end, per tenant; consumer lag at the Sunday-evening peak; the count of consumers still reading producer tables directly; dead-letter rates per consumer; the share of `ListingUpdated` events that change nothing (a sign that hash suppression has broken); and schema changes per quarter, which should be small and boring.

The long-term consideration. The event stream makes copies cheap, and cheap copies multiply. Every consumer holds a read model of listings, and each read model is a place where stale data can be served and personal data can linger. The stream's success is the reason System of Record and Read Models: Listing Data Management exists as a separate discipline.

Questions to Ask

1. Which system emits each listing lifecycle transition, and is there any code path — admin tools, cron jobs, bulk operations — that changes listing state without emitting an event?
2. How does a committed change become an event: outbox, CDC with a translator, or publish-after-commit? What happens if the process crashes between the two?
3. Is every event keyed by listing ID, and does every bridge, mirror and republisher preserve that key?

4. Does every consumer apply events by listing version, and what happens when it receives the same event twice, or an older one after a newer one?
5. Which consumers read the producer's database or raw change stream directly, and when will each move to the contract?
6. What compatibility mode does the registry enforce, and who reviews changes in a field's meaning that the registry cannot detect?
7. How would we rebuild the search index from scratch today, from what source, and how long would it take at peak traffic?
8. What is the end-to-end latency from "seller marks sold" to "listing gone from search, alerts and partner feeds", and is it measured continuously?
9. Which personal data fields are on the stream, and how would an erasure reach every retained copy of them?
10. Which artifacts are computed from listing data and delivered hours later, and do they re-check availability at delivery?

From Events to Copies

The event stream moves changes; it does not decide where the resulting copies live, how stale each may be, or how a seller's erasure request reaches all seven of them. System of Record and Read Models: Listing Data Management takes up those questions with the same bike, now held in seven places at once.

Sources and Evidence

Company engineering and investor sources

- Make data migration easy with Debezium and Apache Kafka — Adevinta Tech Blog (Daniil Roman, Kleinanzeigen), 2025. Table-level CDC into a compacted topic; throughput and troubleshooting lessons.
- Inferring Kafka topic format during JSON deserialization — Adevinta Tech Blog (Rodrigo Molina), 2022. Kafka as the data-platform backbone; mixed Avro and JSON topics (older evidence).
- Creating source-aligned data products in Adevinta Spain — Adevinta Tech Blog (Sergio Couto), 2024. Immutable data contracts, quarantine table, five days to thirty minutes.
- Vinted Search Scaling Chapter 7: Rebuilding search indexing pipeline — Vinted Engineering, 2023. CDC and Flink read models; reindexing merged with organic changes; seven minutes to about one.

- Building a Global, Event-Driven Platform, Part 1 and Part 2 — Vinted Engineering, 2026. Business events replacing synchronous calls; sagas; late and out-of-order events.
- Scalable and non-blocking event ingestion pipeline — Vinted Engineering, 2019. Client analytics events, schema-generated tracking code, server-side repair (older evidence).
- What I learned in my first year at AutoScout24 — AutoScout24 Tech Blog (Martin Lechner), 2018. Mostly synchronous services with some Kafka (older evidence).
- Seller-API documentation — mobile.de, n.d. Separation of `renewalDate` from `creationDate`.

Standards, regulation and documentation

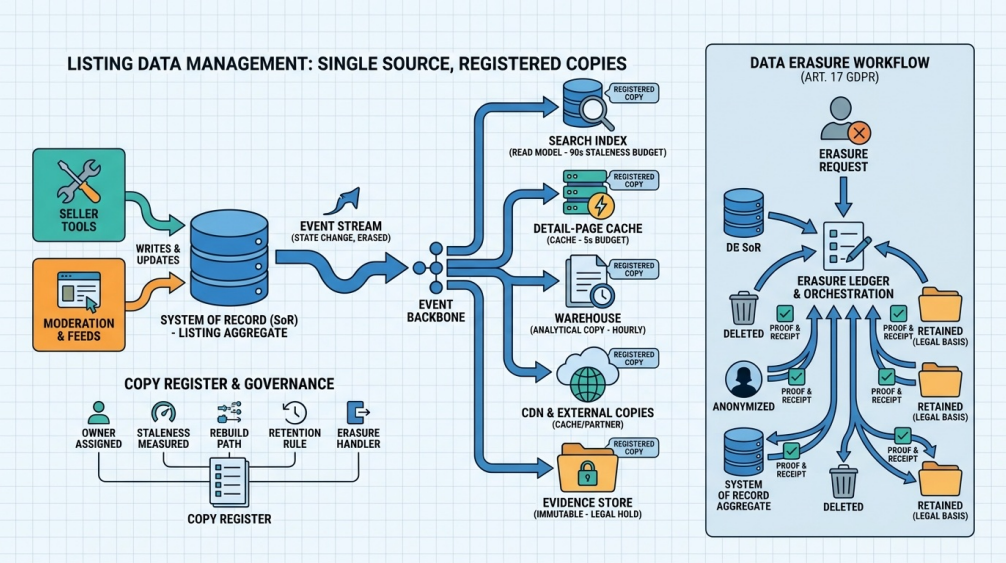
- Outbox Event Router — Debezium documentation, n.d. Outbox table columns, aggregate ID as key, per-aggregate order.
- Schema Evolution and Compatibility — Confluent documentation, n.d. Compatibility modes and recommendations.
- Log compaction — Confluent documentation, n.d. Compaction guarantees, tombstones, `delete.retention.ms` default.
- Job posting structured data — Google Search Central, updated 2026. Expired-job removal and manual actions.

Research and practitioner sources

- What do you mean by "Event-Driven"? — Martin Fowler, 2017. Event notification, event-carried state transfer, event sourcing, CQRS.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

34. System of Record and Read Models: Listing Data Management



IN THIS CHAPTER, YOU WILL: Learn to decide where listing data lives, which store is authoritative, how read models are built and kept within a staleness budget, how to partition listing data by country and region, how to retain and archive closed listings, and how corrections and GDPR erasure reach every copy. The chapter follows Veldmark's seven copies of one sold bike.

WHY THE BUSINESS CARES: Buyers judge a marketplace by whether what they see is true: the price, the availability, the seller. Regulators judge it by whether it can prove what it did with personal data. Both depend on knowing where every copy of a listing is and how current it is.

WHY ARCHITECTS SHOULD CARE: An event-driven marketplace makes copies cheap, and cheap copies multiply. Without a registry of copies, a staleness budget per read path and an erasure path per store, every new feature adds a place where stale or unlawful data can be served.

KEY POINTS:

- **One system of record per listing, and everything else is a registered copy.** The system of record is where a change becomes true; search indices, caches, warehouses, evidence stores and partner exports are read models with an owner, a source, a staleness budget and a rebuild path. A copy that cannot be rebuilt or cleaned is a liability, not an asset.
- **Consistency is a per-read-path promise, not a database setting.** Sellers need to read their own writes, buyers need bounded staleness and no flapping, and removals need tighter limits than additions. Write those promises down as budgets and measure them, because users experience staleness, not replication lag.
- **Retention and erasure are designed per data class, not per database.** A closed listing's content, its seller's personal data, its photos, its trust-and-safety evidence and its payment records have different lawful lifetimes. An erasure request becomes a workflow with a receipt from every copy owner: deleted, anonymized, or retained with a stated legal basis.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Search as System of Record** — the search index holds fields and corrections that exist nowhere else, so it can be neither rebuilt nor trusted as a read model.
- **Ghost Inventory** — read models without staleness budgets and caches without removal paths keep serving sold and removed listings.
- **Deal-Shaped Architecture** — acquired brands bring their own databases, warehouses and retention rules, so the group cannot say where a listing or a person's data lives.

The previous chapter, Every Change Is an Event: Event-Driven Listing Architecture, made it easy for any system to keep its own copy of a listing. That was the point: search, alerts, trust and safety, SEO, analytics and partners each need listing data in a different shape, and a published event stream lets them build it without calling the listing service at the Sunday-evening peak. **The price of that independence is copies**, and copies raise three questions that events alone do not answer: which copy is right, how wrong may the others be, and how does a deletion reach all of them?

Listing data management answers those questions, and the answers decide whether buyers see a price that changed ten minutes ago, whether a sold car appears in tomorrow's alert, whether the group can rebuild its search index after a bad mapping change, and whether a private seller's erasure request can be completed, and proven, within the month the law allows.

This chapter owns the **listing system of record**, **read models** and **erasure propagation**. It assumes the lifecycle from The Listing Aggregate: Identity, Lifecycle and Invariants and the event contract from Every Change Is an Event: Event-Driven Listing Architecture. The search index as a read model is designed in Search Architecture for Listings: Indices, Freshness and Scale; the analytical copies in Data Platform for Marketplaces: Events, Lakehouse and Data Products; the legal obligations behind retention in Rules of the Marketplace: DSA, DAC7 and Vertical Regulation; regional failover in Sunday Evening Peaks: Reliability, Performance and Cost. Later chapters build on its rules: Found by Google: Web, App and SEO Architecture for Listings keeps a sold page's URL without the private seller's data, and Changing the Engine in Flight: Modernizing a Live Marketplace carries the copy register and the erasure ledger through Yardstick's retirement.

Setting the Scene at Veldmark: Seven Copies of One Bike

On **Monday 6 September 2027**, Femke, the Rotterdam seller whose €240 city bike sold for €225 in March, closes her Tradeyard account and asks Tradeyard to erase her personal data under Article 17 of the General Data Protection Regulation (GDPR). Lucía Ortega, Veldmark's group data protection officer, forwards the request to Ingrid Halvorsen, Group Chief Architect at Veldmark — the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build). Her note has one line: *"Before we answer her, I need to know every place her bike listing exists."*

Ingrid gives the question to Daan Visser, who owns Yardstick, Tradeyard's PHP monolith, and to Sven Lindqvist from the event platform. Their answer, two days later, lists **seven** places that still hold the listing six months after the sale:

#	Copy	Store	What it holds about the bike	Owner
1	Listing system of record	Yardstick MySQL, primary and two replicas	Listing row in state *sold*, 9 versions of history, seller ID, approximate location, title, description	Tradeyard listing team

#	Copy	Store	What it holds about the bike	Owner
2	Sold-listing page	Rendered HTML in the page cache and CloudFront	Public “sold” page with photos, title, price, Femke’s first name and her neighborhood; kept because sold pages for popular categories rank in search engines	Tradeyard web team
3	Media	Tradeyard image service and CloudFront	Six original photos and their derivatives; the derivatives were stripped of metadata, but the stored originals still carry EXIF data, including the GPS position of her home	Tradeyard media team
4	Warehouse	Snowflake	Listing history, seller ID, postcode, contact and conversation counts	Tradeyard data team
5	Trust-and-safety case	T&S case store	The 7 March phishing case: the scam message, Femke’s report, a snapshot of her listing	Group T&S (Aisha Rahman)

#	Copy	Store	What it holds about the bike	Owner
6	Safe Deal records	Tradeyard order database and Adyen	The €225 sale, the €11.95 buyer fee, payout to Femke's bank account	Tradeyard payments team
7	Alert digest archive	Digest store and the e-mail provider's logs	The 9 March digest that featured the bike after it sold	Tradeyard notifications team

Four observations make this a chapter rather than a ticket. Nobody had this list before Femke asked; it was assembled from memory and code search. Two copies (2 and 3) are **public** and still show personal data six months after the sale. Two copies (5 and 6) **must not** simply be deleted, because the law requires or allows the platform to keep them. And copy 7 is held partly by a third party, the e-mail delivery provider. The chapter's question follows directly: **where does listing data live, who owns it, and how do the copies stay consistent — including when the right state of a copy is "gone"?**

Figure 1: *Six months after the sale, Femke's bike still existed in seven places: two public copies still showed personal data, two had to be kept by law, and one sat with a third party, and nobody had the list until she asked.*

Concepts: System of Record, Read Models and Copies

The system of record

The **listing system of record** (SoR) is the one store in which a change to a listing becomes true. A price is the new price when the SoR commits it, not when search shows it. The SoR enforces the listing's invariants (see The Listing Aggregate: Identity, Lifecycle and Invariants), assigns its version, and emits its events. Three properties define it:

- **It accepts writes; nothing else does.** Feeds, sellers, moderators and expiry jobs all write through the listing context. A search index that receives corrections directly has become a second system of record.
- **It is complete for the listing's own data.** Every field a read model shows must originate in the SoR or in another named system of record (seller accounts, entitlements, enrichment outputs with provenance). If a field exists only in a read model, that read model is no longer rebuildable.
- **It is not necessarily one database.** "One system of record per listing" means one authoritative owner per fact. During a migration, Tradeyard listings live in Yardstick and Motorvane listings in Motorvane's listing service; what must never happen is that one listing is authoritative in two places at once.

The SoR is authoritative for the **listing**, but a listing references facts with their own systems of record — the seller account (Private Sellers, Dealers and Agencies: Modeling Who Lists), the paid products (Packages, Bumps and Quotas: Products, Pricing and Entitlements), the canonical property behind a real-estate listing (One Property, Three Listings: Duplicates and Entity Resolution), the moderation decisions (Trust and Safety as a Platform). Listing data management covers the listing's own copies and the references it carries, not those neighbors' data.

Read models and the kinds of copy

A **read model** is a copy of listing data shaped for one way of reading it, derived from the SoR — usually from the event stream — and rebuildable from it. Separating the write model from read models is the pattern known as command query responsibility segregation (CQRS). In listing marketplaces CQRS is not an architectural fashion; it is the only way one record can serve full-text search, map tiles, alert matching, detail pages at peak and a warehouse. Martin Fowler's 2017 description of event-carried state transfer, the mechanism that feeds most of these read models, names the price plainly: "lots of data schlepped around and lots of copies" (martinfowler.com, 2017).

Not every copy is a read model, and the differences decide how each is governed.

Kind of copy	Definition	Rebuildable from the SoR?	Example at Tradeyard	Governing rule
Replica	Byte-for-byte copy of the SoR for availability or read scaling	Yes, by the database	MySQL read replicas	Same retention as the SoR, automatically
Read model	Projection shaped for a query	Yes, from events or a backfill	Solr index, detail-page cache, pro statistics	Owner, staleness budget, rebuild time
Cache	Short-lived copy that expires	Yes, by expiry	CloudFront pages, Redis	Expiry plus explicit purge on removal
Archive	Closed listings kept for a purpose after leaving the SoR's active set	Not after the SoR purges	Sold listings kept for relisting and disputes	Purpose, retention period, reduced fields
Evidence store	Immutable record kept on purpose so that it cannot be rewritten	No, by design	T&S cases, statements of reasons	Legal basis, retention per data class, access control
Analytical copy	History for analysis and model training	Partly: history before the stream is not	Snowflake	Personal-data split, retention, erasure job
External copy	Held by a third party	No	E-mail provider logs, partner feeds, search engines' caches	Contract, notification duty, removal requests

The rule I use: **every copy must be either rebuildable or governed**. A rebuildable copy can be dropped and recreated, so its correctness problem reduces to freshness. A copy that cannot be rebuilt — an archive, an evidence store, an external copy — needs its own owner, purpose, retention and deletion path, written down before it exists.

Consistency, as users experience it

Users do not see database consistency models; they see **staleness**, the gap between the SoR's current state and what a screen shows. Four promises cover nearly every listing read path.

- **Read-your-writes**. A seller who changes the price sees the new price on their own listing page and in "my listings" immediately. Violating it produces support tickets and duplicate edits.
- **Monotonic reads**. A buyer who has seen the price drop does not see the old price again on the next refresh because a different replica answered. Violating it produces distrust.
- **Bounded staleness**. Search results, recommendations and alerts may lag the SoR, but by a known, measured maximum.
- **Removal priority**. A removal — sold, removed for fraud, erased — has a tighter bound than an addition, because showing what should be gone costs more than missing what is new.

A **staleness budget** is the written maximum lag for one read path, with the measurement that proves it. It is the requirement that connects the event architecture to the user experience.

Architecture: One System of Record, Registered Read Models

Polyglot persistence for one listing

Listing marketplaces are polyglot by necessity. The generic domain model this book draws on assigns the listing a relational system of record (PostgreSQL) with a search-index projection (OpenSearch) and a `listing-changed` event — one owner, many readers, search via projection — which matches the shape most marketplaces converge on. Around that core sit the stores in the table below.

Store type	What it holds for listings	Role	Typical staleness budget at Veldmark	How a removal or erasure reaches it
Relational database	The listing aggregate, version history, references	**System of record**	—	State change; purge job after retention
Event log (Kafka)	Listing events and latest-state topic	Transport and replay source	Seconds	Retention expiry; tombstone on the state topic
Search index	Denormalized listing documents	Read model	Additions 90 s p95 (2028 target); removals 30 s p99	Delete on `ListingSold`, `ListingRemoved`, `ListingErased`
Key-value cache	Detail-page data, counters	Cache	5 s for price and status	Invalidate on any event for the listing
CDN and page cache	Rendered pages, image derivatives	Cache	60 s for page HTML	Purge by surrogate key (listing ID)
Object storage	Original photos, documents, floor plans	Owned by the media context	—	Lifecycle rule plus explicit deletion
Alert and saved-search store	Match candidates, pending digests	Read model	Re-checked at send time	Drop on removal events; send-time availability check
Warehouse or lakehouse	History for analytics and models	Analytical copy	Hourly	Personal-data split; erasure job; anonymization

Store type	What it holds for listings	Role	Typical staleness budget at Veldmark	How a removal or erasure reaches it
Evidence store	T&S cases, statements of reasons	Evidence	—	Retention schedule; never edited in place
Third-party systems	Partner feeds, e-mail logs, search engines	External	Contractual	Delete instruction, Art. 19 notification, removal request

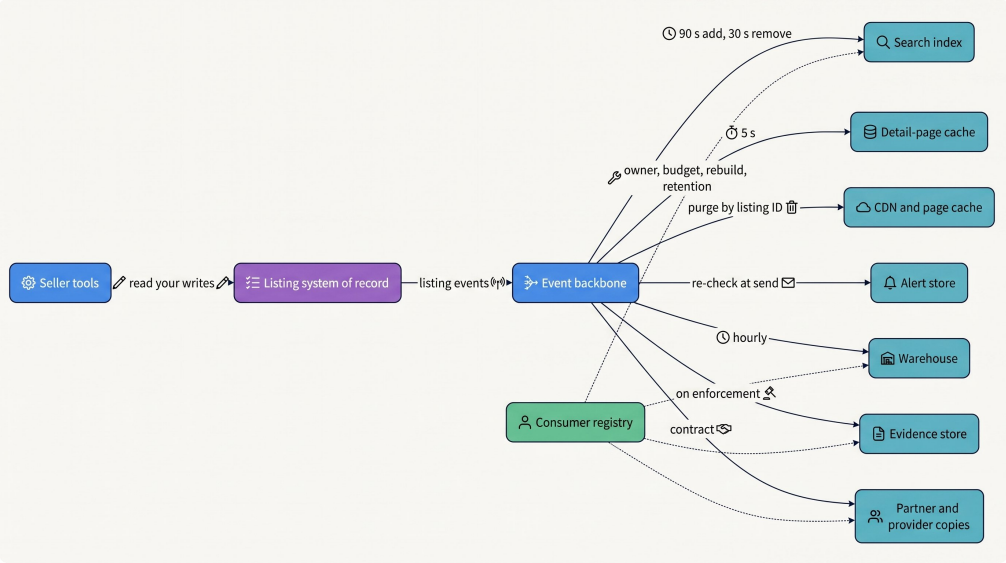


Figure 2: *One listing, many copies, each with a promise. Read models are fed from the event stream with a staleness budget; seller tools read from the system of record to guarantee read-your-writes; the consumer registry gives every store an owner, a rebuild path and a retention rule.*

Implementing the consistency guarantees

Each consistency promise has a standard implementation.

Read-your-writes for sellers is cheapest when seller-facing pages read from the SoR or a synchronous replica. Where they must use a read model — a dealer's inventory screen

backed by search because it filters 3,000 cars — the write returns the new `listing_version`, the client sends it with the next read, and the read model answers only if it has applied that version; otherwise the request falls back to the SoR for the affected listings. The fallback is rare and keeps the promise exact.

Monotonic reads for buyers come from session affinity to one replica set or from version-aware caches that never replace a newer version with an older one. The second is better: it also protects against backfills that deliver old state late, the failure Every Change Is an Event: Event-Driven Listing Architecture describes.

Bounded staleness is measured, not assumed. Veldmark's search team runs a synthetic listing per tenant that is created, repriced and removed every minute, and alarms when the observed lag exceeds the budget. The same probe measures the CDN purge.

Removal priority is implemented by putting removal events on the fast path everywhere: higher consumer priority, separate alarms, and caches that treat removals as purges rather than updates. A sold page that should stay public for SEO (copy 2 in Femke's case) is a *different page* — a sold-state page whose content is chosen by policy — not the old listing page left in cache.

The index is a read model

The search index is the copy most likely to stop being a read model. Teams add fields to the index that are computed nowhere else (a quality score, a manual boost, a moderator's correction), run one-off update scripts against it, and eventually discover that they cannot rebuild it without losing those changes. That is **Search as System of Record**. The test: could you delete the index today and rebuild it from the SoR and the event stream, with no loss? If not, find the fields that live only in the index and give each a system of record. Vinted's 2024 migration from Elasticsearch to Vespa illustrates the payoff of the opposite discipline: a "search contract" in a Go gateway let the engine be replaced without changing product applications (Vinted Engineering, 2024). Search Architecture for Listings: Indices, Freshness and Scale owns the index design; this chapter owns the rule that it is rebuildable.

In Practice: What Marketplaces Have Documented

Public, first-hand evidence on listing data management is sparse; the most useful sources are Vinted and Adevinta.

- **Single writer, replicated projections.** In a 2026 post, Vinted stated that "all writes happen in the primary site and read-only projections are replicated around the world", that teams decide what their projection looks like, how it is optimized and where it is hosted, and that "eventual consistency is being built into how the

system works, not an edge case we try to hide" (Vinted Engineering, 2026). The post says global consistency "would always lag by at least a few moments" and that projections must tolerate replication delays. It does not describe read-your-writes handling or user-visible staleness; how those are handled is not public.

- **Read models as external stores**. Vinted's 2023 indexing post describes Redis hashes, built by Flink jobs from CDC, as the read models from which search documents are assembled; large Flink state was avoided because it would make redeploys take up to about thirty minutes (Vinted Engineering, 2023). The lesson generalizes: keep read-model state in a store you can inspect, rebuild and size, not hidden inside a stream processor.
- **A compacted topic as a cheap materialized view**. Kleinanzeigen's search team described a compacted Kafka topic holding the latest aggregated per-user object for re-consumption (Adevinta Tech Blog, 2025).
- **Partitioning under growth**. Vinted's 2023 series on moving from MySQL to Vitess describes an autumn peak that strained MySQL, manual vertical sharding that was irreversible after cut-over, 42 hand-managed servers, and 23 vertical shards running the largest portal after the migration, with one table of about 3 TB needing its own shard and later horizontal sharding (Vinted Engineering, chapter 1 and chapter 3, 2023).
- **Personal data split in the analytical copy**. Adevinta Spain's 2024 post on data contracts describes removing personal fields with no analytical value at ingestion and putting those with value into a separate, partitioned personal-data table, which makes "the right to be forgotten" easier to handle; Delta Lake time travel is used to keep the two writes atomic (Adevinta Tech Blog, 2024).

What none of these sources documents is a marketplace's full copy inventory, retention schedule or erasure workflow. The rest of this chapter is therefore reference-architecture judgement, tested against Veldmark.

Partitioning, Regions and Tenancy

Partition by country, then by listing

Classifieds are country businesses: listings, buyers, laws and peaks are local. That makes **country the natural tenancy boundary** for listing data, and listing ID the natural partition key within it.

Partitioning choice	Benefits	Costs	Right when
One database for all countries, country as a column	Simple; cross-country features are easy	One blast radius; the biggest country's peak sets everyone's capacity; per-country retention is a filter, not a boundary	Small brands, early stage
Database or schema per country	Isolation of failures, peaks and maintenance; per-country retention and residency are structural	Uneven sizes; cross-country queries need the warehouse	Most multi-country classifieds
Hash-sharded by listing ID across countries	Even load; scales a single huge market	Country isolation lost unless combined with tenancy	One very large market, or a cross-border catalog
Country first, then hash within large countries	Isolation plus scale where needed	Two levels to operate	Large groups with one or two dominant markets

Tradeyard illustrates the costs of the first row. Yardstick holds all four countries in one MySQL cluster, so the Netherlands' 6.1 million live listings and its Sunday-evening peak set the capacity and maintenance windows for Portugal's 1.1 million. When the insertion flow moves out of Yardstick in Q2 2028, the new Tradeyard listing service will use one schema per country on a shared cluster, with country in every key, so that a country can later move to its own cluster without a data migration.

Vinted is a counter-case: a marketplace whose members widely buy across borders, and whose 2026 posts describe a single write primary with replicated projections rather than country partitions. The choice follows the business, not a technology preference.

Regions and residency

Each Veldmark brand, all of them EU businesses with EU users, writes its listing SoR in one primary location — Tradeyard's Yardstick in its co-located Amsterdam data

center until the strangler moves it to AWS eu-west-1, Motorvane on AWS eu-central-1, Hearthly on Google Cloud europe-west1 — and keeps its read models close to it. Multi-writer replication across regions is not on the roadmap. Vinted's 2026 account gives the reasoning a classifieds group can borrow: a single write site "avoided the complexity and operational cost of a fully sharded, multi-writer system", complexity the post calls "hard to justify unless the business absolutely demands it". Disaster recovery to a second EU region is a Sunday Evening Peaks: Reliability, Performance and Cost decision; the data-management rule is that every replica and backup in that region inherits the same retention and erasure obligations as its source.

Retention, Archives and the Life of a Listing Record

Data classes, not databases

A listing record contains several classes of data with different lawful lifetimes. GDPR's storage-limitation principle requires that personal data be kept no longer than necessary for its purpose (Art. 5(1)(e)); its accountability principle requires the controller to be able to demonstrate compliance (Art. 5(2)) (GDPR Art. 5, unofficial reproduction). Other laws require the opposite for some classes: the EU Digital Services Act (DSA) gives users at least six months to complain about a moderation decision (Art. 20), which implies keeping the decision evidence at least that long, and requires platforms that let consumers conclude distance contracts with traders to store trader traceability data for six months after the relationship ends and then delete it (Art. 30(5)) (DSA, Official Journal text, 2022). Dutch rules generally require businesses to keep their administration, including transaction records, for seven years (Ondernemersplein, Netherlands government, n.d.). Rules of the Marketplace: DSA, DAC7 and Vertical Regulation owns these obligations; the architecture consequence is that **retention is set per data class, and one listing touches several classes.**

Veldmark's group retention schedule, adopted in October 2027 after Femke's request, sets the classes for listing data:

Data class	Examples	Retention after the listing closes	Then
Public listing content	Title, description, attributes, price, category	13 months (relisting, disputes, sold pages)	Reduced to an anonymized analytical record

Data class	Examples	Retention after the listing closes	Then
Seller personal data in the listing	Seller name shown, contact options, approximate location	Until the sold page is retired, at most 13 months; immediately on erasure	Deleted
Media	Original photos, derivatives	Derivatives: until the page is retired. Originals: 90 days	Deleted, unless an evidence hold applies
Listing history	Versions, price history	13 months with seller ID; anonymized beyond	Price and attribute history kept without seller identity, for valuation and market statistics
Trust-and-safety evidence	Case records, snapshots, decisions	24 months after case closure	Deleted; hashed identifiers on block lists follow their own schedule
Transaction records	Safe Deal orders, fees, payouts	7 years	Deleted
Behavioral data about the listing	Views, contacts, alert deliveries	Per the tracking plan in Data Platform for Marketplaces: Events, Lakehouse and Data Products	Aggregated

Two rows in the table trade business value against privacy. **Anonymized price history survives** because valuation models — Valuo at Hearthly, Motorvane's market-value estimate — depend on it, and history without a seller identity is no longer personal data if the anonymization holds. In real estate the address makes that "if" hard, which the vertical section returns to. **Sold pages expire** after 13 months at most even when they rank, because a public page showing a private seller's name and neighborhood is the most visible copy of all.

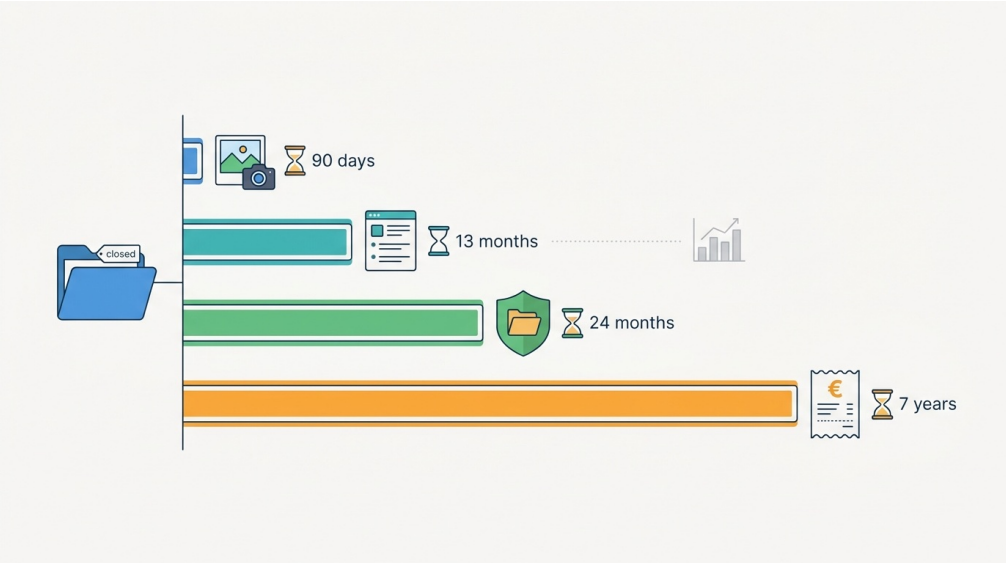


Figure 3: *Retention follows data classes, not databases: a closed listing's original photos, public content, trust-and-safety evidence and payment records each run on their own lawful clock.*

The listing record's data lifecycle

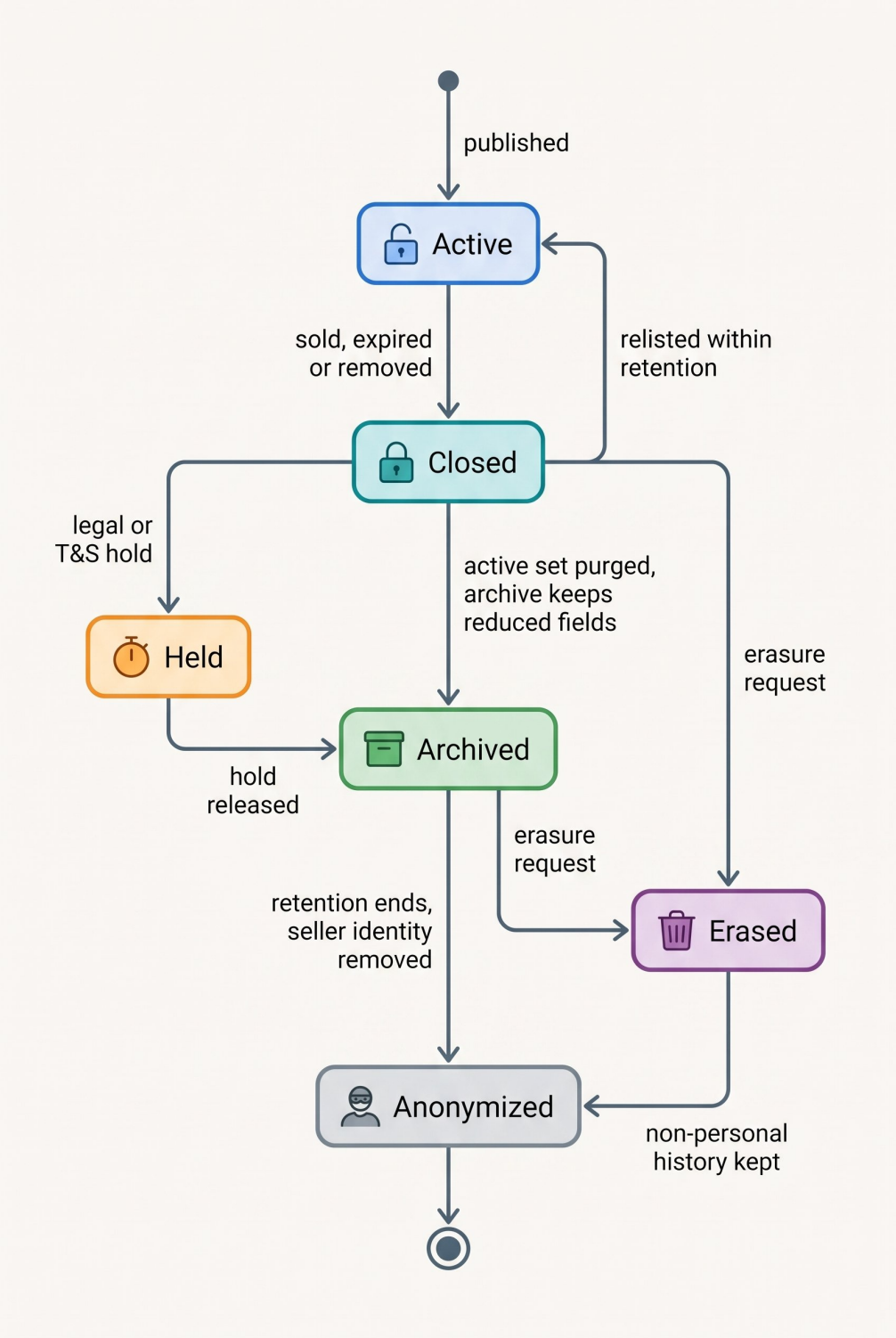

published
Active
sold, expired or removed
relisted within retention
Closed
legal or T&S hold
active set purged, archive keeps reduced fields
erasure request
Held
Archived
hold released
erasure request
retention ends, seller identity removed
Erased
Anonymized
non-personal history kept

Figure 4: *The record's data lifecycle runs after the business lifecycle ends. Closing a listing starts retention clocks; archiving reduces fields; erasure and anonymization remove identity; a hold pauses the clocks for evidence. Every transition is an event that copy owners act on.*

The lifecycle is distinct from the business lifecycle in The Listing Aggregate: Identity, Lifecycle and Invariants. A listing is *sold* once; its record then passes through several data states over months. Each transition is emitted as an event — `ListingErased` is part of the v1 contract — so that copies follow the record without a separate integration.

Erasure Propagation

Erasure is a workflow with receipts

GDPR requires a controller to act on an erasure request "without undue delay" (Art. 17(1)), and to respond to the data subject within one month of receipt, extendable by two further months for complex requests (Art. 12(3)) (GDPR Art. 17; Art. 12, unofficial reproductions). Erasure does not apply where processing is necessary for compliance with a legal obligation or for the establishment, exercise or defense of legal claims (Art. 17(3)). Where data has been made public, the controller must take reasonable steps to inform other controllers processing it (Art. 17(2)), and it must communicate erasure to each recipient to whom the data was disclosed unless that proves impossible or involves disproportionate effort (Art. 19) (GDPR Art. 19).

Read as architecture, those articles describe a workflow, not a `DELETE` statement:

1. **Intake**. The privacy request service records the request, verifies identity and opens an **erasure case** with a deadline.
2. **Resolution of scope**. The case resolves the subject to identifiers — account IDs per brand, listing IDs, hashed contact identifiers — using the identity and listing systems of record.
3. **Fan-out**. The account context emits a subject-level erasure event; the listing context emits `ListingErased` for each affected listing after purging personal fields in the SoR.
4. **Execution per copy**. Each registered copy owner executes its handler and returns an outcome: **deleted**, **anonymized**, **retained with legal basis** (naming the basis and the end date), or **not held**.
5. **External copies**. Partners and processors receive delete instructions; recipients are notified under Art. 19; public copies indexed by search engines get a removal path (a 404 or 410 page and a removal request where warranted).
6. **Proof**. The **erasure ledger** records every outcome. The case closes only when every registered copy has answered, and the response to the data subject is generated from the ledger.

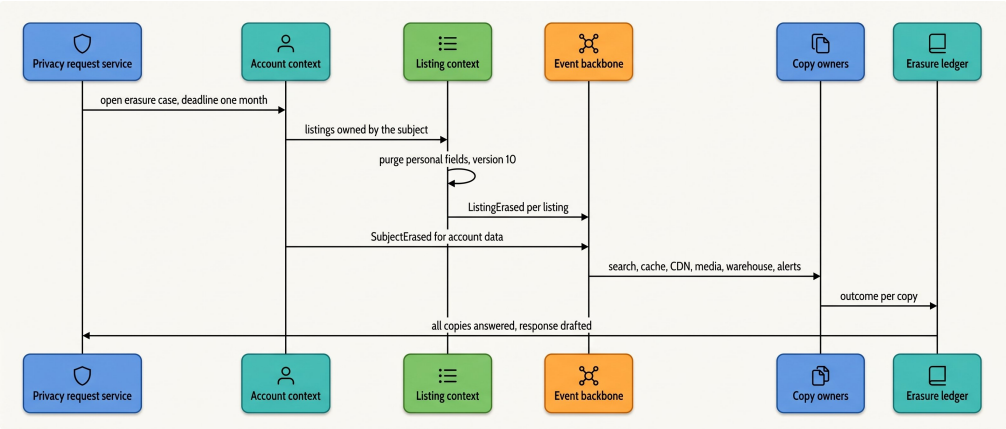


Figure 5: *Erasure as a workflow with receipts. The systems of record act first and emit events; every registered copy reports an outcome to the erasure ledger; the case closes only when the ledger is complete.*

Femke's request, copy by copy

Applying the workflow to the seven copies, with the October 2027 schedule:

#	Copy	Outcome	Basis or mechanism	Done by
1	Yardstick SoR	**Anonymized**: seller ID replaced by a tombstone ID; name, location and description cleared; price and category history kept	Erasure; non-personal history retained	Day 1
2	Sold-listing page	**Deleted**: page returns 410; CDN purged by listing ID; removal requested from search engines	Erasure; Art. 17(2)	Day 1

#	Copy	Outcome	Basis or mechanism	Done by
3	Media	**Deleted**: originals and derivatives; CDN purged	Erasure	Day 2
4	Warehouse	**Anonymized**: rows in the personal-data table deleted; listing facts keyed by tombstone ID	Erasure; personal-data split	Day 3, next erasure job
5	T&S case	**Retained with legal basis** until 24 months after case closure (March 2029): Femke's report is evidence in a fraud case against another account	Art. 17(3)(e), legal claims; fraud prevention as a legitimate interest	Recorded day 1
6	Safe Deal records	**Retained with legal basis** for 7 years; access restricted to finance and audit	Art. 17(3)(b), legal obligation	Recorded day 1
7	Digest archive and e-mail logs	**Deleted** internally; delete instruction to the e-mail provider, confirmed day 9	Erasure; processor contract	Day 9

Lucía sends Femke the response on **Wednesday 15 September 2027**, nine days after the request, listing what was erased and what is kept, for how long and why. The table also shows the cost of not having the registry: the first answer took two days of investigation, and the second time — for the next of the roughly 1,900 erasure requests Tradeyard receives a month — it must take none.

Backups, logs and crypto-shredding

Three stores resist erasure by design.

- **Backups**. Editing backups is impractical and risky. The common practice is to let backups expire on a short schedule (35 days at Veldmark) and to keep the erasure ledger outside the backups, so that any restore re-applies every erasure recorded since the backup was taken. A restore procedure that does not replay the erasure ledger is an erasure failure waiting for an incident.
- **Event logs**. Listing events carry public state only, without seller contact data, which is the main defense (see Every Change Is an Event: Event-Driven Listing Architecture). The events topic expires after 14 days; the compacted state topic receives a tombstone on `ListingErased`.
- **Long-lived immutable stores**. For data that must be immutable but may contain personal fields — a lake's raw zone, an evidence store — **crypto-shredding** encrypts each subject's personal fields with a per-subject key and erases by destroying the key. It works, but it moves the problem to key management and makes every read depend on a key service; I use it only where immutability is a requirement, not as a general erasure strategy.

Corrections, Backfills and Data Contracts

Correct at the source

Listing data goes wrong in bulk: an enricher version writes a wrong value, a feed mapping misreads a field, a taxonomy migration lands listings in the wrong leaf. The temptation is to patch the read model that shows the error. **The rule is to correct at the system of record and let the correction flow**. In August 2027, a Motorvane VIN-decoder release mapped one engine family's fuel type wrongly on about 41,000 listings. The fix took three steps: correct the enrichment output in the listing service (writing new versions with `cause` = `correction`), publish them through the backfill topic, and let search, the market-value model and the warehouse apply them with the usual version gate. Patching the Elasticsearch index directly would have been faster by a day and would have been undone by the next feed update.

The consumer registry as a data contract

The **consumer registry** from From Submitted to Searchable: The Publication Pipeline turns the rules in this chapter into something a team can be held to. This chapter extends each entry beyond removal: every store holding listing data, inside or outside the group, records its source, staleness budget, rebuild path, retention and erasure handler:

Field	Example: Tradeyard search index NL
Owner	Group search platform (Tomasz Wiśniewski); Solr 8 until the 2028 move to OpenSearch
Source	`listing.events.v1`, bootstrap from `listing.state.v1`
Fields held and personal-data fields	Public listing state; seller display name
Staleness budget	Additions 90 s p95 (2028 target); removals 30 s p99
Rebuild path and time	Full rebuild from state topic, 5 h 40 min for NL at off-peak
Retention	Active and reserved listings only
Erasure handler	Delete on `ListingErased`; reconciliation weekly
External recipients	None

The registry is the data-contract idea applied to read models. Adevinta Spain's contracts for analytical ingestion carry a comparable set — schema, landing topic, personal-data fields, owner, service levels, retention — and are immutable and versioned in GitHub (Adevinta Tech Blog, 2024). Veldmark extends the idea from the warehouse to every copy, and makes registration a precondition for receiving access to the listing topics, so the registry cannot drift from reality: a consumer group without an entry gets no credentials. By December 2027 the registry lists **64 copies** of listing data across the group.

Trade-offs

Decision	Option A	Option B	When A is right	When B is right
Seller-facing reads	**From the SoR or a synchronous replica**	From read models with version tokens	Moderate seller traffic; simple screens	Pro tools with heavy filtering on thousands of listings

Decision	Option A	Option B	When A is right	When B is right
Detail pages	From a read model or cache	From the SoR	Peak buyer traffic (nearly always)	Very low traffic, or fields that must be exact (none should be)
Partitioning	**Country first**	Global hash	Country businesses with local law and peaks	Cross-border catalogs
Regions	**Single writer, regional read models**	Multi-writer	No strong business need for local writes (Vinted's documented reasoning)	Write latency or sovereignty requires local writes and the cost is accepted
Closed listings	Keep in the SoR with a closed state	**Move to an archive with reduced fields**	Short retention; relisting is frequent	Long retention; the active store must stay small and fast
Erasure in immutable stores	**Delete or anonymize, let backups expire**	Crypto-shredding	Stores can be rewritten or expire quickly	Immutability is mandated or the store is replayed for years
Analytical personal data	**Split into a separate personal table**	Keep inline with masking	Erasure volume is high (private-seller market-places)	Few personal fields, low request volume
Sold pages	Keep public while they rank	**Retire within a fixed period**	Professional sellers, no personal data on the page	Private sellers whose name or area appears

The most contested row in practice is the last. SEO teams can show that sold pages bring traffic; Found by Google: Web, App and SEO Architecture for Listings explains how to keep the value of a sold page's URL without keeping the private seller's data on it. The data-management position is that the page's *content* is governed by the retention schedule even when its *URL* lives on.

Vertical Differences

Aspect	General classifieds	Real estate	Automotive	Jobs
Share of personal data in listings	High: private sellers, homes, faces in photos	High for private landlords; addresses are sensitive	Low for dealers; plates and VINs can link to owners	Low in the ad; very high in applications and CVs
What outlives the listing	Little, besides transactions and evidence	The property: canonical record, price history, valuation inputs	The vehicle: VIN history, price history	The employer; applications follow their own retention
Hardest erasure problem	Volume of requests	Address-level history that re-identifies an owner even without a name	Photos showing plates; VIN-linked history	Applicant data shared with employers
Read model that matters most	Search and alerts	Map tiles and the property cluster view	Search and the market-value model	Search and external job indexes
Archive value	Low	High: market statistics, valuation training	Medium: residual values, price ratings	Low

In real estate, removing names does not anonymize price history. Hearthly keeps anonymized price history for Valuo's estimates, but a flat's address, surface and price history can identify its owner without a name. For property data, "**anonymized" must be tested against re-identification, not assumed from the removal of names**; aggregation to the building or the street, or retention under a documented legitimate interest with access controls, are the realistic options. Models in Production: ML Platforms and Use Cases owns the model side of that decision.

Failure Modes

- **The unregistered copy**. A team exports listings to a spreadsheet, a partner, a test environment or a notebook "just once". It holds personal data, it is never cleaned and it is found during an erasure request or a breach. Registration as a precondition for access is the control.
- **Search as System of Record**. Fields or corrections that live only in the search index. The index cannot be rebuilt, so migrations stall and corrections are lost on the next reindex.
- **Read-your-writes broken**. A seller edits a price, sees the old one on the next page, edits again, and creates two price-drop alerts for watchers. Sellers' own reads must come from the SoR or carry a version token.
- **Replica flapping**. Load-balanced reads across replicas with different lag show a buyer prices going back and forth. Version-aware caches or session affinity prevent it.
- **Caches without removal paths**. Pages and images cached with long expiry and no purge by listing ID: the sold bike keeps loading for days. Every cache of listing data needs a surrogate key.
- **Retention by database**. One retention rule per store, so evidence is deleted too early or personal data kept too long. Retention must follow data classes.
- **Restore without replay**. A backup restore brings back erased data, because the erasure ledger was inside the backup or the procedure forgot it.
- **External copies forgotten**. E-mail providers, ad platforms, partner feeds and search engines hold listing data that no internal job reaches.
- **Patching the read model**. Bulk corrections applied directly to search or the warehouse, overwritten by the next event, and leaving the SoR wrong.
- **Anonymization that is not**. Address-level or VIN-level histories released as "anonymized" that can be joined back to a person.

The Chief Architect's View

The decisions in listing data management are few, but they are group decisions, because every brand has the same obligations and the same copies.

What to decide centrally. The rule of one system of record per fact; the consumer registry and the rule that registration precedes access; the consistency promises per read-path class (seller, buyer, removal); the group retention schedule by data class; the erasure workflow, its event, its handlers and its ledger; and the backup-restore procedure that replays erasures. At Veldmark these are owned jointly: the event platform team runs the register, Lucía Ortega owns the retention schedule and the erasure ledger, and each copy owner owns its handler. The architecture advice forum reviews every new copy type — not every consumer, but every new *kind* of store.

What to leave local. Each brand's choice of SoR technology during migration, its read-model designs, and its partitioning scheme within the group's tenancy rule.

Who decides on conflicts. When SEO value, model accuracy or fraud prevention argue for keeping data and privacy argues for deletion, the decision belongs to a named owner with the data protection officer's sign-off, recorded as a decision record. The architecture's job is to make each such decision executable per data class, rather than to make it once for a whole database.

What to watch. Staleness per read path against budget, especially removals; the count of copies in the registry and of consumer groups without an entry (which should be zero); erasure cases closed within deadline and the slowest copy owner; the age of the oldest closed listing in each active store; rebuild times for each read model, tested at least twice a year; and the number of fields that exist only in a read model.

The long-term consideration. Copies outlive the systems that created them. Yardstick will be gone by the end of 2029; its listings, their archives, their evidence and their erasure history must not go with it, and they must not be copied into the new platform with their old retention problems intact. A migration plan that does not include the consumer registry and the erasure ledger is not finished; Changing the Engine in Flight: Modernizing a Live Marketplace takes that up.

Questions to Ask

1. For each listing field shown anywhere, which system of record owns it — and are there fields that exist only in the search index or a cache?
2. Can we delete the search index today and rebuild it without loss? How long would it take, and when did we last do it?
3. What is the staleness budget for each read path — seller pages, search, detail pages, alerts, partner feeds — and how is each measured?
4. Do sellers always read their own writes? What do they see in the second after an edit?
5. How do removals reach caches, CDNs and pages, and what is the measured time from "sold" to "gone everywhere"?
6. Is there a registry of every copy of listing data, including exports, test environments and third parties, and does access depend on registration?

7. What retention applies to each data class in a closed listing, and which law or purpose justifies each period?
8. When an erasure request arrives, can we list every copy that holds the subject's data within an hour, and prove each outcome?
9. Does our backup-restore procedure replay erasures recorded after the backup?
10. Which "anonymized" datasets could be re-identified by address, VIN or photo?

From Internal Copies to Search Engines

The copies in this chapter are not only read by Veldmark's own systems. The most valuable read model of all is the one that search engines and AI answer engines build from Tradeyard's 40 million indexed URLs, and its consistency promises are set partly by Google. Found by Google: Web, App and SEO Architecture for Listings takes up the customer-facing web and the pages that represent listings to the outside world.

Sources and Evidence

Company engineering and investor sources

- Building a Global, Event-Driven Platform, Part 2 — Vinted Engineering, 2026. Single-writer primary, replicated projections owned by teams, eventual consistency, rejection of multi-writer.
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- Vinted Vitess Voyage, Chapter 1 and Chapter 3 — Vinted Engineering, 2023. Seasonal peak, manual vertical sharding, 23 shards, a 3 TB table.
- Make data migration easy with Debezium and Apache Kafka — Adevinta Tech Blog (Daniil Roman, Kleinanzeigen), 2025. Compacted topic as a re-consumable materialized view.
- Creating source-aligned data products in Adevinta Spain — Adevinta Tech Blog (Sergio Couto), 2024. Data contracts; personal-data split for erasure; quarantine.

Standards, regulation and documentation

- Art. 5 GDPR, Art. 12 GDPR, Art. 17 GDPR, Art. 19 GDPR — Regulation (EU) 2016/679, unofficial reproduction at gdpr-info.eu, n.d. Storage limitation and accountability; response deadline; erasure and its exceptions; notification of recipients.

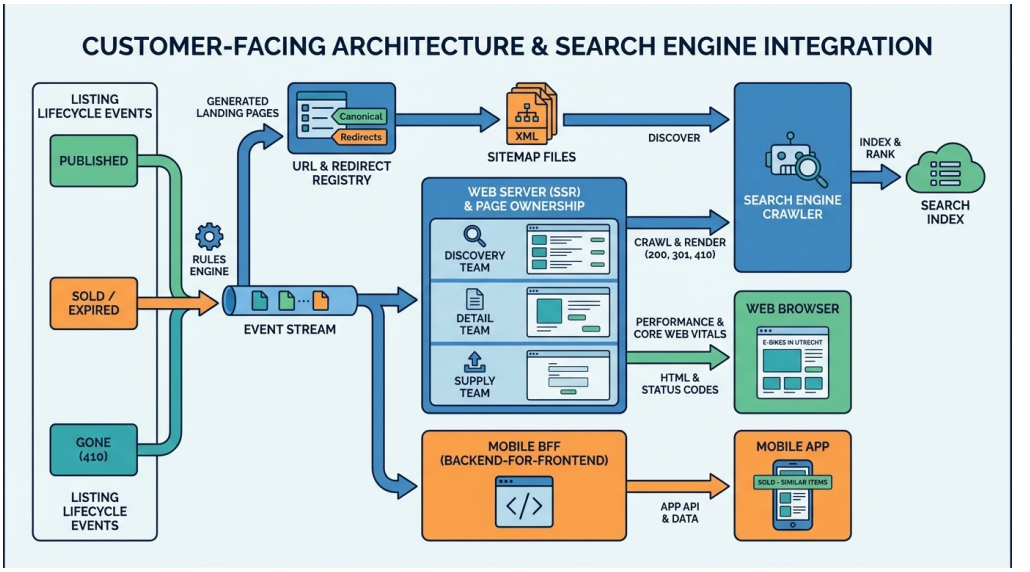
- Regulation (EU) 2022/2065 (Digital Services Act) — Official Journal text via EUR-Lex and the EU Publications Office, 2022. Art. 20(1) (six-month complaint window) and Art. 30(5) (trader data stored six months, then deleted), checked against the Official Journal text in October 2026.
- Administratie bijhouden en bewaren — Ondernemersplein (Netherlands government), n.d. General seven-year retention of business records.

Research and practitioner sources

- What do you mean by "Event-Driven"? — Martin Fowler, 2017. Event-carried state transfer and the cost of copies.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

35. Found by Google: Web, App and SEO Architecture for Listings



IN THIS CHAPTER, YOU WILL: Learn how search engines crawl, render and index a listing site and why that makes URLs, status codes and sitemaps part of the architecture. You will design the landing-page generator, the lifecycle rules for live, sold and removed listings, the rendering and composition model for the web, and the backends-for-frontends behind the apps. Then you will follow Tradeyard as it cuts 40 million indexed URLs down to the ones that earn traffic.

WHY THE BUSINESS CARES: For a horizontal classifieds site, organic search is often the largest free acquisition channel, and it is the one the business controls least. A rendering change, a category rename or a careless expiry rule can remove a large share of that traffic in a few weeks, and recovering it takes months.

WHY ARCHITECTS SHOULD CARE: Search engines are a client of the platform with their own protocol: URLs, HTTP status codes, sitemaps,

structured data and response times. That client never files a bug report. It quietly crawls less, indexes less and ranks less, so its contract has to be designed and tested like any other interface.

KEY POINTS:

- **Landing pages, not listing pages, are the durable SEO asset.** Listings live for days; a page for "e-bikes in Utrecht" can rank for years. Generate landing pages from measured demand and live supply under explicit rules, and give everything else a `noindex`, a `404` or no URL at all.
- **URLs, status codes and sitemaps belong to the listing lifecycle.** A sold listing that still answers `200` with "no longer available" is Ghost Inventory in Google's index. Drive canonical URLs, redirects, sitemap entries and removal signals from the same listing events that drive search and alerts.
- **Render on the server, own pages per team, and give each client its own backend.** Server-side rendering with edge caching serves both crawlers and phones. Page-level ownership with build-time shared components has a better documented record than runtime fragment composition. Apps that ship every two weeks need a backend-for-frontend and server-driven forms, or every schema change waits for app adoption.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Ghost Inventory** — sold and expired listing pages that keep answering `200`, stay in sitemaps and keep collecting crawls put gone inventory in front of buyers through the search engine.
- **The Category Straitjacket** — when category paths are baked into URLs with no redirect registry, every taxonomy change becomes an SEO migration that nobody dares to schedule.
- **Vanity Liquidity** — counting indexed URLs and organic sessions, instead of contacts from organic sessions, rewards generating pages that search engines treat as low value.

Part VII designs the platform as a set of published contracts; this chapter writes the one the marketplace keeps with the outside world. Every page has two audiences. One is a person on a phone who wants a used e-bike near home. The other is a crawler that decides whether that person will ever see the page in a search result. The crawler reads the HTML, follows links, respects status codes, reads sitemaps and measures how fast the server answers. **For a business whose inventory changes by the minute, how the site talks to crawlers is a design decision with a revenue line attached.**

Most marketplaces discover this late. The web front end is built for users, SEO is treated as a set of tags added afterwards, and the first serious conversation about crawl budget happens after a redesign that cost a third of organic traffic. This chapter takes the opposite view. **SEO architecture** — in this book, the set of components and rules that decide which URLs exist, what each URL returns over its lifetime, how pages are rendered and how search engines learn about changes — is part of the customer-facing architecture from the start, alongside rendering, composition and the app backends.

The chapter builds on the two before it. The listing events of Every Change Is an Event: Event-Driven Listing Architecture drive its URLs, sitemaps and cache purges, and the index a search engine builds from Tradeyard's pages is the outermost of the read models in System of Record and Read Models: Listing Data Management. It does not re-teach search indexing, which Search Architecture for Listings: Indices, Freshness and Scale owns. Later chapters use what it sets up: Data Platform for Marketplaces: Events, Lakehouse and Data Products governs the organic-contact metric the worked example ends on, and Changing the Engine in Flight: Modernizing a Live Marketplace starts Yardstick's strangler at the web layer.

Setting the Scene at Veldmark: Forty Million URLs and a Sold Bike

It is late March 2027 at Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build). Ingrid Halvorsen, the Group Chief Architect, is reviewing Tradeyard's web estate with Daan Visser, Tradeyard's engineering director, and the brand's SEO lead. Tradeyard has about **40 million** URLs in Google's index across its four country domains, and **46%** of its web sessions start from organic search. Those two numbers appear in every board pack as evidence of strength. Ingrid wants to know what is behind them.

The answer comes from a crawl audit the SEO team finished in February. Of the 40 million indexed URLs:

URL type	Indexed URLs	Live listings or results behind them	Organic visits in the last 90 days
Live listing detail pages	11.4M	One live listing each	31% of organic sessions
Sold, expired or removed listing pages	6.3M	None; page answers `200` with "no longer available"	6% of organic sessions, 0.4% of organic contacts

URL type	Indexed URLs	Live listings or results behind them	Organic visits in the last 90 days
Search and landing pages (category × location × filters)	19.8M	Varies; 7.1M have fewer than three live results	58% of organic sessions, from about 1.1M URLs
Seller shops, profiles, help and guides	2.5M	—	5% of organic sessions

Three findings stand out. **Most organic value comes from a small set of landing pages**: about 1.1 million of 19.8 million search URLs received an organic visit in 90 days. **A sixth of the index is gone inventory**: 6.3 million pages for listings that no longer exist still return `200`. And Googlebot fetches about 14 million Tradeyard URLs a day, 41% of them parameterized facet URLs or dead listing pages, while a new Tradeyard NL listing waits a median of **2.6 days** for its first crawl. Most private goods listings are sold or withdrawn within a week, so for them that crawl comes too late to matter.

Femke's city bike shows the problem at the scale of one item. She listed it in Rotterdam for **€240** on Sunday **7 March 2027**, and it sold on Monday **8 March** for **€225**. Tradeyard marked it sold at **18:10** and removed it from search at **18:11**. Its listing page kept answering `200` with a "this ad is no longer available" banner, stayed in the listings sitemap with an unchanged `lastmod`, and was still in Google's results nine days later, where it collected 37 clicks from people who could not buy it. The alert digest assembled that evening, which still featured it when it went out the next morning, is a problem Tell Me When It Appears: Saved Searches and Alerts owns; the search engine result is a problem this chapter owns.

At the same time, Daan's team is preparing two changes that touch the same layer. Tradeyard's search pages are rendered by Yardstick, the 2006 PHP monolith, and score poorly on mobile performance. A team has proposed rebuilding them as a client-rendered single-page application (SPA). And the taxonomy team plans to split the single bikes category into city bikes, e-bikes, cargo bikes and kids' bikes on **1 November 2027** (see Category Trees That Can Change: Designing Marketplace Taxonomies), which will change thousands of landing-page URLs. Ingrid's question for the rest of the chapter: *what should the customer-facing architecture look like so that changes like these are routine instead of risky?*

Concepts: How a Listing Site Earns Organic Traffic

Crawl, render, index, rank

A search engine processes a site in four stages. It **crawls** URLs it discovers through links and sitemaps, **renders** the HTML (and, for JavaScript-heavy pages, executes scripts, often later), **indexes** what it decides is worth keeping, and **ranks** indexed pages against queries. Each stage filters. A URL that is never crawled cannot be indexed; a page that renders as an empty shell may be indexed without its content; an indexed page that duplicates thousands of others ranks for nothing.

Google's documentation names the sites that need to manage the first stage explicitly: "large sites (1 million+ unique pages) with content that changes moderately often (once a week)" or medium sites with "very rapidly changing content (daily)" (Google Search Central, crawl budget, updated 2026). Every sizable classifieds site meets both conditions. The same page explains that crawl capacity falls when responses are slow or return `5xx` or `429` errors, and that soft-404 pages keep being crawled and waste budget. **Crawl budget** is therefore not a fixed allowance. It is a function of how valuable the engine believes the URLs are and how healthy the server looks while it fetches them.

The page types of a listing site

Each page type of a listing site has its own lifetime and its own SEO role.

Page type	Example	Typical lifetime	SEO role	Rendered from
Landing page (search result page eligible for indexing)	"E-bikes in Utrecht", "Used Volkswagen Golf in Essen", "3-room apartments for sale in Lyon 7e"	Years	Main durable organic asset; matches recurring queries	The search engine, with curated copy and links
Listing detail page	One bike, one car, one apartment	Days (goods) to weeks (cars, property)	Long-tail queries; shares on social and messaging	The listing read model

Page type	Example	Typical lifetime	SEO role	Rendered from
Faceted result page (non-curated filter combination)	Bikes, Rotterdam, red, €100–€150, sorted by price	Seconds to days	Usually none; a crawl trap if exposed	The search engine
Seller or agency page	A dealer's stock, an agency's branch	Years	Brand and local queries	Seller profile and search
Editorial and tool pages	Buying guides, price checkers, valuation pages	Years	Informational queries; links into landing pages	Content management system (CMS) and data services
Gone listing page	A sold car, an expired ad, a removed scam	Should be short	Little, except where demand persists	Archive or nothing

In this book a **landing page** is a search result page that the platform has deliberately made indexable: it has a canonical URL, curated title and copy, internal links, and a rule that says when it may exist. The definition matters because every faceted search page *could* be crawled, and most of them should not be. **Landing page generation** is the process that decides which category, location and attribute combinations get a canonical, indexable URL, keeps that set current as demand and supply change, and retires pages cleanly.

Why landing pages carry the weight

The landing page is the page that can accumulate links, history and ranking signals, because its URL stays stable while its contents change. Classifieds inventory is short-lived and unique: a specific bike exists on the site for a few days, while a query like "tweedehands e-bike Utrecht" exists for years. This is industry practice inferred from inventory lifetime, not documented fact; no classifieds operator in the evidence published its organic traffic by page type. Detail pages matter more where items live longer and searches are specific, such as a dealer's car or an apartment.

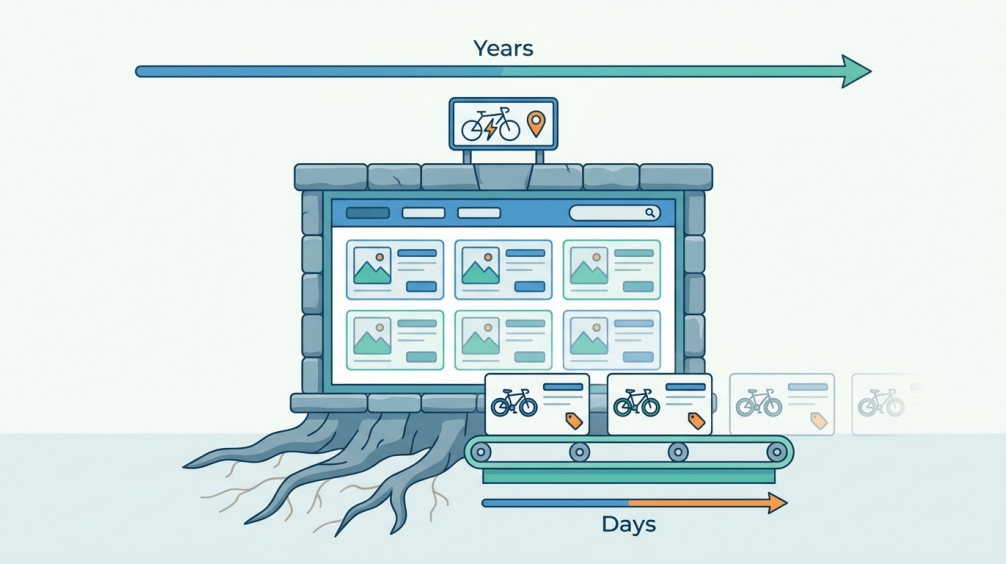


Figure 1: *Listings live for days, but a landing page such as e-bikes in Utrecht keeps a stable URL for years while its contents change, which is why it accumulates the links and ranking history.*

The backend-for-frontend

A **backend-for-frontend (BFF)** is a server-side component owned by the team that owns a client (the iOS and Android apps, or the web front end), which aggregates domain APIs into the shape that client needs and absorbs differences in release cadence. Sam Newman described the pattern in 2015, with the advice that the team building the user interface should own its backend (Newman, 2015). Its relevance in classifieds comes from two facts. Pages and screens draw on many domains at once — listing, search, seller, media, ads, entitlements, trust signals — and apps cannot be updated on demand. Vinted's engineers wrote in 2019 (older evidence; current release practice may differ) that mobile releases take about two weeks to reach users, which is why they built server-side repair for malformed analytics events (Vinted Engineering, 2019). The same delay applies to every API change.

Architecture: The Customer-Facing Stack

The customer-facing stack has five layers: the edge, the page applications, the backends-for-frontends, the SEO services and the domain APIs beneath them.

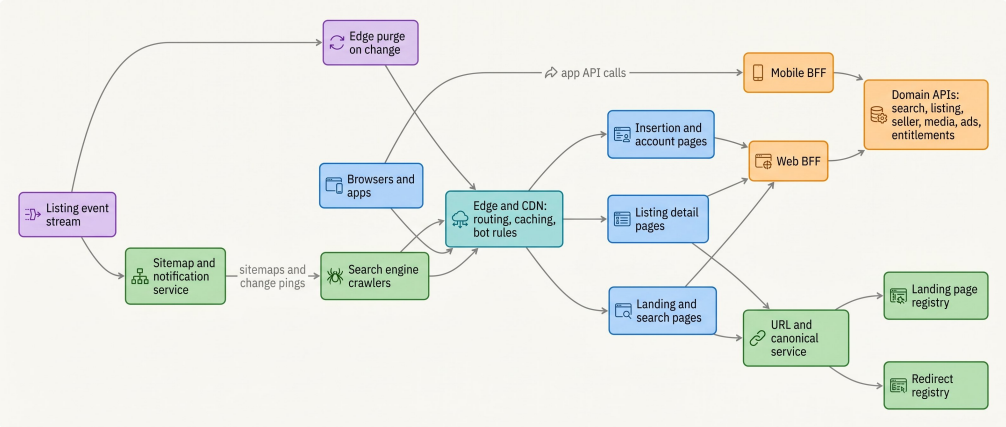


Figure 2: *The customer-facing stack. Pages and apps reach domain data through their own backends-for-frontends; the SEO services decide which URLs exist and what they return, and the listing event stream keeps sitemaps and edge caches in step with the inventory.*

The edge

The edge — a content delivery network (CDN) with programmable routing — does four jobs in a listing site. It **routes** URL paths to the page application that owns them, which makes it the natural seam for migration (Subito, below, configures route ownership on Akamai). It **caches** rendered pages with time-to-live (TTL) values per page type and purges them on change. It **applies bot rules**, separating verified search engine crawlers from scrapers (see Sunday Evening Peaks: Reliability, Performance and Cost). And it **terminates** redirects and simple status responses without touching the origin, which matters when millions of retired URLs must answer `301` or `410` cheaply.

Caching policy follows page type. Landing pages tolerate a few minutes of staleness and are a large share of read traffic, so they should be served from cache rather than live search. Listing detail pages can be cached longer, provided a listing event (price change, sold, removed) purges the cached copy by a surrogate key such as the listing ID. Pages with personal data — messages, account, saved searches — are never cached at the edge.

Rendering: what the crawler and the phone receive

The rendering strategy is the most consequential decision in the web layer. Five are in common use.

Strategy	What happens	Crawler sees	Phone performance	Fits
Client-side rendering (CSR)	Server sends a shell; JavaScript fetches data and renders	Content only after rendering, which may be deferred	Slow first content on mid-range phones	Logged-in tools: messaging, seller dashboards
Server-side rendering (SSR) with hydration	Server renders HTML per request; JavaScript attaches interactivity	Full content in the HTML	Fast first content; interactivity cost depends on the JavaScript shipped	Landing, search and listing pages
Streaming SSR	Server sends the page in chunks as data arrives	Full content	Faster first bytes when some data is slow	Pages with slow components such as facets or ads
Static or incremental generation	Pages built ahead of time and regenerated on a schedule or event	Full content	Fastest	Editorial pages, stable landing pages with few changes
Dynamic rendering	Server detects crawlers and serves them pre-rendered HTML	Pre-rendered HTML	Unchanged for users	Nothing new; Google calls it "a workaround and not a recommended solution" (Google Search Central, updated 2025)

For public pages of a listing site the decision is rarely in doubt: **landing, search and listing detail pages should be server-rendered**, with as little client JavaScript

as interaction requires. Instead of dynamic rendering, Google recommends "server-side rendering, static rendering, or hydration". Client-side rendering remains right for authenticated tools where crawlers never go. The proposed client-rendered rebuild of Tradeyard's search pages would have solved a real problem, slow Yardstick templates, by handing the most valuable pages to the crawler's deferred rendering queue.

Google also indexes the mobile rendering. Its mobile-first indexing guidance asks sites to "make sure that your mobile site contains the same content as your desktop site" and to keep the same structured data on both (Google Search Central, updated 2025). A mobile layout that hides seller details or attribute tables behind tabs that load on demand may hide them from the index too.

Composition: who owns which page

Once more than one team serves a site, someone has to decide how pages are composed. The documented record in marketplaces is unusually clear.

- **Runtime fragments**. Zalando's Mosaic assembled pages at request time from fragments served by independent services. In a 2018 post, Zalando described the costs: inconsistent user experience, "an issue most customer-facing websites cannot ignore", and a high contribution barrier because each fragment needed its own full stack. Its replacement, the Interface Framework, combined a GraphQL aggregation layer with renderers built from the design system in a monorepo (Zalando Engineering, 2018). The post is old and describes a system that has since evolved; the lesson about consistency has aged well.
- **Server-side fragment stitching**. AutoScout24 described in 2018 how services served HTML fragments that an nginx-based service stitched together, while newer pages were written in React behind a GraphQL backend-for-frontend (AutoScout24 Tech Blog, 2018). The current state is not public.
- **Microsites with page-level ownership**. In a 2023 post, Subito (Adevinta, Italy) described splitting its web monolith into "many small independent sites, where each site has, at most, a few pages", each a Next.js application with its own repository, pipeline and team. Routing is configured on Akamai at page level, and shared components come from a monorepo at build time, wrapped in error boundaries (Adevinta Tech Blog, 2023).
- **Framework migration**. A leboncoin engineer pointed in 2021 to the team's write-up of a migration from a legacy React 15 front end to Next.js, describing about 30 million monthly users and about 40 web developers (GitHub discussion, 2021). The article itself could not be retrieved for this book.

The direction of travel in this evidence is **away from runtime composition of fragments and toward page-level ownership with shared components linked at build time**. That fits listing sites well, because their page types map cleanly onto teams:

discovery owns landing and search pages, the listing team owns detail pages, the supply team owns insertion, the messaging team owns conversations. My rule is that **one team owns each URL pattern end to end** — rendering, performance budget, status codes and structured data — and the design system, not a runtime composer, keeps the pages consistent.

The SEO services

Four small services hold the rules that pages must not each reimplement.

1. **The URL and canonical service** builds every public URL from domain identifiers: listing ID, category ID, location ID and attribute values. It enforces one canonical form — filter order, separators, lowercase, no duplicate filters — and returns the canonical URL for any combination. Pages never assemble URLs by string concatenation.
2. **The landing page registry** holds the set of indexable combinations, each with its state, rule, copy template and history (next section).
3. **The redirect registry** maps every retired URL to its successor or to a gone status, with the reason and date. Category splits, city renames, brand migrations and listing merges all write to it. It is served from the edge.
4. **The sitemap and notification service** maintains sitemaps and change notifications from listing and landing-page events, with accurate `lastmod` values.

Putting category and location *identifiers* behind human-readable slugs, rather than treating the slug as the identifier, is what makes the redirect registry tractable. When the bikes category splits, the service knows which old landing URLs map to which new ones because both are derived from category IDs, not from strings typed into templates.

Landing Pages: Generating the Durable SEO Inventory

The rule set

A landing page should exist when three conditions hold: **people search for it**, **the marketplace has inventory for it**, **and it is meaningfully different from the pages that already exist**. Each condition becomes a rule.

Rule	Measured by	Typical threshold (illustrative)	Action when it fails
Demand	External keyword data, internal search logs, saved searches for the combination	Recurring monthly searches above a floor per country	Not created; existing page reviewed
Supply	Live listings matching the combination, averaged over weeks	At least a minimum number of results (for example five) on most days	`noindex, follow` while thin; `404` when empty for a sustained period
Distinctness	Overlap of results with the parent page	Less than a set share of results identical to the parent	Not created; canonical to parent
Depth	Number of filters in the combination	At most two or three dimensions beyond category and location	Not indexable; reachable only by users
Quality	Curated title, unique intro copy, internal links, structured data	Template complete and reviewed	Not published

The thresholds are judgement calls that differ by vertical. What matters is that indexability is a **computed state with explicit rules**, recalculated as inventory and demand change, not a side effect of a filter being clickable.

Google's guidance on faceted navigation gives the outer boundary. If filter combinations need not be indexed, block them with `robots.txt` or keep the filters in URL fragments. If they must be indexed, use standard `&` separators, keep the filter order consistent, prevent duplicate filters, and return `404` for combinations with no results rather than redirecting them to a generic page (Google Search Central, updated 2025).

Google's spam policies define the inner boundary. **Scaled content abuse** is when "many pages are generated for the primary purpose of manipulating search rankings", and **doorway abuse** includes "multiple domain names or pages targeted at specific regions or cities that funnel users to one page" (Google Search Central, spam policies, updated 2026). A classifieds landing page for "e-bikes in Utrecht" with real e-bikes for sale in Utrecht is not a doorway. A page for "e-bikes in Wijk bij Duurstede" with three listings

from Utrecht, wrapped in generated text, comes close. The supply rule is what keeps generated landing pages on the right side of that line.

The lifecycle of a landing page

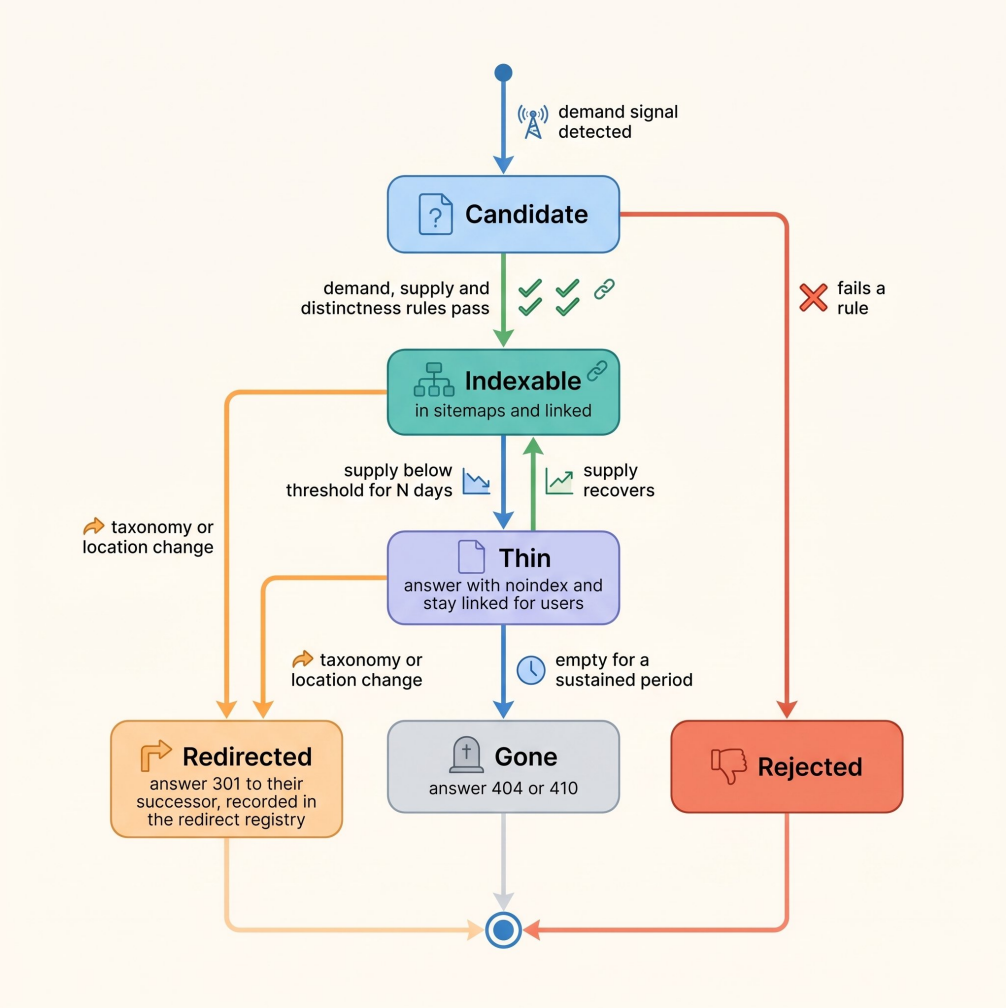


Figure 3: *A landing page is a managed object with states. Indexable pages are in sitemaps and linked; thin pages answer with noindex and stay linked for users; gone pages answer 404 or 410; redirected pages answer 301 to their successor, recorded in the redirect registry.*

Each state maps to a response: **Indexable** pages return `200`, appear in sitemaps and receive internal links; **Thin** pages return `200` with `noindex, follow`, so users can still use them and crawlers still follow their links to listings; **Gone** pages return `404`

or `410`; **Redirected** pages return `301` to the successor. Hysteresis — a page must stay thin for some days before it changes state, and recover for some days before it returns — prevents seasonal or weekend swings from flipping pages in and out of the index.

Internal linking and copy

Landing pages rank partly because the site links to them. The links that matter are structural: breadcrumbs from category to location, "nearby cities" and "related searches" blocks, links from listing detail pages to the landing pages they belong to, and links from editorial pages. In a 2023 post, Adevinta's SEO lead described generating internal links at scale using semantic similarity and search volume, classifying keywords and generating related searches with newer AI models, and using AI-generated meta titles and descriptions to improve click-through (Adevinta Tech Blog, 2023). Generated copy is useful when it describes the actual inventory — price ranges, common brands, counts — from data. It becomes a liability when it is filler; that is the behavior the scaled-content policy targets.

Taxonomy changes are URL migrations

Every category change touches landing pages. The taxonomy chapter (Category Trees That Can Change: Designing Marketplace Taxonomies) owns the change process; this chapter owns what the change does to URLs. When Tradeyard splits bikes into four subcategories, each old landing page — bikes in a city, bikes in a city with a brand, bikes under a price — has one of three fates: it maps to one successor (`301`), it maps to a parent that still makes sense (`301` to the broader page), or it has no meaningful successor (`410`). The redirect registry records each decision, the sitemap service publishes the new pages, and the old ones leave the sitemaps the same day. A taxonomy service that emits a `CategorySplit` event with an old-to-new mapping makes this a pipeline. A taxonomy held in templates makes it a project, which is how The Category Straitjacket shows up in the SEO layer.

Listing Pages Across the Lifecycle

What each state should return

A listing passes through states that The Listing Aggregate: Identity, Lifecycle and Invariants defines: draft, pending, published, paused, sold, expired, removed. The public page has to answer for each of them. No classifieds operator in the evidence for this book published its expired-listing handling, so the policy below is a reference-architecture inference built on Google's documented rules: return `404` or `410` for permanently

removed pages, avoid soft 404s, use `unavailable_after` (as a robots meta tag or `X-Robots-Tag` header) when a page should leave results after a date, and keep `lastmod` accurate (Google Search Central, robots meta tag, updated 2026; sitemaps, updated 2026).

Listing state	Response	Page content	Sitemap	Rationale
Published	`200`, indexable	Full listing	Present, `lastmod` = last material change	The live asset
Paused by seller	`200`, `noindex`	Listing marked paused, related live listings	Removed	May return; should not attract searchers now
Sold or let, low residual demand (most private goods)	`410` after a short grace period; during grace, `200` with `noindex`	"Sold" notice, related live listings, link to the landing page	Removed at once	Links from messages and alerts still land somewhere useful; the index learns quickly that the page is gone
Sold or let, high residual demand (specific car models, popular buildings)	`200` with `unavailable_after` set at sale, for a fixed window, then `410`	"Sold" notice, prominent similar live listings, price context if allowed	Removed at once	The URL may still rank for a specific query; use it to route searchers to live supply, not to pretend the item is available
Expired without sale	As for sold, per vertical	"No longer listed", related live listings	Removed at once	Same logic

Listing state	Response	Page content	Sitemap	Rationale
Removed for policy or fraud	`410` immediately	Minimal page, no listing content	Removed at once	Nothing worth keeping; content may be harmful
Merged into a canonical listing (duplicates)	`301` to the canonical listing	—	Removed	One URL per item; see One Property, Three Listings: Duplicates and Entity Resolution

Two details matter in practice. **The grace period is for people, not crawlers.** Sellers share links in messaging apps and buyers follow links from alerts; a sold page that explains what happened and offers alternatives is better than an error for those visitors. That does not require the page to stay indexable. **Removal latency is a metric.** The time from `ListingSold` to the page returning its new status, leaving the sitemap and being purged from the edge should be measured like publication latency is measured in From Submitted to Searchable: The Publication Pipeline.

Sitemaps from events

Sitemaps should be built from the listing event stream, not regenerated nightly from a database scan. Google's limits shape the design: a sitemap holds at most 50,000 URLs or 50 MB uncompressed, a sitemap index holds up to 50,000 sitemaps, `priority` and `changefreq` are ignored, and `lastmod` is used only if it is "consistently and verifiably" accurate (Google Search Central, updated 2026; sitemap index files, updated 2025).

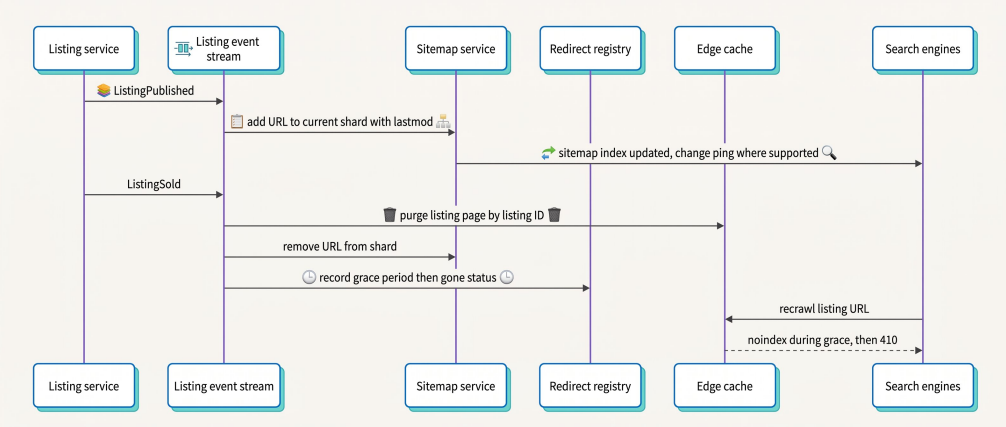


Figure 4: *Sitemaps and removal signals driven by listing events. The same ListingSold event that removes the listing from search purges the edge cache, drops the URL from its sitemap shard and schedules the gone status, so the search engine learns about the sale through every channel at once.*

A practical shard layout for a large site is by page type, country and creation date: new listings append to the current day's shard, so recently changed shards are small and their `lastmod` is meaningful. Landing pages live in their own shards, updated when the registry changes state. Search Console's per-sitemap reporting then shows indexing coverage by page type, which is the diagnostic the SEO team needs.

Push notification beyond sitemaps varies by engine. **IndexNow** lets a site notify participating engines — the protocol site lists Bing, Naver, Seznam.cz, Yandex, Yep and Amazon — when URLs are added, updated or deleted, while warning that submission "does not guarantee immediate indexing" (IndexNow FAQ, n.d.). Google is not listed. Google's Indexing API is documented for job postings: its job-posting guide recommends the Indexing API over sitemaps because it prompts a faster recrawl, and says "we don't allow expired job postings", which must be removed by setting `validThrough` in the past, returning `404` or `410`, or removing the markup (Google Search Central, job postings, updated 2026). For a jobs section such as Tradeyard's in Poland and Portugal, expiry is a compliance rule, not only hygiene.

Structured Data: Saying What the Page Is

Structured data is machine-readable markup — usually JSON-LD using the schema.org vocabulary — that describes what a page is about. For listing sites it has three uses: rich results in search engines, consistent machine-readable facts for other consumers, and a check that the visible page and the data agree.

The available types and the search features they enable change, and the architecture should not depend on any one of them.

- **Product and Offer**. Google distinguishes **product snippets**, for pages where people cannot buy the product directly, from **merchant listings**, for "pages where customers can purchase products from you" (Google Search Central, product structured data, updated 2025). Most classifieds pages fall in the first group. Whether a page with an in-platform payment option, such as Tradeyard Safe Deal, qualifies for the second is a question for the SEO team and counsel, not a default.
- **Vehicle listings**. Google added vehicle listing markup in October 2023, announced on 12 June 2025 that "we will be phasing out support for a few structured data features in Search", and on 9 September 2025 removed the vehicle listing documentation because the type was "no longer shown in Google Search results" (Google Search Central, documentation updates, 2025). An automotive marketplace that built a pipeline for that feature in 2024 found its rich result gone within two years.
- **Real estate**. schema.org has a `RealEstateListing` type, with properties such as `datePosted` and `leaseLength`, in its "new" area, which means the definition is still open to change (schema.org, n.d.).
- **Jobs**. `JobPosting` remains a documented Google feature with strict expiry rules (above).

Generate structured data from the same read model that renders the visible page, in one builder per page type, so a search feature can be added or dropped by changing the builder, not every template. Never put data in the markup that the page does not show, and never put data in it that the page must not show: the precise address of a private seller's home, which Where Is It? Location, Geocoding and Privacy keeps out of every public surface, includes JSON-LD.

Performance: Core Web Vitals as an Architecture Budget

Google's **Core Web Vitals** measure three aspects of user experience at the 75th percentile of page loads, separately for mobile and desktop: **Largest Contentful Paint (LCP)**, good at 2.5 seconds or less; **Interaction to Next Paint (INP)**, good at 200 milliseconds or less; and **Cumulative Layout Shift (CLS)**, good at 0.1 or less. INP replaced First Input Delay as a stable Core Web Vital in 2024 (web.dev, updated 2024).

For listing pages the main threats to those numbers are predictable: large hero images, third-party advertising and tracking scripts, heavy client-side state for filters, and layout shifts from late-loading ad slots. The best-documented marketplace case is QuintoAndar, the Brazilian rental and real-estate marketplace. In a 2025 case study it reported that mobile INP fell from 1,006 ms to 216 ms during 2024 and the share of pages rated "good" rose from 42% to 78%. It linked a 36% year-over-year rise in conversions — scheduled property visits — to the work "strongly, but not solely". Its techniques included yielding

in long tasks, removing third-party pixels and dropping CSS-in-JS, governed through alarms and canary rollbacks (web.dev, 2025).

Three architectural consequences follow.

1. **Performance budgets belong to page owners**. Each URL pattern has an LCP and INP budget in the owning team's objectives, measured from field data. A shared dashboard nobody owns does not move.
2. **Advertising is a budgeted guest**. Display advertising pays for much of a free classifieds site, and header bidding delays ad decisions by design (see Selling Attention: Display, Native and Retail-Media Advertising). Reserve ad slot dimensions to prevent layout shift, load ads after the main content, and give the ads team an explicit share of the performance budget rather than an open line.
3. **Images come from the media pipeline at the right size**. Responsive derivatives, modern formats and explicit dimensions are a Photos Sell: Media Pipelines for Listings responsibility that shows up as LCP.

Performance and crawl capacity are the same budget seen from two sides. A landing page that renders in 400 ms at the origin serves users faster and lets the crawler fetch more pages per day; a page that takes four seconds under load does the opposite on both counts.

Apps: Backends-for-Frontends, Server-Driven UI and Deep Links

Apps and the web do different jobs. At Tradeyard in 2027, the apps account for **57%** of sessions and the web for **43%**. My reading, which no source in this chapter quantifies for other brands, is that apps carry the most engaged users while the web, through organic search, carries most of the acquisition. The architecture must serve both without building every capability twice.

One backend per client family

Tradeyard's apps call Yardstick and about 40 newer services directly, through an API gateway, with versioning negotiated case by case. Every screen makes several calls, and every backend change has to stay compatible with app versions that some users will not update for months. A **mobile BFF** owned by the app teams changes that: the apps call one backend shaped for their screens, and the BFF calls the domain APIs. The web has its own BFF, used by the server-rendered pages. The BFFs hold presentation logic and aggregation, not business rules: if the BFF decides whether a seller may bump a listing, the entitlement rule now lives in two places, which is how Billing by Guesswork starts (see Packages, Bumps and Quotas: Products, Pricing and Entitlements).

The trade-off is duplication between BFFs. I accept some, because the alternative — one general-purpose API for all clients — tends to grow into a lowest-common-denominator interface that every client works around. The rule I use is that **domain APIs are shared and stable**, **BFFs are per client family and allowed to change fast**, and a BFF that starts holding domain rules gets those rules pushed down.

Server-driven forms

Insertion forms in classifieds are defined by the category's attribute schema: a bike has frame size and gears, a car has mileage and first registration, an apartment has rooms and energy class (see Attributes Without Chaos: Schemas, Dictionaries and Validation). If the form is compiled into the app, every schema change waits for app adoption, and a taxonomy change like the bikes split cannot go live on a date. My recommendation, a reference-architecture inference rather than documented practice, is to render insertion forms from the attribute schema served by the backend, so the app ships a form engine rather than forms. The same mapping problem appears in the app-first photo-to-listing features covered in AI Across the Listing Lifecycle.

Deep links and web–app parity

A listing URL shared in a messaging app or clicked in a search result should open the app if it is installed and the web page otherwise. Both platforms support this through verified associations between the domain and the app: Android App Links verify the app's signing certificate against an `assetlinks.json` file on the domain (Android Developers, n.d.), and Apple's universal links rely on an `apple-app-site-association` file (Apple Developer, n.d.). The requirement that is often missed: **the app must understand the same canonical URLs the web uses**, including retired ones, so the redirect registry serves both. An app that cannot resolve a URL the web redirected last month opens on an error screen.

Localization and Multi-Country Domains

Multi-country groups usually run one domain per country; Tradeyard runs four, and Tradeyard Belgium serves Dutch and French speakers with the same inventory. Google's `hreflang` annotations tell the engine which pages are language or regional versions of the same content. They can be declared in HTML, HTTP headers or sitemaps; every version must list itself and all other versions, and "if two pages don't both point to each other, the tags will be ignored" (Google Search Central, updated 2026). For a bilingual landing page — "e-bikes in Brussels" in Dutch and in French — that reciprocity is only reliable when both URLs come from the URL service, keyed on the same category and location identifiers, rather than from translated templates maintained separately.

The landing page registry should be per country, sharing rules and code but not data, because category names, location hierarchies and useful combinations differ by market. One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture owns the tenancy model that makes that configuration rather than code.

AI Search and Answer Engines: A New Channel and a New Crawler

Between 2024 and 2026 the search results page itself changed. Google introduced AI Overviews in Search in May 2024, documented them as replacing the Search Generative Experience, and from June 2025 counted AI Mode data in Search Console's performance totals (Google Search Central, documentation updates, 2024–2025). Assistants such as ChatGPT, Claude and Perplexity answer shopping and housing questions with links to sources.

For a listing site, three points matter.

- **There is no special markup**. Google's guidance says "there are no additional requirements to appear in AI Overviews or AI Mode, nor other special optimizations necessary", and that no special schema.org structured data is needed. Pages must be indexed and eligible for a snippet; `nosnippet`, `data-nosnippet`, `max-snippet` and `noindex` control what is shown (Google Search Central, AI features, updated 2025).
- **Opting out of Google's AI training does not opt out of Search**. The same page says the `Google-Extended` token applies to "AI training and grounding in some of Google's other systems", and that Search crawling is controlled through `Googlebot` rules.
- **Clicks per crawl are falling**. On 1 July 2025 Cloudflare announced that it would block AI crawlers by default for its customers unless they pay for access, and claimed that getting traffic had become far harder: by its estimate, publishers get about ten times less traffic from Google for the same content than a decade earlier, and much less again from AI assistants (Cloudflare, 2025). These are a vendor's figures with a commercial interest, but the direction is consistent with what publishers report.

Classifieds have an advantage that informational publishers lack: **an answer engine cannot complete the job**. Whoever wants the bike still has to contact Femke, and whoever wants the apartment still has to book a viewing. The defensible asset is fresh, contactable inventory, which an answer cannot copy without going stale. Classifieds evidence is thin: Adevinta's 2023 post saw no drastic click loss after Bing's AI snippets, but that predates AI Overviews.

The architectural moves are practical: keep landing and listing pages fast, accurate and crawlable for the engines that send buyers; decide per crawler, as policy, which automated agents may read the site (see Sunday Evening Peaks: Reliability, Performance and Cost for bot management); measure referrals from AI assistants as their own channel; and treat structured, licensed access for agents as a product question that AI Across the Listing Lifecycle owns. Tradeyard's 2027 numbers are a reminder of scale: assistants refer about **0.7%** of its web sessions, while self-declared AI crawlers make about **11%** of requests for public HTML.

Veldmark Worked Example: Rebuilding Tradeyard's SEO Inventory

Ingrid does not approve the client-rendered rebuild. In May 2027 the architecture advice forum adopts a decision record for Tradeyard's web layer with five parts, and the web and SEO work is scheduled ahead of the Yardstick search extraction planned for Q4 2028, because the web layer is where the strangler starts (see Changing the Engine in Flight: Modernizing a Live Marketplace).

1. **Server-rendered page applications per URL pattern**, React with server-side rendering and streaming, owned by the discovery, listing and supply teams, behind CloudFront route rules. Yardstick keeps rendering any page type that has not moved.
2. **A web BFF and a mobile BFF**, owned by the web and app teams, over the domain APIs.
3. **The four SEO services** — URL and canonical, landing page registry, redirect registry, sitemap and notification — built once and configured per country.
4. **Listing lifecycle rules** as in the table above, driven by the Common Ground listing event contract v1, which goes live on 1 July 2027 (see Every Change Is an Event: Event-Driven Listing Architecture).
5. **Performance budgets** for landing and listing pages: mobile LCP p75 at or under 2.5 s, INP p75 at or under 200 ms, owned by the page teams.

The landing page registry starts from data. The SEO team takes 14 months of internal search logs, saved searches and external keyword data and scores every category × location × attribute combination with recurring demand. It applies the supply rule — at least five live results on most days over the previous eight weeks — and the depth rule, at most two attributes beyond category and location. About **2.4 million** combinations across the four countries qualify. They become the indexable set. The remaining 17.4 million search URLs that were in the index get `noindex, follow` if they are reachable from the user interface, `404` if they have no results, and drop out of all sitemaps.

The old gone-listing pages are handled in one pass and then by the event-driven rules: about 6.3 million pages move to `410`, except roughly 180,000 that the SEO team identifies as still ranking for specific queries — mostly sold cars and cargo bikes — which keep a `200` page with similar live listings and an `unavailable_after` date. Femke's bike falls under the general rule: under the new policy its page would have answered `noindex` from 18:11 on 8 March and `410` from 22 March.

Measure	February 2027 (before)	February 2028 (after)
Indexed URLs, all Tradeyard domains	40.0M	15.9M
Sold, expired or removed listing pages indexed	6.3M	0.4M
Share of Googlebot fetches on facet or dead URLs	41%	9%
Median time to first crawl, new NL listing	2.6 days	0.6 days
Organic sessions, all domains (index, Feb 2027 = 100)	100	104
Contacts from organic sessions (index, Feb 2027 = 100)	100	117
Mobile LCP p75, landing pages	3.4 s	2.2 s
Mobile INP p75, landing pages	310 ms	180 ms

The number that went down — indexed URLs, by 60% — is the one the board pack used to celebrate. The number that matters went up: **contacts from organic sessions rose 17%** while organic sessions rose only 4%, because searchers landed on pages with live inventory. The forum also changes the board pack: indexed URLs disappear from it, and organic contacts per thousand organic sessions replaces organic sessions as the headline SEO metric. That is the Vanity Liquidity correction applied to SEO.

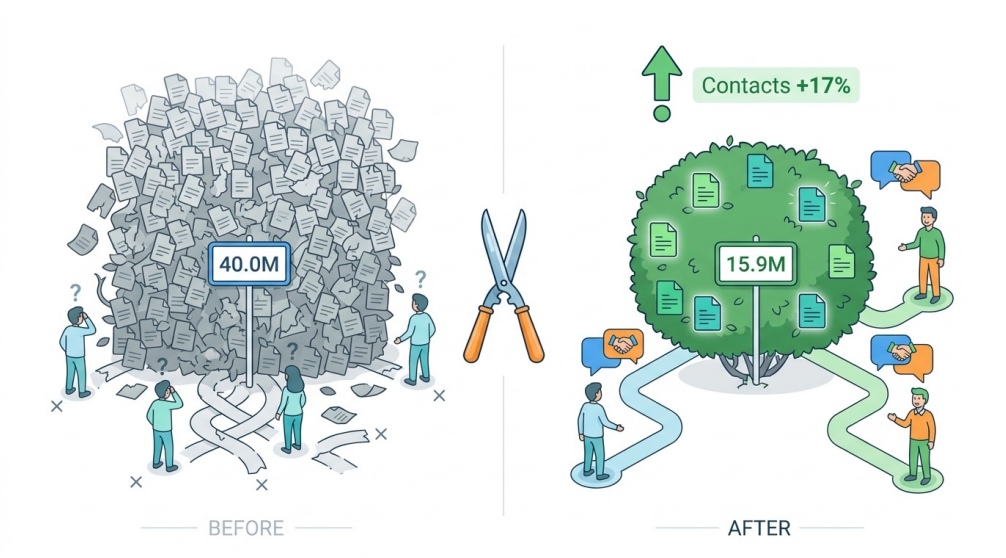


Figure 5: *Tradeyard cut its indexed URLs from 40.0 million to 15.9 million, and contacts from organic sessions rose 17%, because searchers now land on pages with live inventory.*

The bikes split on **1 November 2027** is the first test. The taxonomy service emits the split with its mapping; the URL service derives about 18,400 old bike landing URLs in the Netherlands and Belgium and their candidates; the SEO team reviews about 1,100 ambiguous cases by hand (a "bikes under €200 in Utrecht" page has no single successor and is redirected to the broader city page); and the redirect registry, sitemaps and new landing pages go live with the category. Organic sessions to bike landing pages dip by about 8% in the first three weeks and recover by mid-December. Under the old architecture the same change would have needed template edits in Yardstick for each country and a spreadsheet of redirects.

Trade-offs

Decision	Option A	Option B	Choose A when	Choose B when
Rendering of public pages	**SSR with hydration**	Client-side rendering	Pages must rank or be shared; mid-range phones matter	Logged-in tools with no SEO role

Decision	Option A	Option B	Choose A when	Choose B when
Composition	**Page-level ownership, build-time shared components**	Runtime fragment composition	Page types map to teams; consistency matters	A single page genuinely needs independently deployed parts from many teams, and you can fund the consistency work
Landing page breadth	**Rule-based allow-list**	Index every reachable combination	Large inventory; crawl budget is binding	Small site where crawl budget is not a constraint
Gone listing pages	**`410` after grace; `200` only for proven demand**	Permanent `200` "sold" pages	Most verticals with short-lived, unique items	Verticals where sold-item pages are a product (price history, model pages) with clear unavailability signals
App backend	**BFF per client family**	One shared public API	Several clients with different release cycles	One client, or an external API product for partners
Structured data	**Builder per page type from the read model**	Markup in templates	Search features change and several page types share data	A very small site

Decision	Option A	Option B	Choose A when	Choose B when
Edge caching of landing pages	**Cached with minutes of TTL and event purges**	Rendered live per request	High read volume, inventory tolerant of minutes of staleness	Very low traffic or results that must be exact
AI crawlers	**Explicit per-agent policy**	Default allow or default block	Always; the default should be a decision	—

The trade-off that generates the most argument is the gone-listing policy, because a `200` sold page does capture some organic traffic. The question is what that traffic converts into. At Tradeyard, sold and expired pages drew 6% of organic sessions and 0.4% of organic contacts. In automotive and property, where people search for specific models and buildings, the numbers can justify keeping a curated subset; the evidence should decide, page group by page group.

Vertical Differences

Concern	General classifieds	Real estate	Automotive	Jobs
Main landing page dimensions	Category × city, with one attribute (brand, type)	Transaction type × property type × area, with rooms or price band	Make × model × city, with fuel or body type	Role × city, with contract type
Listing lifetime and detail page value	Days; detail pages matter little after sale	Weeks to months; detail pages earn long-tail traffic	Weeks; specific model searches persist after sale	Weeks; expiry must be prompt

Concern	General classifieds	Real estate	Automotive	Jobs
Gone listing rule	`410` after short grace	`410` or sold page per market practice and law; the same property may return with a new listing	Curated sold pages for proven demand, then `410`	Required removal: `validThrough`, `404` or `410`
Duplicates	Reposts; canonicalize to the newest	Several agencies list one property: one canonical URL per cluster (see One Property, Three Listings: Duplicates and Entity Resolution)	Dealer feeds across brands	Same job via several job boards
Structured data	Product snippet	`RealEstateListing` (schema.org "new" area)	Vehicle listing feature retired by Google in 2025	`JobPosting`, with strict policy
Location precision on public pages	Neighborhood	Address only with seller consent; otherwise area	Dealer address public; private seller area only	Employer location
Performance pressure	Ads-heavy pages	Many large images, maps	Many images, finance calculators	Light pages

Real estate needs one more rule: when several agencies advertise the same apartment,

the cluster gets one canonical URL, through canonical tags or a property page. Otherwise the Lyon apartment listed by three agencies in May 2027 would produce three detail pages competing for the same queries. Choosing which listing is canonical is commercially sensitive, because agencies pay for visibility. The rule should be published and neutral; Hearthly's rule picks the listing with the highest completeness and media-quality score, never paid status, and it is owned with the clustering rules in One Property, Three Listings: Duplicates and Entity Resolution.

Failure Modes

- **The soft-404 graveyard**. Gone listings answer `200` with "no longer available". Crawlers keep fetching them, sitemaps keep listing them, and searchers keep landing on them. This is Ghost Inventory at the search-engine layer, and it hides because no user-facing metric moves.
- **The infinite facet space**. Every filter combination has a crawlable URL with parameters in arbitrary order. Crawlers spend their budget on red bikes between €110 and €115, and new listings wait days to be discovered.
- **The SPA cliff**. A redesign moves public pages to client-side rendering. Indexing of content lags, rankings drop over several weeks, and the cause is found only after the traffic is gone.
- **The template redirect spreadsheet**. Category and location changes are handled by hand-maintained redirect lists in templates. Old URLs break, chains of redirects build up, and teams avoid taxonomy changes.
- **Doorway drift**. Landing page generation is tuned for coverage. Pages for small towns with borrowed inventory and generated filler accumulate until the site's quality signals suffer.
- **The app that cannot follow**. A taxonomy or schema change ships on the web, while app versions in the field cannot render the new form or resolve the new URLs.
- **The vanity index count**. Indexed URLs and organic sessions are reported as success; the share of organic sessions that produce a contact is not reported at all.

The Chief Architect's View

What to share. The four SEO services, the event-driven sitemap pipeline, the redirect registry pattern, the structured-data builders and the lifecycle rules are good group capabilities: they are rules plus data, configured per brand and country. The page applications, design and copy are not. Under Common Ground's principle — share

platforms, not products — Tradeyard, Motorvane and Hearthly keep their own pages, brands and SEO strategies, while running on shared services that make the rules explicit.

What to decide centrally. Three decisions should be group policy recorded in decision records: the rendering standard for public pages (server-rendered), the gone-listing rules per vertical, and the per-agent crawler policy, which has commercial and legal implications beyond SEO. Everything else — landing page dimensions, thresholds, copy — belongs to each brand's SEO and product teams.

Who owns what. Each URL pattern has one owning team. The SEO team owns the rules, the registry and the measurement, not the pages. The platform team owns the edge, the URL service and the sitemap pipeline. The taxonomy team must not be able to ship a category change without the redirect registry being updated in the same release; that dependency is enforced in the pipeline, not by memory.

Migrations are SEO events. Brand migrations, domain consolidations and monolith extractions all change URLs or responses at scale. After the Deal: Consolidating Platforms After Acquisitions and Changing the Engine in Flight: Modernizing a Live Marketplace, in Part VIII — LEADERSHIP: The Chief Architect's Agenda, own those programs; the Chief Architect's job is to require a URL migration plan, a redirect registry and a traffic baseline before any of them starts.

What to watch. Organic contacts per thousand organic sessions, by brand and page type; the share of crawl spent on indexable URLs; time to first crawl for new listings; removal latency from sale to gone status; Core Web Vitals at the 75th percentile by page type; and AI-assistant referrals as a separate channel. Indexed URL count belongs on an operational dashboard, never in a board pack.

Questions to Ask

1. Which URL patterns exist, which team owns each one end to end, and where is that ownership recorded?
2. What rule decides whether a search result page is indexable, how often is it recomputed, and how many pages are in each state today?
3. What does a listing URL return in each lifecycle state, and how long after a sale does the page reach its final status, leave the sitemap and leave the edge cache?
4. Are sitemaps generated from listing events with accurate `lastmod`, and can we see indexing coverage by page type?
5. What share of crawler fetches land on indexable URLs, and what is the median time to first crawl of a new listing?
6. Which public pages are rendered on the client, and what would they look like to a crawler that does not execute JavaScript?
7. How is a category split or a city rename turned into redirects, and can the taxonomy change ship without them?

8. Who owns each page type's Core Web Vitals, and what share of the performance budget is reserved for advertising and third-party scripts?
9. Can the apps resolve every canonical and retired URL the web uses, and how long does a form change take to reach most app users?
10. What is our policy for each AI crawler and assistant, who decided it, and how do we measure the traffic they send?

From Pages to Data

The customer-facing layer is where a marketplace meets its two most demanding clients: people on phones and the crawlers that send them. Both are served best by the same architecture — fast server-rendered pages owned by clear teams, URLs derived from domain identifiers, and lifecycle rules driven by the listing event stream. The decisive change at Tradeyard was not a framework but a metric, organic contacts instead of indexed URLs. Defining such metrics consistently across brands — what a "contact" is, which session counts as organic, how page views are tracked with consent — is the work of the data platform, which Data Platform for Marketplaces: Events, Lakehouse and Data Products takes up next.

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Research and practitioner sources

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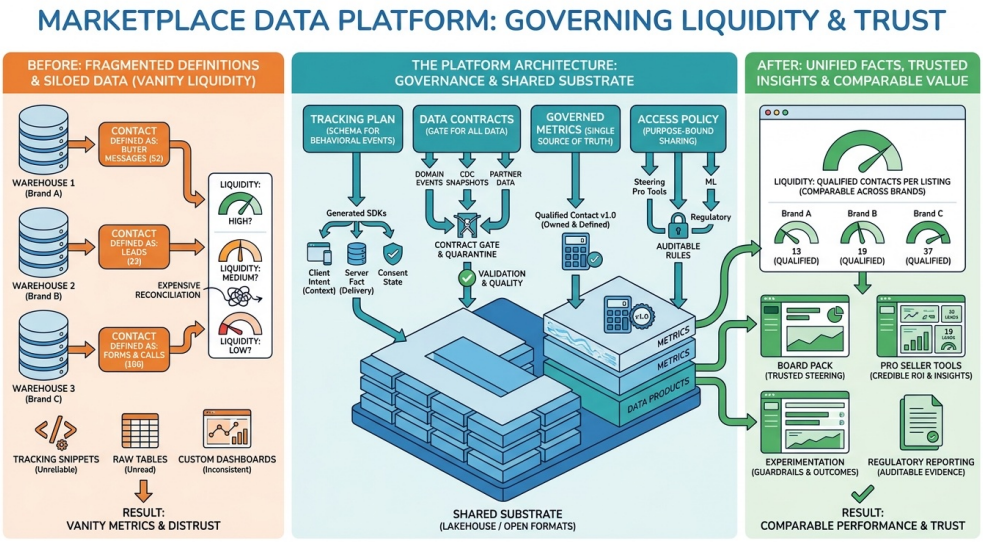
Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

36. Data Platform for Marketplaces: Events, Lakehouse and Data Products



IN THIS CHAPTER, YOU WILL: Follow Veldmark's attempt to answer one board question, "how many contacts did we create last week?", across three warehouses and three definitions. Along the way you will see how a marketplace data platform should collect behavioral and domain data, govern it with contracts and consent, organize it into data products, publish liquidity metrics with one definition, serve data to professional sellers, and stay inside the legal limits on sharing data across brands.

WHY THE BUSINESS CARES: Every decision a marketplace general manager makes, from price rises to category launches to the next acquisition, is argued with liquidity numbers. If each brand counts a "contact" differently, the group cannot compare its marketplaces, cannot sell credible return on investment (ROI) to dealers and agents, and cannot tell whether a change created matches or only moved them around.

WHY ARCHITECTS SHOULD CARE: The data platform is the one place where every bounded context's data meets, so it is where inconsistent models, missing consent and unowned definitions become visible and expensive. Its contracts (the tracking plan, the data contracts, the metric definitions and the access policy) decide what experiments, models, pro products and regulatory reports can be built, and how cheaply.

KEY POINTS:

- **The tracking plan is the real schema of a marketplace**. Behavioral events from web and apps carry the evidence of liquidity (impressions with their ranking context, views and contacts by channel), but they ship in app binaries nobody can recall. Generate them from one governed plan, carry the listing ID, the ranking and experiment context and the consent state on every event, and repair on the server what clients get wrong.
- **Liquidity metrics must be governed metrics**. A "contact" needs one owner, one written definition, one computation and one published version, served to dashboards, experiments and pro reports alike. Until then each brand optimizes its own number, and the group optimizes nothing it can compare.
- **Federate ownership, centralize the substrate and the rules**. Domains own data products on a shared lakehouse with contracts, quality gates and a separate store for personal data. Cross-brand sharing is a legal and commercial decision about purpose, so it belongs in access policy that can be audited, not in the goodwill of whoever has warehouse credentials.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Vanity Liquidity** — Without a governed definition of a contact, brands report message counts, page views and listings, and the group steers on the numbers that are easiest to grow.
- **Deal-Shaped Architecture** — Three warehouses with three event vocabularies, inherited from three acquisitions, make every cross-brand question a reconciliation project.

Most marketplace data platforms are built backwards. A warehouse arrives first, because finance needs revenue reporting. Tracking arrives next, because product managers want funnels, and it is usually a vendor snippet that records whatever the front-end team names an event that sprint. Machine learning arrives third and discovers that impressions were never logged with their positions. Professional-seller statistics arrive last, computed in yet another place, and dealers notice that the leads in their dashboard

do not match the leads in their inbox. **By then the platform holds a great deal of data and very few facts**.

This chapter takes the opposite route, applying Part VII's rule to data: a shared platform earns trust through contracts. It starts from what a listing marketplace needs to know: whether fresh, trustworthy supply is turning into matches with demand (Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics defines the metrics). Part VI priced, billed and tested against contact counts, and Found by Google: Web, App and SEO Architecture for Listings steered Tradeyard on organic contacts; this chapter makes "contact" mean the same thing in every brand. It works back to the contracts that make such facts trustworthy: a **tracking plan** for behavioral events, **data contracts** for everything that enters the platform, **governed metrics** for what the business steers on, and **purpose-bound access policy** for what may be combined. Technology choices (lakehouse or warehouse, one engine or two) matter less than those contracts and come after them.

Two neighbors own adjacent ground. Every Change Is an Event: Event-Driven Listing Architecture owns the listing event stream and its schema evolution; this chapter consumes that stream. System of Record and Read Models: Listing Data Management owns the system of record, read models and erasure propagation; this chapter applies erasure to analytical copies. The next chapter, Testing a Two-Sided Market: Experimentation Platforms, builds on the exposure and metric layers designed here, and Models in Production: ML Platforms and Use Cases builds on its training data.

Setting the Scene at Veldmark: Three Warehouses, Three Contacts

On Monday 15 March 2027, Henrik de Boer, Veldmark's Group CEO, sends a short e-mail to his leadership team. Veldmark is the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build). The board pack for the first quarter will, for the first time, show "contacts per live listing" for every brand, because the investor's operating partners want one liquidity number that can be compared across Tradeyard, Motorvane and Hearthly. He asks for last week's figures by Wednesday.

Priya Raman, Head of Data & ML, already knows what will happen. Tradeyard and Motorvane run Snowflake, in separate accounts with separate models. Hearthly runs BigQuery. Keybridge runs Azure Synapse. Each brand has a "contacts" table and each table is correct by its own definition:

Brand	What its dashboard calls a contact (March 2027)	Source
Tradeyard	Every message a buyer sends, plus every phone-number reveal	Messaging database export and web tracking
Motorvane	A "lead": an e-mail form, a tracked call of any length, or a test-drive booking, counted per channel	Lead service and call-tracking vendor
Hearthly	A contact form or phone reveal, de-duplicated per user, listing and day, excluding new-build campaign landing pages	Web and app tracking in BigQuery

On Wednesday, Ingrid Halvorsen, the Group Chief Architect, puts two of the book's recurring examples on the screen. Femke's €240 Rotterdam city bike, listed on Sunday 7 March, received **14 conversations within 24 hours**; Tradeyard's dashboard counts **52 contacts** for it, because it counts buyer messages. One of the 14 "buyers" was the phisher who sent a fake Safe Deal payment link at 20:11 and was suspended at 20:26. The Essen Volkswagen Golf, listed by Autohaus Ruhrblick on 14 April, will receive **23 leads in 10 days** (14 e-mails, 7 calls, 2 test-drive bookings); three of the e-mail senders will also call, and one test-drive booking will come from a buyer who had already e-mailed. Motorvane will report 23, the dealer will feel he heard from 19 people, and a fraud review would want to know whether any of them should not count at all.

"We do not have a data problem," Ingrid tells the room. "We have three definitions of the most important event in the business, three event vocabularies, and nobody who owns the word." The rest of the chapter is the design that came out of that meeting.

What a Marketplace Data Platform Is For

A **data platform** is the shared set of pipelines, storage, compute, metadata and access controls through which a group's data is collected, governed and used for anything other than serving the live product. It is not the system of record for listings, accounts or payments; those live in their bounded contexts (Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models). It is where their facts meet, and its value is the questions no single context can answer.

A listing marketplace needs six kinds of output from its data platform, each with its own requirement:

Consumer	What it needs	The requirement it imposes
Steering (board, general managers, finance)	Liquidity, revenue and cost metrics comparable across brands and periods	Governed metric definitions; restatement rules; reconciliation with finance
Product analytics and experiments	Funnels, cohorts, exposure and outcomes per user	A tracking plan with stable identifiers, experiment assignment on events, low latency for guardrails (Testing a Two-Sided Market: Experimentation Platforms)
Machine learning	Training sets with features as they were at decision time, labels such as contact, sale or fraud	Impression and ranking-context logging; point-in-time correctness (Models in Production: ML Platforms and Use Cases)
Professional-seller data products	Listing performance, market prices, demand by area, ROI	Externally served, tenant-isolated, aggregated data with service levels (Serving the Pros: Dealer, Agent and Merchant Tooling)
Regulatory and compliance reporting	Notice-and-action statistics, statements of reasons, seller reporting under DAC7, erasure evidence	Auditability and lineage; data held for legal retention periods (Rules of the Marketplace: DSA, DAC7 and Vertical Regulation)
Group data and partner products	Valuation data, price indices, advertising audiences	Purpose limitation and consent; contracts with buyers of data (Selling Attention: Display, Native and Retail-Media Advertising)

The table is the argument for treating the platform as architecture rather than as tooling. A warehouse that serves only steering can tolerate a weekly batch and a loose event vocabulary. One that must also train ranking models, monitor experiment guardrails in near real time, serve 21,400 dealers and prove to a regulator what happened to a listing cannot.

Three kinds of data, three different contracts

Data reaches the platform in three forms, and confusing them is the root of most marketplace data quality problems.

Kind	Example	Producer	What it can tell you	Contract owner
Domain events	`ListingPublished`, `PriceChanged`, `ListingSold`	Backend services, through the outbox (Every Change Is an Event: Event-Driven Listing Architecture)	Facts about state changes, authoritative and ordered per listing	The producing bounded context
Change data capture (CDC) snapshots	Row changes from Yardstick's MySQL	Database log, via Debezium or similar	The current state of legacy tables, including fields nobody modeled as events	Nobody, by default; the schema becomes a de facto public contract
Behavioral events	Search performed, results shown, listing viewed, phone reveal tapped	Web and app clients, plus server-side search and contact services	What people saw and did, which no domain database records	The tracking plan's owner

Domain events tell you that a listing went live at 19:48; only behavioral events tell

you that it was shown at position 3 to a buyer in Rotterdam who opened it and tapped "message". **Liquidity lives in the join between the two**. A data platform whose behavioral events do not carry the same listing identifier, category version and experiment context as its domain events cannot compute liquidity at all; it can only count.

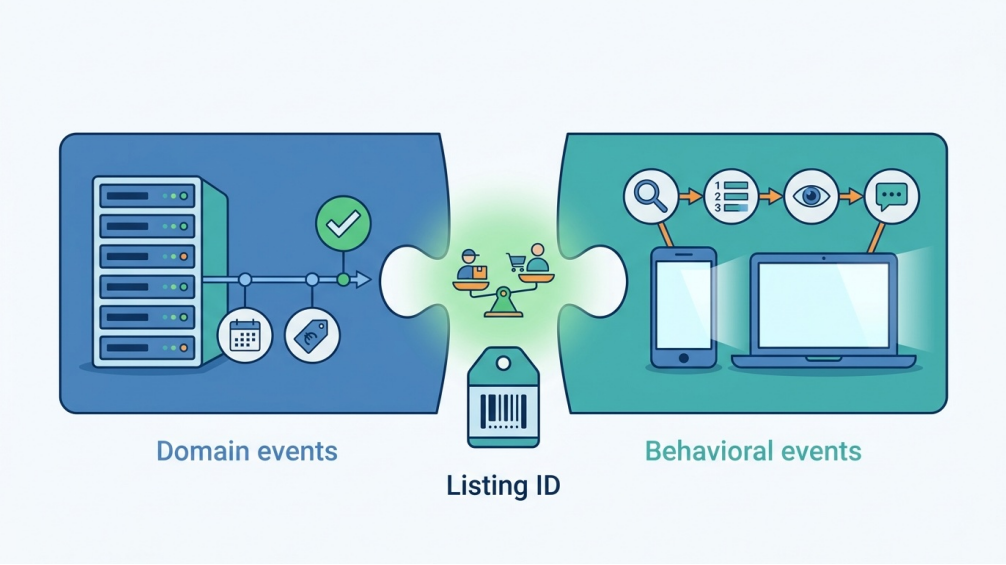


Figure 1: *Liquidity lives in the join: domain events say what happened to a listing, behavioral events say who saw it and contacted its seller, and only a shared listing identifier and context turn the two into a liquidity measure.*

CDC should be temporary in the data platform too. It gets Yardstick's history into the lakehouse without touching the monolith, but every CDC-fed table should carry a planned replacement by a domain-event-fed one, because the monolith's schema was never designed to be read by analysts. Every Change Is an Event: Event-Driven Listing Architecture makes the same case for operational consumers: raw-table CDC is the strangler's tool and the outbox its successor.

The Tracking Plan: The Schema Nobody Can Recall

A **tracking plan** is the governed catalog of behavioral events a product emits: each event's name, trigger, required properties and their types, identifiers, consent category and owner, versioned like any other contract. It is the most important and the most neglected schema in a marketplace, for one structural reason: **client events ship inside app binaries**, and a mobile release that emits a malformed event cannot be called back.

Vinted and Zalando both found tracking schemas to be a precondition for trustworthy

data. In Vinted's 2019 description of its ingestion pipeline, mobile releases took about two weeks to reach users, so Vinted generated Kotlin and Swift tracking code from schemas to stop malformed events at the source, while web events, which lacked that safeguard, were patched in flight by the relay (Vinted Engineering, 2019; the current pipeline is not described publicly). Zalando's account of its experimentation platform names inconsistent tracking data as a cause of corrupted analyses and credits a dedicated team that "unified the schema" (Zalando Engineering, 2021).

At group scale, Adevinta's 2022 engineering blog describes its data platform carrying nearly a billion events a day classified as User, Behavioural, Content and Transaction (UBCT) events, with Kafka as "the backbone of the data platform" and topics in both Avro and JSON (Adevinta Tech Blog, 2022; older evidence, and the current platform may differ). It is the only cross-marketplace event classification from a classifieds group found in the research for this book, and the mixed formats show how hard a single standard is to impose on many marketplaces at once.

What a marketplace tracking plan must contain

A marketplace's tracking plan is not a web-analytics plan with listings added. It must record the two-sided mechanics: what supply was shown to whom, in what order and why, and what contact followed. The core events below are the minimum I would accept for a group plan; brands extend them, but may not redefine them.

Event	Fired by	Required properties beyond the common envelope	Why it is core
`search_performed`	Search service (server side)	Query text or filter set, category and taxonomy version, location and radius, sort, result count, `search_id`	Denominator of search-to-contact; zero-result monitoring

Event	Fired by	Required properties beyond the common envelope	Why it is core
`results_impression`	Client, per rendered batch	`search_id`, listing IDs with positions, `ranking_id` and model version, paid placement flags	Position bias correction, ranking training, paid-organic exposure audit (Ranking for Two Sides: Relevance, Freshness and Paid Visibility)
`listing_viewed`	Client and server	Listing ID, source surface (search, alert, recommendation, SEO landing page, external), `search_id` if any	Attribution of views to discovery surfaces
`contact_initiated`	Client	Listing ID, channel (chat, form, phone reveal, call, booking), `contact_attempt_id`	Intent; drop-off before delivery
`contact_delivered`	Messaging and lead services (server side)	`contact_attempt_id`, conversation or lead ID, seller ID, channel, delivery time	The authoritative contact fact (From Interest to Contact: Messaging and Lead Management)

Event	Fired by	Required properties beyond the common envelope	Why it is core
`saved_search_created`, `alert_opened`	Server and client	Saved-search ID, alert ID, listing IDs in the alert	Alert contribution to liquidity (Tell Me When It Appears: Saved Searches and Alerts)
`listing_submitted`	Insertion flow (client)	Draft ID, step reached, category prediction accepted or not	Insertion funnel conversion
`experiment_exposure`	Assignment service	Experiment ID, variant, unit type and ID	Joins every event to the experiments that could have influenced it

Every event carries a **common envelope**: event ID (for de-duplication), event time and receive time, the brand and country tenant, an anonymous device or session identifier, the pseudonymous user identifier when signed in (after 2028, the Veldmark ID subject), the app or web build, the **consent state** at the moment of the event, and the IDs of active experiment assignments. Two design rules follow from the table.

1. **Facts are server-side; intentions are client-side.** A contact is counted when the messaging or lead service delivers it, not when a client says a button was tapped. The client event records intent and context; the server event records the fact. This one rule removed the largest disagreement between Veldmark's brands, because clients lose events to ad blockers, crashes and missing consent, while servers do not.
2. **Impressions are expensive and indispensable.** Logging every rendered listing with its position multiplies event volume, which tempts teams to sample or drop impressions. Do not, at least for search and recommendation surfaces: without impressions there is no position-bias correction, no unbiased ranking training set and no way to show a regulator or a dealer how paid placements were exposed. Compact them instead (one event per rendered batch, listing IDs as an array) and tier their retention.

Consent on the event, not in the job

The consent state must travel with the event, because EU law limits both what may be collected and what it may be used for. In the European Union, storing or reading information on a user's device (cookies, local storage, device identifiers) requires consent unless it is strictly necessary for a service the user requested, under Article 5(3) of the ePrivacy Directive 2002/58/EC as amended; analytics cookies are generally not "strictly necessary", although national regulators differ on narrow audience-measurement exemptions. The General Data Protection Regulation's purpose-limitation principle (GDPR Art. 5(1)(b)) then limits what the collected data may be used for. The consent mechanics for advertising, including the industry's Transparency and Consent Framework, belong to Selling Attention: Display, Native and Retail-Media Advertising.

Adevinta Spain's data contracts apply the same discipline to personal fields in production: each contract lists the personal fields and flags which have analytical value, fields without analytical value are never ingested, and analytical personal data is split into a separate per-site personal table, an arrangement the 2024 post says makes right-to-be-forgotten requests easier to handle (Adevinta Tech Blog, 2024).

Consent makes client-side liquidity counts biased. If 30% of web sessions decline analytics consent, client-side funnels are biased samples, and the bias differs by country, device and audience. Server-side facts (searches performed, contacts delivered, listings published) can usually be counted in aggregate without device storage, subject to legal review per country, so **the governed liquidity metrics should be built from server-side facts**, and client-side events used for behavior and attribution, with their coverage reported next to every number.

Architecture: One Substrate, Many Owners

The reference architecture below has five layers and one rule: every arrow into the platform passes through a contract. It describes no company's actual systems.

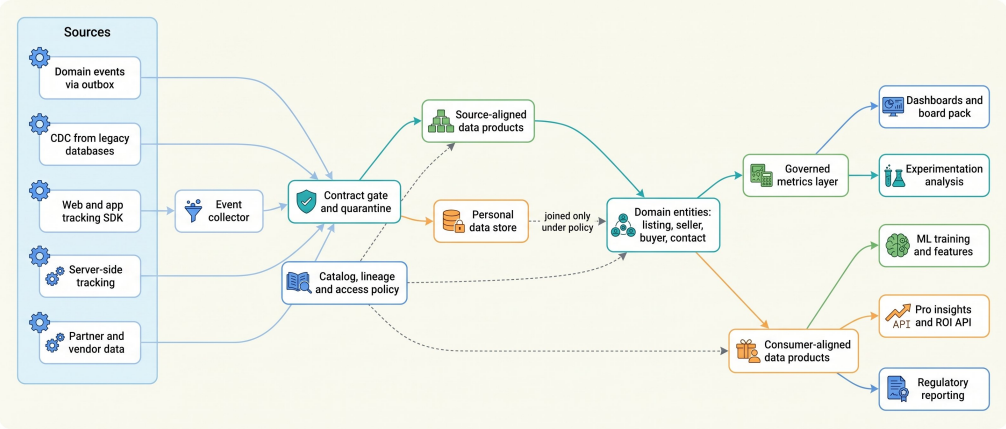


Figure 2: *A reference data platform for a listing marketplace. Every source passes a contract gate; personal data is stored apart and joined only under policy; governed metrics are computed once and consumed by dashboards, experiments and pro reporting alike.*

Layer	Responsibility	Owner	Interface
Collection	SDKs generated from the tracking plan; an event collector that never blocks the product's request path; server-side emitters	Platform team (collector, SDK generator); product teams (instrumentation)	Tracking plan versions; collector endpoint
Contract gate	Validate every record against its contract version; route failures to quarantine with the raw payload and error; split personal fields	Platform team	Data contracts in a repository; quarantine tables per source

Layer	Responsibility	Owner	Interface
Source-aligned data products	Faithful, deduplicated, typed copies of each source, partitioned by time and tenant	The producing domain	Tables in open formats; freshness and completeness service levels
Domain entities and metrics	Cross-source entities (listing, seller, buyer, contact, search session) and the governed metrics computed from them	Domain data owners, with a central metrics steward	Entity tables; a metrics layer with versioned definitions
Consumer-aligned data products	Purpose-built datasets for business intelligence (BI), ML, pro products, regulatory reports and partners	The consuming team, under policy	Tables, APIs, feature pipelines, exports

The layering mirrors the bronze, silver and gold "medallion" layout. Adevinta Spain's 2024 account shows a marketplace mapping those layers onto data products: bronze as "a space for source-aligned data products" fed by declarative contracts, silver as aggregated "one big tables" for domain entities such as user, buyer, seller and publications, and gold as consumer-aligned products for BI and AI, with the verticals (real estate, jobs, motor, generalist) as domains (Adevinta Tech Blog, 2024). The same post warns that wide, sparse domain tables bring "additional privacy and governance challenges as the table grows", a warning for anyone tempted to build one big listing table for every vertical.

The contract gate

A **data contract** is a versioned, machine-readable agreement between a data producer and the platform: schema, semantics, personal-data classification, ownership, freshness and retention. The most concrete public example in classifieds is again Adevinta Spain's. Its contracts are immutable JSON files in GitHub, each naming the schema by URL with an X.Y version, the validity window, the list of personal fields with an analytical flag, the key that links personal to non-personal rows, and service levels covering owner, support contact, periodicity, recovery time and retention. Airflow pipelines are generated from

the contracts, one Spark job runs per contract, and records that fail validation go to a quarantine table with the raw event, the error and the contract version, used for debugging and never re-ingested. Generating the pipelines from contracts cut the time from contract definition to available data, which the team calls "Time to Effective Data", from about five days to about 30 minutes (Adevinta Tech Blog, 2024).

Three properties of that design are worth copying. **Contracts are immutable**, so a change is a new version with an overlap window, and consumers can see what changed. **Quarantine is a feedback channel to producers**, not a dumping ground, which is the data equivalent of the feed error reports Many Doors In: Forms, APIs and Bulk Feeds recommends for partner feeds. And **personal data is separated at the gate**, so erasure and access control apply to one well-known store instead of every derived table.

Lakehouse, warehouse or both

For a marketplace group the decision is less "which product" than **where the durable copy of the data lives and in what format**. A **data warehouse** stores data in an engine's managed format and couples metadata, storage and query optimization tightly (Snowflake, BigQuery, Synapse in Veldmark's landscape). A **lakehouse**, in the definition of its originators, is an architecture "based on open direct-access data formats, such as Apache Parquet", with "first-class support for machine learning and data science" and warehouse-grade performance (Armbrust, Ghodsi, Xin and Zaharia, CIDR 2021). In practice the line has blurred: open table formats such as Apache Iceberg and Delta Lake add transactions and schema evolution to files in object storage, and the major warehouses now read and write some of them.

Option	Strengths	Weaknesses	When it is right
One managed warehouse	Simplest operations; strong SQL and BI; governance features built in	Storage format tied to the vendor; ML and large event volumes can be expensive; acquisitions on other clouds must migrate or copy	A single-brand or single-cloud group with modest ML needs

Option	Strengths	Weaknesses	When it is right
Lakehouse on open table formats	Durable copy in open formats; several engines over one copy; cheaper storage for high-volume events; portable through carve-outs	More platform engineering; governance must be assembled; query performance depends on table maintenance	Multi-cloud groups, heavy behavioral and ML workloads, groups that expect to buy and sell brands
Open storage plus a warehouse as main compute	Open, portable storage with the warehouse's SQL and governance for most users	Two sets of permissions to keep aligned; cost visibility split across storage and compute	Most marketplace groups in 2027, including Veldmark
One warehouse per brand, federated queries	No migration; brand autonomy	Every cross-brand question is a federation or copy problem; three definitions of everything	Only as a transitional state, with an end date

The portability argument deserves more weight in marketplaces than in most industries, because marketplace groups change shape. The 2025 strategic report of Adevinta's parent company, Aurelia UK Feederco Ltd, describes the group moving to "a decentralised, standalone operating model" with "the localisation of intellectual property" and a temporary cost spike from "parallel running of central and local services during a transitional period" (Strategic Report 2025, published 2026); the Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform chapter covers that history. **A data platform whose durable copy is in open formats, partitioned by tenant, is one that can be split when a brand is sold**, and one that can absorb an acquired brand without first migrating its warehouse.

Data Products: Federate Ownership, Not Substrate

A **data product** is a dataset (or a small family of them) that a named team owns and publishes for others to use, with a contract, documentation, quality and freshness

service levels, access policy and a support channel. The term comes from Zhamak Dehghani's 2019 essay on the data mesh, which argued that domains should "host and serve" their datasets, treat "their data assets as their products" with other teams as customers, rely on a platform that provides "the data infrastructure components in a self-service manner", and agree on standards that "should belong to a global governance, to enable interoperability between polyglot domain datasets" (Dehghani, martinfowler.com, 2019).

Marketplaces map onto this naturally, because their bounded contexts already exist: supply, catalog, search, trust and safety, messaging, monetization. The documented classifieds cases show a consistent shape, though not a uniform outcome.

- **Adevinta, group level**. The 2019 annual report described data and analytics as an explicitly centralized domain, "built once; used many times" (Adevinta Annual Report 2019, published 2020). By 2023 Adevinta Spain was describing a move "from a data platform to a data products platform", defining data products as autonomous, read-optimized, standardized units (Adevinta Tech Blog, 2023), and by December 2024 a three-year path from a Databricks lakehouse to data mesh, with in-house frameworks for distributed pipelines, non-distributed pipelines, SQL and dbt transformation ("Transform It Yourself") and prototyping, and "governance as code" built into the frameworks (Adevinta Tech Blog, 2024). The author's conclusion is that "the data platform must also be treated as a complete and integral product".
- **Zalando** moved from a central warehouse and lake to distributed domain ownership around 2019 while keeping "a central governance layer", as reported in 2021 secondary coverage of its conference talks (SiliconANGLE, 2021). Zalando is a fashion e-commerce platform, cited here for the pattern.
- **AutoScout24** consolidated business intelligence from MicroStrategy and several other tools to Amazon QuickSight between November 2024 and March 2025, cutting more than 500 dashboards to about 150 "meaningful" ones after usage analysis, and planning a standard document for each dashboard with "ownership, purpose, and data definitions" (AutoScout24 Tech Blog, 2025).

My reading of this record is that **ownership federates but the substrate and the rules do not**. Every documented case keeps one platform, one catalog and one set of governance mechanisms, and distributes the ownership of datasets to domains. None of them describes a mesh in which each domain chooses its own storage, formats and access model. That distinction matters for a group like Veldmark, where "data mesh" can easily become a polite name for keeping three warehouses.

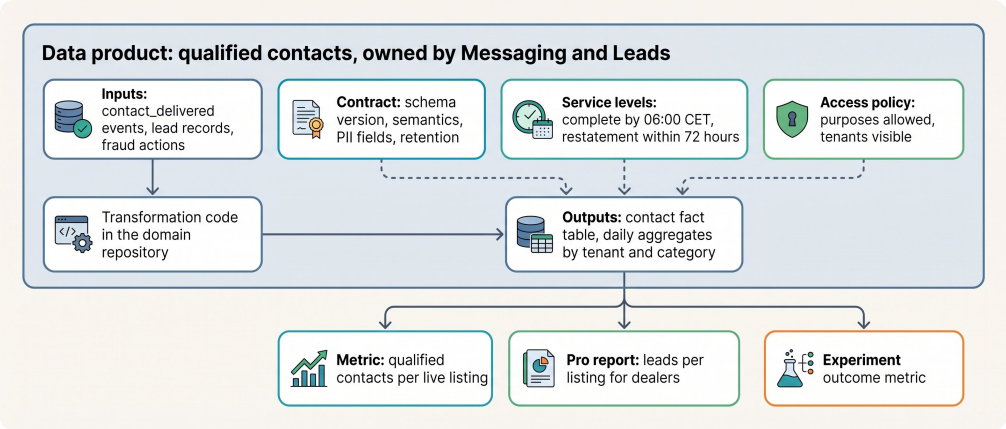


Figure 3: *Anatomy of a data product. The domain that owns contacts publishes them once, with a contract, service levels and access policy; the board metric, the dealer report and the experiment outcome all read the same table.*

Who owns which data product

The ownership question is where data mesh meets Who Owns What: Organizing Marketplace Engineering. The rule I use: **the team that owns the fact in production owns its source-aligned and domain data products; the team that owns the decision owns the consumer-aligned product**. In a marketplace that rule gives this matrix:

Data product	Owner	Main consumers	Note
Listing history (every version, state and price)	Listing domain	Everyone	Built from domain events, not CDC, once available
Search sessions and impressions	Search platform	Ranking, experiments, pro exposure reports	Highest volume; needs retention tiers
Qualified contacts and leads	Messaging and leads	Board metrics, pro ROI, experiments	The governed contact definition lives here

Data product	Owner	Main consumers	Note
Moderation and enforcement actions	Trust and safety	DSA reporting, fraud models, contact restatement	Must link to statements of reasons (Trust and Safety as a Platform)
Orders and fees	Monetization, payments	Finance, take-rate analysis	Reconciled to the ledger, not to tracking
Market prices and demand by area	Data & ML (consumer-aligned)	Pro insights, automated valuation models (AVMs), SEO pages	Aggregated, with minimum cell sizes

Governed Metrics: One Definition of a Contact

A **governed metric** is a business measure with a named owner, a written definition, a single computation in code, a version, and rules for how it is restated when the definition or the underlying data changes. A **metrics layer** (sometimes called a semantic layer) is the component that holds those definitions and computes or serves the metrics to every consumer, so that a dashboard, an experiment readout and a dealer report do not each reimplement "contacts". The idea is old in BI; what makes it urgent in marketplaces is that liquidity metrics are **two-sided and channel-dependent**, so small definitional choices change the numbers by large factors, as Veldmark's replay below shows.

The contact definition card

The decision record Veldmark's architecture advice forum approved in May 2027 reads, in its essential part, as follows.

Field	Decision
Name	Qualified contact (`qualified_contact`, version 1.0)
Owner	Messaging and leads domain; metrics steward in Data & ML
Definition	The first contact a distinct buyer makes with a listing's seller, through any channel, counted once per buyer and listing within a rolling 30-day window
Channels counted	Delivered chat message; delivered e-mail or contact form; tracked call connected for at least 30 seconds; booked viewing or test drive; phone-number reveal by a signed-in buyer
Buyer identity	Signed-in account; after Veldmark ID, the group identity subject; anonymous phone reveals counted as a separate, non-comparable metric
Exclusions	Contacts from accounts actioned for fraud or spam within 72 hours; the seller's own accounts; internal test traffic; known bots
Timing	Counted at delivery to the seller, in the listing's market time zone
Restatement	Daily figures final after 72 hours; definition changes create a new version, published side by side for one quarter
Denominator for "per live listing"	Average daily live listings, from the listing history product, same tenant and category

Each row closes an argument the brands had been having implicitly. Counting **once per buyer and listing** removes the difference between a chatty marketplace and a terse one; Tradeyard's buyers write several messages per conversation, Hearthly's send one form. Counting **at delivery** puts the fact on the server. The **72-hour fraud restatement** is the most consequential choice: it means Femke's phishing "buyer", suspended at 20:26 on Sunday, never reaches the final figure for that day, which closes on Wednesday, and it ties the liquidity metric to the trust-and-safety data product, so a marketplace cannot improve its contact rate by tolerating scammers. That is **Vanity Liquidity** prevented by

definition rather than by exhortation.

Billing and steering must both be derived from the same contacts data product, even when billing applies a stricter rule, as Motorvane's proposed billable lead does with its 14-day window (How Listing Marketplaces Make Money). Otherwise the dealer's invoice and the board pack will disagree about the same car.

Replaying one week under the new definition

Priya's team replayed the week of 8–14 March 2027 under the new definition for all three listing brands, and added the Essen Golf's first 10 days once they were in. Keybridge, which has no buyers, was out of scope.

Example or aggregate	Old brand figure	Qualified contacts v1.0	Why it moved
Femke's city bike, first 24 hours	52 "contacts" (buyer messages)	**13**	14 conversations from 14 buyers; the phishing account was actioned within 72 hours and excluded
The Essen Golf, first 10 days from 14 April (replayed in April)	23 leads	**19**	Three e-mail senders also called and one test-drive booker had already e-mailed; each buyer counts once
Tradeyard, all four countries, week total	Index 100	**37**	Messages replaced by first contacts per buyer and listing
Motorvane, week total	Index 100	**86**	Cross-channel duplicates removed; calls under 30 seconds dropped

Example or aggregate	Old brand figure	Qualified contacts v1.0	Why it moved
Hearthly, week total	Index 100	**106**	Contacts from new-build campaign landing pages now included

The headline was not that Tradeyard's number fell by 63%; nobody had believed it. It was that **for the first time the group could see that Motorvane and Hearthly were closer in contacts per live listing than their dashboards had claimed**, and that Tradeyard Portugal's contact rate sat far below the Netherlands' once both were measured the same way, which Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics treats as a local-liquidity problem rather than a data artifact.

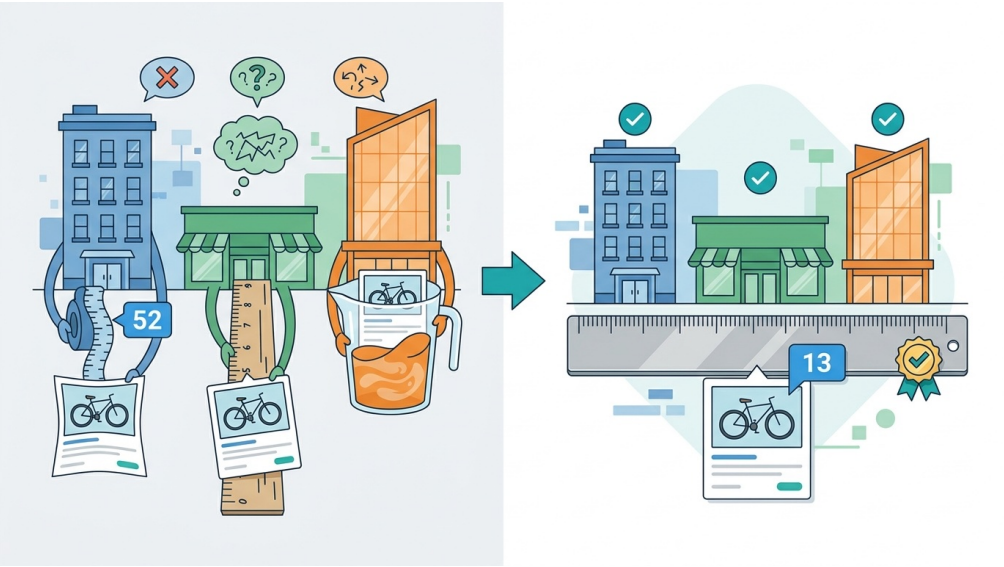


Figure 4: *One governed definition replaces each brand's own ruler: counted once per buyer and listing, with the phishing account excluded, Femke's bike drops from 52 messages to 13 qualified contacts, and the brands can finally be compared.*

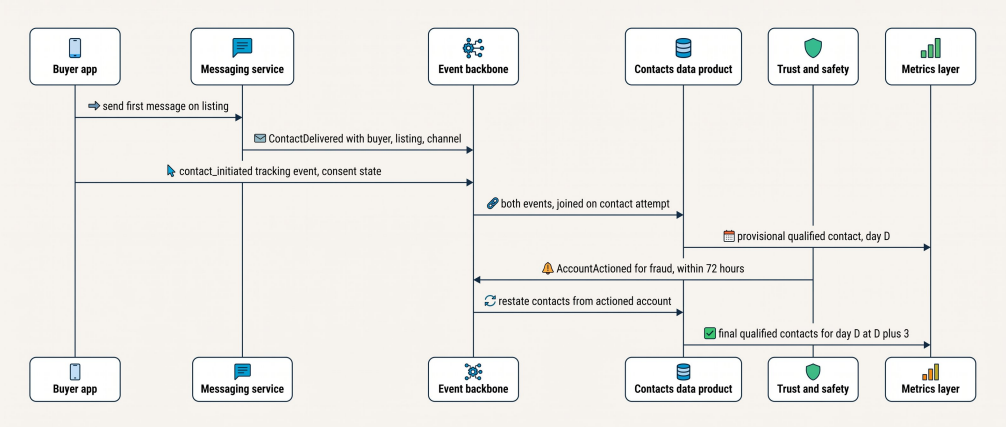


Figure 5: *How a qualified contact is computed. The server-side delivery event is the fact; the client event adds context; a fraud action within 72 hours restates the day before it becomes final.*

Which liquidity metrics to govern

Govern only the metrics that the board, pricing and experiments use to compare marketplaces or to decide money. A governed metric costs a steward's time, a versioning discipline and the political work of changing a brand's history. Veldmark's first catalog has twelve: live listings, new listings, publication time (p95 submit to searchable, linked to the 90-second target in From Submitted to Searchable: The Publication Pipeline), qualified contacts, contacts per live listing, search-to-contact rate, time to first contact, share of new listings with a qualified contact within the vertical's window (48 hours for goods, longer for cars and property, as Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics sets out), sell-through where outcomes are known, alert-attributed contacts, duplicate rate (listings per canonical property or vehicle, One Property, Three Listings: Duplicates and Entity Resolution), and removals for policy or fraud per thousand new listings. Every other metric can live in a team's own models, labeled as such.

Governed metrics also need monitoring, because a definition is only as good as the data under it. Adevinta Benelux published an example in 2024: analysts built "Artemis", which forecasts bounds for more than 140 metrics with Meta's Prophet library on Databricks and alerts teams in Slack; it helped detect a drop of roughly 100,000 leads a day on iOS, worth about €41,000 a day in revenue, which was resolved in less than a day (Adevinta Tech Blog, 2024). A broken app release that stops emitting contact events looks, in a dashboard, exactly like a liquidity collapse; **metric monitoring is how the platform tells the two apart**.

Data for Professionals: Insights as a Product

Dealers and agents buy visibility, but they renew because of evidence. A **pro data product** is any dataset the marketplace serves to professional sellers: listing performance (impressions, views, contacts by channel), market benchmarks (price position against comparable listings, days on market, demand by area), and ROI reporting for paid products. Real-estate portals go further and sell valuation data: in France, Meilleurs Agents reported a median estimate error of 6% against final transaction values for 2021, a self-reported figure that is now several years old and may have changed (Online Marketplaces, 2022), and Scout24 reports a Transaction Enablement line, covering customer relationship management (CRM), data and valuation businesses, of €106.6 million for 2025 (Scout24 preliminary results, 2026; see How Listing Marketplaces Make Money). Veldmark's Valuo plays this role in France.

Pro data products differ from internal analytics in four ways that change the architecture.

1. **They are served, not queried.** Twenty-one thousand dealers opening Motorvane Dealer Hub on Monday morning is a production workload with a latency target. Serve pro statistics from a read store fed by the platform, not from the warehouse.
2. **They are tenant-isolated.** A dealer sees its own listings' performance and aggregated market data, never another dealer's leads. The isolation belongs in the data product's access policy and is tested like any other security control.
3. **They must be causal where they claim value.** OLX described in 2026 how naive dashboards made premium visibility packages appear to perform worse than cheaper ones, because professional sellers self-select into packages; propensity-score matching on vehicle attributes estimated uplifts of +17.6% to +97.4% between package levels, and the results were validated against a randomized test (OLX Tech Blog, 2026). An ROI report that shows correlations as value will eventually be contradicted by a dealer's own spreadsheet. Testing a Two-Sided Market: Experimentation Platforms covers the methods.
4. **They aggregate with minimum cell sizes.** Market price per square meter for a street with three sales identifies those sales. Set a minimum number of listings or transactions per published cell, and coarsen the geography until it is met.

The same pro data products feed Serving the Pros: Dealer, Agent and Merchant Tooling and the sales conversations in Packages, Bumps and Quotas: Products, Pricing and Entitlements. They should read the same governed contact metric as the board; when a dealer's dashboard and the group's investor presentation disagree about leads, the marketplace loses an argument it cannot afford to lose.

Sharing Data Across Brands: The Legal and Commercial Limits

The synergy case for a marketplace group usually assumes that data can flow freely between brands: Tradeyard's car-section buyers could be recommended Motorvane listings, Hearthly could use Keybridge's agency data to prioritize sales calls, Valuo could learn from Tradeyard Portugal's property section. Some of those flows are legitimate, some need consent or a new legal basis, and some are forbidden by contract. **The architecture must make the difference enforceable.**

Four regimes limit sharing. This is an architect's summary, not legal advice; Rules of the Marketplace: DSA, DAC7 and Vertical Regulation owns the regulatory detail.

Limit	What it constrains	Architectural consequence
Purpose limitation (GDPR Art. 5(1)(b))	Personal data collected to run Tradeyard may not be reused for an incompatible purpose, such as Motorvane marketing, without a valid basis	Purpose tags on data products; access granted per purpose, not per person; logs of which purpose each query served
Consent and ePrivacy	Device-based tracking needs consent, often per brand and per purpose; consent given to one brand does not cover another	Consent state on every event; consumer products filter by consent at read time
Controller boundaries	Each brand's legal entity is typically the controller for its users' data; group-wide use may need joint-controller arrangements or processor agreements	Tenant and legal-entity attributes on every table; cross-entity joins only through approved products

Limit	What it constrains	Architectural consequence
Contracts and competition	Agency and dealer data supplied under contract, and data from a neutral SaaS business, may not be used to favor a sister brand	Hard isolation: separate accounts, no shared credentials, independent review

The last row is Veldmark's most sensitive case. In March 2027 the group decided that **Keybridge stays a separate, neutral SaaS**: no Keybridge agency data is used by Hearthly for ranking or sales targeting, enforced by access controls and an annual independent review. In data-platform terms that means Keybridge's Synapse environment does not federate into the group platform; only aggregated financial figures and SaaS usage metrics required for group reporting leave it, through a contract that names each field. Emma Laurent, Keybridge's founder, asked for one more control, which Ingrid accepted: the access policy and its audit log are readable by the independent reviewer without asking Veldmark's engineers. **Neutrality that cannot be audited is a promise; neutrality expressed as access policy is an architecture.**

In Poland, where Tradeyard's cars section and Motorvane compete for the same dealers, the line falls between aggregated and dealer-level data. Using aggregated, non-personal market data across the two (price levels, demand by region) is a group analytics question. Using one brand's dealer-level performance data to target the other brand's sales calls is a commercial question with contract and competition implications, and it should go to the architecture advice forum and legal as a decision record, not be settled by whoever builds the dashboard.

Quality, Freshness and Cost

Quality is a producer's obligation

The documented lesson from Adevinta Spain's contracts, Vinted's generated tracking code and Artemis is the same: **quality is cheapest at the source**. The platform's job is to make the producer's obligations explicit and visible.

Control	Where it runs	What it catches
Schema validation and quarantine	Contract gate	Malformed events, missing required properties, wrong types
Volume and freshness checks per tenant	After ingestion	A country's app release that stopped emitting events; a late CDC feed
Referential checks	Domain entities	Contacts on listing IDs that do not exist; impressions with unknown ranking IDs
Metric anomaly detection	Metrics layer	Liquidity drops that are really instrumentation drops
Reconciliation with systems of record	Daily	Orders and fees that disagree with finance; listings counts that disagree with the listing service

Freshness targets should follow the consumer, not a platform-wide default. Board metrics can be final at D+3. Experiment guardrails, such as a sudden drop in contacts for a variant, need hours, which is why Adevinta built streaming results computation for its experimentation platform (covered in Testing a Two-Sided Market: Experimentation Platforms). Fraud features need seconds and should not run through the analytical platform at all; Scams, Fakes and Account Takeovers: Fraud Prevention for Listings and Models in Production: ML Platforms and Use Cases own those paths.

Cost has an owner or it grows

Behavioral data is the largest and least valuable-per-byte part of a marketplace's data estate, and it grows with traffic, not with revenue. Adevinta's 2024 logging post shows how much small format choices matter at volume: trimming fields such as identifiers and hostnames, and moving to Vector, cut its OpenSearch logging cost by 35%, more than €30,000 a year (Adevinta Tech Blog, 2024). The same arithmetic applies to tracking events. The levers, in the order I would pull them:

1. **Ownership metadata on every dataset and every compute workload**, so cost is attributed to a team and a purpose. Adevinta's multi-cloud governance post makes the general point that accurate ownership and cost-center metadata underpins governance and cost allocation (Adevinta Tech Blog, 2024).
2. **Retention tiers by event class**. Raw impressions in hot storage for 90 days, compacted for two years, aggregated beyond; contacts and domain events kept longer because they are facts.

3. **Compact envelopes**. Arrays of listing IDs per impression batch, short identifiers, no repeated user agents.
4. **Kill unused datasets and dashboards**. AutoScout24's cut from more than 500 dashboards to about 150 was a governance gain as much as a cost one.

In 2026 Veldmark spent about **€21 million** on its data estate: about €11.5 million on Snowflake across Tradeyard and Motorvane, €4.2 million on BigQuery at Hearthly, €1.1 million on Synapse at Keybridge, and €4.2 million on tracking vendors and BI licenses. Group events run at about **2.6 billion** behavioral events a day, 1.9 billion of them from Tradeyard. The Common Ground target is **€13 million a year by 2030** at higher volume, which counts toward the program's €38 million run-cost reduction.

In Practice: Veldmark's Common Ground Data Platform

The design Priya and Ingrid took to the architecture advice forum in September 2027 had five parts, each a decision record.

1. **Open storage, one main engine**. Group data products are stored as Apache Iceberg tables on Amazon S3, partitioned by tenant (brand and country) and time, with Snowflake as the main compute engine for Tradeyard and Motorvane. Hearthly keeps BigQuery while it stays on Google Cloud through 2028 and publishes its governed data products to the group store under the same contracts; the engine question for Hearthly returns at the 2029 cloud review. Keybridge stays outside, as described above.
2. **One tracking plan, generated SDKs**. Tracking plan v1 has **64 core events** with the common envelope, replacing about 1,140 distinct event names in Tradeyard's apps and web. SDKs for iOS, Android and web are generated from the plan; brands may add extension events with a brand prefix but may not redefine core ones. Server-side events cover every governed fact.
3. **Contracts at the gate**. Every source, including CDC from Yardstick, enters through a contract with personal fields declared; failures land in per-source quarantine tables that the producing team owns and monitors.
4. **A metrics catalog with twelve governed metrics**, starting with qualified contacts v1.0, computed once in the metrics layer and served to dashboards, the experimentation platform and Motorvane Dealer Hub. Old brand metrics remain for one quarter side by side, labeled "legacy definition".
5. **Purpose-bound access**. Data products carry purpose tags (operations, steering, product analytics, ML, pro reporting, advertising, regulatory). Lucía Ortega, the Group Data Protection Officer, owns the purpose taxonomy; access requests name a purpose; Keybridge's isolation is a policy with an external audit trail.

The sequence matters as much as the target. The forum approved the **tracking plan and the contact definition first**, because they could be delivered on the existing warehouses and would pay back immediately in the board pack, and left the storage migration for 2028. Sven Lindqvist's event backbone, live since 1 July 2027, gave the platform a domain-event source for listings from day one, so the listing-history data product was built on Listing event contract v1 rather than on Yardstick's tables, with CDC kept only for history before July. Felix Brandt, Motorvane's CTO, accepted the shared contact definition on one condition: dealer reports switch at the same time as the board pack, so his sales team never has to explain two numbers.

Trade-offs

Decision	Option A	Option B	How to choose
Where metrics are defined	In each BI tool and team model	In a central metrics layer with stewards	Central for metrics that compare brands or decide money; local for everything else
Behavioral collection	Vendor analytics SDK	Own collector with generated SDKs	Vendor for a single brand starting out; own collector once events feed ranking, experiments and pro reports
Facts for liquidity	Client-side events	Server-side events	Server-side for counted facts; client-side for context and intent
Durable storage	Warehouse-managed format	Open table formats in object storage	Open formats when the group spans clouds or expects acquisitions and disposals

Decision	Option A	Option B	How to choose
Ownership	Central data team builds everything	Domains own data products on a shared platform	Domains own facts; the center owns substrate, contracts, catalog and the governed metrics' stewardship
Personal data	In every table, masked by views	Split at ingestion into a personal store	Split, unless volumes are trivial; erasure and access become tractable
Cross-brand data	Open by default, restricted on request	Closed by default, opened per purpose	Closed by default, with fast purpose-based approval

Vertical Differences

Concern	General classifieds	Real estate	Automotive	Jobs (where relevant)
Dominant contact channel	Chat	Forms, phone, viewings	Calls, forms, test drives	Applications
Hardest metric to define	Contact (many messages per buyer)	Lead quality and attribution to agencies	Call quality (connected, duration)	Application completion versus click-out
Outcome data	Sale often unknown off-platform; known for on-platform payments	Transaction prices from public sources (France's DVF) and agencies	Sale signals from dealer feeds and removals	Hire rarely known

Concern	General classifieds	Real estate	Automotive	Jobs (where relevant)
Pro data product	Seller statistics	Valuations, market prices, seller leads	Price rating, days to sell, stock benchmarks	Candidate pool and salary benchmarks
Sensitive data	Addresses at pickup, payment data	Exact addresses, owner identity, tenant files	Vehicle identification numbers linked to owners	CVs and special-category data
Data volume driver	Behavioral events from very high traffic	Images and rich attributes per listing	Feeds and price changes	Applications and documents

In real estate, **outcome data from outside the marketplace** (public transaction records, agencies' reported sales) is a core input, so the platform must model external datasets with their own contracts and licensing. In general classifieds, the sale usually happens off-platform, so liquidity metrics stop at the contact unless on-platform payments such as Tradeyard Safe Deal supply an outcome; the data platform should report the share of listings with known outcomes beside any sell-through figure.

Failure Modes

- **Counting messages as contacts**. The easiest number to grow becomes the steering metric. The fix is a governed definition based on distinct buyers and delivered contacts.
- **Tracking as an afterthought in app releases**. A new app version renames or drops an event, and the dashboard shows a liquidity collapse. Generated SDKs, volume checks per release and metric monitoring catch it.
- **No impressions, no ranking evidence**. The ML team cannot correct position bias, the experiment team cannot measure exposure, and the trust team cannot show how paid placements were shown. Log impressions from the start.
- **Consent applied downstream**. Events without consent state are collected "for later filtering", and later never comes. Put consent on the event and enforce it at read time.
- **CDC as the permanent contract**. The analytics model mirrors a legacy schema, and every refactor of the monolith breaks reporting. Treat CDC-fed tables as temporary and plan their replacement by domain events.

- **The mesh as an excuse**. "Each domain owns its data" becomes three warehouses, three vocabularies and no shared metrics. Federate ownership, not substrate.
- **Pro dashboards from the warehouse**. Monday-morning dealer traffic hits analytical compute, costs spike and dashboards time out. Serve pro products from a read store.
- **Unaudited cross-brand access**. A well-meaning analyst joins a neutral SaaS business's data with a portal's, and the group's neutrality promise is broken without anyone deciding to break it.

The Chief Architect's View

At group scale, the data platform is less a technology choice than a set of **contracts the Chief Architect must make someone own**. The decisions I would take to the architecture forum, in order:

1. **Who owns the governed metrics, and which ones are governed**. A short list, a steward per metric, a definition card per metric, and a restatement policy. This decision is cheap in technology and expensive in politics, because it changes brands' histories. Make it early, while the numbers are new to the board.
2. **The tracking plan's ownership and the "server-side for facts" rule**. One plan, generated SDKs, brand extensions without redefinition.
3. **Where the durable copy lives**. Open formats, tenant-partitioned, so that acquisitions can be absorbed and disposals separated. Pick the main compute engine for the next five years, not for ever.
4. **The purpose taxonomy and access model**, with the Data Protection Officer as co-owner, and neutrality commitments expressed as policy with external audit.
5. **Cost attribution**, so the data estate's growth is a decision some team makes, not a property of traffic.

Who decides: the Group Chief Architect owns the platform's shape and the contract mechanisms; the Head of Data & ML owns the substrate and the metrics stewardship; domains own their data products; the Chief Product Officer owns which metrics are steering metrics; the Data Protection Officer owns purposes and legal bases.

What to watch: share of governed metrics computed from server-side facts; share of sources entering through contracts; quarantine rates by producer; time from a new event in the plan to its availability in the platform; consent coverage by country; data estate cost per thousand live listings; and, most telling, how often two dashboards disagree about the same number.

Questions to Ask

1. Who owns the definition of a contact in each of our marketplaces, and could they write it down on one page today?
2. Are our governed liquidity metrics computed from server-side facts, and what share of each figure depends on client events and consent?
3. Do our impression events record position, ranking model version, paid placement and experiment assignment?
4. Which sources enter the platform without a data contract, and who sees their quarantine?
5. Where is personal data in our analytical estate, and how long does an erasure request take to reach every copy?
6. If we sold one brand next year, how many weeks would it take to separate its data, and what would we have to copy?
7. Which cross-brand data flows exist today, under which purpose and legal basis, and who approved them?
8. Do our dealer and agent reports use the same definitions as our board pack?
9. What does our data estate cost per thousand live listings, and which team owns each of the ten largest cost items?
10. When a liquidity metric drops, how quickly can we tell whether the market or the instrumentation changed?

From Measurement to Causation

With one definition of a contact, exposure logged with its ranking context, and data products that every team reads in the same way, Veldmark can start asking the question that every product team asks next: did our change cause that? In a two-sided market, where buyers and sellers affect each other, the answer is harder to get than an A/B testing tool suggests, and Testing a Two-Sided Market: Experimentation Platforms takes it up.

Sources and Evidence

Company engineering and investor sources

- Inferring Kafka topic format during JSON deserialization, Adevinta Tech Blog (Rodrigo Molina), 2022 — Kafka as backbone of the data platform; UBCT event classification; mixed Avro and JSON topics (older evidence).
- Evolving from a Data Platform to a Data Products Platform, Adevinta Tech Blog (Marta Diaz Lopez), 2023 — data products as autonomous, read-optimized units at Adevinta Spain.

- Creating source-aligned data products in Adevinta Spain, Adevinta Tech Blog (Sergio Couto), 2024 — immutable data contracts, quarantine, personal-data split, Time to Effective Data from about five days to about 30 minutes.
- From Lakehouse architecture to data mesh, Adevinta Tech Blog (Marc Planagumà), 2024 — medallion layers as data products, vertical domains, governance as code, platform as product.
- Artemis: fostering autonomy in monitoring data quality, Adevinta Tech Blog (Scotti, Paulvannan and Yaşa), 2024 — metric monitoring for more than 140 metrics; the iOS leads drop.
- The 300 Bytes That Saved Millions: Optimising Logging at Scale, Adevinta Tech Blog (Paco Orozco), 2024 — format-level cost savings at volume.
- Multi-cloud governance at scale, Adevinta Tech Blog (Pablo Rodriguez), 2024 — ownership metadata as the basis of cost allocation.
- Scalable and non-blocking event ingestion pipeline, Vinted Engineering, 2019 — schema-generated mobile tracking code, server-side repair, two-week app rollout.
- Experimentation Platform at Zalando: Part 1 – Evolution, Zalando Engineering (Shan Huang), 2021 — unified tracking schema as a precondition for trustworthy analysis.
- From Migration to Innovation: AutoScout24's BI Modernisation Journey, AutoScout24 Tech Blog, 2025 — dashboard consolidation from more than 500 to about 150; ownership and definitions per dashboard.
- Correlation lied to us: rethinking product impact with causal inference, OLX Tech Blog, 2026 — selection bias in package dashboards; propensity-score matching validated by a randomized test.
- Adevinta Annual Report 2019, Adevinta, 2020 — data and analytics as the centralized domain, "built once; used many times".
- Strategic Report 2025, Aurelia UK Feederco Ltd, Adevinta's parent company, 2026 — decentralization, localization of IP, parallel running of central and local services.
- Scout24 preliminary FY2025 results (IR News, German), Scout24 SE, 2026 — Transaction Enablement revenue of €106.6 million.
- Meilleurs Agents launches new campaign, Online Marketplaces, 2022 — self-reported 6% median valuation error for 2021.

Standards, regulation and documentation

- Regulation (EU) 2016/679 (General Data Protection Regulation), EUR-Lex, 2016 — purpose limitation (Art. 5(1)(b)), controller responsibilities.
- Directive 2002/58/EC (ePrivacy Directive), EUR-Lex, 2002, as amended in 2009 — consent for storing or accessing information on terminal equipment (Art. 5(3)).

Research and practitioner sources

- How to Move Beyond a Monolithic Data Lake to a Distributed Data Mesh, Zhamak Dehghani, martinfowler.com, 2019 — domain ownership, data as a product, self-serve platform, global governance for interoperability.
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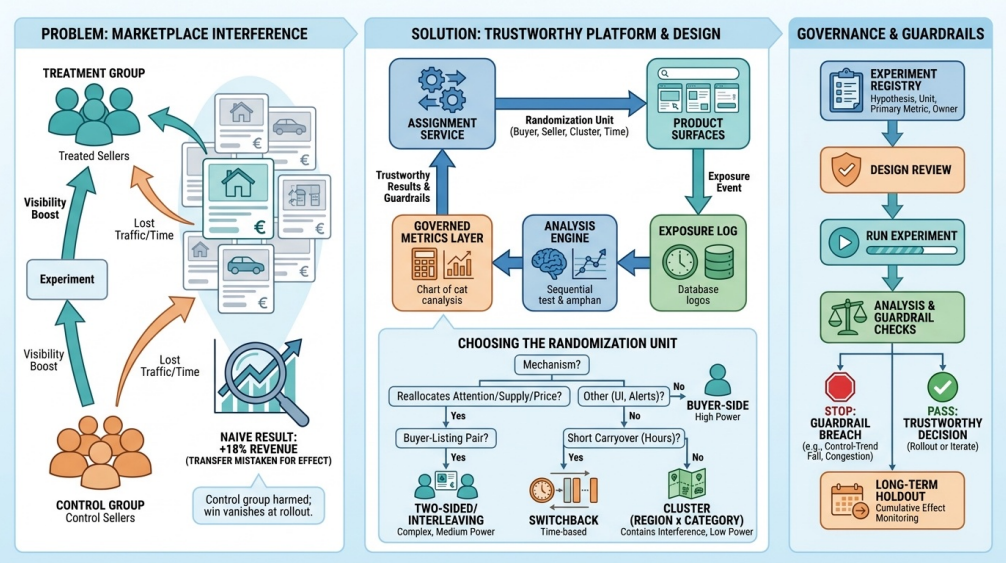
- Data warehousing problems? Data mesh solution, SiliconANGLE, 2021 — secondary account of Zalando's move to domain ownership with central governance.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

37. Testing a Two-Sided Market: Experimentation Platforms



IN THIS CHAPTER, YOU WILL: Take apart an experiment at Veldmark that reported an 18% revenue gain and delivered a loss. You will see the four components of an experimentation platform, why interference between buyers and sellers biases ordinary A/B tests, which designs (listing-side, two-sided, cluster, switchback) fix which problems, how to choose metrics and guardrails for a liquidity business, and how to govern experiments on ranking and prices.

WHY THE BUSINESS CARES: Price changes, ranking changes and new paid products are the levers that move a marketplace's revenue, and they are decided on experiment results. A biased experiment does not just waste a quarter; it commits the business to a change whose real effect is the opposite of the one in the slide deck.

WHY ARCHITECTS SHOULD CARE: Whether a marketplace can run the right experiment is decided by architecture: which units the assignment

service can randomize, whether exposure is logged where the treatment acts, whether metrics come from governed definitions, and whether the analysis engine understands clusters. A platform that can only split users cannot test most of what a marketplace needs to know.

KEY POINTS:

- **An experimentation platform is four contracts**. Assignment, exposure logging, metrics and analysis must agree on the unit, the time and the definitions. Centralize the statistics engine and the metric definitions first; standardizing the interface and the flag tooling can follow, as Adevinta's path from one marketplace's package to a group platform shows.
- **In a two-sided market, the randomization unit is a design decision**. Treated and control units compete for the same buyers' attention or the same scarce listings, so a buyer- or seller-randomized test can measure a transfer between groups instead of an effect. Choose listing-side, two-sided, cluster or switchback designs by the mechanism the change works through, and accept the loss of statistical power that comes with it.
- **Guardrails protect the side you are not testing**. Revenue experiments need liquidity guardrails, ranking experiments need seller-exposure guardrails, and every experiment needs a check that control units were not harmed. Pricing and ranking experiments are monetization and regulatory decisions, so they need an owner, pre-registered metrics and review before they start.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Vanity Liquidity** — Experiments judged on clicks, views or bump purchases rather than qualified contacts approve changes that move activity without creating matches.
- **Multiplier Creep** — Ranking and visibility experiments evaluated on revenue alone, with buyer-randomized designs, keep approving more paid prominence, because the cost lands on control sellers and on relevance outside the test's window.

An A/B test rests on an assumption so basic that most tools never state it: **what happens to one unit does not depend on which group the other units are in**. Statisticians call it the stable unit treatment value assumption (SUTVA). In a search engine or a news site it is usually close enough to true. In a listing marketplace it is often false, because the units compete. Buyers compete for the same apartment in Lyon. Sellers compete for the

first screen of a bike search in Rotterdam. A paid bump helps one listing only by moving another one down.

When SUTVA fails, the difference between treatment and control is not the effect of the change. It is the effect plus a **transfer** between the groups, and the transfer can be larger than the effect. This is called **interference**: the outcome of a unit depends on the treatment of other units. The consequence is not a small statistical correction: experiments can report the wrong sign.

Data Platform for Marketplaces: Events, Lakehouse and Data Products gave Veldmark one definition of a contact and a log of exposures. This chapter asks the next question: did a change cause the movement? The answer decides which ranking, pricing and paid-product changes ship, and a wrong answer spends liquidity on changes that only looked like wins. Ranking for Two Sides: Relevance, Freshness and Paid Visibility, Recommendations for Inventory That Disappears, How Listing Marketplaces Make Money and Models in Production: ML Platforms and Use Cases send their changes here to be tested. The chapter owns three terms: **interference**, **cluster randomization** and **guardrail metric**. Offline evaluation of models belongs to Ranking for Two Sides: Relevance, Freshness and Paid Visibility and Models in Production: ML Platforms and Use Cases.

Setting the Scene at Veldmark: The Bump That Ate Itself

Veldmark is the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build). In January 2027 Olivier Petit, VP Monetization, proposed a price test for Tradeyard Netherlands. A **bump**, the paid product that re-dates a listing so it reappears near the top of newest-first results (Ranking for Two Sides: Relevance, Freshness and Paid Visibility describes the mechanics), cost €2.49. Olivier's hypothesis was that the bump was overpriced for low-value goods: many sellers of a €40 lamp or a €240 city bike would bump if it cost less, and volume would more than compensate.

The test ran from Monday 1 February to Sunday 28 February 2027. Half of Tradeyard NL's private sellers, chosen at random by account, saw a bump price of €1.99; the other half saw €2.49. The results looked decisive:

Metric, per seller (February 2027)	Control (€2.49)	Treatment (€1.99)	Relative change
Bumps bought	1.00 (index)	1.48	+48%

Metric, per seller (February 2027)	Control (€2.49)	Treatment (€1.99)	Relative change
Bump revenue	1.00 (index)	1.18	**+18%**
Contacts per bumped listing	1.00 (index)	0.99	−1% (not significant)
Sellers who bought at least one bump	11.2%	15.9%	+42%

Treatment sellers bought 48% more bumps at a 20% lower price, so bump revenue per seller rose by 18% (1.48 × 1.99 ÷ 2.49 ≈ 1.18), and their bumped listings did as well as control's. Tradeyard NL's bump revenue was about €38 million a year; the forecast added about €6.8 million. The new price went live for all NL sellers on Tuesday 6 April 2027.

Eight weeks after rollout, bump revenue was down about 2.5%, not up 18%: bump purchases were up only 22% on the same weeks of the previous year, not 48% (1.22 × 1.99 ÷ 2.49 ≈ 0.975), so revenue ran roughly €950,000 a year below the old price instead of €6.8 million above it. Contacts per bumped listing had fallen 31%. The median time a bumped listing stayed on the first page of its category's newest-first results had dropped from 2.6 hours to 1.5 hours. Repeat bump purchases were falling.

Ingrid Halvorsen, the Group Chief Architect, asked Priya Raman's team one question: what happened to the control group's bumps in February? The answer was in the February data. **Control sellers' contacts per bump had fallen 9% during the test** compared with January, while treatment's stayed flat. The treatment did not make bumps work better; it let treatment sellers take first-page time from control sellers' bumps. The test measured a transfer, not an effect, and the transfer disappeared at full rollout because there was no control group left to take from. Everyone was now bumping more, each bump bought less time at the top, and sellers noticed.

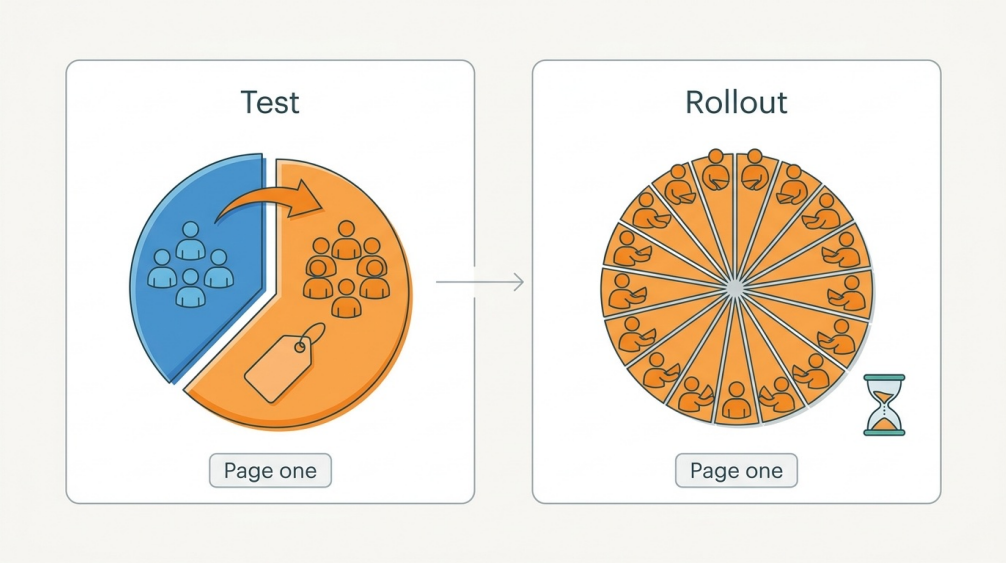


Figure 1: *The bump test measured a transfer, not an effect: treatment sellers gained first-page time that control sellers lost, and at rollout the same page was simply cut into thinner slices.*

The rest of this chapter is the platform and the method Veldmark built so that the next price test would not repeat it.

Concepts: What an Experiment Must Get Right

An **online controlled experiment** randomly assigns units (users, sellers, listings, regions, time periods) to variants, exposes them to the variant's experience, measures outcomes, and attributes the difference to the variant. The standard reference, Kohavi, Tang and Xu's *Trustworthy Online Controlled Experiments* (Cambridge University Press, 2020), covers the general practice. Five terms carry the marketplace-specific argument.

Term	Definition	Why it matters in a marketplace
Randomization unit	The entity assigned to a variant	Buyers, sellers, listings, region × category cells and time slots give different biases and different power

Term	Definition	Why it matters in a marketplace
Analysis unit	The entity over which metrics are averaged	Must match or be nested in the randomization unit; analyzing per buyer a test randomized per region understates uncertainty
Interference	A unit's outcome depends on other units' assignments	The default condition for ranking, pricing and visibility changes, not an edge case
Cluster randomization	Assigning groups of units that interact mainly with each other (a city's bike market, a district's rentals) as one unit	Contains most interference inside clusters, at the cost of fewer effective units
Guardrail metric	A metric that must not degrade beyond a set threshold for a change to ship, regardless of the primary result	Protects liquidity, the other side of the market, and the control group itself

The **primary metric** (or overall evaluation criterion) is the one metric a test is designed and powered to move; choosing it before the test starts is the main protection against picking the metric that happened to win. **Exposure** is the moment a unit encounters the treatment, which is often much later than assignment: a seller assigned to a new bump price is exposed only when the price is shown.

Architecture: The Four Contracts

An experimentation platform has four components. Each is simple; the hard part is making them agree on units, timestamps and definitions.

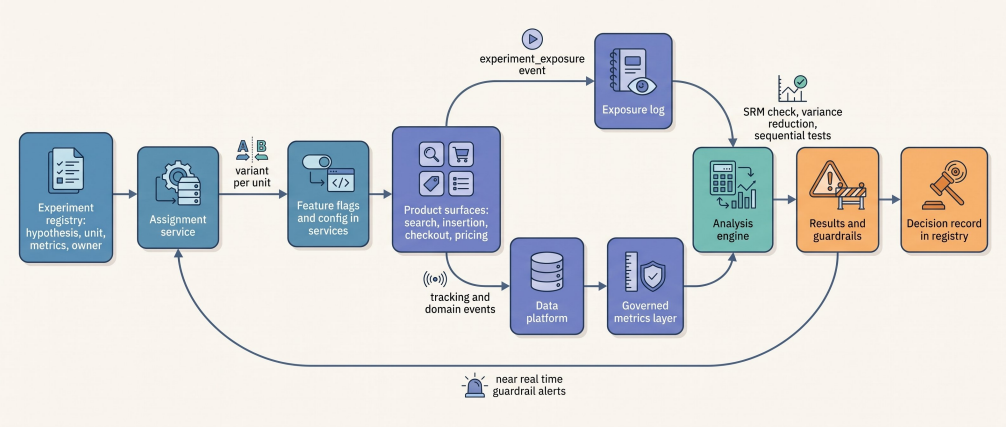


Figure 2: *The four contracts of an experimentation platform. The registry fixes unit and metrics before launch; assignment and exposure share a unit identifier; metrics come from the governed layer; the analysis engine checks the experiment's health before it reports an effect.*

1. Assignment

The **assignment service** decides, deterministically, which variant a unit receives. The usual mechanism is a hash of the unit identifier and an experiment-specific salt, mapped to buckets; deterministic hashing means any service can compute the assignment without a lookup and the same unit always gets the same variant. Three marketplace requirements go beyond what generic tools offer.

- **Several unit types.** The service must assign by buyer, by seller account, by listing, by cluster (a region × category cell from a published cluster map) and by time slot. A platform that knows only users and devices forces every experiment into the design that marketplaces most need to avoid.
- **Layers for mutual exclusion.** Experiments that touch the same mechanism (two ranking changes, a ranking change and a bump-price change) must run in the same layer, so that a unit is in at most one of them, or be analyzed for interaction. Veldmark learned this the hard way: the September 2027 ranking-policy experiment in Ranking for Two Sides: Relevance, Freshness and Paid Visibility and a bump-price retest could not share Tradeyard NL's clusters.
- **Server-side evaluation for anything priced or ranked.** Assignment for prices, fees and ranking runs in the back end, never in the client, so a seller cannot see a different price by reinstalling the app and the treatment can be enforced where the money moves.

2. Exposure logging

The **exposure log** records when a unit encountered its variant. It is the `experiment_exposure` event defined in Data Platform for Marketplaces: Events, Lakehouse and Data Products: experiment ID, variant, unit type and unit ID, time. Analyses should be restricted to exposed units (with the assignment as the randomization), because including the many assigned units that never reached the changed surface dilutes the effect. In marketplaces, exposure is two-sided. A ranking change exposes buyers who saw the page **and** sellers whose listings were moved; a test that logs only buyer exposure cannot compute seller-side outcomes.

3. Metrics

Experiments should consume the **governed metrics** of Data Platform for Marketplaces: Events, Lakehouse and Data Products, computed by the same code as the board pack, with experiment-specific metrics added in the registry. Shared code keeps tests and the business aligned. When the primary metric of a ranking test is "qualified contacts per searcher" and the board steers on "qualified contacts per live listing", both must use the same definition of a contact, including the 72-hour fraud restatement; otherwise a ranking change that surfaces scam-prone listings could win a test and lose the business.

4. The analysis engine

The **analysis engine** turns exposures and metric values into effects, intervals and decisions. Four capabilities separate a trustworthy engine from a dashboard.

Capability	What it does	Evidence
Sample ratio mismatch (SRM) check	Tests whether the observed split matches the designed split; a mismatch means broken assignment or logging, and the result should not be read	Zalando's platform added SRM alerts after early analyses were corrupted (Zalando Engineering, 2021)

Capability	What it does	Evidence
Variance reduction	Uses pre-experiment data to remove predictable variation; CUPED reduced variance by about 50% at Bing, "effectively achieving the same statistical power with only half of the users, or half the duration"	Deng, Xu, Kohavi and Walker, WSDM 2013
Valid early looks	Sequential methods give p-values and intervals that stay valid when results are checked continuously	"Always valid" inference (Johari, Pekelis and Walsh, 2015, revised 2019); safe testing on 162 Vinted experiments needed fewer samples than classical tests (Beasley, 2023)
Cluster-aware inference	Computes uncertainty at the randomization unit (cluster or time slot), not per user	Required for every design in the next section

The Vinted study adds a marketplace-specific nuance: agreement between sequential and classical tests was lower for long-delay metrics such as transactions and cancellations than for searches and sessions (Beasley, 2023). Contacts, sales and seller retention are exactly those delayed metrics, so **early stopping should be allowed on guardrails that signal harm, and used with care on the primary metric**.

What the documented classifieds platforms show

Adevinta has published the most complete record of a classifieds group's experimentation platform. In 2021 its 22 marketplaces used ten different A/B testing tools; the 2024 account says this setup brought "inefficiencies, redundancy and escalating costs", and describes a cross-functional working group, a scoring matrix asking whether a tool would "support our needs in 3-5 years", and a hybrid decision: build the in-house platform and use a third-party tool in some marketplaces during the transition (Adevinta Tech Blog, 2024). A 2023 benchmark of 21 companies by experiments per tech full-time employee placed Adevinta in the "crawling" band and concluded that it should "keep building" its platform, Houston (Adevinta Tech Blog, 2023).

The statistics came from one marketplace. Fisher, a Python analysis package built at Marktplaats with CUPED and multiple-testing corrections, was used in more than 90%

of Marktplaats experiments within a year, cutting hands-on analyst time to about three hours per experiment, and Nightingale, built on Fisher, was positioned to become the engine for all of Houston's statistical calculations (Adevinta Tech Blog, 2023). Adevinta also described computing results incrementally with Kafka Streams, keeping user-level aggregates and updating means and variances with a weighted Welford algorithm, so that rollout problems are visible in near real time without re-reading billions of events (Adevinta Tech Blog, 2023). All of these posts predate Adevinta's 2025 decentralization; whether Houston remains a group platform is not public.

My reading of that path is the first key point's recommendation: **centralize the statistics and the metric definitions before the user interface and the flag tooling**. Two marketplaces using different flag systems but the same engine and the same contact metric produce comparable results; two marketplaces using the same tool with different metric code do not.

Interference: Why Marketplace Tests Report the Wrong Number

Interference in a listing marketplace comes from three mechanisms, and each suggests a different design.

Mechanism	Example	What a naive test measures
Competition for attention	Paid bumps, ranking boosts, highlight badges: a treated listing gains position only by displacing others	Treated sellers' gain, which includes what they took from control sellers
Competition for scarce supply	A ranking change sends treated buyers to the best Lyon apartments sooner; control buyers find them taken	Treated buyers' gain, which includes the listings control buyers lost
Market-level responses	A price badge or price hint shifts asking prices, so the comparison set changes for everyone	Neither group's effect; the market is not the one that will exist at rollout

Johari, Li, Liskovich and Weintraub analyzed the first two mechanisms formally. In

their model, randomizing customers is unbiased when the market is demand-constrained (buyers are scarce relative to listings, so treated and control buyers rarely compete for the same listing) and biased when it is supply-constrained; randomizing listings shows the reverse; and **two-sided randomization**, which treats a buyer-listing interaction only when both are treated and compares the four cells, can be close to unbiased at both extremes (Johari et al., 2020, revised 2021). The practical reading for marketplaces is that **the bias depends on which side is scarce, and that differs by category and city**: buyers are scarce for second-hand furniture in a small town and listings are scarce for rentals in central Barcelona.

The bias can be large. In a 2020 meta-experiment on Airbnb's pricing tools, Holtz, Lobel, Liskovich and Aral ran the same treatment under listing-level randomization and under cluster randomization of listings, and estimated that "32.60% of the Bernoulli-randomized treatment effect estimate is due to interference bias", which they describe as a lower bound (Holtz et al., 2020). Airbnb is a booking marketplace, not a classifieds site, but the mechanism, listings competing for the same guests, is the same one that undid Veldmark's bump test.

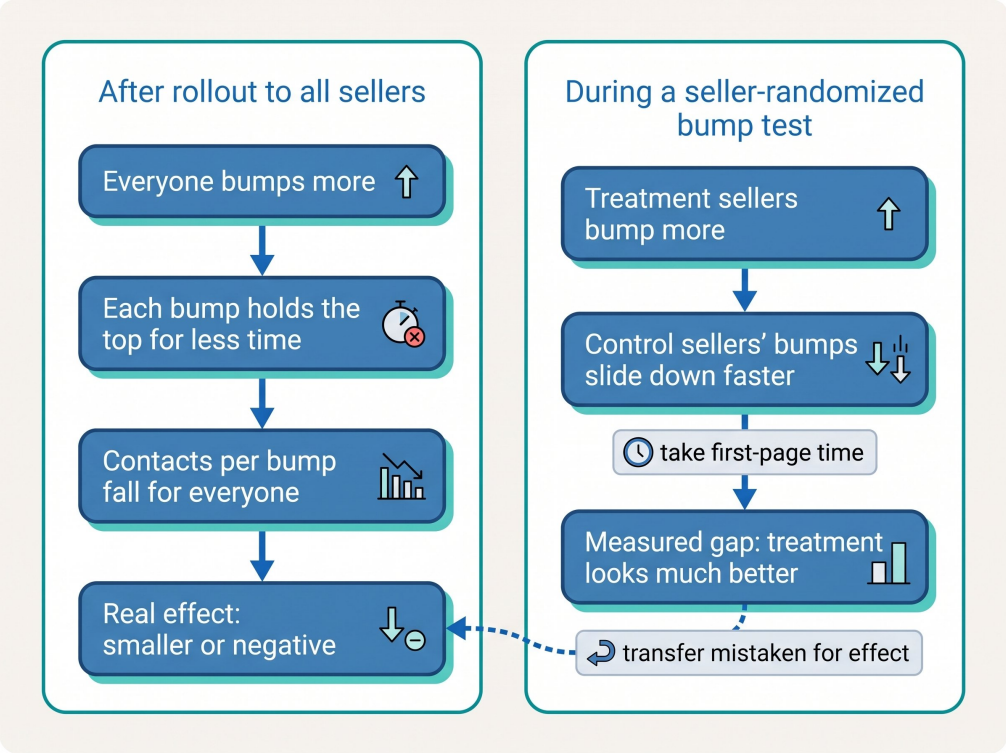


Figure 3: *How the bump test misled. Under seller randomization, treatment sellers gained partly at control sellers' expense; at full rollout there was no one left to take from, and the transfer turned into congestion.*

Designs for Two-Sided Markets

Each design trades bias for power, cost or speed; the right one depends on the mechanism the change works through.

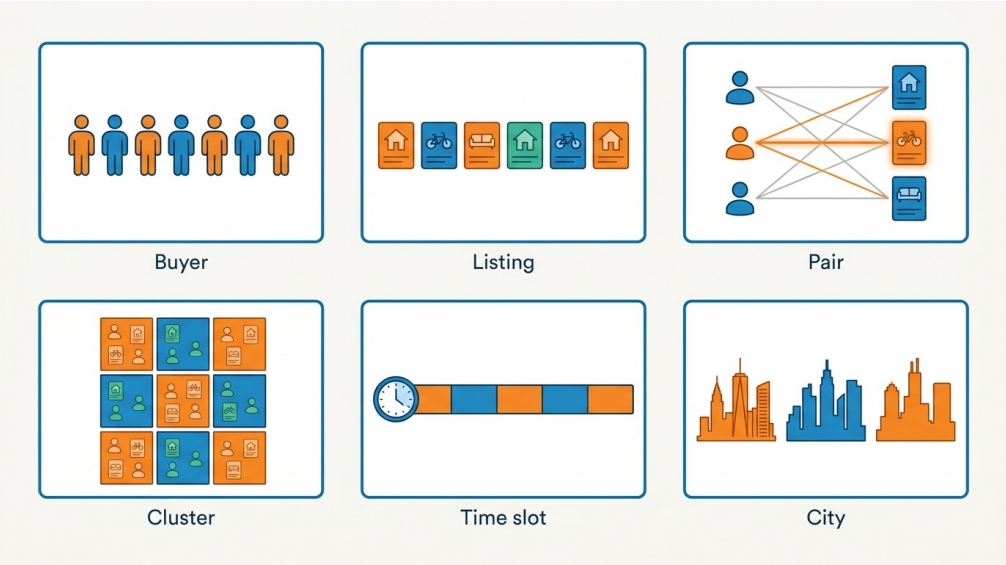


Figure 4: *Each design splits the market along a different unit; the coarser the unit, the more interference it contains inside and the fewer units it leaves to learn from.*

Listing-side or seller-side randomization

Assign listings or seller accounts. It suits changes that act on supply and whose effect on buyers is diffuse: a new insertion flow, a photo-quality hint, a seller dashboard. It is biased when treated listings compete with control listings for the same buyers, which is exactly the case for anything that changes visibility. The bump test was seller-randomized, and that was its error.

Buyer-side randomization

Assign buyers (signed-in users or devices). It suits changes on the demand side when listings are plentiful relative to buyers: query understanding, filters, page layout, alert formatting. It is biased for ranking changes in supply-constrained cells, where treated buyers take the best listings first. Recommendations for Inventory That Disappears reports a Veldmark carousel test randomized by user, which is reasonable because a recommendation surface redistributes less attention than the main ranking.

Two-sided randomization

Assign both buyers and listings, and apply the treatment only to interactions in which both are treated, analyzing the four cells together. It is the design the Johari et al. analysis recommends when it is unclear which side is scarce. Its costs are complexity in the ranking service (treatment depends on the pair) and fewer units in each cell.

Cluster randomization

Assign whole clusters of units that interact mostly with one another, such as a region × category cell for goods or a city district for rentals, and analyze at the cluster level. It contains most interference inside clusters, at the price of statistical power: fifty clusters are fifty units, however many buyers they contain. Cluster maps should be published reference data, built from observed interaction (where buyers who search a cell also contact) rather than from administrative boundaries alone, and they should be checked for spillover: buyers near a cluster boundary see both treatments.

Cluster randomization is also the only practical design for changes that cannot be split by user. **Search-engine optimization changes** are the clearest case: the crawler is the "user", so the test randomizes groups of landing pages (for example city × category templates) and measures organic sessions per group; Found by Google: Web, App and SEO Architecture for Listings owns those pages. No classifieds company has documented its SEO testing design publicly; this is a reference practice, not a reported one.

Switchback designs

Alternate the whole market (or a region) between variants over time slots, and compare slots. Delivery and ride-hailing marketplaces use them because their supply and demand reset every hour; DoorDash described region-by-time switchbacks in 2018 (widely reported; the original post could not be retrieved for this book). Bojinov, Simchi-Levi and Zhao give the design and analysis theory, including how to handle **carryover**, where a slot's outcome depends on the previous slot's treatment (Bojinov et al., 2020, revised 2025). In listing marketplaces carryover is long: a listing bumped at 19:00 is still on page one at 21:00, and a seller who saw a cheaper bump price yesterday remembers it today. Switchbacks are therefore suited to short-lived mechanisms only, such as a ranking change for a peak hour or a notification send-time policy, and poorly suited to prices.

Geographic experiments

Assign whole cities or countries, a coarse form of clustering. It suits market-level changes such as fees, policies or TV campaigns, but with so few units that the

analysis relies on synthetic-control or difference-in-differences methods and on matched markets. Treat their results as estimates with wide uncertainty, not as A/B tests.

Design	Bias from interference	Power	Implementation cost	Fits
Buyer-side	Low when listings are plentiful; high for ranking in scarce cells	High	Low	UI, query understanding, alerts
Listing- or seller-side	High for visibility changes	High	Low	Insertion flow, seller tools
Two-sided	Low at both extremes	Medium	High (pair-level treatment)	Ranking where scarcity varies
Cluster (region × category, district)	Low inside clusters; spillover at edges	Low to medium	Medium (cluster maps, cluster-aware analysis)	Prices, paid visibility, badges, ranking policy, SEO
Switchback	Low if carryover is short	Medium	Medium	Short-lived mechanisms
Geographic	Low	Very low	Low to medium	Fees, policies, campaigns

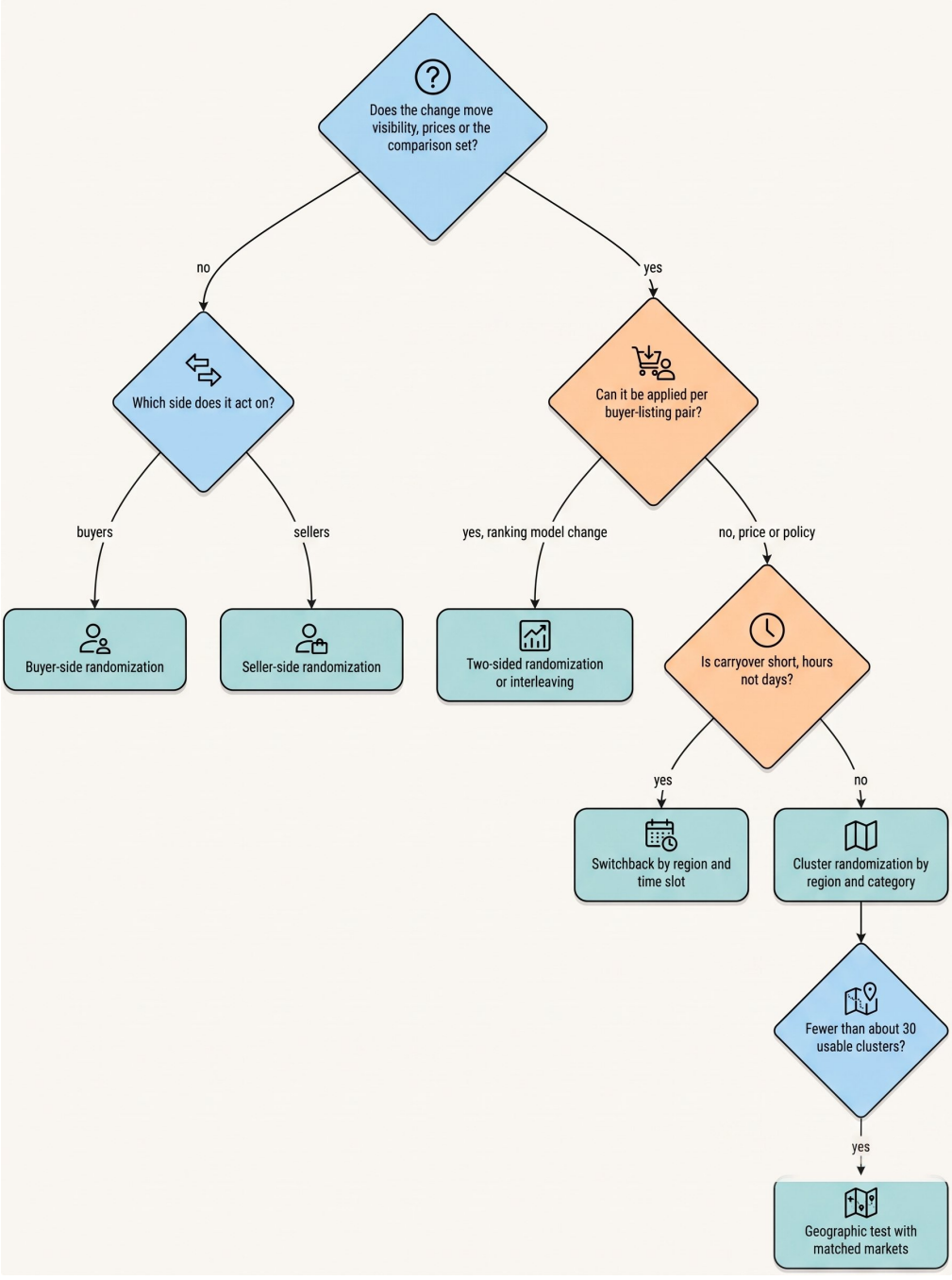


Figure 5: *Choosing a randomization unit. The first question is whether the change works by reallocating something units compete for; if it does, user-level randomization is the wrong default.*

Interleaving: power for ranking models

For comparing two ranking models on the same traffic, **interleaving** merges both rankers' results on one page and credits each contact to the ranker that contributed the listing. Airbnb reported in 2025 that interleaving and counterfactual methods increased experiment sensitivity by up to 100×, depending on method and metric (Zhang et al., 2025). Because both rankers serve the same buyer on the same page, interleaving sidesteps buyer-side interference for model comparisons. It does not answer policy questions such as how many paid slots to show, whose effect is on the whole page; Ranking for Two Sides: Relevance, Freshness and Paid Visibility uses it to pick candidates and a cluster test to decide policy.

Metrics: Liquidity First, Revenue Second, Both Watched

The choice of primary metric decides what a marketplace learns to do. Revenue is the easiest primary metric to defend in a monetization review and the most dangerous one to optimize alone, because a marketplace can raise revenue for months by spending liquidity it will need later.

A metric hierarchy for marketplace experiments

Level	Examples	Use in experiments
Liquidity outcomes	Qualified contacts per searcher; share of new listings with a qualified contact within the window; time to first contact	Primary metric for discovery, supply and trust changes; guardrail for everything
Monetization outcomes	Paid-product revenue per seller; delivery of paid exposure; repeat purchase	Primary metric for pricing and product tests, never alone
Behavior	Searches, views, saves, bump purchases	Diagnostics; explains why an outcome moved
Health	Latency, errors, crash rate, unsubscribes, complaints, fraud reports	Guardrails

A marketplace experiment's metrics must be **observed on both sides**. A bump-price test's primary metric must include what happened to sellers who did not bump, and a ranking test must report seller-side outcomes such as new listings contacted within the window. Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics defines the liquidity metrics and Data Platform for Marketplaces: Events, Lakehouse and Data Products governs them.

Guardrails, including the control group

A **guardrail metric** is a metric that must not degrade beyond a pre-set threshold for a change to ship. Ranking for Two Sides: Relevance, Freshness and Paid Visibility lists the guardrails Veldmark applies to every ranking experiment: search-to-contact rate, new listings contacted, seller exposure concentration, paid-product delivery, paid share of first-page positions, contacts on unavailable listings, and ranking latency. Two more belong on every marketplace experiment, and Veldmark's bump test would have been stopped by either.

1. **The control-trend guardrail.** Compare the control group's outcome with its own pre-period and with an untouched holdout. If control sellers' contacts per bump fall while treatment's hold, the treatment is taking from control. In the bump test this signal (−9%) was visible by the second week.
2. **The congestion guardrail.** For any paid visibility product, track the resource it sells, here first-page time per bump, at the market level. A product whose unit value falls when more of it is sold must be tested by cluster.

Guardrails need near-real-time computation for harm (a broken checkout, a spike in fraud reports) and final computation for the rest. That is the reason Adevinta's streaming computation exists, and it is why Data Platform for Marketplaces: Events, Lakehouse and Data Products separates provisional from final metric values.

Long-term effects and holdouts

Marketplace effects compound. A ranking change that favors professional sellers may lift contacts this month and reduce private-seller supply next quarter. Two practices help. **Long-running holdouts**, a small share of units kept on the old experience for months, measure the cumulative effect of many changes. And **surrogate or long-term models** estimate long-run outcomes from short-run signals; Airbnb's 2025 publications include a causal framework for long-term ranking effects estimated from A/B data (Airbnb Tech, 2025). Both cost traffic and patience; both are cheaper than discovering a supply decline a year later.

Experimenting on Ranking and Prices

Ranking and pricing experiments are where marketplace experimentation becomes governance, because they change who gets exposure and who pays what.

Ranking. Every ranking change reallocates exposure among sellers, and some of those sellers have paid for exposure. eBay's 2020 description (the current system is not public) of its unified ranker for promoted listings makes the coupling explicit: promoted and organic listings compete for the top slots in one ranker that balances relevance against cannibalization of gross merchandise bought (eBay, 2020). For a classifieds marketplace, a ranking experiment is therefore also a monetization experiment and must report paid-product delivery as a guardrail. Veldmark's ranking-policy experiment, run from 6 September to 17 October 2027 in Tradeyard Netherlands with category × region clusters, is described in Ranking for Two Sides: Relevance, Freshness and Paid Visibility; its design followed the standard below.

Prices. Price experiments add three concerns.

- **Supply elasticity**. Prices for sellers change supply as well as revenue. A higher insertion fee reduces listings; a cheaper bump changes the congestion of the first page. Measure supply and liquidity, not just revenue.
- **Fairness and law**. Showing different prices to different consumers raises legal questions. The Omnibus Directive (Directive (EU) 2019/2161) inserted a new point (ea) into Article 6(1) of the Consumer Rights Directive: traders must inform consumers, "where applicable, that the price was personalised on the basis of automated decision-making". Its recital 45 excludes dynamic or real-time pricing that does not involve personalization, and the Commission's guidance on the amended directive covers the topic (European Commission, n.d.). Whether a randomized price test falls under that duty is a question for counsel, not for the experimentation team. Rules of the Marketplace: DSA, DAC7 and Vertical Regulation owns the detail. Veldmark's practice is to test prices by cluster (a whole category in a whole region sees one price) and to publish the price list per cluster during the test, which avoids individual price differences for comparable sellers.
- **Professional contracts**. Dealers and agencies have contracts, and price tests for them are usually negotiation designs, not randomized tests. Causal methods on observational data are often the honest alternative. OLX described in 2026 how propensity-score matching corrected selection bias in package dashboards, estimating uplifts of +17.6% to +97.4% between package levels and validating them against a randomized test (OLX Tech Blog, 2026); How Listing Marketplaces Make Money discusses the commercial implications.

Price badges and valuations. A badge such as "good price" or "below the Valuo estimate" changes how buyers judge the neighbors of the badged listing, which is

market-level interference. Hearthly's pricing hint therefore goes live in Lyon and Bordeaux on 22 November 2027 as an experiment randomized by city district rather than by user, as Models in Production: ML Platforms and Use Cases describes.

In Practice: Veldmark's Experimentation Standard

The bump post-mortem reached the architecture advice forum on Wednesday 9 June 2027, two months after rollout. Ingrid framed it not as a statistics error but as a platform gap: Tradeyard's tooling could randomize only users and seller accounts, its analysis computed per-user t-tests, and nobody had been asked to watch the control group. In 2026, Tradeyard, Motorvane and Hearthly ran about 640 experiments between them on three tools (a vendor tool at Tradeyard, a home-grown framework at Motorvane and Hearthly's own scripts in BigQuery), with three different contact metrics.

The decision record, **Experimentation standard v1**, adopted on 30 June 2027, has six parts.

1. **One analysis engine first**. A group statistics service, owned by Priya Raman's Data & ML team, with SRM checks, CUPED, sequential testing for guardrails and cluster-aware inference. Brands keep their flag tools until 2028 but must send assignments and exposures to the group exposure log.
2. **Unit types as a platform capability**. The assignment service supports buyer, seller, listing, cluster and time-slot units. Cluster maps for Tradeyard (region × category group, built from observed search-to-contact flows) and for Hearthly (city districts) are published reference data with versions.
3. **Governed metrics only** for primary metrics and guardrails, from the Data Platform for Marketplaces: Events, Lakehouse and Data Products metrics catalog, starting with qualified contacts v1.0.
4. **Design review by risk**. Any experiment that changes visibility, prices, fees or trust thresholds needs a registry entry with hypothesis, unit, primary metric, guardrails and owner before launch, reviewed by the experimentation guild, and, for prices, by Olivier Petit's team and Lucía Ortega's regulatory team.
5. **Mandatory marketplace guardrails**: the control-trend guardrail, the congestion guardrail for paid visibility, and the ranking guardrail set.
6. **A 2% long-term holdout** per brand, rotated annually, for cumulative effects.

The bump retest ran under the new standard from Monday 6 September to Sunday 31 October 2027 in **Tradeyard Belgium**, because Tradeyard NL's clusters were reserved for the ranking-policy experiment in the same weeks. Belgium's private sellers were split into 96 region × category-group clusters; 48 saw €1.99 and 48 saw €2.49, with the price shown on the published price page for each category and region.

Metric (cluster level, 8 weeks)	Control clusters (€2.49)	Treatment clusters (€1.99)	Relative change	95% interval
Bumps bought per seller	1.00 (index)	1.21	+21%	+14% to +28%
Bump revenue per seller	1.00 (index)	0.967	**−3.3%**	−6.2% to −0.4%
Contacts per bumped listing	1.00 (index)	0.74	−26%	−33% to −19%
Median first-page time per bump	2.4 h	1.5 h	−38%	—
Qualified contacts per live listing (all listings)	1.00 (index)	1.00	0%	−2% to +2%

The cluster estimate (−3.3%) was close to what NL had experienced after rollout (about −2.5%) and nowhere near the February estimate (+18%). The intervals are much wider than in February, because 96 clusters carry far less information than hundreds of thousands of sellers; that loss of power is the price of an answer with the right sign. Olivier's team withdrew the €1.99 price in the Netherlands on Monday 6 December 2027 and started designing a different product, a bump with guaranteed first-page time, whose value does not collapse as more sellers buy it.

Trade-offs

Decision	Option A	Option B	How to choose
Platform	Buy a commercial experimentation tool	Build an in-house platform	Buy for flagging and UI if the vendor supports custom units and exports raw assignments; build or own the analysis engine once marketplace designs and governed metrics matter
Centralization order	Standardize tools and UI first	Standardize the stats engine and metrics first	Engine and metrics first; they make results comparable, which tools alone do not
Randomization	User-level by default	Unit chosen per experiment	User-level only for changes that do not reallocate attention, supply or prices
Stopping	Fixed horizon	Sequential tests	Sequential for guardrails and harm; fixed horizon or carefully bounded sequential for delayed primary metrics
Sensitivity	More traffic, longer tests	Variance reduction, interleaving	Variance reduction always; interleaving for ranking model comparisons

Decision	Option A	Option B	How to choose
Prices for pros	Randomized price tests	Matched observational analysis and staged rollouts	Observational methods where contracts and negotiation make randomization impractical
Long-term effects	Rely on short tests	Holdouts and surrogate models	Holdouts for any brand running dozens of ranking and visibility changes a year

Vertical Differences

Concern	General classifieds	Real estate	Automotive	Jobs (where relevant)
Usually scarce side	Buyers for long-tail goods; listings for underpriced popular items	Listings in tight rental and city markets	Varies by model and region; buyers scarce for high-value cars	Candidates for skilled roles; jobs for entry roles
Main interference risk	Paid visibility and ranking among private sellers	Buyers competing for the same rentals; agencies competing within a district	Dealers competing for the same model searches	Applicants competing for the same jobs
Natural cluster	Region × category	City district	Region × make or segment	Region × occupation

Concern	General classifieds	Real estate	Automotive	Jobs (where relevant)
Outcome delay	Hours to days	Weeks to months (sale), days (rental contact)	Days to weeks	Weeks
Pricing experiments	Bumps and insertion fees for consumers; legal review of personalization	Agency packages by negotiation; observational methods	Dealer packages; slot pricing; private-seller fees testable by cluster	Employer posting prices
Sample size	Large; power rarely the problem	Small per district; power often the problem	Medium	Medium

The real-estate row matters most for design. A district in Lyon has few listings and few buyers per week, outcomes arrive months later, and agencies compare notes. Hearthly's experiments will often be underpowered as A/B tests; **its honest options are longer cluster tests, stronger variance reduction, and leading indicators validated against outcomes**, plus restraint about calling small effects.

Failure Modes

- **The transfer mistaken for an effect.** Seller- or buyer-randomized tests of visibility, ranking or price changes report the gain treated units took from control units. Watch the control trend; cluster when units compete.
- **Revenue as the only primary metric.** Monetization tests approve changes that spend liquidity. Use liquidity guardrails on every monetization test.
- **Sample ratio mismatch ignored.** A bot filter, a client crash or a redirect drops units from one variant, and the result is read anyway. Block results when SRM fires.
- **Peeking with fixed-horizon statistics.** Daily looks at a test designed for one look inflate false positives. Use sequential methods or commit to the horizon.
- **Analysis unit finer than the randomization unit.** A cluster test analyzed per user reports intervals that are far too narrow.
- **Overlapping experiments on the same mechanism.** Two ranking tests or a ranking and a pricing test on the same units interact. Use layers.

- **Assignment in the client for prices**. Sellers see different prices on different devices; complaints follow. Assign prices and fees on the server.
- **Metrics that disagree with the board**. An experiment's "contact" is not the governed contact, so a winning test does not move the number the business steers on.
- **No exposure logging for the passive side**. A ranking test logs buyers but not the sellers it moved, so seller outcomes cannot be computed.

The Chief Architect's View

The Chief Architect's job is to make sure the experimentation platform can test what the business actually needs to decide: prices, visibility and paid products, not only page layouts. The decisions, in the order I would take them:

1. **Fund one analysis engine and one exposure log for the group**, before standardizing flag tools or interfaces, and tie it to the governed metrics of Data Platform for Marketplaces: Events, Lakehouse and Data Products.
2. **Make the randomization unit a platform capability**. Buyer, seller, listing, cluster and time slot, with published cluster maps owned as reference data.
3. **Classify experiments by risk** and require pre-registration and review for anything that changes visibility, prices, fees or trust thresholds. Review is not bureaucracy when the alternative is a forecast that missed by nearly €8 million a year.
4. **Mandate the marketplace guardrails**: control trend, congestion, and the ranking set.
5. **Keep a long-term holdout** and report cumulative effects annually.

Who decides: the Head of Data & ML owns the engine, the methods and the cluster maps; product and monetization owners own hypotheses and primary metrics; the experimentation guild reviews designs; legal and regulatory review price and personalization tests; the Group Chief Architect owns the assignment and exposure contracts across brands. What to watch: experiments per year per brand and the share using non-user units; SRM rate; share of experiments with pre-registered primary metrics; the gap between experiment estimates and post-rollout effects, measured for every major launch; and how often a decision was reversed after rollout. The last metric is the platform's real report card. If it is high, the platform is producing confident answers to the wrong questions.

Questions to Ask

1. Which units can our assignment service randomize today, and which experiments have we avoided because it cannot?
2. For our last five pricing or visibility experiments, what happened to the control group's outcomes compared with its own pre-period?
3. Which side of the market is scarce in our largest categories and cities, and does our default design reflect that?
4. Do our experiments use the same definition of a contact as our board pack and our pro reports?
5. Do we log exposure for sellers whose listings a ranking change moved, or only for buyers who saw the page?
6. How do we stop tests early, and are our statistics valid for the way we look at results?
7. When did we last compare an experiment's estimate with the measured effect after rollout, and how large was the gap?
8. Who reviews price experiments for legal exposure, and do we test prices per seller or per cluster?
9. Which experiments overlap on the same mechanism today, and how are they isolated?
10. What share of our traffic sits in a long-term holdout, and what has it told us?

From Experiments to Models

The experiment platform and the data platform give Veldmark trustworthy measurements; most of the changes it measures are now made by models, from ranking and recommendations to valuation and fraud scoring. How those models are trained, served, monitored and governed is the subject of Models in Production: ML Platforms and Use Cases.

Sources and Evidence

Company engineering and investor sources

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- Experimentation Platform at Zalando: Part 1 – Evolution, Zalando Engineering (Shan Huang), 2021 — SRM alerts; analysis rebuilt for concurrent tests; unified tracking.
- eBay makes Promoted Listings in search results more relevant and dynamic, eBay, 2020 — unified ranking of promoted and organic listings, balancing cannibalization.
- Correlation lied to us: rethinking product impact with causal inference, OLX Tech Blog, 2026 — propensity-score matching for package effects, validated against an A/B test.
- Academic publications: Airbnb Tech 2025 year in review, Airbnb, 2025 — long-term ranking effects from A/B data.

Standards, regulation and documentation

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- Consumer Rights Directive, European Commission, n.d. — guidance on the amended directive covers personalized prices.

Research and practitioner sources

- Trustworthy Online Controlled Experiments: A Practical Guide to A/B Testing, Kohavi, Tang and Xu, Cambridge University Press, 2020 — general practice of online experiments.
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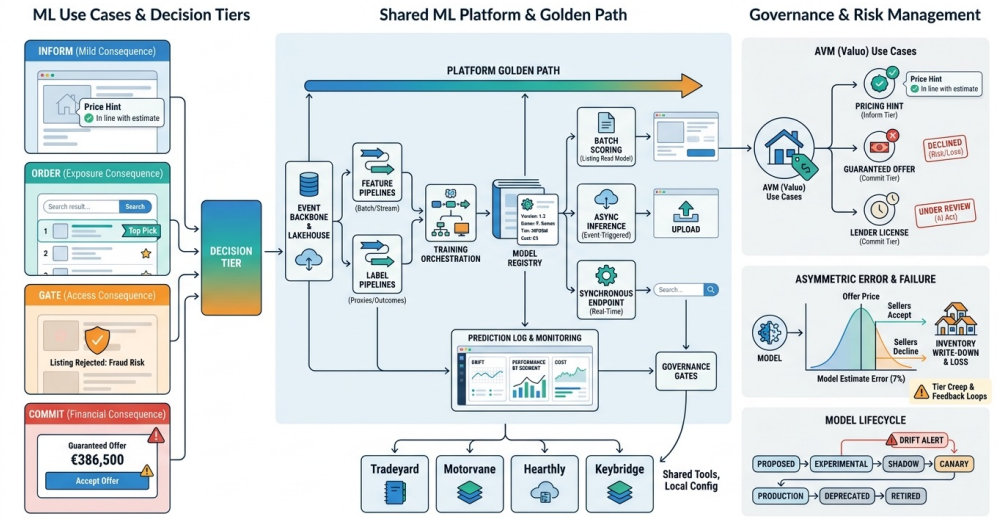
- Safe Testing for Large-Scale Experimentation Platforms, Daniel Beasley, 2023 — 162 Vinted experiments; sequential versus classical tests on delayed metrics.

- Experimental Design in Two-Sided Platforms: An Analysis of Bias, Johari, Li, Liskovich and Weintraub, 2020 (revised 2021) — customer-side, listing-side and two-sided randomization bias.
- Reducing Interference Bias in Online Marketplace Pricing Experiments, Holtz, Lobel, Liskovich and Aral, 2020 — Airbnb meta-experiment; at least 32.60% of the naive estimate due to interference.
- Design and Analysis of Switchback Experiments, Bojinov, Simchi-Levi and Zhao, 2020 (revised 2025) — switchback designs and carryover.
- Harnessing the Power of Interleaving and Counterfactual Evaluation for Airbnb Search Ranking, Zhang, Deng, Du, Gao, He and Katariya, 2025 (KDD 2025) — up to 100× sensitivity gains.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

38. Models in Production: ML Platforms and Use Cases



IN THIS CHAPTER, YOU WILL: Map the machine-learning (ML) use cases of a listing marketplace by the kind of decision each model makes, design the platform that trains, serves and monitors them, and follow Veldmark's Valuo valuation model from a price estimate to a "pricing hint" on every Hearthly listing, and to two proposals that would have turned it into something far riskier.

WHY THE BUSINESS CARES: Models already decide which listings buyers see, which sellers get blocked, what price a seller asks and which homeowner becomes a lead. A GM gains from them only if they are accurate, owned and cheap enough to run in every market. A model that quietly degrades costs liquidity long before it shows up in revenue.

WHY ARCHITECTS SHOULD CARE: Most ML failures in marketplaces are architecture failures: features computed differently in training and serving, labels nobody owns, taxonomy changes that break models silently,

and estimates built to inform that end up being used to commit money. The platform and the governance around it decide whether a group runs 150 models or 150 liabilities.

KEY POINTS:

- **Classify every model by the decision it makes, not by its algorithm.** A model that *informs* (a price estimate), *orders* (ranking), *gates* (moderation, fraud) or *commits* (an offer, a credit input) needs a different depth of validation, monitoring and approval. The decision tier, not the technique, sets the governance.
- **The platform is a golden path, and the hard parts are features and labels.** Training and serving are commodity in 2026. Point-in-time-correct features, labels from moderation decisions, contacts and sales, and a prediction log joined to outcomes are what marketplaces get wrong, and what a shared platform should provide first.
- **An estimate good enough to inform is not good enough to commit.** Zillow's 2021 wind-down of Zillow Offers is the documented case: the valuation model was used to buy homes, forecast error became inventory write-downs, and the business closed. Promote a model up a tier only with a new validation regime, new owners and a capital limit.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Set-and-Forget AI** — Valuation, pricing, ranking and fraud models make consequential decisions, and in most groups nobody can list them all, let alone name an owner and a monitor for each.
- **The Forced Fit** — A central ML platform imposed on teams with working stacks gets routed around; one adopted for its features, models registered first and migrated later, gets used.
- **Vanity Liquidity** — Models trained on clicks and views learn to produce clicks and views; the label decides what the marketplace optimizes, so it needs the same scrutiny as the model.

Every listing marketplace above a modest size runs dozens of models, and few of them were planned as a portfolio. A category predictor arrived with a new insertion flow, a fraud model after a bad quarter, a learning-to-rank model when the search team hired its first data scientist, a price estimate because a competitor had one. Each was built on whatever stack its team knew. A few years later the group has more models than anyone can list, **training data that nobody can reproduce**, and a valuation model being pulled into business decisions it was never validated for.

This chapter turns that accumulation into a portfolio with a platform under it. It builds on the two chapters before it: Data Platform for Marketplaces: Events, Lakehouse and

Data Products supplies the events, lakehouse and governed metrics that models train on, and Testing a Two-Sided Market: Experimentation Platforms supplies the online tests that judge them. Its decision on which models to share across brands feeds One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture. It does not re-teach the use cases that other chapters own: Ranking for Two Sides: Relevance, Freshness and Paid Visibility owns learning to rank, Recommendations for Inventory That Disappears owns candidate generation, Scams, Fakes and Account Takeovers: Fraud Prevention for Listings owns risk scoring, From Raw Supply to Usable Listings: Normalization and Enrichment owns attribute extraction, and Testing a Two-Sided Market: Experimentation Platforms owns online evaluation. It owns the things they share: the **ML platform**, the **feature store** question, **model governance**, and the use case that raises governance questions most sharply, the **automated valuation model (AVM)**. Generative AI and agents, which reuse much of this platform but raise their own questions, are the subject of AI Across the Listing Lifecycle.

The argument runs through one model. Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), owns Valuo, Hearthly's AVM in France. In 2027, Valuo is asked to do three things: put a price-position badge on every listing, power a guaranteed-offer product, and feed a bank's mortgage decisions. The model is the same in all three. The decisions are not.

Setting the Scene at Veldmark: 146 Models and One Price Badge

On **Monday 20 September 2027**, Priya Raman, Veldmark's Head of Data & ML, presents the first group model inventory to the architecture advice forum. It took her team six weeks to compile, mostly by asking around. The inventory counts **146 models in production**: 71 at Tradeyard, 38 at Motorvane, 29 at Hearthly and 8 at Keybridge. They run on four training stacks (SageMaker notebooks and scheduled jobs at Tradeyard, a Kubeflow installation at Motorvane, Vertex AI at Hearthly and Azure Machine Learning at Keybridge). **23 models have no named owner**, 41 have no monitoring beyond service uptime, and 12 could not be retrained from code and data that still exist. None of these figures was known before the inventory.

Three requests arrive the same month, and all three involve Valuo.

1. **The pricing hint**. Claire Fontaine, Hearthly's CEO, wants every French sale listing to carry a badge that compares the asking price with Valuo's estimate: *below the estimate*, *in line*, or *above*. Agencies would see the same comparison in the insertion flow. Hearthly's product team expects higher contact rates on well-priced listings and fewer over-priced listings that sit for months.
2. **The guaranteed offer**. A Hearthly product manager proposes "Hearthly Direct": homeowners who request a Valuo estimate would receive a guaranteed purchase

offer at 94% of the estimate, funded by a property-investment partner, with Hearthly taking a share of the resale margin.

3. **The lender license.** A French bank asks to license Valuo estimates as an input to mortgage pre-approval.

Ingrid Halvorsen, the Group Chief Architect, asks the forum one question about each request before anyone discusses technology: **if the estimate is 8% wrong, who loses money, and how much?** For the badge, a buyer may contact a listing that is less of a bargain than it looked. For the guaranteed offer, Veldmark's partner buys a home for more than it can sell it for, and the loss comes back through the contract. For the lender, a household may borrow against a value that is not there. The model is identical; the consequence of its error is not, and the rest of this chapter is organized around that consequence, not the algorithm.

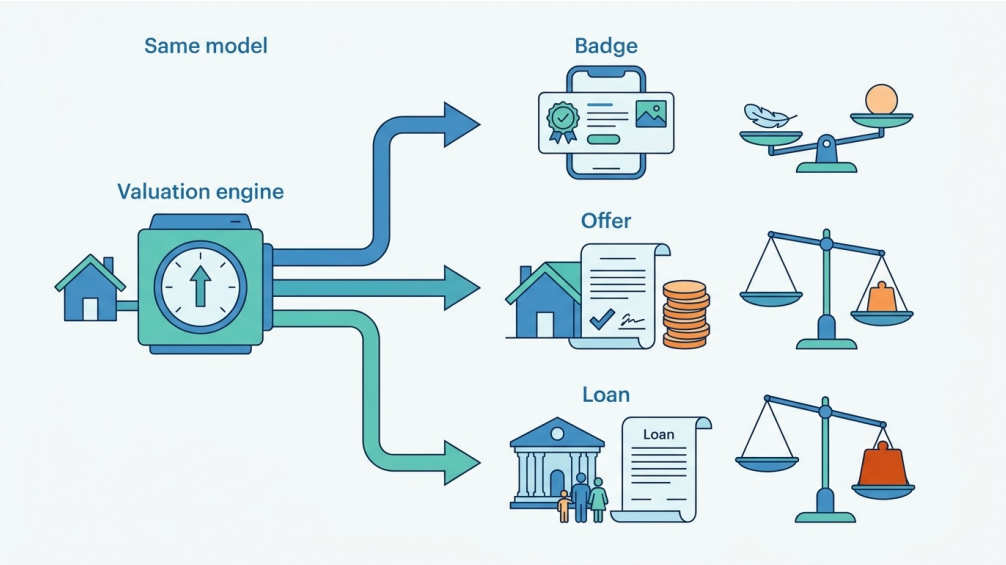


Figure 1: *Valuo is the same model in all three requests; what changes is who loses money when it is 8% wrong, from a mildly misled buyer to a partner's balance sheet to a household's mortgage.*

Concepts: Use Cases, Decision Tiers and the Platform

Terms: from ML capability to model governance

- **ML capability** — a business capability whose output comes from a learned model: category prediction, price estimation, fraud scoring. The capability is owned by

a product team; the model is one implementation of it, and it can be replaced by rules or a vendor.

- **Model** — a trained artifact plus its feature definitions, training data snapshot, evaluation results and serving configuration. A model without its data lineage cannot be audited or reproduced, so the lineage is part of the model.
- **ML platform** — the shared tooling that lets teams build, deploy and operate models without rebuilding infrastructure: data access, feature pipelines, training orchestration, a model registry, serving, a prediction log, and monitoring. In a group it is a platform product with internal customers (see Who Owns What: Organizing Marketplace Engineering).
- **Feature store** — a system that defines features once and serves them consistently to training (offline, point-in-time-correct history) and to inference (online, low-latency current values). Uber's 2017 description of Michelangelo is the reference account: online features precomputed into Cassandra by batch or streaming jobs, using the same pipelines for training and serving to keep them consistent (Uber Engineering, 2017; the system has evolved since).
- **Training–serving skew** — a difference between the features a model saw in training and those it receives in production. It is the most common silent ML bug.
- **Drift** — change in the input distribution (data drift), in the relationship between inputs and outcomes (concept drift), or in what a label means (label drift).
- **Automated valuation model (AVM)** — a model that estimates the market value of a specific asset, usually a property, from its characteristics, location, comparable transactions and market trends. In vehicles the equivalent is a market-value or price-rating model.
- **Model governance** — the rules that decide who may deploy a model, what evidence its deployment requires, how it is monitored, and who answers for it.

Four decision tiers

A model's governance follows the decision it makes, and this chapter sorts those decisions into four tiers.

Tier	What the model does	Marketplace examples	Who bears the error	Governance it needs
Inform	Shows a number or suggestion that a person may act on	Price estimate, pricing hint, category suggestion, demand insight for pros	The user, mildly; trust in the brand	Offline accuracy by segment, display rules for uncertainty, drift monitoring
Order	Decides what people see first	Ranking, recommendations, alert relevance, query categorization	Sellers who lose exposure; buyers who miss listings	Online experiments, exposure and fairness metrics, transparency duties under the P2B Regulation (see Ranking for Two Sides: Relevance, Freshness and Paid Visibility)
Gate	Allows, blocks or delays an action	Moderation, fraud scoring, account suspension, message withholding	A blocked legitimate seller, or a victim of a missed scam	Precision and recall targets, human review of the grey zone, appeal routes, statements of reasons under the Digital Services Act (DSA) (see Trust and Safety as a Platform)

Tier	What the model does	Marketplace examples	Who bears the error	Governance it needs
Commit	Sets a price or term the company or a partner is bound to	Guaranteed offers, instant buying, credit inputs, insurance pricing, automated pro-package pricing	The company's balance sheet, a partner, or a consumer's finances	Separate validation regime, forward-looking back-tests, capital limits, independent review, regulatory obligations

Three properties of the tiers matter.

1. **Tiers attach to uses, not models.** Valuo's estimate is an *inform* model on a listing page and a *commit* model inside a purchase offer. Governance has to follow the use, which means the model registry must record every consumer.
2. **Moving up a tier is a new launch.** A model validated for informing has not been validated for committing, because the error that matters changes (see the Zillow section below).
3. **Gate models generate their own labels.** A moderation model that rejects a listing removes the evidence of whether the rejection was right. Gate-tier models need sampling of passed and blocked items to keep the labels honest (see Scams, Fakes and Account Takeovers: Fraud Prevention for Listings).

The use-case portfolio

The table maps the common ML use cases in listing marketplaces to tier, serving mode and label source. Labels are listed because marketplace ML differs most from e-commerce there: a classifieds marketplace often never sees the sale.

Use case	Lifecycle stage	Tier	Typical serving mode	Label source	Owning chapter
Category prediction	Insertion	Inform	Synchronous, in the insertion flow	Seller's final category; moderator corrections	Category Trees That Can Change: Designing Marketplace Taxonomies
Attribute extraction (text, image)	Enrichment	Inform	Asynchronous, event-triggered	Seller confirmations; audit samples	From Raw Supply to Usable Listings: Normalization and Enrichment
Image classification and quality	Media ingestion	Inform or gate	Asynchronous, batch on upload	Moderator decisions; human labeling	Photos Sell: Media Pipelines for Listings
Duplicate detection	Publication	Order or gate	Asynchronous, pairwise scoring	Confirmed merges; seller disputes	One Property, Three Listings: Duplicates and Entity Resolution
Moderation and fraud scoring	Insertion, messaging	Gate	Synchronous fast path plus asynchronous enrichment	Moderator decisions, user reports, chargebacks	Scams, Fakes and Account Takeovers: Fraud Prevention for Listings

Use case	Lifecycle stage	Tier	Typical serving mode	Label source	Owning chapter
Ranking (learning to rank)	Search	Order	Synchronous, in the search engine or a re-ranker	Contacts, favorites, clicks with position correction	Ranking for Two Sides: Relevance, Freshness and Paid Visibility
Recommendations	Discovery	Order	Batch candidates plus online re-ranking	Views, contacts	Recommendations for Inventory That Disappears
Query understanding	Search	Order	Synchronous, sub-50 ms	Reformulations, click-through	What Did They Mean? Query Understanding and Autocomplete
Price estimation (goods, vehicles)	Insertion, listing page	Inform	Request-time score cached on the listing	Sold prices where known; delisting after contact as proxy	this chapter
AVM (property)	Listing page, owner funnel	Inform, sometimes commit	Batch over the property base, refresh on events	Registered transaction prices	this chapter

Use case	Lifecycle stage	Tier	Typical serving mode	Label source	Owning chapter
Lead scoring	Lead routing for pros	Order	Asynchronous on lead creation	Pro-reported outcomes, CRM status	From Interest to Contact: Messaging and Lead Management
Demand and market insight	Pro tooling	Inform	Batch aggregates	Market data itself	Serving the Pros: Dealer, Agent and Merchant Tooling

Public evidence shows how concentrated such portfolios are. In a 2020 overview, OLX reported "more than 40 machine learning services" in production and that "more than half of the models we use in production are in the trust and safety and search and recommendations areas" (OLX Tech, 2020; the 2020 figures may be superseded, and the current portfolio is not public). That matches what Priya's inventory found at Veldmark: 58 of the 146 models are gate-tier trust and safety models, and 39 are order-tier search and recommendation models.

Architecture: What an ML Platform for Listings Contains

Components and boundaries

The platform has seven parts. The first four are mostly generic; the last three are where listing marketplaces need specific design.

1. **Data access** over the event backbone and the lakehouse (see Every Change Is an Event: Event-Driven Listing Architecture and Data Platform for Marketplaces: Events, Lakehouse and Data Products): listing events, behavioral tracking events with impression position and ranking context, moderation decisions, contacts and transactions.

2. **Training orchestration**: pipelines, compute (including GPUs), experiment tracking.
3. **Model registry**: versions, lineage, evaluation reports, model cards, approvals, and the list of consumers per model.
4. **Serving**: batch scoring, event-triggered asynchronous inference, synchronous endpoints, and in-engine scoring (models executed inside the search engine).
5. **Features**: shared definitions, offline history, and online values where real-time features are needed.
6. **Labels**: pipelines that turn operational outcomes into training labels, with their delay, noise and ownership recorded.
7. **Prediction log and monitoring**: every consequential prediction logged with its model version and features, joined later to outcomes; drift, performance by segment and cost dashboards.

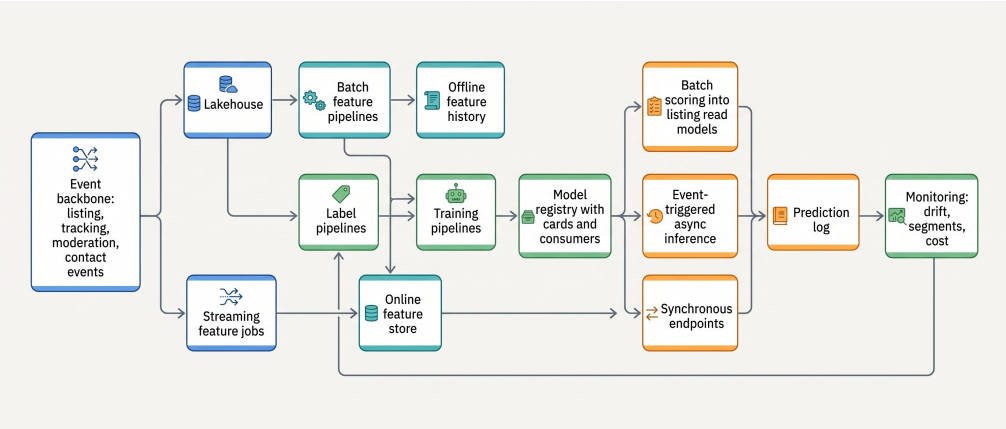


Figure 2: *The listing-specific parts of an ML platform are the label pipelines, the feature definitions shared between training and serving, and the prediction log that joins every consequential score to its outcome.*

Serving modes: most listing models do not need real-time inference

A listing changes rarely compared with how often it is viewed. A Tradeyard listing is published once, edited a few times and viewed hundreds of times. That makes **batch or event-triggered scoring, with the result stored on the listing**, the default serving mode for most listing-level models. The score travels with the listing document into search, the listing page and alerts, and it is recomputed when a `ListingUpdated` event changes an input.

Serving mode	When it fits	Listing examples	Main risk
Batch scoring over a population	Inputs change slowly; the population is known	AVM over all French properties; demand indices	Stale scores after market moves
Event-triggered asynchronous inference	Inputs change on listing events; a few seconds of delay are acceptable	Price rating after a price cut; image classification on upload; duplicate scoring	Event lag; missed events leave old scores
Synchronous endpoint	The input exists only at request time	Category prediction while typing; fraud fast path at submit; query understanding	Latency budget; capacity at peak
In-engine scoring	The model ranks documents already in the index	Learning to rank in Solr, OpenSearch or Vespa	Feature availability in the index; engine upgrades

Zillow's 2020 account of floor-plan detection shows the second mode: when a 3D tour finishes, its panoramas land in object storage, a workflow launches batch-transform jobs that run window and door detection, and the results are stored as JSON. The team chose batch jobs because a live endpoint took about ten minutes to create while a batch job had a fixed startup cost of one to two minutes (Zillow Tech Hub, 2020; the post describes 2020 systems and the current state is not public). **Inference attached to the media and listing lifecycle, not online serving, is the pattern to copy.**

In-engine scoring is the documented pattern for ranking. In 2022, OLX described a LambdaMART model trained with XGBoost and converted to run inside Solr's learning-to-rank plugin, with some features computed at query time and others "refreshed periodically (e.g. CTR)" (OLX Tech, 2022; the current setup may differ). The model lives in the engine; the platform's job is to keep its features fresh in the index and its version in the registry.

The feature store question

Feature stores are the most oversold part of ML platforms in this industry. In the public record I could find, **no classifieds company has documented a feature store.**

Adevinta's 2024 description of its ML Golden Path, built with Adevinta Spain, leboncoin, Marktplaats and Subito, uses Kubeflow Pipelines for workflows and MLflow as model registry, and does not mention one (Adevinta Tech Blog, 2024). OLX's 2025 recommendations service reads precomputed data from ScyllaDB and DynamoDB, which play the role of an online store without the name (OLX Tech, 2025).

Many listing models do not need one, because they consume **listing-intrinsic features**: category, attributes, price, location, image count. These already live on the listing record and in the search document. A feature store earns its cost for two other kinds of feature:

Feature kind	Examples	Where it lives without a feature store	When a feature store pays
Listing-intrinsic	Make, model, mileage, surface, energy class, price	Listing record, search document	Rarely; use the listing event as the source
Aggregated behavioral (batch)	Listing contact rate, seller response time, category demand	Lakehouse tables, refreshed daily	When several models share them and must agree on definitions
Real-time behavioral (streaming)	Messages sent by an account in the last ten minutes, session intent, new-device logins	Ad-hoc caches per team	Almost always: fraud and session ranking need point-in-time-correct history to train on

The rule I use: **build or buy an online feature store when real-time behavioral features drive a gate-tier or order-tier model**; **until then, standardize feature definitions in the lakehouse and log the features at prediction time**. Logged features solve the training–serving skew problem for the models that do not need a store: train on what the model actually saw. Airbnb's open-source Chronon shows the full version of the idea, with features defined once, point-in-time-correct backfills, and a consistency job that compares backfilled values with logged fetches (Chronon repository README, n.d.).

Labels: where marketplace ML is different

A transactional marketplace knows when an item sold. A classifieds marketplace usually does not. Labels therefore come from **proxies**, and each proxy has a bias.

- **Contacts as relevance labels**. OLX trained its 2022 ranking model on purchases "or messages/calls to sellers", because "transactions aren't always confirmed" (OLX Tech, 2022). Contacts are better than clicks, but they reward listings that attract questions, including scams.
- **Delisting as a sale label**. A listing removed soon after contacts probably sold; one that expired probably did not. The seller's "mark as sold" button is the better label, and a reason to design the removal flow to ask for it (see The Listing Aggregate: Identity, Lifecycle and Invariants).
- **Moderator decisions as abuse labels**. They are accurate on what moderators saw and blind to what they did not.
- **Registered transaction prices as valuation labels**. France's *Demandes de valeurs foncières* (DVF) open data publishes property transactions; it is the label set any French AVM is graded against, and it lags the market by months.

The label pipeline is therefore a product in its own right, with an owner, a definition and a delay, and it deserves the same contract discipline as any event (see Data Platform for Marketplaces: Events, Lakehouse and Data Products). **Choosing the label is choosing what the marketplace optimizes**: a ranking model trained on clicks learns to produce clicks, which is the mechanism behind **Vanity Liquidity**.

Monitoring and the drifts specific to listings

Beyond generic drift monitoring, which compares input distributions over time, listing marketplaces have four drift sources that deserve explicit monitors.

1. **Taxonomy and schema changes**. When Tradeyard's "Bikes" category splits into city, e-bikes, cargo and kids bikes on **Monday 1 November 2027**, every model with category as a feature sees a new value and loses an old one. The category predictor, the price-hint model and the fraud model all need a migration plan, which is why the taxonomy service publishes category changes as versioned events with mappings (see Category Trees That Can Change: Designing Marketplace Taxonomies).
2. **Seasonality and market shifts**. Property prices, rental demand and used-car values move with interest rates and seasons. A valuation model must be monitored against fresh transactions, not against last year's error.
3. **Adversarial drift**. Fraudsters adapt to gate models within days. Gate-tier models need monitors on the share of passed listings later reported, not only on input distributions.
4. **Feedback loops**. A price hint changes the prices sellers ask, and the model then trains on prices it influenced. Ranking changes which listings get contacts, and the next ranking model trains on those contacts. D. Sculley and colleagues named this "hidden feedback loops" in their 2015 paper on technical debt in ML

systems, alongside "undeclared consumers" (Sculley et al., NeurIPS 2015). Both are marketplace defaults, not edge cases.

What Marketplaces Have Documented

The public record on marketplace ML platforms is thin and uneven. Adevinta and OLX publish the most; Zillow, Rightmove and the motor portals publish outcomes but little about how. The evidence below is dated because platform posts age quickly.

Adevinta's golden path (2023–2024). Adevinta redesigned its ML Golden Path in January 2023 on top of its shared Kubernetes platform: Kubeflow Pipelines, MLflow as registry, a project template and a command-line tool that hides CI and infrastructure code, plus templates for Spark and distributed training. Two of its stated lessons are worth carrying into any group platform: "Do not hide external tools", and the motto "Use it quickly first, learn it progressively later" (Adevinta Tech Blog, 2024). The same group runs GPU serving on its Kubernetes platform, with custom-metric autoscaling and scale-up under three minutes (Adevinta Tech Blog, 2023), and an image text-detection service on SageMaker endpoints that grew to 330 million requests a month, about twelve times cheaper than the external OCR API it replaced (Adevinta Tech Blog, 2023). **A group platform does not have to mean one serving runtime**; it means one way of registering, deploying and observing models.

OLX's consolidation of ranking models (2025). OLX's learning-to-rank work spread across teams and repositories and caused "duplicated efforts", so the company centralized "all LTR projects within a single repository", with hierarchical configuration per brand (OLX Tech, 2025). That is the shared-model-with-local-configuration pattern discussed below. Its 2023 delivery-time model shows the platform pieces: MLflow, an internal deployment tool called FrejaML, Airflow and batch prediction on AWS (OLX Tech, 2023). Its 2025 recommendations service moved into a data-team-owned Python service with p99 latency under 10 milliseconds, a deliberate choice of team ownership over runtime efficiency that the author openly questions (OLX Tech, 2025).

OLX's model-selection discipline (2025). When choosing an embedding model, OLX filtered public leaderboards by memory, latency and language constraints, built an offline evaluation set from consecutive user interactions in its own markets, fine-tuned the winner, and still lost the first A/B test because the model ignored location and price. It added a re-ranking layer and learned that "similarity is not relevance" (OLX Tech, 2025). **Offline evaluation is necessary and never sufficient** for order-tier models.

Valuation as a product. Zillow's FY2025 annual report calls the Zestimate "our patented proprietary automated valuation model" and gives a 2025 median error of 1.8% for homes listed for sale and 7.2% for off-market homes (Zillow Group 10-K, filed February 2026). The gap has a simple cause: a model that sees the asking price, which is the agent's own valuation, is graded on an easier test. In France, Meilleurs Agents (part of AVIV

Group) reported a median estimate error of 6% against final transaction values for 2021; the figure is self-reported (Online Marketplaces, 2022). Zillow's 10-K also warns that changes in laws, contracts or data-provider practices could limit uses of licensed data "such as model training". **Listing licenses now govern training, not only display.**

Pricing guidance outside property. In 2020, OLX described an ML model that suggests an optimal price range to Otomoto sellers (the current product may differ) (OLX Tech, 2020). Airbnb's KDD 2018 paper described a three-stage pricing system (a booking-probability model, a regression to a suggested price with a custom loss, and personalization) behind Price Tips and Smart Pricing (KDD 2018). The methodology behind the price ratings of mobile.de, AutoScout24 and Auto Trader UK is not public in a form I could verify; a plausible reference architecture, not a description of any of them, is a gradient-boosted or hedonic model per market over make, model, trim, year, mileage, region and options, served as a score stored on the listing and refreshed by listing events.

The Zillow Offers lesson

Zillow Offers was Zillow's iBuying business: it bought homes directly from sellers, renovated them and resold them, launched in 2018. On 2 November 2021 Zillow's board decided to "wind down Zillow Offers operations", citing "home pricing unpredictability, capacity constraints and other operational challenges", with a third-quarter "$304.4 million write-down of inventory" and a reduction of about 25% of the workforce (Zillow Group 8-K, 2 November 2021). The shareholder letter is unusually direct: "We have been unable to accurately forecast future home prices at different times in both directions", and "further scaling up Zillow Offers is too risky, too volatile to our earnings and operations". It also records that Zillow converted "only about 10% of the serious sellers who ask for a Zillow Offer" (Zillow shareholder letter, 2 November 2021).

Four architectural lessons follow. They are my interpretation of the documented statements, not Zillow's own analysis.

1. **The binding problem was the forward forecast, not the present estimate.** Buying, renovating and reselling takes months. A model that estimates today's value is used outside its validated horizon when it prices a resale.
2. **Error becomes asymmetric under selection.** Sellers whose homes the model overvalues accept the offer; the others walk away. A 7% median error on a population becomes a systematic loss on the accepted subset.
3. **Operational capacity couples to model error.** Forecast misses turned into inventory, renovation capacity and demand-planning problems, which a model monitor cannot see.
4. **The commit tier needs its own validation regime.** Back-tests of resale prices over the holding period, error on the accepted population, capital limits that cap

exposure while the model proves itself, and an owner outside the team that built the model.

Figure 3: *Selection makes error one-sided: owners accept the offers the model overvalued and decline the rest, so an error that averages out across all homes becomes a systematic loss on the homes actually bought.*

In Practice: Valuo and the Pricing Hint

Valuo estimates the value of French residential properties from DVF transactions and Hearthly listing data. It runs on Hearthly's Google Cloud stack, scores the property base weekly, and rescores a property when a listing event changes its inputs. In the October 2027 back-test, Valuo's **median absolute error is 7.4% for off-market properties** against subsequent DVF sales and **2.3% for listed properties**, for the same reason Zillow's on-market error is lower: the model sees the asking price. Each estimate comes with an **80% prediction interval**.

Take the canonical Lyon apartment: a three-room, 68 m² flat in Lyon 7e, energy class D, listed in May 2027 at **€389,000** by three agencies, which entity resolution clusters into one canonical property (see One Property, Three Listings: Duplicates and Entity Resolution). Valuo's interval is **€372,000–€401,000**, a width of €29,000, or 7.5% of its midpoint of €386,500. The asking price sits inside the interval, so the badge reads *in line with the estimate*.

Before anyone builds the badge, the forum turns it into rules, recorded on **Monday 4 October 2027**:

Rule	Decision	Reason
When to show a badge	Only when the interval width is at most 12% of the midpoint and the property has at least 8 comparable DVF sales within 18 months	Thin data produces wide intervals; showing them as precise misleads buyers
Thresholds	*Below* if price < lower bound; *above* if price > upper bound; otherwise *in line*	Uses the model's own uncertainty, not an arbitrary percentage
Coverage	Expected on about 71% of French sale listings; none for new builds, rentals or properties without resolved addresses	New builds have no comparable resale history in DVF
Explanation	Badge links to the estimate range, the number of comparables and their date range	Buyers and agencies can see why
Labels	Valuo trains on DVF transaction prices only, never on asking prices of listings that carried a badge	Prevents the model learning from prices it influenced
Data exclusion	Training pipelines reject any dataset tagged as Keybridge agency data	Enforces the March 2027 Keybridge neutrality decision in the pipeline, not in a policy document
Monitoring	Weekly error against new DVF sales by arrondissement and price band; monthly report of asking-price convergence toward estimates	Detects drift and the feedback loop

Rule	Decision	Reason
Owner	Hearthly's valuation product manager owns the badge; Priya's team owns the platform; Julien Marchand signs off model versions	Every consumer and version has a name

Two of these rules are enforced by the architecture, not by policy. The **Keybridge exclusion** is implemented as a lineage tag checked by the training pipeline, so a data scientist cannot violate the neutrality commitment by joining the wrong table. The **label rule** cuts the feedback loop at its source: badges will move asking prices, but Valuo learns only from what buyers paid.

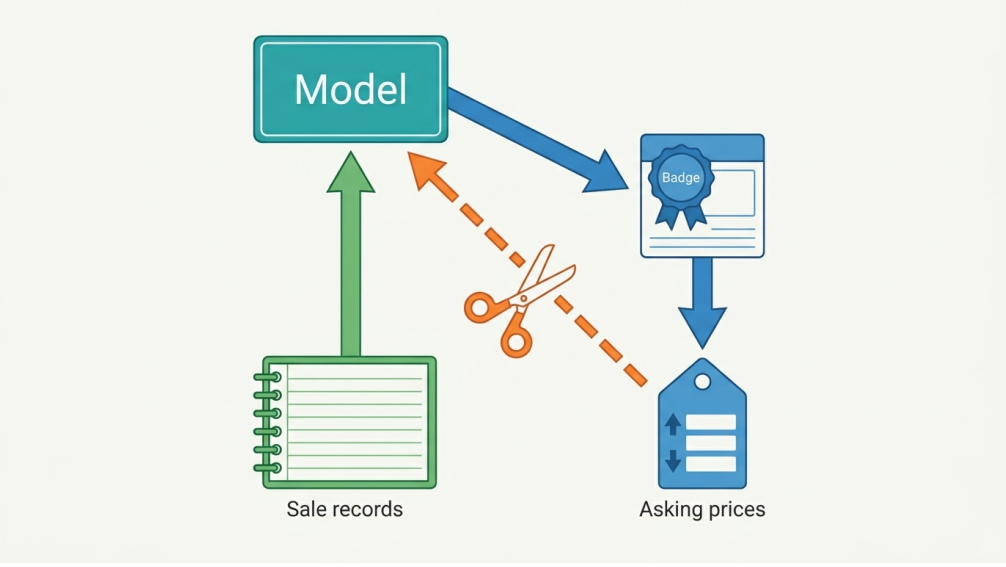


Figure 4: *The label rule cuts the feedback loop at its source: badges will move asking prices, but Valuo trains only on the prices buyers actually paid in the DVF records.*

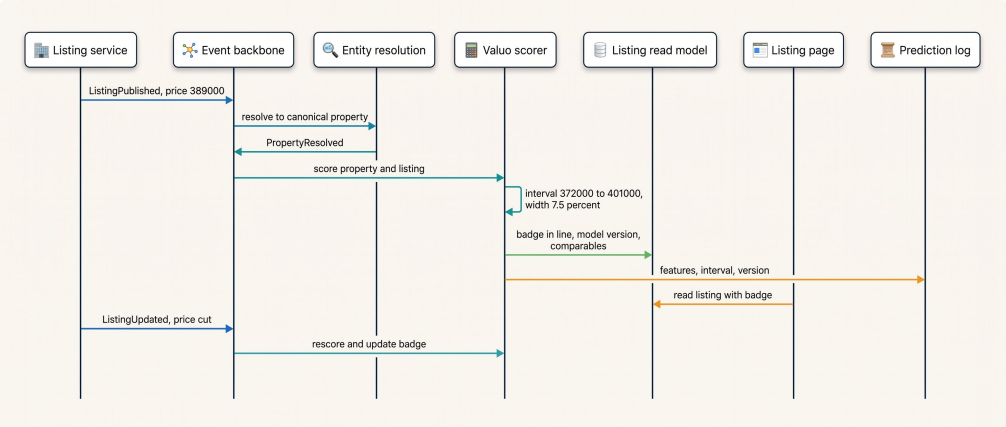


Figure 5: *The pricing hint is event-triggered inference stored on the listing read model; every score is logged with its model version so that later DVF sales can grade it.*

The badge goes live in Lyon and Bordeaux on **Monday 22 November 2027** as an experiment randomized by city district rather than by user, because a badge on one listing changes how buyers judge its neighbors (see Testing a Two-Sided Market: Experimentation Platforms).

The other two requests take a different path. **Hearthly Direct is declined in 2027.** Ingrid's written decision record says that a guaranteed offer is a commit-tier use; that Valuo has never been back-tested on resale prices over a holding period or on the subset of owners who would accept 94% of its estimate; and that the partner's losses would return to Veldmark through the revenue share. If the idea returns, it returns as a referral to an independent buyer who prices its own offers, with Veldmark as neither principal nor price-setter.

The lender license goes to review, not to a yes or no. Lucía Ortega, the Group Data Protection Officer and regulatory lead, points out that under the EU Artificial Intelligence Act (AI Act), "AI systems intended to be used to evaluate the creditworthiness of natural persons" are high-risk under Annex III, with obligations for such stand-alone systems applying from 2 December 2027 after the 2026 amendment (see the regulation note below). Whether a property estimate used inside a credit decision falls under that point is a question for counsel. The architecture does not wait for the answer: the lender would receive a **separate model version with its own model card**, its own segment-level error report and a documented bias test across neighborhoods, so that whichever way counsel rules, the evidence exists.

Motorvane: the same pattern in vehicles

Motorvane's market-value model rates the canonical Essen Volkswagen Golf (2021, 1.5 TSI, 48,000 km) at **€22,800** when the dealer's feed lists it at **€21,900** on **Wednesday 14 April 2027**, so the listing shows "good price". When the dealer cuts the price to **€21,400**

on **Thursday 22 April**, the `ListingUpdated` event triggers a rescore and the rating moves up a band, while price-drop alerts go to 1,180 watchers. The model is inform-tier for buyers, but it is order-tier in practice, because Motorvane's ranking uses the rating as a feature: a well-priced car rises. That second consumer is exactly what Priya's registry did not record in September; **undeclared consumers turn inform models into order models without anyone deciding it**.

Shared, Per-Brand or Per-Market Models

A group has to decide, model by model, whether to share it.

Option	When it fits	Examples	Main risk
One shared model	The task is language- and market-independent, or labels are scarce everywhere	Image quality, NSFW detection, perceptual-hash matching, document OCR	Lowest common denominator; one failure everywhere
Shared model, per-market configuration or fine-tuning	Same task, different language, taxonomy or behavior	Ranking per brand on one code base (the OLX pattern); multilingual text classifiers fine-tuned per language	Configuration sprawl; markets diverge silently
Per-brand model on the shared platform	The task depends on vertical data and economics	AVM, vehicle price rating, dealer lead scoring	Duplicated effort across similar brands
Per-brand model on a local stack	A working model and team exist and migration has no payoff yet	Valuo on Vertex AI until Hearthly leaves Google Cloud	Invisible to group governance unless registered

My rule for Veldmark: **share the platform and the registry everywhere, share models where the task is generic, and never force a vertical model onto a shared code base for the sake of it**. A fraud model for Tradeyard's payment-link phishing and one for Hearthly's fake rentals look for different things (see Scams, Fakes and Account Takeovers: Fraud Prevention for Listings). They should share features, labels infrastructure and monitoring, not weights.

Trade-offs

Platform strategy

Option	Strengths	Weaknesses	Right when
Managed cloud ML suite (SageMaker, Vertex AI, Azure ML) per brand	Fast start; little platform staff	Three clouds, three registries; governance fragments along cloud lines	A single-brand company on one cloud
One managed suite group-wide	One registry and one deployment path	Forces migrations off other clouds; vendor coupling	The group is mostly on one cloud already
Open-source golden path on the shared Kubernetes platform (Kubeflow or similar, MLflow)	Portable across clouds; matches the Adevinta pattern	Needs a platform team; upgrades are yours	The group runs a shared Kubernetes platform and has several clouds
Hybrid: one registry and monitoring layer over heterogeneous training and serving	Governance first, migration later; low disruption	Two kinds of support; standards must be enforced at the registry	Groups after acquisitions, which is to say most groups

Veldmark chooses the hybrid. The decision of **Tuesday 9 November 2027** makes one group model registry (MLflow) and one prediction-log and monitoring standard mandatory for every production model, regardless of where it trains. New models at Tradeyard and Motorvane use a golden path on AWS: SageMaker for training jobs, serving on the shared EKS platform, features from the event backbone. Hearthly keeps Vertex AI while it stays on Google Cloud, and Keybridge keeps Azure Machine Learning, but their models are registered in the group registry and log predictions to the same standard. **Registration first**, **migration later** is how a central ML platform avoids becoming **The Forced Fit**: teams adopt it because registering takes a day, and migration happens when the golden path is better than what they have.

Build or buy for individual capabilities

Capability	Build	Buy or use a model API	Judgement
Ranking	Core differentiation; depends on your behavior data	Vendor relevance engines exist	Build; it is the marketplace's matching engine
AVM and price rating	Depends on licensed transaction data and your listings	Third-party AVMs exist in many markets	Build if valuation is a product or lead source; buy if it is a display feature
Image OCR, NSFW, generic vision	Commodity tasks	Cloud APIs or open models	Buy or use open models, but watch per-call cost at billions of images (Adevinta's OCR replacement was about 12 times cheaper than the external API)
Fraud scoring	Depends on your abuse patterns and labels	Vendor device and network signals	Build the model; buy signals
Text extraction and classification	Increasingly done with large language models	Model APIs	See AI Across the Listing Lifecycle

What ML costs: GPUs, rescoring and unused models

ML cost in marketplaces is dominated by three items: GPU training and serving for image and language models, scoring the whole inventory repeatedly, and the people who maintain models nobody uses. At Veldmark, Priya's inventory estimates **€11.2 million a year** in compute, licenses and tooling across the four stacks in 2027, of which

about €3.1 million is image inference. The most effective cost controls are architectural: score on events instead of on schedules, store scores on listings instead of recomputing per view, retire models with no consumers, and make the registry show cost per model next to its owner. A model whose cost nobody sees will never be retired.

Vertical Differences

Dimension	General classifieds	Real estate	Automotive	Jobs
Highest-consequence models	Fraud and moderation (gate)	AVM (inform, sometimes commit)	Price rating; dealer lead scoring	Candidate matching and ranking
Label quality	Weak: sales rarely observed	Strong but delayed: registered transactions	Medium: dealer-reported sales, delisting	Medium: applications, employer responses
Typical inputs	Text and photos, sparse attributes	Rich attributes (hundreds per property), location, public data	Decoded vehicle data, mileage, options	Structured taxonomies, CVs
Price model use	Seller hint; scam signal	Product, lead source and data revenue	Buyer badge and dealer tooling	Salary estimates

Dimension	General classifieds	Real estate	Automotive	Jobs
Regulatory exposure	DSA for gate models	AVM rules once used for credit; possible AI Act high-risk in credit contexts	Consumer-law exposure of price claims	**AI Act high-risk**: Annex III covers systems "to place targeted job advertisements, to analyse and filter job applications, and to evaluate candidates"
Data rights	Own user data	Licensed feeds may restrict training	Feed and DMS terms	Candidate consent; CV data minimization

The jobs row matters for Veldmark because Tradeyard runs small jobs sections in Poland and Portugal. Any model that ranks candidates or filters applications there is a high-risk system under the AI Act from 2 December 2027 for stand-alone systems, which brings risk management, data governance, logging, human oversight and documentation obligations. The cheapest option may be not to build such models at all in a minor section.

Regulation note (verified October 2026). The AI Act's high-risk obligations for Annex III systems were postponed by the Digital Omnibus on AI, published in the Official Journal on 24 July 2026 as Regulation (EU) 2026/1744: stand-alone high-risk systems from **2 December 2027**, high-risk systems embedded in regulated products from **2 August 2028** (European Parliament Legislative Train, updated September 2026). Annex III point 5(b) covers creditworthiness evaluation of natural persons "with the exception of AI systems used for the purpose of detecting financial fraud" (AI Act, Annex III). Check the dates and the scope with counsel before relying on them.

Failure Modes

- **The model nobody can list.** No registry, so nobody knows which models exist, who owns them or what they cost. Priya's 23 ownerless models are the measurable form of **Set-and-Forget AI**.

- **The undeclared consumer**. A price model built for a badge is used as a ranking feature, a fraud feature and a sales-targeting input, and a change for one consumer breaks the other three.
- **Tier creep**. An inform-tier model is pulled into a commit-tier use because it exists and looks accurate. Zillow Offers is the documented extreme; smaller versions include automatic pricing of pro packages from a demand model.
- **Training on your own influence**. A price hint shifts asking prices, ranking shifts contacts, and the next model learns from the shift. Without a clean label, the model converges on itself.
- **Taxonomy breaks models silently**. A category split or attribute rename changes feature values; models keep scoring with defaults, and accuracy drops without an alert.
- **Gate models without sampled labels**. Rejected listings are never reviewed, so the model's precision is unknown and creeping false positives drive legitimate sellers away.
- **Monitoring the service, not the model**. Endpoints are up, latency is green, and the model has been wrong for a month because a feature pipeline stopped updating.
- **Licensed data in the wrong pipeline**. Feed data whose license excludes model training ends up in a training set because lineage was not tracked; at Veldmark the Keybridge commitment would be the casualty.
- **Platform built before demand**. A feature store and a serving layer arrive before any team needs them, and teams continue with what they had.

The Chief Architect's View

Own the inventory and the tiers. The group architect does not own models, but owns the rule that every production model is registered with an owner, a tier per consumer, a label definition, a monitor and a cost. At Veldmark, registration became a condition of production access on 9 November 2027, with a deadline of 31 March 2028 for the existing 146 models. A model that is not registered by then loses its production credentials, which is the only enforcement that works.

Scale governance with the tier, not with the model's sophistication. A gradient-boosted price badge needs a light review; a simple rule that suspends accounts needs a heavy one. The tiers translate into concrete gates: inform-tier models need segment-level offline evaluation and display rules for uncertainty; order-tier models need online experiments with two-sided metrics; gate-tier models need sampled labels, appeal routes and DSA statements of reasons; commit-tier models need independent validation, back-tests on the population that will accept, capital limits and sign-off by someone who does not report to the model's builder. Model cards, in the sense proposed by Margaret Mitchell and colleagues in 2018, are the right documentation format for all four tiers (Mitchell et al., 2018).

The registry enforces the same lifecycle for every model, with the tier deciding how heavy each gate is.

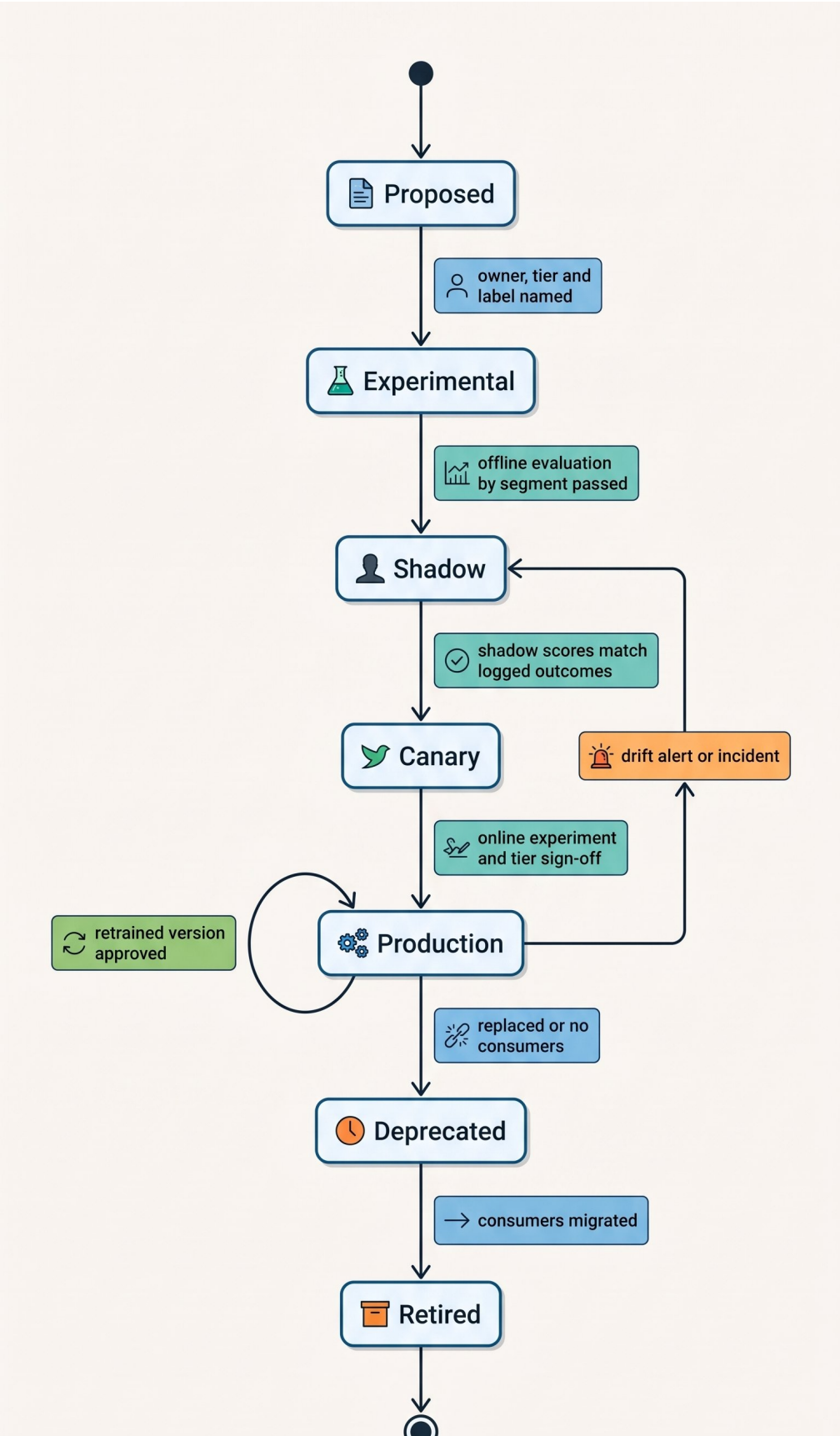

Proposed
owner, tier and label named
Experimental
offline evaluation by segment passed
Shadow
shadow scores match logged outcomes
Canary
drift alert or incident
online experiment and tier sign-off
retrained version approved
Production
replaced or no consumers
Deprecated
consumers migrated
Retired

Figure 6: *Every model passes the same states; what changes with the tier is the evidence each transition requires, and a drift alert sends a production model back to shadow rather than leaving it to decide unwatched.*

Decide who owns labels. Labels come from moderation, messaging, transactions and feeds, which are all owned by other teams. Make label events part of those teams' published contracts, so that a change in how moderators record decisions is reviewed as a change to every model that learns from them.

Fund the platform as a product, measured by adoption. Track the share of production models registered, deployed through the golden path, and monitored to standard. If adoption stalls, the platform is not yet better than the alternatives, and a mandate will not fix that.

Treat promotion up a tier as a board-level decision when money is committed. The Hearthly Direct decision record is the template: what the model was validated for, what the new use requires, who absorbs the loss, and what limit applies while the model proves itself.

Watch these indicators: models without owners; models without monitors; share of predictions logged with versions; median days from training to production; label delay per model; cost per model and cost per thousand predictions; error by segment for every inform-tier model shown to consumers; and, for each commit-tier use, realized loss against the limit.

Questions to Ask

1. Can we list every production model, its owner, its consumers and the decision tier of each consumer, today?
2. For each model, what is the label, who owns the data it comes from, and how long after a prediction does the label arrive?
3. Which models consume features computed differently in training and serving, and do we log features at prediction time?
4. Which of our real-time models actually need an online feature store, and which would be served better by scores stored on the listing?
5. What happens to each model when a category is split or an attribute renamed, and who is notified?
6. Which models influence the data they are later trained on, and how do we break that loop?
7. Is any inform-tier model being used, or proposed, for a commit-tier decision, and what validation would that require?
8. Which licensed data sources restrict model training, and does the training pipeline enforce it?

9. Which of our models could fall under the AI Act's high-risk categories, especially in jobs and credit-adjacent uses, and what evidence would we need by December 2027?
10. What does each model cost per year, and which models have no consumer left?

From Models to Many Markets

This chapter treated models as a portfolio: four decision tiers, a platform that makes features, labels and prediction logs first-class, and governance that grows with the consequence of error. Valuo showed the pattern in one model: an inform-tier badge built with uncertainty rules, a clean label and an enforced data exclusion, and two commit-tier proposals declined or diverted until the evidence exists. Generative models and agents reuse this platform and add their own questions, which AI Across the Listing Lifecycle takes up. The next chapter in this part, One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture, turns to the question that the shared-versus-per-brand model table only touched: how one platform serves many countries, brands and verticals without forcing them into one product.

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- Zillow Group Form 8-K and shareholder letter, 2 November 2021: Zillow Offers wind-down, write-down and quoted reasons.
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Standards, regulation and documentation

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Research and practitioner sources

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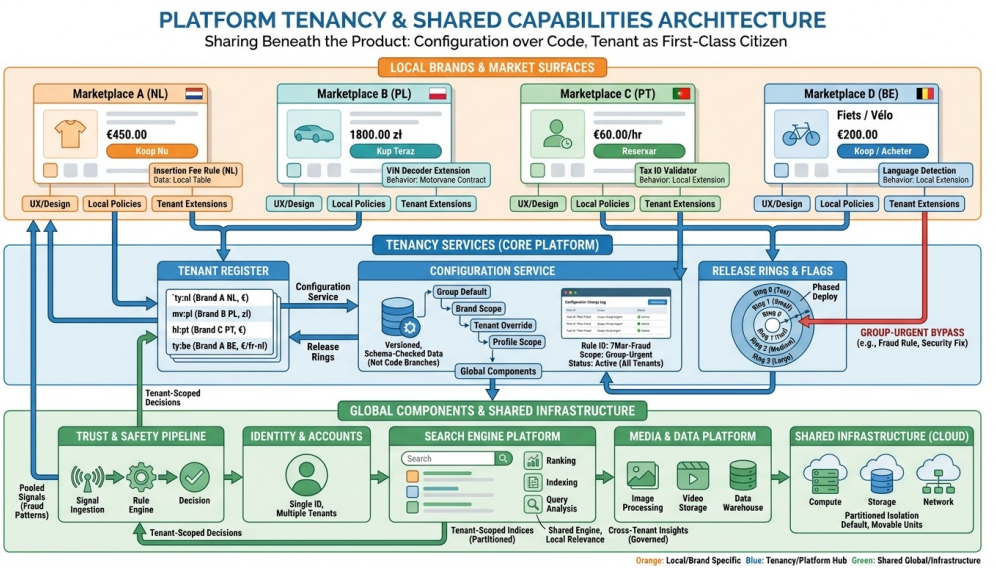
- Model Cards for Model Reporting, Margaret Mitchell et al., 2018: model documentation by intended use and segment performance.
- Customized Regression Model for Airbnb Dynamic Pricing, KDD 2018: three-stage pricing guidance system.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

39. One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture



IN THIS CHAPTER, YOU WILL: Learn the platform models that marketplace groups use to serve many countries and brands, from the fork per country to global components beneath local products. You will design a tenancy model, a configuration hierarchy and a tenant-by-tenant release process, and decide what to share first.

WHY THE BUSINESS CARES: A group with ten marketplaces pays for one fraud rule, one search upgrade or one consent change ten times when its markets are forks. When the platform is forced on them as one product, it pays in lost local fit instead. The tenancy model sets how much of every future improvement is bought once.

WHY ARCHITECTS SHOULD CARE: "Multi-country" is not a deployment detail. It decides how every record, event, configuration value, release and

cost line is scoped. It also decides whether one market can be changed, moved or sold without touching the others, and that decision is far cheaper to make early than to retrofit.

KEY POINTS:

- **The tenant is a first-class architectural element, not a column added later.** Define what a tenant is (at a marketplace group, usually brand × country), give it a stable code, and make every shared capability carry it in data, events, configuration, releases, logs and cost. Separate isolation from identity: tenants can share a cluster and still be separable.
- **Configuration over code, with a limit.** Market differences that are data (locales, currencies, categories, fees, legal texts, feature switches) belong in versioned, schema-checked, effective-dated configuration resolved along a hierarchy. Differences that are behavior go into owned extension points. A fork is allowed only with an expiry date.
- **Share beneath the product, starting with what is uniform and pooled.** The documented public record, Adevinta's 2019 rule above all, supports sharing infrastructure, data, trust signals, media and engines first, and treating the convergence of whole product platforms as a separate, much riskier decision. The centralization pendulum swings; design shared layers so that it can swing back.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Copy-Paste Country** — Copying a codebase per market turns every fraud rule, consent change and search fix into several projects, which this chapter replaces with tenancy and configuration.
- **The Forced Fit** — A shared platform that cannot express a market's real differences gets routed around, which the configure–extend–fork ladder and adoption-led sharing prevent.
- **Deal-Shaped Architecture** — Without a tenancy model, each acquired marketplace stays a separate stack because nothing exists for it to join.

Most marketplace groups grew one country at a time, by launching a site or buying a national leader, and each step was rational: local teams knew local buyers, law and competitors, and copying what worked in the last market was the fastest way into the next. The bill comes later. **Every market that was copied must later be changed separately**, and a group with ten copies of its insertion flow pays for each new rule, regulation and improvement ten times.

The opposite mistake, one platform imposed on every market, is just as common. A group sees the duplication, builds one platform and asks every market to use it as is. The platform fits the market that designed it and nobody else. Local teams add workarounds,

then shadow systems, then a case for leaving. **A platform that cannot express a market's real differences is not shared**; **it is imposed**, and imposed platforms are routed around.

This chapter designs the architecture between those two failures, and applies Part VII's argument to markets: a shared platform earns its tenants through contracts, not mandates. It owns three terms: the **tenancy model** (what a tenant is and how every shared capability scopes, isolates and separates by it), the **global component** (a capability run once for the group beneath local products) and **configuration over code** (expressing market differences as governed data instead of code branches).

The chapter builds on the `tenant` field that the listing event envelope already carries (Every Change Is an Event: Event-Driven Listing Architecture) and on the marketplace profiles of Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy. Sunday Evening Peaks: Reliability, Performance and Cost then runs the shared platform at its peak, and After the Deal: Consolidating Platforms After Acquisitions moves acquired stacks onto the tenancy model; the organization that runs a shared platform belongs to Who Owns What: Organizing Marketplace Engineering. The evidence comes from the public record of Adevinta, eBay Classifieds Group, OLX, AVIV and Vinted.

Setting the Scene at Veldmark: One Rule, Four Countries, Three Days

On **Sunday 7 March 2027** at 20:11 CET, a fake "buyer" sent Femke, a private seller on Tradeyard Netherlands, a phishing link disguised as a Tradeyard Safe Deal payment page. She reported it at 20:19, the account was suspended at 20:26, and a new rule written that evening withheld **37** similar messages to other sellers. Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), had stopped the attack in one market in under an hour.

It had not stopped it in the others. Tradeyard runs four countries on **Yardstick**, its PHP monolith, and the rule reached Tradeyard Belgium only on **Wednesday 10 March**, with the next Yardstick deploy. It never reached Poland or Portugal, whose moderation runs through outsourced queues with their own rule sets. The suspended account's phone number had also registered a private-seller account on Motorvane, Veldmark's automotive marketplace, and nothing connected the two.

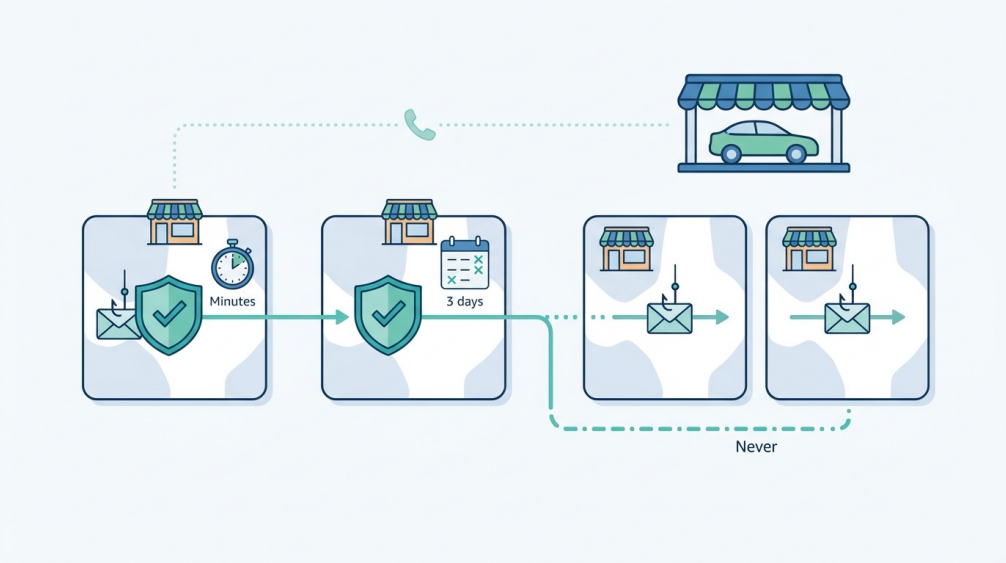


Figure 1: *One rule, four countries, three outcomes: the phishing rule protected Tradeyard Netherlands within the hour, reached Belgium three days later with the next deploy, and never reached Poland, Portugal or Motorvane.*

Ingrid Halvorsen, the Group Chief Architect, used the incident to open a question that Common Ground, the group's €140 million platform program approved in January 2027, had so far answered only with a slogan: "**share platforms, not products.**" What, exactly, is a "market" in a shared platform? If one rule should reach four countries in minutes, but a Polish fee or a Belgian bilingual form should stay local, where does each difference live?

Her team's inventory in May 2027 made the problem concrete. Veldmark runs **ten marketplace tenants** across eight countries: Tradeyard in the Netherlands, Belgium, Poland and Portugal; Motorvane in Germany, the Netherlands, Poland and Austria; Hearthly in France and Spain. Keybridge, the agency software business, is a SaaS company with its own customers, not a marketplace. The tenants span **nine locales** (Dutch in the Netherlands and Belgium, French in Belgium and France, Polish, Portuguese, German in Germany and Austria, Spanish) and **two currencies**, because Poland prices in złoty. Yardstick alone contained about **2,600** conditional branches on the country code and four separate category trees (1,420, 1,380, 980 and 760 leaf categories). Motorvane, by contrast, already ran four countries on one Kotlin stack with configuration per country. Hearthly had rebranded its Spanish acquisition in 2026 and was moving it onto its French platform piece by piece; the Spanish login still ran on Casavista's code.

The three brands had each solved "multi-country" inside their own walls, in three different ways. **Common Ground had to solve it across brands**, and the decision record Ingrid's team brought to the architecture advice forum on **Wednesday 9 June 2027**, "Tenancy model v1", is the thread of this chapter.

Concepts: Tenants, Global Components and Configuration

Market, brand, tenant and profile

Marketplace groups use four words interchangeably, and the confusion causes defects. This book uses them as follows.

Term	Definition	Veldmark example
Brand	The consumer-facing marketplace identity: name, domain, design, go-to-market	Tradeyard, Motorvane, Hearthly
Market	One country (or region) in which a brand operates, with its own law, language, currency and competitors	Poland
Tenant	The unit a shared capability scopes, configures, isolates, releases and bills by; at a marketplace group, usually **brand × country**	`ty:pl` (Tradeyard Poland)
Marketplace profile	A configuration scope below the tenant for a category group with its own endpoint and monetization (see Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy)	Tradeyard Poland vehicles: contact endpoint, insertion fees

A **tenancy model** is the set of decisions that answer, for every shared capability: what is a tenant, how is it identified, how strongly is it isolated, how is its configuration resolved, how are changes released to it, how are its costs attributed, and how could it leave. The questions are the same for search, identity, trust and safety, media and events; the answers may differ by capability, but the tenant codes may not.

The tenant is usually brand × country rather than brand alone, because the things that

differ most between markets (law, language, currency, competitors, partner ecosystems) follow the country. It is usually not smaller, because a marketplace profile or a category group shares its market's users, accounts and legal entity. Profiles are configuration scopes inside a tenant, not tenants.

Isolation is not identity

Tenant identity and tenant isolation are separate properties, although many teams read "multi-tenant" as "shared database".

- **Tenant identity** means every record, event, request, log line, metric and cost line carries the tenant code. It is mandatory everywhere; without it nothing else in this chapter works.
- **Tenant isolation** means how strongly one tenant's data and load are separated from another's. It is a per-capability choice with a cost.

Isolation tier	What is separate	Typical use	Cost
Logical	Nothing physical; a tenant key on every row, message and document, enforced in queries and access policies	Event topics, configuration, catalogs with small tenants	Cheapest; a missing filter leaks data; noisy neighbors
Partitioned	Schema, index, topic or bucket per tenant on shared infrastructure	Search indices, listing schemas, data products	Moderate; a tenant can be moved or exported as a unit
Dedicated	Cluster, database, cloud account or region per tenant	The largest tenant's peak workload; a tenant with contractual or regulatory demands; a tenant that may be sold	Highest; fleet economics lost for that tenant

The right default for a marketplace group is **partitioned**: shared fleets, with each tenant's data in units that can be moved, rebuilt or handed over. The search chapter reaches the same conclusion for index topology: one index per brand and country, on shared clusters sized by load, with dedicated clusters only where a workload or market needs isolation (Search Architecture for Listings: Indices, Freshness and Scale). One design rule follows: **moving a tenant from partitioned to dedicated must not change any client**. Aliases, routing tables and per-tenant connection configuration make that possible.

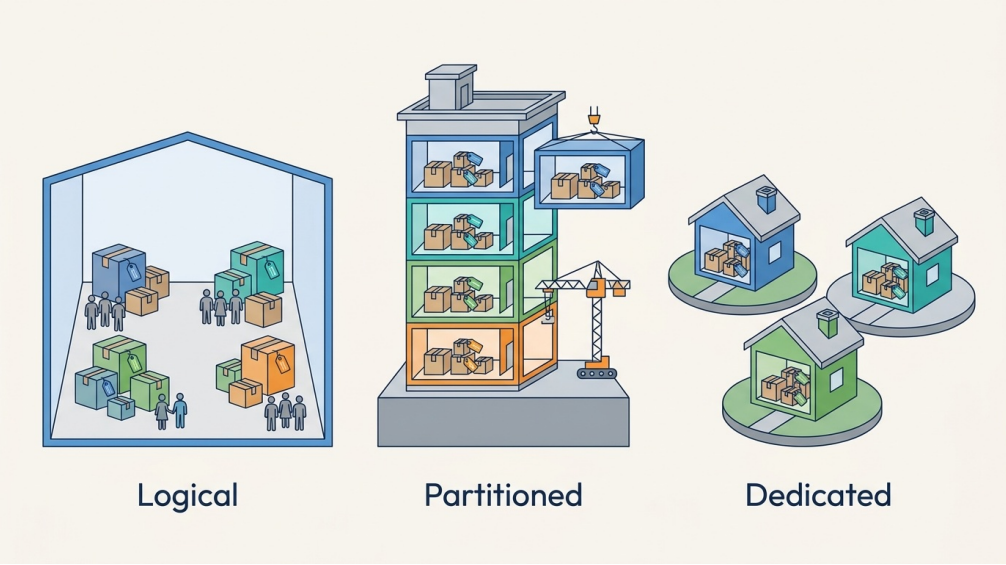


Figure 2: *Identity is mandatory at every tier; isolation is a choice. The partitioned default keeps shared fleets while each tenant's data lives in units that can be moved, rebuilt or handed over.*

Global component

A **global component** is a capability built and run once for the group, with a stable contract, serving many tenants beneath their local products. Examples are an image pipeline, an identity service, a moderation decision pipeline, a search engine platform or an experimentation statistics engine. A global component is not a shared product: the tenant's product team decides how, and whether, the capability appears to its users.

Adevinta's first annual report as a listed company gave the clearest public test for choosing global components. It adopted "a centralised strategy to data and analytics" because this was the area "where investment requirements are large and the technology is uniform, and where the maxim 'built once; used many times' clearly applies", while keeping "local autonomy and decentralised capacity in product and technology"

elsewhere (Adevinta Annual Report 2019, 2020). The rule has two conditions, large investment and uniform technology, and a capability must meet both. Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform shows how well the components that passed it held up over the next five years.

Configuration over code

Configuration over code is the principle that differences between tenants which are *data* are expressed as governed configuration, not as branches in code. "Data" here means values a tenant owner can change without a developer: locales, currencies, category trees, attribute requirements, fee tables, legal texts, consent purposes, moderation thresholds, feature switches, endpoints per profile.

Configuration that can express anything becomes an unreviewed programming language, so the principle comes with three companions: a **schema** for every configuration type, so invalid values fail at review rather than in production; a **resolution hierarchy**, so a value is set once at the right level and inherited below; and **change control** (versioning, review, effective dates, audit), because a configuration change in a shared platform is a production change for a tenant.

When a difference is behavior — a vehicle identification number (VIN) decoder for cars, an energy-certificate validator for French property, a Polish tax rule — it does not belong in configuration. It belongs in an **extension point**: a defined interface in the shared capability where a tenant-owned component plugs in. The ladder is configure, then extend, and only then fork.

Platform Models: From the Fork to the Product Platform

Marketplace groups use six platform models, and they are not a maturity ladder: most groups run several at once, for different capabilities.

Model	What is shared	Documented example	Where it works	Where it fails
Fork per country	A starting codebase, copied and then evolved locally	The Schibsted-era model is widely described as local sites on a Blocket-derived code base; primary documentation is not available (see Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform)	Fast market entry when nothing else exists	Every later improvement is built per copy: **the Copy-Paste Country**

Model	What is shared	Documented example	Where it works	Where it fails
Federation with coordinated choices	Technology decisions and sometimes tools; each brand implements	In 2015 Marktplaats described how "Marktplaats and other eBay Classifieds Group (eCG) companies" jointly evaluated replacements for eBay's legacy search engine; Marktplaats then migrated itself in three months with two or three developers (Marktplaats Technology, 2015, republished 2017)	Strong national brands with capable teams	Duplication of every implementation; slow convergence

Model	What is shared	Documented example	Where it works	Where it fails
Shared core with local configuration	One codebase per vertical or region, configured per brand and country	OLX's 2026 frontend talk, "One-product, three brands", on "serving different countries from a single repo" (OLX Tech, 2026); Marktplaats, 2dehands and 2ememain "hosted on a multi-tenant platform, operated from our Amsterdam location" (Adevinta job posting, read October 2026)	Markets in one vertical with similar products	Markets whose products have really diverged

Model	What is shared	Documented example	Where it works	Where it fails
Global components beneath local products	Capabilities with contracts; products stay local	Adevinta's 2019 rule and its shared infrastructure, data, OCR, media and experimentation services (see Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform)	Multi-brand, multi-vertical groups	Components whose contracts leak one market's assumptions
Platform as product	An internal platform that tenants choose to use	Adevinta's Common Platform, voluntary, onboarding in about ten minutes, grown from about 1,500 engineers in 2020 to more than 4,000 developers by 2023 (Navarro, 2020; see Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform)	Infrastructure and developer tooling	Capabilities where one tenant's opt-out breaks the pooled value

Model	What is shared	Documented example	Where it works	Where it fails
Converged product platform	The product itself: one platform for several brands' user journeys	Adevinta's 2023 plan to turn the leboncoin and Kleinanzeigen platforms into "a unified global platform"; AVIV's 2021 ambition of "aligning our tech roadmap across the organization to create a European real estate marketplace"	Markets that share language, law and product strategy, under stable ownership	Both public examples were reversed or rowed back by 2025

Most successful multi-country sharing is within a vertical. OLX's evidence points to one codebase per vertical serving several country brands: one frontend repository for three brands, a centralized learning-to-rank repository with brand-level configuration, and a "unified platform" in real estate credited with growth in Prosus's 2025 results (Prosus FY25 results, 2025). Which OLX brands share which codebase is not documented; that reading is an inference. eCG's motors vertical published a partner-facing listing API and webhook contract in 2020 (the "MoVe" reference repositories), which suggests a shared motors platform for several national sites, though the tenants are not named (eBay Classifieds Group on GitHub, 2020).

Across verticals, the shared layers sit lower. A car marketplace and a property portal share infrastructure, identity, media, messaging, trust signals and data far more easily than they share an insertion flow. Same Platform, Different Physics: How Listing Verticals Differ explains why the listing models differ.

Both public attempts at a converged product platform were rowed back. Adevinta's is told in Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform; AVIV's is the second. In August 2021, announcing a new chief technology officer, AVIV's chief

executive said the group was "aligning our tech roadmap across the organization to create a European real estate marketplace" (Online Marketplaces, 2021). In December 2025 an interviewer asked the head of Immowelt about AVIV having rowed "back on its plans for a fully integrated platform for all its European portals" (the interviewer's phrase, not AVIV's). The answer was that AVIV had "strengthened the autonomy of its local businesses" while continuing "to operate group-level functions where they create clear advantages… sometimes local and sometimes cross-group" (Online Marketplaces, 2025). What the integrated platform consisted of is not public.

Architecture: A Tenancy Model for a Marketplace Group

Shared and local, layer by layer

The reference architecture below is the author's model, not a description of any company's systems. It separates what is shared from what is local by layer, and puts the tenancy services, which every layer depends on, in the middle.

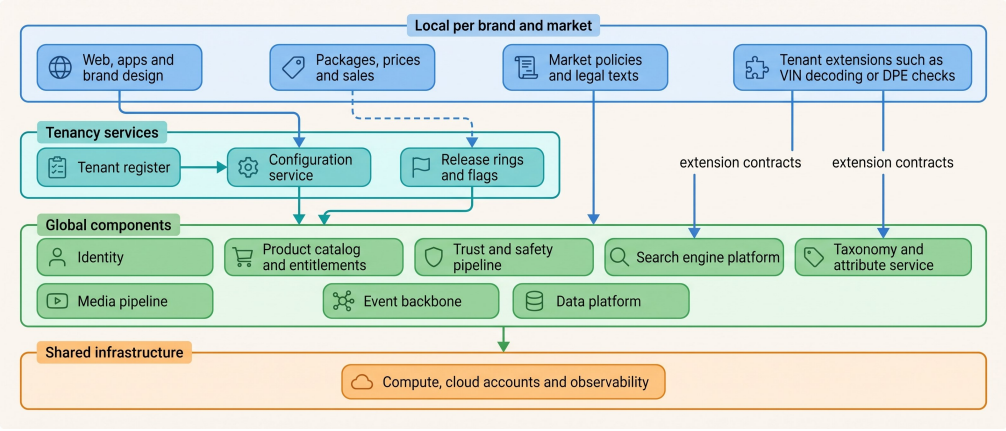


Figure 3: *Shared and local by layer. Brands own the product surfaces, commercial terms and market policies; the group owns global components and infrastructure; the tenancy services in the middle carry every market difference down into the shared layers as configuration, extensions and release scope.*

Local does not mean "outside the platform". A Polish fee table, a Belgian bilingual form or a French energy-label rule is local content, but it is applied by shared components through configuration and extension contracts. The brand owns the content; the platform owns the mechanism.

The tenant register

The tenant register is the authoritative list of tenants and their attributes. It is small, changes rarely, and every shared capability reads it. A useful register entry holds:

- **Tenant code** — stable, never reused, short enough to appear in identifiers and log lines (`ty:pl`).
- **Brand**, **country**, **legal entity** — the entity that contracts with users and sellers, which matters for terms, invoices, data protection and statements of reasons under the EU Digital Services Act (DSA) (Rules of the Marketplace: DSA, DAC7 and Vertical Regulation).
- **Locales** — as BCP 47 language tags (RFC 5646, 2009), with a default; a tenant may have several (`nl-BE`, `fr-BE`).
- **Currency** — as an ISO 4217 code with its minor unit (ISO 4217, n.d.); prices are stored in minor units with the code.
- **Time zone**, **territorial scope and regulator** — for scheduling, statutory deadlines and the Member State whose rules apply.
- **Hosting and isolation profile** — region, cloud account, and the isolation tier per capability.
- **Profiles** — the marketplace profiles inside the tenant, with their endpoints.
- **Lifecycle** — onboarding, live, frozen, leaving, retired.

The last attribute is the one most groups omit. A tenant that is *leaving* (sold, closed or moved to another platform) needs export jobs, transitional services and a switch-off plan, and capabilities should be able to see that state. After the Deal: Consolidating Platforms After Acquisitions covers the deal side; the register is where the platform learns about it.

The configuration hierarchy

Configuration is resolved along a hierarchy, from the most general scope to the most specific. A value set at a higher level is inherited unless a lower level overrides it, and the configuration schema says at which levels each key may be set.

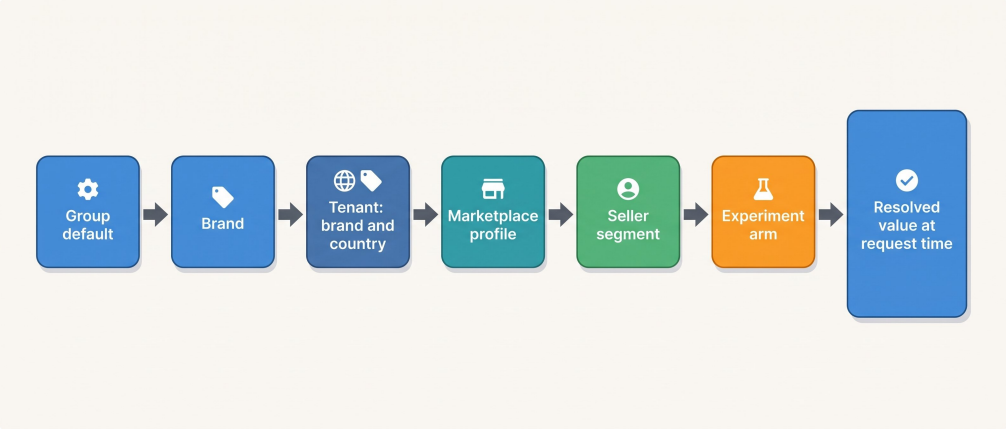


Figure 4: *The configuration hierarchy. Most keys are set at one or two levels; the schema forbids the rest. Experiment arms sit last so that an experiment can override any value for its population without forking it.*

OLX published a small, concrete version of this idea in 2025. Its learning-to-rank projects had spread across teams and repositories and caused "duplicated efforts", so it centralized them in one repository and used the Hydra library for hierarchical configuration, with brand-level configs such as `brand: olx` and `brand: real_estate` (OLX Tech, 2025). The scope there is one capability, but the shape generalizes: one code path, configuration composed by scope.

Four rules keep a configuration hierarchy healthy.

1. **Every key has a schema, an owner and a level.** The schema states type, allowed values and the levels at which the key may be set. The owner is the team that decides its value at each level. A fee table is owned by the brand's monetization team at tenant level; a moderation threshold for phishing is owned by the trust and safety platform at group level, with tenant overrides allowed only upward (stricter).
2. **Changes are versioned, reviewed and effective-dated.** A configuration change that alters what users see is a production change. It gets a reviewer, an effective time and an audit record. Effective dating matters for regulation: a new disclosure rule applies from a date, and listings created before and after it may be validated differently.
3. **Configuration is data, not logic.** If a key starts to need conditions ("if category is bikes and the price is above €500 and the seller is new…"), it has become code. Move it into a rule engine that is designed for rules (with tests and simulation), or into an extension.
4. **Resolution is observable.** For any request, an engineer can see which value was resolved and where it came from. Without this, a multi-tenant incident becomes guesswork.

Extension points

When configuration is not enough, a shared capability offers **extension points**: interfaces at defined places in its processing where a tenant-owned component runs. Each extension point has a contract (inputs, outputs, latency budget, failure behavior) and an owner on each side.

Adevinta's 2023 dispute back office is the clearest published example of the shape. Transaction events from each marketplace arrived via Kafka, SQS or REST and were normalized into one model; the user interface was a micro-frontend that marketplaces could extend; and the stated constraint was "We couldn't hardcode anything… or make any assumptions" (Adevinta Tech Blog, 2023). Per-tenant adapters on the way in, a canonical model in the middle and an extension surface on the way out: that is the general pattern for a global component that serves different products.

The configure–extend–fork ladder decides which mechanism a difference gets:

Rung	When	Who owns it	Exit
Configure	The difference is a value	Brand (content) on a group schema	—
Extend	The difference is behavior at a known point	Brand (extension) on a group contract	Promote to the core if a second tenant needs it
Fork	Neither works, and the market cannot wait	Brand, under a decision record	Expiry date and a plan to fold back or to leave the platform

The fork rung exists because pretending it does not is how shadow systems start. A fork that is declared, dated and reviewed is a managed exception. An undeclared fork is the Copy-Paste Country in miniature.

Release by tenant

A shared platform changes every tenant at once unless its releases are scoped by tenant, with two mechanisms.

- **Release rings**. Deployments of a global component progress through rings of tenants: synthetic test tenants first, then small and low-risk tenants, then the largest. The ring definitions live in the tenant register.
- **Feature flags scoped by tenant**. A deployed change stays dark until a flag turns it on for a tenant, a profile or an experiment population. Pete Hodgson's

taxonomy distinguishes release toggles, experiment toggles, operational toggles and permission toggles, each with a different lifetime (martinfowler.com, 2017); in a multi-tenant platform, release toggles must expire or they become permanent configuration nobody owns.

Rings and flags solve opposite problems. Rings limit the blast radius of a defect; flags let a tenant adopt a change on its own schedule. A platform needs both, and it needs one more rule: **some changes must not wait for a tenant's schedule**. A fraud rule, a security fix or a legal takedown applies everywhere it is relevant, at once. The tenancy model should name these "group-urgent" change classes in advance.

Data residency and data boundaries

For an EU marketplace group, data residency (the requirement that certain data is stored or processed only in certain places) does not mean keeping data in the user's country. The General Data Protection Regulation (GDPR) restricts transfers of personal data to countries outside the European Economic Area unless one of the transfer mechanisms in Chapter V applies (Regulation (EU) 2016/679, 2016). Hosting every tenant in EU regions, and governing support and vendor access from outside the EU through those mechanisms, therefore covers most marketplace needs. Some sectors and contracts impose stricter national rules; the register's hosting profile is where such a constraint is recorded.

The harder boundary in a multi-brand group is not geographic but legal and commercial: **which tenant may use which data**. Purpose limitation under the GDPR, competition law, platform-to-business commitments and the group's own promises to customers all restrict sharing between tenants, even when the data sits in the same warehouse. A shared platform therefore needs tenant-scoped access policies by default and explicit, recorded exceptions (a shared fraud block list, for example). Data Platform for Marketplaces: Events, Lakehouse and Data Products covers governed data sharing across brands; System of Record and Read Models: Listing Data Management covers erasure across copies.

Multi-region deployment is a separate decision. Classifieds are mostly country-scoped businesses, so "multi-country" usually means many tenants on a platform rather than one global dataset. Vinted is the documented exception: it runs one cross-border catalog and chose "not to shard our primary data models across regions", with writes going to a primary site and read projections replicated, accepting freshness lag because "Eventual consistency is being built into how the system works, not an edge case we try to hide" (Vinted Engineering, 2026). A group whose markets do not trade across borders rarely needs that design.

Localization is more than translation

Localization in a listing platform touches at least six layers, and only the first is text:

1. **Interface text** in each locale, owned by the brand, translated through a managed workflow, with plural and gender rules from the Unicode Common Locale Data Repository (CLDR) (Unicode CLDR, n.d.).
2. **Formats**: numbers, dates, prices, units and addresses. Address shapes and postcode rules belong in the gazetteer configuration (Where Is It? Location, Geocoding and Privacy).
3. **Taxonomy and attribute labels** per locale, on one shared concept model (Category Trees That Can Change: Designing Marketplace Taxonomies, Attributes Without Chaos: Schemas, Dictionaries and Validation).
4. **Search analysis**: analyzers per language, and per-language fields where a country is multilingual (Search Architecture for Listings: Indices, Freshness and Scale).
5. **Legal and policy content**: terms, consent texts, prohibited-item lists, mandatory disclosures, statement-of-reasons templates.
6. **Trust content**: scam patterns, phishing domains and message templates differ by language; a rule written in Dutch does not catch a Polish phishing message.

Belgium is the test case. When eBay bought 2dehands.be and 2ememain.be in 2013, its brand page described one marketplace with "bilingual" Dutch and French search and listing under two names (eBay Classifieds Group, 2013 snapshot). One tenant, one inventory, two locales and two consumer-facing names: a tenancy model that equates tenant with locale, or brand with domain, cannot represent it.

In Practice: Veldmark's Tenancy Model v1

The decision record

The decision record adopted on **9 June 2027** had nine parts, summarized below with the reason for each.

#	Decision	Reason
1	A tenant is **brand × country**; ten tenants with codes `ty:nl`, `ty:be`, `ty:pl`, `ty:pt`, `mv:de`, `mv:nl`, `mv:pl`, `mv:at`, `hl:fr`, `hl:es`	Law, language, currency and competitors follow the country; the 20 marketplace profiles are configuration scopes inside tenants

#	Decision	Reason
2	Tenant codes appear in every group identifier (`vm:<brand>:<country>:<id>`), event envelope, API call, log line, metric label and cost tag	Identity is mandatory everywhere; the listing event contract adopted it on 1 July 2027 (Every Change Is an Event: Event-Driven Listing Architecture)
3	**Keybridge is not a tenant** of Common Ground's marketplace capabilities; it integrates with Hearthly through the same public supply contract as every other vendor, and reaches the shared seller registry only as an integration principal with scoped grants (Private Sellers, Dealers and Agencies: Modeling Who Lists)	The March 2027 neutrality commitment: no Keybridge agency data is used for Hearthly's advantage
4	Default isolation is **partitioned**; logical isolation is allowed for configuration and the event backbone; dedicated isolation is decided per capability and recorded	Fleet economics with movable units
5	One **configuration service** with schemas per key; resolution order group → brand → tenant → profile → seller segment → experiment; every change versioned, reviewed and effective-dated	Removes country branches from code as capabilities move out of Yardstick
6	Extension points defined per global component; brand-owned extensions must meet the component's latency and failure contract	VIN decoding (Motorvane), energy-certificate checks (Hearthly) and Polish insertion-fee rules (Tradeyard) are behavior, not values
7	**Release rings**: ring 0 synthetic tenants; ring 1 `ty:pt` and `mv:at`; ring 2 `ty:be`, `ty:pl`, `mv:nl`, `mv:pl`, `hl:es`; ring 3 `ty:nl`, `mv:de`, `hl:fr`	Smallest tenants first; the three largest tenants last
8	**Group-urgent change classes** bypass rings: fraud and scam rules, security fixes, legal takedowns	The 7 March phishing rule would have taken minutes, not three days

#	Decision	Reason
9	Every shared capability documents its **tenant exit path**: export format, handover, transitional run and switch-off	Veldmark is majority-owned by a private-equity investor; any tenant could be sold

Two choices drew argument. Felix Brandt, Motorvane's chief technology officer, objected that Motorvane already ran four countries on one stack and did not need a group configuration service to tell it how. The forum's answer was decision 5's scope: the configuration service is mandatory for **global components**, not for a brand's internal services. Motorvane keeps its own configuration for its own services and uses the group service only where it consumes a group capability. Katarzyna Nowak, Tradeyard Poland's general manager, asked why her market sat in ring 2 rather than ring 1 when it had the least to lose from an early defect. The answer was traffic: with **3.6 million** live listings and 8.9 million monthly users, Poland is too large to be a canary.

The phishing rule, replayed

Before the shared trust-and-safety platform went live for Tradeyard Netherlands and Belgium in **November 2027** (Trust and Safety as a Platform), Ingrid's team replayed the March incident against it as a design test of the tenancy model.

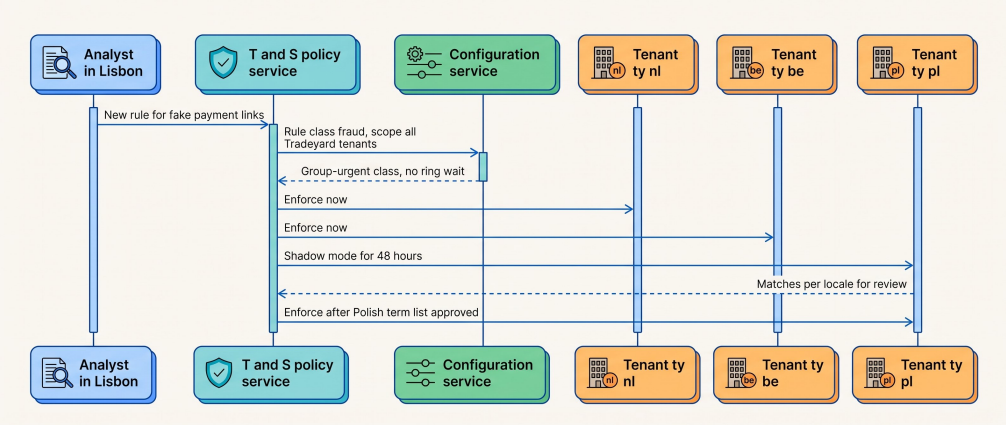


Figure 5: *The 7 March rule replayed on the shared platform. The rule is policy content with a tenant scope, not a deployment. Tenants whose language the rule was written for enforce at once; tenants in other languages run it in shadow mode until a locale-specific term list is reviewed.*

The replay produced three findings that went back into the tenancy model.

- **The rule's structure was global, its content was not.** The domain-age signal (a payment domain registered that day) and the link pattern applied everywhere. The message wording that triggered the rule was Dutch. The platform split rules into a group-owned **detector** and tenant-owned **locale packs**, so a rule reaches every tenant at once, and a Polish or Portuguese pack can follow within the 48-hour shadow window.
- **Cross-brand signals need an explicit boundary.** The phone number that linked the Tradeyard account to a Motorvane account is exactly the kind of signal that pooled trust data is for. The Group Data Protection Officer had approved a shared block list holding only hashed identifiers of accounts terminated for fraud (Scams, Fakes and Account Takeovers: Fraud Prevention for Listings). The tenancy model records it as the one standing exception to tenant-scoped access in trust and safety.
- **Statements of reasons are tenant-scoped.** Each moderation decision carries its tenant's legal entity, locale and territorial scope, because the DSA statement of reasons is issued by the provider to the user in the relevant Member State (Rules of the Marketplace: DSA, DAC7 and Vertical Regulation). A global rule produces local decisions.

Configuration replacing branches

The second test, on the Tradeyard insertion flow that moves out of Yardstick in the second quarter of 2028, found no country branch that justified a fork. Daan Visser's team sampled 300 of the roughly 2,600 country branches in Yardstick and classified them. About 70% were values (field lengths, mandatory attributes per category, fee amounts, legal text keys, date formats): configuration. About 20% were behavior at a small number of points (Polish insertion-fee calculation, Belgian language detection on titles, Portuguese tax identifier validation): extensions. About 10% were dead, referring to features retired years earlier.

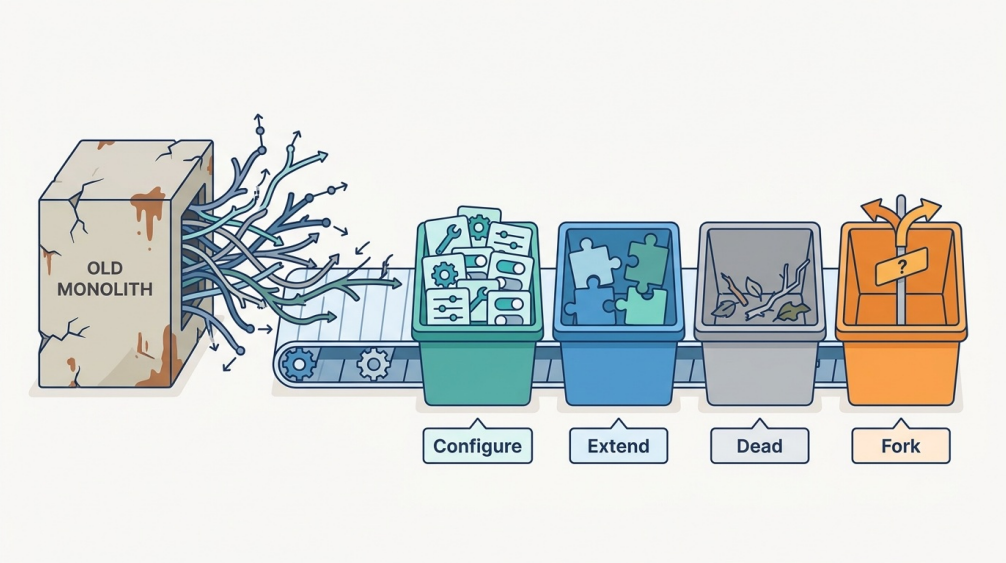


Figure 6: *Sorting 300 sampled Yardstick country branches: about 70% were values for configuration, about 20% behavior for extensions, about 10% dead code, and none justified a fork.*

The Polish vehicles section shows how the layers combine. Tradeyard Poland's vehicle listings keep classifieds-style insertion fees for private sellers, priced in złoty from the brand's fee table (configuration at the `ty:pl` vehicles profile). They consume Motorvane's VIN decoding through the same extension contract Motorvane uses, as Same Platform, Different Physics: How Listing Verticals Differ recommends. And dealers above a small listing count are offered Motorvane packages, with their stock syndicated to Tradeyard through Motorvane's feed; the shared listing core records which brand owns the commercial relationship for each listing (Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy). **One market, two brands, one shared capability, two commercial owners:** the tenancy model has to represent all four facts.

Release rings in numbers

The rings were used for the first time for the group taxonomy and attribute service, which went live in October 2027 with Tradeyard first. Ring 1 (`ty:pt`) received the new category resolution on Monday 4 October 2027, ring 2 a week later and ring 3 (`ty:nl`) on Monday 25 October, one week before the bikes split went live on **Monday 1 November 2027** (Category Trees That Can Change: Designing Marketplace Taxonomies). The ring-1 week found two defects that would have mattered at scale: a Portuguese label falling back to Dutch where a translation was missing, and a cache key that omitted the tenant code. **The second defect is the classic multi-tenant bug**, and it is the reason decision 2 makes the tenant code mandatory in every cache key, not only in data.

Trade-offs

Choosing a platform model per capability

Capability	Fork per country	Shared core, local config	Global component	Converged product	Veldmark 2027 choice
Compute, cloud accounts, observability	Wasteful	—	**Yes**	—	Global; AWS primary, Hearthly on Google Cloud through 2028
Identity	Splits fraud signals	Possible per brand	**Yes**, with brand profiles	—	Veldmark ID, approved June 2027, rollout 2028
Trust and safety	Rules built four times	—	**Yes**, local policy as content	—	Shared pipeline, local policies
Search	Upgrades done three times	—	**Yes**: engine, indexer, ranking service; relevance per brand	—	Shared OpenSearch for Tradeyard and Hearthly; Motorvane stays on Elasticsearch until 2029

Capability	Fork per country	Shared core, local config	Global component	Converged product	Veldmark 2027 choice
Taxonomy and attributes	The Category Straitjacket	—	**Yes**, trees per tenant on one concept model	—	Group service, October 2027
Product catalog and entitlements	Billing by Guesswork risk	—	**Yes**, shared model and ledger; packages and prices per brand	—	Group catalog by Q2 2028
Listing insertion flow	—	**Per brand**, configured per country	—	Possible within one brand only	Per brand
Web and apps	—	**Per brand**	Shared SEO services (Found by Google: Web, App and SEO Architecture for Listings)	—	Per brand
Packages, prices, sales	—	**Per brand and market**	—	—	Local

Isolation choices

Choice	Benefit	Cost	When it is right
Logical isolation everywhere	Lowest cost; simplest operations	Leaks from a missing filter; noisy neighbors; hard to separate a tenant	Configuration, low-risk metadata, small tenants
Partitioned by default	Movable, exportable units; shared fleet	Many objects to manage; per-tenant schema changes	Most listing, search and data workloads
Dedicated for the largest tenant	Protects the peak; independent maintenance	Pooled economics lost; two operating modes	When one tenant's peak sets the fleet's capacity
Dedicated per tenant	Strongest separation; easiest sale	Fleet economics lost entirely	Regulated data, contractual isolation, an imminent sale

When the guidance does not hold

The configure-and-share approach assumes that markets are similar enough to be described by one model with parameters. It fails in three situations. When a market's product has diverged for structural reasons — a transactional model where others are lead-based, or a regulatory regime with no equivalent elsewhere — a shared core turns its differences into extensions that carry most of the logic, and a separate product is more honest. When one tenant is so much larger than the others that its needs dominate every design review, the "shared" platform becomes that tenant's platform with passengers, and the others are better served by components than by a core. And when ownership is likely to change soon, deep sharing creates separation work that the group will pay for at exit (Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform).

Vertical Differences

Concern	General classifieds	Real estate	Automotive	Jobs
What varies most by country	Language, payment and shipping options, prohibited items, consumer law	Partner feed ecosystems, mandatory disclosures (energy labels), address and parcel systems, agency monetization norms	Vehicle data sources (VIN and registration databases), dealer management systems, financing partners	Labor law, salary disclosure rules, job classification standards
Easiest to share across countries	Identity, trust, media, messaging, search engine, ads	Media, identity, data and ML tooling, search engine	Media, VIN decoding where sources overlap, lead management, search engine	Matching infrastructure, identity, data
Hardest to share	Transactional journeys (PSP and carrier contracts are national)	Ingestion adapters, pro tools, valuation data sources	Dealer packages and price ratings	Classification and compliance
Documented signal	OLX's one-repo, three-brand frontend (2026)	AVIV integrate-then-localize (2021–2025); OLX "unified platform" in real estate (2025)	eCG motors vertical partner API (2020)	—

Real estate is the hardest vertical to share across countries. National professional-software ecosystems, mandatory disclosures and valuation data sources differ more between countries than anything in general classifieds, and the abstraction cost falls on exactly the features that make a portal win locally. A shared engine with local

ingestion, disclosure and pro-tooling edges fits the evidence; a single property product across countries does not. Same Platform, Different Physics: How Listing Verticals Differ owns the profiles; AVIV Group: Building a European Real-Estate Platform from Acquisitions examines AVIV's record.

Failure Modes

1. **The tenant added as an afterthought**. A capability built for one brand gains a `tenant` column when a second brand joins. Caches, search queries, exports, metrics and cost reports miss it for months, and the first cross-tenant leak is found by a customer.
2. **Brand equated with domain or locale**. Belgium breaks it (one tenant, two locales, two names); Poland's vehicles section breaks it (one tenant, two commercial owners); a rebrand breaks it (one tenant, two domains during migration).
3. **Configuration that became code**. Conditional expressions in configuration values, with no tests, no simulation and no owner. The shared platform's real behavior is now spread across hundreds of untested configuration files.
4. **Release toggles that never expire**. Every release flag left in place becomes permanent per-tenant configuration with no schema, and the number of possible platform behaviors grows with the product of all of them.
5. **Rings without a group-urgent lane**. A fraud rule waits for the largest tenant's release window, which is the Yardstick problem rebuilt on a modern platform.
6. **The largest tenant as the design authority**. Tradeyard Netherlands carries about 45% of Tradeyard's search traffic. If every shared component is designed in its review meetings, the smaller tenants get a platform built for someone else.
7. **Pooled data without recorded boundaries**. Signals pooled for fraud detection quietly become inputs to ranking, pricing or sales targeting, in breach of the purpose they were pooled for and of commitments such as Keybridge's.
8. **No exit path**. A shared capability that cannot export or hand over one tenant's data turns any future sale or carve-out into a rebuild under deadline.

The Chief Architect's View

Decide the tenancy model once, early, at group level. The tenant definition, the code format, the mandatory carriage of the tenant code and the exit-path requirement are group decisions recorded in one decision record and changed only through the architecture forum. Everything else in this chapter can vary by capability; those cannot.

Decide the platform model per capability, not for the group. "Are we a shared platform or a federation?" is the wrong question. The right one is asked capability by capability: does it pass both halves of Adevinta's 2019 test, large investment *and* uniform

technology? Does its value grow when data is pooled, as it does for fraud signals and experimentation statistics? Can it be shared without converging the product above it? I would score every Common Ground capability on those three questions and publish the scores, because the scores make the disagreements discussable.

What to share first. In order: infrastructure and cloud governance, because they are uniform and invisible to users; the event backbone and data platform, because every later capability depends on them; trust signals and identity, because their value grows with pooling and fraud does not respect brand boundaries; media and the search engine, because they are expensive and technically uniform; then the taxonomy service and the catalog and entitlements model, which need the tenancy services to exist. Product surfaces come last and are often never shared. The sequence after an acquisition follows similar logic with deal-specific constraints; After the Deal: Consolidating Platforms After Acquisitions develops it.

Who decides. The group decides schemas, contracts and the tenancy model; the brand decides configuration values, extensions and product. The platform team decides how a capability is built and run, and is accountable to its tenants for service levels and cost. When a brand asks for a fork, the forum decides, and the decision record carries an expiry date.

What to watch.

- **Adoption without mandate**. For each global component, the share of eligible tenants and workloads using it, and the reasons of those that do not. A platform that only grows by mandate is heading for the Forced Fit.
- **Country branches in code**. The count should fall as capabilities move to configuration; a rising count is an early sign of a new fork.
- **Configuration health**. Keys without an owner, release toggles older than their expiry, values set at forbidden levels.
- **Lead time from one tenant to all**. For a fraud rule, a consent change or a search improvement: the time from first tenant to last. On 7 March 2027 this was three days to Belgium and never to Poland or Portugal.
- **Cost per tenant**. Every shared capability reports cost by tenant, so tenants can see what they pay for and the group can see whether pooling saves money.
- **Separability drills**. Once a year, export one tenant's data from each shared capability and prove it can run on its own. The drill is cheap; discovering at signing that it cannot be done is not.

The pendulum. Marketplace groups swing between centralization and local autonomy, and the public record shows both Adevinta (2023 convergence, 2025 decentralization) and AVIV (2021 integration, 2025 localization) swinging within a few years. I do not think a Chief Architect can stop the pendulum; ownership, strategy and leadership change faster than platforms. What the architecture can do is make the swing cheap in

both directions: global components with contracts that a tenant can leave, configuration that a tenant can take with it, and products that never depended on another brand's code. **The goal is not the right position on the pendulum; it is a platform that survives the next swing**. Built to Be Replaced: Evolving a Marketplace Platform Over a Decade returns to this as the cost to reverse each sharing decision.

Questions to Ask

1. What is a tenant in our platform, what is its code, and does every record, event, cache key, log line and cost line carry it?
2. For each shared capability, what is its isolation tier, and could we move one tenant to dedicated infrastructure without changing any client?
3. Which market differences live in code today, and how many of them are values that belong in configuration?
4. Does every configuration key have a schema, an owner, permitted levels and an effective date? Who reviewed the last change to it?
5. Which extension points does each global component offer, and which brand-owned extensions run in them under what contract?
6. How long does a fraud rule, a security fix or a consent change take to reach every relevant tenant? Which change classes bypass release rings?
7. Which data do our tenants share with each other, under which recorded exception, and for which purpose only?
8. For each shared capability, what is the exit path for one tenant: export, handover, transitional run, switch-off?
9. Which shared components grew by adoption and which by mandate, and what would happen if the mandate were withdrawn?
10. Is any current initiative converging whole product platforms rather than sharing capabilities beneath them? If so, where is its separate investment case?

From Sharing to Staying Up

A tenancy model decides how the platform is shared; it does not decide whether the shared platform stays up when everyone uses it at once. The next chapter, Sunday Evening Peaks: Reliability, Performance and Cost, takes the shared platform to its Sunday-evening peak and asks how to keep it fast, available and affordable.

Sources and Evidence

Company engineering and investor sources

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- eBay Classifieds Group: 2dehands.be – 2ememain.be — eBay Classifieds Group, 2013 (archived snapshot). One bilingual marketplace under two names.
- Adevinta job posting describing the Benelux multi-tenant platform — Adevinta via TCV portfolio jobs, n.d. (read October 2026). Marktplaats, 2dehands and 2ememain on one multi-tenant platform; recruiter copy, not a technical account.
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- Building a global event-driven platform, part 2 — Vinted Engineering, 2026. Single-writer primary with replicated read projections across regions.
- Ex-eBay Classifieds and Domain CTO Brent McLean switches to AVIV Group — Online Marketplaces, 2021. AVIV's integration ambition across its European portals.
- Interview: new Immowelt boss Theo Mseka — Online Marketplaces, 2025. AVIV rowing back from a fully integrated platform toward local autonomy with selective group functions.

Standards, regulation and documentation

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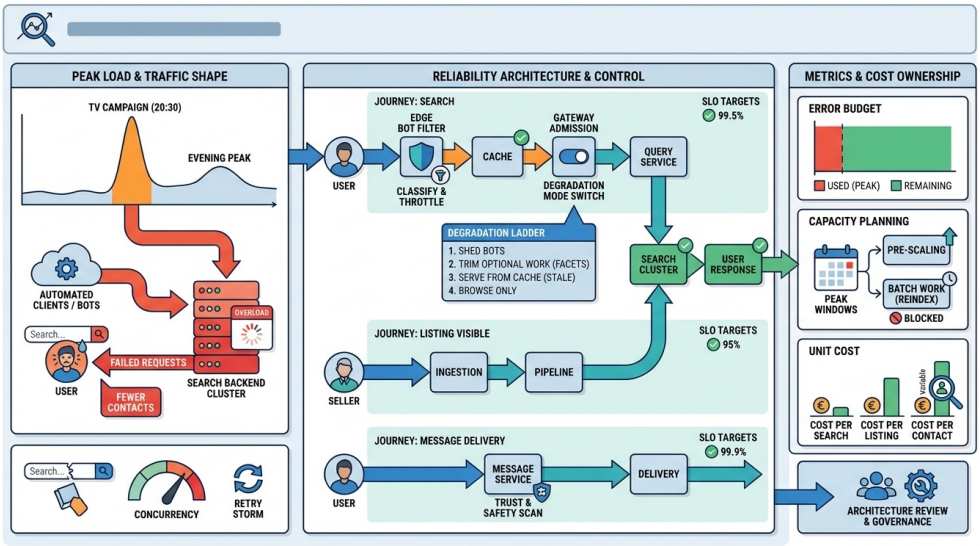
- Regulation (EU) 2022/2065 (Digital Services Act) — European Union, 2022. Statements of reasons and Member State scope of moderation decisions.
- RFC 5646: Tags for Identifying Languages — IETF, 2009. BCP 47 locale tags such as `nl-BE` and `fr-BE`.
- ISO 4217 currency codes — ISO, n.d. Currency codes and minor units.
- Unicode Common Locale Data Repository — Unicode Consortium, n.d. Locale formats and plural rules.

Research and practitioner sources

- Feature Toggles (aka Feature Flags) — Pete Hodgson, martinfowler.com, 2017. Toggle categories and lifetimes.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

40. Sunday Evening Peaks: Reliability, Performance and Cost



IN THIS CHAPTER, YOU WILL: Take apart one 48-minute search brown-out and rebuild the reliability architecture it exposed. You will define service level objectives for the journeys that create liquidity, plan capacity for evening and campaign peaks, design degradation modes, decide how far to go with multiple regions, manage bots and scrapers, and measure cost per search and per listing.

WHY THE BUSINESS CARES: A marketplace's busiest hour is also its most valuable: it is when buyers browse, sellers list and contacts happen. An outage at 20:30 on a Sunday costs more contacts, more seller trust and more advertising revenue than the same outage at 04:00 on a Tuesday, and the infrastructure that prevents it is one of the largest lines in the technology budget.

WHY ARCHITECTS SHOULD CARE: Reliability problems in listing platforms are rarely exotic. They come from predictable peaks meeting scheduled jobs, caches, retries, bots and shared clusters. The fixes are architectural:

journey-level objectives, isolation of batch from interactive work, degradation designed in advance, and cost made visible per unit of work.

KEY POINTS:

- **Reliability is defined per journey, not per service.** Search, viewing a listing, getting a new listing searchable, delivering a message and sending an alert are the journeys that create liquidity. Each needs a service level indicator measured where the user is, an objective, and an error budget that changes what teams do when it runs low.
- **Most peaks are predictable, and many failures are self-inflicted.** Evening leisure browsing, TV campaigns, scheduled reindexing and bot surges are known in advance. Plan capacity with the marketing calendar, keep batch work out of peak windows, and design degradation modes that shed bots and optional features before searches fail.
- **Cost is an architecture metric.** Cost per thousand searches, per live listing and per contact, allocated through an ownership catalog, shows where scale is cheap and where it is not. Bots, duplicated stacks and unowned telemetry inflate those costs more than any single service does.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Search as System of Record** — when the search index cannot be rebuilt cheaply and safely from upstream data, every reindex is a risky event that someone schedules for a quiet hour that turns out not to be quiet.
- **Vanity Liquidity** — uptime percentages per service look healthy while the evening journeys that produce contacts degrade; objectives must be set on the journeys that create matches.
- **Deal-Shaped Architecture** — four stacks with four search clusters, four observability bills and four on-call rotations pay for peak capacity and reliability engineering four times.

The earlier chapters of Part VII — PLATFORM: The Distributed System Underneath built the shared platform beneath Veldmark's brands; this one asks whether that platform holds at the hour when liquidity is made. **Reliability engineering in a marketplace is mostly the discipline of knowing the traffic shape and designing for it**, rather than reacting to it. The shape is familiar: quiet nights, a lunchtime bump and a long evening peak when people browse from the sofa. On top come the spikes the business creates itself — a television campaign, a push notification to millions of alert subscribers, a bulk feed from a large dealer group — and the traffic nobody invited, from scrapers and bots.

The stakes are asymmetric: a minute of failed search at the weekly peak loses more contacts than an hour at night, and contacts are the currency of a listing marketplace

(see Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics). Peaks are also when sellers check whether their new listing is visible, when scammers count on moderators being stretched, and when advertising inventory is worth the most. **The peak hour is where reliability, revenue and trust meet**.

The chapter builds on its neighbors: Search Architecture for Listings: Indices, Freshness and Scale owns index design and reindexing mechanics, Every Change Is an Event: Event-Driven Listing Architecture the event backbone, and One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture the shared-platform model whose blast radius this chapter has to manage. What it adds — journey objectives, a degradation ladder, a regional ambition and unit costs per brand — is what Part VIII — LEADERSHIP: The Chief Architect's Agenda weighs when it decides which of Veldmark's stacks to consolidate.

Setting the Scene at Veldmark: Forty-Eight Minutes on a Sunday Evening

On **Sunday 17 January 2027**, at **20:35 CET**, search on Tradeyard Netherlands slowed down. Tradeyard is the general classifieds brand of Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build). The 99th-percentile latency of search requests rose from about **400 ms** to about **7 seconds** and stayed there for **48 minutes**. About **9%** of search requests failed outright. The apps showed spinners, then error messages; the web showed partial result pages without facets.

The post-incident review, which Ingrid Halvorsen reads in her first week as Group Chief Architect in February 2027, finds no single cause. Three ordinary things happened at once.

1. **A television campaign**. Tradeyard's marketing team ran a New Year "declutter" spot in the 20:30 ad break on a national channel. Search traffic on Tradeyard NL rose about 30% above the usual Sunday peak within four minutes.
2. **A scheduled reindex**. A weekly full reindex of the NL Solr collections had been scheduled for Sunday at 20:00 years earlier, when the Sunday evening was quieter and the job shorter. It now ran for almost three hours and competed with queries for CPU and disk.
3. **Cache eviction at peak**. Solr's caches were flushed as the reindex committed new segments, so the most frequent queries — category browse pages with facets — hit cold caches at the moment traffic peaked.

Two amplifiers made it worse. The apps retried failed search requests twice with no backoff, turning slow responses into more load. And about a third of the requests

reaching the search backend came from automated clients — search engine crawlers, price monitors, scrapers — none of which were throttled when the cluster struggled.

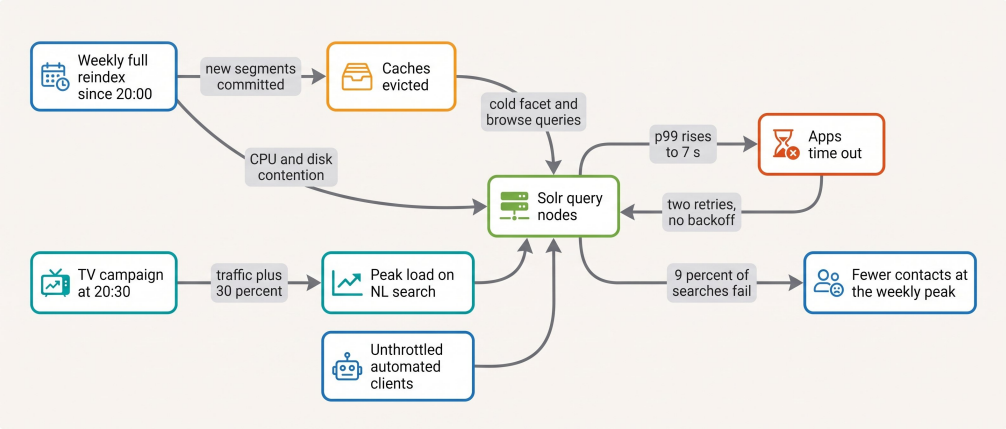


Figure 1: *The brown-out's causal chain. No component failed on its own; a predictable peak met a scheduled batch job, cold caches, unthrottled bots and client retries that fed the overload back into the cluster.*

The review estimates about **3.4 million** failed search requests in the 48 minutes and about **9,800** fewer first contacts than on a comparable Sunday evening. Its action items are familiar: move the reindex, warm caches, add capacity. Ingrid asks a different question, which the rest of this chapter answers. *Why did nothing in the architecture know that Sunday at 20:35 matters more than Tuesday at 04:00, that a campaign was coming, that bots could wait, or that facets were optional?*

The "Sunday-evening peak" is a Veldmark illustration; the documented public example of a marketing-driven peak is Rightmove's, which in 2021 named its busiest minute as "8:48 pm on Wednesday, during the final ad break on prime-time TV" (Google Cloud press release, 2021).

The Traffic Shape of a Listing Platform

Predictable, sharp and partly self-made

Four kinds of load shape a listing platform, and three of them can be predicted.

Load	Shape	Predictability	Example
Leisure browsing	Daily evening peak, weekly peak on one or two evenings, seasonal swings	High	Veldmark's search peaks at **31,000 queries per second** on Sunday 20:00–22:00 CET, against an average of **9,000**
Marketing and notifications	Steps of tens of percent within minutes	High, if marketing and engineering share a calendar	TV ad breaks; push campaigns; alert digests sent at a fixed hour
Supply events	Bursts of writes and index updates	Medium	Bulk dealer feed uploads; price changes across a dealer group; the morning after a holiday
External shocks	Sudden shifts from outside	Low	Vinted reported that the October 2021 Facebook outage added about 15% load to its databases (Vinted Engineering, 2023)

Seasonality can be strong enough to drive architecture. Vinted described its autumn — "the most extreme growth and workload period of the whole year" — as the pressure that pushed it from manually sharded MySQL to Vitess (Vinted Engineering, 2023). Real estate has spring and back-to-school rental peaks; automotive has registration-plate and bonus-season effects in some markets. None of this is surprising, yet architectures often treat it as if it were.

The traffic nobody invited

Automated traffic is a large share of load on any popular public site. Imperva's 2025 Bad Bot Report states that automated traffic has overtaken human activity, at 51% of web traffic, and that bad bots alone make up 37% (Imperva, 2025). The figures are a

security vendor's and cover the whole web, not classifieds specifically. For a listing site, automated clients fall into recognizable groups: **search engine crawlers** that the business wants; **AI crawlers and assistants** whose value is contested (see Found by Google: Web, App and SEO Architecture for Listings); **partners and aggregators**, some contracted and some not; **scrapers** that copy listings, prices and contact details; and **attack traffic** such as credential stuffing against logins and automated listing creation by scammers.

Each group costs capacity, and each needs a different policy. At Tradeyard in 2027, about a third of requests reaching the search backend on a normal day come from automated clients, and only about 6% of the total from verified search engine crawlers.

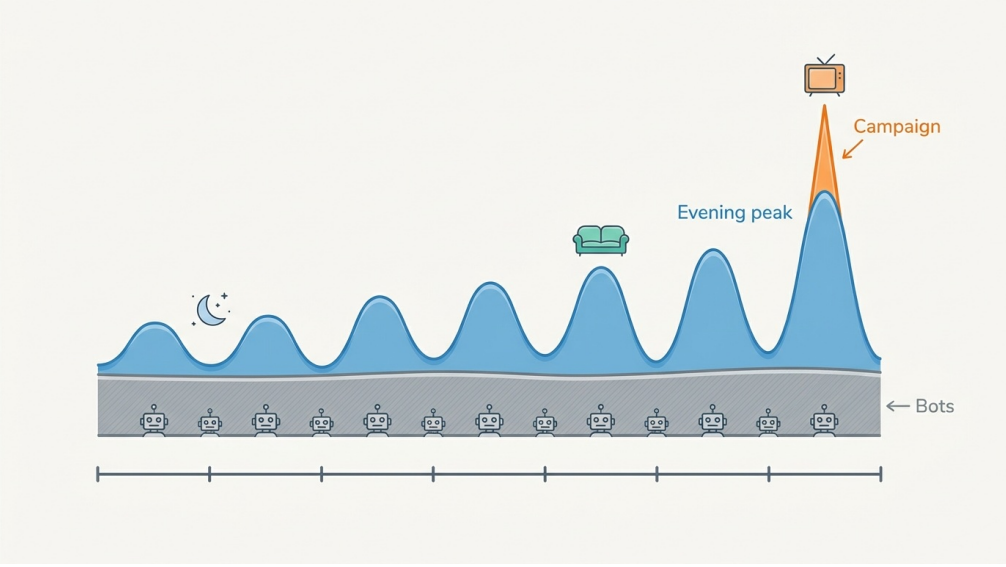


Figure 2: *The traffic shape is known in advance: evening peaks highest on Sunday, campaign spikes the business schedules itself, and a steady band of automated traffic that takes about a third of the search backend's requests.*

Journey SLOs: Defining Reliability Where Liquidity Is Made

Indicators, objectives and budgets

The vocabulary comes from site reliability engineering. A **service level indicator (SLI)** is best expressed as "the number of good events divided by the total number of events"; a **service level objective (SLO)** "sets a target level of reliability for the service's customers"; and the **error budget** is "100% minus the SLO" (Thurgood, Ferguson,

Hidalgo and Beyer, *The Site Reliability Workbook*, 2018). The same chapter recommends writing objectives around **critical user journeys** — "a sequence of tasks that is a core part of a given user's experience" — rather than around individual components.

In this book a **journey SLO** is an objective set on a marketplace journey that creates or protects liquidity, measured from the user's side of the system, with a named owner and an error budget policy. Component objectives still exist, but they serve the journey objectives, not the other way around.

The liquidity journeys

Journey	SLI: a good event is…	Measured at	Objective (Tradeyard, 2027 targets)	Why it matters
Search	A human search request answered successfully within 1 s	Edge and app telemetry	99.5% over 28 days	The entry point to almost every match
Listing view	A listing detail page or screen served within 1.5 s	Edge and app telemetry	99.9%	Where contact decisions are made
Listing visible	A newly submitted, approved listing searchable within 90 s	Publication pipeline events	95% (today's p95 is 6 minutes; target by end of 2028)	Sellers check; buyers find fresh supply first
Listing gone	A sold or removed listing out of search, alerts and pages within 60 s	Lifecycle events	99%	Ghost Inventory wastes buyer contacts

Journey	SLI: a good event is…	Measured at	Objective (Tradeyard, 2027 targets)	Why it matters
Message delivered	A message delivered, or deliberately held by safety checks, within 10 s	Messaging service	99.9%	Contacts are the currency; see From Interest to Contact: Messaging and Lead Management
Alert sent	An instant alert sent within 5 minutes of the matching listing becoming visible	Alert pipeline	95%	Speed to contact; see Tell Me When It Appears: Saved Searches and Alerts
Insertion	A listing submission accepted without error	Insertion service	99.9%	Supply; sellers who fail once often do not return

Three design choices in that table matter more than the numbers.

- **Measure humans, classify bots.** The search SLI counts human requests as the valid events. Automated traffic is measured and has its own, lower objective. Otherwise a scraper surge can burn the error budget that protects buyers, or a scraper blocked at the edge can look like an improvement in reliability.
- **Count deliberate holds as good events.** A message withheld because it contains a phishing link — the 37 messages withheld after Femke's report on 7 March 2027 — is the system working. A messaging SLO that counts holds as failures puts reliability and trust and safety in conflict (see Trust and Safety as a Platform).
- **Removal is a journey.** The "listing gone" objective is rarely set, but it is the reliability side of Ghost Inventory: the Rotterdam city bike left search one minute after sale, while the morning alert digest still featured it.

Error budgets that change behavior

An error budget is useful only if spending it changes decisions. Veldmark's policy, adopted by the architecture advice forum in April 2027: when a journey has spent more than half of its 28-day budget, changes to the components on that journey's critical path need the journey owner's approval; when the budget is gone, only reliability fixes ship to those components until it recovers. The brown-out alone spent about **30%** of Tradeyard NL's 28-day search budget in 48 minutes, counting failed and slow human requests against about 7 billion search requests in a 28-day period.

Peak-hour weighting is the marketplace-specific refinement. A failure at the weekly peak costs more contacts than the same failure at night, so Veldmark reports search reliability both overall and for the peak window, Sunday 18:00–23:00, with the peak window as the one the brand general managers see. The alternative — weighting events by expected contacts — is more precise and harder to explain; I would start with the separate peak window.

Architecture: Control Points in the Search Path

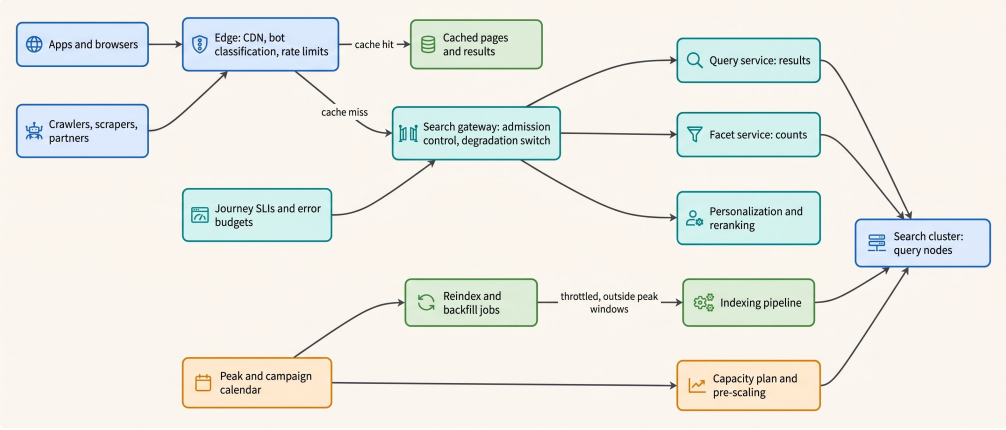


Figure 3: *Control points in the search path. The edge classifies and throttles automated traffic and serves cached pages; the gateway admits, sheds and degrades requests; facets and personalization are separate, optional calls; batch reindexing is throttled and kept out of peak windows by the same calendar that drives pre-scaling.*

The same control points apply to every journey, not only to search, the one most exposed to peaks: classify at the edge, admit and degrade at a gateway, separate optional work from essential work, and keep batch work under a calendar.

Isolation: separating what can wait from what cannot

The brown-out happened because interactive queries, reindexing and cache warm-up shared the same nodes at the same time. Five standard rules keep them apart.

1. **Batch work never shares capacity with peak interactive work by default**. Reindexing, backfills, sitemap rebuilds and model scoring run on separate capacity, or are throttled and scheduled outside peak windows by a calendar the platform enforces. Vinted's indexing team switched its connectors to "idle indexing" after catch-up to reduce cluster pressure (Vinted Engineering, 2021).
2. **A reindex is a deployment**. Building a new index beside the old one and switching an alias, as Search Architecture for Listings: Indices, Freshness and Scale describes, lets the new index warm before it takes traffic. An in-place full reindex that evicts caches is the Search as System of Record failure in operational form: the index cannot be rebuilt cheaply from upstream data, so it is rebuilt in place.
3. **Optional calls are separate calls**. Vinted split facet requests from item search in 2023, accepting about 20% more cluster CPU to make item search about 71 ms faster, and cached facet responses at about a 50% hit ratio (Vinted Engineering, 2023). Separate calls can be degraded separately.
4. **Bulk writers are rate-limited separately from consumers**. A dealer group's 40,000-listing price update should queue behind interactive edits, not beside them (see Many Doors In: Forms, APIs and Bulk Feeds).
5. **Tenants are isolated where blast radius matters**. On a shared platform, one brand's surge should not slow another's search. Separate clusters, or separate node pools with quotas, per brand or per country are the usual answer.

Retries, timeouts and admission control

Google's SRE book gives the rules that would have shortened the brown-out: "always use randomized exponential backoff when scheduling retries", "consider having a server-wide retry budget", and "avoid amplifying retries by issuing retries at multiple levels" (Ulrich, in *Site Reliability Engineering*, 2016). The same chapter recommends that overloaded servers shed load and that degraded modes be tested regularly, because rarely used paths break.

In a marketplace, apps retry search at most once, with jitter, and only on errors that a retry can fix. The gateway enforces a retry budget across clients. Timeouts shrink as requests move down the stack, so a slow search node fails a request quickly instead of holding it. Admission control at the gateway rejects or degrades requests when concurrency passes a limit, rather than letting queues grow until everything is slow. OLX is reported to have published a two-part post in 2021 on load shedding with NGINX using adaptive concurrency control; it could not be retrieved or verified for this book, so treat it as a pointer, not evidence.

Caching in three layers

Caching in a listing platform has three layers, each with a different staleness tolerance.

Layer	What is cached	Staleness tolerated	Invalidation
Edge	Landing pages, listing pages, images, static assets	Minutes for landing pages; until change for listing pages	Time-to-live plus event-driven purge by listing or page key
Gateway and service	Normalized search results for frequent queries, facet counts	Seconds to a minute	Short time-to-live; keys normalized so equivalent queries share entries
Engine	Filter, query and field caches inside the search engine	Until the next segment refresh or commit	Engine-managed; warmed before a new index takes traffic

Two lessons from the brown-out apply. **Warm before you switch**: any operation that resets engine caches — a new index, a node replacement, a restart — must warm them with replayed production queries before taking peak traffic. And **normalize cache keys**: browse queries for "bikes in Rotterdam, newest first" arrive in many equivalent forms, and only a normalized key lets them share a cache entry. Vinted reported that removing a per-user filter from facet queries raised its facet cache hit ratio by about five percentage points, at the cost of slightly inaccurate counts.

Capacity Planning for Known Peaks

Sizing from the weekly peak

Capacity planning in a marketplace starts from the weekly peak, not the average. Veldmark's group-wide search load averages 9,000 queries per second and peaks at 31,000, a ratio of about 3.4. Tradeyard carries about 70% of that, roughly **21,700** queries per second at the Sunday peak, and Tradeyard NL a little under half of Tradeyard's.

Figures here count queries, meaning results-page requests; each query fans out to about 2.6 engine requests at Tradeyard (see Search Architecture for Listings: Indices, Freshness

and Scale). A simple model for the capacity a search backend must hold at peak:

> required capacity = expected peak × campaign factor × failure headroom × (1 − share served from cache or shed)

For Tradeyard NL in 2027, before the changes in this chapter, the model reads: an expected Sunday peak of about **10,000** human and automated queries per second; a campaign factor of **1.3**, from the brown-out; failure headroom of **1.5**, so that the loss of one of three availability zones still leaves enough capacity; and almost nothing served from cache or shed. That is about **19,500** queries per second of backend capacity, close to twice the expected peak, and the cluster was sized for about 13,000.

Two levers change the result more than buying nodes. Serving landing-page and browse traffic from edge and gateway caches, and shedding unverified automated traffic at peak, together take about **35%** of peak requests off the backend in Veldmark's 2027 plan. The required backend capacity falls to about **12,700** queries per second — roughly what the cluster already has — with headroom for a campaign and a zone failure. The cheapest capacity is the request that never reaches the cluster.

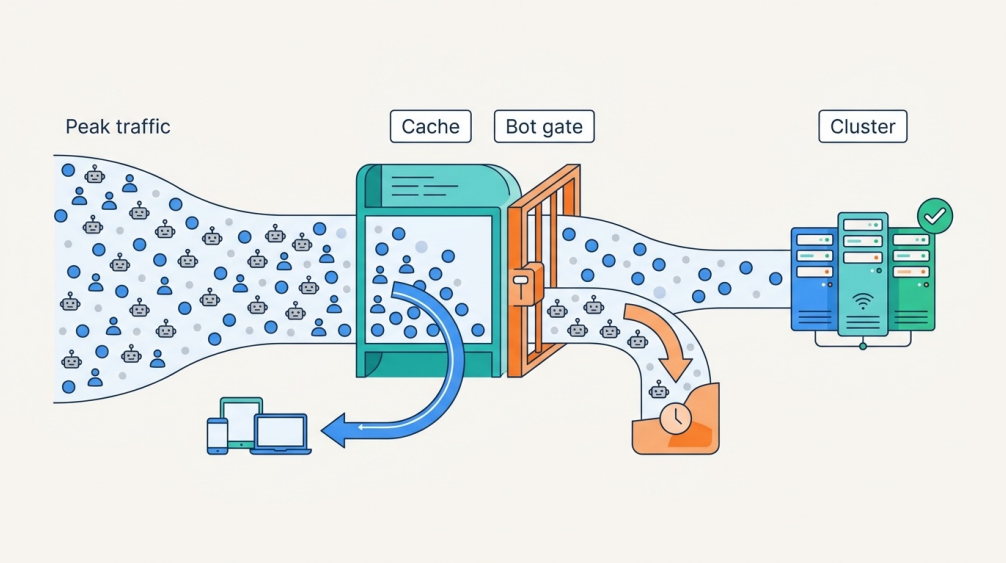


Figure 4: *The cheapest capacity is the request that never reaches the cluster: edge and gateway caches plus bot shedding take about 35% of peak requests off the backend in Veldmark's 2027 plan.*

A shared calendar of planned load

The campaign factor is only knowable if engineering knows about campaigns. Veldmark's fix is organizational: marketing campaigns, scheduled push and e-mail sends,

bulk partner imports and large batch jobs go into one **peak and campaign calendar**, owned by platform engineering, that drives pre-scaling and blocks heavy batch work in peak windows. The calendar is also where alert digests are scheduled, so that 38 million daily alert notifications do not all fire into the same evening minute.

Peak rehearsals

The useful load test for a listing platform is a **peak rehearsal**: production traffic replayed at 1.3 times the weekly peak, during a quiet window, with a deliberate cache reset and the reindex job running. It replays production query mixes, including facets and deep pagination, against production-like data volumes; synthetic queries that hit warm caches prove little.

Graceful Degradation: Deciding in Advance What to Give Up

In this book a **degradation mode** is a predefined, tested reduction of functionality that a journey can switch to under load or partial failure, so that the essential part of the journey keeps working. Degradation modes are designed as product decisions, with the product owner, before an incident. During an incident nobody should be deciding whether facets matter.

Level	Trigger (search journey)	What changes	What users notice	Who switches
0 Normal	—	Full results, exact facets, personalization	Nothing	—
1 Shed bots	Gateway concurrency above 70% of limit	Unverified automated clients rate-limited hard; verified crawlers slowed through `429` responses	Nothing	Automatic

Level	Trigger (search journey)	What changes	What users notice	Who switches
2 Trim optional work	p99 above 1 s for 3 minutes	Facet counts approximate ("500+"); personalized reranking off; "related searches" and recommendation blocks hidden	Slightly less tailored results	Automatic, with on-call override
3 Serve from cache	p99 above 2 s, or error rate above 1%	Frequent queries and landing pages served from cache up to 10 minutes stale; deep pagination capped; sort options limited	Results a few minutes old; fewer options	On-call, with incident commander
4 Browse only	Search cluster unavailable	Category and landing pages from edge cache; free-text search box replaced by category browse; insertion accepted and queued	Browse works, search does not, new listings delayed	Incident commander

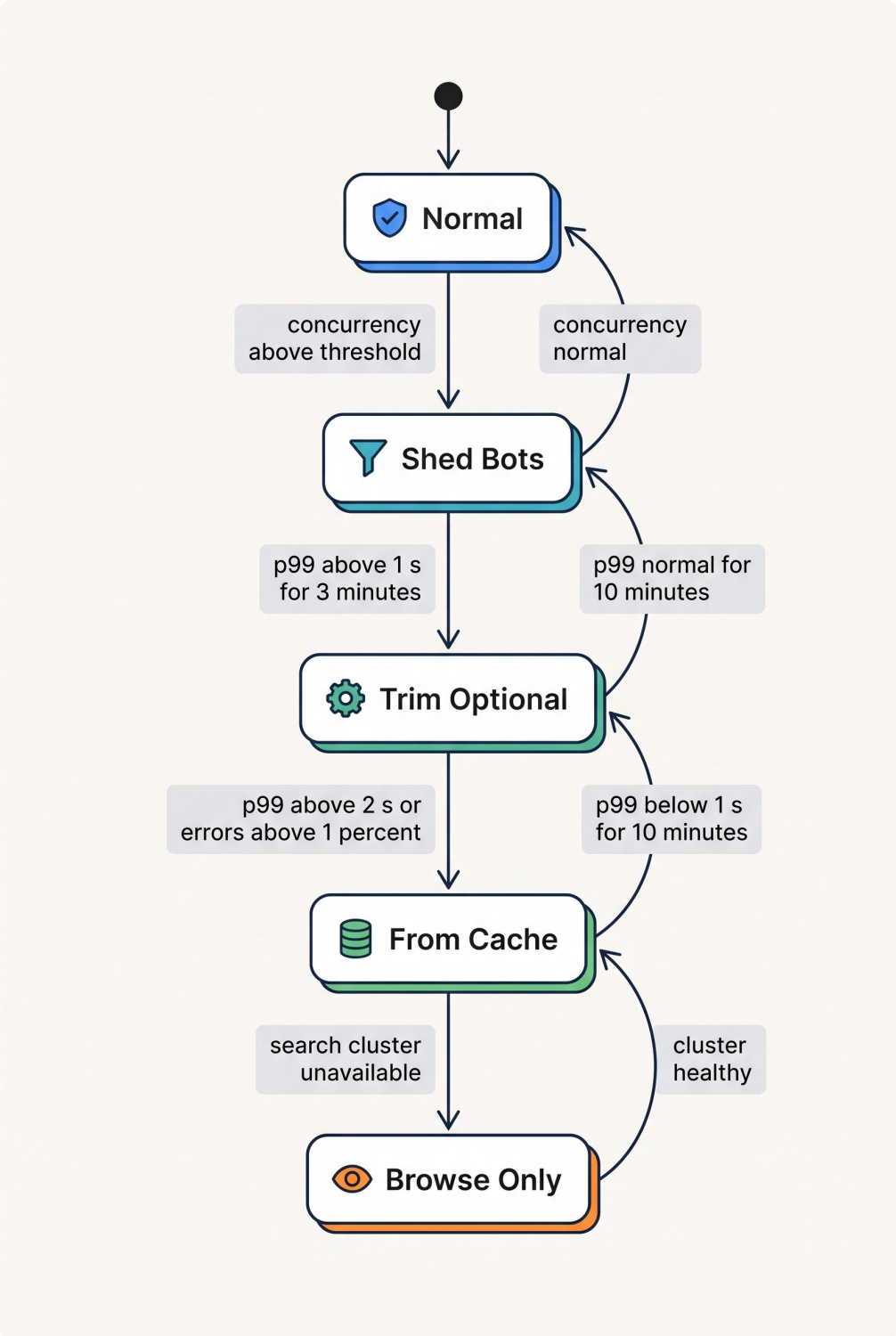

Normal
concurrency above threshold
concurrency normal
Shed Bots
p99 above 1 s for 3 minutes
p99 normal for 10 minutes
Trim Optional
p99 above 2 s or errors above 1 percent
p99 below 1 s for 10 minutes
From Cache
search cluster unavailable
cluster healthy
Browse Only

Figure 5: *The search degradation ladder. Each step gives up something less essential than the step after it, and recovery steps down only after a sustained period of health, so the system does not oscillate at the peak.*

The marketplace-specific part of the ladder is its order: **bots first**, **optional features second**, **freshness third**, **search itself last**. Two pieces of documented practice support the steps. Vinted's facet work showed that approximate counts are acceptable to users — counts above 500 are shown as "500+" and computed with an early-termination limit — and that capping facet cardinality lowers latency (Vinted Engineering, 2023). And the SRE guidance on cascading failures recommends cheaper fallback work under load and regular testing of degraded paths. Had level 1 and level 2 existed on 17 January 2027, about a third of the backend load and the facet traffic would have gone first, and the 9% failure rate would most likely have been a few minutes of approximate counts.

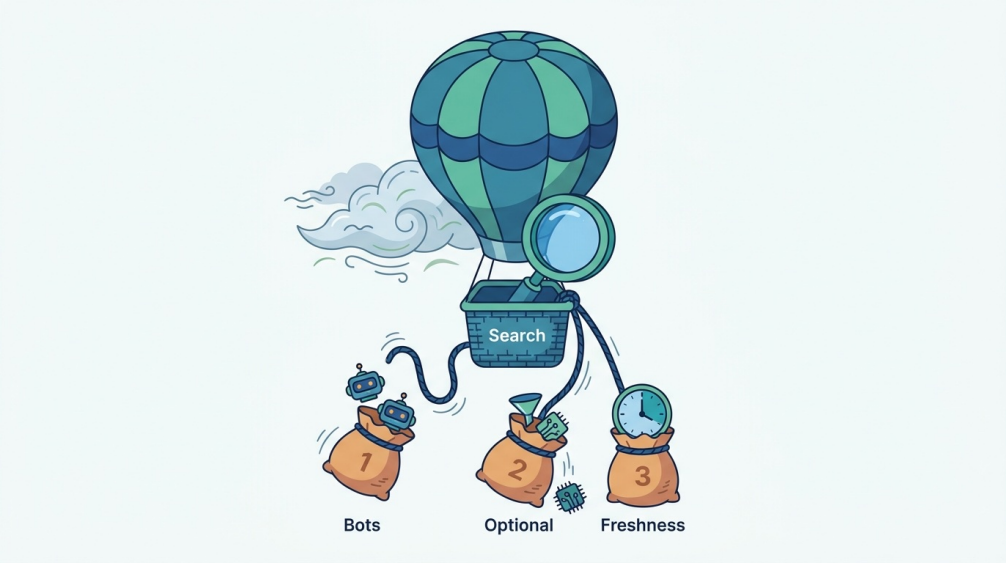


Figure 6: *Degradation is planned ballast: under load the search journey gives up bots first, optional features second and freshness third, so that search itself is the last thing to go.*

Each journey has its own ladder. For messaging, degraded mode delays attachments and read receipts but not the delivery of text, and never skips safety scanning: a messaging system that turns off scam detection under load turns a reliability incident into a fraud incident, as the Fast Mail for Fraudsters pattern warns. For alerts, degraded mode delays digests and keeps instant alerts. For insertion, degraded mode accepts and queues submissions, tells the seller that publication is delayed, and keeps moderation in the path.

Multiple Regions: How Far to Go

What the documented record shows

Classifieds are mostly country-scoped businesses, which shapes the regional question. Vinted, which runs one cross-border catalog, described in 2026 a design in which "all writes happen in the primary site and read-only projections are replicated around the world", and in which a multi-writer design was avoided because its operational cost is hard to justify "without a strong business need" (Vinted Engineering, 2026). Adevinta's shared Kubernetes platform reduced blast radius by running many clusters and shifting traffic between them with weighted DNS records and a controller that watches pod health (Adevinta Tech Blog, 2023).

Regional failures of a cloud provider happen, and they take shared dependencies down together. AWS's post-event summary of the 19–20 October 2025 disruption in its N. Virginia (us-east-1) region describes a latent defect in DynamoDB's automated DNS management that left the regional endpoint with an empty DNS record; the effects spread over about 14.5 hours to EC2 instance launches, Network Load Balancer, Lambda, container services, the Security Token Service and console sign-in (AWS, 2025). Two lessons transfer to any platform. **Regional control planes are shared dependencies**: identity tokens, instance launches and load balancers can fail together. And **autoscaling is not capacity during a regional event**: if new instances cannot launch, only capacity that is already running counts.

Five multi-region options

Option	What it gives	What it costs	Fits
Multi-zone in one region	Survives loss of a zone; standard for managed services	Little beyond headroom	Every production system
Backup and restore to a second region	Recovery in hours to days	Cheap; slow recovery	Data that must survive; non-critical tools

Option	What it gives	What it costs	Fits
Read-only warm standby	Browse and listing pages served from a second region within minutes; writes queued or paused	Replicated read models and search, a second edge origin, regular failover tests	Marketplaces where browsing is most of the value and writes can wait
Active-active with a single writer	Reads served everywhere, writes in one region (Vinted's documented model)	Replication, staleness handling, operational complexity	Cross-border catalogs with global users
Active-active multi-writer	Writes in every region	Conflict resolution and the highest complexity	Rarely justified for classifieds

For a country-scoped classifieds group the right answer for most journeys is **multi-zone plus a tested read-only mode**. A read-only marketplace — browse, search and view listings work; insertion and messaging queue — keeps the most valuable traffic, protects advertising revenue and loses relatively few contacts for a short outage. It also doubles as a degradation mode for incidents inside a region, such as a failed primary database. The costs are real: the search index and listing read models must be continuously replicated, and the failover must be exercised, or it will not work when needed.

Veldmark's starting point is more fragile than its cloud estate suggests. Tradeyard's newer services run across three zones in AWS eu-west-1, but Yardstick and its MySQL database run in a single co-located data center in Amsterdam, and Solr is self-hosted. Yardstick's single site is a larger risk than any cloud region. The 2027 reliability plan therefore puts **regional read-only standby for Tradeyard's search and listing pages** after the Yardstick search extraction in Q4 2028, and in the meantime funds a tested backup of Yardstick's database to AWS and a static "browse from cache" fallback at the edge.

Incident Management: When It Goes Wrong Anyway

Incident management in a marketplace has the usual parts — detection, an incident commander, communication, mitigation, review — with three marketplace-specific emphases.

- **Severity follows journeys and time.** An incident's severity is set by which journey is affected, how much of its error budget is burning, and whether it is the peak window. A search failure at 20:35 on Sunday is a top-severity incident even at a lower error rate than a 04:00 failure that nobody notices.
- **Professional customers need their own communication.** Dealers, agencies and feed partners run businesses on the platform. An outage of feed ingestion or lead delivery needs a status page and direct notices to professional accounts, with the time window, so that disputes about missed leads can be settled from records rather than arguments (see Serving the Pros: Dealer, Agent and Merchant Tooling).
- **Shared platforms have shared blast radius.** Adevinta's engineers have published incident reviews from their shared platform with unusual candor. In one, a GitOps controller treated a source-control outage's malformed responses as "0 applications" and began deleting applications across more than 30 clusters (Adevinta Tech Blog, 2024). In another, renaming Kubernetes node groups replaced all of them at once, removing about 40 nodes per group within a minute; one critical cluster took 8 minutes to recover (Adevinta Tech Blog, 2023). The lesson for a group platform is that its failure modes reach every tenant at once, so changes to the platform need the same staged rollouts and peak-window freezes as changes to a product.

Reviews should be blameless and should produce architectural actions, not only operational ones. The brown-out review's first draft listed "move the reindex job" and "add two nodes". The version the forum accepted added the degradation ladder, the peak calendar, the retry budget, bot classification at the edge and the journey SLOs, each with an owner and a date.

Bots, Scrapers and Crawlers: Managing Automated Demand

Why classifieds attract scrapers

Listing data is valuable to others. Competitors and aggregators copy inventory; price-monitoring services track dealers' stock; data brokers harvest phone numbers and e-mail addresses; and scammers copy real listings, photos and prices to create convincing fakes. The Barcelona fake-rental ring of July 2027 posted 180 listings in Eixample using photos copied from real listings (see Scams, Fakes and Account Takeovers: Fraud Prevention for Listings). Scraping is therefore a capacity problem, a commercial problem and a trust-and-safety problem at once.

The control layers

Layer	Control	Notes
Policy	`robots.txt` and terms of use	The Robots Exclusion Protocol is standardized as RFC 9309 (2022), which states plainly that "these rules are not a form of access authorization" (IETF, 2022). Well-behaved crawlers follow them; scrapers do not
Verification	Confirm that a claimed search engine crawler is genuine	Google documents reverse and forward DNS checks and published IP range files for its crawlers (Google Search Central, updated 2026)
Classification	Score every client at the edge by behavior, fingerprint, network and reputation	Usually a CDN or bot-management service; the classifier's output travels with the request so downstream services can apply quotas
Rate limits and quotas	Per client, per network, per account, per data type	Strictest on contact data (phone reveal, seller e-mail) and on deep pagination, which is how bulk copying shows up
Product design	Put high-value data behind login or friction	Masked phone numbers and in-platform messaging reduce harvesting (see From Interest to Contact: Messaging and Lead Management); contact reveal per account is rate-limited

Layer	Control	Notes
Contracted access	A partner API with keys, quotas and terms	Gives legitimate aggregators and partners a cheaper route than scraping, and makes the rest easier to block
Legal	Database rights, terms of use, enforcement	In the EU the sui generis database right of Directive 96/9/EC protects substantial investment in databases; enforcement is slow and complements, not replaces, technical controls

AI crawlers add a commercial question that is not only technical. On 1 July 2025, Cloudflare announced that it was changing its default to block AI crawlers unless they pay for content (Cloudflare, 2025). Whether a marketplace allows, blocks or licenses each AI agent is a group policy decision covered in Found by Google: Web, App and SEO Architecture for Listings and AI Across the Listing Lifecycle; this chapter's concern is that the policy is enforced at the edge and that automated traffic has its own capacity class.

Cost: Making Scale Visible per Unit of Work

Three unit costs

The FinOps Foundation describes unit economics as bringing together "what an organization spends on technology and the value that technology spending creates" (FinOps Foundation, n.d.). For a listing platform, three unit costs are worth computing and publishing.

- **Cost per thousand searches** — search infrastructure cost divided by search requests. In this book, **cost per search** means the fully allocated run cost of the search path (engine, gateway, caches, observability for that path) divided by search requests, reported separately for human and automated traffic.
- **Cost per live listing per month** — platform run cost attributable to a brand, divided by its average live listings.

- **Cost per contact** — the run cost of the journeys that lead to a contact, divided by contacts. It is the hardest to compute and the most useful, because it relates infrastructure to liquidity.

Veldmark's 2027 baseline shows why the first two belong in architecture reviews.

Brand	Search infrastructure cost (2027 run rate)	Share of group queries	Cost per 1,000 queries	Platform run cost	Live listings	Cost per live listing per month
Tradeyard	€6.0M	70%	€0.030	€44M	13.2M	€0.28
Motorvane	€3.2M	16%	€0.070	€27M	1.45M	€1.55
Hearthly	€2.6M	14%	€0.065	€24M	2.6M	€0.77
Group	**€11.8M**	**100%**	**€0.042**	**€104M** incl. Keybridge €5M and group €4M	17.25M	—

The figures are run rates for infrastructure, not people, and exclude the data estate, which Data Platform for Marketplaces: Events, Lakehouse and Data Products covers; query costs are computed on about 284 billion queries a year (9,000 per second on average). Three readings follow. **Search is cheap per query and expensive in total**: a few euro cents per thousand queries adds up to €11.8 million a year across three engines. **Motorvane's costs per query and per listing are the highest**, partly because Elasticsearch on a managed service with dealer-feed update storms is expensive for 1.45 million listings, and partly because its listings carry more revenue; the unit cost should be read against revenue per listing, not alone. And **about a third of Tradeyard's search cost serves automated clients**, most of which bring no buyers. Shedding and caching them is a cost decision as much as a reliability one.

Where the documented savings come from

The best-documented FinOps work in classifieds is Adevinta's, and it points to platform-level levers rather than team-by-team optimization.

- **An ownership catalog**. Adevinta reported 732 AWS accounts and 425 Google Cloud projects in 2024, linked to cost centers and teams through a catalog API that also handles account creation and access (Adevinta Tech Blog, 2024). Without ownership metadata, no cost can be allocated to a journey or a brand.
- **Telemetry as a cost line**. Trimming log fields and moving to a different log shipper cut OpenSearch logging cost by 35% (Adevinta Tech Blog, 2024); aggregating metrics cut ingested metrics by more than 33% and the observability bill by more than 20% (Adevinta Tech Blog, 2024).
- **Platform defaults**. Changing node provisioning made instance-family savings possible (Intel to AMD could save up to €30,000 a month), but a "cheapest first" policy hurt CPU-bound workloads (Adevinta Tech Blog, 2024). A webhook that schedules workloads on ARM nodes from image metadata brought ARM to thousands of applications without manifest changes; no savings figure was published (Adevinta Tech Blog, 2023).

The platform team holds the largest cost levers, because its defaults apply to everyone.

Cost and Deal-Shaped Architecture

Veldmark's three search engines, three clouds and four observability stacks are the cost face of Deal-Shaped Architecture. Each brand pays for its own peak headroom, its own on-call rotation and its own telemetry. The September 2027 decision to move Tradeyard and Hearthly to a shared search platform on Amazon OpenSearch Service, with Motorvane staying on Elasticsearch until a 2029 review, is justified in the decision record partly by reliability — one team that does peak rehearsals and degradation ladders well instead of three that do them partly — and partly by cost. The record also carries the counter-argument: a shared cluster concentrates blast radius, so tenancy isolation per brand and country is a requirement of the decision, not an option.

Veldmark Worked Example: The Brown-Out, Redesigned

By the end of 2027, the changes from the review have been built for Tradeyard NL and BE. On **Sunday 16 January 2028**, Tradeyard runs its next New Year campaign in the same 20:30 ad break. The comparison is what the architecture advice forum asked for.

Measure	Sunday 17 January 2027	Sunday 16 January 2028
Peak search requests at origin, Tradeyard NL	~13,000/s (cluster saturated)	~8,900/s after edge cache and bot shedding
Share of peak requests shed or served from cache	~2%	~37%
Reindex or backfill running in peak window	Yes, full in-place reindex	No; blocked by the peak calendar
Degradation level reached	None defined	Level 2 for 6 minutes (approximate facets, personalization off)
Search p99 latency at peak	~7 s for 48 minutes	~1.3 s for 6 minutes, then ~450 ms
Failed human search requests	~9% for 48 minutes	under 0.2%
Peak-window search error budget spent	~30% of the 28-day budget	~2%
App retries	Two, no backoff	One, with jitter, within a gateway retry budget

The redesign did not need a new search engine; Tradeyard NL is still on Solr in January 2028, ahead of the move to OpenSearch. It needed the architecture to know four things it did not know a year earlier: when the peak is, what is optional, who the bots are and how much budget remains.

Trade-offs

Decision	Option A	Option B	Choose A when	Choose B when
Reliability objectives	**Journey SLOs**	Component uptime targets	Several services contribute to one user outcome	A single service is the whole journey

Decision	Option A	Option B	Choose A when	Choose B when
Peak capacity	**Cache, shed and pre-scale on a calendar**	Provision for peak × headroom at all times	Peaks are predictable and much higher than average	Load is flat, or cache hit ratios are low
Reindexing	**New index beside old, warm, switch**	In-place reindex in a quiet window	Always for large indices	Small indices with no peak exposure
Degradation	**Predefined ladder, automatic first steps**	Manual decisions during incidents	Peaks are frequent and the order of sacrifice is known	Rarely loaded systems
Regions	**Multi-zone plus read-only standby**	Active-active	Country-scoped business; browsing is most of the value	Cross-border catalog with global users and a business need for writes everywhere
Automated traffic	**Classify and give a separate capacity class**	Treat all traffic alike	Always for public listing sites	—
Search platform	**Shared platform with tenant isolation**	One cluster per brand	Several brands with similar workloads and a strong platform team	Very different workloads, or no team to run a shared platform well
Cost visibility	**Unit costs per brand and journey**	One infrastructure line in the budget	More than one brand or a shared platform	A single small product

The most contested trade-off is the shared search platform. Sharing reduces duplicated

headroom and spreads reliability practice, but a bad change can now affect four countries at the Sunday peak. The answer is not to avoid sharing but to price the blast radius in: per-tenant clusters or node pools, staged rollouts country by country, and peak-window change freezes for the platform itself.

Vertical Differences

Concern	General classifieds	Real estate	Automotive
Traffic shape	Evening and weekend peaks; seasonal goods	Evening peaks; TV-driven spikes; spring and rental seasons	Evening and weekend peaks; month-end and registration effects in some markets
Most demanding journey	High-volume search and browse; message delivery	Map search and alerts in tight rental markets	Lead delivery to dealers; feed ingestion
Write load	Many small private edits	Agency feed syncs from software vendors	Dealer feed updates and bulk price changes
Main scraper interest	Inventory copying; contact harvesting; scam templates	Listing and price data; photos for fake rentals	Dealer stock and price monitoring
Professional SLAs	Few; pro sellers on subscriptions	Agencies expect lead delivery and feed freshness	Dealers expect lead delivery, feed freshness and status notices
Cost driver	Query volume and images	Images (Hearthly stores 1.8 billion) and map tiles	Update storms in the index; lead and call tracking

Real estate's peculiarity is that the scarcest moments are not always the busiest. In a tight rental market the first alert wins the viewing, so the alert journey's objective matters more than search latency; a Hearthly outage of five minutes in alerts can cost more trust

than a slower search page. Automotive's peculiarity is the professional relationship: a dealer paying for a package of listings expects leads to arrive, and an hour of lost lead delivery becomes a commercial dispute.

Failure Modes

- **The reindex at the peak**. Batch jobs scheduled years ago for a quiet hour that is no longer quiet compete with interactive traffic and reset caches.
- **Retry storms**. Clients and services retry at several layers without backoff, turning slowness into overload.
- **The unclassified bot load**. Automated clients consume a third of backend capacity and are not throttled when humans need it.
- **Degradation invented during the incident**. Nobody knows whether facets can be turned off, who may decide, or whether the switch works.
- **Uptime theater**. Every service reports 99.9% while the evening search journey fails one request in ten.
- **The untested failover**. A second region or a read-only mode exists on paper, has never carried production traffic and fails when needed.
- **Platform changes at the peak**. A shared platform change reaches every brand and country at once, without staged rollout or freeze windows.
- **Trust turned off for speed**. Safety scanning on messages or listings is bypassed under load, and a reliability incident becomes a fraud incident.
- **Unowned telemetry**. Logging and metrics grow without owners until observability is one of the largest infrastructure lines.

The Chief Architect's View

Decide the journeys and their owners. The group defines the liquidity journeys and their indicators once; each brand sets its own objectives on them and names an owner per journey. The Chief Architect's role is to make sure the objectives exist, are measured from the user's side, and have error budget policies that change behavior.

Make the peak a platform concept. The peak and campaign calendar, peak-window change freezes, batch scheduling rules and peak rehearsals are platform capabilities. They cost little and prevent the most common incidents. Marketing joins the calendar; that is a governance decision, not a technical one.

Fund degradation and read-only modes deliberately. They are features used a few times a year that save the most valuable hours, and they will not be built unless they are on someone's roadmap with a date.

Set the regional ambition explicitly. For a country-scoped group, multi-zone plus a tested read-only standby is usually enough. Active-active should need a business case. Single-site legacy systems, like Yardstick, are the larger regional risk and deserve attention first.

Treat automated traffic as a policy area. Which crawlers, partners and AI agents get access, at what rate, under which terms, is a cross-functional decision involving SEO, legal, trust and safety and commercial teams. Engineering enforces it.

Publish unit costs. Cost per thousand searches, per live listing and per contact, by brand, belong in the quarterly architecture review next to the journey SLOs. They are how the €38 million run-cost target of Common Ground becomes specific work rather than a cut applied evenly.

What to watch. Peak-window error budget consumption per journey; time spent in degraded modes; share of load from automated clients; cache hit ratios at the edge; cost per thousand human searches; and the number of batch jobs scheduled inside peak windows, which should be zero.

Questions to Ask

1. Which journeys create liquidity on this platform, and does each have an indicator measured from the user's side, an objective and an owner?
2. How much of each journey's error budget is spent in the weekly peak window, and who sees that number?
3. What batch work — reindexing, backfills, sitemap rebuilds, model scoring — can run during the peak, and what stops it?
4. Who tells engineering about TV campaigns, push campaigns and bulk partner imports, and how far in advance?
5. What share of backend load comes from automated clients, how is it classified, and what happens to it when humans need the capacity?
6. What are the degradation modes for search, messaging, alerts and insertion, in what order do they give things up, and when were they last tested?
7. How many retries can one user action cause across the stack, and is there a retry budget?
8. If our primary region or our oldest data center were unavailable for four hours, what could buyers still do?
9. What are our cost per thousand searches and cost per live listing by brand, and how much of each is driven by automated traffic?
10. Which changes to shared platforms are frozen during peak windows, and how are platform changes rolled out across tenants?

From Platform to Leadership

This chapter closes Part VII — PLATFORM: The Distributed System Underneath. The platform beneath a listing marketplace — events, data management, web and SEO, data, experimentation, ML, the multi-country platform and the reliability practices in this chapter — is what makes liquidity dependable at the moments that matter. Building it across brands that a group has acquired, with teams that were never designed to work together, is a leadership problem as much as an engineering one. Part VIII — LEADERSHIP: The Chief Architect's Agenda takes up the Chief Architect's agenda, beginning with the question every acquisitive group faces: after the deal, what should be consolidated, when and how (After the Deal: Consolidating Platforms After Acquisitions).

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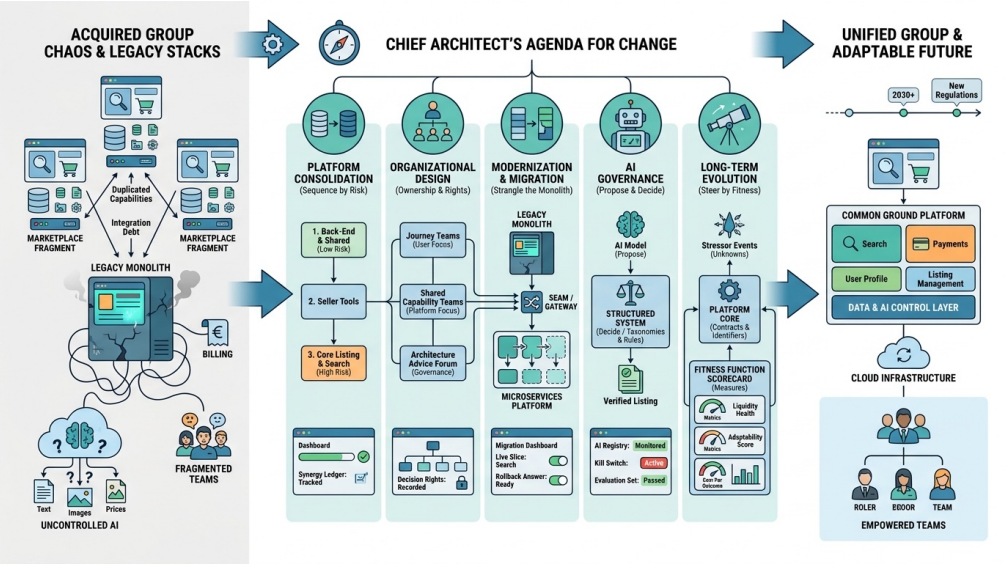
Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

Part VIII — LEADERSHIP: The Chief Architect's Agenda



IN THIS PART, YOU WILL: Set the Chief Architect's agenda for a marketplace group: consolidate platforms after acquisitions, organize engineering around journeys and shared capabilities, modernize a live marketplace slice by slice, put AI across the listing lifecycle under control, and evolve the platform over ten years.

WHY THE BUSINESS CARES: Acquisitions are priced on synergies and platform programs on run-cost savings; both are paid for in liquidity when they go wrong. Sequencing, ownership and gates decide whether the group keeps the traffic, sellers and partners it bought.

WHY ARCHITECTS SHOULD CARE: Parts I to VII describe the systems; this part describes the decisions, teams and migrations that move a real group

toward them. A design with no owning team, no stage gates and no measure of success stays half-built.

KEY POINTS:

- **Sequence change by risk to liquidity, not by the size of the saving.** Consolidate what users never see first and the core listing and search experience last, strangle the monolith slice by slice behind a seam, and gate every step on measured outcomes with a rollback answer.
- **Ownership and decision rights are part of the architecture.** Every journey, bounded context and shared capability needs exactly one owning team, platform teams earn their tenants, and cross-brand decisions go through an open, recorded advice process rather than a review board that owns nothing.
- **Design for a future you cannot predict, and measure whether the design still works.** Let AI propose while the structured systems decide, invest in the few contracts and identifiers that survive many stressors, and steer by marketplace fitness functions instead of program milestones.

FAILURE PATTERNS THIS PART ADDRESSES:

- **Deal-Shaped Architecture** — Gives every acquired marketplace an integration option and a consolidation sequence, so the group stops paying for each capability several times.
- **The Forced Fit** — Makes shared platforms internal products with owners, contracts and migration help, so marketplaces adopt them instead of routing around them.
- **Set-and-Forget AI** — Registers every AI use with an owner, an evaluation set, a cost per outcome and a kill switch.

Part VII built the distributed system underneath the marketplaces. Part VIII turns to **the Chief Architect's agenda**: the work that decides whether that system ever exists outside a diagram. It asks in what order the earlier chapters' decisions are carried out, who owns them, how a live business moves onto them, and how a group knows, years later, that they still serve their purpose.

The argument of this part is that **the hard problems of a marketplace group are problems of change, not of design.** The group was assembled by acquisition, so its platforms come in several copies; its teams follow old org charts, its monolith still bills the bumps, and its future owners and regulators are unknown. The architect changes all of this while buyers keep finding things and sellers keep getting contacts.

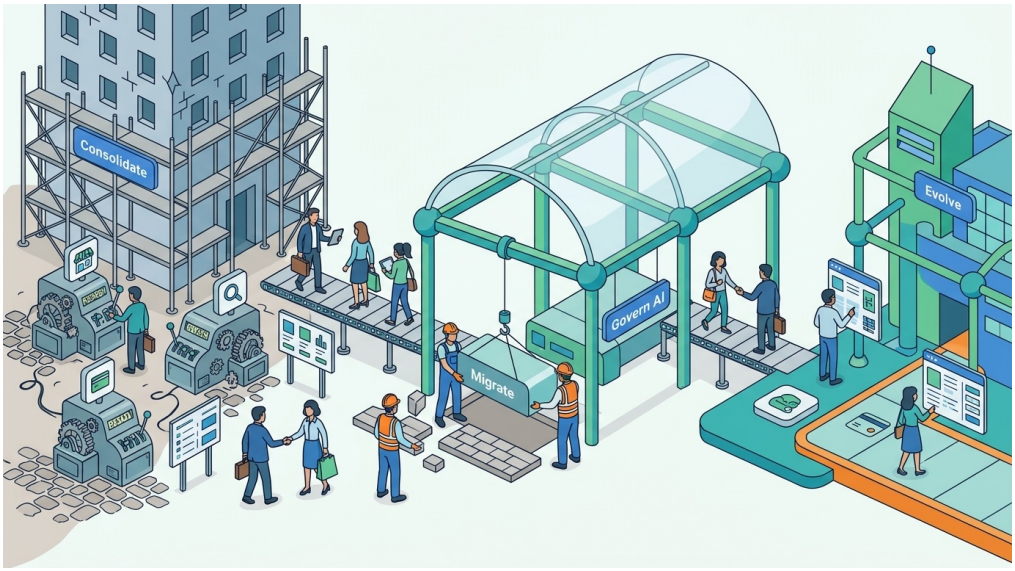


Figure 1: *Part VIII treats the group's hard problems as problems of change: consolidating, assigning owners, migrating, governing AI and evolving the platform while buyers keep finding things and sellers keep getting contacts.*

Chapters in this part

Chapter	The question it answers	Read it when…
After the Deal: Consolidating Platforms After Acquisitions	When a group acquires marketplaces, what should it consolidate, when, and how?	The board asks where the synergies are, or a rebrand is planned alongside a replatform.
Who Owns What: Organizing Marketplace Engineering	Which organizational design makes a multi-marketplace architecture work?	A fix reaches one country and not the next, or a shared platform has no team that owns it.
Changing the Engine in Flight: Modernizing a Live Marketplace	How do you move a live marketplace off legacy systems without losing traffic, sellers or revenue?	Your monolith deploys twice a week, or a rewrite is being proposed.

Chapter	The question it answers	Read it when…
AI Across the Listing Lifecycle	Where do generative AI and agents create value, and what architecture keeps them controlled?	Several AI proposals arrive at once, or external assistants start querying your search.
Built to Be Replaced: Evolving a Marketplace Platform Over a Decade	How do you keep a marketplace platform coherent over ten years of change?	An investor asks what a sale would cost, or the program reports milestones instead of matches.

How this part connects

Each chapter builds on a design from an earlier part. The tenancy model and release rings of One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture make consolidation possible. The Every Change Is an Event: Event-Driven Listing Architecture contract and the Search Architecture for Listings: Indices, Freshness and Scale platform are the seams a migration routes through. The context map from Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models becomes the list of owning teams. AI Across the Listing Lifecycle extends Models in Production: ML Platforms and Use Cases and depends on the taxonomy and attribute schemas of Part II. Built to Be Replaced: Evolving a Marketplace Platform Over a Decade closes the book's argument. Part IX — PRACTICE: How Others Built It then tests that argument against publicly documented marketplace groups, several of which lived through the reversals this part warns about.

The Veldmark thread

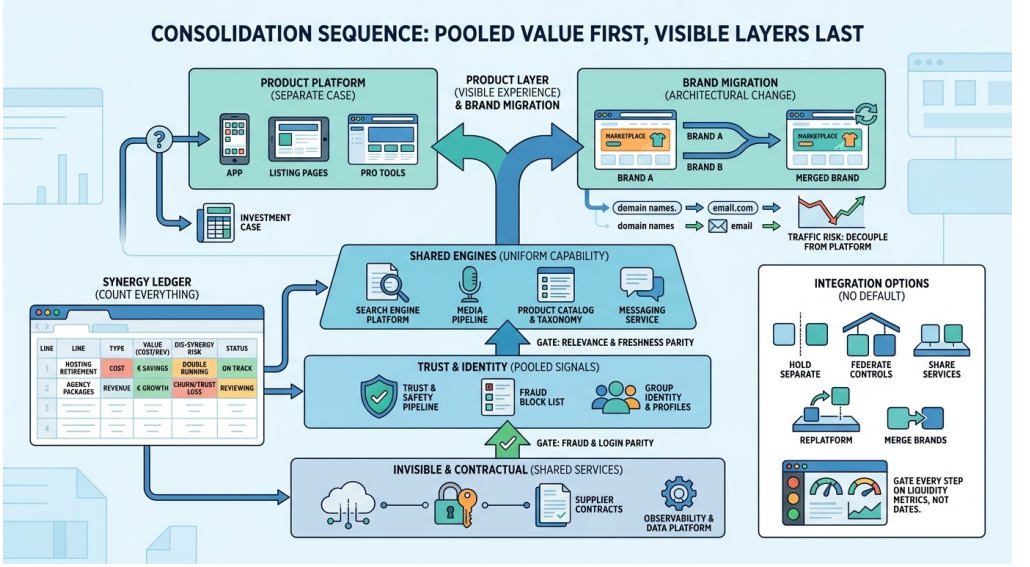
Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), has approved Common Ground: €140 million over 2027–2030 to share platforms and retire Yardstick by the end of 2029. By October 2027 the Casavista deal has delivered €9.6 million of the €18 million in promised synergies. Ingrid Halvorsen, the Group Chief Architect, shapes the decisions that follow, though the CTO, the board and the advice forum take them:

- **Common Ground organization v1** (22 June 2027, effective 1 September), with a group platform organization under Yara Mendes, journey-aligned brand teams and the weekly architecture advice forum;

- the **Yardstick Exit plan** (October 2027): six waves that end in decommissioning by 31 December 2029, when the Amsterdam colocation contract expires;
- the **consolidation review** (9 November 2027): a synergy ledger in place of spreadsheets, the Keybridge cross-selling line withdrawn, and Keybridge federated under group controls permanently;
- the **photo-to-listing assistant pilot** (15 November 2027), in which the model proposes and the taxonomy, attribute schema and trust rules decide;
- a **stressor workshop** (18–19 November 2027) and the **marketplace fitness scorecard v1** of twelve measures (8 December 2027), ahead of the board's question about selling Hearthly.

By the end of Part VIII you should be able to say what to change first, who owns it, how you will know it worked, and what must stay true when the group's owners, laws and channels change. Begin with After the Deal: Consolidating Platforms After Acquisitions.

41. After the Deal: Consolidating Platforms After Acquisitions



IN THIS CHAPTER, YOU WILL: Choose an integration option for an acquired marketplace, sequence its consolidation from the first hundred days to the product layer, build a synergy case that survives contact with the migration, and plan a brand migration that protects organic traffic, accounts and partner integrations.

WHY THE BUSINESS CARES: Acquisitions in classifieds are priced on synergies, and most of those synergies are platform decisions. A consolidation sequenced badly loses the traffic, sellers and partner feeds that the deal was paid for; one never started leaves the group paying for every capability several times.

WHY ARCHITECTS SHOULD CARE: After a deal, the architect is asked to turn a spreadsheet of synergies into a migration plan with dates. The order of that plan, and the gates between its steps, decide whether the acquired marketplace keeps its liquidity while its platform changes underneath it.

KEY POINTS:

- **Not every acquisition should be integrated, and none should be integrated all at once.** Choose an integration option per acquired business (hold separate, federate under group controls, share services beneath it, replatform onto a sibling, merge brands) and recognize that regulators, commitments to customers and the chance of a later sale can each rule options out.
- **Consolidate from the bottom up: invisible and pooled first, product last.** Controls and infrastructure first, then data, ads and payment contracts, then trust signals and identity, then shared engines; the core listing and search experience last and only with its own investment case. Gate every step on liquidity metrics, not on the plan's dates.
- **A synergy case is a hypothesis, so run it as a ledger.** Count dis-synergies (traffic loss, churn, double running, integration cost) and the cost of separating the business again, track each line against dated evidence, and decouple brand migrations from platform migrations so that one risk lands at a time.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Deal-Shaped Architecture** — Acquired marketplaces that keep their whole stacks indefinitely leave the group paying for each capability several times; the consolidation sequence is the way out.
- **The Forced Fit** — Forcing an acquired marketplace onto the acquirer's product platform on the deal's timetable is how local teams learn to route around the group.
- **Ghost Inventory** — Brand and platform migrations that break redirects, alerts and partner removals leave sold and moved listings live in search engines, inboxes and feeds.

A marketplace acquisition is usually justified in two sentences: the target is a market leader, and owning it alongside the group's other marketplaces will save money and grow revenue. The first sentence is about liquidity, which the acquirer must not damage. The second is about platforms, which the acquirer must change. **Consolidation is the work of changing the platform without damaging the liquidity**, and its difficulty is that the two compete for the same months, teams and risk budget.

The public record of classifieds groups is unusually rich on this topic, because the industry has been assembled almost entirely by acquisition. eBay bought national leaders and kept their brands; Axel Springer combined property portals in several countries into AVIV; Naspers rebranded acquired sites as OLX; Adevinta bought eBay Classifieds Group, integrated part of it and then, under new owners, decentralized and

sold businesses. **The record shows more reversals than finished integrations**, which is itself the main lesson.

This chapter opens Part VIII — LEADERSHIP: The Chief Architect's Agenda, the Chief Architect's agenda, by asking how an acquired marketplace joins the shared platform that Part VII built; it relies on the tenancy model of One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture. It owns three terms: the **consolidation sequence** (the order and gates in which an acquired marketplace's capabilities move onto group platforms), the **synergy case** (the quantified hypothesis of savings and growth that justifies the integration, with its costs) and the **brand migration** (moving users, sellers, partners and search engines from one consumer brand to another). The later chapters of Part VIII build on it: Who Owns What: Organizing Marketplace Engineering decides who owns each moved capability, Changing the Engine in Flight: Modernizing a Live Marketplace supplies the techniques for moving a live system, and Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform tells Adevinta's full record.

Setting the Scene at Veldmark: Where Are the Casavista Synergies?

On **Thursday 14 October 2027**, Henrik de Boer, Veldmark's Group CEO, forwarded Ingrid Halvorsen a question from the private-equity owner's operating partner ahead of the 2028 budget. Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), had bought two businesses into its real-estate portfolio in two years. **The question was blunt: where were the synergies that had been paid for?**

The first business was **Casavista**, a Spanish property portal bought in 2025 and rebranded **Hearthly.es** in 2026. Its deal case had promised **€18 million** a year of run-rate synergies by the end of 2027: **€8 million** from retiring Casavista's platform and hosting, **€4 million** from overheads and **€6 million** of revenue from selling Hearthly's agency packages and new-build campaigns in Spain. In October 2027 the synergy tracker showed **€9.6 million**: €3.9 million of technology savings, €3.6 million of overheads and €2.1 million of revenue. One-off integration costs had reached **€11 million** against **€6 million** planned. The listings and search had moved onto Hearthly's platform on Google Cloud in 2026, but the Spanish login, the Casavista review tool used by moderators and the agency billing still ran on Casavista's code, in Casavista's hosting, at about **€2.6 million** a year.

The second business was **Keybridge**, the agent CRM bought in 2024. Its deal case had included **€7 million** a year of revenue synergies from using Keybridge's insight into its 9,800 agency customers to sell them Hearthly packages. In March 2027 Veldmark had committed that **no Keybridge agency data would be used by Hearthly for ranking**

or sales targeting, because agencies would otherwise leave a CRM that publishes their listings to Hearthly's competitors. That line of the synergy case was now not delayed but forbidden.

Then there was July. In the **Barcelona fake-rental ring**, 26 accounts posted 180 fake listings on Hearthly.es, and 63 people reported losses of about €71,000 before the takedown (Scams, Fakes and Account Takeovers: Fraud Prevention for Listings). None of the 26 accounts had proved more than an e-mail address, because the Spanish login was still Casavista's (Who Is on the Other Side? Identity and Verification). The parts of the acquisition that had not been consolidated were exactly the parts that mattered most for trust.

Ingrid's answer to the board, the **consolidation review** presented on **Tuesday 9 November 2027**, is the thread of this chapter. It made three arguments: the synergy case had been sequenced in the wrong order, it had counted only savings, and one of its lines should never have been written.

Concepts: Options, Sequences, Synergies and Brands

Integration options

An acquirer has five integration options for an acquired marketplace, plus the option to sell it. They differ in what is shared and in how reversible the result is.

Option	What changes	What stays	Reversibility	Documented example
Hold separate	Ownership, finance, board reporting	Brand, product, platform, organization	High	eBay's 2004 statement that Marktplaats would be operated "as a separate business" (eBay, 2004)

Option	What changes	What stays	Reversibility	Documented example
Federate under group controls	Security baseline, access, finance close, data protection, cloud governance	Product and platform	High	Common in practice; Adevinta's account-to-cost-center catalog is a documented control layer (see Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform)
Share services beneath	Global components: infrastructure, data, trust, media, engines	Brand, product surfaces, local commercial terms	Medium; depends on separability	Adevinta 2019–2023: shared infrastructure, data, OCR, media and experimentation beneath local marketplaces
Replatform onto a sibling	The acquired marketplace moves onto another brand's product platform	Brand, sometimes	Low	eBay's Mexican iBazar retired in 2015 and its users migrated to the acquired Vivanuncios "over the next several weeks" (eBay, 2015)

Option	What changes	What stays	Reversibility	Documented example
Merge brands	One consumer brand replaces another	Inventory and accounts, migrated	Very low	Naspers rebranded acquired local sites as OLX in several countries in 2014 (widely reported, Wikipedia)
Divest	The business leaves the group	—	—	The UK competition authority required the sale of Gumtree UK and Shpock as a condition of Adevinta's eCG acquisition, with transitional services offered for both (CMA, 2021)

Options are chosen per business and may be combined over time: a marketplace can be held separate for a year, federated, and then moved onto shared services capability by capability. What should not happen is a choice by default. **An acquisition without a recorded integration option becomes Deal-Shaped Architecture by inertia**.

Commercial integration can run years ahead of technical integration, and sometimes should. Two documented cases show how long "hold separate" can last at the brand level while the companies are combined. Immowelt and Immonet were combined under one holding company in 2015, with two sites in Nuremberg and Hamburg (Immowelt, 2015); in 2024 Immowelt GmbH was renamed AVIV Germany and still ran both immowelt and immonet.de (Online Marketplaces, 2024). In France, AVIV launched **Ultimmo** in 2022, a single professional offer across SeLoger, Logic-Immo and Meilleurs Agents (MySweetImmo, 2022).

Consolidation sequence

A **consolidation sequence** is the ordered set of stages through which an acquired marketplace's capabilities move onto group platforms, with a gate before each stage. The sequence answers three questions: which capability moves next, what must be true before it moves, and what would stop or reverse it.

Two rules set the order, and the core listing and search experience fails both. First, **move what users and sellers cannot see before what they can**, because invisible changes do not put liquidity at risk. Second, **move what gains value from pooling before what merely saves cost**, because pooled capabilities (fraud signals, identity, experimentation statistics, advertising audiences) make the whole group better, while cost-only moves can wait for a convenient moment. The core experience is the most visible part of the marketplace and gains little from pooling, so it comes last, and often never.

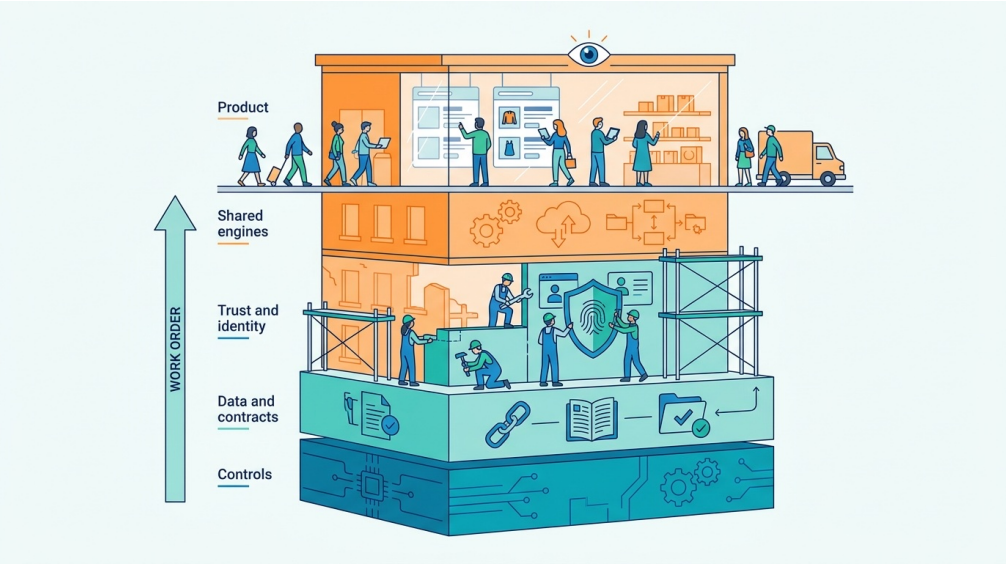


Figure 1: *Consolidate from the bottom up: what users never see and what gains value from pooling moves first, while the core listing and search experience, the most visible layer, comes last and often never.*

Synergy case

A **synergy case** is the quantified hypothesis that combining a business with the group will raise the combined profit beyond what each would earn alone. It has four parts, of which deal models usually show two:

Part	Typical lines	Usually in the deal model?
Cost synergies	Retired platforms, hosting and licenses; duplicated teams; procurement (cloud, payment, map and ad-tech contracts)	Yes
Revenue synergies	Cross-selling packages, pricing, advertising yield, new products from shared data	Yes, often optimistic
Dis-synergies	Organic traffic lost in migrations, seller and agency churn, partner integrations broken, key people leaving, double running during migration	Rarely, and then as a lump
Separation cost	Cost of carving the business out again if it is sold: transitional services, data separation, re-platforming	Almost never

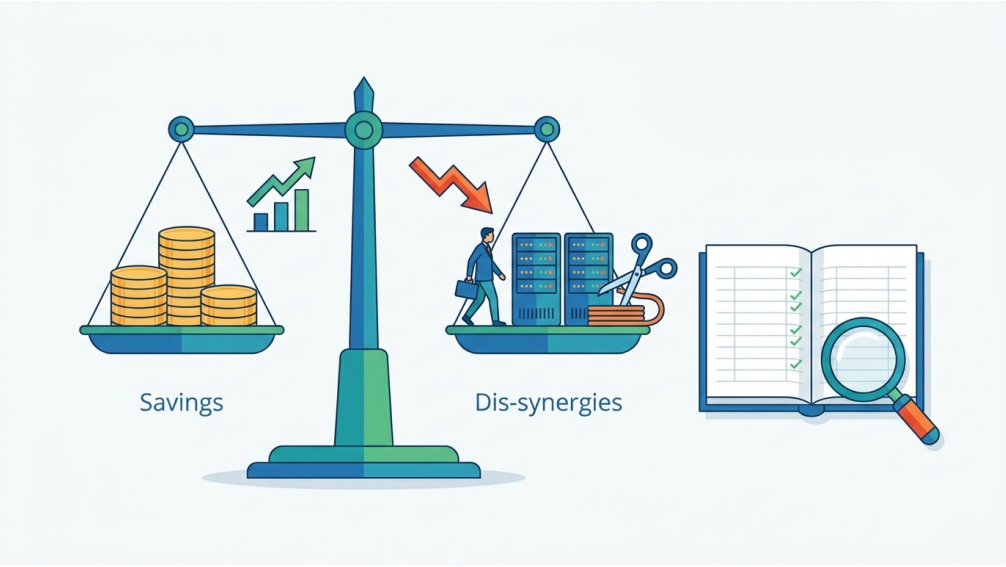


Figure 2: *A synergy case is a hypothesis: run it as a ledger that weighs traffic loss, churn,*

double running and the cost of separating the business again against the savings.

Adevinta's eCG deal shows the parts a deal model leaves out arriving later. It targeted run-rate cost synergies of **€130–165 million** after three years (Adevinta, 2021) and recorded **€69 million** of eCG integration expenses in 2023 alone, plus €15 million for the eBay Kleinanzeigen rebrand (Adevinta Annual Report 2023, 2024). Whether the target was met is not public. Two years after the take-private, the 2025 strategic report described "parallel running of central and local services during a transitional period" as the group decentralized (Adevinta Strategic Report 2025, 2026): separation cost, arriving after the synergies it was never netted against. Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform traces the whole cycle.

Brand migration

A **brand migration** moves a marketplace's users, sellers, partners and search-engine presence from one consumer brand to another: a rebrand to a group brand, the removal of a former parent's name, or the merger of two brands into one. It is an architectural event, not a marketing one, because the brand is embedded in domains, URLs, app identities, e-mail sender reputations, partner endpoints, legal entities and contracts.

The best-documented case in the industry is a forced one. Under the acquisition agreement with eBay, Adevinta could not use the "eBay Kleinanzeigen" brand beyond 2024, and the rebrand cost €15 million in 2023 (Adevinta Annual Report 2023, 2024). The site became Kleinanzeigen on 16 May 2023, with a new domain, while "the previous domain [would] remain accessible for several more years" (Adevinta, 2023). The platform migration came separately: a 2025 Kleinanzeigen engineering post describes migrating "the whole platform to a new system", running the new system in parallel with the old one and switching users incrementally behind a reverse proxy, with rollback possible (Berlin Tech Blog, 2025). The post does not say whether the two programs were planned together; the public record shows them two years apart.

Architecture: The Consolidation Sequence

Choosing the option

The integration option is the first architectural decision after signing, and Figure 3 makes it with four questions.

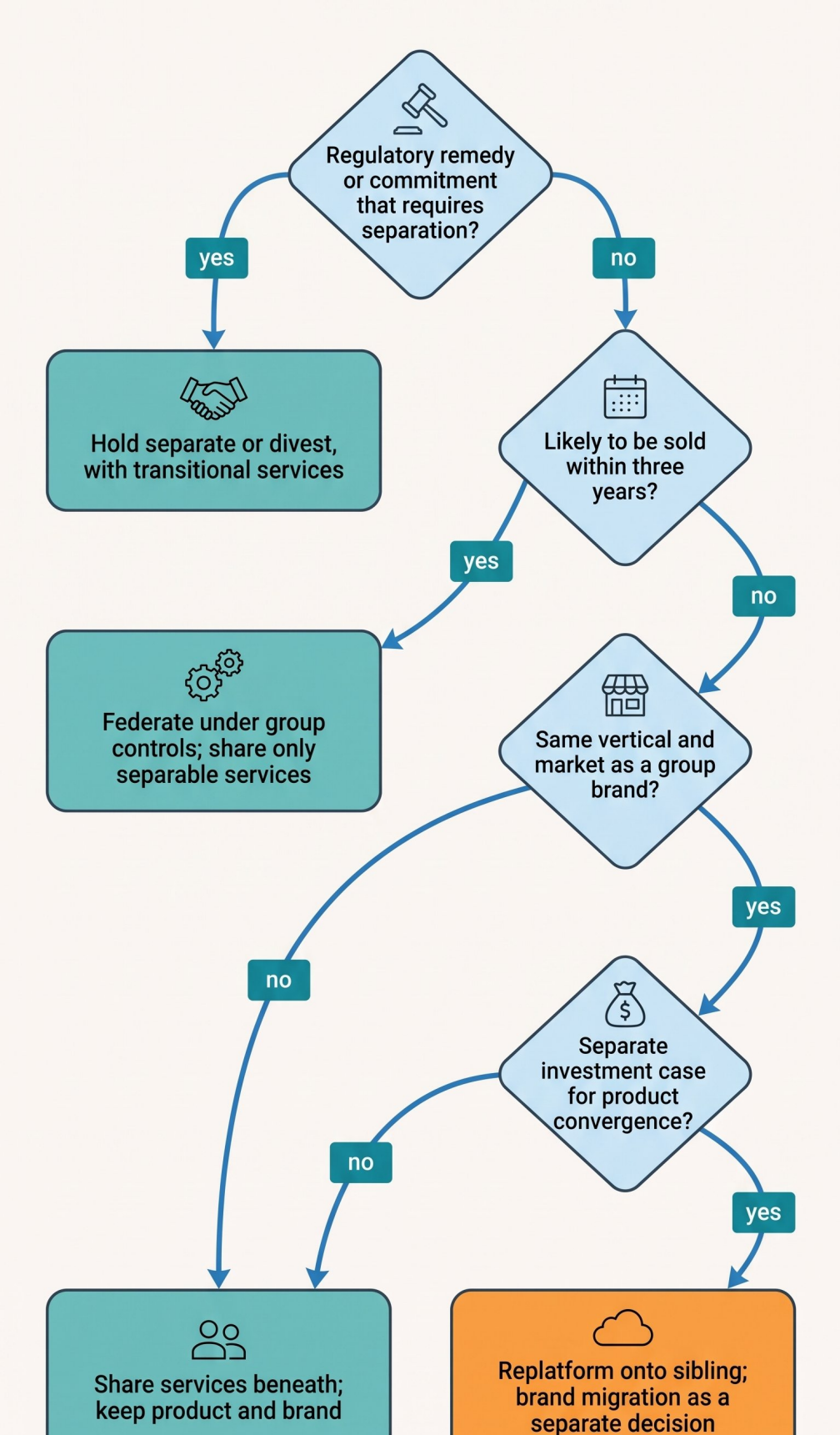

Regulatory remedy or commitment that requires separation?
yes
no
Hold separate or divest, with transitional services
Likely to be sold within three years?
yes
no
Federate under group controls; share only separable services
Same vertical and market as a group brand?
yes
no
Separate investment case for product convergence?
no
yes
Share services beneath; keep product and brand
Replatform onto sibling; brand migration as a separate decision

Figure 3: *Choosing an integration option. Regulation, commitments and the expected holding period remove options before technology is considered; replatforming onto a sibling needs its own investment case, and a brand merger is a separate decision even then.*

The questions are ordered by how absolute they are. A merger-control remedy or a commitment to customers, such as Keybridge's neutrality, rules out integration whatever its savings. A likely sale within a few years rules out deep sharing, because every shared layer becomes separation work. Only then does the question of fit arise.

Stages and gates after an acquisition

Figure 4 is the reference sequence for a marketplace acquired into a group that already has shared platforms; its gates matter more than its exact stage boundaries.

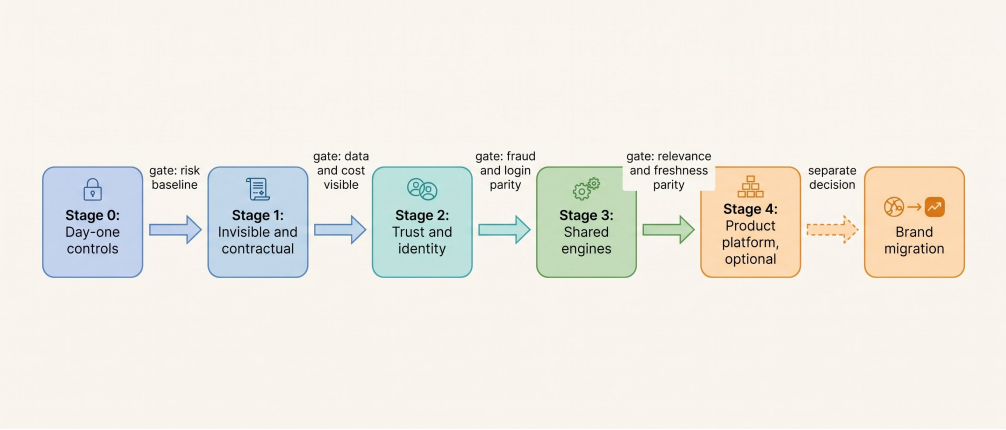


Figure 4: *The consolidation sequence. Each stage moves capabilities that are more visible to users than the last; each gate is a measured condition, not a date. Product convergence and brand migration are optional and decided separately.*

Stage	Capabilities	Why at this point	Gate to the next stage
0. Day-one controls (first 100 days)	Security baseline and incident response, privileged access, finance close, data-protection inventory, legal entities and terms, capability map entry (Map What the Marketplace Does Before What Runs It), transitional services if the deal is a carve-out	Risk the acquirer now owns; no user impact	A risk register, a capability and system map, a liquidity baseline for the acquired marketplace
1. Invisible and contractual	Cloud accounts and cost tagging, observability, the data platform (governed copies, metric definitions), advertising stack, payment and other supplier contracts, ML model registry	Pooled value or procurement savings without touching user journeys	The acquired marketplace's cost, data and models are visible in group tooling

Stage	Capabilities	Why at this point	Gate to the next stage
2. Trust and identity	Shared trust-and-safety pipeline with local policy (Trust and Safety as a Platform), shared fraud block lists, group identity with brand profiles (Who Is on the Other Side? Identity and Verification)	Fraud crosses brands; pooled signals protect everyone; acquired marketplaces are often the weakest link	Detection rates, time to action and login success at or above the baseline
3. Shared engines	Media pipeline, search engine platform, taxonomy and attribute service, product catalog and entitlements, messaging service, event backbone integration	Expensive and technically uniform; the product layer above can stay local	Relevance parity, publication freshness, billing parity in shadow mode
4. Product platform (optional)	Insertion flow, search and listing pages, apps, pro tools	Only when markets share language, law and product strategy and ownership is stable	Its own investment case and stage gates

Payments appear in stage 1 as a contract, not as code. Consolidating payment-provider relationships (pricing, risk models, onboarding of sub-merchants) is invisible to users and often worth more than merging checkout code. The eCG brands illustrate the pattern: Worldline's 2022 release on taking a stake in Online Payment Platform named both Marktplaats and eBay Kleinanzeigen as clients (Worldline, 2022), so two marketplaces shared a provider while running their own buyer-protection flows.

Identity appears in stage 2, not stage 1, because although it is foundational, a login change is visible to every user and must be planned with the trust capabilities that depend on it.

Hygiene before consolidation

Every stage depends on a prerequisite that acquisitions rarely satisfy: the acquired estate must be described well enough to move. OLX's 2026 account of a cross-functional program at OLX Motors is a useful public checklist. Among its outcomes were user identification standardized across 71 services, "with 73% now using modern UUIDs instead of legacy user IDs", 45 components moved to a central secrets manager, backup compliance for tier-1 and tier-2 databases and one deployment-notification service across more than 60 services (OLX Tech, 2026). None of these is a consolidation step. Each is a precondition for one.

The minimum hygiene I would require before stage 2 is: one identifier per user and per seller within the acquired marketplace, an inventory of every copy of listing and user data (System of Record and Read Models: Listing Data Management), the marketplace's events mapped to the group's listing event contract (Every Change Is an Event: Event-Driven Listing Architecture), and a list of every partner integration with its owner.

Brand migration as a separate workstream

The governing rule is to **decouple the brand migration from the platform migration**. Each is a traffic risk on its own; together, a ranking drop cannot be attributed, rolled back or fixed quickly, because nobody knows which change caused it.

A brand migration touches more systems than its sponsors expect. Figure 5 lists the workstreams and their dependencies.

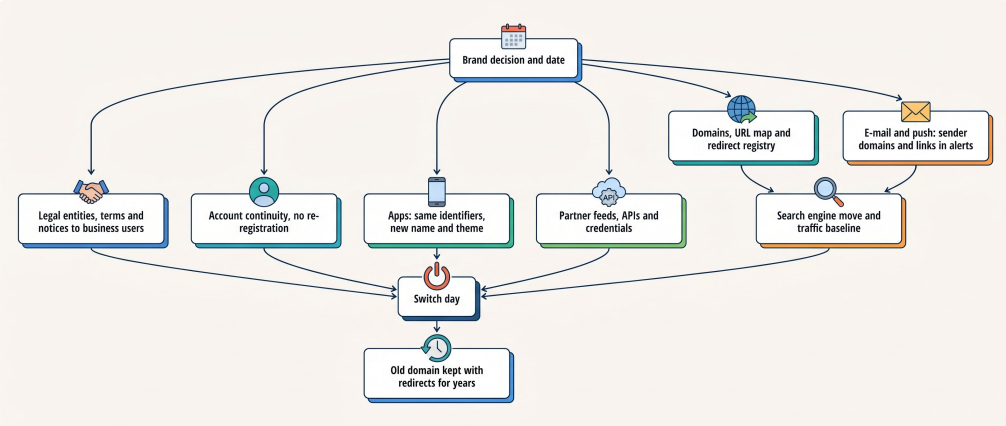


Figure 5: *The workstreams of a brand migration. Most are invisible in the brand plan and visible on switch day: a broken redirect, a forced re-registration, an alert e-mail linking to a retired domain or a partner feed sent to an endpoint that no longer answers.*

Each workstream has its own rules:

- **Domains and URLs**. Google's site-move guidance asks for server-side permanent redirects from each old URL to its final destination, short redirect chains, no mass redirects to an unrelated page such as the new home page, redirects kept "for as long as possible", generally at least one year, and the Change of Address tool when moving between domains; it warns to expect "temporary fluctuation" in rankings while the site is recrawled (Google Search Central, updated 2026). Kleinanzeigen's decision to keep its old domain accessible for several more years is consistent with that guidance. The redirect registry described in Found by Google: Web, App and SEO Architecture for Listings is the mechanism.
- **Accounts**. Users must not re-register. German press reporting on the Kleinanzeigen rebrand said existing accounts and payment options carried over without re-registration (widely reported; not verified against a primary source). The tenant code should not change with the brand either: the tenancy model in One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture keys data, events and cost by tenant, not by brand name.
- **Apps**. Keep the app's store identity and update its name and theme, rather than publishing a new app that users must find. A 2025 mobile.de engineering post, about a later rebrand of mobile.de's own Android app, reports that "the rebranding itself was the final step and took only two weeks to implement once the base refactoring was done", the refactoring being token-based theming and the removal of legacy components (Berlin Tech Blog, 2025). Design tokens make a brand a configuration value.
- **E-mail and notifications**. New sender domains need authentication and gradual volume ramp-up before they carry alert digests; every link in saved-search alerts must point to a domain that will answer for years (Tell Me When It Appears: Saved Searches and Alerts).
- **Partners**. Feed endpoints, API hostnames and credentials must keep working, or be migrated with each software vendor and dealer system on a published schedule (Many Doors In: Forms, APIs and Bulk Feeds).
- **Legal**. The contracting entity, terms, privacy notices and the EU Digital Services Act points of contact may change. For business users, the EU platform-to-business regulation requires changes to terms and conditions to be notified at least 15 days before they take effect (Regulation (EU) 2019/1150, Article 3).

In Practice: Veldmark's Consolidation Review

What went wrong with Casavista

The review reconstructed the Casavista integration from the 2025 deal case and the 2026–2027 program records.

Period	What happened	Effect
2025	Acquisition; deal case of €18M run-rate synergies by end of 2027; integration planned as "replatform onto Hearthly France, then rebrand"	Technology savings depended on moving everything, so the plan moved the most visible parts first
Monday 9 March 2026	Rebrand: casavista.es redirected to hearthly.es; app renamed in place; accounts carried over	Organic sessions down **19%** in the first six weeks, back to **–5%** by September 2026
September 2026	Listings and search moved onto Hearthly's platform on Google Cloud	Second traffic dip, smaller, on a site whose baseline had not recovered; the cause of each dip could not be separated
Late 2026	Remaining migration (login, moderation review tool, agency billing) deferred to 2027–2028	Casavista hosting kept running; the €8M technology line stalled at €3.9M
July 2027	Barcelona fake-rental ring, on accounts that had proved no more than an e-mail address	Trust and identity, the capabilities with the most pooled value, were the ones still on Casavista's code

The sequence was inverted. The integration moved the product first (search, listings, brand), because that is what a "replatform" plan naturally starts with and what makes the acquisition visible. It left trust and identity for last, because they were technically awkward and invisible to the brand plan. The order this chapter recommends is the reverse: trust and identity early, because their value grows with pooling and their

absence is where fraud concentrates; the product late, because it carries the liquidity.

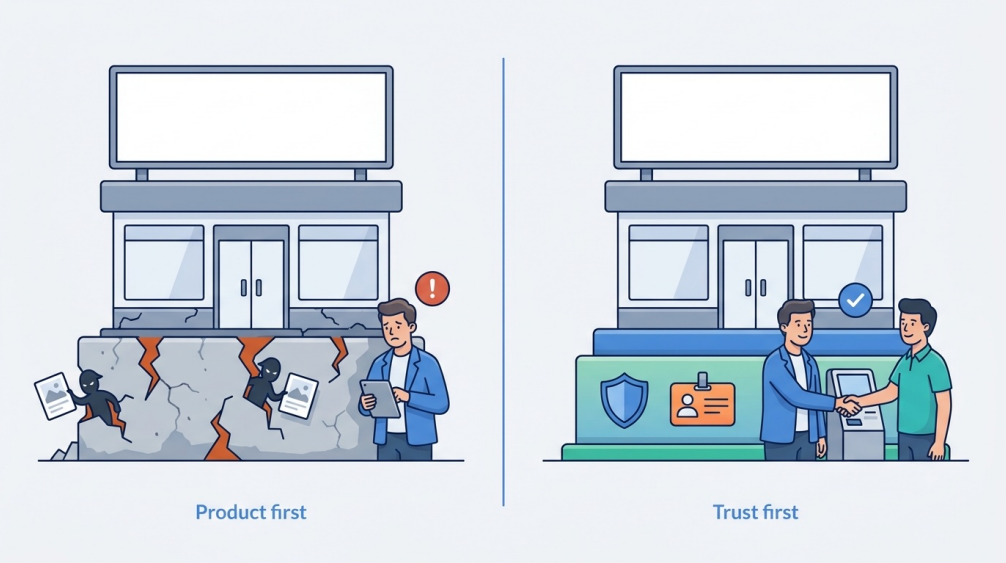


Figure 6: *Casavista's integration moved search, listings and the brand first and left trust and identity on the old code, which is where the Barcelona fake-rental ring found its gap; the recommended order is the reverse.*

The rebrand and the platform move were six months apart, which was better than doing both at once, but the second move started before the first had recovered. The review recorded a rule for future migrations: **a second traffic-affecting change may start only when organic sessions are back within 3% of the pre-migration baseline**, or when the board accepts the compounded risk in writing.

The revised sequence for Hearthly.es

The review resequenced the remaining work on one principle: **migrate to the target, not to the sibling**. It used Common Ground's dates rather than those of the original Casavista plan. Moving the Spanish login onto Hearthly France's login in 2027 would have been a migration to a system that Veldmark ID was due to replace. Instead:

Capability	Old plan	Revised plan	Source of the date
Moderation review tool	Hearthly FR tool, 2028	Shared trust-and-safety platform, **Q1 2028** (brought forward after Barcelona)	Scams, Fakes and Account Takeovers: Fraud Prevention for Listings
Private-landlord verification	Not planned	Stronger verification level for private landlords in Spain, live with the T&S move	Who Is on the Other Side? Identity and Verification
Login	Hearthly FR login, 2027	Federated with Veldmark ID in **2028**, migrated in **2029**	Who Is on the Other Side? Identity and Verification
Agency billing	Hearthly FR billing, 2027	Group catalog and entitlement service, Hearthly in the **second half of 2028**, after a full billing cycle in shadow mode	Packages, Bumps and Quotas: Products, Pricing and Entitlements
Casavista hosting	Switch-off 2027	Switch-off after the login migration, end of **2029**	Consequence of the above

The revised plan delays the last €4.1 million of technology savings by about two years. Ingrid's argument to the board was that the original date had never been real: a plan that moved the Spanish login twice would have cost more, and it would have left Hearthly.es on a sibling's login for a year in which it was the group's most-attacked marketplace.

Rewriting the synergy case as a ledger

The review replaced the Casavista and Keybridge synergy spreadsheets with a **synergy ledger**: one line per synergy or dis-synergy, each with an owner, a mechanism, a dependency on a consolidation stage and dated evidence.

Line	Type	Planned run rate	Oct 2027 evidence	Depends on	Status
Casavista hosting and licenses retired	Cost	€4.0M	€1.4M (search and listing infrastructure retired 2026)	Login migration, 2029	Re-dated
Casavista search and media on Hearthly platform	Cost	€4.0M	€2.5M	Done; remainder from shared OpenSearch by end of 2028	On track
Overheads (finance, HR, office)	Cost	€4.0M	€3.6M	Stage 0	Nearly done
Agency packages and new-build campaigns in Spain	Revenue	€6.0M	€2.1M	Catalog and entitlements, 2028	Re-dated
Organic traffic loss from rebrand and migration	Dis-synergy	not in case	about €1.3M revenue in 2026 (estimate)	—	Recorded
Double running of Casavista hosting 2027–2029	Dis-synergy	not in case	€2.6M a year	Login migration	Recorded

Line	Type	Planned run rate	Oct 2027 evidence	Depends on	Status
Keybridge cross-selling from agency data	Revenue	€7.0M	—	Forbidden by the March 2027 commitment	**Withdrawn**
Keybridge neutral synergies (finance, security operations, procurement)	Cost	—	€1.9M	Stage 0 and 1	Added
Hearthly.es separation cost if sold	Option cost	not in case	estimated per shared capability	Exit paths per capability	Recorded

The Keybridge line is the instructive one. A revenue synergy that depends on using one customer group's data to sell to them through another business is not a synergy but a breach of trust waiting for a date. The review's rule for future deal cases is that every revenue synergy names the data it uses and the commitment, contract or law that permits that use; Lucía Ortega, the Group Data Protection Officer, signs off the list before the deal model goes to the board.

Keybridge: the case for holding separate

Keybridge is the group's deliberate exception to consolidation. It runs on Azure, publishes listings for 9,800 agencies to Hearthly and to Hearthly's competitors, and integrates with Hearthly only through the same public supply contract that every software vendor uses (Serving the Pros: Dealer, Agent and Merchant Tooling). Its engineering organization is not part of the Common Ground platform organization, and group security standards are the only shared engineering rules (Who Owns What: Organizing Marketplace Engineering).

The review classified Keybridge's integration option as **federate under group controls**, permanently: finance, security operations, procurement and the annual independent review of the neutrality commitment, and nothing that touches agency data. It also recorded the corollary. Because Keybridge shares almost nothing, it is the group's most

separable business; if an owner ever wanted to sell it, the separation cost would be close to zero. **Neutrality and separability turn out to require the same architecture**. The real-estate evidence points the same way: a portal that owns the software agents use to originate listings holds a strong position, and the agencies and competition authorities know it (AVIV Group: Building a European Real-Estate Platform from Acquisitions examines AVIV's portals and professional offers).

The overlap inside the group

The one commercial overlap between Veldmark's own brands needed a portfolio decision, not a platform one. Tradeyard Poland's cars section competed with Motorvane Poland for the same dealers: about **2,900** Polish dealers had accounts on both, and about **40%** paid both. The decision had already been made in the marketplace profile register: Tradeyard Poland's vehicles section stays a classifieds section with insertion fees for private sellers, while dealers above a small listing count are offered Motorvane packages and their stock is syndicated to Tradeyard through Motorvane's feed (Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy). The consequence for consolidation is one field: the shared listing core records which brand owns the commercial relationship for each listing, so the overlap can be managed without merging the brands.

Trade-offs

Integration options compared

Option	Savings	Liquidity risk	Time to value	Separation cost later	Right when
Hold separate	None beyond overheads	None	Immediate	Lowest	Remedies, neutrality, likely sale, very different business
Federate under controls	Small; risk reduction	Very low	Months	Low	Default for the first year of any acquisition

Option	Savings	Liquidity risk	Time to value	Separation cost later	Right when
Share services beneath	Large over time, pooled value	Low to medium, gated	One to three years	Medium, if capabilities are tenant-separable	Long-term holdings in any vertical
Replatform onto a sibling	Largest technology savings	High	Two to four years	High	Same vertical and market, stable ownership, own investment case
Merge brands	Marketing and SEO concentration	Very high	Years	Highest	Two weak brands in one market, or a contractual obligation

Speed against safety

Approach	Benefit	Cost	When it is right
Big-bang replatform and rebrand	One program, one cutover	Compounded traffic risk; causes cannot be attributed; no partial rollback	Almost never in a liquid marketplace

Approach	Benefit	Cost	When it is right
Rebrand first, platform later	Brand value captured early	Two traffic events; the platform migration starts on a weakened baseline	When the brand change is forced by contract
Platform first, rebrand later	Users see one change at a time; rebrand becomes configuration	Two brands run on one platform for a while	The default for voluntary rebrands
Capability by capability beneath an unchanged brand	Lowest risk; synergies arrive per stage	Slower; requires shared platforms to exist	The Common Ground approach

When the guidance does not hold

Bottom-up sequencing assumes the acquirer has shared platforms worth joining. A group whose only real asset is one brand's product platform has nothing beneath to share, and replatforming onto that product may be the only way to gain anything. It also assumes the acquisition is meant to be kept. When the purpose of an acquisition is to remove a competitor in a market where the acquirer already leads, merging brands and inventory early can be right, subject to competition law, because the value lies in concentrating liquidity, not in keeping two products alive.

Vertical Differences

Concern	General classifieds	Real estate	Automotive
What the deal usually buys	Audience and private-seller supply in a new market	Agency relationships and listing feeds in a market	Dealer relationships and vehicle data

Concern	General classifieds	Real estate	Automotive
Easiest early consolidation	Ads, trust signals, media, identity	Media, data platform, ad stack	Media, data platform, vehicle data services
Hardest consolidation	Transactional journeys (national PSP, carrier and consumer-law set-ups)	Ingestion adapters, pro tools, valuation data sources, disclosures	Dealer packages, price ratings, dealer management system integrations
Brand migration risk	Organic search and app habit	Agencies' feeds and portal choice; multi-listing means agencies can shift spend quickly	Dealers' feed integrations and package contracts
Documented signal	Kleinanzeigen rebrand (2023) and separate platform migration (published 2025)	Immowelt and immonet combined in 2015, both brands still running in 2024; AVIV described in 2025 as having rowed back on plans for a fully integrated platform (the interviewer's phrase; see AVIV Group: Building a European Real-Estate Platform from Acquisitions)	OLX bought La Centrale in France, closed November 2025 (OLX Group, 2025); whether it will move onto OLX's motors platform is not public

Real estate and automotive carry an extra risk: supply arrives through professional software, so a consolidation that changes an endpoint, a feed format or an error code is felt by every agency or dealer at once. The German competition authority's October 2026 clearance of mobile.de's acquisition of Carwow's German business noted

that dealers multi-home and that "many platforms offer dealers data export to other platforms" (Bundeskartellamt, 2026). In feed-driven verticals, professionals can move their inventory; a migration that makes the integration worse invites them to.

Failure Modes

1. **The synergy case written before the architecture**. Technology savings are counted on the assumption that the acquired platform can be switched off on a date nobody has planned, and the switch-off slips while the savings remain in the budget.
2. **Product first, trust last**. The visible migration happens because it shows progress; the pooled, invisible capabilities wait, and fraud finds the gap, as it did on Hearthly.es.
3. **Two migrations, one traffic curve**. A rebrand and a replatform, or two replatform steps, overlap, and a ranking drop cannot be attributed or reversed.
4. **Migrating to the sibling, then to the target**. An acquired capability moves onto another brand's system that is itself due to be replaced, and the work is done twice.
5. **Revenue synergies built on forbidden data**. Cross-selling plans assume data use that customers, contracts or law do not allow; the line is withdrawn after trust has been damaged.
6. **Integration scope assumed at signing**. Competition remedies remove the businesses with the most overlap before anyone integrates them, as the CMA's 2021 conditions did for Gumtree UK and Shpock; the integration plan was drawn for a portfolio the group did not end up owning.
7. **The forgotten acquired organization**. The acquired marketplace's engineers are kept "until integration", which never comes, and their best capabilities are never promoted to group level (Who Owns What: Organizing Marketplace Engineering). Adevinta's documented counter-examples, such as an experimentation statistics package built at Marktplaats and adopted group-wide, show the value of the opposite (Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform).
8. **No exit path**. Deep integration under one owner becomes separation cost under the next; Adevinta's parallel running in 2025 is the public example.

The Chief Architect's View

Decide the integration option in the first month, by decision record. The record names the option, the reasons, the regulatory and contractual constraints, the expected holding period and the first review date. It goes to the architecture forum and to the deal sponsor. An acquisition without one will be integrated by whoever moves first.

Own the consolidation sequence; let the deal team own the synergy targets. The Chief Architect cannot promise a synergy, but can say which stage each synergy depends on, what gate it must pass and what it will cost to run two systems in the meantime. The synergy ledger is where those statements live, and it should be reviewed quarterly with the finance team that reports the synergies.

Sign off the deal model's technology lines before signing. In a group that buys marketplaces regularly, the architecture function should review every technology synergy and every revenue synergy that depends on data, before the price is agreed. This is the cheapest point at which to remove a forbidden line or to add a missing dis-synergy.

Make liquidity the gate. Every stage gate includes the acquired marketplace's own liquidity metrics: time to first contact, contacts per listing, organic sessions, pro feed success rate (Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics). A stage whose gate fails is paused, not pushed through to meet a date.

Keep the brand decision separate from the platform decision. Brands are commercial assets with their own investment case; platforms are cost and capability decisions. Merging the two decisions produces programs that are too big to stop.

Design every integration so it can be undone. Tenant-separable shared capabilities, exportable data, contracts that can be split and code whose ownership is clear cost little when built in from the first stage (One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture). Veldmark's owner is a private-equity investor; any of its brands could be sold. The same discipline that makes a consolidation safe makes a later separation cheap.

What to watch.

- Synergy ledger lines whose dependent stage has slipped while the line has not.
- Double-running costs, reported as their own line, with an end date.
- The acquired marketplace's liquidity metrics against its pre-deal baseline, monthly for two years.
- Organic sessions and partner-feed error rates around every migration.
- Capabilities of the acquired business that have been adopted elsewhere in the group: the measure of whether the acquisition brought anything to the platform, not only took from it.

Questions to Ask

1. Which integration option did we choose for each acquired business, in which decision record, and when is it next reviewed?

2. Which regulatory remedies, customer commitments or contracts rule out integration options for this business?
3. In what order will its capabilities move onto group platforms, and what measured gate stands before each stage?
4. Which synergies depend on which stage, and which of them assume a platform switch-off date that nobody has planned?
5. What dis-synergies have we counted: traffic loss, churn, double running, partner rework, key people?
6. Which revenue synergies use data, and what permits each use?
7. Is any brand migration planned in the same period as a platform migration? Who decided that the risks may compound?
8. For a brand migration: how long will the old domain answer, how are accounts carried over, which partner endpoints change, and when are business users notified?
9. Are we migrating any capability onto a sibling system that is itself scheduled to be replaced?
10. If this business were sold in three years, what would we hand over, switch off and run twice, and what would it cost?

From Sequence to Organization

A consolidation sequence describes what moves and when; it does not say who owns each capability once it has moved, or how the acquired engineers become part of the group. The next chapter, Who Owns What: Organizing Marketplace Engineering, takes up the organization that makes a consolidated platform work.

Sources and Evidence

Company engineering and investor sources

- eBay acquires Dutch company Marktplaats.nl (8-K exhibit) — eBay Inc., 2004. Marktplaats to be operated "as a separate business".
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- Acquisition of eBay Classifieds Group complete — Adevinta, 2021. €130–165 million run-rate cost synergy target; remedy sales.
- Adevinta Annual Report 2023 — Adevinta, 2024. Integration expenses, rebrand cost and the contractual end of the eBay brand.

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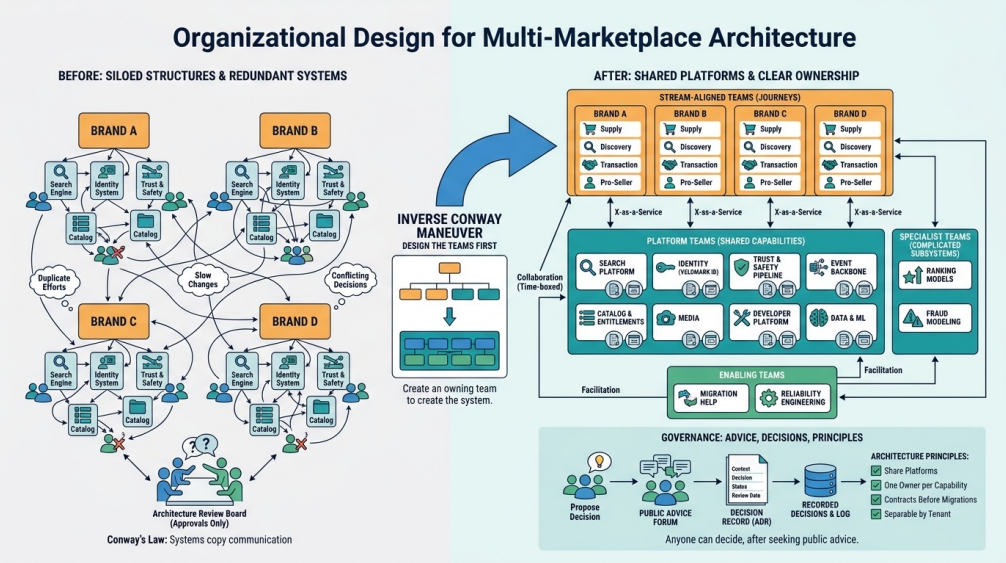
- Regulation (EU) 2019/1150 on platform-to-business relations — European Union, 2019. Notice period for changes to business users' terms.
- Site moves with URL changes — Google Search Central, updated 2026. Redirect rules, duration and expected ranking fluctuation in domain moves.

Research and practitioner sources

- OLX — Wikipedia, n.d. (read October 2026). 2014 rebranding of acquired local sites to OLX (widely reported, not primary).

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

42. Who Owns What: Organizing Marketplace Engineering



IN THIS CHAPTER, YOU WILL: Design the team topology and the architecture governance for a group of marketplaces: which teams are aligned to which journeys, which shared capabilities get platform teams, where specialist search, ranking and valuation teams sit, how platform work is funded, and how cross-brand decisions are made, recorded and revisited. You will follow Veldmark's Common Ground organization design and its architecture advice forum through their first year.

WHY THE BUSINESS CARES: A marketplace group pays for its organization twice: once in salaries, and again in the time it takes for one improvement to reach every brand and country. A fraud rule that takes three days to reach Belgium and never reaches Poland is an organizational defect, not a technical one.

WHY ARCHITECTS SHOULD CARE: Systems follow the communication structure of the teams that build them. A shared platform without an

owning team decays, a bounded context split between two teams becomes a permanent negotiation, and an architecture function that reviews without owning anything is routed around. The org chart is part of the architecture.

KEY POINTS:

- **Assign owners before you assign services.** Every buyer and seller journey, every bounded context and every shared capability needs exactly one owning team, and the team's cognitive load must fit. Stream-aligned teams own marketplace journeys; platform teams own shared capabilities as internal products; specialist teams own the few subsystems, such as ranking models and valuation, that need rare expertise.
- **A platform team earns its tenants.** Shared capabilities become platforms when an accountable team offers them with a contract, a roadmap and migration help, funded so that the cheapest choice for a marketplace is to use them. Mandates without those things produce shadow copies; voluntary adoption with them is the best-documented success in the public record.
- **Govern with decisions, not with gates.** An advice process — anyone affected may decide, after seeking advice from those affected and from experts, and must record the decision — plus a small set of principles and an open weekly forum scales better across brands than an architecture review board. Decision rights still need to be explicit: what the group decides, what platforms decide and what each marketplace decides.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Forced Fit** — a central platform imposed by deadline, without a team that treats marketplaces as customers, is the failure that organization design is most often asked to prevent and most often causes.
- **The Copy-Paste Country** — when each country or brand has its own full-stack teams for the same journey, every improvement is built, staffed and maintained once per team, and the code forks to follow the org chart.
- **Deal-Shaped Architecture** — acquired marketplaces arrive with their own engineering organizations, and unless ownership of shared capabilities is assigned deliberately, the group keeps four of everything because four teams each own one.

Marketplace groups are usually assembled, not designed. A founding marketplace grows its own engineering organization; acquisitions bring theirs; a group function appears to run finance, security and perhaps an advertising stack. Each organization works.

Together they produce the familiar picture of four identity systems and four trust-and-safety stacks, because **each capability exists as many times as there are teams that believe they own it**.

The technical chapters of this book argue for shared platforms beneath local products: one event backbone, one search platform, one trust-and-safety pipeline, one entitlement ledger, configured per brand and country. None of that survives contact with an organization that has no team to own it. After the Deal: Consolidating Platforms After Acquisitions set the order in which the shared capabilities move after acquisitions; this chapter covers the other half of the architecture: **who owns which journey**, **which capability and which decision**, how those teams interact, how they are paid for, and how a group makes architectural decisions that bind several brands without a central committee that slows all of them. Changing the Engine in Flight: Modernizing a Live Marketplace then moves the code to the owners named here.

This chapter owns two terms. A **marketplace team topology** is the assignment of team types to a marketplace group's journeys, bounded contexts and shared capabilities, together with the interaction mode between each pair of teams. **Architecture governance** is the set of decision rights, forums, records, principles and standards through which architectural decisions are made, communicated and revisited. Where boundaries come from is the subject of Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models; what to share across countries and brands is One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture.

Setting the Scene at Veldmark: The Rule That Reached Belgium Three Days Late

On Sunday 7 March 2027, Femke, a private seller in Rotterdam, listed a city bike on Tradeyard Netherlands. At 20:11 a "buyer" sent her a fake Safe Deal payment link. She reported it at 20:19, the account was suspended at 20:26, and Tradeyard NL's trust-and-safety team wrote a new rule that withheld 37 similar messages to other sellers that evening. Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), handled the incident well in the Netherlands.

The rule reached Tradeyard Belgium on **Wednesday 10 March**, because Belgian rules shipped with Yardstick, Tradeyard's PHP monolith, which deploys twice a week. It never reached Tradeyard Poland or Portugal, where moderation was run by two outsourced vendors with their own rule sets. And the phone number behind the suspended account had also registered a private-seller account on Motorvane, Veldmark's automotive marketplace, which nobody at Motorvane learned about.

Ingrid Halvorsen, Veldmark's Group Chief Architect since February, does not read this as a trust-and-safety failure. She reads it as an organization chart. She asks each brand

for its team list and gets four documents in four formats:

Brand	Engineers (2027)	Teams	How teams are cut	Architecture function
Tradeyard	about 430, of whom about 140 work in teams whose changes ship only with a Yardstick deploy	61	Mixed: Yardstick component teams, country teams in Poland and Portugal, a few journey teams on the AWS services	Two architects in Daan Visser's office
Motorvane	about 290	38	Journeys (dealer supply, discovery, leads, financing)	Principal engineers' guild
Hearthly	about 250	33	France and Spain as separate organizations since the 2025 acquisition	One architect, in France
Keybridge	about 70	10	Product modules of the CRM	The founder's CTO
Group functions (Veldmark Ads, group IT)	about 110	16	Functions	None
Total	**about 1,150**	**158**		

The team list explains the incident better than the rule engine does. Trust and safety for Tradeyard was owned by an in-house team for the Netherlands and Belgium and by vendors for Poland and Portugal; nobody owned *trust and safety for Tradeyard*, let alone for the group. The capability map that Ingrid presented on 31 March 2027 (see Map What the Marketplace Does Before What Runs It) found the same shape in identity, search, catalog and analytics: **each overlap existed because several teams each owned a**

piece of it, and no team owned the whole.

Figure 1: *The March 2027 phishing rule reached Belgium three days late and Poland and Portugal never, because each country's trust and safety had a different owner and nobody owned trust and safety for Tradeyard.*

Mateus Carvalho, the Group CTO, asks Ingrid the question this chapter answers: *Common Ground has €140 million and four years to build shared platforms. Who will own them, who will use them, and who decides when the two disagree?*

Concepts: Teams as Part of the Architecture

Conway's law, read as a design tool

Melvin Conway observed in 1968 that organizations which design systems produce designs that copy their own communication structures (Conway, 1968). In a marketplace group the observation is easy to verify: four brands with four engineering organizations have four identity systems. The useful reading is the reverse one, sometimes called the *inverse Conway maneuver*: **if you want a shared search platform, create a team whose job is the search platform**, and give the brands' discovery teams a reason to depend on it.

The maneuver has two limits. First, changing teams does not change code; the old structure persists in the systems until someone migrates them, which is why Changing the Engine in Flight: Modernizing a Live Marketplace follows this chapter. Second, teams have a limited capacity for complexity. The Team Topologies authors, Matthew

Skelton and Manuel Pais, put it as a principle: "Teams can only handle so much complexity before breaking down" (Team Topologies, key concepts, n.d.). A team asked to own Tradeyard's insertion flow in four countries, the pro-seller API and the image pipeline will own all three badly.

Four team types and three interaction modes

Team Topologies names four fundamental team types and three ways teams interact (Team Topologies, key concepts, n.d.). The tables below map each onto a listing marketplace.

Team type	Definition (Team Topologies)	In a marketplace group	Example at Veldmark (2027 design)
Stream-aligned	Organized around a flow of work, usually a slice of the business domain	A buyer or seller journey of one marketplace: supply and insertion, discovery, conversation and transaction, pro-seller tooling, monetization	Tradeyard Discovery; Motorvane Dealer Supply; Hearthly Agency Experience
Platform	Provides an internal product that speeds up stream-aligned delivery	A shared capability with a contract: event backbone, search platform, identity, trust-and-safety pipeline, media, entitlement ledger, taxonomy service, developer platform	Event Platform (Sven Lindqvist); Search Platform (Tomasz Wiśniewski)

Team type	Definition (Team Topologies)	In a marketplace group	Example at Veldmark (2027 design)
Complicated subsystem	Owns work that needs deep specialist expertise	Ranking models, automated valuation models (AVMs), fraud models, query understanding for one language family	Valuo AVM team; group fraud-modeling team
Enabling	Helps stream-aligned teams overcome obstacles and adopt new capabilities	Migration help, reliability engineering, experimentation practice, accessibility	Yardstick Exit team; Search Migration enablement

Interaction mode	Meaning	When it fits in a marketplace
Collaboration	Two teams work together for a set period to discover an API, practice or technology	A platform team and its first tenant designing a contract (the event platform and Tradeyard in 2027)
X-as-a-Service	One team provides, another consumes	The steady state for every mature platform
Facilitation	One team helps and mentors another	An enabling team moving a brand's alerts onto the new matcher

The mode is a property of a pair of teams at a point in time. **Collaboration that never ends is a sign that the platform has no contract**; X-as-a-Service from the first day is a sign that the platform was designed without its tenants.

The marketplace team topology

Every marketplace organization design must choose the axis that cuts its stream-aligned teams: country, journey or capability layer.

Axis	Teams look like	Strength	Weakness
Country	"Tradeyard Poland" with full-stack teams	Close to the local market, language and law	Every journey is built once per country; the Copy-Paste Country in organizational form
Journey	"Tradeyard Discovery" serving all four countries	One implementation per journey; local needs carried by market product leads and configuration	Needs strong configuration and localization; local managers feel distant
Capability layer	"Tradeyard Search Backend", "Tradeyard Android"	Deep technical skill	Every product change needs several teams; nobody owns an outcome

For marketplaces with one product in several countries, I would cut stream-aligned teams **by journey within a brand**, and carry the country through product roles and configuration rather than through separate teams. The exception is a country whose product differs, such as Tradeyard Poland's vehicles and property sections with insertion fees, which may justify a market team for those sections. The rule I use: **a country team is justified by a different product, not by a different language**.

Across brands, the same journey stays separate. Motorvane's dealer supply and Hearthly's agency supply are different products with different customers, feeds and contracts, and the shared part — the supply contract, the validator library, the taxonomy — is a platform, not a shared team (see Many Doors In: Forms, APIs and Bulk Feeds).

Ownership is the unit of design

Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models establishes the rule that each bounded context has exactly one owning team. This chapter extends it to three kinds of things a marketplace must own:

1. **Journeys** — the flow a buyer or seller experiences, with an outcome metric: time to first contact, listings published within the publication SLA, contacts per search session.

2. **Contexts** — the models and their truth: the listing catalog, entitlements, trust decisions, the search read model.
3. **Shared capabilities** — the platforms that several journeys use.

An owner is the team that is woken up when it breaks, decides its roadmap, and answers for its cost. A team that is consulted is not an owner. A committee is never an owner.

Architecture governance

Governance is often treated as a synonym for review. In a marketplace group it has five parts, and review is the least important:

Part	What it is	Failure when it is missing
Decision rights	Who may decide what: group, platform, brand, team	Every decision escalates, or none does
Decision records	Short documents that capture context, decision, status and consequences, kept with the code	The same argument is held every year; nobody knows why the system is shaped the way it is
Principles	A small set of testable rules against which options are judged	Decisions are argued from taste
Forum	A regular, open place where proposed decisions get advice in public	Advice is private, inconsistent and slow
Standards and golden paths	The few things every team must do the same way, ideally made easy by a platform	Either everything is a standard or nothing is

Michael Nygard's 2011 proposal for architecture decision records (ADRs) is still the common shape: a title, the context ("the forces at play"), the decision in active voice, a status (proposed, accepted, deprecated, superseded) and the consequences, positive and negative (Nygard, 2011).

Andrew Harmel-Law's **advice process**, described in 2021, replaces approval with one rule: "anyone can make an architectural decision", provided they first seek advice from everyone meaningfully affected and from people with relevant expertise, and

record the advice they received; they need not follow it. It is supported by decision records, a weekly open architecture advisory forum, team-sourced principles and a local technology radar (Harmel-Law, martinfowler.com, 2021).

Architecture: A Topology for a Group of Marketplaces

Stream-aligned teams on top of platform teams

A group of marketplaces needs stream-aligned teams per brand and journey on top, platform teams per shared capability underneath, a few complicated-subsystem teams beside them, and enabling teams that move between them. Figure 2 shows the version Veldmark adopted for Common Ground.

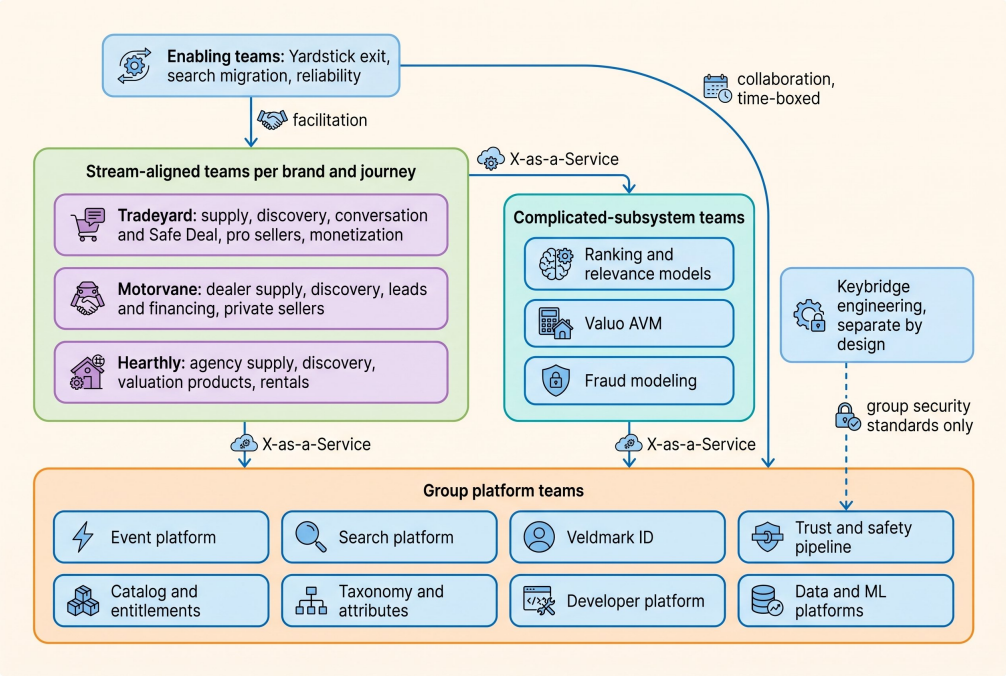


Figure 2: *Veldmark's Common Ground team topology (target 2028). Brands keep their journeys and their products; the group owns the capabilities beneath them; specialists and enablers sit beside both. Keybridge stays outside the platform organization because its neutrality commitment requires it.*

Three properties of the figure matter more than its boxes:

- **Arrows point down**. Brands depend on platforms; platforms never depend on a brand's internals. When a platform needs something from a brand — Tradeyard's listing events, for example — it gets it through a published contract, not a call into Yardstick.
- **Specialists serve brands through platforms**. A ranking model is trained by a specialist team, served through the search platform's ranking service, and configured by each brand's discovery team. The relevance objective stays with the brand, as Ranking for Two Sides: Relevance, Freshness and Paid Visibility argues; the model mechanics stay with the specialists.
- **Enabling teams are temporary by design**. The Yardstick Exit team exists to make itself unnecessary by the end of 2029.

Platform teams as internal products

A platform team differs from a shared-services department in five things it must have:

A platform team has	Why it matters in a marketplace group	Evidence it is missing
A **product owner** with a tenant-facing roadmap	Brands plan around it; the roadmap is where conflicts between brands become visible	Priorities set by whichever brand escalated last
A **contract** (API, event schema, service levels)	Brands can depend on it without reading its code	Tenants call internal endpoints; upgrades break them
Self-service onboarding	A new tenant or country costs days, not a project	Every onboarding is a ticket to the platform team
Migration help	Moving onto the platform is the tenant's cost; reducing it is how the platform wins	Adoption stalls at the second tenant
Run responsibility and cost transparency	The team owns incidents and can show cost per tenant	Brands blame the platform; the platform blames brands' usage

The best-documented example in the classifieds record is Adevinta's internal platform. A 2019 talk, written up in January 2020, described a "Common Platform" for about 1,500 engineers in about 20 marketplace companies, with voluntary adoption, about ten minutes of self-service onboarding, and platform engineers embedded with large tenants

during migration; by 2023 the platform, by then called SCHIP, served "more than 4,000 developers" (see Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform for the sources and the later decentralization). **Voluntary adoption with migration help is the pattern; a mandate is what you use only after the platform is clearly cheaper and the last holdouts are cost, not need.**

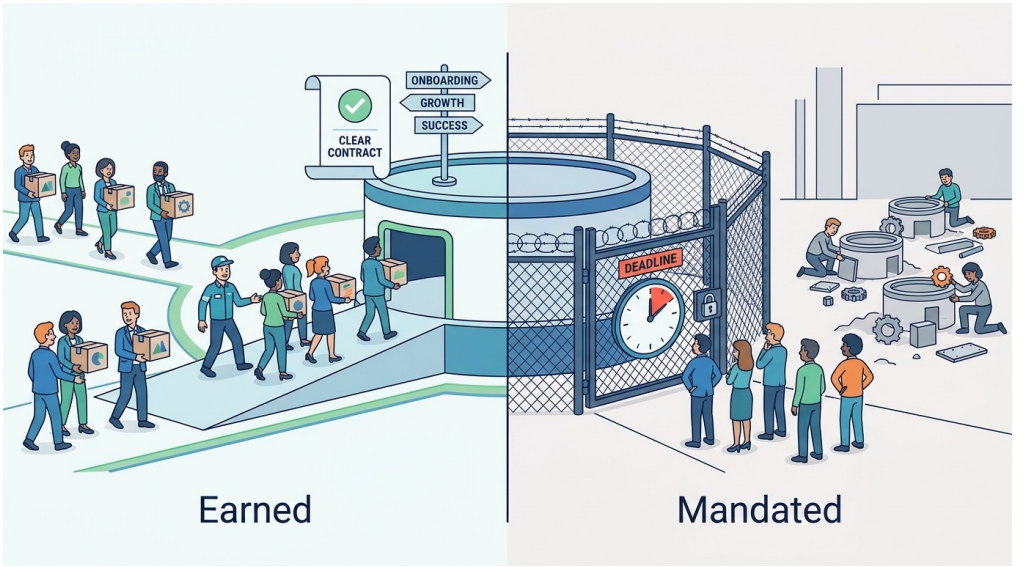


Figure 3: *A platform team earns its tenants with a contract, self-service onboarding and migration help; a mandate without those things produces compliance on paper and shadow copies behind it.*

Decision rights across group, platform and brand

Governance starts with a table, not a forum. The decision-rights matrix below is the one that Veldmark's forum adopted in April 2027 and revised in September. *D* means decides, *A* means must be asked for advice, *I* means informed.

Decision	Group (CTO, Chief Architect)	Platform team	Brand engineering	Brand product
Whether a capability is shared (enters Common Ground)	**D** (CTO, after forum advice)	A	A	A
Platform contract and roadmap	A	**D**	A	A
Engine, store or vendor behind a platform	I	**D**	I	—
Brand product features, UX, ranking objective, packages and prices	—	A (feasibility)	A	**D**
Brand-internal architecture (services, stores) within principles	I	—	**D**	—
Exception to a principle or standard	A	A	**D**, with a decision record	—
Personal data crossing brands	Group DPO **D**	A	A	A

Decision	Group (CTO, Chief Architect)	Platform team	Brand engineering	Brand product
Funding shifts between platforms and brands	**D** (CTO and CFO)	A	A	A

The matrix has two deliberate features. Brands decide exceptions themselves, provided they record them; the forum's job is to make the trade-off visible, not to veto it. And the decision whether a capability becomes shared sits with the CTO, because it moves money and people, but only after the forum has heard the brands.

The advice process in a marketplace group

Under the advice process, most cross-brand decisions stay with the team that proposes them. Figure 4 shows the path as Veldmark runs it.

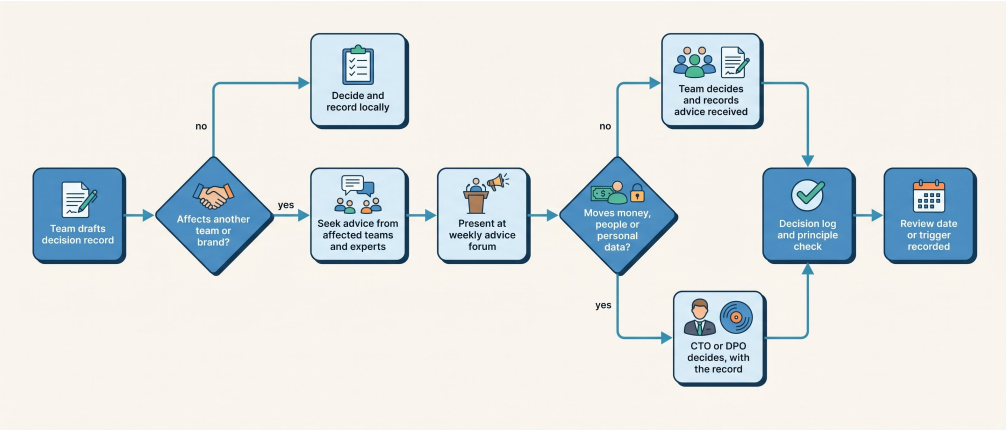


Figure 4: *The advice process applied to cross-brand decisions. Most decisions stay with the team that proposes them; the forum makes advice public and consistent; only decisions that move funding, people or personal data across brands go to an accountable executive.*

The process gives up approval and keeps three things that matter: **advice is public**, so the Motorvane team hears what the search platform told Hearthly; **decisions are recorded**, so the reasons survive the people; and **every record carries a review date or trigger**, so decisions are revisited when the facts change rather than when someone new arrives.

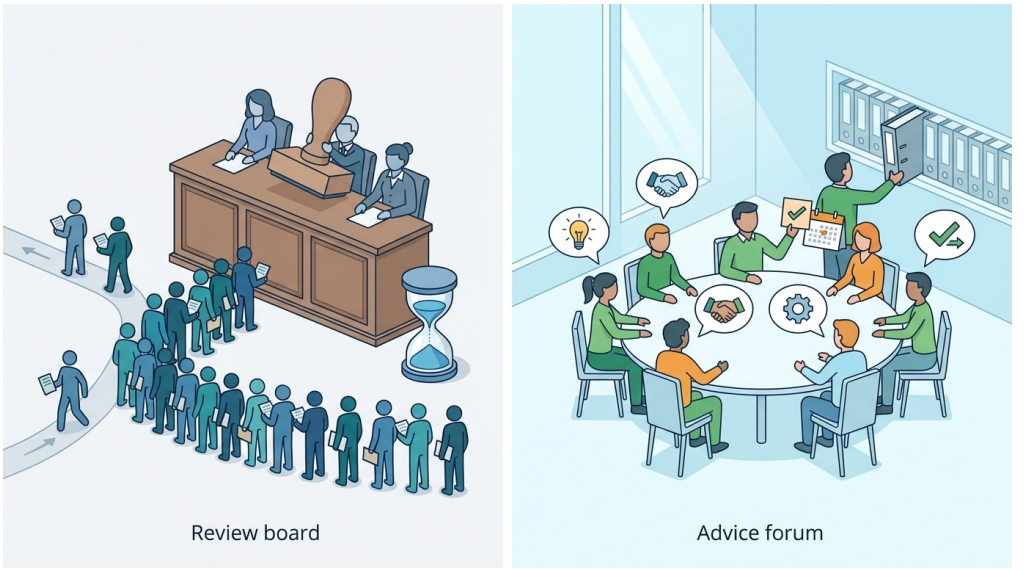


Figure 5: *An approval board becomes a queue that teams learn to route around; under the advice process the team decides after public advice, records why, and sets a date to revisit.*

Funding models

How platform work is paid for shapes behavior more than any principle. Five models are common:

Model	How it works	Behavior it produces	When it fits
Central program budget	A board-approved program funds platform teams directly	Fast start; platforms can be built before tenants pay	The build phase of a consolidation program
Fixed allocation ("platform tax")	Platform cost split by a fixed key, such as revenue share	Predictable; no incentive for tenants to use the platform efficiently or for the platform to serve them well	Stable, uncontested infrastructure

Model	How it works	Behavior it produces	When it fits
Showback	Usage-based cost reported per tenant, not charged	Visibility without budget fights; a rehearsal for chargeback	The first year after a platform goes live
Chargeback by usage	Tenants pay per unit (per thousand searches, per million events, per stored image)	Efficient use; risk that tenants avoid a platform to save budget, or build shadow copies	Mature platforms with stable unit costs
Internal market	Tenants may buy outside if the platform is not competitive	Strong pressure on the platform	Rarely right for capabilities that must be shared for policy reasons, such as trust and safety or identity

The trap is to pick one model for all platforms. Trust and safety and identity are shared because the group needs one answer for fraud and for login; charging per decision would tempt a brand to skip checks. Search and media are shared for cost; usage pricing makes their cost visible and fair.

Cost attribution needs an ownership catalog before it needs a pricing model. Adevinta's cloud governance team stated the principle for accounts and infrastructure in a December 2024 post as "You build it, you run it (and you pay for it)", backed by an ownership catalog mapping more than a thousand cloud accounts and projects to teams and cost centers (Adevinta Tech Blog, 2024).

The roles of architects

A marketplace group needs fewer architects than it thinks, placed where decisions are made.

Role	Owns	Does not own
Group Chief Architect	The principles, the decision-rights matrix, the forum, the capability and context maps, the consolidation sequence, and the architectural case for funding shifts	Brand designs; platform internals; any approval gate
Brand lead architect (one per brand)	The brand's target architecture and migration plan; the brand's voice in group decisions	Group platforms' contracts
Platform architect (for large platforms)	Contract design, tenancy model, evolution and deprecation of the platform	Brand product decisions
Principal engineers in teams	Most decision records; the code that proves or disproves them	Group-level trade-offs alone

A 2018 post on the Berlin Tech Blog, shared by the eBay Kleinanzeigen and mobile.de engineering teams, stated the rule I would put on the wall: "A Software Architect should never make a decision that the team can effectively make on their own", and described the architect's job as replicating success stories across teams and finding the weakest link (Berlin Tech Blog, 2018). That is old evidence, from a period before the brands' later replatforming, but it describes the role rather than a system, and it has aged well.

In Practice: What the Public Record Shows

Few marketplace groups publish their organization designs; each dated fragment below is evidence about one organization at one time.

Marktplaats: from business-unit teams to Journey Teams (2017). In a November 2017 post, Marktplaats described replacing product teams aligned to business units, each product owner with a "huge backlog", by cross-functional **Journey Teams** organized around the customer journey, which included product, technology, marketing, customer relations, product analytics and data science. The company moved to objectives and key results, with at most three company objectives and four key results each, refreshed quarterly. The head of technology wrote that in this way of working "the platform

has been changed from a set of monolithic services to microservices, allowing teams to deliver 10x faster each year" — a self-reported claim (Marktplaats Technology, 2017). The post is nine years old, and the current organization is not public. It is still the clearest classifieds account of cutting teams by journey rather than by business unit, and of the architecture following the teams.

Vinted: domains first, then teams. In January 2026, Vinted described how a few engineers introduced domain-driven design to map responsibilities in its Rails monolith: "It took us at least two years to understand the domains, as we identified almost 300 of them." The domains "gave teams clarity about what they owned" and "eventually, it guided how we restructured teams and how we planned the future of the architecture" (Vinted Engineering, 2026). The order is the lesson: the domain map came before the team restructure, and both came before the services.

AutoScout24: growing the new organization one team at a time. A 2016 conference report described AutoScout24's move from a .NET monolith to microservices on AWS, "Project Tatsu", which began in November 2014 with one team and grew to eight, adding teams every two to three months, with no big upfront plan (InfoQ, 2016). The evidence is ten years old; it shows a migration staffed by growing the new organization rather than reassigning the old one in one move.

OLX: squads and chapters, then a framework for cross-functional work. A 2021 post from OLX Autos, a business OLX has since closed, described squads owning end-to-end journeys and chapters (Android, backend, quality) led by chapter leads who also line-managed (OLX Tech, 2021). In March 2026, OLX Motors described its "Work Organization Framework", introduced because "by 2024, at OLX Motors, we had a classic case of 'too much of everything, not enough of anything' ": a governance model with champions, drivers and outcome tracking that completed 29 initiatives, including standardized user identification across 71 services and one deployment-notification center across more than 60 services (OLX Tech, 2026). Two lessons: journey squads alone do not deliver cross-cutting hygiene; and a lightweight, outcome-tracked structure for cross-team initiatives is a governance tool in its own right.

A third OLX fragment concerns ownership: a July 2025 post explains that recommendations serving moved out of "a shared backend service" to the data team "to gain greater autonomy and flexibility" (OLX Tech, 2025) — a complicated subsystem taking ownership of its own runtime, with the efficiency trade-offs the post describes.

Adevinta: voluntary platforms, harvested inventions, and a swing back. Adevinta's record covers the full pendulum and is told in Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform. For organization design, three fragments matter. The deployment abstraction that ran across its marketplaces, FIAAS, came from FINN.no, and its experimentation analysis package, Fisher, came from Marktplaats; both were local inventions promoted to group level (2017–2023 sources). In 2023 the group "redesigned the operating model and created a robust structure organised by verticals", and in 2025 it "evolved from a centralised structure to a decentralised, standalone

operating model", with "the ramp-down of central functions, the transfer of key capabilities, the localisation of intellectual property" (Adevinta Annual Report 2023; Strategic Report 2025). In a September 2026 post about an infrastructure self-service bot, leboncoin's engineering blog described about ten infrastructure engineers supporting about 70 feature teams (leboncoin tech blog, 2026; figures from the blog's feed summary, the full post not retrieved).

> **INFERENCE, NOT DOCUMENTED:** Read as Conway's law in reverse, Adevinta's 2023 reorganization by verticals preceded its plan to converge the leboncoin and Kleinanzeigen product platforms, and its 2025 decentralization preceded the localization of intellectual property and the parallel running of central and local services. In both directions the organization moved first and the platforms were expected to follow. The public record does not describe the internal team designs, so this is the author's reading of the sequence, not a description of Adevinta's organization.

eBay Classifieds Group: a federation with group functions. The public traces of eCG's organization, mostly job postings from 2009–2011 and a 2018 conference talk, show national businesses with their own platforms, group roles asked to "leverage existing technology components" across the portfolio, and a "Global Motors vertical" with a shared partner API by 2018–2020 (see eBay Classifieds Group: Shared Platforms, Local Brands and a Sale). It is a federation that coordinated choices — the eCG companies jointly evaluated replacements for eBay's Voyager search engine in 2015 — while each brand migrated itself.

The common thread is modest: **journey-aligned teams, domain maps before reorganizations, platforms that grew by adoption, and specialists owning their runtime.** No classifieds group has published a complete topology, and anyone who claims to know one should be asked for the source.

Veldmark Worked Example: The Common Ground Organization Design

Three options

In May 2027 Ingrid and Yara Mendes, the Head of Platform Engineering, bring three organization options to the advice forum.

Option	Description	For	Against
A. Federated guilds	Platforms built by brand teams "on behalf of the group", coordinated by cross-brand guilds	No reorganization; brand budgets keep control	Nobody owns a platform; the strongest brand's version wins by default; the 2027 overlaps exist because this is the current model
B. Central platform organization, product-run	Group platform teams under the Head of Platform Engineering, each with a product owner and tenant-facing contract; brands keep journeys	One owner per shared capability; fleet-level economics	Brands lose engineers and some control; risk of the Forced Fit if platforms are slow
C. Central product and technology	One group organization owning all products and platforms, with brand "market teams" for local needs	Maximum reuse	Breaks "share platforms, not products"; the 2023-style convergence bet; expensive to reverse if a brand is sold

The forum's advice is option B, with conditions. Mateus Carvalho decides it on **Tuesday 22 June 2027**, effective **1 September 2027**, in the decision record "Common Ground organization v1". It has five parts:

1. **One owning team per shared capability**. The group platform organization, under Yara Mendes, owns the Common Ground platforms: event backbone, search platform, identity (Veldmark ID), trust-and-safety pipeline engineering, media, messaging, catalog and entitlements, taxonomy and attributes, the listing platform (publication pipeline and consumer registry) and the developer platform. Data and ML platforms stay in Priya Raman's Data & ML organization, which follows the same platform rules.
2. **Brands keep their journeys, products and teams**. Stream-aligned teams stay in Tradeyard, Motorvane and Hearthly, cut by journey. Tradeyard's Polish and Portuguese country teams become journey teams serving all four Tradeyard

countries, with market product leads per country; Tradeyard Poland keeps one market team for its vehicles and property sections.

3. **Policy stays local, mechanism goes central.** Aisha Rahman, Group Head of Trust & Safety, owns group policy standards; each brand's trust operations own local policy and moderation; the platform team owns the pipeline, rule engine and case tooling. This is the split the capability map recommended.
4. **People move with work, not with a date.** Engineers move to a platform team when their capability moves, not on 1 September. About **150** engineers transfer from brands in 2027 and about **80** are hired, giving the platform organization about **230** engineers in **24** teams by the end of 2027.
5. **Keybridge stays outside.** Keybridge's engineering organization is not merged into the platform organization and is not a tenant of the shared trust-and-safety platform or the listing event stream; group security standards apply. The March 2027 neutrality decision makes this a structural choice, not a postponement.

Measure	March 2027	December 2027
Engineers	about 1,150	about 1,200
Teams	158	152
Common Ground capabilities with exactly one owning team	0 of 9	8 of 9 (the data platform migration is still split with the brands)
Teams whose changes ship only with a Yardstick deploy	19	14
Stream-aligned teams cut by country	17	4
Teams with more than three owned contexts or capabilities (cognitive-load flag)	23	9

The platform contract per team

Each platform team publishes a one-page **team API**: what it offers (contracts, service levels), how to get help, its roadmap for the next two quarters, and its current interaction mode with each tenant. For the second half of 2027 the search platform's page reads, in part: *collaboration* with Tradeyard Portugal (index design, until the Q2 2028 migration), *facilitation* with Motorvane (moving to Listing Search API v1 on Elasticsearch), *X-as-a-Service* with nobody yet. **The page makes visible that the platform has no steady-state tenant yet**, which is the honest state of a platform in its first year.

Collaboration is time-boxed to two quarters per tenant. When the box runs out, the forum asks one question: what is missing from the contract that still requires the two teams to work together?

Funding

The decision record sets funding per phase:

- **2027–2028: Common Ground funds platform build.** The €140 million program pays for platform teams' build work and for the enabling teams.
- **From Q1 2028: showback.** Each platform publishes monthly cost per tenant with a unit: per thousand searches, per million events, per thousand stored images, per moderation decision.
- **From 2029: usage allocation for cost-driven platforms** (search, media, event backbone, developer platform), and **fixed allocation for policy-driven platforms** (identity, trust and safety), so that no brand saves money by skipping a fraud check.

Olivier Petit, VP Monetization, adds one condition that turns out to matter: the catalog and entitlement service, Tally, is funded centrally through 2029, because its first tenants (Motorvane in Q1 2028, Tradeyard in Q2 2028) are migrating revenue-critical billing and should not also be negotiating its price.

The architecture advice forum, one year in

Veldmark set up the forum in April 2027, together with its architecture principles, and made decision records mandatory for cross-brand decisions. It meets weekly, for an hour, and anyone may attend; about 60 people do on a typical week. Its first year can be read from its decision log.

Month (2027)	Decision records the forum advised on, as told in other chapters
April	Architecture principles; decision-rights matrix
May	Common Ground context map (Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models); data platform design (Data Platform for Marketplaces: Events, Lakehouse and Data Products); Tradeyard web layer (Found by Google: Web, App and SEO Architecture for Listings)

Month (2027)	Decision records the forum advised on, as told in other chapters
June	Ranking policy v1 (Ranking for Two Sides: Relevance, Freshness and Paid Visibility); Veldmark ID (Who Is on the Other Side? Identity and Verification); Common Ground organization v1 (this chapter)
July	Listing event contract v1 live (Every Change Is an Event: Event-Driven Listing Architecture)
September	Supply Contract v1 (Many Doors In: Forms, APIs and Bulk Feeds); search platform decision (Search Architecture for Listings: Indices, Freshness and Scale)
October	Group taxonomy release and bikes split (Category Trees That Can Change: Designing Marketplace Taxonomies); Tally (Packages, Bumps and Quotas: Products, Pricing and Entitlements)
November	ML platform decision (Models in Production: ML Platforms and Use Cases); shared trust-and-safety platform live for Tradeyard NL and BE

By 31 December 2027 the log holds **74 cross-brand decision records**. The median time from first draft to decision is **nine working days**; **six** went to the CTO because they moved funding or people, and **two** to the data protection officer because they moved personal data across brands. Three were superseded within the year, which Ingrid counts as a sign of health: the forum changes its mind when facts change, and the records say why.

Veldmark's principles fit on one page. Four of the twelve do most of the work in the log:

Principle	Test it implies	A 2027 decision it shaped
Share platforms, not products	Is the proposal a capability beneath several products, or a product feature?	Ranking policy: shared ranking service, brand-owned objective
One owner per context and capability	Can the proposer name the single owning team?	Removal propagation to alerts given to the Alerts team
Contracts before migrations	Does a published contract exist before any tenant moves?	Every application on Listing Search API v1 before its engine changes

Principle	Test it implies	A 2027 decision it shaped
Separable by tenant	Can a brand leave this capability within 18 months with its data?	Per-brand topics and schemas on the event backbone

The fourth principle comes from Veldmark's ownership: a private-equity owner might one day sell a brand, and Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform shows what unwinding a shared platform costs when separability was not designed in.

The phishing rule, replayed

In November 2027 the shared trust-and-safety platform goes live for Tradeyard Netherlands and Belgium. The forum's acceptance test is the March incident. A rule written by Tradeyard's trust operations team in Lisbon is a configuration change in the platform, reviewed in the platform's rule workflow, and effective for every tenant that subscribes to the rule family within an hour, without any Yardstick deploy. The phone number of a terminated fraud account enters the shared hashed block list, which the data protection officer approved with a two-year retention, so Motorvane sees the match at its next account check. Poland and Portugal are not tenants yet; their move is on the platform's roadmap, and the gap is listed on the team API page rather than hidden.

The rule engine did not change. **The ownership did**: there is now one team that owns "a rule reaches every tenant", and its service level is written down.

Trade-offs

Decision	Option	When it is right	When it is wrong
Stream-aligned axis	By journey within a brand	One product in several countries; strong configuration	Countries with genuinely different products, or a brand with one country

Decision	Option	When it is right	When it is wrong
Stream-aligned axis	By country	Distinct local products, regulation or ownership; a brand that may be sold separately per country	One product in several countries: it rebuilds the Copy-Paste Country
Shared capability ownership	Central platform team	Uniform technology, pooled volume, group-wide policy need	Capability is really a product feature; only one brand uses it
Shared capability ownership	"Lead brand" builds for the group	Early discovery, one dominant tenant	Long term: the lead brand's roadmap always wins
Specialists	Central complicated-subsystem team	Rare skills (ranking ML, AVMs, fraud models), shared tooling	The model is inseparable from one brand's product decisions; embed instead
Governance	Advice process and open forum	Many autonomous teams; decisions must be fast and visible	Regulated decisions that require formal sign-off (handled by named executives, not the forum)
Governance	Architecture review board with approval	Small organizations with few architects; safety-critical systems	Multi-brand groups: it becomes a queue and teams route around it
Funding	Central program budget	Build phase	Run phase: hides cost, removes pressure to be efficient

Decision	Option	When it is right	When it is wrong
Funding	Usage chargeback	Mature, cost-driven platforms	Policy-driven platforms (trust, identity) where skipping use is the risk

Vertical Differences

Topic	General classifieds	Real estate	Automotive	Professional SaaS (agent or dealer tools)
Dominant stream-aligned teams	Private-seller supply, discovery, conversation and transaction	Agency supply and feeds, discovery, new-build campaigns, valuation products	Dealer supply and feeds, discovery, leads, financing	Product modules of the customer's workflow
Specialist teams	Fraud and scam models, category prediction, image models	AVM and price estimates; entity resolution for multi-agency listings	Market-value models, VIN decoding, lead scoring	Integrations with portals and back-office systems
Sales and account management	Small; pro sellers mostly self-serve	Large; agency relationships drive revenue	Large; dealer contracts and packages	Customer success is the product

Topic	General classifieds	Real estate	Automotive	Professional SaaS (agent or dealer tools)
Local-team pressure	Low to medium: one product, many countries	High: national legal regimes, mandates, energy labels, fee rules	Medium: national registries and dealer markets	High: per-country agency practices
Fit with group platforms	Strong for most capabilities	Strong for infrastructure, media and events; weaker for supply, which is vendor-driven	Strong; may stay on its own search engine for a while	Often separate, as Keybridge is, for neutrality or because it is not a marketplace

Two consequences for organization design. In real estate and automotive, the **supply journey has a commercial side** — agency and dealer contracts, feeds, packages — so the supply team needs close links to sales and account management, which a central platform cannot provide. And a SaaS business that serves the marketplaces' competitors, like Keybridge, needs an organization boundary that customers can see, not only an access control list.

Failure Modes

- **The platform without a product owner**. A team builds what the loudest brand asks for, has no roadmap that tenants can read, and measures itself by tickets closed. Tenants start building around it within a year. The fix is a product owner, a team API page and a tenant review each quarter.
- **The mandate before the platform**. A deadline for all brands to move onto a platform that is not yet cheaper, faster or safer than what they have. The brands comply on paper and keep their copies running. This is the Forced Fit; it often follows a board-level synergy target.
- **Reorganizing on a date**. Hundreds of engineers move to new teams on the first of the month, before contexts are mapped and before the systems they will own are separable. For months the new teams own code they cannot deploy independently. Move people when their work moves.

- **Country teams for language**. A brand creates a full-stack team per country because the countries speak different languages. Within two years the four teams have four insertion flows.
- **The review board**. A central board approves designs. Its queue becomes the critical path for every cross-brand change, and teams learn to describe changes as local. Decisions get slower and less visible at the same time.
- **Governance without records**. Decisions are made in meetings and remembered by the people who attended. When those people leave, the same questions return, and inherited constraints become folklore.
- **Specialists with no runtime**. A ranking or valuation team that hands models to another team to deploy cannot own model quality in production, and nobody else can either. This is how Set-and-Forget AI begins (see Models in Production: ML Platforms and Use Cases).
- **Enabling teams that never end**. A migration team becomes a permanent second owner of the systems it migrated. Give each enabling team an end condition at creation.
- **The forgotten acquired organization**. An acquired marketplace keeps its engineering organization intact "until integration", which never comes. Its people are never asked to own a group capability, and its best inventions never become one.

The Chief Architect's View

Design ownership first, and draw it on the same map as the architecture. The capability map, the context map and the team topology are three views of one model. If a capability or context has no owning team on the map, it has none in reality. I would review the three together twice a year and treat any orphan or any split ownership as an architectural defect with a named fix and date.

Measure platforms by adoption and by tenant cost, not by output. The questions I ask of every platform team each quarter: how many tenants use it in production; what it costs a new tenant to onboard; how many brands still run a shadow copy and why; and what share of its roadmap came from tenants. A platform that cannot answer these is a shared-services department.

Keep the forum open and the executive decisions few. The advice forum works because it is open and because most decisions stay with teams. The executive escalation path must exist — money, people and personal data need an accountable decision-maker — but if more than about one decision in ten escalates, the decision rights are wrong, not the teams.

Expect the pendulum. Groups centralize for synergy and decentralize for focus, often after a change of owner or strategy. The organization will swing; the architecture should not have to be rebuilt each time. That is why separability by tenant belongs

in the principles from the start, and why platform contracts, not team structures, are the durable assets. Built to Be Replaced: Evolving a Marketplace Platform Over a Decade returns to this.

Who decides, and what to watch. The CTO decides the topology and funding, on the forum's advice; the Group Chief Architect owns the principles, the decision-rights matrix and the maps; platform heads own contracts; brand leaders own journeys. Watch: orphaned or split capabilities; teams over the cognitive-load threshold; teams blocked by another team's deploy (Yardstick's count is the headline number at Veldmark); median time from decision draft to decision; decisions with an expired review date; and shadow copies of platform capabilities in brands.

Questions to Ask

1. For each shared capability, which single team is woken up when it breaks, and who decides its roadmap?
2. Which bounded contexts or journeys are owned by more than one team, and which by none?
3. Are stream-aligned teams cut by journey, country or technical layer, and what product difference justifies each country team?
4. What does it cost a brand, in engineering weeks, to adopt each platform, and who pays for that migration?
5. Which platforms are funded by program budget, allocation, showback or charge-back, and does each model fit why the capability is shared?
6. Where do ranking, valuation and fraud specialists sit, and do they own their models in production?
7. Who may decide an exception to a principle, and where is the record?
8. How many cross-brand decisions were made last quarter, how long did they take, and how many escalated?
9. Which decisions are past their review date or trigger?
10. If a brand were sold in 18 months, which teams and platforms would have to split, and is that written down?

From Organization to Migration

The organization assigns owners; it does not move code. Tradeyard's teams still deploy through Yardstick, and the new platform teams own capabilities that the monolith still implements. Changing the Engine in Flight: Modernizing a Live Marketplace takes up the work of moving a live marketplace from the old system to the new one without losing traffic, sellers or revenue.

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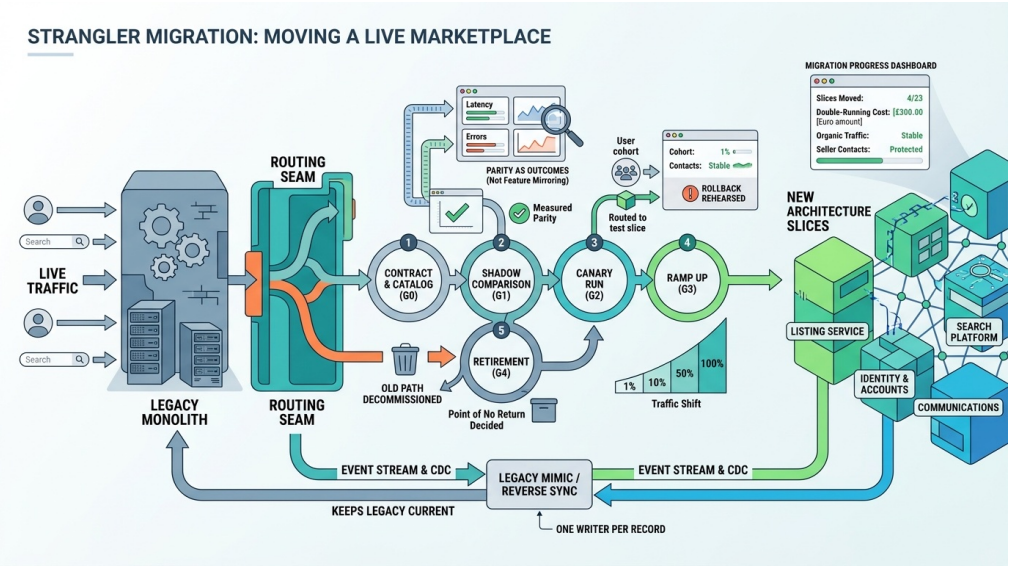
Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

43. Changing the Engine in Flight: Modernizing a Live Marketplace



IN THIS CHAPTER, YOU WILL: Plan the retirement of a marketplace monolith while it keeps serving millions of buyers and sellers. You will choose slices and their order, put a routing seam in front of the old system, intercept its events, run old and new in parallel, migrate listings and accounts, move search with measured relevance parity, keep URLs stable for search engines, and decide when rollback stops being possible. The worked example is Veldmark's plan to retire Tradeyard's PHP monolith, Yardstick, by the end of 2029.

WHY THE BUSINESS CARES: A marketplace cannot close for a weekend. A migration that drops organic traffic by a third, loses dealers' feeds, mis-bills paid listings or confuses private sellers for a month costs more than the old system ever did. The business case for modernization is only as good as the plan for not losing liquidity during it.

WHY ARCHITECTS SHOULD CARE: Legacy marketplaces hide behavior in places nobody documented: cron jobs, admin tools, templates and search

configuration. A migration surfaces all of it, one surprise at a time. The architecture of the migration — seams, transitional components, parity tests and rollback — decides whether those surprises are caught in a shadow comparison or by sellers.

KEY POINTS:

- **Migrate by slice behind a seam, never by rewrite.** Put routing in front of the monolith (at the edge, in the backends-for-frontends, in the event stream), move one slice at a time, and keep the old path warm until the new one has proven itself. Read paths first, write paths later, the system of record last.
- **Parity means equal outcomes, not equal features.** A new search engine will never return the same results; a new insertion flow should not copy every legacy quirk. Define parity per slice as measured outcomes — contacts per search session, publication outcomes, billing, organic traffic — compared in shadow and canary runs, and catalog legacy behavior deliberately, retiring what nobody needs.
- **Every slice passes stage gates, and every gate has a rollback answer.** Contract, shadow, canary, ramp and retire are separate decisions with exit criteria. Rollback is cheap while the old system stays current through reverse synchronization; the point at which it stops being possible must be a decision, recorded in advance, not a discovery.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Search as System of Record** — a search migration is where a group learns whether its index can be rebuilt from anything other than itself; if it cannot, the migration has to build that source first.
- **Ghost Inventory** — while two systems run side by side, every removal must reach both, and every copy that listens to only one of them keeps serving listings that are gone.

Part VIII's agenda turns here from decisions to code. After the Deal: Consolidating Platforms After Acquisitions chose what Veldmark shares and Who Owns What: Organizing Marketplace Engineering gave each capability an owner, but Tradeyard's teams still ship through Yardstick, the monolith that launched the site in 2006. It still holds the listing table, renders the pages that search engines index, runs the jobs that send alerts and bills the bumps. It is also the reason a fraud rule takes three days to reach another country and why 140 engineers wait for the same deploy. **Modernization is not optional for such a marketplace; the only choice is how much liquidity it puts at risk.**

The industry has a settled answer to the first-order question. Martin Fowler's **strangler fig** describes gradual replacement that "begins with small additions, often new features", and argues that "the reduced risk and earlier value from the gradual approach outweigh its costs" (Fowler, page dated 2024). The hard questions are the second-order ones: which slice first, how to run two systems on one dataset, how to know a new search engine is good enough, how to keep 40 million URLs from breaking, and when to stop being able to go back.

Figure 1: *Modernization by strangler: new components grow around the monolith one slice at a time while the marketplace keeps trading, and the old path stays warm until each new slice has proven itself.*

This chapter owns three terms. A **migration stage gate** is a decision point in moving one slice of a system, with measured exit criteria and a defined rollback, that must be passed before more traffic or responsibility moves. **Relevance parity** is the condition that a new search path produces outcomes for buyers and sellers that are not worse than the old path's, within agreed tolerances, measured on real queries; it is not identical results. **URL migration** is the controlled change of public URLs, or the deliberate decision not to change them, with mappings, redirects, sitemaps and monitoring that preserve search-engine signals. The event mechanisms this chapter uses belong to Every Change Is an Event: Event-Driven Listing Architecture, the search platform to Search Architecture for Listings: Indices, Freshness and Scale, and the URL and redirect services to Found by Google: Web, App and SEO Architecture for Listings. Earlier chapters already assume the plan's dates, such as the alert cut-over in Tell Me When It Appears: Saved Searches and Alerts, and Built to Be Replaced: Evolving a Marketplace Platform Over a Decade asks how the platform that replaces Yardstick avoids becoming another one.

Setting the Scene at Veldmark: Yardstick in 2027

Yardstick is the core of Tradeyard, the horizontal classifieds brand of Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build). It is a PHP monolith written from 2006 onwards, running on MySQL in a co-located data center in Amsterdam. Since 2019 about 40 microservices have grown around it on AWS, but Yardstick still owns the listing table, insertion, most page rendering, alerts, the pro-seller tools, the admin back office and the integration with Tradeyard Safe Deal. It serves Tradeyard's 13.2 million live listings in four countries, about 2.3 million new listings a week and 31 million monthly users.

In July 2027 Daan Visser, Tradeyard's engineering director, who owns Yardstick, and Ingrid Halvorsen, the Group Chief Architect, set up the **Yardstick Exit** enabling team to produce the retirement plan. Its inventory, presented to the architecture advice forum in September 2027, is the starting point for everything that follows.

Inventory item (September 2027)	Count	Why it matters
Level-2 business capabilities implemented by Yardstick (from the March capability map)	23	Each is a slice that must move or be retired
Columns in the `ads` table	214, read by 38 Yardstick modules	No module can move without deciding which columns it owns
Scheduled jobs (cron)	47, of which 9 had no known owner	Hidden behavior; several write listing state
Admin back-office actions	312	Moderators, support and sales depend on them daily
Mobile app endpoints served directly by Yardstick	118	Old app versions keep calling them for months after any change
Systems reading Yardstick's MySQL tables or raw change stream directly	14 (down from 26 in March, after Listing event contract v1)	Each is a veto on changing the schema

Inventory item (September 2027)	Count	Why it matters
Teams whose changes ship only with a Yardstick deploy	16 (19 in March)	The organizational cost, two deploys a week
Hard deadline	The Amsterdam colocation contract ends 31 December 2029	The board's date for decommissioning is a contract date, not an aspiration

The board's Common Ground program fixed three milestones in January 2027: the **insertion flow moves out in Q2 2028**, **search moves out in Q4 2028**, and **Yardstick is decommissioned at the end of 2029**. The inventory turns those dates into a question that this chapter answers: *what moves in which order, how do we know each move worked, and how do we go back if it did not?*

Some groundwork already exists. Since 1 July 2027, Tradeyard publishes Listing event contract v1 through **Yardstick Relay**, a translator that reads Yardstick's tables by change data capture (CDC) and emits domain events (see Every Change Is an Event: Event-Driven Listing Architecture). Since May 2027, the architecture advice forum's web-layer decision puts server-rendered page applications behind CloudFront route rules, with Yardstick rendering any page type that has not moved (see Found by Google: Web, App and SEO Architecture for Listings). Two seams are in place. Neither moves a single write.

Concepts: The Vocabulary of Displacement

Re-architect, re-platform or rewrite

Three strategies are often confused:

Strategy	What changes	Typical duration and risk	Documented classifieds example
Rewrite (big bang)	A new system replaces the old on a cut-over date	Long build with no value until the end; parity defined by the old system's undocumented behavior	None documented as successful in the research for this book

Strategy	What changes	Typical duration and risk	Documented classifieds example
Re-platform ("move first, then optimize")	Runtime and hosting change; code changes as little as possible	Fast; deadline-friendly; leaves the architecture as it was	ImmobilienScout24's move from its data center to AWS, begun in 2018 under the motto "Erst umziehen, dann optimieren" (Computerwoche, 2019)
Re-architect incrementally (strangler)	The system is replaced slice by slice behind seams	Years; value from the first slice; complexity of running two systems	AutoScout24's move from a .NET monolith to microservices on AWS, begun in November 2014; the same 2019 press account describes it as successful but "very time-intensive"

Scout24 ran both, a decade apart, which makes the contrast unusually clean. My reading of the evidence: **re-architect when the runtime or the data model blocks the target; re-platform first when the code already fits the target runtime and a contract deadline dominates**. Yardstick has both problems — a co-located data center with a fixed end date, and a 214-column table that blocks every bounded context — which is why its plan combines a strangler for the core with a lift of whatever is left in 2029.

Seams and transitional architecture

A **seam** is a place where traffic or data can be diverted from the old system to the new one without either knowing much about the other. In a listing marketplace there are four natural seams:

Seam	What it diverts	Mechanism
Edge routing	Page requests by URL pattern, country or cohort	CDN rules, as Subito documented in 2023 for its Next.js microsites, with routing configured on Akamai and ownership set per page (Adevinta Tech Blog, 2023)
Backends-for-frontends (BFFs)	App and web API calls by endpoint or user	The BFF decides which backend serves a call
Event interception	State changes, routed to new consumers	CDC or an outbox; the new component listens instead of the old one being called
Branch by abstraction	Calls inside the code base	An abstraction layer that allows "multiple implementations to co-exist" while the change is in progress (Fowler, 2014)

In their 2024 catalog of legacy displacement patterns, Ian Cartwright, Rob Horn and James Lewis name several of the components this chapter uses: **Event Interception** (intercepting updates to system state and routing some of them to a new component), **Legacy Mimic** (letting the new system interact with the old one so that the old system is unaware of the change), **Transitional Architecture** (software added only to ease displacement and removed afterwards) and **Feature Parity** (replicating legacy functionality on a new stack), which they present as a common cause of failure: "even defining and agreeing current 'as is' functionality was a huge effort" (Cartwright, Horn and Lewis, martinfowler.com, 2024).

Yardstick Relay is transitional architecture: it exists to be switched off. So is every reverse-synchronization job in this chapter. **Transitional components need owners, budgets and end dates like any other system**, and they need them more, because nobody wants to own them.

Parallel run, shadow traffic and canary

Three ways of running old and new side by side differ in what the new system is allowed to do:

Mode	New system receives	New system's output	Use
Shadow (dark launch)	A copy of real requests or events	Discarded after comparison	Find differences without user impact
Parallel run	Real requests; both systems compute	One system's output is authoritative; the other is compared	Prove equivalence of computed outcomes, such as billing
Canary	A small share of real traffic	Served to users	Find problems that only appear with real users and real side effects

Shadow traffic has a well-known weakness in listings: the data changes underneath it. In a 2015 account, Marktplaats's search team chose to **replay recorded user queries** against indices built from a copy of the production database, in small random batches every five minutes, rather than mirroring live traffic, because a live index that changes during the comparison makes differences impossible to reproduce (Marktplaats Technology, 2015, republished 2017). **Reproducibility beats realism** for the first comparisons; live shadow traffic comes after.

The migration stage gate

A slice moves through five gates. The names are Veldmark's; the shape is general.

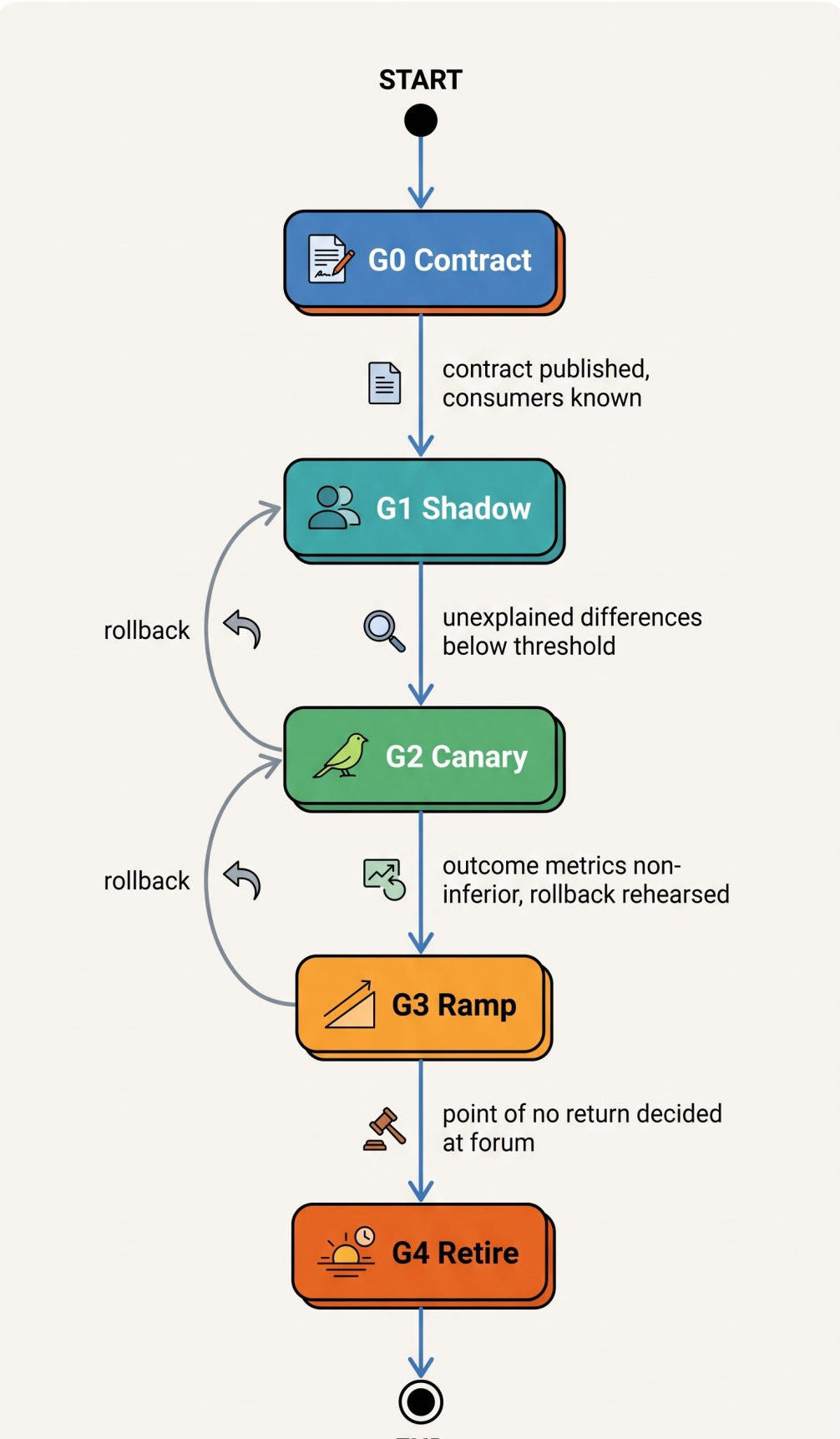

START
G0 Contract
contract published, consumers known
G1 Shadow
rollback
unexplained differences below threshold
G2 Canary
rollback
outcome metrics non-inferior, rollback rehearsed
G3 Ramp
point of no return decided at forum
G4 Retire

Figure 2: *The five stage gates for one migration slice. Rollback is a normal transition until G3; the move to G4 is the recorded decision that the old path will no longer be kept current.*

Gate	Exit criteria (examples)	Rollback
G0 Contract	The new slice's API or event contract is published; every consumer of the old behavior is listed; the behavior catalog is reviewed	Nothing to roll back
G1 Shadow	Shadow or replay comparison runs on real inputs; unexplained differences below an agreed rate; performance within budget	Switch off the shadow
G2 Canary	A small cohort is served by the new path; outcome metrics non-inferior; rollback rehearsed with real data	Route the cohort back; reverse sync keeps the old system current
G3 Ramp	Majority of traffic on the new path; old path warm; reconciliation clean for a defined period	Route back, at the cost of losing features only the new path has
G4 Retire	Old path removed; data archived under the retention schedule; transitional components deleted	None: re-building the old path is a new project

Two rules make the gates useful. **Exit criteria are numbers agreed before the gate, not after.** And **the move from G3 to G4 is a decision with an owner**, because it is the moment rollback stops being cheap.

Architecture: Running Two Systems on One Marketplace

The shape of a strangler for a listing site

Figure 3 shows the transitional architecture for Tradeyard in mid-2028, after the insertion flow has moved and before search has. It generalizes to any listing monolith.

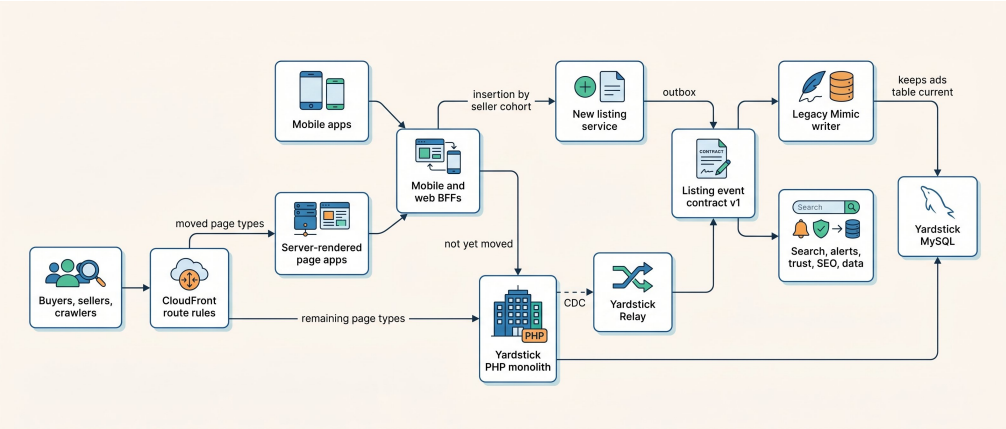


Figure 3: *Tradeyard's transitional architecture in mid-2028. Three seams — edge routes, BFF routing and the event stream — decide which system serves what. The new listing service is the system of record for listings it created; the Legacy Mimic writer keeps Yardstick's table current so that modules still in the monolith keep working and rollback stays possible.*

Four principles follow from it.

Read paths before write paths. Pages and read APIs can be fed from read models built from the event stream while the monolith remains the only writer. Moving them first delivers visible value (performance, SEO, deploy independence) with low data risk. The platform research for this book describes this order as a reference-architecture inference for listing sites: extract read-heavy, SEO-critical page types first from CDC-fed projections, and write paths such as insertion, payments and messaging last.

One writer per listing at any moment. During the move of the insertion flow, a listing is owned either by Yardstick or by the new listing service, never by both. Ownership is decided per listing at creation (by seller cohort) and recorded on the listing. **Dual writes from application code are the classic way to corrupt two systems at once**; the event stream and a reverse writer replace them.

The old system is kept current, not kept authoritative. Modules still inside Yardstick — messaging, pro tools, admin — need to see listings created by the new service. The

Legacy Mimic writer consumes the event contract and writes those listings into the `ads` table in the shape Yardstick expects, flagged as externally owned so Yardstick's own write paths refuse to change them. This is what keeps rollback cheap until G4.

Loops are broken by version, not by time. When changes flow in both directions — Relay out of Yardstick, Mimic into it — an update can echo back and forth. Both directions carry the listing's `listing_version`, and each side ignores events whose version it already holds or whose owner is the other side.

Data migration: listings and accounts

Listings and accounts are migrated differently because listings are short-lived and accounts are not.

Listings are short-lived. Most Tradeyard listings live days or weeks, so the cheapest migration for most of them is no migration at all: new listings are created in the new service from the cut-over date of their cohort, and existing listings stay in Yardstick until they end. Only long-lived listings (pro sellers' stock, listings renewed many times) and the archive need a bulk move. A backfill reads Yardstick's tables through Relay's snapshot mode, writes into the new service, and publishes on the backfill topic, so that downstream copies are corrected rather than duplicated (see System of Record and Read Models: Listing Data Management).

Accounts are long-lived and carry identity, credentials, payment links and history. They need a per-user migration. The best public description in the research is Kleinanzeigen's. In a June 2025 post, an engineer described migrating "the whole platform to a new system" that ran in parallel with the legacy one: all user data was migrated, and users were then switched incrementally, with rollback possible; a reverse proxy routed each user's traffic by migration state; core user data moved first, then listings, transactions and messages; and, instead of dual writes, the team streamed changes asynchronously with Debezium CDC, accepting a synchronization delay of up to a minute as "a business decision". They rejected the transactional outbox because millions of user updates a day would overwhelm polling and because of the risk of missing write paths in a legacy code base. A per-user state machine (identified, eligible, migration requested, migrated, transitioned) recorded progress, and bi-directional sync loops were broken with a version field rather than timestamps. The post does not name either platform (Berlin Tech Blog, 2025). It also records a constraint every monolith owner should check early: the production MySQL cluster used statement-based binary logs without global transaction IDs, so the team added row-format replicas for CDC.

At Veldmark, accounts move with **Veldmark ID**, the group identity rolled out in 2028, so the per-user migration is the identity team's slice, not Yardstick Exit's (see Who Is on the Other Side? Identity and Verification and Private Sellers, Dealers and Agencies: Modeling Who Lists).

Search migration with relevance parity

A search migration changes the component buyers notice first. Three facts shape it:

1. **Results will differ.** Different analyzers, scoring functions and tie-breaking change the order even on identical documents. A migration that demands identical results will never finish.
2. **The index must be rebuildable from a source other than itself.** If the old index is the only complete projection of listings — Search as System of Record — the migration must first build that source, typically a replayable event stream or a compacted state topic. eBay Kleinanzeigen (now Kleinanzeigen) described in 2019 rebuilding its Elasticsearch cluster from a compacted Kafka topic in under 30 minutes (Berlin Tech Blog, 2019).
3. **Analyzers carry history.** Synonyms, stopwords, decompounding and spelling tables encode years of tuning that nobody remembers adding.

Relevance parity is therefore measured in layers:

Layer	Method	What it catches	Typical gate
Analyzer parity	A test set of real queries; tokens before and after must match	Lost synonyms, broken decompounding, accent handling	100% on the curated set; reviewed exceptions
Offline result comparison	Recorded queries replayed against frozen copies of both indices; overlap measures such as rank-biased overlap; zero-result rate	Large regressions, missing documents, broken filters	Zero-result rate within a tolerance; every large divergence explained
Shadow traffic	Live queries mirrored; latency and errors measured	Performance under real load, timeouts	Latency within budget at peak
Online experiment	Users randomized between engines	The outcome: contacts, saves, sessions with a contact	**Non-inferiority** on the primary outcome metric

The last row is the definition that matters. The question is not "are the results the same?" but "do buyers find and contact sellers at least as well?" — in classifieds, contacts are the observable success signal, as Ranking for Two Sides: Relevance, Freshness and Paid Visibility argues. Ranking experiments in two-sided markets have interference problems that Testing a Two-Sided Market: Experimentation Platforms covers; for an engine swap, user-level randomization is usually adequate because the engine does not change sellers' incentives.

Two documented migrations show the sequence. Marktplaats replaced eBay's Voyager search engine with Elasticsearch in 2015: new services behind the same interfaces, several hundred reused integration and acceptance tests, recorded-query replay, then a ramp of 1% of traffic on a Friday, 50% by the end of the following week and then 100%, with Voyager kept warm. It took two to three developers about three months with zero downtime, and many test failures came from "hidden functionality in other parts of the system that we couldn't detect in the code of the legacy services" (2015 post, republished 2017).

At Vinted, **the indexing path was modernized before the engine**, which is the order Veldmark follows. Its rebuild of the indexing pipeline, from delta jobs to CDC-fed Flink read models, cut item visibility latency from up to seven minutes to about one (Vinted Engineering, 2023). Vinted then moved from six Elasticsearch clusters to one Vespa deployment between May 2023 and April 2024, using shadow traffic first and then A/B tests of relevance, and reports search latency improved 2.5 times and change-to-visible time down from 300 seconds to 5 (Vinted Engineering, 2024).

URL migration: the inventory you cannot see

A listing marketplace's URL space is its search-engine inventory. Tradeyard has about 40 million indexed URLs at the start of 2027, and 46% of its web sessions start from organic search. Every URL the migration changes is a signal that search engines must re-learn.

Google's site-move guidance is explicit: map old URLs to new ones, use server-side permanent redirects (301 or 308) directly to the final destination, keep redirects "generally at least 1 year" and from a user's perspective indefinitely, submit sitemaps with the new URLs, and "expect temporary fluctuation in site ranking during the move", with larger sites taking longer to settle (Google Search Central, updated 2026). When eBay Kleinanzeigen became Kleinanzeigen in May 2023, the previous domain was to "remain accessible for several more years" (Adevinta, 2023).

The rule I use for platform migrations: **do not change URLs because the platform changed**. Change them only when the URL design itself is wrong, and then do it in a separate, measured step, not on the day the new system takes over. Concretely:

- **Keep listing identifiers in URLs stable.** If the new listing service issues new identifiers, keep the public numeric form, for example by continuing the legacy sequence, so that `/a/123456789` still means the same listing.

- **Route by pattern at the edge**, so a page type moves from old to new renderer with no URL change, and back again if needed.
- **Put every unavoidable change in a redirect registry** driven by domain identifiers (category, location, listing), as Found by Google: Web, App and SEO Architecture for Listings describes, and monitor crawl errors, indexed pages and organic contacts per page type during and after the move.
- **Status codes are part of parity**. A new renderer that answers `200` where the old one answered `410` for removed listings, or `404` for empty facets, changes the crawl budget, even if every page looks right to a user.

Feature parity and the behavior catalog

Feature parity is a trap because nobody knows the full feature set of a fifteen-year-old marketplace. The alternative is a **behavior catalog** per slice: every observable behavior of the old system — including those found in cron jobs, templates, admin tools and search configuration — with a decision for each: *carry*, *change* or *retire*, and an owner. The catalog is built from code reading, from logs of what actually runs, and from shadow comparisons, which find the behaviors that reading missed.

Retiring is the step most migrations skip. Every carried behavior is cost on the new platform; every behavior retired deliberately is cost removed. A behavior that is retired by accident, on the other hand, is a seller-facing incident.

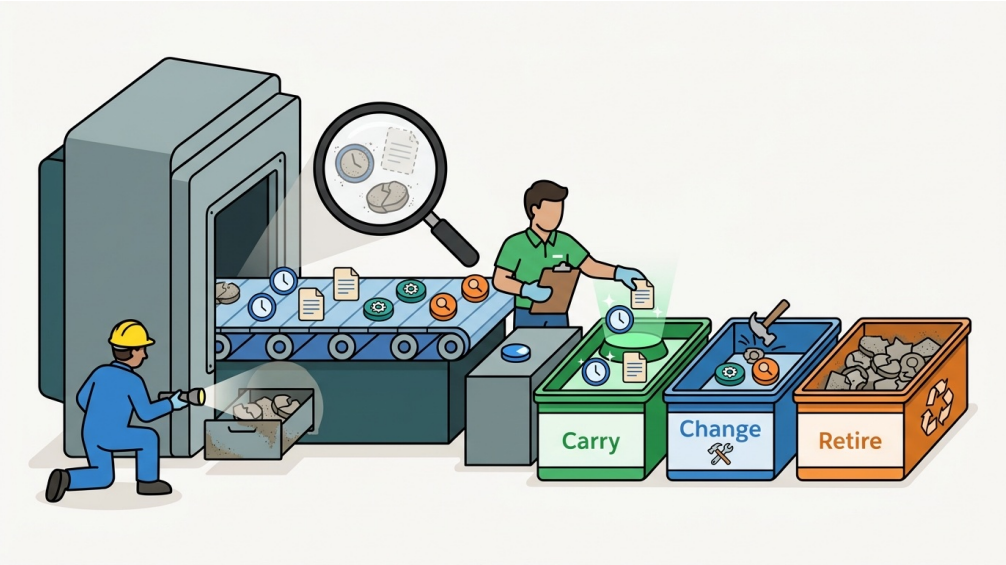


Figure 4: *A behavior catalog replaces the feature-parity trap: every observable behavior of the old system, including the ones hidden in cron jobs and admin tools, gets an owner and a decision to carry, change or retire it.*

Kill switches and rollback

Each seam needs a switch an on-call engineer can flip in minutes without a deploy:

Switch	Scope	Rollback effect
Edge route weight per page type and country	Page rendering	Pages served by the old renderer again
BFF routing flag per endpoint and seller cohort	Writes and reads through APIs	New listings for the cohort created in the old system again
Consumer switch per event-driven component	Search indexer, alerts, sitemap	Old consumer resumes; new one pauses
Search engine selection in the search contract	Queries	Old engine serves queries

A switch is only as good as the data behind it. Routing writes back to Yardstick works only if Yardstick knows about every listing the new service created, which is what the Legacy Mimic writer guarantees. **Rehearse rollback with real data at G2, and again before every ramp step**; a rollback that has not been rehearsed is a hope.

Measuring progress

Lines of code moved is the wrong measure. The right ones are outcomes of the migration itself:

Measure	What it shows
Capabilities still implemented only in the legacy system	The scope remaining (23 at Yardstick in 2027)
Share of page views, API calls and listing writes served by the new paths	Traffic actually moved
Systems still reading legacy tables directly	Hidden dependencies that block the schema
Teams whose changes still ship with a legacy deploy	The organizational benefit
Transitional components alive, with their end dates	Debt the migration created

Measure	What it shows
Business guardrails: contacts per search session, publication outcomes, organic contacts, billing reconciliation	Liquidity protected during the move
Double-running cost per month	What delay costs

Double running is real money. Adevinta's 2025 strategic report attributed a more-than-doubling of cloud-provider emissions to "parallel running of central and local services during a transitional period" of its decentralization (see Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform). Any migration plan should carry its double-running cost as a line item, so that delay has a visible price.

In Practice: Veldmark's Yardstick Exit Plan

The sequence

The Yardstick Exit team's plan, approved by the advice forum in October 2027, moves 23 capabilities in six waves. The ordering principle is read paths first, then the write path that unblocks the most teams, then search, then the long tail, and the system of record's archive last.

Wave	When	Slices	Seam
0	Jul – Dec 2027	Listing event contract v1 via Yardstick Relay; event-driven indexer on Solr; landing pages, listing detail pages and the four SEO services	Event interception; edge routes
1	Q4 2027 – Q1 2028	Alert matching in comparison mode; web and mobile BFFs in front of all 118 app endpoints	Event stream; BFF

Wave	When	Slices	Seam
2	Q2 2028	**Insertion flow** to the new listing service (outbox); Tradeyard VAS on Tally with one shadow billing month; Legacy Mimic writer	BFF by seller cohort; events
3	Q2 – Q4 2028	**Search** to the group OpenSearch platform: Portugal in Q2, then Poland, then Belgium and the Netherlands in Q4; alerts switched off in Yardstick	Search contract (Listing Search API v1)
4	2028	Accounts to Veldmark ID, per user (identity team's slice)	Per-user routing
5	H1 2029	Messaging, Safe Deal orchestration, pro-seller tools and API, admin back office (rebuilt for 168 of the 312 actions; 144 retired)	BFF; events
6	H2 2029	Archive of closed listings under the retention schedule; Relay and Mimic switched off; MySQL and the Amsterdam racks decommissioned by 31 December 2029	—

The plan deliberately leaves Yardstick running some slices until 2029. Messaging and Safe Deal are revenue- and fraud-critical, and they depend on accounts, which move with Veldmark ID during 2028. Moving them before their dependencies would mean migrating them twice.

Wave 2: the insertion flow

The insertion flow is the slice that unblocks the most teams, because publication, moderation, entitlement checks and media all start there (see From Submitted to Searchable: The Publication Pipeline). It is also a write path, so its gates are strict.

G0, contract and catalog (Q4 2027). The new listing service, built by Tradeyard's supply teams on the group developer platform, publishes the insertion API and emits Listing event contract v1 through a transactional outbox. It uses one schema per country, as the 2027 data-management decisions record (see System of Record and Read Models: Listing Data Management). The behavior catalog for insertion lists **146** behaviors. The supply teams decide to carry 82, change 23 and retire 41. Retired behaviors include a free relisting button that duplicated listings older than 30 days (a source of Seeing Double) and a Belgian title template that appended the municipality in capitals.

G1, shadow (January – March 2028). Every Yardstick submission is replayed into the new service in dry-run mode, and the two outcomes are compared on category, normalized attributes, moderation route, entitlement result, fee and published document. In the first week **6.8%** of submissions differ. Most differences are expected (changed or retired behaviors). Nine behaviors appear that the catalog missed, among them a cron job that silently converted a price of €0 into "price on request" in Portugal, and a Polish rule that waived the vehicle insertion fee for accounts created before 2015. Each gets an owner and a decision. The gate threshold — unexplained differences below **0.5%** of submissions for two consecutive weeks — is met in mid-March.

G2, canary (April 2028). Tradeyard Portugal goes first, because it is the smallest market (about 210,000 new listings a week) and has no Safe Deal. The BFF routes 1% of Portuguese sellers' new listings to the new service, then 10%. The new service's listings reach Yardstick's `ads` table through the Legacy Mimic writer within about 3 seconds p95, so messaging and admin work unchanged. Rollback is rehearsed twice with real cohorts: the BFF flag routes the cohort back, and Yardstick continues to serve their existing listings, which it already holds. Guardrails over the canary weeks: share of submissions published, time from submission to searchable, first-contact rate within 48 hours, and support tickets per thousand listings — all within tolerance.

G3, ramp (May – June 2028). Poland follows in May, Belgium in early June and the Netherlands from mid-June, each ramped by seller cohort over about three weeks: 1%, 10%, 50%, 100%. Pro sellers move last in each country, after their API clients have been tested against the new endpoints. By 30 June 2028, all new Tradeyard listings are created in the new service; Yardstick still holds the listings created before each cohort's switch, until they end.

Insertion flow, Tradeyard (2028)	Portugal	Poland	Belgium	Netherlands
New listings per week	0.21M	0.62M	0.42M	1.05M
Canary start	7 April	5 May	2 June	16 June
100% of new listings on the new service	28 April	26 May	23 June	30 June
Rollbacks during ramp	1 (rehearsal)	1 (fee rule bug, 40 minutes)	0	1 (image upload timeouts at the Sunday peak, 2 hours)

G4 for insertion is not declared in 2028. It waits until the last listing created in Yardstick has ended and the Mimic writer is no longer needed by insertion, which the plan sets for Q1 2029. Until then the forum keeps the option of routing writes back, and the point of no return is explicit: the first new feature that exists only in the new insertion flow, VAS purchase through Tally at insertion, cannot be rolled back to Yardstick without losing it. The forum accepts that loss of rollback in May 2028, for that feature only, in a recorded decision.

Wave 3: search, with relevance parity

The search slice follows the September 2027 search platform decision: every application on Listing Search API v1 before its engine changes, documents built only from the event contract, and Tradeyard Portugal first (see Search Architecture for Listings: Indices, Freshness and Scale). Yardstick's Solr index is not the source of the new one; the compacted listing state topic is, so Search as System of Record does not arise.

The parity gates for Tradeyard Portugal, set before the migration began:

Gate measure	Threshold	Result (Portugal, Q2 2028)
Analyzer test set (2,400 Portuguese queries)	All tokens match, or difference reviewed	31 differences, all reviewed; 6 synonyms restored

Gate measure	Threshold	Result (Portugal, Q2 2028)
Zero-result rate, replayed queries	Not more than 0.2 percentage points worse	3.1% ⟶ 2.9%
Median rank-biased overlap, top 10, head queries	Reported, not gated	0.71
p95 latency at the Sunday-evening peak, shadow traffic, four weeks	≤ 250 ms	180 ms
Search sessions with a contact, two-week 50/50 experiment	Non-inferior: not more than 1% relative below the old engine	8.8% vs 8.7% for Solr (difference not significant)

The overlap line shows why identical results are the wrong target: fewer than three-quarters of the top ten agreed, and buyers contacted sellers at the same rate. The relevance parity gate passed; the ramp to 100% in Portugal took one week, with Solr kept warm for four more. Poland followed on the same gates; Belgium and the Netherlands moved in Q4 2028 together with Yardstick's search extraction, when the Yardstick alert cron job also stopped sending (see Tell Me When It Appears: Saved Searches and Alerts).

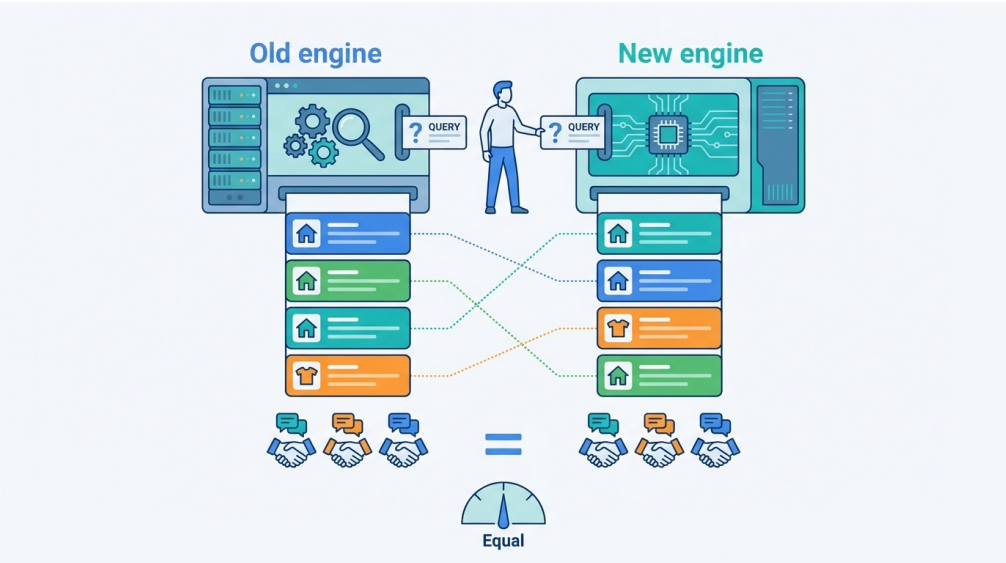


Figure 5: *Relevance parity is about outcomes, not identical results: on Tradeyard Portugal fewer than three-quarters of the top ten results matched, and buyers contacted sellers at the same rate.*

URLs: the decision not to change them

No listing URL changed in 2028. Tradeyard's listing URLs carry the numeric listing identifier, and the new listing service issues group listing identifiers (`vm:tradeyard:<country>:<id>`) whose local part continues Yardstick's numeric sequence from a reserved block, so a listing created in the new service has a URL of exactly the same shape as one created in Yardstick. Landing pages had already moved to the URL and canonical service in 2027, and the legacy search URLs that Yardstick generated before then are permanent redirects in the redirect registry, which Veldmark keeps without an end date.

During each wave the SEO team watches four numbers per page type and country: indexed pages, crawl errors, organic sessions and organic contacts. In the insertion ramp none moved outside normal weekly variation.

Progress, end of 2028

Measure	Sep 2027	Dec 2028
Capabilities implemented only in Yardstick	23	6
Share of Tradeyard page views served by new page apps	31%	88%
Share of new listings created outside Yardstick	0%	100%
Systems reading Yardstick tables directly	14	3 (all transitional)
Teams whose changes ship only with a Yardstick deploy	16	5
Yardstick deploys per week	2	1
Transitional components alive	2 (Relay, CDC connectors)	3 (Relay, Mimic writer, archive exporter), all with 2029 end dates

Trade-offs

Decision	Option	When it is right	When it is wrong
Strategy	Incremental strangler	Live marketplace; complex hidden behavior; value needed early	The runtime itself must go by a near date and the code would run elsewhere as is
Strategy	Re-platform first	Fixed hosting deadline; code fits the target runtime	The architecture, not the hosting, is the problem
Strategy	Big-bang rewrite	Small, well-understood systems with little traffic	Any marketplace with SEO, feeds and paid products
Slice order	Read paths first	SEO and performance gains; low data risk	The write path is the bottleneck for most teams and cannot wait
Synchronization	Asynchronous CDC and events, one writer per record	Most marketplace data; tolerable seconds of lag	Money movements, where a parallel run and reconciliation are needed
Synchronization	Dual writes from application code	Almost never	Partial failures leave the systems silently inconsistent
Parity	Outcome parity (non-inferiority)	Search, ranking, insertion outcomes	Billing and payments: amounts must match exactly, so use a parallel run
Parity	Feature parity	Narrow, well-specified slices	Old systems whose behavior nobody can enumerate

Decision	Option	When it is right	When it is wrong
Shadow method	Recorded-query replay on frozen data	First comparisons; reproducible differences	Performance testing, which needs live shadow traffic
URLs	Keep stable	Platform migrations	The URL design itself is harmful (unbounded facets, unstable slugs) — change it separately

Vertical Differences

Topic	General classifieds	Real estate	Automotive	Jobs
Main migration risk	Private sellers' insertion experience; long tail of app versions	Feed integrations from many software vendors; agency trust	Dealer feeds and the seller API; package billing	Expiry and structured data for search engines
Who must change their integration	Few external parties	Up to 160 vendors at Hearthly; each must test the new endpoint	DMS vendors and dealers' integrators	Applicant-tracking system vendors
Natural migration cohort	Seller or country	Software vendor (all agencies of one vendor move together)	Dealer or DMS vendor	Employer or feed
Parity that matters most	Contacts per search; publication outcomes	Duplicate clustering and listing counts per agency	Lead delivery and billing per dealer	Applications per posting; expired jobs removed

Topic	General classifieds	Real estate	Automotive	Jobs
SEO sensitivity	High: landing pages	High: area and building pages	High: model and dealer pages	Very high: job-posting structured data and expiry

The professional verticals have one advantage: cohorts are natural. Moving a software vendor's or a DMS vendor's customers together, with the vendor's integration tested once, is easier than moving millions of private sellers whose apps are three versions behind. Their disadvantage is that **external parties must change their integrations**, and their timetable, not yours, sets the pace. A migration that keeps the external contract stable — the Supply Contract v1 at Veldmark (see Many Doors In: Forms, APIs and Bulk Feeds) — removes that dependency.

Failure Modes

- **The rewrite that never ships**. A new platform is built beside the old one for two years with no traffic, chasing a feature list that grows as the old system keeps changing. When it finally launches, parity disputes take another year.
- **Dual writes**. Application code writes to both systems "temporarily". A timeout after the first write leaves them inconsistent, and nobody knows which one is right. Use one writer and asynchronous propagation.
- **Ghost Inventory between two systems**. A component listens only to the old system's removals, or only to the new one's. Sold listings stay in alerts, recommendations or partner feeds. Every consumer must read the contract stream, which carries removals from both.
- **Search as System of Record revealed**. The team discovers mid-migration that the old index holds fields that exist nowhere else: counters, boosts, manual fixes. The new index cannot be built until those fields get a source and an owner.
- **Analyzer amnesia**. A new engine launches with default analyzers; fifteen years of synonyms and stopwords disappear, and long-tail queries return nothing. Port analyzers with a test set of real queries.
- **The SEO cliff**. URLs change "while we are at it", redirects are incomplete or chained, status codes for removed listings change, or pages move to client-side rendering. Organic traffic falls over weeks, after the old system is gone.
- **Parity by spreadsheet**. Parity is asserted by a checklist of features, not measured on outcomes. The new flow has every button and publishes fewer listings.
- **Transitional components that become permanent**. The relay, the mimic writer and the CDC connectors outlive the migration because nobody owns their removal. Give each an end date and a line on the progress dashboard.

- **The unrehearsed rollback**. The switch exists but has never been used with real data. When it is needed, the old system is missing the last day's listings.
- **The forgotten consumer**. A partner feed, a finance export or a moderator's report reads the old database. It breaks on the day the table stops being updated. The consumer registry finds these before the cut-over.

The Chief Architect's View

Treat the migration as a portfolio of slices, each with an owner and a business case. A monolith retirement is not one project; it is twenty or thirty migrations with different risks, customers and teams. Each slice needs an owning team from the target organization (see Who Owns What: Organizing Marketplace Engineering), a behavior catalog, stage gates with numbers, and a rollback answer. The enabling team plans and helps; it does not own the slices.

Spend early on seams. The edge routes, the BFFs and the event contract cost months in 2027 and delivered little visible value on their own. They are what made every later wave reversible. I would defend that spend against any pressure to "just start rebuilding pages".

Decide the points of no return in advance. Every migration has moments after which going back means rebuilding: the first feature only the new path has, the day the reverse sync stops, the day the archive is exported. Each should be a forum decision with a recorded rationale, taken when the evidence is in, not discovered during an incident.

Fund double running explicitly, and shorten it. Running two systems costs infrastructure, on-call time and attention. Put the cost on the dashboard. The pressure to shorten it is healthy; the pressure to skip gates to shorten it is not.

Protect liquidity, and say so in the gates. The business does not care which system serves a page; it cares that buyers keep finding and contacting sellers, that sellers keep publishing, that dealers are billed correctly and that search engines keep sending traffic. Put those metrics into every gate. A migration that passes every technical check and loses contacts has failed.

Who decides and what to watch. The owning team decides its slice's design and proposes gate criteria; the advice forum reviews the criteria before the slice starts and the G3-to-G4 decisions; the CTO decides the sequence when it moves funding or dates; finance signs off billing parallel runs. Watch the progress table above, the business guardrails per wave, rollback rehearsals performed, transitional components past their end date, and the double-running cost.

Questions to Ask

1. What are the slices, in what order, and which team in the target organization owns each one?
2. Where are the seams (edge, BFF, events, code abstraction), and can each one route back in minutes without a deploy?
3. Is there exactly one writer for each record at any moment, and how do changes reach the other system?
4. What does the behavior catalog for this slice say, and which behaviors are we deliberately retiring?
5. What are the numerical exit criteria for each gate, and were they agreed before the slice started?
6. How is relevance parity defined for the search move, and what outcome metric decides it?
7. Can the new search index be built from a source other than the old index?
8. Which URLs, status codes or rendering modes change, and where are the redirects and the monitoring?
9. When was rollback last rehearsed with real data, and what is the point of no return?
10. Which consumers still read the legacy database, and which transitional components have no end date?

From Migration to AI

Yardstick's retirement is a 2029 event; the platform that replaces it will itself be replaced, in parts, during the next decade. Before that question, the next chapter, AI Across the Listing Lifecycle, looks at the technology shift most likely to reshape listing creation, search and moderation on the new platform, and at the architecture that keeps it under control.

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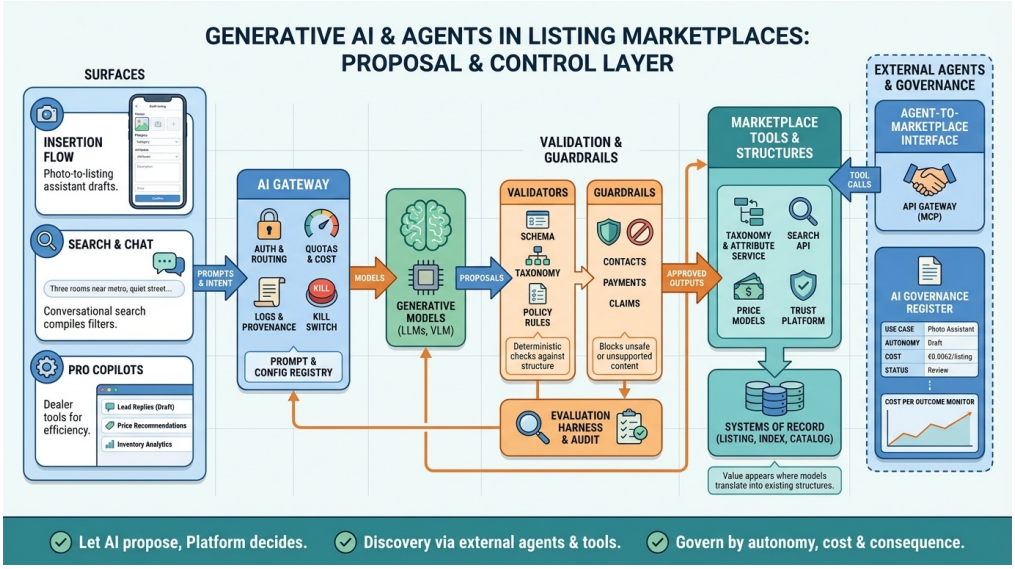
Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

44. AI Across the Listing Lifecycle



IN THIS CHAPTER, YOU WILL: Walk the listing lifecycle from creation to search, moderation, pro tooling and valuation, see where AI and agents have actually shipped at marketplaces between 2023 and 2026, design the layer that controls them, and follow Veldmark's photo-to-listing assistant from a cargo-bike photo to a published, validated listing.

WHY THE BUSINESS CARES: AI can cut the effort of listing an item from minutes to seconds, turn vague buyer wishes into searches that find something, and help pros answer leads faster, all of which is liquidity. It also creates new costs per call, a new channel where buyers search inside other companies' assistants, and new ways for fraudsters to produce convincing fakes at scale.

WHY ARCHITECTS SHOULD CARE: Generative models produce plausible output, not valid output. Without a layer that validates every suggestion against the taxonomy, the attribute dictionary and the trust rules, AI fills the catalog with confident errors. And once external agents call the marketplace's search, search becomes an API product with contracts, quotas and attribution.

KEY POINTS:

- **Let AI propose and the platform decide.** Every documented success in classifieds uses the model to translate intent into existing structures (category and attributes from a photo, filters from a sentence, a policy label from a message) while the structured systems remain the system of record. Validate every proposal against the schema, record its provenance, and keep humans in the loop where the law or the consequence requires it.
- **Discovery is leaving the marketplace's own surfaces.** In 2026 marketplaces ship their search as apps inside ChatGPT and as tools for external agents, while AI crawlers harvest listing data without asking. Structured, API-grade inventory, a bot and licensing policy, and attribution for leads that arrive through assistants are now architecture.
- **Govern AI uses by autonomy, cost and consequence.** Keep a register of every AI use with its autonomy level, owner, evaluation set, cost per outcome and kill switch. The EU AI Act adds dated obligations: disclosure for conversational systems and marking of generated content from 2026, and high-risk duties for recruitment and credit uses from 2 December 2027.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Attribute Junkyard** — Generative listing assistants that write free text into attributes, unchecked against the dictionary, produce the junkyard faster than any seller could.
- **Moderation as an Afterthought** — An assistant that makes listing easier also makes fake listings easier, so trust and safety has to sit inside the assisted flow, not after it.
- **Set-and-Forget AI** — Prompts, model versions and providers change weekly; an AI feature without an evaluation set, a cost monitor and an owner degrades without anyone noticing.

For most of their history, listing marketplaces asked people to do the work of structure. A seller had to choose a category from a tree of 1,400 leaves, fill twenty attribute fields and write a description. A buyer had to translate "a family car that fits a dog and costs less than twenty thousand" into make, body type, price and mileage filters. Every chapter in Parts II to IV of this book is, in part, about the cost of that translation. **Generative AI is the first technology that can do much of the translation for them**, and that is where its value in listing systems lies.

The same property makes it dangerous. A language model asked to fill attributes will fill them, whether or not the photo shows what it claims. A conversational assistant will answer, whether or not the inventory supports the answer. This chapter's argument is

that the architecture must treat AI output as **a proposal to be validated**, never as data, and that the marketplace's structured assets (the taxonomy, the attribute dictionary, the index, the trust rules) become more valuable, not less, when AI arrives.

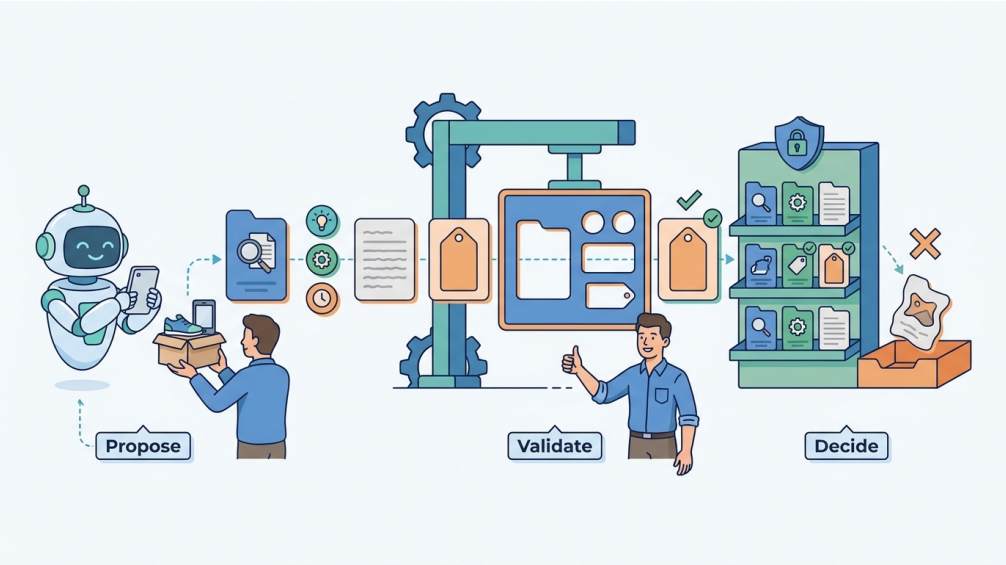


Figure 1: *The model proposes and the platform decides: every AI output is a proposal checked against the taxonomy, the attribute schema and the trust rules before it becomes listing data.*

The chapter builds on two earlier designs: Models in Production: ML Platforms and Use Cases, which owns model governance, decision tiers and the platform for predictive models, and the taxonomy and attribute schemas of Part II (Category Trees That Can Change: Designing Marketplace Taxonomies, Attributes Without Chaos: Schemas, Dictionaries and Validation), which AI output compiles into. It adds what generative models and agents change, and links to the chapters that own LLM-based enrichment (From Raw Supply to Usable Listings: Normalization and Enrichment), natural-language search (What Did They Mean? Query Understanding and Autocomplete), moderation (Trust and Safety as a Platform) and SEO for AI answer engines (Found by Google: Web, App and SEO Architecture for Listings). The next chapter, Built to Be Replaced: Evolving a Marketplace Platform Over a Decade, treats AI as one of the stressors the platform must absorb over a decade. The worked example belongs to Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build).

Setting the Scene at Veldmark: Six AI Requests in One Quarter

By **November 2027**, Noor Hassan, Veldmark's Group Chief Product Officer, has six AI proposals on her desk, each from a different team and each with a vendor demo behind it.

1. **Tradeyard** wants a photo-to-listing assistant: a private seller takes photos and the app proposes category, attributes, title, description and price. Tradeyard publishes about **2.3 million** new listings a week, **78%** from private sellers, and its insertion flow is its biggest conversion funnel.
2. **Hearthly** wants conversational search in France and Spain ("three rooms near a metro, under €400,000, quiet street") and an app inside a third-party AI assistant.
3. **Motorvane** wants a dealer copilot in Dealer Hub: drafted replies to leads, price recommendations and descriptions in the dealer's tone, for its **21,400** dealers.
4. **Trust and safety** wants large language models (LLMs) in the moderation gray zone, after the July 2027 Barcelona fake-rental ring showed how fast fraudsters reuse stolen photos.
5. **Hearthly's valuation team** wants Valuo to explain its estimates in plain language.
6. **Tradeyard's SEO team** wants to know what to do about AI crawlers: about **46%** of Tradeyard's web sessions start from organic search, and bot traffic from AI companies has risen through 2027.

Aisha Rahman, Group Head of Trust & Safety, asks what happens when fraudsters use the listing assistant. Lucía Ortega, the Group Data Protection Officer and regulatory lead, asks which of the six fall under the EU AI Act. Tomasz Wiśniewski, head of the group search platform, asks who pays for an external assistant's queries at the Sunday-evening peak of 31,000 queries per second. Ingrid Halvorsen, the Group Chief Architect, asks a fourth question of each proposal: **what structured thing does the model produce, and what checks it?** The answer to that question decides the architecture, and it turns out to be the same for all six.

Concepts: Generative AI, Agents and Autonomy

- **Generative AI** — models that produce text, images, structured output or actions from prompts, typically large language models (LLMs) and vision-language models. Predictive models that score or classify are covered in Models in Production: ML Platforms and Use Cases; generative models reuse that platform for registration, logging and monitoring.

- **AI listing assistant** — a feature in the insertion flow that proposes listing content (category, attributes, title, description, price) from photos, voice or short text, for the seller to accept, edit or reject.
- **Structured-output compiler** — the pattern in which a model translates unstructured input into a structure the marketplace already has: an attribute set, a filter query, a policy label. The structure is validated deterministically before use. It is the central pattern of this chapter.
- **Agentic search** — search in which a model plans steps and calls tools (keyword or vector search, filters, comparisons, external lookups) over several turns, rather than returning one ranked list for one query.
- **Agent-to-marketplace interface** — an interface through which external AI agents call the marketplace's capabilities as tools. The most widely used such protocol as of 2026 is the Model Context Protocol (MCP). Anthropic introduced it in November 2024 as an open standard for connecting AI assistants to data sources (Anthropic, 2024) and donated it to the Agentic AI Foundation under the Linux Foundation in December 2025 (Anthropic, 2025).
- **Autonomy level** — how far an AI feature acts without a person: it *suggests*, *drafts* for confirmation, *acts with confirmation*, or *acts autonomously*.
- **AI governance for marketplaces** — the register, rules and controls that decide which AI uses may run, at what autonomy, with which data, evaluated how, at what cost, and with whom accountable.

The autonomy ladder

Level	What the AI does	Listing example	Who confirms	Typical controls
Suggest	Offers options the user may pick	Category suggestions; query suggestions	User, implicitly	Offline accuracy; acceptance rate
Draft	Produces a complete artifact the user reviews	Description and attributes from photos; reply draft for a dealer	User, explicitly, before publication	Schema validation; provenance per field; audit sampling

Level	What the AI does	Listing example	Who confirms	Typical controls
Act with confirmation	Performs a change after a person approves	Copilot applies a price change to 40 dealer listings after the dealer clicks "apply"	User, per action or batch	Preview of effects; undo; rate limits
Act autonomously	Performs a change or decision without approval	Auto-withholding a suspicious message; auto-renewing listings	Nobody at the time; reviewed later	Gate-tier governance (see Models in Production: ML Platforms and Use Cases); appeals; legal duties

The rule I use for listing systems: **anything that changes what buyers see about a seller's listing stays at "draft" or below unless the seller has opted in**, and anything that restricts a user's rights follows the trust-and-safety rules for automated decisions.

Where AI Has Shipped: The Lifecycle in Public Evidence

Between 2023 and 2026, marketplaces shipped AI at every stage of the listing lifecycle. Most entries are company announcements rather than engineering accounts; where a source is trade press or a third-party summary, it is labeled.

Lifecycle stage	Documented example	Date	What it shows
Listing creation	eBay's "magical listing": generative descriptions from title and category, then photo-to-listing filling title, description, category and item specifics with price and shipping suggestions; iOS beta first; built on Azure OpenAI (eBay, 2023)	September 2023	Text assist first, photo second
Listing creation	Kleinanzeigen: AI description helper (August 2025), then photo-to-listing proposing title, description and category across almost all categories except vehicles, jobs and services (mobiflip, trade press, 2025)	November 2025	High-structure verticals excluded at first

Lifecycle stage	Documented example	Date	What it shows
Listing creation	OLX Auto Description: draft from a photo plus basic details, reviewed by the seller; human evaluation of 98% for style and relevance and 89% for faithfulness (OLX Group, 2026)	August 2026	Faithfulness is measured separately from fluency
Enrichment	OLX extracted job roles from job-ad text with an LLM service subscribed to ad events; about 4,000 requests a day cost "approximately 15K per month" (currency unstated), which prompted evaluation of self-hosted models (OLX Tech, 2024)	January 2024	Event-driven enrichment; cost per call matters

Lifecycle stage	Documented example	Date	What it shows
Search	coches.net (Adevinta Spain): natural-language car search converted to the site's existing filters by Claude Haiku on Amazon Bedrock, after a cheap moderation pre-classifier; about 20,000 calls a day for about €19 a day (Adevinta Tech Blog, 2024)	November 2024	The LLM compiles filters; it does not run the search
Search	leboncoin: conversational assistant turning preferences into JSON for its product search API; guardrail prompts replaced by fine-tuning for latency; a hosted assistant service rejected because conversations would be stored by the vendor (Adevinta Tech Blog, 2024)	March 2024	Data residency of conversations is a design input

Lifecycle stage	Documented example	Date	What it shows
Agentic search	OLX Magic: an agent calling text, visual and web search tools over hybrid retrieval, with latency of about 5–10 seconds against 300–500 ms for classic search; a blank chat interface was replaced by the familiar search bar and filters with AI built in (ZenML LLMOps database, third-party summary of a 2024 talk)	2024	"Only as good as your search"
Off-platform agents	OLX CompassGPT (property, Imovirtual) and AutoGPT (cars, Otomoto first) run as apps inside ChatGPT, built with OpenAI; OLX reports buyers finding a vehicle "at least 20% faster" (OLX Group, 2026)	February and April 2026	Discovery on someone else's surface; company-reported outcome

Lifecycle stage	Documented example	Date	What it shows
Off-platform agents	ImmoScout24 shipped an MCP server for its assistant in November 2025 and an app in ChatGPT; the MCP server was designed to be "completely stateless", and cached tool descriptors forced a rotation of production environments (Scout24 Engineering, 2026)	January and March 2026	Agent channels have their own release mechanics
Moderation	OLX "Chat Images GenAI" detects suspicious images and QR codes in chat in real time; fraudulent images are "around 30% of all fraud cases on the platform in Poland" (OLX Group, 2026)	January 2026	Generative models in the gate tier, on a narrow task

Lifecycle stage	Documented example	Date	What it shows
Pro copilots	OLX AutoIQ for Otomoto dealers: listing creation, lead dashboard, follow-up and pricing recommendations, descriptions in the dealer's tone; dealers publish "up to 36% faster" (OLX Group, 2026)	February 2026	Copilots sit on pro tooling, not on buyer surfaces
SEO	Adevinta's SEO team used AI to classify keywords, generate related searches and write meta titles and descriptions (Adevinta Tech Blog, 2023)	June 2023	Predates Google's AI answers; do not extrapolate

Three patterns run through the table. **Every implementation keeps the structured system of record**: the filters, the category tree, the index and the ranking model remain, and the model translates into them. **Rollouts are staged by structure**: text help first, photos second, high-structure verticals such as vehicles last. And **the biggest 2026 change is distribution, not modeling**: marketplaces now publish their search into assistants they do not control, which reverses the long-held assumption that discovery happens on the marketplace's own surfaces.

Architecture: The Control Layer Between Models and Listings

Four components the ML platform lacks

An AI feature in a listing system has four parts that the predictive-ML platform does not already provide, plus the guardrails that sit around them.

1. **An AI gateway** through which every model call passes: authentication, routing between providers and self-hosted models, rate limits, cost metering per feature and per tenant, logging of prompts and outputs (with personal data handled per the retention rules), and a kill switch per feature.
2. **A prompt and configuration registry**: prompts, few-shot examples, output schemas and model choices versioned like code and registered like models, so that "the assistant got worse on Tuesday" can be traced to a change.
3. **Tools over the marketplace's own capabilities**: the taxonomy and attribute service, the search API, the price models, the trust-and-safety checks. Models call these rather than guessing.
4. **An evaluation harness**: golden sets per language, category and use, run before every prompt or model change and continuously on sampled production traffic.

Around them sit **validators**: deterministic checks that an output conforms to the attribute dictionary, the filter schema or the policy taxonomy, and **guardrails** that block outputs containing contact details, payment instructions, prohibited items or unsupported claims.

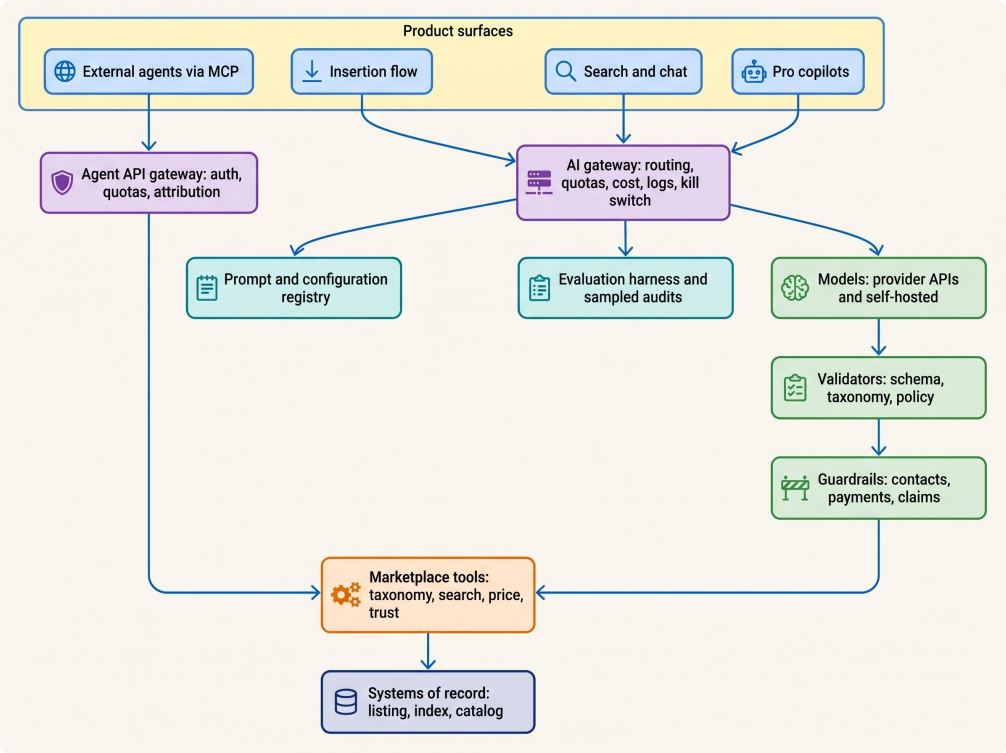


Figure 2: *Models never write to systems of record directly: every output passes a deterministic validator and a guardrail, and reaches listings, search or catalog only through the same tools and contracts that the rest of the platform uses.*

Boundaries that matter

- **Model output is never a listing field until a validator and a person accept it.** The listing aggregate's invariants (see The Listing Aggregate: Identity, Lifecycle and Invariants) apply to AI-suggested values exactly as to typed ones.
- **Every field carries provenance.** The attribute-provenance model from From Raw Supply to Usable Listings: Normalization and Enrichment gains values for `ai_suggested`, `seller_confirmed` and `seller_edited`, so that search, trust and analytics can tell what a seller asserted from what a model inferred. This matters legally too: a seller is responsible for a description they confirmed, and the marketplace for one it published unconfirmed.
- **The search index stays the retrieval layer.** Conversational and agentic search compile into the same filter schema and call the same search API; generated answers never describe listings the index did not return (see What Did They Mean? Query Understanding and Autocomplete).
- **Trust and safety runs on assisted content too.** An assisted listing goes through the same moderation pipeline as any other, with the assistant's own signals added

(see below).

Model choice and hosting

Option	Strengths	Weaknesses	Fits
Frontier model through a provider API	Best quality on open-ended tasks; no infrastructure	Cost per call; data-processing terms; latency; provider changes	Low-volume, high-value tasks: pro copilots, explanation, conversational search pilots
Small or mid-sized model through an API	Low cost; fast	Weaker on hard cases	High-volume structured tasks: filter compilation, category proposals
Self-hosted open model, possibly fine-tuned	Cost control at volume; data stays in-house	GPU operations; evaluation is yours	Steady, high-volume extraction and classification once the task is stable
Distilled classical model	Cheapest at scale; predictable	Needs labels; narrow	Tasks where LLM outputs have produced enough labels to train on

The documented trajectory matches the table: leboncoin moved guardrails from prompts into fine-tuning for latency, and OLX's job-role extraction cost prompted an evaluation of self-hosted models. My rule: **start with an API to learn the task, then move volume down the table as the task stabilizes and labels accumulate**, keeping the gateway so that the move is a routing change, not a rewrite.

Veldmark Worked Example: The Photo-to-Listing Assistant

Five steps before the seller confirms

Tradeyard's assistant was approved for a pilot on **Monday 15 November 2027**: Tradeyard Netherlands, iOS first, goods categories only, excluding vehicles, property, jobs and services, for 5% of private sellers. The exclusions follow the documented sequence at eBay and Kleinanzeigen and Tradeyard's own constraint: vehicles need decoded vehicle data, not photos, and property needs addresses and mandatory disclosures.

It depends on two Common Ground capabilities that went live just before it: the **group taxonomy and attribute service** (October 2027), which publishes the category tree and each category's attribute schema as machine-readable reference data, and the **shared trust-and-safety platform** for Tradeyard Netherlands and Belgium (November 2027). Without the first, the model would have nothing to compile into; without the second, nothing would check what it produced.

1. **Screen the photos before generating anything.** Perceptual hashes are compared against live listings and known scam images, and an image classifier checks for prohibited items. A match stops the assistant: the seller continues manually and the listing is flagged for review. The assistant never makes a stolen photo easier to list.
2. **Propose a category** from the photos with a vision-language model, constrained to the current leaf categories returned by the taxonomy service. The model ranks candidates; it cannot invent a category.
3. **Propose attributes** for that category's schema only, each with a confidence, returning `unknown` rather than a guess where the photos do not show the value. Values must come from the attribute dictionary's enumerations or pass its type and unit rules.
4. **Propose a price range** from Tradeyard's existing price-hint model (a predictive model registered in the group registry; see Models in Production: ML Platforms and Use Cases), not from the language model. **The language model never produces a number that a buyer will rely on.**
5. **Draft a title and description** from the confirmed category and attributes, in the seller's language, with guardrails that strip contact details, URLs, payment instructions and claims not supported by the attributes.

The seller sees all five as an editable form and must confirm before publishing. The published listing carries a small label, "Description drafted with AI and checked by the seller", and per-field provenance.

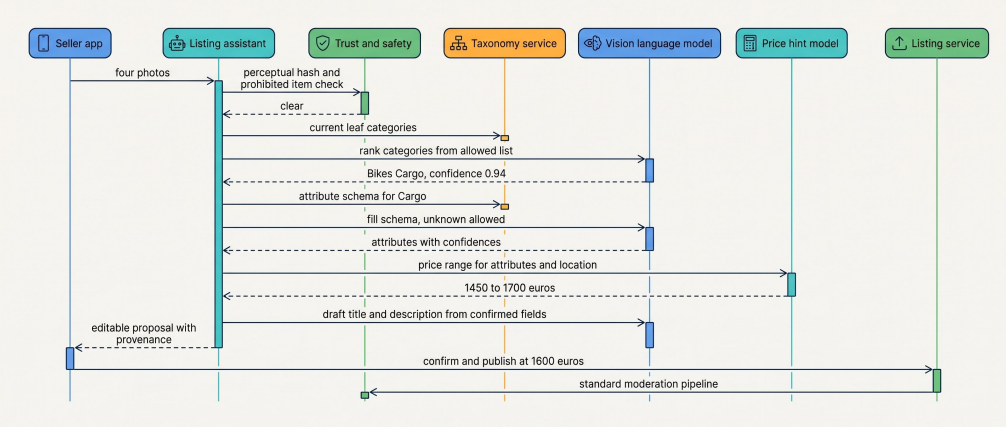


Figure 3: *The language model ranks categories from an allowed list, fills a schema it is given, and writes prose last; the price comes from a registered predictive model and the photos are screened before any generation starts.*

One listing: a Utrecht cargo bike

On **Monday 6 December 2027 at 19:05 CET**, a private seller in Utrecht photographs his electric cargo bike four times. The trust check finds no hash matches. The assistant proposes **Bikes › Cargo**, one of the canonical leaves of the bikes split released on 1 November 2027, with a confidence of 0.94. For the schema it fills seven of nine attributes: brand (read from the frame decal), electric assist (yes), box position (front), number of wheels (two), color, frame size and condition (good). It returns `unknown` for battery capacity and model year, which the photos do not show; the seller adds the year. The price-hint model proposes **€1,450–€1,700** for the attributes and Utrecht; the seller sets **€1,600**. He publishes at 19:07, **2 minutes 10 seconds** after the first photo, and the listing enters the standard moderation pipeline like any other.

The pilot’s numbers

By **Monday 17 January 2028**, after nine weeks, the pilot has produced about **154,000 assisted listings**. The evaluation report Priya Raman’s team presents to the architecture advice forum shows:

Measure	Result	How it is measured
Uptake	62% of eligible sellers used the assistant	Assistant started / eligible insertion starts
Category accepted unchanged	92.4%	Final category equals top suggestion
Attribute suggestions accepted unchanged	78%; edited 15%; removed 7%	Per-field provenance
Unsupported claims in descriptions	1.9% of a 2,000-listing audit sample	Human reviewers compare text with photos and attributes
Median insertion time	2 min 50 s assisted, against 6 min 40 s manual in the same categories	Insertion-flow events
Started-to-published rate	+9 percentage points against the control group	Seller-randomized experiment
Contacts per listing in the first 7 days	+4%	Same experiment; a weak signal, since sellers who finish faster may differ
Model cost	€0.0062 per assisted listing	AI gateway metering

Two findings change the design. First, **assisted listings fill more attributes**, which improves filter coverage: the share of cargo-bike listings with electric assist recorded rose from 61% to 88%. That is the Attribute Junkyard running in reverse, but only because values were constrained to the dictionary; an earlier internal prototype that let the model write free-text attributes had produced 1,100 distinct "colors" in two weeks. Second, the 1.9% of unsupported claims cluster in a few phrases ("barely used", "like new") that the model infers from clean photos. The guardrail is changed to strip condition superlatives unless the seller selects the matching condition value.

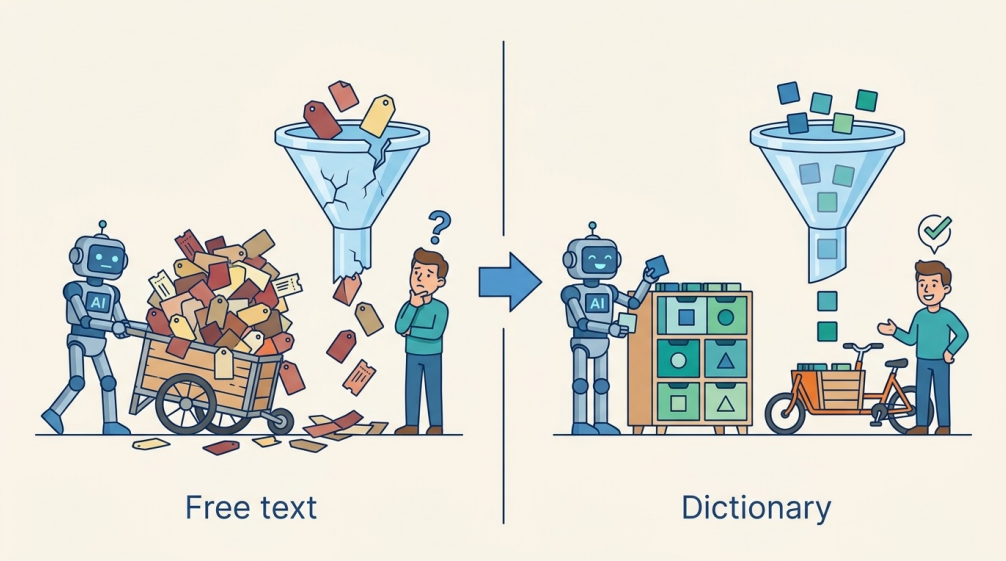


Figure 4: *Assisted listings filled more attributes only because values were constrained to the dictionary; a prototype that let the model write free-text attributes produced 1,100 distinct colors in two weeks.*

At full rollout, model cost is not the constraint: the gateway forecasts about **€230,000 a year** in model calls across Tradeyard's four countries, before evaluation and platform costs, which is small against the conversion gain on the insertion funnel. The constraint is **evaluation in four languages across the group tree's 1,640 leaf categories**. The forum approves the extension to 25% of Netherlands sellers and Belgium on both iOS and Android from **Monday 24 January 2028**, with Poland and Portugal once Polish and Portuguese evaluation sets exist, and keeps the category exclusions.

What fraudsters did with it

Fraudsters used the assistant like everyone else, and **an assistant that writes good descriptions removes a weak scam signal** (bad spelling, generic text). Aisha's team tracked accounts later removed for fraud: they had used the assistant at the same rate as other sellers, and their assisted listings were as polished as anyone's. The trust model must therefore rely less on text quality and more on account, device, image and price signals (see Scams, Fakes and Account Takeovers: Fraud Prevention for Listings). The photo screen in step 1 blocked 2,300 attempts to list with photos matching existing listings during the pilot. It cannot catch what the Barcelona ring would do next: generated photos of apartments that exist nowhere, which have no hash to match. For generated imagery the defense is behavioral and economic (new accounts, below-market prices, requests for payment outside the platform), not visual.

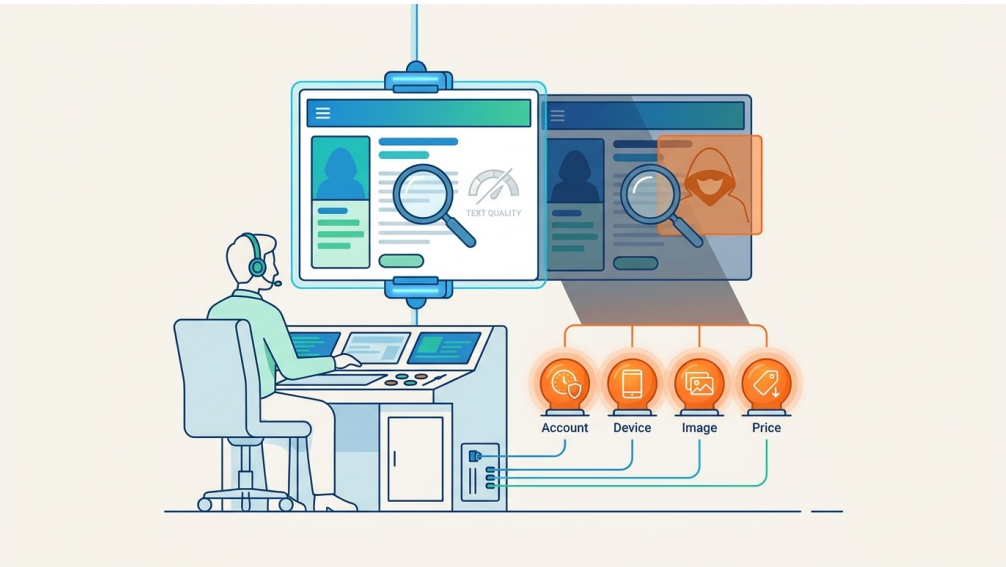


Figure 5: *An assistant that writes good descriptions for everyone removes a weak scam signal, so the trust model must lean less on text quality and more on account, device, image and price signals.*

Search, Agents and the New Channel

Conversational search as a query compiler

Hearthly's conversational search follows the coches.net and leboncoin pattern: the model translates a sentence into Hearthly's existing filter schema, the filters are validated against the schema and shown to the user, and the normal search runs. The user can edit any filter, so the search stays explainable and the result set is the index's, not the model's. The compiler pattern also keeps cost low (one short call per query, cacheable by normalized output) and keeps the LLM off the hot path for the majority of keyword queries. The design and evaluation of query understanding belong to What Did They Mean? Query Understanding and Autocomplete.

Natural-language input is useful mainly where filters are many and vocabulary is vague. "Quiet", "bright" and "near a good school" have no filter, and the honest answer is to say so. **A conversational layer that cannot say "we have no filter for that" will invent one.**

Agentic search, in which the model plans several steps and calls tools, costs an order of magnitude more latency. OLX's account puts it at 5–10 seconds against 300–500 ms for classic search, and records that a blank chat interface failed and was replaced by the familiar search page with AI built in. My judgement for Veldmark: **agentic flows**

belong in high-consideration journeys (a home, a car) where buyers spend weeks, not in the default search of a general classifieds site where they spend seconds.

External agents: search as an API product

When an external assistant calls the marketplace, search becomes an API product with its own contract. ImmoScout24's account of its ChatGPT app shows that the mechanics differ from web releases: the assistant platform caches tool descriptors, so a deploy can break references that the platform still holds. Rightmove's FY2025 results, published in February 2026, described conversational search going live and an app submitted to the ChatGPT ecosystem (Rightmove plc, 2026). In both cases the assistant does not get the website; it gets **tools**: search with a structured filter schema, listing summaries, and links back.

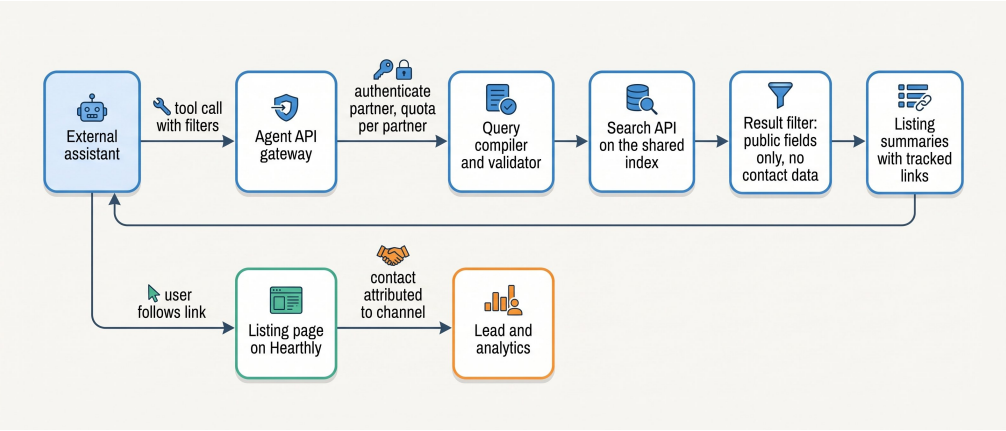


Figure 6: *An external assistant gets tools, not pages: authenticated, rate-limited search over the same index, public fields only, and tracked links that bring contacts back to the marketplace and attribute them to the channel.*

The decisions Veldmark records for the channel in its **February 2028** decision record are the ones every group will face:

Question	Veldmark's answer	Reason
Which brands first	Hearthly France and Motorvane Germany	High-consideration journeys where conversational shortlists help; Tradeyard's buyers mostly browse

Question	Veldmark's answer	Reason
What the tool returns	Public listing fields, price, location at the listing's published precision, a tracked link	Contact happens on Veldmark's surfaces, where messaging safety runs (see From Interest to Contact: Messaging and Lead Management)
Contact through the assistant	Not offered	Moving contact off-platform recreates Fast Mail for Fraudsters outside Veldmark's controls
Quotas	Per partner, with lower limits during Sunday 20:00–22:00 CET	Agent traffic must not compete with the peak (see Sunday Evening Peaks: Reliability, Performance and Cost)
Attribution	Leads from assistant links tagged by channel and partner	Pros pay for leads and will ask where they came from
Ranking	The same organic ranking as on site; paid placements labeled as such	Ranking transparency duties do not disappear in another company's interface (see Ranking for Two Sides: Relevance, Freshness and Paid Visibility)
Data use by the partner	Contract forbids training on returned listings and caching beyond a short window	Listings, photos and descriptions are the sellers' and agencies' content as well as Veldmark's asset

AI crawlers and data licensing

The other half of the channel is involuntary. AI companies crawl listing sites to train models and to answer questions about inventory, prices and neighborhoods. For a listing marketplace, its inventory is its product, and a crawler that copies it delivers no traffic in return.

The architecture has three levers. **Technical**: robots directives per crawler, bot management at the CDN, and rate limits by behavior rather than by user agent, since

user agents lie. On 1 July 2025 Cloudflare began blocking AI crawlers by default for new domains and announced a pay-per-crawl model in beta (Cloudflare, 2025), which shows the infrastructure market moving the same way. **Legal**: in the EU, the text-and-data-mining exception for commercial use applies only where rights holders have not reserved their rights in a machine-readable way, and the AI Act obliges general-purpose AI model providers to have a policy that identifies and respects such reservations under Article 4(3) of Directive (EU) 2019/790 (AI Act, Article 53). A machine-readable reservation in robots directives and terms is therefore worth publishing; how strongly listings data is protected by database rights depends on the case and is a question for counsel. **Commercial**: licensed feeds or APIs for answer engines and partners who want the data, with attribution and freshness terms.

Crawler policy trades data protection against search traffic: the search engines that bring 46% of Tradeyard's web sessions also build AI answers, so blocking the crawler that feeds those answers may also cost classic search traffic. The decision belongs to the brand's GM with the SEO and legal leads, and it should be revisited each quarter; the architecture's job is to make each lever a configuration, not a project (see Found by Google: Web, App and SEO Architecture for Listings).

AI in Trust and Safety

Generative models are useful in moderation where policy changes faster than labels accumulate. A new scam script can be described in a prompt the same day and later distilled into a cheap classifier once moderators have labeled enough cases. Three rules keep this controlled.

1. **Use LLMs in the gray zone only**. Run the cheap classifier on all traffic and the LLM on the uncertain band and the human-review queue. Tradeyard sends about 138,000 new-listing decisions a week to human review; that is where an LLM's triage and summarization pays, not on the 94% decided automatically.
2. **Keep a deterministic floor**. Hash lists, blocked domains and hard rules run before and regardless of any model, because fraudsters probe the gate and prompt injection can manipulate a model reading a listing or a message.
3. **Respect the law on automated decisions**. Under the Digital Services Act, decisions on users' complaints must not be taken "solely on the basis of automated means" (Article 20(6)), and statements of reasons must explain each restriction (see Rules of the Marketplace: DSA, DAC7 and Vertical Regulation). An LLM can draft the explanation from the decision facts; a template or a person must constrain it, because the statement is a legal document.

OLX's chat-image model (2026) shows a well-scoped gate-tier use: one fraud vector (images and QR codes in chat), real-time, measured precision. Aisha's gray-zone design for Veldmark follows the same scope discipline, and every model in it is registered at the gate tier with sampled labels (see Models in Production: ML Platforms and Use Cases).

Pro Copilots and Valuation Explanations

Motorvane's dealer copilot sits inside Dealer Hub, where dealers already manage inventory and leads (see Serving the Pros: Dealer, Agent and Merchant Tooling). It drafts replies to leads from the listing and the dealer's past replies, suggests price changes using Motorvane's registered market-value model, and writes descriptions in the dealer's tone. Its autonomy is "act with confirmation": a dealer can apply a suggested price change to forty cars in one action, after a preview that shows each car's new price rating. Two constraints are specific to marketplaces. The copilot may use only the dealer's own data and Motorvane's market data, never another dealer's leads or prices; and price suggestions must be explainable in the same terms as the public price rating, or dealers will see the marketplace as steering them. The equivalent for Keybridge is stricter: any copilot in Keybridge's agent CRM runs inside Keybridge's tenancy, under the March 2027 neutrality decision, and nothing it sees reaches Hearthly.

Valuo explanations follow a retrieve-then-narrate pattern. The explanation for the canonical Lyon apartment (68 m², Lyon 7e, energy class D, estimate €372,000–€401,000) is generated from structured inputs that Valuo already produces: the comparable sales used, their dates and prices per square meter, and the effect of the energy class. The language model writes sentences around those facts; it may not introduce a comparable, a number or a trend that is not in the input, and a validator checks every number in the output against the input. **An explanation that invents a comparable is worse than no explanation**, because it turns an estimate into a misrepresentation.

Cost and Evaluation

Cost per outcome

Generative AI cost is per call, which makes it visible in a way predictive ML rarely is. The public numbers are few and old by model-pricing standards.

Source	Volume	Reported cost	Year	Note
coches.net natural-language car search	~20,000 calls a day	~€19 a day	2024	About 6,000 input and 500 output tokens per call

Source	Volume	Reported cost	Year	Note
OLX job-role extraction	~4,000 calls a day	"approximately 15K per month", currency unstated	2024	Prompted evaluation of self-hosted models
Veldmark photo-to-listing (illustration)	~154,000 listings in the pilot	€0.0062 per assisted listing	2027–2028	Forecast €230,000 a year at four-country rollout

The unit matters more than the figures, which fall as model prices fall: **cost per outcome**, such as cost per published listing, per successful search or per qualified lead. The AI gateway should report it per feature and per brand, with a budget and an alert, so that a prompt change that triples tokens is noticed the same day.

Evaluation before and after launch

Stage	What it checks	Listing example
Golden sets before launch	Accuracy and faithfulness per language, category and edge case	2,000 photographed items per country with known categories and attributes
Regression on every change	No degradation from a prompt, model or provider change	Golden sets rerun automatically; the change is blocked if category accuracy drops more than 1 point
Adversarial testing	Guardrails hold under abuse	Photos with phone numbers in the image, prompt-injection text in descriptions, scam scripts
Online experiments	Effect on liquidity, not only on usage	Started-to-published rate, contacts, removals for policy reasons

Stage	What it checks	Listing example
Sampled production audit	Drift in real traffic	Weekly human review of assisted listings and conversational queries

Faithfulness, whether the output is supported by the input, deserves its own metric, separate from fluency; OLX's reporting of 98% for style and 89% for faithfulness shows why. **Fluent and wrong is the characteristic failure of generative models**, and it is invisible to usage metrics.

Regulation: The EU AI Act for Marketplaces

The EU Artificial Intelligence Act (Regulation (EU) 2024/1689) applies in phases, and the Digital Omnibus on AI, published as Regulation (EU) 2026/1744 on 24 July 2026, moved some of them (European Parliament Legislative Train, updated September 2026; implementation timeline, updated August 2026). As verified in October 2026:

Obligation	Applies from	Marketplace relevance
Prohibited practices and AI literacy	2 February 2025	Manipulative or exploitative designs in assistants; staff training
General-purpose AI model obligations, including the copyright policy	2 August 2025	Applies to model providers; relevant to crawler and licensing policy
Article 50 transparency: people must be informed they are interacting with an AI system unless it is obvious; generated content must be marked in a machine-readable way	2 August 2026; marking for generative systems placed on the market before that date by 2 December 2026	Conversational search, chat assistants, generated descriptions and images
High-risk obligations for stand-alone Annex III systems	2 December 2027	Candidate selection and job-ad targeting; creditworthiness of natural persons

Obligation	Applies from	Marketplace relevance
High-risk systems embedded in regulated products	2 August 2028	Rarely relevant to listings

Annex III point 4(a) lists AI systems "intended to be used for the recruitment or selection of natural persons", in particular "to place targeted job advertisements, to analyse and filter job applications, and to evaluate candidates" (AI Act, Annex III). Any jobs marketplace that ranks candidates or targets job ads with AI is in scope, and so is Tradeyard's small jobs section in Poland and Portugal if it builds such features. Point 5(b) covers creditworthiness evaluation of natural persons, which matters if a valuation or affordability model feeds a lender's decision (see Models in Production: ML Platforms and Use Cases).

Article 50's marking duty applies to providers of systems that generate synthetic text, images, audio or video, with an exception for systems that only assist standard editing or do not substantially alter the input (AI Act, Article 50). Whether a marketplace that builds a listing assistant on a third-party model is a provider of that system, and whether a seller-confirmed description falls under the exception, are questions for counsel. Veldmark's architectural answer is not to depend on the answer: assisted descriptions and AI-edited images carry machine-readable provenance metadata and a visible label from launch, and every conversational surface says it is an AI assistant.

The AI Act sits alongside the rules this book covers elsewhere: the DSA's limits on automated complaint handling and its statements of reasons, the P2B Regulation's ranking transparency, consumer law on misleading claims, and the GDPR for prompts and conversations that contain personal data (see Rules of the Marketplace: DSA, DAC7 and Vertical Regulation).

Trade-offs

Decision	Option A	Option B	When A is right	When B is right
Where AI writes	Into a draft the user confirms	Directly into the listing	Almost always for seller-owned content	Internal metadata the seller never sees, such as search keywords, with provenance
Output form	Constrained to schema and enumerations	Free text	Anything searchable, filterable or used by trust rules	Descriptions only, behind guardrails
Search interface	Query compiler over existing search	Agentic multi-step assistant	Default search; latency-sensitive journeys	High-consideration journeys with long sessions
External channel	Own surfaces only	Tools for external assistants	Brands whose buyers browse and whose SEO is strong	Verticals where buyers ask assistants for shortlists
Model hosting	Provider API	Self-hosted or distilled	Learning a new task; low or variable volume	Stable, high-volume tasks with labels
Crawler policy	Allow and license	Block and reserve rights	Partners that bring traffic or pay	Crawlers that copy inventory without return
Moderation	LLM in gray zone	LLM on all traffic	Large platforms with millions of decisions a week	Small markets where volume is low and policy changes fast

Vertical Differences

Dimension	General classifieds	Real estate	Automotive	Jobs
Listing assistant value	Highest: private sellers, sparse attributes, many categories	Moderate: agents list via feeds; value is in descriptions and disclosures	Low from photos; vehicle data comes from VIN decoding and DMS feeds	Moderate: structuring job ads; risky when applied to candidates
Main AI risk in creation	Unsupported condition claims; prohibited items	Wrong mandatory disclosures (energy class, fees)	Wrong specification or history claims	Discriminatory wording
Conversational search fit	Low for browsing; useful for vague goods queries	High: many filters, vague preferences	High: many filters, comparisons	Moderate
External agent channel	Weak fit	Strong fit (documented: ImmoScout24, OLX CompassGPT)	Strong fit (documented: OLX AutoGPT)	Emerging
Copilots	Pro sellers in some categories	Agency CRM, listing descriptions	Dealer lead replies and pricing	Recruiter tools
AI Act exposure	Article 50 transparency	Article 50; credit-adjacent valuation uses to review	Article 50; finance referrals to review	**High-risk** for candidate selection and job-ad targeting (Annex III point 4)

Failure Modes

- **Free-text attributes from a model**. Values outside the dictionary pour into the catalog; filters, facets and feeds disagree. The Attribute Junkyard, automated.
- **The assistant that helps scammers**. Generation runs before the trust checks, so stolen photos become polished listings. Screening photos first and moderating assisted listings like any other prevents it; **Moderation as an Afterthought** is the default if nobody designs it in.
- **Numbers from the language model**. A price, mileage, surface or valuation that a model generated rather than retrieved. Prices come from registered predictive models; explanations narrate retrieved facts.
- **Prompt changes without regression tests**. Quality drops after a "small" prompt edit or a provider's model update, and nobody notices for weeks: **Set-and-Forget AI** in its generative form.
- **Agentic search as the default**. Five-to-ten-second responses on a browse-heavy site, with costs that scale with sessions instead of outcomes.
- **Contact through the assistant**. Leads that start in an external assistant and move to e-mail or phone outside the marketplace, where messaging safety and attribution do not reach.
- **Conversation data at the vendor**. Buyers' preferences, budgets and sometimes addresses stored by a model provider under terms nobody reviewed; leboncoin's documented rejection of a hosted assistant API for this reason is the right instinct.
- **Crawler policy by accident**. A CDN default, a robots line copied from a blog, or no decision at all decides whether the inventory trains other companies' models.
- **AI-washed explanations**. Valuation or ranking explanations that sound plausible but do not match what the model did, which is worse for trust and transparency duties than a plain table of inputs.

The Chief Architect's View

Extend the model register into an AI use register. Every AI use (not every model) is registered with its autonomy level, decision tier, owner, model and prompt versions, data sources and their use rights, evaluation sets, cost per outcome, AI Act classification and kill switch. At Veldmark this extends the registry decided on 9 November 2027 for predictive models rather than creating a parallel one. A feature that cannot fill in the register is not ready to launch.

Make the structured assets the AI strategy. The taxonomy and attribute service, the search API with a clean filter schema, the price models and the trust platform are what AI features compile into. Groups that skipped that work will find their AI features produce the mess faster. At Veldmark, the photo-to-listing assistant was possible in November

2027 only because the taxonomy service went live in October; **the order of Common Ground's deliveries was also the order of what AI could do**.

Own the gateway centrally, own the features locally. One AI gateway and evaluation harness for the group, with routing, metering, logging and kill switches; features built by brand teams on top. That splits the work the same way as the rest of Common Ground: share platforms, not products.

Decide the channel strategy per brand, explicitly. Whether to publish search as tools for external assistants, which partners, with which data, and how leads are attributed are product and commercial decisions with architectural consequences. Put them in decision records with a review date, because the channel is moving faster than any other part of the business.

Write the data-use rules down and enforce them in pipelines. Which data may be sent to which provider, which data may train which model, which content external partners may cache or train on, and what the Keybridge neutrality commitment means for every AI feature. Rules that live only in policy documents are broken by the first prototype.

Budget AI by outcome. Approve features with a cost per outcome and a monthly budget per brand, reported by the gateway. Move high-volume stable tasks to cheaper hosting as labels accumulate.

Watch these indicators: share of AI uses registered; faithfulness on audit samples; attribute acceptance and edit rates; share of AI-suggested values outside the dictionary (should be zero); cost per outcome per feature; latency at peak; share of leads from external assistants and their quality; crawler traffic by declared and undeclared agents; and the number of AI features with an AI Act classification still marked "to review".

Questions to Ask

1. For each AI feature, what structured output does the model produce, and which deterministic validator checks it before anything is stored or shown?
2. Can any model output reach a listing field, a buyer-visible number or a user restriction without a person or a validator in between?
3. Do listing fields record whether a value was AI-suggested, seller-confirmed or seller-edited, and do search, trust and analytics use that provenance?
4. Which trust checks run before generation, and how does the assistant change the signals our fraud models rely on?
5. What is the cost per outcome of each AI feature, who sees it daily, and what happens when it doubles?
6. Which evaluation sets guard each feature, in which languages and categories, and do they run on every prompt and model change?

7. Which brands expose search to external assistants, what data do the tools return, and how are leads from that channel attributed and protected?
8. What is our crawler and licensing policy per brand, and is it a decision or a default?
9. Which of our AI uses are covered by Article 50 of the AI Act, and which could become high-risk from 2 December 2027, especially in jobs and credit-adjacent uses?
10. Which data may we send to which model provider, and where is that rule enforced?

From AI to the Long View

AI arrives in listing systems as a translator between people and the structures the marketplace already has. Where those structures are clean (a versioned taxonomy, an attribute dictionary, a search API with a filter schema, registered price models, a trust platform) AI makes listing faster, search more forgiving and pros more productive, and the control layer keeps every proposal checked, attributed and costed. Where they are not, AI accelerates the mess. The final chapter of this part, Built to Be Replaced: Evolving a Marketplace Platform Over a Decade, takes the longer view: how a Chief Architect keeps a marketplace platform coherent over a decade in which AI will be only one of the stressors, alongside acquisitions, regulation and the swing between central and local control.

Sources and Evidence

Company engineering and investor sources

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- How OLX uses AI to reduce repetitive work — OLX Group, 2026. Auto Description and its human evaluation.
- Extracting job roles in job ads: a journey with generative AI — OLX Tech, 2024. Event-driven LLM enrichment, volumes and cost.
- From Filters to Phrases: Our AI Revolution in Car Search — Adevinta Tech Blog (coches.net), 2024. Filter compilation, pre-classifier, tokens and cost.
- Building a Smarter Shopping Experience: Conversational Search in E-Commerce — Adevinta Tech Blog (leboncoin), 2024. JSON output, fine-tuning for latency, rejection of a hosted assistant API.

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- Scout24 Engineering — Scout24, 2026. "Shipping the first MCP in HeyImmo" (January) and "What we learned from launching ImmoScout24 app in ChatGPT" (March).
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Standards, regulation and documentation

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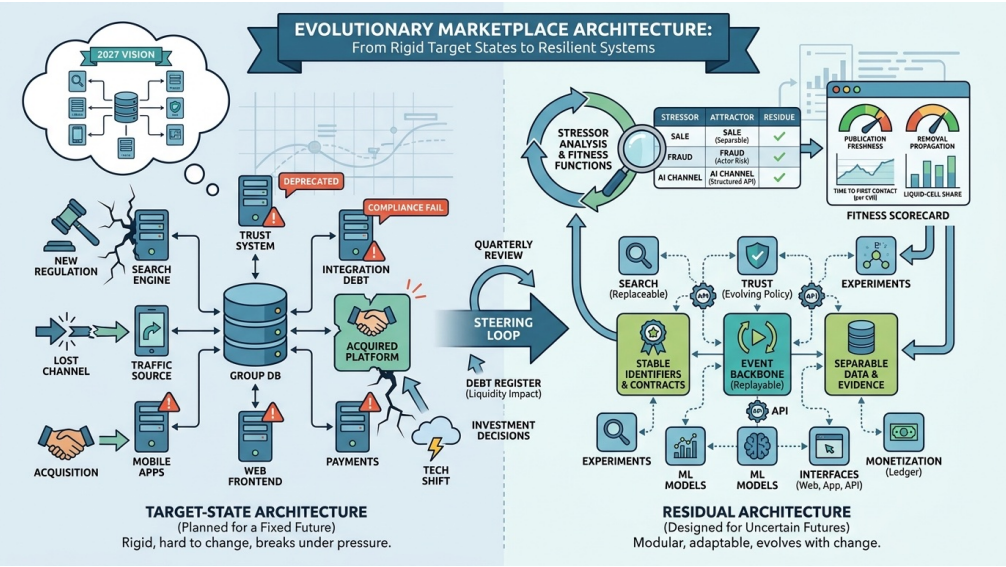
- Donating the Model Context Protocol and establishing the Agentic AI Foundation — Anthropic, 2025. MCP governance under the Linux Foundation.
- Article 50: Transparency obligations — EU AI Act (Regulation (EU) 2024/1689), 2024. Disclosure and marking duties.
- Article 53: Obligations for providers of general-purpose AI models — EU AI Act, 2024. Copyright policy and Article 4(3) reservations.
- Annex III: High-risk AI systems — EU AI Act, 2024. Employment and creditworthiness points.
- Digital Omnibus on AI — European Parliament Legislative Train, updated September 2026. Regulation (EU) 2026/1744 and postponed dates.
- AI Act implementation timeline — Future of Life Institute, updated August 2026. Application dates.

Research and practitioner sources

- Llama Guard: LLM-based input-output safeguard for human-AI conversations — Meta researchers, 2023. Policy-as-prompt safety classification with a customizable taxonomy.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

45. Built to Be Replaced: Evolving a Marketplace Platform Over a Decade



IN THIS CHAPTER, YOU WILL: Restate the book's argument in one page, then use it to plan for a decade. You will run a stressor analysis on a marketplace group, separate the assets that must outlive every system from the systems that should be replaced, design for separability as well as sharing, manage the swing between central and local control, and define the marketplace fitness functions that tell a group whether its architecture is still doing its job.

WHY THE BUSINESS CARES: Over ten years a marketplace group will almost certainly buy and sell businesses, face new laws, lose a traffic channel and change owners. The platform decides how much each of those events costs: a carve-out that takes nine months or two years, a new regulation absorbed by configuration or by a rewrite, a channel shift met with an API or with a panic.

WHY ARCHITECTS SHOULD CARE: Every system in this book will be replaced at least once within a decade: the monolith, the search engine, the cloud, the models and probably the org chart. What a Chief Architect can make durable is a small set of contracts, identifiers and measures. Choosing them, and proving every quarter that they still hold, is the long-term job.

KEY POINTS:

- **A listing marketplace is a liquidity machine, so judge its architecture by its throughput**. Every part of this book is a stage of one pipeline: supply comes in, becomes a well-formed listing, is published, found, trusted, contacted and paid for. The architecture is good if it turns fresh, trustworthy supply into matches reliably, in every cell, at a cost the business can carry, and keeps doing so while everything underneath it changes.
- **Design against stressors, not toward a target state**. A target-state diagram assumes the future is known. A stressor analysis asks what could happen (a sale, a law, a fraud wave, a lost channel), what the business would do, and what structure would have to exist for the platform to survive it. The structures that survive many stressors (the listing event contract, stable identifiers, tenant-separable data, the entitlement ledger) are where long-term investment belongs.
- **Steer by marketplace fitness functions, not by program milestones**. A dozen measured properties, such as freshness, removal latency, ghost share, time to first contact, paid share of the first page, cost per live listing and time to carve out a tenant, say whether the machine still works. Automate them where possible, drill the rest, and give each one an owner and a failure pattern it guards against.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **The Forced Fit** — A ten-year plan that fixes what is central and what is local turns into a mandate the moment the group's strategy or owner changes, and local teams route around it.
- **Deal-Shaped Architecture** — Without durable group contracts, every acquisition over a decade adds another island, and the integration debt compounds faster than any program can pay it down.
- **Vanity Liquidity** — Long programs drift toward reporting milestones, listings and visits; fitness functions tied to matches keep a decade of investment pointed at what the marketplace actually sells.

Architecture books usually end with a target state: the clean diagram of what the platform will look like once the program is done. This one ends differently, because the

documented record of marketplace groups argues against target states. In the ten years from 2016 to 2026, Schibsted pitched "one global product and tech organization" (widely reported), demerged its international marketplaces as Adevinta, Adevinta bought eBay Classifieds Group, set out to merge two of its largest product platforms, was taken private, and then **moved to "a decentralised, standalone operating model"** and sold businesses. Prosus exited OLX Autos. Zillow closed Zillow Offers. Every one of those groups had a credible architecture plan at the start of the decade. None of them had the decade they planned for.

So the question for a Chief Architect is not "what should the platform look like in ten years?" It is "**what must stay true while everything else changes, and how will we know?**" This chapter answers it in three moves. First, it restates the argument of this book, because the argument defines what must stay true: a listing marketplace is a liquidity machine, and its architecture should be judged by how reliably it turns fresh, trustworthy supply into matches with demand. Second, it introduces **stressor analysis**, a way to find the structures that survive an unknown future, adapted from Barry O'Reilly's residuality theory. Third, it defines the **marketplace fitness functions** that tell a group, every quarter, whether its architecture still does its job.

The chapter closes Part VIII. It builds on After the Deal: Consolidating Platforms After Acquisitions, Who Owns What: Organizing Marketplace Engineering, Changing the Engine in Flight: Modernizing a Live Marketplace and AI Across the Listing Lifecycle, and on the platform chapters of Part VII, especially One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture, which owns the tenancy model and the documented history of the centralization pendulum. The worked example belongs to Veldmark, the fictional marketplace group this book follows (Meet Veldmark: Four Brands, One Platform to Build). Its numbers come from the earlier chapters; this chapter puts them on one page. The cases of Part IX — PRACTICE: How Others Built It then test the argument against what real groups did.

Setting the Scene at Veldmark: The Investor's Question

On **Tuesday 9 November 2027**, ten months after approving Common Ground, Veldmark's board reviewed the integration of Casavista and Keybridge (After the Deal: Consolidating Platforms After Acquisitions). A week later the private-equity owner's operating partner sends the group CEO, Henrik de Boer, a follow-up for the board meeting on **Tuesday 14 December 2027**, in two lines. *Common Ground will cost €140 million and is meant to save €38 million a year from 2030. What does it make harder? In particular: if we decide to sell Hearthly, or Hearthly Spain alone, in 2030, what will it cost us to separate it from the platforms you are building now?*

Henrik passes the question to Ingrid Halvorsen, the Group Chief Architect, with a second one of his own. Veldmark has spent 2027 deciding things: the capability map in March,

the context map in May, Veldmark ID in June, the listing event contract in July, the search platform in September, Tally and the taxonomy service in October, the ML platform and the trust-and-safety platform in November. *"In 2037, nobody here will remember why we decided any of this. What stops it from turning into another Yardstick?"*

Ingrid sees that both questions ask the same thing: how Common Ground holds up in a world it was not designed for. It was designed against the 2027 problems: four of everything, a six-minute publication time, duplicates, payment-link phishing, a monolith that deploys twice a week. Nobody has yet tested it against the problems of 2030 or 2035. The 2027 decisions are sound for the world they assumed; the investor is asking about a world in which the group's shape changes.

She does three things in the four weeks before the board meeting:

1. She writes down **the argument behind Common Ground** in one page, so that the board and the next Chief Architect can tell which decisions serve it and which only served 2027.
2. She runs a two-day **stressor workshop** in Rotterdam on **Thursday 18 and Friday 19 November 2027**, with the brand CTOs, the platform leads, Aisha Rahman (trust and safety), Lucía Ortega (data protection and regulation), Priya Raman (data and ML), Olivier Petit (monetization) and a partner from the investor's portfolio team. They work through fourteen stressors and the combinations that matter.
3. She asks the architecture advice forum to adopt a **marketplace fitness scorecard**: twelve measured properties, each with an owner, a baseline and a target, reviewed every quarter. The forum adopts it on **Wednesday 8 December 2027**.

The answer to the investor's question is in the stressor table later in this chapter. The answer to Henrik's is the scorecard. Both rest on the argument.

The Argument: A Liquidity Machine

A listing marketplace sells one thing: **matches**. A seller wants a buyer, a dealer wants leads, an agency wants mandates, a buyer wants the right item, car or home before somebody else gets it. Listings, search, trust and monetization are not separate products; they are **stages of one pipeline whose throughput is liquidity**. Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics defines liquidity precisely: the probability that a listing finds a buyer, or a search finds a listing worth contacting, within the time window that matters in its category. It is local, measured per cell (region × category), and it is the only thing that compounds into network effects.

Read that way, every part of this book describes one stage of the machine, the guarantee that stage must give, and the way it fails.

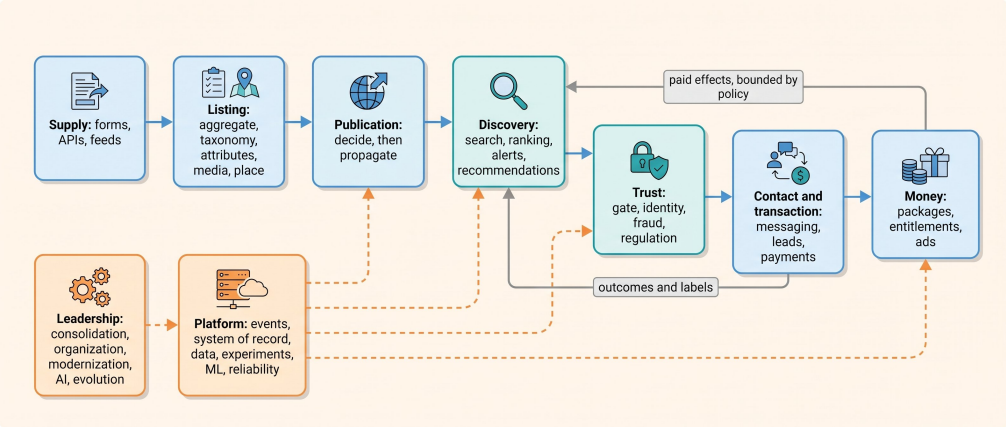


Figure 1: *The liquidity machine. Supply becomes listings, listings are published and found, trust decides which contacts are safe, contacts become matches and money. Outcomes flow back into ranking and trust; paid effects flow back into discovery under policy. The platform carries every stage; leadership decides how the platform evolves.*

The table states the argument part by part: each stage protects a liquidity metric and has a named way to fail.

Part	Stage of the machine	What the architecture must guarantee	Liquidity metric it protects	Failure pattern when it does not
I — Marketplace	The business model	The endpoint (contact, lead, transaction) is chosen per category, and capabilities and contexts follow it (Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy, Map What the Marketplace Does Before What Runs It, Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models, Same Platform, Different Physics: How Listing Verticals Differ)	Matches per cell, measured with governed definitions	Vanity Liquidity

Part	Stage of the machine	What the architecture must guarantee	Liquidity metric it protects	Failure pattern when it does not
II — Listing	The thing being sold	A listing is an offer with one owner, a lifecycle and invariants; categories and attributes are versioned data (The Listing Aggregate: Identity, Lifecycle and Invariants, Category Trees That Can Change: Designing Marketplace Taxonomies, Attributes Without Chaos: Schemas, Dictionaries and Validation, Photos Sell: Media Pipelines for Listings, Where Is It? Location, Geocoding and Privacy, Private Sellers, Dealers and Agencies: Modeling Who Lists)	Findability: a listing missing the filtered attribute does not exist for that buyer	The Category Straitjacket; The Attribute Junkyard

Part	Stage of the machine	What the architecture must guarantee	Liquidity metric it protects	Failure pattern when it does not
III — Supply	Getting inventory in	Many doors, one write path; feeds under contract; duplicates resolved; publication fast and removal faster (Many Doors In: Forms, APIs and Bulk Feeds, From Raw Supply to Usable Listings: Normalization and Enrichment, One Property, Three Listings: Duplicates and Entity Resolution, From Submitted to Searchable: The Publication Pipeline, Serving the Pros: Dealer, Agent and Merchant Tooling)	Fresh, available, deduplicated supply	Feed Roulette; Seeing Double; Ghost Inventory

Part	Stage of the machine	What the architecture must guarantee	Liquidity metric it protects	Failure pattern when it does not
IV — Discovery	Matching intent to inventory	The index is a rebuildable projection behind a contract; relevance and policy are separate; alerts re-check at send time (Search Architecture for Listings: Indices, Freshness and Scale, What Did They Mean? Query Understanding and Autocomplete, Ranking for Two Sides: Relevance, Freshness and Paid Visibility, Recommendations for Inventory That Disappears, Searching by Place: Map, Polygon and Commute Search, Tell Me When It Appears: Saved Searches and Alerts)	Search-to-contact, time to first contact	Search as System of Record; Multiplier Creep

Part	Stage of the machine	What the architecture must guarantee	Liquidity metric it protects	Failure pattern when it does not
V — Trust	Keeping matches safe	Policy, detection, decision, enforcement and review as a platform; actor-level risk; identity as claims; evidence by design (Trust and Safety as a Platform, Scams, Fakes and Account Takeovers: Fraud Prevention for Listings, Who Is on the Other Side? Identity and Verification, Rules of the Marketplace: DSA, DAC7 and Vertical Regulation)	Safe contacts; prevalence of harm	Moderation as an Afterthought; Fast Mail for Fraudsters

Part	Stage of the machine	What the architecture must guarantee	Liquidity metric it protects	Failure pattern when it does not
VI — Money	Charging for liquidity	Conversations and leads are separate objects; the unit of charge has a system of record; what was sold is proven delivered (From Interest to Contact: Messaging and Lead Management, Beyond the Contact: Payments, Shipping and Buyer Protection, How Listing Marketplaces Make Money, Packages, Bumps and Quotas: Products, Pricing and Entitlements, Selling Attention: Display, Native and Retail-Media Advertising)	Revenue that does not spend more liquidity than it earns	Billing by Guesswork; Multiplier Creep

Part	Stage of the machine	What the architecture must guarantee	Liquidity metric it protects	Failure pattern when it does not
VII — Platform	The machinery underneath	One event contract, one system of record per fact, governed metrics, two-sided experiments, tiered model governance, journey SLOs and unit costs (Every Change Is an Event: Event-Driven Listing Architecture, System of Record and Read Models: Listing Data Management, Found by Google: Web, App and SEO Architecture for Listings, Data Platform for Marketplaces: Events, Lakehouse and Data Products, Testing a Two-Sided Market:	Every metric above, at the Sunday-evening peak	The Copy-Paste Country; Set-and-Forget AI

Part	Stage of the machine	What the architecture must guarantee	Liquidity metric it protects	Failure pattern when it does not
VIII — Leadership	Keeping the machine coherent over time	Consolidation in the right order, an organization that matches the context map, migration by stage gates, AI that proposes while the platform decides, and evolution against stressors (After the Deal: Consolidating Platforms After Acquisitions, Who Owns What: Organizing Marketplace Engineering, Changing the Engine in Flight: Modernizing a Live Marketplace, AI Across the Listing Lifecycle)	All of them, over a decade	Deal-Shaped Architecture; The Forced Fit

Four ideas run through all eight parts, and they are the ones worth carrying past any particular system:

1. **The listing is an offer, not the thing, and it has one owner**. Every other context (search, alerts, trust, monetization, analytics) holds a registered copy built from one published event. Ghost Inventory, Search as System of Record and most privacy failures are copies that nobody registered.
2. **Content is data; engines are shared**. Categories, attributes, ranking policies, packages, prices, trust policies and precision rules change weekly and belong to the marketplaces that know them. Engines (search, events, identity, media, the entitlement ledger) change rarely and can serve every brand. *Share platforms, not products*, Veldmark's Common Ground principle, is this idea applied to a group.
3. **Every consequential decision leaves evidence**. Moderation decisions, ranking reasons, paid-slot deliveries, model predictions and experiment exposures are logged with versions, because regulators, professionals and the group's own analysts will ask months later what happened and expect the same answer twice.
4. **Measure matches, not activity**. Listings, visits, sends, removals and queue sizes are inputs. Contacts, leads, sales, safe conversations and outcomes per cell are what the marketplace sells. A team that optimizes an input will eventually spend liquidity to grow it.

These four ideas are the criteria for the rest of the chapter. A decision that serves them should survive the decade. A decision that served only the 2027 problem list should be easy to replace.

Concepts: Stressors, Residues and Fitness Functions

Why target states fail in marketplaces

A target-state architecture is useful for communication and dangerous as a plan, for three reasons specific to marketplace groups.

- **Ownership changes the question**. The documented record above shows capital events (mergers, take-privates, disposals, carve-outs) arriving every two or three years in the large groups. Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform tells that story in detail: shared platforms valued in a 2021 synergy case became separation costs in 2025. A target state rarely includes "half of this may be sold".

- **Regulation arrives on fixed dates but with uncertain scope**. The Digital Services Act, DAC7, the Platform-to-Business Regulation, the AI Act and the General Product Safety Regulation all changed what marketplaces must store, show and report between 2019 and 2026 (Rules of the Marketplace: DSA, DAC7 and Vertical Regulation). Each one added structure (decision records, trader records, disclosure surfaces) that no 2018 target state contained.
- **Demand channels move**. Organic search brought 46% of Tradeyard's web sessions in 2026. In 2026 marketplaces began to publish their search as tools inside AI assistants, and AI crawlers began to harvest listings (AI Across the Listing Lifecycle). A channel that carries half the traffic can shrink faster than a platform program can finish.

The alternative is not to stop planning. It is to plan for **properties** that must hold in many possible futures, and to treat any specific system as replaceable.

Stressors, attractors and residues

Residuality theory, developed by Barry M. O'Reilly in a series of articles from 2020 and in his 2024 book *Residues: Time, Change, and Uncertainty in Software Architecture*, offers a vocabulary for planning against an unknown future. I use it in a simplified, practical form; readers who want the theory should read O'Reilly.

- A **stressor** is anything that could happen to the business or its environment: a law, a competitor's move, a fraud technique, a supplier's failure, a sale. It is deliberately not limited to likely events or to technical risks.
- An **attractor** is the state the business is pushed into when the stressor hits: the way people, partners and the market will actually behave. A carve-out pushes the business into transitional service agreements; a fraud wave pushes it into blanket friction.
- A **residue** is what remains of the architecture after the stressor: which parts still work, which break and what structural change would let them survive. Designing for residues means adding the separations, contracts and records that the stressed business will need.
- **Naive architecture** is the design that works under normal conditions. **Residual architecture** is the naive design plus the residues that survive the stressors considered.

The two Productscape domain models from which this book started (generic models of a general-listings and a real-estate marketplace, not descriptions of any company) include exactly this exercise. Their general-listings model starts from a naive design ("a seller creates an account and listing, the marketplace classifies and ranks it, buyers discover it, message the seller") and stresses it with a product-safety law, synthetic-identity fraud

and a ban on opaque behavioral ranking. Its residual architecture "separates actor, identity evidence, account, role, listing, physical item, product identity, safety status, jurisdiction, policy, moderation decision, interaction, channel, transaction protection, ranking candidate, ranking reason, paid placement, and consent". The real-estate model, stressed with climate remapping, a credit shock and title fraud, ends with a property whose identity persists "while risk, value, financing, listing authority, and transaction readiness change independently". Those lists are a fair summary of Parts II to VI of this book, reached from the other direction.

Two properties of the method make it useful at group scale. First, **residues recur**: the same few structures (an event contract, layered identity, versioned policy, an evidence record) survive many unrelated stressors. Second, **combined stressors test the decomposition**: when two or three stressors hit together and the platform survives by composing existing residues rather than by needing new ones, the decomposition is probably right. Both models end with such a combined test, and both pass it by composing a case record from residues they already had.

Figure 2: *Design against stressors, not toward a target state: many unrelated stressors are survived by the same few residues, such as the listing event contract, stable identifiers, tenant-separable data and the entitlement ledger.*

I define **stressor analysis** for this book as the practice of listing stressors for a marketplace group, recording for each the attractor, what breaks in the current architecture and the residue that would survive it, and then counting which components and contracts carry the most residues. Those components deserve the most care, the best owners and the slowest change.

Fitness functions

The second concept, the **fitness function**, comes from evolutionary architecture: an objective, preferably automated check that an architecture still has a property it was designed to have. Neal Ford, Rebecca Parsons, Patrick Kua and Pramod Sadalage, in *Building Evolutionary Architectures* (first edition 2017, second edition 2022), give examples such as a dependency rule in the build, a latency limit in production and a resilience test on a schedule. Auto Trader's engineers described in a 2026 post how they run ArchUnit tests in the build to enforce layer boundaries and conventions, a code-level example of the idea.

A **marketplace fitness function**, as this book uses the term, is a fitness function whose property is a stage of the liquidity machine: a measured, owned, regularly evaluated statement such as "a sold listing disappears from every registered consumer within two minutes at p95" or "no more than four paid listings appear in the first twenty results". It differs from a business KPI in having an architectural owner and an architectural remedy, and from a service-level objective in measuring a property of the whole machine rather than of one service. The scorecard later in this chapter is the set Veldmark adopted.

Built to be replaced

The chapter's title states its stance: **systems are built to be replaced**; **contracts, identifiers and measures are built to last**. Yardstick was a reasonable design in 2006 and is the problem of 2027 not because PHP aged but because its tables became the group's contract with everything else: 214 columns in the `ads` table read by 38 modules, a digest that read a MySQL replica, ranking boosts that lived in Solr configuration. When a system's internals become everyone's interface, replacing the system means replacing the interface, and nobody can afford that. The long-term architecture work is to keep interfaces deliberate, so that the systems behind them can be replaced one by one.

What Lasts and What Gets Replaced

The first output of Ingrid's paper is a sorting of the 2027 decisions by how long they are meant to last. It is the most useful single page for the next Chief Architect, because it says which decisions are load-bearing and which are bets that should be revisited on schedule.

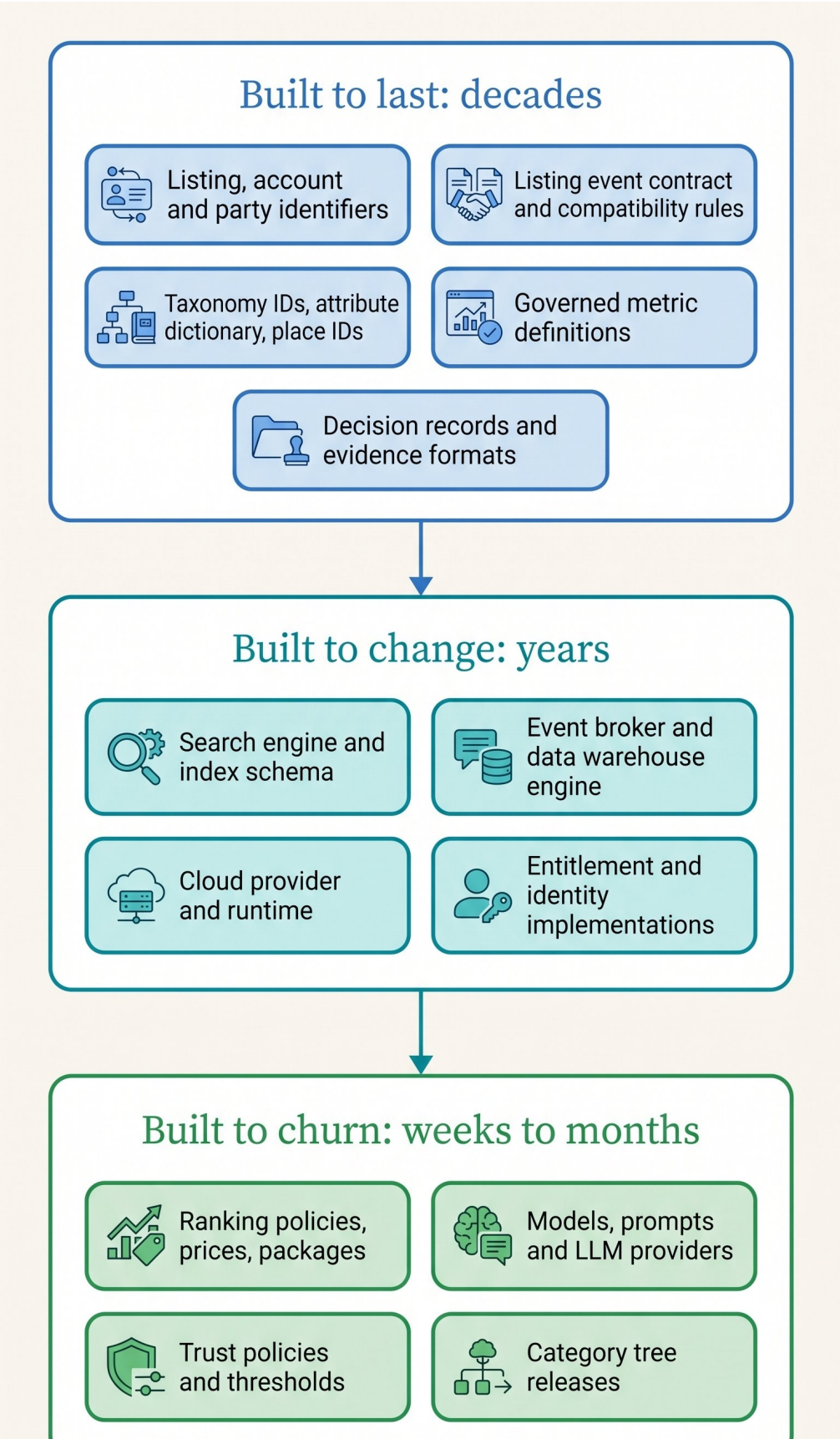

Built to last: decades
Listing, account and party identifiers
Listing event contract and compatibility rules
Taxonomy IDs, attribute dictionary, place IDs
Governed metric definitions
Decision records and evidence formats
Built to change: years
Search engine and index schema
Event broker and data warehouse engine
Cloud provider and runtime
Entitlement and identity implementations
Built to churn: weeks to months
Ranking policies, prices, packages
Models, prompts and LLM providers
Trust policies and thresholds
Category tree releases

Figure 3: *Three rates of change. The top layer is what other systems, partners and regulators depend on; it changes only by versioned contract. The middle layer is replaceable behind those contracts. The bottom layer is content that brands change as data, through review rather than release.*

Layer	Veldmark 2027 decisions	Expected life	How it changes	Owner
Contracts and identifiers	Group listing ID `vm:<brand>:<country>:<id>`; Listing event contract v1 (FULL_-TRANSI-TIVE compatibil-ity); lifecycle states and enforce-ment overlay; taxonomy IDs and calendar-versioned releases; Veldmark Place IDs; provenance vocabulary; Supply Contract v1; Listing Search API v1; Qualified contact v1.0 and the governed metrics catalog	Ten years or more	New versions beside old ones; deprecation windows; architecture advice forum	Group architecture with the owning platform team

Layer	Veldmark 2027 decisions	Expected life	How it changes	Owner
Engines and implementations	OpenSearch shared search platform (Motorvane on Elasticsearch until the 2029 review); Kafka on Confluent Cloud; Apache Iceberg on S3 with Snowflake compute; Veldmark ID; Tally; MLflow registry; AWS as primary cloud	Three to eight years	Replaced behind the contract, with parallel run and parity checks (Changing the Engine in Flight: Modernizing a Live Marketplace)	Platform teams under Yara Mendes, Tomasz Wiśniewski, Sven Lindqvist, Priya Raman
Content and policy	Ranking policy v1; Tradeyard prices and packages; trust policies and thresholds; category tree 2027.11; Valuo pricing-hint rules; AI assistant prompts	Weeks to two years	Released as data, with experiments and review	Brands, with T&S, monetization and data protection

The top layer deserves a disproportionate share of design attention, even though it is the smallest. A wrong field in the listing event envelope will cost more over a

decade than a wrong search engine, because the engine can be replaced behind the search contract in a year, while the envelope is copied into 23 consumer groups at go-live and many more later.

The middle layer should be chosen for replaceability as much as for fit. Veldmark's September 2027 decision to put every product application on Listing Search API v1 *before* any engine changes is the clearest example: it made the engine a middle-layer decision. The deferred Vespa evaluation, the 2029 review of Motorvane's Elasticsearch and the 2029 cloud review for Hearthly are scheduled revisits, not open questions.

A useful test for any new proposal at the forum: *which layer does this belong in, and does its design let the layer below it change without touching the layer above?* A proposal that puts a brand's ranking rule into the event envelope fails the test; so does one that lets a consumer read the search engine's internal document format.

Stressor Analysis for a Marketplace Group

The method

The workshop's value comes from the breadth of the stressors and the honesty of the "what breaks" column, not from its loop, which is simple.

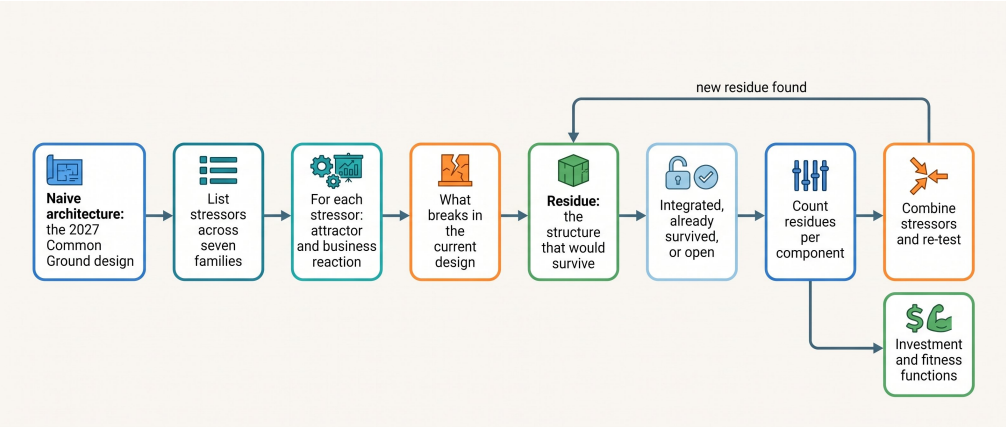


Figure 4: *Stressor analysis as Veldmark runs it. Combined stressors are the check: if the platform survives them by composing residues it already has, the decomposition holds; if not, the loop adds a residue.*

The stressors are deliberately drawn from seven families, because architects left alone list mostly technical risks:

Family	Examples for a listing marketplace group
Capital and portfolio	A brand is sold or carved out; a competitor is acquired; the owner changes; a joint venture is formed or unwound
Regulation	New platform duties (safety, ranking transparency, AI); new tax reporting; data localization; a competition case
Demand channel	Organic search traffic shrinks; AI assistants become a main discovery channel; an app store changes its terms
Supply ecosystem	A dominant feed vendor is bought by a competitor; professionals boycott a price rise; multi-homing becomes free
Adversarial	Fraud with synthetic identities and generated media; mass scraping; account takeover waves
Technology and suppliers	A managed service changes price or license; a cloud region becomes unacceptable; a model provider withdraws a model
Physical world and markets	Climate remapping changes property values; a credit shock; a used-car price collapse; a pandemic-style shift in demand

Veldmark's stressor table

Fourteen stressors were worked through over the two days; the table shows ten, each written as the workshop wrote it. "Status" follows the Productscape convention: *integrated* means a 2027 decision already provides the residue, *already survived* means existing residues compose to cover it, and *open* means a new decision is needed.

#	Stressor	Attractor (what the business would do)	What breaks in the 2027 design	Residue that survives	Status
S1	**The owner sells Hearthly (or Hearthly Spain alone) in 2030**	Carve-out with transitional service agreements; the buyer wants clean data, accounts and contracts on day one	Shared trust-and-safety evidence, the hashed block list, Tally ledgers and data products pool data from several tenants; Hearthly.es login still runs on Casavista code	Tenant codes (`hl:es`) on every identifier, event and log under Tenancy model v1; pairwise identifiers in Veldmark ID; tenant- and time-partitioned Iceberg tables; a documented and tested exit path for every shared platform; contributions to shared lists tagged with their source tenant	Partly integrated; T&S and Tally exports open

#	Stressor	Attractor (what the business would do)	What breaks in the 2027 design	Residue that survives	Status
S2	**Veldmark acquires a Portuguese property portal**	Keep the brand; integrate under pressure to show synergies within 18 months	Nothing breaks immediately; the risk is a fifth island of everything	Capability map in the first month; Supply Contract v1 for its feeds; an anti-corruption layer into the listing event contract; the consolidation sequence of After the Deal: Consolidating Platforms After Acquisitions	Already survived

#	Stressor	Attractor (what the business would do)	What breaks in the 2027 design	Residue that survives	Status
S3	**Product-safety duties tighten for C2C goods**: a recalled e-bike battery must be withdrawn from every listing that offers it	Withdraw affected listings, notify buyers who contacted them, keep evidence	Goods listings have no product identity; recall matching relies on free text	Enforcement overlay with outcome codes; exposure and contact records per listing; recall matching on attributes plus a classifier, routed to T&S cases. *Not* a canonical goods entity, which One Property, Three Listings: Duplicates and Entity Resolution advises against	Partly integrated

#	Stressor	Attractor (what the business would do)	What breaks in the 2027 design	Residue that survives	Status
S4	**A ban on opaque behavioral ranking**, with paid placement explained per result	Offer a non-personalized ranking mode; explain paid influence	Nothing structural, since Ranking policy v1 separated relevance from policy and logs reason codes	Ranking policy layer; reason codes and paid-slot delivery events; non-personalized mode per surface (Ranking for Two Sides: Relevance, Freshness and Paid Visibility, Recommendations for Inventory That Disappears)	Integrated

#	Stressor	Attractor (what the business would do)	What breaks in the 2027 design	Residue that survives	Status
S5	**AI assistants carry a third of discovery by 2032**, and organic web sessions halve	Publish search to assistants; compete for attribution; renegotiate data licensing	Landing pages and SEO lose value; leads arrive without session context; partners may cache listings	Search as an API product with quotas and per-channel attribution; listing data licensing terms; structured, API-grade inventory (AI Across the Listing Lifecycle)	Open in November 2027; closed by the February 2028 decision record on the external assistant channel

#	Stressor	Attractor (what the business would do)	What breaks in the 2027 design	Residue that survives	Status
S6	**Fraud rings use synthetic identities and generated photos and video**	More friction on new accounts; pressure to verify everyone at the door	Photo hashes miss generated images; account-level scoring misses rings that replace accounts	Actor and graph scoring; media provenance per asset; identity as claims with step-up at the moment of risk (Scams, Fakes and Account Takeovers: Fraud Prevention for Listings, Who Is on the Other Side? Identity and Verification, Photos Sell: Media Pipelines for Listings)	Integrated in design; provenance open

#	Stressor	Attractor (what the business would do)	What breaks in the 2027 design	Residue that survives	Status
S7	**Motorvane's largest DMS vendor (about 1,900 dealers) is bought by a competitor** and throttles its feed	Move dealers to another channel within weeks	The vendor holds integration credentials and the dealers' stock flow	Integration principals with per-dealer grants; the Seller API as a direct channel; Supply Contract v1 with field ownership and reconciliation (Private Sellers, Dealers and Agencies: Modeling Who Lists, Many Doors In: Forms, APIs and Bulk Feeds)	Partly integrated; dealer migration playbook open

#	Stressor	Attractor (what the business would do)	What breaks in the 2027 design	Residue that survives	Status
S8	**The managed search service doubles its price or changes license**	Move engine or renegotiate under time pressure	Nothing, if every application uses the search contract and the index can be rebuilt	Listing Search API v1; quarterly rebuild drills; index as a projection (Search Architecture for Listings: Indices, Freshness and Scale)	Integrated
S9	**A climate risk remap in France reclassifies flood zones in Lyon**	Correct estimates and disclosures for affected properties; answer agencies' disputes	Valuo estimates and pricing hints are recomputed without recording which map version they used	Dated, versioned facts on the canonical property; estimate versions with stated assumptions; event-triggered rescoring (Models in Production: ML Platforms and Use Cases)	Open

#	Stressor	Attractor (what the business would do)	What breaks in the 2027 design	Residue that survives	Status
S10	**Keybridge faces a competition complaint** from a rival portal	Prove that Hearthly never used Keybridge data	Nothing structural; the evidence must be producible quickly	Separate tenancy; contract-named data flows; access policy and audit log readable by the independent reviewer (Serving the Pros: Dealer, Agent and Merchant Tooling, Data Platform for Marketplaces: Events, Lakehouse and Data Products)	Integrated

The four stressors not shown were a deeper transactional pivot at Tradeyard (survived by the profile register and the transaction context of Beyond the Contact: Payments, Shipping and Buyer Protection), a model provider withdrawing the model behind the photo-to-listing assistant (survived by the AI gateway), a data-localization requirement for Polish data (open, low likelihood), and a professional boycott of Motorvane's 2028 price event (survived by price-as-data and the lead ledger, but with real commercial cost).

Three things about the table matter more than any single row. First, **most residues already existed**. Eight of the ten stressors are fully or partly covered by 2027 decisions taken for other reasons, which is evidence that Common Ground's decomposition is sound. Second, **the open residues cluster**: tenant-scoped exports for shared platforms, media provenance, and dated facts on the canonical property. Those are the investments that the 2027 problem list never asked for. Third, **S1 is the stressor the 2027 design**

made worse. Before Common Ground, selling Hearthly would have been cheap, because the group shared almost nothing. Every shared platform adds separation cost. That is not an argument against sharing; it is an argument for paying the small, early price of separability.

Counting residues: where long-term investment belongs

The second exercise counts, for each Common Ground component, how many of the fourteen stressors require it to change or to carry a residue. The count is crude, but it ranks components by how much of the group's future they carry.

Component	Stressors that touch it (of 14)	Reading
Listing event contract and listing core	11	The group's most load-bearing artifact; change it only by versioned contract, give it the best owner
Identity (Veldmark ID and the business party registry)	8	Separability and fraud both run through it; pairwise identifiers are not optional
Data platform and governed metrics	8	Every carve-out, regulator and dispute needs it; tenant partitioning is the residue
Trust-and-safety platform	7	Shares evidence across brands, so it needs tenant provenance on everything it pools
Tally (entitlements)	5	Price events, carve-outs and new models all land here
Search platform and its contract	5	Protected by the contract; the engine itself is replaceable
Taxonomy, attributes and Veldmark Place	4	Recalls, AI channels and acquisitions depend on stable IDs
AI gateway	3	Provider churn is contained here

Component	Stressors that touch it (of 14)	Reading
Media service	2	Mostly a cost question; safe to optimize for price

The ranking matches the layering of Figure 3: the components carrying most residues are the ones whose contracts sit in the top layer. It also gives Ingrid an argument she uses at the board. **The places to spend Common Ground's quality budget are the event contract, identity, the data platform and the T&S platform's tenancy model.** The media service, by contrast, can be optimized for cost without much long-term risk.

A combined stressor: selling Hearthly Spain in a bad month

On the second afternoon the workshop combines three stressors into one scenario, set in **July 2030**: the investor agrees to sell Hearthly Spain; in the same month a fake-rental ring of the Barcelona 2027 kind returns, this time with generated photos that no perceptual hash matches; and an AI assistant partner sends a fifth of Hearthly Spain's search traffic.

The group walks the scenario through the residues it has:

- **The sale.** Hearthly Spain's accounts in Veldmark ID are separable because each brand sees only pairwise identifiers; the buyer receives brand profiles and credentials, not group identities. Its data products sit in tenant-partitioned Iceberg tables and can be exported with their contracts. Its listings already carry `vm:hearthly:es:<id>` identifiers that the buyer can keep.
- **The fraud.** The shared T&S platform links the new ring to accounts, devices and IBANs via the actor graph rather than via photo hashes. The ring is stopped through actor-level scoring and the bank liaison feed.
- **The channel.** Leads from the assistant are attributed per channel and partner under the 2028 contract; the contract is per brand, so it transfers with Hearthly Spain.

One residue is missing. The shared hashed block list holds identifiers from all brands, including those contributed by Hearthly Spain's cases, and nothing records which tenant contributed which entry. After the sale, Veldmark may not keep data that the buyer's business generated, and the buyer has no legal basis to receive identifiers contributed by Tradeyard. **The block list needs source-tenant provenance on every entry**, and the shared T&S platform needs a tested per-tenant export of cases and evidence. Both were added to the T&S platform backlog for 2028.

This is what the combined test is for. The scenario needed no new components, only one field and one export on an existing platform: a cheap residue found in 2027 instead of an expensive one discovered during a carve-out.

The answer to the investor

Ingrid's paper answers the investor's question with an estimate, labeled as one. With the separability residues in place (pairwise identities, tenant-partitioned data, tenant-scoped exports from the shared T&S platform and Tally, brand-owned contracts), a 2030 carve-out of Hearthly would need transitional services for **seven** shared capabilities (Veldmark ID, the T&S platform, the search platform, the event backbone, the data platform, Tally, and the taxonomy and place services) for about **nine to twelve months**. Without those residues, her team estimates twenty months or more, dominated by untangling identities and pooled evidence. The incremental cost of the residues is small compared with Common Ground's €140 million, because most of them are design rules rather than systems.

The board accepts the paper with one addition from the investor: a **separability tabletop** for Hearthly Spain in the third quarter of 2028, repeated every year, in which each shared platform's owner shows how its tenant export and switch-off would work.

Keeping Listing Data and Events Portable

Portability is the practical form of "built to be replaced". It means that the group's listings, accounts, events and evidence can move to a new system, a new owner or a new cloud without being reverse-engineered from the old one. The rules come from the earlier chapters; the table collects them with the stressors they answer.

Rule	Where it is designed	Stressors it answers
Own the identifiers. Listing, party, account, place and taxonomy IDs are minted by the group, never by an engine, a vendor or a cloud service	The Listing Aggregate: Identity, Lifecycle and Invariants, Private Sellers, Dealers and Agencies: Modeling Who Lists, Where Is It? Location, Geocoding and Privacy, Category Trees That Can Change: Designing Marketplace Taxonomies	Engine change, acquisition, carve-out

Rule	Where it is designed	Stressors it answers
Publish one event contract and keep it replayable. Current public state on every event, a compacted state topic, a backfill path	Every Change Is an Event: Event-Driven Listing Architecture	Engine change, new consumers, carve-out, regulator questions
One system of record per fact; every copy registered with its owner, purpose and rebuild path	System of Record and Read Models: Listing Data Management	Erasure, carve-out, Ghost Inventory
Store the durable copy in open formats, partitioned by tenant and time	Data Platform for Marketplaces: Events, Lakehouse and Data Products	Cloud change, warehouse change, carve-out
Pairwise identifiers per brand in the group identity	Who Is on the Other Side? Identity and Verification	Carve-out, privacy, cross-brand profiling rules
Entitlements as an append-only ledger with delivery evidence	Packages, Bumps and Quotas: Products, Pricing and Entitlements	New pricing models, disputes, carve-out of a billing relationship
Contracts in front of every engine: search, media, AI gateway, notification	Search Architecture for Listings: Indices, Freshness and Scale, Photos Sell: Media Pipelines for Listings, AI Across the Listing Lifecycle	Supplier price or license changes, model withdrawal
Configuration over code for per-market differences: endpoints, Safe Deal gates, precision levels, ranking policies	One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture, Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy	New countries, regulation by country, transactional pivots

Portability does not mean avoiding managed cloud services; Veldmark runs Kafka on Confluent Cloud, search on Amazon OpenSearch Service and compute on Snowflake. It means **accepting lock-in deliberately, with an exit estimate**. For each managed

service, the forum records what the group would do if it had to leave within a year, what that would cost, and which contract makes it possible. A managed service behind a group-owned contract with open-format data is a reasonable bet. A managed service whose proprietary API is called directly from forty product services is a decision nobody made.

Separability is the strongest form of portability, and the one most groups discover late. Veldmark wrote it into its rules early: Tenancy model v1, adopted at the forum on Wednesday 9 June 2027, makes the tenant code (brand × country, such as `hl:es`) mandatory in every identifier, event, API call, log, metric and cost tag, and requires every shared capability to document a tenant exit path (One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture). The stressor workshop's contribution is to test those paths rather than trust the documents.

Adevinta's is the documented case of what separation costs. Its 2025 strategic report states that the group "evolved from a centralised structure to a decentralised, standalone operating model", including "the ramp-down of central functions, the transfer of key capabilities, the localisation of intellectual property", and that cloud emissions more than doubled, which it expected to be "a temporary spike linked to our ongoing decentralisation, which required parallel running of central and local services during a transitional period". The parallel running is the cost of separation. It is paid in money, in carbon and in engineering attention, at exactly the moment the group least wants to spend them.

Managing the Centralization Pendulum

One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture describes the pendulum: groups swing between local autonomy and central platforms, and the documented record shows the swing in Schibsted's 2016 "one global product and tech organization", in Adevinta's 2019 rule to centralize where "investment requirements are large and the technology is uniform", in its 2023 decision to converge the leboncoin and Kleinanzeigen platforms into "a unified global platform", and in its 2025 decentralization. This chapter adds the long-term view: **the pendulum is driven mostly by things architecture does not control** (capital strategy, market maturity, leadership) and the architecture's job is to make each swing cheaper.

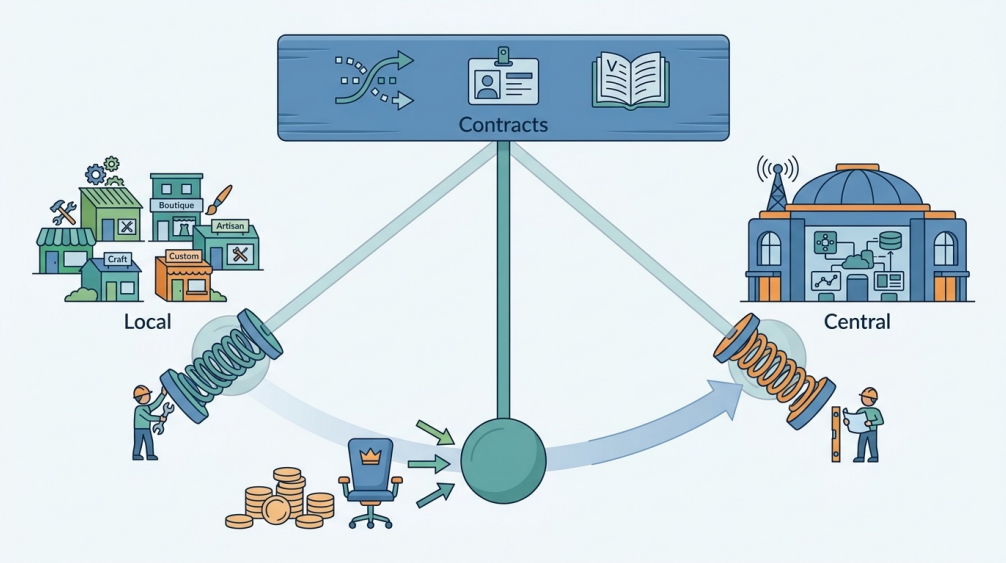


Figure 5: *The centralization pendulum is driven mostly by capital strategy, market maturity and leadership; the architecture's job is to make each swing cheaper, which starts with sharing contracts and vocabularies that survive either direction.*

At the forum, the question changes from "should this be central or local?" to "**if we have to reverse this decision in four years, what will it cost?**" The answer depends on what is being shared.

What is shared	Example at Veldmark	Value of sharing	Cost to reverse	Long-term stance
Contracts and vocabularies	Listing event contract, taxonomy IDs, governed metrics, provenance vocabulary	High: every later integration is cheaper	Low: a contract can be kept by separated businesses	Share early, almost always
Engines as multi-tenant platforms	Search platform, T&S platform, Tally, Veldmark ID	High where the technology is uniform and investment large	Medium: tenant exports, transitional services, parallel running	Share where the stressor count is low or separability is designed in

What is shared	Example at Veldmark	Value of sharing	Cost to reverse	Long-term stance
Infrastructure and tooling	Kubernetes golden path, CI, observability, AI gateway	Medium: scale and skills	Low to medium: workloads can be redeployed	Share by adoption, not by mandate
Product platforms	One insertion flow or one search UX for several brands	Uncertain; depends on how similar the markets are	High: brands lose their product capability, and getting it back takes years	Share only with a scoped investment case and stage gates

Veldmark's *share platforms, not products* sits deliberately in the first three rows. The fourth row is where Adevinta's 2023 convergence decision sat, and the public record does not show that it was completed before the 2025 reversal (Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform). The lesson I draw is not that convergence is wrong. It is that **the cost to reverse should be part of every sharing decision**, written in the decision record next to the expected savings, and that the most reversible forms of sharing should be done first.

Three practices keep the pendulum cheap at Veldmark:

- **Every shared platform has a tenant-exit test**. The annual separability tabletop is the fitness function for it.
- **Brands keep the capability to build product**. The stream-aligned teams of Who Owns What: Organizing Marketplace Engineering stay in the brands, so a swing toward local control does not start from zero.
- **Platform adoption is measured, not mandated, below the contract layer**. The contract layer is mandatory: every brand publishes the listing event contract and uses the search contract. Above it, a brand that keeps its own implementation must meet the same fitness functions and explain the duplicate in the capability map. Reasons are recorded; Keybridge's distribution stack is the deliberate duplicate on Veldmark's map, and Motorvane's Elasticsearch is a dated one.

Technical Debt as a Liquidity Tax

Over a decade, technical debt is the main way architecture loses coherence. The usual framing, debt as code quality, is too narrow for a marketplace. The debt that matters is

the debt that **slows a stage of the liquidity machine or hides its true state**. Classifying it by the fitness function it hurts turns a debt register into an investment ranking.

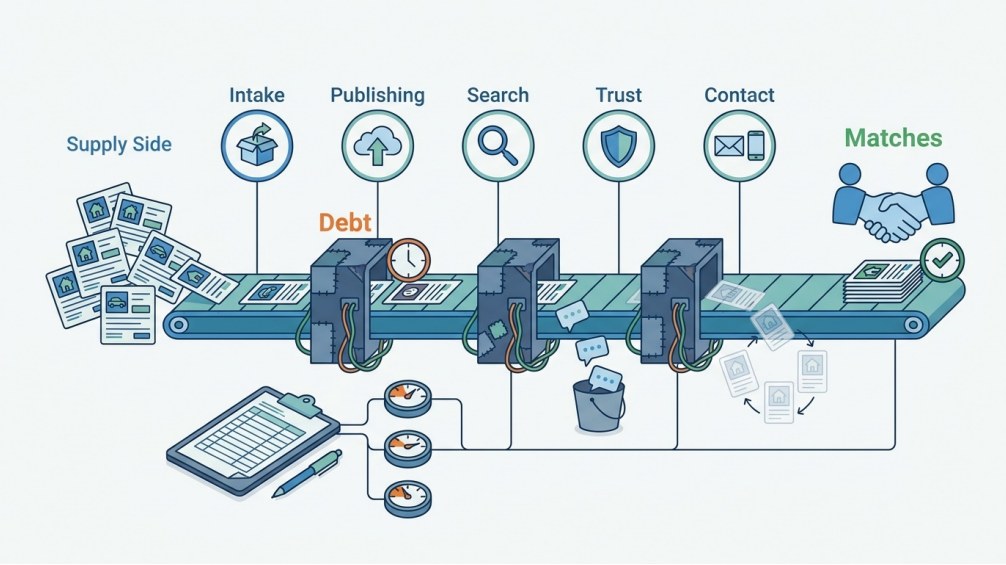


Figure 6: *Technical debt in a marketplace is a liquidity tax: the debt that matters is the debt that slows a stage of the liquidity machine or hides its true state, so each item names the fitness function it hurts.*

Kind of debt	Veldmark example (2027)	What it taxes	Fitness function that exposes it
Structural	Yardstick: one deployable for 140 engineers, two deploys a week, 14 systems still reading its tables in September 2027; a phishing rule reached Tradeyard BE three days late and never reached PL/PT	Speed of every change, including trust responses	Change lead time for policy and trust rules

Kind of debt	Veldmark example (2027)	What it taxes	Fitness function that exposes it
Copy debt	The bike digest read a MySQL replica snapshot; about 41,000 Tradeyard NL listings a week still present in a downstream copy 24 hours after ending	Buyer trust; contacts wasted on gone listings	Removal propagation; ghost share
Policy debt	Five layers of paid boosts accumulated in Yardstick's Solr configuration since 2016; paid listings held 38% of Tradeyard NL's first-page positions	Search-to-contact and new-listing exposure	Paid share of the first page; new-listing contact rate
Definition debt	Three definitions of "contact" across brands; 1,140 distinct event names in Tradeyard's apps	Every comparison and experiment	Share of steering metrics that are governed
Schema debt	3,900 attribute keys, 41% used on fewer than 0.1% of listings; a new category took six weeks	Findability and taxonomy change	Category change lead time; filterable-attribute coverage
Model debt	23 production models without a named owner, 41 without monitoring	Silent decline of ranking, fraud and pricing quality	Registry and monitoring coverage

Two rules make the register useful. First, **every item names the fitness function it hurts and an estimate of the effect**, in the same currency as the business uses: contacts, ghost contacts, minutes of delay, euros. "Yardstick is old" is not an item; "the phishing rule took three days to reach Tradeyard BE and never reached PL/PT" is. Second, **debt items compete for funding against features through the same liquidity case**. Ranking policy v1 is the example: it was framed as debt repayment (boost archaeology) and justified as a liquidity improvement (search sessions with a contact up 5.3% in the cluster-randomized test), at a cost of 2.5% of VAS revenue per search.

Debt never reaches zero, and should not. A decade-long platform will always carry some deliberate debt: a dated duplicate, a legacy adapter, a migration not yet finished. The fitness functions are what keep deliberate debt from becoming invisible debt.

Marketplace Fitness Functions: The Scorecard

The scorecard adopted by the architecture advice forum on Wednesday 8 December 2027 has twelve functions in four groups. Each has a definition, a 2027 baseline from the measurements earlier chapters describe, a target, an owner, and the failure pattern it guards against. Some are production monitors, some are build-time or replay checks, and some are drills, because not every property can be tested continuously.

#	Fitness function	Definition	Yeldmark baseline (2027)	Target	Kind	Owner	Guards against
Supply							

#	Fitness function	Definition	Yeldmark baseline (2027)	Target	Kind	Owner	Guards against
F1	**Publication freshness**	p95 platform latency from submission receipt to searchable, for listings cleared by the automated gate	Tradeyard 6 min (about 2 min on Solr with the event-driven indexer by end of 2027)	90 s by end of 2028	Monitor	Listing platform team	Vanity Liquidity
F2	**Removal propagation**	p95 from a terminal transition to removal in search, and in every registered consumer	About 1 min in search; unknown for most consumers; ~41,000 lingering copies a week (Tradeyard NL)	Search 10 s; every registered consumer 2 min; nothing sent for a listing that ended before the send	Monitor	Listing platform team	Ghost Inventory

#	Fitness function	Definition	Yeldmark baseline (2027)	Target	Kind	Owner	Guards against
F3	**Ghost share**	Share of alert and digest items, and of recommendation clicks, that point to unavailable listings	~4% of Tradeyard digest items; 3.8% of Motorvane carousel clicks before the rebuild (0.3% after)	Under 0.5% on every surface	Monitor	Search platform and alerts	Ghost Inventory
F4	**Duplicate exposure**	Listings per canonical entity (tracked) and share of first-page cards that duplicate an earlier card (targeted)	Hearthly FR 1.7 listings per property	One card per cluster; abuse duplicates routed to T&S	Monitor and sample	Hearthly listing data	Seeing Double

Matching

#	Fitness function	Definition	Yeldmark baseline (2027)	Target	Kind	Owner	Guards against
F5	**Time to first contact**	Median from searchable to the first qualified contact, per cell	NL 2.6 h; PT 19 h	Per-cell targets set by brands; trend reviewed quarterly	Monitor	Data & ML with brands	Vanity Liquidity
F6	**Liquid-cell share**	Share of cells with at least 25 active listings, 50 searches a week and median time to first contact under 24 h	NL 38% of cells (86% of listings); PT 11% (52%)	Rising per market; no target imposed from the center	Monitor	Group CPO	Vanity Liquidity

#	Fitness function	Definition	Yeldmark baseline (2027)	Target	Kind	Owner	Guards against
F7	**Paid share of the first page**	Share of first-page positions held by paid listings, with a relevance floor in place	Tradeyard NL 38% (17% in the Ranking policy v1 test)	At most 4 paid in the first 20 positions	Monitor	Ranking council per brand	Multiplier Creep
Trust							
F8	**Harm prevalence and time to action**	Prevalence of policy-violating live listings from blind sampling; median time to action on fraud reports	Tradeyard NL 0.6% prevalence; 3 h 10 min to action	Prevalence falling; under 30 min to action by end of 2028	Sample and monitor	Group T&S	Moderation as an Afterthought; Fast Mail for Fraudsters

#	Fitness function	Definition	Yeldmark baseline (2027)	Target	Kind	Owner	Guards against
Economics and evolvability							
F9	**Unit cost**	Platform run cost per live listing per month, and search cost per 1,000 queries, by brand	Tradeyard €0.28, Motorvane €1.55, Hearthly €0.77 per listing; group €0.042 per 1,000 queries	Falling at constant SLOs; €38M a year run-cost reduction by 2030	Monthly report	CTO, with platform owners	Deal-Shaped Architecture
F10	**Change lead time for content**	Time from an approved category change, price event or trust rule to production in every market of the brand	New category about 6 weeks; phishing rule 3 days to BE, never to PL/PT	Additive category change 2 days (met by "Fatbikes" in Dec 2027); trust rules same day in all markets	Monitor	Taxonomy team, T&S	The Category Straitjacket; The Copy-Paste Country

#	Fitness function	Definition	Yeldmark baseline (2027)	Target	Kind	Owner	Guards against
F11	**Rebuild and replay**	Each market's search index is rebuilt from the event stream and system of record, and a new consumer is bootstrapped from the state topic, on schedule	Not tested before 2027	Quarterly rebuild drill passed per market index	Drill	Search platform; event platform	Search as System of Record

#	Fitness function	Definition	Yeldmark baseline (2027)	Target	Kind	Owner	Guards against
F12	**Separability and registration**	Share of copies and production models registered with an owner; annual tenant-exit tabletop for one brand per shared platform	64 registered copies; 23 of 146 models without owner, 41 without monitoring; no tabletop	100% registration (models by 31 Mar 2028); first tabletop for Hearthly ES in Q3 2028	Drill and audit	Group architecture	Set-and-Forget AI; The Forced Fit

Five design choices in the scorecard are the parts other groups tend to get wrong.

Fewer functions, each with an owner. Twelve is about as many as a quarterly review can discuss. Each function has one owner who can act on it; F6, the liquid-cell share, belongs to the Group CPO rather than to architecture, because the remedies are commercial as often as technical.

Targets are set where the remedy lives. F5 and F6 have no central target. Time to first contact in a Portuguese rural district depends on supply, marketing and seller acquisition more than on platform latency; a central target would push teams to game the measure. The center sets the definition and the cell key, the brands set the targets.

Counter-metrics sit next to the metrics. F1 (publication freshness) is reported beside F2 (removal propagation) and F8 (harm), so that nobody can speed up publication by

weakening the gate. F7 is reported beside VAS revenue per search, so that the trade-off is visible rather than hidden.

Drills count as fitness functions. Rebuildability and separability cannot be monitored; they can only be shown. A drill that has not been run in two quarters is a failed fitness function, not a missing data point.

The scorecard is in the architecture repository, versioned. Definitions change through decision records, and the governed metrics catalog of Data Platform for Marketplaces: Events, Lakehouse and Data Products computes F5, F6 and the contact counts, so the scorecard and the business dashboards use the same numbers.

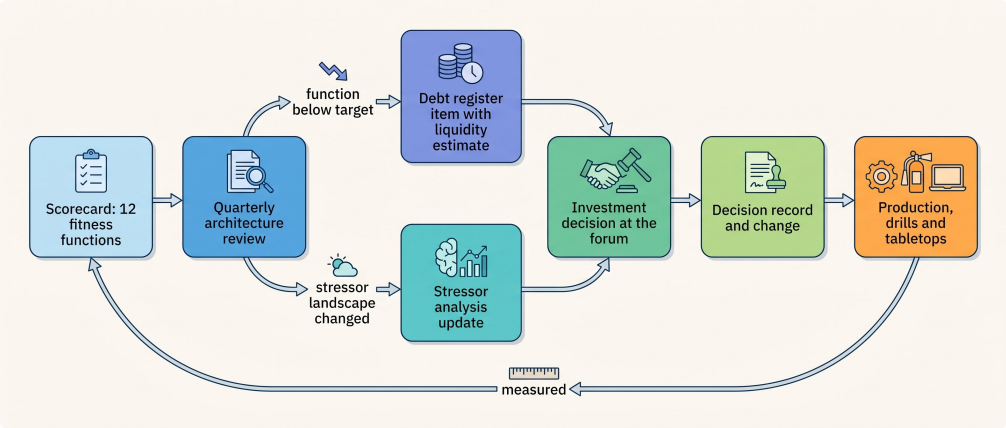


Figure 7: *The steering loop. Fitness functions feed the quarterly review; failures become debt items with a liquidity estimate; changes in the outside world update the stressor analysis; both compete for investment through decision records. The loop, not a target diagram, is the long-term plan.*

The Architecture Function's Own Evolution

An architecture function that is right for a group's founding year is usually wrong for its fifth. Veldmark's illustrates the arc. In 2027 Ingrid's small group architecture team did work that only a central function could do: the capability map, the context map, the profile register, twelve principles (among them *share platforms, not products* and *separable by tenant*), the weekly advice forum and a year of decisions: 74 cross-brand decision records by 31 December 2027, with a median of nine working days from draft to decision. That is the **founding phase**, and its products are mostly the top layer of Figure 3.

From 2028 to 2030, the **building phase**, the work shifts to the platform teams that own the shared engines, and the architecture function's job becomes holding the contracts

steady while those engines are built and migrated. The danger in this phase is the Forced Fit: a center that has made many good decisions keeps making more of them, deeper into the brands' territory.

After 2030, if Common Ground succeeds, the **stewardship phase** begins. The function should become *smaller in decisions and larger in evidence*: owning the contracts, the scorecard, the stressor analysis and the separability drills, and leaving most design decisions to the teams within the rules. Who Owns What: Organizing Marketplace Engineering describes the advice process that makes this possible: anyone can decide after seeking advice from those affected, and the forum records the decision.

The architecture function should measure itself too, with a short list of its own fitness functions:

- **Decision lead time**: from a proposal reaching the forum to a recorded decision; nine working days at the median in 2027. Rising lead times mean the center has become a bottleneck.
- **Share of decisions taken by teams**, with advice, rather than by the forum. It should rise over the building phase.
- **Exceptions and deliberate duplicates**: their number and age. A growing list means the platform does not fit; a list of zero means nobody admits to the duplicates that exist.
- **Revisits on schedule**: the share of dated decisions (Motorvane's Elasticsearch, Hearthly's Google Cloud, the Vespa deferral) reviewed when they said they would be.
- **Survivability of the record**: whether a new architect can reconstruct why a top-layer contract looks as it does from the decision records alone. Henrik's question, "what stops it from turning into another Yardstick?", is answered by this one.

Trade-offs

Choice	Option A	Option B	When A is right	When B is right
How to plan	Target-state architecture with a roadmap	Stressor analysis plus fitness functions, revisited quarterly	A single business, stable ownership, a migration with a clear end state (retiring Yardstick)	A portfolio group with changing ownership, regulation and channels; any horizon beyond three years
How much to share	Maximize synergy: shared engines and product platforms	Share contracts first, engines where uniform, products rarely	Markets that really are alike and an owner committed to holding them for a long time	Different markets, private-equity ownership, likely disposals
Separability	Design it in from the start (pairwise IDs, tenant partitions, exports)	Add it when a sale is announced	Any group whose owner could sell a business within the platform's life	A single-brand company with no realistic carve-out; even then, privacy needs much of it
Fitness functions	Automated monitors and build checks	Reviewed indicators and drills	Properties that can be measured continu-ously (latency, ghost share, paid share)	Properties that can only be shown (rebuild, carve-out, recovery)

Choice	Option A	Option B	When A is right	When B is right
Who sets targets	Central targets per function	Central definitions, local targets	Platform properties with one remedy owner (removal latency)	Market outcomes whose remedies are local (time to first contact, liquid cells)

The main tension is between synergy and separability. Every shared platform saves money while the group holds together and costs money when it comes apart. A Chief Architect cannot know which world will arrive, so the defensible position is to take the synergy where it is large and the technology uniform, and to buy separability insurance cheaply, at design time, for all of it.

Vertical Differences

The long-term stressors and the fitness functions that matter most differ by vertical, in Veldmark's four businesses and in the industry generally.

Dimension	General classifieds (Tradeyard)	Automotive (Motorvane)	Real estate (Hearthly)	Professional SaaS (Keybridge) and jobs
Dominant long-term stressors	Fraud evolution; transactional pivots and retreats per category; product-safety duties; AI-generated listings	Dealer concentration and DMS vendor power; pricing pushback; EV transition changing attributes and valuations; retail and financing moves	Climate and hazard remapping; credit shocks; title and rental fraud; agency multi-homing; data licensing	Neutrality and competition law (SaaS); AI Act high-risk duties for recruitment (jobs)

Dimension	General classifieds (Tradeyard)	Automotive (Motorvane)	Real estate (Hearthly)	Professional SaaS (Keybridge) and jobs
Most important fitness functions	Publication freshness, removal propagation, harm prevalence, time to first contact	Feed removal latency, lead reconciliation with dealer CRMs, unit cost per listing	Duplicate exposure, estimate accuracy and versioning, removal of let or sold properties	Evidence of data separation; API parity
Entity that must outlive listings	Mostly none; product identity only for recall matching	The vehicle (VIN)	The property, with dated facts	The agency and its mandates
What separability protects	Brand carve-outs by country	Sale or listing of the vertical (a common capital move in the industry)	Sale of a country portal	The neutrality promise itself
Typical time windows	Hours to days	Days to weeks	Weeks to months	Contract years

The table explains why a single group scorecard needs vertical parameters. A 90-second publication target is worth a great deal in Tradeyard NL's dense goods cells and much less in Hearthly's property cells, a point Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics makes; a 1.7 listings-per-property ratio means nothing for city bikes.

Failure Modes

The frozen target state. A program plan drawn in its first year becomes the architecture's definition of success. When the owner, the law or the channel changes, the plan is defended rather than revised, and teams route around it. *Remedy:* plan in stressors and fitness functions; treat the target diagram as a picture of the current bet.

Synergy without separability. Shared platforms are built with tenants mixed in tables, logs, block lists and ledgers, because the business case counted only savings. The first carve-out reveals two years of untangling. *Remedy:* tenant-exit tests on every shared platform, run before they are needed.

Contracts that drift into internals. The listing event contract slowly acquires brand-specific fields, ranking hints and convenience copies of seller data, because adding a field is easier than adding a topic. In five years the contract is as hard to change as Yardstick's tables. *Remedy:* review every contract change at the forum against the layering rule; keep brand-specific events out of the group contract.

Scorecard decay. Fitness functions are defined once and then reported without owners, or quietly redefined to look better. The scorecard becomes Vanity Liquidity with better graphics. *Remedy:* owners, versioned definitions, counter-metrics beside every metric, drills that fail when not run.

The permanent founding phase. The architecture function keeps deciding everything because it decided well in its first year. Decision lead times grow, exceptions grow, and the platform becomes a mandate. *Remedy:* measure the function's own fitness and shift decisions to teams on purpose.

The Chief Architect's View

Over a decade, the Chief Architect of a marketplace group owns very few systems and a handful of things that outlast all of them. The decisions I would hold at group level are these.

Own the argument. Write down, in one page, what the platform is for: turning fresh, trustworthy supply into matches, cell by cell, at a cost the business can carry. Every top-layer decision should cite it, and every new Chief Architect should be able to read it on their first day.

Own the top layer, and only the top layer by default. Identifiers, the listing event contract, the taxonomy and attribute vocabulary, the governed metrics, the decision and evidence formats. The engines belong to platform teams; the content belongs to brands. Draw that line explicitly and defend it in both directions.

Run a stressor analysis every year and after every capital event. Keep the stressor table in the architecture repository next to the decision records. Count residues per component and spend quality where the count is high. Re-run the combined scenarios when the owner, the portfolio or the law changes.

Make separability a review criterion, like security. Every shared platform answers the question: *if this tenant leaves in eighteen months, what do we hand over, what do we switch off, what runs twice?* The answer goes into the decision record before the platform is built, and the tabletop proves it every year.

Write the cost to reverse into every sharing decision, next to the synergy case. It is what makes the centralization pendulum a managed choice rather than a periodic upheaval.

Steer by the scorecard. A dozen fitness functions, each with an owner, a baseline, a target, a counter-metric and a failure pattern. Quarterly review, versioned definitions, drills for the properties that cannot be monitored.

Who decides. The Group Chief Architect owns the argument, the top layer, the stressor analysis and the scorecard's definitions. The CTO owns the investment trade-offs and unit costs. The Group CPO owns the market-outcome functions (time to first contact, liquid cells). Platform owners own their engines' functions. The board, or the owner, sees the stressor table and the separability estimate once a year, because they are the ones who will create the stressors.

What to watch. The trend of the twelve fitness functions, especially removal propagation and paid share, which degrade quietly; the number of open residues in the stressor table; drills not run on schedule; the age of deliberate duplicates; and the architecture function's own decision lead time. If the scorecard is green while sellers and buyers complain, the scorecard is measuring the wrong thing, and fixing that comes first.

Questions to Ask

1. Can we state in one page what our platform is for, in terms of matches rather than systems, and does every top-layer decision cite it?
2. Which of our decisions are meant to last ten years, which three to eight, and which are content? Is that written down?
3. When did we last run a stressor analysis, and did it include capital, regulatory, channel, supply, adversarial, supplier and physical-world stressors?
4. Which components carry the most residues, and do they have our best owners and our slowest change process?
5. If our owner sold one brand in eighteen months, how many shared capabilities would need transitional services, for how long, and when did we last test that?
6. Do our shared block lists, evidence stores, ledgers and data products record which tenant each entry came from?
7. For each managed service, what is our exit estimate, and which contract makes the exit possible?
8. What are our marketplace fitness functions, who owns each, and which counter-metric sits beside each one?
9. Which fitness functions are drills, and when were they last run?
10. Does every item in our debt register name the fitness function it hurts and an estimate of the liquidity effect?

From Argument to Evidence

This book has argued that a listing marketplace is a liquidity machine, and that its architecture should be judged by how reliably it turns fresh, trustworthy supply into matches with demand, in every country, category and vertical, while each marketplace evolves at its own pace. Parts I to VII built the machine stage by stage; Part VIII has been about keeping it coherent while its owners, its markets and its technology change. Veldmark has been the test bench throughout, and it is fictional: its decisions show how the argument plays out, not that it is right.

Whether it is right is a question for evidence, and the best evidence is what real groups did over the decades this book has compressed. Part IX — PRACTICE: How Others Built It turns to them: Adevinta's journey from Schibsted's copy-paste model to a portfolio platform and back toward standalone marketplaces, OLX's regional platforms and retreats, eBay Classifieds Group's national leaders, thin shared layer and sale, AVIV's acquisitions, the property portals and the automotive marketplaces. Each case separates what is documented, with dates, from what is interpretation, and each ends by testing the patterns in this book against what happened. Read them as a set of stressors that actually arrived, and as a check on which residues survived.

Sources and Evidence

Company engineering and investor sources

- Strategic Report 2025, Aurelia UK Feederco Ltd (Adevinta) — Adevinta, 2026. The move to "a decentralised, standalone operating model" and the parallel running of central and local services.
- Annual Report 2019 — Adevinta, 2020. Centralizing where "investment requirements are large and the technology is uniform".
- Adevinta Annual Report 2023 — Adevinta, 2024. The decision to converge the leboncoin and Kleinanzeigen platforms into "a unified global platform".
- Adevinta completes acquisition of eBay Classifieds Group — Adevinta, 2021. Completion of the eCG acquisition.
- Completion of the voluntary offer for Adevinta ASA — Adevinta, 2024. The take-private by the Permira and Blackstone-led consortium.
- Prosus looks to ditch OLX Autos business as second-hand car market slows — News24 (trade press), 2023. Prosus's move to exit OLX Autos (the exit itself is traced in OLX Group: Regional Platforms, Verticalization and Retreat).
- Zillow Group Form 8-K and shareholder letter, Q3 2021 — Zillow Group, 2021. The wind-down of Zillow Offers.
- Why you should be testing your architecture — Auto Trader UK Engineering, 2026. ArchUnit tests as build-time architecture checks.

- Schibsted's 2016 "one global product and tech organization" investor slide is cited as widely reported (secondary summary of the Q3 2016 investor presentation); the primary slide was not retrieved. See From Newspaper Columns to Portfolios: How the Classifieds Industry Evolved.

Standards, regulation and documentation

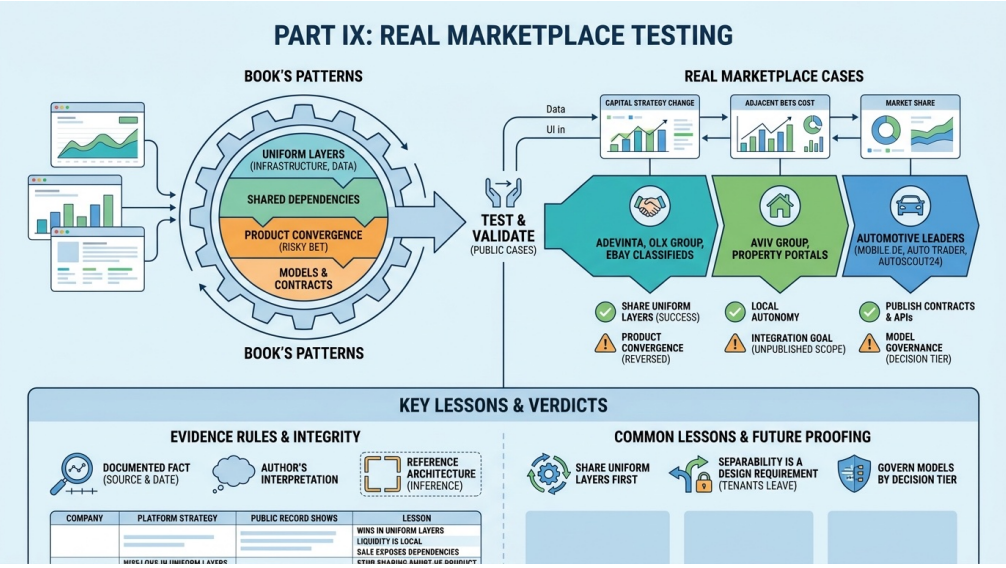
- General Product Safety Regulation, procedure summary — European Parliament Legislative Observatory, 2023. Regulation (EU) 2023/988, applicable from 13 December 2024, including duties for online marketplaces.

Research and practitioner sources

- Residues: Time, Change, and Uncertainty in Software Architecture — Barry M. O'Reilly, Leanpub, 2024. Stressors, residues, attractors and random simulation.
- Applying Residuality Theory for Better Software Design — Barry M. O'Reilly, Cutter Consortium, 2020. Early statement of residuality as design practice.
- Building Evolutionary Architectures, 2nd edition — Neal Ford, Rebecca Parsons, Patrick Kua and Pramod Sadalage, O'Reilly Media, 2022 (first edition 2017). Fitness functions for architectural governance.
- Productscape general-listings and real-estate domain models (author's sibling repository, 2026), summarized in this book's research digest: naive and residual architectures and their stressor tests.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

Part IX — PRACTICE: How Others Built It



IN THIS PART, YOU WILL: Test the book's patterns against six public cases: Adevinta, OLX Group, eBay Classifieds Group, AVIV Group, the property portals Zillow, Rightmove, Idealista and REA, and the automotive marketplaces mobile.de, Auto Trader and AutoScout24.

WHY THE BUSINESS CARES: These groups bought, merged, shared, sold and retreated in public. Their records show what platform decisions were worth when the capital strategy changed, and which adjacent bets cost more than they earned.

WHY ARCHITECTS SHOULD CARE: Parts I–VIII argue from principles and a fictional group. Here the argument meets evidence that the author did not design, including evidence that cuts against it.

KEY POINTS:

- **Share the uniform layers, and treat product convergence as a separate, riskier bet.** Infrastructure, data, experimentation and narrow machine-learning services were shared with documented success. Adevinta's 2023 plan to turn two generalist platforms into one was reversed by 2025, and AVIV's 2021 goal of "a European real estate marketplace", whose scope was never published, gave way to local autonomy.
- **Every shared dependency must survive a change of owner.** eBay's 2020–2021 sale and Adevinta's 2025 break-up turned shared platforms into transitional-service terms and separation costs. Separability is a design requirement, not a contingency plan.
- **Contracts and models carry the vertical lessons.** Automotive leaders publish their dealer-feed contracts, property portals depend on licensed supply, and Zillow showed in 2021 what happens when an estimate built to inform is used to commit capital.

FAILURE PATTERNS THIS PART ADDRESSES:

- **Deal-Shaped Architecture** — Four of the six cases are portfolios assembled by acquisition, and the record shows what each kept, shared and paid for.
- **The Forced Fit** — Adevinta's planned product-platform convergence and AVIV's 2021 integration goal both gave way to local autonomy with selected group functions.
- **Set-and-Forget AI** — Zillow Offers and automotive price ratings show models making consequential decisions about money and exposure.

Built to Be Replaced: Evolving a Marketplace Platform Over a Decade closed the argument of this book. Part IX asks **whether it survives contact with what real groups did.** The cases are not company profiles. Each asks what was shared, at which layer, and what it cost when the strategy changed.

How to read the cases

Part IX follows stricter evidence rules than Parts I–VIII: every statement is one of three kinds.

Kind of statement	How you recognize it
Documented fact	Source and publication year in the sentence or the timeline. Old sources describe systems that may since have been replaced: "true at the date of its source, and no later".
The author's interpretation	Marked "my reading", or placed in the Chief Architect's view.
Reference-architecture inference	Only in a callout labeled `INFERENCE, NOT DOCUMENTED`. It describes how a group like this *could* be built, never how it *is* built.

The evidence is uneven, and each case says where it stops. Adevinta published a great deal, eBay Classifieds almost nothing as a group, and AVIV no engineering material at all. **The depth of each section follows the evidence, not the importance of the company.**

Veldmark appears only in each case's final comparison. That keeps the fiction from being read as a disguised real group, and the real record from being stretched to fit the fiction.

Chapters in this part

Chapter	The question it answers	Read it when…
Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform	What does a full swing from federation to shared platform and back teach?	You are deciding what a group platform should share.
OLX Group: Regional Platforms, Verticalization and Retreat	What happens to a platform as a portfolio shrinks to its local leaders?	You are weighing an inventory or transactional bet.
eBay Classifieds Group: Shared Platforms, Local Brands and a Sale	How thin can a shared layer be, and what does a sale expose?	A brand may be sold, or a parent may leave.

Chapter	The question it answers	Read it when…
AVIV Group: Building a European Real-Estate Platform from Acquisitions	How much of a property portal can be shared across countries?	You are integrating acquired national portals.
Zillow, Rightmove, Idealista and REA: What Property Portals Teach	What do licensed supply, durable properties and valuation models demand?	Your models start to price, not describe.
mobile.de, Auto Trader and AutoScout24: Lessons from Automotive Marketplaces	What do dealer-fed listing systems reveal through their public APIs?	Most of your supply arrives through vendor feeds.

The cases side by side

Company	Platform strategy	What the public record shows	Lesson
Adevinta	Selective centralization (2019), product convergence (2023), decentralization (2025)	Wins in uniform layers; no public source shows the 2023 convergence completed before the reversal	Share uniform layers first; let tenants leave
OLX Group	Roll-up and rebrand, then retreat to a European core	From 45 markets (2018) to seven (FY to March 2026); OLX Autos exited from 2023	Liquidity is local; build adjacent bets to be switched off

Company	Platform strategy	What the public record shows	Lesson
eBay Classifieds	Federation of national leaders with a thin shared layer	Shared search choice migrated brand by brand; sale completed June 2021 with transitional services	A sale turns shared dependencies into contract terms
AVIV Group	Acquired portals, shared valuation and professional offer	2021 integration intent of unpublished scope; by December 2025, local autonomy with group functions where they "create clear advantages"	In real estate, stop sharing well short of the product
Property portals	Four economic engines (per advertiser, per listing, per lead)	Zillow Offers wound down in November 2021; supply licensed from MLSs and agents	Govern models by decision tier; carry license terms as data
Automotive	Dealer-fed platforms with published APIs	Feed contracts, change feeds and five-band price ratings in public	Price ratings need ranking-model governance

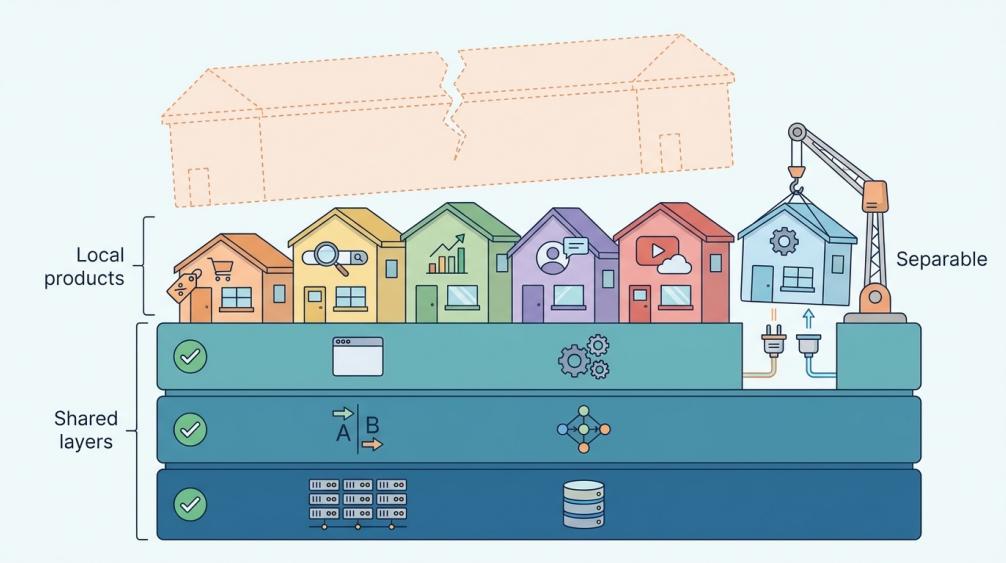


Figure 1: *The cases' common lesson: sharing the uniform layers held up, converging products into one was the riskier bet, and every shared dependency had to let a tenant leave when its owner changed.*

How this part connects

Each case links back to the chapters it tests, chiefly One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture, After the Deal: Consolidating Platforms After Acquisitions, Models in Production: ML Platforms and Use Cases and Many Doors In: Forms, APIs and Bulk Feeds, and ends with a verdict table. The Field Kit: Templates for the Marketplace Architect then turns the book's decisions into templates.

The Veldmark thread

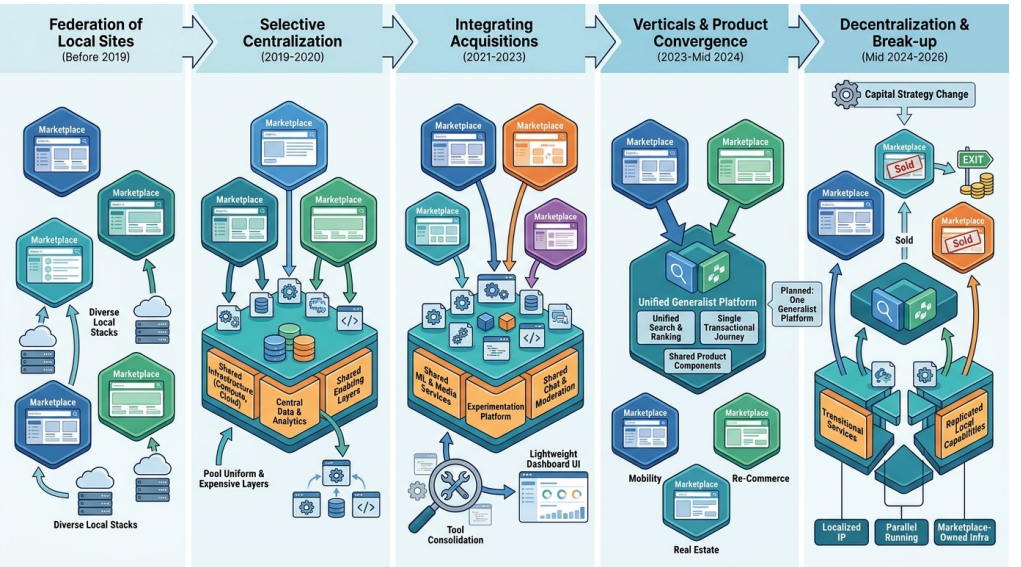
Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), is tested here, not extended. Ingrid Halvorsen's team holds its 2027 choices against the record:

- "**share platforms, not products**" in Common Ground, with every shared capability tenant-separable from its first release;
- the **shared OpenSearch platform** for Tradeyard and Hearthly, with Motorvane on Elasticsearch until a 2029 review (14 September 2027);
- **Supply Contract v1** (September 2027): one contract and validator for the group, adapters kept brand-local;

- **"Hearthly Direct" declined** in 2027 as a commit-tier use of Valuo without validation.

By the end of Part IX you should be able to say which of the book's patterns the public record supports, which it only fails to contradict, and where it is silent. Begin with Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform.

46. Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform



IN THIS CHAPTER, YOU WILL: Trace Adevinta's public record from the 2019 demerger out of Schibsted through the eBay Classifieds acquisition, the 2023 platform-convergence decision and the 2025 decentralization. You will separate what Adevinta documented from what the author infers, and test the case against the book's patterns and Veldmark's choices.

WHY THE BUSINESS CARES: Adevinta's shared platforms were valued as synergy in a 2021 deal and became a separation cost in a 2025 break-up. A group that may buy or sell marketplaces is making a capital decision every time it decides what to share.

WHY ARCHITECTS SHOULD CARE: Few marketplace groups have published this much engineering evidence across a full centralize-and-decentralize cycle. The record shows which shared layers paid for

themselves, which bets were never shown to land, and what it costs to unwind a platform that many marketplaces depend on.

KEY POINTS:

- **Adevinta's 2019 rule for what to centralize held up best**. It centralized where "investment requirements are large and the technology is uniform" (infrastructure, data, narrow machine-learning services) and kept product surfaces local. The documented wins (a voluntary Kubernetes platform for thousands of developers, ten experimentation tools reduced to one, shared optical character recognition (OCR) and media services) all sit in that zone.
- **Converging two product platforms is a different class of bet from sharing infrastructure**. In 2023 Adevinta set out to turn the leboncoin and Kleinanzeigen platforms into "a unified global platform" for its generalist marketplaces. The last public word, in May 2024, was "progressing well"; no public source shows it completed before the reversal.
- **Platform topology must survive a change in capital strategy**. Under private-equity ownership the group moved to "a decentralised, standalone operating model", localized intellectual property, ran central and local services in parallel, and sold Spain as a carve-out. Separability (tenant-scoped accounts, data products with contracts, portable deployment) is a design requirement whenever ownership can change.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Deal-Shaped Architecture** — The eBay Classifieds acquisition brought a second federation of comparable size, and the public record shows both the consolidation work it took and where it stopped.
- **The Forced Fit** — Adevinta's platform team grew adoption without a mandate, while the 2023 product-platform convergence moved toward the kind of central imposition that the 2025 reorganization undid.
- **The Copy-Paste Country** — The Schibsted-era model of local sites on a shared starting code base is the historical root of the duplication that every later phase tried to remove.

Adevinta changed its platform strategy several times in seven years, in public. It was demerged from Schibsted in 2019 as a federation of locally built classifieds sites, bought eBay Classifieds Group in 2021, and announced in 2023 that two of its largest generalist platforms would become one. From mid-2024, under new private-equity owners, it reversed course: central functions were ramped down, capabilities and intellectual property moved to the marketplaces, and businesses were sold. **The case is valuable not**

because Adevinta was right or wrong, but because it ran the whole centralization pendulum and wrote much of it down.

This chapter asks three questions of each phase: *what was shared, at which layer, and what did the sharing cost when the strategy changed?* The answers support the argument of One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture and After the Deal: Consolidating Platforms After Acquisitions: **share the uniform, expensive layers first, treat product-platform convergence as a separate decision, and design every shared layer so a tenant can leave**.

How to Read This Case

Part IX cases follow stricter rules than the chapters in Parts I–VIII (see Part IX — PRACTICE: How Others Built It). A **documented fact** carries its source and publication year in the sentence. **The author's interpretation** is marked "my reading" or sits in the Chief Architect's view. A **reference-architecture inference** appears in a callout labeled `INFERENCE, NOT DOCUMENTED`.

The case is strongest on infrastructure, data and organization, and weakest exactly where the 2023 convergence bet was placed. The engineering blog covers platform layers well and core product layers poorly: there is almost no public detail on how leboncoin, Kleinanzeigen, Marktplaats or mobile.de run search, ranking, listings or messaging. Since the 2024 take-private, group accounts appear only as the annual strategic report of a holding company, Aurelia UK Feederco Ltd, without the revenue mix by type. And most of the engineering detail, the most any classifieds group in the book's research published between 2019 and 2025, describes central platforms that the 2025 decentralization may since have split or retired. Every "Adevinta runs X" statement below is true at the date of its source, and no later.

Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), appears only in the final comparison. It is a composite of industry patterns, not a disguised description of Adevinta or any other group.

Timeline: Adevinta in Dated Events

Dates are those of the cited releases and reports. Ownership and portfolio facts are as of October 2026 and may change.

Date	Event	Source type
2006	leboncoin (France) launched by Schibsted; willhaben (Austria) founded as a 50/50 joint venture with Styria	Wikipedia (leboncoin); Adevinta Annual Report 2023 (willhaben)
Feb–Apr 2019	Schibsted shareholders approve the demerger of its non-Nordic marketplaces as Adevinta (16 countries, about 1.5 billion monthly visits); listing on Oslo Børs on 10 April, Schibsted keeping about 60%	Schibsted/Vend release, 2019; law-firm deal notes, 2019
2020	First annual report (for 2019) adopts a "centralised strategy to data and analytics" while keeping local product autonomy; 2019 revenue €680 million	Adevinta Annual Report 2019
Jan 2020	"Common Platform" talk: an internal platform-as-a-service for about 1,500 engineers across about 20 marketplaces	Galo Navarro, 2020
21 Jul 2020	Agreement to acquire eBay Classifieds Group (eCG) for US$2.5 billion in cash plus 540 million shares	Trade press, 2020
25 Jun 2021	Acquisition completes; #1 positions claimed in 16 countries; run-rate cost synergies of €130–165 million targeted after three years	Adevinta release, 2021
Jun–Dec 2021	UK competition remedies: Shpock sold to Russmedia Equity Partners (2 June), Gumtree UK and Motors.co.uk sold (December); eCG Denmark sold to Schibsted for about €278 million	Adevinta releases, 2021

Date	Event	Source type
2022	Australian and Mexican businesses sold; focus narrowed to "five core European markets"	Adevinta release, 2022; trade press, 2022
2023	Operating model reorganized by verticals (Mobility, Re-Commerce, Real Estate & Emerging Verticals); eBay Kleinanzeigen renamed Kleinanzeigen; most EU5 on-premise platforms moved to AWS; leboncoin and Kleinanzeigen platforms to become "a unified global platform"	Adevinta Annual Report 2023
21 Nov 2023	Permira- and Blackstone-led consortium offers NOK 115 per share; settled 29 May 2024 at about 94.8% of capital, followed by delisting	Schibsted/Vend release, 2023; Adevinta release, 2024
28 May 2024	Last public quarterly report: "Platform convergence project progressing well and according to plan"	Adevinta Q1 2024 release
Jul 2024	Work starts on a new Target Operating Model, "empowering local marketplaces"	Strategic Report 2024 (published 2025)
Mar 2025	Sale of the 50% willhaben stake announced (reported valuation about €500 million)	Trade press, 2025
21 Jul 2025	Sale of Adevinta Spain (Milanuncios, Fotocasa, Habitaclia, InfoJobs, coches.net, motos.net) to EQT announced; in September Scout24 agrees to buy Fotocasa and Habitaclia from EQT	Adevinta release, 2025; Scout24 release, 2025

Date	Event	Source type
Oct 2025	Financial press reports an IPO-or-sale "dual track" for mobile.de; no completed transaction found as of October 2026	Trade press relaying the Financial Times, 2025
27 Feb 2026	Spain sale completes, with a gain of about €1.1 billion	Strategic Report 2025 (published 2026)
3 Mar 2026	A Spanish branch is set up to provide software development "for the account of Kleinanzeigen.de GmbH"	Strategic Report 2025

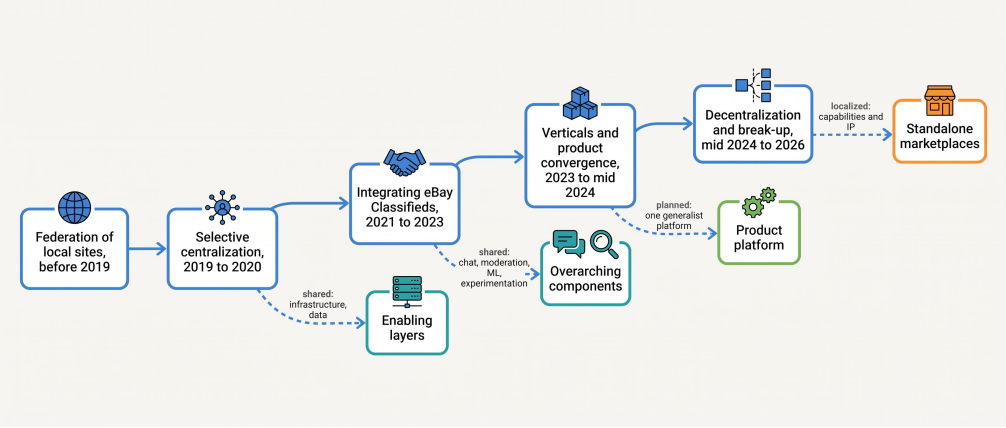


Figure 1: *Each phase moved the line between shared and local to a different layer: infrastructure and data first, overarching components next, the product platform itself in 2023, and then back to standalone marketplaces.*

Phase 1: The Schibsted Federation (Before 2019)

The public record documents the Schibsted-era portfolio but not how it was built. The portfolio was national generalist sites such as leboncoin (launched 2006), Subito, InfoJobs and Segundamano, plus verticals, many of which later formed Adevinta. The mechanism is widely repeated but not verified from a primary source: Schibsted is said to have launched or bought a generalist site in each country and run it on a shared code base derived from Blocket, its Swedish site. A 2012 Nieman Lab article is commonly cited for "28 countries on the Blocket platform"; the research for this book could not retrieve it, so it stays industry lore here.

My reading is that a shared starting code base, copied and then evolved locally, is the classic origin of the Copy-Paste Country: every site begins identical and diverges with every local change, until improvements must be built once per country. The later phases read as attempts to recover the economics that copying gave away, without giving up the local autonomy that made the sites win their markets.

The group's most durable shared abstraction was a local invention, promoted to group level. **FIAAS** ("FINN Infrastructure as a Service") was built at FINN.no, Schibsted's Norwegian marketplace. FINN's 2017 KubeCon talk describes about 350 microservices and 1,200 deployments per week, with FIAAS replacing roughly 147 lines of Kubernetes configuration with a short application file. FIAAS was open-sourced in 2019 and became the deployment abstraction of Adevinta's platform; a 2023 Adevinta post still calls it the "Golden Path". FINN.no itself stayed with Schibsted and never joined Adevinta.

Phase 2: Selective Centralization (2019–2020)

The 2019 rule: large investment, uniform technology

Adevinta's first annual report as a listed company, for 2019, named the trade-off directly:

> "While one of our historic success formulas has been local autonomy and decentralised capacity in product and technology, we are adopting a centralised strategy to data and analytics. This one area where investment requirements are large and the technology is uniform, and where the maxim 'built once; used many times' clearly applies."

The same report commits to "scalable components" developed centrally "aligned closely with local on-the-ground technical expertise", and attributes rising central losses to "investments in scalable tech and data". The 2019 revenue was €680 million across France (€357 million, mostly leboncoin), Spain (€182 million), Brazil (an equity-accounted joint venture) and a long tail of "Global Markets".

I read this as the strongest decision rule in the whole record. It does not say "centralize what is shared". It names two conditions, *large investment* and *uniform technology*, and accepts that product surfaces fail them. That is the same test One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture proposes for choosing global components, and it explains why the documented successes of the next five years cluster where they do.

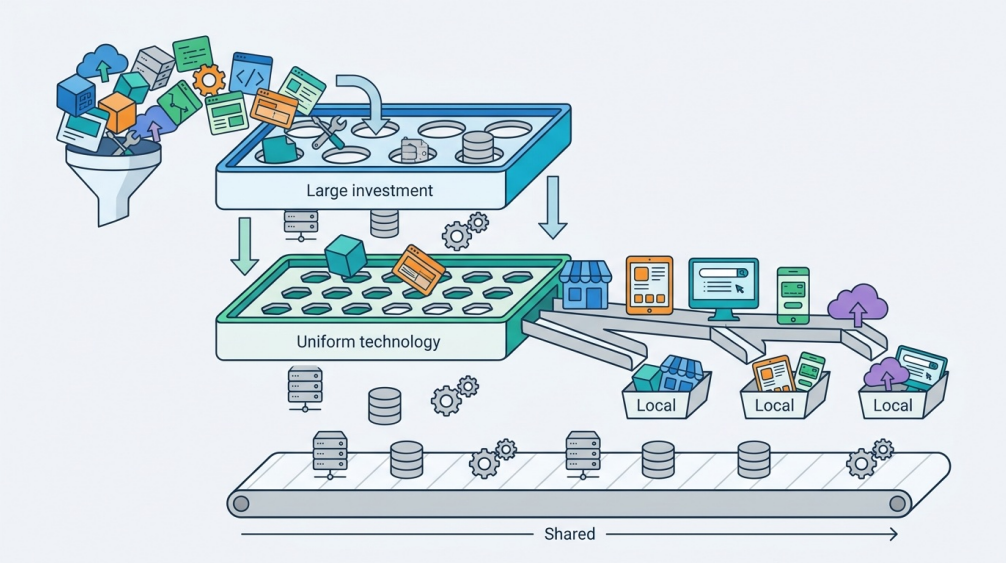


Figure 2: *The 2019 rule as a two-stage test: a layer earns central ownership only when it needs large investment and its technology is uniform across markets, which is why product surfaces stayed local.*

A platform that grew without a mandate

The "Common Platform", described in Galo Navarro's 2019 talk (written up in January 2020), was a platform-as-a-service for about 1,500 engineers in about 20 marketplace companies. It was built from ordinary tools: Spinnaker, FIAAS, self-run Kubernetes on EC2, Prometheus and others.

Its design choices were less ordinary, starting with voluntary adoption ("You prefer to do the plumbing yourself? It's your call, we respect that."). Onboarding took about ten minutes through a web form that set up delivery pipelines, a namespace, metrics and logging. For large migrations, such as the Spanish marketplaces' roughly 200 microservices, the platform team embedded one or two engineers with the local team. A controller called "Eurek8s" bridged the services' existing service discovery into Kubernetes so they could move without code changes (described in 2020).

The platform grew from about 1,500 to more than 4,000 developers, through the eCG integration, and the public record shows no mandate. By June 2023 it was called **SCHIP**, a multi-tenant Kubernetes distribution, and served "more than 4,000 developers" and about 5,000 applications. It is the clearest documented counter-example to the Forced Fit in the book's research: a central platform that won its tenants by being cheaper to use than the alternative.

Phase 3: Absorbing eBay Classifieds (2021–2023)

The deal and what regulators removed

The eBay Classifieds Group acquisition, completed on 25 June 2021, roughly doubled the group. Adevinta's completion release claims number-one positions in 16 countries, about three billion monthly visits and pro forma 2019 revenue of about €1.6 billion, and targets **run-rate cost synergies of €130–165 million after three years**. The acquired brands included mobile.de, eBay Kleinanzeigen, Marktplaats, 2dehands/2ememain, Kijiji, Gumtree, Automobile.it and others, each with its own stack.

Regulators and strategy removed part of the integration scope before integration began, so the scope is not known at signing. The UK Competition and Markets Authority required the sale of Shpock and of Gumtree UK with Motors.co.uk, both completed in 2021; eCG's Danish businesses went to Schibsted. Further sales in 2022 (Australia, South Africa, Mexico) and the 2023 exit from Hungary were strategic, following a review that focused the group on "five core European markets: Benelux, France, Germany, Italy and Spain". The platforms that overlap most are the likeliest to be divested before anyone integrates them.

The 2023 annual report shows the integration's cost: €69 million of eCG integration expenses in that year alone, €15 million for the eBay Kleinanzeigen rebrand (under the acquisition agreement, Adevinta could not use the "eBay Kleinanzeigen" brand beyond 2024), and €22 million for the verticalization project. Whether the €130–165 million synergy target was achieved, and how much of it came from technology, is not public.

What was consolidated, layer by layer

The engineering blog (about 90 posts, August 2022 to March 2026) documents consolidation at four layers.

Infrastructure and cloud. The 2023 annual report states that infrastructure for the "EU5" businesses (the five core European markets) was consolidated "to a single public cloud vendor" (AWS), and Canada moved to Google Cloud. The plan for 2024 was to "close three of four data centre footprints" and "onboard most product platforms onto Adevinta's common internal development platform", with a target of being 100% cloud-native by 2025. SCHIP moved to Amazon EKS: a September 2023 post describes more than 30 production clusters, the largest about 200 nodes, and by October 2024 it ran "over 2,000 Kubernetes nodes" in 30 clusters "across 25 marketplaces". A December 2024 post counts **732 AWS accounts and 425 Google Cloud projects** (fewer than 150 AWS accounts in 2019), each mapped to teams and cost centers under the principle "You build it, you run it (and you pay for it)".

Data. The central data platform carried nearly a billion behavioral, content and transaction events a day over Kafka, according to a 2022 post. In parallel, Adevinta

Spain built a Databricks lakehouse that, by its December 2024 account, had evolved over three years "to a lakehouse that supports data mesh principles". Domain data products were owned by the verticals and ingested through immutable data contracts, and the time from source to usable data fell from about five days to about 30 minutes (see Data Platform for Marketplaces: Events, Lakehouse and Data Products).

Overarching components. The 2023 annual report says the group's focus "historically" was on "consolidating enabling capabilities such as cloud and data, and overarching elements such as chat and moderation". No architecture post on the shared messaging or moderation services was found. Several narrower shared services are documented:

Shared service	What it did	Scale, with date	Source
"Text in Image" OCR (Cognition team)	Finds insults, phone numbers, e-mail addresses and URLs inside listing images, for moderation and search	100 million → 330 million requests per month; about 12× cheaper than an external OCR API (2023)	Tech blog, March 2023
Image models (Cognition team)	Text detection, car classification, background removal, image embeddings on marketplace images	Not stated (2023)	Tech blog, August 2023
YAMS media service	Image, document and video delivery "used by all marketplaces"	"Over 800 million" objects served daily (2024)	Tech blog, July 2024
Dispute management back office	Case handling for transaction disputes, pluggable "into any marketplace"	Demonstrated with Subito and Milanuncios data (2023)	Tech blog, February 2023

Shared service	What it did	Scale, with date	Source
Houston experimentation platform	Assignment, analysis and a shared statistics engine	Ten A/B tools across 22 marketplaces reduced to one platform (2021–2024)	Tech blog, October 2024

Experimentation is the best-documented consolidation. In 2021 the 22 marketplaces used ten different A/B testing tools. A cross-functional group scored build and buy options against core and growth requirements over a three-to-five-year horizon and chose a hybrid: keep building the in-house platform, Houston, and use a third-party tool to bridge the transition in some markets. A 2023 benchmark of 21 companies, measured in experiments per technology employee, supported the build side. Like FIAAS, the shared capability started as a local one: the statistics package at Houston's core, **Fisher**, was built at Marktplaats, used in more than 90% of its experiments within a year, and then adopted group-wide (see Testing a Two-Sided Market: Experimentation Platforms).

Machine learning followed the same path: the August 2024 "ML Golden Path" replaced what its team called "a set of scattered tooling" with one workflow on SCHIP and a central catalog to which marketplace teams could promote their own components (see Models in Production: ML Platforms and Use Cases).

The dispute back office is the only cross-marketplace *product* component with a published architecture. Each marketplace's transaction events (via Kafka, SQS or REST) were normalized into a common model behind a Kotlin API, with a micro-frontend user interface that marketplaces could extend. The stated constraint was "We couldn't hardcode anything… or make any assumptions."

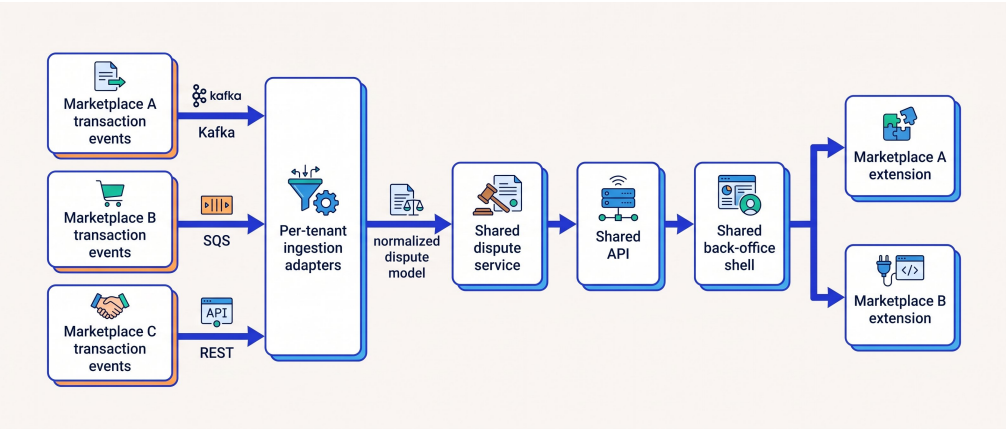


Figure 3: *The documented shape of Adevinta's 2023 dispute back office: per-tenant ingestion into a canonical model, a shared core, and a front end that each marketplace can extend. The shared component inherits the quality of every marketplace's events.*

> **INFERENCE, NOT DOCUMENTED:** The dispute tool is the canonical shape for any cross-marketplace product capability: an anti-corruption layer per tenant, a canonical event model, and an extension surface in the user interface. Its weakness is that the global component can only be as good as the worst tenant's event stream, so the real integration work moves into each marketplace's event contracts. This is the author's reading of the published design, not a description of how the service was operated.

Phase 4: Verticals and the Convergence Bet (2023 to Mid-2024)

Verticals and the 2023 revenue mix

The 2023 annual report, published in April 2024, describes the "Growing at Scale" strategy (five core markets, high-quality verticals, "fully transactional in Re-Commerce", technology and advertising) and an operating model "organised by verticals": Mobility, Re-Commerce, and Real Estate & Emerging Verticals. Group revenue was €1,826 million and EBITDA €651 million.

2023, € million	France	mobile.de	European Marketplaces	International (Kijiji)	Other/HQ	Total
Classifieds	413	408	510	69	–	1,399
Advertising	62	24	204	21	2	312
Transactional	73	0	30	–	–	103
Total	**548**	**432**	**744**	**90**	**12**	**1,826**

Source: Adevinta Annual Report 2023, segment note. "European Markets" covers Kleinanzeigen, Marktplaats, 2dehands/2ememain, Spain, Italy and Ireland.

mobile.de, monetized almost entirely through dealer listings, earned about as much classifieds revenue as the whole French segment. Transactional revenue, after years of investment, was about 6% of the group.

From sharing components to sharing the product

In 2023 Adevinta decided to share the product itself, not only the components beneath it. In its words:

> "Historically, the team's focus was on consolidating enabling capabilities such as cloud and data, and overarching elements such as chat and moderation. However, the compelling case for stronger consolidation prompted a change in this strategy. The core rationale lies in a pronounced overlap in customer needs across all of Adevinta's markets and the technical complexity inherent in expected user experiences. Rather than individually addressing local market customer needs — such as the full transactional journey, AI-assisted search and modern approaches to first-party advertising and personalised marketing — the team has opted to avoid duplicating these investments… In collaboration, Kleinanzeigen, leboncoin and global teams embarked on transforming the two platforms into a unified global platform to support Adevinta's generalist marketplaces."

The CEO letter in the same report speaks of "the creation of a multi-tenant platform" and of "scaling the best elements of each marketplace across all our platforms". The May 2024 first-quarter release, the last public quarterly report, says: "Platform convergence project progressing well and according to plan." No later public source describes its delivery, its architecture or its fate.

My reading is that this was the boldest step in the record, and a different kind of step from everything before it. The earlier layers (infrastructure, data, OCR, experimentation) pass the 2019 test: large investment, uniform technology. A generalist product platform for France and Germany fails the second condition. Its capabilities — listing creation, search and ranking, messaging, the transactional journey — are exactly where local language, consumer law, payment and carrier contracts, and years of local product decisions live.

Adevinta's rationale is real, but it argues for shared *capabilities behind* two products more than for one product platform. Building transactional checkout, AI-assisted search and first-party advertising twice is expensive. Sharing them is what One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture calls a global component. The public record does not say which of the two Adevinta was building, and the chapter does not guess.

The transactional journey: similar flows, national contracts

The capability named first in the convergence rationale, "the full transactional journey", is also the best documented at marketplace level.

Marketplace	What is documented	Source and date
leboncoin	A transactional model in some generalist categories from 2019; transactional revenue €53 million (2022) and €73 million (2023); Adyen reported as payment partner	Annual Reports 2019 and 2023; Wikipedia
Kleinanzeigen	"Sicher bezahlen": offer or "Direkt kaufen" from the chat; buyer fee; integrated carrier labels and tracking; buyer protection. The help page read in October 2026 says Adyen holds funds for up to 14 days until the buyer confirms receipt. At the 2020 launch, and still in a July 2023 report, the provider was Online Payment Platform; when the switch to Adyen happened is not public (see eBay Classifieds Group: Shared Platforms, Local Brands and a Sale)	Kleinanzeigen help page, undated, read October 2026; onlinemarketing.de, 2020; t3n, 2023
Marktplaats	Transactional services from 2017, shipping from 2019; the escrow-style "Gelijk Oversteken" was withdrawn in April 2022 and replaced by "Kopersbescherming" (buyer protection) through the payment provider Online Payment Platform, with a 5% buyer fee, as reported in 2022; the provider in 2026 was not verified	Annual Report 2023; AVROTROS Radar, 2022
mobile.de	Transactional revenue of about zero in 2023; dealer listings plus financing and leasing services	Annual Report 2023

Transactions grew revenue and costs together. The 2023 report notes rising "direct costs from transactional services (such as payment and shipping service provider costs)", and a two-point fall in France's EBITDA margin "mainly reflecting the evolution of the business mix with an increasing share of transaction services". Transactional revenue rose 31% year on year in the first quarter of 2024.

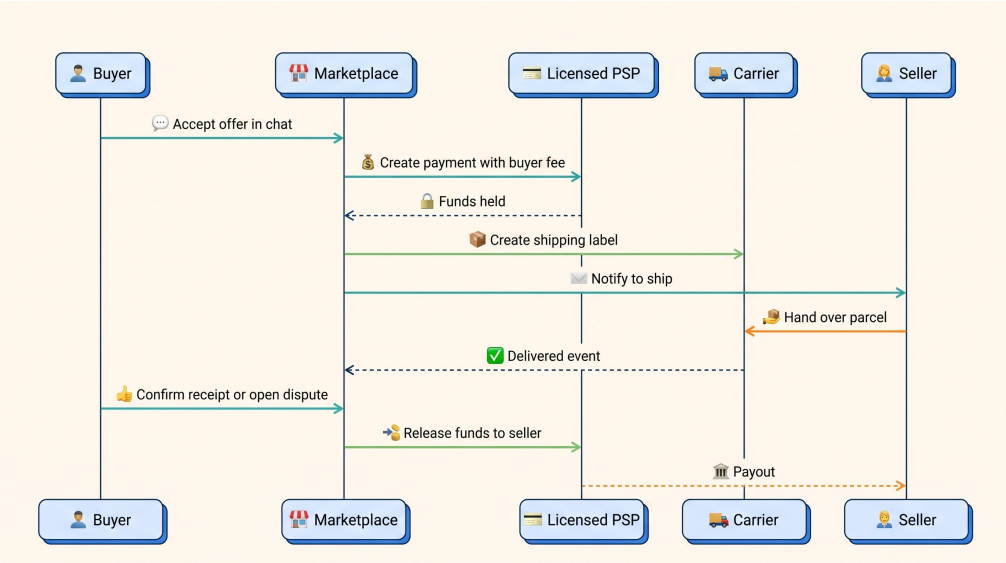


Figure 4: *A plausible reference flow for C2C buyer protection consistent with the public descriptions of Kleinanzeigen and Marktplaats — not a description of either system. The provider holds the money; the marketplace owns the offer state, logistics, release timer and disputes.*

INFERENCE, NOT DOCUMENTED: In this pattern the marketplace never holds funds; a licensed payment service provider (PSP) does, under its marketplace product. What the marketplace owns is the state machine around the money: offer and acceptance, fee pricing, label creation and tracking, the release timer and dispute handling. That split explains both halves of the Adevinta story. It looked centralizable in 2023, because the state machine is similar everywhere. It argued for local ownership again in 2025, because the PSP contracts, carriers and consumer-law obligations underneath it are national. Beyond the Contact: Payments, Shipping and Buyer Protection develops the design, and the sibling book's Adyen: One Platform from Checkout to Settlement chapter covers the provider side.

Phase 5: Decentralization and Break-Up (Mid-2024 to 2026)

The stated move to standalone marketplaces

In July 2024, two months after the private-equity consortium's offer settled on 29 May, work started on a new Target Operating Model, "empowering local marketplaces", according to the holding company's 2024 strategic report. The 2025 report, published in May 2026, describes the result:

> "The Group has evolved from a centralised structure to a decentralised, standalone operating model, empowering each marketplace to operate with greater independence, accountability and focus. This transition has included the ramp-down of central functions, the transfer of key capabilities, the localisation of intellectual property, and the realignment of reporting structures."

The group created "standalone teams across core functions to strengthen local market ownership", and it had foreseen the risks: the 2024 risk register had already listed "operational decentralisation (which may affect areas such as… trust and fraud) and data decentralisation" as risks.

What decentralizing cost

The strategic reports, not the engineering posts, hold the most concrete evidence of what decentralizing a shared platform costs.

- **Parallel running**. Cloud-provider emissions in purchased services "more than doubled" in 2025, which the report expects "to mainly reflect a temporary spike linked to our ongoing decentralisation, which required parallel running of central and local services during a transitional period". In the same year, business-travel emissions fell 42%, attributed to less travel "between teams across global hubs".
- **Intellectual property and legal entities**. Capabilities were transferred and intellectual property "localised". On 3 March 2026 the group set up a Spanish branch to provide software development and product research "under the instruction and for the account of Kleinanzeigen.de GmbH", one week after the Spanish marketplaces were sold. **My reading** is that engineering capacity in Barcelona was re-homed under a marketplace rather than a group function: decentralization carried out in company law, not only in org charts.
- **A carve-out with advisors**. EQT's July 2025 release on the Spanish purchase promised to support Adevinta Spain "as it transforms into a fully independent company", with EY as carve-out advisor. The sale completed on 27 February 2026

with a gain of about €1.1 billion; the group then paid a €1,550 million dividend and returned €470 million of capital. Fotocasa and Habitaclia moved on to Scout24.

- **Smaller**. Headcount was 4,439 in December 2025. A March 2026 engineering post counts "over 2,000 software engineers", against "more than 4,000 developers" on SCHIP in 2023; the definitions differ, so the figures should not be subtracted.
- **Marketplace-owned infrastructure**. leboncoin's own engineering blog in 2026 describes its infrastructure organization on AWS, with about ten infrastructure engineers supporting about 70 feature teams. It does not say whether any workloads still run on SCHIP.

The 2025 accounts report revenue from continuing operations of €1,891 million and adjusted EBITDA of €865 million (€970 million including Spain). The remaining portfolio, as of October 2026, is Germany (mobile.de, Kleinanzeigen), France (leboncoin and professional verticals), the Benelux (Marktplaats, 2dehands/2ememain), Italy (Subito, Automobile.it), Canada (Kijiji) and the OLX Brasil joint venture.

Why the economics flipped

My reading is that the reversal was economic before it was technical. Under the 2021 synergy case, a shared platform was an asset: every marketplace that moved onto it removed a duplicate cost. Under a strategy that realizes value by selling or listing individual marketplaces (Spain, sold with a gain of about €1.1 billion; willhaben, at a reported valuation of about €500 million; mobile.de, with a reported valuation of up to €10 billion), the same platform is a liability. At exit, each shared layer becomes a transitional services agreement, an intellectual-property question, a data-separation project and a period of double running. **The lesson is not that centralization was a mistake. It is that platform topology must match the capital strategy, and capital strategies change faster than platforms**.

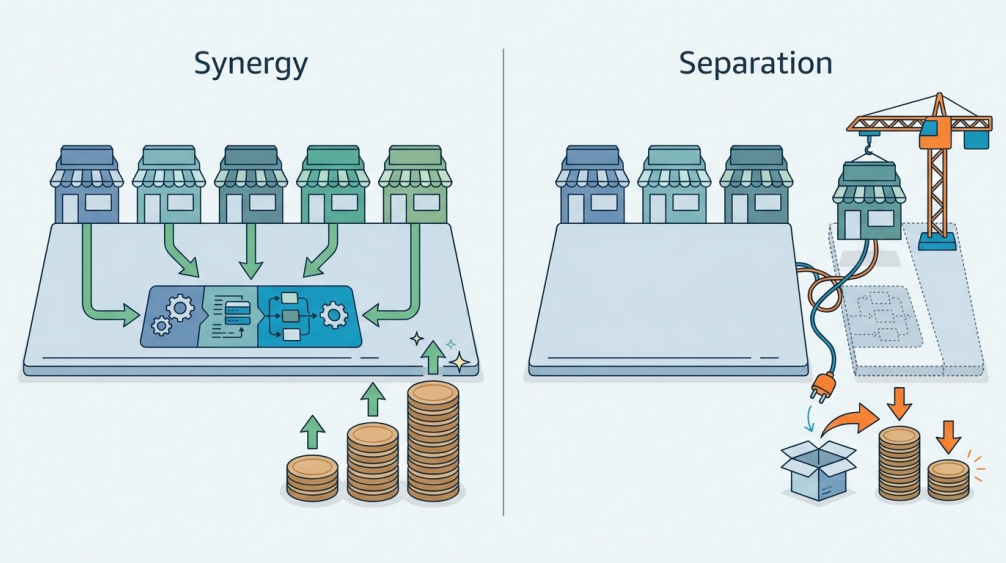


Figure 5: *The same shared platform is a synergy asset while the group stays together and a separation cost when value is realized by selling individual marketplaces: platform topology has to match the capital strategy.*

The Layer Map: What Was Shared, When

The table pulls the five phases into one view, at the layers Figure 1 introduced. "Shared" means documented as run centrally for several marketplaces at the date given; a blank means no public evidence either way.

Layer	2019–2020	2021–2023	2023–mid-2024	2025–2026
Compute and deployment (Common Platform, SCHIP, FIAAS)	Shared, voluntary	Shared, 4,000+ developers (2023)	Shared, 25 marketplaces (Oct 2024)	Not documented; leboncoin runs its own infrastructure team (2026)
Cloud accounts and governance	Under 150 AWS accounts	Consolidated on AWS for EU5 (2023)	1,157 accounts and projects mapped to cost centers (Dec 2024)	Not documented

Layer	2019–2020	2021–2023	2023–mid-2024	2025–2026
Data and analytics	Centralized by decision (2019)	Central event platform; Spain builds a data mesh	Federated data products on shared substrate	"Data decentralisation" named as a risk (2024)
Narrow ML and media services (OCR, images, YAMS)	—	Shared, high volume	Shared	Not documented
Experimentation (Houston, Fisher)	Ten tools in 22 marketplaces (2021)	Consolidation under way	One platform (Oct 2024)	Not documented
Chat and moderation	—	Consolidated, per AR 2023	Consolidated	Trust and fraud named as decentralization risk
Generalist product platform	Local	Local	**Convergence of leboncoin and Kleinanzeigen announced**	Not documented; marketplaces "standalone"

A Reference Architecture for the Shared Platform

The public posts describe parts of the platform in detail but never the whole. The inference below fills that gap only to reason about coupling and separability.

> **INFERENCE, NOT DOCUMENTED**: A plausible reference architecture — not a description of Adevinta's systems — is a hub-and-spoke internal developer platform. The hub is a central team that owns a Kubernetes distribution, a golden-path deployment abstraction, ingress, DNS, logging, identity and policy, each with platform service-level objectives. The spokes are marketplace product teams, isolated by namespace and cloud account but sharing multi-tenant clusters across regions. The economics are fleet-level: moves such as Adevinta's documented shift to ARM nodes, or Intel

> to AMD instances worth "up to €30,000 per month" (October 2024), pay off only because roughly 2,000 nodes are pooled. The coupling is fleet-level too: thousands of applications depend on incidental platform behavior, so every upgrade or policy change carries a blast radius across marketplaces.

The engineering posts confirm the fleet-level coupling. A September 2023 post on upgrades describes a renamed metrics component that broke tenants' dashboards; a May 2024 post describes an Argo CD ApplicationSet that began deleting applications across more than 30 clusters when a source-control outage returned an empty list. The 2025 PlatformCon retrospective frames its lessons around Hyrum's Law: with enough users, every observable behavior of a platform becomes a dependency.

My reading is that most of the separability the 2025 transition needed was already present, built for other reasons. Per-tenant cloud accounts mapped to cost centers existed for FinOps; data products with contracts existed in Spain for data quality; FIAAS deployment files were portable by design. A platform built for chargeback and tenant isolation is a platform from which a tenant can be carved. What the public record does not show is how long the carve-outs took or what they cost beyond the "temporary spike" in cloud emissions.

Trade-offs the Case Illustrates

Option	What Adevinta's record shows	When it is right	When it fails
Share enabling infrastructure (compute, cloud accounts, deployment)	Voluntary growth to 4,000+ developers; fleet savings from pooling; upgrade blast radius	Almost always in a group with several marketplaces on one cloud	When tenants are likely to leave soon, or when one tenant's scale dwarfs the rest
Share narrow, uniform services (OCR, image models, media delivery, experimentation statistics)	High volume at low unit cost; harvested from local inventions	Where the task is the same in every market and volume is pooled	When the service needs market-specific behavior that its contract cannot express

Option	What Adevinta's record shows	When it is right	When it fails
Share cross-marketplace product components (disputes, chat, moderation)	Pluggable design with per-tenant adapters and extensible UI	When a canonical model fits most markets and local events are good	When the canonical model leaks local assumptions, or when trust and fraud regimes diverge
Converge whole product platforms	Announced 2023; no public evidence of completion; reversed in 2025	When markets share language, law and product strategy, and ownership is stable for years	When markets differ in law, payments and product history, or when ownership may change
Decentralize to standalone marketplaces	Parallel running, IP localization, carve-out advisors, capacity re-homed in legal entities	When value is realized by selling or listing individual businesses	When the group expects to stay together and keep buying

The right line between shared and local depends on the layer, on how uniform the technology is, and on how long the group expects to own each tenant. The rows are not a maturity ladder. The third condition is the one Adevinta's record adds to the usual platform discussion.

Vertical Differences in the Record

Adevinta's generalist, automotive and real-estate marketplaces behaved differently under the same group strategy.

Vertical	Adevinta examples	What the record shows	Implication for platform sharing
Generalist (recommerce)	leboncoin, Kleinanzeigen, Marktplaats, Subito	Large audiences (Kleinanzeigen more than 58 million live ads, leboncoin more than 88 million, 2025); transactional journeys with PSP-held funds and shipping; the target of the 2023 convergence	Strongest case for shared transactional and AI-search capabilities; also where local law, carriers and payment contracts bite hardest
Automotive	mobile.de, Automobile.it, coches.net (sold 2026)	mobile.de: 1.6 million listings, more than 140 million monthly visits (2025); revenue almost entirely dealer classifieds; the most valuable single asset, with a reported dual-track process	High-value, dealer-driven businesses are the likeliest to be separated, so separability matters most here
Real estate	Fotocasa, Habitaclia (sold), A Vendre A Louer	Spanish property portals moved to Scout24 in 2026, a buyer with its own platforms	Acquirers bring their own stacks; anything shared must be easy to hand over or switch off

A capability built inside a marketplace leaves with that marketplace; one built centrally has to be split or licensed. coches.net's natural-language car search (November 2024), in which a language model turns free text into the site's existing filters for about €19 a day, was built locally on group infrastructure and went with Spain to EQT. AI Across the Listing Lifecycle and What Did They Mean? Query Understanding and Autocomplete cover the design.

Failure Modes the Record Exposes

The case exposes five risks a Chief Architect can name in advance.

1. **Synergy cases that count only the entry cost**. Nothing in the public record suggests the 2021 case priced the later cost of separating a tenant, which the 2025 report shows to be real.
2. **Convergence announced without a visible exit criterion**. The 2023 plan had a rationale and a progress statement but no public milestone after May 2024. A product-platform convergence that is not finished when strategy changes leaves two platforms, both partly rebuilt.
3. **Fleet coupling through incidental behavior**. The upgrade and Argo CD incidents show that a shared platform's blast radius crosses marketplace boundaries. Adevinta's published mitigations (serialized rebuilds, platform service-level objectives, validating upstream responses before deleting) apply anywhere.
4. **Trust and fraud as decentralization casualties**. The 2024 risk register names "trust and fraud" among the areas decentralization may affect. Moderation and fraud signals improve with data pooled across markets; splitting them can weaken detection exactly when organizations are changing (see Trust and Safety as a Platform).
5. **Tool sprawl after acquisitions**. Ten A/B tools across 22 marketplaces is a measurable symptom of Deal-Shaped Architecture; the build-buy-bridge decision that fixed it is reusable.

Where the Public Evidence Stops

No public source found for this book answers the following questions, and the chapter does not fill them with inference.

Question	What is public	What is not
Did the leboncoin–Kleinanzeigen "unified global platform" ship?	The 2023 decision; "progressing well and according to plan" (May 2024)	Its architecture (shared backend, one code base, or shared components), its scope, whether any market went live on it, and its fate after 2025

Question	What is public	What is not
What happened to SCHIP after decentralization?	Its state up to the June 2025 PlatformCon talk	Whether it was split per marketplace, handed to a marketplace, or retired; how Spain was carved out of shared clusters and accounts
How did chat and moderation work centrally?	That they were consolidated (AR 2023)	Any architecture, data model or operating model
How do the marketplaces run search and ranking?	A Kleinanzeigen search data pipeline built on change data capture (2025); LLM front ends at coches.net and leboncoin (2024)	Engines, index topology, ranking models and their ownership
Were the eCG synergies achieved?	The €130–165 million target (2021); €69 million integration cost in 2023	The outcome and the technology share
What did separation cost?	A "temporary spike" in cloud emissions from parallel running (2025); a carve-out advisor	Duration, cost, transitional service agreements
What is the revenue mix after 2023?	Group totals for 2025	Classifieds, advertising and transactional split
What did the Schibsted-era shared code base look like?	Portfolio and dates	Primary documentation of the "Blocket platform" model

Two further gaps are volatile as of October 2026: the outcome of the reported mobile.de process, and the status of the OLX Brasil joint venture. Check both before relying on the portfolio list above.

The Chief Architect's View

Adevinta's record gives me five decision rules.

Decide what to share by layer, not by enthusiasm. The 2019 test — large investment, uniform technology — sorted Adevinta's shared layers correctly for five years. I would write it into a group's architecture principles and require every proposal for a new shared capability to answer both halves.

Grow platforms by adoption, and harvest local inventions. The documented successes — the voluntary Common Platform, FIAAS from FINN, Fisher from Marktplaats, an ML tooling catalog that accepts promoted components — share one shape: the center made the good local thing easy to use everywhere. To avoid the Forced Fit, measure a platform by how many tenants choose it, and fund migration engineers rather than issue deadlines.

Treat product-platform convergence as its own investment case. It needs a scope (which capabilities, which markets), stage gates with exit criteria, and a plan that survives a change of owner or strategy. If those cannot be written down, the group should share capabilities behind the products instead. After the Deal: Consolidating Platforms After Acquisitions develops the consolidation sequence; Adevinta's record is the strongest public argument for putting core product convergence last.

Make separability a design requirement, reviewed like security. When a group has investors who might sell a business, every shared capability should answer: *if this tenant leaves in 18 months, what do we hand over, what do we switch off, and what runs twice in the meantime?* Concretely: one cloud account boundary per tenant and capability, tenant-scoped data with contracts, deployment descriptors that run outside the platform, and clear legal ownership of code and models. These properties cost little when designed in and a great deal when retrofitted.

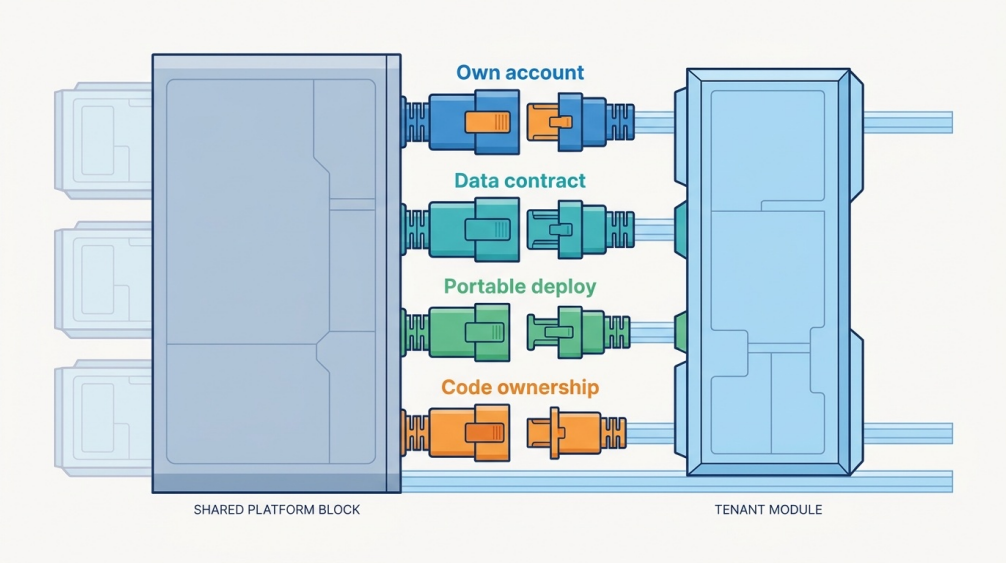


Figure 6: *Separability designed in: one account boundary per tenant, tenant-scoped data with contracts, deployment descriptors that run outside the platform, and clear code ownership let a tenant leave without a rebuild.*

Expect owners to reverse platform strategy within a year. The 2023 convergence and the 2025 decentralization were owner-level decisions about a year apart. An architecture function cannot prevent that, but it can make the platform cheap to reshape in either direction — the practical meaning of optionality in Built to Be Replaced: Evolving a Marketplace Platform Over a Decade.

Comparing the Case to the Book's Patterns and to Veldmark

Adevinta's record supports every book pattern it touches, and adds capital strategy as a stressor.

Book pattern or claim	Adevinta's record	Verdict
Share global components beneath local products (One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture)	Infrastructure, data, narrow ML, experimentation shared; products local until 2023	Supported for 2019–2024
Consolidate enabling layers first, core product last (After the Deal: Consolidating Platforms After Acquisitions)	Followed until 2023; then product convergence announced; reversed 2025	Supported, by both the success and the reversal
Platform as product with voluntary adoption (Who Owns What: Organizing Marketplace Engineering)	1,500 ⟶ 4,000+ developers without a documented mandate	Strongly supported

Book pattern or claim	Adevinta's record	Verdict
Data products with contracts on a shared substrate (Data Platform for Marketplaces: Events, Lakehouse and Data Products)	Adevinta Spain's data mesh and contract-driven ingestion	Supported; also eases separation
Keep funds with a licensed PSP and own the state machine (Beyond the Contact: Payments, Shipping and Buyer Protection)	Adyen holds funds at Kleinanzeigen (help page, 2026) and Online Payment Platform at Marktplaats (as reported in 2022)	Consistent with public descriptions
Put language models in front of structured search (AI Across the Listing Lifecycle)	coches.net and leboncoin map text to filters	Supported; adoption slow
Design for portability and stressors (Built to Be Replaced: Evolving a Marketplace Platform Over a Decade)	Ownership change turned shared platforms into separation costs	Supported; adds capital strategy as a stressor

Veldmark's 2027 program sits close to Adevinta's 2019 rule and further from its 2023 plan. Common Ground, approved in January 2027 at €140 million over 2027–2030, follows the principle "**share platforms, not products**". The case sharpens these Veldmark choices:

Veldmark choice (2027)	Adevinta parallel	What the case suggests
Share identity, trust and safety, media, search platform, messaging, catalog and entitlements, events, data and taxonomy; keep brands, UX and go-to-market local	2019 rule; 2021–2023 shared components	The right layers; but Adevinta named trust and fraud as a decentralization risk (2024), so the trust and messaging contracts must allow a brand to leave

Veldmark choice (2027)	Adevinta parallel	What the case suggests
Shared OpenSearch for Tradeyard and Hearthly by end of 2028; **Motorvane stays on Elasticsearch** until a 2029 review	mobile.de stayed the most distinct, most monetized and most separable asset	Keeping the automotive brand loosely coupled is cheap insurance, not a failure of ambition
Keybridge kept as a separate, neutral SaaS on Azure (March 2027)	Kijiji moved to Google Cloud in 2023 while the EU5 businesses consolidated on AWS	Not every tenant has to sit on the main platform; neutrality and separability both call for tenant-scoped data and access controls
Retire Yardstick by end of 2029 through a strangler sequence (Changing the Engine in Flight: Modernizing a Live Marketplace)	No public evidence of a completed product-platform convergence	Stage gates with exit criteria matter more than the target end state
Majority-owned by a private-equity investor since 2025	Take-private in 2024, then break-up	Veldmark's own capital structure makes separability a live requirement, not a hypothetical

The last row turns separability into a costed question: in November 2027 Veldmark's investor asks what a 2030 sale of Hearthly would cost. By then, under the scenario's timeline, Hearthly would use Veldmark ID (rolled out 2028), the shared OpenSearch search platform (by end of 2028), the Confluent event backbone (from July 2027) and the group catalog and entitlement service (Q2 2028), while running its own services on Google Cloud (through at least 2028). Each would need a handover plan, a transitional service or a switch-off date — what Adevinta's 2025 report calls "transfer of key capabilities" and "localisation of intellectual property".

The architectural answer is not to share less, but to make each shared capability tenant-separable from its first release. That means identity with per-brand profile stores and an export path; a search platform with per-tenant clusters or indices and portable index definitions; event topics and schemas owned per brand; and an entitlement service whose data can be extracted per brand. Veldmark's tenancy model, adopted in June 2027, already requires every shared capability to document a tenant exit path (One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture). Ingrid Halvorsen, Veldmark's fictional Group Chief Architect, puts a number on it for the

board in December 2027. With those rules, a Hearthly carve-out would need transitional services for seven shared capabilities for about nine to twelve months; without them, for more than twenty months. Built to Be Replaced: Evolving a Marketplace Platform Over a Decade develops this labeled estimate.

Questions to Ask

1. For each capability we share or plan to share, does it pass both halves of the 2019 test: large investment *and* uniform technology across markets?
2. Which of our shared capabilities grew by adoption, and which by mandate? What happens to the mandated ones if the mandate is withdrawn?
3. Is any current initiative converging whole product platforms rather than sharing capabilities beneath them? What are its scope, stage gates and exit criteria?
4. If a given marketplace left the group in 18 months, what would we hand over, switch off and run twice, and what would that cost?
5. Are cloud accounts, data, deployment descriptors and code ownership scoped per tenant clearly enough to support that exit?
6. Which local inventions in our marketplaces deserve promotion to group level, and what is the path to promote them?
7. Does our synergy case price the cost of separation as well as the savings from consolidation?
8. Which signals in trust and fraud depend on pooled data across markets, and how would we keep them if the organization were decentralized?
9. What is the blast radius of a change to our shared platform, and which tenants depend on behavior we never promised?

Adevinta shows one group running the whole pendulum in public. The next case, OLX Group: Regional Platforms, Verticalization and Retreat, follows a group that built regional platforms across several continents, verticalized its businesses and then retreated from many markets — a different route through the same questions.

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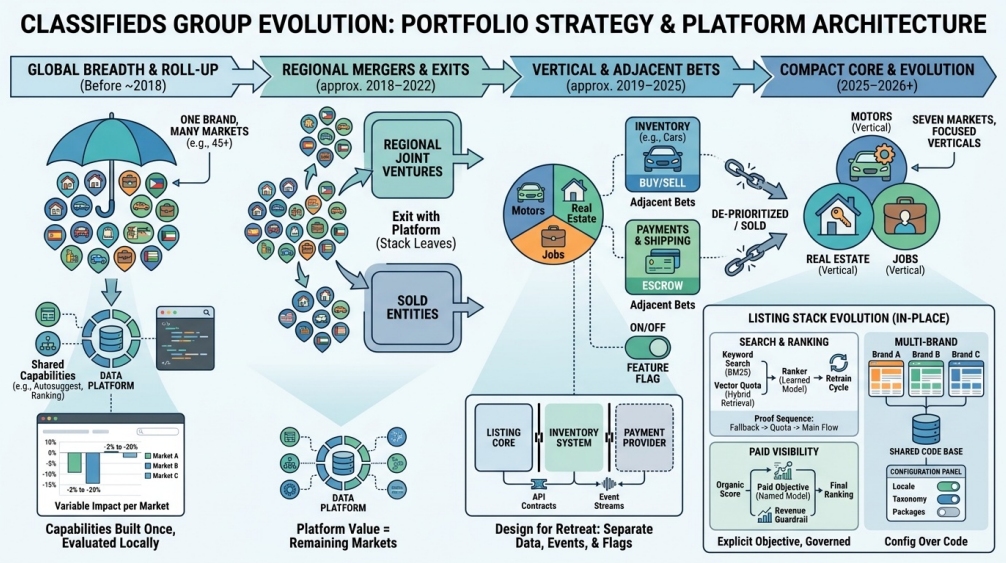
Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

47. OLX Group: Regional Platforms, Verticalization and Retreat



IN THIS CHAPTER, YOU WILL: Trace OLX Group's public record from its 2006 founding and Naspers' roll-up, through the regional mergers and the OLX Autos bet, to the compact European business that Prosus reported in 2026. You will read eight years of OLX engineering posts by era, separate what is documented from the author's reading, and test the case against the book's patterns and Veldmark's choices.

WHY THE BUSINESS CARES: OLX's reported margin roughly doubled in two years after it left most of its markets and closed an inventory business that had made up just over half of its revenue. A group deciding where to compete and which adjacent businesses to enter is deciding what its listing platform will have to carry, and later shed.

WHY ARCHITECTS SHOULD CARE: OLX has published unusually concrete search, ranking, trust-and-safety and multi-brand engineering work from 2018 to 2026. The record shows how one listing stack evolved in place

under portfolio upheaval: lexical search kept and extended, paid visibility encoded in the ranker, and several country brands served from one code base by configuration.

KEY POINTS:

- **OLX converged on local leadership in a few markets.** From "45 markets" in 2018 the group shrank to seven in its year to March 2026, with US$992 million of revenue, a 48% adjusted EBITDA margin, and 71% of revenue from motors, real estate and jobs. Exits went through sales, mergers and joint ventures, and each departing market took its stack with it.
- **Owning inventory did not combine with listing economics.** OLX Autos reached US$1.6 billion of revenue in the year to March 2022, just over half of the classifieds business, while losing money; Prosus exited it from 2023. Pay-and-ship was de-prioritized in Europe and the credit broker and services marketplace were sold. Adjacent bets should be built so that they can be switched off.
- **The listing stack improved by evolution, not replacement.** OLX kept Solr and BM25, moved ranking into the engine as learning to rank, gave paid visibility its own explicit objective, and added vector retrieval as a fixed quota, proving it first in the zero-results fallback of a smaller market. Multi-brand work moved to one repository with per-brand configuration.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Deal-Shaped Architecture** — OLX was assembled from dozens of acquired and rebranded local sites, and its engineering record shows the identity and service-estate clean-up that cross-brand work still needed in 2024–2025.
- **Multiplier Creep** — OLX is the clearest public case of paid visibility handled as an explicit ranking objective, and of a retrieval change that quietly penalized paid listings until the ranker was retrained.
- **The Copy-Paste Country** — The documented move to one front-end repository for three brands and one learning-to-rank repository with per-brand configuration is the remedy for per-country copies.

OLX Group has made almost every portfolio move a classifieds group can make. It rolled up local sites across four continents and rebranded many of them to one name. It merged regional businesses into joint ventures in which it kept a minority stake. It built an inventory-owning car business that briefly doubled its revenue, then sold or closed it. It left Russia after the 2022 invasion of Ukraine. And by 2026 it was a **compact**

European business of seven markets, reported by its parent Prosus with margins near 50%.

Part IX tests the book's patterns against real groups, and OLX answers the question Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform leaves open: **what happens to a platform when the portfolio, not the platform strategy, changes?** This chapter asks of each phase: *which markets and businesses did the platform carry, what did the group bet on beyond listings, and what did the engineering teams publish about the stack while the portfolio changed around them?* The answers support three arguments of this book: **liquidity is local and so is the platform's value** (Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics); **adjacent transactional and inventory bets must be designed to be retired** (Beyond the Contact: Payments, Shipping and Buyer Protection); and **a listing stack is improved by evolving search, ranking and configuration in place** (Ranking for Two Sides: Relevance, Freshness and Paid Visibility, One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture).

OLX's engineering blog spans 2018 to 2026, but it describes three different organizations: the global OLX of India, Indonesia, Pakistan, South Africa and Latin America; OLX Autos; and today's OLX Europe. **Many posts describe markets OLX no longer owns.** Every "OLX runs X" statement below is true at the date of its source and for the market it names, and no further.

How to Read This Case

Part IX cases follow stricter rules than the chapters in Parts I–VIII (see Part IX — PRACTICE: How Others Built It). A **documented fact** carries its source and publication year in the sentence. **The author's interpretation** is marked "my reading" or sits in the Chief Architect's view. A **reference-architecture inference** appears only in a callout labeled `INFERENCE, NOT DOCUMENTED`.

Three limits shape the case. The financial evidence is unusually good, because Prosus reports OLX separately and, since 2026, by vertical. The engineering evidence is strong on search, ranking, experimentation and trust and safety, and almost silent on platform consolidation. And the history before 2019 rests largely on secondary sources, marked as widely reported.

Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), appears only in the final comparison. It operates in some of the same countries as OLX, but it is a composite of industry patterns and is not a disguised description of OLX or any other group.

Timeline: OLX in Dated Events

The chronology follows. Dates are those of the cited releases and reports; ownership facts are as of October 2026 and may change. Prosus's fiscal year ends on 31 March, so "FY26" is the year to March 2026.

Date	Event	Source type
2006	OLX founded by Fabrice Grinda and Alec Oxenford as a classifieds site for markets outside the United States	Wikipedia (widely reported)
2014	Naspers raises its stake to about 95%; acquired local brands in Central and Eastern Europe and Asia are rebranded to OLX; OLX Brazil becomes a joint venture with Schibsted (whose stake passed to Adevinta at the 2019 demerger)	Wikipedia (widely reported)
2016	Naspers' classifieds businesses consolidated as OLX Group, headquartered in Amsterdam	Wikipedia (widely reported)
Nov 2018	OLX describes itself as "a network of leading classifieds platforms in 45 markets", including OLX, Avito, letgo and dubizzle	OLX tech blog, 2018
Jan 2019	Naspers takes full control of Avito (Russia)	Wikipedia (widely reported)
Apr 2019	Businesses in Kenya, Ghana, Uganda, Tanzania and Nigeria sold to Jiji; OLX Philippines to Carousell, with OLX taking a 10% stake	Wikipedia (widely reported)

Date	Event	Source type
2019	Controlling stake in Frontier Car Group, the instant-offer car-buying business that becomes OLX Autos	TechCrunch, 2021
Mar 2020	OfferUp and letgo combine their US businesses; OLX owns 40%	OLX Group release, 2020
Jun 2020	OLX contributes dubizzle and its Middle East and South Asia businesses to Emerging Markets Property Group (later Dubizzle Group); OLX holds 39%	Wamda, 2020
Oct 2020	OLX Brazil, a 50/50 joint venture of Adevinta and Prosus, completes the R$2.9 billion acquisition of Grupo ZAP	OLX Group release, 2020
Mar 2021	Frontier Car Group's Berlin entity closed; OLX Autos focuses on Latin America and Asia	TechCrunch, 2021
Oct 2022	Avito sold for RUB 151 billion; Russian exit complete	Naspers release, 2022
Mar 2023	Prosus announces its exit from OLX Autos	News24/Fin24, 2023
Aug 2023	OLX India sold to CarTrade; Indonesian business sold to Astra	Wikipedia (widely reported)
Aug–Oct 2025	Kiwi Finance sale closed; agreements to sell OLX Uzbekistan, OLX Kazakhstan and Fixly (Polish services marketplace)	OLX Group releases, 2025–2026
17 Nov 2025	La Centrale (France, motors) acquired for about €1.1 billion	OLX Group release, 2025

Date	Event	Source type
May 2026	Otodim launched in beta in Ukraine, a real-estate search platform drawing on OLX Ukraine's inventory	OLX Group release, 2026
Jun 2026	FY26 results: seven markets (Bulgaria, France, Poland, Portugal, Romania, South Africa, Ukraine), US$992 million revenue	OLX Group and Prosus releases, 2026

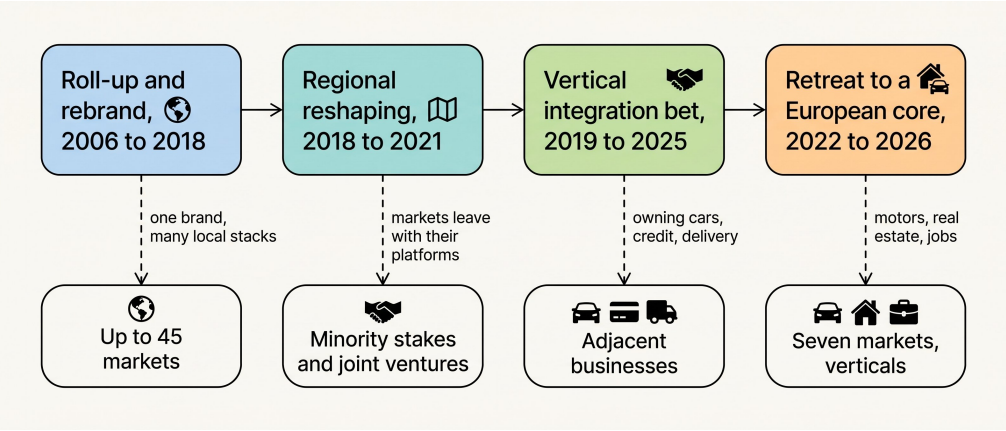


Figure 1: *Each phase changed what the listing platform had to carry: many markets under one brand, then fewer markets as regions left through mergers, then inventory and payment businesses beside the listings, and finally a small set of markets organized by vertical.*

Phase 1: Roll-Up and Rebrand (2006–2018)

The secondary sources behind the early history agree on its shape: Naspers bought into OLX in 2010, raised its stake to about 95% in 2014, and in April 2014 rebranded acquired local leaders to OLX in Romania, Bulgaria, Kazakhstan, Belarus, Hungary and Poland, while vertical brands such as Otomoto, Otodom, Storia, Autovit, Standvirtual and Imovirtual kept their names.

My reading is that the 2014 rebranding was brand consolidation, not platform consolidation. A shared name says nothing about a shared code base, and nothing public describes how many stacks the 45 markets ran. The engineering record of this era does

show **capabilities built once and run across regional markets**, and a habit of fitting new methods into the existing search engine rather than beside it.

- **Ranking inside the engine (2019).** A May 2019 post on the global-markets search team describes a logistic-regression ranking model whose coefficients were "plugged into the ranking function of the existing search engine", to avoid new serving infrastructure. It reports about 10% improvement on the key search metric, and states that "search generates about 80% of total conversions".
- **Synonyms from sessions (2018).** For OLX South Africa, a 2018 post describes "query2vec": a skip-gram model over chains of queries in the same session, used to expand searches. Null searches fell by 27%. An early version returned "shoes" for "clothes", which led the team to a stronger evaluation process.
- **Autosuggest across regions (2020).** A December 2020 post describes an autosuggest corpus built from user queries, served through Solr's suggest component, and reports results per market:

Market (2020)	Null searches	Search conversion
Indonesia	−5.1%	+1.2%
Pakistan	−20%	+4.1%
South Africa	−5%	+2.4%
India	−2%	not reported

Source: OLX tech blog, "AutoSuggest Retrieval & Ranking (Part 2)", December 2020. All four markets have since been sold, merged or reduced to a minority stake except South Africa.

The same capability produced a tenfold spread in null-search reduction across markets. **My reading** is that this is the strongest early evidence in the record for the book's argument that a shared capability must be evaluated per market (Testing a Two-Sided Market: Experimentation Platforms): the code was shared, the liquidity and language were not.

The data platform of this era was AWS-based: a 2018 GOTO Berlin talk by OLX's chief technology officer describes a data lake "collecting and storing billions of events and records per day", and a 2019 post describes ML pipelines on SageMaker and Step Functions. Both are likely replaced; a 2023 post shows Airflow and MLflow (see Data Platform for Marketplaces: Events, Lakehouse and Data Products).

Phase 2: Regional Reshaping Through Mergers and Joint Ventures (2018–2021)

From 2018 the group began to concentrate. It left Nigeria and Venezuela in 2018, sold its East and West African businesses to Jiji and OLX Philippines to Carousell in 2019, and then made three structural moves in 2020:

- **United States**. OfferUp and letgo combined their US businesses in March 2020, with OfferUp operating the merged company and OLX owning 40% (OLX Group release, 2020).
- **Middle East, North Africa and South Asia**. In June 2020 OLX contributed dubizzle and its businesses in Egypt, Lebanon, Pakistan and several Gulf countries to Emerging Markets Property Group for a 39% stake in a company valued at about US$1 billion (Wamda, 2020).
- **Brazil**. OLX Brazil, owned 50/50 with Adevinta, bought Grupo ZAP for about R$2.9 billion, creating a business with about 1,600 staff, about 400 of them in technology; the OLX, ZAP and Viva Real brands kept their names (OLX Group release, 2020).

My reading is that these deals swapped operating control for equity. They are portfolio decisions, but they have a direct architectural consequence that the public record makes visible by omission: **a market contributed to a merger takes its platform with it**. No source describes OLX technology being shared with OfferUp, Dubizzle Group or OLX Brazil after the deals. Whatever regional capabilities had been built centrally either were already separable or had to be made so. That is the same lesson Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform draws from Adevinta's 2025 carve-outs, reached five years earlier and by a different route.

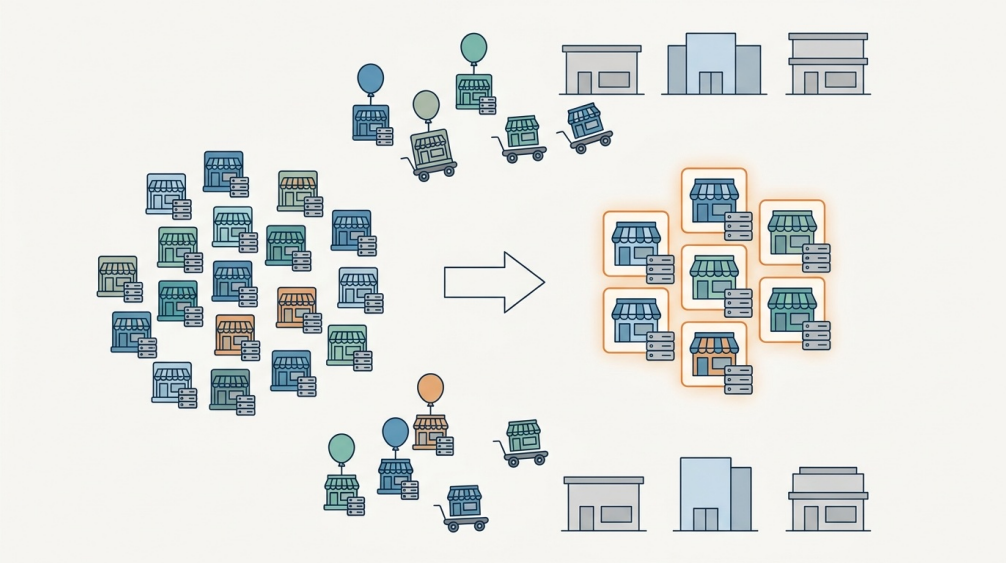


Figure 2: *From 45 markets in 2018 to seven in 2026: every market that left through a sale, merger or joint venture took its stack with it, so a shared capability is worth only the markets that stay.*

Phase 3: The Vertical Integration Bet (2019–2025)

OLX Autos

In 2019 OLX took a controlling stake in Frontier Car Group, investing about US$400 million (TechCrunch, 2021). The business, renamed OLX Autos, bought cars from consumers through an instant offer, reconditioned and resold them. In March 2021 OLX closed its Berlin entity, moved core product development to India and focused OLX Autos on Latin America and Asia (TechCrunch, 2021).

Revenue then grew while losses widened. According to News24/Fin24's March 2023 report on the Prosus results:

- OLX Autos revenue was about **US$1.6 billion** in the year to March 2022, up 158%, "just over half of the classifieds business's revenue", with about **175,000 cars** transacted (98,000 the year before).
- In the first half of the following year revenue grew about 60% to about US$1 billion, while **trading losses widened to US$206 million**, which Prosus linked to spending on retail infrastructure.
- Prosus announced the exit, citing "higher cost of capital, high inflation and a reversal of pandemic trends" and a slowdown in used cars.

In June 2023 OLX cut about 800 roles and stopped buying cars in Argentina, Mexico and Colombia (TechCrunch, 2023); OLX India and the Indonesian business were sold in August 2023. Prosus's FY25 results record that all OLX Autos operations were "disposed of, classified as held for sale or closed by 31 March 2025", with a discontinued-operations loss of US$128 million in that year. South Africa's AutoTrader, a listing business, was never part of OLX Autos.

> **INFERENCE, NOT DOCUMENTED:** A plausible reading of why the bet strained the architecture as well as the balance sheet — not a description of OLX Autos' systems — is that an inventory business needs capabilities a listing platform does not have: inspection and condition records, a pricing authority that commits capital, reconditioning workflow, vehicle logistics and stock accounting, a finance book, and a retail front end that sells owned stock. A classifieds platform influences price and approximates condition; a retailer owns both. The two share a vehicle taxonomy and an audience, and little else. Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy places this "managed" model on the ladder and mobile.de, Auto Trader and AutoScout24: Lessons from Automotive Marketplaces compares it with other automotive groups.

The other adjacent bets

Beside OLX Autos, its largest bet beyond listings, OLX entered several smaller adjacent businesses and has left most of them.

Bet	Entered	What it was	Fate	Source
OLX Autos	2019	Instant-offer car buying and resale	Exit announced 2023; all operations gone by March 2025	TechCrunch, 2021; Fin24, 2023; Prosus FY25
Kiwi Finance (Romania)	Dec 2020	Credit broker	Sale closed 1 August 2025	OLX Group releases, 2020 and 2026

Bet	Entered	What it was	Fate	Source
Pay&Ship (Europe)	by 2023	"OLX holds the money of the transaction and transfers it to the seller when the package is delivered" (the post does not name the payment provider or licensed party)	"De-prioritisation of pay-and-ship in Europe" in the second half of FY25	OLX tech blog, 2023; Prosus FY25
Fixly (Poland)	not verified	Services marketplace	Sale agreed 29 October 2025	OLX Group release, 2026
Inmokey (Argentina)	Oct 2019	Software for real-estate agents	Not reported after the acquisition	OLX Group release, 2019
Obido (Poland)	Feb 2021	New-build platform matching developers and buyers	Brand retained; current status not reported	OLX Group release, 2021

The only engineering view of the transactional layer is a May 2023 post on predicting delivery-time ranges for Pay&Ship with two quantile-regression models. Two years later the product line was de-prioritized. **My reading** is that, from outside, this is what a cheap retreat looks like: a transactional product scaled back with no visible disruption to the listing business. Whether that reflects how it was built is not public. Beyond the Contact: Payments, Shipping and Buyer Protection develops the design of such a flow and why the money should sit with a licensed provider.

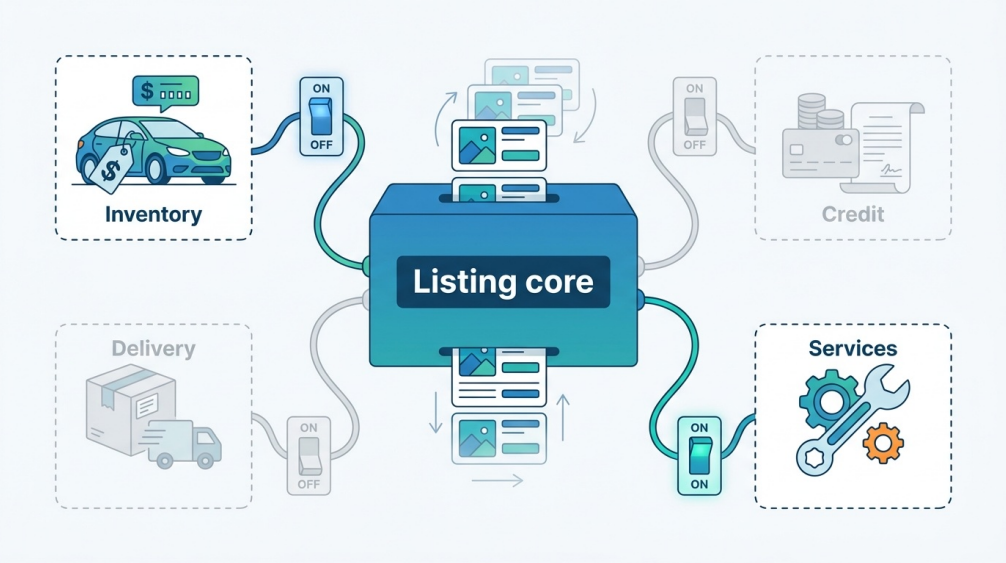


Figure 3: *Adjacent bets designed to be retired: inventory, credit, delivery and services kept outside the listing core, each behind its own boundary and switch, turn a retreat into a configuration change rather than a migration.*

Phase 4: Retreat to a European Core (2022–2026)

Exits and one acquisition

Russia went first. On 14 October 2022 Naspers announced the sale of Avito, in which Prosus held an effective 99%, for RUB 151 billion in cash, about US$2.46 billion according to Euronews/Reuters, against earlier valuation estimates of about US$6 billion.

Central Asia followed. In August 2025 OLX agreed to sell OLX Uzbekistan to a joint venture led by TBC Bank Group, to "sharpen our focus on core product categories and geographies" in the words of chief executive Christian Gisy, and in October 2025 OLX Kazakhstan to VEON's Beeline Kazakhstan for US$75 million. The FY26 release gives only signing dates, so this chapter treats completion as not confirmed.

The same period brought the one large acquisition. OLX bought **La Centrale**, a French motors marketplace, from Providence Equity for about €1.1 billion; the deal closed on 17 November 2025. The FY26 release reports 13% traffic growth and 30% more leads at La Centrale, with integration "ahead of plan". It was OLX's first market in Western Europe, entered through a vertical, not a horizontal brand.

The economics of focus

Prosus reports OLX separately, so the retreat shows directly in its margins.

Period	Revenue	Profit measure	Margin	Source
FY24 (to March 2024)	about US$660m (implied)	aEBIT	25%	Prosus FY25 results, 2025
FY25	US$777m (+18%)	aEBIT US$270m	35%	OLX Group FY25 release, 2025
FY26	US$992m (+28%; +16% in local currency excluding M&A)	aEBITDA US$481m (+53%)	48%	Prosus FY26 media release, 2026

aEBIT is adjusted earnings before interest and tax ("trading profit"); aEBITDA adds back depreciation and amortization. The measure changed between FY25 and FY26, so the 35% and 48% are not the same metric; Prosus describes the FY26 margin as an eight-point expansion, which implies an aEBITDA margin of about 40% in FY25.

From FY26 Prosus also reports OLX by vertical:

Vertical, FY26	Revenue	Growth	aEBITDA margin
Motors	US$429m	+42% (+20% excluding M&A and currency)	59%
Real estate	US$185m	+26%	46%
Jobs	US$87m	+14%	43%
Remainder (general classifieds and other), by subtraction	about US$291m	not reported	not reported

Source: Prosus FY2026 media release, 29 June 2026. The remainder row is the author's arithmetic (US$992m minus the three verticals), not a reported figure.

Motors, including La Centrale from November 2025, is the largest and most profitable line, and the three verticals are 71% of revenue although the horizontal OLX sites carry most of the audience. Prosus's FY25 booklet ties the growth to vertical products: in motors, "dealer tools" plus vehicle-history transparency and dealer ratings; in real estate, "the rollout of a unified platform, fully reimagined app and tools optimised for agents".

My reading is that the horizontal brand works as a funnel and the verticals monetize it. Otodim, launched in beta in Ukraine in May 2026, shows the move in its plainest form: a dedicated real-estate search platform that "draws on OLX Ukraine's extensive listings inventory", with filters for proximity to shelters and power autonomy, and with independent listing and in-platform messaging still "in development". **A vertical was launched first as a new lens on horizontal inventory**, not as a new supply system. Same Platform, Different Physics: How Listing Verticals Differ discusses what a vertical needs beyond such a lens, and Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform shows Adevinta's parallel reorganization by vertical in 2023.

The Engineering Record: How the Listing Stack Evolved

The posts of 2022–2026 describe OLX Europe, the business that remains. They are concentrated in search, ranking, recommendations, experimentation, trust and safety, and the supply-side use of language models.

Search: five stages on one engine

OLX's search evolved through five stages on one engine, Apache Solr, and is the best-documented search evolution in the book's research.

Stage	When	What OLX described	Source
Sort only	before about 2021	"It wasn't possible to sort search results by any criteria other than ad age or price"	OLX tech blog, 2022

Stage	When	What OLX described	Source
Hand-tuned formula	about 2021	A "Recommended" sort combining text similarity, ad statistics such as click-through rate, and time, "hand-crafted into a simple formula"; making click-through data available to ranking "was no joke"	OLX tech blog, 2022
Learning to rank in the engine	2022	A LambdaMART model trained with XGBoost and run through Solr's learning-to-rank plugin; labels are purchases or "messages/calls to sellers" "because transactions aren't always confirmed"; +12% conversions on page one, +5% purchases, +4% unique buyers; unique contacted sellers also rose; offline NDCG correlated with online results	OLX tech blog, 2022

Stage	When	What OLX described	Source
Two objectives	by 2025	"Vanilla LTR", "trained using seller contact events, which we call successful events", and "VAS LTR", which "balances two objectives: maximizing relevance for users and ensuring visibility for Value Added Services (VAS), ads that have paid for additional exposure"	OLX tech blog, 2025
Hybrid retrieval	2025	BM25 keyword retrieval plus a fixed number of vector results, re-ranked	OLX tech blog, 2025

Learning to rank (LTR) trains a model to order results from labeled outcomes; NDCG (normalized discounted cumulative gain) is the standard offline ranking metric. Both are defined in Ranking for Two Sides: Relevance, Freshness and Paid Visibility.

The success label is a contact, not a sale, because a classifieds platform rarely observes the sale, and the check that unique contacted sellers *rose* guards against a ranker that concentrates demand on a few sellers (Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics). And **paid visibility is an explicit, separate objective** rather than a multiplier hidden in a scoring formula. That is the opposite of Multiplier Creep: whoever changes the ranker has to see the paid objective.

Hybrid search: proved in the fallback, then in the main flow

The September 2025 hybrid-search post is the most instructive single document in the record, because it describes the order of proof as well as the design. Keyword matching "often led to low recall or even zero results pages (ZRPs)", and the "chain of fallback

retrieval steps designed to address edge cases and mitigate ZRPs", the extension chain, had become hard to improve. What followed:

1. **Keep BM25 and add a quota of vectors.** Vector similarity scores "did not consistently separate relevant from non-relevant ads", so OLX used a fixed allocation instead of a threshold: "We began with 35 results from semantic search", later "from 35 to 70", which "improved the quality of the hybrid recall set". Embeddings came from one multilingual text model for all markets.
2. **Notice that the ranker had never seen vector candidates.** The LTR model, trained only on keyword results, "consistently assigned lower relevance scores to vector search results", which "would be particularly problematic for VAS ads retrieved via semantic search". OLX retrained it with query–ad similarity as a feature.
3. **Bootstrap training data with a deliberately modest ranker.** Because vector results had no interaction history, OLX introduced "a linear re-ranking model to temporarily replace LTR", built on its earlier BM25-weighted function plus recency. Recency was protected because paid listings receive recency resets when sellers buy services.
4. **Prove value where risk is lowest.** Vector search first ran as a step in the extension chain in a smaller market, where it more than doubled seller-contact events per user. A main-flow test failed to converge because of a web tracking bug; re-runs showed "an approximately 3% increase in the share of users who completed a successful event" on apps and "a ~5% increase" on mobile web, after which OLX decided to "proceed with a full rollout of Hybrid Search on all platforms in this market".

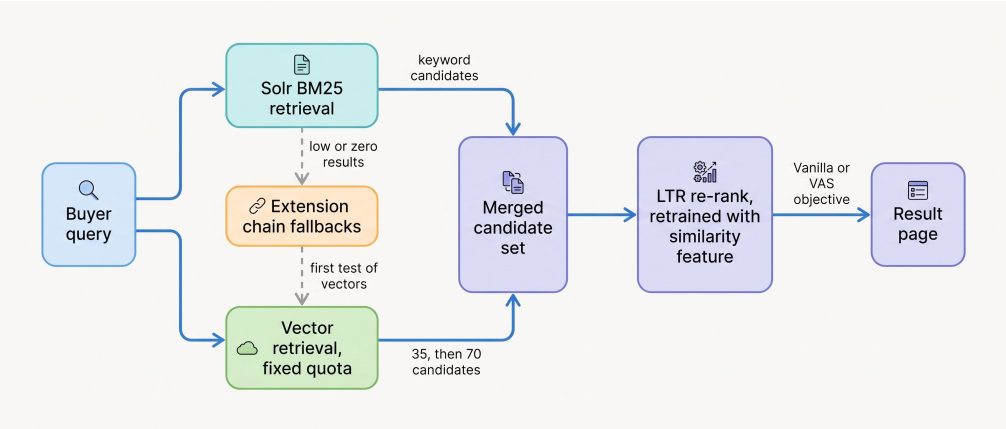


Figure 4: *The hybrid design as OLX described it in 2025: lexical retrieval kept, vectors added as a fixed quota, and one ranker retrained to judge both, with the zero-results*

fallback as the first place vectors were tested. The diagram shows the documented shape, not internal components.

Similarity is not relevance in local classifieds, as a companion post in October 2025 shows. OLX chose the embedding model by memory, latency and language constraints, with an offline dataset that weighted favorites and chat replies above clicks. The online test for similar items still lost to the incumbent until location and price were brought back through re-ranking. What Did They Mean? Query Understanding and Autocomplete and Search Architecture for Listings: Indices, Freshness and Scale develop hybrid retrieval in general.

Measuring paid visibility honestly

OLX treats its paid products as claims to be measured. They are packages with a number of "boosts" per 30 days and Value Added Services (VAS), paid exposure sold per listing. In August 2026 OLX described dashboards on which premium motors packages "appeared to perform worse than ads using cheaper packages", because "professional sellers self select into a package". Propensity-score matching, validated against a randomized experiment, estimated uplifts of 17.6% to 97.4% between package levels. An August 2025 post on group sequential testing shows the same rigor in method: with daily peeking, "despite a significance level of 5%, 27.9% of the simulations yielded a significant result". Packages, Bumps and Quotas: Products, Pricing and Entitlements and Testing a Two-Sided Market: Experimentation Platforms develop both methods.

One code base, many brands

The record has three explicit statements about multi-brand engineering, all recent.

- **Front end**. A February 2026 Poznań masterclass was titled "One-product, three brands — Our strategy for serving different countries from a single repo", on managing "multiple websites from a single repository, balancing localisation, configuration, and design".
- **Ranking**. A June 2025 post explains that LTR projects spread across teams and repositories caused "duplicated efforts", so OLX centralized "all LTR projects within a single repository", using the Hydra configuration framework for brand-level settings across the general marketplace, motors and real estate.
- **Service-estate hygiene**. A March 2026 post on OLX Motors' Work Organization Framework lists what cross-brand work required first: "standardized user identification across 71 services, with 73% now using modern UUIDs instead of legacy user IDs", 45 or more components "migrated to AWS Secrets Manager", deployment notifications unified "across 60+ services", and pipeline times cut "by up to 44% (saving teams 94 hours per week)". It opens with the diagnosis that by 2024 OLX Motors had "a classic case of 'too much of everything, not enough of anything' ".

An August 2026 post on coding agents mentions, in passing, a change across "three codebases": two Go services and a PHP service.

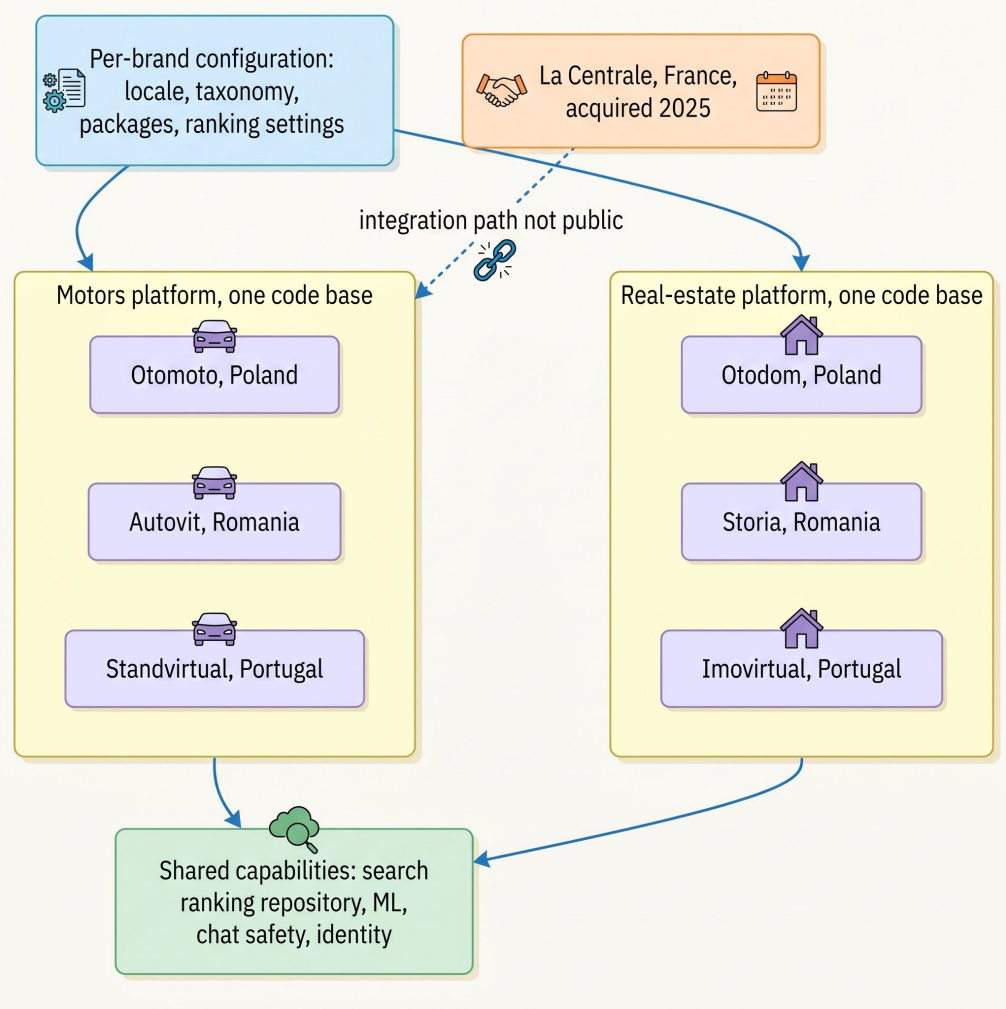


Figure 5: *A reference reading of OLX Europe's multi-brand shape, assembled from the documented fragments. Everything in this diagram except the brand names, the ranking repository and the La Centrale acquisition is the author's inference.*

INFERENCE, NOT DOCUMENTED: The fragments — one front-end repository for three brands, one ranking repository with brand configuration, a "unified platform" credited for real-estate growth, user identifiers standardized across 71 services — are consistent with a common classifieds pattern: one code base per vertical, serving several country brands, with locale, taxonomy and commercial packages held as configuration, and a

legacy PHP core being replaced by Go services. That is a plausible reference architecture, not a description of OLX's systems. OLX has not published which brands share which code base, when the real-estate platform was unified, or whether La Centrale will move onto the motors platform. One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture treats this tenancy model in general.

Trust and safety, media and recommendations

The rest of the record belongs to the chapters that own each topic; the table summarizes it and links to them.

Area	What OLX published	Date and era	Chapter
Moderation	An automatic moderation system of many models that accepts, rejects or escalates listings to a human moderation panel; "more than 40 machine learning services" in production, more than half in trust and safety and in search and recommendations	2020; global and Europe	Trust and Safety as a Platform

Area	What OLX published	Date and era	Chapter
Duplicates and fraud rings	Two-step duplicate detection (cheap candidates, then a pairwise classifier); outlier detection plus graphs of accounts sharing phones, e-mails or message text	2020 (also PyCon DE 2019)	One Property, Three Listings: Duplicates and Entity Resolution, Scams, Fakes and Account Takeovers: Fraud Prevention for Listings
Chat fraud	"Chat Images GenAI" blocks suspicious images and QR codes in chat in real time; fraudulent images "around 30% of all fraud cases" in Poland; QR detection at 80% precision; live in Poland, Portugal, Bulgaria, Romania and other countries	January 2026; Europe	From Interest to Contact: Messaging and Lead Management

Area	What OLX published	Date and era	Chapter
Media	CDN serving about 80% of image traffic; an on-demand processor rewritten from thumbor (about 6,000 pods at peak) to Dali, a Rust service on libvips, at about twice the load per instance	2020; current state not public	Photos Sell: Media Pipelines for Listings
Recommendations	Item2Vec embeddings with 10–20% uplift in clicks and contacts from recommendation sections (2021); a data-team-owned Python service on ScyllaDB and DynamoDB reaching "a p99 latency under 10ms" after async I/O, batching and garbage-collection tuning (2025)	2021, 2025; Europe	Recommendations for Inventory That Disappears

Language models on the supply side and off the platform

OLX's 2025–2026 releases describe generative AI mostly on the supply side and in new channels; the outcomes are company-reported. Auto-filled attributes and descriptions cut posting time by 35–55% (2025). Auto Description drafts a listing from a photo for the seller to review, with human evaluation of 98% for style and relevance and 89% for faithfulness (2026). A January 2024 post describes a service "subscribed to ad events"

that extracted job roles with an LLM for the search index, at "approximately 15K per month" (currency unstated) for about 4,000 calls a day. AutoIQ is a copilot for Otomoto dealers (2026). And CompassGPT (property, Imovirtual) and AutoGPT (cars, Otomoto first) run as apps inside ChatGPT, built with OpenAI (2026).

My reading is that the last item is the architecturally significant one. A marketplace that serves its inventory to an assistant it does not control turns structured attributes, availability and API-grade access into the product, and needs the channel controls that AI Across the Listing Lifecycle sets out.

Trade-offs the Case Illustrates

Option	What OLX's record shows	When it is right	When it fails
Global breadth under one brand	45 markets in 2018; shared capabilities with very different effects per market	When a market can be won early and cheaply, and capabilities are portable	When local leadership cannot be reached, or the group's attention and capital are spread thin
Exit by merger or joint venture	US, Middle East and South Asia, Brazil: minority stakes, no shared technology documented after the deals	When a stronger local partner exists and the group wants exposure without operations	When shared group capabilities cannot be separated from the departing market
Owning inventory beside listings	OLX Autos: half of revenue, growing losses, exit within four years	Rarely for a classifieds group; only with cheap capital, pricing data and a retail operation	When used-car prices or capital costs move, as they did in 2022–2023

Option	What OLX's record shows	When it is right	When it fails
One code base per vertical with brand configuration	One front-end repository for three brands; one LTR repository with brand configs; identity clean-up across 71 services	When brands share a vertical's data model and differ mainly in locale and packages	When a brand's product genuinely differs, or the configuration surface grows into a second code base
Evolve the search stack in place	Solr kept; LTR in the engine; vectors as a quota proved in the fallback path	When the incumbent engine is healthy and the team can retrain the ranker	When the engine cannot host the new retrieval or ranking method at the needed latency

The rows are not a maturity ladder. **What the OLX record adds to the usual platform discussion is that the portfolio decision comes first**: which markets and businesses the group keeps decides what a shared platform is worth, and every exit in the record went through a deal structure that took the market's platform with it.

Vertical Differences in the Record

Vertical	OLX examples (2026)	What the record shows	Implication
General classifieds	OLX.pl, OLX.ro, OLX.pt, OLX.bg, OLX.ua	Most of the audience; contact as the success label; VAS and boosts in ranking; chat fraud is the main trust problem; pay-and-ship de-prioritized	The horizontal site is the funnel, so search quality, chat safety and listing creation matter most

Vertical	OLX examples (2026)	What the record shows	Implication
Motors	Otomoto, Autovit, Standvirtual, La Centrale	US$429m revenue at 59% margin (FY26); dealer tools, vehicle history, dealer ratings; AutoIQ; package boosts measured causally; the inventory-owning model abandoned	Dealer products and lead quality carry the economics; the listing model, not the retail model, is the one that paid
Real estate	Otodom, Storia, Imovirtual, Obido, Otodim	US$185m at 46% margin; "unified platform" and agent tools; a new vertical launched as a lens on horizontal inventory	A shared vertical platform across countries is credited with growth; its mechanics are not public
Jobs	OLX jobs sections in Poland, Romania, Ukraine	US$87m at 43% margin; job-role extraction with an LLM; AI matchmaking and CV parsing; blue- and grey-collar focus	Taxonomy normalization of free text is the core enrichment problem

Failure Modes the Record Exposes

1. **Revenue that hides a different business**. OLX Autos made reported classifieds revenue look twice as large while losing money. A platform that books listing revenue and principal revenue from selling cars in the same metrics will mislead its own investment decisions (How Listing Marketplaces Make Money).
2. **Shared capabilities assumed to have one effect**. The 2020 autosuggest results ranged from −2% to −20% null searches across four markets. A group that rolls out

a shared capability and reports one average learns nothing about where it works.

3. **A retrieval change that silently changes revenue**. The keyword-trained ranker under-scored vector candidates, including paid ones. Without an explicit paid objective and revenue guardrails, a search improvement can become a monetization regression (Ranking for Two Sides: Relevance, Freshness and Paid Visibility).
4. **Identity fragmentation after a roll-up**. Needing to standardize user identification across 71 services in 2024–2025, a decade after the rebranding, shows how long acquired identity models survive. Identity is the first thing to consolidate, and the last to finish (Who Is on the Other Side? Identity and Verification).
5. **Adjacent bets without an exit design**. OLX Autos, Kiwi Finance, Pay&Ship and Fixly were all entered and left within about six years. The record does not show what the exits cost in engineering terms; the frequency alone argues for designing adjacent businesses as separable from the start.

Where the Public Evidence Stops

The following questions matter to an architect and are not answered by any public source found for this book. The chapter does not fill them with inference beyond the labeled callouts above.

Question	What is public	What is not
How do OLX Europe's country brands share code?	One front-end repository for three brands (2026); one LTR repository with brand configuration (2025); a "unified platform" in real estate (FY25)	Which brands share which code base, when unification happened, and what remains local
Will La Centrale move onto the motors platform?	Integration "ahead of plan" (FY26)	Whether "integration" means technology, commercial or organizational integration
What is the event backbone?	Kinesis used for search ingestion of enriched ad events (2024)	Whether Kafka, Kinesis or both carry listing events; schema governance
How far has the PHP-to-Go migration gone?	Go and PHP services coexist (2026)	Scope, timeline and remaining legacy share

Question	What is public	What is not
What runs chat in Europe today?	Real-time chat screening of images and QR codes (2026); a Cassandra chat store described by the India-era chat team (2021)	The current European messaging stack
What did the exits cost technically?	Deal terms and dates	Any account of separating shared systems from departing markets, or of transitional services

Three facts are volatile as of October 2026: completion of the Uzbekistan, Kazakhstan and Fixly sales; the status of the OLX Brasil joint venture with Adevinta; and the "OLX Magic" assistant described in a 2024 talk, which later releases do not mention. Readers should check them before relying on the portfolio list above.

The Chief Architect's View

What I take from OLX is a set of decision rules, each conditioned on the evidence.

Let the portfolio decide what the platform is worth. A shared platform's value is the sum of the markets it serves for as long as the group keeps them. OLX served 45 markets in 2018 and seven in 2026, and every exit took its market's stack along. Before funding a shared capability, I would ask which of today's markets will still be in the group in five years, and design the capability so that the others can leave.

Price adjacent bets as businesses that may be closed. Inventory, credit, delivery and services were all entered and left. The architectural defense is cheap: keep the money with a licensed provider, keep inventory and retail systems outside the listing core, give the adjacent business its own data boundary and its own events, and make its user-facing entry points feature-flagged per category. Then a retreat is a configuration change and a contract notice, not a migration.

Change the search stack by evolution, with a proof sequence. OLX's hybrid-search rollout is the sequence I would adopt as a standard: keep the incumbent retrieval, add the new method as a bounded quota, retrain the ranker so that it can judge the new candidates, ship a deliberately modest ranker to collect data, test first in the fallback path of a smaller market, and only then in the main flow. Each step limits what can go wrong.

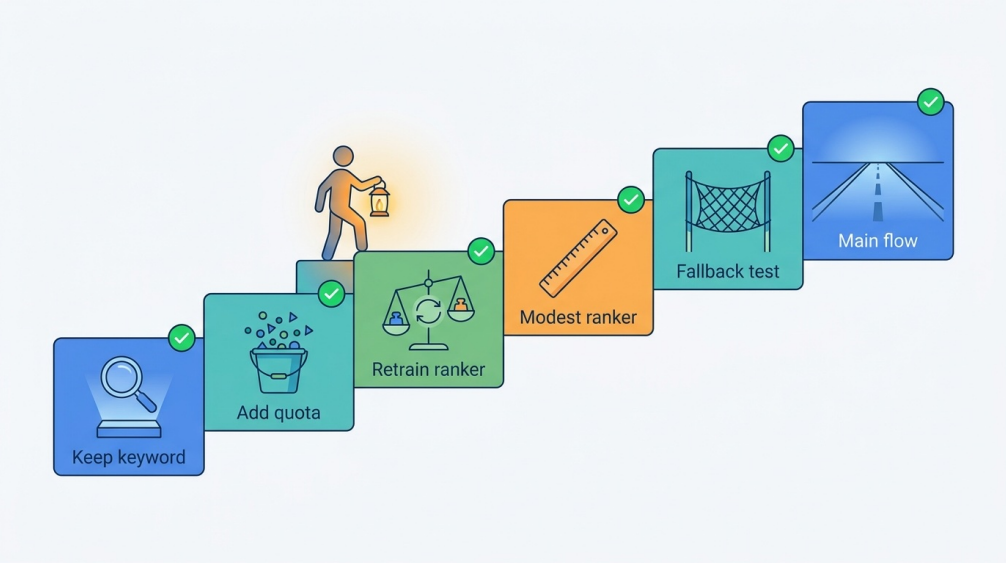


Figure 6: *Evolution with a proof sequence: keep the incumbent retrieval, add vectors as a bounded quota, retrain the ranker, collect data with a modest ranker, prove value in the fallback path of a smaller market, and only then change the main flow.*

Make paid visibility a named objective with an owner. A separate paid-visibility model, as OLX describes, or a policy layer with explicit caps, as Ranking for Two Sides: Relevance, Freshness and Paid Visibility proposes, both force the trade-off into the open. What fails is the multiplier nobody remembers adding. Every ranking or retrieval change should carry a revenue guardrail metric next to its relevance metric.

Treat identity and service-estate hygiene as the start of consolidation. OLX Motors' framework spent its early effort on user identifiers, secrets, pipelines and deployment notifications. That is unglamorous work and the precondition for every cross-brand capability. I would schedule it first in any group and measure it, as OLX did, in services migrated rather than in plans approved (After the Deal: Consolidating Platforms After Acquisitions).

Comparing the Case to the Book's Patterns and to Veldmark

The book's patterns, as OLX's record tests them:

Book pattern or claim	OLX's record	Verdict
Liquidity is local; groups are portfolios of local leaders (Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics)	From 45 markets to seven, with margin rising as the portfolio narrowed	Strongly supported
Contacts as the success event where sales are invisible (Ranking for Two Sides: Relevance, Freshness and Paid Visibility)	LTR labels are purchases or seller contacts; unique contacted sellers checked	Supported
Paid visibility belongs in an explicit, governed ranking layer (Ranking for Two Sides: Relevance, Freshness and Paid Visibility, How Listing Marketplaces Make Money)	"VAS LTR" as a separate objective; retraining after vector retrieval penalized paid ads	Supported, with a documented failure mode
Measure paid products causally (Testing a Two-Sided Market: Experimentation Platforms)	Propensity matching validated by a randomized test	Supported
Hybrid retrieval as an extension of lexical search (What Did They Mean? Query Understanding and Autocomplete)	BM25 kept; vectors as a fixed quota; ranker retrained	Strongly supported
Configuration over code across brands (One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture)	One front-end repository for three brands; one LTR repository with brand configs	Supported for the layers documented; the rest is inference

Book pattern or claim	OLX's record	Verdict
Category-gated, reversible transactional pivot (Beyond the Contact: Payments, Shipping and Buyer Protection, Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy)	Pay&Ship de-prioritized; OLX Autos exited	Supported, from the retreat side
Language models first on the supply side (AI Across the Listing Lifecycle)	Attribute filling, Auto Description, job-role extraction, dealer copilot	Supported; outcomes company-reported

Now Veldmark, the fictional group, as of its 2027–2028 decisions in the scenario. Veldmark's Tradeyard is a horizontal brand in the Netherlands, Belgium, Poland and Portugal, with car, property and job sections in Poland and Portugal, beside a separate automotive brand, Motorvane. That structure resembles the horizontal-plus-vertical shape OLX runs in Poland and Portugal; it is a common industry pattern, and the ledger's details are invented. The comparison sharpens five of Veldmark's choices.

Veldmark choice (2027–2028)	OLX parallel	What the case suggests
Tradeyard keeps Safe Deal (€75 million of 2026 revenue, about 6% of group revenue) in NL, BE and PL goods categories, on Adyen for Platforms	OLX de-prioritized pay-and-ship in Europe in FY25	Safe Deal earns its place only while it beats the same investment in the listing core; the category gates and the provider-held money in the ledger make a retreat cheap if it stops doing so

Veldmark choice (2027–2028)	OLX parallel	What the case suggests
"Hearthly Direct", a guaranteed-offer proposal at 94% of the Valuo estimate, declined in 2027	OLX Autos: half of revenue, rising losses, exit	The decision is consistent with the strongest lesson in the record; the Models in Production: ML Platforms and Use Cases reasoning (a commit-tier use without validation) is the architectural form of it
Ranking policy v1: a relevance model with no entitlement features, paid slots above a relevance floor, caps on paid listings	OLX's separate "VAS LTR" objective	Two defensible designs for the same problem; Veldmark's caps are easier to audit, OLX's combined objective can trade off more finely. Either needs revenue guardrails on every retrieval change
Shared OpenSearch platform for Tradeyard and Hearthly by end of 2028, replacing Tradeyard's Solr and Elasticsearch estates	OLX evolved ranking and retrieval on Solr for years	An engine migration is a choice, not a precondition: Veldmark should run its hybrid-retrieval work with OLX's proof sequence whichever engine hosts it
Feb 2028 external assistant channel, Hearthly FR and Motorvane DE first, with quotas, lead attribution and no training on returned listings	CompassGPT and AutoGPT inside ChatGPT	Same direction; Veldmark's contractual and quota controls address the questions OLX's releases leave unanswered

The Polish overlap is the Veldmark rule the case can point toward but not validate. In the ledger, Tradeyard Poland's car section competed with Motorvane Poland for the same dealers, and Veldmark resolved it in early 2027: private car listings stay on Tradeyard with insertion fees; dealers above a small listing count are offered Motorvane

packages, with their stock syndicated to Tradeyard through Motorvane's feed; and the shared listing core records which brand owns the commercial relationship for each listing. OLX runs a horizontal site and a motors vertical in the same country too, but **it has not published how OLX.pl and Otomoto divide dealers, listings or revenue**, so the case cannot validate Veldmark's rule. What the record does support is the direction: OLX's money is in the verticals, and Otodim shows a vertical being built on horizontal inventory. Ingrid Halvorsen, Veldmark's fictional Group Chief Architect, should therefore make sure the "commercial owner per listing" field and Motorvane's syndication feed are first-class parts of the Listing event contract, since they are what would let Tradeyard's cars or homes become a vertical lens later, or be handed to Motorvane entirely.

Questions to Ask

1. Which of our markets and businesses do we expect to own in five years, and is our shared platform sized and designed for that set rather than today's?
2. For each adjacent business (payments, shipping, financing, inventory, services), what would it take to switch it off in one category or one country, and how long?
3. Is any of our reported revenue principal revenue from inventory or pass-through, and do our dashboards keep it apart from listing revenue?
4. When we roll out a shared capability, do we measure its effect per market and per category, or only on average?
5. Where does paid visibility enter our ranking, who owns it, and which metric would catch a retrieval change that lowers paid exposure or relevance?
6. Does our search roadmap have a proof sequence (bounded quota, retrained ranker, fallback path, smaller market) or a cut-over date?
7. How many user identifier schemes do our services still use, and what is the plan and metric for converging them?
8. Which of our listing data would an external assistant need, and are its attributes, availability and access controls good enough to expose?

OLX shows a group that reached for global breadth and adjacent businesses, then concentrated on a few markets and verticals while its engineers kept improving one listing stack. The next case, eBay Classifieds Group: Shared Platforms, Local Brands and a Sale, follows a group that built shared platforms under local brands across many countries and was then sold whole — the integration question seen from the other side of the deal.

Sources and Evidence

Company engineering and investor sources

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- OLX Group reports strong FY 2025 results — OLX Group, 2025. FY25 revenue and aEBIT, 64 million daily listings, nine brands in nine markets, unified real-estate platform (re-verified).
- Prosus financial results FY25 booklet — Prosus, 2025 (copy on the Bundestag lobby register). FY24 margin, de-prioritization of pay-and-ship, OLX Autos disposals and discontinued-operations loss.
- OLX Group H1 FY26 results — OLX Group, 2025. La Centrale closing; GenAI ad-posting outcomes.
- Disposal of Avito Group — Naspers, 2022. Buyer, price and rationale (re-verified).
- OfferUp and letgo combine US marketplaces — OLX Group, 2020. 40% stake; non-US letgo retained.
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Standards, regulation and documentation

- No standards or regulatory sources are specific to this case; the regulatory context for chat moderation and trader information is in Rules of the Marketplace: DSA, DAC7 and Vertical Regulation.

Research and practitioner sources

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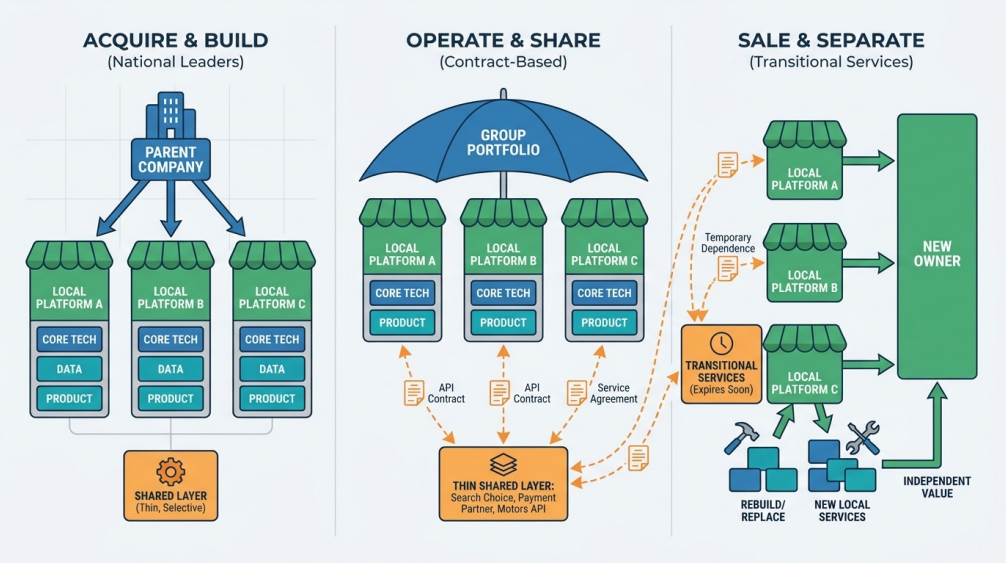
Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

48. eBay Classifieds Group: Shared Platforms, Local Brands and a Sale



IN THIS CHAPTER, YOU WILL: Trace eBay Classifieds Group (eCG) from eBay's first classifieds purchases in 2004 to the 2021 sale to Adevinta and the separation work that followed. You will see what the group shared, what it left to each brand, and what the sale exposed, with documented fact, the author's reading and labeled inference kept apart.

WHY THE BUSINESS CARES: eCG was sold for about US$9.2 billion, and much of that value sat in a few national brands that ran their own platforms. Those brands could be moved to a new owner. Brands that depended on shared group services needed transitional services from the seller.

WHY ARCHITECTS SHOULD CARE: eCG never published a group engineering blog, so its architecture has to be pieced together from brand blogs, open-source repositories, job postings and a competition regulator's decision. Read together, these sources show a federation that shared a few layers: a coordinated search-engine choice, a group-built platform-as-a-service, a motors platform behind partner contracts and one external payments partner. Each brand kept its own core platform.

KEY POINTS:

- **The value was in national leaders on their own platforms**. eBay bought Marktplaats, mobile.de, Gumtree, dba.dk and 2dehands and ran them as separate businesses. Several of its greenfield launches under group brands were closed, renamed or sold. In 2021 the large, self-sufficient platforms became Adevinta's core markets.
- **The shared layer was thin and selective, and brands migrated themselves**. eCG companies chose a replacement for an abandoned eBay search engine together, but each brand migrated on its own (Marktplaats took three months with two or three developers). The group built a platform-as-a-service (PaaS) and a motors-vertical platform with published partner contracts. Payment holding was bought from one external payment institution, not built.
- **A sale turns every shared dependency into a contract term**. The UK regulator's remedy required eBay to offer transitional services to the buyer of Gumtree UK. Adevinta promised to support the Australian businesses "during the transition". On the brands it kept, removing the eBay name took years of domain, design-system and platform work.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Deal-Shaped Architecture** — eCG kept almost every acquired stack and coordinated them only by agreement. The record shows what that cost (a search engine nobody could maintain) and what it bought (brands that could be sold or kept on their own).
- **Search as System of Record** — Kleinanzeigen's 2019 practice of rebuilding its whole search index from a compacted event log in under 30 minutes is the clearest public counter-example in the book's research.
- **Feed Roulette** — eCG's motors-vertical platform gave dealers and integrators a published listing API, a sandbox, partner identifiers and typed webhook contracts. That is the opposite of ingesting partner feeds without contracts.

eBay entered classifieds in 2004 by buying national leaders, and over sixteen years it built one of the largest classifieds portfolios in the world: Marktplaats in the Netherlands, mobile.de and eBay Kleinanzeigen in Germany, Gumtree in the United Kingdom, Australia and South Africa, Kijiji in Canada, dba.dk and Bilbasen in Denmark, 2dehands and 2ememain in Belgium, Vivanuncios in Mexico, and car sites in Italy, the UK and Australia. In July 2020 it agreed to sell the whole group to Adevinta. The sale completed in June 2021.

The case is the mirror image of Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform. Adevinta published a great deal about its central platforms. eCG

published almost nothing as a group, and **that silence is itself evidence about how it was organized**. What does exist, scattered across brand blogs in Amsterdam and Berlin, a GitHub organization, conference talks, archived job postings and a regulator's decision, shows a **federation of strong national businesses with a thin, selective shared layer**. The sale then showed which dependencies on that shared layer had to be written into contracts.

Among Part IX's cases, eCG tests how thin a shared layer can be and what a sale exposes, the stressor One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture and After the Deal: Consolidating Platforms After Acquisitions leave open. *What did the group share, at which layer, and through what kind of interface? What stayed local? And what did the sale reveal about the coupling?* The answers feed the closing test of Veldmark's Common Ground program. The chapter covers eCG's own history and the separation from eBay. For Adevinta's strategy after 2021 — integration, convergence and decentralization — it links to Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform and does not repeat it.

How to Read This Case

Part IX cases follow stricter evidence rules than the chapters in Parts I–VIII (see Part IX — PRACTICE: How Others Built It). A **documented fact** carries its source and publication year in the sentence. **The author's interpretation** is marked "my reading" or appears in the Chief Architect's view. A **reference-architecture inference** appears only in a callout labeled `INFERENCE, NOT DOCUMENTED`.

Three limits shape the case.

1. **No group engineering voice.** The research for this book found no eCG group blog. The engineering record comes from the brands (Marktplaats in Amsterdam; Kleinanzeigen and mobile.de through their shared Berlin Tech Blog), from the `eBayClassifiedsGroup` GitHub organization and from a few talks.
2. **Old evidence.** Most group-era sources date from 2009–2020. They describe systems that have since been replaced, so every "eCG ran X" statement is true at the date of its source and no later.
3. **Unverified platform names.** Names circulate for internal eCG platforms said to have served several brands. No public source confirmed any of them, so none is printed here. Where the case suggests that a shared generalist platform existed, it says so as inference.

Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), appears only in the final comparison. It is a composite of industry patterns. It is not a disguised description of eCG or of any of its brands.

Timeline: eBay Classifieds in Dated Events

Dates are those of the cited releases, filings and archived group web pages. Brand and ownership facts are given as of October 2026 and may change.

Date	Event	Source type
2004	eBay buys **mobile.de** (Hamburg, vehicle classifieds, founded 1996)	eCG brand page, 2014 snapshot
10 Nov 2004	eBay buys **Marktplaats.nl** for about €225 million in cash, to be run "as a separate business"; more than one million active listings	eBay SEC filing, 2004
Feb 2005	**Kijiji** launches in Montreal and Quebec City, Italy and Taiwan, then across Canada in November 2005	eCG brand page, 2014 snapshot
2005	eBay buys **Gumtree** (founded in London in 2000), **LoQUo** (Spain) and OpusForum (Germany)	eCG brand pages, 2014 snapshots; Wikipedia
6 Oct 2008	eBay buys **dba.dk** and **Bilbasen** (Denmark) for about US$390 million; its Global Classifieds unit then has "five major brands" in more than 20 countries	eBay Inc. release, 2008
9 Sep 2009	**eBay Kleinanzeigen** launches in Germany, absorbing Kijiji's German listings; free listings, revenue from paid highlighting and advertising	IT Reseller, 2009

Date	Event	Source type
Jul 2013	eBay buys **2dehands.be / 2ememain.be** (Belgium): 5.5 million monthly unique visitors, 3.2 million listings, bilingual search	eCG brand page, 2013 snapshot
28 Jan 2015	eCG buys **Vivanuncios** (Mexico); its own Mexican site iBazar is retired and users migrated "over the next several weeks"	eBay Inc. release, 2015
2015	Marktplaats replaces eBay's legacy search engine Voyager with Elasticsearch; Kleinanzeigen introduces Kafka	Marktplaats post, 2015; Berlin Tech Blog, 2019
6 Dec 2016	eBay inventory to appear on Gumtree Australia, Gumtree UK and eBay Kleinanzeigen; shared motors seller tools for Kijiji and Gumtree South Africa	eBay Inc. post, 2016
2020	eCG buys CarsGuide and Autotrader Australia; publishes reference clients for its motors-vertical platform, **MoVe**	Trade press, 2022; GitHub, 2020
21 Jul 2020	eBay agrees to sell eCG to Adevinta: about US$9.2 billion, of which US$2.5 billion in cash plus about 540 million shares	eBay SEC filing, 2020

Date	Event	Source type
2 Mar 2021	UK competition authority (CMA) decides that undertakings might be accepted: sale of Gumtree UK (with Motors.co.uk) and Shpock, with transitional services	CMA decision, 2021
25 Jun 2021	Sale completes; Danish businesses go to Schibsted for about €278 million	Adevinta release, 2021
Dec 2021 – Sep 2022	Gumtree UK and Motors.co.uk, Gumtree South Africa, Gumtree Australia with CarsGuide and Autotrader Australia, and Vivanuncios sold	Adevinta releases, 2021–2022; trade press
16 May 2023	eBay Kleinanzeigen becomes **Kleinanzeigen**	Adevinta release, 2023
Jun 2025	Kleinanzeigen describes migrating “the whole platform to a new system”, user by user	Berlin Tech Blog, 2025

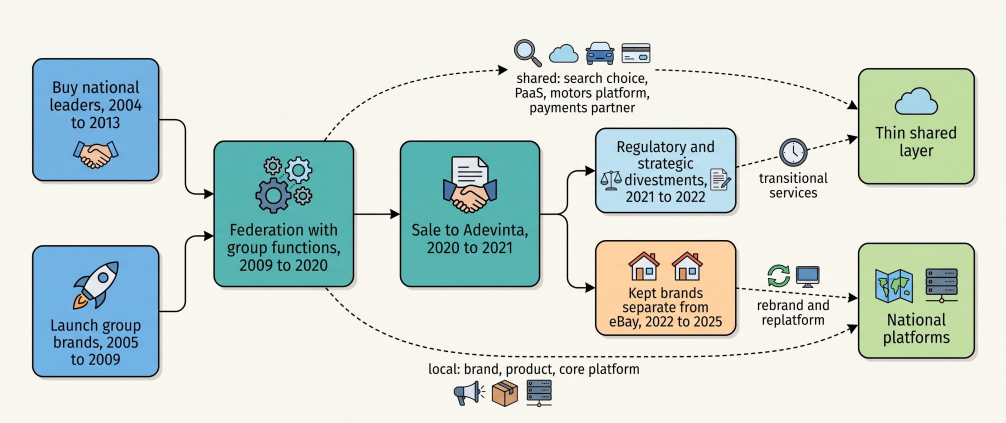


Figure 1: *eCG grew in two ways: it bought national leaders and it launched group*

brands. Both ended up as a federation with a thin shared layer. The sale split the brands into those that were sold, which needed transitional services from the shared layer, and those Adevinta kept, which then had to remove the eBay name from their own platforms.

Phase 1: Building a Portfolio Two Ways (2004–2015)

eBay assembled the portfolio by two routes, and the outcomes differ sharply.

The first route was **buying the national leader and keeping its brand**. eBay's November 2004 filing on Marktplaats is explicit: the company would be run "as a separate business". mobile.de, Gumtree, dba.dk with Bilbasen, 2dehands/2ememain and Vivanuncios followed the same pattern. Each kept its name, its local management and, as far as the record shows, its own platform.

The second route was **launching new sites under a group brand** in markets with no leader to buy. Kijiji started in 2005 in Canada, Italy and Taiwan. Wikipedia reports (not verified from a primary source) that it was incubated "under Marktplaats BV", which would make the Amsterdam organization the group's first platform builder. Kijiji later appeared in US cities, China and India. In Germany, eBay Kleinanzeigen absorbed Kijiji's German listings in 2009. In Mexico, eCG launched iBazar in 2013.

Route	Examples	What happened by 2022
Buy the national leader	Marktplaats (2004), mobile.de (2004), Gumtree (2005), dba.dk and Bilbasen (2008), 2dehands/2ememain (2013), Vivanuncios (2015)	Each still trading under its own name at its last documented sale or report (2021–2026), several under new owners in Denmark, the UK, Australia and Mexico
Launch a group brand	Kijiji (2005 onward), eBay Kleinanzeigen (2009), iBazar (2013)	Kijiji in China and India renamed or spun off (2008), Kijiji US rebranded (2010), Hong Kong and Taiwan closed (2015), iBazar retired (2015); Kijiji Canada and Kleinanzeigen became market leaders

Sources: eBay Inc. releases 2004, 2008 and 2015; eCG brand pages 2009–2020 (archived); Wikipedia for Kijiji's non-Canadian history.

My reading is that the table is about liquidity as much as strategy. A bought leader comes with local supply and demand already in balance (see Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics). A greenfield site has to create that balance, and most of eBay's attempts did not. The two greenfield sites that succeeded both had a head start. eBay Kleinanzeigen used the eBay name in Germany, eBay's strongest European market. The record does not explain why Kijiji Canada succeeded where other Kijiji launches did not. The architectural consequence followed from the business one. **The bought leaders brought platforms good enough to win their markets. The group had little reason to replace them, and every reason not to risk their liquidity by trying.**

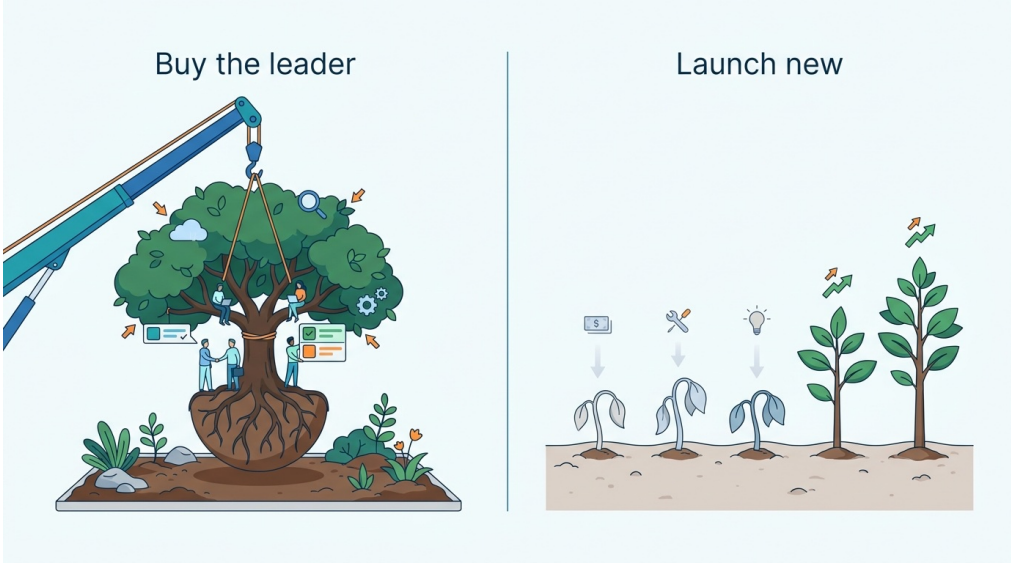


Figure 2: *A bought leader arrives with local supply and demand already in balance; a greenfield brand has to grow that balance from bare soil, and most of the group's launches did not.*

A brand absorbed a hundredfold growth in daily inserts on its own engineering path, not through a group platform. eBay Kleinanzeigen was built in 2009 "to process 10K new ads per day" and launched with about 700,000 live ads; by 2019 it held more than 34 million live ads and took up to a million new ads a day (Berlin Tech Blog, 2019). The group's archived pages give self-reported scale figures too, but they use different definitions (visits, unique visitors, users) and should not be compared across brands.

Phase 2: A Federation with Group Functions (2009–2020)

What the organization looked like from outside

Four kinds of indirect source show group functions beside brand-run platforms; eCG never described its operating model in public.

- **Job postings, 2009–2011** (archived from the group's careers pages). A 2010 post sought a "Senior Developer – Kijiji Platform" in Italy (PHP and Java). A 2011 post for Kijiji US asked engineers to "collaborate with development teams of sister Classifieds platform" and build solutions "that can be reused by other Classifieds platforms". A 2011 Gumtree post for a technology director described "a variety of eCommerce platforms that form the portfolio of the group". The careers site listed "eBay Classifieds Global" as a hiring entity beside the brands.
- **A Berlin hub shared by two brands.** Kleinanzeigen and mobile.de publish one engineering blog, "written by developers, engineers, testers and many more from Kleinanzeigen and mobile.de in Berlin". Its predecessor domain was `ebaytech.berlin`.
- **A group motors function.** At HashiConf in November 2018, Rick Rackow introduced himself as a site reliability engineer at eCG's "Global Motors vertical", "just an SRE at a central team", and placed its "technology center" in Berlin.
- **Brand-level operating models.** Marktplaats's November 2017 post describes its own reorganization into cross-functional "Journey Teams" with company OKRs. A 2018 Berlin post states the local architecture culture: "A Software Architect should never make a decision that the team can effectively make on their own."

My reading is that eCG ran **several platforms, with reuse encouraged and not mandated**, at least in 2009–2011; the plural is in eCG's own words. Group functions existed, but the strongest brands made their own architecture and organization decisions. How group and brand engineering reported to each other is not public.

What was shared, layer by layer

Five shared layers are documented, each behind a different kind of interface, and the interface type is the most useful thing the case teaches.

1. **Search: a coordinated choice, migrated brand by brand.** In a December 2015 post (republished on Medium in 2017), Frank Laan of Marktplaats described replacing **Voyager**, a search engine "developed internally by eBay in the USA" and introduced at Marktplaats around 2008–2009. By 2015, "Ebay itself has moved on from Voyager several years ago". Development had stopped, few people still knew the system, and

it had "almost become a black box for us". Marktplaats and "other eBay Classifieds Group (eCG) companies facing similar problems" evaluated replacements together and chose Elasticsearch, a choice the group's data-and-architecture director also presented at Elastic{ON} in Amsterdam in 2015. **The choice was shared. The migration was not.** Marktplaats migrated itself:

Step	What Marktplaats did (2015)
Interface	Built new services behind the *same interfaces* as the legacy search services, in a service-oriented platform that communicated over Apache Thrift and asynchronous messaging on ActiveMQ or Kafka
Scope	Replicated existing Voyager behavior first and deferred new Elasticsearch features
Tests	Reused several hundred existing integration and acceptance tests; most failures traced to "hidden functionality in other parts of the system"
Comparison	Replayed recorded real user queries, in small random batches every five minutes, against indices built from a copy of the production database
Rollout	Feature switch: 1% on a Friday, 50% by the end of the next week, then 100% with the old platform kept running
Result	Two or three developers, about three months, zero downtime; "At no point did we consider going back"

Replaying recorded queries against frozen copies made every difference explainable. Shadowing live traffic against a live index would not, because the index changes every second and differences cannot be reproduced. Search Architecture for Listings: Indices, Freshness and Scale treats this as the default way to compare engines during a migration.

2. **Infrastructure: a group-built platform-as-a-service.** The `eBayClassifieds-Group` GitHub organization holds the group's open-source infrastructure work. **PanteraS** (created November 2014, "Platform as a Service in a box") combined Mesos and Marathon for scheduling, Consul for service discovery and Fabio for load balancing; around it sit deployment, release-recording, Cassandra and NetScaler tooling. Rackow's 2018 talk described the Global Motors stack on top: Terraform on an **OpenStack private cloud**, a small production-like tenant for every developer, stateless applications on PanteraS, and databases and Kafka on Nomad "after having poor results with stateful applications on PanteraS".

My reading is that this is a familiar first generation of a group platform: good at stateless services, weaker for stateful ones, and built on a private cloud the brands later left. By 2026 mobile.de described ML serving on Kubernetes on AWS, and Kleinanzeigen's home feed ran on Vespa Cloud on AWS. The PanteraS repository's last push was in 2021.

3. **A motors-vertical platform with published contracts.** Two repositories created in September 2020 document **MoVe**, the "eCG Motors Vertical" platform, from the outside. The reference client targets a **Listing API** whose specification was published at `api.ecgmove.com` (the host no longer resolves). Integrators needed sandbox credentials and an assigned `partnerId`, and could call the public APIs only from "well-known, whitelisted IPs". The reference webhook receiver describes the outbound side. MoVe "is able to hand over a number of different events around the main business entities: listings, dealers and promotions", plus "enriched listings" (a listing with its promotions and dealer document, resent whenever any part changes) and "listing-url" events. For every listing event, MoVe expects a synchronous reply with the recipient's own listing ID and "one consumer facing VIP URL per supported language" (a VIP is a listing's detail page). And: "MoVe Listings are dispatched to a variety of different receivers. In the Canadian tenant, this can include Kijiji.ca, kijijiautos.ca and others."

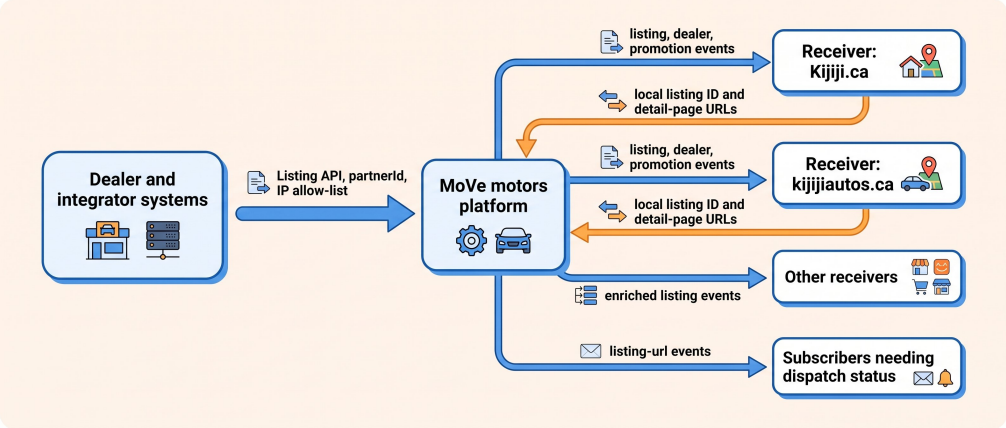


Figure 3: *The documented shape of MoVe, from its 2020 reference repositories: dealers send stock to one platform, which pushes typed events to several marketplaces in a tenant and records each marketplace's own listing ID and URL. The repositories name only the Canadian tenant's receivers.*

> **INFERENCE, NOT DOCUMENTED:** MoVe reads as a shared supply hub for the automotive vertical. It holds the canonical dealer listing, accepts stock once, and syndicates it to several consumer-facing marketplaces per "tenant", which is probably a country. The word "tenant" and the multi-receiver dispatch suggest it was built to serve more than Canada; the Australian car sites bought in 2020 are plausible candidates. The repositories do not name any tenant other than Canada, and nothing public says whether mobile.de used MoVe. This is the author's reading of two README files, not a description of how the platform was deployed.

MoVe is the one shared product platform in the eCG record, and it sits in the

vertical where sharing pays most clearly. Dealer stock is structured, arrives through integrators, and benefits from being entered once and shown in several places. The interface was a contract with versioned payloads, a sandbox and partner identity, not a shared database. This is the pattern Many Doors In: Forms, APIs and Bulk Feeds and Serving the Pros: Dealer, Agent and Merchant Tooling recommend for professional supply.

4. **The parent's marketplace as a supply source**. In December 2016, eCG's general manager Alessandro Coppo announced that eBay marketplace inventory, tested on Gumtree Australia that year, would appear on Gumtree UK and eBay Kleinanzeigen, and that motors seller tools from eBay's Cargigi acquisition were "being used by Kijiji in Canada and Gumtree in South Africa". He spoke of "integrating and cooperating at our technology infrastructure level", without architecture. **My reading** is that this was the most parent-dependent layer in the record, and the first to be exposed by a sale.

5. **Payments: one external partner, two brands**. Marktplaats launched person-to-person payments in June 2017 through **Online Betaalplatform**, a Dutch payment institution that verified users' bank and identity data (Emerce, 2017). Its escrow-style "Gelijk Oversteken" flow (November 2017) held the buyer's iDEAL payment in an intermediary account until the parcel arrived, released it automatically after seven days, and charged the buyer 2% plus €0.40 (RetailTrends, 2017). eBay Kleinanzeigen's "Sicher bezahlen" (October 2020) held the price for 14 days, charged the buyer a fee and mediated disputes through "Online Payment Plattform" (onlinemarketing.de, 2020); t3n still named Online Payment Platform as Kleinanzeigen's payment provider in July 2023. Worldline's September 2022 release on buying 40% of **Online Payment Platform B.V.** names Marktplaats and eBay Kleinanzeigen among its clients. **My reading** is that the group bought escrow-style payment holding (the platform holds the buyer's money until delivery is confirmed) from one licensed provider rather than building or licensing it twice. Both flows have changed since. Marktplaats withdrew Gelijk Oversteken in April 2022 in favor of buyer protection through Online Payment Platform, and Kleinanzeigen's help page, read in October 2026, names Adyen as the party holding funds. When that switch happened, some time after July 2023, is not public; Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform covers the current flows and Beyond the Contact: Payments, Shipping and Buyer Protection the design.

A sixth layer is regional rather than group-wide. A recent Adevinta job posting says Marktplaats, 2dehands and 2ememain "are hosted on a multi-tenant platform, operated from our Amsterdam location". It is recruiter copy, not a technical account, and it does not say when the Belgian sites moved onto that platform. It is the only public statement that two eCG countries shared one product platform.

The layer map

Layer	Shared across brands?	Interface	Evidence and date
Brand, domain, product decisions	Local	—	eBay filing, 2004; brand pages, 2009–2020
Organization and architecture practice	Local, with group functions	Job descriptions, a central SRE team	Job postings, 2009–2011; Marktplaats, 2017; Berlin Tech Blog, 2018
Search engine	Coordinated choice, local migration	Agreement among brands	Marktplaats, 2015; Elastic{ON}, 2015
Infrastructure and PaaS	Group-built tools, adopted at least by the motors vertical	Open-source tooling, private cloud tenants	GitHub, 2011–2021; HashiConf, 2018
Motors supply platform	Shared within a tenant (documented for Canada)	Published Listing API and webhook contracts	GitHub, 2020
Parent marketplace inventory and tools	Group programs	Not documented	eBay Inc., 2016
Payment holding and payouts	One external provider	Provider's platform product	Emerce, 2017; onlinemarket-ing.de, 2020; Worldline, 2022
Core generalist product platform	Regional in the Benelux; otherwise not documented	Multi-tenant platform	Adevinta job posting, undated (2025–2026)

INFERENCE, NOT DOCUMENTED: The pattern of evidence is consistent with a shared generalist platform serving some of the smaller and non-European horizontal sites. The greenfield brands were launched from one organization, iBazar was retired in weeks in favor of an acquired platform, a central motors team existed, and the regulator needed eBay to offer transitional services to the buyer of Gumtree UK. No public source confirms such a platform, names it, or lists its tenants. The chapter treats it as a plausible reading, not a fact.

Inside One National Platform: Kleinanzeigen, 2009–2025

The best-documented eCG platform belongs to a brand, not to the group, and shows what a national platform does on its own engineering path before and after the parent leaves. Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform covers the group strategy around it after 2021; this section covers only what Kleinanzeigen's engineers published.

An event backbone that started with alerts

Kleinanzeigen's event backbone began with one job the database could not do: saved-search alerts. Andre Charton, Kleinanzeigen's head of search, summarized ten years of the platform in an October 2019 post. The 2009 system was "a plain old database backend", "a search backend based on Apache Solr (with a Master/Slave setup)" and web servers. The core system, "the monolith", was Java. In 2015 the team "introduced Kafka to break out from the monolith". The first use case was saved searches with push notifications, which they could not run "in real time with MySQL" at scale. Every important event, command or entity update became a message, and a separate saved-search backend consumed them. By 2019 the platform held 45 million saved searches and processed up to a million new ads a day. "By introducing Kafka with a truly scalable use case, we won big trust."

Two later uses followed from that trust. **Kafka became storage**. The search cluster, by then Elasticsearch with more than 34 million live ads, could be rebuilt from a compacted topic that keeps the latest full entity per key: "We are able to fully index 34 million documents in our search cluster in less than 30 minutes." **Kafka became a stream**, feeding a personalized home feed "consumed like a stream".

My reading is that this is the case's most transferable engineering lesson, and it owes nothing to the group. Kleinanzeigen picked an alerting problem that the old architecture visibly could not solve, solved it with events, and then reused the same log to make the index disposable. A search index that can be rebuilt from a durable log in half an hour cannot quietly become the system of record. That is the defense against Search as System of Record that Every Change Is an Event: Event-Driven Listing Architecture and System of Record and Read Models: Listing Data Management build on, and the reason Tell Me When It Appears: Saved Searches and Alerts treats alerts as a first-class consumer of listing events.

Replacing the whole platform, one user at a time

In June 2025, MD Sayem Ahmed of Kleinanzeigen's "Team Red" wrote that "KA recently migrated the whole platform to a new system". The team built the new system, "ran

it parallel to our legacy system, migrated all users' data to the new system, and then incrementally switched users to the new one", with the option to "revert the users to the old system". The post does not name either platform. It does not say whether the new system is an Adevinta group platform, and this chapter does not connect it to the 2023 convergence plan described in Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform. The post gives no basis for doing so.

The post documents six design choices.

- **Order by data dependency**. Core user data (identifiers, authentication, profile) moved first, and dependent migrations (ads, transactions, messages) started only when told a user had moved.
- **Route by user**. A reverse proxy "with just enough business logic" forwarded each user's traffic to the legacy or the new system according to that user's migration state.
- **No dual writes**. The team chose asynchronous streaming. A sync delay of up to a minute was a business decision: "our business was fine with a slight syncing delay of up to a minute".
- **Change data capture rather than an outbox**. A polling outbox could not keep up with user data updated "millions of times per day", event listeners risked out-of-order publication, and every team would have repeated the work. The team ran Debezium against MySQL, working around a two-data-center cluster with statement-based binary logs and no global transaction identifiers (GTIDs) by adding row-format replicas.
- **Back-sync for rollback**. Updates by users already on the new system flowed back to the legacy system, so most users still saw their ads and anyone could be switched back.
- **A state machine per user**, and loops broken by version, not time. Each user moved through IDENTIFIED, ELIGIBLE, MIGRATION_REQUESTED, MIGRATED and TRANSITIONED. Update cycles between the two systems were detected with the JPA `@Version` field as a logical clock. The coordination logic lived in one service built as a modular monolith, so it could use MySQL transactions.

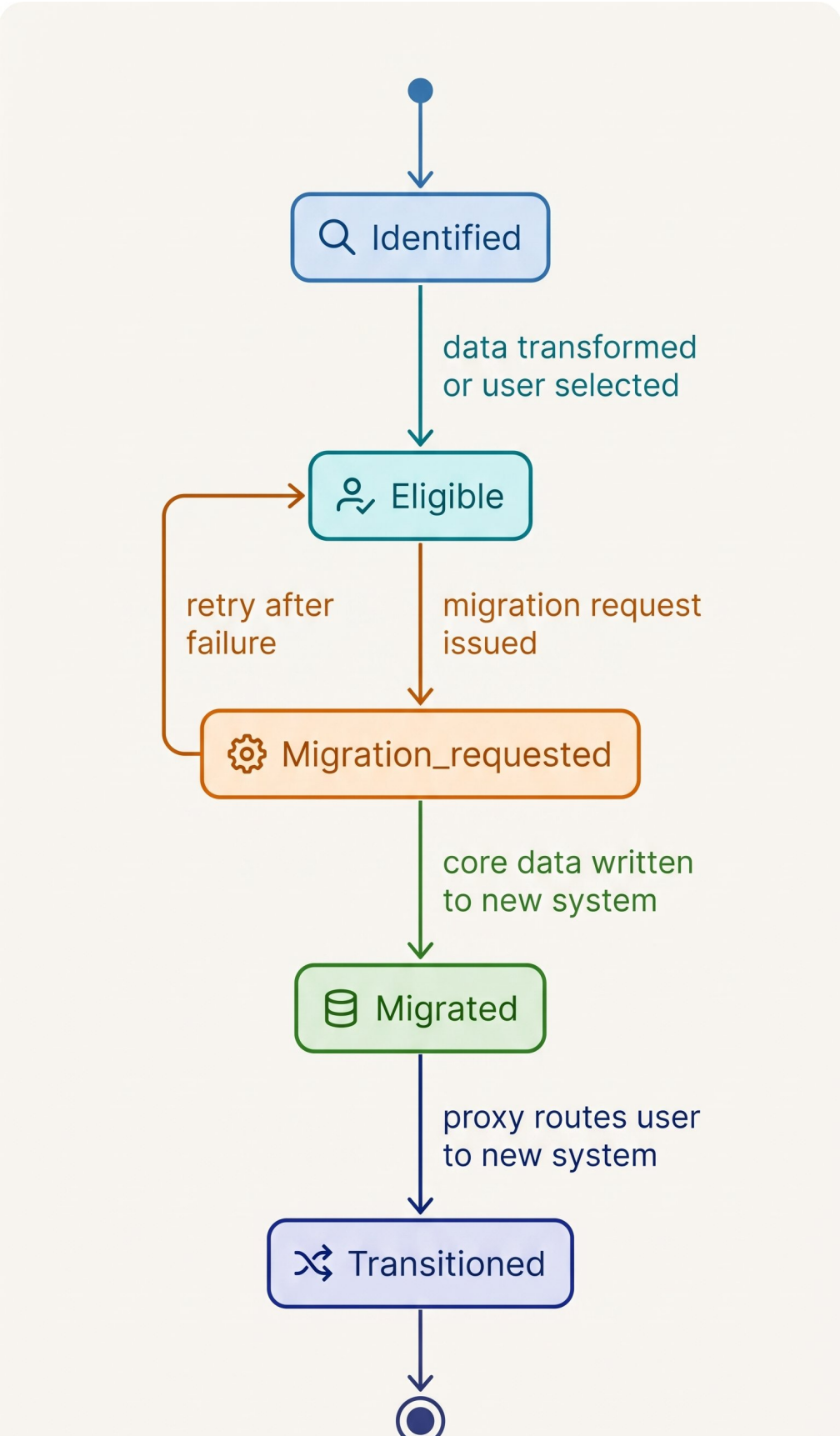

Identified
data transformed
or user selected
Eligible
retry after
failure
migration request
issued
Migration_requested
core data written
to new system
Migrated
proxy routes user
to new system
Transitioned

Figure 4: *The per-user migration states Kleinanzeigen published in 2025. The retry edge reflects the post's description of resetting a user to ELIGIBLE. Replatforming became a sequence of small, reversible per-user moves instead of a cut-over date.*

My reading is that the migration's ownership matters as much as its algorithm. The new system "was being developed by a whole separate team", so the transformation logic could not simply live inside it. A migration between two systems owned by different teams needs its own owner and its own anti-corruption layer, and Kleinanzeigen gave it one. Changing the Engine in Flight: Modernizing a Live Marketplace develops the per-user strangler pattern, and Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models the translation layer.

mobile.de: commercial commitments inside ranking

mobile.de, the other Berlin brand, published less about its platform and more about models. In a February 2024 post, Manish Saraswat described search ranking over more than a million listings. It had to find the top 100 "within a few milliseconds" while honoring "exposure guarantees made to our premium dealers". The old learning-to-rank (LTR) model ran inside Elasticsearch as a plugin, which made upgrades and model deployment slow. The team moved it into a separate Python ranking service, where a pairwise XGBoost model beat the earlier pointwise model in an A/B test.

My reading is that the ranking post is the clearest public statement in the book's research that a dealer package's promise of exposure becomes a ranking constraint. The ranking service has to know about entitlements, which is the boundary Ranking for Two Sides: Relevance, Freshness and Paid Visibility and Packages, Bumps and Quotas: Products, Pricing and Entitlements try to keep explicit.

Phase 3: The Sale (2020–2021)

Terms and the technology clause

eBay's 21 July 2020 release valued eCG at about **US$9.2 billion**: US$2.5 billion in cash plus about 540 million Adevinta shares, giving eBay about 44% of Adevinta's equity and a 33.3% voting stake. Schibsted, then Adevinta's majority owner, would buy the Danish business. The release estimated synergies of **US$150–185 million a year within three years**. At completion on 25 June 2021, Adevinta guided **€130–165 million** of run-rate cost synergies after three years.

On technology, the release says only that eCG innovates "across its product portfolio and advertising technology platform", and that the two companies "will continue to partner in key areas post-closing to ensure a seamless transition". Nothing public describes which eBay systems eCG depended on at signing, or how long that dependence lasted.

What the regulator wrote down

The UK Competition and Markets Authority (CMA) found that Gumtree and Adevinta's Shpock competed closely in UK generalist classifieds. It also found that Gumtree "has significant potential to compete with eBay Marketplace in the UK" if it were independent of eBay. The remedy, accepted on 2 June 2021, was the sale of Gumtree UK with Motors.co.uk and of Shpock, as **undertakings in lieu** (commitments that replace an in-depth investigation). Gumtree UK was to be sold with its "platforms, related tangible and intangible assets, employees, leaseholds, contracts and goodwill". The decision of 2 March 2021, published in April 2021, adds one sentence an architect should read twice:

> "Adevinta and eBay have offered to provide transitional services in relation to Shpock and Gumtree UK respectively as part of the Proposed Undertakings."

The offer to provide transitional services for Gumtree UK came from eBay, the seller, not from Adevinta. A transitional services agreement (TSA) is a contract under which the seller keeps running some systems or functions for the divested business for a limited period. The CMA text does not say what the services covered. **My reading** is that a business being sold with "its platforms" still needed something from eBay to operate while separating. The something was not in the package because it was shared.

Shpock went to Russmedia Equity Partners on the day the undertakings were accepted. Gumtree UK and Motors.co.uk, about 7% of eCG's 2020 revenue, were sold to O3 Industries and Novum Capital on 1 December 2021. Adevinta then sold the Australian businesses to The Market Herald for US$60 million in August 2022, saying it "will continue to support the Australian business and its employees during the transition". The Mexican business went to Navent in September 2022, and Gumtree South Africa to Impresa Capital in July 2022. Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform covers the strategic review behind these sales, which focused the group on five core European markets.

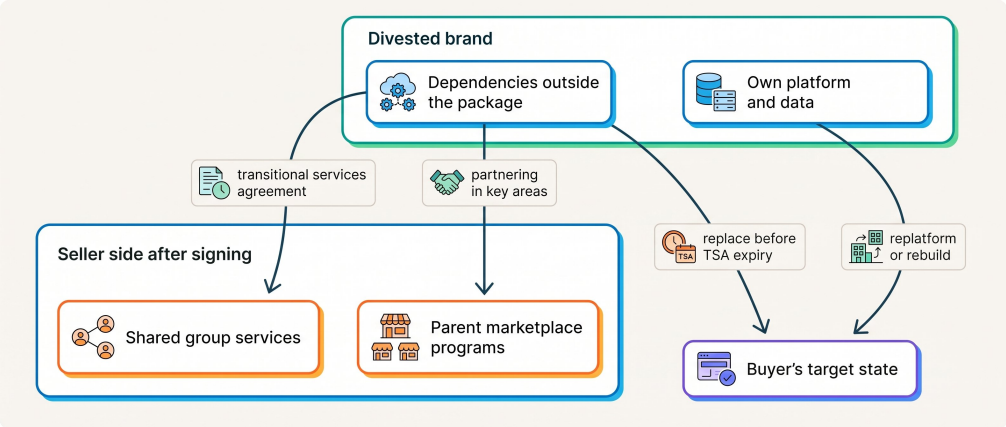


Figure 5: *A plausible reference view of a carve-out, consistent with the CMA decision and Adevinta's 2022 release but not a description of any specific separation. Everything a divested brand uses but does not own becomes a TSA line with an expiry date.*

Which brands needed help

The record supports one pattern, offered here as **my reading**. Gumtree UK is the eCG brand documented as needing transitional services, and the Australian businesses were promised transition support. The large national platforms with their own engineering hubs — Marktplaats in Amsterdam, Kleinanzeigen and mobile.de in Berlin — became the core of the combined group. **Self-sufficiency kept options open**: these businesses could be integrated, kept standalone or, as Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform shows, prepared for sale later. The record cannot show the counterfactual. It shows only that separation cost appeared where sharing had been.

Aftermath: Removing a Parent from a Platform

The brands Adevinta kept had to stop being eBay. Three pieces of evidence show what that took.

- **The name and the domain**. On 16 May 2023 eBay Kleinanzeigen became Kleinanzeigen, with a new logo and domain. Adevinta's release said "the previous domain [will] remain accessible for several more years". Adevinta's 2023 annual report, cited in Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform, put the rebrand cost at €15 million and noted that the acquisition agreement did not allow use of the eBay Kleinanzeigen brand beyond 2024.
- **The apps**. A July 2025 Berlin Tech Blog interview describes a later rebrand of mobile.de's Android app. It needed a refactor first: dissolving legacy components, adopting Material Components and token-based colors. "The rebranding itself was

the final step and took only two weeks to implement once the base refactoring was done." The work removed "technical debt from 8 years ago". It concerns mobile.de, not Kleinanzeigen, but it is the best public analogue for what a brand change costs a mobile estate.

- **The platform**. Kleinanzeigen's 2025 whole-platform migration, described above, is not documented as a consequence of the sale, and this chapter does not claim it was one. It does show that a platform built in 2009 for 10,000 ads a day was still being replaced sixteen years later, by a team that had inherited statement-based binary logs from an older era of operations.

Away from the core, Adevinta recorded a €147 million impairment on Kijiji Canada in its 2023 results, citing competition and a new mobile app rolled out in 2023 (OnlineMarketplaces, 2024). Kijiji is still listed among Adevinta's brands as of October 2026.

My reading is that separating from a parent takes work at three layers below the logo, and the logo change, the cheapest part, is the one most often budgeted. Domains need redirects that last years, for search engines and for bookmarks (see Found by Google: Web, App and SEO Architecture for Listings). Identity must carry over without forcing users to register again. Design systems need themes driven by tokens before a rebrand can be fast. The visible logo change is the last two weeks of a much longer job.

Trade-offs the Case Illustrates

Option	What eCG's record shows	When it is right	When it fails
Buy the leader and keep its platform	Marktplaats, mobile.de, Kleinanzeigen-scale platforms evolved locally and became the valuable core	The acquired platform already wins its market and the group's advantage is capital, not code	When the platform is too weak to scale and the group defers the fix indefinitely
Launch greenfield brands on a group platform	Several group brands were closed, renamed or sold; Kijiji Canada and Kleinanzeigen succeeded	Markets with no leader to buy and a strong parent brand or early start	When the shared platform is cheaper to launch on than to win on; liquidity, not code, decides

Option	What eCG's record shows	When it is right	When it fails
Coordinate a technology choice, migrate locally	Elasticsearch chosen together; Marktplaats migrated in three months with two or three developers	A common problem (an abandoned engine) and capable brand teams	When brands diverge so far that "the same choice" means different products, with no shared expertise
Build a group platform-as-a-service	PanteraS and the HashiCorp stack served stateless services well and stateful ones poorly; later replaced by public cloud	When the group has no good shared runtime and enough volume to justify one	When the platform's generation ages faster than the brands' needs, or stateful workloads are forced onto it
Share a vertical supply platform behind contracts	MoVe: Listing API, partner IDs, typed webhooks, dispatch to several marketplaces in a tenant	Structured, professional supply that benefits from one entry and many outlets	When local marketplaces need fields or behavior the canonical model cannot express
Buy a shared capability from one external provider	One licensed payment institution for Marktplaats and Kleinanzeigen escrow	Regulated capabilities where the provider holds the license and the risk	When the provider's product cannot follow local law or local product changes; switching then costs a migration
Depend on the parent's systems and programs	eBay inventory and tools on classifieds sites; TSAs offered by eBay at the sale	When the parent is a long-term owner and the integration creates real supply	When the parent may sell; every dependency becomes a separation project

The rows differ mainly in **the kind of interface they create**. Agreements and external

providers are easy to unwind. Published contracts such as MoVe's can be handed over or re-pointed. Undocumented dependence on a parent's systems is the hardest to separate, and in this case it is the one the regulator had to write into the remedy.

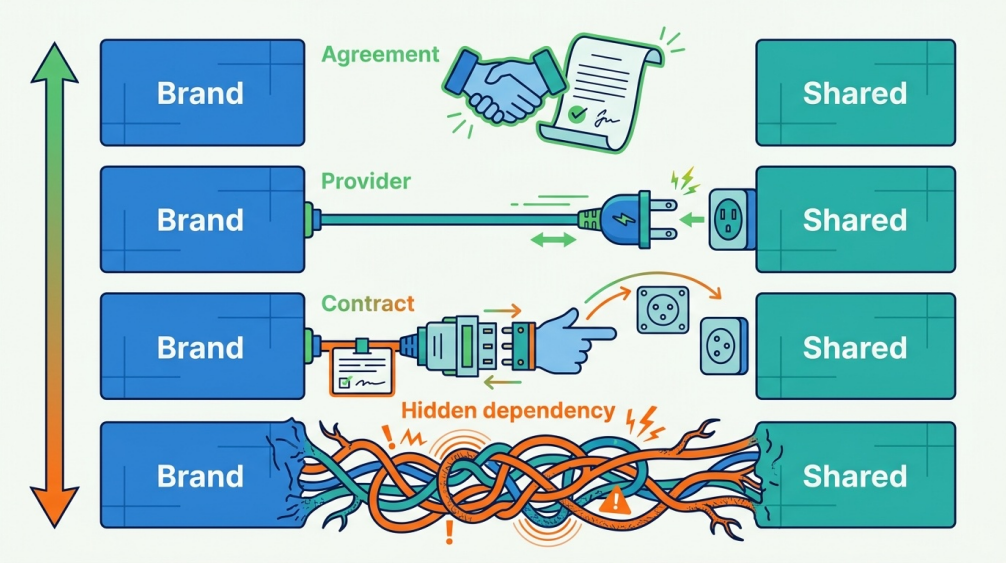


Figure 6: *Shared layers unwind as easily as their interfaces allow: agreements and external providers come apart cleanly, published contracts can be re-pointed, and undocumented dependence on a parent is what the regulator had to write into the remedy.*

Vertical Differences in the Record

eCG's portfolio was mostly generalist and automotive, with little real estate and jobs. The record shows the two main verticals being treated differently.

Vertical	eCG examples	What the record shows	Implication for sharing
Generalist (C2C goods)	Marktplaats, eBay Kleinanzeigen, Gumtree, Kijiji, 2dehands, Vivanuncios	Strong national brands on local platforms; one regional multi-tenant platform (Benelux); shared payment provider; parent inventory programs	Share capabilities beneath the product (search choice, payments provider); the product itself stayed local except where language and market overlapped
Automotive	mobile.de, Automobile.it, Motors.co.uk, Kijiji Autos, Autotrader and CarsGuide Australia	A group "Global Motors vertical" with a central SRE team in Berlin; a shared supply platform with partner contracts; mobile.de ranking that honors dealer exposure guarantees	Dealer supply and integrator contracts are the most shareable layer; ranking and packages stay tied to each market's commercial terms
Real estate and jobs	Largely outside the portfolio (property and jobs sections inside generalist sites)	No documented vertical platform	The case says little; see AVIV Group: Building a European Real-Estate Platform from Acquisitions and Zillow, Rightmove, Idealista and REA: What Property Portals Teach

My reading is that the automotive pattern generalizes. Where supply is professional

and structured, and arrives through integrators, a group can share the supply side (listing API, dealer model, syndication) and still leave consumer marketplaces separate. Same Platform, Different Physics: How Listing Verticals Differ explains why automotive supply has that shape.

Failure Modes the Record Exposes

1. **Inheriting a parent's abandoned technology**. Voyager was eBay's own engine. Once eBay moved on, Marktplaats ran a system with stopped development and dwindling knowledge. A shared platform owned by a distant parent can become orphaned legacy faster than local code, because nobody nearby is paid to keep it alive.
2. **Dependencies that only appear at the sale**. The transaction release promised "partnering in key areas". The regulator's remedy needed eBay's transitional services. Neither says what was shared. A group that cannot list each brand's dependencies on shared services cannot price a sale.
3. **A first-generation group platform outlived by its tenants**. PanteraS on a private cloud suited 2014. By the mid-2020s the brands' documented workloads ran on public clouds. A group runtime with no exit path becomes a migration every tenant has to fund.
4. **Hidden behavior in legacy systems**. Marktplaats traced most search-migration failures to "hidden functionality in other parts of the system"; Kleinanzeigen's replication settings blocked standard change data capture. The cost of leaving a system is set by behavior nobody wrote down.

Where the Public Evidence Stops

No public source found for this book answers the following questions. The chapter does not fill them with inference beyond the two labeled callouts above.

Question	What is public	What is not
Did eCG run a shared generalist platform, and for which brands?	Job postings mentioning a "Kijiji Platform" (2010) and "sister Classifieds platform" (2011); a Benelux multi-tenant platform (undated job posting)	Its name, architecture, tenants and fate after 2021

Question	What is public	What is not
How was eCG engineering organized?	Group hiring entity, a Global Motors vertical, a central SRE team, Berlin and Amsterdam brand hubs	Reporting lines, headcount, which hub served which brand
What did MoVe serve?	The Canadian tenant's receivers (Kijiji.ca, kijijiautos.ca); public API and webhook contracts (2020)	Other tenants, whether mobile.de used it, and what replaced `api.ecgmove.com`
What did eCG depend on from eBay at signing?	"Partner in key areas post-closing" (2020); eBay inventory programs (2016)	The systems, data and contracts involved, and how long they lasted
What did the Gumtree UK transitional services cover?	That eBay offered them (CMA, 2021)	Scope, duration and how the buyer replaced them
How were the Australian, Mexican and South African sites separated?	Sale dates and Adevinta's promise of transition support (2022)	Any engineering account
What is Kleinanzeigen's 2025 "new system"?	The migration method in detail (2025)	Its name, its owner, its relation to any Adevinta platform
What did Marktplaats build after 2017?	Organization (2017); Fisher, its A/B testing package, later adopted group-wide (see Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform)	Search, ranking, moderation or cloud architecture after the Elasticsearch move
Were the deal synergies achieved?	US$150–185 million estimated (2020); €130–165 million guided (2021)	The outcome and the technology share

Ownership of Gumtree UK, Gumtree Australia and Kijiji Canada after 2024 is volatile; readers should check it before relying on the portfolio facts above.

The Chief Architect's View

What I take from eCG is a set of decision rules for a group that buys marketplaces and may one day sell them.

Prefer interfaces you can hand over. The shared layers that left the cleanest record were agreements (a common engine choice), external providers (one payment institution) and published contracts (MoVe's API and webhooks). The layer that needed a regulator's attention was undocumented dependence on a parent. I would require every shared capability to have a written interface that a third party could operate against, even when no third party is in sight.

Keep a dependency register per brand. For each marketplace, list every system, data flow and contract it uses but does not own, with an owner and a replacement path. The register makes a sale, a merger or a reorganization cheaper. Without one, transitional services are negotiated at signing, under time pressure, by people who have to discover what is shared.

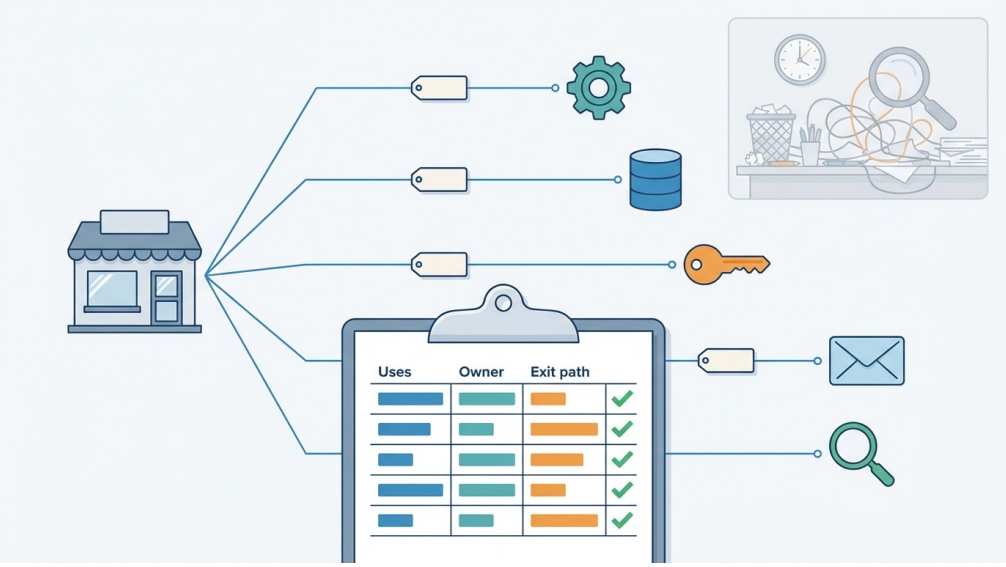


Figure 7: *A dependency register per brand lists everything it uses but does not own, with an owner and a replacement path, so a sale's transitional services are read from a list rather than discovered at signing.*

Share the supply side of professional verticals first. Dealer stock entered once and shown in several marketplaces is the clearest case for a shared product platform in this record. Consumer-facing generalist products stayed local, and the strongest of them won their markets that way.

Coordinate choices, fund local migrations. Marktplaats's search migration worked because a group decision removed the evaluation burden and a small local team owned the cut-over. The Chief Architect's role is the decision record, the shared expertise and the migration method — recorded-query replay, unchanged interfaces, gradual traffic. It is not the migration itself.

Treat the parent's technology as a supplier with an end date. Voyager shows what happens when a parent stops investing in a system its subsidiaries run. When a group runs technology it does not control, it needs an exit plan before the supplier moves on, not after.

Comparing the Case to the Book's Patterns and to Veldmark

The book's patterns, as eCG's record tests them:

Book pattern or claim	eCG's record	Verdict
Share global components beneath local products (One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture)	Shared engine choice, PaaS, payments provider, motors supply platform; generalist products local	Supported, with a thinner shared layer than the book recommends
Consolidate enabling layers first, core product last (After the Deal: Consolidating Platforms After Acquisitions)	eCG largely never consolidated core products, except regionally in the Benelux	Consistent; shows the cost of never doing it (a parent's engine nobody could maintain)
Contracts at the supply boundary (Many Doors In: Forms, APIs and Bulk Feeds)	MoVe's Listing API, sandbox, partner identity and typed webhooks	Strongly supported

Book pattern or claim	eCG's record	Verdict
The search index is a rebuildable read model (Search Architecture for Listings: Indices, Freshness and Scale, Every Change Is an Event: Event-Driven Listing Architecture)	Kleinanzeigen rebuilt 34 million documents from a compacted topic in under 30 minutes (2019)	Strongly supported
Per-user strangler migration out of a monolith (Changing the Engine in Flight: Modernizing a Live Marketplace)	Kleinanzeigen's proxy routing, state machine and change-data-capture sync (2025)	Strongly supported
Ranking must keep commercial commitments explicit (Ranking for Two Sides: Relevance, Freshness and Paid Visibility)	mobile.de's exposure guarantees for premium dealers shaped its ranking service (2024)	Supported
Design for portability and stressors (Built to Be Replaced: Evolving a Marketplace Platform Over a Decade)	Transitional services at the sale; years of rebrand work afterwards	Supported; adds "the parent sells" as a stressor

Now Veldmark, the fictional group, as of its 2027 decisions in the scenario. Its Common Ground program, approved in January 2027 at €140 million over 2027–2030, follows the principle "**share platforms, not products**". That principle is close to what eCG did in practice, with a much thicker shared layer.

Veldmark choice (2027)	eCG parallel	What the case suggests
Tradeyard Poland's car dealers above a small listing count are offered Motorvane packages, with their stock syndicated to Tradeyard through Motorvane's feed; the shared listing core records which brand owns the commercial relationship	MoVe dispatched dealer listings to Kijiji.ca and kijijiautos.ca and recorded each receiver's listing ID and URL	The right shape. Give the syndication an explicit contract (listing ID and URL per receiver, dispatch status events) so that either brand could be separated
Tradeyard moves from Solr to the shared OpenSearch platform by end of 2028; Motorvane stays on Elasticsearch until a 2029 review	eCG brands chose Elasticsearch together and each migrated itself	Use Marktplaats's method: unchanged interfaces behind the Listing Search API, recorded-query replay against frozen copies, gradual traffic
Yardstick retired through a strangler sequence by end of 2029; Tradeyard publishes listing events via Debezium CDC on Yardstick's MySQL	Kleinanzeigen's per-user migration with CDC, a reverse proxy and back-sync	Check the binary-log format and replication setup before planning CDC; give the migration its own owning team
Group event backbone (Kafka on Confluent Cloud) and a 5-minute alert target for 46 million saved searches	Kleinanzeigen's Kafka adoption began with saved-search alerts and earned "big trust"	Lead with a use case that visibly fails today; alerts are a good one
Tradeyard Safe Deal runs on Adyen for Platforms; Veldmark holds no payment license	Two eCG brands used one licensed payment institution	Consistent; keep the provider edge behind an anti-corruption layer so a provider switch is a migration, not a rewrite
Majority-owned by a private-equity investor since 2025	eCG's parent sold it; buyers then sold several brands within 18 months	Keep a dependency register per brand from the first Common Ground release

The last row deserves one worked question. Suppose Veldmark's owner decided in 2029 to sell **Tradeyard Portugal**, the group's least liquid Tradeyard country, with 1.1 million live listings, about 210,000 new listings a week and no Safe Deal. Under the ledger's timeline, Tradeyard Portugal would by then use the group taxonomy and attribute service (Tradeyard first, from October 2027), the Confluent event backbone and Listing event contract v1 (from July 2027), Veldmark ID (rolled out in 2028), the shared OpenSearch platform (all Tradeyard countries by end of 2028) and the group entitlement service Tally (Tradeyard from the second quarter of 2028). It would still depend on Yardstick until its decommissioning at the end of 2029. Each of these would become a line in a transitional services agreement, the same kind of line the CMA recorded for Gumtree UK.

The architectural answer is not to share less but to make each shared capability hand-over-able: country-scoped taxonomy releases that can be exported, listing topics and schemas filtered by the brand and country in the group listing ID, identity with per-brand profiles and an export path, per-country indices with portable definitions, and an entitlement ledger whose Portuguese accounts can be extracted. Veldmark's tenancy model (June 2027) already asks every shared capability to document a tenant exit path (One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture); a per-brand dependency register is the list that makes those exit paths checkable. Ingrid Halvorsen, Veldmark's fictional Group Chief Architect, can make it a standing item at the architecture advice forum in 2027, at little cost, and test it in the separability tabletops that Built to Be Replaced: Evolving a Marketplace Platform Over a Decade describes.

Questions to Ask

1. For each marketplace in our group, which systems, data flows and contracts does it use but not own, and who could operate them if the marketplace were sold?
2. Which of our shared capabilities sit behind a written, versioned interface, and which are reached through shared databases or undocumented calls?
3. Are we running any technology whose original owner has stopped investing in it? What is our exit plan and date?
4. When we coordinate a technology choice across brands, who owns each migration, and is the comparison method (recorded traffic, frozen copies, unchanged interfaces) written down?
5. In our professional verticals, is supply entered once and syndicated through a contract that records each receiver's identifiers, or copied between systems?
6. Can our search indices be rebuilt from a durable log, and how long does a full rebuild take today?
7. If we replaced a core platform tomorrow, could we route by user and roll back by user, or only cut over by date?
8. If we removed a brand name, what would domains, identity and design systems each need, and how long would the old domain have to stay alive?

eCG shows a federation whose value lay in strong national platforms and whose shared layer was tested by a sale. The next case, AVIV Group: Building a European Real-Estate Platform from Acquisitions, follows a group that took the opposite route in one vertical: it assembled a European real-estate platform from acquired portals and set out to make them work as one.

Sources and Evidence

Company engineering and investor sources

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- Acquisition of eBay Classifieds Group complete — Adevinta, 2021. Completion, share issue, €130–165 million synergy guidance, Denmark sale.
- Adevinta sells Gumtree UK and Motors.co.uk to O3 Industries and Novum Capital — Adevinta, 2021. Buyers and revenue share.
- Adevinta announces sale of Gumtree Australia, CarsGuide and Autotrader Australia — Adevinta, 2022. Price and transition-support commitment.
- Adevinta's eBay Kleinanzeigen is now Kleinanzeigen — Adevinta, 2023. Rebrand and old domain kept "several more years".
- eBay Classifieds Group website: brand and "About us" pages — eBay Inc., archived 2009–2020. Brand lists, founding and acquisition dates, self-reported scale.
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Standards, regulation and documentation

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Research and practitioner sources

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- Kleinanzeigen.de ermöglicht Sofortkauf – doch die Funktion hat einen Haken — t3n, 2023. Online Payment Platform still Kleinanzeigen's payment provider in July 2023.
- Gelijk Oversteken rollout — RetailTrends, 2017. Escrow flow, release timer and fee.
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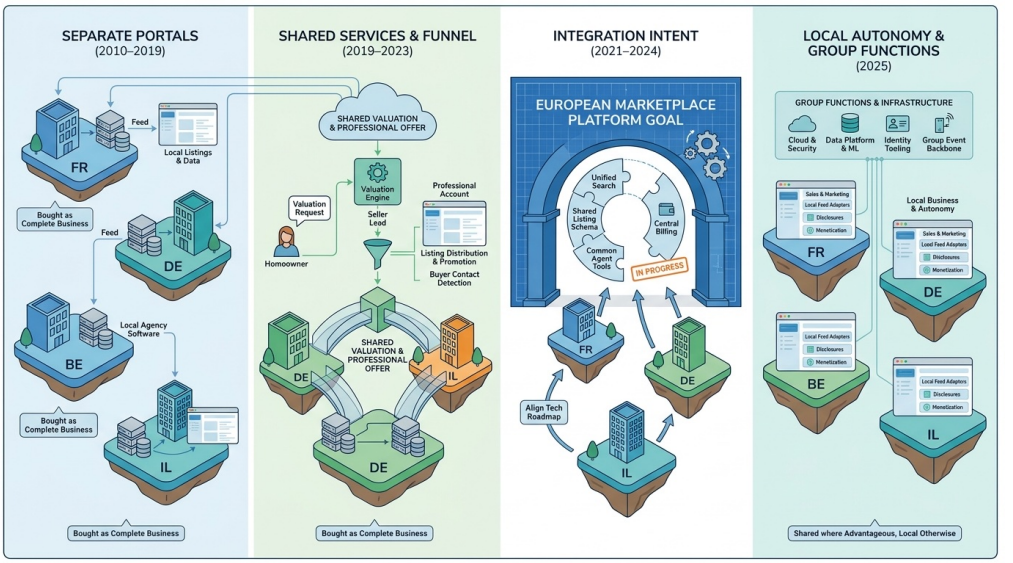
- Adevinta grows revenue and progresses verticalization plan in 2023 — OnlineMarketplaces, 2024. Kijiji impairment.
- Adevinta sells Vivanuncios and Segundamano to Navent Group — OnlineMarketplaces, 2022. Mexico exit.
- Kijiji and Gumtree — Wikipedia, n.d. (read October 2026). Kijiji's origin under Marktplaats BV and non-Canadian history; Gumtree South Africa sale.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

49. AVIV Group: Building a European Real-Estate Platform from Acquisitions



IN THIS CHAPTER, YOU WILL: Trace AVIV Group's public record from Axel Springer's first property-portal acquisitions in 2010–2012 to the 2025 carve-out under KKR and CPP Investments and the move to local autonomy. You will separate the few documented facts from the author's reading and from labeled reference-architecture inferences, and test the case against the book's patterns and Veldmark's choices.

WHY THE BUSINESS CARES: AVIV's portals were bought as national leaders and challengers, and their value came from local agent relationships, local data and local monetization. A group that tries to turn such portals into one platform spends against the very assets it bought.

WHY ARCHITECTS SHOULD CARE: AVIV is the clearest public example of a multi-country real-estate group stating an integration goal and then,

four years later, reversing it. The case shows which layers of a property portal travel across borders and which do not, even though AVIV itself has published almost nothing about its systems.

KEY POINTS:

- **AVIV is a portfolio built by acquisition, and its public record is corporate, not technical.** SeLoger, Immoweb, Immowelt and Immonet, Logic-Immo, Yad2 and Meilleurs Agents joined between 2010 and 2019 as separate businesses with separate stacks. AVIV publishes no engineering blog, talks or public code, so every statement about its systems below is either dated press evidence or a labeled inference.
- **The group's documented integration moves are commercial and data-driven, not a shared listing platform.** The 2019 Meilleurs Agents purchase, the 2022 Ultimmo bundle across SeLoger, Logic-Immo and Meilleurs Agents, and the 2023 opening of SeLoger to private sellers all serve one funnel: capture the future seller early and route them to a paying agent. Valuation science and the professional offer were shared before any platform was.
- **The 2021 integration intent gave way in 2025 to local autonomy with selected group functions.** In 2021 AVIV's CEO spoke of "aligning our tech roadmap across the organization to create a European real estate marketplace". In December 2025 the press described AVIV as having rowed back "on its plans for a fully integrated platform", and the new head of Immowelt described "group-level functions where they create clear advantages". The shape fits a "shared core, local edge" model; what the integrated platform contained is not public.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Deal-Shaped Architecture** — AVIV's portals arrived over nine years as complete businesses, and the public record shows brands such as Immowelt and Immonet still operating side by side nine years after their 2015 combination.
- **The Forced Fit** — A fully integrated European platform for portals with different feed standards, disclosures and pricing norms is the kind of central plan the 2025 shift toward local autonomy undid.
- **Vanity Liquidity** — AVIV's documented strategy measures supply acquisition (sellers identified, seller leads, commission share) rather than listing counts, which is the cure for this pattern in a lead-generation vertical.

Most marketplace groups in this book's Part IX either built their platforms or bought marketplaces that already shared one. AVIV Group did neither. Axel Springer bought

national property portals one at a time (SeLoger in France, Immoweb in Belgium, Immowelt in Germany, Logic-Immo in France again) together with an Israeli horizontal site, Yad2, and a French valuation business, Meilleurs Agents. It then gave them a common name and a common technology leadership, and in 2021 announced the goal of a European real-estate marketplace. **The case is a test of one question: how much of a property portal can be shared across countries?**

The evidence is thin. AVIV's GitHub organization has no public repositories, its Medium publication had no posts when the book's research fetched it in October 2026, and no conference talk by AVIV engineers was found. What exists is corporate history, regulatory decisions, trade-press interviews with AVIV executives and a handful of product announcements. **That evidence is enough to show the direction of the strategy and the funnel it served; it is not enough to describe a single system.**

The record tests the argument of Same Platform, Different Physics: How Listing Verticals Differ and One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture: in real estate, ingestion formats, mandatory disclosures, address systems, agent software and monetization norms are national, so a group platform has to stop well short of the product. **AVIV's documented moves (shared valuation science, a bundled professional offer, then local autonomy with group functions) trace that boundary from the outside.** The comparison with Veldmark near the end applies the same boundary to Hearthly, Veldmark's property portal in France and Spain.

How to Read This Case

Part IX cases follow stricter rules than the chapters in Parts I–VIII (see Part IX — PRACTICE: How Others Built It). A **documented fact** carries its source and publication year in the sentence or in the timeline. **The author's interpretation** is marked "my reading" or sits in the Chief Architect's view. A **reference-architecture inference** appears in a callout labeled `INFERENCE, NOT DOCUMENTED`, and describes how a group like AVIV could plausibly be built, never how AVIV is built.

The limits are tighter than in Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform. AVIV has not been listed: before 2025 it was a segment inside Axel Springer, which itself delisted in 2020, and since April 2025 it has been a privately held joint venture. Trade press notes that AVIV "does not publish" a financial breakdown (Online Marketplaces, July 2023). Most of the evidence below comes from one trade publication, Online Marketplaces, and the French real-estate trade site MySweetImmo, both quoting AVIV executives; from Axel Springer, KKR and CPP Investments releases; and from the German and French competition authorities.

Three kinds of evidence were found and deliberately not used as fact: job advertisements seen only as search snippets (one mentions a team managing networking for hundreds of AWS accounts), a tech-stack aggregator's profile, and former employees'

CVs. They are weak signals about technology choices and say nothing about architecture.

Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), appears only in the final comparison. Hearthly's overlap with AVIV's French business is a teaching device, not a disguised description of AVIV or SeLoger.

Timeline: AVIV in Dated Events

Dates are those of the cited releases, decisions and articles; ownership and portfolio facts are as of October 2026 and may change.

Date	Event	Source
Sep 2010	Axel Springer offers €34 per share for SeLoger.com, valuing it at about €566 million, after agreeing to buy 12.4% from the co-founders and other shareholders	Tradingsat (BFM), 2010
Nov 2012	Axel Springer Digital Classifieds, a joint venture with General Atlantic, buys 80% of Immoweb, Belgium's leading property portal, for €127.5 million; the founding family keeps 20%	FrenchWeb, 2012
2014	Axel Springer buys Yad2 (Israel; real estate, cars, jobs and general goods) from Walla for about NIS 800 million	Online Marketplaces, 2025

Date	Event	Source
11 Feb 2015	Immowelt and Immonet combine under a new Immowelt Holding AG, valued together at €420 million; Axel Springer Digital Classifieds holds 55% and pays former Immowelt shareholders about €131 million; the stated aim is to be a "starke Nummer zwei" (strong number two)	Immowelt press release, 2015
21 Apr 2015	The Bundeskartellamt clears the combination of Germany's second- and third-largest property portals, expecting ImmoScout24 to remain the leader	Bundeskartellamt, 2015
1 Feb 2018	France's Autorité de la concurrence clears, without conditions, Axel Springer's purchase of Concept Multimédia (Logic-Immo), after its first in-depth review of a merger between two online platforms	Autorité de la concurrence, 2018
2018–2019	AVIV Group described in later company profiles as founded in 2018; in 2019 SeLoger and Logic-Immo get the shared corporate brand "Groupe SeLoger"	Company profiles (search snippets); MySweetImmo, 2019
27 Sep 2019	AVIV completes the purchase of Meilleurs Agents (valuation and owner-to-agency matching) for €200 million, to "élargir la chaîne de valeur" (widen the value chain)	FrenchWeb, 2019

Date	Event	Source
23 Aug 2021	Ex-eBay Classifieds and Domain CTO Brent McLean becomes AVIV CTO; CEO Ait Voncke: "we are aligning our tech roadmap across the organization to create a European real estate marketplace"	Online Marketplaces, 2021
Oct 2022	AVIV France's vice-president of client operations describes bringing SeLoger and Meilleurs Agents together "sous l'ombrelle AVIV" (under the AVIV umbrella)	MySweetImmo, 2022
3 Nov 2022	AVIV launches **Ultimmo**, one professional offer across SeLoger, Logic-Immo and Meilleurs Agents	MySweetImmo, 2022
Mar 2023	AVIV France plans 300 Tech, Product and Data hires in 2023 and aims to double visits to its French sites by 2025; it monetizes seller leads "through a commission-share model"	Online Marketplaces, 2023
Late Jun 2023	SeLoger opens free sale listings to private sellers, with the stated goal of routing undecided sellers to professionals	MySweetImmo, 2023
Jul 2023	Brand list: SeLoger, Logic-Immo, Meilleurs Agents (France); Immoweb (Belgium); Immowelt, Immonet, AllMyHomes, Homeday (Germany); Yad2 (Israel); Housell (Spain); a stake in Zumper (US). H1 revenue up about 3%, about 6% adjusted for consolidation effects	Online Marketplaces, 2023

Date	Event	Source
Jan 2024	Immoweb agrees to buy Realo, a Belgian portal and valuation-data specialist; both services "unaffected" in the short term	Online Marketplaces, 2024
Feb 2024	Immowelt GmbH renamed **AVIV Germany**; it runs Immowelt, immonet.de and umzugspreisvergleich.de	Online Marketplaces, 2024
19 Dec 2024	Axel Springer signs the split: Stepstone and AVIV to become joint ventures majority-owned by KKR and CPP Investments, Axel Springer keeping about 10%, "following significant platform and technology investments over the past five years"	Axel Springer/CPP Investments release, 2024
Jan 2025	AVIV Germany takes majority stakes in two German tax-depreciation services, to be integrated into Immowelt and Immoweb	Online Marketplaces, 2025
Feb–Mar 2025	Ait Voncke leaves after four years as CEO; Vladimir Pravdivy, former CEO of Avito, takes over in March	Online Marketplaces, 2025
29 Apr 2025	The new structure becomes effective; AS Classifieds GmbH, led by Julian Deutz from 1 June, supports the management of Stepstone and AVIV	KKR release, 2025

Date	Event	Source
2 Dec 2025	Apax Partners signs a deal to buy Yad2 from KKR for about $950 million; sale documents reportedly projected 2025 revenue of about €83 million (real estate 56%, autos 27%, jobs 15%). No closing announcement found as of October 2026	Online Marketplaces, 2025; Globes, 2025
4 Dec 2025	New Immowelt head Theo Mseka, asked about AVIV rowing back "on its plans for a fully integrated platform for all its European portals", describes strengthened local autonomy plus group-level functions	Online Marketplaces, 2025

A correction worth stating once. Immowelt and Immonet were combined as companies in **2015**, not in 2023. The 2015 release describes a holding and two sites (Nuremberg and Hamburg) and says nothing about merging the portals' technology. In February 2024 both brands were still operated by AVIV Germany. No public source found for this book documents a platform merger or a shutdown of immonet; readers should check before relying on either brand's current status.

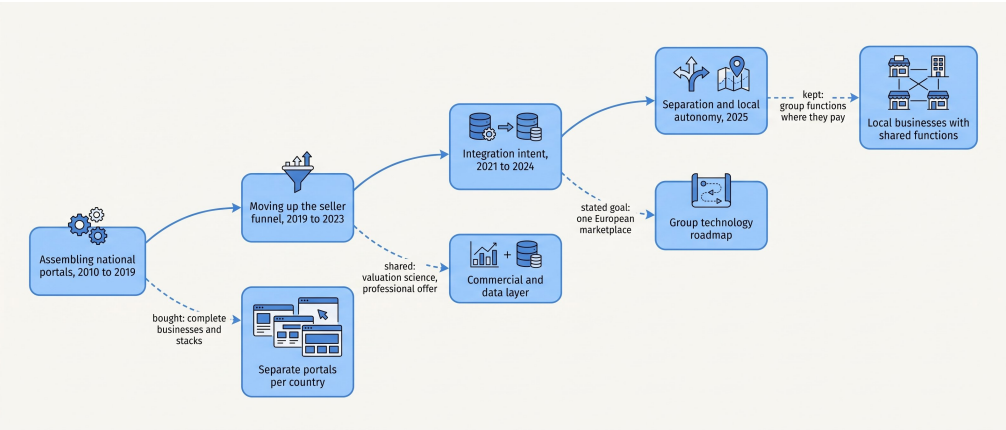


Figure 1: *AVIV's public record moves from separate portals to a shared commercial and data layer, then to a stated group technology goal, and back to local businesses with*

selected group functions. Only the second step is documented in any product detail.

Phase 1: Assembling National Portals (2010–2019)

What was bought

Every AVIV portal was bought as a working business with its own customers, brand and technology. The deals came through Axel Springer and, until General Atlantic's exit, through Axel Springer Digital Classifieds, of which General Atlantic held 15% in 2015 (Immowelt release, 2015). SeLoger was a listed French company when Axel Springer bid for it in 2010; Immoweb was a family-founded Belgian portal from 1996; Immowelt and Immonet were Germany's second and third portals; Logic-Immo came with print magazines; Yad2 was Israel's horizontal classifieds leader; Meilleurs Agents was a ten-year-old valuation start-up.

None of the acquisition announcements found for this book describes a technology integration plan. The 2015 Immowelt–Immonet release is typical: it describes the shareholdings, the leadership, the two sites and the Immowelt broker-software products (estateOffice, estatePro), and says nothing about merging platforms. **My reading** is that the group was built as a holding of national leaders, each kept whole because its value was local: agency relationships, brand traffic, local data and local pricing power.

What the regulators saw

Two competition decisions give the clearest public description of how these markets work, and both bear on architecture.

The **Bundeskartellamt** (April 2015) cleared the Immowelt–Immonet combination with an argument about network effects: where many small portals compete, property providers and seekers both drift to the leader, so with the merger "customers will have another large portal", in the words of its president, Andreas Mundt. The combination was justified as liquidity aggregation, not as cost synergy.

The **Autorité de la concurrence** (February 2018) cleared Logic-Immo's acquisition after its first in-depth review of a merger between two online platforms. Its reasoning was that agencies facing a price rise at SeLoger or Logic-Immo would shift their listings to other portals, mainly leboncoin, and it noted the rise of Bien'ici, a portal created on the initiative of the property industry. In the book's terms, French agencies **multi-home** (list on several portals at once, see Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics), and the regulator counted on that.

My reading of what this means for systems: a French or German portal does not own its supply. The agency's own software holds the mandate and pushes the same listing to several portals, so every portal in the group competes for the same feed and must

accept the agency software's formats. That is why Many Doors In: Forms, APIs and Bulk Feeds treats real-estate feeds as contracts with external systems of record, and why a group that owns two portals in one country still runs two ingestion edges facing the same vendors.

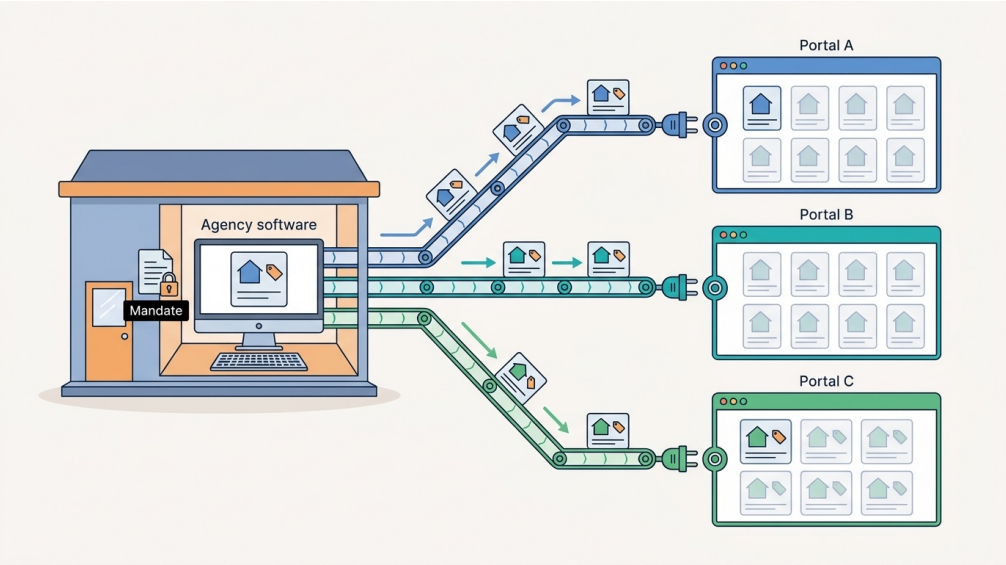


Figure 2: *Reference illustration, not AVIV's systems: in a multi-homing market the agency's software holds the mandate and pushes the same listing to several portals, so every portal competes for the same feed.*

Two portals in one country

AVIV kept both of its same-country pairs, Immowelt and Immonet in Germany and SeLoger and Logic-Immo in France, as separate brands for years. Immonet was still listed in AVIV's brand list in July 2023 and run by AVIV Germany in February 2024; SeLoger and Logic-Immo shared a corporate brand from 2019 and a professional offer from 2022.

> **INFERENCE, NOT DOCUMENTED:** When two portals in one country keep their brands, the cheapest technical integration is usually below the brand: one ingestion pipeline that accepts each agency's feed once and publishes to both sites, one media store, one agency account and contract system, with two front ends and two search experiences on top. Whether AVIV did this for either pair is not public. "Two brands" says nothing about "two stacks" in either direction, so an architect assessing such a group has to ask.

Phase 2: Moving Up the Seller Funnel (2019–2023)

Meilleurs Agents: valuation as supply acquisition

Valuation is AVIV's best-documented business, because it was a product, not a platform.

When Meilleurs Agents launched in 2008, FrenchWeb described its method: a historical base of "les informations enregistrées par les notaires depuis 1990" (the information recorded by notaries since 1990), recent sales by partner agencies added in real time, an algorithm that re-indexes past transactions to current market conditions, and a price per square meter that the user then adjusts for condition, exposure, layout and extras (FrenchWeb, 2008).

Fourteen years later, Meilleurs Agents' VP Data and Science, Thomas Lefevre, said that "it is not complicated to make an estimation tool. What is complicated, however, is to make a fair estimate", that the data team had grown to 45 people, and that "the median estimate error of our valuations against final transaction value in 2021 was just 6%" (Online Marketplaces, June 2022). The 6% figure is self-reported; no independent audit or methodology note was found.

The 2019 purchase carries the earliest public statement of something AVIV intended to share across countries, and it is a capability (valuation science), not a system. AVIV bought the company in September 2019 for €200 million; FrenchWeb reported that Axel Springer wanted to widen the value chain of its French portals and **share Meilleurs Agents' valuation expertise with its foreign sites** (FrenchWeb, 2019).

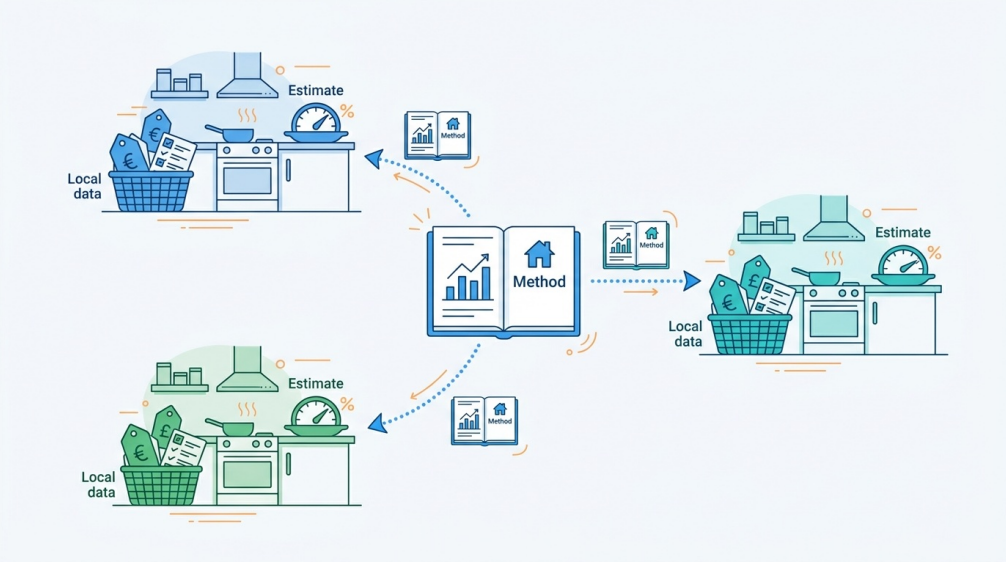


Figure 3: *What AVIV said in 2019 it would share was valuation expertise, a method that can travel between countries while each country's model still has to be trained on its own transaction data.*

My reading is that the valuation tool is a **supply-acquisition product**. A homeowner who asks what their flat is worth is a likely seller months before they choose an agent. A portal that captures that moment can sell it to agents as a seller lead. This is the same logic as Rightmove's Online Agent Valuation and Zillow's Zestimate, compared in Zillow, Rightmove, Idealista and REA: What Property Portals Teach; the modeling side, including why on-market and off-market accuracy must be reported separately, is in Models in Production: ML Platforms and Use Cases.

Ultimmo: the professional offer as the first integration

In November 2022 AVIV launched **Ultimmo**, one offer for agents across SeLoger, Logic-Immo and Meilleurs Agents. The launch material cites "plus d'1 million de visites par jour" (more than one million visits a day) across the three brands' platforms, 630,000 sellers identified each month, and the claim that two sales in three begin on one of the group's platforms. It bundles listing distribution and promotion, logo display on result pages, an estimation tool called Inventory, **detection of seller prospects among buyer contacts**, and an optional prospecting map. AVIV France's managing director, Caroline Evans de Gantès, framed it: "Les professionnels ne sont plus seulement en demande de mandats" (professionals no longer just want mandates; they also want visibility with buyers) (MySweetImmo, November 2022).

A month earlier, AVIV France's vice-president of client operations, Franck Le Tendre, described the goal of combining SeLoger's relationship with professionals and Meilleurs Agents' work on mandate acquisition, and said Meilleurs Agents' prospecting map had already been rolled out across the group's entities (MySweetImmo, October 2022).

This is the only cross-brand integration in AVIV's public record described at the level of features, and it is commercial: one contract and one set of tools sold to an agent, over brands that stay separate. Its features need shared data (sellers identified across brands, buyer contacts classified as potential sellers) and a shared professional account, but nothing in the record says they required a shared listing or search platform.

Opening SeLoger to private sellers

In late June 2023 SeLoger, until then an agent-only portal for sales, opened free sale listings to private sellers. Le Tendre explained the reasoning: professionals' share of the market had been flat for 15 to 20 years, about a third of sellers who try to sell alone give up, and "notre objectif est de capter ces indécis et de les diriger vers des professionnels" (our goal is to capture these undecided sellers and direct them to professionals). He said the group had 800 people working on innovation, and that many

agents were angry about the move. At the end of September 2023 about 6,000 private sale listings were online, against just under a million professional listings for sale and rent (MySweetImmo, October 2023).

The revenue model behind the funnel was stated the same year: AVIV France monetizes seller leads "through a commission-share model" (Online Marketplaces, March 2023).

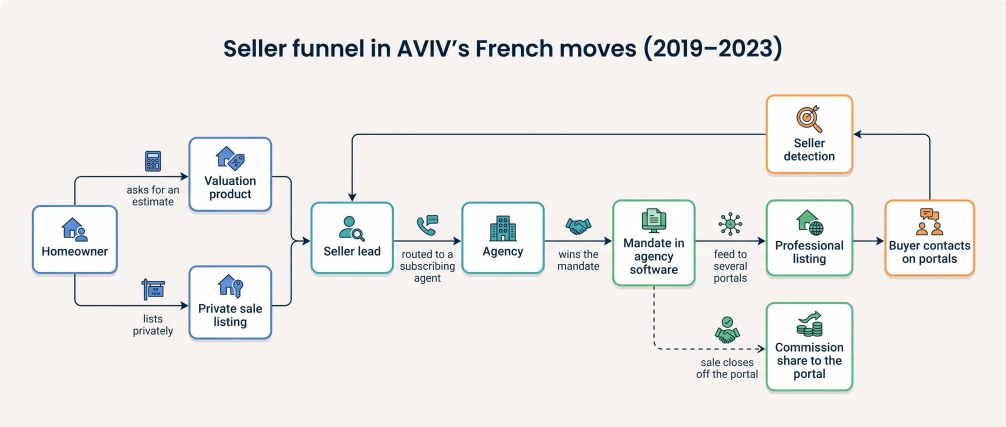


Figure 4: *The seller funnel that AVIV's documented French moves serve, drawn from the public product descriptions (2019–2023) — not a description of AVIV's systems. Every entry point produces a seller lead; the revenue event, a commission share, happens outside the portal.*

> **INFERENCE, NOT DOCUMENTED:** A commission-share model makes the portal's revenue depend on a sale recorded by an agency and a notary, weeks or months after the lead. A plausible reference architecture — not a description of AVIV's systems — needs a lead ledger with stable identifiers for the homeowner, the property and the agency, homeowner consent to routing and outcome tracking, and a reconciliation process against agent-reported or contractually reported closings. That is closer to the attribution systems of How Listing Marketplaces Make Money and Packages, Bumps and Quotas: Products, Pricing and Entitlements than to a classifieds listing service, and it is local by nature: French mandates, fees and notarial practice do not exist in the same form in Germany or Belgium.

Germany and Belgium

Each market built its own adjacencies around its own transaction: valuation and commission share in France, tax and financing services in Germany, valuation data in Belgium. The German and Belgian records are thinner but point the same way. In

December 2025 Theo Mseka said "the subscription model remains the core model in the German market today", that Immowelt offers AI features including "a tool for creating high-quality property descriptions", and that in 2025 Immowelt bought Kaufpreis-Aufteilung.com, nutzungsdauer.com and BauDarlehen24 to build an ecosystem around the property transaction (Online Marketplaces, December 2025). In Belgium, Immoweb bought Realo, a portal that also sells "data as a service" and valuations, in January 2024.

Phase 3: The Integration Intent (2021–2024)

What was said

The integration phase is documented almost entirely by statements of intent.

- In August 2021, announcing Brent McLean as CTO with responsibility for technology at Immoweb, SeLoger, Yad2 and Immowelt, CEO Ait Voncke said: "we are aligning our tech roadmap across the organization to create a European real estate marketplace" (Online Marketplaces, 2021). McLean came from eBay Classifieds Group and Domain, both groups that ran platforms across several marketplaces.
- In March 2023 AVIV France planned 300 technology, product and data hires against an industry trend of cuts; its managing director, Caroline Evans de Gantès, said AVIV would "write the future of real estate" (Online Marketplaces, 2023).
- In February 2024 the German company's managing director, Felix Kusch, explained the rename to AVIV Germany as combining a well-known brand with "the know-how and innovative strength of an international tech company", while the services Immowelt was known for would stay (Online Marketplaces, 2024).
- In December 2024 the signing release for the Axel Springer split referred to "significant platform and technology investments over the past five years" at Stepstone and AVIV (Axel Springer/CPP Investments, 2024).

What was not said

No public source found for this book describes what the European marketplace or the "fully integrated platform" was meant to contain: a shared listing service, a shared search engine, shared agent tools, shared billing, or only shared infrastructure and data. No source describes a portal moving onto a group platform, a migration milestone or a cost. The phrase "fully integrated platform" itself comes from the interviewer's question in December 2025, not from an AVIV statement.

My reading is that the integration intent is documented as a direction of travel, with leadership and hiring behind it, and its scope is unknown. The book therefore treats it as a goal set and later changed, and draws lessons only from that sequence and the structure of the markets, not from any assumed architecture.

Phase 4: Separation and Local Autonomy (2025)

The ownership change

The split signed in December 2024 became effective on 29 April 2025. AVIV and Stepstone became "independent joint venture companies with KKR and CPP Investments as majority owners", with Axel Springer holding "ten percent in both classifieds companies"; a holding company, AS Classifieds GmbH, led by Julian Deutz, supports their management (KKR, 2025). The signing release said the classifieds businesses "will continue to independently pursue their respective growth strategies". AVIV's CEO changed the same spring.

The first visible portfolio move under the new owners was the signed sale of the most separable asset. In December 2025 Apax Partners signed to buy Yad2 from KKR for about $950 million. The trade report names KKR, not AVIV, as the seller and says KKR gained full control of Yad2 in March 2025 through the carve-out; whether Yad2 still sat inside AVIV's structure at the time is not clear from public sources (Online Marketplaces, December 2025). Yad2 is a horizontal marketplace in a non-European market, with real estate only 56% of its projected 2025 revenue; it never fitted a European real-estate platform.

The reversal, in an executive's words

The clearest statement of AVIV's current model comes from the December 2025 interview with Immowelt's new head. The interviewer's question records the reversal: "We recently saw AVIV Group row back on its plans for a fully integrated platform for all its European portals." Mseka answered that "AVIV has strengthened the autonomy of its local businesses", making them faster in marketing and sales, while "we continue to operate group-level functions where they create clear advantages"; depending on the task, the group uses "sometimes local and sometimes cross-group" approaches. He said he had worked on multi-country platforms and seen their efficiency and scale, but was convinced by the localization strategy because it lets each business "run at their own pace" (Online Marketplaces, December 2025).

My reading has three parts, each conditional on thin evidence.

1. **The reversal is real but its size is unknown**. "Row back on plans" can mean a platform half-built and stopped, or a plan never far advanced. The record does not say which.
2. **The timing coincides with the ownership change, and no source links the two**. Under owners who had just sold Yad2 separately, separable businesses are worth more, as Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform shows for Adevinta's 2025 break-up. That is a plausible motive, not a documented one.

3. **The stated end state is a recognizable architecture.** "Group-level functions where they create clear advantages" plus local autonomy in "marketing and sales" is the "shared core, local edge" model of One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture. What AVIV includes in its group functions is not public.

Why Real Estate Resists One Platform

National supply software, ingestion standards, disclosures, monetization and property data explain AVIV's record, and they are documented outside AVIV. Same Platform, Different Physics: How Listing Verticals Differ develops them; the table keeps the points relevant to AVIV's markets.

Dimension	France (SeLoger, Logic-Immo, Meilleurs Agents)	Germany (Immowelt, Immonet)	Belgium (Immoweb)	What a group platform must absorb
Supply source	Agency transaction software pushing to many portals	Broker software exporting OpenImmo XML, typically over FTP	Agency software and portal-specific feeds	One adapter set per national software ecosystem
Ingestion standard	Portal- or vendor-defined export formats (no public common standard found)	OpenImmo since 2001; REST and JSON version 2.0 expected at the end of 2026	Not documented in the research	Two generations of standards at once in some markets

Dimension	France (SeLoger, Logic-Immo, Meilleurs Agents)	Germany (Immowelt, Immonet)	Belgium (Immoweb)	What a group platform must absorb
Mandatory disclosures	DPE (diagnostic de performance énergétique) energy and climate classes, estimated annual energy costs, warning for F and G homes	Energy-certificate data (national rules; not verified for this book)	Regional rules (not verified for this book)	Country-specific fields, validation and display rules
Main monetization	Agency subscriptions plus commission-share seller leads	Subscriptions remain "the core model" (2025)	Not documented in the research	Different products, contracts and billing per market
Property data	Notary transaction history since 1990, agency sales, open DVF (demandes de valeurs foncières) transaction data	Different sources and no free national address-ID register widely used by portals	Valuation data via Realo (2024)	Valuation models trained per country

Sources: Service-Public.fr on DPE (page verified January 2026); OpenImmo e.V. (accessed October 2026); Mseka interview (2025); FrenchWeb (2008); Immoweb–Realo (2024). The France ingestion row, the Germany address entry and the unverified disclosure entries are the book's research findings and judgement, not AVIV statements.

OpenImmo e.V. lists AVIV Group as its member from France (accessed October 2026). **My reading** is that AVIV has at least looked at a German standard as a possible common ingestion format beyond Germany; no detail of how it uses the standard was found.

> **INFERENCE, NOT DOCUMENTED:** A plausible reference architecture for a federated multi-country property group — not a description of AVIV's systems — puts cloud landing zones, security, identity tooling, the data platform and machine-learning tooling at group level; offers a shared ingestion engine and a core listing schema (location, price, surface, rooms, media) that each country extends with its own disclosure fields and property-type taxonomy; may share a search engine while keeping ranking policy and paid visibility local; shares valuation *methods* while training on local data; and keeps monetization, billing, agent CRM and professional tools local. In that design the group platform ends at the ingestion adapters on one side and the ranking policy on the other.

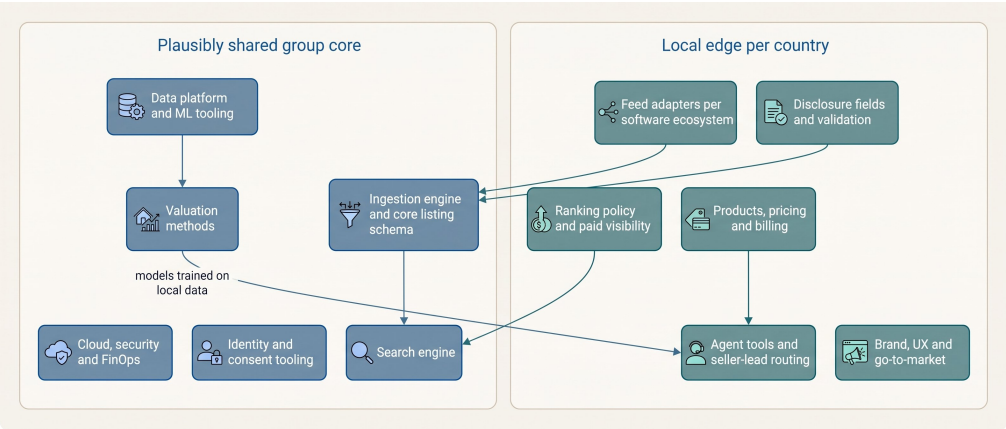


Figure 5: *A labeled reference architecture for a multi-country property group, not AVIV's: the shared core is infrastructure, data, identity, engines and methods; the local edge is everything an agent, a regulator or a homeowner sees.*

Trade-offs the Case Illustrates

Option	What AVIV's record shows	When it is right	When it fails
Keep acquired portals whole	2010–2019: each portal bought and kept as a business; two brands per country in Germany and France	Early in a roll-up, when each portal's value is local relationships and brand	When the same capability (valuation, agent tools, media) is bought or built four times
Share a capability, not a system (valuation science)	Meilleurs Agents bought in 2019 partly to share valuation expertise with foreign sites	When the method transfers and the data does not	When a group expects the French model to work on German or Belgian data unchanged
Bundle the professional offer within a country	Ultimmo (2022) across three French brands	When several brands in one market sell to the same agents	When the bundle hides which brand delivers the value, or invites competition scrutiny
Integrate into one European platform	Stated in 2021; reversed by 2025; scope never published	When markets share standards, disclosures and monetization, and ownership is stable	When feeds, disclosures, address systems and pricing norms are national, as in real estate
Local autonomy with group functions	Stated in 2025	When the businesses may be sold or run at different paces, and group functions pass a clear-advantage test	When "group functions" is left undefined and each local business rebuilds them

The table is not a ladder. **The line that AVIV's record draws runs through the capability map, not between countries**: methods, data science and infrastructure can be shared; the agent-facing product, the feed edge and the monetization stay local.

Vertical Differences in the Record

AVIV was a real-estate group with one horizontal asset, and the two behaved differently.

Asset type	AVIV examples	What the record shows	Implication
Agency-supplied property portals	SeLoger, Logic-Immo, Immowelt, Immonet, Immoweb	Kept as brands; national monetization (commission share in France, subscriptions in Germany); local adjacencies (tax services, valuation data)	The feed edge and the professional product are the least shareable parts
Valuation and seller-lead services	Meilleurs Agents, Realo	Bought for data and seller acquisition; valuation expertise to be shared across countries (2019)	Methods and teams travel; training data and lead routing do not
Hybrid brokerage	Homeday (Germany), Housell (Spain)	In the 2023 brand list; later status not documented	A portal owning brokers competes with its own customers, a trust constraint on data use
Horizontal classifieds	Yad2 (Israel)	Real estate 56%, autos 27%, jobs 15% of projected 2025 revenue; sale signed December 2025	The asset least aligned with a property platform was the first to be separated

A portal group that also owns brokerages, or that routes private sellers to agents, holds data its agency customers would not want a competitor to see. That is the same neutrality problem Veldmark faces with Keybridge, discussed below, and it is one reason seller-lead and agent data are usually kept inside the local business.

Failure Modes the Record Exposes

1. **Integration goals without a published scope**. The 2021 goal had a leader, a budget signal and hiring, but no public statement of which layers it covered. A plan whose scope is not written down is easy to reverse and hard to evaluate afterwards.
2. **Two portals in one market, two of everything**. Keeping brands is a sound commercial choice; keeping two ingestion edges, two media stores and two agent account systems behind them is a cost that the brand decision does not require. The public record does not show which AVIV did.
3. **Treating valuation as a feature**. Meilleurs Agents shows valuation driving seller acquisition and revenue; a group that files it under "data science" will underinvest in its governance, accuracy reporting and data rights. Models in Production: ML Platforms and Use Cases covers the governance.
4. **Funnel moves that anger the supply side**. The 2023 private-seller opening was explained as routing sellers to agents, and many agents were still angry. In a lead-generation vertical, agents control supply and multi-home; any product that looks like disintermediation is a liquidity risk (see Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics).
5. **Group functions defined after the fact**. "Where they create clear advantages" is a sound test, but if each function's owner, interface and exit path are not explicit, local businesses rebuild them and the group pays twice.

Where the Public Evidence Stops

The following questions matter to an architect and are not answered by any public source found for this book. The chapter does not fill them with inference.

Question	What is public	What is not
What was the "European real estate marketplace" or "fully integrated platform"?	The 2021 intent; the 2025 interviewer's description of rowing back	Scope, architecture, how far it got, cost, which portals moved
Which group-level functions remain?	That they exist "where they create clear advantages" (2025)	The list, their owners and interfaces

Question	What is public	What is not
Do Immowelt and Immonet, or SeLoger and Logic-Immo, share a stack?	Corporate combinations (2015, 2019) and a shared French professional offer (2022)	Whether ingestion, search, media or agent accounts are shared
How do AVIV portals ingest agency feeds?	AVIV's OpenImmo membership as a French member	Formats, adapters, validation, reconciliation
How does Meilleurs Agents estimate prices?	The 2008 method description; a self-reported 6% median error for 2021	Model family, features, intervals, retraining, how agency and notary records are de-duplicated
How does the commission-share model attribute revenue?	That it exists (2023)	Lead identifiers, consent, outcome tracking, reconciliation
What do AVIV's engineers run?	Job ads, an aggregator profile and CVs, seen as search snippets	Any first-party technical description; no blog posts, talks or public repositories were found
What happened to Housell, Homeday and the Zumper stake?	Their presence in the 2023 brand list	Current status

Two facts are volatile as of October 2026: the closing of the Yad2 sale to Apax, and the status of immonet as a separate brand. Readers should check both before relying on the portfolio described here.

The Chief Architect's View

What I take from AVIV is less about systems, which are not public, than about where the boundary of a property group's platform sits.

Draw the shared boundary by capability, and expect it to stop short of the agent. In real estate the agent's software, the national feed format, the disclosure rules and the pricing model meet in one place: the professional edge. I would share infrastructure, data, identity tooling and engines without hesitation, share valuation and ranking *methods* as group science, and keep the feed adapters, disclosure validation, professional products and lead routing local. If a proposal for a group platform crosses that line, it needs a separate case.

Write down the scope of any integration goal before staffing it. A goal such as "one European marketplace" can mean anything from shared cloud accounts to one listing service. I would require a layer-by-layer scope, the markets in each stage, and an exit criterion per stage. That is what makes a later change of direction cheap rather than a write-off, and it is what After the Deal: Consolidating Platforms After Acquisitions calls a consolidation sequence.

Treat valuation as a supply product with its own governance. The AVIV and Meilleurs Agents record, together with Zillow's and Rightmove's in Zillow, Rightmove, Idealista and REA: What Property Portals Teach, says that the estimate is how a portal reaches the seller first. It deserves an owner, published accuracy definitions (on-market and off-market separately), data-rights checks on every source, and a clear line between an estimate that informs and one that commits money.

Keep the attribution chain local and explicit. Commission-share revenue depends on events outside the platform. The lead ledger, consent and reconciliation are local, legal and auditable work; they should be designed as such, not bolted onto a listing service.

Design every group function to be handed over. Yad2's sale was signed about seven months after the carve-out became effective. Any function a business might take with it (identity, data products, valuation models) should be tenant-scoped from its first release, as Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform argues at greater length.

Comparing the Case to the Book's Patterns and to Veldmark

The book's patterns, as AVIV's record tests them:

Book pattern or claim	AVIV's record	Verdict
Real estate is the least portable vertical across countries (Same Platform, Different Physics: How Listing Verticals Differ)	National feeds, disclosures and monetization; 2025 shift to local autonomy	Supported, though AVIV never stated its reasons

Book pattern or claim	AVIV's record	Verdict
Share global components beneath local products (One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture)	Group functions "where they create clear advantages" (2025)	Consistent with the stated end state; contents not public
Consolidate enabling layers first, core product last (After the Deal: Consolidating Platforms After Acquisitions)	A "fully integrated platform" planned and rowed back	Consistent, as a cautionary data point
Valuation is a supply-acquisition engine (Models in Production: ML Platforms and Use Cases, How Listing Marketplaces Make Money)	Meilleurs Agents, Ultimmo seller detection, commission-share seller leads	Strongly supported for France
Real-estate feeds are contracts with external systems of record (Many Doors In: Forms, APIs and Bulk Feeds)	Multi-homing agencies cited by the French regulator (2018); OpenImmo membership	Supported by regulators' descriptions; AVIV's ingestion not public
Measure matches, not listings (Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics)	"630,000 sellers identified each month"; seller leads monetized	Supported: AVIV's public metrics are funnel metrics

Now Veldmark, the fictional group, as of its 2027 decisions in the scenario. Hearthly, Veldmark's real-estate portal, operates in France (acquired 2023) and Spain (acquired 2025), with **€395 million** of 2026 revenue, **84%** of listings arriving through feeds from about **160** real-estate software vendors, and **Valuo**, its valuation product, trained on France's open DVF transaction data and Hearthly listing data. Veldmark's Common Ground program follows the principle "**share platforms, not products**". The compari-

son sharpens four choices.

Veldmark choice (2027)	AVIV parallel	What the case suggests
Supply Contract v1 (September 2027): one envelope, error catalog and validator library for the group, but **adapters stay brand-local**; ingestion is not a Common Ground shared platform	National software ecosystems; OpenImmo as one possible common standard	The right line: share the contract and the validator, keep the adapters where the vendors are
Hearthly stays on Google Cloud through 2028 and moves to the shared OpenSearch platform by end of 2028, keeping its own ranking policy	Group functions plus local autonomy (2025)	Sharing the engine and keeping the relevance policy local is the "shared core, local edge" the case points to
Valuo seller leads: flat-fee product, plus a **commission-share pilot in the Lyon metropolitan area in 2028**, with homeowner consent and no use of Keybridge mandate data	AVIV France's commission-share seller leads (2023)	The attribution chain must be local, consented and auditable before the pilot starts; it is not a group listing feature
Keybridge kept a separate, neutral SaaS (March 2027); no Keybridge agency data used by Hearthly	SeLoger's private-seller opening angered agents (2023); hybrid brokerages in AVIV's portfolio	Agent trust is a supply constraint; neutrality has to be enforced in data access, not promised

The worked question is the one AVIV faced in 2021. Suppose Claire Fontaine, Hearthly's fictional CEO, proposed in 2027 a single Hearthly platform for France and Spain, to be extended to any future property acquisition. Under the ledger, Hearthly France already de-duplicates **1.9 million** listings into about **1.12 million** distinct properties (**1.7** listings per property), with clustering in a Hearthly property registry; Hearthly Spain was rebranded only in 2026; and Valuo's Spanish version is planned for 2028. Following AVIV's record and the trade-off table above, I would advise Ingrid Halvorsen,

Veldmark's fictional Group Chief Architect, to approve the parts that pass the clear-advantage test (the shared search engine, the property-registry *design*, the Valuo *method* and its governance, the group event backbone) and to reject a single product platform until the Spanish feed vendors, disclosure fields and pricing norms have been profiled against the French ones. **The decision record should state which layers are shared, which are local, and the evidence that would change the split** — the scope that the public AVIV record never shows.

Questions to Ask

1. For each capability we plan to share across countries, is it a method, an engine or a product, and does it pass a written clear-advantage test?
2. Where exactly does our group platform stop on the supply side: at the ingestion engine, the validator library, or the feed adapters?
3. Which mandatory disclosure fields differ by country, and who owns their validation and display rules?
4. Is our valuation product governed as a supply-acquisition product, with an owner, accuracy definitions and data-rights checks per source?
5. If we monetize seller leads by commission share, what identifiers, consent records and reconciliation process prove the outcome?
6. When two of our brands serve one market, which layers behind the brands are duplicated, and is that duplication a decision or an accident?
7. What is the written scope, stage plan and exit criterion of any integration goal we have announced?
8. Which product moves could look like disintermediation to agents, and how will we detect supply withdrawal early?
9. If one business were sold within a year, which group functions would it take, and are they tenant-scoped today?

AVIV shows how little of a property portal travels across borders, seen from the outside. The next case, Zillow, Rightmove, Idealista and REA: What Property Portals Teach, turns to the portals that publish the most about their economics and engineering — Zillow, Rightmove, Idealista and REA — and to what their valuation, ingestion and monetization records teach.

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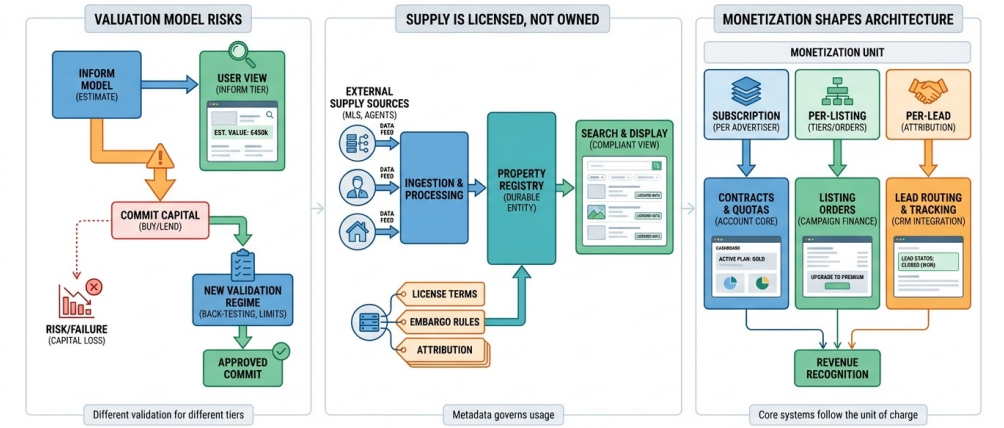
Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

50. Zillow, Rightmove, Idealista and REA: What Property Portals Teach



IN THIS CHAPTER, YOU WILL: Read the public record of Zillow, Rightmove, REA Group and Idealista, with CoStar as a counterpoint, from Zillow's iBuying bet to Rightmove's completed cloud migration. You will separate what each company documented from what the author infers, see where the evidence runs out, and test the cases against the book's patterns and Veldmark's Hearthly decisions.

WHY THE BUSINESS CARES: Property portals are among the most profitable listing businesses in the world. Rightmove reported a 70% underlying operating margin for 2025 and REA a 66% operating EBITDA margin for its 2026 fiscal year. Zillow also wrote down $304.4 million of homes in one quarter when it used its valuation model to buy them. The same data assets can earn rents or destroy capital, depending on the decision they are allowed to drive.

WHY ARCHITECTS SHOULD CARE: Real estate is the vertical where the listing comes from a system the portal does not own, where the property

outlives every listing of it, and where models move into lending and buying. The documented cases show which design choices follow from those facts, and which failures follow from ignoring them.

KEY POINTS:

- **An estimate built to inform is not fit to commit.** In February 2021 Zillow made the Zestimate an initial cash offer in 23 markets. By November 2021 it was winding down Zillow Offers, writing that it had been "unable to accurately forecast future home prices at different times in both directions". The model did not suddenly get worse. It was moved from describing present values to pricing purchases resold months later, and no validation regime for that new use is visible in the public record.
- **A portal's supply is licensed, not owned.** Zillow depends on multiple listing service (MLS) memberships and data licenses that may restrict "the training of models". Rightmove and REA depend on agents who also feed rival portals. Zillow's 2025 listing policy started a lawsuit, and Zillow changed the policy in March 2026. License terms, embargoes and agent relations therefore belong in the architecture as data and controls, not only in contracts.
- **The unit of charge shapes the architecture.** Rightmove charges per advertiser (£1,621 a month on average in 2025), REA per listing through vendor-paid depth tiers, and Zillow per lead through share of voice or a fee at closing. Each needs a different core: contracts and quotas, per-listing orders, or lead routing with attribution over up to two years.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Set-and-Forget AI** — Zillow Offers is the documented extreme: a valuation model was given a commit-tier decision without visible horizon, selection or capital controls.
- **Seeing Double** — In property the same home is listed by several agencies and portals, and Idealista's 2021 duplicate-control tool is a public example of clustering rather than deleting.
- **Feed Roulette** — Rightmove's published feed contract shows the opposite discipline: a schema, an idempotency key, tenancy, and snapshot reconciliation for lost deletes.

AVIV Group: Building a European Real-Estate Platform from Acquisitions asked how much of a property portal a group can share across countries. This case asks the question underneath it: what do licensed supply, durable properties and valuation models demand of one portal? Agents pay to advertise homes, buyers and renters search for free, and the

leader in each country takes most of the profit. Underneath, property portals carry three properties that the rest of this book treats as hard cases. **The listing usually comes from an external system of record**: an MLS in the US, an agency CRM almost everywhere else. **The property is a durable entity** that is listed, sold, let and valued many times. And **the portal's data is good enough to tempt it up the value chain**, into valuation, mortgages, agent software and, in Zillow's case, buying homes. The record therefore tests three earlier answers: the decision tiers of Models in Production: ML Platforms and Use Cases, the property model of One Property, Three Listings: Duplicates and Entity Resolution and the feed contracts of Many Doors In: Forms, APIs and Bulk Feeds. The comparison with Veldmark near the end holds Hearthly, Veldmark's property portal, to the same record.

This case reads four portals as an architect would. It asks what each one charges for, where its supply comes from, what its models are allowed to decide, and what it has published about how its systems work. The public evidence is uneven, and the chapter says so. Zillow and Rightmove publish detailed filings and, at times, engineering material. REA publishes rich financial disclosures and almost no engineering material that I could retrieve. Idealista publishes very little. **The depth of each section below follows the depth of the evidence, not the importance of the company.**

How to Read This Case

Part IX cases follow stricter rules than the chapters in Parts I–VIII (see Part IX — PRACTICE: How Others Built It). A **documented fact** carries its source and publication year in the sentence. **The author's interpretation** is marked "my reading" or sits in the Chief Architect's view. A **reference-architecture inference** appears in a callout labeled `INFERENCE, NOT DOCUMENTED`.

Two cautions apply throughout: the engineering material is dated, and several facts are still moving. The engineering material is old or vendor-written (Zillow posts from 2020–2021, a 2015-era REA session, a 2025 Elastic case study on Rightmove) and may describe replaced systems. And several facts are volatile as of October 2026: litigation against Rightmove and around Zillow, CoStar's plans for Domain, and Idealista's ownership after the 2024 deal. Check them before relying on them.

Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), appears only in the final comparison. Its real-estate portal Hearthly is a composite of industry patterns. It is not a disguised description of any portal in this case.

Timeline: Property Portals in Dated Events

Dates are those of the cited filings, releases and reports. Ownership and litigation facts are as of October 2026.

Date	Event	Source type
Feb 2006	Zillow launches with the free Zestimate home-value estimate	Encyclopedia (secondary)
2014	Rightmove moves search to Elasticsearch, citing map-centric results and "instant property alerts"	Elastic vendor case study, 2025
c. 2015	REA describes a gradual migration of realestate.com.au search to Elasticsearch at an Elastic{ON} Tour session in Sydney	Elastic event page, n.d.
Apr 2018	Zillow Offers, Zillow's iBuying business (buying homes directly, renovating and reselling them), launches	Encyclopedia (secondary)
25 Feb 2021	The Zestimate becomes an initial cash offer for eligible homes in 23 Zillow Offers markets	CNN Newsource, 2021
11 Apr 2021	Idealista launches a duplicate-control tool that groups listings of the same property	Trade press, 2021
15 Jun 2021	Zillow launches the neural Zestimate; median off-market error 6.9%	Trade press, 2021
2 Nov 2021	Zillow's board decides to wind down Zillow Offers; $304.4 million inventory write-down; workforce cut by about 25%	Zillow 8-K and shareholder letter, 2021

Date	Event	Source type
8 Dec 2021	Rightmove announces a move of "a number of its systems" to Google Cloud, starting with containers on GKE	Google Cloud press release, 2021
28 Jun 2024	Cinven agrees to buy 70% of Idealista in a deal valuing it at about €2.9–3 billion; EQT keeps 18%	Trade press, 2024
2025	Zillow introduces its Listing Access Standards, excluding homes publicly marketed for more than 24 hours before MLS submission; Compass sues in June	Trade press, 2026
Feb–Aug 2025	CoStar completes its Matterport acquisition (28 February) and its Domain acquisition in Australia (27 August, A$4.43 per share)	CoStar 10-K, 2026; CoStar release, 2025
6 Feb 2026	A US federal judge denies Compass a preliminary injunction against Zillow's policy	Trade press, 2026
11 Feb 2026	Zillow's FY2025 10-K: Zestimate median error 1.8% on-market, 7.2% off-market; a living database of about 173 million homes	Zillow 10-K, 2026
19–20 Mar 2026	Zillow stops barring listings first marketed on Redfin or Compass sites; Compass drops its suit without prejudice	Trade press, 2026
Apr 2026	An application for a collective claim of up to £1.56 billion against Rightmove is filed with the UK Competition Appeal Tribunal	Rightmove H1 2026 results

Date	Event	Source type
31 Jul 2026	Rightmove: "100% of applications are in the cloud", more than 5 PB on a unified data platform, 99.99% uptime	Rightmove H1 2026 results
6 Aug 2026	REA FY26: Buy yield +13% on flat listings; 66% operating EBITDA margin	REA ASX announcement, 2026

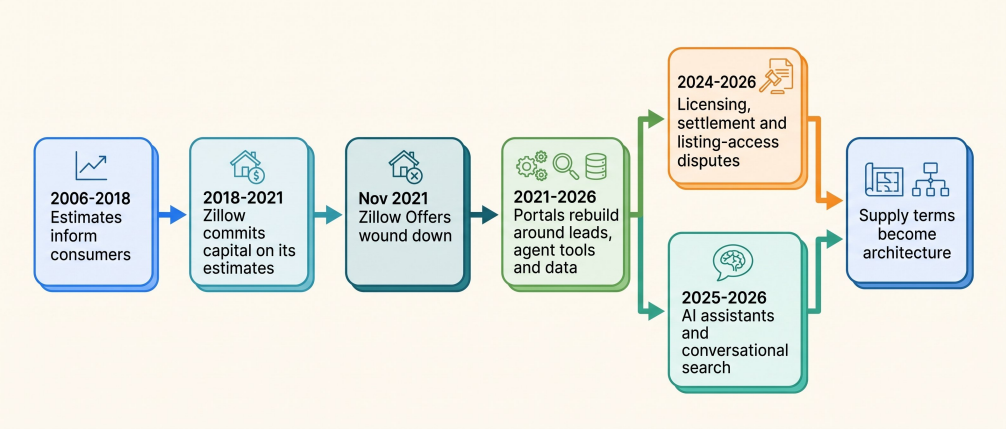


Figure 1: *The arc of the public record. Valuation moved from informing consumers to committing capital and back. The current phase sits under two pressures: who controls supply, and how AI assistants reach listings.*

Four Portals, Four Economic Engines

The four leading portals in this case report different unit KPIs, and **the choice of denominator reveals the model** (see How Listing Marketplaces Make Money). The figures cannot be compared across rows: currencies, periods and definitions differ.

Portal	What it charges for	Published unit KPI	Period	What the architecture must hold
Rightmove (UK)	Monthly subscriptions per agency branch or development, plus products above the package	Average revenue per advertiser (ARPA) £1,621 a month; agency branches 16,385	Calendar 2025	Account hierarchy, package commitments with metered extras, lead and performance reporting per branch
REA Group (Australia)	Per-listing depth tiers (Premiere+ at the top), mostly paid by the vendor through the agent, plus subscriptions and add-ons	Residential Buy yield +13% with national listings flat	Year to June 2026	Per-listing orders and upgrades, tier as a listing attribute, campaign finance for vendors
Zillow (US)	Buyer leads to agents, by share of voice per zip code or as a fee at closing; rentals per lead, lease, listing or impression	Residential revenue per visit $0.178	Calendar 2025	Lead routing and allocation per geography, attribution from lead to closing, revenue estimation and true-ups
Idealista (Spain, Portugal, Italy)	Not disclosed in sources found for this book	None public	—	Not inferable from public evidence

Portal	What it charges for	Published unit KPI	Period	What the architecture must hold
Homes.com (CoStar, US)	Listing-agent advertising on a "your-listing your lead basis"	None in the sources used	2025	Leads routed to the listing agent, not resold to buyer agents

Rightmove's is a subscription with a commitment plus overage: in 2025 "more than 50% of independent agents purchased incremental product above their package commitment", and 62% of agency ARPA growth came from product rather than price (Rightmove FY2025 results, 2026). **REA's is per-listing depth**: its fiscal 2026 Buy yield rose on "a 7% average Premiere+ price rise, growth in add-on products, higher subscription revenues and increased depth penetration", and a subsidiary, Campaign Agent, provides "vendor paid advertising finance solutions" (REA FY26 results, 2026). **Zillow's is lead generation**: connections are distributed "in proportion to their share of voice", and in Zillow Preferred, its invite-only pay-when-you-close program, agents pay "when a real estate transaction is closed with one of the leads, generally within two years", which Zillow accounts for as variable consideration (Zillow 10-K for 2025, 2026).

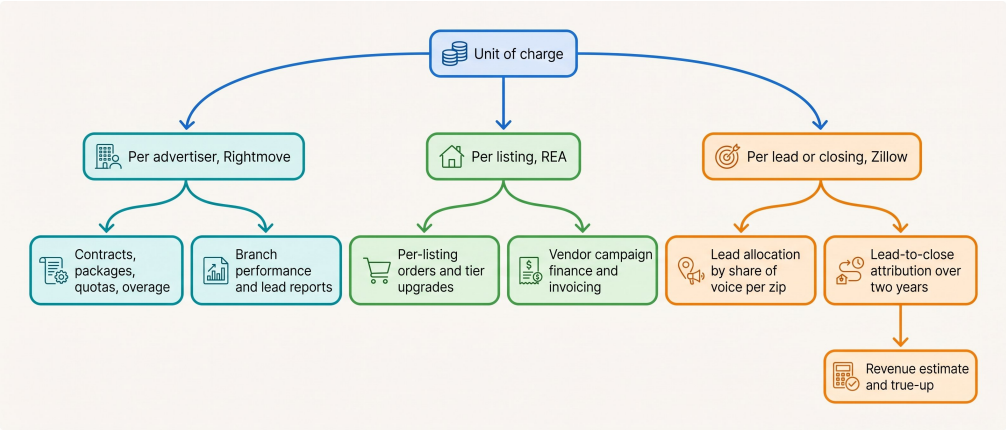


Figure 2: *The unit a portal charges for determines the core of its monetization architecture. A subscription portal centers on contracts and quotas, a depth portal on per-listing orders, and a lead portal on allocation and attribution.*

My reading is that these engines also explain each portal's appetite for valuation. Rightmove and REA both sell **seller leads**, homeowners identified before they choose an

agent, and both report growth in them: Rightmove's AI-enabled Online Agent Valuation supported "c.50% total increase in unique valuation leads to estate agents" in the first half of 2026, and REA reported a 22% increase in seller leads for fiscal 2026. Valuation in property is a supply-acquisition product: it reaches the homeowner at the moment a listing is born. Zillow took the same insight one step further and bought the house.

Zillow: When the Estimate Started to Commit

The Zestimate as an information product

Zillow launched in 2006 with the Zestimate, a free automated valuation model (AVM) estimate for homes whether or not they were for sale. In June 2021 it launched a neural-network version and reported a national median error of 6.9% for more than 104 million off-market homes, about a full percentage point better than before (Online Marketplaces, 2021). A secondary account from the same year says the neural model replaced roughly 1,000 market-specific algorithms (The Batch, 2021). Zillow's annual report for 2025 calls the Zestimate "our patented proprietary automated valuation model" and gives "a median error rate of 1.8% for homes listed for sale and 7.2% for off-market homes". It describes "our living database of approximately 173 million U.S. homes" (Zillow 10-K, 2026).

The on-market figure measures an easier test: a model that sees the asking price, the agent's own valuation, is mostly predicting the gap between asking and selling price. The off-market figure is closer to the real performance of the AVM. Models in Production: ML Platforms and Use Cases uses the same two numbers to explain why on-market and off-market accuracy must be reported separately.

The off-market error in the 2025 filing (7.2%) is higher than the 2021 launch figure (6.9%), but it should not be read as model regression without Zillow's methodology notes. The public sources do not say whether methodology, market volatility or the measured population changed.

The "living database" also shows a **property-first data model**: the home is the durable entity and listings are events on it, with user-generated updates to tens of millions of property records (Zillow 10-K, 2026). One Property, Three Listings: Duplicates and Entity Resolution develops that model.

From estimate to offer

Zillow Offers launched in 2018. On 25 February 2021, CNN reported that for certain homes the Zestimate "will now represent an initial cash offer from the company to buy the property", available "in 23 markets to start" and depending on how much data Zillow had on the home and nearby comparables. After acceptance, "a Zillow employee

inspects the home, provides a list of needed repairs and an adjusted final offer". Zillow's chief operating officer said that for homeowners who declined, the gap between the offer and the eventual sale price was "typically less than 1%" (CNN Newsource via KVIA, 2021).

That announcement is the documented moment at which an **inform-tier model**, one that shows a number to a consumer, became the opening price of a **commit-tier decision**, one that spends the company's capital. The terms come from Models in Production: ML Platforms and Use Cases's four decision tiers (inform, order, gate, commit). The model now drove both. The quoted accuracy figure was measured on homeowners who *declined*, which says nothing about the homes Zillow bought.

Figure 3: *The same estimate that informs a homeowner becomes a commitment when it opens a purchase: the outcome then depends on a resale price months later, which the present-value accuracy figures never measured.*

The wind-down in Zillow's words

On 2 November 2021, "the Board of Directors of Zillow Group made the determination to wind down Zillow Offers operations", citing "home pricing unpredictability, capacity constraints and other operational challenges". The third quarter carried a "$304.4 million" write-down of inventory. Zillow expected a further "approximately $240 million to $265 million" in the fourth quarter on homes it was already under contract to buy, plus $175–230 million of other wind-down charges. It expected to reduce its workforce by about 25% and to keep selling inventory through 2022 (Zillow 8-K, 2021).

The shareholder letter published the same day is unusually direct about the causes (Zillow Q3 2021 shareholder letter):

- "We have been unable to accurately forecast future home prices at different times in both directions", with unit economics "swinging approximately 1,200 basis points from Q2".
- Scaling further was "too risky, too volatile to our earnings and operations, provides too little opportunity for return on equity, and serves too narrow a portion of our customers".
- "This home price forecasting volatility has also contributed to significant capacity and demand planning challenges."
- Zillow could "convert only about 10% of the serious sellers who ask for a Zillow Offer".
- The quarter ended "with 9,790 homes in inventory and 8,172 homes under contract".
- The letter also names "our ability to continually improve the Zestimate" as foundational to the company's top-of-funnel strategy, which continued after the wind-down.

What the record supports

The following lessons are my interpretation of those statements, not Zillow's own analysis. Models in Production: ML Platforms and Use Cases states the first four briefly; this case adds the evidence and the fifth.

1. **The binding problem was the forward forecast, not the present estimate**. Buying, repairing and reselling takes months. The letter's own words are about forecasting "future home prices". A model validated on present value was used to price an outcome at a later date.
2. **Error turns asymmetric under selection**. Sellers whose homes the model overvalues tend to accept; others walk away. A small median error over the whole population can become a systematic loss on the accepted subset. The 10% conversion figure is a reminder that the accepted population was a narrow, self-selected slice.
3. **Operational capacity couples to model error**. "Capacity and demand planning challenges" mean that forecast misses became renovation queues, holding time and inventory. A model monitor watching prediction error alone would not have seen those.
4. **Accuracy was reported on the wrong population**. "Typically less than 1%" was measured on declined offers. The decision-relevant metric was the error on purchased homes at resale, net of holding and repair costs.

5. **Exposure grew faster than evidence.** By the end of the quarter, almost 18,000 homes were in inventory or under contract. The public record shows no capital limit tied to validated performance. That does not prove none existed. It shows that nothing in the published reasoning stopped exposure from outrunning the evidence.

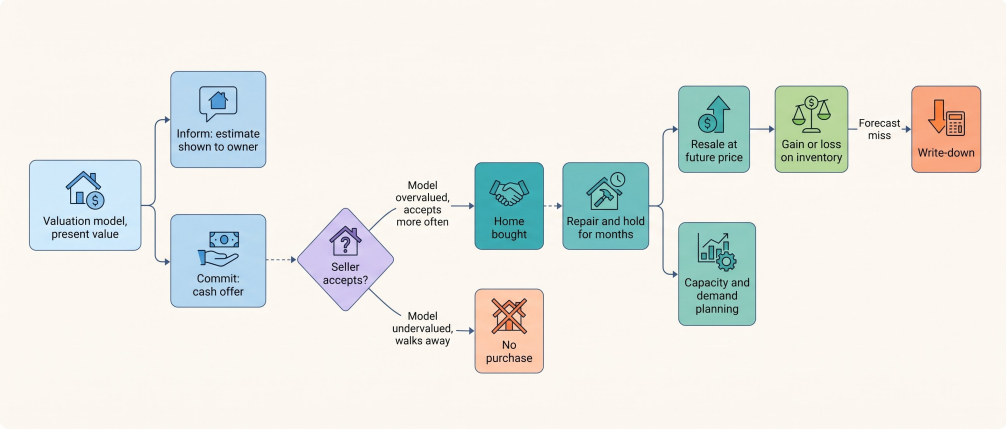


Figure 4: *Why a commit-tier use needs its own validation. The same model feeds an information product and a purchase. In the purchase path, selection biases the accepted homes, and the outcome depends on a future price after months of holding.*

INFERENCE, NOT DOCUMENTED: A validation regime fit for a commit-tier valuation use — a reference design, not a description of anything Zillow ran — would have five parts. (1) Back-tests on **resale price over the holding period**, not on present value. (2) Error measured on the **accepted population**, with a correction for selection. (3) **Capital limits** per market that grow only as realized margins confirm the back-tests. (4) **Operational capacity** (repair crews, days to resale) modeled as an input to pricing, not only as a cost. (5) An **owner outside the model team** with authority to pause buying when realized error leaves an agreed band. Each part is standard practice in credit risk, which is a closer analogy for iBuying than consumer valuation.

After Offers: leads, agent tools and enhanced markets

Zillow kept the Zestimate and the "top of the seller funnel" and rebuilt around agent leads. The 2025 annual report describes residential revenue as 66% of the total. It covers the two Premier Agent pricing models above, Zillow Preferred introduced as "the next

chapter for our Flex program", and Enhanced Markets: "geographies where Zillow is investing to provide an integrated home-buying experience". It also lists the agent tools Zillow owns: Follow Up Boss (a CRM), dotloop, ShowingTime (tour scheduling), and Zillow Showcase, a premium listing tier with interactive floor plans and 3D tours (Zillow 10-K, 2026).

> **INFERENCE, NOT DOCUMENTED:** A pay-at-close model implies an attribution pipeline. Its parts would be a stable lead identity, deduplication, assignment to an agent, synchronization with the agent's CRM, and matching to transaction records for up to two years. It would also need a revenue estimate per open lead that is trued up when outcomes arrive. Owning the agent's CRM shortens that pipeline. This is a plausible reading of why lead-generation portals buy agent software, not a description of Zillow's systems.

Supply is licensed, not owned

Zillow states that it obtains listings through "relationships with real estate brokerages, real estate listing aggregators, MLSs" and other providers, with subsidiaries holding MLS memberships whose rules they must follow. Its risk factors warn that law, regulation or data licenses could limit data uses "including the training of models, generation of predictive or derivative outputs" (Zillow 10-K, 2026). **A listing license now governs machine learning, not only display**.

The listing-access dispute shows the same dependence from the other side. In 2025 Zillow introduced its Listing Access Standards, which keep homes off Zillow when they "have been marketed publicly for more than 24 hours" without first being submitted to an MLS. Compass, a brokerage, sued in June 2025. On 6 February 2026 a federal judge denied Compass a preliminary injunction in a 50-page decision. It was a procedural ruling, not a judgment on the merits (Online Marketplaces, 2026). On 19 March 2026, Zillow said it would no longer bar sellers and agents from marketing listings on Redfin or Compass sites before listing them on Zillow, and the next day Compass dropped its suit without prejudice (The Mortgage Point, 2026). The 2025 annual report names complaints by "the Federal Trade Commission, certain state attorneys general, Compass, Inc., CoStar Group, Inc." among its legal proceedings.

Industry rules add visibility terms of their own. NAR's "Multiple Listing Options for Sellers" policy, effective 25 March 2025, allows "office exclusive" and "delayed marketing" listings, whose IDX (Internet Data Exchange) and syndication distribution is held back (NAR policy pages). **The same record can therefore carry an MLS embargo, a brokerage's marketing plan and a portal's access rule at once**. Rules of the Marketplace: DSA, DAC7 and Vertical Regulation and Many Doors In: Forms, APIs and Bulk Feeds treat visibility and license flags as data that ingestion must carry.

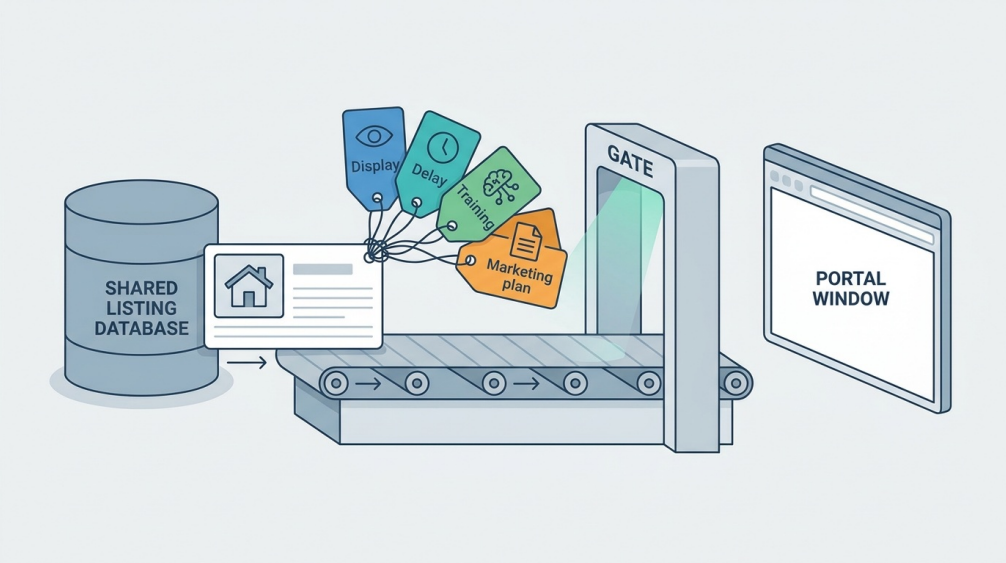


Figure 5: *Supply is licensed, not owned: one listing can carry an embargo, a brokerage's marketing plan, a portal's access rule and limits on model training at once, and ingestion has to keep every one of them attached.*

Since the 2024 NAR settlement, buyers must also sign a written agreement before touring with an agent (NAR settlement FAQs), which puts an agreement step in the middle of any portal's tour-request flow.

What Zillow published about its engineering

The Zillow engineering posts I could read come from the Zillow Tech Hub publication on Medium and date from 2020–2021. They may describe systems that have since been replaced (Zillow Tech Hub feed):

Post (date)	What it documents	Pattern
Guided Search (September 2020)	Refinement suggestions ranked by filter co-occurrence and filter-change models per region; nearly 50 filters and 200+ attributes per property; a 33% lift in refinement engagement in an A/B test	Real-estate schemas are far richer than goods schemas (What Did They Mean? Query Understanding and Autocomplete)

Post (date)	What it documents	Pattern
How to Pitch Kafka (August 2020)	Kafka chosen over Kinesis for schema registry, Connect and consumer decoupling; topics provisioned through Terraform with ownership, schema and privacy metadata	Topic metadata as governance (Every Change Is an Event: Event-Driven Listing Architecture)
Floor-plan ML serving (November 2020)	When a 3D tour finishes, a Step Functions workflow launches SageMaker batch jobs that detect windows and doors; batch chosen over live endpoints for cost and startup time	Inference attached to the media lifecycle (Photos Sell: Media Pipelines for Listings)

Zillow's search engine, the architecture of its natural-language search (launched in 2023 and extended in September 2024 to commute, affordability and schools, per HousingWire, 2024) and the neural Zestimate's serving design are not public in any source I could retrieve.

Rightmove: A Subscription Machine with a Published Feed Contract

Subscription economics

Rightmove is the best-documented property portal economically. For 2025 it reported revenue of £425.1 million, a 70% underlying operating margin and total ARPA of £1,621 a month across 19,272 advertisers. Retention of existing agency partners was above 90%. Its top agency package, Optimiser Edge, had 35% penetration among independents (Rightmove FY2025 results, 2026). In the first half of 2026 it reported "90% Comscore; 75% SimilarWeb/Sensor Tower" share of time spent on UK property portals, "over 85% of traffic direct and organic", revenue of £225.8 million (+7%) and ARPA of £1,726 (Rightmove H1 2026 results, 2026). The two audience panels disagree by 15 points, so always name the panel. **Direct traffic is the margin**: a portal that does not buy its audience does not see acquisition cost grow with revenue (Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics).

The feed contract

Rightmove's Real Time Datafeed specification (version 1.4.1, last updated November 2023) is the most detailed public ingestion contract of any portal in this book's research (Rightmove RTDF specification). It gives each software provider a network ID and each office a branch ID with sales and lettings channels, authenticates with Rightmove-issued client certificates, keys listings on the agent's own `Agent_Ref`, pulls media by URL with change detection, and offers `GetBranchPropertyList`, a branch snapshot for finding properties "still live on Rightmove which should have been removed". Its version history tracks regulation: ground rent, service charge and council-tax band in 2022, then utilities, flood risk and accessibility in 2023. Earlier chapters use it in detail (Many Doors In: Forms, APIs and Bulk Feeds, Private Sellers, Dealers and Agencies: Modeling Who Lists, Photos Sell: Media Pipelines for Listings, Attributes Without Chaos: Schemas, Dictionaries and Validation). They pass over two of its calls: the feed also carries the paid products `AddPremiumListing` and `AddFeaturedProperty`.

My reading: **when upsells travel through the same feed as listings, an agent's software can buy visibility without anyone opening the portal**, and the entitlement must be applied by the ingestion path, not by a checkout (Packages, Bumps and Quotas: Products, Pricing and Entitlements).

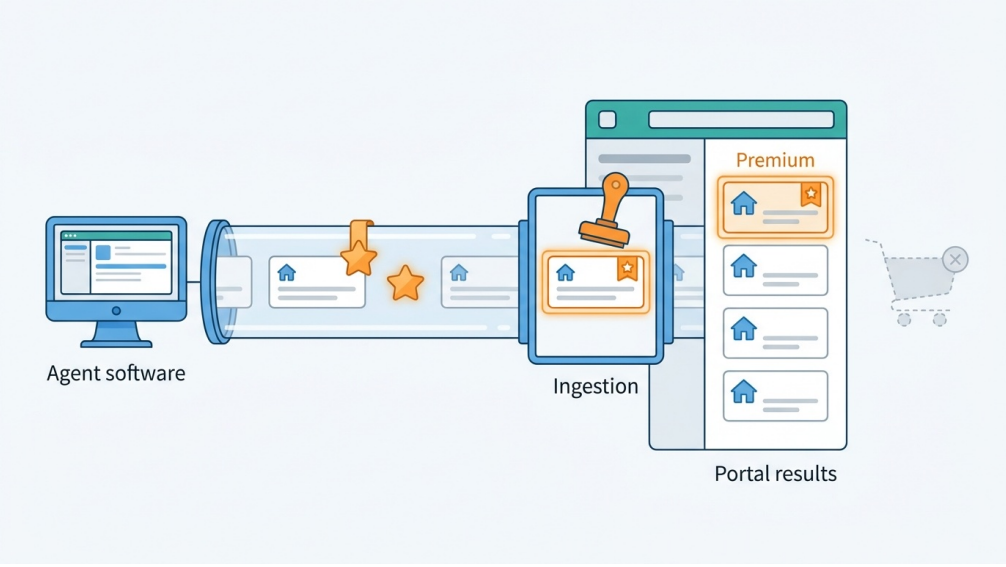


Figure 6: *When upsells travel in the same feed as listings, the agent's software buys visibility without anyone opening the portal, so the ingestion path, not a checkout, must apply the entitlement.*

Search, alerts and geography

A 2025 Elastic customer story says Rightmove moved to Elasticsearch in 2014 with three priorities, among them "rich, map-centric results" and "instant property alerts, made possible by Elasticsearch's percolators". It also mentions vector search for suggestions and plans to derive attributes from photos (Elastic, 2025). It is vendor marketing, so treat its performance claims accordingly. Percolator-based alerts are the documented building block that Tell Me When It Appears: Saved Searches and Alerts generalizes.

A TravelTime case study, also undated vendor material, says Rightmove's "My Places" travel-time feature runs on TravelTime's API. Rightmove chose it partly for a fixed annual license with unlimited requests, which avoids pay-per-call cost growth (TravelTime). **The routing vendor's pricing model became a design constraint** (Where Is It? Location, Geocoding and Privacy).

A five-year cloud migration

In December 2021 Google Cloud announced that Rightmove would move "a number of its systems" from a multi-data-center setup to Google Cloud, starting with containerized applications on GKE. The announcement cited more than 180 million visits a month and a busiest minute at "8:48 pm on Wednesday, during the final ad break on prime-time TV" (Google Cloud, 2021). In February 2026 Rightmove described "4 petabytes (PB) of historic and live data" on a unified data platform, "c.200 proprietary models", more than 6,000 technology releases in 2025, and a multi-year collaboration with Google Cloud. By 31 July 2026: "100% of applications are in the cloud as of 31 July, with >5 petabytes of historical and live data now on Rightmove's unified data platform while maintaining 99.99% uptime in the period". Headcount rose by 37 to 935, with over 80% of new roles in technology, and a nearshore engineering partnership raised contractor costs (Rightmove H1 2026 results).

My reading: the migration of a 70%-margin portal took about five years from announcement to "100% of applications", and it finished with the data platform, not with compute. The public sources do not say how the migration was sequenced, whether data centers were closed, or what it cost beyond the "additional cloud hosting and migration costs" noted in 2026. Changing the Engine in Flight: Modernizing a Live Marketplace uses the case only for its duration.

The AI turn

Rightmove's 2026 disclosures put AI at the center of its product story. "Ask Rightmove" is a conversational experience across search and listings. Early data from its AI Search Assistant (to 30 June 2026) "suggest a c.40% uplift in average time on site, and roughly double the propensity to send leads". Online Agent Valuation is AI-enabled, 46 strategic

AI initiatives are in flight (31 in December 2025), and the company frames its "cloud-based tech and data platform" as "fully AI-enabled" (Rightmove H1 2026 results). The February 2026 results also mention a planned app in ChatGPT. The same results guided the 2026 margin down to 67% "reflecting this period of investment". Engagement figures from self-selected early users are not causal estimates. Testing a Two-Sided Market: Experimentation Platforms explains why, and AI Across the Listing Lifecycle covers external assistant channels.

The collective claim

An application for a Collective Proceedings Order was filed with the Competition Appeal Tribunal in April 2026. The proposed claim "seeks damages of up to £1.56 billion"; certification is scheduled for November 2026, and Rightmove considers the claim "without merit" (Rightmove H1 2026 results). The fetched text does not identify the proposed class or its theory, and this chapter does not guess.

REA Group: Selling Depth per Listing

REA Group operates realestate.com.au, realcommercial.com.au and property.com.au. It owns PropTrack (property data), Mortgage Choice (broking), Campaign Agent (vendor-paid advertising finance) and Realtair (transaction software), and holds 20% of Move, Inc., operator of realtor.com in the US. Its fiscal 2026 release, published 6 August 2026, is the main source for this section (REA FY26 results, 2026):

- Revenue from core operations A$1,793 million (+7%; Australia +11%); operating EBITDA margin 66%.
- Residential revenue A$1,290 million (+12%), with Buy yield +13% on flat national listings. The fiscal 2027 outlook expects "low double-digit" Buy yield growth "driven by an 8% Premiere+ price rise and growth in add-ons".
- Audience: 12.7 million average monthly visitors, 6.3 million of them using realestate.com.au exclusively; 146.4 million monthly visits, "104.5 million more monthly visits than the nearest competitor".
- Engagement: 5.2 million "unique properties tracked by their owner" (+15%); 2.5 million monthly buyer enquiries (+8%); seller leads +22%.
- Product: a consumer AI Assistant available to all members; Campaign Assist for customers on REA's Ignite agent platform; the acquisition of Planitar (iGUIDE, 3D tours and floor plans); PropTrack revenue "driven by customer data contracts".

My reading of REA is that it is the cleanest documented case of a per-listing depth model. Revenue is roughly listings times yield, and yield is a mix of tiers, prices and add-ons. That is why REA could grow residential revenue 12% with flat listings, and

why its architecture must treat the depth tier as a per-listing order that can be upgraded mid-campaign, financed by the vendor and attributed to an agent. A subscription portal like Rightmove does not need any of that. The 5.2 million owner-tracked properties are the same property-first asset as Zillow's living database: owners who track a home are tomorrow's sellers.

The engineering evidence is thin, and I will not pad it. REA's technology blog now sits on rea-group.com, which refused automated fetches for this book. An undated Elastic event page (its URL places it at the 2015 Elastic{ON} Tour in Sydney) says realestate.com.au "has been slowly migrating its search function to Elasticsearch", with millions of listings and sales histories, thousands of agencies, "thousands of simultaneous updates" and more than 100 queries per second (Elastic). A 2022 Elastic community event listing describes percolation matching new listings against millions of saved searches at REA, but the page redirected and could not be read (Elastic Community, 2022). Both are more than three years old and describe a system whose current state is not public.

Idealista: What Little Is Public

Idealista is the leading property portal in Spain and Portugal and a challenger in Italy, according to trade press. On 28 June 2024 Cinven agreed to buy a 70% stake in a deal reported at €2.9 billion; reports differ on whether that figure is the price of the stake or the value of the whole business (Online Marketplaces reports a valuation of a little over €3 billion, Bloomberg a valuation of €2.9 billion). Either way it was more than double the €1.3 billion valuation at which EQT took control in September 2020. EQT kept 18%, and founder Jesús Encinar continued to lead the company (Online Marketplaces, 2024). The article gives no revenue, margin or audience figures, and I found no primary financial disclosure.

Idealista's one useful public product fact is about duplicates. In April 2021 it launched a duplicate-control feature in its agent tools that groups ads for the same property, "listed by multiple agencies or individuals on idealista and other portals". It notifies agents of price changes, removals and new duplicates, is "updated daily", and shows agents duplicates only of their own published properties (Online Marketplaces, 2021). **The design treats legitimate multi-agency duplicates as a cluster to present, not as abuse to remove**, the distinction One Property, Three Listings: Duplicates and Entity Resolution builds on. Idealista also offers a Search API to approved partners on request (idealista developers, n.d.).

That is where the evidence ends. I found no engineering blog, conference talk or filing that describes Idealista's search, ingestion, data platform or ranking. A web search for its engineering material returned nothing relevant. Any statement about how Idealista builds its systems would be invention, so this case makes none.

CoStar: A Curated-Data Counterpoint

CoStar Group is useful as a contrast because it reached property marketplaces from the other direction: from commercial real-estate data. Its 2025 annual report describes researchers who gather data by phone, e-mail, field inspection, public records and news monitoring. Field teams photograph and measure buildings and capture Matterport tours, and direct feeds come from apartment sites, owners, brokers and builders. Homes.com connects owners and agents with buyers on a "your-listing your lead basis" and is powered by "Homes AI". CoStar completed the Matterport acquisition on 28 February 2025 and the Domain acquisition in Australia on 27 August 2025 (CoStar 10-K, 2026). Announcing Domain, CoStar's chief executive Andy Florance said: "We are the agent's ally, and we will never operate at their expense" (CoStar release, 2025).

My reading is that **CoStar is a curated data business with a marketplace on top, and Homes.com competes on whom the lead belongs to**. Zillow sells buyer leads to buyer agents; Homes.com gives the lead to the listing agent. The difference is commercial, but it shows up in the architecture as a lead-routing rule, and it is aimed at the agents who control supply.

A Reference Architecture for a Property Portal

None of the companies above has published a whole-system view, so the synthesis below connects the documented pieces and is labeled as inference.

> **INFERENCE, NOT DOCUMENTED:** A plausible reference architecture for a national property portal — a composite derived from the public record, not a description of Zillow, Rightmove, REA, Idealista or CoStar — has five load-bearing parts. (1) A **licensed-supply ingestion tier**: adapters per source (MLS replication, agent-software feeds), each record carrying license, embargo and attribution metadata, with snapshot reconciliation. (2) A **property registry**: a durable property entity resolved from address and parcel identifiers, with listings, sales, rentals and valuations as events on it. (3) **Valuation and market models** whose outputs are registered by decision tier, so that the same AVM can feed an estimate page, a seller-lead product or a lender only through separately validated versions. (4) A **monetization core** shaped by the unit of charge: contracts and quotas, per-listing orders, or lead allocation with attribution. (5) **Agent workflow surfaces** (CRM, tour scheduling, valuation tools) that both serve the supply side and feed outcome data back.

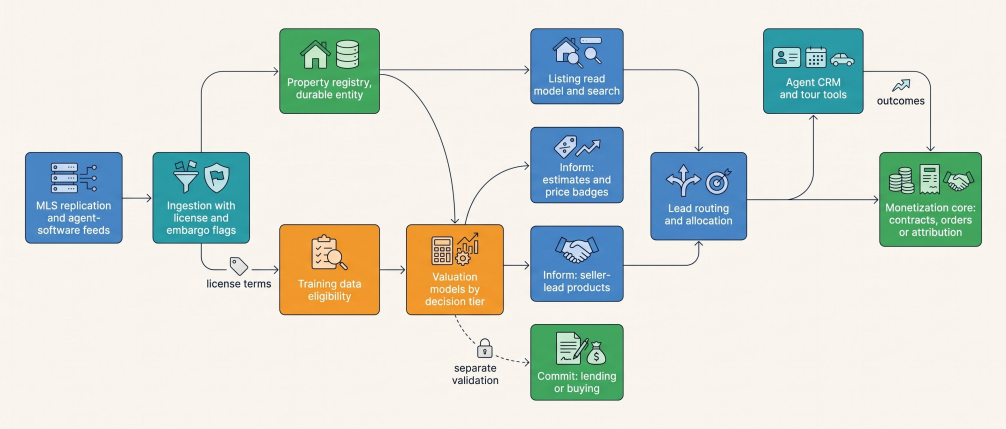


Figure 7: *A reference architecture inferred from the public record. The property registry, not the listing, is the durable entity. License terms flow from ingestion to training, and any commit-tier use of valuation sits behind a separate validation path.*

Trade-offs the Cases Illustrate

Choice	Option A	Option B	What the record shows	When A is right
Unit of charge	Per advertiser subscription (Rightmove)	Per listing depth (REA) or per lead (Zillow)	Rightmove 70% margin; REA yield +13% on flat listings; Zillow revenue per visit $0.178	When agents value steady presence and branch count is stable; B when vendors fund marketing per sale (REA) or when the portal can measure lead outcomes (Zillow)

Choice	Option A	Option B	What the record shows	When A is right
Use of valuation	Inform only (estimates, seller leads)	Commit (iBuying, lending)	Rightmove and REA grow seller leads; Zillow Offers wound down in 2021	Almost always A, unless the business accepts balance-sheet risk with a credit-style validation regime
Supply posture	Neutral aggregator of all supply	Access rules that favor MLS or portal-first supply	Zillow's 2025 standards, the Compass suit and the March 2026 change	A when agents can withhold supply cheaply; B only with legal review and a measured view of supply share
Lead ownership	Listing agent receives the lead (Homes.com)	Leads sold to buyer agents (Zillow Premier Agent)	CoStar's "agent's ally" positioning; Zillow's share-of-voice and pay-at-close models	A when courting supply is the priority; B when the audience is dominant and buyer-agent spend is the revenue base

Rightmove and REA, with almost opposite pricing units, report the two highest margins in the case, so the rows are not a scorecard. The architecture follows the unit of charge; the profit follows the audience position.

Vertical Differences in the Record

Concern	Real estate (this case)	Automotive	General classifieds
System of record	Outside the portal: MLS or agency CRM	Dealer management system feeds	The portal itself
Durable entity	Property, with parcel or address identifiers	Vehicle, with a VIN	Usually none; the ad is the item
Valuation's role	Supply acquisition (seller leads) and a temptation to commit capital	Price ratings on listings; trade-in and instant-offer products	Price hints for private sellers
License constraints	MLS rules, embargoes, attribution, model-training limits	Dealer contracts and data-provider licenses	Platform terms
Commit-tier failure	iBuying (Zillow Offers)	Instant buying and retail of cars	Rare

The commit-tier row is the one other verticals should study. Instant-offer and car-retail businesses in automotive carry the same structural risk as Zillow Offers whenever a marketplace prices an asset it will hold and resell; mobile.de, Auto Trader and AutoScout24: Lessons from Automotive Marketplaces follows the dealer-centric version of the vertical.

Failure Modes the Record Exposes

1. **Promoting an inform-tier model by announcement**. In February 2021 the Zestimate became an opening offer in 23 markets. The accuracy claim that accompanied it was measured on declined offers. A tier change that arrives as a product launch, not as a model-governance decision, skips the validation the new tier needs.
2. **Mistaking present-value accuracy for forecast accuracy**. The wind-down letter is about "future home prices". Any marketplace product that holds inventory,

guarantees a price or underwrites a loan is a forecasting product.

3. **Ignoring operational coupling**. "Capacity and demand planning challenges" turned model error into physical work queues. Monitoring must include the operations the model drives.
4. **Treating supply as owned**. License changes can remove the right to train on listings. Access rules can provoke litigation. Agents can redirect stock. A portal that cannot measure supply share by source and agent will learn about a supply problem from its revenue.
5. **Reporting engagement as impact**. Early AI-assistant figures such as "roughly double the propensity to send leads" come from users who chose to engage. They are a reason to run an experiment, not its result.

Where the Public Evidence Stops

Question	What is public	What is not
How did Zillow Offers price homes and set exposure limits?	The February 2021 Zestimate-offer launch; the November 2021 reasons, write-downs and inventory counts	The pricing model, adjustment rules, capital limits, back-tests and governance
How does the neural Zestimate run today?	Error figures for 2021 and 2025; 173 million homes	Architecture, retraining cadence, the reason for the change in off-market error
How does Zillow attribute pay-at-close fees?	Variable-consideration accounting; "generally within two years"	Matching method, data sources, true-up process
How did Rightmove sequence its cloud migration?	2021 start; July 2026 completion; data volumes; uptime	Sequence, data-center closure, cost, architecture
What runs Rightmove search today?	Elasticsearch since 2014, percolators, vector search for suggestions (vendor, 2025)	Index topology, ranking models, how paid products affect order
What does REA's platform look like?	Financial and product disclosures to June 2026; a 2015-era Elasticsearch migration	Anything current about search, ingestion, data or ML architecture

Question	What is public	What is not
How does Idealista build its systems?	Ownership deal (2024); duplicate control (2021); a partner Search API	Everything technical, plus revenue and margin

Also open as of October 2026: the class and theory of the Rightmove collective claim (certification hearing in November 2026), the remaining government complaints named in Zillow's 2025 annual report, and CoStar's plans for Domain's platform.

The Chief Architect's View

These cases leave me with five decision rules.

Register models by the decision they drive, and make tier changes a governance event. The Zestimate could inform; the Zestimate as an offer committed. The tier, not the algorithm, should decide the validation, the owner and the capital limit. A product launch that changes a model's tier should not be able to ship without a decision record signed by someone outside the model team.

Treat supply terms as architecture. Every listing should carry its license, embargo, attribution and training-eligibility terms from ingestion to every consumer: search, syndication, feeds to AI assistants, and training pipelines. Zillow's risk factor on "the training of models" is the clearest public warning that these terms change, and that a data platform unable to filter by them will have to be rebuilt.

Design the monetization core for the unit of charge you have, and keep the next one possible. Rightmove, REA and Zillow show that subscription, depth and lead models each need a different core. A group with portals in several countries will run more than one. A shared catalog and entitlement model can carry all three, but only if listing-level orders and lead-level attribution are designed in from the start (Packages, Bumps and Quotas: Products, Pricing and Entitlements).

Make the property, not the listing, the durable entity. Zillow's living database, REA's 5.2 million owner-tracked properties and Idealista's duplicate clusters are all property-first assets. Valuation, seller leads, price history and duplicate presentation hang off that entity.

Measure AI assistants as a lead channel. Rightmove and Zillow both now treat conversational search as a route to leads. It needs the same attribution, experiment discipline and ranking rules as any other channel (AI Across the Listing Lifecycle).

What to watch at group level: models whose tier has changed in the past year; the share of listings carrying complete license metadata; supply share by source and by top agencies; the ratio of on-market to off-market valuation error; and, for any commit-tier use, realized loss against its limit.

Comparing the Cases to the Book's Patterns and to Veldmark

The book's patterns, as the property-portal record tests them:

Book pattern or claim	The record	Verdict
Govern models by decision tier (Models in Production: ML Platforms and Use Cases)	Zestimate as information product versus Zestimate as offer	Strongly supported, by failure
Model property and listing separately (One Property, Three Listings: Duplicates and Entity Resolution)	Zillow's living database; REA's owner-tracked properties; Idealista's duplicate clusters	Supported
Ingest professional supply through versioned, reconciled contracts (Many Doors In: Forms, APIs and Bulk Feeds)	Rightmove RTDF	Supported
Carry license and visibility rules as data (Rules of the Marketplace: DSA, DAC7 and Vertical Regulation)	MLS policies, Zillow's training-use risk factor, the listing-access dispute	Supported; the training clause is the newest pressure
The unit of charge shapes monetization architecture (How Listing Marketplaces Make Money)	Rightmove, REA and Zillow differ in denominator and in core systems	Supported

Hearthly, Veldmark's fictional real-estate portal in France and Spain, took several 2027 decisions that this case tests directly. Every figure below comes from the scenario ledger.

Veldmark choice (2027)	Parallel in the record	What the case suggests
"Hearthly Direct" declined — a partner-funded guaranteed offer at 94% of the Valuo estimate, rejected as a commit-tier use without validation	Zillow Offers, from Zestimate offer (February 2021) to wind-down (November 2021)	The right call. If the idea returns, it needs the five-part validation regime above, not a better model
Valuo reported as 7.4% off-market and 2.3% on-market median error (October 2027 back-test), trained on DVF transaction prices, never on asking prices	Zillow's 7.2% and 1.8% (2025)	Reporting both figures, and refusing asking prices as labels, avoids the circularity the on-market number hides
Valuo commission-share pilot on seller leads in the Lyon metropolitan area in 2028, with homeowner consent to routing and outcome tracking	Zillow Preferred's pay-at-close; Rightmove's and REA's seller-lead growth	A commission share is a pay-at-close model, so it needs a lead-to-outcome attribution pipeline and revenue true-ups before the pilot bills anyone
Hearthly agency subscriptions (€270 million in 2026; derived ARPA about €616 per agency per month)	Rightmove ARPA £1,621	The same subscription core, but the denominators differ (Hearthly counts agencies, Rightmove branches and developments). Do not benchmark one against the other directly
Keybridge stays neutral: its agency data is never used by Hearthly for ranking or sales targeting	Zillow's listing-access dispute; CoStar's "agent's ally" positioning	Agents control supply, so neutrality controls are supply protection as well as compliance

Veldmark choice (2027)	Parallel in the record	What the case suggests
Supply Contract v1 (September 2027) with snapshot reconciliation for Hearthly's feed-sourced supply, and **canonical property clustering** of the Lyon apartment's three agency listings	Rightmove's `GetBranchPropertyList`; Idealista's duplicate control	Consistent; never let sold status choose a cluster's representative

The Lyon apartment shows the difference between the tiers in one listing. The 68 m² T3 in Lyon 7e is listed at €389,000, and Valuo's 80% interval is €372,000–€401,000. As an inform-tier output, that interval supports an "in line" badge, and Hearthly shows it. As a commit-tier output it would have to answer different questions. What will the flat sell for after three to six months of holding? How wrong is Valuo on the owners who would accept 94% of its midpoint, about €363,000? How many such flats can Hearthly carry before one bad quarter wipes out a year of Valuo's €40 million in data and lead revenue? A present-value back-test answers none of them, which is the whole of Zillow's lesson. Ingrid Halvorsen, Veldmark's fictional Group Chief Architect, recorded the same reasoning when she declined Hearthly Direct (see Models in Production: ML Platforms and Use Cases).

Questions to Ask

1. Which of our models drive a commit-tier decision (buying, guaranteeing, lending, pricing contracts), and which validation, owner and capital limit applies to each?
2. Has any model changed tier in the past year through a product launch rather than a governance decision?
3. Do we report valuation error separately for listed and unlisted properties, and on the population that acts on the estimate?
4. Does every listing carry its license, embargo, attribution and training-eligibility terms, and do our training pipelines enforce them?
5. What share of our supply comes from each source and from our ten largest agencies, and how quickly would we notice it moving?
6. What is our unit of charge, and does our monetization core support per-listing orders and lead attribution if we need them?
7. Do our feed contracts reconcile deletes from external systems of record, and how often?

8. How are we measuring AI assistant channels: by engagement, or by experiment on leads and outcomes?

Property portals show what happens when a listing business sits on valuable data and a professional supply base it does not control. The next case, mobile.de, Auto Trader and AutoScout24: Lessons from Automotive Marketplaces, follows mobile.de, Auto Trader and AutoScout24 through the same questions in a vertical where the durable entity is the vehicle and the professional is a dealer.

Sources and Evidence

Company engineering and investor sources

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- Form 10-K for fiscal 2025 — Zillow Group, 2026. Zestimate errors, living database, listing-data sources and MLS dependence, training-use risk, Premier Agent and Zillow Preferred, Enhanced Markets, agent tools, legal proceedings.
- Zillow Tech Hub (Medium RSS feed) — Zillow, posts 2020–2021. Guided Search, Kafka, floor-plan ML serving, listing embeddings.
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- Real Time Datafeed specification v1.4.1 — Rightmove, 2023 (last updated 30 November 2023). Tenancy, authentication, idempotency, reconciliation, media, paid products, schema history.
- Rightmove sets up home with Google Cloud — Google Cloud Press Corner, 2021. Migration start, visits, busiest minute.
- REA Group delivers excellent FY26 results — REA Group (ASX announcement), 2026. Revenue, margin, Buy yield drivers, audience, owner-tracked properties, seller leads, AI products, portfolio.
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- idealista launches new tool to help organize duplicate listings — Online Marketplaces, 2021. Duplicate-control feature.
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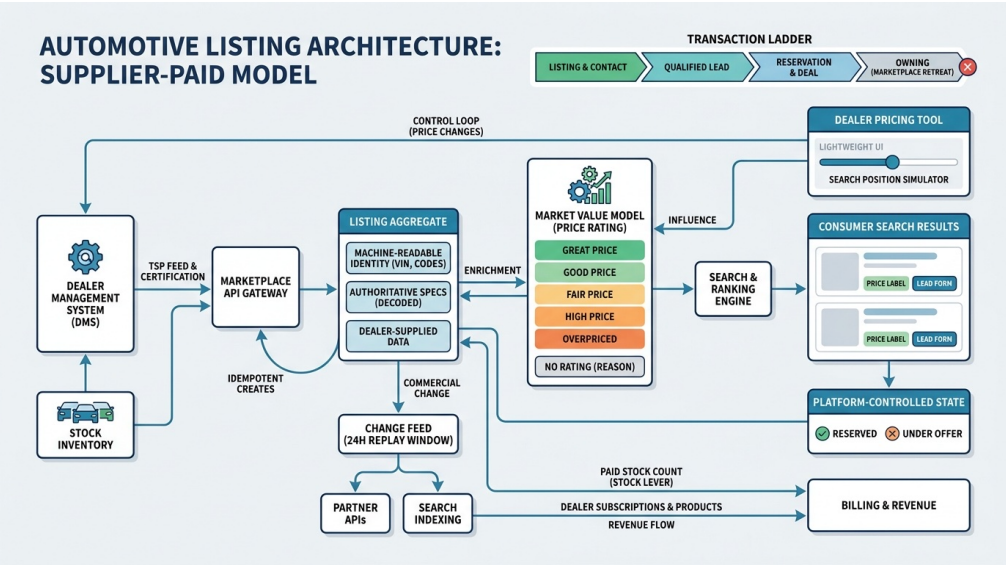
Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

51. mobile.de, Auto Trader and AutoScout24: Lessons from Automotive Marketplaces



IN THIS CHAPTER, YOU WILL: Read the public record of mobile.de, Autotrader UK, AutoScout24 and CarGurus, with OLX Autos as a contrast, through their integration APIs, investor reports and engineering posts. You will separate what each company documented from what the author infers, and test the case against the book's patterns and Veldmark's Motorvane.

WHY THE BUSINESS CARES: Automotive classifieds are among the most profitable listing businesses there are. Autotrader reported a 70% operating margin for its year to March 2026, and its revenue moves with dealer prices, dealer packages and the number of paid cars on the site. Each of those levers is set by a system.

WHY ARCHITECTS SHOULD CARE: The automotive leaders publish their integration contracts in unusual detail, and those contracts expose

design decisions that engineering blogs rarely state: idempotent dealer writes, change feeds with replay windows, price ratings with explicit band boundaries, and a breaking-change policy for a data business. These are some of the best public evidence on listing architecture outside real estate.

KEY POINTS:

- **In automotive, the public API is the architecture document.** mobile.de's Seller API and Ad Stream and Autotrader's developer platform show the shape of a dealer-fed listing system: transfer service providers as integration principals, idempotent creates, one write at a time per ad, taxonomy keyed to a vehicle derivative, and a change feed with a 24-hour catch-up window. A vehicle listing is mostly enriched from authoritative data rather than written by the seller.
- **A price rating is a control loop, not a label.** mobile.de, CarGurus and Autotrader all publish five-band price ratings or price flags. Dealers are shown how a price change moves the band, and at mobile.de how it moves search position. Once a rating feeds ranking and dealer tools, the model shapes the prices it learns from, and it needs the governance of a ranking model.
- **When the supplier is the paying customer, platform-controlled state on its stock is a commercial-model change.** Autotrader's Deal Builder reached 137,000 full reservations in a year while forecourts fell in the second half and dealers cut paid stock. One step further, holding inventory, is where CarGurus' CarOffer and OLX Autos retreated. Transactional states should be opt-in, configurable per dealer and reversible.

FAILURE PATTERNS THIS CHAPTER ADDRESSES:

- **Feed Roulette** — Nine-tenths of automotive supply arrives through dealer feeds, and the published contracts (TSP certification, idempotency keys, serialized writes, stable error keys) are the documented antidote.
- **Set-and-Forget AI** — Price ratings decide what a buyer sees as a good deal and, where they feed ranking, which car a buyer sees first, so they make consequential decisions about exposure.
- **Billing by Guesswork** — Autotrader's "stock lever" shows paid stock counts flowing straight into revenue, so a disagreement between the feed importer and billing becomes a dealer dispute.

This last case of Part IX tests the book's patterns in the vertical whose marketplaces are the clearest about their architecture, because they have to be. **A car carries a machine-readable identity, most cars are listed by professional dealers through software,**

and the dealer who supplies the stock is also the customer who pays. So the leaders publish integration contracts in detail for the vendors who upload on dealers' behalf, and the listed ones report the levers of their revenue to investors. Read together, those documents describe a listing system more precisely than most conference talks. Where Zillow, Rightmove, Idealista and REA: What Property Portals Teach showed supply licensed from agents, this case shows supply written by dealer software, and it tests the feed contracts of Many Doors In: Forms, APIs and Bulk Feeds, the change feeds of Every Change Is an Event: Event-Driven Listing Architecture and the model governance of Models in Production: ML Platforms and Use Cases against public documents.

This chapter reads that record for four companies. **mobile.de** (Germany) documents its dealer write path, its price rating and its change feed in public API manuals. **Autotrader UK** discloses its unit economics, its data platform and the friction caused by its move into online reservations. **AutoScout24** (pan-European) documents its engineering platform. **CarGurus** (United States and international) builds its proposition around price transparency, and its 2025 annual report records the wind-down of a wholesale business that held price risk. **OLX Autos** appears briefly, as the group that went furthest toward owning cars and then exited.

The chapter asks three questions of each: *what does the public contract reveal, what do the reported numbers say about the systems behind them, and where does the marketplace's responsibility end?*

How to Read This Case

Part IX cases follow stricter rules than the chapters in Parts I–VIII (see Part IX — PRACTICE: How Others Built It). A **documented fact** carries its source and publication year in the sentence. **The author's interpretation** is marked "my reading" or sits in the Chief Architect's view. A **reference-architecture inference** appears in a callout labeled `INFERENCE, NOT DOCUMENTED`.

The evidence is uneven: each company documents a different layer.

Company	Strongest public evidence	Weakest
mobile.de	Seller API, Search API and Ad Stream manuals (undated, read October 2026); newsroom releases (2026); a 2024 conference abstract on ranking	No engineering blog found; no description of internal services, search engine or cloud

Company	Strongest public evidence	Weakest
Autotrader UK	Full-year results to March 2026 (published May 2026); developer portal; engineering blog (2026 posts)	Platform topology after about 2019; valuation methodology
AutoScout24	Engineering blog on platform and data (2015–2026); careers page with platform figures	Dealer-facing APIs, pricing model, product architecture
CarGurus	Form 10-K for 2025 (filed February 2026); help center	Engineering detail; model thresholds and error bands
OLX Autos	Trade press and Prosus results (2021–2025)	Systems; the business is closed

The API manuals carry no publication dates. They describe the contract as of the day they were read (19 October 2026), not when it was designed. Ownership facts are volatile and stated "as of" a date.

Veldmark, the fictional marketplace group this book follows (see Meet Veldmark: Four Brands, One Platform to Build), appears only in the final comparison. Its automotive brand, Motorvane, is a composite of industry patterns and is not a disguised description of any company in this chapter.

Timeline: Automotive Marketplaces in Dated Events

The chronology below draws on the cited releases, filings and reports. Ownership is as of October 2026 and may change.

Date	Event	Source type
1996	mobile.de founded; by 1998, 50,000 listings from 1,200 dealers	mobile.de newsroom history, n.d.
April 2004	eBay acquires mobile.de	eBay Germany, n.d.
28 June 2013	Auto Trader's last print edition in the UK; print circulation had fallen from 368,000 (January 2000) to 27,000 (March 2013)	Press Gazette, 2013

Date	Event	Source type
November 2014	AutoScout24 begins "Project Tatsu", moving from a .NET monolith in its own data center to JVM microservices on AWS	InfoQ, 2016
March 2015	Auto Trader Group lists on the London Stock Exchange	Autotrader FY26 results, 2026
c. 2019	Auto Trader runs on Google Kubernetes Engine with the Istio service mesh, about 400 microservices	Istio and Google Cloud case studies, undated
2019	OLX takes control of Frontier Car Group, which becomes OLX Autos	TechCrunch, 2021
April 2020	Hellman & Friedman acquires AutoScout24 from Scout24 (reported at €2.84 billion)	Wikipedia (German) citing Handelsblatt
January 2021	CarGurus acquires 51% of CarOffer, a digital wholesale business	CarGurus 10-K, 2026
25 June 2021	Adevinta completes the acquisition of eBay Classifieds Group, including mobile.de	Adevinta release, 2021
March 2022	mobile.de acquires Null-Leasing.com	Adevinta release, 2022
2 March 2023	Prosus announces it will exit OLX Autos	Fin24, 2023
May 2024	Adevinta taken private by a Permira- and Blackstone-led consortium	Adevinta release, 2024
1 April 2025	Autotrader bundles its generative-AI tool Co-Driver into retailer packages	Autotrader FY26 results, 2026
6 August 2025	CarGurus' board decides to wind down CarOffer	CarGurus 10-K, 2026

Date	Event	Source type
21 August 2025	AutoScout24 describes migrating 110 AWS accounts from Trader (Canada) into its AWS organization	AutoScout24 Tech Blog, 2025
October 2025	Financial press reports an IPO-or-sale "dual track" for mobile.de	PYMNTS relaying the Financial Times, 2025
November 2025	Autotrader retailers start cutting advertised stock, citing in part the Deal Builder rollout	Autotrader FY26 results, 2026
15 July 2026	mobile.de "Plattform-Update Sommer 2026": AI lead tools, call assistant, Lead API, price planner	mobile.de newsroom, 2026
8 October 2026	The Bundeskartellamt clears the acquisition of Carwow's German business by Oak Germany Buyer GmbH, which it names as mobile.de's parent	Bundeskartellamt, 2026

The four marketplaces and the contrast case sit at different points on a single line: how far the marketplace reaches into the transaction.

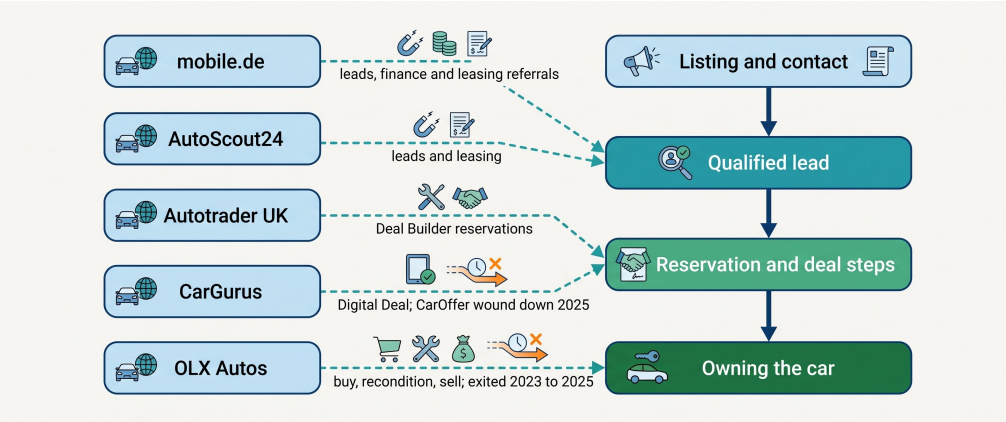


Figure 1: *Where each case sits on the transaction ladder, as of 2026. The documented*

retreats are all from the last step, where the marketplace takes price risk on the car itself.

Why Automotive Behaves Differently

Four properties of the automotive profile, which Same Platform, Different Physics: How Listing Verticals Differ develops in full, matter for reading this case.

Property	What it means for the architecture	Where the case shows it
Strong item identity	A vehicle identification number (VIN), a UK registration plate, or Germany's type-approval codes (HSN/TSN) resolve to a manufacturer specification, so most attributes are *decoded*, not typed	mobile.de's `hsn`, `tsn` and Schwacke codes; Autotrader's derivative-level taxonomy
Professional, software-mediated supply	Dealers list through their dealer management system (DMS) or an inventory tool, so the write path is an API for vendors	mobile.de's transfer service providers; Autotrader's Stock API
The supplier pays	Revenue is a dealer subscription, often scaled by stock, plus prominence products; inventory state is billing state	Autotrader's ARPR levers; CarGurus' tiers by inventory size
High ticket, uncertain value	Buyers need price guidance, history and finance; each is a data product	Price ratings at all four; CARFAX at mobile.de; finance and leasing everywhere

Price evaluation — rating an asking price against a model of market value — is the vertical's signature data product, and the cross-case section below treats it separately.

mobile.de: An Architecture Written in Its APIs

The business, as far as it is public

mobile.de was founded in 1996, bought by eBay in 2004 and became part of Adevinta when Adevinta completed its purchase of eBay Classifieds Group on 25 June 2021. In Adevinta's last public annual report, for 2023, the mobile.de segment earned €432 million, of which €408 million was classifieds revenue, €24 million advertising and essentially no transactional revenue. Its newsroom reported more than 1.6 million listed vehicles and more than 140 million visits a month as 2025 averages (release of 25 August 2026). **Ownership as of October 2026 is not fully clear from public sources**. In October 2025 the financial press reported an IPO-or-sale process. In August 2026 mobile.de still called itself an Adevinta subsidiary. On 8 October 2026 the Bundeskartellamt named "Oak Germany Buyer GmbH" as the parent of mobile.de GmbH when it cleared the Carwow acquisition. No completed sale or listing was found.

In a dealer-fed vertical, inventory is not exclusive. The same clearance decision records this in the regulator's words: mobile.de has a "starke Marktposition" in used-car brokerage, AutoScout24 is a "starker Wettbewerber", consumers and dealers "häufig mehrere Plattformen parallel" use several platforms at once, and many platforms offer dealers "Datenexport zu anderen Plattformen". The same car is on several marketplaces, so differentiation has to come from audience, data products and workflow, not from supply lock-in.

No mobile.de engineering blog was found, so the architectural evidence is the integration documentation, which is unusually complete.

The write path: the Seller API

The Seller API manual (undated, read October 2026) defines who may write and how.

- **Integrators are principals**. "Seller API Access can be granted to Transfer Service Providers and Self-Uploading Dealers registered with mobile.de." A transfer service provider (TSP) is a DMS or inventory-tool vendor uploading on dealers' behalf. TSPs that want to serve more than ten customers must supply contractual and technical contacts, software details, a VAT ID and customer numbers before access is granted. An API user sees only the sellers linked to it.
- **Creates are idempotent; writes to one ad are serialized**. A create call can carry an `X-Mobile-Insertion-Request-Id` header; a retry with the same value returns HTTP 303 pointing to the ad already created. GET, PUT and DELETE are idempotent and POST is not. The manual is blunt: "Do not send multiple requests for the same ad at the same time." An update is a full-document PUT.

- **Freshness is a separate field**. An ad carries `creationDate`, `modificationDate` and `renewalDate`, the date the ad "was last marked as new, which makes it appear new in search" (see The Listing Aggregate: Identity, Lifecycle and Invariants).
- **Vehicle identity is carried as codes**. Ads carry the German type-approval codes `hsn` and `tsn` and a `schwackeCode`, a make-and-model classification "given by Eurotax/Schwacke", and enumerated fields (gearboxes, colors, makes, models) come from reference-data endpoints whose valid values depend on the seller's site.
- **Bulk upload exists alongside the REST API**. The XML upload interface identifies each vehicle by the dealer's `seller-inventory-key` and requires TSPs to name the dealer's `customerNumber` on every ad.
- **The resource list is a product map**: price ratings, statistics, dealer ratings, leasing rates, multi-location sellers and battery-health certificates for electric vehicles.

Taken together, mobile.de treats the integration vendor as an accountable party, makes retries safe, and pushes ordering responsibility onto the client in writing. Each of those choices removes a class of Feed Roulette incident. Many Doors In: Forms, APIs and Bulk Feeds and Private Sellers, Dealers and Agencies: Modeling Who Lists use the individual details as evidence.

The price rating as an API resource

The price rating is part of the same contract. The manual lists five labels, `VERY_GOOD_PRICE`, `GOOD_PRICE`, `REASONABLE_PRICE`, `INCREASED_PRICE` and `HIGH_PRICE`, and a preview endpoint that returns the euro range of each band for a vehicle "that has not been created yet". Where no rating is available, "a reason will be provided", for example that the evaluation is still in progress. The rating depends on images: "If images are uploaded later (at any time), the price rating will be recalculated including the image data," and the manual recommends uploading images first.

Three design choices are visible. The rating is a *band with published boundaries*, not a point estimate. There is an explicit *no-rating state with a reason.* And the rating is *recomputed when its inputs change*, including inputs (photos) that a dealer might not think of as price-relevant. From Raw Supply to Usable Listings: Normalization and Enrichment generalizes the last point into enrichers that declare their dependencies.

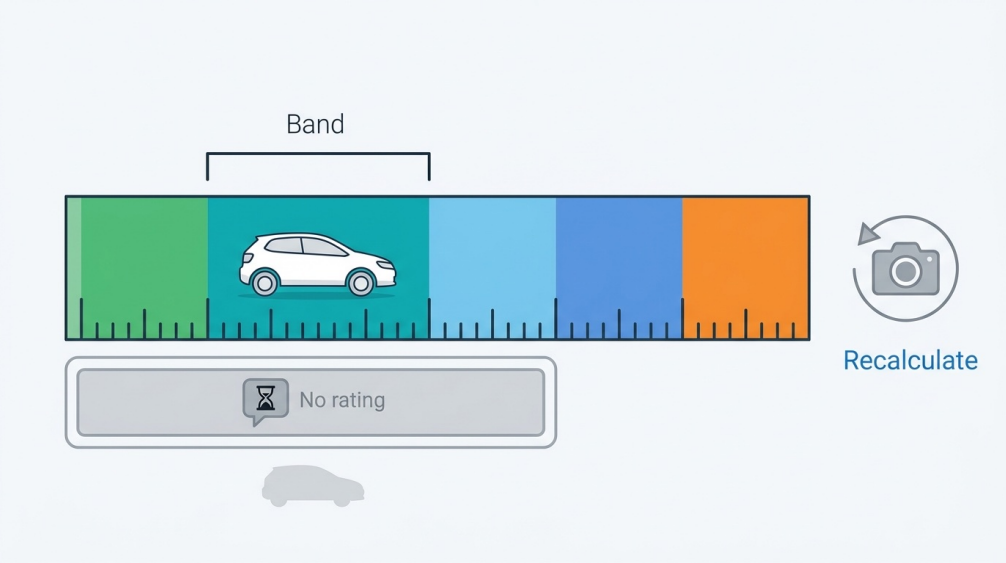


Figure 2: *A price rating designed as a contract: a band with published boundaries rather than a point estimate, an explicit no-rating state with a reason, and a recalculation whenever an input such as the photos changes.*

The read path: Search API and Ad Stream

Two further interfaces let others consume mobile.de's inventory. The Search API lets dealers embed their mobile.de listings in their own websites, with data types that include WLTP (the EU test procedure for light vehicles) consumption values, emissions and CO2 cost models, the information German labeling rules require (see Rules of the Marketplace: DSA, DAC7 and Vertical Regulation).

Ad Stream is a change feed. Its manual (undated, read October 2026) specifies a WebSocket interface with two event types, `AD_CREATE_OR_UPDATE` and `AD_DELETE`, delivered after a delay that depends on the client's contract. It allows "one connection per user"; a new connection closes the old one. With `catchup=true`, a reconnecting client first receives the events it missed, and "the historical events are kept for 24 hours." A test endpoint delivers "only 10% of all events" with 60 seconds of extra delay.

The 24-hour replay window is a public benchmark: it tells consumers how long they may be disconnected before they need a full resynchronization, which is the contract question Every Change Is an Event: Event-Driven Listing Architecture asks of every listing stream.

> **INFERENCE, NOT DOCUMENTED:** The external contracts imply an internal design: a log of listing state changes that feeds Ad Stream (and plausibly search indexing), a price-rating service invoked synchronously for

previews and asynchronously when images or market data change, and per-ad serialization of writes. A plausible reference flow — not a description of mobile.de's systems — is shown in Figure 3. mobile.de has not published how these components are built.

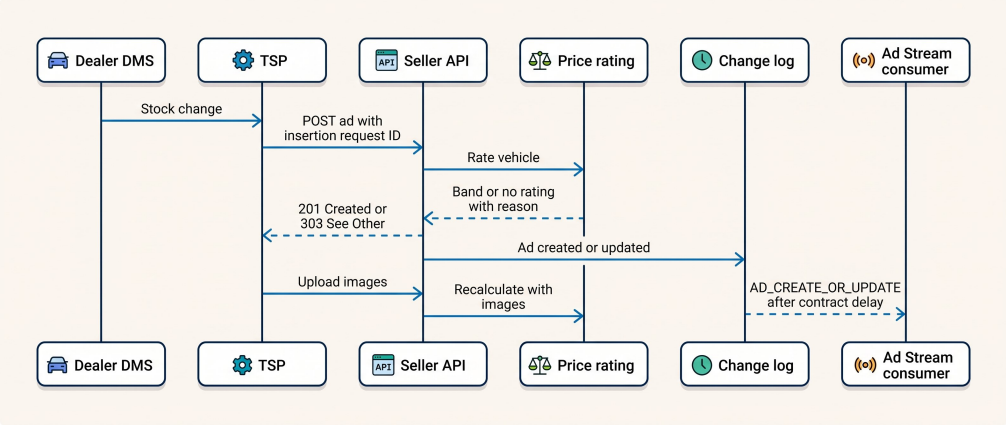


Figure 3: *A plausible reference flow consistent with mobile.de's public API documentation — not a description of its internal systems. Retries are safe, the rating is recomputed when photos arrive, and downstream consumers read a delayed change feed with a 24-hour replay window.*

Ranking and the 2026 product layer

mobile.de's ranking is documented only by a 2024 conference abstract: learning to rank with XGBoost, "millions of searches daily" and a "strict latency SLA <30ms" (Berlin Buzzwords, June 2024). It does not say which features the model uses.

The "Plattform-Update Sommer 2026" (15 July 2026) describes the dealer side: AI reply suggestions produce 45% of written replies in the Lead Manager, a Lead API sends leads "directly and securely into existing CRM systems", and an AI phone assistant's pilot found that 91% of answered calls came outside opening hours. It also announces a "**Preis-Planer**" that simulates how price changes in 2% steps affect days on market, the price rating and *search position*. Other 2026 releases describe Stellantis pushing dealers' leasing offers automatically (June) and VIN-based CARFAX history checks (September).

My reading is that the Preis-Planer is the most revealing sentence in the public record. It confirms that, at mobile.de, price relative to the market model affects where a car ranks, and that the marketplace now shows dealers that relationship as a tool. That makes the price-rating model part of the ranking function, with the consequences discussed in the cross-case section.

Autotrader UK: The Economics Disclosed

A listed company that explains its levers

Autotrader (the group now styles itself "Autotrader Group plc") has been listed in London since March 2015 and published its last print magazine in June 2013. Its full-year results to 31 March 2026, published 21 May 2026, report group revenue of £624.3 million, an Autotrader operating margin of 70%, an average of 13,942 retailer forecourts and over nine million monthly unique visitors. More than 80% of visits arrive directly, 13% from organic search and 4% from paid traffic, and "currently less than 1% of audience comes from generative AI chat assistants."

The results decompose the growth in **average revenue per retailer (ARPR)**, per forecourt per month, into three levers.

ARPR lever	FY2026	FY2025	Autotrader's explanation	What a system has to get right
Price	+£117	+£78	"the annual 1 April 2025 pricing and product event, which combined a like-for-like price increase with additional products"	An effective-dated catalog and price list; contract migration on one date
Stock	−£48	−£22	From November 2025, retailers "reduced the number of vehicles advertised", contributing to lower paid stock	Paid-stock counts derived from the same inventory state that search shows
Product	+£72	+£77	"the inclusion of Co-Driver within retailer packages", more new-car paying retailers, "offset by lower average package penetration"	Entitlements per package tier; features switched on by package

ARPR lever	FY2026	FY2025	Autotrader's explanation	What a system has to get right
ARPR	**£2,995**	**£2,854**	+£141 (+5%)	

Source: Autotrader full-year results to 31 March 2026, published 21 May 2026.

The arithmetic closes: £2,995 × 13,942 forecourts × 12 months ≈ £501 million, the reported retailer revenue of £501.1 million. **The stock lever is the architecturally interesting one**. A dealer can cut spending without churning by advertising fewer cars, and the marketplace's revenue then depends on the accuracy of its count of paid cars. Packages, Bumps and Quotas: Products, Pricing and Entitlements builds its entitlement design on exactly this property.

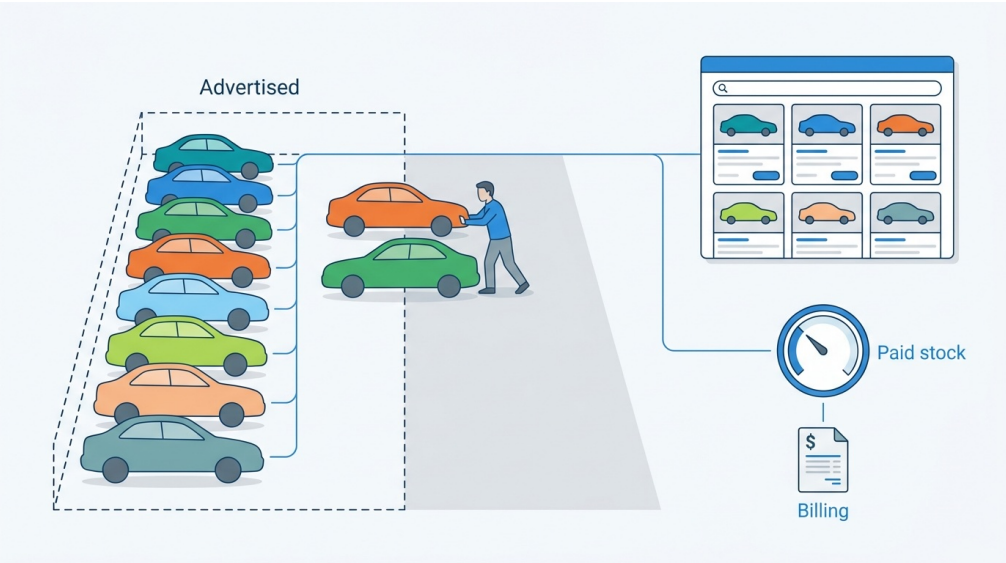


Figure 4: *The stock lever: a dealer can cut spending without leaving by advertising fewer cars, so the marketplace's revenue depends on a paid-stock count taken from the same inventory state that search shows.*

Autotrader Connect: data as a governed product

The results state that Autotrader's data is "integrated with over 220 technology partners" and that calls on its technology and data services averaged **155 million a month** (91 million the year before). The developer portal (undated, release notes into 2026) lists the APIs behind that figure: vehicle lookup and checks, a **taxonomy** that descends from vehicle types through makes, models and generations to derivatives, stock management across advertising channels, images, search for third-party sites, current, future and

historic valuations, vehicle metrics including a "Retail Rating", and the deal, message, part-exchange, delivery and call objects behind its online deal product.

Three details show how the contract is run.

- **Policy is encoded as validation**. Documented error responses include a price rejected for "not meeting minimum price of 75" and "This vehicle has been classified as a category A/B vehicle by the DVLA. These are not permitted for sale on Autotrader."
- **Breaking changes have a notice period**. Breaking changes are announced at least 30 days ahead, and the policy lists what counts as breaking (removing a property, adding mandatory parameters, new validation errors) and what does not (adding properties or endpoints), which clients must tolerate without notice.
- **Stock changes are pushed**. Stock Notifications webhooks mirror the Stock API, and a release note describes adding fields "to gain parity between the Stock API and the Stock Notifications".

My reading is that 155 million calls a month from 220 partners make Autotrader's data a B2B platform, not a by-product of the consumer site; its change policy is the contract Attributes Without Chaos: Schemas, Dictionaries and Validation recommends for any schema partners depend on.

The platform from 2019 to 2026

The engineering evidence spans seven years and grows thinner toward the present. Istio's case study, from about 2019, describes Google Kubernetes Engine with the Istio service mesh, about 400 microservices and 200–250 deployments a day. The 2026 results say only that the company has "transitioned fully to a modern cloud-based delivery and data platform based largely on open-source technologies", now with "an AI platform built using the same principles". An August 2026 engineering post describes streaming personalization: Snowplow events in Google Cloud Storage scored by Python models in Spark Structured Streaming, with visitor profiles written to Kafka.

The results also claim "more than 50 proprietary AI and machine learning models" and argue that general-purpose language models cannot match them because "publicly available vehicle listings represent only a small part of the required data". That is an incumbent's defensibility claim, not a neutral finding. The products it describes are concrete: Co-Driver, which writes descriptions and orders images and "anecdotally" cuts listing time from 28 to 5 minutes; "specification adjusted valuations underpinning our price flags"; Buying Signals, a lead-intent model whose high-intent leads "convert at twice the average rate"; and a ChatGPT app integration via the Model Context Protocol (see AI Across the Listing Lifecycle).

Deal Builder: platform state on dealer stock

Deal Builder lets a buyer start a deal online: part-exchange, finance, delivery and a reservation. By March 2026 more than 6,700 retailers were on it (2,000 a year earlier), with 175,000 vehicles live and 137,000 deals with a full reservation in the year, which Autotrader calls "the very best enquiries in terms of conversion to sale".

The same results describe what the rollout cost. Retailer numbers "softened", reflecting trading conditions "and concerns with the speed and nature of our Deal Builder product roll-out". Average forecourts were 236 (1.7%) lower in the second half. The stock lever turned more negative as retailers cut advertised stock from November 2025. Employee engagement fell from 91% to 72%. Autotrader's response was "pausing auto-rollout; holding open listening sessions; establishing customer advisory groups; and introducing 'request a reservation' ". Trade press reported in December 2025 that the option lets a buyer ask for a reservation without paying the £99 fee.

My reading is that Deal Builder put a platform-controlled state, "reserved", on inventory the dealer owns and manages in its own systems, which changes who decides that a car is off the market and which lead gets it. The documented response amounts to making the state opt-in and dealer-controlled. The Listing Aggregate: Identity, Lifecycle and Invariants states the general rule: **who may trigger a transition is part of the business model**.

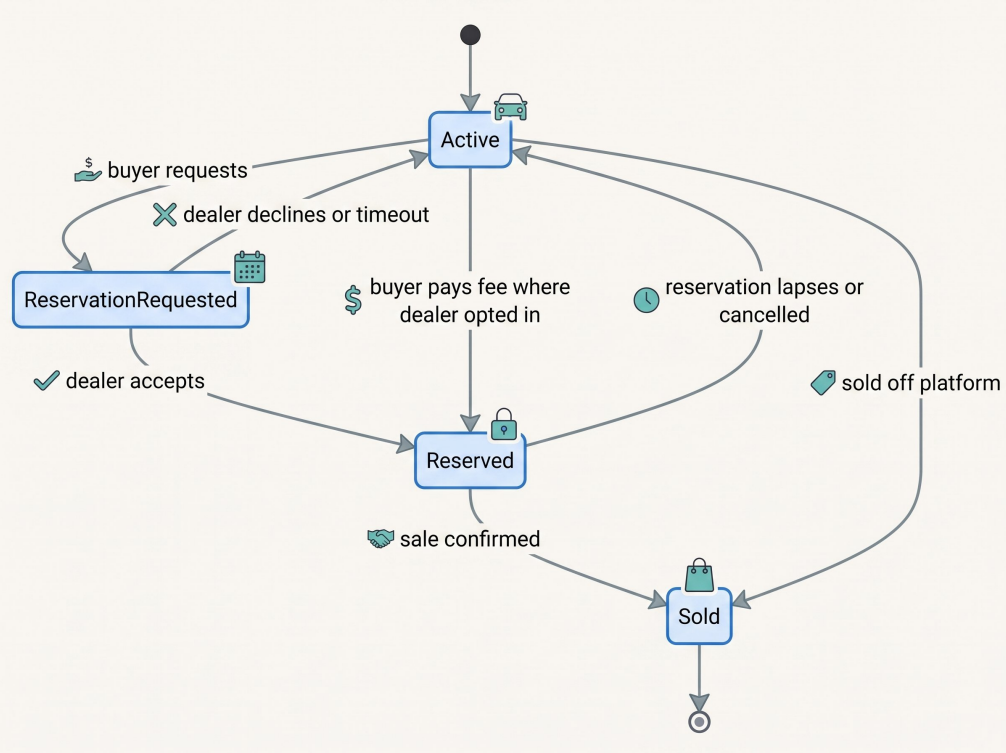


Figure 5: *A reference state model for reservations on dealer stock — not Autotrader's design. The paid, platform-set path exists only where the dealer opted in; otherwise a reservation is a request the dealer accepts, and every reserved state can lapse back to active.*

AutoScout24: The Platform Story

AutoScout24 calls itself "the largest pan-European online car market". It was part of Scout24 until 2020, when Hellman & Friedman bought it (reported at €2.84 billion). Its public evidence is almost entirely about engineering platform and data, and it is the best-dated platform history among the four.

- **Re-architecture, 2014–2018.** "Project Tatsu" began in November 2014 with one team under the principles "AWS first" and "shared nothing", moving from .NET in its own data center to JVM services on AWS (InfoQ, 2016). By 2018 the result was microservices aligned to bounded contexts, mostly synchronous, "some" using Kafka (AutoScout24 Tech Blog, 2018).
- **Consolidation, 2025.** Platform engineers migrated "110 AWS accounts" across four business units from Trader, "the biggest automotive market source in Canada", into AutoScout24's AWS organization, with "zero downtime", using Terraform-managed organization structure, IAM Identity Center roles provisioned from

a central file through pull requests, baseline StackSets and batch-wise account moves with tested rollback (August 2025). The post does not state the corporate relationship between the two companies.

- **Current platform, 2026.** The careers page (undated, read October 2026) reports 994 services in production, 107 APIs serving 2,700 requests per second, 30 TB of vehicle picture data and more than 270 engineers, "100% cloud". A July 2026 post describes Amazon EKS clusters, Argo CD and Backstage, and a read-only AI debugging agent.

The record says little about AutoScout24's listing, ranking or pricing systems, and this chapter does not fill the gap. **What it does show is an acquirer's platform absorbing another marketplace at the account and identity layer** — the enabling layers After the Deal: Consolidating Platforms After Acquisitions puts first — without any public sign that the two product platforms were merged.

CarGurus: Price Transparency as the Product

CarGurus' Form 10-K for 2025 (filed 19 February 2026) makes price evaluation the center of the proposition. Eligible used listings receive one of five deal ratings, from Great to Overpriced, based on **Instant Market Value (IMV)**, "our proprietary algorithm that assesses the market value of a used vehicle in a local market", calculated from "tens of millions of data points". The help center (updated May 2026) says IMV is updated "daily", calls it pricing guidance rather than an appraisal, and says the best deals appear first "based on our algorithm, not how much a dealer pays us."

The 10-K documents the dealer side of the loop. "Next Best Deal Rating" tells dealers the smallest price cut that reaches the next band; "Maximize Margin" shows price *increases* that keep the rating; PriceVantage is pricing software using "VIN-level activity, turn-time predictions, lead potential, market days' supply". Monetization follows the dealer: listing subscriptions priced by "a dealer's inventory size and region", a free "Restricted Listings" tier with "capped anonymous leads and connections", and add-ons such as Highlight. CarGurus reported revenue of $907.0 million for 2025, 34,409 paying dealers at year end, and a quarterly average revenue per subscribing dealer (QARSD) of $7,938 in the United States in the fourth quarter.

CarGurus also records the retreat that bounds the case. It acquired 51% of CarOffer, a digital wholesale business, in January 2021, and on 6 August 2025 its board decided to wind it down, including its transactions business, because it had "proven less effective in today's more volatile and unpredictable pricing environment". CarGurus will focus on "technology and analytics that will enable smarter sourcing and pricing decisions rather than facilitating the transactions themselves." The wind-down completed by 31 December 2025.

My reading is that CarGurus shows both ends of price data: a market-value model is a strong *information* product that sorts results and justifies a subscription, while facilitating transactions on volatile prices proved a different business.

OLX Autos: The Contrast

OLX Group went furthest toward owning cars, and then exited. After taking control of Frontier Car Group in 2019, it built OLX Autos as a "buy, recondition, sell" business in Latin America and Asia. Prosus reported OLX Autos revenue of $1.6 billion for its fiscal 2022, up 158% and "just over half" of its classifieds revenue, with about 175,000 cars transacted. On 2 March 2023 it announced that it would seek to exit, saying that "higher cost of capital, high inflation and a reversal of pandemic trends" had led to "a significant and persistent slowdown in the second-hand car market", after first-half trading losses of $206 million (Fin24, 2023). By 31 March 2025 all OLX Autos operations had been "disposed of, classified as held for sale or closed" (Prosus results, 2025). OLX Group: Regional Platforms, Verticalization and Retreat covers the group story.

A business that owns cars is a retailer with an e-commerce front end, needing inspection, reconditioning, logistics, pricing authority over its own stock and a finance book; a marketplace can only approximate condition and influence price. The platform built for one does not carry the other.

Across the Cases: Price Evaluation as a Control Loop

Three of the four companies publish enough to compare their price products directly.

	mobile.de	**CarGurus**	**Autotrader UK**
Consumer output	Five labels, euro boundaries per band	Five deal ratings from IMV	Price flags from "specification adjusted valuations"
No-rating path	`NO_RATING` with a reason	Ratings for "eligible" listings only	Not documented
Recalculation	On create and update; again when images arrive	Daily	Not stated

	mobile.de	CarGurus	Autotrader UK
Dealer tool	Preview endpoint; Preis-Planer in 2% steps showing search position	Next Best Deal Rating; Maximize Margin; PriceVantage	Retail Rating; price-adjusted days to sell; valuations API
Link to ordering	Search position responds to price (2026 release)	Deals sorted by the algorithm, "not how much a dealer pays"	Not documented
Source and date	Seller API manual, n.d.; newsroom, 2026	10-K, 2026; help center, 2026	FY26 results, 2026; developer portal, n.d.

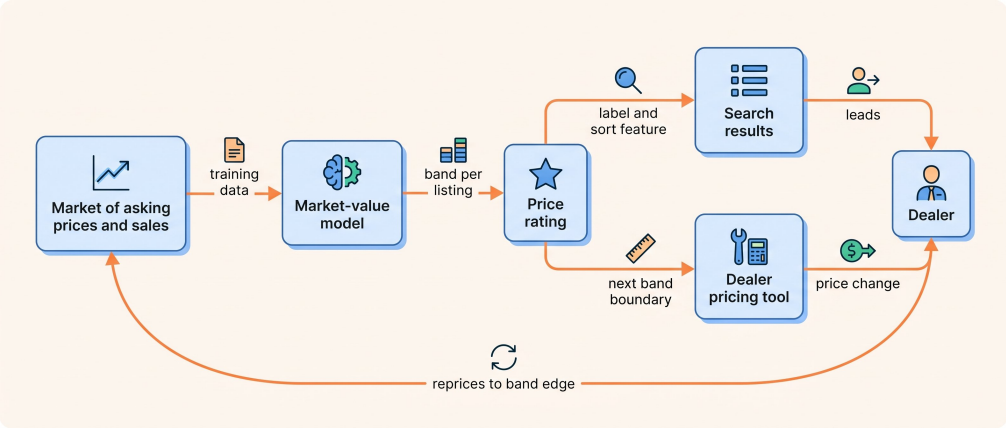


Figure 6: *The loop the public record implies. Once a rating feeds ordering and dealer tools, dealers price toward band boundaries, and the distribution of prices the model learns from shifts. The model is measuring a market it is helping to set.*

INFERENCE, NOT DOCUMENTED: None of the companies publishes its valuation methodology in a form verified for this book. A plausible reference architecture — not a description of any of them — is a gradient-boosted or hedonic model per market over make, model, derivative, year, mileage, options, condition signals and region; served as a band stored on the listing and refreshed by listing events and periodic market recomputation; with the band boundaries derived from a prediction interval rather than a fixed percentage. This matches the statement in From Raw Supply to Usable Listings: Normalization and Enrichment and the governance in Models in Production: ML Platforms and Use Cases.

My reading is that the loop needs four things: **bands rather than point estimates**, which survive model error and give dealers an actionable boundary; **an explicit no-rating state** for unusual cars; **monitoring for clustering at band boundaries**, the first sign that dealers price to the model rather than to the car; and **a declared list of every consumer of the rating**, because an inform-only label becomes an exposure model the day ranking reads it.

Trade-offs the Case Illustrates

Option	What the record shows	When it is right	When it fails
Publish integration contracts in full (TSP access, idempotency, error keys, change policy)	mobile.de and Autotrader document them; Autotrader serves 155 million calls a month to 220 partners	Feed-driven verticals where vendors write most listings	When the contract is published but not versioned; partners then build on undocumented behavior
Offer a change feed with a replay window	mobile.de Ad Stream: WebSocket, 24-hour catch-up; Autotrader Stock Notifications	When dealers' websites and partners mirror inventory	When consumers can be offline longer than the window and no resync path exists
Price rating as a label only	All four show ratings; CarGurus states ordering is not by payment	Low-trust markets where buyers need price guidance	When the label is wrong often enough to anger dealers and has no no-rating path
Price rating as a ranking feature and dealer tool	mobile.de links price to search position; CarGurus sorts by deal quality	When the model is governed like ranking, with monitoring and an owner	When the loop is unmonitored and dealers price to band edges

Option	What the record shows	When it is right	When it fails
Subscription scaled by stock	Autotrader's stock lever; CarGurus prices by inventory size	Dealer-supplied verticals with high lead value	When feed, search and billing count stock differently
Platform-controlled transaction states	Deal Builder: 137,000 reservations, forecourts down in H2, rollout paused	Opt-in, per-dealer configuration, reversible states	Auto-rollout onto dealers' stock without their control
Holding inventory or guaranteeing transactions	CarOffer wound down (2025); OLX Autos exited (2023–2025)	Rarely for a classifieds company; it is a different business	When prices turn volatile and capital costs rise

The rows are not a maturity ladder. **The leading automotive marketplaces moved deeper into data and workflow, and stepped back from price risk.**

Vertical Differences in the Record

Lesson from automotive	Real estate	General classifieds	Jobs
Identity decoded from authoritative data	Partly: cadastral and address registers, energy certificates; no VIN-strength identifier for a listing, so entity resolution does the work (One Property, Three Listings: Duplicates and Entity Resolution)	Weak: user-typed, image similarity	Probabilistic: titles mapped to occupation taxonomies

Lesson from automotive	Real estate	General classifieds	Jobs
Vendor-mediated writes with contracts	Strong: agency software vendors and portal feed formats	Weak: mostly app forms; pro feeds for a minority	Strong: applicant-tracking systems and career-site feeds
Price rating as a control loop	Valuations and price estimates; loop risk where they feed ranking or seller leads	Price suggestions at listing time; weaker loop	Salary estimates; transparency rules apply
Billing tied to stock	Agency subscriptions and per-listing depth products	Per-item paid visibility	Per-job packages or pay per click or application
Platform state on supplier stock	"Under offer" and viewing bookings; agencies resist portal control	Buyer-protected checkout on private items	Applications live in the employer's systems

Failure Modes the Record Exposes

1. **Feed truncation and duplicate creates**. The documented defenses (insertion request IDs, serialized writes, dealer-owned inventory keys) exist because dealer feeds retry, overlap and arrive incomplete.
2. **Inventory, search and billing counting different things**. With a stock lever, a counting error is a revenue error, and the dealer has the evidence.
3. **The undeclared consumer**. A price rating added to ranking without being re-registered as an exposure model escapes the evaluation ranking changes get.
4. **Rollout as architecture**. Deal Builder's cost came from the "speed and nature" of the rollout as much as from the product.
5. **Model error becoming balance-sheet risk**. A valuation good enough to label a listing is not necessarily good enough to buy a car on.
6. **New intermediaries**. Autotrader's register of risks names "AI being used to disintermediate marketplaces", while less than 1% of its audience came from AI assistants in its 2026 year. Its ChatGPT integration is a bet that the marketplace stays the source of truth inside those assistants.

Where the Public Evidence Stops

No public source found for this book answers the questions below, and the chapter does not fill them with inference.

Question	What is public	What is not
How does mobile.de build its listing, search and price-rating systems?	API contracts; XGBoost ranking under 30 ms (2024)	Services, engine, cloud, event backbone, ranking features
Who owns mobile.de as of October 2026?	Bundeskartellamt names Oak Germany Buyer GmbH as parent (October 2026); reported dual track (2025)	Whether a sale or listing completed, and the holding structure
What happens when a full XML upload omits a car?	Bulk XML with dealer inventory keys	Full-versus-delta semantics (documented on a page not read)
How are valuations and price ratings computed?	Inputs and outputs at all four; recalculation cadence at two	Model families, error bands, band thresholds, anti-gaming measures
Does price rating feed ranking at Autotrader and AutoScout24?	mobile.de: yes, by its 2026 release; CarGurus: deal quality orders results	Autotrader and AutoScout24 ranking features
What is Autotrader's current platform topology?	2019 GKE and Istio; 2026 posts on Google Cloud Storage, Spark and Kafka	Provider commitment, scale, service count today
How do AutoScout24's dealer APIs and product platform work?	Platform engineering and data	Dealer import contracts, listing model, pricing
What did Deal Builder change in Autotrader's systems?	Products, figures, rollout response	Reservation state model, payment flow, dealer configuration
How are licensed vehicle taxonomies redistributed?	Codes carried in APIs (Schwacke, KBA)	License terms for JATO, Schwacke/Autovista, DAT or CAP HPI data

The Chief Architect's View

What I take from the automotive leaders is six decision rules, each tied to the evidence above.

Write the dealer contract before the importer. mobile.de's manual reads like a list of incidents someone decided never to repeat. I would publish the same commitments for any professional supply channel — accountable integrators, idempotent creates, one writer per listing, stable error keys, a notice period for breaking changes — and treat vendor certification as a release gate.

Publish change, not only state. A change feed with a stated delay and replay window keeps dealers' websites and partners consistent without polling; 24 hours is a defensible default window.

Govern the price rating as ranking. Whatever the model is called, once it labels a result or contributes to its position it decides exposure. It needs an owner, an evaluation on band accuracy and on boundary clustering, a no-rating path, and the same review that a ranking change gets (Ranking for Two Sides: Relevance, Freshness and Paid Visibility, Models in Production: ML Platforms and Use Cases). Dealers will contest it, so the explanation shown to them must use the same terms as the public label.

Make inventory state the single source for search and billing. If a dealer pays per car, the count that billing uses must come from the same state that search shows, with the same timestamps. This is cheap to design and expensive to reconcile later.

Move up the transaction ladder only with the dealer's consent. Reservations can lift lead quality, as Autotrader's conversion claims suggest, but they need opt-in, configuration and reversal. Owning cars or guaranteeing prices is a different business with a different balance sheet; the record shows two well-funded retreats.

Treat data APIs as a business line. At 155 million calls a month, data services carry partners' operations and need their own service levels, versioning and owner.

Comparing the Case to the Book's Patterns and to Veldmark

The book's patterns, as the automotive record tests them:

Book pattern or claim	What the record shows	Verdict
Feed contracts with idempotency, ordering and actionable errors (Many Doors In: Forms, APIs and Bulk Feeds)	mobile.de insertion request IDs, per-ad serialization, TSP certification; Autotrader validation errors	Strongly supported
Separate identity, specification and listing; record provenance (From Raw Supply to Usable Listings: Normalization and Enrichment)	HSN/TSN and Schwacke codes; derivative-level taxonomy; rating recomputed when images arrive	Supported
Publish listing changes as a contract with retention (Every Change Is an Event: Event-Driven Listing Architecture)	Ad Stream with 24-hour catch-up; Stock Notifications parity with the Stock API	Supported
Price and quality models are exposure models once ranking reads them (Ranking for Two Sides: Relevance, Freshness and Paid Visibility, Models in Production: ML Platforms and Use Cases)	mobile.de Preis-Planer links price to search position; CarGurus sorts by deal quality	Supported, with the loop visible in public
Billing derives from entitlement and inventory state (Packages, Bumps and Quotas: Products, Pricing and Entitlements)	Autotrader's stock lever; CarGurus pricing by inventory size	Supported
The endpoint is a business decision (Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy)	Deal Builder friction; CarOffer and OLX Autos retreats	Strongly supported

Now Veldmark, the fictional group, as of its 2027 decisions in the scenario. Motorvane earns €360 million, 91% of its 1.45 million listings come from 21,400 dealers, and 88% of those arrive through DMS feeds — the same physics as mobile.de's.

Veldmark choice or event (2027)	Automotive parallel	What the case suggests
The truncated-snapshot incident of 8 March 2027: one transfer service provider's short files unpublished 38,400 listings, and about a fifth came back with new IDs	mobile.de's dealer inventory keys and TSP accountability	Keep the external key mapping immutable and make the vendor a certified principal; the importer's diff must never purge identity
The Essen Golf's sale reached Motorvane 26 hours late because status traveled in a daily snapshot; the fix sends status changes as Seller-API deltas within 15 minutes	Ad Stream and Stock Notifications publish changes, not only snapshots	Correct; the delta channel needs its own idempotency and replay window
Motorvane's market-value model rates the Golf "good price" (€21,900 against €22,800) and is also a ranking feature	mobile.de Preis-Planer; CarGurus deal sort	Register it as an exposure model, publish band boundaries to dealers, and monitor clustering at the edges
The "Pro 150" overbilling: the importer skipped over-quota cars while billing counted feed rows; 1,140 dealers overbilled about €212,000	Autotrader's stock lever makes paid stock revenue	Derive billing from the entitlement ledger that search also reads, as Tally does from 2028
Platform-set reservations only for dealers who opt in	Deal Builder's paused auto-rollout and "request a reservation"	The right default, reached without the backlash

Veldmark choice or event (2027)	Automotive parallel	What the case suggests
Per-lead billing approved for design only: a six-month lead ledger counting 19 billable leads where the Golf had 23 contacts	Autotrader's Buying Signals; CarGurus capped leads in its free tier	Measure before charging; dealers will reconcile every lead
Motorvane stays on Elasticsearch until a 2029 review	AutoScout24 consolidated accounts and identity, not visibly product platforms	Keeping the automotive brand's search loosely coupled is consistent with the record

One worked comparison makes the price-rating point concrete. On Wednesday 14 April 2027 the Golf is listed at €21,900 and rated "good price" against a market value of €22,800. On Thursday 22 April the dealer cuts it to €21,400; the listing event triggers a rescore, the rating moves up a band and the car rises in Motorvane's ranking while price-drop alerts go to 1,180 watchers. Nothing in that sequence is wrong. But if Motorvane, like mobile.de, showed dealers a planner of price steps and search position, a dealer with 150 cars would soon price many of them just inside a band boundary. **The model would then be learning from prices it induced**. Priya Raman's registry, in the scenario's 9 November 2027 decision, is the place to record that the market-value model has two consumers, and Felix Brandt, Motorvane's CTO, the person to own its band-boundary monitoring.

One caution on numbers: Motorvane's derived subscription revenue of about €927 per dealer a month and Autotrader's £2,995 per forecourt a month use different denominators, currencies and package scopes, and should not be ranked against each other.

Questions to Ask

1. Is every party that writes listings on a professional's behalf a named, certified principal with its own credentials and scope?
2. Are creates idempotent and writes serialized per listing, and does our documentation say so to clients?
3. Do we publish listing changes as a feed with a stated delay and replay window, and what happens to a consumer offline longer than that window?
4. Which systems read our price or quality model, and is it registered and evaluated as an exposure model wherever ranking or sorting uses it?

5. Do we show dealers how price affects position, and if so, do we monitor clustering at band boundaries?
6. Does billing count paid stock from the same state that search shows, with the same timestamps?
7. Which states on a professional's stock can the platform set without the professional, and can each be switched off per account and reversed?
8. What is our breaking-change policy for partner APIs, and how many days' notice does it promise?
9. Where on the transaction ladder do we stop, and what evidence would make us move — or step back?

The automotive leaders show a vertical that publishes its contracts and its levers, and that learned in public where its responsibility ends. That closes the cases of Part IX. The Field Kit: Templates for the Marketplace Architect turns the book's recurring decisions, including the feed contract, the listing event specification and the entitlement matrix used in this chapter's comparison, into templates a team can fill in.

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Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

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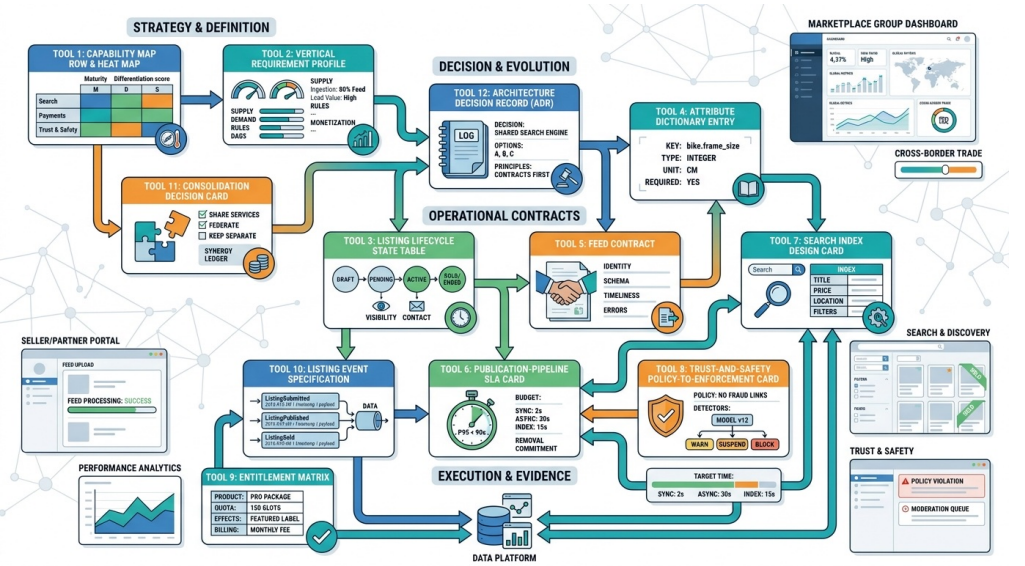
Field Kit: Templates for the Marketplace Architect



IN THIS SECTION, YOU WILL: Find the template that fits the design or decision in front of you, copy it, and see it filled in once for Veldmark, with the mistakes that most often make it useless and the chapter that explains the reasoning behind every field.

KEY POINTS:

- **Every template names an owner, a stable identifier and a measurable promise.** The records cite each other by identifier (`vm:tradeyard:nl:…`, `bike.frame_size_cm`, `POL-FRAUD-PAYLINK` v3, `listing.events.v1`, `pro-150@v5`, tenant `hl:es`), and that chain is what lets a new architect, an auditor or an AI tool follow a stale listing in a digest back to the contract that allowed it.
- **The removal path is written down before the publication path is celebrated.** Seven of the twelve templates have a field for what happens when a listing ends: the lifecycle table, the feed contract, the SLA card, the index card, the enforcement card, the entitlement matrix and the

event specification. A template with that field blank is the cheapest Ghost Inventory finding you will ever get.
- **Start with the record that changes the decision on your desk this week**. Fill the fields that change the decision, write "unknown" rather than guess, and add fields only when a later stage needs them. A template earns its place when it changes a decision or keeps evidence, not when every cell is full.

FAILURE PATTERNS THIS SECTION ADDRESSES:

- **Ghost Inventory** — the lifecycle table, the SLA card and the event specification each force a named removal objective per consumer, so ended listings stop surviving in copies nobody registered.
- **Feed Roulette** — the feed contract turns an import into an agreement on identity, semantics, field ownership, errors and reconciliation, with a mass-change guard that holds a truncated file instead of applying it.
- **Billing by Guesswork** — the entitlement matrix puts every product's effects, enforcement points, failure behavior and billing basis on one page, so what was sold can be compared with what was delivered.

These are **working templates for a group of listing marketplaces**: the records that Parts I to VIII ask a Chief Architect to keep, from the capability map of Map What the Marketplace Does Before What Runs It to the decision record of Who Owns What: Organizing Marketplace Engineering, each tested against Veldmark, the fictional marketplace group the book follows. They are not a certification scheme or a vendor method. Each is the smallest form I have found that makes a design or decision explicit, assigns an owner, and leaves evidence someone else can check. **Adapt the wording to your marketplaces; keep the fields that carry ownership, removal behavior and dates.** Those are the fields that get dropped first, and the ones whose absence the next incident finds.

Three conventions run through all twelve.

- **They are maintained data, not documents.** The capability map and vertical profiles live in the architecture repository; the attribute dictionary in the taxonomy and attribute service; the event specification beside the registered schema; policy records in the trust-and-safety policy registry; the entitlement matrix in the catalog; decision records in one Git register indexed by the architecture advice forum. A template copied into a slide deck stops being true within a quarter.
- **They reference each other by identifier.** A decision record cites the capability row and the profiles it changes; an index design card cites the attribute dictionary's purposes; an SLA card cites the consumer registry; an enforcement card cites the policy version. Use the same identifier everywhere, so a reader can move from any record to the next.

- **Each has a Veldmark example consistent with the chapter that teaches it.** All Veldmark names, figures and dates are fictional and illustrative. The examples are deliberately short; the chapter linked at the end of each tool works the full case.

Find the Template for the Question on Your Desk

Tool	Use it when	Chapter
Tool 1: Capability map row and heat map	You must show where the group pays several times for the same ability, or who owns a capability	Map What the Marketplace Does Before What Runs It
Tool 2: Vertical requirement profile and share-layer row	A vertical or brand section is asked to move onto a shared engine, or a new vertical is proposed	Same Platform, Different Physics: How Listing Verticals Differ
Tool 3: Listing lifecycle state table	You design or change listing states, or two systems disagree about whether a listing is live	The Listing Aggregate: Identity, Lifecycle and Invariants
Tool 4: Attribute dictionary entry	A new attribute is proposed, an attribute changes meaning, or facets and feeds disagree	Attributes Without Chaos: Schemas, Dictionaries and Validation
Tool 5: Feed contract	A partner, dealer system or agency software vendor will write listings	Many Doors In: Forms, APIs and Bulk Feeds
Tool 6: Publication-pipeline SLA card	You promise sellers a time to searchable, or buyers that ended listings disappear	From Submitted to Searchable: The Publication Pipeline
Tool 7: Search index design card	A market index is designed, migrated, reindexed or moved to another engine	Search Architecture for Listings: Indices, Freshness and Scale

Tool	Use it when	Chapter
Tool 8: Trust-and-safety policy-to-enforcement card	A policy is written or changed, or a new detector or action is proposed	Trust and Safety as a Platform
Tool 9: Entitlement matrix	A paid product or package is launched, repriced or migrated to a shared catalog	Packages, Bumps and Quotas: Products, Pricing and Entitlements
Tool 10: Listing event specification	A listing change must reach other contexts, or a field's meaning is in doubt	Every Change Is an Event: Event-Driven Listing Architecture
Tool 11: Consolidation decision card	An acquired marketplace, brand or SaaS business must be integrated, or deliberately left alone	After the Deal: Consolidating Platforms After Acquisitions
Tool 12: Architecture decision record	A decision crosses a team, brand or tenant, moves money or personal data, or needs an exception	Who Owns What: Organizing Marketplace Engineering

The tenant register and configuration hierarchy of One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture, the qualified-contact definition of Data Platform for Marketplaces: Events, Lakehouse and Data Products and the journey SLOs of Sunday Evening Peaks: Reliability, Performance and Cost are not repeated here; Tools 6, 7, 11 and 12 link to them where a field depends on them.

How the Records Connect

Most changes in a marketplace group pass through several of the tools, and each record points to the next.

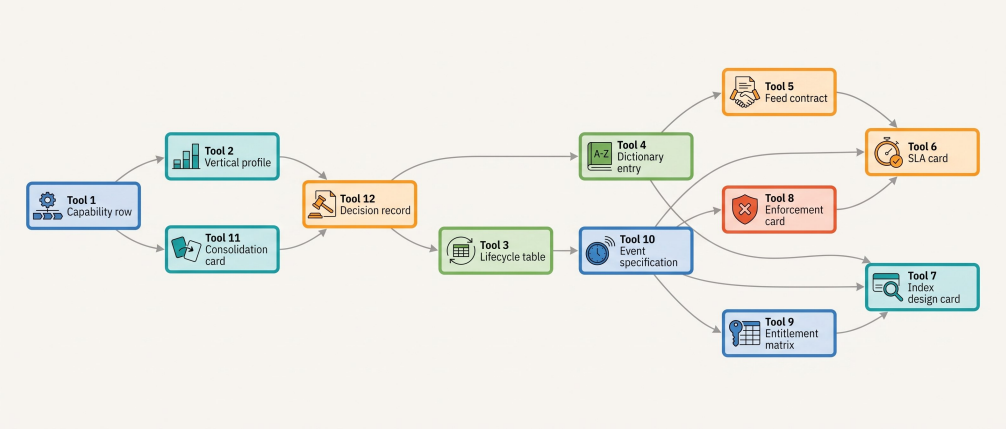


Figure 1: *How the templates reference each other. Understanding records feed decisions; decisions produce the listing's model and its contracts; and nearly every operational record rests on the event specification and the SLA card, because in a listing marketplace a change is finished only when every copy of the listing reflects it.*

Tool 1: Capability Map Row and Heat Map

Purpose. A capability map describes what the marketplace must be able to do, independent of which system, team or brand does it. The heat map scores each level-2 capability on three separate dimensions (differentiation, maturity and duplication) so that sharing, investment and ownership questions are argued from the same evidence. It is the frame every later record cites, because it does not move when an engine, a vendor or an organization chart does.

Use it when you prepare a Common Ground–style sharing decision, a consolidation case or a portfolio discussion about overlap, and whenever a capability has no named owner. Refresh it before every annual planning round and after every acquisition.

The template. One row per level-2 capability; level 3 only where a decision needs it.

Field	What to record	Guidance
ID and level-1 area	e.g. `3.1`, Discovery	Ten level-1 areas: Supply Acquisition, Listing Management, Discovery, Matching and Contact, Trust and Safety, Transactions, Monetization, Professional Services, Data and Insights, Platform
Capability name	A stable noun phrase	Never a system ("Yardstick search"), a team or a brand
Definition	One sentence a product manager can read aloud	States what must be true, not which technology does it
Value streams served	List-to-sell, search-to-contact, lead-to-deal, pro onboarding	Mark the stage where this capability is the bottleneck, if any
Owner	A named person, accountable for the outcome	Split ownership is allowed only when the level-3 split is recorded
Implementations	Count and names, per brand and country	From the application-to-capability mapping, verified with brand engineering
Differentiation	Low / Medium / High, per vertical	Scored by product leaders, not engineers
Maturity	1 absent to 5 leading, **per implementation**, with dated evidence	"2: p95 6 minutes to searchable, January brown-out" is evidence; "the team feels it is weak" is not
Duplication verdict	Consolidate / Promote strongest / Split at level 3 / Invest locally / Protect / Keep separate on purpose	Use the verdict table in Map What the Marketplace Does Before What Runs It; "keep separate on purpose" needs a reason and a review date

Field	What to record	Guidance
Annual run cost	Infrastructure and team cost allocated	Ranks attention; it is not a savings estimate
Linked decision	The decision record that chose the verdict, or "undecided"	Duplication is a problem only when nobody chose it
Reading	One sentence for the group leadership team	The sentence that goes on the slide

Veldmark example (map v1, presented 31 March 2027; run cost from the 2027 unit-cost review):

Field	3.1 Search and filtering
Definition	Returning the live listings that match a buyer's query, filters and location, with counts, within the journey's latency objective
Value streams	Search-to-contact (bottleneck at the Sunday-evening peak); list-to-sell (time to searchable)
Implementations	3 engines: Solr 8 (Tradeyard NL, BE, PT), Elasticsearch (Tradeyard PL on 7, Motorvane on 8), OpenSearch (Hearthly)
Differentiation	Engine low; relevance high, per brand
Maturity	Tradeyard 2 (6-minute p95 to searchable; 48-minute brown-out on 17 January 2027); Motorvane 4; Hearthly 3
Verdict	**Overlap 3: split at level 3.** Shared engine and indexing; brand-owned relevance and ranking policy
Annual run cost	About €11.8M (Tradeyard €6.0M, Motorvane €3.2M, Hearthly €2.6M)
Linked decision	The 14 September 2027 search platform decision (Tool 12)
Reading	Share the machinery that breaks at peak; never share the ranking that decides how each brand competes

Common mistakes.

- Scoring one maturity per capability. "Trust and safety: 3" averages a strong Tradeyard NL implementation with weak ones elsewhere and hides both; score per implementation, then aggregate.
- Letting one group score everything. When architecture scores differentiation, every capability looks like a platform candidate; separate the scorers.

- Forgetting the deliberate duplicate. Keybridge's listing distribution and Hearthly's feed ingestion are duplicated on purpose under the March 2027 neutrality decision; a map without a "keep separate" verdict invites the next cost program to merge them.

Learn more: Map What the Marketplace Does Before What Runs It

Tool 2: Vertical Requirement Profile and Share-Layer Row

Purpose. A vertical requirement profile describes a vertical, or a brand section that behaves like one, by the forces that act on its systems rather than by its features: fourteen dimensions, filled with numbers where numbers exist. The share-layer row is its companion: for each part of a capability, it records whether that part is a shared platform, a shared engine with vertical content, a vertical capability with one group owner, or brand-local product, and on what condition a vertical moves onto it.

Use it when a vertical or brand section is asked to move onto a shared engine, when a new vertical or country section is proposed, and when a vertical's business model changes (a transactional pivot, pay-per-lead, an acquisition). The profile goes into the decision record (Tool 12) as evidence.

The template. One page per vertical, numbers first.

Group	Dimension	Record
Supply	**Supply mix**	Share of new listings from private sellers, professionals, partners
	Ingestion path	Form, API, bulk feed, replication; share per path
	External system of record	Where the authoritative listing lives; who wins a conflict
Item	**Item identity**	Identity outside the listing (VIN, address, parcel, occupation code), or none
	Listing lifetime	Typical and implied lifetime; what ends a listing
	Attribute density	Required, recommended and optional attributes per typical category
	Media profile	Images per listing, other media, pushed or pulled

Group	Dimension	Record
	Location semantics	What location means; display precision allowed
Demand	**Decision horizon**	Minutes, days or months from first search to decision
	Lead value	What one qualified contact is worth to the seller
Money	**Monetization unit**	Who pays, per what unit
	Transaction locus	Where the money of the sale moves
Rules	**Regulatory load**	Fields, displays and prohibitions the law requires
	Fraud typology	What the typical scam looks like

```
SHARE-LAYER ROW
Capability part:   <e.g. vehicle identity, specification and price rating>
Layer:             1 shared platform | 2 shared engine, vertical content |
                   3 vertical capability, group-owned | 4 brand-local
Owner:             <team>                 Consumers: <brands, sections>
Reason:            <which profile dimensions put it in this layer>
Stage gate:        <the vertical's own metric the shared engine must meet
                    before the vertical moves onto it>
Extension points:  <named, with latency and failure contract>
Review trigger:    <profile change, new consumer, acquisition>
```

Veldmark example (Motorvane DE profile, March 2027 scoping workshop, ratified April 2027; abbreviated):

Dimension	Motorvane DE
Supply mix	91% of Motorvane listings from 21,400 dealers
Ingestion path	88% of dealer listings by DMS (dealer management system) feed, about 1.16 million listings group-wide, mostly nightly full snapshots
External system of record	The dealer's DMS; status changes now contracted as deltas within 15 minutes (Tool 5)
Item identity	VIN; decoding filled 31 attributes on the Essen Golf
Listing lifetime	Implied about 6.9 weeks; the Golf sold in 15 days, the feed delete arrived 26 hours later

Dimension	Motorvane DE
Attribute density	Used-car schema with 14 required and 26 recommended attributes; 212 vehicle attributes in `vehicle.*`
Lead value	High: the Golf drew 23 raw leads in 10 days, 19 qualified
Monetization unit	Dealer packages with slots (Pro 150: 150 slots, 20 boosts a month, €1,450 a month)
Regulatory load	Trader status and the consumer conformity guarantee for dealers
Fraud typology	Vehicle deposit scams, clocked mileage, VIN reuse

Share-layer row: *vehicle identity, specification and price rating*, **layer 3**, owned by Motorvane, consumed by Tradeyard Poland's cars section (about 410,000 live listings, about 40% from dealers). Stage gate for Motorvane on the shared ingestion engine: **feed success rate and delete latency no worse than Motorvane reports today**.

Common mistakes.

- Profiling features instead of forces. "Has a dealer portal" says nothing; "88% of dealer listings by nightly snapshot" decides the removal SLA.
- Classifying a whole capability into one layer. "Search" is the engine at layer 1, mappings and ranking policies at layer 2, a price-rating signal at layer 3 and the result page at layer 4.
- Omitting the stage gate. Without it, a vertical moves onto a shared engine when the program plan says so, which is how the Forced Fit starts.

Learn more: Same Platform, Different Physics: How Listing Verticals Differ

Tool 3: Listing Lifecycle State Table

Purpose. A listing lifecycle is the set of states a listing can be in, the transitions between them, who may trigger each one, and what each state means for visibility, contact, alerts, URLs and retention. The state table makes all of it explicit in one place. It separates the **publication state**, owned by the listing, from the **enforcement overlay**, owned by trust and safety, and it derives visibility from both instead of storing it in a dozen places.

Use it when you design a listing service, add a state ("under offer", "let agreed", "reserved"), integrate a vertical whose sellers use other words, or investigate why a listing is live in one system and gone in another.

The template. Two tables and a list.

Transitions

From	To	Triggered by	Command	Event emitted	Outcome code required?
Draft	PendingReview	Seller or feed	`Submit`	`ListingSubmitted`	No
…	…	Seller / platform clock / order / trust and safety	…	…	Yes for every terminal transition

Derived behavior per state

Publication state (overlay)	Searchable	Detail page	Contactable	Alertable and recommendable	URL behavior	Retention clock starts?
…	…	…	…	…	200 / 404 / 410	…

Rules to record

- **Owners**: which states the seller, the platform clock, an order and trust and safety may set; nobody else may.
- **Timestamps**: `created_at` (never moves), `published_at` (once), `ranked_at` (moved by renewals and bumps, which are not transitions).
- **Outcome vocabulary** for terminal transitions: sold on platform (with order reference and final price), sold elsewhere, no longer available, unknown.
- **Vertical mappings**: how local words map onto the canonical states.

Veldmark example (Common Ground lifecycle, Listing event contract v1, Tradeyard goods):

Publication state (overlay)	Searchable	Contactable	Alertable	URL behavior
Active (none or restored)	Yes	Yes	Yes	200
Active (restricted)	Demoted or hidden per country; statement of reasons required	Yes	No	200
Active (held or blocked)	No	No	No	404, or 410 when blocked
Reserved	Yes, labeled	Existing conversations	No	200
Sold	No	No	No	200 with related live inventory, then 410
Withdrawn, Archived	No	No	No	410

The Rotterdam bike exercises the table: active at 19:48 on Sunday 7 March 2027; `ListingReserved` by the Safe Deal order at 14:02 on Monday 8 March; `ListingSold` at 18:10 with outcome *sold on platform, €225*, removed from search at 18:11. The €225 goes into the outcome, not into the price, so no "price reduced" alert fires for an item that is gone. The table's "Alertable: No" for Sold is the rule the digest assembled at 17:30 that day broke. Vertical mappings: Hearthly's "under offer" and "let agreed" map onto Reserved and Sold; Motorvane allows platform-set reservations only for dealers who opt in.

Common mistakes.

- One `status` column for three owners. A seller's "resume" then clears a block that trust and safety set; keep the overlay separate.
- Treating a bump or renewal as a new listing. It must move `ranked_at` only; resetting `created_at` turns recency ranking into a paid lottery and breaks every time-to-sell metric.
- Ending listings without an outcome. Without it, analytics recorded Femke's sale as "expired", and sell-through and price models train on noise.

Learn more: The Listing Aggregate: Identity, Lifecycle and Invariants

Tool 4: Attribute Dictionary Entry

Purpose. An attribute dictionary entry is a small contract for one attribute: its key, meaning, type, unit, allowed values, scope, the purposes it serves, its provenance precedence, and who owns it. Category schemas bind entries to categories; the schema compiler turns them into forms, validators, index mappings and feed mappings. The entry is what keeps facets, filters, feeds and trust rules agreeing about what a listing is.

Use it when a new attribute is proposed, an attribute is about to become required, a value list changes, two keys turn out to mean the same thing, or one key turns out to mean two things.

The template. Builds on the public models of eBay's Taxonomy API and Amazon's product type definitions, as Attributes Without Chaos: Schemas, Dictionaries and Validation explains.

Field	What to record	Guidance
Key	Namespaced, language-neutral (`bike.battery_capacity_wh`)	Never a label; never reused after retirement
Tier	Group core / vertical (`vehicle.*`, `property.*`, `goods.*`) / country extension	Promotion to core needs a decision record
Labels and help text	Per locale	Localization is a property of the entry, not its identity
Meaning	Precise definition, including how it is measured	The sentence a feed vendor and a moderator both read
Type and unit	Integer, decimal, boolean, date, enumeration, reference, text, range; canonical unit and accepted input units	Conversion rule stated, with rounding
Allowed values	Range, pattern, enumeration or reference-data set	Closed enumerations power facets
Mode and cardinality	Selection only or free text with suggestions; single or multi-valued	

Field	What to record	Guidance
Scope	Product attribute or instance attribute	A catalog match may suggest product attributes, never overwrite instance ones
Editability	May the value change after publication?	A VIN should not; a price can
Requiredness by category	Required / recommended / optional, with an expected-required date	Enforce on create and revise, never retroactively
Purposes	Facet, filter, sort, ranking feature, trust rule, ads targeting, feed, SEO, duplicate detection	Drives which compiled artifacts include it; the index card (Tool 7) promotes typed fields from this flag
Provenance precedence	Order among seller, partner_feed, registry, catalog, model, moderator, platform	Model values carry confidence; who wins a conflict is decided here
Sensitivity	Personal data, special-category risk, legally mandated	Steers retention, display and ads use
Mappings	External standards per version	OpenImmo, RESO, schema.org, Google Merchant, partner formats
Owner and status	Steward; proposed / active / deprecated / retired	Retire only when no producer writes it and no reader depends on it

Veldmark example (Tradeyard goods stewardship, schema release 2027.11, the bikes split):

Field	`bike.battery_capacity_wh`
Tier and meaning	Vertical `goods.*` bikes family; nominal battery energy as stated by the manufacturer
Type and unit	Integer; canonical unit Wh; Ah accepted only with a stated voltage
Scope, editability	Product attribute (a model match may suggest it); editable
Requiredness	E-bikes leaf: **recommended** from 1 November 2027, **required on create and revise from 1 March 2028**, a four-month runway for dealer feeds
Purposes	Facet (bucketed), filter, outbound feed, query-understanding target
Provenance precedence	seller, then partner_feed, then model (above the confidence threshold); a model value is shown as a suggestion until the seller confirms
Owner, status	Tradeyard goods steward; active from release 2027.11

The rollout followed the producer-breaking rule: the dictionary published the field and a readiness report per dealer feed, feeds validated with warnings only during the runway, and only then did a missing value return a stable error code.

Common mistakes.

- Changing meaning, unit or type in place. Turning kilometers into miles "for one launch" corrupts every stored value and every reader at once; a change of meaning is a new attribute.
- Converting what cannot be converted. Letter frame sizes are brand-specific, so `bike.frame_size_class` stays a separate enumeration beside `bike.frame_size_cm`.
- Leaving purposes blank. Then the only way to find an attribute's readers is to break them; Tradeyard's cleanup found 31 of 990 retired keys still read by something.

Learn more: Attributes Without Chaos: Schemas, Dictionaries and Validation

Tool 5: Feed Contract

Purpose. A feed contract is everything both sides must agree on for a supply channel to be operable: scope, identity, semantics, field ownership, schema, timeliness, media, errors, limits, reconciliation and change management. Schema validity is perhaps a third of it. The contract is a product artifact as much as a technical one, because partners judge a marketplace by how predictable its feed is.

Use it when a dealer system, agency software vendor, partner or professional seller will write listings through a file, an API or replication, and whenever a channel's removal behavior is unknown. Every channel gets one, including the insertion flow, which is a channel too.

The template.

```
feed_contract: <channel id>            # e.g. motorvane.dms.<vendor>
version: <n>      effective: <date>     owner: <ingestion team>
partner:
  integration_principal: <vendor>       grants: [listings.write,
  ↪ statistics.read]
  scope: <sellers, branches, categories this principal may write>
identity:
  external_key: <field>                 unique_per: seller | integration
  idempotency: <how a resend is recognized>
semantics:
  mode: full_snapshot | delta | hybrid
  absence_means: <removal | nothing>    empty_field_means: unknown | remove |
  ↪ unchanged
field_ownership:                        # who may overwrite what
  identity_facts: feed (re-moderate if material)
  price_availability_text_media: feed, unless seller lock in pro tool
  visibility_products: platform entitlements, via product calls only
  moderation_state: platform, never the feed
  enriched_values: platform, with provenance
schema: <compiled category schema versions accepted>; retirement notice: <days>
timeliness:
  creates_updates: <objective from receipt>
  status_changes: <sold | reserved | withdrawn within N minutes>
media: pushed | pulled URLs; change detection: <hash | etag>; limits: <n, size>
errors:
  envelope: RFC 9457 problem details; codes: <stable catalog>
  outcomes: success | warning | error; run report: <API, webhook, pro tool>
limits: rate <n/s>, batch <n>, concurrency per listing <1>, quota policy <ref>
reconciliation:
  mass_change_guard: hold run if removals > <share or count, by seller size>
  delta_reconciliation: <weekly | daily> full key list
change_management: breaking changes announced <N> days ahead; what counts as
↪ breaking
```

Veldmark example (Motorvane DMS channel under Supply Contract v1, September 2027; the vendor that serves Autohaus Ruhrblick and about 1,900 other Motorvane dealers):

Clause	Contract
Partner	Integration principal with listings-write and statistics-read grants; scope: its dealers' own accounts and sites (Ruhrblick: Essen and Bochum)
Mode	**Hybrid**: one full snapshot daily at 17:00 for completeness; sold, reserved and withdrawn as Seller-API deltas within **15 minutes** of entry in the DMS
Change detection	Content hash per vehicle; about 91% of snapshot records unchanged and skipped
Field ownership	Price and description feed-owned unless the dealer sets a lock in Dealer Hub; visibility products only through product calls
Quota policy	Stored with the entitlement: dealer priority, then newest first, no overage without consent (Tool 9)
Errors	One RFC 9457 error catalog with stable codes, shared validator library, run report per run
Reconciliation	Mass-change guard mandatory on the snapshot; deltas reconciled at least weekly

The contract replaces two incidents with two controls. The truncated files of Monday 8 March 2027, which unpublished 38,400 cars for 286 dealers by 02:55, become 286 held runs. The Golf's 26-hour-late removal becomes a 15-minute status delta.

Common mistakes.

- Leaving "absence means removal" implicit. A daily snapshot is a 24-hour removal objective by construction; choose it, or add status deltas.
- Purging the external-key mapping on removal. When Motorvane's cars came back, about a fifth returned with new listing IDs and lost their history; keep the mapping through the retention window.
- Giving one partner a private door. Keybridge integrates with Hearthly through the same public contract, limits and error catalog as the other 160 or so vendors; anything else breaks the neutrality commitment.

Learn more: Many Doors In: Forms, APIs and Bulk Feeds

Tool 6: Publication-Pipeline SLA Card

Purpose. The SLA card turns "listings should go live fast" into a contract with three parts (coverage, clock and removal) and a budget per stage that adds up to the target. It names what the target does not cover, so that the fast path cannot be met by holding more listings for review, and it treats removal as a separate, stricter promise.

Use it when you set or change a time-to-searchable target, accept a new channel or vertical onto the pipeline, add a stage (a new risk check, an enrichment), or register a new consumer of listing data.

The template.

```
SLA CARD  <brand / channel>                    Owner: <team>     Version: <n>
Liquidity metric protected: <e.g. time to first contact from searchable>
COVERAGE   Applies to:   <listings cleared by the automated gate: share %>
           Excludes:     <held for review, own objective: p90 ...>
           Published beside both: held share
CLOCK      Starts:       <receipt of submission | receipt of feed record>
           Ends:         <visible in search results, caches included>
           Kind:         platform latency | source latency
TARGET     p95 <n s>  by <date>      Interim: <value, date, engine>
BUDGET     Stage                          p95     How it is met
           Synchronous path               <n s>
           Outbox to topic                <n s>
           Async checks and media         <n s>   (parallel; longest counts)
           Publication decision           <n s>
           Indexing and refresh           <n s>   (see index design card)
           Result and count caches        <n s>
           Slack                          <n s>
REMOVAL    Search:          p95 <n s>
           Every registered consumer: p95 <n min>   (consumer registry ref)
           Partner feeds:   within <n min>
           Rule: nothing sent for a listing that ended before the send
EVIDENCE   Freshness monitor, synthetic listings per tenant, per-consumer acks
```

Veldmark example (Tradeyard SLA card, 2027; owned by the listing platform team in Yara Mendes's platform engineering organization):

Part	Value
Coverage	Listings cleared by the automated gate, 94% of new listings; held listings p90 under 30 minutes from submission to decision during staffed hours; held share published beside both
Clock	Platform latency, receipt of submission to searchable, for Tradeyard's own apps and web; feeds have their own card
Target	**p95 90 seconds by end of 2028**, from a 6-minute baseline; interim about 2 minutes by end of 2027 on Solr with an event-driven indexer
Budget (p95)	Synchronous path 2 s; outbox to topic 2 s; async checks and media 30 s (parallel); publication decision 1 s; indexing 15 s; result-page cache 20 s; slack 20 s

Part	Value
Removal	Search p95 10 s; every registered consumer p95 2 min; partner feeds within 15 min; nothing sent for a listing that ended before the send
Liquidity metric	Time to first contact, measured from the searchable event

The removal row is the one the Rotterdam bike tested: removed from search one minute after the sale, yet in 214 digests the next morning because the digest service was in no registry and re-checked nothing at send time.

Common mistakes.

- A single p95 that hides the held share. Tradeyard's six minutes said nothing about the 6% of listings in human review, whose queue reported backlog, not time.
- Budgets that do not add up. Stacked timers (a 60-second batch pull, a 2-minute indexer cycle, a 60-second commit) each add up to their full interval before any work is done.
- Removal objectives for the index only. A removal must reach every copy; a consumer that is not registered has no objective and will be the next Ghost Inventory incident.

Learn more: From Submitted to Searchable: The Publication Pipeline

Tool 7: Search Index Design Card

Purpose. The index design card records how one market's search index is built and operated: where each field comes from, which attributes become typed fields, how the index is partitioned, how fast writes and removals become visible, what the read path may spend, how the index is rebuilt, and who owns it. Its central rule is that **no field exists in the index without a source outside it**, which is the test against Search as System of Record.

Use it when you design a new market index, change mappings or analyzers, migrate an index to another engine or cluster, or plan capacity for a peak.

The template.

Section	Field	Guidance
Scope	Tenant(s), index name, engine and cluster	One index per brand and country is the typical default
Document	One document per listing, built from: event stream, read models, taxonomy and attribute services	List each field group's source; a field with no source is a finding
Field groups	Identity and version; lifecycle; taxonomy; typed attributes; long-tail attributes; price; location; text per language; ranking features; paid effects; embeddings	Ranking features and paid effects kept separately updatable
Promotion rule	Typed field only when the dictionary marks filter, facet or sort (Tool 4); the rest in one flattened field	Count typed fields against the engine's mapping limits
Analyzers	Per language; synonyms and stopwords owned by query understanding	Real-query test set per market; tokens compared before and after any change
Partitioning	Per country, language field, workload, time, vertical	State which queries cross the boundary
Write path	External version from `listing_version`; refresh interval; update-storm controls	Counters and high-churn features never on the text path
Freshness budget	Index share of the SLA card; removal lane	Removals on a priority lane with their own objective
Read path budget	p99 per step; facets on a separate request	The table that makes a new 80 ms step justify itself
Capacity	Engine requests per results page; peak sizing factor; campaign calendar	Size from engine requests, not user queries

Section	Field	Guidance
Reindex	Alias swap with live stream; validation (counts, samples, shadow queries); rollback window	Never at peak
Rebuild drill	Frequency; differences explained	A field that differs and cannot be explained exists only in the index
Owner	Platform team (engine, operations) and brand search team (relevance)	Two owners, one card

Veldmark example (Tradeyard Belgium market index on the shared platform, target design after the 14 September 2027 decision; migrates with Yardstick's search extraction in Q4 2028):

Field	Tradeyard BE
Scope	Tenant `ty:be`; Amazon OpenSearch Service behind Listing Search API v1; about 2.4 million live listings
Document	Built only from Listing event contract v1 plus the taxonomy and attribute service; `paid.*` fields from the effect projector
Text	One index, `title_nl` and `title_fr` with their own analyzers; a language detector at ingest picks the field
Promotion	Tradeyard's first group schema version promoted about 610 of 3,900 original keys to typed fields; the rest flattened
Write path	Event about 2 s, document build about 3 s, refresh 5 s; **index share 15 s p95**; removals **10 s p95** on a priority lane; view counters no longer written to the index
Read path	Search API p99 250 ms for the result list; facets on a separate request
Capacity	2.6 engine requests per results page; sized at 1.5× trailing peak
Reindex and drill	Throttled alias-swap reindex outside the published peak window; **quarterly rebuild drill** from the event stream
Analyzers	Ported Solr Dutch and French analyzers checked against a 5,000-query test set

Common mistakes.

- Writing into the live index at peak. The 17 January 2027 brown-out combined an in-place reindex, cache eviction on every commit and a TV campaign at the Sunday-evening peak.
- Letting the junkyard into the mappings. Free-string attributes turn one brand into three facet values; the index should see dictionary values only.
- Rebuilding effects by copying the old index. Paid fields and ranking features are re-projected from their ledgers and pipelines, never inherited.

Learn more: Search Architecture for Listings: Indices, Freshness and Scale

Tool 8: Trust-and-Safety Policy-to-Enforcement Card

Purpose. The card traces one policy from its text to every place it acts: the detectors that find violations, the rungs of the enforcement ladder it allows, the consumers that must apply a decision, the user notice and statement of reasons, the appeal path and the metrics that show it works. It treats the policy as versioned reference data, so a decision stays explainable under the version in force when it was taken.

Use it when a policy is written or changed, a new detector, rule or model is proposed, a new action (demotion, payment freeze) is considered, or a new tenant joins the shared platform. Legal reviews the policy and the user-facing reasons; engineering reviews the detectors; neither needs to read the other's code.

The template.

```
POLICY      <POL-ID> v<n>     effective <date>     supersedes v<n-1>
Owner:      <group policy lead>; local owners per country
Scope:      <listings | messages | accounts>; tenants; account types
Statement:  <one sentence a reviewer and a seller can both read>
Ground:     contractual (<terms section>) | illegal content (<law>, <country>)
DSA category: <transparency-database category>
DETECTORS   <rule / model / hash list>, version, owner, precision measured on
  ↪ <sample>
            Gate placement: synchronous (budget <ms>) | asynchronous
ACTIONS     Allowed rungs: warn | nudge | friction | demote | hold | remove |
            payment freeze | suspend; evidence threshold per rung
            Statement of reasons required for: <rungs>
PROPAGATION EnforcementApplied consumers and time to last acknowledgement:
            search | alerts | recommendations | messaging | SEO (410) |
            ads | payments | partner feeds | analytics
ROLLOUT     Change class: normal (release rings) | group-urgent (bypass rings)
            Tenants subscribed: <list>    Gaps: <tenants not yet on platform>
SHARING     Signals shared across brands: <hashed identifiers only>; DPO
  ↪ approval
```

```
            <date>; retention <period>
NOTICE      Template ID per locale; appeal route; reinstatement restores ranking
METRICS     Prevalence (blind sample), time to action, appeal overturn rate,
            reviewer agreement
```

Veldmark example (the payment-link phishing policy after 7 March 2027):

Field	Value
Policy	`POL-FRAUD-PAYLINK` **v3**, effective 18 March 2027 (the 7 March suspension cited v2)
Statement	Messages may not contain links or QR codes that imitate Tradeyard payment, shipping or Safe Deal pages
Ground and category	Contractual (terms section 7.2); in NL also an illegal-content ground (fraud); DSA category scams and fraud
Detectors	Domain-age rule, platform-brand-in-URL rule, QR decoder, message model v12; message safety gate at **150 ms p99**, fail-closed for links or payment details from accounts under 24 hours old
Allowed actions	Withhold message (silent to sender); suspend account; freeze Safe Deal payouts pending review
Propagation	`EnforcementApplied` to search, messaging (warn counterparties), alerts and payments, each acknowledging
Rollout	Group-urgent change class; on 7 March 2027 the rule reached BE only on Wednesday 10 March and never reached PL or PT
Sharing	Terminated fraud accounts' phone, e-mail, IBAN and device identifiers in the shared block list, hashed, 2-year retention, approved by the DPO
Metrics	Median time to action on fraud reports under 30 minutes by end of 2028 (baseline 3 h 10 min)

The card replays Femke's evening: report at 20:19, suspension at 20:26, then a rule that withheld 37 similar messages. The rollout row is the one the card exists for: the same phone number had registered a Motorvane private-seller account, and no field in the March process said who should tell Motorvane.

Common mistakes.

- Demoting silently. Under the Digital Services Act a demotion is a restriction that needs a statement of reasons; actor-trust down-ranking belongs behind the decision service, not inside a ranking feature.

- Treating a decision as finished when the status changes. Without per-consumer acknowledgements, nobody can prove when a takedown took effect in digests, ads and partner feeds.
- Rolling fraud rules out on the product release train. A scam rule is a group-urgent change; three days to reach Belgium is a design choice, not bad luck.

Learn more: Trust and Safety as a Platform

Tool 9: Entitlement Matrix

Purpose. The entitlement matrix lists, for every product and package a brand sells, who holds it, what quota and effects it grants, which enforcement points read those effects, how each point behaves when the entitlement service is unavailable, which ledger entries record its life, what evidence proves delivery, and what billing counts. It is the one page that lets a seller, a finance controller and an engineer agree on what was sold and what was delivered.

Use it when a product or package is launched, repriced or retired, a brand moves onto the shared catalog and entitlement service, a dispute or overbilling finding arrives, or a ranking or rendering change touches paid effects.

The template. One row per product version; the failure column is a commercial decision written as engineering.

Product version	Holder	Quota and validity	Effects (fields)	Enforcement points	Grant service unavailability	Ledger entries	Delivery evidence	Billing basis
`<product>`	Account / branch / listing	Quantity, window, overage rule	`paid.*` fields, account quota effects	Insertion, feed import, ranking, rendering, lead routing, pro tools, scheduler	Fail open / fail closed / queue and retry, per point	Grant, Consume, Release, Expire, Suspend, Revoke, Adjust, PublishRefused	Slots used, paid impressions, boosts executed, leads routed	Contract line, metered quantity from the ledger, one-off

Record beside the matrix: the price-list version and contract that produced each quote, the idempotency key of every producer, and the daily granted-delivered-billed reconciliation and its owner.

Veldmark example (Tally design, October 2027; rollout Motorvane Q1 2028, Tradeyard Q2 2028):

Product version	Holder	Quota and validity	Effects	Enforcement points (failure mode)	Delivery evidence	Billing basis
Motorvane `pro-150@v5`	Dealer account	150 slots, 20 boosts a month; €1,450 a month; overage €9 per car per month **only with consent**	Account quota effect	Feed import (queue and retry; every refused car a `PublishRefused` entry in the run report); scheduler (retry with the same key)	Slots consumed per day; boosts executed	Package fee plus overage from `Consume` entries, never from feed rows
Tradeyard Listing `top-listing-7d@v4`	Vehicle	7 days; extended automatically on underdelivery	`paid.top_eligible_until`, `paid.top_scope`, `paid.label`, `paid.product_version`	Ranking (serve last projected effects; served slots only above the relevance floor); rendering (never a paid style without a label)	`PaidImpression` events with positions and queries	One-off purchase; recognized over the display period

Product version	Holder	Quota and validity	Effects	Enforcement points (failure mode)	Delivery evidence	Billing basis
Tradeyard bump	Listing	One bump	`paid.bump_time` (moves `ranked_at`, never `created_at`)	Ranking (bounded recency term); insertion check-out (fail closed for new consumer charges)	Bump executed; first-page time per bump	One-off purchase

The first row is the overbilling finding of late September 2027 turned into a rule. Ruhrblick's August feed carried 163 cars, 150 went live, and €117 of overage was billed for 13 cars that were never published, because billing counted feed rows. The 14-month audit found 1,140 dealers overbilled by about €212,000, all credited in October 2027. Under the matrix, billing reads 150 `Consume` entries and bills €1,450.

Common mistakes.

- Paid state as a column in the listing payload. Then a feed, a moderator or a migration can grant visibility nobody sold; effects come only from the ledger, through the projector.
- A paid effect that outlives its listing. Sold, removed or blocked listings carry no `paid.*` fields, whatever the ledger says about remaining days; the projector consumes lifecycle events too.
- Billing that recomputes entitlements on its own. Every metered line comes from the ledger and metering, or the next audit finds another €212,000.

Learn more: Packages, Bumps and Quotas: Products, Pricing and Entitlements

Tool 10: Listing Event Specification

Purpose. A listing event specification defines one event in the listing event contract (see Every Change Is an Event: Event-Driven Listing Architecture): it states one listing fact in the group's published language, in a shape that search, alerts, trust and safety, monetization, SEO and the data platform can rely on for years. It sits beside the

registered schema and says what the event means, when it is emitted, what makes it unique and ordered, what it carries, who consumes it, how fast it must arrive and how it may change. The producer states what happened; each consumer decides what it means for its own copy.

Use it when a new event type is proposed for the listing stream, a field's meaning is in doubt, a consumer registers, or a breaking change is being considered. The default answer to a new type is no: every type is a cost for every consumer.

The template.

Section	Field	Guidance
Identity	`event_type`, contract version	Past tense, a fact (`ListingSold`), never an instruction
	Definition	One paragraph: the business moment and when it is emitted
Ownership	Producing context and team; catalog owner	The listing context owns lifecycle events; trust and safety owns enforcement events
Key and order	Message key; `listing_version` rule	Group listing ID as key; strictly increasing version; consumers apply only newer versions
Envelope	`event_id`, `event_type`, `listing_id`, `listing_version`, `occurred_at`, `published_at`, `tenant`, `cause`, `correlation_id`, `schema_version`, `state`, `change`	All mandatory; `cause` says why, not only what
State and change	Public state after the change; type-specific delta	Both, so a consumer that missed an event is repaired by the next one
Excluded data	Seller contact data, exact private locations, internal performance data	Fetched through access-controlled APIs at the moment of use

Section	Field	Guidance
Topics and retention	Events topic; compacted state topic; backfill topic	State the tombstone rule for closed and erased listings
Objectives	Commit to topic; topic to consumer	Removal events get their own, stricter objective
Consumers	Registered consumer groups and what each does	A consumer not in the registry has no objective
Compatibility	Registry mode; enumeration rule; semantic review	Registries check syntax, not meaning
Replay and dead letters	Backfill behavior; parking per key	Backfills carry current versions, so they never resurrect a sold listing
Change	Additive, new type with notice, breaking (v2 by parallel run)	

Veldmark example (Listing event contract v1, live 1 July 2027):

Field	**`ListingSold`** in **`listing.events.v1`**
Definition	The seller marked the listing sold, or a transaction completed
Producer	Tradeyard listing context via Debezium CDC and Yardstick Relay until Q2 2028; Motorvane by transactional outbox; Hearthly by Pub/Sub outbox and bridge
Key and version	`vm:tradeyard:nl:<id>`; the bike's sale was version 9
Change	Final price and channel: €225, Safe Deal; plus the outcome code (Tool 3)
Topics and retention	Events 14 days; compacted `listing.state.v1` keeps the final state 30 days, then a tombstone; erased listings tombstoned at once; corrections on `listing.backfill.v1`
Objectives	Removal events p99 under 5 s to topic and under 30 s to search deletion, alarmed per consumer; a synthetic listing created and removed every minute per tenant

Field	`ListingSold` in `listing.events.v1`
Consumers	Search removal, alert matcher and digests, SEO page state, recommendations, warehouse (23 consumer groups at go-live)
Compatibility	Avro, `FULL_TRANSITIVE` in Confluent Schema Registry; enumerations as validated strings; every change through the forum's one-week event-catalog review
Exclusions	No seller contact data; Keybridge neither publishes to nor consumes the stream

The digest that featured the sold bike read a MySQL replica snapshot. Under the specification it subscribes to `ListingSold`, purges the bike from assembled digests and re-checks state before sending.

Common mistakes.

- Putting bumps or views on the listing stream. A bump is `VisibilityChanged`; views and contacts are behavioral tracking events with their own volume and consent rules, and billions of them would delay the removals the stream exists to carry.
- Changing a field's meaning within a version. If `surface_m2` silently switches from total to habitable surface, every compatibility check passes and every valuation is wrong.
- Emitting no-op updates. Without Motorvane's content hash, about a million unchanged snapshot records a day would reach every consumer.

Learn more: Every Change Is an Event: Event-Driven Listing Architecture

Tool 11: Consolidation Decision Card

Purpose. The consolidation decision card records, for one acquired marketplace, brand or business, its **integration option** (hold separate, federate under group controls, share services beneath, replatform onto a sibling, merge brands, or divest), which capabilities move onto group platforms in which stage of the consolidation sequence and behind which gate, and which are deliberately left alone. It replaces one yes-or-no on "replatforming" with a staged, gated plan. Beside the plan it records a synergy ledger whose lines each have an owner, evidence and a dependency, the brand and URL risk, the constraints that bind the deal, and the exit path that keeps the business separable.

Use it when a deal closes or is being priced, a synergy target is announced, a brand or domain migration is proposed, or the next cost program proposes merging something that was kept separate on purpose. Public cases show why the constraints row matters:

the UK Competition and Markets Authority accepted undertakings on 2 June 2021 that required Gumtree UK and Shpock to be sold as part of Adevinta's acquisition of eBay Classifieds Group, with transitional services offered by the sellers (CMA decision, published 2021). And when eBay Kleinanzeigen became Kleinanzeigen on 16 May 2023, the press release said the previous domain would "remain accessible for several more years" (Adevinta press release, 2023): a brand migration is measured in years of redirects.

The template.

```
CONSOLIDATION CARD   <business>   Deal: <acquired / closed, date>   Owner: <name>
Tenant(s):        <codes>          Profiles: <ref to Tool 2>
Capability rows:  <ref to Tool 1 rows and verdicts>
INTEGRATION OPTION  hold separate | federate under group controls |
                  share services beneath | replatform onto a sibling |
                  merge brands | divest                    Reversibility: <high / low>
CONSTRAINTS       Regulatory conditions:   <remedies, undertakings, clearances>
                  Commercial commitments:  <neutrality, customer contracts>
                  Transitional services:   <what, from whom, until when>
                  Likely sale within 5 years? <yes / no: limits deep sharing>
SEQUENCE (stage, capability, target date, gate; migrate to the target, not to a sibling)
  0 Day-one controls       security, access, finance close, data-protection inventory
  1 Invisible/contractual  cloud accounts, observability, data platform, ads, supplier contracts
  2 Trust and identity     shared T&S pipeline, block lists, group identity
  3 Shared engines         media, search engine, taxonomy, catalog and entitlements, messaging
  4 Product platform       optional; its own investment case
  Kept separate:           <capability: reason>
SYNERGY LEDGER    Line | type (cost, revenue, dis-synergy, option cost) | planned run rate |
                  dated evidence | depends on (stage) | owner | status
                  Every revenue line names the data it uses and what permits that use
BRAND AND URLS    Rebrand? Domain change? Redirect plan and duration; organic-traffic
                  guardrail: next traffic-affecting change only within <x>% of baseline
PEOPLE            Teams that change owner; retention risks; decision rights
EXIT PATH         Export format, transitional run, switch-off per shared layer
REVIEW            Date or trigger
```

Veldmark example (Hearthly.es, the former Casavista, acquired 2025 and rebranded 2026; with Keybridge as the deliberate counter-case), as recorded in the consolidation review that went to the board on Tuesday 9 November 2027. Every decision below is

recorded in the chapter named in brackets; After the Deal: Consolidating Platforms After Acquisitions works the full case.

Field	Hearthly.es (`hl:es`)	Keybridge
Integration option	**Share services beneath**: stages 0–3; a stage-4 product platform not approved	**Federate under group controls**, permanently: finance, security operations, procurement and the annual neutrality review; nothing that touches agency data
Constraints	Spanish market rules and feed vendors profiled separately from France	March 2027 neutrality commitment: no Keybridge agency data used for Hearthly ranking or sales targeting; annual independent review
Stage 2: trust and identity	Moderation review tool onto the shared T&S platform in **Q1 2028**, brought forward after the July 2027 Barcelona ring (Scams, Fakes and Account Takeovers: Fraud Prevention for Listings); private landlords publishing rentals must reach L4 from Q4 2027; Casavista login **federated into Veldmark ID in 2028, migrated in 2029**, not moved onto Hearthly France's login first (Who Is on the Other Side? Identity and Verification)	**Kept separate**: outside Veldmark ID; not a T&S tenant and does not consume block lists

Field	Hearthly.es (`hl:es`)	Keybridge
Stage 3: shared engines	Listings and search moved onto Hearthly's Google Cloud platform in September 2026; shared OpenSearch engine by end of 2028, Hearthly last in the migration order, **ranking policy stays Hearthly's** (Search Architecture for Listings: Indices, Freshness and Scale); shared perceptual-hash index from Q1 2028 (Photos Sell: Media Pipelines for Listings); agency billing onto Tally in H2 2028 after a shadow billing cycle (Packages, Bumps and Quotas: Products, Pricing and Entitlements)	**Kept separate**: own billing through 2028; Keybridge-handled media excluded from the hash index
Data	BigQuery through 2028, publishing governed data products under group contracts (Data Platform for Marketplaces: Events, Lakehouse and Data Products)	Synapse not federated; only contract-named aggregates leave

Field	Hearthly.es (`hl:es`)	Keybridge
Stage 4: listing UX and supply	A single France-plus-Spain product platform **not approved** until Spanish feed vendors, disclosures and pricing norms are profiled against the French ones (AVIV Group: Building a European Real-Estate Platform from Acquisitions)	Integrates with Hearthly only through the public supply contract (Tool 5)
Synergy ledger	Deal case €18M run rate by end of 2027 (€8M platform and hosting, €4M overheads, €6M revenue); October 2027 evidence €9.6M (technology €3.9M, overheads €3.6M, revenue €2.1M); one-off integration cost €11M against €6M planned; dis-synergies recorded: about €1.3M of 2026 revenue lost to the traffic dips (estimate), €2.6M a year of double running	€7M a year of cross-selling from agency data **withdrawn** (forbidden by the commitment); €1.9M of neutral synergies (finance, security operations, procurement) added

Field	Hearthly.es (`hl:es`)	Keybridge
Brand and URLs	Rebrand on Monday 9 March 2026: organic sessions −19% in six weeks, −5% by September 2026; listings and search moved six months later, before traffic had recovered. Rule since: a second traffic-affecting change starts only when organic sessions are within 3% of baseline	No rebrand; Keybridge keeps its own brand
Hosting and exit path	Casavista hosting switched off at the **end of 2029**, after the login migration; tenant exit path documented for every shared layer (Tenancy model v1, decision 9)	Separate tenancy; the group's most separable business, with a separation cost close to zero

The two columns are the point of the card. The same group, in the same year, decided to share almost every layer beneath one acquired portal and almost none beneath another, and each decision has its reason in the constraints row.

Common mistakes.

- Consolidating the visible layer first. Listing UX and search relevance are where traffic and seller trust live; move what users cannot see and what gains value from pooling (controls, data, ads, trust and identity) first, as Casavista's inverted sequence showed.
- A synergy case with one number. Each ledger line needs an owner, dated evidence and the stage it depends on; dis-synergies (double running, traffic loss during a URL migration) and the separation cost if the business is sold belong on the same page.
- Forgetting separability. A private-equity owner may sell any brand; a shared layer without an exit path turns a future sale into a multi-year carve-out.

Learn more: After the Deal: Consolidating Platforms After Acquisitions

Tool 12: Architecture Decision Record

Purpose. An architecture decision record (ADR) captures one decision, the options rejected and its consequences, in about a page. A marketplace group needs five fields that common formats lack: the **tenants and brands** affected, the **principles** the decision was tested against, the **liquidity metric** it protects, the **advice received** under the advice process, and the **separability** of what it shares. Records are indexed by the architecture advice forum's decision log and cited by every other tool.

Use it when a decision crosses a team, brand or tenant boundary, changes a contract or a system of record, moves money, people or personal data between brands, or makes an exception to a principle or standard. Team-local decisions may use a lighter record in the same register.

The template. Builds on Nygard's 2011 record and Harmel-Law's advice process, as Who Owns What: Organizing Marketplace Engineering explains.

```
# DR-YYYY-NNN — <title that states the decision, not the topic>

Status: Proposed | Accepted <date> | Superseded by DR-... | Rejected
Proposer: <name, team>     Decides: <team | Group CTO | Group DPO> per decision-rights matrix
Tenants affected: <codes>  Capability rows: <Tool 1 ids>   First drafted: <date>

## Context        Facts, constraints, deadlines; the incident or finding behind it.
## Drivers        What a good answer must achieve; the liquidity metric it protects.
## Options        At least two real options plus "do nothing".
                  | Option | For | Against | Cost | Reversibility |
## Decision       "We will ...", with scope, tenants and effective date.
## Conditions     What must be true for the decision to hold (contracts first, gates).
## Principles     Tested against: share platforms not products; one owner per context;
                  contracts before migrations; separable by tenant. Exceptions recorded.
## Consequences   Better: ...  Worse: ...  New work carried by: ...
## Data and regulation  Personal data crossing brands? DSA, DAC7 or vertical rules touched?
## Advice received  Who said what; followed or not, and why.
## Confirmation   How conformance is checked (contract tests, drills, monitors).
## Review         Date, or the trigger that reopens it.
```

Veldmark example. *The group search platform runs OpenSearch for Tradeyard and Hearthly behind one search contract; Motorvane stays on Elasticsearch.* Accepted 14

September 2027, proposed by Tomasz Wiśniewski's group search platform team.

Section	Content
Context	Three engines; Tradeyard's Solr tied to Yardstick; 6-minute p95 to searchable; the 17 January 2027 brown-out
Options	A keep three engines; B Elasticsearch on Elastic Cloud for all; **C OpenSearch on Amazon OpenSearch Service for Tradeyard and Hearthly, Motorvane stays on Elasticsearch**; D Vespa for all
Decision	Option C; Tradeyard PT first (Q2 2028, four weeks of shadow traffic), then PL, then BE and NL with Yardstick's search extraction (Q4 2028), Hearthly last
Conditions	Every product application on **Listing Search API v1** before its engine changes; documents built only from Listing event contract v1 and the taxonomy and attribute service; quarterly rebuild drill
Principles	Contracts before migrations; share platforms, not products (ranking stays with the brands)
Consequences	Two engines remain; Hearthly calls the platform across clouds until it leaves Google Cloud
Advice received	Motorvane's case to stay on Elasticsearch, where its search scored maturity 4, was followed
Review	Engine review in 2029; Vespa reopened if Tradeyard's update storms or ranking depth miss their budgets on OpenSearch

The forum logged 74 cross-brand records by 31 December 2027, with a median of nine working days from first draft to decision; six went to the CTO because they moved funding or people, and two to the data protection officer because they moved personal data across brands.

Common mistakes.

- A record with one option. It is a justification, not a decision; include "do nothing" and the cheap answers so the record shows why they lost.
- No review trigger. The Vespa clause is what keeps a deferred option from being re-argued every quarter, or forgotten.
- Editing an accepted record instead of superseding it. Three of Veldmark's 2027 records were superseded within the year, and the register says why; an edited record says nothing.

Learn more: Who Owns What: Organizing Marketplace Engineering

Using the Toolkit Over Time

The templates stay true only when they are revisited, and three habits make that happen.

- **Review them on a calendar, not ad hoc.** The capability map and vertical profiles before annual planning and after every acquisition; decision records on their review dates and triggers; SLA cards and index design cards before the peak and campaign calendar's busiest weeks and after every rebuild drill; feed contracts at each partner's schema retirement notice; the entitlement matrix before every price event; policy cards whenever a transparency report shows a category moving.
- **Let incidents add fields.** Veldmark's mass-change guard came from 38,400 vanished cars, the send-time re-check from one sold bike in 214 digests, the `PublishRefused` entry from €212,000 of credits and the group-urgent change class from a rule that took three days to cross a border. When a post-incident review finds a gap, the fix usually includes a new field in one of these templates.
- **Keep the set small.** Every field should change a decision or preserve evidence. When a field is filled in by habit and read by nobody, remove it; when a review keeps asking a question no field answers, add it.

Built to Be Replaced: Evolving a Marketplace Platform Over a Decade explains how fitness functions for freshness, duplicates, time to contact and cost keep these records honest over a decade; Glossary: The Language of Listing Systems defines the terms the templates use, and Bibliography: Sources and Evidence lists the sources behind them.

Sources and Evidence

The templates are derived from the chapters linked in each tool, which cite their sources in full. The external formats and documented cases they build on:

Company engineering and investor sources

- Adevinta's eBay Kleinanzeigen is now Kleinanzeigen, Adevinta, 2023 — rebrand date and the previous domain kept accessible for years, cited in Tool 11.
- eBay Taxonomy API specification, eBay, n.d. — aspect constraints (mode, cardinality, applicability, expected-required date) behind Tool 4.
- Product Type Definitions use-case guide, Amazon Selling Partner API, updated 2026 — JSON Schema–based attribute requirements behind Tool 4.

Standards, regulation and documentation

- Decision that undertakings in lieu may be acceptable: Adevinta / eBay Classifieds Group, UK Competition and Markets Authority, 2021 — divestment remedies and transitional services, cited in Tool 11.

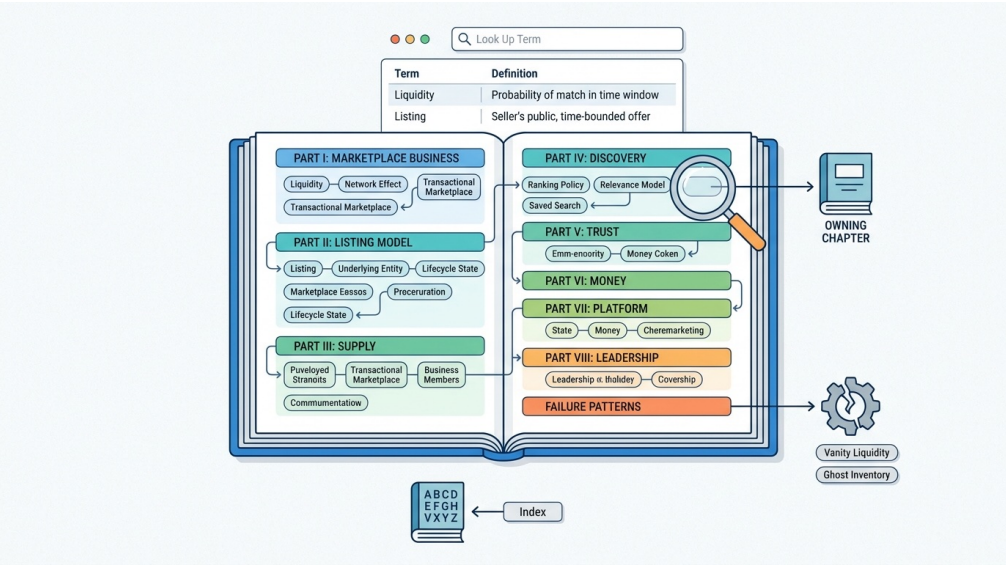
- Regulation (EU) 2022/2065 (Digital Services Act), European Union, 2022 — statements of reasons for removal and demotion (Article 17), used in Tools 3 and 8.
- RFC 9457: Problem Details for HTTP APIs, IETF, 2023 — the error envelope in Tool 5.
- Schema Evolution and Compatibility, Confluent documentation, n.d. — compatibility modes behind Tools 4 and 10.
- Mapping limit settings, Elastic documentation, n.d. — field-count limits behind the promotion rule in Tool 7.

Research and practitioner sources

- Documenting Architecture Decisions, Michael Nygard, 2011 — the record anatomy Tool 12 extends.
- Scaling the Practice of Architecture, Conversationally, Andrew Harmel-Law, martinfowler.com, 2021 — the advice process and advice-received field in Tool 12.
- Core Domain Charts, DDD Crew, n.d. — separating business differentiation from model complexity, behind the scorer rule in Tool 1.
- Idempotent requests, Stripe documentation, n.d. — the idempotency principle behind the ledger entries in Tool 9.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.

Glossary: The Language of Listing Systems



IN THIS SECTION, YOU WILL: Find the book's vocabulary in one place: read a plain-language definition grouped with the other terms of the same part, follow the link to the chapter that owns the term, or look a term or abbreviation up in the alphabetical index at the end.

KEY POINTS:

- **Look a term up, then read its neighbors.** Each section collects one part's terms, so the rows around a term carry the contrasts that cause real design errors: listing versus underlying entity, publication state versus enforcement overlay, full snapshot versus delta feed, contact versus conversation versus lead, entitlement versus effect, system of record versus read model.
- **One term, one meaning, one owning chapter.** Every row links to the chapter that owns or best explains the term, and the chapters are written to follow these definitions. When a chapter and this page disagree, one of them needs correcting, and the chapter's worked example is the better test of which.

- **Regulatory and company terms date quickly.** Regulations, standards, product names and their status are given as of 2026, following the sources cited in the chapters. Check the current source before a definition here goes into a contract, a policy or a design decision.

FAILURE PATTERNS THIS SECTION ADDRESSES:

- **Vanity Liquidity** — when "contact" means a message in one warehouse, a lead in another and a phone reveal in a third, the group steers on numbers that cannot be compared; an owned definition is the first step out.
- **The Attribute Junkyard** — the same failure at the level of words: one concept under several names, or one name for several concepts, until filters, feeds, models and teams disagree about what a listing is.

How to Use This Glossary

The sections follow the book's structure. The first collects the **evidence labels** every chapter uses and the **fifteen failure patterns**; the next eight follow **Parts I to VIII**, from the marketplace business through the listing model, supply, discovery, trust, money and the platform underneath to the Chief Architect's leadership agenda; the last collects the recurring **Veldmark names**. Part IX's case studies apply these terms rather than adding new ones. The **alphabetical index** at the end lists every term, abbreviation and common alias with a link to the section that defines it.

Each section opens with the chapters that own its terms, then a `| Term | Definition | Chapter |` table. The definitions are short teaching descriptions in plain language, not legal text, standards or vendor documentation; the chapter in the last column gives the full treatment, the sources and the worked example. Rows that combine two or more terms with a slash define all of them, because the difference between them is the point. Veldmark, named in some rows and in the last section, is the fictional marketplace group this book follows (Meet Veldmark: Four Brands, One Platform to Build); its numbers are illustrations, not facts about any real company.

Reading the Book: Evidence Labels and Failure Patterns

The book marks every statement as one of five kinds, and it names fifteen recurring ways listing systems fail. Introduction: Architecture for Liquidity explains the labels; Unlisted: Fifteen Ways Listing Systems Go Wrong tells each failure pattern in full, with

its warning signs, mechanism, cost and the chapters that address it. The last column below names the guide and the chapter that does most of the repair work.

Term	Definition	Chapter
Documented fact	A company's public statement, a standard, a regulation, a date or a figure from a filing, linked to its source with the year of publication. Company engineering evidence older than about three years may describe systems that have since been replaced.	Introduction: Architecture for Liquidity
Industry practice	What marketplaces commonly do, phrased as "commonly" or "a typical pattern", without invented percentages.	Introduction: Architecture for Liquidity
Reference-architecture inference	A plausible design derived from engineering principles. When it concerns a real company it is labeled "INFERENCE, NOT DOCUMENTED" and is never a description of that company's systems.	Introduction: Architecture for Liquidity
Recommendation (evidence label)	The author's architectural judgement, given with its reasoning and the conditions under which it does not hold.	Introduction: Architecture for Liquidity

Term	Definition	Chapter
Illustration	Anything about Veldmark, the fictional marketplace group the book follows. Its numbers are invented for teaching and agree with one shared scenario ledger.	Meet Veldmark: Four Brands, One Platform to Build
Failure pattern	A named, recurring way listing systems go wrong, with warning signs that can be checked in a review. The book uses fifteen.	Unlisted: Fifteen Ways Listing Systems Go Wrong
The Category Straitjacket	Adding, splitting or renaming a category needs a code release, a data migration and a cross-team meeting, because the tree is entangled with code.	Unlisted: Fifteen Ways Listing Systems Go Wrong, Category Trees That Can Change: Designing Marketplace Taxonomies
The Attribute Junkyard	Attributes are free-form strings with no dictionary, so filters, facets, feeds and trust rules disagree about what a listing is.	Unlisted: Fifteen Ways Listing Systems Go Wrong, Attributes Without Chaos: Schemas, Dictionaries and Validation
Ghost Inventory	Sold, expired or removed listings stay searchable, alertable, recommendable or contactable long after they are gone, usually because one copy never heard about the removal.	Unlisted: Fifteen Ways Listing Systems Go Wrong, From Submitted to Searchable: The Publication Pipeline
Seeing Double	The same item or property appears many times, from reposts, multiple feeds or multiple agencies, and crowds out everything else.	Unlisted: Fifteen Ways Listing Systems Go Wrong, One Property, Three Listings: Duplicates and Entity Resolution

Term	Definition	Chapter
Feed Roulette	Partner and professional feeds are ingested without contracts, validation or feedback, so every import is a surprise.	Unlisted: Fifteen Ways Listing Systems Go Wrong, Many Doors In: Forms, APIs and Bulk Feeds
Search as System of Record	The search index has quietly become the system of record, holding facts that exist nowhere else, so nobody can rebuild it.	Unlisted: Fifteen Ways Listing Systems Go Wrong, Search Architecture for Listings: Indices, Freshness and Scale
Multiplier Creep	Paid boosts and promotions accumulate in the ranking until relevance degrades and buyers stop trusting the order.	Unlisted: Fifteen Ways Listing Systems Go Wrong, Ranking for Two Sides: Relevance, Freshness and Paid Visibility
Moderation as an Afterthought	Trust and safety is bolted on after launch, as a queue that grows faster than the team that empties it.	Unlisted: Fifteen Ways Listing Systems Go Wrong, Trust and Safety as a Platform
Fast Mail for Fraudsters	Messaging and contact features ship without safety signals and become the main channel for fraud.	Unlisted: Fifteen Ways Listing Systems Go Wrong, From Interest to Contact: Messaging and Lead Management
The Copy-Paste Country	Each market runs its own copy of the codebase or reference data, and every improvement has to be built several times.	Unlisted: Fifteen Ways Listing Systems Go Wrong, One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture
Deal-Shaped Architecture	Acquired marketplaces keep their entire stacks indefinitely, and the group pays for every capability several times.	Unlisted: Fifteen Ways Listing Systems Go Wrong, After the Deal: Consolidating Platforms After Acquisitions

Term	Definition	Chapter
The Forced Fit	A central platform is imposed on markets it does not fit, and local teams route around it.	Unlisted: Fifteen Ways Listing Systems Go Wrong, Who Owns What: Organizing Marketplace Engineering
Vanity Liquidity	The marketplace measures listings and visits instead of matches, contacts and sales, and optimizes the wrong thing.	Unlisted: Fifteen Ways Listing Systems Go Wrong, Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics
Billing by Guesswork	What a seller has paid for is checked ad hoc in many services, so nobody can say what was sold versus what was delivered.	Unlisted: Fifteen Ways Listing Systems Go Wrong, Packages, Bumps and Quotas: Products, Pricing and Entitlements
Set-and-Forget AI	A ranking, pricing, valuation or fraud model makes consequential decisions without evaluation, monitoring or an owner.	Unlisted: Fifteen Ways Listing Systems Go Wrong, Models in Production: ML Platforms and Use Cases

Part I — Marketplace

The business vocabulary: what kind of marketplace this is, what it sells, how it is measured and how it is decomposed. Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy owns the business models; Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics owns liquidity and its metrics; From Newspaper Columns to Portfolios: How the Classifieds Industry Evolved owns the industry's history; Map What the Marketplace Does Before What Runs It and Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models own the two decompositions; Same Platform, Different Physics: How Listing Verticals Differ owns what differs between verticals.

Term	Definition	Chapter
Listing marketplace	An intermediary that publishes offers from many suppliers and helps buyers find them. Its architecture should be judged by how reliably it turns fresh, trustworthy supply into matches with demand.	Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy
Endpoint	The last step of the transaction for which the platform takes responsibility: a contact, a delivered lead, a completed payment and delivery, or a guaranteed outcome. The most useful single axis for classifying a marketplace.	Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy
Classifieds	A marketplace whose job ends when a buyer contacts a seller. The platform never sees the money, so it monetizes visibility and access: insertion fees, paid visibility, professional subscriptions and advertising.	Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy

Term	Definition	Chapter
Lead-generation marketplace	A marketplace (or portal) whose job ends when a qualified enquiry is delivered to a professional such as a dealer, estate agent or employer. The professional pays because the lead has value, so the platform must qualify, count, route and attribute leads.	Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy
Transactional marketplace	A marketplace whose job ends when payment and delivery are complete on the platform. Revenue becomes a take rate on gross merchandise value, and the platform takes on payments, disputes and buyer-protection losses.	Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy
Managed marketplace	A marketplace that takes inventory, quality or outcome risk itself, for example by buying the items it sells. It carries price and inventory risk on its balance sheet.	Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy

Term	Definition	Chapter
Horizontal / vertical marketplace	A horizontal marketplace lists many unrelated categories under one brand and one category tree; a vertical marketplace specializes in one domain, such as property, vehicles or jobs, with data, tools and monetization built for it.	Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy
C2C / B2C / B2B	Consumer to consumer, business to consumer and business to business: the seller-type axis, which decides supply channels, the account model, trader obligations and the unit of monetization.	Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy
Marketplace profile	The combination of answers on six axes (endpoint, scope, seller type, item identity, money flow, risk ownership) that describes one marketplace in one country and category group. A brand is not a profile: one brand can run several.	Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy
Gross merchandise value (GMV)	The total value of goods sold through a marketplace in a period. Only a transactional marketplace observes its true GMV; a classifieds site can at best estimate it.	How Listing Marketplaces Make Money

Term	Definition	Chapter
Two-sided market	A market in which a platform serves two distinct groups whose value depends on each other: buyers and sellers, renters and landlords, job seekers and employers.	Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics
Network effect	A participant's value from the platform depends on how many others use it. Cross-side effects link the two sides (more buyers attract sellers); same-side effects link participants on one side, and can be negative (more sellers compete for the same buyers).	Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics
Liquidity	The probability that, within a time window that matters to users, a listing receives a qualified contact and a buyer finds a suitable, available listing, measured for a given place and category. What the marketplace actually sells.	Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics
Cell (liquidity cell)	A geographic area that buyers will travel or search across, crossed with a category whose items substitute for each other. Liquidity is a property of a cell, not of a site, and leadership is won cell by cell.	Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics

Term	Definition	Chapter
Chicken-and-egg problem / seeding	Buyers do not come without supply and sellers do not list without buyers. Seeding breaks the loop with free private listings, professional feeds, one dense cell first, or distribution from an existing network.	Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics
Multi-homing	Participating in several competing platforms at once. Professional sellers multi-home cheaply through feeds; buyers multi-home at first and then concentrate on the site with the freshest, most complete supply.	Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics
Disintermediation (leakage)	Parties who met on the platform complete their deal, or their next one, outside it. It costs fees, hides outcomes and is the route scammers use to escape the platform's protections.	Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics, From Interest to Contact: Messaging and Lead Management
Liquidity metrics	The measurements that tell a marketplace whether matches are happening, per cell, fast enough: fresh available supply, early exposure, contact rate, sell-through, seller return rate, each paired with a counter-metric.	Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics

Term	Definition	Chapter
Time to first contact	The time from a listing becoming searchable (not from submission) to its first qualified contact. Measuring from searchable separates publication speed from demand.	Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics
Contact rate / sell-through	Contact rate is the share of new listings with at least one qualified contact within a window (48 hours for goods, longer for cars and property). Sell-through is the share of listings sold within N days, hard to measure because most sellers never mark an item sold.	Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics
Counter-metric / ghost share	A counter-metric guards a liquidity metric against dishonest improvement. Ghost share, the share of contacts and alerts that reach listings already sold or removed, is the counter-metric for supply.	Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics

Term	Definition	Chapter
Portfolio model	A marketplace group's strategy of owning many local marketplaces, usually one leader per country and vertical, bought or launched one by one, and sharing some capabilities beneath their brands. The group's value is the sum of its local leaderships plus whatever the shared layer saves or enables, which is why groups inherit several stacks of everything.	From Newspaper Columns to Portfolios: How the Classifieds Industry Evolved
Copy-paste market entry	Launching a marketplace in a new country by deploying a copy of an existing marketplace's codebase and operating model, then localizing it; widely associated with Schibsted's international expansion. Fast to start and expensive later, because the copies diverge: the classic origin of the Copy-Paste Country.	From Newspaper Columns to Portfolios: How the Classifieds Industry Evolved, Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform

Term	Definition	Chapter
Carve-out / transitional services agreement (TSA)	A carve-out separates a business from its group for sale; the TSA is the contract under which the seller keeps providing shared services (hosting, platforms, back office) to the sold business for a period. Every shared layer becomes a line in it.	From Newspaper Columns to Portfolios: How the Classifieds Industry Evolved, After the Deal: Consolidating Platforms After Acquisitions
Business capability	What the business does, described independently of how, where and by whom: "Saved searches and alerts", not "the alert cron". Capability labels contain no system, vendor, team or channel names.	Map What the Marketplace Does Before What Runs It
Capability map	A stable, implementation-free hierarchy of capabilities, comparable across brands. The book's marketplace map has ten level-1 areas: Supply Acquisition, Listing Management, Discovery, Matching and Contact, Trust and Safety, Transactions, Monetization, Professional Services, Data and Insights, and Platform.	Map What the Marketplace Does Before What Runs It

Term	Definition	Chapter
Value stream	The end-to-end sequence of stages that delivers an outcome to a customer: list-to-sell, search-to-contact, lead-to-deal and professional onboarding. Streams carry outcome metrics; capabilities carry activity.	Map What the Marketplace Does Before What Runs It
Heat map (differentiation, maturity, duplication)	Scoring each capability for how much it makes customers choose us (scored by product), how well it works today with dated evidence (scored by engineering) and how many implementations the group runs (counted by architecture). The combination suggests consolidate, promote, buy, split, invest, protect or keep separate.	Map What the Marketplace Does Before What Runs It
Core / supporting / generic subdomain	Core subdomains are where the business competes and the best people should work; supporting subdomains are specific and necessary but not where it wins; generic subdomains are needed by every business. Discovery is core everywhere; lead management is core in verticals.	Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models

Term	Definition	Chapter
Bounded context	The boundary within which one model and its vocabulary apply consistently. In a marketplace "the listing" is a different model in feed ingestion, the catalog, search, trust and safety, monetization and analytics.	Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models
Ubiquitous language	The vocabulary used consistently inside one bounded context, in conversation, documents and code.	Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models
Context map	A map of the bounded contexts and the relationships between them: who is upstream, who adapts to whom, and where translation happens.	Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models
Partnership / shared kernel / customer–supplier / conformist	Context relationships: two teams planning an integration together; a small model two teams co-own (listing identifiers and tenant codes); an upstream that plans for its downstream's needs; a downstream that adopts the upstream model as it is (a checkout conforming to its payment provider).	Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models

Term	Definition	Chapter
Anti-corruption layer (ACL)	A translator a downstream context builds so that a foreign model does not leak into its own: feed ingestion in front of 160 vendors' formats, or a translator in front of a monolith.	Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models
Open host service / published language	An open host service offers one well-defined protocol to many consumers; a published language is the documented shared vocabulary it speaks. The listing event contract is the marketplace's main published language.	Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models
Aggregate	A cluster of domain objects treated as one unit for changes, with one root and one owner; transactions do not cross aggregate boundaries.	Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models, The Listing Aggregate: Identity, Lifecycle and Invariants
Domain event	A past-tense, immutable statement in business language that something meaningful happened in a context, such as `ListingSold`, published so other contexts can react.	Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models, Every Change Is an Event: Event-Driven Listing Architecture

Term	Definition	Chapter
Vertical requirement profile	A one-page description of a vertical's physics, written with numbers, against which architecture decisions can be checked. It covers fourteen dimensions, including supply mix, item identity, listing lifetime, attribute density, decision horizon, lead value, monetization unit, regulatory load and fraud typology.	Same Platform, Different Physics: How Listing Verticals Differ
Shared-versus-specific decision	Placing a capability, or part of one, in one of four layers with stated reasons: shared platform; shared engine with vertical content; vertical capability, group-owned; or brand-local product. "The machinery is generic, the content is specific."	Same Platform, Different Physics: How Listing Verticals Differ
External system of record	Where the authoritative listing lives outside the marketplace, in a dealer management system, an agency CRM or a multiple-listing service, so the marketplace holds a mirror and the source wins.	Same Platform, Different Physics: How Listing Verticals Differ, Many Doors In: Forms, APIs and Bulk Feeds

Part II — Listing

The domain model of the thing being sold. The Listing Aggregate: Identity, Lifecycle and Invariants owns the listing, its identity and lifecycle; Category Trees That Can Change: Designing Marketplace Taxonomies and Attributes Without Chaos: Schemas, Dictionaries and Validation own classification and structured facts; Photos Sell: Media Pipelines for Listings and Where Is It? Location, Geocoding and Privacy own photos and place; Private Sellers, Dealers and Agencies: Modeling Who Lists owns who lists.

Term	Definition	Chapter
Listing	A seller's public, time-bounded offer about a specific thing: an item to sell, a vehicle, a property to sell or let, a job to fill. It references the underlying thing; it is never the thing itself.	The Listing Aggregate: Identity, Lifecycle and Invariants
Listing aggregate	The listing treated as one unit of consistency: identity, owner, category, attributes, price, location, media, contact options and state, with one owning service and a set of invariants that hold after every change. Paid visibility and moderation decisions are held as references, not as flags.	The Listing Aggregate: Identity, Lifecycle and Invariants
Underlying entity	The physical or legal thing an offer is about, such as a specific apartment or a specific car, identified by VIN, address or parcel where one exists. It outlives every listing of it.	The Listing Aggregate: Identity, Lifecycle and Invariants, One Property, Three Listings: Duplicates and Entity Resolution

Term	Definition	Chapter
Listing ID / public ID	The listing ID is the aggregate's internal identity: globally unique across brands and countries, never reused, encoding nothing that can change. The public ID is the short, stable identifier shown in URLs and support conversations, mapped one-to-one to it.	The Listing Aggregate: Identity, Lifecycle and Invariants
External key / request key / entity reference	Three further keys: the partner's own reference, unique per integration and the idempotency key for feeds; a short-lived key that stops a flaky mobile connection creating a listing twice; and an optional link to the canonical property, vehicle or product.	The Listing Aggregate: Identity, Lifecycle and Invariants
Relisting	Putting an item on the market again. Renewals and a sale falling through keep the same listing ID; a delete-and-repost or a property returning a year later gets a new ID, linked to the old one through entity resolution.	The Listing Aggregate: Identity, Lifecycle and Invariants

Term	Definition	Chapter
Invariant / policy	An invariant is a rule that holds after every successful command on the aggregate, such as classification in a valid leaf. A policy is a rule enforced elsewhere or later, such as the seller's slot quota or same-seller duplicates.	The Listing Aggregate: Identity, Lifecycle and Invariants
Listing lifecycle	The states a listing can be in, the transitions between them, who may trigger each, and what each state means for visibility, contact and retention. Seller intent, the platform clock and trust and safety change it for different reasons.	The Listing Aggregate: Identity, Lifecycle and Invariants
Publication state / enforcement overlay	The book models the lifecycle as two dimensions: a publication state owned by the listing (draft, pending review, active, paused, reserved, sold, expired, withdrawn, archived) and an enforcement overlay owned by trust and safety (none, held, restricted, blocked, restored). Visibility, contactability and alerting are derived from both, never stored separately.	The Listing Aggregate: Identity, Lifecycle and Invariants

Term	Definition	Chapter
Outcome record	The reason attached to every terminal transition: sold on platform, sold elsewhere, sold through a contact here but paid off-platform, no longer available, or unknown. Without it, sell-through cannot be measured.	The Listing Aggregate: Identity, Lifecycle and Invariants
Listing version	An immutable snapshot of the aggregate's seller-visible content after one accepted change, numbered with an integer that increases per listing. It orders events, guards concurrent writes, preserves evidence and drives re-moderation.	The Listing Aggregate: Identity, Lifecycle and Invariants
Material change	An edit that alters what was approved (category, price beyond a threshold, images, title, payment terms or contact details) and must therefore pass the moderation gate again.	The Listing Aggregate: Identity, Lifecycle and Invariants, From Submitted to Searchable: The Publication Pipeline
Sold-to-gone latency	The time each consumer of the listing stream takes to remove a listing after it leaves the active state, declared in that consumer's contract and monitored per consumer.	The Listing Aggregate: Identity, Lifecycle and Invariants

Term	Definition	Chapter
Category taxonomy	A controlled hierarchy of classes into which every listing is placed, with the rules that hang off each class. The category selects the contract a listing must meet: schema, fees, moderation rules, facets and ad segments.	Category Trees That Can Change: Designing Marketplace Taxonomies
Strict tree / polyhierarchy / faceted classification	A strict tree gives each node one parent; a polyhierarchy allows several; faceted classification avoids deep trees by classifying along independent dimensions. The recommended default is a strict, leaf-only classification tree, with polyhierarchy only in navigation.	Category Trees That Can Change: Designing Marketplace Taxonomies
Classification tree / navigation tree	The classification tree is canonical, strict and leaf-only, with stable IDs; navigation trees are derived per brand and country, with aliases, curated groupings and localized labels and slugs that are properties of a node, never its identity.	Category Trees That Can Change: Designing Marketplace Taxonomies

Term	Definition	Chapter
Canonical tree with country views	One canonical set of nodes with stable IDs, of which each country enables a subset with its own labels, slugs and navigation. It shares meaning where it exists and keeps local fit where it does not.	Category Trees That Can Change: Designing Marketplace Taxonomies
Reference data	Data that other data points at to give it meaning, that changes rarely and must be identical wherever it is used: currencies, country codes, categories, vehicle makes and models, energy classes.	Category Trees That Can Change: Designing Marketplace Taxonomies, Attributes Without Chaos: Schemas, Dictionaries and Validation
Taxonomy versioning	Publishing the taxonomy as immutable, numbered releases with stable, meaningless node IDs, each release shipping its migration; listings and saved searches record the version they were classified under.	Category Trees That Can Change: Designing Marketplace Taxonomies
Category migration	Any taxonomy change that alters which node an existing listing, saved search, URL, rule or metric points at: rename, move, merge, split, retire or add. Merges are deterministic; splits are lossy and need reclassification.	Category Trees That Can Change: Designing Marketplace Taxonomies

Term	Definition	Chapter
Category prediction	Suggesting or assigning a listing's category from its title, description, photos or structured data. Best used to suggest the top three, taxonomy-version-aware, and measured by seller acceptance and later correction.	Category Trees That Can Change: Designing Marketplace Taxonomies
Attribute	A structured, typed fact about the item, such as frame size, mileage or energy class, that drives filters, facets, sorting, ranking, validation and outbound feeds.	Attributes Without Chaos: Schemas, Dictionaries and Validation
Attribute definition / attribute dictionary	A definition states what a fact means: stable key, data type, canonical unit, allowed values, cardinality, scope, editability, purposes, sensitivity, mappings, owner and status. The dictionary is the governed collection of definitions.	Attributes Without Chaos: Schemas, Dictionaries and Validation
Category schema	The binding of attribute definitions to a category: which attributes a listing must, should or may carry, which values apply, which validation rules run and how the form shows them. Schema versions are immutable.	Attributes Without Chaos: Schemas, Dictionaries and Validation

Term	Definition	Chapter
Required / recommended / optional	Three levels of requiredness. Required blocks publication; recommended is requested and scored but does not block; new mandatory attributes start as recommended with an effective date and are enforced only on create and revise.	Attributes Without Chaos: Schemas, Dictionaries and Validation
Product attribute / instance attribute	A product attribute belongs to the catalog type (the color of a phone model) and can be inherited from a matched product; an instance attribute belongs to this item (the scratches on this phone) and only the seller knows it.	Attributes Without Chaos: Schemas, Dictionaries and Validation
Entity–attribute–value (EAV)	A storage model of rows `(listing, attribute, value)` that needs no schema changes and many self-joins. It is where junkyards grow fastest, because a new attribute is an insert no reviewer sees. Alternatives are wide tables, schema-validated documents and typed per-vertical models.	Attributes Without Chaos: Schemas, Dictionaries and Validation

Term	Definition	Chapter
Hybrid attribute storage	The recommended model for a group: typed core fields on the aggregate, schema-validated documents for long-tail attributes, and typed vertical models where a vertical's depth justifies them.	Attributes Without Chaos: Schemas, Dictionaries and Validation
Schema compiler / validator library	Treating schemas as source code: a compiler turns the dictionary into JSON Schema documents, form definitions, index mappings and feed mappings per release, and one validator library embedded in every write path makes all channels validate with the same code.	Attributes Without Chaos: Schemas, Dictionaries and Validation
Backward / forward / full compatibility	Schema-evolution rules: consumers on the new schema can read old data; consumers on the old schema can read new data; or both. In attribute schemas a change of type, unit or meaning is a new attribute, never an edit.	Attributes Without Chaos: Schemas, Dictionaries and Validation, Every Change Is an Event: Event-Driven Listing Architecture

Term	Definition	Chapter
Canonical unit	The single unit each attribute is stored in (centimeters, kilometers, square meters, minor currency units), with the value as entered kept beside it for display, audit and disputes.	Attributes Without Chaos: Schemas, Dictionaries and Validation
Energy performance certificate (DPE)	France's *diagnostic de performance énergétique*: an energy and climate class with an estimated cost range, mandatory in French residential listings, carried with its date and method because a recalculation can change the class.	Attributes Without Chaos: Schemas, Dictionaries and Validation, Rules of the Marketplace: DSA, DAC7 and Vertical Regulation
Media asset	One piece of listing media: a photo, video, floor plan, 3D tour reference or document. A listing stores media IDs and an order, never URLs.	Photos Sell: Media Pipelines for Listings
Original / master / derivative	The original is the file as received; the master is the sanitized version the platform keeps (re-encoded, oriented, metadata stripped); a derivative or rendition is a version of the master at a given size, crop, format and quality.	Photos Sell: Media Pipelines for Listings

Term	Definition	Chapter
Image service	A service that generates derivatives on request from a URL encoding the transformation, behind a content delivery network (CDN). Its variant space should be bounded and its URLs signed and immutable.	Photos Sell: Media Pipelines for Listings
Media pipeline	The end-to-end system for listing media: upload or fetch, validation, sanitization, hashing, classification and moderation, master storage, derivative delivery, lifecycle and deletion.	Photos Sell: Media Pipelines for Listings
Perceptual hash	A short fingerprint computed from an image's visual content, so re-compressed, resized or lightly edited copies produce similar fingerprints. PDQ is a widely used open algorithm; embeddings catch heavier edits.	Photos Sell: Media Pipelines for Listings
EXIF stripping	Removing camera metadata, including GPS coordinates, from uploaded images before they are stored or served, because a phone photo can reveal a seller's home.	Photos Sell: Media Pipelines for Listings

Term	Definition	Chapter
Media provenance	Recording whether an image is a photograph, an edited photograph or generated, so that virtual staging and AI-generated media can be labeled; Article 50 of the EU AI Act requires marking of synthetic content from 2 August 2026.	Photos Sell: Media Pipelines for Listings
Item / seller / handover location; service area	Four different locations: where the item is, where the seller is based, where buyer and seller meet, and the region a seller serves (for services and jobs, where a point is the wrong shape).	Where Is It? Location, Geocoding and Privacy
Geocoding / reverse geocoding	Geocoding turns an address or place name into coordinates; reverse geocoding turns coordinates into an address or place. Every result should carry its match level, confidence and provenance.	Where Is It? Location, Geocoding and Privacy
Gazetteer	A reference dataset of named places with stable identifiers, names in several languages, geometries and parent-child relations. It holds administrative places from official sources and marketplace places curated for how buyers search.	Where Is It? Location, Geocoding and Privacy

Term	Definition	Chapter
Place ID	A stable identifier for an administrative or named area (country, region, city, district, neighborhood); listings carry the place IDs that contain them, so named-area search is a term filter.	Where Is It? Location, Geocoding and Privacy, Searching by Place: Map, Polygon and Commute Search
Location precision	The level of detail at which a location is known, stored or shown, applied three times: capture precision, stored precision and display precision per audience.	Where Is It? Location, Geocoding and Privacy
Precise location / public point	The precise point is restricted data (for private sellers, personal data) kept in a separate store; the public point is the coarser coordinate a listing may expose, derived by snapping to a grid cell or by a deterministic, seeded offset that cannot be averaged away.	Where Is It? Location, Geocoding and Privacy
Geo index	A data structure that makes spatial queries cheap: points in a circle, points in a polygon, counts per map tile. Search engines index `geo_point` and `geo_shape` fields natively; grids such as geohash, H3 and S2 give cell keys.	Where Is It? Location, Geocoding and Privacy

Term	Definition	Chapter
Geohash / H3 / S2	Three cell systems: geohash is a base-32 string interleaving latitude and longitude bits; H3 is Uber's hexagonal hierarchical grid with 16 resolutions; S2 is Google's library that orders cells on a sphere along a Hilbert curve.	Where Is It? Location, Geocoding and Privacy
Private / professional seller	A private seller sells their own goods outside any trade or business; a professional seller (dealer, agency, merchant, developer, employer) sells in the course of a business.	Private Sellers, Dealers and Agencies: Modeling Who Lists
Trader	In EU consumer law, a person or company acting for purposes of their trade, business, craft or profession. Whether a seller is a trader decides which consumer rules apply and what the platform must collect and show.	Private Sellers, Dealers and Agencies: Modeling Who Lists, Rules of the Marketplace: DSA, DAC7 and Vertical Regulation
Trader status	The platform's record of whether a seller account acts as a trader, held as a declaration, evidence, a verification result and the platform's own assessment, with a reclassification process when signals contradict the declaration.	Private Sellers, Dealers and Agencies: Modeling Who Lists

Term	Definition	Chapter
Trader traceability	The Digital Services Act duty on online platforms that let consumers conclude distance contracts with traders: collect and check the trader's identity, contact, register and payment details before the trader can offer, and keep them for six months after the relationship ends.	Private Sellers, Dealers and Agencies: Modeling Who Lists, Rules of the Marketplace: DSA, DAC7 and Vertical Regulation
Organization hierarchy	The structure of professional supply: legal entities that sell, branches (physical places of business with their own stock and lead routing), networks or franchises above several organizations, and storefronts shown to buyers.	Private Sellers, Dealers and Agencies: Modeling Who Lists
Integration principal	A software client acting on behalf of sellers, such as a real-estate CRM, a dealer management system vendor or a feed aggregator, authorized per seller, branch and channel.	Private Sellers, Dealers and Agencies: Modeling Who Lists

Term	Definition	Chapter
Seller account model	The separation of who acts (user), who is responsible (legal entity), who sells on this brand (seller account), where (branch), who pays (contract and billing account), who is shown (storefront) and which software acts (integration principal).	Private Sellers, Dealers and Agencies: Modeling Who Lists

Part III — Supply

How inventory gets in and becomes searchable. Many Doors In: Forms, APIs and Bulk Feeds owns channels and feed contracts; From Raw Supply to Usable Listings: Normalization and Enrichment owns canonical values and provenance; One Property, Three Listings: Duplicates and Entity Resolution owns duplicates and canonical entities; From Submitted to Searchable: The Publication Pipeline owns freshness and removal; Serving the Pros: Dealer, Agent and Merchant Tooling owns what professionals use.

Term	Definition	Chapter
Ingestion channel	A distinct path by which listings enter, defined by who uses it, how data moves and its rhythm: the insertion flow, a partner API, a bulk feed, or replication from an external system of record. Every channel should end in one write path.	Many Doors In: Forms, APIs and Bulk Feeds

Term	Definition	Chapter
Insertion flow	The web and app forms through which private sellers and small professionals create listings, one at a time, with server-side drafts. It is a conversion funnel whose real metric is published-and-contacted, not form completion.	Many Doors In: Forms, APIs and Bulk Feeds
Partner API / bulk feed	A partner API takes synchronous per-listing writes from software vendors and large professionals; a bulk feed takes files pushed to the marketplace or pulled from a partner on a schedule, hourly to weekly.	Many Doors In: Forms, APIs and Bulk Feeds
Multiple-listing service (MLS)	A cooperative database of property listings shared among brokers, mainly in North America; portals downstream of it replicate a licensed copy rather than receive listings from sellers.	Many Doors In: Forms, APIs and Bulk Feeds
Feed contract	The agreement that governs a channel: scope and tenancy, identity, snapshot or delta semantics, field ownership, schema and version, timeliness, media, errors and feedback, limits, reconciliation and change management.	Many Doors In: Forms, APIs and Bulk Feeds

Term	Definition	Chapter
Full snapshot / delta feed	A full snapshot sends a partner's complete current inventory every time, so absence means removal; a delta feed sends only creates, updates and explicit deletes, so a lost delete leaves a ghost unless the sides reconcile on a schedule.	Many Doors In: Forms, APIs and Bulk Feeds
Ingestion record	The key internal abstraction of ingestion: one durable, append-only record per partner listing per send, with partner, seller, channel, external key, payload, received time and run ID.	Many Doors In: Forms, APIs and Bulk Feeds
Field ownership / lock	The contract rule for each field saying whether the feed, the seller in the marketplace's tools, or the platform owns it. A lock set by the seller in the pro tool stops the feed overwriting that field.	Many Doors In: Forms, APIs and Bulk Feeds
Mass-change guard	A control that holds a snapshot run for a seller when its effect exceeds a threshold, such as removing a large share of live stock, until it is confirmed. It prevents a truncated file from unpublishing a dealer's inventory.	Many Doors In: Forms, APIs and Bulk Feeds

Term	Definition	Chapter
Command gateway	The ingestion component that turns accepted changes into intent-named commands (`Submit`, `Edit`, `MarkSold`, `Withdraw`) with idempotency keys and expected versions, so feeds pass through the same invariants as the insertion flow.	Many Doors In: Forms, APIs and Bulk Feeds
Run report	Per-run and per-record outcomes, with stable error codes and three outcomes (success, warning, error), delivered back to the partner by API, file and notification. RFC 9457 "Problem Details for HTTP APIs" is a standard envelope.	Many Doors In: Forms, APIs and Bulk Feeds
Dealer management system (DMS) / transfer service provider	A DMS is a dealer's own software for stock, sales and service, and the usual source of automotive feeds; a transfer service provider is a vendor that uploads listings to portals on dealers' behalf.	Many Doors In: Forms, APIs and Bulk Feeds, Private Sellers, Dealers and Agencies: Modeling Who Lists
OpenImmo / RESO	Real-estate exchange standards: OpenImmo is an XML format used in German-speaking markets; the RESO Data Dictionary and Web API are the US real-estate standards maintained by the Real Estate Standards Organization.	Many Doors In: Forms, APIs and Bulk Feeds, Attributes Without Chaos: Schemas, Dictionaries and Validation

Term	Definition	Chapter
Normalization	Turning a received value into the canonical representation the attribute dictionary defines, without changing what it means: encoding, text, numbers and locale, units, enumerations, named entities, dates, prices, addresses and phone numbers.	From Raw Supply to Usable Listings: Normalization and Enrichment
Enrichment	Adding facts the seller did not state, derived from reference data, registries, models or other listings: VIN decoding, geocoding, category prediction, attribute extraction from text and images, translation, market value and quality scores.	From Raw Supply to Usable Listings: Normalization and Enrichment
Canonical value / original value	Each attribute value is stored twice: the canonical value in the dictionary's type and unit, used for comparing and filtering, and the original value as received, with its unit and parsing locale, used for display, audit and re-normalization.	From Raw Supply to Usable Listings: Normalization and Enrichment

Term	Definition	Chapter
Attribute provenance	The record, per value, of where it came from, when, by which process and version, and with what confidence. The book's source classes are seller, partner feed, registry, catalog, model, moderator and platform.	From Raw Supply to Usable Listings: Normalization and Enrichment
Vehicle identification number (VIN)	The 17-character code that identifies a vehicle. Decoding it fills specification attributes, sometimes as a family of candidates that the dealer's statement resolves. Germany's HSN/TSN type-approval codes play a similar role.	From Raw Supply to Usable Listings: Normalization and Enrichment, Attributes Without Chaos: Schemas, Dictionaries and Validation
Listing quality score	A summary of how usable a listing is for discovery and trust, built from completeness, validity, consistency and confidence of its attribute values.	From Raw Supply to Usable Listings: Normalization and Enrichment
Duplicate detection	Finding listings that are copies or near-copies of each other. Detection can be shared across purposes; the decisions about what to do cannot.	One Property, Three Listings: Duplicates and Entity Resolution
Entity resolution (ER)	Deciding which listings refer to the same real-world thing and giving that thing an identity; also called record linkage.	One Property, Three Listings: Duplicates and Entity Resolution

Term	Definition	Chapter
Canonical entity / canonical property	The resulting record of one real thing, with a stable identifier and attributes reconciled from its listings. In real estate it is the canonical property, which outlives every listing of it and never overwrites the listing, because the listing is the seller's statement.	One Property, Three Listings: Duplicates and Entity Resolution
Duplicate types	Five kinds with different remedies: same-seller repost; relisting to game freshness; cross-feed duplicate (one seller, two channels); multi-agency listing (legitimate, several mandates on one property); and stolen or cloned listing (fraud).	One Property, Three Listings: Duplicates and Entity Resolution
Blocking	Generating a small set of candidate pairs with cheap, high-recall keys (category and place cell, normalized address, VIN, text and image fingerprints, account keys) so that expensive scoring runs only on plausible pairs.	One Property, Three Listings: Duplicates and Entity Resolution

Term	Definition	Chapter
Probabilistic record linkage / pairwise classifier	Two ways to score a candidate pair: Fellegi–Sunter weighting of how much each field's agreement says about a match, which can work without labels, or a supervised classifier trained on pairs people labeled. Scores must be calibrated so thresholds mean something.	One Property, Three Listings: Duplicates and Entity Resolution
Field collapsing / representative rule	Field collapsing returns only the top document per cluster in search, with the others available as "also listed by". The representative rule that picks which listing shows is a published ranking parameter, not an engineer's tuning choice.	One Property, Three Listings: Duplicates and Entity Resolution
Publication pipeline	The sequence of stages between a seller's or feed's submission and the listing's visibility on every surface, and the reverse sequence for removal. It has a decision path (validate, media ready, risk gate, entitlements) and a propagation path (index, caches, alerts, feeds).	From Submitted to Searchable: The Publication Pipeline

Term	Definition	Chapter
Freshness SLA	The target, per vertical and channel, for how quickly a change becomes visible and how quickly an ended listing disappears. SLA stands for service-level agreement; the insertion and removal targets belong on one page.	From Submitted to Searchable: The Publication Pipeline
Platform latency / source latency	Platform latency runs from the marketplace receiving a change to it being visible; source latency runs from the real-world change (a car sold at the dealership) to the marketplace receiving it. Feeds hide most delay in source latency.	From Submitted to Searchable: The Publication Pipeline
Pre-moderation / post-moderation	Pre-moderation checks a listing before it is visible; post-moderation publishes first and checks afterwards. Mature platforms choose by risk tier, not globally.	From Submitted to Searchable: The Publication Pipeline, Trust and Safety as a Platform
Tombstone	A record that a listing has ended, kept with its version instead of simply deleting a copy, so a late, older event cannot resurrect the listing.	From Submitted to Searchable: The Publication Pipeline, Every Change Is an Event: Event-Driven Listing Architecture

Term	Definition	Chapter
Zombie listing	A listing that comes back to life in a copy because an out-of-order or replayed event was applied after its removal.	From Submitted to Searchable: The Publication Pipeline
Unpublish propagation	The mechanism that carries the end of a listing, whether sold, expired, withdrawn or taken down, to every system that holds a copy, with evidence that each one did. Fraud takedowns travel on a priority lane.	From Submitted to Searchable: The Publication Pipeline
Consumer registry	A list of every system that holds a copy of listing data, the events it consumes, its removal objective and its owner; the starting point for unpublish propagation.	From Submitted to Searchable: The Publication Pipeline
Pro tooling	The products a marketplace gives professional sellers to run their business on it: inventory and visibility operations, performance and ROI reporting, lead management, pricing and market insight, team administration, and the APIs behind them.	Serving the Pros: Dealer, Agent and Merchant Tooling

Term	Definition	Chapter
Lead management for pros	The professional's side of a lead: receiving, assigning, answering, qualifying and recording the outcome of buyer contacts, inside the portal or in the professional's own CRM (customer relationship management system).	Serving the Pros: Dealer, Agent and Merchant Tooling
Pro SaaS	Software sold to professionals as a product in its own right, such as an agent CRM or a dealer website builder, usable whether or not they list on the group's portals. When a group owns it, neutrality towards competing portals becomes an architecture requirement.	Serving the Pros: Dealer, Agent and Merchant Tooling
Lead ledger	One store of deduplicated leads across channels (messages, tracked calls, bookings, reservations), with a stable lead identifier and a buyer-deduplication key; the base for billing, reporting and CRM delivery.	Serving the Pros: Dealer, Agent and Merchant Tooling, From Interest to Contact: Messaging and Lead Management
Multi-diffusion	Publishing an agency's listings to many portals at once from one CRM, common in French real estate.	Serving the Pros: Dealer, Agent and Merchant Tooling

Term	Definition	Chapter
Neutrality controls	Four checkable controls that make a group-owned pro SaaS neutral: separate tenancy, API parity with other vendors, a data boundary limited to what the agency chose to publish, and ranking blindness to the publishing software.	Serving the Pros: Dealer, Agent and Merchant Tooling

Part IV — Discovery

Matching intent to inventory. Search Architecture for Listings: Indices, Freshness and Scale owns the index; What Did They Mean? Query Understanding and Autocomplete owns the step from query string to structured query; Ranking for Two Sides: Relevance, Freshness and Paid Visibility owns the order of results; Recommendations for Inventory That Disappears owns unrequested suggestions; Searching by Place: Map, Polygon and Commute Search owns place-centric search; Tell Me When It Appears: Saved Searches and Alerts owns standing demand.

Term	Definition	Chapter
Search document	The denormalized representation of one listing in the index, assembled from several sources and shaped for queries, not for writes.	Search Architecture for Listings: Indices, Freshness and Scale

Term	Definition	Chapter
Index / index alias	An index (a collection in Solr) is a named set of documents with one mapping and one layout of shards and replicas; an alias is a stable name pointing at one or more physical indices, which makes zero-downtime switches possible.	Search Architecture for Listings: Indices, Freshness and Scale
Search index design	The decisions about the search document, typed versus flattened attribute fields, index topology (per country, vertical or tenant), shard sizing and the write and read paths. The rule underneath: the search index is a read model, never the system of record.	Search Architecture for Listings: Indices, Freshness and Scale
Facet (aggregation)	The counts per value shown beside results: brands, sizes, makes, price bands, rooms. Facets are a large share of query cost and can be approximate, capped, cached and served on a separate request.	Search Architecture for Listings: Indices, Freshness and Scale
Insertion latency / removal latency	The two freshness measures of a search layer: time from a listing becoming publishable to being findable, and time from it being sold, expired or taken down to disappearing from results, facets, counts and alerts.	Search Architecture for Listings: Indices, Freshness and Scale

Term	Definition	Chapter
Update storm	A burst of document updates caused by something other than seller edits: bulk price changes, seller-level changes that fan out, bump campaigns, or counters written into ranking fields.	Search Architecture for Listings: Indices, Freshness and Scale
External versioning	Writing each search document with the listing's monotonic version as its version, so the engine rejects older writes; the reliable defense against out-of-order updates resurrecting sold listings.	Search Architecture for Listings: Indices, Freshness and Scale
Reindexing	Rebuilding an index's documents, usually into a new physical index with a new mapping, then switching readers to it through an alias. Done safely it is versioned, throttled, validated before the swap and never run at peak.	Search Architecture for Listings: Indices, Freshness and Scale
Search contract	The API product applications use to search, expressed in marketplace terms (category, attribute keys, geo, sort keys, ranking profile) rather than engine terms, so the engine can change behind it.	Search Architecture for Listings: Indices, Freshness and Scale

Term	Definition	Chapter
Rebuild drill	A scheduled rebuild of an index from the event stream and read models into a fresh index, compared with the live one; the practical test against Search as System of Record.	Search Architecture for Listings: Indices, Freshness and Scale
Query understanding	The steps that turn a raw query string into a structured query: normalized and analyzed text, a predicted category, extracted attribute values and places, corrections, expansions and, increasingly, a semantic representation.	What Did They Mean? Query Understanding and Autocomplete
Query categorization	Predicting the category, or a ranked list of categories with probabilities, that a query is about. It is versioned with the taxonomy, calibrated and monitored like any decision model.	What Did They Mean? Query Understanding and Autocomplete
Query tagging (attribute extraction)	Recognizing spans of a query that name attribute values or places (“gazelle” as brand, “T3” as three rooms, “Lyon 7” as an arrondissement).	What Did They Mean? Query Understanding and Autocomplete

Term	Definition	Chapter
Interpretation	The structured result of query understanding: categories, filters, boosts, corrected text and optionally an embedding, each with a confidence that decides whether it is applied as a filter, a boost or only a suggestion.	What Did They Mean? Query Understanding and Autocomplete
Null and low-result recovery	What the system does when an interpretation returns nothing or too little: correct, relax, widen, broaden, or offer a saved search.	What Did They Mean? Query Understanding and Autocomplete
Autocomplete / query suggestions	Completing or proposing queries as the buyer types, ideally only queries that return live inventory and with the interpretation attached ("e-bike in Bikes › E-bikes").	What Did They Mean? Query Understanding and Autocomplete
Hybrid retrieval	Running lexical, structured and dense (embedding-based) retrieval for the same request and fusing the results, often with reciprocal rank fusion (RRF).	What Did They Mean? Query Understanding and Autocomplete

Term	Definition	Chapter
Approximate nearest neighbor (ANN) / HNSW / IVF	ANN search finds similar vectors quickly without comparing all of them. HNSW (hierarchical navigable small world) graphs and IVF (inverted-file) indexes are two common structures, with different behavior under heavy filtering and frequent updates.	What Did They Mean? Query Understanding and Autocomplete, Recommendations for Inventory That Disappears
LLM query compiler	A large language model (LLM) that turns a natural-language request into the marketplace's existing structured query, which the classic engine then executes.	What Did They Mean? Query Understanding and Autocomplete, AI Across the Listing Lifecycle
Ranking	In the EU Omnibus Directive, "the relative prominence given to products … irrespective of the technological means used". Reserved slots, carousels, highlighted cards and map pins are all ranking, even when no score is involved.	Ranking for Two Sides: Relevance, Freshness and Paid Visibility
Sort order / default sort	The ordering mode a user can choose (newest, price, distance, best match). The default sort is the most political decision in a classifieds product, because it decides who wins without anyone choosing.	Ranking for Two Sides: Relevance, Freshness and Paid Visibility

Term	Definition	Chapter
Relevance model / ranking policy	The relevance model estimates how likely a buyer is to act on a listing for a query; the ranking policy is the versioned set of rules applied to its output: freshness, paid slots and floors, per-seller caps, diversity, exploration and trust demotions.	Ranking for Two Sides: Relevance, Freshness and Paid Visibility
Exposure	The attention a listing receives, usually modeled as impressions weighted by position; treated in fair-ranking work as a resource the ranking allocates among sellers.	Ranking for Two Sides: Relevance, Freshness and Paid Visibility
Bump	A paid product that re-dates a listing so it reappears near the top of date-sorted results. It is a ranking event, not a lifecycle transition, and its timestamp is kept separate from the creation time.	Ranking for Two Sides: Relevance, Freshness and Paid Visibility, How Listing Marketplaces Make Money

Term	Definition	Chapter
Learning to rank (LTR)	Using supervised machine learning, trained on logged outcomes, to order candidates. For a marketplace the default primary label is a contact, with clicks and saves as auxiliary tasks; a gradient-boosted tree (GBDT) ranker with a LambdaMART-style objective is a common starting point.	Ranking for Two Sides: Relevance, Freshness and Paid Visibility
Position bias	Higher positions get more clicks and contacts regardless of relevance, so logged outcomes overstate the quality of whatever was shown on top; corrected with randomized exploration or propensity weighting.	Ranking for Two Sides: Relevance, Freshness and Paid Visibility
Paid-organic blending	How paid placements compete with organic results: fixed slots, a score multiplier, a separate auction merged into floating slots, or unified ranking with a relevance floor and a cap on paid positions. The relevance model never reads entitlements.	Ranking for Two Sides: Relevance, Freshness and Paid Visibility

Term	Definition	Chapter
Relevance floor / guaranteed exposure	A floor makes paid listings eligible only above a relevance threshold for the query; guaranteed exposure sells a number of impressions in relevant contexts, with an extension when delivery falls short, instead of an unconditional position.	Ranking for Two Sides: Relevance, Freshness and Paid Visibility
Ranking transparency	The obligations and the machinery to disclose the main ranking parameters to business users (P2B Regulation, Art. 5) and consumers (Omnibus Directive), label paid ranking, and explain recommender systems (DSA, Art. 27), generated from the same policy file the ranking runs on.	Ranking for Two Sides: Relevance, Freshness and Paid Visibility
Interleaving	Comparing two rankers on the same traffic by merging both rankers' results on one page and crediting each contact to the ranker that contributed the listing; far more sensitive than a standard A/B test.	Ranking for Two Sides: Relevance, Freshness and Paid Visibility, Testing a Two-Sided Market: Experimentation Platforms

Term	Definition	Chapter
Recommendation	A list of listings, searches or sellers the user did not explicitly request, chosen from what is known about the context, the user or both. Surfaces include similar listings, the home feed, recommended searches, low-result fallbacks and recommendation e-mails.	Recommendations for Inventory That Disappears
Similar listings	Recommendations anchored on one listing, usually on its detail page, built from attribute and geography similarity, embeddings and co-view or co-contact neighbors.	Recommendations for Inventory That Disappears
Content-based / collaborative filtering	Content-based methods find nearest neighbors on attributes, price, location, text and images and work from minute one; collaborative methods (item-to-item co-views, matrix factorization, session models) learn from behavior and are absent for new listings.	Recommendations for Inventory That Disappears
Candidate generation	The stage that selects a few hundred plausible items from the whole live inventory, from several sources, before filtering and re-ranking.	Recommendations for Inventory That Disappears

Term	Definition	Chapter
Serving-time filter / send-time check	The availability, eligibility, trust, duplicate, already-contacted and consent checks applied to every candidate when it is served, using the same status read model as search; for e-mail and push, applied again when the message is sent.	Recommendations for Inventory That Disappears, Tell Me When It Appears: Saved Searches and Alerts
Cold start	Recommending without the interaction history collaborative methods need, in three forms: a new listing, a new or anonymous user (where the session is the user), and a new market or category.	Recommendations for Inventory That Disappears
Map search	Discovery in which the result set is displayed on, and scoped by, a map. It is a separate search client with its own latency budget and query mix: dozens of queries per session, counts and clusters at every zoom.	Searching by Place: Map, Polygon and Commute Search
Bounding box (viewport) / drawn polygon / named area	Three geo scopes: the rectangle of the visible map, cheap but cacheable only if snapped to tiles; a polygon the user traces, which must be normalized before querying; and a place ID filter, the cheapest and most cacheable.	Searching by Place: Map, Polygon and Commute Search

Term	Definition	Chapter
Isochrone / isochrone search	An isochrone is a polygon enclosing every point reachable from an origin within a time budget for a mode of transport and departure time; isochrone search filters listings by travel time instead of distance, alternatively by computing per-listing travel times after retrieval.	Searching by Place: Map, Polygon and Commute Search
Geo clustering	Grouping nearby results into a marker with a count that splits as the user zooms in, by server-side grid aggregation, client-side clustering or pre-rendered vector tiles. Aggregations can leak precise locations that pins hide.	Searching by Place: Map, Polygon and Commute Search
Saved search	A persisted, normalized search query with an owner, a notification policy and a lifecycle, stored with the taxonomy and gazetteer versions it was written against, never as a URL or engine string.	Tell Me When It Appears: Saved Searches and Alerts

Term	Definition	Chapter
Percolation	Matching a document against stored queries rather than a query against stored documents; also called reverse or prospective search. At scale it needs partitioning and pre-filter keys (category, place cells, price band) to select candidates.	Tell Me When It Appears: Saved Searches and Alerts
Match set	For one saved search, the set of live, alertable listings that satisfy its query right now. Listings enter, stay in and leave it.	Tell Me When It Appears: Saved Searches and Alerts
Notify on entry, retract on exit	The core alert rule: send a notification when a listing enters a match set, cancel pending notifications and digest items when it leaves, and ignore changes that keep it in.	Tell Me When It Appears: Saved Searches and Alerts
Watcher / price-drop alert	A watcher is a user who has favorited a specific listing; a price-drop alert notifies watchers (and saved searches whose price limit the new price now meets) when a listing's price falls.	Tell Me When It Appears: Saved Searches and Alerts

Term	Definition	Chapter
Alert fan-out	Turning matches into notifications across push, e-mail and in-app channels, with deduplication against a match ledger, frequency caps, quiet hours, consent and provider quotas.	Tell Me When It Appears: Saved Searches and Alerts
Digest	A scheduled message collecting matches since the last digest; it needs retraction and a send-time availability check, or it becomes the most visible form of Ghost Inventory.	Tell Me When It Appears: Saved Searches and Alerts
Alert latency / alert-to-contact rate / ghost-alert rate	Alert metrics: time from a listing becoming searchable to the notification being accepted by the provider; contacts within 48 hours per alert sent; and the share of alert items whose listing is unavailable when opened.	Tell Me When It Appears: Saved Searches and Alerts
Alert hold / risk gate	A short delay before a new account's first listings fan out, and a rule that listings above a risk threshold are not alertable until cleared, because alerts amplify whatever they carry, including scams.	Tell Me When It Appears: Saved Searches and Alerts

Part V — Trust

Keeping the marketplace safe and lawful. Trust and Safety as a Platform owns the moderation pipeline, enforcement and T&S metrics; Scams, Fakes and Account Takeovers: Fraud Prevention for Listings owns risk scoring and friction; Who Is on the Other Side? Identity and Verification owns identity levels and badges; Rules of the Marketplace: DSA, DAC7 and Vertical Regulation owns the legal obligations and compliance by design. Regulatory names and status are given as of 2026; the regulation's own text wins over these summaries.

Term	Definition	Chapter
Trust and safety (T&S)	The capability that keeps listings, users, messages and transactions safe and lawful, in five functions (policy, detection, decision, enforcement, review and redress) plus measurement and transparency across all five.	Trust and Safety as a Platform
Policy / policy registry	A policy is a named, versioned rule of conduct with a scope, a rationale and a legal or contractual ground. The registry holds policies as data, with country overlays, so that rules, models, reviewer guidelines, notices and reports cite the same policy ID and version.	Trust and Safety as a Platform

Term	Definition	Chapter
Moderation pipeline	The components that evaluate content and behavior against policies and produce decisions: synchronous gates, asynchronous detectors, random sampling, human review queues and a decision service.	Trust and Safety as a Platform
Synchronous gate / asynchronous detectors	The gate runs rules, list lookups and a light model inside the user's action (publish, edit, send, log in) and returns allow, hold or reject within a strict budget; detectors run on events afterwards (image models, stolen-photo search, LLM classifiers, price anomalies, graph features).	Trust and Safety as a Platform
Risk-tiered moderation	Choosing pre- or post-moderation, and which checks run where, by category risk, seller trust level and listing risk score; a risk decision, not a technical one.	Trust and Safety as a Platform
Random sampling	Drawing unbiased samples of live listings and messages for blind human labeling; the only way to measure prevalence and recall, and to correct the selection bias of labels from flagged items.	Trust and Safety as a Platform

Term	Definition	Chapter
Decision service / decision record	The only component that writes moderation decisions, combining rule hits, scores, actor context and human input into an action; the decision record holds the decision, evidence, policy and model versions and reviewer, and every regulatory output is a view of it.	Trust and Safety as a Platform
Case / case management	A case is the unit of human work: the items, evidence, decisions and communications that belong together. Case management provides queues, prioritization by expected harm times uncertainty, reviewer tooling and service-level timers.	Trust and Safety as a Platform
Enforcement action	The concrete restriction a decision imposes: warning, reach limit, demotion, hold, removal, suspension or payment freeze, ordered on an enforcement ladder.	Trust and Safety as a Platform
Demotion	Pushing content down in ranking for trust reasons. Under DSA Article 17 it is a restriction that needs a statement of reasons, so actor-trust down-ranking belongs behind the decision service, not inside a ranking feature.	Trust and Safety as a Platform, Rules of the Marketplace: DSA, DAC7 and Vertical Regulation

Term	Definition	Chapter
Enforcement propagation	Publishing `EnforcementApplied` and `EnforcementLifted` events and tracking each consumer's acknowledgement, so a removal reaches search, alerts, caches and SEO pages, and a reversal restores the listing and its ranking state.	Trust and Safety as a Platform
Notice and action	The DSA mechanism by which anyone can notify a platform of illegal content, and the platform must process the notice and inform the notifier and the affected user. A notice is an entity with a lifecycle, not an e-mail.	Trust and Safety as a Platform, Rules of the Marketplace: DSA, DAC7 and Vertical Regulation
Trusted flagger	An organization granted that status by a national Digital Services Coordinator, whose notices platforms must process with priority.	Trust and Safety as a Platform
Appeal (internal complaint)	A user's challenge to a decision through the free, electronic complaint system the DSA requires for at least six months after the decision. An appeal creates a new decision linked to the original, which is never overwritten.	Trust and Safety as a Platform

Term	Definition	Chapter
Statement of reasons	The clear and specific explanation a platform must give a user for each restriction (removal, demotion, payment restriction, suspension), including facts, automation used, ground and redress, and submit without personal data to the DSA Transparency Database.	Rules of the Marketplace: DSA, DAC7 and Vertical Regulation, Trust and Safety as a Platform
Prevalence / exposure-weighted prevalence	The share of live listings or delivered messages that violate a policy, estimated from a blind random sample; and the same share weighted by views or contacts, which measures harm as buyers experience it.	Trust and Safety as a Platform
Time to action	Time from first availability or first report to an enforcement acknowledged by all consumers, not merely decided.	Trust and Safety as a Platform
Precision / recall (T&S)	Precision is the share of enforcement actions that were correct, from second review and appeals; recall is the share of violating items the pipeline caught, which only the random sample can estimate.	Trust and Safety as a Platform

Term	Definition	Chapter
Appeal overturn rate	Reversed decisions as a share of appealed decisions, reported per policy and per detector. A reversed automated decision is a confirmed false positive and the best label the platform gets.	Trust and Safety as a Platform
Fraud signal	An observable property of an item, actor, session or relationship that changes the probability of fraud: device, network, account, behavior and velocity, content, price, media reuse, graph links and external reports.	Scams, Fakes and Account Takeovers: Fraud Prevention for Listings
Risk score	A calibrated estimate of the probability of fraud for a specific action, such as publishing a listing, sending a message, logging in or releasing a payout.	Scams, Fakes and Account Takeovers: Fraud Prevention for Listings
Risk scoring service	A shared service that answers "how risky is this action, by this actor, right now?" for many callers, reading precomputed features and combining rules (fast, backtested, shadowed before enforcement) with models (durable).	Scams, Fakes and Account Takeovers: Fraud Prevention for Listings

Term	Definition	Chapter
Step-up friction	An additional check imposed in proportion to risk, which genuine users pass easily and fraudsters do not, such as phone verification for a new account sending many first contacts. The threshold is set by an explicit cost model owned by the business.	Scams, Fakes and Account Takeovers: Fraud Prevention for Listings
Identity graph	Links between accounts through shared devices, phone numbers, payment accounts, IP ranges, image hashes and near-identical text, used to propagate known fraud, detect rings and scope enforcement to a whole ring.	Scams, Fakes and Account Takeovers: Fraud Prevention for Listings
Authorized push payment (APP) fraud / purchase scam	Fraud in which the victim willingly sends money, usually by bank transfer, to someone who is not who they claim to be. A purchase scam, paying for goods that never arrive, is the most common APP type, which is why classifieds controls are mostly pre-payment.	Scams, Fakes and Account Takeovers: Fraud Prevention for Listings

Term	Definition	Chapter
Payment-link phishing	A fake buyer sends the seller a link or QR code to a counterfeit "payment received" or courier page that harvests card or bank details, often impersonating the platform's own payment product.	Scams, Fakes and Account Takeovers: Fraud Prevention for Listings, From Interest to Contact: Messaging and Lead Management
Account takeover (ATO)	Credentials phished or stuffed so that an aged, reputable account posts scam listings or redirects payouts. Defended with login risk, phishing-resistant authentication and step-up on sensitive actions.	Scams, Fakes and Account Takeovers: Fraud Prevention for Listings, Who Is on the Other Side? Identity and Verification
Verification of Payee	A check, mandatory for euro-area payment providers before credit transfers since 9 October 2025 under the Instant Payments Regulation, that the payee name matches the account. It helps but does not stop a victim paying a real person acting as a mule.	Scams, Fakes and Account Takeovers: Fraud Prevention for Listings

Term	Definition	Chapter
Identity / account / profile / claim	The identity is the person or organization the platform has evidence about; the account is the login; a profile is a brand-specific persona on an account; a claim is a statement about an identity with its method, source, assurance, time and expiry.	Who Is on the Other Side? Identity and Verification
Identity assurance / authenticator assurance	Two separate confidences, after NIST SP 800-63-4 (2025): how sure the service is that a person is who they claim to be, and how sure it is that the person presenting the credential is the account holder. A single "verified" flag hides which one is missing.	Who Is on the Other Side? Identity and Verification
Identity level	A named combination of claims that a policy can require, derived and recomputed from claims rather than stored. The book's ladder runs from anonymous and contactable (confirmed e-mail) through reachable, protected (passkey or second factor) and identified (proof of person) to payee-verified.	Who Is on the Other Side? Identity and Verification

Term	Definition	Chapter
KYB / KYC	Know your business: verification that an organization exists and is who it claims, on a business ladder from declared trader to registered business to accredited professional. Know your customer: the payments-industry checks a regulated payment provider must run before a user can receive money.	Who Is on the Other Side? Identity and Verification
Verified badge	A public statement the marketplace makes to one user about another, such as "identity verified" or "registered business". It must say what was checked and disappear when the claim expires or is revoked.	Who Is on the Other Side? Identity and Verification
Passkey	A phishing-resistant authentication credential based on FIDO standards that the user unlocks with a biometric, PIN or pattern on their device; bound to the site, with no shared secret to steal.	Who Is on the Other Side? Identity and Verification

Term	Definition	Chapter
Pre-hijacking	An attack in which someone creates an account with a victim's e-mail address before the victim does and keeps access afterwards. Merging accounts across brands by matching e-mail addresses recreates it.	Who Is on the Other Side? Identity and Verification
eIDAS 2.0 / EU Digital Identity Wallet	The revised EU regulation on electronic identification and its wallet, through which citizens can prove identity attributes; private platforms must accept it only in defined cases and after transition periods. Bank-based schemes such as BankID play the same role in some countries.	Who Is on the Other Side? Identity and Verification
Digital Services Act (DSA)	Regulation (EU) 2022/2065, applicable to all platforms since 17 February 2024: notice and action, statements of reasons and the Transparency Database, trader traceability, internal complaints, transparency reports, trusted flaggers, ad and recommender transparency, and extra duties for very large platforms.	Rules of the Marketplace: DSA, DAC7 and Vertical Regulation

Term	Definition	Chapter
Hosting service / online platform / very large online platform	The DSA's nested categories: a service storing information for users; one that also disseminates it to the public; and one with more than 45 million average monthly EU users, which carries additional risk-assessment and audit duties.	Rules of the Marketplace: DSA, DAC7 and Vertical Regulation
DSA Transparency Database	The European Commission's public database of statements of reasons, accepting single statements or batches of up to 100 that fail as a whole if one is invalid.	Rules of the Marketplace: DSA, DAC7 and Vertical Regulation
P2B Regulation	Regulation (EU) 2019/1150 on fairness for business users of online intermediation services: clear terms, disclosure of the main ranking parameters and of any differentiated treatment of the provider's own services, and access-to-data descriptions.	Rules of the Marketplace: DSA, DAC7 and Vertical Regulation, Ranking for Two Sides: Relevance, Freshness and Paid Visibility

Term	Definition	Chapter
Omnibus Directive / UCPD	Directive (EU) 2019/2161 amended EU consumer law, including the Unfair Commercial Practices Directive (UCPD): undisclosed paid ranking is always unfair, ranking parameters must be disclosed, review authenticity claims need checks, and price reductions must show the prior 30-day lowest price.	Rules of the Marketplace: DSA, DAC7 and Vertical Regulation
General Data Protection Regulation (GDPR)	Regulation (EU) 2016/679. For listing systems its main architectural principles are data minimization, storage limitation and accountability, plus erasure rights and limits on solely automated decisions.	Rules of the Marketplace: DSA, DAC7 and Vertical Regulation
DAC7	Council Directive (EU) 2021/514 on tax reporting by digital platforms: a reporting platform operator collects and verifies seller information and reports each reportable seller's consideration per quarter to the tax authority by 31 January of the following year.	Rules of the Marketplace: DSA, DAC7 and Vertical Regulation

Term	Definition	Chapter
Reporting platform operator / excluded seller	Under DAC7, an operator that connects sellers with users for relevant activities (sale of goods, rental of property or transport, personal services) beyond mere listing; a seller with fewer than 30 sales of goods and no more than €2,000 consideration in the period is excluded.	Rules of the Marketplace: DSA, DAC7 and Vertical Regulation
Obligation / control / evidence	The three artifacts of compliance: a duty in a legal source with its article, jurisdiction and conditions; the system behavior that meets it; and the data and query that prove the control worked.	Rules of the Marketplace: DSA, DAC7 and Vertical Regulation
Compliance by design / compliance by project	Building controls into platform components with the evidence query written at the same time, versus building one exporter per new obligation. The first is cheaper from the second law onward.	Rules of the Marketplace: DSA, DAC7 and Vertical Regulation

Term	Definition	Chapter
UK Online Safety Act / DMCC Act / US INFORM Act	Non-EU rules with marketplace consequences: the UK's online-safety duties, the UK Digital Markets, Competition and Consumers Act 2024's consumer-protection powers, and the US INFORM Consumers Act's verification and disclosure duties for high-volume third-party sellers.	Rules of the Marketplace: DSA, DAC7 and Vertical Regulation
Fair housing rules	US rules (24 CFR Part 100) that prohibit housing advertising indicating a preference on protected grounds; they constrain listing text, ad targeting, recommendations and neighborhood data.	Rules of the Marketplace: DSA, DAC7 and Vertical Regulation, Searching by Place: Map, Polygon and Commute Search

Part VI — Money

Conversion and revenue. From Interest to Contact: Messaging and Lead Management owns contacts, conversations and leads; Beyond the Contact: Payments, Shipping and Buyer Protection owns the move into payments; How Listing Marketplaces Make Money owns revenue models and their KPIs; Packages, Bumps and Quotas: Products, Pricing and Entitlements owns what is sold and delivered; Selling Attention: Display, Native and Retail-Media Advertising owns the ad stack. Payment-industry depth is in the sibling book's Payments for Finance Architects: Eleven Reasons the Numbers Disagree.

Term	Definition	Chapter
Contact	Any action by which a buyer reaches a seller about a listing: a chat message, an e-mail form, a phone call, a callback request, a booking or an application.	From Interest to Contact: Messaging and Lead Management
Conversation / message	A conversation is the platform-held thread of messages between a fixed set of participants about one listing; a message is one immutable unit of content within it, with mutable metadata (delivery state, read receipts, safety verdicts, redactions).	From Interest to Contact: Messaging and Lead Management
Lead	A record that a specific prospective buyer expressed interest in a specific professional's listing or service, created by a defined rule, delivered to that professional and tracked to an outcome. The creation rule is a business decision encoded in software.	From Interest to Contact: Messaging and Lead Management
Lead routing	Deciding which professional entity (dealer, rooftop, agency branch, individual agent) receives a lead, through which delivery channel and with what service-level expectation.	From Interest to Contact: Messaging and Lead Management

Term	Definition	Chapter
Message safety gate	A synchronous verdict on each message before delivery, within a fixed latency budget, with a vocabulary richer than allow and block: deliver, deliver with warning, withhold, block.	From Interest to Contact: Messaging and Lead Management
Warn in context	Showing safety advice at the moment a risky pattern appears in a conversation (payment keywords, a move to another app, a flagged counterpart) when the gate is not confident enough to withhold.	From Interest to Contact: Messaging and Lead Management, Scams, Fakes and Account Takeovers: Fraud Prevention for Listings
Lead contract	One versioned event and API schema for leads across a group: stable lead ID, source channel, timestamps, listing and seller references, consented contact data, typed qualification fields and the routing decision, with each delivery adapter implementing it.	From Interest to Contact: Messaging and Lead Management
Lead qualification / lead attribution	Qualification adds what the professional needs to judge a lead (financing, trade-in, pre-qualification); attribution connects a lead to a sale, mandate, lease or hire and reports it in a way both sides accept.	From Interest to Contact: Messaging and Lead Management

Term	Definition	Chapter
Transactional pivot	A classifieds marketplace moving its endpoint from contact to completed payment and delivery, typically starting with shipping and protected payment for private sales; leboncoin, Kleinanzeigen and Marktplaats have all added transactional layers on top of classifieds.	Beyond the Contact: Payments, Shipping and Buyer Protection
Marketplace payments	Taking payment from a buyer, holding it and paying it out to a seller, usually through a payment service provider's marketplace product so that the provider, not the marketplace, is the licensed party.	Beyond the Contact: Payments, Shipping and Buyer Protection, Payments for Finance Architects: Eleven Reasons the Numbers Disagree
Payment service provider (PSP)	A company that processes payments for merchants and platforms; its marketplace product handles onboarding and KYC of payees, split payments, holds and payouts.	Beyond the Contact: Payments, Shipping and Buyer Protection

Term	Definition	Chapter
Merchant of record	The party whose name is on the buyer's payment statement and which carries the payment-side obligations, refunds and chargebacks. In protected private sales the seller usually remains the seller and the platform an intermediary.	Beyond the Contact: Payments, Shipping and Buyer Protection, Payments for Finance Architects: Eleven Reasons the Numbers Disagree
Buyer protection	A promise that the buyer gets their money back if the item does not arrive or is not as described, funded by a buyer fee. It is a liability to provision, not only a fee to collect.	Beyond the Contact: Payments, Shipping and Buyer Protection, How Listing Marketplaces Make Money
Hold and release (escrow-like hold) / payout	Keeping the buyer's payment in a held balance at the licensed provider until a release event (buyer confirmation, a delivery scan plus a waiting period, a handover code), then making it available for payout to the seller's bank account. Often called escrow informally, though legally it usually is not; payouts and payout-account changes are prime fraud targets.	Beyond the Contact: Payments, Shipping and Buyer Protection

Term	Definition	Chapter
Order state machine	The states of a protected sale, held on the order rather than the listing: placed, paid (listing reserved), awaiting hand-in or pickup, in transit, delivered, completed, disputed, refunded, cancelled or returned, with a card chargeback possible after completion. Every waiting state has a deadline, and each transition emits events for the ledger, trust and safety and messaging.	Beyond the Contact: Payments, Shipping and Buyer Protection
Dispute / chargeback	A dispute is opened with the platform under its buyer protection; a chargeback is opened by the buyer's card issuer under card-scheme rules. Both can occur for the same order, on different clocks.	Beyond the Contact: Payments, Shipping and Buyer Protection
Monetization model	The combination of who pays, for which unit, at which moment in the funnel and on what pricing basis. Each part becomes a system, and each model spends a different part of liquidity.	How Listing Marketplaces Make Money

Term	Definition	Chapter
Price structure / unit of charge	Two-sided markets set a price structure, which side pays and for what, not just a price level. The unit of charge is the thing whose quantity sets the bill: a listing, a slot, a lead, a sale, an impression.	How Listing Marketplaces Make Money
Average revenue per advertiser (ARPA)	Rightmove's KPI: monthly revenue from agency and new-homes advertisers divided by the number of advertisers, averaged over the year. Autotrader's equivalent per retailer forecourt is ARPR (average revenue per retailer).	How Listing Marketplaces Make Money
Take rate	Marketplace revenue divided by gross merchandise value, as eBay defines it: the share of transaction value the platform keeps.	How Listing Marketplaces Make Money
Value-added services (VAS)	The industry name for paid options bought on top of a basic listing: bumps, top or featured positions, highlights and galleries, longer durations. The core of the freemium paid-visibility model.	How Listing Marketplaces Make Money

Term	Definition	Chapter
Insertion fee	A charge to publish a listing, usually keyed by category, region and seller type. It spends supply, because some sellers stop listing.	How Listing Marketplaces Make Money
Paid visibility / top slot / highlight	Free listing with paid prominence: a top or featured slot reserves positions above or within organic results; a highlight or gallery changes how the card looks. Every promoted result must carry a paid label.	How Listing Marketplaces Make Money
Professional subscription	A monthly contract granting listing slots, boosts, tools, statistics and a level of prominence; the center of gravity of vertical portals' revenue.	How Listing Marketplaces Make Money
Depth product	Per-listing tiers (standard, premium) that change a listing's prominence and presentation, so revenue is listings times yield; exposed to supply leaving.	How Listing Marketplaces Make Money

Term	Definition	Chapter
Pay per lead / share of voice / pay at close	Lead-based models: a price per billable lead; connections distributed in proportion to each professional's share of advertising spend in an area; and a fee only when a transaction closes with a lead. Each needs new systems: billable-lead rules, an allocation engine, an outcome pipeline.	How Listing Marketplaces Make Money
Billable lead	The precise, contractual definition of a lead the professional pays for, with deduplication (one buyer, several channels, one lead), validity rules and a dispute and credit process.	How Listing Marketplaces Make Money
Product catalog	The versioned definition of everything sellable: products, packages and add-ons, and the effects each grants. Reference data shipped as releases.	Packages, Bumps and Quotas: Products, Pricing and Entitlements
Product / package / add-on	A product is one sellable thing with a defined effect ("Bump", "Top listing (7 days)"); a package bundles products and quotas on a contract ("150 slots + 20 boosts per month"); an add-on is sold on top of a package, possibly metered.	Packages, Bumps and Quotas: Products, Pricing and Entitlements

Term	Definition	Chapter
Pricing engine	The component that computes the price of a catalog item for a specific buyer, context and date from effective-dated price lists, discounts and promotions.	Packages, Bumps and Quotas: Products, Pricing and Entitlements
Entitlement	A right, held by an account or listing for a period, to receive a defined effect, such as "may publish up to 150 vehicles" or "appears in a top slot until Sunday 23:59", with a holder, product version, validity window, optional quantity and source.	Packages, Bumps and Quotas: Products, Pricing and Entitlements
Quota	A countable entitlement: listing slots, boosts, leads, credits.	Packages, Bumps and Quotas: Products, Pricing and Entitlements
Entitlement ledger	The append-only record of what each account or listing holds and has used, with idempotent entries and explainable balances; the system of record for "what was sold versus what was delivered".	Packages, Bumps and Quotas: Products, Pricing and Entitlements

Term	Definition	Chapter
Effect / effect projector / enforcement point	An effect is the projection of active entitlements that one place in the platform needs; the projector publishes effect changes from ledger events; enforcement points (insertion, ranking, display) read effects, tolerating bounded staleness, and never the ledger live.	Packages, Bumps and Quotas: Products, Pricing and Entitlements
Delivery evidence / delivery reconciliation	The record of what the platform actually did (listings under a slot, paid impressions, boosts executed, leads routed), compared daily with what was granted and what was billed.	Packages, Bumps and Quotas: Products, Pricing and Entitlements
Ad stack	The systems that decide, serve, measure and bill advertising on a marketplace's surfaces: consent, ad server, header bidding, supply-side connections, audience services, creative rendering, measurement and policy.	Selling Attention: Display, Native and Retail-Media Advertising
Ad server	The publisher's decision engine that chooses, for each ad slot, among direct campaigns, deals and programmatic bids, renders the creative and reports delivery.	Selling Attention: Display, Native and Retail-Media Advertising

Term	Definition	Chapter
Supply-side / demand-side platform (SSP / DSP)	An SSP is an exchange that sells a publisher's impressions to many buyers in real-time auctions; a DSP is the buyer's system that bids in them.	Selling Attention: Display, Native and Retail-Media Advertising
Header bidding / Prebid	Running an auction among several SSPs before the ad server decides, so programmatic demand competes with direct campaigns. Prebid is the open-source implementation most publishers use, in the browser or server-side.	Selling Attention: Display, Native and Retail-Media Advertising
Sponsored listing	A seller's own listing promoted into prominent positions in search results or recommendations for payment per click, per sale or per day, served by the marketplace's search stack under relevance floors and slot caps, and labeled as paid.	Selling Attention: Display, Native and Retail-Media Advertising
Retail media	Advertising sold to brands, such as car makers, banks and insurers, against the marketplace's own audiences and placements, often with closed-loop measurement against on-platform outcomes; closer to a data product than to a banner.	Selling Attention: Display, Native and Retail-Media Advertising

Term	Definition	Chapter
First-party audience	A segment of users defined from the marketplace's own data (searches, saved searches, viewed listings, enquiries), built under the consent and purposes users granted, and activated on or off the platform. It is a data product with purpose limits.	Selling Attention: Display, Native and Retail-Media Advertising
Consent management platform (CMP) / TCF	The CMP asks for and records consent; in Europe it usually implements the IAB Transparency and Consent Framework (TCF), which encodes choices in a TC string passed to ad vendors.	Selling Attention: Display, Native and Retail-Media Advertising
Ad load / CPM / CPC / viewability	Ad load is the share of positions given to advertising; CPM is price per thousand impressions (eCPM the effective figure after all sources); CPC is price per click; viewability is whether an ad had a chance to be seen.	Selling Attention: Display, Native and Retail-Media Advertising

Term	Definition	Chapter
Consent or pay / consent or contextual	Consent or pay asks users to consent to tracking or pay, which the European Data Protection Board found in 2024 rarely valid for large platforms; consent or contextual shows users who decline ads targeted by page context, which a listing page states unusually well.	Selling Attention: Display, Native and Retail-Media Advertising

Part VII — Platform

The distributed system underneath. Every Change Is an Event: Event-Driven Listing Architecture owns listing events, the outbox and schema evolution; System of Record and Read Models: Listing Data Management owns the system of record, read models and erasure; Found by Google: Web, App and SEO Architecture for Listings owns the customer-facing stack; Data Platform for Marketplaces: Events, Lakehouse and Data Products owns tracking, data products and governed metrics; Testing a Two-Sided Market: Experimentation Platforms owns two-sided testing; Models in Production: ML Platforms and Use Cases owns models in production; One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture owns tenancy; Sunday Evening Peaks: Reliability, Performance and Cost owns journey objectives and cost.

Term	Definition	Chapter
Listing event	A domain event about one listing aggregate, emitted by the listing's owning context on every invariant-preserving transition (submitted, published, updated, price changed, paused, reserved, sold, expired, removed); the published language other contexts use to learn about listings.	Every Change Is an Event: Event-Driven Listing Architecture
Change event (change data capture, CDC)	A row-level record of an insert, update or delete, read from a database's transaction log. It reflects the table, not the business intent, so it is allowed as a source but never as a contract.	Every Change Is an Event: Event-Driven Listing Architecture
Event notification / event-carried state transfer / event sourcing	Three uses of events: saying only that something changed so consumers call back; carrying the state consumers need to keep their own copy; and making the events the system of record. Listing events should carry state.	Every Change Is an Event: Event-Driven Listing Architecture
At-least-once delivery / idempotent consumer	Practical pipelines may deliver an event twice after a retry or restart; a consumer is idempotent if processing an event twice has the same effect as once.	Every Change Is an Event: Event-Driven Listing Architecture

Term	Definition	Chapter
Version-gated apply	Each event carries the listing's aggregate version, and a consumer applies it only if the version is newer than the one it holds, which makes consumers idempotent and order-safe and lets missed events self-heal.	Every Change Is an Event: Event-Driven Listing Architecture
Transactional outbox	Writing the event into an outbox table in the same database transaction as the state change, then relaying it to the broker. It solves the dual-write problem (a change committed without its event, or the reverse) and lets domain code choose the event's shape.	Every Change Is an Event: Event-Driven Listing Architecture
Translation layer (CDC translator)	An anti-corruption layer that consumes a legacy system's raw change stream, infers the business transition and emits contract events, so that only the translator reads the monolith's tables.	Every Change Is an Event: Event-Driven Listing Architecture

Term	Definition	Chapter
Schema registry / schema evolution	The registry stores every version of an event schema and rejects incompatible changes at publish time; schema evolution is the discipline of changing contracts without breaking consumers. For a shared listing stream the book recommends full transitive compatibility, plus human review of changes in meaning.	Every Change Is an Event: Event-Driven Listing Architecture
Replay / backfill	Re-delivering state to consumers that already processed it: replaying from the log or a compacted topic, or backfilling from the system of record after a correction or schema addition, without starving live traffic.	Every Change Is an Event: Event-Driven Listing Architecture
Dead-letter queue (DLQ)	A place for events a consumer could not process. For consumers that need every delta, a failed event parks that listing until the failure is resolved while other listings continue.	Every Change Is an Event: Event-Driven Listing Architecture

Term	Definition	Chapter
Listing system of record (SoR)	The one store in which a change to a listing becomes true: it accepts writes, nothing else does, and it is complete for the listing's own data. It need not be one database, but each fact has one authoritative owner.	System of Record and Read Models: Listing Data Management
Read model	A copy of listing data shaped for one way of reading it, derived from the system of record, usually from the event stream, and rebuildable from it: the search index, a detail-page cache, pro statistics. The read side of CQRS (command query responsibility segregation).	System of Record and Read Models: Listing Data Management
Copy kinds	The book distinguishes replicas, read models, caches, archives, evidence stores, analytical copies and external copies, each with its own rebuild and retention rules. Every copy must be either rebuildable or governed.	System of Record and Read Models: Listing Data Management

Term	Definition	Chapter
Staleness / staleness budget	Staleness is the gap between the system of record and what a screen shows; a staleness budget is the written maximum lag for one read path, with the measurement that proves it.	System of Record and Read Models: Listing Data Management
Read-your-writes / monotonic reads / bounded staleness / removal priority	The consistency guarantees users notice: sellers see their own change at once; buyers never see an older state after a newer one; derived surfaces lag by a measured maximum; and removals have a tighter bound than additions.	System of Record and Read Models: Listing Data Management
Erasure propagation	Carrying a GDPR erasure request to every copy of a person's data, where each owner deletes, anonymizes, retains with a named legal basis and end date, or reports not held, and an erasure ledger records every outcome as proof.	System of Record and Read Models: Listing Data Management
Crypto-shredding	Encrypting each subject's personal fields with a per-subject key and erasing them by destroying the key, for stores that must stay immutable, such as raw data zones and evidence stores.	System of Record and Read Models: Listing Data Management

Term	Definition	Chapter
SEO architecture	The components and rules that decide which URLs exist, what each URL returns over its lifetime, how pages are rendered and how search engines learn about changes. SEO is search engine optimization.	Found by Google: Web, App and SEO Architecture for Listings
Crawl budget	How much a search engine is willing and able to crawl on a site; not a fixed allowance, and wasted on facet combinations and dead listing pages.	Found by Google: Web, App and SEO Architecture for Listings
Landing page	A search result page the platform has deliberately made indexable, with a canonical URL, curated title and copy, internal links, and a rule saying when it may exist; the main durable organic asset of a listing site.	Found by Google: Web, App and SEO Architecture for Listings
Landing page generation	The process that decides which category, location and attribute combinations get an indexable URL, using demand, supply, distinctness, depth and quality rules, keeps the set current and retires pages cleanly. Indexability is a computed state: indexable, thin (`noindex, follow`), gone or redirected.	Found by Google: Web, App and SEO Architecture for Listings

Term	Definition	Chapter
Backend-for-frontend (BFF)	A server-side component owned by the team that owns a client (iOS and Android apps, or the web front end), aggregating domain APIs into the shape that client needs and absorbing differences in release cadence.	Found by Google: Web, App and SEO Architecture for Listings
Rendering modes	Client-side rendering (CSR) builds pages in the browser; server-side rendering (SSR) with hydration sends full HTML and then attaches interactivity; streaming SSR sends chunks as data arrives; static or incremental generation builds ahead; dynamic rendering for crawlers only is a discouraged workaround.	Found by Google: Web, App and SEO Architecture for Listings
Redirect registry / URL and canonical service	The URL service builds every public URL from domain identifiers; the redirect registry maps every retired URL to its successor or a gone status (`301`, `410`), with reason and date, for web and apps alike.	Found by Google: Web, App and SEO Architecture for Listings

Term	Definition	Chapter
Structured data	Machine-readable markup, usually schema.org vocabulary, that tells search engines what a page is: product snippets versus merchant listings, `JobPosting`, real-estate types. Search engines' support changes; Google removed vehicle-listing documentation in 2025.	Found by Google: Web, App and SEO Architecture for Listings
Core Web Vitals	Google's user-experience measures at the 75th percentile of page loads: Largest Contentful Paint (LCP, good at 2.5 s or less), Interaction to Next Paint (INP, 200 ms or less) and Cumulative Layout Shift (CLS). Treated as budgets owned by page owners.	Found by Google: Web, App and SEO Architecture for Listings
Answer engine	An AI assistant or AI search mode that answers questions directly, sometimes citing or linking to sources; a new referral channel for marketplaces and a new class of crawler.	Found by Google: Web, App and SEO Architecture for Listings, AI Across the Listing Lifecycle

Term	Definition	Chapter
Data platform	The shared pipelines, storage, compute, metadata and access controls through which a group's data is collected, governed and used for anything other than serving the live product.	Data Platform for Marketplaces: Events, Lakehouse and Data Products
Tracking plan	The governed catalog of behavioral events a product emits: each event's name, trigger, required properties and types, identifiers, consent category and owner, versioned like any contract. Every event carries a common envelope including tenant, identifiers and consent state.	Data Platform for Marketplaces: Events, Lakehouse and Data Products
Data contract	A versioned, machine-readable agreement between a data producer and the platform: schema, semantics, personal-data classification, ownership, freshness and retention. Records that fail it go to quarantine, which feeds back to the producer.	Data Platform for Marketplaces: Events, Lakehouse and Data Products

Term	Definition	Chapter
Data product	A dataset or small family of datasets that a named team owns and publishes for others, with a contract, documentation, quality and freshness service levels, access policy and support. Source-aligned products mirror sources; consumer-aligned products serve one purpose.	Data Platform for Marketplaces: Events, Lakehouse and Data Products
Data mesh	An organizational approach in which domains own and publish their data as products on a shared self-service platform under federated governance. The book's reading of the record: ownership federates, the substrate and the rules do not.	Data Platform for Marketplaces: Events, Lakehouse and Data Products
Data warehouse / lakehouse	A warehouse stores data in an engine's managed format with tightly coupled metadata, storage and query; a lakehouse keeps data in open direct-access formats (such as Parquet with an open table format like Apache Iceberg) with warehouse-grade performance and first-class machine-learning support.	Data Platform for Marketplaces: Events, Lakehouse and Data Products

Term	Definition	Chapter
Governed metric	A business measure with a named owner, a written definition, a single computation in code, a version, and rules for restatement when the definition or data changes. Liquidity metrics must be governed because small definitional choices change them by large factors.	Data Platform for Marketplaces: Events, Lakehouse and Data Products
Metrics layer (semantic layer)	The component that holds governed metric definitions and computes or serves them to every consumer, so dashboards, experiment readouts and dealer reports do not each reimplement "contacts".	Data Platform for Marketplaces: Events, Lakehouse and Data Products
Online controlled experiment	Randomly assigning units (users, sellers, listings, regions, time periods) to variants, exposing them, measuring outcomes and attributing the difference to the variant.	Testing a Two-Sided Market: Experimentation Platforms

Term	Definition	Chapter
Interference	The outcome of one unit depends on the treatment of others, as when treated and control sellers compete for the same buyers' attention. The default condition for ranking, pricing and visibility tests, not an edge case: the measured effect includes a transfer between groups.	Testing a Two-Sided Market: Experimentation Platforms
Cluster randomization	Assigning groups of units that interact mainly with each other, such as a region × category cell, as one unit, which contains most interference at the cost of fewer effective units.	Testing a Two-Sided Market: Experimentation Platforms
Switchback / two-sided / geo experiment	Designs for marketplaces: alternating treatment over time slots (with carryover between slots to handle); randomizing buyers and listings together; and comparing whole regions.	Testing a Two-Sided Market: Experimentation Platforms
Primary metric	The one metric a test is designed and powered to move, also called the overall evaluation criterion, chosen before the test starts.	Testing a Two-Sided Market: Experimentation Platforms

Term	Definition	Chapter
Guardrail metric	A metric that must not degrade beyond a set threshold for a change to ship, whatever the primary result: liquidity outcomes, the other side of the market, and the control group's own trend.	Testing a Two-Sided Market: Experimentation Platforms
Sample ratio mismatch (SRM)	A difference between the observed and the designed split of units, which signals broken assignment or logging; the result should not be read.	Testing a Two-Sided Market: Experimentation Platforms
Variance reduction (CUPED) / sequential testing	Using pre-experiment data to remove predictable variation, roughly halving required samples in published cases; and methods whose p-values and intervals stay valid when results are checked continuously.	Testing a Two-Sided Market: Experimentation Platforms
ML platform	The shared tooling that lets teams build, deploy and operate models without rebuilding infrastructure: data access, feature pipelines, training orchestration, a model registry, serving, a prediction log and monitoring.	Models in Production: ML Platforms and Use Cases

Term	Definition	Chapter
Feature store	A system that defines features once and serves them consistently to training (offline, point-in-time-correct history) and inference (online, low-latency values); justified once real-time behavioral features drive decisions.	Models in Production: ML Platforms and Use Cases, Scams, Fakes and Account Takeovers: Fraud Prevention for Listings
Point-in-time correctness / training–serving skew	Training must use feature values as they were when the decision was taken; skew is any difference between the features a model saw in training and those it receives in production.	Models in Production: ML Platforms and Use Cases, Ranking for Two Sides: Relevance, Freshness and Paid Visibility
Drift	Change in the input distribution (data drift), in the relationship between inputs and outcomes (concept drift), or in what a label means (label drift); in marketplaces caused by taxonomy changes, seasons, adversaries and feedback loops.	Models in Production: ML Platforms and Use Cases
Automated valuation model (AVM)	A model that estimates the market value of a specific asset, usually a property, from its characteristics, location, comparable transactions and market trends, such as Zillow's Zestimate.	Models in Production: ML Platforms and Use Cases

Term	Definition	Chapter
Model governance / decision tiers	The rules for who may deploy a model, what evidence its deployment requires, how it is monitored and who answers for it, scaled to the decision it makes: inform, order, gate or commit. Tiers attach to uses, not models; moving up a tier is a new launch.	Models in Production: ML Platforms and Use Cases
Proxy label	An observable stand-in for the outcome a model should predict, such as contacts for relevance, delisting for sale or moderator decisions for abuse; each has a bias, and choosing it is choosing what the marketplace optimizes.	Models in Production: ML Platforms and Use Cases
Brand / market / tenant	A brand is the consumer-facing marketplace identity; a market is one country in which a brand operates; a tenant is the unit a shared capability scopes, configures, isolates, releases and bills by, usually brand × country.	One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture

Term	Definition	Chapter
Tenancy model	The decisions that answer, for every shared capability: what a tenant is, how it is identified, how strongly it is isolated, how its configuration is resolved, how changes are released to it, how its costs are attributed and how it could leave.	One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture
Tenant isolation (logical / partitioned / dedicated)	A tenant key on every row and message; a schema, index, topic or bucket per tenant on shared infrastructure; or a cluster, database or account per tenant. The recommended default is partitioned, so a tenant can be moved or handed over as a unit.	One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture
Global component	A capability built and run once for the group, with a stable contract, serving many tenants beneath their local products, such as identity, the trust-and-safety pipeline or the search engine.	One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture

Term	Definition	Chapter
Configuration over code	Expressing differences between tenants that are data as governed configuration with a schema per key, a resolution hierarchy (group, brand, tenant, profile, segment, experiment) and change control, not as branches in code.	One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture
Extension point	A defined interface in a shared capability where a tenant-owned component plugs in, used when a difference is behavior rather than a value: configure, then extend, and fork only under a decision record with an expiry date.	One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture, Same Platform, Different Physics: How Listing Verticals Differ
Release rings / group-urgent change	Rolling changes to a global component through rings of tenants, smallest first; some classes of change (fraud rules, security fixes, legal takedowns) bypass the rings and apply everywhere at once.	One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture
Tenant exit path / separability drill	Every shared capability documents how a tenant's data is exported, handed over, run transitionally and switched off; once a year a drill proves it can be done.	One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture

Term	Definition	Chapter
Platform models	Ways to share across markets, from fork per country through federation with coordinated choices, shared core with local configuration, global components beneath local products and platform as product, to a converged product platform. Decided per capability, not for the group.	One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture
Centralization pendulum	The tendency of marketplace groups to swing between centralizing for synergy and decentralizing for focus, often after a change of owner or strategy; the architecture should survive the swing without being rebuilt.	One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture, Who Owns What: Organizing Marketplace Engineering
Service level indicator / objective (SLI / SLO) / error budget	An SLI is good events divided by total events; an SLO is the target for it; the error budget is 100% minus the SLO, the unreliability a team may spend.	Sunday Evening Peaks: Reliability, Performance and Cost

Term	Definition	Chapter
Journey SLO	An objective set on a marketplace journey that creates or protects liquidity (search, listing view, listing visible, listing gone, message delivered, alert sent, insertion), measured from the user's side, with an owner and an error-budget policy.	Sunday Evening Peaks: Reliability, Performance and Cost
Degradation mode	A predefined, tested reduction of functionality that a journey switches to under load or partial failure so its essential part keeps working, in steps from shedding bots, trimming optional work and serving from cache to browse only.	Sunday Evening Peaks: Reliability, Performance and Cost
Read-only warm standby	A multi-region posture in which a second region serves browse and listing pages within minutes while writes queue or pause; with multi-zone deployment, the right answer for most journeys of a country-scoped classifieds group.	Sunday Evening Peaks: Reliability, Performance and Cost

Term	Definition	Chapter
Bot management	Policy, crawler verification, edge classification, quotas, product design and contracted access that separate wanted crawlers from scrapers and contested AI crawlers. In the EU the sui generis database right adds a legal layer.	Sunday Evening Peaks: Reliability, Performance and Cost
Cost per search	The fully allocated run cost of the search path (engine, gateway, caches, observability) divided by search requests, reported separately for human and automated traffic. Companions are cost per live listing per month and cost per contact.	Sunday Evening Peaks: Reliability, Performance and Cost
FinOps	The practice of managing cloud and technology cost through shared engineering, finance and business ownership; in marketplaces made concrete through unit costs per listing, search and contact.	Sunday Evening Peaks: Reliability, Performance and Cost

Part VIII — Leadership

The Chief Architect's agenda. After the Deal: Consolidating Platforms After Acquisitions owns what to consolidate after acquisitions; Who Owns What: Organizing Marketplace Engineering owns team design and governance; Changing the Engine in Flight: Modernizing a Live Marketplace owns moving a live marketplace; AI Across

the Listing Lifecycle owns AI across the lifecycle; Built to Be Replaced: Evolving a Marketplace Platform Over a Decade owns decade-scale thinking.

Term	Definition	Chapter
Consolidation sequence	The ordered, gated stages through which an acquired marketplace's capabilities move onto group platforms: day-one controls; invisible and contractual capabilities (cloud, data, advertising, supplier contracts such as payments); trust and identity; shared engines; and, last and optionally, the product platform. What users cannot see and what gains value from pooling moves first; each gate is a measured condition, not a date.	After the Deal: Consolidating Platforms After Acquisitions
Integration options	The choices for an acquired marketplace, from most to least reversible: hold separate, federate under group controls, share services beneath its own product, replatform onto a sibling, merge brands, or divest. They are chosen per business and may be combined over time; an acquisition without a recorded option drifts into Deal-Shaped Architecture.	After the Deal: Consolidating Platforms After Acquisitions

Term	Definition	Chapter
Synergy case / synergy ledger	The quantified hypothesis that combining a business with the group raises combined profit: cost and revenue synergies, weighed against dis-synergies (traffic loss, churn, double running) and the separation cost if the business is sold again. Deal models usually show only the first two. A synergy ledger tracks it as one line per synergy or dis-synergy, each with an owner, dated evidence and the consolidation stage it depends on.	After the Deal: Consolidating Platforms After Acquisitions
Brand migration	Moving a marketplace's users, sellers, partners and search-engine presence from one consumer brand or domain to another, as in a rebrand, the removal of a former parent's name or a merger of portals. It is an architectural event, planned as a workstream separate from any platform migration; organic search traffic is the main risk.	After the Deal: Consolidating Platforms After Acquisitions

Term	Definition	Chapter
Marketplace team topology	The assignment of team types to a marketplace group's journeys, bounded contexts and shared capabilities, together with the interaction mode between each pair of teams.	Who Owns What: Organizing Marketplace Engineering
Stream-aligned / platform / complicated-subsystem / enabling team	Team Topologies' four types: teams on a flow of work, such as a buyer or seller journey; teams providing an internal product with a contract; teams owning work that needs deep expertise, such as ranking or AVMs; and temporary teams that help others adopt or migrate.	Who Owns What: Organizing Marketplace Engineering
Architecture governance	The decision rights, forums, records, principles and standards through which architectural decisions are made, communicated and revisited.	Who Owns What: Organizing Marketplace Engineering
Decision rights / decision record	Who may decide what (group, platform, brand, team); and the short document capturing a decision's context, decision, status, consequences and review date, kept with the code. An architecture decision record (ADR) is the common form.	Who Owns What: Organizing Marketplace Engineering

Term	Definition	Chapter
Advice process	Andrew Harmel-Law's model in which anyone can make an architectural decision, provided they first seek advice from everyone meaningfully affected and from relevant experts, and record it. Advice is public and decisions are recorded.	Who Owns What: Organizing Marketplace Engineering
Platform funding models	Ways to pay for platforms: a central program budget, fixed allocation (a "platform tax"), showback of usage-based cost without charging, chargeback by usage, or an internal market. Policy-driven platforms such as identity and trust and safety suit fixed allocation.	Who Owns What: Organizing Marketplace Engineering
Conway's law	The observation that systems tend to mirror the communication structure of the organization that builds them, which makes team design an architectural decision.	Who Owns What: Organizing Marketplace Engineering
Strangler fig	Replacing a legacy system piece by piece by routing traffic for one capability at a time to new components, until the old system can be switched off.	Changing the Engine in Flight: Modernizing a Live Marketplace

Term	Definition	Chapter
Branch by abstraction / event interception	Two migration techniques: putting an abstraction in front of a capability so old and new implementations can be swapped behind it; and capturing the legacy system's changes as events so new components can be fed without changing its code.	Changing the Engine in Flight: Modernizing a Live Marketplace
Parallel run / shadow traffic / canary	Three ways of running old and new side by side. Shadow traffic (a dark launch) sends the new system a copy of real requests or events and discards its output after comparison; a parallel run has both systems compute on real requests, with one authoritative and the other compared; a canary serves a small share of real traffic from the new system.	Changing the Engine in Flight: Modernizing a Live Marketplace
Migration stage gate	A decision point in moving one slice of a system, with exit criteria agreed beforehand and a defined rollback, that must be passed before more traffic or responsibility moves. Veldmark uses five per slice: contract, shadow, canary, ramp and retire; the move to retire is the recorded point of no return.	Changing the Engine in Flight: Modernizing a Live Marketplace

Term	Definition	Chapter
Relevance parity	The condition that a new search path produces outcomes for buyers and sellers that are not worse than the old path's, within agreed tolerances, measured on real queries (replayed comparisons of top results, then online tests); not identical results.	Changing the Engine in Flight: Modernizing a Live Marketplace
URL migration	The controlled change of public URLs, or the deliberate decision not to change them, with old-to-new mappings, permanent redirects, consistent canonicals and sitemaps, and monitoring of crawl and traffic that preserve search-engine signals.	Changing the Engine in Flight: Modernizing a Live Marketplace, Found by Google: Web, App and SEO Architecture for Listings
Feature parity trap	The mistake of insisting that a new system reproduce every feature of the old one before launch, including features nobody uses, which delays migration indefinitely. The remedy is a behavior catalog in which each legacy behavior is carried, changed or retired.	Changing the Engine in Flight: Modernizing a Live Marketplace

Term	Definition	Chapter
AI listing assistant	A feature in the insertion flow that proposes listing content (category, attributes, title, description, price) from photos, voice or short text, for the seller to accept, edit or reject.	AI Across the Listing Lifecycle
Structured-output compiler	The pattern in which a model translates unstructured input into a structure the marketplace already has (an attribute set, a filter query, a policy label), so the structured system of record stays in charge.	AI Across the Listing Lifecycle
Agentic search	Search in which a model plans steps and calls tools (keyword or vector search, filters, comparisons, external lookups) over several turns, rather than returning one ranked list for one query; suited to high-consideration journeys.	AI Across the Listing Lifecycle
Agent-to-marketplace interface / Model Context Protocol (MCP)	An interface through which external AI agents call the marketplace's capabilities as tools; MCP is an open protocol, introduced by Anthropic in 2024, for connecting agents to tools and data.	AI Across the Listing Lifecycle

Term	Definition	Chapter
Autonomy level	How far an AI feature acts without a person: it suggests, drafts for confirmation, acts with confirmation, or acts autonomously.	AI Across the Listing Lifecycle
AI governance for marketplaces	The register, rules and controls that decide which AI uses may run, at what autonomy, with which data, evaluated how, at what cost and with whom accountable.	AI Across the Listing Lifecycle
AI gateway / evaluation harness	The gateway is the control point every model call passes (authentication, routing, rate limits, cost metering, logging, a kill switch per feature); the harness runs golden sets per language, category and use before every prompt or model change and on sampled production traffic.	AI Across the Listing Lifecycle
EU AI Act	Regulation (EU) 2024/1689, with obligations phased in from 2025: transparency for generated content (Article 50), duties for general-purpose AI model providers including respecting text-and-data-mining reservations, and stricter rules for high-risk uses.	AI Across the Listing Lifecycle

Term	Definition	Chapter
Stressor analysis	Listing what could happen to a marketplace group (a sale, a law, a fraud wave, a lost channel), recording for each the attractor, what breaks in the current architecture and the residue that would survive it, then counting which components and contracts carry the most residues so they get the most care. Adapted from Barry M. O'Reilly's residuality theory.	Built to Be Replaced: Evolving a Marketplace Platform Over a Decade
Stressor / residue / attractor	A stressor is anything that could happen to the business or its environment, likely or not, technical or not; the attractor is the state the business is pushed into when it hits (how people, partners and the market will actually behave, such as a carve-out pushing it into transitional services); the residue is what remains of the architecture afterwards, and the structural change that would let it survive.	Built to Be Replaced: Evolving a Marketplace Platform Over a Decade

Term	Definition	Chapter
Marketplace fitness function	A measured, owned, regularly evaluated statement about a stage of the liquidity machine, such as "a sold listing disappears from every registered consumer within two minutes at p95". Unlike a business KPI it has an architectural owner and remedy; unlike an SLO it measures the whole machine, not one service. Veldmark's scorecard has twelve.	Built to Be Replaced: Evolving a Marketplace Platform Over a Decade
Optionality / separability	Keeping future choices open: listing data and events in portable, documented forms, and every tenant able to leave the platform, so a sale, a merger or a new vertical does not force a rebuild.	Built to Be Replaced: Evolving a Marketplace Platform Over a Decade, One Platform, Many Marketplaces: Multi-Country and Multi-Brand Architecture

Veldmark Names

Veldmark is fictional, and so is everything below; these names recur across chapters, so they are collected here. Meet Veldmark: Four Brands, One Platform to Build introduces the group, its brands, cast and systems.

Term	Definition	Chapter
Veldmark Group	The fictional multi-country, multi-vertical marketplace group the book follows, headquartered in Rotterdam, with €1.31 billion revenue in 2026 and four brands in eight countries.	Meet Veldmark: Four Brands, One Platform to Build
Tradeyard / Motorvane / Hearthly / Keybridge	Veldmark's brands: horizontal classifieds in the Netherlands, Belgium, Poland and Portugal; an automotive marketplace led by Germany; a real-estate portal in France and Spain; and a neutral CRM and listing-management SaaS for real-estate agents.	Meet Veldmark: Four Brands, One Platform to Build
Veldmark Ads	The group advertising unit selling display, native, sponsored-listing and retail-media advertising across the brands.	Selling Attention: Display, Native and Retail-Media Advertising
Tradeyard Safe Deal	Tradeyard's protected payment with shipping or pay-at-pickup for private sales, run on Adyen for Platforms as the licensed party, with a buyer fee of 5% of the item price plus €0.70.	Beyond the Contact: Payments, Shipping and Buyer Protection

Term	Definition	Chapter
Valuo	Hearthly's automated valuation model and price-estimate product in France, trained on public transaction data (France's DVF open data).	Models in Production: ML Platforms and Use Cases
Common Ground	Veldmark's 2027–2030 program (€140 million) to share platform capabilities beneath the brands and retire Yardstick, under the principle "share platforms, not products".	Meet Veldmark: Four Brands, One Platform to Build
Yardstick / Yardstick Relay	Tradeyard's PHP monolith on MySQL, in service since 2006 and due to be decommissioned at the end of 2029; Yardstick Relay is the translator that reads its tables through change data capture and emits contract events.	Changing the Engine in Flight: Modernizing a Live Marketplace, Bounded Contexts for Marketplaces: Where One Listing Becomes Many Models
Veldmark ID	The group identity approved in June 2027: one login across brands with brand-specific profiles, rolled out from 2028.	Who Is on the Other Side? Identity and Verification
Listing event contract v1	The group's published listing event stream, live 1 July 2027 on Kafka with a schema registry: a stream of domain events and a compacted topic of current public state, keyed by group listing ID.	Every Change Is an Event: Event-Driven Listing Architecture

Term	Definition	Chapter
Ranking policy v1	Veldmark's versioned ranking configuration: top-listing slots only above a relevance floor, at most four paid and three same-seller listings in the first twenty positions, and exploration positions for new listings.	Ranking for Two Sides: Relevance, Freshness and Paid Visibility
Architecture advice forum	Veldmark's weekly, open forum, set up in April 2027, where cross-brand decisions get advice in public and are recorded as decision records.	Who Owns What: Organizing Marketplace Engineering

Alphabetical Index

- **A** — Account · Account takeover (ATO) · ACL · Ad load · Ad server · Ad stack · Add-on · ADR · Advice process · Agent-to-marketplace interface · Agentic search · Aggregate · Aggregation (search) · AI gateway · AI governance for marketplaces · AI listing assistant · Alert fan-out · Alert hold · Alert latency · Alert-to-contact rate · ANN · Answer engine · Anti-corruption layer (ACL) · Apache Iceberg · APP fraud · Appeal · Appeal overturn rate · Approximate nearest neighbor (ANN) · Architecture advice forum · Architecture decision record (ADR) · Architecture governance · ARPA · ARPR · Asynchronous detectors · At-least-once delivery · ATO · Attractor · Attribute · Attribute definition · Attribute dictionary · Attribute extraction (query) · The Attribute Junkyard · Attribute provenance · Attribution · Authenticator assurance · Authorized push payment (APP) fraud · Autocomplete · Automated valuation model (AVM) · Autonomy level · Average revenue per advertiser (ARPA) · AVM
- **B** — B2B · B2C · Backend-for-frontend (BFF) · Backfill · Backward compatibility · BankID · BFF · Billable lead · Billing by Guesswork · Blocking · Bot management · Bounded context · Bounded staleness · Bounding box · Branch by abstraction · Brand · Brand migration · Bulk feed · Bump · Business capability · Business ladder · Buyer protection
- **C** — C2C · Canary · Candidate generation · Canonical entity · Canonical property · Canonical tree with country views · Canonical unit · Canonical value · Capability

map · Carryover · Carve-out · Case · Case management · Category migration · Category prediction · Category schema · The Category Straitjacket · Category taxonomy · CDC · CDN · Cell (liquidity cell) · Centralization pendulum · Change data capture (CDC) · Change event · Chargeback (card) · Chargeback (platform cost) · Chicken-and-egg problem · Claim · Classification tree · Classifieds · Client-side rendering (CSR) · CLS · Cluster randomization · CMP · Cold start · Collaborative filtering · Command gateway · Common Ground · Compliance by design · Compliance by project · Complicated-subsystem team · Configuration over code · Conformist · Consent management platform (CMP) · Consent or contextual · Consent or pay · Consolidation sequence · Consumer registry · Contact · Contact rate · Content delivery network (CDN) · Content-based recommendation · Context map · Control · Converged product platform · Conversation · Conway's law · Copy kinds · The Copy-Paste Country · Copy-paste market entry · Core domain · Core Web Vitals · Cost per lead · Cost per search · Counter-metric · Country overlay · CPC · CPM · CQRS · Crawl budget · CRM (customer relationship management) · Crypto-shredding · CSR · Cumulative Layout Shift (CLS) · CUPED · Customer–supplier

- **D** — DAC7 · Data contract · Data mesh · Data platform · Data product · Data warehouse · Dead-letter queue (DLQ) · Deal-Shaped Architecture · Dealer management system (DMS) · Decision record · Decision record (moderation) · Decision rights · Decision service · Decision tiers (inform, order, gate, commit) · Dedicated isolation · Default sort · Degradation mode · Delivery evidence · Delivery reconciliation · Delta feed · Demand-side platform (DSP) · Demotion · Dense retrieval · Depth product · Derivative (rendition) · Differentiation (heat map) · Digest · Digital Services Act (DSA) · Dis-synergy · Disintermediation · Dispute · DLQ · DMCC Act · DMS · Documented fact · Domain event · DPE · Drawn polygon · Drift · DSA · DSA Transparency Database · DSP · Duplicate detection · Duplicate types · Duplication (heat map) · Dynamic rendering
- **E** — EAV · eCPM · Effect · Effect projector · eIDAS 2.0 · Enabling team · Endpoint · Energy performance certificate (DPE) · Enforcement action · Enforcement overlay · Enforcement point · Enforcement propagation · Enrichment · Entitlement · Entitlement ledger · Entity reference · Entity resolution (ER) · Entity–attribute–value (EAV) · Erasure propagation · Error budget · Escrow-like hold · EU AI Act · EU Digital Identity Wallet · Evaluation harness · Event interception · Event notification · Event sourcing · Event-carried state transfer · Evidence · Excluded seller · EXIF stripping · Exposure (ranking) · Exposure-weighted prevalence · Extension point · External key · External system of record · External versioning
- **F** — Facet · Faceted classification · Failure pattern · Fair housing rules · Fast Mail for Fraudsters · Feature parity trap · Feature store · Featured slot · Federation (platform model) · Feed contract · Feed Roulette · Fellegi–Sunter · Field collapsing · Field ownership · FinOps · First-party audience · The Forced Fit · Fork per country · Forward compatibility · Fraud signal · Freshness SLA · Full compatibility · Full snapshot

- **G** — Gazetteer · GBDT · GDPR · General Data Protection Regulation (GDPR) · Generic subdomain · Geo clustering · Geo experiment · Geo index · Geocoding · Geohash · Ghost Inventory · Ghost share · Ghost-alert rate · Global component · GMV · Governed metric · Gross merchandise value (GMV) · Group-urgent change · Guaranteed exposure · Guardrail metric
- **H** — H3 · Handover location · Header bidding · Hearthly · Heat map · Highlight · HNSW · Hold and release · Horizontal marketplace · Hosting service · HSN/TSN · Hybrid attribute storage · Hybrid retrieval · Hydration
- **I** — Idempotency · Idempotent consumer · Identity · Identity assurance · Identity graph · Identity level · Illustration · Image service · Index · Index alias · Industry practice · INFORM Act · Ingestion channel · Ingestion record · INP · Insertion fee · Insertion flow · Insertion latency · Instance attribute (aspect) · Integration options · Integration principal · Interaction to Next Paint (INP) · Interference · Interleaving · Internal complaint · Internal market · Interpretation · Invariant · Isochrone · Isochrone search · Item location · IVF
- **J** — Journey SLO
- **K** — Keybridge · KYB (know your business) · KYC (know your customer)
- **L** — Lakehouse · LambdaMART · Landing page · Landing page generation · Large language model (LLM) · Largest Contentful Paint (LCP) · LCP · Lead · Lead attribution · Lead contract · Lead ledger · Lead management for pros · Lead qualification · Lead routing · Lead-generation marketplace · Leakage · Learning to rank (LTR) · Liquidity · Liquidity metrics · Listing · Listing aggregate · Listing event · Listing event contract v1 · Listing ID · Listing lifecycle · Listing marketplace · Listing quality score · Listing system of record (SoR) · Listing version · LLM · LLM query compiler · Location precision · Lock (field) · Logical isolation · LTR
- **M** — Managed marketplace · Map search · Market · Marketplace fitness function · Marketplace payments · Marketplace profile · Marketplace team topology · Mass-change guard · Master (media) · Match set · Material change · Matrix factorization · Maturity (heat map) · MCP · Media asset · Media pipeline · Media provenance · Merchant of record · Message · Message safety gate · Metrics layer · Migration stage gate · ML platform · MLS · Model Context Protocol (MCP) · Model governance · Moderation as an Afterthought · Moderation pipeline · Monetization model · Monotonic reads · Motorvane · Multi-diffusion · Multi-homing · Multiple-listing service (MLS) · Multiplier Creep
- **N** — Named area · Navigation tree · Network effect · Neutrality controls · Normalization · Notice and action · Notify on entry, retract on exit · Null and low-result recovery
- **O** — Obligation · Omnibus Directive · Online controlled experiment · Online platform · Open host service · OpenImmo · Optional attribute · Optionality · Order state machine · Organization hierarchy · Original (media) · Original value · Outcome record · Overall evaluation criterion
- **P** — P2B Regulation · Package · Paid visibility · Paid-organic blending · Pairwise classifier · Parallel run · Partitioned isolation · Partner API · Partnership (context

relationship) · Passkey · Pay at close · Pay per lead · Payment service provider (PSP) · Payment-link phishing · Payout · PDQ · Perceptual hash · Percolation · Place ID · Platform as product · Platform funding models · Platform latency · Platform models · Platform tax · Platform team · Point-in-time correctness · Policy (listing rule) · Policy (trust and safety) · Policy registry · Polyhierarchy · Portfolio model · Position bias · Post-moderation · Pre-hijacking · Pre-moderation · Prebid · Precise location · Precision · Prevalence · Price structure · Price-drop alert · Pricing engine · Primary metric · Prior price rule · Private seller · Pro SaaS · Pro tooling · Probabilistic record linkage · Product (catalog item) · Product attribute (aspect) · Product catalog · Professional seller · Professional subscription · Profile · Proxy label · PSP · Public ID · Public point · Publication pipeline · Publication state · Published language · Purchase scam

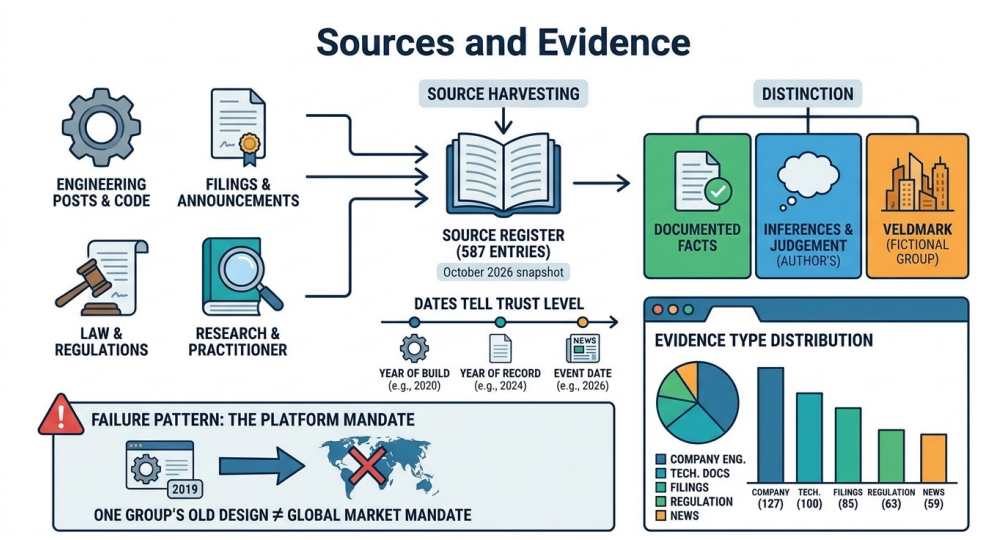
- **Q** — Query categorization · Query suggestions · Query tagging · Query understanding · Quota
- **R** — Random sampling · Ranking · Ranking policy · Ranking policy v1 · Ranking transparency · Read model · Read-only warm standby · Read-your-writes · Rebuild drill · Recall · Reciprocal rank fusion (RRF) · Recommendation · Recommendation (evidence label) · Recommended attribute · Record linkage · Redirect registry · Reference data · Reference-architecture inference · Reindexing · Release rings · Relevance floor · Relevance model · Relevance parity · Relevant activities (DAC7) · Relisting · Removal latency · Removal priority · Rendering modes · Rendition · Replay · Reporting platform operator · Representative rule · Request key · Required attribute · Residuality theory · Residue · RESO · RESO Data Dictionary · Retail media · Reverse geocoding · RFC 9457 (Problem Details) · Risk gate · Risk score · Risk scoring service · Risk-tiered moderation · Run report
- **S** — S2 · Sample ratio mismatch (SRM) · Saved search · Schema compiler · Schema evolution · Schema registry · Search as System of Record · Search contract · Search document · Search index design · Seeded offset · Seeding · Seeing Double · Sell-through · Seller account model · Seller location · Semantic layer · Send-time check · SEO architecture · Separability · Separability drill · Sequential testing · Server-side rendering (SSR) · Service area · Service level indicator (SLI) · Service level objective (SLO) · Serving-time filter · Set-and-Forget AI · Shadow traffic · Share of voice · Shared kernel · Shared-versus-specific decision · Showback · Similar listings · SLA · SLI · SLO · Sold-to-gone latency · SoR · Sort order · Source latency · Sponsored listing · SRM · SSP · SSR · Staleness · Staleness budget · Statement of reasons · Step-up friction · Strangler fig · Stream-aligned team · Streaming SSR · Stressor · Stressor analysis · Strict tree · Structured data · Structured-output compiler · Supply-side platform (SSP) · Supporting subdomain · Switchback experiment · Synchronous gate · Synergy case · Synergy ledger · System of record
- **T** — T&S · Take rate · Taxonomy versioning · TC string · TCF · Team Topologies · Tenancy model · Tenant · Tenant exit path · Tenant isolation · Time to action · Time to first contact · Tombstone · Top slot · Tracking plan · Trader · Trader status · Trader traceability · Tradeyard · Tradeyard Safe Deal · Training–serving skew ·

Transactional marketplace · Transactional outbox · Transactional pivot · Transfer service provider · Transitional services agreement (TSA) · Translation layer (CDC translator) · Transparency and Consent Framework (TCF) · Trust and safety (T&S) · Trusted flagger · TSA · Two-sided market · Two-sided randomization

- **U** — Ubiquitous language · UCPD · UK Online Safety Act · Underlying entity · Unfair Commercial Practices Directive (UCPD) · Unit of charge · Unpublish propagation · Update storm · URL and canonical service · URL migration
- **V** — Validator library · Value stream · Value-added services (VAS) · Valuo · Vanity Liquidity · Variance reduction · VAS · Vehicle identification number (VIN) · Veldmark Ads · Veldmark Group · Veldmark ID · Verification of Payee · Verified badge · Version-gated apply · Vertical marketplace · Vertical requirement profile · Very large online platform (VLOP) · Viewability · Viewport · VIN · VLOP
- **W** — Warn in context · Watcher · Wide table
- **Y** — Yardstick · Yardstick Relay
- **Z** — Zombie listing

The templates that turn many of these terms into working documents (the capability map, the vertical requirement profile, the listing lifecycle state table, the feed contract, the entitlement matrix, the listing event spec) are collected in Field Kit: Templates for the Marketplace Architect, and the sources behind the definitions, with their evidence type and publication date, are listed by topic and chapter in Bibliography: Sources and Evidence.

Bibliography: Sources and Evidence



IN THIS SECTION, YOU WILL: Find where the book's evidence comes from: each source once, with its year, its evidence type and the chapters that rely on it, findable by topic or by chapter.

KEY POINTS:

- **Sources are listed because they were cited.** Every entry was harvested from a page's **Sources and Evidence** section, and its **Cited in** line names those pages. Nothing is added for completeness, and a listing does not mean every page of a long document was read.
- **Dates and evidence types tell you how far to trust a claim.** A company's engineering post records what it said it built in that year, a filing records reported numbers, a regulation records a requirement and trade press records an event. Weigh each claim by the kind and age of the source behind it.
- **Keep documented facts, inferences and Veldmark distinct.** The sources support the documented facts. Reference architectures for real companies are labeled inferences, the recommendations are my judgement, and Veldmark, the marketplace group the book follows, is fictional: no source here describes it.

FAILURE PATTERNS THIS SECTION ADDRESSES:

- **The Forced Fit** — company engineering posts are the largest evidence type on this page, and each describes one group's platform at one date; reading the year and the citing context keeps a published design from becoming a mandate for markets it was not built for.

This page lists the external sources the chapters cite. The research was conducted in **October 2026**, and most entries were consulted then; where a chapter recorded an access date, a version or a "last updated" date, the entry keeps it. Some company pages blocked automated access during the research, and a few facts are known only through **secondary summaries** (a trade-press article or a third-party digest of an engineering post); the entries say so where the citing chapter did. Ownership and brand names follow the sources as of 2026: Schibsted is now Vend, eBay Kleinanzeigen is now Kleinanzeigen, and the former eBay Classifieds Group is part of Adevinta.

How to Read This Register

R marks a consulted source; the numbers identify, they do not rank. The register holds **587 sources** harvested from the **Sources and Evidence** sections of 55 pages, deduplicated by normalized URL (scheme, `www.`, trailing slash, fragment, `.html` suffix and arXiv version ignored); where chapters cited the same legal text or page under two URLs, the entry keeps both. Identifiers were assigned in topic order for this first edition and stay stable: **new sources append from the next free R-number and are never renumbered**, and a removed source's number is retired rather than reused, so a reference to an identifier keeps meaning the same document.

Each entry gives the identifier, the linked title, the author or publisher, the **publication year as the citing chapter gave it** ("n.d." where the source carries no date, with access or update dates kept where the chapter recorded them), one clause on what the source supports, the evidence type and the chapters that cite it. Company engineering evidence carries its year because old posts may describe systems that have since been replaced; a post more than about three years old is a record of that date, not of today. Where chapters cited one source with slightly different metadata, the entry keeps the wording most chapters used, and where they gave different years it keeps the most specific citation.

Evidence type	Sources	What it is	How to read it
Company engineering	127	Engineering blog posts, conference talks and public repositories by a marketplace or technology company	Documents what the company says it built at that date; it may describe systems since replaced, and it is silent on what was not published
Company filing or announcement	85	Annual reports, regulatory filings, results releases, press releases and corporate announcements	Authoritative on the company's reported numbers and events at their date; written for investors and the press, not as design documentation
Company documentation	38	Help-center pages, API and feed documentation, price lists and archived company pages	Describes the product or interface as offered at the time it was read; living pages change without notice
Technical documentation	100	Documentation of search engines, databases, clouds, payment platforms, browsers and open-source projects	Authoritative on what the product does; silent on fit, cost of operation and alternatives
Standard or specification	25	RFCs, schema.org, RESO, ISO and identity, advertising and industry standards	Authoritative on the specification; an architecture applies it, and versions matter
Law, regulator or official source	63	Legal texts, regulator and competition-authority publications, government guidance and official datasets	Authoritative on the official position at its date; interpretation for a specific business belongs to its lawyers

Evidence type	Sources	What it is	How to read it
Research	34	Peer-reviewed papers, preprints and industry research reports	Findings hold within the study's method, data and date; company-authored papers describe that company's setting
Book	7	Books and book chapters	Durable concepts and vocabulary; editions matter
Practitioner	34	Articles, essays, case studies and secondary summaries by practitioners, consultancies, investors and vendors	Describes common practice or interprets primary sources; secondary, so follow it to the primary source where one exists
News and trade press	59	News reporting and marketplace trade press	Records events and claims at a date; not independent verification of a company's statements
Encyclopedia	14	Wikipedia entries, used for founding dates and ownership history	Tertiary; the chapters use it only for dates and ownership, and the primary source wins where one is cited
Author's working model	1	The author's generic domain models (Productscape), digested in the book's research notes	Not public evidence and not a description of any company; it shapes the reference architectures, which are labeled as the author's judgement

The chapters also rely on the book's own research briefs, written in October 2026 to collect and check the company, regulatory and practice facts the chapters use. Those

briefs are working notes, not public evidence, and are not listed; wherever a brief's fact matters, the chapter cites the public source behind it, and that source is listed below. Plain-language definitions of the terms used in the entries are in Glossary: The Language of Listing Systems.

Topic Index

Identifiers run in topic order, so each topic is one contiguous block. A source cited by chapters across several topics sits under the topic it serves best; its **Cited in** line lists every chapter that uses it. The six company sections collect what each group or marketplace published or what was reported about it; generic engine documentation and research papers sit under the technical topics they inform, even when a company wrote them.

Marketplace economics and business models (R1–R8, 8 sources). Two-sided market theory, network effects, marketplace metrics and the history of classifieds as a business. The most widely cited is R3, *13 Metrics for Marketplace Companies*, cited in 4 pages.

Company engineering: Adevinta and Schibsted (R9–R75, 67 sources). Adevinta, its former parent Schibsted (now Vend) and their brands, including FINN, leboncoin, Subito and coches.net: tech-blog posts, annual and strategic reports, transaction announcements and help-center pages. The most widely cited is R10, *Strategic Report 2025, Aurelia UK Feederco Ltd*, cited in 13 pages.

Company engineering: OLX and Prosus (R76–R133, 58 sources). OLX and its owner Prosus (and Naspers): the OLX tech blog, group results, acquisitions and disposals, and press coverage. The most widely cited is R79, *Correlation lied to us: rethinking product impact with causal inference*, cited in 9 pages.

Company engineering: eBay Classifieds, Kleinanzeigen and Marktplaats (R134–R174, 41 sources). The former eBay Classifieds Group and its brands, chiefly Kleinanzeigen and Marktplaats: engineering posts, open-source repositories, archived pages, acquisition records and payment launches. The most widely cited is R135, *Acquisition of eBay Classifieds Group complete*, cited in 7 pages.

Company engineering: AVIV and property portals (R175–R226, 52 sources). AVIV Group and other property portals, including Rightmove, Scout24, Hemnet, REA Group, idealista, Homegate and CoStar: results, datafeed specifications, engineering posts and trade-press history. The most widely cited is R175, *Rightmove FY2025 results*, cited in 14 pages.

Company engineering: automotive marketplaces (R227–R262, 36 sources). Automotive marketplaces, chiefly mobile.de, AutoScout24, Auto Trader UK, CarGurus and Cars.com: seller and search API documentation, results, engineering posts and ownership history. The most widely cited is R228, *Full year results for the year ended 31 March 2026*, cited in 16 pages.

Company engineering: other marketplaces (R263–R327, 65 sources). Other marketplaces the chapters use as comparisons: Vinted, Airbnb, Etsy, eBay, Mercari, Zillow, Amazon, Meta Marketplace, Indeed and Uber. The most widely cited is R269, *Zillow Group Form 10-K for fiscal 2025*, cited in 10 pages.

Search, ranking and recommendations (R328–R390, 63 sources). Search engines, query understanding, learning to rank, recommendations, geo and map search, and saved-search matching: engine documentation and ranking research. The most widely cited is R329, *Applying Embedding-Based Retrieval to Airbnb Search*, cited in 6 pages.

Trust, safety and fraud (R391–R406, 16 sources). Fraud, scams, spam, identity assurance and account security: industry reports, identity standards and safety research. The most widely cited is R392, *Scam ads slipping through the net too easily on second-hand marketplaces*, cited in 5 pages.

Regulation and standards (R407–R489, 83 sources). Laws, regulators, official datasets and technical standards: EU, UK and US law, competition decisions, RFCs, schema.org, RESO, OpenImmo, address registries and advertising standards. The most widely cited is R412, *Regulation (EU) 2022/2065 (Digital Services Act)*, cited in 16 pages.

Platform, data, ML and experimentation (R490–R528, 39 sources). Event streaming, data platforms, media processing, payments platforms, experimentation and machine learning: documentation and research. The most widely cited is R490, *Outbox Event Router*, cited in 3 pages.

Web, SEO and reliability (R529–R565, 37 sources). Search-engine optimization, structured data, feeds to Google, web performance, crawler control, site reliability and incident reports. The most widely cited is R529, *Job posting structured data*, cited in 7 pages.

Architecture method and organization (R566–R587, 22 sources). Domain-driven design, capability modeling, Team Topologies, architecture decision records, migration patterns and long-term architecture method. The most widely cited is R566, *Productscape domain models for general-listings and real-estate marketplaces*, cited in 12 pages.

Marketplace economics and business models

- **R1** · The spawn of craigslist — Andrew Parker, 2010. Unbundling of horizontal classifieds into verticals. *Practitioner.* Cited in: Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy, Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics, From Newspaper Columns to Portfolios: How the Classifieds Industry Evolved.
- **R2** · All Markets Are Not Created Equal — Bill Gurley, 2012. "If you bring the cash, you are king": the economics of being in the payment flow. *Practitioner.* Cited in:

Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy, Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics, How Listing Marketplaces Make Money.

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- **R4** · Platform Competition in Two-Sided Markets — Jean-Charles Rochet and Jean Tirole, *Journal of the European Economic Association*, 2003. Getting both sides on board; price structure. *Research.* Cited in: Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics, How Listing Marketplaces Make Money.
- **R5** · The Network Effects Bible — NFX, 2019 (updated 2024). Asymptotic network effects; harder side; multi-tenanting. *Practitioner.* Cited in: Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics.
- **R6** · Network Effects Aren't Enough — Andrei Hagiu and Simon Rothman, *Harvard Business Review*, 2016. Traps beyond network effects, including disintermediation (summarized; body paywalled). *Practitioner.* Cited in: Liquidity Is the Product: Network Effects, Local Markets and Marketplace Metrics.
- **R7** · New research looks at Craigslist's damaging blow to local newspapers — NYU Stern, 2013. Seamans and Zhu's findings on classified-ad rates. *Research.* Cited in: From Newspaper Columns to Portfolios: How the Classifieds Industry Evolved.
- **R8** · Craigslist — Wikipedia, accessed 2026. Founding dates and ownership history (secondary sources). *Encyclopedia.* Cited in: From Newspaper Columns to Portfolios: How the Classifieds Industry Evolved.

Company engineering: Adevinta and Schibsted

- **R9** · Adevinta Annual Report 2023 — Adevinta, 2024. The plan to turn the leboncoin and Kleinanzeigen platforms into "a unified global platform", and the re-brand of eBay Kleinanzeigen as Kleinanzeigen. *Company filing or announcement.* Cited in: Introduction: Architecture for Liquidity, Unlisted: Fifteen Ways Listing Systems Go Wrong, From Newspaper Columns to Portfolios: How the Classifieds Industry Evolved, Same Platform, Different Physics: How Listing Verticals Differ, Trust and Safety as a Platform, Beyond the Contact: Payments, Shipping and Buyer Protection, After the Deal: Consolidating Platforms After Acquisitions, Who Owns What: Organizing Marketplace Engineering, Built to Be Replaced: Evolving a Marketplace Platform Over a Decade, Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform, mobile.de, Auto Trader and AutoScout24: Lessons from Automotive Marketplaces.

- **R10** · Strategic Report 2025, Aurelia UK Feederco Ltd — Adevinta, 2026. The move to "a decentralised, standalone operating model". *Company filing or announcement.* Cited in: Introduction: Architecture for Liquidity, Unlisted: Fifteen Ways Listing Systems Go Wrong, Classifieds, Leads, Transactions and Managed Models: A Marketplace Taxonomy, From Newspaper Columns to Portfolios: How the Classifieds Industry Evolved, Map What the Marketplace Does Before What Runs It, Same Platform, Different Physics: How Listing Verticals Differ, Beyond the Contact: Payments, Shipping and Buyer Protection, Data Platform for Marketplaces: Events, Lakehouse and Data Products, After the Deal: Consolidating Platforms After Acquisitions, Who Owns What: Organizing Marketplace Engineering, Changing the Engine in Flight: Modernizing a Live Marketplace, Built to Be Replaced: Evolving a Marketplace Platform Over a Decade, Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform.
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- Who Owns What: Organizing Marketplace Engineering (16): R9, R10, R48, R58, R59, R86, R103, R108, R157, R158, R237, R278, R571, R579, R580, R581.
- Changing the Engine in Flight: Modernizing a Live Marketplace (14): R10, R44, R134, R139, R144, R146, R204, R237, R279, R297, R564, R582, R583, R584.

- AI Across the Listing Lifecycle (22): R28, R29, R45, R80, R93, R100, R109, R110, R111, R159, R175, R205, R287, R398, R435, R480, R481, R482, R487, R527, R528, R565.
- Built to Be Replaced: Evolving a Marketplace Platform Over a Decade (14): R9, R10, R11, R17, R21, R83, R135, R247, R267, R488, R566, R585, R586, R587.
- Adevinta: From Schibsted's Copy-Paste Model to a Portfolio Platform (50): R9, R10, R11, R12, R13, R16, R17, R18, R19, R22, R24, R25, R26, R28, R41, R42, R43, R47, R48, R49, R50, R52, R55, R59, R60, R61, R62, R63, R64, R65, R66, R67, R68, R69, R70, R71, R72, R73, R135, R136, R137, R138, R148, R149, R153, R160, R161, R162, R186, R235.
- OLX Group: Regional Platforms, Verticalization and Retreat (49): R76, R77, R78, R79, R80, R81, R82, R83, R85, R86, R87, R88, R89, R90, R91, R92, R93, R94, R96, R103, R104, R105, R106, R107, R109, R110, R111, R112, R113, R114, R115, R116, R117, R118, R119, R120, R121, R122, R123, R124, R125, R126, R127, R128, R129, R130, R131, R132, R133.
- eBay Classifieds Group: Shared Platforms, Local Brands and a Sale (37): R20, R53, R74, R75, R134, R135, R136, R138, R139, R140, R141, R142, R144, R145, R146, R147, R148, R152, R153, R154, R155, R157, R158, R160, R162, R163, R164, R165, R166, R167, R168, R169, R170, R171, R172, R173, R174.
- AVIV Group: Building a European Real-Estate Platform from Acquisitions (28): R179, R184, R185, R189, R190, R191, R195, R202, R203, R206, R207, R208, R209, R210, R211, R212, R213, R214, R215, R216, R217, R218, R219, R220, R221, R222, R417, R422.
- Zillow, Rightmove, Idealista and REA: What Property Portals Teach (27): R175, R176, R177, R180, R183, R192, R193, R196, R198, R223, R224, R225, R226, R267, R268, R269, R280, R305, R322, R323, R324, R325, R326, R327, R390, R442, R489.
- mobile.de, Auto Trader and AutoScout24: Lessons from Automotive Marketplaces (37): R9, R83, R91, R106, R135, R227, R228, R229, R230, R231, R232, R234, R235, R236, R237, R238, R239, R240, R241, R242, R245, R248, R249, R250, R251, R252, R253, R254, R255, R256, R257, R258, R259, R260, R261, R262, R416.
- Field Kit: Templates for the Marketplace Architect (12): R20, R139, R264, R284, R347, R409, R412, R491, R500, R570, R580, R581.

Veldmark is fictional; architectural recommendations and labeled inferences are the author's judgement.